

Publications by Model: ECCO

Total Publications: 2480 (sorted newest to oldest)

1. Attempts to Determine the Velocity Variations in Ocean Geostrophic Currents Based on GRACE Solutions

Authors: Zofia Rzepecka (Faculty of Geoengineering, University of Warmia and Mazury in Olsztyn, Oczapowskiego St. 2, 10-719 Olsztyn, Poland); Monika Birylo (Faculty of Geoengineering, University of Warmia and Mazury in Olsztyn, Oczapowskiego St. 2, 10-719 Olsztyn, Poland)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs17010076

URL: <https://doi.org/10.3390/rs17010076>

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References: 40

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Abstract: The paper shows the idea of using GRACE solutions to determine the velocity of the Atlantic geostrophic currents at a depth of 2000 m, where the assumptions of geostrophic motion are met. The relationship between the water velocity variations and the OBP, determined based on the L2 GRACE solutions, is shown. The velocities obtained were compared with the results of a direct ocean current measurement program, the RAPID AMOC, carried out in the Atlantic area. The upper North Atlantic Deep Water layer transport was used for comparisons performed. An analysis of the obtained results, their summary, and discussion are shown. The research covers the period from November 2004 to March 2022. As a result of flow calculations through the parallel, a satisfactory agreement was obtained between the GRACE and RAPID AMOC fitted functions. The correlation between the GRACE and RAPID AMOC time series observations exceeded the significance level.

2. Upstream and Downstream Wind-Stress Forcing of Seasonal Variability of Luzon Strait Deep Overflow Transport

Authors: Lei Chen (Ocean University of China); Jiayan Yang (Woods Hole Oceanographic Institution); Lixin Wu (Ocean University of China, Laoshan Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024gl112817

URL: <https://doi.org/10.1029/2024gl112817>

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References: 47

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Keywords: N/A

Funding: National Natural Science Foundation of China

License: Creative Commons

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Abstract: The Luzon Strait Deep Overflow (LZDO) transports deep-water masses from the Pacific Ocean (upstream) to the South China Sea (SCS, downstream), playing a vital role in shaping the hydrographic and biogeochemical structure and regulating abyssal circulation within the SCS. Recent studies suggest that the seasonal variability of LZDO transport is primarily wind-driven; however, the specific mechanisms and key wind-forcing areas remain unclear. In this study, we use numerical model experiments to demonstrate that the seasonal variability of the LZDO is driven by wind stress both within the downstream SCS and upstream Pacific Ocean. This finding contrasts with previous views that considered the LZDO hydraulically controlled at the Luzon Strait sill and thus influenced solely by the upstream Pacific Ocean forcing. Additionally, we identify that the LZDO is particularly sensitive to wind forcing in specific North Pacific regions where geostrophic contours connect to the Luzon Strait due to topographic effects.

3. Emerging influence of the Australian Monsoon on Indian Ocean interannual variability in a warming climate

Authors: Mengyan Chen; Matthew Collins; Jin-Yi Yu; Xin Wang; Lei Zhang; Chi-Yung Tam

Journal: npj Climate and Atmospheric Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41612-024-00879-9

URL: <https://doi.org/10.1038/s41612-024-00879-9>

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Abstract: The Indian Ocean Dipole (IOD) and Tripole (IOT) represent primary modes of interannual variability in the Indian Ocean, impacting both regional and global climate. Unlike the IOD, which is closely related to the El Niño-Southern Oscillation (ENSO), our findings unveil a substantial influence of the Australian Monsoon on the IOT. An anomalously strong Monsoon induces local sea surface temperature (SST) variations via the wind-evaporation-SST mechanism, triggering atmospheric circulation anomalies in the eastern Indian Ocean. These circulation changes lead to changes in oceanic heat transport, facilitating the formation of the IOT. Our analysis reveals a strengthening connection between the Australian Monsoon and the IOT in recent decades, with a projected further strengthening under global warming. This contrasts with the diminished coupling between ENSO and IOD in recent decades from observations and model projections, illustrating evolving Indian Ocean dynamics under the warming climate.

4. Record-low Antarctic sea ice in 2023 increased ocean heat loss and storms

Authors: Simon A. Josey; Andrew J. S. Meijers; Adam T. Blaker; Jeremy P. Grist; Jenny Mecking; Holly C. Ayres

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-024-08368-y

URL: <https://doi.org/10.1038/s41586-024-08368-y>

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References: 35

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Language: en

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Funding: N/A

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Open Access: N/A

Abstract: Recent Antarctic sea-ice decline is a substantial source of concern, notably the record low in 2023 (ref. 1). Progress has been made towards establishing the causes of ice loss¹⁻⁵ but uncertainty remains about its consequences for ocean-atmosphere interaction. Resolution of this uncertainty is important as ice decline can substantially alter surface heat loss and thus the ocean and atmosphere⁶. Here we show that the strongest winter 2023 ice-retraction regions provide an important new source of turbulent ocean heat loss to the atmosphere in wintertime. Ice concentration in these regions (located primarily in the Weddell, Bellingshausen and Ross seas) is reduced by up to 80% and is accompanied by an unprecedented doubling of mid-winter ocean heat loss. Also, there is a phase shift in the time of peak heat loss from late April to mid-June, with weaker than normal heat loss in austral autumn. The winter surface-heat-loss intensification is accompanied by substantial changes on both sides of the ocean-atmosphere interface. These include increases in atmospheric-storm frequency and surface-heat-loss-driven dense water formation, although the implications of the densification for broader processes such as Antarctic bottom water formation remain unclear. Our results reveal that the 2023 Antarctic sea-ice loss has substantially modified air-sea interaction in the Southern Ocean and motivate in-depth analysis of the wider climate-system impacts.

5. Global upper ocean dissolved oxygen budget for constraining the biological carbon pump

Authors: Ryohei Yamaguchi; Shinya Kouketsu; Naohiro Kosugi; Masao Ishii

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01886-7

URL: <https://doi.org/10.1038/s43247-024-01886-7>

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Keywords: N/A

Funding: MEXT | Japan Society for the Promotion of Science

License: Creative Commons

Open Access: N/A

Abstract: N/A

6. Water Mass Transformation and Its Relationship With the Overturning Circulation in the Eastern Subpolar North Atlantic

Authors: Yao Fu (Georgia Institute of Technology); M. Susan Lozier (Georgia Institute of Technology); Sudip Majumder (Georgia Institute of Technology); Tillys Petit (National Oceanography Centre)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Keywords: water mass transformation, AMOC, OSNAP

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Open Access: N/A

Abstract: A recent study using the first 21 months of the OSNAP time series revealed that the export of dense waters in the eastern subpolar North Atlantic--as part of the Atlantic Meridional Overturning Circulation (MOC)--can be almost wholly attributed to surface-forced water mass transformation (SFWMT) in the Irminger and Iceland basins, thus suggesting a minor role for other means of transformation, such as diapycnal mixing. To understand whether this result is valid over a period that exceeds the current observational record, we use four different ocean reanalysis products to investigate the relationship between surface buoyancy forcing and dense water production in this region. We also reexplore this relationship with the now available 6-year OSNAP time series. Our analysis finds that although surface transformation in the eastern subpolar gyre dominates the production of deep waters, mixing processes downstream of the Greenland Scotland Ridge are also responsible for the production of waters carried within the AMOC's lower limb both in the observations and reanalyses. Further analysis of the reanalyses shows that SFWMT partly explains MOC interannual variability, the remaining portion can be attributed to basin storage and mixing. Compared to the observations, the reanalyses exhibit stronger MOC variance but comparable SFWMT variance on interannual timescales.

7. Interannual Time-Scale Dynamics of Deep Cross-Equatorial Overturning in the Indian Ocean

Authors: Weiqi Hong (Chinese Academy of Sciences); Gengxin Chen (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jc021740

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References: 63

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Open Access: N/A

Abstract: The Deep Cross-Equatorial Cell (DCEC) is the primary branch of Indian Ocean Meridional Overturning Circulation (MOC) in the tropical Indian Ocean, essential for energy redistribution, water exchange, and diapycnal mixing. However, the mechanisms behind its interannual variability remain limited. This study utilized two reanalysis data sets and a series of ocean model experiments with a Hybrid Coordinate Ocean Model and a Linear Ocean Model to investigate the underlying mechanisms. Model experiments highlight the critical role of direct local wind forcing and eastern boundary waves induced by remote equatorial wind forcing in influencing the DCEC variability. Specifically, through the first mode of baroclinic dynamics, direct wind forcing initiates reverse meridional flow at the DCEC core (around 8 degreesS) in both surface and deep ocean layers, leading to interannual variations of the DCEC. During transitions of climate modes like ENSO and Indian Ocean Dipole from positive to negative phases, both positive and negative DCEC anomalies intensify. In addition to direct local wind forcing, the delayed-time Rossby waves reflected from the eastern boundary excited by the equatorial easterly wind in the previous year make substantial contributions (37.8%). The interplay of faster baroclinic Rossby waves at lower latitudes and slower baroclinic Rossby waves at higher latitudes alters the basin-wide pressure gradient, ultimately amplifying interannual DCEC anomalies in the subsequent year.

8. An Observation-Based Estimate of Atlantic Meridional Freshwater Transport

Authors: Huayi Zheng (Chinese Academy of Sciences); Lijing Cheng (Chinese Academy of Sciences); Feili Li (Xiamen University); Yuying Pan (Chinese Academy of Sciences); Chenyu Zhu (Chinese Academy of Sciences)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: ocean salinity, meridional coherence, Atlantic meridional freshwater transport, ocean freshwater budget

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Open Access: N/A

Abstract: Meridional freshwater transport (MFT) in the Atlantic Ocean (Atlantic meridional freshwater transport (AMFT)) plays a vital role in the Atlantic Ocean circulations, but an accurate estimate of AMFT time series remains challenging. This study uses an indirect approach that combines ocean salinity, surface evaporation and precipitation observations to derive AMFT and its uncertainty by solving the ocean freshwater budget equation. Climatologically, AMFT is southward between 18.5 degreesS and 34.5 degreesS, but northward from 18.5 degreesS to 66.5 degreesN. AMFT also shows substantial inter-annual variability with a clear separation at ~40 degreesN and is more coincident with the Atlantic Meridional Overturning Circulation (AMOC) at 26 degreesN than 47 degreesN across latitudes. The derived time series indicates that throughout the Atlantic Ocean, there is a positive trend in the AMFT from 2004 to 2020, resulting in an AMFT convergence in the tropical Atlantic and an AMFT divergence in the subtropical North Atlantic.

9. Seasonal differences of Wyrтки Jet intraseasonal variabilities

Authors: Guojiao Cao; Tengfei Xu; Zexun Wei

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2024.1517779

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Open Access: N/A

Abstract: This paper examines the intraseasonal variabilities (ISVs) of the Wyrтки Jet in boreal spring and fall and their impacts on the oceanic ISVs along the southern coast of Sumatra-Java Island. The results reveal that the Wyrтки Jet ISVs in spring are significantly stronger than those in fall, with the standard deviation of the 0-30m-averaged zonal current reaching up to 0.25 m/s in spring, while the highest value in fall is only 0.2 m/s. The Wyrтки Jet ISVs are significantly correlated with surface zonal wind anomalies and sea level anomalies (SLAs) in the equatorial Indian Ocean (EIO) at intraseasonal timescale, and are modulated by the propagation of equatorial Kelvin waves. The intraseasonal SLAs along the southern coast of the Sumatra-Java Island are significantly correlated with the Wyrтки Jet ISVs, exhibiting similar seasonal fluctuation characteristics. In spring, the Wyrтки Jet intraseasonal signals initially appear near 75 degreesE at the equator, approximately 10 days before the positive peaks of the intraseasonal SLAs, while in fall, the Wyrтки Jet intraseasonal signals first appear about 15 days before the peaks near 60 degreesE at the equator, which is relatively further west compared to signals in spring. In addition, the composite Wyrтки Jet ISVs in spring are approximately 0.2 m/s stronger than those in fall. The enhanced ISVs of sea surface zonal wind forcing and Wyrтки Jet in spring, relative to those in fall, indicate that the seasonality in the intraseasonal SLAs along the southern coast of Sumatra-Java is attributable to the combined effects of surface wind forcing and current fields.

10. An improved model for air-sea exchange of elemental mercury in MITgcm-ECCOV4-Hg: the role of surfactants and waves

Authors: Ling Li; Peipei Wu; Peng Zhang; Shaojian Huang; Yanxu Zhang

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-8683-2024

URL: <https://doi.org/10.5194/gmd-17-8683-2024>

Publication Date: 2024-12-10

Volume/Issue: Volume 17, Issue 23

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Language: en

Subjects: N/A

Keywords: Mercury|Elemental mercury

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Open Access: N/A

Abstract: . The air-sea exchange of elemental mercury (Hg⁰) plays an important role in the global Hg cycle. Existing air-sea exchange models for Hg⁰ have not considered the impact of sea surfactants and wave breaking on the exchange velocity, leading to insufficient constraints on the flux of Hg⁰. In this study, we have improved the air-sea exchange model of Hg⁰ in the three-dimensional ocean transport model MITgcm (MIT General Circulation Model) by incorporating sea surfactants and wave-breaking processes through parameterization, utilizing the total organic carbon concentration and significant wave height data. The inclusion of these factors results in an increase of 62 %-225 % in the global transfer velocity of Hg⁰ relative to the baseline model. Air-sea exchange flux is increased in mid-latitude to high-latitude regions with high wind and wave-breaking efficiency, while it is reduced by surfactant and concentration change at low latitudes with low wind speeds and in nearshore areas with low wave heights. Compared with previous parameterizations, the updated model demonstrates a stronger dependence of Hg⁰ air-sea exchange velocity on wind speed. Our results also provide a theoretical explanation for the large variances in estimated transfer velocity between different schemes.

11. Exploring the relationship between upper ocean states and the falling ice radiative effects using ECCO product and global climate models

Authors: Jui-Lin F Li; Li-Chiao Wang; Yu-Cian Tsai; Yu-Sung Huang; Wei-Liang Lee; Jonathan H Jiang; Ou Wang; Jia-Yuh Yu; Graeme Stephens; Tzu-Yun Liu

Journal: Environmental Research Communications

Publisher: IOP Publishing

DOI: 10.1088/2515-7620/ad9c1d

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References: 38

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Open Access: N/A

Abstract: This study seeks to explore the relationship between upper ocean current (UOC) anomalies (above 200 meters) and surface wind stress (TAU), focusing on the influence of falling ice (snow) radiative effects (FIREs) over the tropical and subtropical Pacific regions. To achieve this, we conducted sensitivity experiments with the CESM1-CAM5 model, using the Coupled Model Intercomparison Project phase 5 (CMIP5) historical run setting, with FIREs turned off (NOS) and on (SON). The monthly ocean current and temperature of the ocean reanalysis from the NASA Estimating the Circulation and Climate of the Ocean (ECCO) project, which assimilates satellite and in situ measurements, serves as a reference for this study. The spatial patterns of the horizontal UOC anomaly (UOCA) differences between the NOS and SON experiments show a strong correlation with the TAU patterns across the studied domain. When compared to the experiments with NOS, the experiments with SON demonstrate an improvement in the annual mean UOC. The improvement in UOC can be attributed to the enhancements in TAU, specifically in the trade-wind regions. The enhancements in TAU play a significant role in influencing the UOCA patterns and contribute to the overall improvement observed in the experiments with SON. In SON, the average absolute bias of simulated UOCA over the study area is reduced by up to 30% compared to NOS against ECCO. Although biases in UOC are present over the southern and northern flanks of the equator in SON, the improvements in annual mean ocean currents are closely related to enhancements in TAU driven by the inclusion of FIREs. Notably, stronger ocean current magnitudes correspond to more significant changes in TAU due to Coriolis forces. When evaluating the ensemble mean absolute biases of UOC from the CMIP5 models, similarities to NOS, however, are limited over the South Pacific region.

12. Evaluation of MITgcm-based ocean reanalyses for the Southern Ocean

Authors: Yoshihiro Nakayama; Alena Malyarenko; Hong Zhang; Ou Wang; Matthis Auger; Yafei Nie; Ian Fenty; Matthew Mazloff; Armin Köhl; Dimitris Menemenlis

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-8613-2024

URL: <https://doi.org/10.5194/gmd-17-8613-2024>

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Open Access: N/A

Abstract: . Global- and basin-scale ocean reanalyses are becoming easily accessible and are utilized widely to study the Southern Ocean. However, such ocean reanalyses are optimized to achieve the best model-data agreement for their entire model domains and their ability to simulate the Southern Ocean requires investigation. Here, we compare several ocean reanalyses (ECCOV4r5, ECCO LLC270, B-SOSE, and GECCO3) based on the Massachusetts Institute of Technology General Circulation Model (MITgcm) for the Southern Ocean. For the open ocean, the simulated time-mean hydrography and ocean circulation are similar to observations. The MITgcm-based ocean reanalyses show Antarctic Circumpolar Current (ACC) levels measuring approximately 149 +/- 11 Sv. The simulated 2 degreesC isotherms are located in positions similar to the ACC and roughly represent the southern extent of the current. Simulated Weddell Gyre and Ross Gyre strengths are 51 +/- 11 and 25 +/- 8 Sv, respectively, which is consistent with observation-based estimates. However, our evaluation finds that the time evolution of the Southern Ocean is not well simulated in these ocean reanalyses. While observations showed little change in open-ocean properties in the Weddell and Ross gyres, all simulations showed larger trends, most of which are excessive warming. For the continental shelf region, all reanalyses are unable to reproduce observed hydrographic features, suggesting that the simulated physics determining on-shelf hydrography and circulation is not well represented. Nevertheless, ocean reanalyses are valuable resources and can be used for generating ocean lateral boundary conditions for regional high-resolution simulations. We recommend that future users of these ocean reanalyses pay extra attention if their studies target open-ocean Southern Ocean temporal changes or on-shelf processes.

13. Impact of Horizontal Resolution on Simulating Eddy-Induced Meridional Heat Transport in the Southeast Indian Ocean

Authors: Xingshang Qian (Shanghai Jiao Tong University); Lei Zhou (Shanghai Jiao Tong University, Southern Marine Science and Engineering Guangdong Laboratory); Jianhuang Qin (Hohai University, Ministry of Natural Resources)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jc021554

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Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China

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Open Access: N/A

Abstract: The eddy-induced meridional heat transport (EMHT) in the Southeast Indian Ocean (SEIO) plays a crucial role in regional air-sea interactions and climate changes, but it is commonly underestimated in widely-known models and reanalysis data sets. This study uses three experiments of the Regional Ocean Modeling System (ROMS) with varying horizontal resolutions to assess their impact on simulating EMHT in the SEIO. The results show that increasing horizontal resolution enhances the simulated EMHT with a stronger seasonal cycle. The higher horizontal resolution experiments show improvements in the association and phase difference between high-frequency temperature and meridional velocity, both of which are the definition of EMHT. According to turbulent diffusion theory, the magnitude of EMHT is directly correlated with eddy kinetic energy (EKE). Both of them show similar vertical and temporal variations as well as spatial patterns. EKE budget analysis suggests that the improvement in EKE is primarily attributed to the enhanced baroclinic energy conversion, which indicates the EKE originated from eddy available potential energy. It is further found that the amplitude and number of mesoscale eddies are notably large in high-resolution experiments, thereby augmenting the EKE and EMHT. Additionally, the seasonal variation in EMHT in each model experiment is tied to the number of mesoscale eddies, rather than other characteristics of them. Thus, this study provides a valuable insight into the enhancing horizontal resolution to improve the simulation of EMHT.

14. Spatiotemporal similarity of relative dispersion in the Gulf of Mexico

Authors: Yinxiang Ma; Yongxiang Huang; Jianyu Hu

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2024.1446297

URL: <https://doi.org/10.3389/fmars.2024.1446297>

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: How a pair of pollutant parcels deviates from each other with an initial separation distance r_0 , known as relative dispersion or Richardson dispersion, is relevant in many circumstances. This study examines the spatiotemporal similarity of the Richardson relative dispersion in the Gulf of Mexico by reanalyzing the Lagrangian trajectory of the surface drifter provided by two famous field experiments, that is, the Grand Lagrangian Deployment and the Lagrangian Submesoscale Experiment. The experimental dispersion curve indicates a critical separation time. When above this critical time, the dispersion shows an asymptotic power law growth independent of the initial separation distance r_0 . Below it, the dispersion curve shows a strong spatiotemporal dependence with two spatiotemporal similarity regimes that can be identified for both experiments by looking at the isoline of the normalized dispersion curve. A new similarity variable is introduced to successfully collapse measured dispersion curves. However, the observed spatiotemporal similarity cannot be reproduced by the submesoscale preserved model. Thus, our results suggest that small-scale fluctuations play a crucial role in the relative dispersion of oceanic flows.

15. Physical oceanographic factors controlling the ocean circulation-induced magnetic field

Authors: David S. Trossman (University of Maryland); Robert H. Tyler (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Helen R. Pillar (University of Texas at Austin)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rsta.2024.0076

URL: <https://doi.org/10.1098/rsta.2024.0076>

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Keywords: N/A

Funding: Division of Ocean Sciences

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Open Access: N/A

Abstract: Oceanic tidal constituents and depth-integrated electrical conductivity (ocean conductivity content, or OCC) extracted from electromagnetic (EM) field data are known to have a strong potential for monitoring ocean heat content, which reflects the Earth's energy imbalance. In comparison to ocean tide models, realistic ocean general circulation models have a greater need to be baroclinic; therefore, both OCC and depth-integrated conductivity-weighted velocity ($T \sigma$) data are required to calculate the ocean circulation-induced magnetic field (OCIMF). Owing to a lack of $T \sigma$ observations, we calculate the OCIMF using an ocean state estimate. There are significant trends in the OCIMF primarily owing to responses in the velocities to external forcings and the warming influence on OCC between 1993 and 2017, particularly in the Southern Ocean. Despite being depth-integrated quantities, OCC and $T \sigma$ (which primarily determine the OCIMF in an idealized EM model) can provide a strong constraint on the baroclinic velocities and ocean mixing parameters when assimilated into an ocean state estimation framework. A hypothetical fleet of full-depth EM-capable floats would therefore help improve the accuracy of the OCIMF computed with an ocean state estimate, which could potentially provide valuable guidance on how to extract the OCIMF from satellite magnetometry observations. This article is part of the theme issue 'Magnetometric remote sensing of Earth and planetary oceans'.

16. Satellite monitoring of long period ocean-induced magnetic field variations

Authors: Christopher C. Finlay (Technical University of Denmark); Jakub Velínský (Charles University); Clemens Kloss (Technical University of Denmark); Rasmus M. Blangsbøll (Technical University of Denmark)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rsta.2024.0077

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Keywords: N/A

Funding: European Space Agency, Ministry of Education, Youth and Sports of the Czech Republic

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Open Access: N/A

Abstract: Satellite magnetic field observations have the potential to provide valuable information on dynamics, heat content and salinity throughout the ocean. Here, we present the expected spatio-temporal characteristics of the ocean-induced magnetic field (OIMF) at satellite altitude on periods of months to decades. We compare these to the characteristics of other sources of Earth's magnetic field, and discuss whether it is feasible for the OIMF to be retrieved and routinely monitored from space. We focus on large length scales (spherical harmonic degrees up to 30) and periods from one month up to 5 years. To characterize the expected ocean signal, we make use of advanced numerical simulations taking high-resolution oceanographic inputs and solve the magnetic induction equation in three dimensions, including galvanic coupling and self-induction effects. We find the time-varying ocean-induced signal dominates over the primary source of the internal field, the core dynamo, at high spherical harmonic degree with the cross-over taking place at degrees 13-19 depending on the considered period. The ionospheric and magnetospheric fields (including their Earth-induced counterparts) have most power on periods shorter than one month and are expected to be mostly zonal in magnetic coordinates at satellite altitude. Based on these findings, we discuss future prospects for isolating and monitoring long period OIMF variations using data collected by present and upcoming magnetic survey satellites. This article is part of the theme issue 'Magnetometric remote sensing of Earth and planetary oceans'.

17. Influence of the Southern Hemisphere Supergyre on Antarctic Intermediate Water Properties in CMIP6 Models

Authors: Ophélie Meuriot (Imperial College London); Camille Lique (Institut de Recherche pour le Développement (IRD)); Yves Plancherel (Imperial College London)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jc021140

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License: CC BY 4.0

Open Access: N/A

Abstract: The supergyre in the Southern Hemisphere is thought to connect the Atlantic, Indian, and Pacific subtropical gyres together. The aim of the study is to investigate whether the supergyre is identifiable in the Coupled Model Intercomparison Project Phase 6 (CMIP6) models and in the Estimating the Circulation and Climate of the Ocean (ECCO) reanalysis and to evaluate the influence of the supergyre on the properties of Antarctic Intermediate Water (AAIW), the dominant water mass at intermediate depths in the Southern Hemisphere. CMIP6 models and ECCO are in agreement at the surface with supergyres connected across all basins but present some differences at depth in both position and strength. AAIW core properties (temperature and salinity) present a high degree of similarity across basins within the supergyre but not outside of it. By the end of the century, the supergyre reduces in size and intensifies at intermediate depths, and the AAIW core depth warms in all basins and freshens in the Pacific although no clear trend in salinity can be found in the Atlantic and Indian basins in the SSP5-8.5 scenario. The high degree of similarity across basins within the supergyre is maintained in the future scenario. The results suggest that by connecting the basins together at intermediate depth, the supergyre plays a key role in circulating and homogenizing the AAIW core properties. Our results emphasize the role of the supergyre in circulating water masses at the surface and intermediate depths in CMIP6 models and hence its importance to the global circulation.

18. Investigation of Current-Wave Interaction Effect on Ocean Surface Current Retrieval Under DCA Framework Using an Improved Doppler Radar Imaging Model

Authors: Yanlei Du (Chinese Academy of Sciences); Jianing Shao (Chinese Academy of Sciences); Xiaofeng Yang (Nanjing University); Robert Wang (Chinese Academy of Sciences); Jian Yang (Tsinghua University); Xiaofeng Li (Chinese Academy of Sciences)

Journal: IEEE Transactions on Geoscience and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tgrs.2024.3506952

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Language: N/A

Subjects: N/A

Keywords: improved Doppler radar imaging model (IDopRIM), Current-wave interaction effect, Doppler centroid anomaly (DCA) method, ocean surface current (OSC) retrieval

Funding: Hainan Provincial Natural Science Foundation of China, National Natural Science Foundation of China, National Key Research and Development Program of China

License: N/A

Open Access: N/A

Abstract: N/A

19. Enhanced Net Community Production With Sea Ice Loss in the Western Arctic Ocean Uncovered by Machine-Learning-Based Mapping

Authors: Tianyu Zhou (University of Delaware); Yun Li (University of Delaware); Zhangxian Ouyang (University of Delaware); Wei-Jun Cai (University of Delaware); Rubao Ji (Woods Hole Oceanographic Institution)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: O₂/Ar, Arctic ocean, machine learning, net community production, export production

Funding: National Aeronautics and Space Administration

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Abstract: In the Arctic Ocean (AO), net community production (NCP) has displayed spatially heterogeneous responses to sea ice reduction and associated environmental changes. Using a random forest machine learning model trained with >42,000 in situ measurements and concurrent, collocated environmental predictors, we reconstructed 19 years of 8-day, 6-km maps. During 2015-2021, the integrated between late-May and early-September (NCP_{late-May-early-September}) over the western AO was per year, with interannual variations positively tracking open water area. While the relationship between NCP_{late-May-early-September} and open water area was quasi-linear at high latitudes, strong nonlinearity was detected on the inflow shelf. The nonlinearity highlights that the increase resulted from area gain could be compounded by sea-ice loss induced ecosystem adjustments. Additional retrospective analysis for 2003-2014 suggests a potential long-term increase of export production and efficiency in the western AO with sea ice loss.

20. Climate-Induced Saltwater Intrusion in 2100: Recharge-Driven Severity, Sea Level-Driven Prevalence

Authors: Kyra H. Adams (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Brett A. Buzzanga (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Audrey H. Sawyer (Ohio State University); Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

Funding: U.S. Department of Defense, Strategic Environmental Research and Development Program, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Saltwater intrusion is a critical concern for coastal communities due to its impacts on fresh ecosystems and civil infrastructure. Declining recharge and rising sea level are the two dominant drivers of saltwater intrusion along the land-ocean continuum, but there are currently no global estimates of future saltwater intrusion that synthesize these two spatially variable processes. Here, for the first time, we provide a novel assessment of global saltwater intrusion risk by integrating future recharge and sea level rise while considering the unique geology and topography of coastal regions. We show that nearly 77% of global coastal areas below 60 degrees north will undergo saltwater intrusion by 2100, with different dominant drivers. Climate-driven changes in subsurface water replenishment (recharge) is responsible for the high-magnitude cases of saltwater intrusion, whereas sea level rise and coastline migration are responsible for the global pervasiveness of saltwater intrusion and have a greater effect on low-lying areas.

21. Atlantic Meridional Overturning Circulation slowdown modulates wind-driven circulations in a warmer climate

Authors: Mohima Sultana Mimi; Wei Liu

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01907-5

URL: <https://doi.org/10.1038/s43247-024-01907-5>

Publication Date: 2024-11-21

Volume/Issue: Volume 5, Issue 1

References: 52

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Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: Wind-driven and thermohaline circulations, two major components of global large-scale ocean circulations, are intrinsically related. As part of the thermohaline circulation, the Atlantic Meridional Overturning Circulation has been observed and is expected to decline over the twenty-first century, potentially modulating global wind-driven circulation. Here we perform coupled climate model experiments with either a slow or steady Atlantic overturning under anthropogenic warming to segregate its effect on wind-driven circulation. We find that the weakened Atlantic overturning generates anticyclonic surface wind anomalies over the subpolar North Atlantic to decelerate the gyre circulation there. Fingerprints of overturning slowdown are evident on Atlantic western boundary currents, encompassing a weaker northward Gulf Stream and Guiana Current and a stronger southward Brazil Current. Beyond the Atlantic, the weakened Atlantic overturning causes a poleward displacement of Southern Hemisphere surface westerly winds by changing meridional gradients of atmospheric temperature, leading to poleward shifts of the Antarctic Circumpolar Current and Southern Ocean meridional overturning circulations.

22. Evaluation of sea surface temperature from ocean reanalysis products over the North Indian Ocean

Authors: Raheema Rahman; Hasibur Rahaman

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2024.1461696

URL: <https://doi.org/10.3389/fmars.2024.1461696>

Publication Date: 2024-11-13

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Language: N/A

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Ocean and sea ice reanalyses (ORAs or ocean syntheses) are reconstructions of the ocean and sea ice states using an ocean model integration constrained by atmospheric surface forcing and ocean observations via a data assimilation method. Ocean reanalyses are a valuable tool for monitoring and understanding long-term ocean variability at depth, mainly because this part of the ocean is still largely unobserved. Sea surface temperature (SST) is the key variable that drives the air-sea interaction process on different time scales. Despite improvements in model and reanalysis schemes, ocean reanalyses show errors when evaluated with independent observations. The independent evaluation studies of SST from ocean reanalysis over the Indian Ocean are limited. In this study, we evaluated the SST from 10 reanalysis products (ECCO, BRAN, SODA, NCEP-GODAS, GODAS-MOM4p1, ORAS5, CGLORS, GLORYS2V4, GLOSEA, and GREP) and five synthetic observation products (COBE, ERSST, OISST, OSTIA, and HadISST) and from the pure observation-based product AMSR2 for 2012-2017 with 12 in-situ buoy observations (OMNI) over the Arabian Sea and Bay of Bengal. Even though the reanalysis and observational products perform very well in the open ocean, the performance is poorer near the coast and islands. The reanalysis products perform comparatively better than most of the observational products. COBE and OISST perform better among the synthetic observational products in the northern Indian Ocean. GODAS-MOM4p1 and GREP performs best among the reanalysis products, often surpassing the observational products. ECCO shows poorer performance and higher bias in the Bay of Bengal. Comparing the BRAN daily and monthly SST, the monthly SST performance of reanalysis is better than the daily time scale.

23. An optimal transformation method for inferring ocean tracer sources and sinks

Authors: Jan D. Zika; Taimoor Sohail

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-8049-2024

URL: <https://doi.org/10.5194/gmd-17-8049-2024>

Publication Date: 2024-11-12

Volume/Issue: Volume 17, Issue 21

References: 46

Cited by: N/A publications

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Open Access: N/A

Abstract: . The geography of changes in the fluxes of heat, carbon, freshwater and other tracers at the sea surface is highly uncertain and is critical to our understanding of climate change and its impacts. We present a state estimation framework wherein prior estimates of boundary fluxes can be adjusted to make them consistent with the evolving ocean state. In this framework, we define a discrete set of ocean water masses distinguished by their geographical, thermodynamic and chemical properties for specific time periods. Ocean circulation then moves these water masses in geographic space. In phase space, geographically adjacent water masses are able to mix together, representing a convergence, and air-sea property fluxes move the water masses over time. We define an optimisation problem whose solution is constrained by the physically permissible bounds of changes in ocean circulation, air-sea fluxes and mixing. As a proof-of-concept implementation, we use data from a historical numerical climate model simulation with a closed heat and salinity budget. An inverse model solution is found for the evolution of temperature and salinity that is consistent with "true" air-sea heat and freshwater fluxes which are introduced as model priors. When biases are introduced into the prior fluxes, the inverse model finds a solution closer to the true fluxes. This framework, which we call the optimal transformation method, represents a modular, relatively computationally cost-effective, open-source and transparent state estimation tool that complements existing approaches.

24. Submesoscales are a significant turbulence source in global ocean surface boundary layer

Authors: Jihai Dong; Baylor Fox-Kemper; Jacob O. Wenegrat; Abigail S. Bodner; Xiaolong Yu; Stephen Belcher; Changming Dong

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-024-53959-y

URL: <https://doi.org/10.1038/s41467-024-53959-y>

Publication Date: 2024-11-05

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Language: en

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Keywords: N/A

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License: Creative Commons

Open Access: N/A

Abstract: N/A

25. Future directions for deep ocean climate science and evidence-based decision making

Authors: Helen R. Pillar; Elizabeth Hetherington; Lisa A. Levin; Laura Cimoli; Jonathan M. Lauderdale; Jesse M. A. van der Grient; Kristen Johannes; Patrick Heimbach; Leslie Smith; Charles I. Addey; Pavanee Annasawmy; Sandra Antonio; Narissa Bax; Henri F. Drake; Elva Escobar; Laura G. Elsler; Mara A. Freilich; Natalya D. Gallo; Fanny Girard; Matthew J. Harke; Daniel O. B. Jones; Siddhi Joshi; Xinfeng Liang; Paige J. Maroni; Otmane Sarti; Paris V. Stefanoudis; Olivier Sulpis; David Trossman

Journal: Frontiers in Climate

Publisher: Frontiers Media SA

DOI: 10.3389/fclim.2024.1445694

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References: 148

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Funding: N/A

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Abstract: IntroductionA defining aspect of the Intergovernmental Panel on Climate Change (IPCC) assessment reports (AR) is a formal uncertainty language framework that emphasizes higher certainty issues across the reports, especially in the executive summaries and short summaries for policymakers. As a result, potentially significant risks involving understudied components of the climate system are shielded from view. MethodsHere we seek to address this in the latest, sixth assessment report (AR6) for one such component--the deep ocean--by summarizing major uncertainties (based on discussions of low confidence issues or gaps) regarding its role in our changing climate system. The goal is to identify key research priorities to improve IPCC confidence levels in deep ocean systems and facilitate the dissemination of IPCC results regarding potentially high impact deep ocean processes to decision-makers. This will accelerate improvement of global climate projections and aid in informing efforts to mitigate climate change impacts. An analysis of 3,000 pages across the six selected AR6 reports revealed 219 major science gaps related to the deep ocean. These were categorized by climate stressor and nature of impacts. ResultsHalf of these are biological science gaps, primarily surrounding our understanding of changes in ocean ecosystems, fisheries, and primary productivity. The remaining science gaps are related to uncertainties in the physical (32%) and biogeochemical (15%) ocean states and processes. Model deficiencies are the leading cited cause of low certainty in the physical ocean and ice states, whereas causes of biological uncertainties are most often attributed to limited studies and observations or conflicting results. DiscussionKey areas for coordinated effort within the deep ocean observing and modeling community have emerged, which will improve confidence in the deep ocean state and its ongoing changes for the next assessment report. This list of key "known unknowns" includes meridional overturning circulation, ocean deoxygenation and acidification, primary production, food supply and the ocean carbon cycle, climate change impacts on ocean ecosystems and fisheries,

and ocean-based climate interventions. From these findings, we offer recommendations for AR7 to avoid omitting low confidence-high risk changes in the climate system.

26. Large Mixed Layer Salinity Variation in the Southern Tropical Indian Ocean Due To the Blending of Water Masses

Authors: Zhangzhe Zhao (Scripps Institution of Oceanography / University of California San Diego, Chinese Academy of Sciences, Guangdong Key Laboratory of Ocean Remote Sensing Guangzhou China); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego); Yan Du (Chinese Academy of Sciences, Guangdong Key Laboratory of Ocean Remote Sensing Guangzhou China)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: The southern tropical Indian Ocean (TIO) displays large mixed layer salinity (MLS) variation. Circulation in this region is governed by the Indian Ocean tropical gyre (IOTG), where the source water proportion and associated mixing remain unclear. Particles integrating into the IOTG and entering the central southern TIO originate from the Bay of Bengal, Malacca Strait, western Indian Ocean, and Indonesian Throughflow. Surprisingly, cross-equatorial advection is particularly important, implying a significant connection between both the Bay of Bengal and the South China Sea via Malacca Strait into the southern TIO. The anomalous anticlockwise circulation weakens the IOTG during positive Indian Ocean Dipole (IOD). An opposite pattern is observed in the negative IOD. A particle experiment reveals that water masses are modulated by the anomalous circulation that drives the redistribution of MLS by changing the proportion of the different source waters. This represents a potential predictability for the southern TIO MLS variability.

27. The international multi-system OSEs/OSSEs by the UN Ocean Decade Project SynObs and its early results

Authors: Yosuke Fujii; Elisabeth Remy; Magdalena Alonso Balmaseda; Shoichiro Kido; Jennifer Waters; K. Andrew Peterson; Gregory C. Smith; Ichiro Ishikawa; Kamel Chikhar

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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Open Access: N/A

Abstract: "Synergistic Observing Network for Ocean Prediction (SynObs)" was launched in 2022 as a project of the United Nations Decade of Ocean Science for Sustainable Development to evaluate the importance of ocean observation systems and co-design the future evolution of the ocean observing network. SynObs is currently leading the flagship OSEs/OSSEs, an internationally coordinated activity in which observing system experiments (OSEs) and observing system simulation experiments (OSSEs) are conducted using a variety of ocean and coupled atmosphere-ocean prediction systems to evaluate ocean observation impacts consistent across most prediction systems. The flagship OSEs/OSSEs comprises the ocean prediction (OP) OSEs for high-resolution ocean predictions, the subseasonal-to-seasonal (S2S) OSEs for long-term lead-time coupled ocean-atmosphere predictions, and the OP OSSEs for evaluating new and future observing systems. SynObs plans to use the results of the flagship OSEs to contribute to the reports on the ocean observing network design made by international organizations and projects. Here, we introduce this initiative, and we report on some initial results. Some observation impacts consistent across four ocean prediction systems are found by a preliminary analysis of the analysis runs for the OP OSEs. For example, impacts of the altimetry data on the assimilated sea surface height (SSH) field are generally large in the westerly boundary current regions and around Antarctic Circumpolar Currents where SSH has large variability but are small in the tropical regions, despite the relatively large SSH variability there. The analysis also indicates the possibility that there are some characteristic differences in the observation impacts between low-resolution and eddy-resolving ocean prediction systems. Although OSE outputs of only four ocean prediction systems are available now, we will make further investigation, adding OSE outputs of other prediction systems that will be submitted in the near future.

28. Mapping the sources and strands of marine debris via particle modelling and In-situ sampling approaches in archipelagic countries

Authors: Noir P. Purba; Ibnu Faizal; David Christie; Munawir B. Pratama; Darryl A. Valino; Marine K. Martasuganda; Titin Herawati; Ajeng Wulandari; Sanny T. Utami; Noor C.D. Aryanto; Muhammad H. Ilmi; Raffy R. Alfarez; Buntora Pasaribu

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

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References: 63

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: N/A

29. Scale-Dependent Drivers of Air-Sea CO₂ Flux Variability

Authors: Amanda R. Fay (Columbia University); Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); Galen A. McKinley (Columbia University); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Abstract: In climate studies, it is crucial to distinguish between changes caused by natural variability and those resulting from external forcing. Here we use a suite of numerical experiments based on the ECCO-Darwin ocean biogeochemistry model to separate the impact of the atmospheric carbon dioxide (CO₂) growth rate and climate on the ocean carbon sink -- with a goal of disentangling the space-time variability of the dominant drivers. When globally integrated, the variable atmospheric growth rate and climate exhibit similar magnitude impacts on ocean carbon uptake. At local scales, interannual variability in air-sea CO₂ flux is dominated by climate. The implications of our study for real-world ocean observing systems are clear: in order to detect future changes in the ocean sink due to slowing atmospheric CO₂ growth rates, better observing systems and constraints on climate-driven ocean variability are required.

30. Seasonality of Submesoscale Vertical Heat Transport Modulated by Oceanic Mesoscale Eddies in the Kuroshio Extension

Authors: Qingyue Wang (Nanjing University, Jiangsu Provincial Meteorological Service Center Nanjing China); Changming Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory); Jihai Dong (Nanjing University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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References: 82

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Language: en

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Funding: N/A

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Open Access: N/A

Abstract: Energetic mesoscale eddies are often accompanied by strong submesoscale variability, which plays a significant role in connecting mesoscale and turbulent motions in the ocean and leads to strong vertical motions. The product of a high-resolution (1/48 degrees) oceanic numerical model, the LLC4320, is employed to investigate the seasonal variations of vertical heat transport induced by submesoscale processes within multiple mesoscale eddies in the Kuroshio Extension (KE) region. In different seasons, the submesoscale vertical heat transport exhibits a consistent upward pattern, with notably higher magnitudes observed during winter. In winter, the maxima value of submesoscale vertical heat flux (SVHF) can account for approximately 60% of the total vertical heat flux (VHF). This is equivalent to the average net sea surface heat flux in a single eddy region. In summer and autumn, the maxima absolute value of submesoscale vertical heat flux can account for approximately 30% of the total VHF. Energy analysis reveals that baroclinic instability associated with vertical buoyancy flux has a crucial effect on generating submesoscale processes within the eddy region. The submesoscale motions are influenced by the mixed layer instability, strain-induced frontogenesis, turbulent thermal wind and turbulent thermal wind-induced frontogenesis within the upper mixed layer, while they are largely associated with the strain-induced frontogenesis in the ocean interior. Furthermore, the upward low-frequency submesoscale vertical heat transport is generated by submesoscale secondary circulation at eddy peripheries.

31. North Atlantic Heat Transport Convergence Derived from a Regional Energy Budget Using Different Ocean Heat Content Estimates

Authors: B. Meyssignac; S. Fourest; Michael Mayer; G. C. Johnson; F. M. Calafat; M. Ablain; T. Boyer; L. Cheng; D. Desbruyères; G. Forget; D. Giglio; M. Kuusela; R. Locarnini; J. M. Lyman; W. Llovel; A. Mishonov; J. Reagan; V. Rousseau; J. Benveniste

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

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Keywords: N/A

Funding: Global Ocean Monitoring and Observing Program, Austrian Science Fund

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Open Access: N/A

Abstract: This study uses an oceanic energy budget to estimate the ocean heat transport convergence in the North Atlantic during 2005-2018. The horizontal convergence of the ocean heat transport is estimated using ocean heat content tendency primarily derived from satellite altimetry combined with space gravimetry. The net surface energy fluxes are inferred from mass-corrected divergence of atmospheric energy transport and tendency of the ECMWF ERA5 reanalysis combined with top-of-the-atmosphere radiative fluxes from the clouds and the Earth's radiant energy system project. The indirectly estimated horizontal convergence of the ocean heat transport is integrated between the rapid climate change-meridional overturning circulation and heatflux array (RAPID) section at 26.5 degreesN (operating since 2004) and the overturning in the subpolar north atlantic program (OSNAP) section, situated at 53 degrees-60 degreesN (operating since 2014). This is to validate the ocean heat transport convergence estimate against an independent estimate derived from RAPID and OSNAP in-situ measurements. The mean ocean energy budget of the North Atlantic is closed to within +/- 0.25 PW between RAPID and OSNAP sections. The mean oceanic heat transport convergence between these sections is 0.58 +/- 0.25 PW, which agrees well with observed section transports. Interannual variability of the inferred oceanic heat transport convergence is also in reasonable agreement with the interannual variability observed at RAPID and OSNAP, with a correlation of 0.54 between annual time series. The correlation increases to 0.67 for biannual time series. Other estimates of the ocean energy budget based on ocean heat content tendency derived from various methods give similar results. Despite a large spread, the correlation is always significant meaning the results are robust against the method to estimate the ocean heat content tendency.

32. Antarctic climate response in Last Interglacial simulations using the Community Earth System Model (CESM2)

Authors: Mira Berdahl; Gunter R. Leguy; William H. Lipscomb; Bette L. Otto-Bliesner; Esther C. Brady; Robert A. Tomas; Nathan M. Urban; Ian Miller; Harriet Morgan; Eric J. Steig

Journal: Climate of the Past

Publisher: Copernicus GmbH

DOI: 10.5194/cp-20-2349-2024

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References: 67

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Open Access: N/A

Abstract: . We examine results from two transient modeling experiments that simulate the Last Interglacial period (LIG) using the state-of-the-art Community Earth System Model (CESM2), with a focus on climate and ocean changes relevant to the possible collapse of the Antarctic ice sheet. The experiments simulate the early millennia of the LIG warm period using orbital forcing, greenhouse gas concentrations, and vegetation appropriate for 127 ka. In the first case (127ka), no other changes are made; in the second case (127kaFW), we include a 0.2 Sv freshwater forcing in the North Atlantic. Both are compared with a pre-industrial control simulation (piControl). In the 127ka simulation, the global average temperature is only marginally warmer (0.004 degreesC) than in the piControl. When freshwater forcing is added (127kaFW), there is surface cooling in the Northern Hemisphere (NH) and warming in the Southern Hemisphere (SH), consistent with the bipolar seesaw effect. Near the Antarctic ice sheet, the 127ka simulation generates notable ocean warming (up to 0.4 degreesC) at depths below 200 m compared to the piControl. In contrast, the addition of freshwater in the North Atlantic in the 127kaFW run results in a multi-century subsurface ocean cooling that rebounds slowly over multiple millennia near the Antarctic ice sheet. These results have implications for the thermal forcing (and thereby mass balance) of the Antarctic ice sheet. We explore the physical processes that lead to this result and discuss implications for climate forcing of Antarctic ice sheet mass loss during the LIG.

33. GRACE Satellite Observations of Antarctic Bottom Water Transport Variability

Authors: Jemma Jeffree (Australian National University, ARC Centre of Excellence for Climate Extremes Canberra ACT Australia); Andrew McC. Hogg (Australian National University, ARC Centre of Excellence for Climate Extremes Canberra ACT Australia); Adele K. Morrison (University of Tasmania, Australian National University, ARC Centre of Excellence for Climate Extremes Canberra ACT Australia); Aviv Solodoch (Institute of Earth Sciences Hebrew University of Jerusalem Jerusalem Israel); Andrew L. Stewart (University of California Los Angeles); Rebecca McGirr (University of Tasmania, Australian National University)

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License: N/A

Open Access: N/A

Abstract: Antarctic Bottom Water (AABW) formation and transport constitute a key component of the global ocean circulation. Direct observations suggest that AABW volumes and transport rates may be decreasing, but these observations are too temporally or spatially sparse to determine the cause. To address this problem, we develop a new method to reconstruct AABW transport variability using data from the GRACE (Gravity Recovery and Climate Experiment) satellite mission. We use an ocean general circulation model to investigate the relationship between ocean bottom pressure and AABW: we calculate both of these quantities in the model, and link them using a regularized linear regression. Our reconstruction from modeled ocean bottom pressure can capture 65%-90% of modeled AABW transport variability, depending on the ocean basin. When realistic observational uncertainty values are added to the modeled ocean bottom pressure, the reconstruction can still capture 30%-80% of AABW transport variability. Using the same regression values, the reconstruction skill is within the same range in a second, independent, general circulation model. We conclude that our reconstruction method is not unique to the model in which it was developed and can be applied to GRACE satellite observations of ocean bottom pressure. These advances allow us to create the first global reconstruction of AABW transport variability over the satellite era. Our reconstruction provides information on the interannual variability of AABW transport, but more accurate observations are needed to discern AABW transport trends.

34. MAESSTRO: Masked Autoencoders for Sea Surface Temperature Reconstruction under Occlusion

Authors: Edwin Goh; Alice Yepremyan; Jinbo Wang; Brian Wilson

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-20-1309-2024

URL: <https://doi.org/10.5194/os-20-1309-2024>

Publication Date: 2024-10-24

Volume/Issue: Volume 20, Issue 5

References: 35

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . This study investigates the use of a masked autoencoder (MAE) to address the challenge of filling gaps in high-resolution (1 km) sea surface temperature (SST) fields caused by cloud cover, which often result in gaps in the SST data and/or blurry imagery in blended SST products. Our study demonstrates that MAE, a deep learning model, can efficiently learn the anisotropic nature of small-scale ocean fronts from numerical simulations and reconstruct the artificially masked SST images. The MAE model is trained and evaluated on synthetic SST fields and tested on real satellite SST data from the Visible Infrared Imaging Radiometer Suite (VIIRS) sensor on the Suomi NPP satellite. We demonstrate that the MAE model trained on numerical simulations can provide a computationally efficient alternative for filling gaps in satellite SST. MAE can reconstruct randomly occluded images with a root mean square error (RMSE) of under 0.2 degreesC for masking ratios of up to 80 %. A trained MAE model in inference mode is exceptionally efficient, requiring 3 orders of magnitude (approximately 5000x) less time compared to the conventional approaches of cubic radial basis interpolation and Kriging tested on a single CPU. The ability to reconstruct high-resolution SST fields under cloud cover has important implications for understanding and predicting global and regional climates and detecting small-scale SST fronts that play a crucial role in the exchange of heat, carbon, and nutrients between the ocean surface and deeper layers. Our findings highlight the potential of deep learning models such as MAE to improve the accuracy and resolution of SST data at kilometer scales. This presents a promising avenue for future research in the field of small-scale ocean remote sensing analyses.

35. Detecting marine heatwaves below the sea surface globally using dynamics-guided statistical learning

Authors: Xiang Zhang; Furong Li; Zhao Jing; Bohai Zhang; Xiaohui Ma; Tianshi Du

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01769-x

URL: <https://doi.org/10.1038/s43247-024-01769-x>

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Volume/Issue: Volume 5, Issue 1

References: 64

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: N/A

36. Reconstructing the Ocean State Using Argo Data and a Data-Driven Method

Authors: Erwan Oulhen (Institut de Recherche pour le Développement (IRD)); Nicolas Kolodziejczyk (Institut de Recherche pour le Développement (IRD)); Pierre Tandeo (RIKEN Center for Computational Science, Kobe, Japan, IMT Atlantique, Lab-STICC, UMR CNRS 6285, Brest, France, Odyssey, Inria/IMT/CNRS, Rennes, France); Bruno Blanke (Institut de Recherche pour le Développement (IRD)); Florian Sévellec (Institut de Recherche pour le Développement (IRD), Odyssey, Inria/IMT/CNRS, Rennes, France)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-23-0157.1

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Publication Date: 2024-10-23

Volume/Issue: Volume 41, Issue 12

References: 71

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Language: N/A

Subjects: N/A

Keywords: Statistical forecasting, Interpolation schemes, Empirical orthogonal functions, Data assimilation, ENSO (and 1 more)

Funding: Université de Bretagne Occidentale, Région Bretagne, Institut national des sciences de l'Univers

License: N/A

Open Access: N/A

Abstract: Over the past two decades, Argo profiles have provided unprecedented insight into the global patterns of space and time variability of ocean temperature and salinity, significantly reducing associated uncertainties. However, analyzing such assessments during the pre-Argo period remains challenging due to the scarcity of observations in many regions. From the Argo period, a set of dominant three-dimensional patterns can be estimated using empirical orthogonal function (EOF) analysis, which helps to fill in observational gaps. From the associated principal components, temporal fluctuations can be observed, aiming to build a catalog of possible ocean state trajectories. To map pre-Argo observations, EOFs are used in a data assimilation framework that uses this catalog to feed an analog prediction and provide reanalysis. In this study, a new data-driven interpolation method called Reduced-Space Analog Data Assimilation (RedAnDA) was tested in the tropical Pacific Ocean. RedAnDA was first validated through an observing system simulation experiment (OSSE) approach, using synthetic observations extracted from a model simulation. It was subsequently applied to a real historical dataset and compared with other available reanalysis products. Overall, the reconstructed temperature field showed variability consistent with the OSSE true field and other reanalysis products in the real data application. Further improvements are needed to optimally estimate uncertainty, but RedAnDA already combines valuable information about state predictability, observational sampling, and unresolved scale issues.

37. The case for a Framework for UnderStanding Ice-Ocean iNteractions (FUSION) in the Antarctic-Southern Ocean system

Authors: Felicity S. McCormack (Monash University); Sue Cook (University of Tasmania); Daniel N. Goldberg (University of Edinburgh); Yoshihiro Nakayama (Hokkaido University); Hélène Seroussi (Dartmouth College); Isabel Nias (University of Liverpool); Lu An (Tongji University); Donald Slater (University of Edinburgh); Tore Hattermann (Norwegian Polar Institute)

Journal: Elem Sci Anth

Publisher: University of California Press

DOI: 10.1525/elementa.2024.00036

URL: <https://doi.org/10.1525/elementa.2024.00036>

Publication Date: 2024-10-22

Volume/Issue: Volume 12. Issue 1

References: 267

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Ice, Antarctica, Ocean

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Open Access: N/A

Abstract: We are in a period of rapidly accelerating change across the Antarctic continent and Southern Ocean, with land ice loss leading to sea level rise and multiple other climate impacts. The ice-ocean interactions that dominate the current ice loss signal are a key underdeveloped area of knowledge. The paucity of direct and continuous observations leads to high uncertainty in the glaciological, oceanographic and atmospheric fields required to constrain ice-ocean interactions, and there is a lack of standardised protocols for reconciling observations across different platforms and technologies and modelled outputs. Funding to support observational campaigns is under increasing pressure, including for long-term, internationally coordinated monitoring plans for the Antarctic continent and Southern Ocean. In this Practice Bridge article, we outline research priorities highlighted by the international ice-ocean community and propose the development of a Framework for UnderSTanding Ice-Ocean iNteractions (FUSION), using a combined observational-modelling approach, to address these issues. Finally, we propose an implementation plan for putting FUSION into practice by focusing first on an essential variable in ice-ocean interactions: ocean-driven ice shelf melt.

38. Synchronous decadal climate variability in the tropical Central Pacific and tropical South Atlantic

Authors: Chao Liu; Soon-Il An; Soong-Ki Kim; Malte F. Stuecker; Wenjun Zhang; Fei-Fei Jin; Jae-Heung Park; Leishan Jiang; Aoyun Xue; Xin Geng; Hyo-Jin Park; Young-Min Yang; Jong-Seong Kug

Journal: npj Climate and Atmospheric Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41612-024-00806-y

URL: <https://doi.org/10.1038/s41612-024-00806-y>

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Volume/Issue: Volume 7, Issue 1

References: 104

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: N/A

39. Diagnostic Framework Linking Eddy Flux Ellipse with Eddy-Mean Energy Exchange

Authors: Ru Chen (Tianjin Key Laboratory for Marine Environmental Research and Service, School of Marine Science and Technology, Tianjin University, Tianjin 300072, China.); Yi Yang (Tianjin Key Laboratory for Marine Environmental Research and Service, School of Marine Science and Technology, Tianjin University, Tianjin 300072, China.); Qianqian Geng (Tianjin Key Laboratory for Marine Environmental Research and Service, School of Marine Science and Technology, Tianjin University, Tianjin 300072, China.); Andrew Stewart (University of California Los Angeles); Glenn Flierl (Massachusetts Institute of Technology); Junyi Wang (Tianjin Key Laboratory for Marine Environmental Research and Service, School of Marine Science and Technology, Tianjin University, Tianjin 300072, China.)

Journal: Ocean-Land-Atmosphere Research

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.34133/olar.0072

URL: <https://doi.org/10.34133/olar.0072>

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Volume/Issue: Volume 3

References: 46

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: The design of non-eddy-resolving numerical models requires a good understanding and an appropriate representation of the eddy-mean flow feedback. To understand this feedback, we propose a diagnostic framework that links eddy geometry with the eddy-mean energy exchange terms in the Lorenz energy diagram. This framework provides explicit mathematical formulas that link eddy-mean energy exchange rates with both the mean state structure and the properties of eddy momentum ellipses and eddy buoyancy ellipses. Considering that the mean flow contains both along- and cross-stream variations, we decompose the eddy-mean kinetic energy exchange term into 3 components: one associated with the cross-stream variation in mean flow ($\overline{M}^C$), one associated with the along-stream variation in mean flow ($\overline{M}^A$), and one associated with the variation in mean flow ($\overline{M}^R$). We also state the corresponding geometric formulas. The geometric interpretation of $\overline{M}^C$ is consistent with barotropic instability theories and the literature on eddy geometry. As for $\overline{M}^A$, the weakening (strengthening) of mean flow in the along-stream direction corresponds to eddy kinetic energy generation (decay) through $\overline{M}^A$. $\overline{M}^A$ and a portion of $\overline{M}^R$ are related under the quasi-geostrophic assumption. From a global integral perspective, both the along-stream and cross-stream variations in the mean flow contribute considerably to eddy-mean kinetic energy exchange. At the Kuroshio Extension, both the mean state energy level and eddy energy level are key to shaping the spatial pattern of eddy-mean energy exchange. This framework offers a tool for geometrically interpreting eddy-mean energy exchange, which may offer guidance for eddy

parameterizations.

40. Fingerprints of El Niño Southern Oscillation on global and regional oceanic chlorophyll-a timeseries (1997-2022)

Authors: Chinglen Meetei Tensubam; Alexander V. Babanin; Mihir Kumar Dash

Journal: Science of The Total Environment

Publisher: Elsevier BV

DOI: 10.1016/j.scitotenv.2024.176893

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Language: en

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Keywords: N/A

Funding: The University of Melbourne, Indian Institute of Technology Kharagpur

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Open Access: N/A

Abstract: N/A

41. Evaluation of the ArcIOPS sea ice forecasts during 2021-2023

Authors: Xi Liang; Zhongxiang Tian; Fu Zhao; Ming Li; Na Liu; Chunhua Li

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2024.1477626

URL: <https://doi.org/10.3389/feart.2024.1477626>

Publication Date: 2024-10-16

Volume/Issue: Volume 12

References: 35

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Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The operational sea ice forecasts from the Arctic Ice Ocean Prediction System (ArcIOPS) during 2021-2023 are validated against satellite-retrieved sea ice concentration and drift data, in situ and reanalyzed sea ice thickness data. The results indicate that the ArcIOPS has a reliable capacity on the Arctic sea ice forecasts for the future 7 days. Over the validation period, the root mean square error (RMSE) of the ArcIOPS sea ice concentration forecasts at a lead time of up to 168 h ranges between 8% and 20%, and the integrated ice edge error (IIEE) is lower than 1.6×10^6 km² with respect to the Hai Yang 2B (HY-2B) sea ice concentration data. Compared to the Pan-Arctic Ice Ocean Modeling and Assimilation System (PIOMAS), sea ice volume evolution from the ArcIOPS forecasts is closer to that derived from the CS2SMOS sea ice thickness observations, which have been assimilated into the ArcIOPS. Sea ice thickness comparisons at three locations in the Beaufort Sea between the ArcIOPS forecasts and in situ mooring observations also prove that the sea ice thickness forecasts are credible, which sets a solid basis for supporting ice-breaker navigation in the Arctic thick ice zone. The sea ice drift deviations between the ArcIOPS forecasts and the National Snow and Ice Data Center (NSIDC) data are lower than 4 cm/s in most of the months. Future work will emphasize on developing multi-variable data assimilation scheme and fully coupled air-ice-ocean forecasting system for the Arctic sea ice forecasts.

42. Modeling Ocean Heat Transport to the Grounding Lines of Pine Island, Thwaites, Smith, and Kohler Glaciers, West Antarctica

Authors: Andy Dinh (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Matthew Mazloff (Scripps Institution of Oceanography / University of California San Diego); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024gl110078

URL: <https://doi.org/10.1029/2024gl110078>

Publication Date: 2024-10-07

Volume/Issue: Volume 51, Issue 19

References: 45

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: ice shelves, amundsen sea embayment, particle tracking, west Antarctica, antarctic coastal current (and 1 more)

Funding: N/A

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Open Access: N/A

Abstract: Pine Island, Thwaites, Smith, and Kohler glaciers in the Amundsen Sea Embayment (ASE) sector of West Antarctica experience rapid mass loss and grounding line retreat due to enhanced ocean thermal forcing from Circumpolar Deep Water (CDW) reaching the grounding lines. We use simulated Lagrangian particles advected with a looping 1 year output from the Southern Ocean high-resolution model to backtrack the transport and cooling of CDW to these glaciers. For the simulated year 2005-2006, we find that the median time needed to reach the grounding lines from the edge of the ASE is 3 years. In addition, the Antarctic Coastal Current contributes an equal number of particles as off-shelf sources to the grounding lines of Pine Island and Thwaites. For CDW coming from off-shelf, results from SOHi indicate that 25%-66% of the cooling occurs within ice shelf cavities.

43. Internal variability of Arctic liquid freshwater content in a coupled climate model large ensemble

Authors: Victoria L. Ford; Oliver W. Frauenfeld

Journal: Climatic Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10584-024-03808-0

URL: <https://doi.org/10.1007/s10584-024-03808-0>

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References: 69

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

44. Characteristics of Submesoscale Compensated/Reinforced Fronts in the Northern Bay of Bengal

Authors: Wei Duan (Hohai University, Institut de Recherche pour le Développement (IRD)); Xuhua Cheng (Hohai University); Yifei Zhou (Hohai University); Jonathan Gula (Institut Universitaire de France, Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jc021204

URL: <https://doi.org/10.1029/2024jc021204>

Publication Date: 2024-10-05

Volume/Issue: Volume 129, Issue 10

References: 59

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: submesoscale fronts, northern Bay of Bengal, reinforcement effect, LLC4320, temperature inversion (and 1 more)

Funding: National Natural Science Foundation of China, Science and Technology Planning Project of Guangdong Province, Hohai University

License: N/A

Open Access: N/A

Abstract: Fronts in the Bay of Bengal (BoB) are active and can potentially impact the regional dynamics such as temperature variability, salinity distribution and oceanic circulation. Based on the high resolution model output (LLC4320), this study investigates the characteristics of submesoscale fronts in the northern BoB and associated compensation/reinforcement effects. At sea surface, horizontal gradients of salinity and density are remarkable in the northern BoB, and they are nearly 3 times larger than temperature gradients. As the depth deepens, temperature gradients increase and become comparable to salinity gradients, while density gradients decrease a lot due to the increasing effects of compensation at subsurface. Statistical results show the dominance of salinity-controlled fronts over temperature-controlled fronts, and compensated fronts over reinforced fronts. The surface cooling/heating results in significant temporal variation of compensation at surface, but this variation is limited at subsurface by the blocking of the mixed layer base. The submesoscale-selective feature of compensation is much more pronounced at subsurface layer than surface layer. From statistical analysis and idealized numerical model, we found the slump of salinity-controlled compensated fronts are important in generating temperature inversion and maintaining barrier layer. This study validates the compensation theories originating from observations, and further illustrates the importance of subsurface compensated fronts using spatially continuous, regionally extended and longer-term model output. The subsurface-intensified submesoscale-selective compensation is proved for the first time in this study.

45. The Bering Strait Throughflow Component of the Global Mass, Heat and Freshwater Transport

Authors: Xiaoting Yang (Scripps Institution of Oceanography / University of California San Diego); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jc021463

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References: 65

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: As the only oceanic connection between the Pacific and Arctic-Atlantic Oceans, Bering Strait throughflow carries a climatological northward transport of about 1 Sv, contributing to the Atlantic Meridional Overturning Circulation (AMOC). Here, Lagrangian analysis quantifies the global distributions of volume transport, transit-times, thermohaline properties, diapycnal transformation, heat and freshwater transports associated with Bering Strait throughflow. Virtual Lagrangian parcels, released at Bering Strait, are advected by the velocity of Estimating the Circulation and Climate of the Ocean, backward and forward in time. Backward trajectories reveal that Bering Strait throughflow enters the Pacific basin on the southeast side, as part of fresh Antarctic Intermediate Water, then follows the wind-driven circulation to Bering Strait. Median transit time from S in Indo-Pacific to Bering Strait is 175 years. Sixty-four percent of Bering Strait throughflow enters the North Atlantic through the Labrador Sea. The remaining 36% flows through the Greenland Sea, warmed and salinified by the northward flowing Atlantic waters. Deep water formation of water flowing through Bering Strait occurs predominantly in the Labrador Sea. Subsequently, this water joins the lower branch of AMOC, flowing southward in the deep western boundary current as North Atlantic Deep Water. Median transit time from Bering Strait to S in South Atlantic is 160 years. The net heat transport of Bering Strait throughflow is northward everywhere, and net freshwater transport by Bering Strait throughflow is mostly northward. The freshwater transport is largest in the subpolar region of basin sectors: northward in the Pacific and Arctic and southward in the Atlantic.

46. Response of the Indian Ocean Meridional Overturning Circulation to the Subtropical Indian Ocean Dipole

Authors: Linfang Zhang (Sun Yat-Sen University); Marco Yu-Ting Leung (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Fengchao Yao (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Guangli Zhang (Sun Yat-Sen University); Dongxiao Wang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-23-0643.1

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Publication Date: 2024-10-03

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References: 90

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Ocean models, Ekman pumping/transport, Meridional overturning circulation, Indian Ocean, Atmosphere-ocean interaction (and 1 more)

Funding: Innovative Research Group Project of the National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Using the German contribution to Estimating the Circulation and Climate of the Ocean, version 3 (GECCO3), reanalysis data, this work explores the Indian Ocean meridional overturning circulation (IMOC) variability and mechanism during the mature phase of the subtropical Indian Ocean dipole (SIOD). The IMOC is decomposed into the Ekman component, geostrophic component, external mode, and residue. The IMOC exhibits counterclockwise circulation anomaly in 0 degrees-30 degreesS during the mature phase of the SIOD. While the Ekman component dominates in 10 degrees-30 degreesS, the geostrophic component prevails in 5 degrees-20 degreesS. During the mature phase of the positive SIOD events, while an anticyclonic wind anomaly over the southern Indian Ocean causes a convergence and sinking of the seawater near 30 degreesS, a cyclonic wind field anomaly near 10 degreesS induces a divergence and rising, causing a counterclockwise Ekman component anomaly in 10 degrees-30 degreesS. The geostrophic component anomaly in 5 degrees-20 degreesS is caused by the sea level anomaly (SLA) gradient around 10 degreesS related to a Rossby wave. A linear, 1D baroclinic Rossby wave model shows that the negative SLA in the west is caused by local and remote wind forcing, whereas the positive SLA in the east is generated by radiation from the eastern boundary and is slightly contributed by local wind forcing. Further, Parallel Ocean Program, version 2 (POP2), experiments confirmed that the Ekman component anomaly primarily responds to the wind field of the mature phase of the SIOD and revealed that the geostrophic component anomaly is affected by the wind field of both the developing and mature phases of the SIOD.

47. Negative Surface Chlorophyll Concentration Anomalies in the Southeast Arabian Sea During Summer in 2015 and 2019

Authors: Huaming Huang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Ke Huang (Chinese Academy of Sciences); Lei Yang (Chinese Academy of Sciences); Zhanlin Liang (Chinese Academy of Sciences); Wei Song (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Dongxiao Wang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jc021154

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Publication Date: 2024-10-01

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Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Basic and Applied Basic Research Foundation of Guangdong Province, National Key Research and Development Program of China (and 2 more)

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Open Access: N/A

Abstract: Satellite observations revealed two extremely low surface chlorophyll concentration (SCC) events with a warm sea surface temperature anomaly in the southeastern Arabian Sea (SEAS, 6 degrees-15 degreesN, 72 degrees-77 degreesE) during the summer (July-August-September) in 2015 and 2019. We find that the physical processes leading to these two similar low SCC events are remarkably different. The low SCC in the SEAS during summer 2019 is mainly related to the weakened upwelling and deepening of the thermocline depth due to the combined effects of the local wind anomalies and the arrival of westward-propagating downwelling coastal Kelvin wave driven by easterly anomalies near the eastern Sri Lanka during an extreme positive Indian Ocean Dipole (IOD) event. In summer 2015, a weaker positive IOD-induced easterly anomalies in the southern Bay of Bengal also drives downwelling coastal Kelvin waves westward, deepening the thermocline in the SEAS. But unlike that in summer 2019, the local wind stress curl anomalies in the SEAS during summer 2015 favors upwelling, which counteracts the downward motion of the coastal Kelvin waves, leading to weaker downward transport (one-third of that in 2019). Meanwhile, the upper ocean layer in the SEAS experiences extreme warming during summer owing to the development of 2015/2016 super El Niño. This substantial warming enhances upper oceanic stratification, which results in weaker vertical mixing and reduces the SCC to an extremely low level despite the much weaker IOD strength in 2015.

48. Variability in manometric sea level from reanalyses and observation-based products over the Arctic and North Atlantic oceans and the Mediterranean Sea

Authors: Andrea Storto; Giulia Chierici; Julia Pfeffer; Anne Barnoud; Romain Bourdalle-Badie; Alejandro Blazquez; Davide Cavaliere; Noémie Lalau; Benjamin Coupry; Marie Drevillon; Sebastien Fourest; Gilles Larnicol; Chunxue Yang

Journal: State of the Planet

Publisher: Copernicus GmbH

DOI: 10.5194/sp-4-osr8-12-2024

URL: <https://doi.org/10.5194/sp-4-osr8-12-2024>

Publication Date: 2024-09-30

Volume/Issue: Volume 4-osr8

References: 85

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Regional variations in the mass component of sea level (manometric sea level) are intimately linked with the changes in the water cycle, volume transports, and inter-basin exchanges. Here, we investigate the consistency at the regional level of the manometric sea level from the Copernicus Marine Service Global Reanalysis Ensemble Product (GREP) and compare with observation-based products deduced from either gravimetry (GRACE missions) or altimetry and in situ ocean observations (sea level budget, SLB, approach) for some climate-relevant diagnostics such as interannual variability, trends, and seasonal amplitude. The analysis is performed for three basins (the Mediterranean Sea and Arctic and North Atlantic oceans) and indicates very different characteristics across the three. The Mediterranean Sea exhibits the largest interannual variability, the Arctic Ocean the largest trends, and the North Atlantic a nearly linear increase that is highly correlated to global barystatic sea level variations. The three datasets show significant consistency at both the seasonal and the interannual timescales, although the differences in the linear trends are sometimes significant (e.g. GRACE overestimates the trend in the Arctic and underestimates it in the Mediterranean Sea when compared to the other products). Furthermore, the Gravity Recovery and Climate Experiment (GRACE) and GREP data prove to be mutually more consistent than SLB in most cases. Finally, we analyse the main modes of climate variability affecting the manometric sea level variations over the selected ocean basins through regularised regression; the North Pacific Gyre Oscillation, the Arctic Oscillation, and the Atlantic Multidecadal Oscillation are proven to be the most influential modes for the North Atlantic Ocean, Mediterranean Sea, and Arctic Ocean manometric sea levels, respectively.

49. Monitoring global ocean heat content from space geodetic observations to estimate the Earth energy imbalance

Authors: Florence Marti; Benoit Meyssignac; Victor Rousseau; Michaël Ablain; Robin Fraudeau; Alejandro Blazquez; Sébastien Fourest

Journal: State of the Planet

Publisher: Copernicus GmbH

DOI: 10.5194/sp-4-osr8-3-2024

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Language: en

Subjects: N/A

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Funding: Centre National d'Etudes Spatiales, Centre National de la Recherche Scientifique, European Space Agency

License: CC BY 4.0

Open Access: N/A

Abstract: . This study presents an improved space geodetic approach to estimate the global ocean heat content (GOHC) change and the Earth energy imbalance (EEI) over 1993-2022. The EEI exhibits a positive trend of 0.29 W m⁻² per decade, significant at the 90 % confidence level, indicating accelerated ocean warming in line with independent Clouds and the Earth's Radiant Energy System (CERES) data. The study highlights the importance of comparing various estimates (e.g. in-situ-based GOHC) and their uncertainties to reliably assess EEI changes.

50. Constraining the trend in the ocean CO₂ sink during 2000-2022

Authors: Nicolas Mayot; Erik T. Buitenhuis; Rebecca M. Wright; Judith Hauck; Dorothee C. E. Bakker; Corinne Le Quéré

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

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Abstract: The ocean will ultimately store most of the CO₂ emitted to the atmosphere by human activities. Despite its importance, estimates of the 2000-2022 trend in the ocean CO₂ sink differ by a factor of two between observation-based products and process-based models. Here we address this discrepancy using a hybrid approach that preserves the consistency of known processes but constrains the outcome using observations. We show that the hybrid approach reproduces the stagnation of the ocean CO₂ sink in the 1990s and its reinvigoration in the 2000s suggested by observation-based products and matches their amplitude. It suggests that process-based models underestimate the amplitude of the decadal variability in the ocean CO₂ sink, but that observation-based products on average overestimate the decadal trend in the 2010s. The hybrid approach constrains the 2000-2022 trend in the ocean CO₂ sink to 0.42 ± 0.06 Pg C yr⁻¹ decade⁻¹, and by inference the total land CO₂ sink to 0.28 ± 0.13 Pg C yr⁻¹ decade⁻¹.

51. The influence of the Biobio Canyon on the circulation and coastal upwelling/downwelling off central Chile

Authors: Odette A. Vergara; Pedro A. Figueroa; Cristian Salas; Sebastián I. Vásquez; Richard Muñoz; Gonzalo S. Saldías

Journal: Continental Shelf Research

Publisher: Elsevier BV

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Abstract: N/A

52. High-resolution numerical modelling of seasonal volume, freshwater, and heat transport along the Indian coast

Authors: Kunal Madkaiker; Ambarukhana D. Rao; Sudheer Joseph

Journal: Ocean Science

Publisher: Copernicus GmbH

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URL: <https://doi.org/10.5194/os-20-1167-2024>

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Abstract: . Seasonal reversal of winds and equatorial remote forcing influences the circulation of the Arabian Sea (AS) and Bay of Bengal (BoB) basins in the northern Indian Ocean. In this study, we numerically modelled the physical characteristics of the AS and BoB, using the Massachusetts Institute of Technology general circulation model (MITgcm) at a high spatial resolution of 1/20 degrees forced with climatological initial and boundary conditions. The simulated temperature, salinity, and flow fields were validated with satellite and in situ datasets. We then studied the exchange of coastal waters by evaluating transports computed from the model simulations. The alongshore volume transport on the eastern coast is stronger with high seasonal variability due to the poleward-flowing western boundary current and equatorward-flowing East Indian Coastal Current. West coast transport is influenced by large intraseasonal oscillations. The alongshore freshwater transport is an order less than the alongshore volume transport. Out of the net volume transport, freshwater accounts for a maximum of 6.03 % during the southwestern monsoon season, followed by 4.85 % in the post-monsoon season. We observe an inverse relationship between alongshore freshwater and volume transport on the western coast and a direct relationship on the eastern coast. The contribution of eddy-induced heat and freshwater transport was also examined. The relation between net heat transport and net heat flux illustrates the role of coastal currents and equatorial forcing in dissipating heat within the coastal waters. We observed that meridional heat transport over the AS is stronger than over the BoB. Both basins act as a heat source during the summer monsoon and heat sink during the winter. This high-resolution model setup simulates all the important physical climatological patterns, leading to a better understanding of the state of the northern Indian Ocean.

53. Characteristics and dynamics of the interannual eddy kinetic energy variation in the Western Equatorial Pacific Ocean

Authors: Xueqi Liu; Hui Zhou; Hengchang Liu; Wenlong Yang

Journal: Progress in Oceanography

Publisher: Elsevier BV

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Open Access: N/A

Abstract: N/A

54. Southern Ocean Ice Prediction System version 1.0 (SOIPS v1.0): description of the system and evaluation of synoptic-scale sea ice forecasts

Authors: Fu Zhao; Xi Liang; Zhongxiang Tian; Ming Li; Na Liu; Chengyan Liu

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-6867-2024

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Keywords: N/A

Funding: Chinese Polar Environment Comprehensive Investigation and Assessment Programmes, International Cooperation and Exchange Programme

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Open Access: N/A

Abstract: . An operational synoptic-scale sea ice forecasting system for the Southern Ocean, namely the Southern Ocean Ice Prediction System (SOIPS), has been developed to support ship navigation in the Antarctic sea ice zone. Practical application of the SOIPS forecasts had been implemented for the 38th Chinese National Antarctic Research Expedition for the first time. The SOIPS is configured on an Antarctic regional sea ice-ocean-ice shelf coupled model and an ensemble-based localized error subspace transform Kalman filter data assimilation model. Daily near-real-time satellite sea ice concentration observations are assimilated into the SOIPS to update sea ice concentration and thickness in the 12 ensemble members of the model state. By evaluating the SOIPS performance in forecasting sea ice metrics in a complete melt-freeze cycle from 1 October 2021 to 30 September 2022, this study shows that the SOIPS can provide reliable Antarctic sea ice forecasts. In comparison with non-assimilated EUMETSAT Ocean and Sea Ice Satellite Application Facility (OSI SAF) data, annual mean root mean square errors in the sea ice concentration forecasts at a lead time of up to 168 h are lower than 0.19, and the integrated ice edge errors in the sea ice forecasts in most freezing months at lead times of 24 and 72 h maintain around 0.5×10^6 km² and below 1.0×10^6 km², respectively. With respect to the scarce Ice, Cloud, and land Elevation Satellite-2 (ICESat-2) observations, the mean absolute errors in the sea ice thickness forecasts at a lead time of 24 h are lower than 0.3 m, which is in the range of the ICESat-2 uncertainties. Specifically, the SOIPS has the ability to forecast sea ice drift, in both magnitude and direction. The derived sea ice convergence rate forecasts have great potential for supporting ship navigation on a fine local scale. The comparison between the persistence forecasts and the SOIPS forecasts with and without data assimilation further shows that both model physics and the data assimilation scheme play important roles in producing reliable sea ice forecasts in the Southern Ocean.

55. Identification of Internal Tides in ECCO Estimates of Sea Surface Salinity in the Andaman Sea

Authors: Bulusu Subrahmanyam (School of the Earth, Ocean and Environment, University of South Carolina, Columbia, SC 29208, USA); V. S. N. Murty (Physical Oceanography Division, Council of Scientific and Industrial Research (CSIR)--National Institute of Oceanography (NIO), Regional Center, Visakhapatnam 530017, India); Sarah B. Hall (School of the Earth, Ocean and Environment, University of South Carolina, Columbia, SC 29208, USA); Corinne B. Trott (Naval Research Laboratory, Stennis Space Center, Hancock County, MS 39529, USA)

Journal: Remote Sensing

Publisher: MDPI AG

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Keywords: Andaman Sea, Bay of Bengal, internal tides, ECCO, sea surface salinity

Funding: United States Office of Naval Research Award, Council of Scientific and Industrial Research (CSIR), University of South Carolina, Columbia, USA

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Open Access: N/A

Abstract: We used NASA's high-resolution (1/48 degrees or 2.3 km, hourly) Estimating the Circulation and Climate of the Ocean (ECCO) estimates of salinity at a 1 m depth from November 2011 to October 2012 to detect semi-diurnal and diurnal internal tides (ITs) in the Andaman Sea and determine their characteristics in three 2 degrees x 2 degrees boxes off the Myanmar coast (box A), central Andaman Sea (box B), and off the Thailand coast (box C). We also used observed salinity and temperature data for the above period at the BD12-moored buoy in the central Andaman Sea. ECCO salinity data were bandpass-filtered with 11-14 h and 22-26 h periods. Large variations in filtered ECCO salinity (~0.1 psu) in the boxes corresponded with near-surface imprints of propagating ITs. Observed data from the box B domain reveals strong salinity stratification (halocline) in the upper 40 m. Our analyses reveal that the shallow halocline affects the signatures of propagating semi-diurnal ITs reaching the surface, but diurnal ITs propagating in the halocline reach up to the surface and bring variability in ECCO salinity. In box A, the semi-diurnal IT characteristics are higher speeds (0.96 m/s) with larger wavelengths (45 km), that are closer to theoretical mode 2 estimates, but the diurnal ITs propagating in the box A domain, with a possible source over the shelf of Gulf of Martaban, attain lower values (0.45 m/s, 38 km). In box B, the propagation speed is lower (higher) for semi-diurnal (diurnal) ITs. Estimates for box C are closer to those for box A.

56. Why did numerical weather forecasting systems fail to predict the Hurricane Otis's development?

Authors: Juan Carlos Guillermo-Montiel; Benjamín Martínez-López; Carlos Abraham Ochoa-Moya; Ignacio Arturo Quintanar; William David Cabos-Narváez

Journal: *Atmósfera*

Publisher: Universidad Nacional Autonoma de Mexico

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Language: N/A

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Keywords: WRF-PWP., Hurricane Otis development, coupled ocean-atmosphere system, ERA5 reanalysis, cumulus parameterization scheme (and 1 more)

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Open Access: N/A

Abstract: Hurricane Otis (HO) occurred in the eastern tropical Pacific (ETP), intensifying rapidly and unexpectedly, making landfall near Acapulco at 06:25 UTC on October 25, 2023 as a category five hurricane. Official and unofficial national weather forecasts (NWF) failed to predict HO's development, trajectory, and intensification. To analyze the reasons for the failure of the NWF, we conducted two experiments using the Weather Research and Forecasting (WRF) model, with Global Forecast System (GFS) and fifth-generation ECMWF atmospheric reanalysis (ERA5) data as initial condition (IC). Our results showed that some fields in the GFS data, such as relative humidity, convective available potential energy, and even sea surface temperature, were more favorable for the development and intensification of the disturbance compared to ERA5. However, the three-dimensional structure of the wind field in the ETP in GFS did not contribute to the initial development of HO. Additionally, we explored the WRF's sensitivity to different model configurations to simulate the trajectory and intensity of the hurricane using a coupled ocean-atmosphere system composed of WRF and a three-dimensional upper-ocean circulation model based on Price-Weller-Pinkel. Our numerical experiments involve modifications in the IC, cumulus parameterizations (CP), roughness coefficients, spatial resolutions, different time steps, and an idealized coupled model. The sensitivity test reveals the significance of the CP scheme, where the Kain-Fritsch was the only one that helped simulate the HO properly, altogether with increased spatial resolution. Furthermore, ocean-atmosphere coupling improves the prediction of the landfall time and location of the HO. However, no experiment captured the intensity or rapid intensification of HO.

57. A Review on Modeling Environmental Loading Effects and Their Contributions to Nonlinear Variations of Global Navigation Satellite System Coordinate Time Series

Authors: Zhao Li; Weiping Jiang; Tonie van Dam; Xiaowei Zou; Qusen Chen; Hua Chen

Journal: Engineering

Publisher: Elsevier BV

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Abstract: N/A

58. Three Atmospheric Patterns Dominate Decadal North Atlantic Overturning Variability

Authors: Dafydd Stephenson (National Center for Atmospheric Research); Daniel E. Amrhein (National Center for Atmospheric Research); LuAnne Thompson (University of Washington)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: Atlantic meridional overturning circulation, North Atlantic oscillation, decadal AMOC variability, dynamical attribution, modes of climate variability

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Open Access: N/A

Abstract: Atlantic Meridional Overturning Circulation (AMOC) variability originates from a large number of interacting processes with multiple time scales, with dominant processes dependent on both the latitude and timescale of interest. Here, we isolate the optimal atmospheric modes driving climate-relevant decadal AMOC variability using a novel approach combining dynamical and statistical attribution (dynamics-weighted principal component, or DPC analysis). We find that for both the subpolar (55 degreesN) and subtropical (25 degreesN) AMOC, the most effective independent mode of heat flux forcing closely resembles the North Atlantic Oscillation, and drives meridionally coherent AMOC anomalies through western boundary density anomalies. Conversely, established modes of wind stress variability possess limited quantitative similarity to the optimal wind stress patterns, which generate low-frequency AMOC fluctuations by rearranging the ocean buoyancy field. We demonstrate (by running a modified version of the ECCOV4r4 state estimate) that most AMOC variability on decadal time scales can be explained by the DPCs.

59. Consistent Seasonal Hydrography From Moorings at Northwest Greenland Glacier Fronts

Authors: Marie J. Zahn (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of Washington); Kristin L. Laidre (Greenland Institute of Natural Resources, University of Washington); Malene Simon (Greenland Institute of Natural Resources); Kathleen M. Stafford (Oregon State University); Michael Wood (San José State University); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Elizabeth M. Phillips (Fishery Resource Analysis and Monitoring Division Northwest Fisheries Science Center National Marine Fisheries Service National Oceanic and Atmospheric Administration Seattle WA USA); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

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Funding: Office of Naval Research, Cooperative Institute for Climate, Ocean, and Ecosystem Studies, University of Washington

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Open Access: N/A

Abstract: Greenland's marine-terminating glaciers connect the ice sheet to the ocean and provide a critical boundary where heat, freshwater, and nutrient exchanges take place. Buoyant freshwater runoff from inland ice sheet melt is discharged at the base of marine-terminating glaciers, forming vigorous upwelling plumes. It is understood that subglacial plumes modify waters near glacier fronts and increase submarine glacier melt by entraining warm ambient waters at depth. However, ocean observations along Greenland's coastal margins remain biased toward summer months which limits accurate estimation of ocean forcing on glacier retreat and acceleration. Here, we fill a key observational gap in northwest Greenland by describing seasonal hydrographic variation at glacier fronts in Melville Bay using in situ observations from moorings deployed year-round, CTDs, and profiling floats. We evaluated local and remote forcing using remote sensing and reanalysis data products alongside a high-resolution ocean model. Analysis of the year-round hydrographic data revealed consistent above-sill seasonality in temperature and salinity. The warmest, saltiest waters occurred in spring (April-May) and primed glaciers for enhanced submarine melt in summer when meltwater plumes entrain deep waters. Waters were coldest and freshest in early winter (November-December) after summer melt from sea ice, glacier ice, and icebergs provided cold freshwater along the shelf. Ocean variability was greatest in the summer and fall, coincident with increased freshwater runoff and large wind events before winter sea ice formation. Results increase our

mechanistic understanding of Greenland ice-ocean interactions and enable improvements in ocean model parameterization.

60. Why Is Decadal Climate Variability Predominantly Observed in the Niño4 Region?

Authors: Sieu-Cuong San (National Taiwan University); Yu-Heng Tseng (National Taiwan University); Ruiqiang Ding (Beijing Normal University); Emanuele Di Lorenzo (Brown University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 91

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Open Access: N/A

Abstract: This study investigates why observed decadal-scale climate variability is predominantly pronounced in the Niño4 region compared to other equatorial Pacific areas using both observation and model sensitivity experiments. The initial shift to the negative phase of Tropical Pacific Decadal Variability (TPDV) is primarily driven by the upward and eastward migration of isopycnal negative temperature anomalies along the equator. Subsequently, the wind fields associated with the negative phase of the Pacific Meridional Mode (PMM) induce anomalous vertical currents in the equatorial Pacific. This leads to anomalous upwelling and downwelling of mean temperature in the Niño4 and Niño3 regions, respectively, thereby strengthening and weakening the corresponding subsurface-produced sea surface temperature anomalies. Our findings clarify the roles of subsurface temperature anomalies in the phase reversal of TPDV and PMM in amplifying decadal variance, specifically in the equatorial central Pacific.

61. Interannual Variability of the Heat Budget in the Tropical Pacific Ocean and Its Link to the Overturning Circulation

Authors: Oleg A. Saenko (Department of Earth and Space Science and Engineering York University Toronto ON Canada, School of Earth and Ocean Sciences University of Victoria Victoria BC Canada); Neil F. Tandon (Centre for Research in Earth and Space Science York University Toronto ON Canada, Department of Earth and Space Science and Engineering York University Toronto ON Canada)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: tropical Pacific, interannual variability, ocean heat budget, overturning circulation, correlation

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Open Access: N/A

Abstract: Using a suite of coupled climate models and an extensive set of ocean heat budget diagnostics, we address the relative roles of heat convergence and surface heat flux in driving the annual rate of ocean heat content (OHC) change in the tropical Pacific and its interannual variability. The net heat convergence is further separated into convergences associated with the large-scale ocean circulation, (parameterized) mesoscale effects and small-scale mixing. It is found that the heat convergence due to the large-scale ocean circulation provides the dominant contribution to the annual OHC tendency. Interannual variations of heat convergence are larger in the tropical Pacific than in the tropical Atlantic. These heat convergence variations are linked to interannual variations of the Pacific meridional overturning circulation (PMOC), driven by the associated variations in the northward Ekman transport (EkT). Northward variations of the tropical PMOC and EkT are typically associated with heat divergence and negative annual OHC tendency in the central and eastern near-equatorial Pacific along with heat convergence and positive annual OHC tendency in the western and northwestern tropical Pacific. In the Niño3.4 region, interannual variations of the near-surface OHC tendency negatively (positively) correlate with interannual PMOC variations at zero lag (1 year lag, when PMOC leads OHC).

62. An integrated dataset of near-surface Eulerian fields and Lagrangian trajectories from an ocean model

Authors: Shane Elipot; Eli Faigle; Brian K. Arbic; Jay F. Shriver

Journal: Scientific Data

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41597-024-03813-z

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Keywords: N/A

Funding: National Science Foundation, National Science Foundation, United States Department of Defense | United States Navy | Office of Naval Research (and 1 more)

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Open Access: N/A

Abstract: A dataset consisting of numerically simulated oceanic velocities and sea surface height changes, provided conjointly from Eulerian and Lagrangian points of view, is made available as cloud-optimized archives on a cloud storage platform for unrestricted access. The Eulerian component of the dataset comprises oceanic velocity components at 0 m and 15 m depth, as well as total and steric sea surface height changes, obtained at hourly time steps for one year, with an approximate horizontal resolution of 1/25 degree on an irregular global geographical spatial grid, from the HYbrid Coordinate Ocean Model. The Lagrangian component of the dataset comprises the trajectories of particles advected in the Eulerian velocity field of the model. The particles were advected forward and backward for 30 days from a regular 1/4 degree grid in order to achieve 60-day long trajectories at 0 m and 15 m depths, with start times separated by 30 days, in 11 releases. This integrated dataset may help to link Eulerian and Lagrangian observational perspectives.

63. Simulated Impact of Time-Varying River Runoff and Greenland Freshwater Discharge on Sea Level Variability in the Beaufort Gyre Over 2005-2018

Authors: S. Tajouri (Institut de Recherche pour le Développement (IRD)); W. Llovel (Institut de Recherche pour le Développement (IRD)); F. Sévellec (Odyssey Team-Project INRIA CNRS Brest France, Institut de Recherche pour le Développement (IRD)); J.-M. Molines (Institut de Recherche pour le Développement (IRD)); P. Mathiot (Institut de Recherche pour le Développement (IRD)); T. Penduff (Institut de Recherche pour le Développement (IRD)); S. Leroux (Datlas Grenoble France)

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Subjects: N/A

Keywords: Beaufort gyre, Arctic Ocean, salinity variability, regional sea level change, land freshwater discharge

Funding: Université de Bretagne Occidentale

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Open Access: N/A

Abstract: Global mean sea level has been rising at a rate of 3.25 ± 0.4 mm yr⁻¹ over 1993-2018. Yet several regions are increasing at a much faster rate, such as the Beaufort Gyre in the Arctic Ocean at a rate of 9.3 ± 7.0 mm yr⁻¹ over 2003-2014. At interannual to decadal time scales, the Beaufort Gyre sea level is controlled by salinity changes due to sea ice melt and wind-driven lateral Ekman convergence-divergence of freshwater. This study uses recent Greenland discharge and river runoff estimates to isolate and quantify the sea level response to freshwater fluxes variability over the period 1980-2018. It relies on sensitivity experiments based on a global ocean model including sea-ice and icebergs. These sensitivity experiments only differ by the freshwater fluxes temporal variability of Greenland and global rivers which are either seasonal climatologies or fully time varying, revealing the individual and combined impact of these freshwater sources fluctuations. Fully varying Greenland discharge and river runoff produce an opposite impact on sea level trends over 2005-2018 in the Beaufort Gyre region, the former driving an increase, while the latter, a decrease. Their combined impact leads to a fairly weak sea level trend. The sea level response is primarily driven by salinity variations in the upper 300 m, which are mainly caused by salinity advection involving complex compensations between passive, active, and nonlinear advection. This study shows that including the temporal variability of freshwater fluxes in forced global ocean models results in a better representation of regional sea level change.

64. Controls on Upper Ocean Salinity Variability in the Eastern Subpolar North Atlantic During 1992-2017

Authors: Ali H. Siddiqui (Johns Hopkins University); Thomas W. N. Haine (Johns Hopkins University); An T. Nguyen (University of Texas at Austin); Martha W. Buckley (George Mason University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: fresh anomaly, salt budget, decadal variability, North Atlantic, salinity (and 1 more)

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Open Access: N/A

Abstract: The upper ocean salinity in the eastern subpolar North Atlantic undergoes decadal fluctuations. A large fresh anomaly event occurred during 2012-2016. Using the ECCOV4r4 state estimate, we diagnose and compare mechanisms of this low salinity event with those of the 1990s fresh anomaly event. To avoid issues related to the choice of reference salinity values in the freshwater budget, we perform a salt mass content budget analysis of the eastern subpolar North Atlantic. It shows that the recent low salt content anomaly occurs due to the circulation of anomalous salinity by mean currents entering the eastern subpolar basin from its western boundary via the North Atlantic Current. This is in contrast to the early 1990s, when the dominant mechanism governing the low salt content anomaly was the transport of the mean salinity field by anomalous currents.

65. Spatiotemporal Evolution of Slow Slip Events at the Offshore Hikurangi Subduction Zone in 2019 Using GNSS, InSAR, and Seafloor Geodetic Data

Authors: K. Woods (Victoria University of Wellington); L. M. Wallace (Institute for Geophysics University of Texas Austin TX USA, GEOMAR Helmholtz Centre for Ocean Research, Institute of Geosciences Christian-Albrechts-Universität zu Kiel Kiel Germany); C. A. Williams (GNS Science Lower Hutt New Zealand); I. J. Hamling (GNS Science Lower Hutt New Zealand); S. C. Webb (Columbia University); Y. Ito (Kyoto University); N. Palmer (GNS Science Lower Hutt New Zealand); R. Hino (Tohoku University); S. Suzuki (Tohoku University); M. K. Savage (Victoria University of Wellington); E. Warren-Smith (GNS Science Lower Hutt New Zealand); K. Mochizuki (University of Tokyo)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: earthquake, seafloor geodesy, crustal deformation, slow slip events, subduction

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Open Access: N/A

Abstract: Detecting crustal deformation during transient deformation events at offshore subduction zones remains challenging. The spatiotemporal evolution of slow slip events (SSEs) on the offshore Hikurangi subduction zone, New Zealand, during February-July 2019, is revealed through a time-dependent inversion of onshore and offshore geodetic data that also accounts for spatially varying elastic crustal properties. Our model is constrained by seafloor pressure time series (as a proxy for vertical seafloor deformation), onshore continuous Global Navigation Satellite System (GNSS) data, and Interferometric Synthetic Aperture Radar displacements. Large GNSS displacements onshore and uplift of the seafloor (10-33 mm) require peak slip during the event of 150 to >200 mm at 6-12 km depth offshore Hawkes Bay and Gisborne, comparable to maximum slip observed during previous seafloor pressure deployments at north Hikurangi. The onshore and offshore data reveal a complex evolution of the SSE, over a period of months. Seafloor pressure data indicates the slow slip may have persisted longer near the trench than suggested by onshore GNSS stations in both the Gisborne and Hawkes Bay regions. Seafloor pressure data also reveal up-dip migration of SSE slip beneath Hawke Bay occurred over a period of a few weeks. The SSE source region appears to coincide with locations of the March 1947 Mw 7.0-7.1 tsunami earthquake offshore Gisborne and estimated great earthquake rupture sources from paleoseismic investigations offshore Hawkes Bay, suggesting that the shallow megathrust at north and central Hikurangi is capable of both seismic and aseismic rupture.

66. The West Antarctic Ice Sheet may not be vulnerable to marine ice cliff instability during the 21st century

Authors: Mathieu Morlighem (Dartmouth College); Daniel Goldberg (University of Edinburgh); Jowan M. Barnes (University of Edinburgh, School of Geography and Environmental Sciences, Northumbria University, Newcastle upon Tyne, UK.); Jeremy N. Bassis (University of Michigan); Douglas I. Benn (University of St Andrews); Anna J. Crawford (Division of Biological and Environmental Sciences, University of Stirling, Stirling FK9 4LA, UK.); G. Hilmar Gudmundsson (School of Geography and Environmental Sciences, Northumbria University, Newcastle upon Tyne, UK.); H  l  ne Seroussi (Dartmouth College)

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Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The collapse of ice shelves could expose tall ice cliffs at ice sheet margins. The marine ice cliff instability (MICI) is a hypothesis that predicts that, if these cliffs are tall enough, ice may fail structurally leading to self-sustained retreat. To date, projections that include MICI have been performed with a single model based on a simple parameterization. Here, we implement a physically motivated parameterization in three ice sheet models and simulate the response of the Amundsen Sea Embayment after a hypothetical collapse of floating ice. All models show that Thwaites Glacier would not retreat further in the 21st century. In another set of simulations, we force the grounding line to retreat into Thwaites' deeper basin to expose a taller cliff. In these simulations, rapid thinning and velocity increase reduce the calving rate, stabilizing the cliff. These experiments show that Thwaites may be less vulnerable to MICI than previously thought, and model projections that include this process should be re-evaluated.

67. Investigation of North Atlantic Salinity Long-Term Trends Based on Historical Datasets

Authors: Pavel Sukhonos (Institute of Natural and Technical Systems, 299011 Sevastopol, Russia); Anatoly Gusev (Zubov State Oceanographic Institute, Roshydromet, 119034 Moscow, Russia, Russian Academy of Sciences); Nikolay Diansky (Faculty of Physics, Lomonosov Moscow State University, 119991 Moscow, Russia, Zubov State Oceanographic Institute, Roshydromet, 119034 Moscow, Russia, Russian Academy of Sciences)

Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

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References: 66

Cited by: N/A publications

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Funding: Russian Science Foundation

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Open Access: N/A

Abstract: In contrast to fairly good knowledge of seasonal and interannual variability in North Atlantic salinity, its long-term historical changes remain poorly characterized, making it difficult to assess the current state and possible future changes. To fill this gap, we present the results of applying a non-parametric method of regression analysis (quantile regression) to assess long-term changes in North Atlantic salinity (0 degrees-70 degrees N, 8 degrees-80 degrees W) based on multiple datasets. The features of quantile trends in monthly salinity for a median value in two periods (1948-2018 and 1961-2011) are considered. In 1948-2018, salinization was generally detected in North Atlantic tropical and subtropical latitudes, while desalination was found in subpolar latitudes. For the 71-year period under consideration, the median monthly salinity in subtropical latitudes increased by 0.07 ± 0.02 PSU. Over the period 1961-2011, pronounced long-term changes in the North Atlantic salinity are difficult to identify based on the datasets used. A consistency analysis of significant salinity trends across the most used datasets allowed us to detect five small areas with pronounced positive trends in the upper ocean salinity. These include the Guiana Current, the vicinity of 12 degrees N, 48 degrees W, the Canary upwelling area, the region of the Gulf Stream transition to the North Atlantic Current and the western part of the North Atlantic Subpolar Gyre. In these areas, over a 51-year period, salinity in the 10-400 m layer increased by an average of 0.10 ± 0.04 PSU.

68. The integration of diel vertical migration and hydrodynamic process influences the transport of swimming crab zoea (*Portunus trituberculatus*)

Authors: Huidong Jiang (Ocean University of China); Zhenjiang Ye (Ocean University of China); Yixiao Zhang (Ocean University of China); Wenchao Zhang (Ocean University of China); Yongjun Tian (Ocean University of China); Jianchao Li (Ocean University of China); Yang Liu (Ocean University of China); Haiqing Yu (Institute of Marine Science and Technology Shandong University Qingdao China); Xingui Zhang (Ocean University of China)

Journal: Fisheries Oceanography

Publisher: Wiley

DOI: 10.1111/fog.12695

URL: <https://doi.org/10.1111/fog.12695>

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References: 66

Cited by: 1 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Vertical migration and dispersal processes during the marine crab larval stage markedly affect transport, habitat selection, population connectivity, and resource replenishment success rates. However, not much is known of the reproductive ecology of swimming crabs in the nearshore waters of the northwest Pacific shelf. Here, we investigated the diel vertical migration (DVM) characteristics and transport patterns of the swimming crab zoea (*Portunus trituberculatus*) in this area. A Lagrangian particle-tracking algorithm coupled with a hydrodynamic model, incorporating a DVM pattern of zoeae based on observations from a field survey of the diurnal distribution of swimming crab zoea, was used to simulate the transport of zoeae, and the impact of zoeal transport on population connectivity was explored. The results revealed that particles were predominantly transported in a nearshore direction from the particle release point, with short dispersal distances during the zoeal stages. In nearshore waters on the continental shelf, the swimming crab zoeae are exposed to shoreward-moving currents with the aid of prolonged daytime locations in the lower water column, whereas larvae migrate upward to the middle and upper layers of the water column at night rather than the most superficial layer, potentially avoiding surface offshore-moving currents that may be responsible for the retention and shoreward transport of larvae. Most zoeae are transported to shallow waters, and the contribution of transport to population connectivity during the zoeal stages is relatively limited. The findings here have considerable implications for understanding the mechanisms governing the early recruitment dynamics of this species, as well as for fisheries management and conservation of marine biodiversity.

69. Responses of the Arctic sea ice drift to general warming and intraseasonal oscillation in the local atmosphere

Authors: Ming Li; Xi Liang; Na Liu; Fu Zhao; Zhongxiang Tian

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-024-07395-9

URL: <https://doi.org/10.1007/s00382-024-07395-9>

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References: 54

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Keywords: N/A

Funding: National Natural Science Foundation of China, National Key R&D; Program of China

License: N/A

Open Access: N/A

Abstract: N/A

70. An optimal transformation method applied to diagnose the ocean carbon budget

Authors: Neill Mackay; Taimoor Sohail; Jan David Zika; Richard G. Williams; Oliver Andrews; Andrew James Watson

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-5987-2024

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Publication Date: 2024-08-13

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References: 53

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . The ocean carbon sink plays a critical role in climate, absorbing anthropogenic carbon from the atmosphere and mitigating climate change. The sink shows significant variability on decadal timescales, but estimates from models and observations disagree with one another, raising uncertainty over the magnitude of the sink, its variability, and its driving mechanisms. There is a need to reconcile observation-based estimates of air-sea CO₂ fluxes with those of the changing ocean carbon inventory in order to improve our understanding of the sink, and doing so requires knowledge of how carbon is transported within the interior by the ocean circulation. Here we employ a recently developed optimal transformation method (OTM) that uses water-mass theory to relate interior changes in tracer distributions to transports and mixing and boundary forcings, and we extend its application to include carbon using synthetic data. We validate the method using model outputs from a biogeochemical state estimate, and we test its ability to recover boundary carbon fluxes and interior transports consistent with changes in heat, salt, and carbon. Our results show that the OTM effectively reconciles boundary carbon fluxes with interior carbon distributions when given a range of prior fluxes. The OTM shows considerable skill in its reconstructions, reducing root-mean-squared errors from biased priors between model "truth" and reconstructed boundary carbon fluxes by up to 71 %, with the bias of the reconstructions consistently ≤ 0.06 molCm⁻²yr⁻¹ globally. Inter-basin transports of carbon also compare well with the model truth, with residuals < 0.25 Pg C yr⁻¹ for reconstructions produced using a range of priors. The OTM has significant potential for application to reconcile observational estimates of air-sea CO₂ fluxes with the interior accumulation of anthropogenic carbon.

71. Lagrangian Versus Eulerian Spectral Estimates of Surface Kinetic Energy Over the Global Ocean

Authors: Xinwen Zhang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Xiaolong Yu (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Aurélien L. Ponte (Institut de Recherche pour le Développement (IRD)); Zoé Caspar-Cohen (Scripps Institution of Oceanography / University of California San Diego, Institut de Recherche pour le Développement (IRD)); Sylvie Le Gentil (Institut de Recherche pour le Développement (IRD)); Lu Wang (Institut de Recherche pour le Développement (IRD), Peking University); Wenping Gong (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Publication Date: 2024-08-12

Volume/Issue: Volume 129, Issue 8

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Lagrangian-Eulerian comparison, kinetic energy, internal tide, near-inertial waves, global ocean

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In this study, we conducted a novel massive Lagrangian simulation experiment based on a global 1/48 degrees tide-resolving numerical simulation of the ocean circulation. This first-time twin experiment enables a comparison between Eulerian (fixed-point) and Lagrangian (along-flow) estimates of kinetic energy (KE) across the global ocean, and the quantification of systematic differences between both types of estimations. This comparison represents an important step forward for the mapping of upper ocean high-frequency variability from Lagrangian observations of the Global Drifter Program. Eulerian KE rotary frequency spectra and band-integrated energy levels (e.g., tidal and near-inertial) serve as references and are compared to Lagrangian estimates. Our analysis reveals that, except for the near-inertial band, Lagrangian velocity spectra are systematically smoother, for example, with wider and lower spectral peaks compared to Eulerian counterparts. On average, Lagrangian KE levels derived from spectral band integrations tend to underestimate Eulerian KE levels at low-frequency and tidal bands, especially in regions with strong low-frequency KE. Better agreement between Lagrangian and Eulerian low-frequency and tidal KE levels is generally found in regions with weak low-frequency KE and/or convergent surface circulation, where Lagrangian particles tend to accumulate. Conversely, Lagrangian and Eulerian near-inertial spectra and energy levels are comparable. Our results demonstrate that Lagrangian estimates may provide a distorted view of low-frequency and tidal variance. To accurately map near-surface velocity climatology at these frequencies from drifter database, conversion methods accounting for the Lagrangian bias need to be

developed.

72. Fundamental Properties of Adjoint Model and Adjoint Sensitivity Under Fully Nonlinear Hydrostatic Internal Gravity Waves

Authors: Kenji Shimizu (Kobe University Kobe Japan, RPS Australia West Pty Ltd Subiaco WA Australia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: internal wave, data assimilation, adjoint model, adjoint sensitivity, internal tide (and 1 more)

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License: N/A

Open Access: N/A

Abstract: Adjoint sensitivity modeling plays an important role in inverse problems, including four-dimensional variational data assimilation or state estimation, by providing the sensitivity of the objective (or cost) function to the model input data. Although data assimilation has become common in regional ocean modeling, and the model resolution has become high enough to resolve internal tides, it remains unclear whether physically sensible and stable adjoint sensitivity modeling is feasible in the presence of internal waves. As a first step to tackle this problem, this study investigates fundamental properties of the adjoint of a time-dependent circulation model, and the resulting sensitivity under internal waves. The theoretical and numerical results, based on simple-internal-wave theory and MITgcm modeling, show that stable adjoint sensitivity modeling is feasible under fully nonlinear (but stable) hydrostatic internal waves. However, it requires a careful choice of the objective function to obtain sensitivity which has the same propagation properties as (real-world) internal waves, because the definition primarily controls the directionality and vertical-mode content of internal-wave signals in the adjoint model. Also, it should be noted that the internal-wave signals lack indirect sensitivity through the dependence of the wave-propagation speed on the model's prognostic variables. An important implication of the results is that the standard form of the objective function, which has been used in data assimilation studies for quasi-geostrophic flows and barotropic tides, could be problematic for internal tides.

73. Dispersal limits poleward expansion of mangroves on the west coast of North America

Authors: Kyle C. Cavanaugh (University of California Los Angeles); Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); Rémi Bardou (University of Michigan); Tom Van der Stocken (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Vrije Universiteit Brussel)

Journal: Ecography

Publisher: Wiley

DOI: 10.1111/ecog.07288

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Publication Date: 2024-08-07

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References: 77

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Funding: N/A

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Open Access: N/A

Abstract: While much attention has been paid to the climatic controls of species' range limits, other factors such as dispersal limitation are also important. Temperature is an important control of the distribution of coastal mangrove forests, and mangrove expansion at multiple poleward range limits has been linked to increasing temperatures. However, mangrove abundances at other poleward range limits have been surprisingly insensitive to climate change, indicating other drivers of range limitation. For example, along the west coast of North America, the poleward mangrove range limits are found on the Baja California and mainland coasts of Mexico, between 26 degrees48■ and 30 degrees18■N. Non-climatic factors may play an important role in setting these range limits as 1) the abundance of range limit populations has been relatively insensitive to climate variability and 2) an introduced population of mangroves has persisted hundreds of kilometers north of the natural range limits. We combined a species distribution model with a high-resolution oceanographic transport model to identify the roles of climate and dispersal limitation in controlling mangrove distributions. We identified estuarine habitat that is likely climatically suitable for mangroves north of the current range limits. However, propagules from current mangrove populations are unlikely to reach these suitable locations due to prevailing ocean currents and geomorphic factors that create a patchy distribution of estuarine habitat with large between-patch distances. Thus, although climate change is driving range shifts of mangroves in multiple regions around the world, dispersal is currently limiting poleward mangrove expansion at several range limits, including the west coast of North America.

74. Bathymetry-constrained ocean geostrophic currents play a key role in shaping the sea ice circulation in the Canada Basin, Arctic Ocean

Authors: Hengling Leng; Hailun He; Dake Chen; Peigen Lin; Yang Yang; Zhaomin Wang

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/ad6baa

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Keywords: N/A

Funding: National Natural Science Foundation of China, Shanghai Frontiers Science Center of Polar Science, Scientific Research Fund of the Second Institute of Oceanography, MNR (and 2 more)

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Open Access: N/A

Abstract: Satellite-based observations and a pan-Arctic coupled sea ice-ocean model are utilized to study the effect of ocean geostrophic currents on large-scale sea ice circulation in the Canada Basin, Arctic Ocean. We find that surface winds primarily drive sea ice drifts in the west-east direction, while the geostrophic currents in the Beaufort Gyre promote north-south ice drifts. Wind fluctuations can create variable ice drifts, yet geostrophic currents respond more slowly due to their larger vertical scale, serving as a slowly-evolving conveyor belt for maintaining the anticyclonic ice circulation. It is further demonstrated that the bathymetry can regulate the movement of sea ice via constraining the expansion of ocean circulation. This mechanism is indirect in the sense that the ice is far from the seafloor. Our research underscores the necessity of considering the bathymetry-constrained geostrophic currents in understanding Arctic sea ice dynamics. With the rapid retreat of Arctic sea ice, the multi-scale interactions between ice drifts and ocean currents may have significant implications for the Arctic ecosystem, climate, and shipping corridors.

75. Pathways and timescales of Southern Ocean hydrothermal iron and manganese transport

Authors: Antony J. Birchill; Chelsey A. Baker; Neil J. Wyatt; Katsiaryna Pabortsava; Hugh J. Venables; C. Mark Moore; Isobel Turnbull; Angela Milne; Simon J. Ussher; Sophy Oliver; Adrian P. Martin

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01564-8

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References: 83

Cited by: N/A publications

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Subjects: N/A

Keywords: N/A

Funding: RCUK | Natural Environment Research Council

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Open Access: N/A

Abstract: Scarcity of iron and manganese limits the efficiency of the biological carbon pump over large areas of the Southern Ocean. The importance of hydrothermal vents as a source of these micronutrients to the euphotic zone of the Southern Ocean is debated. Here we present full depth profiles of dissolved and total dissolvable trace metals in the remote eastern Pacific sector of the Southern Ocean (55-60 degrees S, 89.1 degrees W), providing evidence of enrichment of iron and manganese at depths of 2000-4000 m. These enhanced micronutrient concentrations were co-located with ³He enrichment, an indicator of hydrothermal fluid originating from ocean ridges. Modelled water trajectories revealed the understudied South East Pacific Rise and the Pacific Antarctic Ridge as likely source regions. Additionally, the trajectories demonstrate pathways for these Southern Ocean hydrothermal ridge-derived trace metals to reach the Southern Ocean surface mixed layer within two decades, potentially supporting a regular supply of micronutrients to fuel Southern Ocean primary production.

76. Challenges, Advances and Opportunities in Regional Sea Level Projections: The Role of Ocean-Shelf Dynamics

Authors: Svetlana Jevrejeva (National Oceanography Centre); Francisco M. Calafat (National Oceanography Centre); Michela De Dominicis (National Oceanography Centre); Joël J.-M. Hirschi (National Oceanography Centre); Jennifer V. Mecking (National Oceanography Centre); Jeff A. Polton (National Oceanography Centre); Bablu Sinha (National Oceanography Centre); Anthony Wise (National Oceanography Centre); Jason Holt (National Oceanography Centre)

Journal: Earth's Future

Publisher: American Geophysical Union (AGU)

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References: 155

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Open Access: N/A

Abstract: Future sea level rise and changes in extreme weather will increase the frequency of flooding and intensify the risks for the millions of people living in low-lying coastal areas. Concerns about coastal adaptation have been broadened due to societal awareness of the threat from rising seas, leading to a large set of potential adaptation users with diverse needs for adequate sea level projections in coastal areas beyond the current state of the art regional projections. In this paper, we provide an overview of the potential steps for improvement of regional sea level projections along the global coastline, with specific focus on the contribution from ocean dynamics to seasonal-decadal variability of coastal sea level, and its implications for changes in frequency and magnitude of extreme sea levels. We discuss the key gaps in our knowledge and predictive capability of these dynamics as they relate to sea level variability on seasonal to decadal timescales, and conclude by suggesting ways in which these knowledge gaps could be addressed.

77. Quantifying the Role of Ocean Dynamics in SST Variability across GCMs and Observations

Authors: L. Siqueira (University of Miami); B. P. Kirtman (University of Miami); L. C. Laurindo (University of Miami); J. T. Fasullo (National Center for Atmospheric Research); A. Hu (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-23-0686.1

URL: <https://doi.org/10.1175/jcli-d-23-0686.1>

Publication Date: 2024-07-30

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References: 122

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Climate models, Sea surface temperature, Surface fluxes, Climate variability, Oceanic mixed layer (and 1 more)

Funding: Climate Program Office, Climate Program Office, Division of Ocean Sciences (and 8 more)

License: N/A

Open Access: N/A

Abstract: Midlatitude SSTs forced by mesoscale oceanic processes can affect the large-scale atmosphere, pointing to the ocean's crucial role outside the tropics. Previous studies have shown oceanic mesoscale processes' effect on global and regional climate variability. This study quantifies the local contribution of ocean dynamics to mixed-layer temperature across the globe by directly estimating the ocean heat flux divergence resolved by state-of-the-art ocean reanalysis, eddy-resolving, and eddy-parameterized versions of two U.S. national climate models and indirectly from air-sea flux satellite-based estimates. Our results show that the eddy-resolving climate simulations resolve mixed-layer temperature variances that are larger and closer to those inferred from observations than both their eddy-parameterized counterparts and ECCO over much of the extratropics. The observations and the eddy-resolving models indicate a more significant role of ocean dynamics in the mixed-layer temperature variability than the surface fluxes over most extratropics compared to their eddy-parameterized versions. A frequency domain analysis shows that the better-resolved ocean mesoscale and thermal gradients enhance the variance over a time scale from 2 months to 30 years. Results show agreement in the ocean's contribution among satellite-based estimates, ocean reanalysis products, and ocean eddy-resolving simulations. At the same time, differences emerge for ECCO and the eddy-parameterized models, suggesting that surface fluxes account for a larger fraction of the mixed-layer temperature variability in most of the extratropics.

78. How Well Do We Know the Seasonal Cycle in Ocean Bottom Pressure?

Authors: R. M. Ponte (Atmospheric and Environmental Research Inc.); M. Zhao (Atmospheric and Environmental Research Inc.); M. Schindelegger (University of Bonn)

Journal: Earth and Space Science

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024ea003661

URL: <https://doi.org/10.1029/2024ea003661>

Publication Date: 2024-07-29

Volume/Issue: Volume 11, Issue 7

References: 46

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: seasonal cycle, gravitational attraction and loading, ocean state estimates, space gravimetry, ocean bottom pressure

Funding: National Science Foundation

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Open Access: N/A

Abstract: We revisit the nature of the ocean bottom pressure (pb) seasonal cycle by leveraging the mounting GRACE-based pb record and its assimilation in the ocean state estimates produced by the project for Estimating the Circulation and Climate of the Ocean (ECCO). We focus on the mean seasonal cycle from both data and ECCO estimates, examining their similarities and differences and exploring the underlying causes. Despite substantial year-to-year variability, the 21-year period studied (2002-2022) provides a relatively robust estimate of the mean seasonal cycle. Results indicate that the pb annual harmonic tends to dominate but the semi-annual harmonic can also be important (e.g., subpolar North Pacific, Bellingshausen Basin). Amplitudes and short-scale phase variability are enhanced near coasts and continental shelves, emphasizing the importance of bottom topography in shaping the seasonal cycle in pb. Comparisons of GRACE and ECCO estimates indicate good qualitative agreement, but considerable quantitative differences remain in many areas. The GRACE amplitudes tend to be higher than those of ECCO typically by 10%-50%, and by more than 50% in extensive regions, particularly around continental boundaries. Phase differences of more than 1 (0.5) months for the annual (semiannual) harmonics are also apparent. Larger differences near coastal regions can be related to enhanced GRACE data uncertainties and also to the absence of gravitational attraction and loading effects in ECCO. Improvements in both data and model-based estimates are still needed to narrow present uncertainties in pb estimates.

79. Impact of ocean heat transport on sea ice captured by a simple energy balance model

Authors: Jake R. Aylmer; David Ferreira; Daniel L. Feltham

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01565-7

URL: <https://doi.org/10.1038/s43247-024-01565-7>

Publication Date: 2024-07-29

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References: 197

Cited by: N/A publications

Language: en

Subjects: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: Future projections of Arctic and Antarctic sea ice suffer from uncertainties largely associated with inter-model spread. Ocean heat transport has been hypothesised as a source of this uncertainty, based on correlations with sea ice extent across climate models. However, a physical explanation of what sets the sea ice sensitivity to ocean heat transport remains to be uncovered. Here, we derive a simple equation using an idealised energy-balance model that captures the emergent relationship between ocean heat transport and sea ice in climate models. Inter-model spread of Arctic sea ice loss depends strongly on the spread in ocean heat transport, with a sensitivity set by compensation of atmospheric heat transport and radiative feedbacks. Southern Ocean heat transport exhibits a comparatively weak relationship with Antarctic sea ice and plays a passive role secondary to atmospheric heat transport. Our results suggest that addressing ocean model biases will substantially reduce uncertainty in projections of Arctic sea ice.

80. Trends and Variability in Earth's Energy Imbalance and Ocean Heat Uptake Since 2005

Authors: Maria Z. Hakuba; Sébastien Fourest; Tim Boyer; Benoit Meyssignac; James A. Carton; Gaël Forget; Lijing Cheng; Donata Giglio; Gregory C. Johnson; Seiji Kato; Rachel E. Killick; Nicolas Kolodziejczyk; Mikael Kuusela; Felix Landerer; William Llovel; Ricardo Locarnini; Norman Loeb; John M. Lyman; Alexey Mishonov; Peter Pilewskie; James Reagan; Andrea Storto; Thea Sukianto; Karina von Schuckmann

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-024-09849-5

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References: 124

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Open Access: N/A

Abstract: Earth's energy imbalance (EEI) is a fundamental metric of global Earth system change, quantifying the cumulative impact of natural and anthropogenic radiative forcings and feedback. To date, the most precise measurements of EEI change are obtained through radiometric observations at the top of the atmosphere (TOA), while the quantification of EEI absolute magnitude is facilitated through heat inventory analysis, where ~ 90% of heat uptake manifests as an increase in ocean heat content (OHC). Various international groups provide OHC datasets derived from in situ and satellite observations, as well as from reanalyses ingesting many available observations. The WCRP formed the GEWEX-EEI Assessment Working Group to better understand discrepancies, uncertainties and reconcile current knowledge of EEI magnitude, variability and trends. Here, 21 OHC datasets and ocean heat uptake (OHU) rates are intercompared, providing OHU estimates ranging between 0.40 +/- 0.12 and 0.96 +/- 0.08 W m⁻² (2005-2019), a spread that is slightly reduced when unequal ocean sampling is accounted for, and that is largely attributable to differing source data, mapping methods and quality control procedures. The rate of increase in OHU varies substantially between - 0.03 +/- 0.13 (reanalysis product) and 1.1 +/- 0.6 W m⁻² dec⁻¹ (satellite product). Products that either more regularly observe (satellites) or fill in situ data-sparse regions based on additional physical knowledge (some reanalysis and hybrid products) tend to track radiometric EEI variability better than purely in situ-based OHC products. This paper also examines zonal trends in TOA radiative fluxes and the impact of data gaps on trend estimates. The GEWEX-EEI community aims to refine their assessment studies, to forge a path toward best practices, e.g., in uncertainty quantification, and to formulate recommendations for future activities.

81. Loading-induced stress variation on active faults and seismicity modulation in the Kuril Islands-Japan region

Authors: Yiting Cai; Maxime Mouyen

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2024.118904

URL: <https://doi.org/10.1016/j.epsl.2024.118904>

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References: 65

Cited by: N/A publications

Language: en

Subjects: N/A

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Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

82. The Potential Role of Seasonal Surface Heating on the Chaotic Origins of the El Niño/Southern Oscillation Spring Predictability Barrier

Authors: Dakuan Yu (Shanghai Jiao Tong University); Meng Zhou (Shanghai Jiao Tong University, Polar Research Institute of China); Chaoxun Hang (Shanghai Jiao Tong University)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: spring predictability barrier, nonlinear dynamics, ENSO theoretical model, El Niño/Southern Oscillation, chaotic regime transition

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Open Access: N/A

Abstract: The Spring Predictability Barrier (SPB) phenomenon is characterized by the reduced accuracy of El Niño/Southern Oscillation (ENSO) forecasts during the spring, which substantially limits our ability to predict ENSO events. By investigating the nonlinear dynamic characteristics of ENSO systems simulated by a box model, we found that the strong surface heating process in spring may contribute to the SPB by regulating the different coupling processes between the ocean and atmosphere. Specifically, the intensified springtime surface heating increases the Sea Surface Temperature (SST), further amplifying the thermal damping effect of SST anomalies and reducing the dynamic connection between zonal SST gradient and upwelling process, and finally increasing the chaotic degree of ENSO systems simulated by the box model. The enhanced chaotic degree of ENSO systems leads to a more rapid growth of initial errors in the forecast model in spring, potentially leading to the SPB phenomenon.

83. Future changes in coastal upwelling and biological production in eastern boundary upwelling systems

Authors: Tianshi Du; Shengpeng Wang; Zhao Jing; Lixin Wu; Chao Zhang; Bihan Zhang

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

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URL: <https://doi.org/10.1038/s41467-024-50570-z>

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Abstract: Upwelling along oceanic eastern boundaries has attracted significant attention due to its profound effects on ocean productivity and associated biological and socioeconomic implications. However, uncertainty persists regarding the evolution of coastal upwelling with climate change, particularly its impact on future biological production. Here, using a series of state-of-the-art climate models, we identify a significant seasonal advancement and prolonged duration of upwelling in major upwelling systems. Nevertheless, the upwelling intensity (total volume of upwelled water) exhibits complex changes in the future. In the North Pacific, the upwelling is expected to attenuate, albeit with a minor magnitude. Conversely, in other basins, coastal upwelling diminishes significantly in equatorward regions but displays a slight decline or even an enhancement at higher latitudes. The climate simulations also reveal a robust connection between changes in upwelling intensity and net primary production, highlighting the crucial impact of future coastal upwelling alterations on marine ecosystems.

84. Dynamics of submesoscale processes and their influence on vertical heat transport in the southeastern tropical Indian Ocean

Authors: Yifei Zhou; Wei Duan; Xuhua Cheng

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-024-01628-5

URL: <https://doi.org/10.1007/s10236-024-01628-5>

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Keywords: N/A

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Abstract: N/A

85. Partial-convolution-implemented generative adversarial network for global oceanic data assimilation

Authors: Yoo-Geun Ham; Yong-Sik Joo; Jeong-Hwan Kim; Jeong-Gil Lee

Journal: Nature Machine Intelligence

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s42256-024-00867-x

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Abstract: N/A

86. Influence of Grid Resolution and Assimilation Window Size on Simulating Storm Surge Levels

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Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

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Abstract: Grid resolution and assimilation window size play significant roles in storm surge models. In the Bohai Sea, Yellow Sea, and East China Sea, the influence of grid resolution and assimilation window size on simulating storm surge levels was investigated during Typhoon 7203. In order to employ a more realistic wind stress drag coefficient that varies with time and space, we corrected the storm surge model using the spatial distribution of the wind stress drag coefficient, which was inverted using the data assimilation method based on the linear expression $C_d = (a + b \times U_{10}) \times 10^{-3}$. Initially, two grid resolutions of $5' \times 5'$ and $10' \times 10'$ were applied to the numerical storm surge model and adjoint assimilation model. It was found that the influence of different grid resolutions on the numerical model is almost negligible. But in the adjoint assimilation model, the root mean square (RMS) errors between the simulated and observed storm surge levels under $5' \times 5'$ and $10' \times 10'$ grid resolutions were 11.6 cm and 15.6 cm, and the average PCC and WSS values for 10 tidal stations changed from 89% and 92% in E3 to 93% and 96% in E4, respectively. The results indicate that the finer grid resolution can yield a closer consistency between the simulation and observations. Subsequently, the effects of assimilation window sizes of 6 h, 3 h, 2 h, and 1 h on simulated storm surge levels were evaluated in an adjoint assimilation model with a $5' \times 5'$ grid resolution. The results show that the average RMS errors were 11.6 cm, 10.6 cm, 9.6 cm, and 9.3 cm under four assimilation window sizes. In particular, the RMS errors for the assimilation window sizes of 1 h and 6 h at RuShan station were 3.9 cm and 10.2 cm, a reduction of 61.76%. The PCC and WSS values from RuShan station in E4 and E7 separately showed significant increases, from 85% to 98% and from 92% to 99%. These results demonstrate that when the assimilation window size is smaller, the simulated storm surge level is closer to the observation. Further, the results show that the simulated storm surge levels are closer to the observation when using

the wind stress drag coefficient with a finer grid resolution and smaller temporal resolution.

87. The Modeled Seasonal Cycles of Surface N₂O Fluxes and Atmospheric N₂O

Authors: Qing Sun (University of Bern); Fortunat Joos (University of Bern); Sebastian Lienert (University of Bern); Sarah Berthet (CNRM Université de Toulouse Météo France CNRS Toulouse France); Dustin Carroll (San José State University); Cheng Gong (Max Planck Institute for Biogeochemistry); Akihiko Ito (University of Tokyo, National Institute for Environmental Studies); Atul K. Jain (University of Illinois); Sian Kou-Giesbrecht (Canadian Centre for Climate Modelling and Analysis, Environment and Climate Change Canada, Dalhousie University); Angela Landolfi (Institute of Marine Sciences); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Naqing Pan (Department of Earth and Environmental Sciences Schiller Institute for Integrated Science and Society Boston College Chestnut Hill MA USA, Auburn University); Michael Prather (University of California Irvine); Pierre Regnier (Department Geoscience, Environment & Society - BGEOSYS Université Libre de Bruxelles Brussels Belgium); Laure Resplandy (Princeton University); Roland Séférian (CNRM Université de Toulouse Météo France CNRS Toulouse France); Hao Shi (Chinese Academy of Sciences); Parvatha Suntharalingam (University of East Anglia); Rona L. Thompson (NILU Kjeller Norway); Hanqin Tian (Department of Earth and Environmental Sciences Schiller Institute for Integrated Science and Society Boston College Chestnut Hill MA USA, Auburn University); Nicolas Vuichard (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL (CEA-CNRS-UVSQ) Université Paris-Saclay Gif-sur-Yvette France); Sönke Zaehle (Max Planck Institute for Biogeochemistry); Qing Zhu (Climate and Ecosystem Sciences Division Lawrence Berkeley National Lab Berkeley CA USA)

Journal: Global Biogeochemical Cycles

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Keywords: terrestrial biosphere model, seasonal cycle, tropospheric N₂O, ocean biogeochemistry model, surface N₂O emissions

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Abstract: Nitrous oxide (N₂O) is a greenhouse gas and stratospheric ozone-depleting substance with large and growing anthropogenic emissions. Previous studies identified the influx of N₂O-depleted air from the stratosphere to partly cause the seasonality in tropospheric N₂O (aN₂O), but other contributions remain unclear. Here, we combine surface fluxes from eight land and four ocean models from phase 2 of the Nitrogen/N₂O Model Intercomparison Project with tropospheric transport modeling to simulate aN₂O at eight remote air sampling sites for modern and pre-industrial periods. Models show general agreement on the seasonal phasing of zonal-average N₂O fluxes for most sites, but seasonal

peak-to-peak amplitudes differ several-fold across models. The modeled seasonal amplitude of surface aN₂O ranges from 0.25 to 0.80 ppb (interquartile ranges 21%-52% of median) for land, 0.14-0.25 ppb (17%-68%) for ocean, and 0.28-0.77 ppb (23%-52%) for combined flux contributions. The observed seasonal amplitude ranges from 0.34 to 1.08 ppb for these sites. The stratospheric contributions to aN₂O, inferred by the difference between the surface-troposphere model and observations, show 16%-126% larger amplitudes and minima delayed by ~1 month compared to Northern Hemisphere site observations. Land fluxes and their seasonal amplitude have increased since the pre-industrial era and are projected to grow further under anthropogenic activities. Our results demonstrate the increasing importance of land fluxes for aN₂O seasonality. Considering the large model spread, in situ aN₂O observations and atmospheric transport-chemistry models will provide opportunities for constraining terrestrial and oceanic biosphere models, critical for projecting carbon-nitrogen cycles under ongoing global warming.

88. *Ziphius cavirostris* presence relative to the vertical and temporal variability of oceanographic conditions in the Southern California Bight

Authors: Clara M. Schoenbeck (Scripps Institution of Oceanography / University of California San Diego); Alba Solsona-Berga (Scripps Institution of Oceanography / University of California San Diego); Peter J. S. Franks (Scripps Institution of Oceanography / University of California San Diego); Kaitlin E. Frasier (Scripps Institution of Oceanography / University of California San Diego); Jennifer S. Trickey (National Oceanic and Atmospheric Administration Pacific Islands Fisheries Science Center Honolulu Hawaii USA); Catalina Aguilar (Scripps Institution of Oceanography / University of California San Diego); Isaac D. Schroeder (University of California Santa Cruz, Institute of Marine Sciences); Ana Širović (Department of Biology Norwegian University of Science and Technology Trondheim Norway); Steven J. Bograd (Department of Biology Norwegian University of Science and Technology Trondheim Norway, Environmental Research Division, Southwest Fisheries Science Center, National Marine Fisheries Service National Oceanic and Atmospheric Administration Monterey California USA); Ganesh Gopalakrishnan (Scripps Institution of Oceanography / University of California San Diego); Simone Baumann-Pickering (Scripps Institution of Oceanography / University of California San Diego)

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Abstract: The oceanographic conditions of the Southern California Bight (SCB) dictate the distribution and abundance of prey resources and therefore the presence of mobile predators, such as goose-beaked whales (*Ziphius cavirostris*). Goose-beaked whales are deep-diving odontocetes that spend a majority of their time foraging at depth. Due to their cryptic behavior, little is known about how they respond to seasonal and interannual changes in their environment. This study utilizes passive acoustic data recorded from two sites within the SCB to explore the oceanographic conditions that goose-beaked whales appear to favor. Utilizing optimum multiparameter analysis, modeled temperature and salinity data are used to identify and quantify these source waters: Pacific Subarctic Upper Water (PSUW), Pacific Equatorial Water (PEW), and Eastern North Pacific Central Water (ENPCW). The interannual and seasonal variability in goose-beaked whale presence was related to the variability in El Niño Southern Oscillation events and the fraction and vertical distribution of the three source waters. Goose-beaked whale acoustic presence was highest during the winter and spring and decreased during the late summer and early fall. These seasonal increases occurred at times of increased fractions of PEW in the California Undercurrent and decreased fractions of ENPCW in

surface waters. Interannual increases in goose-beaked whale presence occurred during El Niño events. These results establish a baseline understanding of the oceanographic characteristics that correlate with goose-beaked whale presence in the SCB. Furthering our knowledge of this elusive species is key to understanding how anthropogenic activities impact goose-beaked whales.

89. A New WENO-Based Momentum Advection Scheme for Simulations of Ocean Mesoscale Turbulence

Authors: Simone Silvestri (Massachusetts Institute of Technology); Gregory L. Wagner (Massachusetts Institute of Technology); Jean-Michel Campin (Massachusetts Institute of Technology); Navid C. Constantinou (Australian National University); Christopher N. Hill (Massachusetts Institute of Technology); Andre Souza (Massachusetts Institute of Technology); Raffaele Ferrari (Massachusetts Institute of Technology)

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Keywords: eddy-permitting simulations, numerical modeling, mesoscale ocean turbulence

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Abstract: Current eddy-permitting and eddy-resolving ocean models require dissipation to prevent a spurious accumulation of enstrophy at the grid scale. We introduce a new numerical scheme for momentum advection in large-scale ocean models that involves upwinding through a weighted essentially non-oscillatory (WENO) reconstruction. The new scheme provides implicit dissipation and thereby avoids the need for an additional explicit dissipation that may require calibration of unknown parameters. This approach uses the rotational, "vector invariant" formulation of the momentum advection operator that is widely employed by global general circulation models. A novel formulation of the WENO "smoothness indicators" is key for avoiding excessive numerical dissipation of kinetic energy and enstrophy at grid-resolved scales. We test the new advection scheme against a standard approach that combines explicit dissipation with a dispersive discretization of the rotational advection operator in two scenarios: (a) two-dimensional turbulence and (b) three-dimensional baroclinic equilibration. In both cases, the solutions are stable, free from dispersive artifacts, and achieve increased "effective" resolution compared to other approaches commonly used in ocean models.

90. Changes in Oceanic Radiocarbon and CFCs Since the 1990s

Authors: J. G. Lester (Imperial College London); H. D. Graven (Imperial College London); S. Khatiwala (University of Oxford); A. P. McNichol (Woods Hole Oceanographic Institution)

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Keywords: circulation, radiocarbon, tracer, chlorofluorocarbon, ocean

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Abstract: Anthropogenic perturbations from fossil fuel burning, nuclear bomb testing, and chlorofluorocarbon (CFC) use have created useful transient tracers of ocean circulation. The atmospheric $^{14}\text{C}/\text{C}$ ratio ($\Delta^{14}\text{C}$) peaked in the early 1960s and has decreased now to pre-industrial levels, while atmospheric CFC-11 and CFC-12 concentrations peaked in the early 1990s and early 2000s, respectively, and have now decreased by 10%-20%. We present the first analysis of a decade of new observations (2007 to 2018-2019) and give a comprehensive overview of the changes in ocean $\Delta^{14}\text{C}$ and CFC concentration since the WOCE surveys in the 1990s. Surface ocean $\Delta^{14}\text{C}$ decreased at a nearly constant rate from the 1990-2010s (20‰/decade). In most of the surface ocean $\Delta^{14}\text{C}$ is higher than in atmospheric CO_2 while in the interior ocean, only a few places are found to have increases in $\Delta^{14}\text{C}$, indicating that globally, oceanic bomb ^{14}C uptake has stopped and reversed. Decreases in surface ocean CFC-11 started between the 1990 and 2000s, and CFC-12 between the 2000-2010s. Strong coherence in model biases of decadal changes in all tracers in the Southern Ocean suggest ventilation of Antarctic Intermediate Water was enhanced from the 1990 to the 2000s, whereas ventilation of Subantarctic Mode Water was enhanced from the 2000 to the 2010s. The decrease in surface tracers globally between the 2000 and 2010s is consistently stronger in observations than in models, indicating a reduction in vertical transport and mixing due to stratification.

91. Gap-filling techniques applied to the GOCI-derived daily sea surface salinity product for the Changjiang diluted water front in the East China Sea

Authors: Jisun Shin; Dae-Won Kim; So-Hyun Kim; Gi Seop Lee; Boo-Keun Khim; Young-Heon Jo

Journal: Earth System Science Data

Publisher: Copernicus GmbH

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Abstract: . The spatial and temporal resolutions of contemporary microwave-based sea surface salinity (SSS) measurements are insufficient. Thus, we developed a gap-free gridded daily SSS product with higher spatial and temporal resolutions, which can provide information on short-term variability in the East China Sea (ECS), such as the front changes by Changjiang diluted water (CDW). Specifically, we conducted gap-filling for daily SSS products based on the Geostationary Ocean Color Imager (GOCI) with a spatial resolution of 1 km (0.01 degrees), using a machine learning approach during the summer seasons from 2015 to 2019. The comparison of the Soil Moisture Active Passive (SMAP), Copernicus Marine Environment Monitoring Service (CMEMS), and Hybrid Coordinate Ocean Model (HYCOM) SSS products with the GOCI-derived SSS over the entire SSS range showed that the SMAP SSS was highly consistent, whereas the HYCOM SSS was the least consistent. In the < 31 psu range, the SMAP SSS was still the most consistent with the GOCI-derived SSS ($R^2=0.46$; root mean squared error: RMSE = 2.41 psu); in the > 31 psu range, the CMEMS and HYCOM SSS products showed similar levels of agreement with that of the SMAP SSS. We trained and tested three machine learning models - the fine trees, boosted trees, and bagged trees models - using the daily GOCI-derived SSS as output, including the three SSS products, environmental variables, and geographical data. We combined the three SSS products to construct input datasets for machine learning. Using the test dataset, the bagged trees model showed the best results (mean $R^2=0.98$ and RMSE = 1.31 psu), and the models that used the SMAP SSS as input had the highest level. For the dataset in the > 31 psu range, all the models exhibited similarly reasonable performances (RMSE = 1.25-1.35 psu). The comparison with in situ SSS data, time series analysis, and the spatial SSS distribution derived from models showed that all the models had proper CDW distributions with reasonable RMSE levels (0.91-1.56 psu). In addition, the CDW front derived from the model gap-free daily SSS product clearly demonstrated the daily oceanic mechanism during the summer season in the ECS at a detailed spatial scale. Notably, the CDW front in the zonal direction, as captured by the Jeodo Ocean Research Station (I-ORS), moved approximately 3.04 km d⁻¹ in 2016, which is very fast compared with the cases in other years. Our model yielded a gap-free gridded daily SSS product with reasonable accuracy and enabled the successful recognition

of daily SSS fronts at the 1 km level, which was previously not possible with ocean color data. Such successful application of machine learning models can further provide useful information on the long-term variation of daily SSS in the ECS. The gridded gap-free SSS dataset at 0.01 degreesx0.01 degrees spatial resolution is freely available at <https://doi.org/10.22808/DATA-2023-2> (Shin et al., 2023).

92. Evaluation of a 4DVAR Assimilation System in the California Current at the SWOT Calibration/Validation Site

Authors: Babette C. Tchonang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Matthew R. Archer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ganesh Gopalakrishnan (Scripps Institution of Oceanography / University of California San Diego); Bruce Cornuelle (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Atmospheric and Oceanic Technology

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Abstract: This study evaluates the feasibility of applying the Estimating the Circulation and Climate of the Ocean (ECCO) data assimilation (DA) framework to a submesoscale-resolving model [Massachusetts Institute of Technology General Circulation Model (MITgcm), at 1-km grid resolution]. This is in preparation for future studies to understand and assimilate the novel swath measurements of sea surface height from the Surface Water and Ocean Topography (SWOT) satellite mission. The model domain is centered at the SWOT calibration/validation (CalVal) site, located 300 km offshore of Monterey Bay, California. We assimilate vertical profiles of temperature and salinity from a linear array of three SWOT prelaunch CalVal moorings, between September and December 2019. Two model solutions are analyzed: 1) a nonassimilating forward simulation termed "first-guess" and 2) an optimized solution that assimilates the in situ observations. Both runs are nested within the global 1/12 degrees Hybrid Coordinate Ocean Model that uses Navy Coupled Ocean Data Assimilation (HYCOM/NCODA) analysis. We evaluate the performance by comparing the two model solutions against assimilated and withheld in situ observations. We show that by assimilating hydrographic data, the model performance over the first-guess solution is improved with an error-to-observation reduction

of 30%-45% for temperature, 14%-41% for salinity, and 44%-56% for steric height. Over the study period, the average steric height error at the three assimilated moorings was 1.27 cm for the optimized solution versus 2.6 cm for the first-guess solution. A comparison to withheld glider observations shows that the optimized solution outperforms the first-guess solution with an error-to-observation reduction of approximately 38% in steric height (1.5 versus 2.4 cm). This study indicates, for the first time, that the MITgcm-ECCO framework can be successfully applied to the reconstruction of submesoscale ocean variability, via the nesting of a high-resolution regional domain into a global outer domain.

93. Observational and Numerical Modeling Constraints on the Global Ocean Biological Carbon Pump

Authors: Scott C. Doney (University of Virginia); Kayla A. Mitchell (University of Virginia, University of California Irvine); Stephanie A. Henson (National Oceanography Centre); Emma Cavan (Imperial College London); Tim DeVries (University of California Santa Barbara); Nicolas Gruber (ETH Zurich); Judith Hauck (Alfred Wegener Institute); Colleen B. Mouw (University of Rhode Island); Jens D. Müller (ETH Zurich); Francois W. Primeau (University of California Irvine)

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Keywords: biogeochemistry, carbon, marine

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Abstract: This study characterized ocean biological carbon pump metrics in the second iteration of the REgional Carbon Cycle Assessment and Processes (RECCAP2) project. The analysis here focused on comparisons of global and biome-scale regional patterns in particulate organic carbon (POC) production and sinking flux from the RECCAP2 ocean biogeochemical model ensemble against observational products derived from satellite remote sensing, sediment traps, and geochemical methods. There was generally good model-data agreement in mean large-scale spatial patterns, but with substantial spread across the model ensemble and observational products. The global-integrated, model ensemble-mean export production, taken as the sinking POC flux at 100 m (6.08 ± 1.17 Pg C yr⁻¹), and export ratio defined as sinking flux divided by net primary production (0.154 ± 0.026) both fell at the lower end of observational estimates. Comparison with observational constraints also suggested that the model ensemble may have underestimated regional biological CO₂ drawdown and air-sea CO₂ flux in high productivity regions. Reasonable model-data agreement was found for global-integrated, ensemble-mean sinking POC flux into the deep ocean at 1,000 m (0.65 ± 0.24 Pg C yr⁻¹) and the transfer efficiency defined as flux at 1,000 m divided by flux at 100 m (0.122 ± 0.041), with both variables exhibiting considerable regional variability. The RECCAP2 analysis presents standard ocean biological carbon pump metrics for assessing biogeochemical model skill, metrics that are crucial for further modeling efforts to resolve remaining uncertainties involving system-level interactions between ocean physics and biogeochemistry.

94. The Eurasian Arctic Ocean along the MOSAiC drift in 2019-2020: An interdisciplinary perspective on physical properties and processes

Authors: Kirstin Schulz (Alfred Wegener Institute, University of Texas at Austin); Zoe Koenig (University of Bergen, Norwegian Polar Institute, Bjerknes Centre for Climate Research, 3UiT The Arctic University of Norway, Tromsø, Norway); Morven Muilwijk (Norwegian Polar Institute); Dorothea Bauch (University of Kiel, GEOMAR Helmholtz Centre for Ocean Research); Clara J. M. Hoppe (Alfred Wegener Institute); Elise S. Droste (Alfred Wegener Institute, University of East Anglia); Mario Hoppmann (Alfred Wegener Institute); Emelia J. Chamberlain (Scripps Institution of Oceanography / University of California San Diego, Woods Hole Oceanographic Institution); Georgi Laukert (University of Bristol, Woods Hole Oceanographic Institution); Tim Stanton (12Oceanography Department, Naval Postgraduate School, Monterey, CA, USA); Alejandra Quintanilla-Zurita (Alfred Wegener Institute); Ilker Fer (University of Bergen, Bjerknes Centre for Climate Research); Céline Heuzé (University of Gothenburg); Salar Karam (University of Gothenburg); Sebastian Mieruch-Schnülle (Alfred Wegener Institute); Till M. Baumann (Institute of Marine Research, University of Bergen, Bjerknes Centre for Climate Research); Myriel Vredenburg (Alfred Wegener Institute); Sandra Tippenhauer (Alfred Wegener Institute); Mats A. Granskog (Norwegian Polar Institute)

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Abstract: The Multidisciplinary drifting Observatory for the Study of Arctic Climate (MOSAIC, 2019-2020), a year-long drift with the Arctic sea ice, has provided the scientific community with an unprecedented, multidisciplinary dataset from the Eurasian Arctic Ocean, covering high atmosphere to deep ocean across all seasons. However, the heterogeneity of data and the superposition of spatial and temporal variability, intrinsic to a drift campaign, complicate the interpretation of observations. In this study, we have compiled a quality-controlled physical hydrographic dataset with best spatio-temporal coverage and derived core parameters, including the mixed layer depth, heat fluxes over key layers, and friction velocity. We provide a comprehensive and accessible overview of the ocean conditions encountered along the MOSAiC drift, discuss their interdisciplinary implications, and compare common ocean climatologies to these new data. Our results indicate that, for the most part, ocean variability was dominated by regional rather than seasonal signals, carrying potentially strong implications for ocean biogeochemistry, ecology, sea ice, and even atmospheric conditions. Near-surface ocean properties were strongly influenced by the relative position of sampling, within or

outside the river-water influenced Transpolar Drift, and seasonal warming and meltwater input. Ventilation down to the Atlantic Water layer in the Nansen Basin allowed for a stronger connectivity between subsurface heat and the sea ice and surface ocean via elevated upward heat fluxes. The Yermak Plateau and Fram Strait regions were characterized by heterogeneous water mass distributions, energetic ocean currents, and stronger lateral gradients in surface water properties in frontal regions. Together with the presented results and core parameters, we offer context for interdisciplinary research, fostering an improved understanding of the complex, coupled Arctic System.

95. Non-Tidal Ocean Loading Signals of the North and Baltic Sea From Terrestrial Gravimetry, GNSS, and High-Resolution Modeling

Authors: C. Voigt (Helmholtz Centre Potsdam); R. Sulzbach (Helmholtz Centre Potsdam, Institute of Meteorology Freie Universität Berlin (FUB) Berlin Germany); H. Dobsław (Helmholtz Centre Potsdam); A. Weise (Institute of Geodesy Leibniz Universität Hannover (LUH) Hannover Germany); L. Timmen (Institute of Geodesy Leibniz Universität Hannover (LUH) Hannover Germany); Z. Deng (Helmholtz Centre Potsdam); M. Reich (Helmholtz Centre Potsdam); N. Stolarczuk (Helmholtz Centre Potsdam); H. Peters (Alfred Wegener Institute); M. Fietz (Alfred Wegener Institute); M. Thomas (Helmholtz Centre Potsdam, Institute of Meteorology Freie Universität Berlin (FUB) Berlin Germany); F. Flechtner (Helmholtz Centre Potsdam, Institute of Geodesy and Geoinformation Science Technische Universität Berlin (TUB) Berlin Germany)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024gl109262

URL: <https://doi.org/10.1029/2024gl109262>

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References: 33

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Leibniz-Gemeinschaft

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Open Access: N/A

Abstract: Non-tidal ocean loading (NTOL) signals are known to be a significant source of geophysically induced noise in gravimetric and geodetic observations also far-away from the coast and especially during extreme events such as storm surges. Operationally available corrections suffer from a low temporal and spatial resolution and reveal too small amplitudes on continental stations. Dedicated high-resolution sea-level modeling of the North and Baltic Sea provides an improved prediction of NTOL signals. Superconducting gravimeter and Global Navigation Satellite Systems observations on the small offshore island of Heligoland in the North Sea are used for an evaluation of the model values revealing largely increased correlations of up to 0.9 and signal reductions of up to 50% during a storm surge period of one month in January and February 2022. Evaluations on additional continental superconducting gravimeter stations also show significant improvements through the recommended high-resolution modeling for improved signal separation further away from the coast.

96. Seasonal Variation of Submesoscale Ageostrophic Motion and Geostrophic Energy Cascade in the Kuroshio

Authors: Zihao Peng (Institute of Marine Science, Shantou University, Shantou 515063, China); Shuwen Zhang (Guangdong Provincial Key Laboratory of Marine Disaster Prediction and Protection, Shantou University, Shantou 515063, China)

Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

DOI: 10.3390/jmse12071121

URL: <https://doi.org/10.3390/jmse12071121>

Publication Date: 2024-07-04

Volume/Issue: Volume 12, Issue 7

References: 50

Cited by: 1 publications

Language: en

Subjects: N/A

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Open Access: N/A

Abstract: The study of submesoscale ageostrophic motion is crucial for enhancing our comprehension of ocean dynamics. This paper employs global sea surface velocity reanalysis data and mixed layer depth data to examine the factors influencing submesoscale ageostrophic energy in the Kuroshio region as well as the energy transition between ageostrophic and geostrophic energy. The findings indicate that submesoscale ageostrophic kinetic energy in the Kuroshio region peaks during winter and spring. Mixed layer depth and geostrophic strain significantly boost ageostrophic kinetic energy, especially in strong current area. Analysis of kinetic energy spectral density reveals how energy distribution and transition scale vary across strong and slow current zones during different seasons, highlighting that submesoscale kinetic energy is susceptible to seasonal variations. In summer and autumn, the transition scale of kinetic energy is generally larger compared to those in spring and winter. Submesoscale ageostrophic motion predominantly gains kinetic energy from the release of available potential energy (APE) and horizontal shear production (HSP) while losing a small portion of its kinetic energy through vertical shear production (VSP) in the Kuroshio.

97. Mitigating Masked Pixels in a Climate-Critical Ocean Dataset

Authors: Angelina Agabin (University of California Santa Cruz); J. Xavier Prochaska (University of California Santa Cruz); Peter C. Cornillon (University of Rhode Island); Christian E. Buckingham (National Oceanography Centre)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs16132439

URL: <https://doi.org/10.3390/rs16132439>

Publication Date: 2024-07-03

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Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA, Simons Foundation Pivot Fellowship

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Open Access: N/A

Abstract: Clouds and other data artefacts frequently limit the retrieval of key variables from remotely sensed Earth observations. We train a natural language processing (NLP)-inspired algorithm with high-fidelity ocean simulations to accurately reconstruct masked or missing data in sea surface temperature (SST) fields--one of 54 essential climate variables identified by the Global Climate Observing System. We demonstrate that the resulting model, referred to as Enki, repeatedly outperforms previously adopted inpainting techniques by up to an order of magnitude in reconstruction error, while displaying exceptional performance even in circumstances where the majority of pixels are masked. Furthermore, experiments on real infrared sensor data with masked percentages of at least 40% show reconstruction errors of less than the known uncertainty of this sensor (root mean square error (RMSE) ≤ 0.1 K). We attribute Enki's success to the attentive nature of NLP combined with realistic SST model outputs--an approach that could be extended to other remotely sensed variables. This study demonstrates that systems built upon Enki--or other advanced systems like it--may therefore yield the optimal solution to mitigating masked pixels in in climate-critical ocean datasets sampling a rapidly changing Earth.

98. Local feedback and ENSO govern decadal changes in variability and seasonal synchronization of the Indian Ocean Dipole

Authors: Hyo-Jin Park; Soon-Il An; Jae-Heung Park; Malte F. Stuecker; Chao Liu; Sang-Wook Yeh

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01525-1

URL: <https://doi.org/10.1038/s43247-024-01525-1>

Publication Date: 2024-07-01

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References: 52

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Research Foundation of Korea

License: CC BY 4.0

Open Access: N/A

Abstract: The Indian Ocean Dipole (IOD), characterized by an interannual fluctuation in zonal dipole pattern of sea surface temperature anomalies along the tropical Indian Ocean, has a large socioeconomic impact on neighboring countries. Here, we investigate the decadal modulations of IOD variability over the last 123 years (1900-2022), by analyzing the observational reanalysis data in conjunction with a low-order IOD model that accounts for both stochastic forcing and the remote impact of ENSO. The observed decadal changes in IOD variability are primarily attributed to the local air-sea coupled feedback, and secondarily to ENSO. The local feedback during the late winter has intensified since the late-1970s due to the Indian Ocean warming, the suppressed westerly winter monsoon in the southeastern Indian Ocean, the shallowing of the mean thermocline in the southeastern Indian Ocean, and the decrease in mean upwelling in the western Indian Ocean. Each of these enhances the convective instability, anomalous evaporative cooling, and oceanic vertical thermal advections, respectively. Intensified local feedback increases the likelihood of the early onset of IOD events in the late winter. Additionally, ENSO, which has strengthened since the mid-twentieth century, has extended the peak phase of IOD into the late fall in recent decades.

99. Explainable El Niño predictability from climate mode interactions

Authors: Sen Zhao; Fei-Fei Jin; Malte F. Stuecker; Philip R. Thompson; Jong-Seong Kug; Michael J. McPhaden; Mark A. Cane; Andrew T. Wittenberg; Wenju Cai

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-024-07534-6

URL: <https://doi.org/10.1038/s41586-024-07534-6>

Publication Date: 2024-06-26

Volume/Issue: Volume 630, Issue 8018

References: 116

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

100. Estimates of the global ocean surface dissolved oxygen and macronutrients from satellite data

Authors: Harish Kumar Kashtan Sundararaman; Palanisamy Shanmugam

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2024.114243

URL: <https://doi.org/10.1016/j.rse.2024.114243>

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Volume/Issue: Volume 311

References: 92

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

101. On the role of seamounts in upwelling deep-ocean waters through turbulent mixing

Authors: Ali Mashayek (University of Cambridge); Jonathan Gula (Institut Universitaire de France, Institut de Recherche pour le Développement (IRD), University of Western Brittany); Lois E. Baker (University of Edinburgh); Alberto C. Naveira Garabato (University of Southampton); Laura Cimoli (University of Cambridge); James J. Riley (University of Washington); Casimir de Lavergne (Institut de Recherche pour le Développement (IRD))

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.2322163121

URL: <https://doi.org/10.1073/pnas.2322163121>

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Volume/Issue: Volume 121, Issue 27

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Language: en

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Keywords: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: Turbulent mixing in the ocean exerts an important control on the rate and structure of the overturning circulation. However, the balance of processes underpinning this mixing is subject to significant uncertainties, limiting our understanding of the overturning's deep upwelling limb. Here, we investigate the hitherto primarily neglected role of tens of thousands of seamounts in sustaining deep-ocean upwelling. Dynamical theory indicates that seamounts may stir and mix deep waters by generating lee waves and topographic wake vortices. At low latitudes, stirring and mixing are predicted to be enhanced by a layered vortex regime in the wakes. Using three realistic regional simulations spanning equatorial to middle latitudes, we show that layered wake vortices and elevated mixing are widespread around seamounts. We identify scalings that relate mixing rate within seamount wakes to topographic and hydrographic parameters. We then apply such scalings to a global seamount dataset and an ocean climatology to show that seamount-generated mixing makes an important contribution to the upwelling of deep waters. Our work thus brings seamounts to the fore of the deep-ocean mixing problem and urges observational, theoretical, and modeling efforts toward incorporating the seamounts' mixing effects in conceptual and numerical ocean circulation models.

102. The Mixed Layer Salinity Balance in the Western Arctic Ocean

Authors: Wenli Zhong (Ocean University of China, Laoshan Laboratory); Youwen Lan (Ocean University of China); Longjiang Mu (Laoshan Laboratory); An T. Nguyen (University of Texas at Austin)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023jc020591

URL: <https://doi.org/10.1029/2023jc020591>

Publication Date: 2024-06-24

Volume/Issue: Volume 129, Issue 6

References: 62

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Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: In this study, we explore the mixed layer salinity (MLS) balance in the western Arctic Ocean based on the Arctic Subpolar gyre sTate Estimate (ASTE) results. The key components of the MLS budgets and their variabilities in response to the Beaufort Gyre (BG) spin-up are identified. Seasonally, the surface forcing (brine rejection plus freshwater input) is the most important dominant contributor to the MLS balance. On the other hand, the entrainment dominates the interannual variability of MLS tendency inside the BG, while the advection dominates that in the Beaufort Sea. The sensitivity test of increased river discharge revealed a greater role of the advection term, along with weakened contributions from the surface forcing and entrainment, in determining the interannual variability of MLS balance. In contrast, the seasonal variabilities remained largely unchanged. The Lagrangian particle tracking reveals that the majority of BG freshwater within the mixed layer exits through the Canadian Archipelago prior to the BG spin-up (2002-2006) and during its relaxation (2012-2017). We found a reduction in mixed layer freshwater sources from the external BG that could feed the gyre during its spin-up (2007-2011), with the major contributions coming from the Beaufort Sea and the BG region itself through Ekman convergence. The mixed layer freshwater pathways are similar in the two versions of ASTE, but with noticeable proportion changes with the increasing river discharge.

103. Coral reef potential connectivity in the southwest Indian Ocean

Authors: Noam S. Vogt-Vincent; April J. Burt; Rosa M. van der Ven; Helen L. Johnson

Journal: Coral Reefs

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00338-024-02521-9

URL: <https://doi.org/10.1007/s00338-024-02521-9>

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References: 80

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

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Open Access: N/A

Abstract: The tropical southwest Indian Ocean is a coral biodiversity hotspot, with remote reefs physically connected by larval dispersal through eddies and a complex set of equatorial and boundary currents. Based on multidecadal, 2 km resolution hydrodynamic and larval dispersal models that incorporate temporal variability in dispersal, we find that powerful zonal currents, current bifurcations, and geographic isolation act as leaky dispersal barriers, partitioning the southwest Indian Ocean into clusters of reefs that tend to consistently retain larvae, and therefore gene flow, over many generations. Whilst exceptionally remote, the Chagos Archipelago can broadcast (and receive) considerable numbers of larvae to (and from) reefs across the wider southwest Indian Ocean, most significantly exchanging larvae with the Inner Islands of Seychelles, but also the Mozambique Channel region. Considering multi-generational dispersal indicates that most coral populations in the southwest Indian Ocean are physically connected within a few hundred steps of dispersal. These results suggest that regional biogeography and population structure can be largely attributed to geologically recent patterns of larval dispersal, although some notable discrepancies indicate that palaeogeography and environmental suitability also play an important role. The model output and connectivity matrices are available in full and will provide useful physical context to regional biogeography and connectivity studies, as well as supporting marine spatial planning efforts.

104. Parameterized Internal Wave Mixing in Three Ocean General Circulation Models

Authors: Nils Brüggemann (Max Planck Institute for Meteorology); Martin Losch (Alfred Wegener Institute); Patrick Scholz (Alfred Wegener Institute); Friederike Pollmann (University of Hamburg); Sergey Danilov (Alfred Wegener Institute); Oliver Gutjahr (Max Planck Institute for Meteorology); Johann Jungclaus (Max Planck Institute for Meteorology); Nikolay Koldunov (Alfred Wegener Institute); Peter Korn (Max Planck Institute for Meteorology); Dirk Olbers (Universität Bremen Bremen Germany, Alfred Wegener Institute); Carsten Eden (University of Hamburg)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023ms003768

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Volume/Issue: Volume 16, Issue 6

References: 114

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: internal waves, parameterization, ocean modeling, vertical mixing

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License: Creative Commons

Open Access: N/A

Abstract: The non-local model of mixing based on internal wave breaking, IDEMIX, is implemented as an enhancement of a turbulent kinetic energy closure model in three non-eddy resolving general circulation ocean models that differ in the discretization and choice of computational grids. In IDEMIX internal wave energy is generated by an energy flux resulting from near-inertial waves induced by wind forcing at the surface, and at the bottom, by an energy flux that parameterizes the transfer of energy between baroclinic and barotropic tides. In all model simulations with IDEMIX, the mixing work is increased compared to the reference solutions without IDEMIX, reaching values in better agreement with finestructure observations. Furthermore, the horizontal structure of the mixing work is more realistic as a consequence of the heterogeneous forcing functions. All models with IDEMIX simulate deeper thermocline depths related to stronger shallow overturning cells in the Indo-Pacific. In the North Atlantic, deeper mixed layers in simulations with IDEMIX are associated with an increased Atlantic overturning circulation and an increase of northward heat transports toward more realistic values. The response of the deep Indo-Pacific overturning circulation and the weak bottom cell of the Atlantic to the inclusion of IDEMIX is incoherent between the models, suggesting that additional unidentified processes and numerical mixing may confound the analysis. Applying different tidal forcing functions leads to simulation differences that are small compared to differences between the different models or between simulations with IDEMIX and without IDEMIX.

105. Continental and oceanic AAM contributions to Chandler Wobble with the amplitude attenuation from 2012 to 2022

Authors: Xue-Qing Xu; Ming Fang; Yong-Hong Zhou; Xin-Hao Liao

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-024-01872-z

URL: <https://doi.org/10.1007/s00190-024-01872-z>

Publication Date: 2024-06-19

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References: 33

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the B-type Strategic Priority Program of the Chinese Academy of Sciences, the National Natural Science Foundation of China, the Youth Innovation Promotion Association of the Chinese Academy of Sciences

License: CC BY 4.0

Open Access: N/A

Abstract: We reconstructed the Chandler Wobble (CW) from 1962 to 2022 by combining the Eigen-oscillator excited by geophysical fluids of atmospheric and oceanic angular momentums (AAM and OAM). The mass and motion terms for the AAM are further divided with respect to the land and ocean domains. Particular attention is placed on the time span from 2012 to 2022 in relation to the observable reduction in the amplitude of the CW. Our research indicates that the main contributor to the CW induced by AAM is the mass term (i.e., the pressure variations over land). Moreover, the phase of the AAM-induced CW remains relatively stable during the interval of 1962-2022. In contrast, the phase of the OAM-induced CW exhibits a periodic variation with a cycle of approximately 20 years. This cyclic variation would modulate the overall amplitude of the CW. The noticeable amplitude deduction over the past ten years can be attributed to the evolution of the CW driven by AAM and OAM, toward a state of cancellation. These findings further reveal that the variation in the phase difference between the CW forced by AAM and OAM, may be indicative of changes in the interaction between the solid Earth, atmosphere, and ocean.

106. Observed change and the extent of coherence in the Gulf Stream system

Authors: Helene Asbjørnsen; Tor Eldevik; Johanne Skrefsrud; Helen L. Johnson; Alejandra Sanchez-Franks

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-20-799-2024

URL: <https://doi.org/10.5194/os-20-799-2024>

Publication Date: 2024-06-18

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References: 116

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . By transporting warm and salty water poleward, the Gulf Stream system maintains a mild climate in northwestern Europe while also facilitating the dense water formation that feeds the deep ocean. The sensitivity of North Atlantic circulation to future greenhouse gas emissions seen in climate models has prompted an increasing effort to monitor the various ocean circulation components in recent decades. Here, we synthesize available ocean transport measurements from several observational programs in the North Atlantic and Nordic Seas, as well as an ocean state estimate (ECCOV4-r4), for an enhanced understanding of the Gulf Stream and its poleward extensions as an interconnected circulation system. We see limited coherent variability between the records on interannual timescales, highlighting the local oceanic response to atmospheric circulation patterns and variable recirculation timescales within the gyres. On decadal timescales, we find a weakening subtropical circulation between the mid-2000s and mid-2010s, while the inflow and circulation in the Nordic Seas remained stable. Differing decadal trends in the subtropics, subpolar North Atlantic, and Nordic Seas warrant caution in using observational records at a single latitude to infer large-scale circulation change.

107. A multi-level parallel approach to increase the computation efficiency of a global ocean temperature dataset reconstruction

Authors: Huifeng Yuan; Lijing Cheng; Yuying Pan; Zhetao Tan; Qian Liu; Zhong Jin

Journal: Journal of Parallel and Distributed Computing

Publisher: Elsevier BV

DOI: 10.1016/j.jpdc.2024.104938

URL: <https://doi.org/10.1016/j.jpdc.2024.104938>

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Volume/Issue: Volume 192

References: 39

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

108. New insights into the eastern subpolar North Atlantic meridional overturning circulation from OVIDE

Authors: Herlé Mercier; Damien Desbruyères; Pascale Lherminier; Antón Velo; Lidia Carracedo; Marcos Fontela; Fiz F. Pérez

Journal: Ocean Science

Publisher: Copernicus GmbH

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020, Centre National de la Recherche Scientifique, Ministerio de Ciencia, Innovación y Universidades

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Open Access: N/A

Abstract: . The Atlantic Meridional Overturning Circulation (AMOC) is a key component of the Earth's climate. However, there are few long time series of observations of the AMOC, and the study of the mechanisms driving its variability depends mainly on numerical simulations. Here, we use four ocean circulation estimates produced by different data-driven approaches of increasing complexity to analyse the seasonal to decadal variability of the subpolar AMOC across the Greenland-Portugal OVIDE (Observatoire de la Variabilité Interannuelle à DÉcennale) line since 1993. We decompose the MOC strength variability into a velocity-driven component due to circulation changes and a volume-driven component due to changes in the depth of the overturning maximum isopycnal. We show that the variance of the time series is dominated by seasonal variability, which is due to both seasonal variability in the volume of the AMOC limbs (linked to the seasonal cycle of density in the East Greenland Current) and to seasonal variability in the transport of the Eastern Boundary Current. The decadal variability of the subpolar AMOC is mainly caused by changes in velocity, which after the mid-2000s are partly offset by changes in the volume of the AMOC limbs. This compensation means that the decadal variability of the AMOC is weaker and therefore more difficult to detect than the decadal variability of its velocity-driven and volume-driven components, which is highlighted by the formalism that we propose.

109. Reassessing the contributions of terrestrial waters to sea level variations in the South China Sea and its response to alternating ENSO events

Authors: Pengfei Yang; Hok Sum Fok; Zhongtian Ma

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-024-1344-8

URL: <https://doi.org/10.1007/s11430-024-1344-8>

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Volume/Issue: Volume 67, Issue 7

References: 57

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

110. Drivers of Seasonal Land-Ice-Flow Variability in the Antarctic Peninsula

Authors: Karla Boxall (University of Cambridge); Frazer D. W. Christie (Airbus Defence & Space Ltd. Newcastle upon Tyne UK); Ian C. Willis (University of Cambridge); Jan Wuite (ENVEO IT GmbH Innsbruck Austria); Thomas Nagler (ENVEO IT GmbH Innsbruck Austria); Stefan Scheiblauer (ENVEO IT GmbH Innsbruck Austria)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Volume/Issue: Volume 129, Issue 6

References: 123

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: seasonal, drivers, velocity, SAR, Antarctica (and 1 more)

Funding: Prince Albert II of Monaco Foundation

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Open Access: N/A

Abstract: Land-ice flow in Antarctica has experienced multi-annual acceleration in response to increased rates of ice thinning, ice-shelf collapse and grounding-line retreat. Superimposed upon this trend, recent observations have revealed that land-ice flow in the Antarctic Peninsula exhibits seasonal velocity variability with distinct summertime speed-ups. The mechanism, or mechanisms, responsible for driving this seasonality are unconstrained at present, yet detailed, process-based understanding of such forcing will be important for accurately estimating Antarctica's future contributions to sea level. Here, we perform time-series analysis on an array of remotely sensed, modeled and reanalysis data sets to examine the influence of potential drivers of ice-flow seasonality in the Antarctic Peninsula. We show that both meltwater presence and ocean temperature act as statistically significant precursors to summertime ice-flow acceleration, although each elicits an ice-velocity response after a distinct lag, with the former prompting a more immediate response. Furthermore, we find that the timing and magnitude of these local drivers are influenced by large-scale climate phenomena, namely the Amundsen Sea Low and the El Niño Southern Oscillation, with the latter initiating an anomalous wintertime ice-flow acceleration event in 2016. This hitherto unidentified link between seasonal ice flow and large-scale climatic forcing may have important implications for ice discharge at and beyond the Antarctic Peninsula in the future, depending upon how the magnitude, frequency and duration of such climate phenomena evolve in a warming world.

111. Response of sea surface temperature to atmospheric rivers

Authors: Tien-Yiao Hsu; Matthew R. Mazloff; Sarah T. Gille; Mara A. Freilich; Rui Sun; Bruce D. Cornuelle

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-024-48486-9

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Volume/Issue: Volume 15, Issue 1

References: 51

Cited by: 2 publications

Language: en

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Keywords: N/A

Funding: Department of Water Resources, Department of Water Resources, Department of Water Resources

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Open Access: N/A

Abstract: Atmospheric rivers (ARs), responsible for extreme weather conditions, are mid-latitude systems that can cause significant damage to coastal areas. While forecasting ARs beyond two weeks remains a challenge, past research suggests potential benefits may come from properly accounting for the changes in sea surface temperature (SST) through air-sea interactions. In this paper, we investigate the impact of ARs on SST over the North Pacific by analyzing 25 years of ocean reanalysis data using an SST budget equation. We show that in the region of strong ocean modification, ocean dynamics can offset over 100% of the anomalous SST warming that would otherwise arise from atmospheric forcing. Among all ocean processes, ageostrophic advection and vertical mixing (diffusion and entrainment) are the most important factors in modifying the SST tendency response. The SST tendency response to ARs varies spatially. For example, in coastal California, the driver of enhanced SST warming is the reduction in ageostrophic advection due to anomalous southerly winds. Moreover, there is a large region where the SST shows a warming response to ARs due to the overall reduction in the total clouds and subsequent increase in total incoming shortwave radiation.

112. Multidisciplinary characterisation of the biodiversity, geomorphology, oceanography and glacial history of Bowditch Seamount in the Sargasso Sea

Authors: Lea-Anne Henry; Igor Yashayaev; Claude Hillaire-Marcel; F. Javier Murillo; Ellen Kenchington; Struan Smith; Jenny Maccali; Jill Bourque; Louis L. Whitcomb; J. Murray Roberts

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2024.104342

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References: 141

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

113. Diurnal Variability of Mixed Layer Overturning Instabilities From Glider Array Observations in the South China Sea

Authors: Ruixi Zheng (Chinese Academy of Sciences); Zhiyou Jing (Chinese Academy of Sciences)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl107694

URL: <https://doi.org/10.1029/2023gl107694>

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Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: diurnal variability, overturning instability, glider observation, nighttime surface cooling

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Abstract: The diurnal variability of mixed layer (ML) overturning instabilities remains poorly understood due to the challenge in capturing their rapid evolutions across large spatiotemporal ranges. Using high-resolution data from 52 gliders in the South China Sea, we examine the diurnal modulations of ML overturning instabilities. The results of the 3-month field observation show that negative potential vorticity occupies ~16% of the ML and facilitates several types of forced overturning instabilities, especially symmetric instability (SI). Surface heat fluxes are identified to primarily modulate the diurnal variability of these overturning cells, where nighttime surface cooling is found to energize SI with an ~2-hr phase lag. As a result, over 60% of forced submesoscale overturning cells tend to restratify the ML at night. These findings quantitatively highlight the modulation of diabatic atmospheric forcing in submesoscale restratification, which should be considered in submesoscale parameterizations of ocean and climate models.

114. Seasonal dynamics of global marine heatwaves over the last four decades

Authors: Yishan Wang; Yuntao Zhou

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2024.1406416

URL: <https://doi.org/10.3389/fmars.2024.1406416>

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Language: N/A

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Marine heatwaves (MHWs), prolonged periods of abnormally high sea temperature, have greater devastating impacts on marine ecosystem services and socioeconomic systems than gradual long-term ocean warming. Despite growing evidence of increases in MHW frequency, duration, and intensity, their interseasonal variations remain unclear. Using satellite-derived daily sea surface temperature (SST) data from 1982 to 2022, this work reveals a strong seasonality in MHWs. Typically, the highest cumulative intensity, characterizing total impacts on ecosystems, occurs during the local warm seasons in most oceans, leading to a significant interseasonal difference between warm and cold seasons. The interseasonal difference is predominantly driven by air-sea heat flux, rather than oceanic horizontal advection and vertical process. An increase in these interseasonal differences is observed in mid and high latitudes, with a significant increase in the warm season and a weaker trend in the cold season. In the Equatorial Pacific and Western Equatorial Indian Ocean, intense MHWs are primarily exacerbated by the El Niño-Southern Oscillation (ENSO), which also determines interseasonal variations in MHWs. Understanding the seasonality of MHWs can help better formulate corresponding policies to reduce economic and ecological losses caused by these events and can improve the accuracy of future predictions.

115. Submesoscale Vertical Heat Transport at the Kuroshio Extension: Characteristics and Mechanisms

Authors: Yang Chen (Shanghai Jiao Tong University, Tianjin Key Laboratory for Marine Environmental Research and Service School of Marine Science and Technology Tianjin University Tianjin China); Ru Chen (Shanghai Jiao Tong University, Tianjin Key Laboratory for Marine Environmental Research and Service School of Marine Science and Technology Tianjin University Tianjin China); Guihua Wang (Fudan University); Xiaozhou Ruan (Boston University); Wei Li (Tianjin Key Laboratory for Marine Environmental Research and Service School of Marine Science and Technology Tianjin University Tianjin China); Qianqian Geng (Shanghai Jiao Tong University, Tianjin Key Laboratory for Marine Environmental Research and Service School of Marine Science and Technology Tianjin University Tianjin China); Junyi Wang (Shanghai Jiao Tong University, Tianjin Key Laboratory for Marine Environmental Research and Service School of Marine Science and Technology Tianjin University Tianjin China)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: causality analysis, filter, Kuroshio Extension, submesoscale vertical heat transport, advection

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Submesoscale processes significantly influence the air-sea interaction, heat budget and the distribution of physical and biogeochemical tracers in the upper-ocean. We study the spatiotemporal characteristics and generation mechanisms of submesoscale vertical heat transport (SVHT) at the Kuroshio Extension using a submesoscale-permitting simulation. Compared with the commonly used Boxcar and Hanning filters, the clean-cut feature of the Butterworth filter in the wavenumber domain makes it a proper filter to diagnose SVHT. SVHT is systematically upward, peaking in winter. Through causality analysis, we find that, among the basic factors (mixed layer depth, strain rate, surface wind stress, and horizontal buoyancy gradient) from the conventional submesoscale generation mechanism scalings, mixed layer depth plays the leading role in modulating the intraseasonal variation of SVHT. Using the budget of submesoscale temperature variance and causality analysis, we find that advection also plays an important role in regulating SVHT. This work suggests that the choice of appropriate spatial filter is important for SVHT diagnosis and including advection may help improve submesoscale parameterization schemes.

116. A Scale-Dependent Analysis of the Barotropic Vorticity Budget in a Global Ocean Simulation

Authors: Hemant Khatri (Princeton University, University of Liverpool); Stephen M. Griffies (Princeton University, NOAA Geophysical Fluid Dynamics Laboratory Princeton NJ USA); Benjamin A. Storer (Department of Mechanical Engineering University of Rochester Rochester NY USA); Michele Buzzicotti (Department of Physics and INFN University of Rome Tor Vergata Roma Italy); Hussein Aluie (Department of Mechanical Engineering University of Rochester Rochester NY USA, Laboratory for Laser Energetics University of Rochester Rochester NY USA); Maike Sonnewald (Princeton University, NOAA Geophysical Fluid Dynamics Laboratory Princeton NJ USA); Raphael Dussin (NOAA Geophysical Fluid Dynamics Laboratory Princeton NJ USA); Andrew Shao (Canadian Centre for Climate Modelling and Analysis)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Abstract: The climatological mean barotropic vorticity budget is analyzed to investigate the relative importance of surface wind stress, topography, planetary vorticity advection, and nonlinear advection in dynamical balances in a global ocean simulation. In addition to a pronounced regional variability in vorticity balances, the relative magnitudes of vorticity budget terms strongly depend on the length-scale of interest. To carry out a length-scale dependent vorticity analysis in different ocean basins, vorticity budget terms are spatially coarse-grained. At length-scales greater than 1,000 km, the dynamics closely follow the Topographic-Sverdrup balance in which bottom pressure torque, surface wind stress curl and planetary vorticity advection terms are in balance. In contrast, when including all length-scales resolved by the model, bottom pressure torque and nonlinear advection terms dominate the vorticity budget (Topographic-Nonlinear balance), which suggests a prominent role of oceanic eddies, which are of km in size, and the associated bottom pressure anomalies in local vorticity balances at length-scales smaller than 1,000 km. Overall, there is a transition from the Topographic-Nonlinear regime at scales smaller than 1,000 km to the Topographic-Sverdrup regime at length-scales greater than 1,000 km. These dynamical balances hold across all ocean basins; however, interpretations of the dominant vorticity balances depend on the level of spatial filtering or the effective model resolution. On the other hand, the contribution of bottom and lateral friction terms in the barotropic vorticity budget remains small and is significant only near sea-land boundaries, where bottom stress and horizontal viscous

friction generally peak.

117. Surface and Sub-Surface Kinetic Energy Wavenumber-Frequency Spectra in Global Ocean Models and Observations

Authors: Joseph K. Ansong (University of Michigan, Now at: Department of Mathematics University of Ghana Accra Ghana); Brian K. Arbic (University of Michigan); Arin D. Nelson (Naval Undersea Warfare Center Newport RI USA, University of Michigan); Matthew H. Alford (Scripps Institution of Oceanography / University of California San Diego); Eric Kunze (NorthWest Research Associates Redmond WA USA); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anna C. Savage (Running Tide Portland ME USA, University of Michigan); Jay F. Shriver (Ocean Dynamics and Prediction Branch, Code 7323, Naval Research Laboratory Stennis Space Center Hancock County, MS USA); Alan J. Wallcraft (Florida State University); Maarten C. Buijsman (Department of Marine Science University of Southern Mississippi Stennis Space Center Hancock County, MS USA)

Journal: Journal of Geophysical Research: Oceans

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Subjects: N/A

Keywords: ratio of high-versus low-frequency surface kinetic energy, vertical wavenumber spectra of internal gravity

Funding: National Science Foundation, Office of Naval Research, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: This paper examines spectra of horizontal kinetic energy (HKE) in the surface and sub-surface ocean, with an emphasis on internal gravity wave (IGW) motions, in global high-resolution ocean simulations. Horizontal wavenumber-frequency spectra of surface HKE are computed over seven oceanic regions from two global simulations of the HYbrid Coordinate Ocean Model (HYCOM) and three global simulations of the Massachusetts Institute of Technology general circulation model (MITgcm). In regions with high IGW activity, high surface HKE variance in the horizontal wavenumber-frequency spectra is aligned along IGW linear dispersion curves. For both HYCOM and MITgcm, and in almost all regions, finer horizontal resolution yields more energetic supertidal IGW continuum spectra. The ratio of high-horizontal-wavenumber variance in semi-diurnal and supertidal motions relative to lower-frequency motions, a quantity of great interest for swath altimetry, depends on the model employed and the horizontal resolution within the model, implying that quantitative predictions of the partition between low- and high-frequency motions taken from particular simulations should be treated with care. The frequency-vertical wavenumber spectra, frequency spectra, and vertical wavenumber spectra from the models are compared to spectra computed from McLane profilers at nine locations. In general, MITgcm spectra match the McLane profiler spectra more closely

at high frequencies ($|\omega| > 4.5$ cpd). In both models, vertical wavenumber spectra roll off more steeply than observations at high vertical wavenumbers ($m > 10^{-2}$ cpm). The vertical wavenumber spectra in such models is an important target for improvement, due to turbulence production and dissipation that takes place at high vertical wavenumbers.

118. Biases in ice sheet models from missing noise-induced drift

Authors: Alexander A. Robel; Vincent Verjans; Aminat A. Ambelorun

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . Most climatic and glaciological processes exhibit internal variability, which is omitted from many ice sheet model simulations. Prior studies have found that climatic variability can change ice sheet sensitivity to the long-term mean and trend in climate forcing. In this study, we use an ensemble of simulations with a stochastic large-scale ice sheet model to demonstrate that variability in frontal ablation of marine-terminating glaciers changes the mean state of the Greenland Ice Sheet through noise-induced drift. Conversely, stochastic variability in surface mass balance does not appear to cause noise-induced drift in these ensembles. We describe three potential causes for noise-induced drift identified in prior statistical physics literature: noise-induced bifurcations, multiplicative noise, and nonlinearities in noisy processes. Idealized simulations and Reynolds decomposition theory show that for marine ice sheets in particular, noise-induced bifurcations and nonlinearities in variable ice sheet processes are likely the cause of the noise-induced drift. We argue that the omnipresence of variability in climate and ice sheet systems means that the state of real-world ice sheets includes this tendency to drift. Thus, the lack of representation of such noise-induced drift in spin-up and transient ice sheet simulations is a potentially ubiquitous source of bias in ice sheet models.

119. Tropical Deep Convection, Cloud Feedbacks and Climate Sensitivity

Authors: Graeme L. Stephens; Kathleen A. Shiro; Maria Z. Hakuba; Hanii Takahashi; Juliet A. Pilewskie; Timothy Andrews; Claudia J. Stubenrauch; Longtao Wu

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

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Open Access: N/A

Abstract: This paper is concerned with how the diabatically-forced overturning circulations of the atmosphere, established by the deep convection within the tropical trough zone (TTZ), first introduced by Riehl and (Malkus) Simpson, in *Contr Atmos Phys* 52:287-305 (1979), fundamentally shape the distributions of tropical and subtropical cloudiness and the changes to cloudiness as Earth warms. The study first draws on an analysis of a range of observations to understand the connections between the energetics of the TTZ, convection and clouds. These observations reveal a tight coupling of the two main components of the diabatic heating, the cloud component of radiative heating, shaped mostly by high clouds formed by deep convection, and the latent heating associated with the precipitation. Interannual variability of the TTZ reveals a marked variation that connects the depth of the tropical troposphere, the depth of convection, the thickness of high clouds and the TOA radiative imbalance. The study examines connections between this convective zone and cloud changes further afield in the context of CMIP6 model experiments of climate warming. The warming realized in the CMIP6 SSP5-8.5 scenario multi-model experiments, for example, produces an enhanced Hadley circulation with increased heating in the zone of tropical deep convection and increased radiative cooling and subsidence in the subtropical regions. This impacts low cloud changes and in turn the model warming response through low cloud feedbacks. The pattern of warming produced by models, also influenced by convection in the tropical region, has a profound influence on the projected global warming.

120. Drivers of Interannual Salinity Variability in the Arctic Ocean

Authors: Antoine Hochet (Institut de Recherche pour le Développement (IRD)); Camille Lique (Institut de Recherche pour le Développement (IRD)); Florian Sévellec (Odyssey Team-project INRIA CNRS Paris France, Institut de Recherche pour le Développement (IRD)); William Llovel (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: salinity, interannual, mechanisms, Arctic

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Open Access: N/A

Abstract: Accurate projections and attribution of Arctic Ocean changes in climate models require a good understanding of the mechanisms underlying interannual salinity variability in the region. Although some mechanisms have been extensively studied in idealized setting, in particular for the dynamics of the Beaufort gyre (BG), it remains unclear how applicable they are to more complex systems. This study introduces a new diagnostic based on salinity variance budget to robustly assess the mechanisms of salinity variations. The diagnostic is then applied to the "Estimating the Circulation and Climate of the Ocean" state estimate. Results indicate that the advection of salinity anomaly in the direction of the mean salinity gradient made by velocity anomalies is the primary source of interannual salinity variability. These velocities are primarily attributed to fluctuating winds via Ekman transports. Fluctuating surface freshwater fluxes from the atmosphere and sea ice are the second most important source of variability and cannot be neglected. The two sinks of interannual salinity variance are associated with the erosion of large scale gradients of the mean circulation by eddies and to a lesser extent to the diffusive terms. Over continental shelves, particularly over the East Siberian Shelf (ESS), ocean surface freshwater fluxes and diffusion play a more important role than in the deep basins. We also report a strong intensification of all sources and sinks of interannual salinity variability in the BG and an opposite weakening in the ESS in the second decade of the analysis (2004-2014) with respect to the first (1993-2003).

121. Model validation and applications of wave and current forecasts from the Hong Kong Observatory's Operational Marine Forecasting System

Authors: Wai Kong; Ching-chi Lam; Dick-shum Lau; Chi-kin Chow; Sze-ning Chong; Pak-wai Chan; Ngo-ching Leung

Journal: Ocean Modelling

Publisher: Elsevier BV

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Open Access: N/A

Abstract: N/A

122. KaRIn, the Ka-Band Radar Interferometer of the SWOT Mission: Design and in-Flight Performance

Authors: Eva Peral (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Daniel Esteban-Fernández (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ernesto Rodríguez (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dalia McWatters (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jan-Willem De Bleser (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Razi Ahmed (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Albert C. Chen (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Slimko (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ruwan Somawardhana (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Knarr (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael Johnson (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Sermak Jaruwatanadilok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Samuel Chan (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Xiaoqing Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Duane Clark (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kenneth Peters (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Curtis W. Chen (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Peter Mao (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Behrouz Khayatian (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jacqueline Chen (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Richard E. Hodges (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dhemetrios Boussalis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bryan Stiles (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Karthik Srinivasan (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Keywords: radar interferometry, surface water, Ocean circulation

Funding: Jet Propulsion Laboratory, California Institute of Technology, through the National Aeronautics and Space Administration

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Open Access: N/A

Abstract: N/A

123. Simulated Sea Surface Salinity Data from a 1/48 degrees Ocean Model

Authors: Frederick M. Bingham; Séverine Fournier; Susannah Brodnitz; Akiko Hayashi; Mikael Kuusela; Elizabeth Westbrook; Karly M. Ulfsax Carlin; Cristina González-Haro; Verónica González-Gambau

Journal: Scientific Data

Publisher: Springer Science and Business Media LLC

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References: 69

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Keywords: N/A

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Abstract: To study the validation process for sea surface salinity (SSS) we have generated one year (November 2011- October 2012) of simulated satellite and in situ "ground truth" data. This was done using the ECCO (Estimating the Circulation and Climate of the Oceans) 1/48 degrees simulation, the highest resolution global ocean model currently available. The ground tracks of three satellites, Aquarius, SMAP (Soil Moisture Active Passive) and SMOS (Soil Moisture and Ocean Salinity) were extracted and used to sample the model with a gaussian weighting similar to that of the spaceborne sensor ground footprint. This produced simulated level 2 (L2) data. Simulated level 3 (L3) data were then produced by averaging L2 data onto a regular grid. The model was sampled to produce simulated Argo and tropical mooring SSS datasets. The Argo data were combined into a simulated gridded monthly 1 degrees Argo product. The simulated data produced from this effort have been used to study sampling errors, matchups, subfootprint variability and the validation process for SSS at L2 and L3.

124. The spatiotemporal characteristics and driving mechanisms of subsurface marine heatwaves in the Xisha Region

Authors: Feng Gao; Cong Liu; Fangguo Zhai; Jinbao Song; Peiliang Li; Yanzhen Gu

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2024.1399096

URL: <https://doi.org/10.3389/fmars.2024.1399096>

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Open Access: N/A

Abstract: Under the background of global climate change, Subsurface Marine Heatwaves (SSMHWs) have become a 'hot-spot' research due to their significant impacts on marine ecosystems. Temperature data from the ECCO2 for the years 1992 to 2021 is used to research the spatiotemporal characteristics of SSMHWs in the upper 500 m over the Xisha region (110 degreesE to 113 degreesE, 15 degreesN to 18 degreesN). This study indicates that SSMHWs of high intensity occur in the Xisha area, with the vertical maximum intensity at approximately 100 m. These events exhibit significant seasonal variations, with the highest intensity occurring in May. The intensity of SSMHW in the analysis region varies interannually. SSMHWs have intensified at a rate of 1.53 degreesC·Days/Year and tend to shift to deeper water over the past three decades. A block-based method for SSMHW identification is proposed, in order to take the vertical extent of the event in consideration. By analyzing the eight most intense events based on spatial cumulative intensity, it suggested that warm mesoscale eddies may play an important role on the spatial distribution of the SSMHWs. Statistical analysis shows that the intensity and coverage of the warm mesoscale eddy in the Xisha zone may influence the intensity and evolution of the SSMHWs. Additionally, only 18.27% of SSMHW events occur with apparent surface marine heatwave signals. These findings are vital for uncovering SSMHW dynamics in the Xisha area, and are important for future monitoring, early warnings and marine conservation.

125. Geolocated fish spawning habitats

Authors: Kimberly L. Oremus; James Rising; Nandini Ramesh; Audrey J. Ostroski

Journal: Scientific Data

Publisher: Springer Science and Business Media LLC

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Abstract: Fish spawning locations are a crucial input into fisheries management and conservation plans, and many stocks are especially sensitive to the environmental conditions within these localized zones. Globally collated data on spawning locations across many species has been unavailable, hindering global stock assessments and analyses of sustainable development and global environmental change. To address this, we created a geocoded fish spawning dataset using qualitative spawning information from FishBase and Science and Conservation of Fish Aggregations (SCRFA). We cleaned and geocoded the spawning locations of 1,045 marine fish species into 2,931 regions. Each spawning region is defined by one or more polygons, and most spawning regions are associated with spawning months. The resulting dataset covers oceans globally. This dataset will be useful to scientists studying marine fish population dynamics and their interactions with the physical environment on regional to large scales.

126. A key role of off-equatorial subsurface temperature anomalies in Tropical Pacific Decadal Variability

Authors: Sieu-Cuong San; Yu-Heng Tseng; Ruiqiang Ding; Emanuele Di Lorenzo

Journal: npj Climate and Atmospheric Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41612-024-00643-z

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References: 86

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Language: en

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Funding: N/A

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Abstract: We demonstrate the key role of off-equatorial subsurface temperature anomalies in driving the phase transition of Tropical Pacific Decadal Variability (TPDV) using observation and model experiments. During the positive phase of TPDV, anomalous atmospheric responses in the off-equatorial northwestern Pacific induce positive Ekman pumping. The resulting negative subsurface temperature anomaly generated then propagates along the North Equatorial Countercurrent pathway towards the central basin, causing a sign reversal of the equatorial sea-surface temperature anomalies around three years later. Moreover, the positive phase of TPDV possibly changes the state of the Kuroshio Extension through tropical-extratropical interaction, which subsequently projects onto the footprint of the Pacific Meridional Mode, thereby amplifying subsurface-produced disturbance 0-12 months before the cold peak phase. The cold phase is completely established after five years. Similarly, the same dynamic applies to the reversed phase, leading to a preferred decadal oscillation driven by off-equatorial subsurface temperature anomalies and extratropical-tropical ocean-atmosphere interaction.

127. A likely role for stratification in long-term changes of the global ocean tides

Authors: Lana Opel; Michael Schindelegger; Richard D. Ray

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01432-5

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Open Access: N/A

Abstract: Stratification--that is, the vertical change in seawater density--exerts a subtle control on the energetics and thus the surface elevation of barotropic (depth independent) flows in the ocean. Changes in stratification therefore provide a plausible pathway to explain some of the puzzling trends in ocean tides evident in tide gauge and, more recently, satellite altimetry data. Using a three-dimensional global ocean model, we estimate that strengthening of stratification between 1993 and 2020 caused open-ocean trends of order 0.1 mm yr⁻¹ in the barotropic M2 tide, similar in structure and magnitude to long-term M2 changes deduced from satellite altimetry. Amplitude trends are predominantly negative, implying enhanced energy transfer to internal tides since the 1990s. Effects of stratification are also a relevant forcing of contemporary M2 trends at the coast, where they may modulate or even overprint the tidal response to sea level rise. Salient examples for such significant near-shore influence of stratification ($\geq 95\%$ confidence) include the Northwest Australian Shelf (- 0.5 mm yr⁻¹) and the coasts of western North America (- 0.1 mm yr⁻¹), commensurate with observed M2 amplitude trends at tide gauges.

128. Global Ocean pCO₂ Variation Regimes: Spatial Patterns and the Emergence of a Hybrid Regime

Authors: Yiming Guo (Yale Center for Natural Carbon Capture New Haven CT USA, Department of Earth & Planetary Sciences Yale University New Haven CT USA); Mary-Louise Timmermans (Yale Center for Natural Carbon Capture New Haven CT USA, Department of Earth & Planetary Sciences Yale University New Haven CT USA)

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License: N/A

Open Access: N/A

Abstract: The variability in ocean surface partial pressure of carbon dioxide (pCO₂), driven by both physical and biological processes, substantially influences air-sea carbon exchange. It is widely recognized that the pCO₂ variation at a specific ocean location is primarily dominated by either thermal or nonthermal effects. However, the distinct pCO₂ variation regimes, their global distribution, and the mechanisms underlying such patterns have yet to be fully determined due to a paucity of global observations. Through the use of observation-based products and an eddy-resolving ocean simulation, this study demonstrates the presence of three distinct regimes in the global ocean: one in which sea surface temperature (SST) is the dominant control on pCO₂ variations; another in which dissolved inorganic carbon (DIC) is the primary control; and a third previously uncharacterized hybrid regime where pCO₂ variations are governed by seasonally-varying factors. This hybrid regime is generally located between SST- and DIC-dominated regimes and occupies approximately 15% of the global ocean. The regimes broadly exist in zonal bands that are closely linked to the relative strengths of SST and DIC variances. Seasonally-varying mixed-layer depth, mesoscale variability (quantified in terms of eddy kinetic energy), and biological processes modulate local pCO₂ variations and play significant roles in shaping the global pattern of regime distributions. Understanding the distribution of pCO₂ regimes, including the hybrid regime revealed in this study, as well as their different oceanic drivers, is essential for future predictions of ocean carbon uptake in response to global warming.

129. Advection surface-flux balance controls the seasonal steric sea level amplitude

Authors: Antoine Hochet; William Llovel; Thierry Huck; Florian Sévellec

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-024-61447-y

URL: <https://doi.org/10.1038/s41598-024-61447-y>

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References: 47

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency

License: CC BY 4.0

Open Access: N/A

Abstract: Along with the mean sea level rise due to climate change, the sea level exhibits natural variations at a large number of different time scales. One of the most important is the one linked with the seasonal cycle. In the Northern Hemisphere winter, the sea level is as much as 20 cm below its summer values in some locations. It is customary to associate these variations with the seasonal cycle of the sea surface net heat flux which drives an upper-ocean thermal expansion creating a positive steric sea level anomaly. Here, using a novel framework based on steric sea level variance budget applied to observations and to the Estimating the Circulation and Climate of the Ocean state estimate, we demonstrate that the steric sea level seasonal cycle amplitude results from a balance between the seasonal sea surface net heat flux and the oceanic advective processes. Moreover, for up to 50% of the ocean surface, surface heat fluxes act to damp the seasonal steric sea level cycle amplitude, which is instead forced by oceanic advection processes. We also show that eddies play an important role in damping the steric sea level seasonal cycle. Our study contributes to a better understanding of the steric sea level mechanisms which is crucial to ensure accurate and reliable climate projections.

130. Impacts of central-Pacific El Niño and physical drivers on eastern Pacific bigeye tuna

Authors: Peng Lian; Le Gao

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-023-3051-3

URL: <https://doi.org/10.1007/s00343-023-3051-3>

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

131. Seasonality and potential generation mechanisms of submesoscale processes in the northern Bay of Bengal

Authors: Yifei Zhou; Wei Duan; Haijin Cao; Guidi Zhou; Rong Cui; Xuhua Cheng

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2024.104318

URL: <https://doi.org/10.1016/j.dsr.2024.104318>

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References: 78

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Hohai University, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

132. Baroclinic Rossby Waves With Phase Lag Cause Seasonal Upward-Propagating Signals in the Mid-Depth Equatorial Pacific Ocean

Authors: Zhengpu Luo (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Dezhou Yang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Lingjing Xu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Yuxuan Li (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Hang Zhang (Chinese Academy of Sciences); Jianing Wang (Chinese Academy of Sciences); Baoshu Yin (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023jc020418

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Publication Date: 2024-04-29

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References: 42

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: Based on OFES outputs verified by mooring observations, the seasonal characteristics in the middepth (1,000-3,000 m) equatorial Pacific Ocean are investigated in detail. The seasonal upward-propagating signals, consisting of one positive and one negative anomaly, are identified at the equator. The harmonic analyses indicate that the seasonal variations in the middepth equatorial Pacific Ocean originate from the downward-propagating energy dominated by the first meridional modes of Rossby waves. The superposition of first and second baroclinic modes of Rossby waves could reproduce the seasonal variations. Furthermore, a series of superposition experiments show that the phase lag between the two modes needs to be in the range of 0 to π to cause upward phase propagation. It implies that the baroclinic modes in the seasonal variations may not be generated simultaneously so that the Rossby waves with specific phase lag can cause upward-propagating signals in the middepth equatorial Pacific Ocean. This new finding will enhance the understanding of seasonal variations in the middepth equatorial Pacific Ocean.

133. Wind-Driven Seasonal Variability of Deep-Water Overflow From the Pacific Ocean to the South China Sea

Authors: Lei Chen (Ocean University of China, Laoshan Laboratory); Jiayan Yang (Woods Hole Oceanographic Institution); Lixin Wu (Ocean University of China, Laoshan Laboratory); Xiaopei Lin (Ocean University of China, Laoshan Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2024gl108322>

Publication Date: 2024-04-26

Volume/Issue: Volume 51, Issue 9

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Woods Hole Oceanographic Institution

License: CC BY 4.0

Open Access: N/A

Abstract: The South China Sea (SCS) is a semi-enclosed marginal sea linked to the broader oceans via various geographically constrained channels. Beneath the main thermocline depth, Luzon Strait is the only conduit for water-mass exchanges. Observations indicate a substantial seasonal variability in the inflow transport of deep water from the Pacific Ocean. This study aims to identify and examine key drivers for such seasonal changes. It is found that seasonal variability of the deep-water transport into the SCS is primarily driven by surface wind stress. An imbalance in wind-driven exchanges of surface water between the SCS and external seas demands compensational transports in subsurface layers so that the net volume transport into the SCS is conserved, resulting in seasonal variations in deep-water overflow. Changes in Karimata Strait exert a particularly influential impact on deep-water inflow, likely due to its unique position as the sole connecting channel across the Equator.

134. Surface current prediction based on a physics-informed deep learning model

Authors: Lu Zhang; Wenyang Duan; Xinmiao Cui; Yuliang Liu; Limin Huang

Journal: Applied Ocean Research

Publisher: Elsevier BV

DOI: 10.1016/j.apor.2024.104005

URL: <https://doi.org/10.1016/j.apor.2024.104005>

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

135. Satellite gravimetry: Methods, products, applications, and future trends

Authors: Mehdi Eshagh; Shuanggen Jin; Roland Pail; Riccardo Barzaghi; Dimitrios Tsoulis; Robert Tenzer; Pavel Novák

Journal: Earth-Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.earscirev.2024.104783

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Publication Date: 2024-04-22

Volume/Issue: Volume 253

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Language: en

Subjects: N/A

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License: N/A

Open Access: N/A

Abstract: N/A

136. The Southern Ocean deep mixing band emerges from a competition between winter buoyancy loss and upper stratification strength

Authors: Romain Caneill; Fabien Roquet; Jonas Nycander

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-20-601-2024

URL: <https://doi.org/10.5194/os-20-601-2024>

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References: 88

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Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The Southern Ocean hosts a winter deep mixing band (DMB) near the Antarctic Circumpolar Current's (ACC) northern boundary, playing a pivotal role in Subantarctic Mode Water formation. Here, we investigate what controls the presence and geographical extent of the DMB. Using observational data, we construct seasonal climatologies of surface buoyancy fluxes, Ekman buoyancy transport, and upper stratification. The strength of the upper-ocean stratification is determined using the columnar buoyancy index, defined as the buoyancy input necessary to produce a 250 m deep mixed layer. It is found that the DMB lies precisely where the autumn-winter buoyancy loss exceeds the columnar buoyancy found in late summer. The buoyancy loss decreases towards the south, while in the north the stratification is too strong to produce deep mixed layers. Although this threshold is also crossed in the Agulhas Current and East Australian Current regions, advection of buoyancy is able to stabilise the stratification. The Ekman buoyancy transport has a secondary impact on the DMB extent due to the compensating effects of temperature and salinity transports on buoyancy. Changes in surface temperature drive spatial variations in the thermal expansion coefficient (TEC). These TEC variations are necessary to explain the limited meridional extent of the DMB. We demonstrate this by comparing buoyancy budgets derived using varying TEC values with those derived using a constant TEC value. Reduced TEC in colder waters leads to decreased winter buoyancy loss south of the DMB, yet substantial heat loss persists. Lower TEC values also weaken the effect of temperature stratification, partially compensating for the effect of buoyancy loss damping. TEC modulation impacts both the DMB characteristics and its meridional extent.

137. Enhancing Scientific Research with FAIR Digital Objects in the National Science Data Fabric

Authors: Michela Taufer (University of Tennessee, Knoxville, TN, 37996, USA); Heberth Martinez (University of Tennessee, Knoxville, TN, 37996, USA); Jakob Luettgau (University of Tennessee, Knoxville, TN, 37996, USA); Lauren Whitnah (University of Tennessee, Knoxville, TN, 37996, USA); Giorgio Scorzelli (University of Utah, Salt Lake City, UT, 84112, USA); Pania Newell (University of Utah, Salt Lake City, UT, 84112, USA); Aashish Panta (University of Utah, Salt Lake City, UT, 84112, USA); Peer-Timo Bremer (Lawrence Livermore National Laboratory); Douglas Fils (Ronin Institute, Montclair, NJ, USA); Christine R. Kirkpatrick (San Diego Supercomputer Center, La Jolla, CA, 92093, USA); Valerio Pascucci (University of Utah, Salt Lake City, UT, 84112, USA)

Journal: Computing in Science & Engineering

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/mcse.2024.3363828

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Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, U.S. Department of Energy

License: Creative Commons

Open Access: N/A

Abstract: N/A

138. Estimation of Spatiotemporal Variability of Global Surface Ocean DIC Fields Using Ocean Color Remote Sensing Data

Authors: Ibrahim Shaik (National Remote Sensing Centre, Hyderabad, India); Kande Vamsi Krishna (Department of Ocean Engineering, Indian Institute of Technology Madras, Chennai, India); P. V. Nagamani (National Remote Sensing Centre, Hyderabad, India); S. K. Begum (Department of Geophysics, College of Science and Technology (AUCST), Andhra University, Visakhapatnam, India); Palanisamy Shanmugam (Department of Ocean Engineering, Indian Institute of Technology Madras, Chennai, India); Reema Mathew (National Remote Sensing Centre, Hyderabad, India); Mahesh Pathakoti (National Remote Sensing Centre, Hyderabad, India); Rajashree V. Bothale (National Remote Sensing Centre, Hyderabad, India); Prakash Chauhan (National Remote Sensing Centre, Hyderabad, India); Mohammed Osama (Department of Commerce, Palamuru University, Mahbubnagar, India)

Journal: IEEE Transactions on Geoscience and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tgrs.2024.3390179

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Publication Date: 2024-04-17

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Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

139. Unraveling the Indian monsoon's role in fueling the unprecedented 2022 Marine Heatwave in the Western North Pacific

Authors: Qianghua Song; Chunzai Wang; Yulong Yao; Hanjie Fan

Journal: npj Climate and Atmospheric Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41612-024-00645-x

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Volume/Issue: Volume 7, Issue 1

References: 46

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: An unprecedented marine heatwave (MHW) event occurred in the middle-high latitudes of the western North Pacific during the summer of 2022. We demonstrate that excessive precipitation thousands of kilometers away fuels this extreme MHW event in July 2022. In the upper atmosphere, a persistent atmospheric blocking system, forming over the MHW region, reduces cloud cover and increases shortwave radiation at the ocean surface, leading to high sea surface temperatures. Atmospheric perturbations induced by latent heat release from the extreme precipitation in the Indian summer monsoon region enhance this atmospheric blocking through the propagation of quasi-stationary Rossby waves. Our hypothesis is verified by using a numerical model that is forced with the observed atmospheric anomalous diabatic heating. This study sheds light on how a subtropical extreme event can fuel another extreme event at middle-high latitudes through an atmospheric teleconnection.

140. Amundsen Sea circulation controls bottom upwelling and Antarctic Pine Island and Thwaites ice shelf melting

Authors: Taewook Park; Yoshihiro Nakayama; SungHyun Nam

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-024-47084-z

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References: 64

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japanese Ministry of Economy, Trade and Industry | Agency for Natural Resources and Energy, Inoue Foundation for Science

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Open Access: N/A

Abstract: The Pine Island and Thwaites Ice Shelves (PIIS/TIS) in the Amundsen Sea are melting rapidly and impacting global sea levels. The thermocline depth (TD) variability, the interface between cold Winter Water and warm modified Circumpolar Deep Water (mCDW), at the PIIS/TIS front strongly correlates with basal melt rates, but the drivers of its interannual variability remain uncertain. Here, using an ocean model, we propose that the strength of the eastern Amundsen Sea on-shelf circulation primarily controls TD variability and consequent PIIS/TIS melt rates. The TD variability occurs because the on-shelf circulation meanders following the submarine glacial trough, creating vertical velocity through bottom Ekman dynamics. We suggest that a strong or weak ocean circulation, possibly linked to remote winds in the Bellingshausen Sea, generates corresponding changes in bottom Ekman convergence, which modulates mCDW upwelling and TD variability. We show that interannual variability of off-shelf zonal winds has a minor effect on ocean heat intrusion into PIIS/TIS cavities, contrary to the widely accepted concept.

141. Surface factors controlling the volume of accumulated Labrador Sea Water

Authors: Yavor Kostov; Marie-José Messias; Herlé Mercier; David P. Marshall; Helen L. Johnson

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-20-521-2024

URL: <https://doi.org/10.5194/os-20-521-2024>

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References: 102

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Keywords: N/A

Funding: Natural Environment Research Council

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Open Access: N/A

Abstract: . We explore historical variability in the volume of Labrador Sea Water (LSW) using ECCO, an ocean state estimate configuration of the Massachusetts Institute of Technology general circulation model (MITgcm). The model's adjoint, a linearization of the MITgcm, is set up to output the lagged sensitivity of the water mass volume to surface boundary conditions. This allows us to reconstruct the evolution of LSW volume over recent decades using historical surface wind stress, heat, and freshwater fluxes. Each of these boundary conditions contributes significantly to the LSW variability that we recover, but these impacts are associated with different geographical fingerprints and arise over a range of time lags. We show that the volume of LSW accumulated in the Labrador Sea exhibits a delayed response to surface wind stress and buoyancy forcing outside the convective interior of the Labrador Sea at important locations in the North Atlantic Ocean. In particular, patterns of wind and surface density anomalies can act as a "traffic controller" and regulate the North Atlantic Current's (NAC's) transport of warm and saline subtropical water masses that are precursors for the formation of LSW. This propensity for a delayed response of LSW to remote forcing allows us to predict a limited yet substantial and significant fraction of LSW variability at least 1 year into the future. Our analysis also enables us to attribute LSW variability to different boundary conditions and to gain insight into the major mechanisms that contribute to volume anomalies in this deep water mass. We point out the important role of key processes that promote the formation of LSW in both the Irminger and Labrador seas: buoyancy loss and preconditioning along the NAC pathway and in the Iceland Basin, the Irminger Sea, and the Nordic Seas.

142. Atmospheric oxygen as a tracer for fossil fuel carbon dioxide: a sensitivity study in the UK

Authors: Hannah Chawner; Eric Saboya; Karina E. Adcock; Tim Arnold; Yuri Artioli; Caroline Dylag; Grant L. Forster; Anita Ganesan; Heather Graven; Gennadi Lessin; Peter Levy; Ingrid T. Lujikx; Alistair Manning; Penelope A. Pickers; Chris Rennick; Christian Rödenbeck; Matthew Rigby

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-24-4231-2024

URL: <https://doi.org/10.5194/acp-24-4231-2024>

Publication Date: 2024-04-09

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References: 74

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Language: en

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Keywords: N/A

Funding: Natural Environment Research Council

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Open Access: N/A

Abstract: . We investigate the use of atmospheric oxygen (O₂) and carbon dioxide (CO₂) measurements for the estimation of the fossil fuel component of atmospheric CO₂ in the UK. Atmospheric potential oxygen (APO) - a tracer that combines O₂ and CO₂, minimizing the influence of terrestrial biosphere fluxes - is simulated at three sites in the UK, two of which make APO measurements. We present a set of model experiments that estimate the sensitivity of APO simulations to key inputs: fluxes from the ocean, fossil fuel flux magnitude and distribution, the APO baseline, and the exchange ratio of O₂ to CO₂ fluxes from fossil fuel combustion and the terrestrial biosphere. To estimate the influence of uncertainties in ocean fluxes, we compare three ocean O₂ flux estimates from the NEMO-ERSEM, the ECCO-Darwin ocean model, and the Jena CarboScope (JC) APO inversion. The sensitivity of APO to fossil fuel emission magnitudes and to terrestrial biosphere and fossil fuel exchange ratios is investigated through Monte Carlo sampling within literature uncertainty ranges and by comparing different inventory estimates. We focus our model-data analysis on the year 2015 as ocean fluxes are not available for later years. As APO measurements are only available for one UK site at this time, our analysis focuses on the Weybourne station. Model-data comparisons for two additional UK sites (Heathfield and Ridge Hill) in 2021, using ocean flux climatologies, are presented in the Supplement. Of the factors that could potentially compromise simulated APO-derived fossil fuel CO₂ (ffCO₂) estimates, we find that the ocean O₂ flux estimate has the largest overall influence at the three sites in the UK. At times, this influence is comparable in magnitude to the contribution of simulated fossil fuel CO₂ to simulated APO. We find that simulations using different ocean fluxes differ from each other substantially. No single model estimate, or a model estimate that assumed zero ocean flux, provided a significantly closer fit than any other. Furthermore, the uncertainty in the ocean contribution to APO could lead to uncertainty in defining an appropriate regional background from the data. Our findings suggest that the contribution of non-terrestrial sources needs to be better accounted for in model simulations of APO in the UK to reduce the potential influence on inferred fossil fuel CO₂ using

APO.

143. Intraseasonal response of marine planktonic ecosystem to summertime Madden-Julian Oscillation in the South China Sea: A model study

Authors: Hengye Ren; Wenfang Lu; Wupeng Xiao; Qing Zhu; Canbo Xiao; Zhigang Lai

Journal: Progress in Oceanography

Publisher: Elsevier BV

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References: 91

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

144. Weakened Seasonality of the Ocean Surface Mixed Layer Depth in the Southern Indian Ocean During 1980-2019

Authors: Shang-Min Long (Hohai University, Second Institute of Oceanography, Ministry of Natural Resources); Shichang Zhao (Hohai University, Ministry of Natural Resources); Zhen Gao (Hohai University, Laoshan Laboratory); Shantong Sun (Laoshan Laboratory); Jia-Rui Shi (Woods Hole Oceanographic Institution); Jun Ying (Second Institute of Oceanography, Ministry of Natural Resources); Guancheng Li (Eco-Environmental Monitoring and Research Center Pearl River Valley and South China Sea Ecology and Environment Administration Ministry of Ecology and Environment Guangzhou China); Lijing Cheng (Chinese Academy of Sciences); Jiajia Chen (Hohai University, Ministry of Natural Resources); Xuhua Cheng (Hohai University, Ministry of Natural Resources); Shaolei Lu (Second Institute of Oceanography, Ministry of Natural Resources)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Publication Date: 2024-04-05

Volume/Issue: Volume 51, Issue 7

References: 73

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: subtropical winds, Southern Annular Mode, mixed layer depth, subtropical ocean gyre, seasonality (and 1 more)

Funding: National Natural Science Foundation of China, Natural Science Foundation of Jiangsu Province, National Key Research and Development Program of China

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Open Access: N/A

Abstract: Temporal and spatial variations in the ocean surface mixed layer are important for the climate and ecological systems. During 1980-2019, the Southern Indian Ocean (SIO) mixed layer depth (MLD) displays a basin-wide shoaling trend that is absent in the other basins within 40 degreesS-40 degreesN. The SIO MLD shoaling is mostly prominent in austral winter with deep climatology MLD, substantially weakening the MLD seasonality. Moreover, the SIO MLD changes are primarily caused by a southward shift of the subtropical anticyclonic winds and hence ocean gyre, associated with a strengthening of the Southern Annular Mode, in recent decades for both winter and summer. However, the poleward-shifted subtropical ocean circulation preferentially shoals the SIO MLD in winter when the meridional MLD gradient is sharp but not in summer when the gradient is flat. This highlights the distinct subtropical MLD response to meridional mitigation in winds due to different background oceanic conditions across seasons.

145. A Deep Learning Approach to Extract Balanced Motions From Sea Surface Height Snapshot

Authors: Zhanwen Gao (Ocean University of China, Ifremer UMR CNRS LOPS Brest France); Bertrand Chapron (Ifremer UMR CNRS LOPS Brest France); Chunyong Ma (Ocean University of China, Laoshan Laboratory); Ronan Fablet (IMT Atlantique UMR CNRS Lab-STICC Brest France); Quentin Febvre (IMT Atlantique UMR CNRS Lab-STICC Brest France); Wenxia Zhao (Ocean University of China); Ge Chen (Ocean University of China, Laoshan Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2023gl106623>

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References: 49

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: Creative Commons

Open Access: N/A

Abstract: Extracting balanced geostrophic motions (BM) from sea surface height (SSH) observations obtained by wide-swath altimetry holds great significance in enhancing our understanding of oceanic dynamic processes at submesoscale wavelength. However, SSH observations derived from wide-swath altimetry are characterized by high spatial resolution while relatively low temporal resolution, thereby posing challenges to extract the BM from a single SSH snapshot. To address this issue, this paper proposes a deep learning model called the BM-UBM Network, which takes an instantaneous SSH snapshot as input and outputs the projection corresponding to the BM. Training experiments are conducted both in the Gulf Stream and South China Sea, and three metrics are considered to diagnose model's outputs. The favorable results highlight the potential capability of the BM-UBM Network to process SSH measurements obtained by wide-swath altimetry.

146. Decadal Evolution of Ice-Ocean Interactions at a Large East Greenland Glacier Resolved at Fjord Scale With Downscaled Ocean Models and Observations

Authors: M. Wood (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); A. Khazendar (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); I. Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); K. Mankoff (National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies, Autonomic Integra LLC New York NY USA, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); A. T. Nguyen (University of Texas at Austin); K. Schulz (University of Texas at Austin); J. K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H. Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: Greenland glaciers, ice-ocean interactions, remote sensing, regional ocean modeling, sea level rise (and 1 more)

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Open Access: N/A

Abstract: In recent decades, the Greenland ice sheet has been losing mass through glacier retreat and ice flow acceleration. This mass loss is linked with variations in submarine melt, yet existing ocean models are either coarse global simulations focused on decadal-scale variability or fine-scale simulations for process-based investigations. Here, we unite these scales with a framework to downscale from a global state estimate (15 km) into a regional model (3 km) that resolves circulation on the continental shelf. We further downscale into a fjord-scale model (500 m) that resolves circulation inside fjords and quantifies melt. We demonstrate this approach in Scoresby Sund, East Greenland, and find that interannual variations in submarine melt at Daugaard-Jensen glacier induced by ocean temperature variability are consistent with the decadal changes in glacier ice dynamics. This study provides a framework by which coarse-resolution models can be refined to quantify glacier submarine melt for future ice sheet projections.

147. Estimating freshwater flux amplification with ocean tracers via linear response theory

Authors: Aurora Basinski-Ferris; Laure Zanna

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esd-15-323-2024

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References: 52

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Subjects: N/A

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Funding: Directorate for Geosciences

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Open Access: N/A

Abstract: . Accurate estimation of changes in the global hydrological cycle over the historical record is important for model evaluation and understanding future trends. Freshwater flux trends cannot be accurately measured directly, so quantification of change often relies on ocean salinity trends. However, anthropogenic forcing has also induced ocean transport change, which imprints on salinity. We find that this ocean transport affects the surface salinity of the saltiest regions (the subtropics) while having little impact on the surface salinity in other parts of the globe. We present a method based on linear response theory which accounts for the regional impact of ocean circulation changes while estimating freshwater fluxes from ocean tracers. Testing on data from the Community Earth System Model large ensemble, we find that our method can recover the true amplification of freshwater fluxes, given thresholded statistical significance values for salinity trends. We apply the method to observations and conclude that from 1975-2019, the hydrological cycle has amplified by 5.04 ± 1.27 % per degree Celsius of surface warming.

148. Gathering Users and Developers to Shape Together the Next-Generation Ocean Reanalyses

Authors: Chunxue Yang (Institute of Marine Science, National Research Council of Italy, Rome, Italy;); Romain Bourdallé-Badie (Mercator Ocean International, Toulouse, France;); Marie Drevillon (Mercator Ocean International, Toulouse, France;); Dillon Amaya (Physical Sciences Laboratory, Earth System Research Laboratory, National Oceanic and Atmospheric Administration, Boulder, Colorado;); Lotfi Aouf (Météo-France, CNRM, Toulouse, France;); Ali Aydogdu (CMCC Foundation); Benjamin Barton (National Oceanography Center, Liverpool, United Kingdom;); Mike Bell (Met Office); Tim Boyer (NOAA/National Centers for Environmental Information, Silver Spring, Maryland;); Anouk Blauw (Deltares, Delft, Netherlands;); James Carton (University of Maryland); Tony Candela (Mercator Ocean International, Toulouse, France;); Gianpiero Cossarini (National Institute of Oceanography and Applied Geophysics - OGS, Trieste, Italy;); Tomasz Dabrowski (Marine Institute, Oranmore, Ireland;); Eric de Boisseson (European Centre for Medium-range Weather Forecast, Reading, United Kingdom;); Lee de Mora (Plymouth Marine Laboratory); Ronan Fablet (IMT Atlantique, Lab-STICC, Brest, France;); INRIA team Odyssey, Brest, France;); Gaël Forget (Massachusetts Institute of Technology); Yosuke Fujii (Meteorological Research Institute); Gilles Garric (Mercator Ocean International, Toulouse, France;); Valentina Giunta (Mercator Ocean International, Toulouse, France;); Peter Salamon (European Commission); Hans Hersbach (European Centre for Medium-range Weather Forecast, Reading, United Kingdom;); Mélanie Juza (Balearic Islands Coastal Forecasting and Observing System (SOCIB), Palma, Spain;); Julien Le Sommer (CNRS, Institut de Geosciences de l'Environnement, Grenoble, France;); Matthew Martin (Met Office); Ronan McAdam (CMCC Foundation); Melisa Menendez Garcia (University of Cantabria, Santander, Spain;); Joao Morim (University of Central Florida, National Centre for Coastal Research, Orlando, Florida;); Dario Nicoli (CMCC Foundation); Antonio Reppucci (Mercator Ocean International, Toulouse, France;); Annette Samuelsen (Nansen Environmental and Remote Sensing Centre, Bergen, Norway;); Raphaëlle Sauzède (Institut de la Mer de Villefranche, Villefranche-sur-Mer, France;); Laura Slivinski (Physical Sciences Laboratory, Earth System Research Laboratory, National Oceanic and Atmospheric Administration, Boulder, Colorado;); Damien Specq (CNRM, Université de Toulouse, Météo-France/CNRS, Toulouse, France;); Andrea Storto (Institute of Marine Science, National Research Council of Italy, Rome, Italy;); Laura Tuomi (Finnish Meteorological Institute); Luc Vandenbulcke (Université de Liège, Liège, Belgium;); Roland Aznar (NOW Systems, Madrid, Spain;); Jonathan Beuvier (CNRM, Université de Toulouse, Météo-France/CNRS, Toulouse, France;); Andrea Cipollone (CMCC Foundation); Emanuela Clementi (CMCC Foundation); Valeria Di Biagio (National Institute of Oceanography and Applied Geophysics - OGS, Trieste, Italy;); Romain Escudier (Mercator Ocean International, Toulouse, France;); Rianne Giesen (Royal Netherlands Meteorological Institute); Eric Greiner (Collecte Localisation Satellites, Ramonville-Saint-Agne, France;); Karen Guihou (Mercator Ocean International, Toulouse, France;); Vasily Korabel (Danish Meteorological Institute); Julien Lamouroux (Mercator Ocean International, Toulouse, France;); Stephane Law Chune (Mercator Ocean International, Toulouse, France;); Jean-Michel Lellouche (Mercator Ocean International, Toulouse, France;); Bruno Levier (Mercator Ocean International, Toulouse, France;); Leonardo Lima (CMCC Foundation); Antoine Mangin (ACRI-ST); Michael Mayer (University of Vienna, Vienna, Austria;); Angelique Melet (Mercator Ocean International, Toulouse, France;); Pietro Miraglio (CMCC Foundation); Charikleia Oikonomou (Hellenic Centre for Marine Research); Julia Pfeffer (Magellium, Toulouse, France); Richard Renshaw (Met Office); Ida Ringgaard (Danish Meteorological Institute); Sulian Thual (Mercator Ocean International, Toulouse, France;); Olivier Titaud (Collecte Localisation Satellites, Ramonville-Saint-Agne, France;); Marina Tonani (Mercator Ocean International, Toulouse, France;); Simon van Gennip (Mercator Ocean International, Toulouse, France;); Karina von Schuckmann (Mercator Ocean International, Toulouse, France;); Yann Drillet (Mercator Ocean International, Toulouse, France;); Pierre-Yves Le Traon (Mercator Ocean International, Toulouse, France;);

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Abstract: N/A

149. Ocean Temperature Profiling Lidar: Analysis of Technology and Potential for Rapid Ocean Observations

Authors: John R. Moisan (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Cecile S. Rousseaux (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Paul R. Stysley (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Gregory B. Clarke (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Demetrios P. Poullos (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Remote Sensing

Publisher: MDPI AG

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Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Development of ocean measurement technologies can improve monitoring of the global Ocean Heat Content (OHC) and Heat Storage Rate (HSR) that serve as early-warning indices for climate-critical circulation processes such as the Atlantic Meridional Overturning Circulation and provide real-time OHC assessments for tropical cyclone forecast models. This paper examines the potential of remotely measuring ocean temperature profiles using a simulated Brillouin lidar for calculating ocean HSR. A series of data analysis ('Nature') and Observational Systems Simulation Experiments (OSSEs) were carried out using 26 years (1992-2017) of daily mean temperature and salinity outputs from the ECCOV4r4 ocean circulation model. The focus of this study is to compare various OSSEs carried out to measure the HSR using a simulated Brillouin lidar against the HSR calculated from the ECCOV4r4 model results. Brillouin lidar simulations are used to predict the probability of detecting a return lidar signal under varying sampling strategies. Correlations were calculated for the difference between sampling strategies. These comparisons ignore the measurement errors inherent in a Brillouin lidar. Brillouin lidar technology and instruments are known to contain numerous, instrument-dependent errors and remain an engineering challenge. A significant decrease in the ability to measuring global ocean HSRs is a consequence of measuring ocean temperature from nadir-pointing instruments that can only take measurements along-track. Other sources of errors include the inability to fully profile ocean regions with deep mixed layers, such as the Southern Ocean and North Atlantic, and ocean regions with high light attenuation levels.

150. Modeling Ocean Circulation and Ice Shelf Melt in the Bellingshausen Sea

Authors: Shuntaro Hyogo (Hokkaido University); Yoshihiro Nakayama (Hokkaido University); Vigan Mensah (Hokkaido University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Abstract: The ice shelves in the Bellingshausen Sea are melting and thinning rapidly due to modified Circumpolar Deep Water (mCDW) intrusions carrying heat toward ice-shelf cavities. Observations are, however, sparse in time and space, and extensive model-data comparisons have never been possible. Here, using a circulation model of the region and ship-based observations, we show that the simulated water mass distributions in several troughs traversing mCDW inflows are in good agreement with observations, implying that our model has the skills to simulate hydrographic structures as well as on-shelf ocean circulations. It takes 7.9 and 11.7 months for mCDW to travel to the George VI Ice Shelf cavities through the Belgica and Marguerite troughs, respectively. Ice-shelf melting is mainly caused by mCDW intrusions along the Belgica and Marguerite troughs, with the heat transport through the former being ~2.8 times larger than that through the latter. The mCDW intrusions toward the George VI Ice Shelf show little seasonal variability, while those toward the Venable Ice Shelf show seasonal variability, with higher velocities in summer likely caused by coastal trapped waves. We also conduct particle experiments tracking glacial meltwater. After 2 years of model integration, ~33% of the released particles are located in the Amundsen Sea, supporting a linkage between Bellingshausen Sea ice-shelf meltwater and Amundsen Sea upper ocean hydrography.

151. Intercomparison of tropical Indian Ocean circulation in ocean reanalysis and evaluation in CMIP6 climate models

Authors: P. Bhanu Deepika; Soumya Mohan; G. Srinivas

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

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Abstract: N/A

152. The linkage between wintertime sea ice drift and atmospheric circulation in an Arctic ice-ocean coupled simulation

Authors: Xi Liang; Haibo Bi; Chengyan Liu; Xichen Li; Dakui Wang; Fu Zhao; Zhongxiang Tian; Ming Li; Na Liu

Journal: Ocean Modelling

Publisher: Elsevier BV

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Abstract: N/A

153. Parameter sensitivity analysis of a sea ice melt pond parametrisation and its emulation using neural networks

Authors: Simon Driscoll; Alberto Carrassi; Julien Brajard; Laurent Bertino; Marc Bocquet; Einar Örn Ólason

Journal: Journal of Computational Science

Publisher: Elsevier BV

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Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

154. Assessing Modeled Mesoscale Stirring Using Microscale Observations

Authors: D. A. Cherian (National Center for Atmospheric Research); Y. Guo (b Yale University, New Haven, Connecticut); F. O. Bryan (National Center for Atmospheric Research)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Language: N/A

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Keywords: Ocean models, Primitive equations model, Diapycnal mixing, Isopycnal mixing, General circulation models (and 2 more)

Funding: NASA Headquarters

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Open Access: N/A

Abstract: We assess the representation of mesoscale stirring in a suite of models against an estimate derived from microstructure data collected during the North Atlantic Tracer Release Experiment (NATRE). We draw heavily from the approximate temperature variance budget framework of Ferrari and Polzin. This framework assumes two sources of temperature variance away from boundaries: first, the vertical stirring of the large-scale mean vertical gradient by small-scale turbulence; and second, the lateral stirring of large-scale mean along-isopycnal gradients by mesoscale eddies. Temperature variance so produced is transformed and on average transferred down scales for ultimate dissipation at the microscale at a rate χ estimated using microstructure observations. Ocean models represent these pathways by a vertical mixing parameterization, and an along-isopycnal lateral mixing parameterization (if needed). We assess the rate of variance production by the latter as a residual from the NATRE dataset and compare against the parameterized representations in a suite of model simulations. We find that variance production due to lateral stirring in a Parallel Ocean Program version 2 (POP2) 1/10 degrees simulation agrees well, to within the estimated error bars, with that inferred from the NATRE estimate. A POP2 1 degrees simulation and the Estimating the Circulation and Climate of the Ocean Version 4 release 4 (ECCOV4r4) simulation appear to dissipate an order of magnitude too much variance by applying a lateral diffusivity, when compared to the NATRE estimate, particularly below 1250 m. The ECCOV4r4-adjusted lateral diffusivities are elevated where the microstructure suggests elevated χ sourced from mesoscale stirring. Such elevated values are absent in other diffusivity estimates suggesting the possibility of compensating errors and caution in interpreting ECCOV4r4's adjusted lateral diffusivities. **Significance Statement** We look at whether microstructure turbulence observations can provide a useful metric for judging the fidelity of representation of mesoscale stirring in a suite of models. We focus on the region of the North Atlantic Tracer Release Experiment (NATRE), the site of a major ocean turbulence observation campaign, and use an approximate variance budget framework for the region with observational estimates from Ferrari and Polzin (2005). The approach provides a novel framework to evaluate the approximate representation of mesoscale stirring in a

variety of models.

155. Wind-steered Eastern Pathway of the Atlantic Meridional Overturning Circulation

Authors: Zhengyu Liu; Sifan Gu; Sijia Zou; Shaoqing Zhang; Yangyang Yu; Chengfei He

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-024-01407-3

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Cited by: 5 publications

Language: en

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

156. Surface Heat Fluxes Drive a Two-Phase Response in Southern Ocean Mode Water Stratification

Authors: Ciara Pimm (University of Liverpool); Richard G. Williams (University of Liverpool); Dani Jones (University of Michigan, British Antarctic Survey NERC UKRI Cambridge UK); Andrew J. S. Meijers (British Antarctic Survey NERC UKRI Cambridge UK)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: subantarctic mode water, adjoint modeling, surface heat flux

Funding: Natural Environment Research Council, UK Research and Innovation

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Open Access: N/A

Abstract: Subantarctic mode waters have low stratification and are formed through subduction from thick winter mixed layers in the Southern Ocean. To investigate how surface forcing affects the stratification in mode water formation regions in the Southern Ocean, a set of adjoint sensitivity experiments are conducted. The objective function is the annual-average stratification over the mode water formation region, which is evaluated from potential temperature and salinity adjoint sensitivity experiments. The analysis of impacts, from the product of sensitivities and forcing variability, identifies the separate effects of the wind stress, heat flux, and freshwater flux, revealing that the dominant control on stratification is from surface heat fluxes, as well as a smaller effect from zonal wind stress. The adjoint sensitivities of stratification to surface heat flux reveal a surprising change in sign over 2 years lead time: surface cooling leads to the expected initial local decrease in stratification, but there is a delayed response leading to an increase in stratification. This delayed response in stratification involves effective atmospheric damping of the surface thermal contribution, so that eventually the oppositely-signed advective haline contribution dominates. This two-phase response of stratification is found to hold over mode water formation regions in the South Indian and Southeast Pacific sectors of the Southern Ocean, where there are strong advective flows linked to the Antarctic Circumpolar Current.

157. 60-YEAR REMOTE SENSING OBSERVATIONS REVEALED CROSS-CYCLE EVOLUTION OF THE BRUNT ICE SHELF IN THE CONTEXT OF GLOBAL WARMING

Authors: Y. Cheng; J. Xue; H. Yu; G. Hai

Journal: The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences

Publisher: Copernicus GmbH

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References: N/A

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Funding: N/A

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Open Access: N/A

Abstract: . The Antarctic Ice Sheet (AIS) holds the potential to raise global sea levels by a staggering 57 m. Ice shelves, extensions of these ice sheets into the ocean, play a crucial role in stabilizing the upstream ice sheet. The Brunt Ice Shelf (BIS), located in the Southern Weddell Sea region and proximal to the continental shelf break, stands as a vital indicator for the region's response to global climate warming. Hence, this study systematically employs a range of remote sensing and field data, including remote sensing data from 1963 to 2023 (ARGON images, Landsat1-9 images, Sentinel-1 images, ICESat-2 altimetry data, etc.) and in-situ ice flow velocity data since 1956, to conduct a comprehensive analysis of the BIS's calving cycle and evolution over nearly six decades. The results reveal a significant exacerbation of both dynamical and structural instabilities in the BIS compared to the calving cycle in the 1970s. Combining analyses of air temperature from meteorological station, water temperature from ocean models and in-situ measurements such as Conductivity-Temperature-Depth profiles, the study suggests that the increased instability of the BIS is likely linked to the intrusion of warmer currents over the continental shelf. This research provides valuable insights into the evolving behavior of the BIS in response to changing climate conditions, underscoring the critical role of remote sensing data in advancing our understanding of these processes on a global scale.

158. Ocean Bottom Distributed Acoustic Sensing for Oceanic Seismicity Detection and Seismic Ocean Thermometry

Authors: Zhichao Shen (Woods Hole Oceanographic Institution); Wenbo Wu (Woods Hole Oceanographic Institution)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: seismic ocean thermometry, DAS, T-wave, curvelet denoising, oceanic seismicity

Funding: N/A

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Open Access: N/A

Abstract: A T-wave is a seismo-acoustic wave that can travel a long distance in the ocean with little attenuation, making it valuable for monitoring remote tectonic activity and changes in ocean temperature using seismic ocean thermometry (SOT). However, current high-quality T-wave stations are sparsely distributed, limiting the detectability of oceanic seismicity and the spatial resolution of global SOT. The use of ocean bottom distributed acoustic sensing (OBDAS), through the conversion of telecommunication cables into dense seismic arrays, is a cost-effective and scalable means to complement existing seismic stations. Here, we systematically investigate the performance of OBDAS for oceanic seismicity detection and SOT using a 4-day Ocean Observatories Initiative community experiment offshore Oregon. We first present T-wave observations from distant and regional earthquakes and develop a curvelet denoising scheme to enhance T-wave signals on OBDAS. After denoising, we show that OBDAS can detect and locate more and smaller T-wave events than regional OBS network. During the 4-day experiment, we detect 92 oceanic earthquakes, most of which are missing from existing catalogs. Leveraging the sensor density and cable directionality, we demonstrate the feasibility of source azimuth estimation for regional Blanco earthquakes. We also evaluate the SOT performance of OBDAS using pseudo-repeating earthquake T-waves. Our results show that OBDAS can utilize repeating earthquakes as small as M3.5 for SOT, outperforming ocean bottom seismometers. However, ocean ambient natural and instrumental noise strongly affects the performance of OBDAS for oceanic seismicity detection and SOT, requiring further investigation.

159. Variability of Eddy Kinetic Energy in the Eurasian Basin of the Arctic Ocean Inferred From a Model Simulation at 1-km Resolution

Authors: Vasco Müller (Alfred Wegener Institute); Qiang Wang (Alfred Wegener Institute); Nikolay Koldunov (Alfred Wegener Institute); Sergey Danilov (Alfred Wegener Institute); Dmitry Sidorenko (Alfred Wegener Institute); Thomas Jung (Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Oceans

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Keywords: Arctic Ocean, ocean model, eddy kinetic energy, mesoscale eddies, high resolution modeling

Funding: Bundesministerium für Bildung und Forschung, Deutsche Forschungsgemeinschaft

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Open Access: N/A

Abstract: Mesoscale eddies play an important role in driving the dynamics of the Arctic Ocean. Understanding their behavior is crucial for comprehending the ongoing changes in the region. In this study, by using a novel decade-long simulation at 1 km resolution with the unstructured-mesh Finite volume Sea ice-Ocean Model, we evaluate the spatial and temporal variability of eddy kinetic energy in the Eurasian Basin. We find that monthly, annual, and interannual variability of EKE near the surface is predominantly influenced by changes in sea ice cover, while the eddy activity at deeper depth, being shielded from the surface by ocean stratification, is more strongly influenced by local baroclinic energy conversion. Moreover, our research demonstrates that eddies in the Eurasian Basin can transport ocean heat from the Atlantic Water layer toward sea ice and cause local basal melting in the order of about 20 cm per month even in wintertime. Our study suggests that eddy activity in the Arctic Ocean will strengthen along with future sea ice decline, and that the impact of ocean heat of the Atlantic Water layer on sea ice retreat may become prominent.

160. Seismic Ocean Thermometry of the Kuroshio Extension Region

Authors: Shirui Peng (California Institute of Technology); Jörn Callies (California Institute of Technology); Wenbo Wu (California Institute of Technology, Woods Hole Oceanographic Institution); Zhongwen Zhan (California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

Funding: Resnick Sustainability Institute for Science, Energy and Sustainability, California Institute of Technology, National Science Foundation, California Institute of Technology

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Open Access: N/A

Abstract: Seismic ocean thermometry uses sound waves generated by repeating earthquakes to measure temperature change in the deep ocean. In this study, waves generated by earthquakes along the Japan Trench and received at Wake Island are used to constrain temperature variations in the Kuroshio Extension region. This region is characterized by energetic mesoscale eddies and large decadal variability, posing a challenging sampling problem for conventional ocean observations. The seismic measurements are obtained from a hydrophone station off and a seismic station on Wake Island, with the seismic station's digital record reaching back to 1997. These measurements are combined in an inversion for the time and azimuth dependence of the range-averaged deep temperatures, revealing lateral and temporal variations due to Kuroshio Extension meanders, mesoscale eddies, and decadal water mass displacements. These results highlight the potential of seismic ocean thermometry for better constraining the variability and trends in deep-ocean temperatures. By overcoming the aliasing problem of point measurements, these measurements complement existing ship- and float-based hydrographic measurements.

161. Marine Heatwaves in the East Asian Marginal Seas Facilitated by Boreal Summer Intraseasonal Oscillations

Authors: Panini Dasgupta (Seoul National University); SungHyun Nam (Seoul National University); J. S. Saranya (Seoul National University); M. K. Roxy (Indian Institute of Tropical Meteorology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: MHWs, mixed layer shoaling, BSISO, heat budget, Southern Yellow Sea (and 1 more)

Funding: Korea Institute of Marine Science and Technology promotion, Ministry of Oceans and Fisheries

License: N/A

Open Access: N/A

Abstract: During the summer of 2016, the northern East China Sea and the southern Yellow Sea (NECS-SYS) experienced one of the most severe and devastating marine heatwaves (MHWs) on record, with a temperature anomaly exceeding 4 degreesC. This shallow semi-enclosed continental shelf region is widely recognized as a significant hotspot for MHWs with associated incidences of harmful algae blooms. Previous studies have highlighted the importance of mixed layer shoaling as a crucial factor in the genesis of MHWs in the global ocean. The current study employed the Hybrid Coordinate Ocean Model reanalysis data set during 1994-2015 to delve into the mechanisms driving mixed layer shoaling during NECS-SYS MHW genesis. Our findings reveal the significant role of the northward propagating boreal summer intraseasonal oscillation in promoting MHW genesis and intensification. Specifically, boreal summer intraseasonal oscillation phases 5, 6, and 7 contribute to the favorable conditions that facilitate MHW formation by inducing mixed layer shoaling and increasing solar influx, with mixed layer shoaling playing a more dominant role. The current study provides insights into the relative influences of wind, salinity, and temperature on mixed layer shoaling. We observe that wind plays the most significant role in mixed layer shoaling, followed by temperature and salinity. The boreal summer intraseasonal oscillation induced wind relaxation, increased shortwave radiation, and freshwater influx lead sea surface temperature by 7, 5, and 4 days, respectively. Importantly, mixed layer shoaling leads SST anomalies by 1-2 days. Therefore, the current study also suggests an intraseasonal predictability source for NECS-SYS MHWs.

162. Numerical investigation of interaction between anticyclonic eddy and semidiurnal internal tide in the northeastern South China Sea

Authors: Liming Fan; Hui Sun; Qingxuan Yang; Jianing Li

Journal: Ocean Science

Publisher: Copernicus GmbH

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Abstract: . We investigate the interaction between an anticyclonic eddy (AE) and semidiurnal internal tide (SIT) on the continental slope of the northeastern South China Sea (SCS), using a high spatiotemporal resolution numerical model. Two key findings are as follows: first, the AE promotes energy conversion from low-mode to higher-mode SIT. Additionally, production terms indicate that energy is also transferred from the SIT field to the eddy field at an average rate of 3.0 mW m^{-2} (accounting for 7 % of the incoming energy flux of SIT when integrated over the eddy diameter). Second, the AE can modify the spatial distribution of tidal-induced dissipation by refracting, scattering, and reflecting low-mode SIT. The phase and group velocities of the SIT are significantly influenced by the eddy field, resulting in a northward or southward shift in the internal tidal rays. These findings deepen our understanding of the complex interactions between AE and SIT, as well as their impacts on energy conversion, wave propagation, and coastal processes.

163. Uncovering the world's largest carbon sink--a profile of ocean carbon sinks research

Authors: Qiang Wang; Feng Ren; Rongrong Li

Journal: Environmental Science and Pollution Research

Publisher: Springer Science and Business Media LLC

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Abstract: N/A

164. Subsurface evolution of three types of surface marine heatwaves over the East Sea (Japan Sea)

Authors: J.S. Saranya; SungHyun Nam

Journal: Progress in Oceanography

Publisher: Elsevier BV

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Keywords: N/A

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Abstract: N/A

165. Processes Driving Subseasonal Variations of Upper Ocean Heat Content in the Equatorial Indian Ocean

Authors: Ashneel Chandra (University of Bergen, Bjerknes Centre for Climate Research); Noel Keenlyside (Nansen Environmental and Remote Sensing Center, University of Bergen, Bjerknes Centre for Climate Research); Lea Svendsen (University of Bergen, Bjerknes Centre for Climate Research); Awnesh Singh (The University of the South Pacific Suva Fiji)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Publication Date: 2024-02-19

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Subjects: N/A

Keywords: ocean heat content, Kelvin waves, Madden-Julian Oscillation, Indian Ocean, subseasonal to seasonal variability (and 1 more)

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License: Creative Commons

Open Access: N/A

Abstract: In the equatorial Indian Ocean, the largest subseasonal temperature variations in the upper ocean are observed below the mixed layer. Subsurface processes can influence mixed layer temperature and consequently air-sea coupling. However, the physical processes driving temperature variability at these depths are not well quantified. During the boreal winter, the Madden-Julian Oscillation (MJO) partly drives upper ocean heat content (OHC) variations. Therefore, to understand processes driving subseasonal OHC variability in the equatorial Indian Ocean, we use an observationally constrained, physically consistent ocean state estimate from the Estimating the Circulation and Climate of the Ocean (ECCO) Consortium. Using a heat budget analysis, we show that the main driver of subseasonal OHC variability in the ECCO ocean state estimate is horizontal advection. Along the equator, OHC variations are driven by zonal advection while the role of meridional advection becomes more important away from the equator. During the active phase of the MJO, net air-sea heat fluxes damp OHC variability along the equator, while away from the equator net air-sea heat fluxes partly drive OHC variability. Equatorial OHC variations are found to be associated with processes driven by Kelvin and Rossby waves consistent with previous studies. By quantifying the physical processes, we highlight the important role of ocean dynamics in contributing to the observed variations of subseasonal OHC in the equatorial Indian Ocean.

166. Physics-based early warning signal shows that AMOC is on tipping course

Authors: René M. van Westen (Utrecht University); Michael Kliphuis (Utrecht University); Henk A. Dijkstra (Utrecht University)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.adk1189

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References: 62

Cited by: 92 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: One of the most prominent climate tipping elements is the Atlantic meridional overturning circulation (AMOC), which can potentially collapse because of the input of fresh water in the North Atlantic. Although AMOC collapses have been induced in complex global climate models by strong freshwater forcing, the processes of an AMOC tipping event have so far not been investigated. Here, we show results of the first tipping event in the Community Earth System Model, including the large climate impacts of the collapse. Using these results, we develop a physics-based and observable early warning signal of AMOC tipping: the minimum of the AMOC-induced freshwater transport at the southern boundary of the Atlantic. Reanalysis products indicate that the present-day AMOC is on route to tipping. The early warning signal is a useful alternative to classical statistical ones, which, when applied to our simulated tipping event, turn out to be sensitive to the analyzed time interval before tipping.

167. Southern Ocean High-Resolution (SOhi) Modeling Along the Antarctic Ice Sheet Periphery

Authors: Andy Dinh (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Matthew Mazloff (Scripps Institution of Oceanography / University of California San Diego); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, ocean-ice interactions, ocean modeling, Antarctica

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The Southern Ocean plays a major role in controlling the evolution of Antarctic glaciers and in turn their impact on sea level rise. We present the Southern Ocean high-resolution (SOhi) simulation of the MITgcm ocean model to reproduce ice-ocean interaction at 1/24 degrees around Antarctica, including all ice shelf cavities and oceanic tides. We evaluate the model accuracy on the continental shelf using Marine Mammals Exploring the Oceans Pole to Pole data and compare the results with three other MITgcm ocean models (ECCO4, SOSE, and LLC4320) and the ISMIP6 temperature reconstruction. Below 400 m, all the models exhibit a warm bias on the continental shelf, but the bias is reduced in the high-resolution simulations. We hypothesize some of the bias is due to an overestimation of sea ice cover, which reduces heat loss to the atmosphere. Both high-resolution and accurate bathymetry are required to improve model accuracy around Antarctica.

168. Tidal effects on dynamics and freshwater transport of a medium-scale river plume with multiple outlets

Authors: Xiaolong Zong; Xuan Cheng; Shuwen Zhang; Qiang Lian; Fangjing Deng; Zhaoyun Chen

Journal: Ocean Modelling

Publisher: Elsevier BV

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Open Access: N/A

Abstract: N/A

169. Submesoscale Variability and Basal Melting in Ice Shelf Cavities of the Amundsen Sea

Authors: Kalyan Shrestha (University of Washington); Georgy E. Manucharyan (University of Washington); Yoshihiro Nakayama (Hokkaido University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Volume/Issue: Volume 51, Issue 3

References: 61

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Melting of ice shelves can energize a wide range of ocean currents, from three-dimensional turbulence to relatively large-scale boundary currents. Here, we conduct high-resolution simulations of the western Amundsen Sea to show that submesoscale eddies are prevalent inside ice shelf cavities. The simulations indicate energetic submesoscale eddies at the top and bottom ocean boundary layers, regions with sharp topographic slopes and strong lateral buoyancy gradients. These eddies play a substantial role in the vertical and lateral (along-isopycnal) heat advection toward the ice shelf base, enhancing the basal melting in all simulated cavities. In turn, the meltwater provides strong buoyancy gradients that energize the submesoscale variability, forming a positive loop that could affect the overall efficiency of heat exchange between the ocean and the ice shelf cavity. Our study implies that submesoscale-induced enhancement of basal melting may be a ubiquitous process that needs to be parameterized in coarse-resolution climate models.

170. Fast Enhancement of the Stratification in the Indian Ocean over the Past 20 Years

Authors: Suqi Peng (Hohai University, Ministry of Natural Resources); Qiang Wang (Hohai University, Ministry of Natural Resources)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-23-0255.1

URL: <https://doi.org/10.1175/jcli-d-23-0255.1>

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References: 81

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Salinity, Climate change, Temperature, Indian Ocean, Atmosphere-ocean interaction (and 1 more)

Funding: the Strategic Priority Research Program of Chinese Academy of Sciences, the National Natural Science Foundation of China, the Fundamental Research Funds for the Central Universities

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Open Access: N/A

Abstract: Indian Ocean (IO) stratification has important effects on the air-sea interaction, ocean dynamics, and ecology. It is, therefore, of significance to investigate the changes in IO stratification. In this study, we use ensemble empirical mode decomposition (EEMD) to extract the nonlinear long-term trend in the upper IO stratification quantified by potential energy anomalies. The results show that the strengthening of the stratification is spatially and temporally nonuniform. Specifically, the trend of stratification intensified gradually before 1996, but accelerated rapidly after 1996. Temperature and salinity changes play a crucial role in the fast enhancement of stratification and its regional differences. Temperature variations dominate the stratification trend in ~90% of the IO area, while the contributions of salinity changes are mainly in the southeast Indian Ocean (SEIO). Vertically, the rapid enhancement of stratification is caused by the trend of temperature and salt in the upper 400 m. We further perform temperature budget analysis and find that the warming trend in the upper 400 m south of the IO is mainly modulated by vertical advection and meridional advection, while the warming in the north of the IO is mainly induced by air-sea heat fluxes. Salinity budget analysis shows that ocean advection has played a primary role in modulating SEIO salinity over the past 20 years.

171. An Optimization Method Based on Decorrelation Scales Analysis for Improving Surface Currents Retrieval From Sea Surface Temperature

Authors: Feng Ye (Nanjing University); Zengzhou Hao (Nanjing University); Delu Pan (Nanjing University)

Journal: IEEE Transactions on Geoscience and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

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Subjects: N/A

Keywords: ocean surface current, Decorrelation scale, optimization, sea surface temperature (SST)

Funding: National Key Research and Development Program of China, Program of Innovation 2030 on Smart Ocean, Zhejiang University

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Open Access: N/A

Abstract: N/A

172. The Global Overturning Circulation and the Role of Non-Equilibrium Effects in ECCOv4r4

Authors: Tatsu Monkman (University of Chicago); Malte F. Jansen (University of Chicago)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Publication Date: 2024-01-27

Volume/Issue: Volume 129, Issue 1

References: 58

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: watermass budgets, global overturning circulation, abyssal circulation, ocean circulation, ECCO

Funding: National Science Foundation

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Open Access: N/A

Abstract: We quantify the volume transport and watermass transformation rates of the global overturning circulation using the Estimating the Circulation and Climate of the Ocean version 4 release 4 (ECCOv4r4) reanalysis product. The ECCO solution shows large rates of intercell exchange between the mid-depth and abyssal cells, consistent with other recent inferences. About 10 Sv of North Atlantic deep water enters the abyssal cell in the Southern Ocean and is balanced by a similar amount of apparent diapycnal upwelling in the Indo-Pacific. However, much of the upwelling in ECCO's deep ocean is not associated with irreversible watermass transformations, as typically assumed in theoretical models. Instead, a dominant portion of the abyssal circulation in ECCO is associated with isopycnal volume tendencies, reflecting a deep ocean in a state of change and a circulation in which transient tendencies play a leading role in the watermass budget. These volume tendencies are particularly prominent in the Indo-Pacific, where ECCO depicts a cooling and densifying deep ocean with relatively little mixing-driven upwelling, in disagreement with recent observations of deep Indo-Pacific warming trends. Although abyssal ocean observations are insufficient to exclude the trends modeled by ECCO, we note that ECCO's parameterized diapycnal mixing in the abyssal ocean is much smaller than observational studies suggest and may lead to an under-representation of Antarctic Bottom Water consumption in the abyssal ocean. Whether or not ECCO's tendencies are realistic, they are a key part of its abyssal circulation and hence need to be taken into consideration when interpreting the ECCO solution.

173. Dynamical downscaling of CMIP6 scenarios with ENEA-REG: an impact-oriented application for the Med-CORDEX region

Authors: Alessandro Anav; Marta Antonelli; Sandro Calmanti; Adriana Carillo; Franco Catalano; Alessandro Dell'Aquila; Roberto Iacono; Salvatore Marullo; Ernesto Napolitano; Massimiliano Palma; Giovanna Pisacane; Gianmaria Sannino; Maria Vittoria Struglia

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

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Keywords: Hindcast|Earth system science

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License: CC BY 4.0

Open Access: N/A

Abstract: In the framework of the coordinated regional modeling initiative Med-CORDEX (Coordinated Regional Climate Downscaling Experiment), we present an updated version of the regional Earth System Model ENEA-REG designed to downscale, over the Mediterranean basin, the models used in the Coupled Model Intercomparison Project phase 6 (CMIP6). The regional ESM includes coupled atmosphere (WRF), ocean (MITgcm), land (Noah-MP, embedded within WRF), and river (HD) components with spatial resolution of 12 km for the atmosphere, 1/12 degrees for the ocean and 0.5 degrees for the river rooting model. For the present climate, we performed a hindcast (i.e. reanalysis-driven) and a historical simulation (GCM-driven) over the 1980-2014 temporal period. The evaluation shows that the regional ESM reliably reproduces the mean state, spatial and temporal variability of the relevant atmospheric and ocean variables. In addition, we analyze the future evolution (2015-2100) of the Euro-Mediterranean climate under three different scenarios (SSP1-2.6, SSP2-4.5, SSP5-8.5), focusing on several relevant essential climate variables and climate indicators for impacts. Among others, results highlight how, for the scenarios SSP2-4.5 and SSP5-8.5, the intensity, frequency and duration of marine heat waves continue to increase until the end of the century and anomalies of up to 2 degreesC, which are considered extreme at the beginning of this century, will be so frequent to become the norm in less than a hundred years under the SSP5-8.5 scenario. Overall, our results demonstrate the improvement due to the high-resolution air-sea coupling for the representation of high impact events, such as marine heat waves, and sea-level height.

174. Influence of Deep-Ocean Warming on Coastal Sea-Level Decadal Trends in the Gulf of Mexico

Authors: Jacob M. Steinberg (Woods Hole Oceanographic Institution); Christopher G. Piecuch (Woods Hole Oceanographic Institution); Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Phillip R. Thompson (University of Hawaii at Mānoa Honolulu HI USA); Sloan Coats (University of Hawaii at Mānoa Honolulu HI USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Cited by: 2 publications

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Subjects: N/A

Keywords: ocean warming, open- and coastal-ocean connections, sea-level

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Based on latest estimates (e.g., <https://sealevel.nasa.gov>), global mean sea level has risen nearly 100 mm since 1993. However, the rate of rise has not been constant in space or time and recent observations (since ~ 2008) reveal pronounced regional acceleration in the Gulf of Mexico (GoM). Here we use model solutions and observational data to identify the physical mechanisms responsible for enhanced rates of coastal sea-level rise in this region. We quantify the effect of offshore subsurface ocean warming on coastal sea-level rise and its relationship to regional hypsometry, the distribution of ocean area with depth. Using an Estimating the Circulation and Climate of the Ocean (ECCO) state estimate, we establish that coastal sea-level changes at the 10-year timescale are largely the result of changes in regional ocean mass, reflected in ocean bottom pressure. These coastal bottom pressure changes reflect both net mass flux into the Gulf, as well as internal mass redistribution within the Gulf, which can be understood as an isostatic ocean response to subsurface warming. We test the relationships among coastal sea-level, bottom pressure, and subsurface warming identified in ECCO using observations from satellite gravimetry, altimetry, tide gauges, and Argo floats. Estimates of mass redistribution explain a significant fraction of coastal sea-level trends observed by tide gauges. For instance, at St. Petersburg, Florida, this mass redistribution mechanism accounts for >50% of the coastal sea-level trend observed between 2008 and 2017. This study thus elucidates a physical mechanism whereby coastal sea-level responds to open-ocean subsurface density change.

175. What Forcing Mechanisms Affect the Interannual Sea Level Co-Variability Between the Northeast and Southeast Coasts of the United States?

Authors: Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Thomas Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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References: 52

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: buoyancy forcing, sea level, adjoint sensitivity, forcing mechanism, wind stress (and 1 more)

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Interannual sea-level variations between the United States (U.S.) Northeast and Southeast Coasts separated by Cape Hatteras are significantly less correlated than those within their respective sectors, but the cause is poorly understood. Here we investigate atmospheric forcing mechanisms that affect the interannual sea-level co-variability between these two sectors using an adjoint reconstruction and decomposition approach in the framework of Estimating the Circulation and Climate of the Ocean (ECCO) ocean state estimate. We compare modeled and observed sea-level changes at representative locations in each sector: Nantucket Island, Massachusetts for the Northeast and Charleston, South Carolina for the Southeast. The adjoint reconstruction and decomposition approach used in this work allows for identification and quantification of the causal mechanisms responsible for observed coastal sea-level variability. Coherent sea-level variations in Nantucket and Charleston arise from nearshore wind stress anomalies north of Cape Hatteras and buoyancy forcing, especially from the subpolar North Atlantic, while offshore wind stress anomalies, in contrast, reduce co-variability. Offshore wind stress contributes much more to interannual sea-level variation at Charleston than at Nantucket, causing incoherent sea level variations between the two locations. Buoyancy forcing anomalies south of Charleston, including over the Florida shelf, the Gulf of Mexico, and the Caribbean Sea, also reduce co-variability because they induce sea-level responses at Charleston but not

Nantucket. However, the relative impact of buoyancy forcing on interannual sea-level co-variability between the two sectors is much smaller than that of offshore wind stress.

176. A Dynamics-Weighted Principal Components Analysis of Dominant Atmospheric Drivers of Ocean Variability with an Application to the North Atlantic Subpolar Gyre

Authors: Daniel E. Amrhein (National Center for Atmospheric Research); Dafydd Stephenson (National Center for Atmospheric Research); LuAnne Thompson (University of Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

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Language: N/A

Subjects: N/A

Keywords: Optimization, North Atlantic Oscillation, Principal components analysis, North Atlantic Ocean, Statistical techniques (and 1 more)

Funding: Earth Sciences Division

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Open Access: N/A

Abstract: This paper describes a framework for identifying dominant atmospheric drivers of ocean variability. The method combines statistics of atmosphere-ocean fluxes with physics from an ocean general circulation model to derive atmospheric patterns optimized to excite variability in a specified ocean quantity of interest. We first derive the method as a weighted principal components analysis and illustrate its capabilities in a toy problem. Next, we apply our analysis to the problem of interannual upper ocean heat content (HC) variability in the North Atlantic Subpolar Gyre (SPG) using the adjoint of the MITgcm and atmosphere-ocean fluxes from the ECCOv4-r4 state estimate. An unweighted principal components analysis reveals that North Atlantic heat and momentum fluxes in ECCOv4-r4 have a range of spatiotemporal patterns. By contrast, dynamics-weighted principal components analysis collapses the space of these patterns onto a small subset--principally associated with the North Atlantic Oscillation--that dominates interannual SPG HC variance. By perturbing the ECCOv4-r4 state estimate, we illustrate the pathways along which variability propagates from the atmosphere to the ocean in a nonlinear ocean model. This technique is applicable across a range of problems across Earth system components, including in the absence of a model adjoint. **Significance Statement** While the oceans have absorbed 90% of the excess heat associated with human-forced climate change, the change in the ocean's heat content is not steady, with peaks and troughs superimposed upon a general increase. These fluctuations come from chaotic changes in the atmosphere and ocean and can be hard to disentangle. We use this case of ocean heat content variability to introduce a new method for determining the patterns of weather and climate in the atmosphere that are most effective at generating fluctuations in the ocean. To do this, we combine the statistics of recent atmospheric activity with output from a state-of-the-art numerical ocean model that reveals physical processes driving changes in ocean quantities including ocean heat content. This approach suggests that the atmospheric patterns

that stimulate the most energetic changes in ocean heat content in the northern North Atlantic are not necessarily the most energetic patterns present in the atmosphere. We test our findings by preventing these patterns from affecting the ocean in our numerical model and measure a strong reduction in ocean heat content fluctuations.

177. Global Estimates of Mesoscale Vertical Velocity Near 1,000 m From Argo Observations

Authors: Katy M. Christensen (University of Washington); Alison R. Gray (University of Washington); Stephen C. Riser (University of Washington)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: Argo floats, vertical velocity, physical oceanography

Funding: National Aeronautics and Space Administration, National Science Foundation, National Oceanic and Atmospheric Administration

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Global estimates of mesoscale vertical velocity remain poorly constrained due to a historical lack of adequate observations on the spatial and temporal scales needed to measure these small magnitude velocities. However, with the wide-spread and frequent observations collected by the Argo array of autonomous profiling floats, we can now better quantify mesoscale vertical velocities throughout the global ocean. We use the underutilized trajectory data files from the Argo array to estimate the time evolution of isotherm displacement around a float as it drifts at 1,000 m, allowing us to quantify vertical velocity averaged over approximately 4.5 days for that depth level. The resulting estimates have a non-normal, high-peak, and heavy-tail distribution. The vertical velocity distribution has a mean value of $(1.9 \pm 0.02) \times 10^{-6} \text{ m s}^{-1}$ and a median value of $(1.3 \pm 0.2) \times 10^{-7} \text{ m s}^{-1}$, but the high-magnitude events can be up to the order of 10^{-4} m s^{-1} . We find that vertical velocity is highly spatially variable and is largely associated with a combination of topographic features and horizontal flow. These are some of the first observational estimates of mesoscale vertical velocity to be taken across such large swaths of the ocean without assumptions of uniformity or reliance on horizontal divergence.

178. A Synthesis of Global Coastal Ocean Greenhouse Gas Fluxes

Authors: L. Resplandy (Princeton University); A. Hogikyan (Princeton University); J. D. Müller (ETH Zurich); R. G. Najjar (Pennsylvania State University); H. W. Bange (GEOMAR Helmholtz Centre for Ocean Research); D. Bianchi (University of California Los Angeles); T. Weber (Department of Earth and Environmental Science University of Rochester Rochester NY USA); W.-J. Cai (University of Delaware); S. C. Doney (University of Virginia); K. Fennel (Dalhousie University); M. Gehlen (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); J. Hauck (Alfred Wegener Institute); F. Lacroix (University of Bern); P. Landschützer (Flanders Marine Institute (VLIZ) Ostend Belgium); C. Le Quéré (University of East Anglia); A. Roobaert (Department Geoscience, Environment and Society--BGEOSYS Université Libre de Bruxelles Brussels Belgium, Flanders Marine Institute (VLIZ) Ostend Belgium); J. Schwinger (Bjerknes Centre for Climate Research); S. Berthet (CNRM Université de Toulouse Météo France CNRS Toulouse France); L. Bopp (École Polytechnique); T. T. T. Chau (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); M. Dai (Xiamen University); N. Gruber (ETH Zurich); T. Ilyina (Max Planck Institute for Meteorology); A. Kock (GEOMAR Helmholtz Centre for Ocean Research, Now at State Office for the Environment of the State of Schleswig-Holstein Flintbek Germany); M. Manizza (Scripps Institution of Oceanography / University of California San Diego); Z. Lachkar (Arabian Center for Climate and Environmental Sciences New York University Abu Dhabi Abu Dhabi United Arab Emirates); G. G. Laruelle (Department Geoscience, Environment and Society--BGEOSYS Université Libre de Bruxelles Brussels Belgium); E. Liao (Princeton University, Now at School of Oceanography Shang Jiao Tong University Shanghai China); I. D. Lima (Woods Hole Oceanographic Institution); C. Nissen (University of Colorado Boulder, Alfred Wegener Institute); C. Rödenbeck (MPI Biogeochemistry Jena Germany); R. Séférian (CNRM Université de Toulouse Météo France CNRS Toulouse France); K. Toyama (Meteorological Research Institute); H. Tsujino (Meteorological Research Institute); P. Regnier (Department Geoscience, Environment and Society--BGEOSYS Université Libre de Bruxelles Brussels Belgium)

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Keywords: RECCAP2, methane, coastal ocean, nitrous oxide, carbon

Funding: Climate Program Office, National Science Foundation, Fonds Wetenschappelijk Onderzoek

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Abstract: The coastal ocean contributes to regulating atmospheric greenhouse gas concentrations by taking up carbon dioxide (CO₂) and releasing nitrous oxide (N₂O) and methane (CH₄). In this second

phase of the Regional Carbon Cycle Assessment and Processes (RECCAP2), we quantify global coastal ocean fluxes of CO₂, N₂O and CH₄ using an ensemble of global gap-filled observation-based products and ocean biogeochemical models. The global coastal ocean is a net sink of CO₂ in both observational products and models, but the magnitude of the median net global coastal uptake is ~60% larger in models (-0.72 vs. -0.44 PgC year⁻¹, 1998-2018, coastal ocean extending to 300 km offshore or 1,000 m isobath with area of 77 million km²). We attribute most of this model-product difference to the seasonality in sea surface CO₂ partial pressure at mid- and high-latitudes, where models simulate stronger winter CO₂ uptake. The coastal ocean CO₂ sink has increased in the past decades but the available time-resolving observation-based products and models show large discrepancies in the magnitude of this increase. The global coastal ocean is a major source of N₂O (+0.70 PgCO₂-e year⁻¹ in observational product and +0.54 PgCO₂-e year⁻¹ in model median) and CH₄ (+0.21 PgCO₂-e year⁻¹ in observational product), which offsets a substantial proportion of the coastal CO₂ uptake in the net radiative balance (30%-60% in CO₂-equivalents), highlighting the importance of considering the three greenhouse gases when examining the influence of the coastal ocean on climate.

179. Ubiquitous acceleration in Greenland Ice Sheet calving from 1985 to 2022

Authors: Chad A. Greene; Alex S. Gardner; Michael Wood; Joshua K. Cuzzone

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-023-06863-2

URL: <https://doi.org/10.1038/s41586-023-06863-2>

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Volume/Issue: Volume 625, Issue 7995

References: 84

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

180. A time-average ocean: Thermal wind and flow spirals

Authors: Carl Wunsch

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2024.103206

URL: <https://doi.org/10.1016/j.pocean.2024.103206>

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Volume/Issue: Volume 221

References: 56

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Crohn's and Colitis Organisation, Massachusetts Institute of Technology

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Open Access: N/A

Abstract: N/A

181. Observations reveal onshore acceleration and offshore deceleration of the Kuroshio Current in the East China Sea over the past three decades

Authors: Haihong Guo; Jinzhuo Cai; Haiyuan Yang; Zhaohui Chen

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/ad1d3b

URL: <https://doi.org/10.1088/1748-9326/ad1d3b>

Publication Date: 2024-01-10

Volume/Issue: Volume 19, Issue 2

References: 53

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Key Research and Development Program of China, Shandong Province Postdoctoral Program for Innovative Talents

License: CC BY 4.0

Open Access: N/A

Abstract: The Kuroshio Current (KC) in the East China Sea is one of the most prominent components of the ocean circulation system in the North Pacific. The onshore intensification of the KC is found to drive nutrient-rich upwelling in the shelf regions, induce anomalous warming that leads to coastal marine heatwaves, and reduce the ability of the oceans to absorb anthropogenic carbon dioxide. Based on altimeter and in situ observations, we find an onshore acceleration and offshore deceleration of the KC over the past three decades. This intensification is characterized by a spatial mean onshore acceleration (offshore deceleration) of 0.39 (-0.63) cm s⁻¹ per decade. This phenomenon can be attributed to changes in wind stress curl (WSC) and oceanic stratification over the subtropical North Pacific. The weakened WSC decreases the vertical extent of the KC by reducing its transport and contributes to the offshore deceleration, whereas the enhanced stratification drives the uplift of the KC and contributes to the onshore acceleration. Our findings underscore the importance of establishing and maintaining a long-term monitoring network for the zonal variations of the KC in the future to obtain a comprehensive understanding of the associated impacts.

182. Can we explain the post-2015 absence of the Chandler wobble?

Authors: Ryuji Yamaguchi; Masato Furuya

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-023-01944-y

URL: <https://doi.org/10.1186/s40623-023-01944-y>

Publication Date: 2024-01-02

Volume/Issue: Volume 76, Issue 1

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Science and Technology Agency

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Open Access: N/A

Abstract: Recent polar motion data do not show a 6-year beat and indicate the absence of the Chandler wobble (CW), whereas we could observe the 6-year beat even in the 1920-40 s when the CW amplitude was known to be smallest. As a free mode, the CW needs excitation one or more sources that were debated decades ago but are now attributed to the atmosphere, ocean, and possibly land water. Here, we show that the anomaly started in 2015, after which two independent estimates of the atmospheric CW excitation became persistently smaller than before. However, the estimates of the oceanic and land-water contributions are too large, suggesting improved estimates are needed. Taking advantage of the recent CW anomaly, we show that the quality factor of CW is not as high as 100 as previously preferred. Although the CW excitation processes have been assumed random, a termination of near-resonant processes would rather be consistent with the present findings. Graphical Abstract

183. Coherent Interannual-Decadal Potential Temperature Variability in the Tropical-North Pacific Ocean and Deep South China Sea

Authors: Yuxin Lin (Southern University of Science and Technology); Jianping Gan (Center for Ocean Research in Hong Kong and Macau Department of Mathematics Department of Ocean Science Hong Kong University of Science and Technology Hong Kong China); Zhongya Cai (State Key Laboratory of Internet of Things for Smart City Department of Ocean Science and Technology University of Macau Macau China); Qi Quan (State Key Laboratory of Estuarine and Coastal Research East China Normal University Shanghai China); Tingting Zu (Chinese Academy of Sciences); Zhiqiang Liu (Southern University of Science and Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl106256

URL: <https://doi.org/10.1029/2023gl106256>

Publication Date: 2024-01-01

Volume/Issue: Volume 51, Issue 1

References: 67

Cited by: N/A publications

Language: en

Subjects: N/A

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Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Climate variability over the Tropical and North Pacific Ocean (TPO and NPO, respectively) modulates marginal sea variability. The South China Sea (SCS), the largest marginal sea in the western NPO, is an outstanding example of a region that responds quickly to climate change. However, there is considerable uncertainty regarding the response of the deep SCS to large-scale climate variability. Multivariate empirical orthogonal function analysis revealed three prominent modes of interconnected temperature anomaly fluctuations within the TPO and NPO. These coherent modes highlight the interactive dynamics among climate variations and reveal their modulation mechanisms for previously less explored potential temperature variabilities in the deep SCS. On the atmospheric bridge, external forces modify the upper-layer Luzon Strait Transport (LST) by adjusting the Ekman transport and Kuroshio intrusion. For the oceanic pathway, climate variations disturb the deep-layer LST by adjusting the barotropic flows in the upper layer.

184. An ARGO and XBT Observing System for the Atlantic Meridional Overturning Circulation and Meridional Heat Transport (AXMOC) at 22.5 degreesS

Authors: I. Pita (Atlantic Oceanographic and Meteorological Laboratory NOAA Miami FL USA, University of Miami); M. Goes (Atlantic Oceanographic and Meteorological Laboratory NOAA Miami FL USA, University of Miami); D. L. Volkov (Atlantic Oceanographic and Meteorological Laboratory NOAA Miami FL USA, University of Miami); S. Dong (Atlantic Oceanographic and Meteorological Laboratory NOAA Miami FL USA); G. Goni (Atlantic Oceanographic and Meteorological Laboratory NOAA Miami FL USA); M. Cirano (Federal University of Rio de Janeiro)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023jc020010

URL: <https://doi.org/10.1029/2023jc020010>

Publication Date: 2023-12-28

Volume/Issue: Volume 129, Issue 1

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Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Atlantic Ocean observing system, Ocean boundary influence on AMOC variability, AX08, AX18, and AX97 high-density XBT transects

Funding: N/A

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Open Access: N/A

Abstract: Changes in the Atlantic Meridional Overturning Circulation (AMOC) and associated Meridional Heat Transport (MHT) can affect climate and weather patterns, regional sea levels, and ecosystems. Direct observations of the AMOC are still limited, particularly in the South Atlantic. This study establishes a cost-effective trans-basin section to estimate for the first time the AMOC and MHT at 22.5 degreesS, using only sustained ocean observations. For this, an optimal mapping method that minimizes the difference between surface in situ dynamic height and satellite altimetry was developed to retrieve monthly temperature and salinity profiles from Argo and XBT data along the 22.5 degreesS section. The mean states, as well as the interannual and seasonal changes of the obtained AMOC and MHT were compared with other products. The mean AMOC and MHT for 22.5 degreesS are 16.3 +/- 3.2 Sv and 0.7 +/- 0.2 PW, respectively, showing stronger transports during austral fall/winter and weaker in spring. The high-density XBT data available at the western boundary were vital for capturing the highly variable Brazil Current (BC), whose mean and variability was improved compared to other products. At 22.5 degreesS, the North Atlantic Deep Water is divided into two cores that flow along both the western and the eastern boundaries near 2,500 m depth. Our results (a) suggest a greater influence of the western boundary current system on the AMOC variability at 22.5 degreesS, (b) highlight the importance of high-density in situ data for AMOC estimates, and (c) contribute to a better understanding of the AMOC and MHT variability in the South Atlantic.

185. The sensitivity of the equatorial pacific ODZ to particulate organic matter remineralization in a climate model under pre-industrial conditions

Authors: Paul Lerner; Anastasia Romanou; David Nicholson; Maxwell Kelley; Reto Ruedy; Gary Russell

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2023.102303

URL: <https://doi.org/10.1016/j.ocemod.2023.102303>

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

186. Solar-Induced Fluorescence Helps Constrain Global Patterns in Net Biosphere Exchange, as Estimated Using Atmospheric CO₂ Observations

Authors: Mingyang Zhang (Johns Hopkins University); Joseph A. Berry (Department of Global Ecology Carnegie Institution for Science Stanford CA USA); Yoichi P. Shiga (Independent Researcher San Francisco CA USA); Russell B. Doughty (University of Oklahoma); Nima Madani (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, UCLA Joint Institute for Regional Earth System Science and Engineering Los Angeles CA USA); Xing Li (Seoul National University); Jingfeng Xiao (University of New Hampshire); Jiaming Wen (School of Integrative Plant Science Soil and Crop Sciences Section Cornell University Ithaca NY USA); Ying Sun (School of Integrative Plant Science Soil and Crop Sciences Section Cornell University Ithaca NY USA); Scot M. Miller (Johns Hopkins University)

Journal: Journal of Geophysical Research: Biogeosciences

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023jg007703

URL: <https://doi.org/10.1029/2023jg007703>

Publication Date: 2023-12-26

Volume/Issue: Volume 128, Issue 12

References: 132

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: carbon cycle, SIF, GPP, OCO-2, inverse modeling

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Solar-induced chlorophyll fluorescence (SIF) shows enormous promise as a proxy for photosynthesis and as a tool for modeling variability in gross primary productivity and net biosphere exchange (NBE). In this study, we explore the skill of SIF and other vegetation indicators in predicting variability in global atmospheric CO₂ observations, and thus global variability in NBE. We do so using a 4-year record of CO₂ observations from NASA's Orbiting Carbon Observatory 2 satellite and using a geostatistical inverse model. We find that existing SIF products closely correlate with space-time variability in atmospheric CO₂ observations, particularly in the extratropics. In the extratropics, all SIF products exhibit greater skill in explaining variability in atmospheric CO₂ observations compared to an ensemble of process-based CO₂ flux models and other vegetation indicators. With that said, other vegetation indicators, when multiplied by photosynthetically active radiation, yield similar results as SIF and may therefore be an effective structural SIF proxy at regional to global spatial scales. Furthermore, we find that using SIF as a predictor variable in the geostatistical inverse model shifts the seasonal cycle of estimated NBE and yields an earlier end to the growing season relative to other vegetation indicators. These results highlight how SIF can help constrain global-scale variability in NBE.

187. Impact of time-dependent data assimilation on ice flow model initialization and projections: a case study of Kjer Glacier, Greenland

Authors: Youngmin Choi; Helene Seroussi; Mathieu Morlighem; Nicole-Jeanne Schlegel; Alex Gardner

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-5499-2023

URL: <https://doi.org/10.5194/tc-17-5499-2023>

Publication Date: 2023-12-22

Volume/Issue: Volume 17, Issue 12

References: 71

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: . Ice sheet models are often initialized with data assimilation of present-day conditions, in which unknown model parameters are estimated using the inverse method. While assimilation of snapshot observations has been widely used for regional- and large-scale ice sheet simulations, data assimilation based on time-dependent data has recently started to emerge to constrain model parameters while capturing the transient evolution of the system. However, this method has been applied only to a few glaciers with fixed ice front positions, using spatially and temporally limited observations, and has not been applied to marine-terminating glaciers of the Greenland Ice Sheet that have been retreating over the last 30 years. In this study, we assimilate time series of surface velocity into a model of Kjer Glacier in West Greenland to better capture the observed acceleration over the past 3 decades. We compare snapshot and transient inverse methods and investigate the impact of initialization procedures on the parameters inferred, as well as model projections. We find that transient-calibrated simulations better capture past trends and better reproduce changes after the calibration period, even when a short period of observations is used. The results show the feasibility and clear benefits of a time-dependent data assimilation for initializing ice sheet models. This approach is now possible with the development of longer observational records, though it remains computationally challenging.

188. Water Stress Dominates 21st-Century Tropical Land Carbon Uptake

Authors: Paul A. Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Victoria Meyer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA); Marcos Longo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Lawrence Berkeley National Laboratory)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gb007702

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Publication Date: 2023-12-20

Volume/Issue: Volume 37, Issue 12

References: 94

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: data assimilation, carbon cycle, water stress

Funding: N/A

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Water stress regulates land-atmosphere carbon dioxide (CO₂) exchanges in the tropics; however, its role remains poorly characterized due to the confounding roles of radiation, temperature and canopy dynamics. In particular, uncertainty stems from the relative roles of plant-available water (supply) and atmospheric water vapor deficit (demand) as mechanistic drivers of photosynthetic carbon (C) uptake variability. Using satellite measurements of gravity, CO₂ and fluorescence to constrain a mechanistic carbon-water cycle model from 2001 to 2018, we found that the interannual variability (IAV) of water stress on photosynthetic C uptake was 52% greater than the combined effects of other factors. Surprisingly, the dominance of water stress on C uptake IAV was greater in the wet tropics (94%) than in the dry tropics (26%). Plant-available water supply and atmospheric demand both contributed to the IAV of water stress on photosynthetic C uptake across the tropics, but the IAV of demand effects was 21% greater than the IAV of supply effects (33% greater in the wet tropics and 6% greater in the dry tropics). We found that the IAV of water stress on C uptake was 24% greater than the IAV of the combination of other factors in the net land-atmosphere C sink in the whole tropics, 26% greater in the wet tropics, and 7% greater in the dry tropics. Given the recent trends in tropical

precipitation and atmospheric humidity, our findings indicate that water stress----from both supply and demand----will likely dominate the climate response of land C sink across tropical ecosystems in the coming decades.

189. Evaluation of Antarctic sea ice thickness and volume during 2003-2014 in CMIP6 using Envisat and CryoSat-2 observations

Authors: Yaqi Hou; Yafei Nie; Chao Min; Qi Shu; Hao Luo; Jiping Liu; Qinghua Yang

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/ad1725

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Publication Date: 2023-12-19

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References: 69

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Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China, the fundamental research funds for the Norges Forskningsråd (and 1 more)

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Open Access: N/A

Abstract: Sea ice thickness (SIT), which is a crucial and sensitive indicator of climate change in the Antarctic, has a substantial impact on atmosphere-sea-ice-ocean interactions. Despite the slight thinning in SIT and reduction in sea ice volume (SIV) in the Antarctic in the recent decade, challenges remain in quantifying their changes, primarily because of the limited availability of high-quality long-term observational data. Therefore, it is crucial to accurately simulate Antarctic SIT and to assess the SIT simulation capability of state-of-the-art climate models. In this study, we evaluated historical simulations of SIT by 51 climate models of the Coupled Model Intercomparison Project Phase 6 (CMIP6) using Envisat (ES) and CryoSat-2 (CS2) observations. Results revealed that most models can capture the seasonal cycles in SIV and that the CMIP6 multimodel mean (MMM) can reproduce the increasing and decreasing trends in the SIV anomaly based on ES and CS2 data, although the magnitudes of the trends in the SIV anomaly are underestimated. Additionally, the intermodel spread in simulations of SIT and SIV was found to be reduced (by 43%) from CMIP5 to CMIP6. Nevertheless, based on the CMIP6 MMM, substantial underestimations in SIV of 57.52% and 59.66% were found compared to those derived from ES and CS2 observations, respectively. The most notable underestimation in SIT was located in the sea ice deformation zone surrounding the northwestern Weddell Sea, coastal areas of the Bellingshausen and Amundsen seas, and the eastern Ross Sea. The substantial bias in the simulated SIT might result from deficiencies in simulating critical physical processes such as ocean heat transport, dynamic sea ice processes, and sea ice-ocean interactions. Therefore, increasing the model resolution and improving the representation of sea ice dynamics and the physical processes controlling sea ice-ocean interactions are essential for improving the accuracy of Antarctic sea ice simulation.

190. A review of global products of air-sea turbulent heat flux: accuracy, mean, variability, and trend

Authors: Ronglin Tang; Yizhe Wang; Yazhen Jiang; Meng Liu; Zhong Peng; Yongxin Hu; Lingxiao Huang; Zhao-Liang Li

Journal: Earth-Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.earscirev.2023.104662

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References: 137

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

191. Influence of rossby wave in southern Indian Ocean on the low frequency variability of eddy kinetic energy within agulhas current system

Authors: Tianyu Wang; Yan Du; Xiaomei Liao; Runjie Zhou; Ogooluwa Samuel Adeagbo

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Chinese Academy of Sciences, National Key Research and Development Program of China

License: N/A

Open Access: N/A

Abstract: N/A

192. Dynamic Response to Ice Shelf Basal Meltwater Relevant to Explain Observed Sea Ice Trends Near the Antarctic Continental Shelf

Authors: Wilma G. C. Huneke (Australian National University); William R. Hobbs (University of Tasmania); Andreas Klocker (NORCE Norwegian Research Centre, Bjerknes Centre for Climate Research); Kaitlin A. Naughten (British Antarctic Survey Cambridge UK)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2023gl105435>

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: ocean-sea ice model, Antarctic melt water, Antarctic slope current, Antarctic sea ice

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Open Access: N/A

Abstract: Observed Antarctic sea ice trends up to 2015 have a distinct regional and seasonal pattern, with a loss during austral summer and autumn in the Bellingshausen and Amundsen Seas, and a year-round increase in the Ross Sea. Global climate models generally failed to reproduce the magnitude of sea ice trends implying that the models miss relevant mechanisms. One possible mechanism is basal meltwater, which is generally not included in the current generation of climate models. Previous work on the effects of meltwater on sea ice has focused on thermodynamic processes. However, local freshening also leads to dynamic changes, affecting ocean currents through geostrophic balance. Using a coupled ocean/sea-ice/ice-shelf model, we demonstrate that basal melting can intensify coastal currents in West Antarctica and the westward transport of sea ice. This change in transport results in sea ice anomalies consistent with observations, and may explain the disparity between climate models and observations.

193. A brief history of physical oceanography with Mediterranean examples

Authors: Paola Malanotte-Rizzoli

Journal: Rendiconti Lincei. Scienze Fisiche e Naturali

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12210-023-01210-z

URL: <https://doi.org/10.1007/s12210-023-01210-z>

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Massachusetts Institute of Technology

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Open Access: N/A

Abstract: Oceanography, or Oceanology, is a very broad name covering all the traditional sciences (physics, chemistry, biology etc.) which can be studied in the ocean context. Hence the distinction must be made using the specific names of Physical, Chemical or Biological Oceanography. This particular review is devoted to the history of Physical Oceanography, a discipline which remained at the empirical stage until roughly the middle of last century, with a slow development of the rigorous, mathematical approach and analysis required to fully understand the processes and dynamics at all space/time scales. This review wants to introduce the uninformed reader to this relatively new science, providing a concise but hopefully comprehensive outline of the milestones of its history, from the adventurous beginnings of the past centuries, to the successive, very recent theoretical and observational advances. In this progress, the scientific "giants" who made possible these advances are also presented. Their choice constitutes not only the author's personal opinion but that of the oceanographic community at large which recognizes their crucial impact in making physical oceanography the quantitative, exact science of today. It is the author's privilege to have personally known most of them, not only as teachers but as mentors and friends.

194. Retrieval of Subsurface Velocities in the Southern Ocean from Satellite Observations

Authors: Liang Xiang (Chinese Academy of Sciences, Laoshan Laboratory); Yongsheng Xu (Chinese Academy of Sciences, Laoshan Laboratory); Hanwei Sun (Spaceborne Radar Research Center, Beijing Institute of Radio Measurement, Beijing 100039, China); Qingjun Zhang (Institute of Remote Sensing Satellite, Chinese Academy of Space Technology, Beijing 100094, China); Liqiang Zhang (Institute of Remote Sensing Satellite, Chinese Academy of Space Technology, Beijing 100094, China); Lin Zhang (Naval Submarine Academy, Qingdao 266199, China); Xiangguang Zhang (Chinese Academy of Sciences, Laoshan Laboratory); Chao Huang (Chinese Academy of Sciences, Laoshan Laboratory); Dandan Zhao (Institute of Oceanographic Instrumentation, Qilu University of Technology (Shandong Academy of Sciences), Qingdao 266061, China)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs15245699

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Keywords: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: Determining the dynamic processes of the subsurface ocean is a critical yet formidable undertaking given the sparse measurement resources available presently. In this study, using the light gradient boosting machine algorithm (LightGBM), we report for the first time a machine learning strategy for retrieving subsurface velocities at 1000 dbar depth in the Southern Ocean from information derived from satellite observations. Argo velocity measurements are used in the training and validation of the LightGBM model. The results show that reconstructed subsurface velocity agrees better with Argo velocity than reanalysis datasets. In particular, the subsurface velocity estimates have a correlation coefficient of 0.78 and an RMSE of 4.09 cm/s, which is much better than the ECCO estimates, GODAS estimates, GLORYS12V1 estimates, and Ora-S5 estimates. The LightGBM model has a higher skill in the reconstruction of subsurface velocity than the random forest and the linear regressor models. The estimated subsurface velocity exhibits a statistically significant increase at 1000 dbar since the 1990s, providing new evidence for the deep acceleration of mean circulation in the Southern Ocean. This study demonstrates the great potential and advantages of statistical methods for subsurface velocity modeling and oceanic dynamical information retrieval.

195. Lateral shift from turbidite- to contourite-dominated continental slope, a case study from southeast Brazil slope

Authors: U. Schattner; C.B. Rocha; R.B. Ramos; N. Shtober-Zisu; F.J. Lobo; M.M. de Mahiques

Journal: Geomorphology

Publisher: Elsevier BV

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References: 63

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

196. Observing Antarctic Bottom Water in the Southern Ocean

Authors: Alessandro Silvano; Sarah Purkey; Arnold L. Gordon; Pasquale Castagno; Andrew L. Stewart; Stephen R. Rintoul; Annie Foppert; Kathryn L. Gunn; Laura Herraiz-Borreguero; Shigeru Aoki; Yoshihiro Nakayama; Alberto C. Naveira Garabato; Carl Spingys; Camille Hayatte Akhoudas; Jean-Baptiste Sallée; Casimir de Lavergne; E. Povl Abrahamsen; Andrew J. S. Meijers; Michael P. Meredith; Shenjie Zhou; Takeshi Tamura; Kaihe Yamazaki; Kay I. Ohshima; Pierpaolo Falco; Giorgio Budillon; Tore Hattermann; Markus A. Janout; Pedro Llanillo; Melissa M. Bowen; Elin Darelus; Svein Østerhus; Keith W. Nicholls; Craig Stevens; Denise Fernandez; Laura Cimoli; Stanley S. Jacobs; Adele K. Morrison; Andrew McC. Hogg; F. Alexander Haumann; Ali Mashayek; Zhaomin Wang; Rodrigo Kerr; Guy D. Williams; Won Sang Lee

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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Funding: N/A

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Open Access: N/A

Abstract: Dense, cold waters formed on Antarctic continental shelves descend along the Antarctic continental margin, where they mix with other Southern Ocean waters to form Antarctic Bottom Water (AABW). AABW then spreads into the deepest parts of all major ocean basins, isolating heat and carbon from the atmosphere for centuries. Despite AABW's key role in regulating Earth's climate on long time scales and in recording Southern Ocean conditions, AABW remains poorly observed. This lack of observational data is mostly due to two factors. First, AABW originates on the Antarctic continental shelf and slope where in situ measurements are limited and ocean observations by satellites are hampered by persistent sea ice cover and long periods of darkness in winter. Second, north of the Antarctic continental slope, AABW is found below approximately 2 km depth, where in situ observations are also scarce and satellites cannot provide direct measurements. Here, we review progress made during the past decades in observing AABW. We describe 1) long-term monitoring obtained by moorings, by ship-based surveys, and beneath ice shelves through bore holes; 2) the recent development of autonomous observing tools in coastal Antarctic and deep ocean systems; and 3) alternative approaches including data assimilation models and satellite-derived proxies. The variety of approaches is beginning to transform our understanding of AABW, including its formation processes, temporal variability, and contribution to the lower limb of the global ocean meridional overturning circulation. In particular, these observations highlight the key role played by winds, sea ice, and the Antarctic Ice Sheet in AABW-related processes. We conclude by discussing future avenues for observing and understanding AABW, impressing the need for a sustained and coordinated observing

system.

197. An evaluation of the LLC4320 global-ocean simulation based on the submesoscale structure of modeled sea surface temperature fields

Authors: Katharina Gallmeier; J. Xavier Prochaska; Peter Cornillon; Dimitris Menemenlis; Madolyn Kelm

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-7143-2023

URL: <https://doi.org/10.5194/gmd-16-7143-2023>

Publication Date: 2023-12-08

Volume/Issue: Volume 16, Issue 23

References: 36

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research, National Aeronautics and Space Administration, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: . We have assembled 2 851 702 nearly cloud-free cutout images (sized 144 km x 144 km) of sea surface temperature (SST) data from the entire 2012-2020 Level-2 Visible Infrared Imaging Radiometer Suite (VIIRS) dataset to perform a quantitative comparison to the ocean model output from the MIT General Circulation Model (MITgcm). Specifically, we evaluate outputs from the LLC4320 (LLC, latitude-longitude-polar cap) 148■ global-ocean simulation for a 1-year period starting on 17 November 2011 but otherwise matched in geography and the day of the year to the VIIRS observations. In lieu of simple (e.g., mean, standard deviation) or complex (e.g., power spectrum) statistics, we analyze the cutouts of SST anomalies with an unsupervised probabilistic autoencoder (PAE) trained to learn the distribution of structures in SST anomaly (SSTa) on ~ 10-80 km scales (i.e., submesoscale to mesoscale). A principal finding is that the LLC4320 simulation reproduces, over a large fraction of the ocean, the observed distribution of SSTa patterns well, both globally and regionally. Globally, the medians of the structure distributions match to within 2sigma for 65 % of the ocean, despite a modest, latitude-dependent offset. Regionally, the model outputs reproduce mesoscale variations in SSTa patterns revealed by the PAE in the VIIRS data, including subtle features imprinted by variations in bathymetry. We also identify significant differences in the distribution of SSTa patterns in several regions: (1) in an equatorial band equatorward of 15■; (2) in the Antarctic Circumpolar Current (ACC), especially in the eastern half of the Indian Ocean; and (3) in the vicinity of the point at which western boundary currents separate from the continental margin. It is clear that region 3 is a result of premature separation in the simulated western boundary currents. The model output in region 2, the southern Indian Ocean, tends to predict more structure than observed, perhaps arising from a misrepresentation of the mixed layer or of energy dissipation and stirring in the simulation. The differences in region 1, the equatorial band, are also likely due to model errors, perhaps arising from the shortness of the simulation or from the lack of high-frequency and/or wavenumber atmospheric forcing. Although we do not yet know the exact causes for these model-data SSTa differences, we

expect that this type of comparison will help guide future developments of high-resolution global-ocean simulations.

198. Impacts of Ice-Ocean Stress on the Subpolar Southern Ocean: Role of the Ocean Surface Current

Authors: Yang Wu; Zhaomin Wang; Chengyan Liu; Liangjun Yan

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-023-3031-8

URL: <https://doi.org/10.1007/s00376-023-3031-8>

Publication Date: 2023-12-06

Volume/Issue: Volume 41, Issue 2

References: 57

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

199. Modelling considerations for research on ocean alkalinity enhancement (OAE)

Authors: Katja Fennel; Matthew C. Long; Christopher Algar; Brendan Carter; David Keller; Arnaud Laurent; Jann Paul Mattern; Ruth Musgrave; Andreas Oschlies; Josiane Ostiguy; Jaime B. Palter; Daniel B. Whitt

Journal: State of the Planet

Publisher: Copernicus GmbH

DOI: 10.5194/sp-2-oae2023-9-2023

URL: <https://doi.org/10.5194/sp-2-oae2023-9-2023>

Publication Date: 2023-11-27

Volume/Issue: Volume 2-oae2023

References: 166

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: ClimateWorks Foundation, Prince Albert II of Monaco Foundation, Natural Sciences and Engineering Research Council of Canada (and 3 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . The deliberate increase in ocean alkalinity (referred to as ocean alkalinity enhancement, or OAE) has been proposed as a method for removing CO₂ from the atmosphere. Before OAE can be implemented safely, efficiently, and at scale several research questions have to be addressed, including (1) which alkaline feedstocks are best suited and the doses in which they can be added safely, (2) how net carbon uptake can be measured and verified, and (3) what the potential ecosystem impacts are. These research questions cannot be addressed by direct observation alone but will require skilful and fit-for-purpose models. This article provides an overview of the most relevant modelling tools, including turbulence-, regional-, and global-scale biogeochemical models and techniques including approaches for model validation, data assimilation, and uncertainty estimation. Typical biogeochemical model assumptions and their limitations are discussed in the context of OAE research, which leads to an identification of further development needs to make models more applicable to OAE research questions. A description of typical steps in model validation is followed by proposed minimum criteria for what constitutes a model that is fit for its intended purpose. After providing an overview of approaches for sound integration of models and observations via data assimilation, the application of observing system simulation experiments (OSSEs) for observing system design is described within the context of OAE research. Criteria for model validation and intercomparison studies are presented. The article concludes with a summary of recommendations and potential pitfalls to be avoided.

200. Reply to RC1

Authors: Yavor Kostov

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-2023-1564-ac1

URL: <https://doi.org/10.5194/egusphere-2023-1564-ac1>

Publication Date: 2023-11-27

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Wind Stress|Forcing (mathematics)|Linearization

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

201. Reply to RC2

Authors: Yavor Kostov

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-2023-1564-ac2

URL: <https://doi.org/10.5194/egusphere-2023-1564-ac2>

Publication Date: 2023-11-27

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Wind Stress|Forcing (mathematics)|Linearization

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

202. Separating Direct Heat Flux Forcing and Freshwater Feedback on AMOC Change Under Global Warming

Authors: Qin Wen (Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China, School of Geography Nanjing Normal University Nanjing China, Chinese Academy of Sciences, Key Laboratory of Virtual Geographic Environment (Nanjing Normal University) Ministry of Education Nanjing China); Chenyu Zhu (Ocean University of China); Deliang Chen (University of Gothenburg); Mengyu Liu (Fudan University); Liang Ning (Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China, School of Geography Nanjing Normal University Nanjing China, Key Laboratory of Virtual Geographic Environment (Nanjing Normal University) Ministry of Education Nanjing China); Mi Yan (Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China, School of Geography Nanjing Normal University Nanjing China, Key Laboratory of Virtual Geographic Environment (Nanjing Normal University) Ministry of Education Nanjing China); Jian Liu (Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China, School of Geography Nanjing Normal University Nanjing China, Key Laboratory of Virtual Geographic Environment (Nanjing Normal University) Ministry of Education Nanjing China); Zhengyu Liu (Ohio State University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl105478

URL: <https://doi.org/10.1029/2023gl105478>

Publication Date: 2023-11-21

Volume/Issue: Volume 50, Issue 22

References: 56

Cited by: 2 publications

Language: en

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Keywords: N/A

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Open Access: N/A

Abstract: The Atlantic meridional overturning circulation (AMOC) is predicted to weaken under global warming. Whether it is caused by heat flux or freshwater flux is under debate. Here we separate these two processes in changing the AMOC under global warming. The simulated AMOC is weakened during the first 600 years and then gradually recovered to its initial state, with heat flux and freshwater feedback dominating at different timescales. Global warming immediately puts freshwater into the Southern Ocean, which triggers the initial AMOC weakening via altering surface temperature. Concurrently, the extensive heat into the ocean surface increases the temperature over the subpolar North Atlantic, reducing the deep convection and thus the AMOC in the subsequent 50-150 years. Meanwhile, the Arctic sea ice melt leads to the AMOC shutdown. Subsequently, the salinity accumulation in the subtropical North Atlantic propagating northward to restart the North Atlantic deep convection is responsible for the AMOC recovery.

203. Frequent marine heatwaves hidden below the surface of the global ocean

Authors: Di Sun; Furong Li; Zhao Jing; Shijian Hu; Bohai Zhang

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-023-01325-w

URL: <https://doi.org/10.1038/s41561-023-01325-w>

Publication Date: 2023-11-20

Volume/Issue: Volume 16, Issue 12

References: 55

Cited by: 18 publications

Language: en

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Keywords: N/A

Funding: National Science Foundation of China, Taishan Scholar Funds, Guangdong Provincial Key Laboratory of Interdisciplinary Research and Application for Data Science

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Open Access: N/A

Abstract: Marine heatwaves are extreme warm water events that can cause devastating impacts on ecosystems and have complex socio-economic ramifications. Surface signals and drivers of marine heatwaves have been extensively investigated based on satellite observations, whereas their vertical structure in the global ocean remains unclear. In this study, we identify marine heatwave events in the epipelagic zone (0-200 m) using a four-dimensional spatio-temporal framework based on three ocean reanalysis datasets. We find that only about half of the marine heatwave events have continuous surface signals during their life cycles and around one-third always reside in the subsurface ocean without any imprint on sea surface temperature. The annual number of these subsurface marine heatwave events shows a significant increase in response to subsurface mean-state warming during the past three decades. Our findings reveal the limitation of identifying marine heatwaves solely based on the sea surface temperature and underscore the necessity of subsurface observations for monitoring marine heatwaves.

204. Environmental Oceanography and Coastal Dynamics

Authors: Swapna Mukherjee; Kaushik Kiran Ghosh; Abhra Chanda

Journal: Unknown Journal

Publisher: Springer International Publishing

DOI: 10.1007/978-3-031-34422-0

URL: <https://doi.org/10.1007/978-3-031-34422-0>

Publication Date: 2023-11-14

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Coastal Pollution, Marine Biogeochemistry, Marine Processes, Coastal Processes, Environmental Oceanography

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

205. An Assessment of CO₂ Uptake in the Arctic Ocean From 1985 to 2018

Authors: Sayaka Yasunaka (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan, Tohoku University); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Jens Terhaar (University of Bern, Woods Hole Oceanographic Institution); Are Olsen (University of Bergen, Bjerknes Centre for Climate Research); Ryohei Yamaguchi (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan); Peter Landschützer (Flanders Marine Institute (VLIZ) Ostend Belgium, Max Planck Institute for Meteorology); Eiji Watanabe (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan); Dustin Carroll (San José State University); Hanani Adiwira (Tohoku University); Jens Daniel Müller (ETH Zurich); Judith Hauck (Alfred Wegener Institute)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gb007806

URL: <https://doi.org/10.1029/2023gb007806>

Publication Date: 2023-11-10

Volume/Issue: Volume 37, Issue 11

References: 131

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: CO₂ flux, sea ice, RECCAP2, pCO₂

Funding: Ministry of Education, Culture, Sports, Science and Technology, Japan Society for the Promotion of Science

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Open Access: N/A

Abstract: As a contribution to the Regional Carbon Cycle Assessment and Processes phase 2 (RECCAP2) project, we present synthesized estimates of Arctic Ocean sea-air CO₂ fluxes and their uncertainties from surface ocean pCO₂-observation products, ocean biogeochemical hindcast and data assimilation models, and atmospheric inversions. For the period of 1985-2018, the Arctic Ocean was a net sink of CO₂ of 116 +/- 4 TgC yr⁻¹ in the pCO₂ products, 92 +/- 30 TgC yr⁻¹ in the models, and 91 +/- 21 TgC yr⁻¹ in the atmospheric inversions. The CO₂ uptake peaks in late summer and early autumn, and is low in winter when sea ice inhibits sea-air fluxes. The long-term mean CO₂ uptake in the Arctic Ocean is primarily caused by steady-state fluxes of natural carbon (70% +/- 15%), and enhanced by the atmospheric CO₂ increase (19% +/- 5%) and climate change (11% +/- 18%). The annual mean CO₂ uptake increased from 1985 to 2018 at a rate of 31 +/- 13 TgC yr⁻¹ dec⁻¹ in the pCO₂ products, 10 +/- 4 TgC yr⁻¹ dec⁻¹ in the models, and 32 +/- 16 TgC yr⁻¹ dec⁻¹ in the atmospheric inversions. Moreover, 77% +/- 38% of the trend in the net CO₂ uptake over time is caused by climate change, primarily due to rapid sea ice loss in recent years. Furthermore, true uncertainties may be larger than the given ensemble standard deviations due to common structural biases across all individual estimates.

206. The Southern Ocean Carbon Cycle 1985-2018: Mean, Seasonal Cycle, Trends, and Storage

Authors: Judith Hauck (Alfred Wegener Institute); Luke Gregor (ETH Zurich); Cara Nissen (University of Colorado Boulder, Alfred Wegener Institute); Lavinia Patara (GEOMAR Helmholtz Centre for Ocean Research); Mark Hague (ETH Zurich); Precious Mongwe (Southern Ocean Carbon-Climate Observatory CSIR Pretoria South Africa); Seth Bushinsky (University of Hawai'i Mānoa Honolulu HI USA); Scott C. Doney (University of Virginia); Nicolas Gruber (ETH Zurich); Corinne Le Quéré (University of East Anglia); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Matthew Mazloff (Scripps Institution of Oceanography / University of California San Diego); Pedro M. S. Monteiro (School for Climate Studies Stellenbosch University Stellenbosch South Africa, Southern Ocean Carbon-Climate Observatory CSIR Pretoria South Africa); Jens Terhaar (University of Bern, Woods Hole Oceanographic Institution)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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Publication Date: 2023-11-10

Volume/Issue: Volume 37, Issue 11

References: 210

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: carbon sink, carbon cycle, polar oceans, seasonality, CO₂ uptake (and 1 more)

Funding: N/A

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Open Access: N/A

Abstract: We assess the Southern Ocean CO₂ uptake (1985-2018) using data sets gathered in the REgional Carbon Cycle Assessment and Processes Project Phase 2. The Southern Ocean acted as a sink for CO₂ with close agreement between simulation results from global ocean biogeochemistry models (GOBMs, 0.75 +/- 0.28 PgC yr⁻¹) and pCO₂-observation-based products (0.73 +/- 0.07 PgC yr⁻¹). This sink is only half that reported by RECCAP1 for the same region and timeframe. The present-day net uptake is to first order a response to rising atmospheric CO₂, driving large amounts of anthropogenic CO₂ (C_{ant}) into the ocean, thereby overcompensating the loss of natural CO₂ to the atmosphere. An apparent knowledge gap is the increase of the sink since 2000, with pCO₂-products suggesting a growth that is more than twice as strong and uncertain as that of GOBMs (0.26 +/- 0.06 and 0.11 +/- 0.03 Pg C yr⁻¹ decade⁻¹, respectively). This is despite nearly identical pCO₂ trends in GOBMs and pCO₂-products when both products are compared only at the locations where pCO₂ was measured. Seasonal analyses revealed agreement in driving processes in winter with uncertainty in the magnitude of outgassing, whereas discrepancies are more fundamental in summer, when GOBMs exhibit difficulties in simulating the effects of the non-thermal processes of biology and mixing/circulation. Ocean interior accumulation of C_{ant} points to an underestimate of C_{ant} uptake and storage in GOBMs. Future work needs to link surface fluxes and interior ocean transport, build long overdue systematic observation networks and push toward better process understanding of drivers of

the carbon cycle.

207. Gravimetry and hydrologic data to constrain the hydrodynamics of a karstic area: The Škocjan Caves study case

Authors: Tommaso Pivetta; Carla Braitenberg; Franci Gabrovšek; Gerald Gabriel; Bruno Meurers

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2023.130453

URL: <https://doi.org/10.1016/j.jhydrol.2023.130453>

Publication Date: 2023-11-10

Volume/Issue: Volume 629

References: 46

Cited by: 2 publications

Language: en

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Keywords: N/A

Funding: Javna Agencija za Raziskovalno Dejavnost RS, European Social Fund Plus, Regione Autonoma Friuli Venezia Giulia

License: N/A

Open Access: N/A

Abstract: N/A

208. Connecting subtropical salinity maxima to tropical salinity minima: Synchronization between ocean dynamics and the water cycle

Authors: Lisan Yu

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2023.103172

URL: <https://doi.org/10.1016/j.pocean.2023.103172>

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Volume/Issue: Volume 219

References: 90

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA

License: N/A

Open Access: N/A

Abstract: N/A

209. Research on Earth rotation and geodynamics in Poland in 2015-2018

Authors: Janusz Bogusz; Aleksander Brzezinski; Jolanta Nastula

Journal: Geodesy and Cartography

Publisher: Polish Academy of Sciences Chancellery

DOI: 10.24425/gac.2019.126094

URL: <https://doi.org/10.24425/gac.2019.126094>

Publication Date: 2023-11-06

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Cited by: 1 publications

Language: pl

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

210. Steric heights of submesoscale processes and internal gravity waves in the subtropical northwestern Pacific and northern South China Sea as revealed by moored observations

Authors: Mingfang Miao; Zhiwei Zhang; Jinchao Zhang; Yue Wang; Wei Zhao; Jiwei Tian

Journal: Progress in Oceanography

Publisher: Elsevier BV

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Publication Date: 2023-11-04

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References: 57

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

211. Concentration and thickness of sea ice in the Weddell Sea from SSM/I passive microwave radiometer data

Authors: FERNANDO LUIS HILLEBRAND (Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul/IFRS, Brazil; Universidade Federal do Rio Grande do Sul/UFRGS, Brazil); MARCOS W.D. DE FREITAS (Universidade Federal do Rio Grande do Sul/UFRGS, Brazil; Universidade Federal do Rio Grande do Sul/UFRGS, Brazil; Universidade Federal do Rio Grande do Sul/UFRGS, Brazil); ULISSES F. BREMER (Universidade Federal do Rio Grande do Sul/UFRGS, Brazil; Universidade Federal do Rio Grande do Sul/UFRGS, Brazil; Universidade Federal do Rio Grande do Sul/UFRGS, Brazil); TALES C. ABRANTES (Universidade Federal do Rio Grande do Sul/UFRGS, Brazil); JEFFERSON C. SIMÕES (Universidade Federal do Rio Grande do Sul/UFRGS, Brazil; Universidade Federal do Rio Grande do Sul/UFRGS, Brazil); CLÁUDIO W. MENDES JÚNIOR (Universidade Federal do Rio Grande do Sul/UFRGS, Brazil; Universidade Federal do Rio Grande do Sul/UFRGS, Brazil); FREDERICO SCHARDONG (Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul/IFRS, Brazil); JORGE ARIGONY-NETO (Universidade Federal do Rio Grande/FURG, Brazil)

Journal: Anais da Academia Brasileira de Ciências

Publisher: FapUNIFESP (SciELO)

DOI: 10.1590/0001-3765202320230342

URL: <https://doi.org/10.1590/0001-3765202320230342>

Publication Date: 2023-11-03

Volume/Issue: Volume 95, Issue suppl 3

References: 59

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

212. The Weakened Atlantic Meridional Overturning Circulation Diminishes Recent Arctic Sea Ice Loss

Authors: Yu-Chi Lee (University of California Riverside); Wei Liu (University of California Riverside)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl105929

URL: <https://doi.org/10.1029/2023gl105929>

Publication Date: 2023-11-02

Volume/Issue: Volume 50, Issue 21

References: 62

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: Creative Commons

Open Access: N/A

Abstract: The Arctic sea ice has been rapidly dwindling over the past four decades, significantly impacting the Arctic region and beyond. During the same period, the Atlantic meridional overturning circulation (AMOC) was also found in a declining trend. Here we investigate the role of the AMOC in the recent Arctic sea ice changes by comparing simulations from the Community Climate System Model version 4 with decelerated and stationary AMOCs under anthropogenic climate change. We find that the weakened AMOC can slow down the decline rates of Arctic sea ice area and volume by 36% and 22% between 1980 and 2020, respectively. The decelerated ocean circulation causes a reduction of northward Atlantic heat transport and hence a general interior ocean cooling in the Arctic Mediterranean, which helps alleviate the Arctic sea ice loss primarily through thermodynamic processes occurring at the base of the sea ice.

213. Subglacial discharge accelerates future retreat of Denman and Scott Glaciers, East Antarctica

Authors: Tyler Pelle (Scripps Institution of Oceanography / University of California San Diego); Jamin S. Greenbaum (Scripps Institution of Oceanography / University of California San Diego); Christine F. Dow (University of Waterloo); Adrian Jenkins (Department of Geography and Environmental Sciences, Northumbria University, Newcastle upon Tyne, NE1 8ST, UK.); Mathieu Morlighem (Dartmouth College)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.adi9014

URL: <https://doi.org/10.1126/sciadv.adi9014>

Publication Date: 2023-10-27

Volume/Issue: Volume 9, Issue 43

References: 58

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Ice shelf basal melting is the primary mechanism driving mass loss from the Antarctic Ice Sheet, yet it is unknown how the localized melt enhancement from subglacial discharge will affect future Antarctic glacial retreat. We develop a parameterization of ice shelf basal melt that accounts for both ocean and subglacial discharge forcing and apply it in future projections of Denman and Scott Glaciers, East Antarctica, through 2300. In forward simulations, subglacial discharge accelerates the onset of retreat of these systems into the deepest continental trench on Earth by 25 years. During this retreat, Denman Glacier alone contributes 0.33 millimeters per year to global sea level rise, comparable to half of the contemporary sea level contribution of the entire Antarctic Ice Sheet. Our results stress the importance of resolving complex interactions between the ice, ocean, and subglacial environments in future Antarctic Ice Sheet projections.

214. Mesoscale Eddy Modulation of Subsurface Chlorophyll Maximum Layers in the South China Sea

Authors: Wenlong Xu (Hohai University, Ministry of Natural Resources, Ocean College Jiangsu University of Science and Technology Zhenjiang China); Guifen Wang (Hohai University, Ministry of Natural Resources); Xuhua Cheng (Hohai University, Ministry of Natural Resources); Xiaogang Xing (Second Institute of Oceanography, Ministry of Natural Resources); Jianhuang Qin (Hohai University, Ministry of Natural Resources); Guidi Zhou (Hohai University, Ministry of Natural Resources); Long Jiang (Hohai University, Ministry of Natural Resources); Bingzhang Chen (Department of Mathematics and Statistics University of Strathclyde Glasgow UK)

Journal: Journal of Geophysical Research: Biogeosciences

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023jg007648

URL: <https://doi.org/10.1029/2023jg007648>

Publication Date: 2023-10-26

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References: 72

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: Subsurface chlorophyll maximum (SCM) layers contribute considerably to the integrated biomass of the water column and can be strongly modulated by mesoscale eddies that are ubiquitous in the global ocean. The mechanisms of eddy-induced surface chlorophyll concentration have been extensively examined in the South China Sea (SCS). However, the potential impact of mesoscale eddies on SCM layers remains unclear. We examined the influence of mesoscale eddies on the depth, thickness and magnitude of SCM layers in the SCS using output from an eddy-permitting biological-physical coupled model over a 22-year period. Our study shows that nutrient distribution is largely driven by eddy dynamics, with cyclonic eddies enhancing the supply of inorganic nutrients in the upper layers by uplifting the thermocline, and downward displacement of the thermocline in anticyclonic eddies, reducing the nutrient supply into the euphotic zone from the depth. We found that anticyclonic (cyclonic) eddies are responsible for increased (decreased) SCM depth and decreased (increased) SCM magnitude; SCM thickness decreased in cyclonic eddies but slightly increased in anticyclonic eddies. The effects of mesoscale eddies depend on eddy amplitude. Maximal anomalies in depth, thickness and magnitude always occur near the center of eddies. Phytoplankton community structure at SCM layers is also affected by eddies, with more diatoms in cyclonic eddies and more coccolithophores in anticyclonic eddies. Our study will advance our understanding of mesoscale physical-biogeochemical interactions.

215. Arctic freshwater impact on the Atlantic Meridional Overturning

Circulation: status and prospects

Authors: Thomas W. N. Haine (Johns Hopkins University); Ali H. Siddiqui (Johns Hopkins University); Wenrui Jiang (Johns Hopkins University)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rsta.2022.0185

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References: 104

Cited by: 9 publications

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License: N/A

Open Access: N/A

Abstract: Arguably, the most conspicuous evidence for anthropogenic climate change lies in the Arctic Ocean. For example, the summer-time Arctic sea ice extent has declined over the last 40 years and the Arctic Ocean freshwater storage has increased over the last 30 years. Coupled climate models project that this extra freshwater will pass Greenland to enter the sub-polar North Atlantic Ocean (SPNA) in the coming decades. Coupled climate models also project that the Atlantic Meridional Overturning Circulation (AMOC) will weaken in the twenty-first century, associated with SPNA buoyancy increases. Yet, it remains unclear when the Arctic anthropogenic freshening signal will be detected in the SPNA, or what form the signal will take. Therefore, this article reviews and synthesizes the state of knowledge on Arctic Ocean and SPNA salinity variations and their causes. This article focuses on the export processes in data-constrained ocean circulation model hindcasts. One challenge is to quantify and understand the relative importance of different competing processes. This article also discusses the prospects to detect the emergence of Arctic anthropogenic freshening and the likely impacts on the AMOC. For this issue, the challenge is to distinguish anthropogenic signals from natural variability. This article is part of a discussion meeting issue 'Atlantic overturning: new observations and challenges'.

216. Buoyancy forcing and the subpolar Atlantic meridional overturning circulation

Authors: Martha W. Buckley (George Mason University); M. Susan Lozier (Georgia Institute of Technology); Damien Desbruyères (Institut de Recherche pour le Développement (IRD)); Dafydd Gwyn Evans (National Oceanography Centre)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

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Keywords: N/A

Funding: Biological and Physical Sciences Division, Agence Nationale de la Recherche, HORIZON EUROPE Climate, Energy and Mobility (and 3 more)

License: N/A

Open Access: N/A

Abstract: The North Atlantic meridional overturning circulation and its variability are examined in terms of the overturning in density space and diapycnal water mass transformation. The magnitude of the mean overturning is similar to the surface water mass transformation, but the density and properties of these waters are modified by diapycnal mixing. Surface waters are progressively densified while circulating cyclonically around the subpolar gyre, with the densest waters and deepest convection occurring in the Labrador Sea and Nordic Seas. The eddy-driven interaction between the convective interior and boundary currents is a key to the export of dense waters from marginal seas. Due to the multitude of pathways of dense waters within the subpolar gyre, as well as mixing with older waters, waters exiting the subpolar gyre have a wide range of ages, with a mean age on the order of a decade. As a result, interannual changes in water mass transformation are mostly balanced locally and do not result in changes in export to the subtropics. Only persistent changes in water mass transformation result in changes in export to the subtropics. The dilution of signals from upstream water mass transformation suggests that variability in export of dense waters to the subtropics may be controlled by other processes, including interaction of dense waters with the energetic upper ocean. This article is part of a discussion meeting issue 'Atlantic overturning: new observations and challenges'.

217. Labrador sea water spreading and the Atlantic meridional overturning circulation

Authors: Isabela Alexander-Astiz Le Bras (Woods Hole Oceanographic Institution)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

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Keywords: N/A

Funding: US National Science Foundation, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: In 1982, Talley and McCartney used the low potential vorticity signature of Labrador Sea Water (LSW) to make the first North Atlantic maps of its properties. Forty years later, our understanding of LSW variability, spreading time scales and importance has deepened. In this review and synthesis article, I showcase recent observational advances in our understanding of how LSW spreads from its formation regions into the Deep Western Boundary Current and southward into the subtropical North Atlantic. I reconcile the fact that decadal variability in LSW formation is reflected in the Deep Western Boundary Current with the fact that LSW formation does not control subpolar overturning strength and discuss hypothesized connections between LSW spreading and decadal Atlantic Meridional Overturning Circulation variability. Ultimately, LSW spreading is of fundamental interest because it is a significant pathway for dissolved gasses such as oxygen and carbon dioxide into the deep ocean. We should hence prioritize adding dissolved gas measurements to standard hydrographic and circulation observations, particularly at targeted western boundary locations. This article is part of a discussion meeting issue 'Atlantic overturning: new observations and challenges'.

218. A high-resolution marine mercury model MITgcm-ECCO2-Hg with online biogeochemistry

Authors: Siyu Zhu; Peipei Wu; Siyi Zhang; Oliver Jahn; Shu Li; Yanxu Zhang

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-5915-2023

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Keywords: Biogeochemistry|Biogeochemical Cycle|Boundary current|Eddy|Ecosystem model

Funding: Fundamental Research Funds for the Central Universities, Frontiers Science Center for Shanghai Municipality, National Natural Science Foundation of China

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Open Access: N/A

Abstract: . Mercury (Hg) is a global persistent contaminant. Modeling studies are useful means of synthesizing a current understanding of the Hg cycle. Previous studies mainly use coarse-resolution models, which makes it impossible to analyze the role of turbulence in the Hg cycle and inaccurately describes the transport of kinetic energy. Furthermore, all of them are coupled with offline biogeochemistry, and therefore they cannot respond to short-term variability in oceanic Hg concentration. In our approach, we utilize a high-resolution ocean model (MITgcm-ECCO2, referred to as "high-resolution-MITgcm") coupled with the concurrent simulation of biogeochemistry processes from the Darwin Project (referred to as "online"). This integration enables us to comprehensively simulate the global biogeochemical cycle of Hg with a horizontal resolution of 1/5°. The finer portrayal of surface Hg concentrations in estuarine and coastal areas, strong western boundary flow and upwelling areas, and concentration diffusion as vortex shapes demonstrate the effects of turbulence that are neglected in previous models. Ecological events such as algal blooms can cause a sudden enhancement of phytoplankton biomass and chlorophyll concentrations, which can also result in a dramatic change in particle-bound Hg (HgaqP) sinking flux simultaneously in our simulation. In the global estuary region, including riverine Hg input in the high-resolution model allows us to reveal the outward spread of Hg in an eddy shape driven by fine-scale ocean currents. With faster current velocities and diffusion rates, our model captures the transport and mixing of Hg from river discharge in a more accurate and detailed way and improves our understanding of Hg cycle in the ocean.

219. Generalized Additive Models for Predicting Sea Level Rise in Coastal Florida

Authors: Hanna N. Vaidya (Wake Forest University); Robert D. Breininger (Department of Mathematics and Systems Engineering, Florida Institute of Technology, Melbourne, FL 32901, USA); Marisela Madrid (Turtle Mountain Tribal College, Belcourt, ND 58316, USA); Steven Lazarus (Department of Ocean Engineering and Marine Sciences, Florida Institute of Technology, Melbourne, FL 32901, USA); Nezamoddin N. Kachouie (Department of Mathematics and Systems Engineering, Florida Institute of Technology, Melbourne, FL 32901, USA)

Journal: Geosciences

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Geoscience REU (Research Experience for Undergraduates)

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Open Access: N/A

Abstract: Within the last century, the global sea level has risen between 16 and 21 cm and will likely accelerate into the future. Projections from the Intergovernmental Panel on Climate Change (IPCC) show the global mean sea level (GMSL) rise may increase to up to 1 m (1000 mm) by 2100. The primary cause of the sea level rise can be attributed to climate change through the thermal expansion of seawater and the recession of glaciers from melting. Because of the complexity of the climate and environmental systems, it is very difficult to accurately predict the increase in sea level. The latest estimate of GMSL rise is about 3 mm/year, but as GMSL is a global measure, it may not represent local sea level changes. It is essential to obtain tailored estimates of sea level rise in coastline Florida, as the state is strongly impacted by the global sea level rise. The goal of this study is to model the sea level in coastal Florida using climate factors. Hence, water temperature, water salinity, sea surface height anomalies (SSHA), and El Niño southern oscillation (ENSO) 3.4 index were considered to predict coastal Florida sea level. The sea level changes across coastal Florida were modeled using both multiple regression as a broadly used parametric model and the generalized additive model (GAM), which is a nonparametric method. The local rates and variances of sea surface height anomalies (SSHA) were analyzed and compared to regional and global measurements. The identified optimal model to explain and predict sea level was a GAM with the year, global and regional (adjacent basins) SSHA, local water temperature and salinity, and ENSO as predictors. All predictors including global SSHA, regional SSHA, water temperature, water salinity, ENSO, and the year were identified to have a positive impact on the sea level and can help to explain the variations in the sea level in coastal Florida. Particularly, the global and regional SSHA and the year are important factors to predict sea level changes.

220. Deep ocean particle flux in the Northeast Atlantic over the past 30 years: carbon sequestration is controlled by ecosystem structure in the upper ocean

Authors: R. S. Lampitt; N. Briggs; B. B. Cael; B. Espinola; P. H  laou  t; S. A. Henson; F. Norrbin; C. A. Pebody; D. Smeed

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

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Volume/Issue: Volume 11

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Cited by: 10 publications

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Funding: N/A

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Open Access: N/A

Abstract: The time series of downward particle flux at 3000 m at the Porcupine Abyssal Plain Sustained Observatory (PAP-SO) in the Northeast Atlantic is presented for the period 1989 to 2018. This flux can be considered to be sequestered for more than 100 years. Measured levels of organic carbon sequestration (average 1.88 gm⁻² y⁻¹) are higher on average at this location than at the six other time series locations in the Atlantic. Interannual variability is also greater than at the other locations (organic carbon flux coefficient of variation = 73%). We find that previously hypothesised drivers of 3,000 m flux, such as net primary production (NPP) and previous-winter mixing are not good predictors of this sequestration flux. In contrast, the composition of the upper ocean biological community, specifically the protozoan Rhizaria (including the Foraminifera and Radiolaria) exhibit a close relationship to sequestration flux. These species become particularly abundant following enhanced upper ocean temperatures in June leading to pulses of this material reaching 3,000 m depth in the late summer. In some years, the organic carbon flux pulses following Rhizaria blooms were responsible for substantial increases in carbon sequestration and we propose that the Rhizaria are one of the major vehicles by which material is transported over a very large depth range (3,000 m) and hence sequestered for climatically relevant time periods. We propose that they sink fast and are degraded little during their transport to depth. In terms of atmospheric CO₂ uptake by the oceans, the Radiolaria and Phaeodaria are likely to have the greatest influence. Foraminifera will also exert an influence in spite of the fact that the generation of their calcite tests enhances upper ocean CO₂ concentration and hence reduces uptake from the atmosphere.

221. Processes Contributing to Bering Sea Temperature Variability in the Late Twentieth and Early Twenty-First Century

Authors: Emily E. Hayden (Oregon State University); Larry W. O'Neill (Oregon State University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-23-0331.1

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Keywords: Air-sea interaction, North Pacific Ocean, Climate variability, Mesoscale processes, Forcing (mathematics)|Mixed layer|Longwave (and 1 more)

Funding: National Defense Science and Engineering Graduate, National Aeronautics and Space Administration, Jet Propulsion Laboratory

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Open Access: N/A

Abstract: Over recent decades, the Bering Sea has experienced oceanic and atmospheric climate extremes, including record warm ocean temperature anomalies and marine heatwaves (MHWs), and increasingly variable air-sea heat fluxes. In this work, we assess the relative roles of surface forcing and ocean dynamical processes on mixed layer temperature (MLT) tendency by computing a closed mixed layer heat budget using the NASA/JPL Estimating the Circulation and Climate of the Ocean (ECCO) Ocean State and Sea Ice Estimate. We show that surface forcing drives the majority of the MLT tendency in the spring and fall and remains dominant to a lesser degree in winter and summer. Surface forcing anomalies are the dominant driver of monthly mixed layer temperature tendency anomalies (MLTa), driving an average of 72% of the MLTa over the ECCO record length (1992-2017). The surface turbulent heat flux (latent plus sensible) accounts for most of the surface heat flux anomalies in January-April and September-December, and the net radiative flux (net longwave plus net shortwave) dominates the surface heat flux anomalies in May-August. Our results suggest that atmospheric variability plays a significant role in Bering Sea ocean temperature anomalies through most of the year. Furthermore, they indicate a recent increase in ocean warming surface forcing anomalies, beginning in 2010. In recent years, the Bering Sea has experienced extremes in ocean temperature, which have had adverse impacts on ocean ecology and marine fisheries and have contributed to increasingly variable sea ice extent. Our results identify anomalous heating by air-sea heat flux anomalies as the process responsible for most of the observed ocean temperature anomalies over the period 1992-2017. We additionally show that there has been an increase in atmosphere-driven ocean warming since 2010. Our work highlights the importance of investigating how ocean-atmosphere interactions might change under future climate change and how this will impact the Bering Sea.

222. Comparison of ADCP and ECCOV4r4 Currents in the Pacific Equatorial Undercurrent

Authors: David Halpern (Scripps Institution of Oceanography / University of California San Diego); Megan K. Le (University of Texas at Austin); Timothy A. Smith (Physical Sciences Laboratory, National Oceanic and Atmospheric Administration, Boulder, Colorado, and Cooperative Institute for Research in Environmental Sciences, Boulder, Colorado, University of Texas at Austin); Patrick Heimbach (University of Texas at Austin)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-23-0013.1

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References: 47

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Language: N/A

Subjects: N/A

Keywords: Ocean models, Model evaluation/performance, El Niño, Tropics, Reanalysis data

Funding: Climate Program Office

License: N/A

Open Access: N/A

Abstract: The Pacific Equatorial Undercurrent (EUC) flows eastward across the Pacific at the equator in the thermocline. Its variability is related to El Niño. Moored acoustic Doppler current profiler (ADCP) measurements recorded at four widely separated sites along the equator in the EUC were compared to currents generated by version 4 release 4 of the Estimating the Circulation and Climate of the Ocean (ECCOV4r4) global model-data synthesis product. We are interested to learn how well ECCOV4r4 currents could complement sparse in situ current measurements. ADCP measurements were not assimilated in ECCOV4r4. Comparisons occurred at 5-m depth intervals at 165 degreesE, 170 degreesW, 140 degreesW, and 110 degreesW over time intervals of 10-14 years from 1995 to 2010. Hourly values of ECCOV4r4 and ADCP EUC core speeds were strongly correlated, similar for the EUC transport per unit width (TPUW). Correlations were substantially weaker at 110 degreesW. Although we expected means and standard deviations of ECCOV4r4 currents to be smaller than ADCP values because of ECCOV4r4's grid representation error, the large differences were unforeseen. The appearance of ECCOV4r4 diurnal-period current oscillations was surprising. As the EUC moved eastward from 170 degrees to 140 degreesW, the ECCOV4r4 TPUW exhibited a much smaller increase compared to the ADCP TPUW. A consequence of smaller ECCOV4r4 EUC core speeds was significantly fewer instances of gradient Richardson number (Ri) less than $1/4$ above and below the depth of the core speed compared to Ri computed with ADCP observations. We present linear regression analyses to use monthly-mean ECCOV4r4 EUC core speeds and TPUWs as proxies for ADCP measurements. Hundreds of scientific papers have used ECCO data products generated with a continually evolving state-of-the-art ocean-model-data synthesis system. We ask, How representative is the latest version of ECCO equatorial ocean currents? We use independent in situ current measurements as the reference dataset to establish the accuracy of ECCO currents in

the tropical Pacific. Attention is focused on the Pacific Equatorial Undercurrent (EUC) because it contributes to the formation of El Niño and La Niña events. ECCO EUC core speeds were smaller in magnitude and less variable in time compared to observations. As a consequence, ECCO currents generated smaller vertical mixing in the EUC compared to that inferred from current measurements. We developed a linear regression model to improve representation of monthly-mean ECCO currents.

223. Ocean 2D eddy energy fluxes from small mesoscale processes with SWOT

Authors: Elisa Carli; Rosemary Morrow; Oscar Vergara; Robin Chevrier; Lionel Renault

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-19-1413-2023

URL: <https://doi.org/10.5194/os-19-1413-2023>

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Cited by: 1 publications

Language: en

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Keywords: N/A

Funding: Centre National d'Etudes Spatiales

License: CC BY 4.0

Open Access: N/A

Abstract: . We investigate ocean dynamics at different scales in the Agulhas Current system, a region of important interocean exchange of heat and energy. While ocean observations and some of the most advanced climate models capture the larger mesoscale dynamics (> 100 km), the smaller-scale fronts and eddies are underrepresented. The recently launched NASA-CNES Surface Water and Ocean Topography (SWOT) wide-swath altimeter mission observes the smaller ocean geostrophic scales down to 15 km in wavelength globally. Here we will analyse different eddy diagnostics in the Agulhas Current region and quantify the contributions from the larger mesoscales observable today and the smaller scales to be observed with SWOT. Surface geostrophic diagnostics of eddy kinetic energy, strain, and energy cascades are estimated from modelled sea surface height (SSH) fields of the Massachusetts Institute of Technology general circulation model (MITgcm) latitude-longitude polar cap (LLC4320) simulation subsampled at $1/10^\circ$. In this region, the smaller scales (< 150 km) have a strong signature on the horizontal geostrophic strain rate and for all eddy diagnostics in the Western Boundary Current and along the meandering Agulhas Extension. We investigate the horizontal cascade of energy using a coarse-graining technique, and we observe that the wavelength range where the inverse cascade occurs is biased towards larger mesoscale wavelengths with today's altimetric sampling. We also calculate the projected sampling of the eddy diagnostics under the SWOT swaths built with the NASA-CNES simulator to include the satellite position and realistic noise. For the swaths, a neural network noise mitigation method is implemented to reduce the residual SWOT random error before calculating eddy diagnostics. In terms of SSH, observable wavelengths of 15 to 20 km are retrieved after neural network noise mitigation, as opposed to wavelengths larger than 40 km before the noise reduction.

224. Contrasts of bimodal tropical instability waves (TIWs)-induced wind stress perturbations in the Pacific Ocean among observations, ocean models, and coupled climate models

Authors: Kai Ma; Chuanyu Liu; Junli Xu; Fan Wang

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-023-2326-z

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License: N/A

Open Access: N/A

Abstract: N/A

225. The Southern Ocean carbon and climate observations and modeling (SOCCOM) project: A review

Authors: Jorge L. Sarmiento; Kenneth S. Johnson; Lionel A. Arteaga; Seth M. Bushinsky; Heidi M. Cullen; Alison R. Gray; Roberta M. Hotinski; Tanya L. Maurer; Matthew R. Mazloff; Stephen C. Riser; Joellen L. Russell; Oscar M. Schofield; Lynne D. Talley

Journal: Progress in Oceanography

Publisher: Elsevier BV

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Language: en

Subjects: N/A

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Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

226. Exploring the AMOC Connectivity Between the RAPID and OSNAP Lines With a Model-Based Data Set

Authors: Lei Han (Xiamen University, Laoshan Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 1 publications

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Keywords: N/A

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Open Access: N/A

Abstract: Two major trans-basin mooring arrays, the Rapid Climate Change-Meridional Overturning Circulation and Heatflux Array (RAPID) at 26.5 degreesN since 2004 and the Overturning in the Subpolar North Atlantic Program (OSNAP) situated at 53 degrees-60 degreesN since 2014, have been continuously monitoring the Atlantic Meridional Overturning Circulation (AMOC). This study explores the connectivity of AMOC across these two mooring lines from a novel adiabatic perspective utilizing a model-based data set. The findings unveil significant in-phase connections facilitated by the adiabatic basinwide redistribution of water between the two lines on a monthly timescale. This adiabatic mode is a possible cause for the observed subpolar AMOC seasonality by OSNAP. Furthermore, the Labrador Sea was identified as a hotspot for adiabatic forcing of the overturning circulations, primarily attributed to its dynamic isopycnal movements.

227. Ecosystem Management Policy Implications Based on Tonga Main Tuna Species Catch Data 2002-2018

Authors: Siosaia Vaihola (School of Agriculture, Geography, Environment, Ocean and Natural Sciences, University of the South Pacific, Suva, Fiji); Stuart Kininmonth (Heron Island Research Station, Faculty of Science, University of Queensland, Gladstone 4680, Australia)

Journal: Diversity

Publisher: MDPI AG

DOI: 10.3390/d15101042

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Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Despite the crucial role played by international and regional tuna fisheries in facilitating the successful implementation of the ecosystem approach to fisheries management, there exist disparities in viewpoints among these stakeholders, resulting in gaps between regional fisheries management and local communities. Nevertheless, the Tongan government, under the Ministry of Fisheries, is dedicated to the efficient management of its tuna resources, aiming to establish it as the preferred and optimal approach for ensuring the long-term sustainability of its tuna fisheries and the ecosystem services they provide to the community. Recognizing that an appropriate legal, policy and institutional framework is in place for sustainable management of tuna, the first part of this paper presents a review of current Tonga fisheries laws and policies for its tuna fisheries. This review reflects the implementation of an information-based management framework, namely the Tonga National Tuna Fishery Management and Development Plan. The tuna fisheries in Tonga mainly catch albacore (*Thunnus alalunga*), bigeye (*Thunnus obesus*), skipjack (*Katsuwonus pelamis*), and yellowfin (*Thunnus albacares*) tuna. These tuna species are caught within Tonga's exclusive economic zones and play a crucial role in the country's economy; hence, it is crucial to examine the spatio-temporal distributions of their catch in relation to their environmental conditions. In pursuit of this goal, the tasks of mapping (i) the spatio-temporal distribution of catch landed at ports and (ii) the spatio-temporal of environmental conditions were performed. The study utilizes longline catch per unit effort data spanning from 2002 to 2018 for albacore, bigeye, skipjack, and yellowfin tuna. It also incorporates data on environmental conditions, including sea surface temperature, sea surface chlorophyll, sea surface current, and sea surface salinity. Additionally, the El Nino Southern Oscillation Index is mapped in relation to catch data to examine the potential effects of climate change on the tuna catch. Results show that bigeye, skipjack, and yellowfin CPUE show a central-northernmost distribution and are primarily caught between latitudes 14 degrees S-22 degrees S, while albacore, shows a central-southern distribution. The highest CPUE for all species are in latitudes 15.5 degrees S-22.5 degrees S and longitudes 172.5

degrees W-176.5 degrees W. The data indicate that sea surface current velocities range from -0.03 to 0.04 ms⁻¹, sea surface salinity ranges from 34.8 to 35.6 PSU, sea surface chlorophyll concentration varies from 0.03 to 0.1 mg m⁻³, and sea surface temperature fluctuates seasonally, ranging from 18 degreesC to 30 degreesC. Mapping also reveals that times of reduced catches in Tonga coincide with periods of moderate to strong El Nino events from 2002 to 2018.

228. Seasonal Tropospheric Distribution and Air-Sea Fluxes of Atmospheric Potential Oxygen From Global Airborne Observations

Authors: Yuming Jin (Scripps Institution of Oceanography / University of California San Diego); Britton B. Stephens (National Center for Atmospheric Research); Ralph F. Keeling (Scripps Institution of Oceanography / University of California San Diego); Eric J. Morgan (Scripps Institution of Oceanography / University of California San Diego); Christian Rödenbeck (Max Planck Institute for Biogeochemistry); Prabir K. Patra (Research Institute for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Matthew C. Long (National Center for Atmospheric Research)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: ocean biogeochemistry, seasonal oceanic oxygen flux, hemispheric assymetry, shallow oxygen maximum, ocean ventilation

Funding: N/A

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Open Access: N/A

Abstract: Seasonal change of atmospheric potential oxygen ($\text{APO} \sim \text{O}_2 + \text{CO}_2$) is a tracer for air-sea O_2 flux with little sensitivity to the terrestrial exchange of O_2 and CO_2 . In this study, we present the tropospheric distribution and inventory of APO in each hemisphere with seasonal resolution, using O_2 and CO_2 measurements from discrete airborne campaigns between 2009 and 2018. The airborne data are represented on a mass-weighted isentropic coordinate ($M\theta_e$) as an alternative to latitude, which reduces the noise from synoptic variability in the APO cycles. We find a larger seasonal amplitude of APO inventory in the Southern Hemisphere relative to the Northern Hemisphere, and a larger amplitude in high latitudes (low $M\theta_e$) relative to low latitudes (high $M\theta_e$) within each hemisphere. With a box model, we invert the seasonal changes in APO inventory to yield estimates of air-sea flux cycles at the hemispheric scale. We found a larger seasonal net outgassing of APO in the Southern Hemisphere ($518 \pm 52.6 \text{ Tmol}$) than in the Northern Hemisphere ($342 \pm 52.1 \text{ Tmol}$). Differences in APO phasing and amplitude between the hemispheres suggest distinct physical and biogeochemical mechanisms driving the air-sea O_2 fluxes, such as fall outgassing of photosynthetic O_2 in the Northern Hemisphere, possibly associated with the formation of the seasonal subsurface shallow oxygen maximum. We compare our estimates with four model- and observation-based products, identifying key limitations in these products or in the tools used to create them.

229. The Four-Dimensional Carbon Cycle of the Southern Ocean

Authors: Alison R. Gray (University of Washington)

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

DOI: 10.1146/annurev-marine-041923-104057

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References: 171

Cited by: 6 publications

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Abstract: The Southern Ocean plays a fundamental role in the global carbon cycle, dominating the oceanic uptake of heat and carbon added by anthropogenic activities and modulating atmospheric carbon concentrations in past, present, and future climates. However, the remote and extreme conditions found there make the Southern Ocean perpetually one of the most difficult places on the planet to observe and to model, resulting in significant and persistent uncertainties in our knowledge of the oceanic carbon cycle there. The flow of carbon in the Southern Ocean is traditionally understood using a zonal mean framework, in which the meridional overturning circulation drives the latitudinal variability observed in both air-sea flux and interior ocean carbon concentration. However, recent advances, based largely on expanded observation and modeling capabilities in the region, reveal the importance of processes acting at smaller scales, including basin-scale zonal asymmetries in mixed-layer depth, mesoscale eddies, and high-frequency atmospheric variability. Assessing the current state of knowledge and remaining gaps emphasizes the need to move beyond the zonal mean picture and embrace a four-dimensional understanding of the carbon cycle in the Southern Ocean.

230. An evaluation of eight global ocean reanalyses for the Northeast U.S. Continental shelf

Authors: Alma Carolina Castillo-Trujillo; Young-Oh Kwon; Paula Fratantoni; Ke Chen; Hyodae Seo; Michael A. Alexander; Vincent S. Saba

Journal: Progress in Oceanography

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231. Ice-Front Retreat Controls on Ocean Dynamics Under Larsen C Ice Shelf, Antarctica

Authors: M. Poinelli (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Delft University of Technology); Y. Nakayama (Hokkaido University); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Vizcaino (Delft University of Technology); R. Riva (Delft University of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: Iceberg A-68 separated from the Larsen C Ice Shelf in July 2017 and the impact of this event on the local ocean circulation has yet to be assessed. Here, we conduct numerical simulations of ocean dynamics near and below the ice shelf pre- and post-calving. Results agree with in situ and remote observations of the area as they indicate that basal melt is primarily controlled by wintertime sea-ice formation, which in turn produces High Salinity Shelf Water (HSSW). After the calving event, we simulate a 50% increase in HSSW intrusion under the ice shelf, enhancing ocean heat delivery by 30%. This results in doubling of the melt rate under Gipps Ice Rise, suggesting a positive feedback for further retreat that could destabilize the Larsen C Ice Shelf. Assessing the impact of ice-front retreat on the heat delivery under the ice is crucial to better understand ice-shelf dynamics in a warming environment.

232. Developing a New Oxygen Atlas of the World's Oceans Using Data Interpolating Variational Analysis

Authors: Christopher J. Roach (a Institute For Marine and Antarctic Studies, Hobart, Tasmania, Australia); Nathaniel L. Bindoff (b Australian Antarctic Program Partnership, Institute For Marine and Antarctic Studies, Hobart, Tasmania, Australia, c Australian Centre of Excellence in Antarctic Science, Institute For Marine and Antarctic Studies, Hobart, Tasmania, Australia, a Institute For Marine and Antarctic Studies, Hobart, Tasmania, Australia, d Australian Research Council in Climate Extremes, Institute For Marine and Antarctic Studies, Hobart, Tasmania, Australia)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-23-0007.1

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Keywords: Ship observations, Variational analysis, Climate change, Interdecadal variability, Ocean (and 1 more)

Funding: Australian Centre for Excellence in Antarctic Science, Antarctic Science Collaboration Initiative

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Abstract: We present a new global oxygen atlas. This atlas uses all of the available full water column profiles of oxygen, salinity, and temperature available as part of the World Ocean Database released in 2018. Instead of optimal interpolation, we use the Data Interpolating Variational Analysis (DIVA) approach to map the available profiles onto 108 depth levels between the surface and 6800 m, covering more than 99% of ocean volume. This 1/2 degrees x 1/2 degrees atlas covers the period 1955-2018 in 1-yr intervals. The DIVA method has significant benefits over traditional optimal interpolation. It allows the explicit inclusion of advection and boundary constraints, thus offering improvements in the representations of oxygen, salinity, and temperature in regions of strong flow and near coastal boundaries. We demonstrate these benefits of this mapping approach with some examples from this atlas. We can explore the regional and temporal variations of oxygen in the global oceans. Preliminary analyses confirm earlier analyses that the oxygen minimum zone in the eastern Pacific Ocean has expanded and intensified. Oxygen inventory changes between 1970 and 2010 are assessed and compared against prior studies. We find that the full ocean oxygen inventory decreased by 0.84% +/- 0.42%. For this period, temperature-driven solubility changes explain about 21% of the oxygen decline over the full water column; in the upper 100 m, solubility changes can explain all of the oxygen decrease; for the 100-600 m depth range, it can explain only 29%, 19% between 600 and 1000 m, and just 11% in the deep ocean. Significance Statement The purpose of this study is to create a new oxygen atlas of the world's oceans using a technique that better represents the effects of ocean currents and topographic boundaries, and to investigate how oxygen in the ocean has changed over

recent decades. We find the total quantity of oxygen in the world's oceans has decreased by 0.84% since 1970, similar to previous studies. We also examine how much of this change can be explained by changes in water temperature; we find that this can explain all the changes in the upper 100 m but only 21% of the oxygen decline over the whole water column.

233. Distribution and Trend of Wind Power Input to Near-Inertial Motions in the Southern Ocean

Authors: Jiangchao Qian (Hohai University, Ministry of Natural Resources); Xiaoming Zhai (University of East Anglia); Zhaomin Wang (Southern Marine Science and Engineering Guangdong Laboratory); Markus Jochum (University of Copenhagen)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

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Open Access: N/A

Abstract: Wind power input to near-inertial motions is an important energy source for generating diapycnal mixing in the ocean. However, the distribution and long-term trend of this input over the Southern Ocean have yet to be quantified. In this study, we investigate the near-inertial wind power input (WPI_i) to the Southern Ocean using a global eddy-permitting coupled ocean-sea ice model forced by a high-resolution atmospheric reanalysis product. Our results reveal a zonally asymmetric distribution of WPI_i in the Southern Ocean, with the strongest input in the South Indian Ocean and the weakest in the South Pacific. The integrated WPI_i between 30 degreesS and 60 degreesS exhibits a significant positive trend over the past four decades due to the intensification of mesoscale weather systems. The surface mixed-layer depth is found to modulate the spatial pattern and trend of WPI_i by altering the surface near-inertial currents.

234. Understanding the Sensitivity of the North Atlantic Subpolar Overturning in Different Resolution Versions of HadGEM3-GC3.1

Authors: T. Petit (University of Reading); J. Robson (University of Reading); D. Ferreira (University of Reading); L. C. Jackson (Met Office)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Publication Date: 2023-09-19

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References: 75

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Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The Atlantic Meridional Overturning Circulation (AMOC) is a key component of the global climate but is not simulated consistently across models or model resolutions. Here, we use a hierarchy of the global coupled model HadGEM3-GC3.1, with ocean resolutions of 1 degrees, 1/4 degrees, and 1/12 degrees, to evaluate the subpolar AMOC and its sensitivity to horizontal resolution. In line with observations, the models show that the mean overturning and surface forced water mass transformation (SFWMT) are concentrated in the eastern subpolar gyre rather than in the Labrador Sea. However, the magnitude of the overturning along the OSNAP line at medium and high resolutions is 25% and 40% larger than in the observations, respectively. This disagreement in overturning strength is noted for both OSNAP East and OSNAP West, and is mainly due to anomalously large SFWMT rather than anomalously large interior mixing or overflow transport from the Nordic Seas. Over the Labrador Sea, the intensification of SFWMT with resolution is explained by a combination of two main biases. Anomalously warm surface water enhances heat loss and reduces the extension of marginal sea ice, which increases the surface density flux over the boundary of the basin. A bias in salinity leads to anomalously dense surface water that shifts the outcropping area of the AMOC isopycnal and results in intense dense water formation along the boundary of the basin at medium and high resolutions. Thus, our analysis sheds light on a range of model biases responsible for large overturning over the Labrador Sea in climate models.

235. Magnitude, Trends, and Variability of the Global Ocean Carbon Sink From 1985 to 2018

Authors: Tim DeVries (University of California Santa Barbara); Kana Yamamoto (University of California Santa Barbara); Rik Wanninkhof (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami FL USA); Nicolas Gruber (ETH Zurich); Judith Hauck (Alfred Wegener Institute); Jens Daniel Müller (ETH Zurich); Laurent Bopp (École Polytechnique); Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); Brendan Carter (Pacific Marine Environmental Laboratory National Oceanic and Atmospheric Administration Seattle WA USA, Cooperative Institute for Climate, Ocean, and Ecosystem Studies Seattle WA USA); Thi-Tuyet-Trang Chau (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Scott C. Doney (University of Virginia); Marion Gehlen (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Lucas Gloege (Open Earth Foundation, Columbia University); Luke Gregor (ETH Zurich); Stephanie Henson (National Oceanography Centre); Ji Hyun Kim (University of California Santa Barbara); Yosuke Iida (Atmosphere and Ocean Department Japan Meteorological Agency Minato City Japan); Tatiana Ilyina (Max Planck Institute for Meteorology, University of Hamburg); Peter Landschützer (Flanders Marine Institute (VLIZ) Ostend Belgium, Max Planck Institute for Meteorology); Corinne Le Quéré (University of East Anglia); David Munro (University of Colorado Boulder, National Oceanic & Atmospheric Administration Global Monitoring Laboratory (NOAA/GML) Boulder CO USA); Cara Nissen (University of Colorado Boulder, Alfred Wegener Institute); Lavinia Patara (GEOMAR Helmholtz Centre for Ocean Research); Fiz F. Pérez (Oceans Department Stanford University Stanford CA USA, Instituto de Investigaciones Marinas (IIM) CSIC Vigo Spain); Laure Resplandy (Princeton University); Keith B. Rodgers (Center for Climate Physics, Institute for Basic Science, Pusan National University Busan South Korea); Jörg Schwinger (Bjerknes Centre for Climate Research); Roland Séférian (CNRM Université de Toulouse Météo-France CNRS Toulouse France); Valentina Sicardi (Barcelona Supercomputing Center); Jens Terhaar (University of Bern, Woods Hole Oceanographic Institution); Joaquin Triñanes (Department of Electronics and Computer Sciences Universidad de Santiago de Compostela Santiago de Compostela Spain, Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami FL USA); Hiroyuki Tsujino (Meteorological Research Institute); Andrew Watson (University of Exeter); Sayaka Yasunaka (Tohoku University, Research Institute for Global Change Japan Agency for Marine-Earth Science and Technology Yokosuka Japan); Jiye Zeng (National Institute for Environmental Studies)

Journal: Global Biogeochemical Cycles

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Keywords: RECCAP2, carbon cycle, anthropogenic carbon, ocean, climate change

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Open Access: N/A

Abstract: This contribution to the RECCAP2 (REgional Carbon Cycle Assessment and Processes) assessment analyzes the processes that determine the global ocean carbon sink, and its trends and variability over the period 1985-2018, using a combination of models and observation-based products. The mean sea-air CO₂ flux from 1985 to 2018 is -1.6 ± 0.2 PgC yr⁻¹ based on an ensemble of reconstructions of the history of sea surface pCO₂ (pCO₂ products). Models indicate that the dominant component of this flux is the net oceanic uptake of anthropogenic CO₂, which is estimated at -2.1 ± 0.3 PgC yr⁻¹ by an ensemble of ocean biogeochemical models, and -2.4 ± 0.1 PgC yr⁻¹ by two ocean circulation inverse models. The ocean also degasses about 0.65 ± 0.3 PgC yr⁻¹ of terrestrially derived CO₂, but this process is not fully resolved by any of the models used here. From 2001 to 2018, the pCO₂ products reconstruct a trend in the ocean carbon sink of -0.61 ± 0.12 PgC yr⁻¹ decade⁻¹, while biogeochemical models and inverse models diagnose an anthropogenic CO₂-driven trend of -0.34 ± 0.06 and -0.41 ± 0.03 PgC yr⁻¹ decade⁻¹, respectively. This implies a climate-forced acceleration of the ocean carbon sink in recent decades, but there are still large uncertainties on the magnitude and cause of this trend. The interannual to decadal variability of the global carbon sink is mainly driven by climate variability, with the climate-driven variability exceeding the CO₂-forced variability by 2-3 times. These results suggest that anthropogenic CO₂ dominates the ocean CO₂ sink, while climate-driven variability is potentially large but highly uncertain and not consistently captured across different methods.

236. Seismic Ocean Thermometry Using CTBTO Hydrophones

Authors: Wenbo Wu (California Institute of Technology, Woods Hole Oceanographic Institution); Zhichao Shen (California Institute of Technology, Woods Hole Oceanographic Institution); Shirui Peng (California Institute of Technology); Zhongwen Zhan (California Institute of Technology); Jörn Callies (California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

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Language: en

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Keywords: N/A

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Open Access: N/A

Abstract: Due to limited observational coverage, monitoring the warming of the global ocean, especially the deep ocean, remains a challenging sampling problem. Seismic ocean thermometry (SOT) complements existing point measurements by inferring large-scale averaged ocean temperature changes using the sound waves generated by submarine earthquakes, called T waves. We demonstrate here that Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO) hydrophones can record T waves with a higher signal-to-noise ratio compared to a previously used land-based T-wave station. This allows us to use small earthquakes (magnitude <4.0), which occur much more frequently than large events, dramatically improving the resulting temporal resolution of SOT. We also find that the travel time changes of T waves at the land-based T-wave station and the CTBTO hydrophone show small but systematic differences, although the two stations are only about 20 km apart. We attribute this feature to their different acoustic mode components sampling different parts of the ocean. Applying SOT to two CTBTO hydrophones in the East Indian Ocean reveals signals from decadal warming, seasonal variations, and mesoscale eddies, some of which are missing or underestimated in previously available temperature reconstructions. This application demonstrates the great advantage of hydrophone stations for global SOT, especially in regions with a low seismicity level.

237. Influencing factors in the stagnation of the Baiu front that induced heavy rainfall in July 2020 over Kyushu, Japan

Authors: Qoosaku Moteki

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-023-41730-0

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Funding: N/A

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Open Access: N/A

Abstract: The influencing factors of the long-term stagnation of the Baiu front that induced heavy rainfall in July 2020 over Kyushu, Japan, were examined. In July 2020, the position of the Baiu front from a weather chart at 130 degrees E remained stationary over Kyushu for 20 days, and that was the longest from 2002 to 2021. By examining a certain index of the Yellow Sea High (YSH), which is a necessary condition for Baiu front stagnation near Kyushu, it was confirmed that in 2020, a positive high-pressure anomaly persisted over the Yellow Sea until the end of July. The YSH maintenance was due to the significant negative sea surface temperature (SST) anomaly over the Yellow Sea. The significant negative SST anomalies in the Yellow Sea were due to vertical mixing in ocean mixed layer in response to the strong winds of the extratropical cyclones. A rapid SST decrease ranging from 0.5 to 1.5 degreesC/day was repeatedly occurred by ocean mixed layer deepening in association with the passage of the extratropical cyclones. The anomalous long-term stagnation of YSH due to the negative SST anomalies could have caused the Baiu front to remain stagnant over Kyushu until the end of July.

238. Weak constraint 4D-Var data assimilation in the Regional Ocean Modeling System (ROMS) using a saddle-point algorithm: Application to the California Current Circulation

Authors: Andrew M. Moore; Hernan G. Arango; John Wilkin; Christopher A. Edwards

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2023.102262

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Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

239. Contribution of Deep Vertical Velocity to Deficiency of Sverdrup Transport in the Low-Latitude North Pacific

Authors: Kun Zhang (Chinese Academy of Sciences, Laoshan Laboratory); Qiang Wang (Hohai University, Ministry of Natural Resources); Baoshu Yin (Chinese Academy of Sciences, Laoshan Laboratory); Dezhou Yang (Chinese Academy of Sciences, Laoshan Laboratory); Lina Yang (Guangdong Ocean University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Subjects: N/A

Keywords: Ocean circulation, Ocean, North Pacific Ocean

Funding: the National Natural Science Foundation of China, the National Natural Science Foundation of China, the National Natural Science Foundation of China

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Abstract: Deep vertical velocity is a critical factor causing deficiencies in Sverdrup theory. However, few studies have focused on its influence in the low-latitude western Pacific. Through multiple analyses of observational, reanalysis, and simulation data, this study explored the contribution of deep nonzero vertical velocity to the Sverdrup transport inaccuracy in the low-latitude North Pacific. The vertical velocities inducing relatively small non-Sverdrup transport exist within 1500-2500 m, which exhibit similar patterns with opposite values to the south and north of 13 degreesN. The zonally integrated meridional volume transport associated with these vertical velocities displays nonnegligible dipolar zonal bands west of approximately 150 degreesW. The positive and negative transport bands, centered at 11 degrees and 17 degreesN, can reach an amplitude of approximately 8.0 Sv (1 Sv $\equiv$ 106 m³ s⁻¹) when integrated from the eastern boundary to 140 degreesE. On average, such integrated meridional transport makes up roughly half of the prominent Sverdrup transport discrepancies in the central-western Pacific. Further investigation indicated that the spatial pattern of these vertical velocities is modulated by ocean topography and deep meridional currents. Moreover, a near-global test suggested that the meridional non-Sverdrup transport related to deep vertical velocity is widespread and undergoes remarkable multidecadal variation. This study reveals the disruptive role of deep vertical velocity in disturbing the Sverdrup balance and emphasizes the consideration of its long-term variation when diagnosing wind-driven circulation changes using Sverdrup theory.

240. A process-based assessment of the sea-level rise in the northwestern Pacific marginal seas

Authors: Hyeonsoo Cha; Jae-Hong Moon; Taekyun Kim; Y. Tony Song

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-023-00965-5

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References: 61

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Research Foundation of Korea, Korea Institute of Marine Science and Technology promotion

License: CC BY 4.0

Open Access: N/A

Abstract: Because regional sea-level rise can threaten coastal communities, understanding and quantifying the underlying process contributing to regional sea-level budget are essential. Here, we assessed whether the regional sea-level rise on the northwestern Pacific marginal seas can be closed with a combination of observations and ocean reanalyses over 1993-2017, as well as with independent observations from in situ profiles including Argo floats and satellite gravity measurements since 2003. The assessment represents that the major contributions come from the land ice melt and stericodynamic components, while the spatial pattern and interannual variability are dominated by stericodynamic effect. The observation-based estimate further shows that along continental shelves, stericodynamic sea-level changes are substantially induced by ocean mass redistribution due to changes in ocean circulation. This result highlights the ocean mass change between the deep ocean and shallow marginal seas, which plays a role in driving regional sea-level rise and variability.

241. Decoupling of the Surface and Bottom-Intensified Antarctic Slope Current in Regions of Dense Shelf Water Export

Authors: Wilma G. C. Huneke (Australian Centre of Excellence for Climate Extremes, Australian National University); Adele K. Morrison (Australian Centre of Excellence for Climate Extremes, Australian National University); Andrew McC. Hogg (Australian Centre of Excellence for Climate Extremes, Australian National University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 71

Cited by: 2 publications

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Subjects: N/A

Keywords: N/A

Funding: National Computational Infrastructure, Australian Government, Australian Research Council (and 1 more)

License: Creative Commons

Open Access: N/A

Abstract: The Antarctic Slope Current is guided by the topographic gradient of the Antarctic continental slope and creates a dynamical barrier between the continental shelf and the open ocean. The current's vertical structure varies around the continent affecting cross-slope water mass exchange with consequences for Antarctic mass loss, ventilation of the deep ocean, and carbon uptake. The Antarctic Slope Current is surface-intensified in many regions but bottom-intensified in regions of dense overflows. This study investigates the role of dense overflows in modifying the dynamics of the bottom-intensified flow using a 0.1 degrees global ocean-sea ice model. The occurrence of bottom-intensification is tightly linked with dense overflows and bottom speeds correlate with dense overflows on interannual time scales. A lack of vertical connectivity between the bottom and surface flow, however, suggests that the along-slope bottom water flows are coincidentally co-located with the Antarctic Slope Current, rather than dynamically a part of the current.

242. Fresh Water and Atmospheric Cooling Control on Density-Compensated Overturning in the Labrador Sea

Authors: Yana Bebieva (Georgia Institute of Technology); M. Susan Lozier (Georgia Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0238.1

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References: 36

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Boundary currents, Mixing, Meridional overturning circulation

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: As they rim the basin from the southern tip of Greenland to the southern Labrador coast, the waters in the Labrador Sea boundary current undergo a significant transformation in salinity and temperature, but much less so in density. Motivated by these observations, a previously developed simple three-layer model is adapted to understand the processes responsible for this density-compensated overturning in the Labrador Sea. From our model simulations, we find that the density-compensating water mass transformation in the boundary current can be largely attributed to the combined effect of 1) direct atmospheric cooling of the relatively warm boundary current and 2) freshening due to mixing with the shallower and fresh waters derived from Greenland meltwater discharge and Arctic Ocean inflow. Freshening of the boundary current waters due to the excess of precipitation over evaporation in the basin has an important, but less impactful, role in the density compensation. Studies examining the sensitivity of the density compensation to the freshwater entry location reveal a larger impact when the freshwater enters the boundary current on the Greenland side of the basin, compared to the Labrador side. These results yield insights into how increasing meltwater in the subpolar North Atlantic will affect the overturning.

243. Widespread slowdown in thinning rates of West Antarctic ice shelves

Authors: Fernando S. Paolo; Alex S. Gardner; Chad A. Greene; Johan Nilsson; Michael P. Schodlok; Nicole-Jeanne Schlegel; Helen A. Fricker

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-3409-2023

URL: <https://doi.org/10.5194/tc-17-3409-2023>

Publication Date: 2023-08-23

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References: 97

Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Antarctica's floating ice shelves modulate discharge of grounded ice into the ocean by providing a backstress. Ice shelf thinning and grounding line retreat have reduced this backstress, driving rapid drawdown of key unstable areas of the Antarctic Ice Sheet, leading to sea-level rise. If ice shelf loss continues, it may initiate irreversible glacier retreat through the marine ice sheet instability. Identification of areas undergoing significant change requires knowledge of spatial and temporal patterns in recent ice shelf loss. We used 26 years (1992-2017) of satellite-derived Antarctic ice shelf thickness, flow, and basal melt rates to construct a time-dependent dataset of ice shelf thickness and basal melt on a 3 km grid every 3 months. We used a novel data fusion approach, state-of-the-art satellite-derived velocities, and a new surface mass balance model. Our data revealed an overall pattern of thinning all around Antarctica, with a thinning slowdown starting around 2008 widespread across the Amundsen, Bellingshausen, and Wilkes sectors. We attribute this slowdown partly to modulation in external ocean forcing, altered in West Antarctica by negative feedbacks between ice shelf thinning rates and grounded ice flow, and sub-ice-shelf cavity geometry and basal melting. In agreement with earlier studies, the highest rates of ice shelf thinning are found for those ice shelves located in the Amundsen and Bellingshausen sectors. Our study reveals that over the 1992-2017 observational period the Amundsen and Bellingshausen ice shelves experienced a slight reduction in rates of basal melting, suggesting that high rates of thinning are largely a response to changes in ocean conditions that predate our satellite altimetry record, with shorter-term variability only resulting in small deviations from the long-term trend. Our work demonstrates that causal inference drawn from ice shelf thinning and basal melt rates must take into account complex feedbacks between thinning and ice advection and between ice shelf draft and basal melt rates.

244. Atlantic meridional overturning circulation increases flood risk along the United States southeast coast

Authors: Denis L. Volkov; Kate Zhang; William E. Johns; Joshua K. Willis; Will Hobbs; Marlos Goes; Hong Zhang; Dimitris Menemenlis

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-023-40848-z

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Subjects: N/A

Keywords: N/A

Funding: United States Department of Commerce | NOAA | Climate Program Office, United States Department of Commerce | National Oceanic and Atmospheric Administration

License: CC BY 4.0

Open Access: N/A

Abstract: The system of oceanic flows constituting the Atlantic Meridional Overturning Circulation (AMOC) moves heat and other properties to the subpolar North Atlantic, controlling regional climate, weather, sea levels, and ecosystems. Climate models suggest a potential AMOC slowdown towards the end of this century due to anthropogenic forcing, accelerating coastal sea level rise along the western boundary and dramatically increasing flood risk. While direct observations of the AMOC are still too short to infer long-term trends, we show here that the AMOC-induced changes in gyre-scale heat content, superimposed on the global mean sea level rise, are already influencing the frequency of floods along the United States southeastern seaboard. We find that ocean heat convergence, being the primary driver for interannual sea level changes in the subtropical North Atlantic, has led to an exceptional gyre-scale warming and associated dynamic sea level rise since 2010, accounting for 30-50% of flood days in 2015-2020.

245. Potential artifacts in conservation laws and invariants inferred from sequential state estimation

Authors: Carl Wunsch; Sarah Williamson; Patrick Heimbach

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-19-1253-2023

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License: CC BY 4.0

Open Access: N/A

Abstract: . In sequential estimation methods often used in oceanic and general climate calculations of the state and of forecasts, observations act mathematically and statistically as source or sink terms in conservation equations for heat, salt, mass, and momentum. These artificial terms obscure the inference of the system's variability or secular changes. Furthermore, for the purposes of calculating changes in important functions of state variables such as total mass and energy or volumetric current transports, results of both filter and smoother-based estimates are sensitive to misrepresentation of a large variety of parameters, including initial conditions, prior uncertainty covariances, and systematic and random errors in observations. Here, toy models of a coupled mass-spring oscillator system and of a barotropic Rossby wave system are used to demonstrate many of the issues that arise from such misrepresentations. Results from Kalman filter estimates and those from finite interval smoothing are analyzed. In the filter (and prediction) problem, entry of data leads to violation of conservation and other invariant rules. A finite interval smoothing method restores the conservation rules, but uncertainties in all such estimation results remain. Convincing trend and other time-dependent determinations in "reanalysis-like" estimates require a full understanding of models, observations, and underlying error structures. Application of smoother-type methods that are designed for optimal reconstruction purposes alleviate some of the issues.

246. A data-assimilative modeling investigation of Gulf Stream variability

Authors: Shun Mao; Ruoying He; John Bane; Glen Gawarkiewicz; Robert E. Todd

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2023.105319

URL: <https://doi.org/10.1016/j.dsr2.2023.105319>

Publication Date: 2023-08-19

Volume/Issue: Volume 211

References: 61

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

247. Impact of Atmospheric Cooling on the High-Frequency Submesoscale Vertical Heat Flux

Authors: Jonathan Aparco-Lara (Physical Oceanography Department Center for Scientific Research and Higher Education at Ensenada Ensenada México); Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jose Gomez-Valdes (Physical Oceanography Department Center for Scientific Research and Higher Education at Ensenada Ensenada México)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023jc020029

URL: <https://doi.org/10.1029/2023jc020029>

Publication Date: 2023-08-19

Volume/Issue: Volume 128, Issue 9

References: 35

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: numerical model, thermal fronts, surface diurnal cooling, submesoscale motions, vertical heat fluxes (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Recent simulations suggest that submesoscale motions with scales smaller than 30 km and frequencies greater than 1 day⁻¹ drive upward vertical heat transport. These simulations have prompted us to revisit the mechanisms that explain high-frequency (HF) vertical heat fluxes (VHFs) within the surface mixed layer (ML). Here, an idealized numerical simulation of a re-entrant channel flow with an unbalanced submesoscale thermal front is used to analyze the impact of surface cooling on HF VHFs. Two types of simulations are analyzed: forced and unforced. The VHFs cospectrum analysis shows that surface diurnal cooling increases VHFs, reaching frequencies larger than 1 day⁻¹. However, the fastest-growing length scale of ML instabilities limits the extension of positive VHFs toward fine scales. Symmetric and gravitational instabilities are the main conduits producing ageostrophic HF and small-scale structures, which in turn enhance upward VHFs across the diurnal frequency. A comparison between forced-idealized simulations with the K-profile parameterization scheme and a realistic regional simulation in the frequency-wavenumber space, reveals that the two simulation types reproduce similar VHFs near the diurnal frequency. However, the realistic simulation displays higher VHFs than the forced-idealized simulation. This study emphasizes that surface diurnal cooling significantly impacts HF VHFs. However, this impact is not sufficient to reach the HF VHFs estimated in realistic submesoscale-permitting and tidal-resolving simulations.

248. Using Probability Density Functions to Evaluate Models (PDFEM, v1.0) to compare a biogeochemical model with satellite-derived chlorophyll

Authors: Bror F. Jönsson; Christopher L. Follett; Jacob Bien; Stephanie Dutkiewicz; Sangwon Hyun; Gemma Kulk; Gael L. Forget; Christian Müller; Marie-Fanny Racault; Christopher N. Hill; Thomas Jackson; Shubha Sathyendranath

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-4639-2023

URL: <https://doi.org/10.5194/gmd-16-4639-2023>

Publication Date: 2023-08-18

Volume/Issue: Volume 16, Issue 16

References: 71

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Simons Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: . Global biogeochemical ocean models are invaluable tools to examine how physical, chemical, and biological processes interact in the ocean. Satellite-derived ocean color properties, on the other hand, provide observations of the surface ocean, with unprecedented coverage and resolution. Advances in our understanding of marine ecosystems and biogeochemistry are strengthened by the combined use of these resources, together with sparse in situ data. Recent modeling advances allow the simulation of the spectral properties of phytoplankton and remote sensing reflectances, bringing model outputs closer to the kind of data that ocean color satellites can provide. However, comparisons between model outputs and analogous satellite products (e.g., chlorophyll a) remain problematic. Most evaluations are based on point-by-point comparisons in space and time, where spuriously large errors can occur from small spatial and temporal mismatches, whereas global statistics provide no information on how well a model resolves processes at regional scales. Here, we employ a unique suite of methodologies, the Probability Density Functions to Evaluate Models (PDFEM), which generate a robust comparison of these resources. The probability density functions of physical and biological properties of Longhurst's provinces are compared to evaluate how well a model resolves related processes. Differences in the distributions of chlorophyll a concentration (mg m^{-3}) provide information on matches and mismatches between models and observations. In particular, mismatches help isolate regional sources of discrepancy, which can lead to improving both simulations and satellite algorithms. Furthermore, the use of radiative transfer in the model to mimic remotely sensed products facilitates model-observation comparisons of optical properties of the ocean.

249. Modeling the Mercury Cycle in the Sea Ice Environment: A Buffer between the Polar Atmosphere and Ocean

Authors: Shaojian Huang (Nanjing University); Feiyue Wang (University of Manitoba); Tengfei Yuan (Nanjing University); Zhengcheng Song (Nanjing University); Peipei Wu (Nanjing University); Yanxu Zhang (Nanjing University)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

DOI: 10.1021/acs.est.3c05080

URL: <https://doi.org/10.1021/acs.est.3c05080>

Publication Date: 2023-08-16

Volume/Issue: Volume 57, Issue 39

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Canada Research Chairs, Fundamental Research Funds for the Central Universities, National Natural Science Foundation of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

250. Quantifying the contribution of temperature, salinity, and climate change to sea level rise in the Pacific Ocean: 2005-2019

Authors: Junlin Ran; Nengfang Chao; Lianzhe Yue; Gang Chen; Zhengtao Wang; Tangting Wu; Chengchun Li

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1200883

URL: <https://doi.org/10.3389/fmars.2023.1200883>

Publication Date: 2023-08-11

Volume/Issue: Volume 10

References: 73

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: In recent decades, Pacific Ocean's steric sea level anomaly (SSLA) has shown prominent patterns among global sea level variations. With ongoing global warming, the frequency and intensity of climate and sea level changes have increased, particularly in the tropical Pacific region. Therefore, it is crucial to comprehend the overall trends and mechanisms governing volumetric sea level changes in the Pacific. To accurately quantify the spatiotemporal evolution characteristics of density-driven sea level change in the Pacific Ocean (PO) from 2005 to 2019, we decomposed temperature and salinity into linear trends, interannual variations, seasonal variations, and residual terms using the STL (seasonal-trend decomposition based on loess) method. To evaluate the influence of ocean temperature, salinity, and climate change on density-driven sea level change and its underlying mechanisms, we decompose temperature as well as salinity changes through into the Heaving (vertical displacements of isopycnal surfaces) and Spicing (density-compensated temperature and salinity change) modes. The findings reveal an average steric sea level rise rate of 0.34 ± 0.16 mm/yr in the PO from 2005 to 2019. Thermosteric sea-level accounts for 82% of this rise, primarily due to seawater temperature rise at depths of 0-700 m caused by Heaving mode changes. Accelerated SSLA increase via the thermosteric effect has been connected to interactions between greater Ekman downwelling from surface winds, radiation forcing linked to global greenhouse gases, and changes in the Pacific warm currents triggered by El Niño-Southern Oscillation (ENSO) episodes. Although salinity is affected by the Subantarctic Mode Water (SAMW) and the Antarctic Intermediate Water (AAIW) in the southern Indian Ocean, however the significance of salinity in sea level change is little compared to the role played by thermocline shift. This study offers a substantial contribution to the field, providing robust data and technical support, and facilitating a deeper understanding of the mechanisms underlying the effects of temperature and salinity on sea level changes during periods of rapid climate change, thus enhancing the accuracy of future predictions regarding sea level rise.

251. Morphological changes and future projections of the Getz Ice shelf, western Antarctica--A statistical and geospatial approach

Authors: Aakriti Srivastava; Avinash Kumar; Juhi Yadav; Dinesh Chandra Gupta; Rahul Mohan

Journal: Regional Studies in Marine Science

Publisher: Elsevier BV

DOI: 10.1016/j.rsma.2023.103144

URL: <https://doi.org/10.1016/j.rsma.2023.103144>

Publication Date: 2023-08-06

Volume/Issue: Volume 66

References: 70

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

252. Interannual variations in salt flux at 80 degreesE section of the equatorial Indian Ocean

Authors: Cuncun Xie; Ruibin Ding; Jiliang Xuan; Daji Huang

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-022-1140-x

URL: <https://doi.org/10.1007/s11430-022-1140-x>

Publication Date: 2023-08-02

Volume/Issue: Volume 66, Issue 9

References: 58

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

253. Assessment of the spatiotemporal variability of seawater temperature and salinity in the Yellow and Bohai seas from multiple high-resolution reanalysis datasets

Authors: Yu Yan; Yuqing Zhou; Yingjun Xu; Wei Gu

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-023-01567-7

URL: <https://doi.org/10.1007/s10236-023-01567-7>

Publication Date: 2023-08-01

Volume/Issue: Volume 73, Issue 9

References: 49

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, 111 project, Fundamental Research Funds for the Central Universities

License: N/A

Open Access: N/A

Abstract: N/A

254. Quasi-Decadal Temperature Variability in the Intermediate Layer of Subtropical South Indian Ocean During the Argo Period

Authors: Lei Huang (Xiamen University, University of Delaware, Southern Marine Science and Engineering Guangdong Laboratory); Wei Zhuang (Xiamen University, Southern Marine Science and Engineering Guangdong Laboratory); Zelun Wu (Xiamen University, University of Delaware); Yang Zhang (University of Delaware); Lingsheng Meng (Xiamen University, University of Delaware); Deanna Edwing (Xiamen University, University of Delaware); Xiao-Hai Yan (Xiamen University, University of Delaware)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Publication Date: 2023-07-28

Volume/Issue: Volume 128, Issue 8

References: 72

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: intermediate layer temperature, South Indian Ocean, quasi-decadal variability, climate modes, water mass

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China, China Scholarship Council (and 2 more)

License: N/A

Open Access: N/A

Abstract: It has been reported that the subtropical South Indian Ocean (SIO) has been rapidly warming over the past two decades and can therefore be characterized as one of the major heat accumulators among the oceanic basins. However, this strong warming is not uniformly distributed in the vertical direction. In comparison to the decade-long warming in the upper layer (0-300 m) in 2004-2013, the intermediate layer (300-1,000 m) displays a shorter warming during 2004-2009 and an intense cooling during 2010-2016. By decomposing temperature variations into heaving and spice components, and performing a heat budget analysis, we show that temperature variations in the intermediate layer during these two periods are primarily contributed by isopycnal migrations driven by local wind forcing. Local wind change in the subtropical SIO can be explained by the Indian Ocean Dipole and El Niño-Southern Oscillation during 2004-2016, while Southern Annular Mode (SAM) favors anomalous wind change in mid-latitudes and the formation of basin-wide wind change in the SIO. Additionally, wind forcing in the Subantarctic Mode Water (SAMW) formation region, which is closely linked to the SAM, modulates the anomalous spreading of SAMW into the interior of the subtropical SIO. This, therefore, leads to the SAMW intrusion being of secondary importance to the quasi-decadal temperature variability. Our findings demonstrate the independence of wind-driven temperature changes on the quasi-decadal scale in the intermediate layer of the subtropical SIO under the overall warming background of SIO waters.

255. Comparison of Freshwater Content and Variability in the Arctic Ocean Using Observations and Model Simulations

Authors: Emma L. Hoffman (School of the Earth, Ocean, and Environment, University of South Carolina, Columbia, SC 29208, USA); Bulusu Subrahmanyam (School of the Earth, Ocean, and Environment, University of South Carolina, Columbia, SC 29208, USA); Corinne B. Trott (Naval Research Laboratory, Stennis Space Center, Kiln, MS 39556, USA); Sarah B. Hall (School of the Earth, Ocean, and Environment, University of South Carolina, Columbia, SC 29208, USA)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs15153715

URL: <https://doi.org/10.3390/rs15153715>

Publication Date: 2023-07-26

Volume/Issue: Volume 15, Issue 15

References: 76

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: United States Office of Naval Research Arctic and Global Prediction, NRL

License: CC BY 4.0

Open Access: N/A

Abstract: Freshwater content (FWC), generally characterized in the Arctic Ocean by salinities lower than 34.8 psu, has shifted in both quantity and distribution in recent decades in the Arctic Ocean. This has been largely driven by changes in the volume and salinity of freshwater sources and the direction and magnitude of major currents. In this study, we analyze the variability in FWC and other physical oceanographic variables from 1993 to 2021 in the Arctic Ocean and Beaufort Gyre (BG) using in situ and remote sensing observations and five ocean models and reanalysis products. Generally, ocean models and reanalysis products underestimate FWC in the BG when compared with observations. Modeled FWC and sea surface height (SSH) in the BG are well correlated during the time period and are similar to correlations of the observational data of these variables. ORAS5 compares best to EN4 salinity over the entire study period, although GLORYS12 agrees well pre-2007 and SODA post-2007. Outside the BG, consistency between modeled SSH, FWC, and limited observations varies between models. These comparisons help identify discrepancies in ocean model and reanalysis products while highlighting areas where future improvements are necessary to further our understanding of Arctic FWC. As observations are scarce in the Arctic, these products and their accuracy are important to studying this dynamic and vulnerable ocean.

256. Uncertainties of monthly ocean bottom pressure from Gravity Recovery and Climate Experiment (GRACE): a case study at the Drake Passage

Authors: Chengcheng Yang; Xuhua Cheng; Jianhuang Qin

Journal: Geoscience Letters

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40562-023-00288-5

URL: <https://doi.org/10.1186/s40562-023-00288-5>

Publication Date: 2023-07-25

Volume/Issue: Volume 10, Issue 1

References: 38

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: Several studies reported some aliasing errors of Ocean bottom pressure (OBP) data from Gravity Recovery and Climate Experiment (GRACE), although this data have been widely used to estimate the oceanic transports. In this study, the performances of monthly OBP data from six GRACE products with two different solutions are evaluated by comparisons with the observed records at the Drake Passage. Result shows that spherical harmonic products have a better ability to capture monthly OBP variability than mascon products at the Drake Passage. In all GRACE products, higher skills occur to the south of Polar Front than those in the northern Drake Passage, and the correlations with observations reach minimum in the Local Dynamics Array (LDA) region. Such spatial differences are mainly attributed to local mesoscale processes, accompanied with high-frequency bottom eddy kinetic energy (EKE). It indicates that the monthly OBP variations from GRACE products are not reliable in the eddy-rich regions.

257. The seasonal variation of shallow meridional overturning circulation in the South China Sea and the related dynamics

Authors: Kun Jiang; Yu Wang; Yan Sun; Jian Lan

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2023.102242

URL: <https://doi.org/10.1016/j.ocemod.2023.102242>

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Volume/Issue: Volume 186

References: 35

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

258. Sources and Pathways of Glacial Meltwater in the Bellingshausen Sea, Antarctica

Authors: Peter M. F. Sheehan (University of East Anglia); Karen J. Heywood (University of East Anglia); Andrew F. Thompson (California Institute of Technology); M. Mar Flexas (California Institute of Technology); Michael P. Schodlok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl102785

URL: <https://doi.org/10.1029/2023gl102785>

Publication Date: 2023-07-18

Volume/Issue: Volume 50, Issue 14

References: 43

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: meltwater, Bellingshausen Sea, Antarctica

Funding: European Research Council, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Meltwater content and pathways determine the impact of Antarctica's melting ice shelves on ocean circulation and climate. Using ocean glider observations, we quantify meltwater distribution and transport within the Bellingshausen Sea's Belgica Trough. Meltwater is present at different densities and with different turbidities: both are indicative of a layer's ice shelf of origin. To investigate how ice-shelf origin separates meltwater into different export pathways, we compare these observations with high-resolution tracer-release model simulations. Meltwater filaments branch off the Antarctic Coastal Current into the southwestern trough. Meltwater also enters the Belgica Trough in the northwest via an extended western pathway, hence the greater observed southward (0.50 mSv) than northward (0.17 mSv) meltwater transport. Together, the observations and simulations reveal meltwater retention within a cyclonic in-trough gyre, which has the potential to promote climactically important feedbacks on circulation and future melting.

259. The Weddell Gyre heat budget associated with the Warm Deep Water circulation derived from Argo floats

Authors: Krissy Anne Reeve; Torsten Kanzow; Olaf Boebel; Myriel Vredenburg; Volker Strass; Rüdiger Gerdes

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-19-1083-2023

URL: <https://doi.org/10.5194/os-19-1083-2023>

Publication Date: 2023-07-18

Volume/Issue: Volume 19, Issue 4

References: 69

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Horizon 2020, Bundesministerium für Bildung, Wissenschaft, Forschung und Technologie

License: CC BY 4.0

Open Access: N/A

Abstract: . The Weddell Gyre plays an important role in the global climate system by supplying heat to underneath the ice shelves and in the formation of deep and bottom water masses, which have been subject to widespread warming over past decades. In this study, we investigate the re-distribution of heat throughout the Weddell Gyre by diagnosing the terms of the heat conservation equation for a 1000 m thick layer of water encompassing the core of Warm Deep Water. The spatial distributions of the different advective and diffusive terms in terms of heat tendencies are estimated using gridded climatologies of temperature and velocity, obtained from Argo floats in the Weddell Gyre from 2002 to 2016. While the results are somewhat noisy on the grid scale and the representation of the effects of eddy mixing is highly uncertain due to the need to parameterise them by means of turbulent diffusion, the heat budget (i.e. the sum of all terms) closes (within the uncertainty range) when integrated over the open inflow region in the southern limb, whereas the interior circulation cell remains unbalanced. There is an overall balance in the southern limb between the mean horizontal advection and horizontal turbulent diffusion of heat, whereas the vertical terms contribute comparatively little to the heat budget throughout the Weddell Gyre. Heat convergence due to mean horizontal advection balances with divergence due to horizontal turbulent diffusion in the open southern limb of the Weddell Gyre. In contrast, heat divergence due to mean horizontal advection is much weaker than convergence due to horizontal turbulent diffusion in the interior circulation cell of the Weddell Gyre, due to large values in the latter along the northern boundary due to large meridional temperature gradients. Heat is advected into the Weddell Gyre along the southern limb, some of which is turbulently diffused northwards into the interior circulation cell, while some is likely turbulently diffused southwards towards the shelf seas. This suggests that horizontal turbulent diffusion plays a role in transporting heat both towards the gyre interior where upwelling occurs and towards the ice shelves. Horizontal turbulent diffusion is also a mechanism by which heat can be transported into the Weddell Gyre across the open northern boundary. Temporal deviations from the mean terms are not included due to study limitations.

In order to appreciate the role of transient eddying processes, a continued effort to increase the spatial and temporal coverage of observations in the eastern Weddell Sea is required.

260. Overturning Pathways Control AMOC Weakening in CMIP6 Models

Authors: Jonathan A. Baker (Met Office); Michael J. Bell (Met Office); Laura C. Jackson (Met Office); Richard Renshaw (Met Office); Geoffrey K. Vallis (University of Exeter); Andrew J. Watson (University of Exeter); Richard A. Wood (Met Office)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl103381

URL: <https://doi.org/10.1029/2023gl103381>

Publication Date: 2023-07-18

Volume/Issue: Volume 50, Issue 14

References: 82

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: general circulation, turbulence, CMIP6, diffusion, weakening (and 7 more)

Funding: Horizon 2020 Framework Programme, Met Office

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Future projections indicate the Atlantic Meridional Overturning Circulation (AMOC) will weaken and shoal in response to global warming, but models disagree widely over the amount of weakening. We analyze projected AMOC weakening in 27 CMIP6 climate models, in terms of changes in three return pathways of the AMOC. The branch of the AMOC that returns through diffusive upwelling in the Indo-Pacific, but does not later upwell in the Southern Ocean (SO), is particularly sensitive to warming, in part, because shallowing of the deep flow prevents it from entering the Indo-Pacific via the SO. The present-day strength of this Indo-Pacific pathway provides a strong constraint on the projected AMOC weakening. However, estimates of this pathway using four observationally based methods imply a wide range of AMOC weakening under the SSP5-8.5 scenario of 29%-61% by 2100. Our results suggest that improved observational constraints on this pathway would substantially reduce uncertainty in 21st century AMOC decline.

261. Reduction of non-tidal oceanographic fluctuations in ocean-bottom pressure records of DONET using principal component analysis to enhance transient tectonic detectability

Authors: Hideto Otsuka; Yusaku Ohta; Ryota Hino; Tatsuya Kubota; Daisuke Inazu; Tomohiro Inoue; Narumi Takahashi

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-023-01862-z

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References: 63

Cited by: 1 publications

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Subjects: N/A

Keywords: Seafloor Spreading

Funding: Fusion Oriented REsearch for disruptive Science and Technology

License: CC BY 4.0

Open Access: N/A

Abstract: Ocean bottom pressure-gauge (OBP) records play an essential role in seafloor geodesy. Oceanographic fluctuations in OBP data, however, pose as a significant noise source in seafloor transient crustal deformation observations, including slow slip events (SSEs), making it crucial to evaluate them quantitatively. To extract the significant fluctuation phenomena common to multiple observation networks, including oceanographic fluctuations and tectonic signals, we applied principal component analysis (PCA) to the 3-year Dense Oceanfloor Network System for Earthquakes and Tsunamis (DONET) OBP time series for 40 stations during 2016-2019. PCA could separate several oceanographic signals based on the characteristics of their spatial distributions, although evident transient tectonic signals could not be confirmed from the observed pressure records during this observed period. The spatial distribution of the first four principal components (PCs) reflected the common component, inclined component along sea depth, longitude component, and parabola-like pattern, respectively. By subtracting each PC (in particular, PC-2 and PC-4) from the time series, we could significantly reduce the sea depth dependence of OBP records, which has been highlighted in several previous studies and is also evident in this region. We interpreted PCs 2-4 as the reflection of the strength and meandering of ocean geostrophic currents based on a comparison with the PC spatial distribution of the numerical oceanographic models. In addition, to evaluate the ability of PCA to separate transient tectonic signal from OBP time series, including oceanographic fluctuations, we conducted a synthetic ramp assuming an SSE by rectangular fault and then applied PCA. The assumed synthetic tectonic signal could be separated from the oceanographic signals and included in the principal component independently depending on its amplitude, suggesting that the spatial distribution of each PC would change if the amplitude of the synthetic signal were sufficiently large. We propose a transient event-detection method based on the spatial distribution difference of a specific PC with or without a tectonic signal. We used the normalized inner product (NIP) between these PCs as

the indicator of their similarities. This method can detect transient tectonic signals more significantly than the moment-magnitude scale of 5.9 from OBP records. Graphical Abstract

262. Variability of volume transport in the Taiwan Strait and its response to tropical MJO convection: A numerical approach

Authors: Haowei Sun; Zhaoyun Chen; Zhonghua Zhao; Mengdi Xu; Yimin Zhang; Xiao-Hai Yan; Xueding Li; Yuwu Jiang

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2023.102240

URL: <https://doi.org/10.1016/j.ocemod.2023.102240>

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China, Ministry of Science and Technology of the People's Republic of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

263. Decadal Reorganization of Subantarctic Mode Water

Authors: Ivana Cerovečki (Scripps Institution of Oceanography / University of California San Diego); F. Alexander Haumann (Ludwig-Maximilian-University Munich Munich Germany, Princeton University, Alfred Wegener Institute)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl102148

URL: <https://doi.org/10.1029/2022gl102148>

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Volume/Issue: Volume 50, Issue 14

References: 46

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, decadal variability, Interdecadal Pacific Oscillation, Subantarctic Mode Water

Funding: Earth Sciences Division, National Science Foundation

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Subantarctic Mode Water (SAMW) is one of the most important water masses globally in taking up anthropogenic heat and carbon dioxide. However, its long-term changes in response to varying climatic conditions are not well understood. We use an ocean state estimate to analyze SAMW volume budgets for the period 1992 to 2017. They reveal a decadal SAMW volume reorganization comparable to the long-term trend in Indian Ocean, and a multi-decadal volume reorganization exceeding the long-term trend in the Pacific. In both sectors, the SAMW reorganization exhibits a two-layer density structure, with compensating volume changes of lighter and denser SAMW, driven by heat flux changes in the Indian sector and central Pacific and freshwater flux changes in the southeast Pacific. This variability is governed by a cumulative effect of surface flux anomalies associated with the Interdecadal Pacific Oscillation. Shorter-term trends observed during the Argo period are largely explained by this variability.

264. Citizen Science and The University of Queensland Seismograph Stations (UQSS)--A Study of Seismic T Waves in S-W Pacific Ocean

Authors: Colin John Lynam (Royal Society of Queensland, P.O. Box 6021, Brisbane, QLD 4067, Australia); Asanka Karunaratne (Science Department, Brisbane State High School, Brisbane, QLD 4101, Australia)

Journal: Sustainability

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Abstract: Seismologists listen to Earth's noise as it rips apart (faulting), exudes (volcanoes) and swallows (subducts) large volumes of rock. Your mobile phone is most likely detecting such noise, right now! This paper is about one such specific noise, the T wave. It summarises an early and successful piece of citizen science, performed within The University of Queensland Seismograph Stations (UQSS) observatory, in cooperation with colleagues at CSIRO. It was designed to encourage young STEM students from Brisbane high schools to engage in "real" research, back in 1995. Bear in mind, this is a time period when science is changing considerably from analog to digital media and operational recording methods. The citizen science students used a pre-prepared decadal collection (1980-1990) of T waves, derived from the Brisbane seismograph (BRS) observatory data catalogue. BRS has been operating since 1937 and is part of the global World-Wide Seismograph Station Network (WWSSN). Fortunately, seismology is a very collaborative field. There is a lot of data analysis involved in the science of recording earthquake signals, with auxiliary definitive catalogues, observers logbooks, housing of the recordings themselves (analog and digital) and the software mediums that change over time. It equally tests housekeeping proficiency, where a maze of record-keeping problems can be encountered in a longitudinal data collection study such as this. Having completed the project report, Earthquake generated T phases on BRS Seismograph (Brisbane, Q'ld) a predictor for Tasman Sea Tsunamis? their (analog) results sat in a cupboard until recently. The project was re-analysed in 2022 for a higher-degree student, discovering a timely climate change implication for the study. The original research question has now been amplified with a brief literature review. We observe that currently in Australia, university and government earth science observatories have diminished, and in their place, public seismic networks (PSN) have evolved, either in backyard sheds or school science labs. We now additionally propose here that the level of expertise required ideally fits the role of advancing citizen science, for a real science advantage. This is already a topical citizen disaster preparedness action area, and we propose that it has applications as a possible educational strategy for citizen engagement

in today's climate emergency. In addition, we are hopeful that other researchers in oceanography will read this paper and decide to explore the ocean's temperature rise phenomenon through the eyes of seismological observers.

265. Seasonal Variability of Eddy Kinetic Energy along the Kuroshio Current

Authors: Xiaomei Yan (Second Institute of Oceanography, Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Dujuan Kang (e Department of Environmental Sciences, Rutgers, The State University of New Jersey, New Brunswick, New Jersey); Enrique N. Curchitser (e Department of Environmental Sciences, Rutgers, The State University of New Jersey, New Brunswick, New Jersey); Xiaohui Liu (Second Institute of Oceanography); Chongguang Pang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Linlin Zhang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Journal of Physical Oceanography

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Keywords: Mesoscale processes, Eddies, North Pacific Ocean, Seasonal variability

Funding: Natural Science Foundation of Shandong Province, open fund of State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, MNR, Strategic Priority Research Program of the Chinese Academy of Sciences (and 1 more)

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Open Access: N/A

Abstract: The seasonal variability of the eddy kinetic energy (EKE) along the Kuroshio Current (KC) is examined using outputs from an eddy-resolving (1/10 degrees) ocean model. Using a theoretical framework for climatological monthly mean EKE, the mechanisms governing the seasonal cycle of upper-ocean EKE are investigated. East of Taiwan, the EKE shows two comparable peaks in spring and summer in the surface layer; only the spring one is evident in the subsurface layer. The seasonality is determined by mixed barotropic (BTI) and baroclinic (BCI) instabilities. Northeast of Taiwan, the EKE is also elevated during spring-summer but with a sole peak in summer, which is dominated by the meridional EKE advection by the KC. In the middle part of the KC in the East China Sea, the mesoscale (>150 km) EKE (EKEMS) is relatively strong during spring-summer, whereas the submesoscale (50-150 km) EKE (EKESM) is significantly enhanced during winter-spring. The seasonal cycles of EKEMS and EKESM are primarily controlled by the external forcing and BCI, respectively. In particular, the higher EKEMS level in summer is mainly due to the increased wind work. West of the Tokara Strait, the EKE exhibits a prominent peak in winter and has its minimum in summer, which is regulated by the BCI. As the submesoscale signals are partially resolved by the model, further studies with higher-resolution simulations and observations are needed for a better understanding of the EKESM seasonality and its contribution to the seasonally modulating EKEMS along the KC.

266. Characteristics of Internal Tides from ECCO Salinity Estimates and Observations in the Bay of Bengal

Authors: Bulusu Subrahmanyam (School of the Earth, Ocean and Environment, University of South Carolina, Columbia, SC 29208, USA); V.S.N. Murty (Physical Oceanography Division, Council of Scientific and Industrial Research (CSIR)--National Institute of Oceanography (NIO), Regional Centre, Visakhapatnam 530017, India); Sarah B. Hall (School of the Earth, Ocean and Environment, University of South Carolina, Columbia, SC 29208, USA)

Journal: Remote Sensing

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Abstract: Internal waves (IWs) are generated in all the oceans, and their amplitudes are large, especially in regions that receive a large amount of freshwater from nearby rivers, which promote highly stratified waters. When barotropic tides encounter regions of shallow bottom-topography, internal tides (known as IWs of the tidal period) are generated and propagated along the pycnocline due to halocline or thermocline. In the North Indian Ocean, the Bay of Bengal (BoB) and the Andaman Sea receive a large volume of freshwater from major rivers and net precipitation during the summer monsoon. This study addresses the characteristics of internal tides in the BoB and Andaman Sea using NASA's Estimating the Circulation and Climate of the Ocean (ECCO) project's high-resolution (1/48 degrees and hourly) salinity estimates at 1 m depth (hereafter written as ECCO salinity) during September 2011–October 2012, time series of temperature, and salinity profiles from moored buoys. A comparison is made between ECCO salinity and NASA's Soil Moisture Active Passive (SMAP) salinity and Aquarius salinity. The time series of ECCO salinity and observed salinity are subjected to bandpass filtering with an 11–14 h period and 22–26 h period to detect and estimate the characteristics of semi-diurnal and diurnal period internal tides. Our analysis reveals that the ECCO salinity captured well the surface imprints of diurnal period internal tide propagating through shallow pycnocline (~50 m depth) due to halocline, and the latter suppresses the impact of semi-diurnal period internal tide propagating at thermocline (~100 m depth) reaching the sea surface. The semi-diurnal (diurnal) period internal tides have their wavelengths and phase speeds increased (decreased) from the central Andaman Sea to the Sri Lanka coast. Propagation of diurnal period internal tide is dominant in the northern BoB and northern Andaman Sea.

267. Southern Ocean polynyas and dense water formation in a high-resolution, coupled Earth system model

Authors: Hyein Jeong; Adrian K. Turner; Andrew F. Roberts; Milena Veneziani; Stephen F. Price; Xylar S. Asay-Davis; Luke P. Van Roekel; Wuyin Lin; Peter M. Caldwell; Hyo-Seok Park; Jonathan D. Wolfe; Azamat Mametjanov

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Abstract: . Antarctic coastal polynyas produce dense shelf water, a primary source of Antarctic Bottom Water that contributes to the global overturning circulation. This paper investigates Antarctic dense water formation in the high-resolution version of the Energy Exascale Earth System Model (E3SM-HR). The model is able to reproduce the main Antarctic coastal polynyas, although the polynyas are smaller in area compared to observations. E3SM-HR also simulates several occurrences of open-ocean polynyas (OOPs) in the Weddell Sea at a higher rate than what the last 50 years of the satellite sea ice observational record suggests, but similarly to other high-resolution Earth system model simulations. Furthermore, the densest water masses in the model are formed within the OOPs rather than on the continental shelf as is typically observed. Biases related to the lack of dense water formation on the continental shelf are associated with overly strong atmospheric polar easterlies, which lead to a strong Antarctic Slope Front and too little exchange between on- and off-continental shelf water masses. Strong polar easterlies also produce excessive southward Ekman transport, causing a build-up of sea ice over the continental shelf and enhanced ice melting in the summer season. This, in turn, produces water masses on the continental shelf that are overly fresh and less dense relative to observations. Our results indicate that high resolution alone is insufficient for models to properly reproduce Antarctic dense water; the large-scale polar atmospheric circulation around Antarctica must also be accurately simulated.

268. Response of phytoplankton functional types to Hurricane Fabian (2003) in the Sargasso Sea

Authors: Dailé Avila-Alonso; Jan M. Baetens; Rolando Cardenas; Bernard De Baets

Journal: Marine Environmental Research

Publisher: Elsevier BV

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Keywords: N/A

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Open Access: N/A

Abstract: N/A

269. Data Assimilation of Range- and Depth-Averaged Sound Speed from Acoustic Tomography Measurements in Fram Strait

Authors: Florian Geyer (Nansen Environmental and Remote Sensing Center); Ganesh Gopalakrishnan (Scripps Institution of Oceanography / University of California San Diego); Hanne Sagen (Nansen Environmental and Remote Sensing Center); Bruce Cornuelle (Scripps Institution of Oceanography / University of California San Diego); François Challet (Nansen Environmental and Remote Sensing Center); Matthew Mazloff (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

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Subjects: N/A

Keywords: Acoustic measurements/effects, Arctic, Ocean, Data assimilation, In situ oceanic observations

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Open Access: N/A

Abstract: The 2010-12 Acoustic Technology for Observing the Interior of the Arctic Ocean (ACOBAR) experiment provided acoustic tomography data along three 167-301-km-long sections in Fram Strait between Greenland and Spitsbergen. Ocean sound speed data were assimilated into a regional numerical ocean model using the Massachusetts Institute of Technology General Circulation Model-Estimating the Circulation and Climate of the Ocean four-dimensional variational (MITgcm-ECCO 4DVAR) assimilation system. The resulting state estimate matched the assimilated sound speed time series; the root-mean-squared (RMS) error of the sound speed estimate (~ 0.4 m s⁻¹) is smaller than the uncertainty of the measurement (~ 0.8 m s⁻¹). Data assimilation improved modeled range- and depth-averaged ocean temperatures at the 78 degrees50'N oceanographic mooring section in Fram Strait. The RMS error of the state estimate (0.21 degreesC) is comparable to the uncertainty of the interpolated mooring section (0.23 degreesC). Lack of depth information in the assimilated ocean sound speed measurements caused an increased temperature bias in the upper ocean (0-500 m). The correlations with the mooring section were not improved as short-term variations in the mooring measurements and the ocean state estimate do not always coincide in time. This is likely due to the small-scale eddying and nonlinearity of the ocean circulation in Fram Strait. Furthermore, the horizontal resolution of the state estimate (4.5 km) is eddy permitting, rather than eddy resolving. Thus, the state estimate cannot represent the full ocean dynamics of the region. This study is the first to demonstrate the usefulness of large-scale acoustic measurements for improving ocean state estimates at high latitudes. Significance Statement Acoustic tomography measurements allow one to observe ocean temperature in large ocean volumes under the Arctic sea ice by measuring sound speed, which is hard

to synoptically observe by other methods. This study has established methods for assimilation of depth- and range-averaged ocean sound speed from an acoustic tomography experiment in Fram Strait. For the first time, a 2-yr time series of ocean sound from acoustic tomography has been assimilated into an ocean state estimate. The results highlight the use of ocean tomography in ice-covered regions to improve state estimates of ocean temperature.

270. Exploring the role of different data types and timescales in the quality of marine biogeochemical model calibration

Authors: Iris Kriest; Julia Getzlaff; Angela Landolfi; Volkmar Sauerland; Markus Schartau; Andreas Oschlies

Journal: Biogeosciences

Publisher: Copernicus GmbH

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Abstract: . Global biogeochemical ocean models help to investigate the present and potential future state of the ocean, its productivity and cascading effects on higher trophic levels such as fish. They are often subjectively tuned against data sets of inorganic tracers and surface chlorophyll and only very rarely against organic components such as particulate organic carbon or zooplankton. The resulting uncertainty in biogeochemical model parameters (and parameterisations) associated with these components can explain some of the large spread of global model solutions with regard to the cycling of organic matter and its impacts on biogeochemical tracer distributions, such as oxygen minimum zones (OMZs). A second source of uncertainty arises from differences in the model spin-up length as, so far, there seems to be no agreement on the required simulation time that should elapse before a global model is assessed against observations. We investigated these two sources of uncertainty by optimising a global biogeochemical ocean model against the root-mean-squared error (RMSE) of six different combinations of data sets and different spin-up times. Besides nutrients and oxygen, the observational data sets also included phyto- and zooplankton, as well as dissolved and particulate organic phosphorus (DOP and POP, respectively). We further analysed the optimised model performance with regard to global biogeochemical fluxes, oxygen inventory and OMZ volume. Following the optimisation procedure, we evaluated the RMSE for all tracers located in the upper 100 m (except for POP, for which we considered the entire vertical domain), regardless of their consideration during optimisation. For the different optimal model solutions, we find a narrow range of the RMSE, between 14 % of the average RMSE after 10 years and 24 % after 3000 years of simulation. Global biogeochemical fluxes, global oxygen bias and OMZ volume showed a much stronger divergence among the models and over time than RMSE, indicating that even models that are similar with regard to local surface tracer concentrations can perform very differently when assessed against the global diagnostics for oxygen. Considering organic tracers in the optimisation had a strong impact on the particle flux exponent (Martin b) and may reduce much of the uncertainty in this parameter and the

resulting deep particle flux. Independent of the optimisation setup, the OMZ volume showed a particularly sensitive response with strong trends over time, even after 3000 years of simulation time (despite the constant physical forcing); a high sensitivity to simulation time; and the highest sensitivity to model parameters arising from the tuning strategy setup (variation of almost 80 % of the ensemble mean). In conclusion, calibration against observations of organic tracers can help to improve global biogeochemical models even after short spin-up times; here especially, observations of deep particle flux could provide a powerful constraint. However, a large uncertainty remains with regard to global OMZ volume and its evolution over time, which can show very dynamic behaviour during the model spin-up, which renders temporal extrapolation to a final equilibrium state difficult if not impossible. Given that the real ocean shows variations on many timescales, the assumption of observations representing a steady-state ocean may require some reconsideration.

271. Mechanism on the Short-Term Variability of the Atlantic Meridional Overturning Circulation in the Subtropical and Tropical Regions

Authors: Lei Han (Xiamen University, Laoshan Laboratory)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-23-0027.1

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Language: N/A

Subjects: N/A

Keywords: Seasonal variability, Atlantic Ocean, Interannual variability, Meridional overturning circulation, Oceanic variability (and 1 more)

Funding: Xiamen University Malaysia Research Fund

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Abstract: The continuous, moored observation revealed significant variability in the strength of the Atlantic meridional overturning circulation (AMOC). The cause of such AMOC variability is an extensively studied subject. This study focuses on the short-term variability, which ranges up to seasonal and interannual time scales. A mechanism is proposed from the perspective of ocean water redistribution by layers. By offering explanations for four phenomena of AMOC variability in the subtropical and tropical oceans (seasonality, meridional coherence, layered-transport compensation as observed at 26.5 degreesN, and the 2009/10 downturn that occurred at 26.5 degreesN), this mechanism suggests that the short-term AMOC variabilities in the entire subtropical and tropical regions are governed by a basinwide adiabatic water redistribution process, or the so-called sloshing dynamics, rather than diapycnal processes. Significance Statement The Atlantic meridional overturning circulation (AMOC) is a key component in the global climate system due to its immense power in redistributing heat meridionally, which contributes to the hospitable climate of the United Kingdom and western Europe. Therefore, any changes in AMOC can have significant impacts on both global and local climate variability. Here I propose a mechanism to explain the short-term (up to interannual) AMOC variability in the subtropical and tropical regions from the perspective of ocean water redistribution. This mechanism suggests that the short-term variability of AMOC strength is dominated by an adiabatic process, and thus, its large-amplitude variation is mostly a reversible process. In other words, AMOC may be more resilient to short-term variability than previously believed, and it could recover autonomously from the abrupt changes.

272. Observed transmissions and ocean-ice-acoustic coupled modelling in the Beaufort Sea

Authors: David R. Barclay (Dalhousie University); Bruce S. Martin (JASCO Applied Sciences 2 , 20 Mount Hope Avenue, Dartmouth, Nova Scotia, B2Y 4S3, Canada); Paul C. Hines (Hines Ocean Science and Technology, Incorporated 3 , 12 Carlson Court, Dartmouth, Nova Scotia, B2W 5X7, Canada); James M. Hamilton (JMH Consulting and Design, Incorporated 4 , 160 Notting Hill Road, Mineville, Nova Scotia, B2Z 1V4, Canada); Mikhail Zykov (JASCO Applied Sciences 2 , 20 Mount Hope Avenue, Dartmouth, Nova Scotia, B2Y 4S3, Canada); Terry Deveau (JASCO Applied Sciences 2 , 20 Mount Hope Avenue, Dartmouth, Nova Scotia, B2Y 4S3, Canada); Pablo Borys (JASCO Applied Sciences 2 , 20 Mount Hope Avenue, Dartmouth, Nova Scotia, B2Y 4S3, Canada)

Journal: The Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

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Keywords: N/A

Funding: Natural Sciences and Engineering Research Council, Defence Research and Development Canada, Defence Research and Development Canada

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Open Access: N/A

Abstract: An ocean-ice-acoustic coupled model is configured for the Beaufort Sea. The model uses outputs from a data assimilating global scale ice-ocean-atmosphere forecast to drive a bimodal roughness algorithm for generating a realistic ice canopy. The resulting range-dependent ice cover obeys observed roughness, keel number density, depth, and slope, and floe size statistics. The ice is inserted into a parabolic equation acoustic propagation model as a near-zero impedance fluid layer along with a model defined range-dependent sound speed profile. Year-long observations of transmissions at 35 Hz from the Coordinated Arctic Acoustic Thermometry Experiment and 925 Hz from the Arctic Mobile Observing System source were recorded over the winter of 2019-2020 on a free-drifting, eight-element vertical line array designed to vertically span the Beaufort duct. The ocean-ice-acoustic coupled model predicts receive levels that reasonably agree with the measurements over propagation ranges of 30-800 km. At 925 Hz, seasonal and sub-seasonal ocean and ice driven variations of propagation loss are captured in the data and reproduced in the model.

273. How Well do Earth System Models Capture Apparent Relationships Between Phytoplankton Biomass and Environmental Variables?

Authors: Christopher Holder (Johns Hopkins University); Anand Gnanadesikan (Johns Hopkins University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: satellite observations, environmental predictors, phytoplankton, Earth system modeling, machine learning

Funding: U.S. Department of Energy, Climate Program Office

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Open Access: N/A

Abstract: As phytoplankton form the base of the marine food web, understanding the controls on their abundance is fundamental to understanding marine ecology and its sensitivity to global climate change. While many Earth System Models (ESMs) predict phytoplankton biomass, it is unclear whether they properly capture the mechanistic relationships that control this quantity in the real ocean. We used Random Forest analysis to analyze the output of 13 ESMs as well as two observational data sets. The target variable was phytoplankton carbon and the predictors included environmental parameters known to influence phytoplankton, including nutrients, light, mixed layer depth, salinity, temperature, and upwelling. We examined the following: (a) What fractions of variability in ESMs and observations can be linked to the large-scale environmental variables simulated by ESMs? (b) What are the dominant predictors and relationships affecting phytoplankton biomass? (c) How well do ESMs simulate phytoplankton carbon and do they simulate the relationships we see in observations? About 88%-96% of the variability in observational data sets and greater than 98% in the ESMs was accounted for by environmental variables known to influence phytoplankton biomass. The dominant predictors in the observational data sets were shortwave radiation and dissolved iron, with temperature and ammonium also relatively important. All the ESMs show that shortwave radiation is the most important variable and most of them predict the right sign of sensitivity to most variables. However, the models predict that biomass reaches maximum levels at unrealistically low levels of iron and unrealistically high levels of light.

274. Anticipated Capabilities of the ODYSEA Wind and Current Mission Concept to Estimate Wind Work at the Air-Sea Interface

Authors: Hector Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alexander Wineteer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Patrice Klein (California Institute of Technology, PSL Research University); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ernesto Rodriguez (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Keywords: N/A

Funding: Jet Propulsion Laboratory, California Institute of Technology, National Aeronautics and Space Administration (NASA), NASA Advanced Supercomputing (NAS) Division

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Open Access: N/A

Abstract: The kinetic energy transfer between the atmosphere and oceans, called wind work, affects ocean dynamics, including near-inertial oscillations and internal gravity waves, mesoscale eddies, and large-scale zonal jets. For the most part, the recent numerical estimates of global wind work amplitude are almost five times larger than those reported 10 years ago. This large increase is explained by the impact of the broad range of spatial and temporal scales covered by winds and currents, the smallest of which has only recently been uncovered by increasingly high-resolution modeling efforts. However, existing satellite observations do not fully sample this broad range of scales. The present study assesses the capabilities of ODYSEA, a conceptual satellite mission to estimate the amplitude of wind work in the global ocean. To this end, we use an ODYSEA measurement simulator fed by the outputs of a km scale coupled ocean-atmosphere model to estimate wind work globally. The results indicate that compared with numerical truth estimates, the ODYSEA instrument performs well globally, except for latitudes north of 40°N during summer due to unresolved storm evolution. This performance is explained by the wide-swath properties of ODYSEA (a 1700 km wide swath with 5 km posting for winds and surface currents), its twice-a-day (daily) coverage at mid-latitudes (low latitudes), and the insensitivity of the wind work to uncorrelated errors in the estimated wind and current.

275. Ocean heat uptake and interbasin redistribution driven by anthropogenic aerosols and greenhouse gases

Authors: Shouwei Li; Wei Liu; Robert J. Allen; Jia-Rui Shi; Laifang Li

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

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Keywords: N/A

Funding: NSF | GEO | Division of Atmospheric and Geospace Sciences, NSF | GEO | Division of Ocean Sciences

License: N/A

Open Access: N/A

Abstract: N/A

276. Review article: Towards strongly coupled ensemble data assimilation with additional improvements from machine learning

Authors: Eugenia Kalnay; Travis Sluka; Takuma Yoshida; Cheng Da; Safa Mote

Journal: Nonlinear Processes in Geophysics

Publisher: Copernicus GmbH

DOI: 10.5194/npg-30-217-2023

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Open Access: N/A

Abstract: . We assessed different coupled data assimilation strategies with a hierarchy of coupled models, ranging from a simple coupled Lorenz model to the state-of-the-art coupled general circulation model CFSv2 (Climate Forecast System version 2). With the coupled Lorenz model, we assessed the analysis accuracy by strongly coupled ensemble Kalman filter (EnKF) and 4D-Variational (4D-Var) methods with varying assimilation window lengths. The analysis accuracy of the strongly coupled EnKF with a short assimilation window is comparable to that of 4D-Var with a long assimilation window. For 4D-Var, the strongly coupled approach with the coupled model produces more accurate ocean analysis than the Estimating the Circulation and Climate of the Ocean (ECCO)-like approach using the uncoupled ocean model. Experiments with the coupled quasi-geostrophic model conclude that the strongly coupled approach outperforms the weakly coupled and uncoupled approaches for both the full-rank EnKF and 4D-Var, with the strongly coupled EnKF and 4D-Var showing a similar level of accuracy higher than other coupled data assimilation approaches such as outer-loop coupling. A strongly coupled EnKF software framework is developed and applied to the intermediate-complexity coupled model SPEEDY-NEMO and the state-of-the-art operational coupled model CFSv2. Experiments assimilating synthetic or real atmospheric observations into the ocean through strongly coupled EnKF show that the strongly coupled approach improves the analysis of the atmosphere and upper ocean but degrades observation fits in the deep ocean, probably due to the unreliable error correlation estimated by a small ensemble. The correlation-cutoff method is developed to reduce the unreliable error correlations between physically irrelevant model states and observations. Experiments with the coupled Lorenz model demonstrate that strongly coupled EnKF informed by the correlation-cutoff method produces more accurate coupled analyses than the weakly coupled and plain strongly coupled EnKF regardless of the ensemble size. To extend the correlation-cutoff method to operational coupled models, a neural network approach is proposed to systematically acquire the observation localization functions for all pairs between the model state and observation types. The following strongly coupled EnKF experiments with an intermediate-complexity

couplemodel show promising results with this method.

277. Shelf slope, estuarine dynamics and river plumes in a z^* vertical coordinate, unstructured grid model

Authors: Giorgia Verri; Ivano Barletta; Nadia Pinardi; Ivan Federico; Jacopo Alessandri; Giovanni Coppini

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2023.102235

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Language: en

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Open Access: N/A

Abstract: N/A

278. Increased tropical South Pacific western boundary current transport over the past century

Authors: Wen-Hui Chen; Haojia Ren; John C. H. Chiang; You-Lin Wang; Ren-Yi Cai-Li; Yi-Chi Chen; Chuan-Chou Shen; Frederick W. Taylor; Thomas M. DeCarlo; Chau-Ron Wu; Horng-Sheng Mii; Xingchen Tony Wang

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

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Abstract: The wind-driven meridional overturning circulation between the tropical and subtropical oceans is important for regulating decadal-scale temperature fluctuations in the Pacific Ocean and globally. An acceleration of the overturning circulation can act to reduce global surface temperature as ocean stores more heat. The equatorward low-latitude western boundary current represents a key component of the meridional circulation cell in the Pacific and a major source of water mass for the Equatorial Undercurrent, yet long-term observations of its transport are scarce. Here we demonstrate that the $^{15}\text{N}/^{14}\text{N}$ ratio recorded by *Porites* spp. corals in the western tropical South Pacific is sensitive to the exchanges of water masses driven by the western boundary transport. Using a 94-year coral record from the Solomon Sea, we report that the $^{15}\text{N}/^{14}\text{N}$ ratio declined as the global surface temperature rose. The record suggests that the South Pacific western boundary current has strengthened in the past century, and it may have contributed to the reported strengthening of the Equatorial Undercurrent. In addition, the $^{15}\text{N}/^{14}\text{N}$ record shows strong decadal variability, indicative of weaker equatorial Pacific upwelling and stronger western boundary transport when the eastern equatorial Pacific is in the warm stage of the Pacific Decadal Oscillation.

279. Balancing Volume, Temperature, and Salinity Budgets During 2014-2018 in the Tropical Pacific Ocean State Estimate

Authors: Ariane Verdy (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Bruce D. Cornuelle (Scripps Institution of Oceanography / University of California San Diego); Aneesh C. Subramanian (Atmospheric and Oceanic Sciences Colorado University Boulder CO USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

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Open Access: N/A

Abstract: A state estimate of the tropical Pacific Ocean is used to analyze regional volume, temperature, and salinity budgets during 2014-2018. The simulated ocean state is constrained by both model dynamics and assimilated observations. Comparisons with independent mooring data show that the state estimate is consistent with the observed variability in temperature and velocity. Budgets are analyzed between 5 degreesS and 5 degreesN in the upper 300 m, inside a box defined to represent the central and eastern equatorial Pacific. Transports through the faces of this box are quantified to understand the processes responsible for variability in box-mean properties. Vertical mixing across 300 m is negligible; temperature and salinity tendencies are balanced by surface fluxes and advective divergence, which is decomposed into geostrophic and ageostrophic components. The onset and recovery of the 2015/2016 El Niño event is found to be dominated by anomalous surface fluxes and horizontal advection. During the onset phase, weaker trade winds cause the shallow meridional overturning circulation to slow down, which reduces the poleward transport of heat and leads to upper ocean warming. Anomalous precipitation and advection of fresh water from the western Pacific drive the net freshening of the region. Relaxation from El Niño conditions is dominated by wind-driven meridional advection at 5 degreesN. As the meridional advection regains strength, Ekman advection efficiently exports the warm, fresh surface water out of the equatorial region. Quantifying the heat and salt transport changes in response to wind variability strengthens our understanding of global ocean heat transport.

280. Unsupervised classification identifies coherent thermohaline structures in the Weddell Gyre region

Authors: Dani C. Jones; Maike Sonnewald; Shenjie Zhou; Ute Hausmann; Andrew J. S. Meijers; Isabella Rosso; Lars Boehme; Michael P. Meredith; Alberto C. Naveira Garabato

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-19-857-2023

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Open Access: N/A

Abstract: . The Weddell Gyre is a major feature of the Southern Ocean and an important component of the planetary climate system; it regulates air-sea exchanges, controls the formation of deep and bottom waters, and hosts upwelling of relatively warm subsurface waters. It is characterised by low sea surface temperatures, ubiquitous sea ice formation, and widespread salt stratification that stabilises the water column. Observing the Weddell Gyre is challenging, as it is extremely remote and largely covered with sea ice. At present, it is one of the most poorly sampled regions of the global ocean, highlighting the need to extract as much value as possible from existing observations. Here, we apply a profile classification model (PCM), which is an unsupervised classification technique, to a Weddell Gyre profile dataset to identify coherent regimes in temperature and salinity. We find that, despite not being given any positional information, the PCM identifies four spatially coherent thermohaline domains that can be described as follows: (1) a circumpolar class, (2) a transition region between the circumpolar waters and the Weddell Gyre, (3) a gyre edge class with northern and southern branches, and (4) a gyre core class. PCM highlights, in an objective and interpretable way, both expected and underappreciated structures in the Weddell Gyre dataset. For instance, PCM identifies the inflow of Circumpolar Deep Water (CDW) across the eastern boundary, the presence of the Weddell-Scotia Confluence waters, and structured spatial variability in mixing between Winter Water and CDW. PCM offers a useful complement to existing expertise-driven approaches for characterising the physical configuration and variability of oceanographic regions, helping to identify coherent thermohaline structures and the boundaries between them.

281. Wind and Heat Forcings of the Seasonal and Interannual Sea Level Variabilities in the Southwest Pacific

Authors: Lina Yang (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Xinyang Zhao (Guangdong Ocean University); Peng Liang (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Tianyu Zhang (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Lingling Xie (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Raghu Murtugudde (University of Maryland, Indian Institute of Technology Bombay)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Language: N/A

Subjects: N/A

Keywords: Seasonal variability, South Pacific Ocean, Interannual variability, Sea level, Planetary waves (and 1 more)

Funding: the National Key Research and Development Program of China, Innovative Research Group Project of the National Natural Science Foundation of China, Innovative Research Group Project of the National Natural Science Foundation of China (and 5 more)

License: N/A

Open Access: N/A

Abstract: Sea level variabilities in the southwest Pacific contribute to the variations of equatorial current bifurcation and the Indonesian Throughflow transport. These processes are closely related to the recharge/discharge of equatorial heat content and dynamic distribution of anthropogenic ocean heating over the Indo-Pacific basin, thus being of profound significance for climate variability and change. Here we identify the major features of seasonal and interannual sea level variabilities in this region, confirming the dominance of the first baroclinic mode in the tropics (contributing 60%-80% of the variances) and higher baroclinic modes in the extratropics (40%-60% of the seasonal variance). Seasonally, except in the western Coral Sea where the Ekman pumping is significant, the wind-driven first-mode baroclinic Rossby waves originating to the east of the date line control the sea level variations over most tropical Pacific regions. In the domain where the 1.5-layer reduced gravity model becomes deficient, the surface heat fluxes dominate, explaining ~40%-80% of sea level variance. For interannual variability, ~40%-60% of the variance are El Niño-Southern Oscillation (ENSO) related. The

wind-driven Rossby and Kelvin waves east of the date line explain ~40%-78% of the interannual variance in the tropical Pacific. Outside the tropics, small-scale diffusive processes are presumed critical for interannual variability according to a thermodynamic analysis using an eddy-permitting ocean model simulation. Further process and predictive understandings can be achieved with the coupled climate models properly parameterizing the subgrid-scale processes.

282. Increased sea ice melt as a driver of enhanced Arctic phytoplankton blooming

Authors: Andrew P. Castagno (University of North Carolina Wilmington); Till J. W. Wagner (University of Wisconsin-Madison, University of North Carolina Wilmington); Mattias R. Cape (University of Washington, Bigelow Laboratory for Ocean Sciences East Boothbay Maine USA); Conner W. Lester (Duke University, University of North Carolina Wilmington); Elizabeth Bailey (Yale University New Haven Connecticut USA, University of North Carolina Wilmington); Catharina Alves-de-Souza (University of North Carolina Wilmington); Robert A. York (University of North Carolina Wilmington); Alyson H. Fleming (University of Wisconsin-Madison, University of North Carolina Wilmington)

Journal: Global Change Biology

Publisher: Wiley

DOI: 10.1111/gcb.16815

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Keywords: N/A

Funding: Office of Polar Programs

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Open Access: N/A

Abstract: Phytoplankton primary production in the Arctic Ocean has been increasing over the last two decades. In 2019, a record spring bloom occurred in Fram Strait, characterized by a peak in chlorophyll that was reached weeks earlier than in other years and was larger than any previously recorded May bloom. Here, we consider the conditions that led to this event and examine drivers of spring phytoplankton blooms in Fram Strait using in situ, remote sensing, and data assimilation methods. From samples collected during the May 2019 bloom, we observe a direct relationship between sea ice meltwater in the upper water column and chlorophyll a pigment concentrations. We place the 2019 spring dynamics in context of the past 20 years, a period marked by rapid change in climatic conditions. Our findings suggest that increased advection of sea ice into the region and warmer surface temperatures led to a rise in meltwater input and stronger near-surface stratification. Over this time period, we identify large-scale spatial correlations in Fram Strait between increased chlorophyll a concentrations and increased freshwater flux from sea ice melt.

283. Typhoons and their upper ocean response over South China Sea using COAWST model

Authors: Anandh Thankaswamy; Tao Xian; Lian-Ping Wang

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2023.1102957

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Open Access: N/A

Abstract: The formation and intensification of typhoons is a complex process where energy and mass exchanges happen between the ocean and the atmosphere. In most typhoon numerical studies, a static ocean and a dynamic atmosphere are used to reduce the complexity of modeling. Using the COAWST model, we studied the air-sea interactions of Typhoon Mujigae in 2015, Typhoon Merbok in 2017, and Typhoon Hato in 2017 over the South China Sea. With different translation speeds, track shapes, and intensities between these cyclones, they act as an excellent case study to analyze the air-sea coupling in the models. The inclusion of coupling between the ocean and atmosphere is found to improve the typhoon track simulation significantly. The bias in the cyclone tracks is reduced by 10%-40% in the coupled model. The upper ocean response to the typhoon was also analyzed using the coupled model output. The coupled simulations show that the major energy extraction occurs to the right of the track, which is consistent with satellite observation and latent heat release analysis. The coupling process shows the air-sea interactions and exchanges in the upper ocean along with the energy released during the passage of typhoons. The heat budget analysis shows that the cooling of the upper ocean is mainly attributed to the advection associated with the typhoon forcing. This study shows that it is necessary to include ocean feedback while analyzing a typhoon, and the application of coupled models can improve our understanding as well as the forecasting capability of typhoons.

284. Zonal circulation in the North Atlantic ocean at 52 degreesW from WOCE-WHP and CLIVAR sections: 1997, 2003 and 2012

Authors: Daniel Santana-Toscano; M. Dolores Pérez-Hernández; Alison M. Macdonald; Cristina Arumí-Planas; Verónica Caínzos; Alonso Hernández-Guerra

Journal: Progress in Oceanography

Publisher: Elsevier BV

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Language: en

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Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

285. Summertime discontinuity of Western Boundary Current in the Bay of Bengal during contrasting Indian Ocean Dipole events of 2008 and 2010

Authors: Bijan Kumar Das; T.S. Anandh; Arun Chakraborty; J. Kuttippurath

Journal: Regional Studies in Marine Science

Publisher: Elsevier BV

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Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

286. Influence of the El Niño-Southern Oscillation on upper-ocean salinity changes in the southeast Indian ocean

Authors: Xunwei Nie; Hao Liu; Tengfei Xu; Zexun Wei

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1181278

URL: <https://doi.org/10.3389/fmars.2023.1181278>

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References: 50

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The interannual-decadal variability in the upper-ocean salinity of the southeast Indian Ocean (SEIO) was found to be highly correlated with the El Niño-Southern Oscillation (ENSO). Based on multisource data, this study revealed that this ENSO-like salinity variability mainly resides in the domain between 13 degreesS-30 degreesS and 100 degreesE-120 degreesE, and at depths above 150 m. This variability is principally driven by meridional geostrophic velocity (MGV), which changes with the zonal pattern of the sea surface height (SSH). Previous studies have reported that the variability in the SSH in the south Indian Ocean is principally driven by local-wind forcing and eastern-boundary forcing. Here the eastern-boundary forcing denotes the influence of SSH anomaly radiated from the western coast of Australia. A recent study emphasized the contribution of local-wind forcing in salinity variability in the SEIO, for its significant role in generation of the zonal dipole pattern of SSH anomaly in the south Indian Ocean, which was considered to be responsible for the anomalous MGV in the SEIO. While our results revealed a latitudinal difference between the domain where the SSH dipole pattern exists (north of 20 degreesS) and the region in which the ENSO-like salinity variability is strongest (20 degreesS-30 degreesS), suggesting that this salinity variability cannot be attributed entirely to the SSH dipole pattern. Our further investigation shows that, the MGV in the SEIO changes with local zonal SSH gradient that principally driven by eastern-boundary forcing. In combination with the strong meridional salinity gradient, the boundary-driven MGV anomalies cause significant meridional salinity advection and eventually give rise to the observed ENSO-like salinity variability. This study revealed the leading role of eastern-boundary forcing in interannual variability of the upper-ocean salinity in the SEIO.

287. Subglacial Freshwater Drainage Increases Simulated Basal Melt of the Totten Ice Shelf

Authors: David E. Gwyther (School of Earth and Environmental Sciences The University of Queensland St Lucia QLD Australia); Christine F. Dow (University of Waterloo); Stefan Jendersie (Victoria University of Wellington); Noel Gourmelen (University of Edinburgh); Benjamin K. Galton-Fenzi (University of Tasmania, Australian Antarctic Division Kingston TAS Australia)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: Totten ice shelf, ice shelves, Antarctica, sea level rise, subglacial discharge (and 1 more)

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Open Access: N/A

Abstract: Subglacial freshwater discharge from beneath Antarctic glaciers likely has a strong impact on ice shelf basal melting. However, the difficulty in directly observing subglacial flow highlights the importance of modeling these processes. We use an ocean model of the Totten Ice Shelf cavity into which we inject subglacial discharge derived from a hydrology model applied to Aurora Subglacial Basin. Our results show (a) discharge increases melting in the vicinity of the outflow region, which correlates with features observed in surface elevation maps and satellite-derived melt maps, with implications for ice shelf stability; (b) the change in melting is driven by the formation of a buoyant plume rather than the addition of heat; and (c) the buoyant plume originating from subglacial discharge-driven melting is far-reaching. Basal melting induced by subglacial hydrology is thus important for ice shelf stability, but is absent from almost all ice-ocean models.

288. Conservation of heat and mass in P-SKRIPS version 1: the coupled atmosphere-ice-ocean model of the Ross Sea

Authors: Alena Malyarenko; Alexandra Gossart; Rui Sun; Mario Krapp

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-3355-2023

URL: <https://doi.org/10.5194/gmd-16-3355-2023>

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Volume/Issue: Volume 16, Issue 11

References: 88

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Business, Innovation and Employment

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Open Access: N/A

Abstract: . Ocean-atmosphere-sea ice interactions are key to understanding the future of the Southern Ocean and the Antarctic continent. Regional coupled climate-sea ice-ocean models have been developed for several polar regions; however the conservation of heat and mass fluxes between coupled models is often overlooked due to computational difficulties. At regional scale, the non-conservation of water and energy can lead to model drift over multi-year model simulations. Here we present P-SKRIPS version 1, a new version of the SKRIPS coupled model setup for the Ross Sea region. Our development includes a full conservation of heat and mass fluxes transferred between the climate (PWRP) and sea ice-ocean (MITgcm) models. We examine open water, sea ice cover, and ice sheet interfaces. We show the evidence of the flux conservation in the results of a 1-month-long summer and 1-month-long winter test experiment. P-SKRIPS v.1 shows the implications of conserving heat flux over the Terra Nova Bay and Ross Sea polynyas in August 2016, eliminating the mismatch between total flux calculation in PWRP and MITgcm up to 922 W m⁻².

289. The Arctic Ocean's Changing Beaufort Gyre System: An Assessment of Current Understanding, Open Questions and Future Research Directions

Authors: Mary-Louise E. Timmermans (Yale University, New Haven, Connecticut;); Robert S. Pickart (Woods Hole Oceanographic Institution)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

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Language: N/A

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Keywords: Beaufort scale|Beaufort sea

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Open Access: N/A

Abstract: N/A

290. Mass, Heat, and Freshwater Transport From Transoceanic Sections in the Atlantic Ocean at 30 degreesS and 24.5 degreesN: Single Sections Versus Box Models?

Authors: Verónica Caínzos (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global, IOCAG Universidad de Las Palmas de Gran Canaria, ULPGC Unidad Asociada ULPGC-CSIC Canary Islands Spain); Alonso Hernández-Guerra (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global, IOCAG Universidad de Las Palmas de Gran Canaria, ULPGC Unidad Asociada ULPGC-CSIC Canary Islands Spain); Riccardo Farneti (The Abdus Salam International Centre for Theoretical Physics (ICTP) Trieste Italy); M. Dolores Pérez-Hernández (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global, IOCAG Universidad de Las Palmas de Gran Canaria, ULPGC Unidad Asociada ULPGC-CSIC Canary Islands Spain); Lynne D. Talley (Scripps Institution of Oceanography / University of California San Diego)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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License: CC BY-NC 4.0

Open Access: N/A

Abstract: The Atlantic meridional overturning circulation (AMOC) is central to the climate of the Atlantic by redistributing mass, heat, and freshwater. Hydrographic sections help monitor its strength at different latitudes, and inverse box models provide estimates of AMOC, heat, and freshwater transports. We have used all available hydrographic zonal sections at 24.5 degreesN and 30 degreesS over the last 30 years to conclude that single section inverse models agree with monthly outputs from an ocean general circulation model at the time of the cruise. In contrast, inverse models using multiple sections at different latitudes and times of the year for each of the last three decades are more consistent with decadal averages from the same model. Therefore, solutions of inverse models with single sections are affected by aliasing and represent the state of the ocean at the time of cruise. However, aliasing is greatly reduced when using multiple sections to assess low-frequency variability.

291. Interannual variations of heat budget in the lower layer of the eastern Ross Sea shelf and the forcing mechanisms in the Southern Ocean State Estimate

Authors: Yuanjie Chen (Shanghai Jiao Tong University); Zhaoru Zhang (Shanghai Jiao Tong University, Polar Research Institute of China, Ministry of Natural Resources); Xuezhu Wang (Hohai University); Xiaohui Liu (Second Institute of Oceanography, Southern Marine Science and Engineering Guangdong Laboratory, Ministry of Natural Resources); Meng Zhou (Shanghai Jiao Tong University, Polar Research Institute of China, Ministry of Natural Resources)

Journal: International Journal of Climatology

Publisher: Wiley

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Keywords: N/A

Funding: National Natural Science Foundation of China, Science and Technology Commission of Shanghai Municipality, National Key Research and Development Program of China

License: Creative Commons

Open Access: N/A

Abstract: The temporal variation of heat budget in the lower layer of the eastern Ross Sea shelf (ERSS) is crucial for understanding the stability of ice shelves in the Ross Sea. In this study, a 6-year (2005-2010) simulation from the Southern Ocean State Estimate (SOSE) is employed to analyse the interannual variations of heat budget in the lower layer of the ERSS and the controlling mechanisms. The results reveal that the annual change in the heat content of the study region is dominated by the horizontal heat advection term, and only in 2015 the vertical advection and diffusion terms also play an important role. The horizontal advection term is affected by the intrusion of warm circumpolar deep water (CDW) onto the shelf across the northern boundary of the ERSS, the transport of cold dense shelf water (DSW) from the western Ross Sea shelf, and the transport of CDW across the western boundary of the ERSS. The contributions of these physical processes to the change in annual heat content vary between years. The interannual variation of the CDW intrusion is associated with the strength of eddy activity over the slope, and the interannual variation of heat transports associated with DSW is affected by both the extent of DSW on the western shelf and the coastal circulations that affect the eastward spread of DSW.

292. Intradecadal variations in length of day: Coherence with models of the Earth's core dynamics

Authors: S. Rosat; N. Gillet

Journal: Physics of the Earth and Planetary Interiors

Publisher: Elsevier BV

DOI: 10.1016/j.pepi.2023.107053

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References: 56

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

293. Generation of Cold Anticyclonic Eddies and Warm Cyclonic Eddies in the Tropical Oceans

Authors: Qinbiao Ni (Second Institute of Oceanography, University of East Anglia, Southern Marine Science and Engineering Guangdong Laboratory, Ministry of Natural Resources); Xiaoming Zhai (University of East Anglia); Zhibin Yang (Ocean University of China); Dake Chen (Second Institute of Oceanography, Southern Marine Science and Engineering Guangdong Laboratory, Ministry of Natural Resources)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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References: 32

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: Eddy|Argo|Anticyclone|Wind Stress|Mixed layer

Funding: Royal Society International Exchanges Award, The Office of China Postdoctoral Council, National Natural Science Foundation of China (and 3 more)

License: N/A

Open Access: N/A

Abstract: Mesoscale eddies are ubiquitous features of the global ocean circulation. Traditionally, anticyclonic eddies are thought to be associated with positive temperature anomalies while cyclonic eddies are associated with negative temperature anomalies. However, our recent study found that about one-fifth of the eddies identified from global satellite observations are cold-core anticyclonic eddies (CAEs) and warm-core cyclonic eddies (WCEs). Here we show that in the tropical oceans where the probabilities of CAEs and WCEs are high, there are significantly more CAEs and WCEs in summer than in winter. We conduct a suite of idealized numerical model experiments initialized with composite eddy structures obtained from Argo profiles as well as a heat budget analysis. The results highlight the key role of relative wind-stress-induced Ekman pumping, surface mixed layer depth, and vertical entrainment in the formation and seasonal cycle of these unconventional eddies. The relative wind stress is found to be particularly effective in converting conventional eddies into CAEs or WCEs when the surface mixed layer is shallow. The abundance of CAEs and WCEs in the global ocean calls for further research on this topic.

294. Reconstructing 3D ocean subsurface salinity (OSS) from T-S mapping via a data-driven deep learning model

Authors: Jiali Zhang; Xuefeng Zhang; Xidong Wang; Pengfei Ning; Anmin Zhang

Journal: Ocean Modelling

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Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China

License: N/A

Open Access: N/A

Abstract: N/A

295. Mixing and air-sea buoyancy fluxes set the time-mean overturning circulation in the subpolar North Atlantic and Nordic Seas

Authors: Dafydd Gwyn Evans; N. Penny Holliday; Sheldon Bacon; Isabela Le Bras

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-19-745-2023

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Open Access: N/A

Abstract: . The overturning streamfunction as measured at the OSNAP (Overturning in the Subpolar North Atlantic Program) mooring array represents the transformation of warm, salty Atlantic Water into cold, fresh North Atlantic Deep Water (NADW). The magnitude of the overturning at the OSNAP array can therefore be linked to the transformation by air-sea buoyancy fluxes and mixing in the region north of the OSNAP array. Here, we estimate these water mass transformations using observational-based, reanalysis-based and model-based datasets. Our results highlight that air-sea fluxes alone cannot account for the time-mean magnitude of the overturning at OSNAP, and therefore a residual mixing-driven transformation is required to explain the difference. A cooling by air-sea heat fluxes and a mixing-driven freshening in the Nordic Seas, Iceland Basin and Irminger Sea precondition the warm, salty Atlantic Water, forming subpolar mode water classes in the subpolar North Atlantic. Mixing in the interior of the Nordic Seas, over the Greenland-Scotland Ridge and along the boundaries of the Irminger Sea and Iceland Basin drive a water mass transformation that leads to the convergence of volume in the water mass classes associated with NADW. Air-sea buoyancy fluxes and mixing therefore play key and complementary roles in setting the magnitude of the overturning within the subpolar North Atlantic and Nordic Seas. This study highlights that, for ocean and climate models to realistically simulate the overturning circulation in the North Atlantic, the small-scale processes that lead to the mixing-driven formation of NADW must be adequately represented within the model's parameterisation scheme.

296. Development of the Operational Oceanography Forecasting System in the Northwest Pacific

Authors: Zhaoyi Wang; Guimei Liu; Wei Li; Hui Wang; Dakui Wang

Journal: Journal of Physics: Conference Series

Publisher: IOP Publishing

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URL: <https://doi.org/10.1088/1742-6596/2486/1/012032>

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Abstract: An operational oceanography forecasting system is established on the base of ROMS and data assimilation method to provide operational predictions and maritime safety applications for the Northwest Pacific. After 10 years spun up through integration, the system reaches a stable field of annual circulation. The model was then integrated from 1990 to 2012 to obtain the assimilation background. After the evaluation of observation data, the ocean forecasting system has been put into operation since 2013 and has been updated for several times. Through the validation of 2018, the system can accurately simulate the main marine environmental characteristics of the northwest pacific. The system can provide high-resolution marine environment fields for red tide, green tide drift and diffusion, oil spill, marine fishery, marine resources development, marine search and rescue, etc.

297. Inter-comparison of global ice sea coupling model based on NEMO and MITgcm

Authors: Xin Liu; Chongji Zeng; Tienan Yin; Zhaoyi Wang; Zhijie Li; Zhongyuan Yao; Yu Zhang; Qihui Yan

Journal: Journal of Physics: Conference Series

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Abstract: Two global high-resolution ice sea coupling model established on basis of NEMO and MITgcm were systematically evaluated, by comparing the climatology simulation results of their sea surface temperature, sea surface salinity, sea surface current and sea ice area with SODA3, EN4, and NSIDC. The simulation accuracy of each element in the global and typical sea areas are quantitatively evaluated. The global RMSE of SST modeled by NEMO and MITgcm are 0.38 degreesC and 0.46 degreesC, respectively. NEMO had lower SST RMSE in strong current areas, especially in Kuroshio and Antarctic circumpolar currents, while MITgcm had better SST simulations in low-middle latitudes and most of Arctic Ocean. The global RMSE of SSS of NEMO and MITgcm is 0.47psu and 0.69psu, and the highest RMSE of SSS of the two models are in the Arctic Ocean. The RMSE of NEMO and MITgcm on surface current simulations are 0.0465m/s and 0.0474m/s, and both NEMO and MITgcm have high RMSE in their simulations of strong currents. The sea ice area simulated by NEMO and MITgcm are basically the same as that of NSIDC in spatial morphology, and the absolute value of the relative error of sea ice area in the Antarctic and Arctic of the two models are within 25%. The inter-comparison and evaluation results provide a reference for the product application and the model improvement of the two high-resolution ice sea coupling models.

298. Characteristics of submesoscale eddy structures within mesoscale eddies in the Gulf of Mexico from 1/48 degrees ECCO estimates

Authors: Paul A. Ernst; Bulusu Subrahmanyam; Corinne B. Trott; Alexis Chaigneau

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Publisher: Frontiers Media SA

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Abstract: Submesoscale oceanic structures (<10-20 km) such as eddies and fronts are often difficult to describe given the influence of the mesoscale. In order to characterize the surface signatures of submesoscale structures, we utilize a custom spatial filtering function to separate the meso- and large-scale sea surface height (SSH) signal from the small scale SSH signal of 1/48 degrees high resolution estimates provided by NASA's Estimating the Circulation and Climate of the Oceans (ECCO) project. In this study, we use ECCO estimates from a 14-month global simulation between September 2011 and November 2012 with a 2 km horizontal grid spacing in the Gulf of Mexico. We then use an eddy detection and tracking algorithm to identify persistent circular features on both scales, giving rise to an atlas of submesoscale eddy-like variabilities (SEVs). We briefly investigate the geographic and temporal variability of SEVs as a whole before collocating SEVs inside mesoscale eddies, allowing us to evaluate the characteristics of internal SEVs and the impact of SEVs on mesoscale eddies. We find that SEVs, both anticyclonic and cyclonic, are ubiquitous inside mesoscale eddies with lifetimes longer than a week, accounting for an average of 10-20% of the spatial area and eddy kinetic energy of mesoscale eddies. We also show that internal SEVs are persistently associated with temperature and salinity anomalies in both eddy centers and edges of up to 0.1 degreesC and 0.05 psu, with anticyclonic internal SEVs being warmer and fresher while cyclonic internal SEVs are colder and saltier. Finally, we examine the life cycle of an anticyclonic Loop Current eddy, demonstrating that the number and intensity of internal SEVs within increases as the eddy approaches separation from the Loop Current until a maximum is obtained just after separation. In light of forthcoming submesoscale SSH observations from NASA's Surface Water and Ocean Topography (SWOT) mission, our results showcase the variability of submesoscale eddy structures and their possible implications for biogeochemical cycling, the inverse energy cascade, and Loop Current prediction techniques.

299. Heat Budget Analysis for the Extended Development of the 2014-2015 Warming Event

Authors: Yinghao Qin (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Huier Mo (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Liying Wan (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Yi Wang (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Yang Liu (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Qinglong Yu (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Xiangyu Wu (National Marine Environmental Forecasting Center, Ministry of Natural Resources)

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Keywords: N/A

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Open Access: N/A

Abstract: In order to figure out the associated underlying dynamical processes of the 2014-2015 warming event, we used the ECCO (Estimating the Circulation and Climate of the Ocean) reanalysis from 1993 to 2016 and two combined scatterometers, QuikSCAT and ASCAT, to analysis hydrodynamic condition and ocean heat budget balance process in the equatorial tropical pacific. The spatiotemporal characteristics of that warming event were revealed by comparing the results with a composite El Niño. The results showed that the significant differences between the 2014 and 2015 warming periods were the magnitudes and positions of the equatorial easterly wind anomalies during the summer months. The abruptly easterly wind anomalies of 2014 that spread across the entire equatorial Pacific triggered the upwelling of the equatorial Kelvin waves and pushed the eastern edge of the warm pool back westward. These combined effects caused abrupt decreases in the sea surface temperatures (SST) and upper ocean heat content (OHC) and damped the 2014 warming process into an El Niño. In addition, the ocean budget of the upper 300 m of the El Niño 3.4 region showed that different dynamical processes were responsible for different warming phases. For example, at the beginning of 2014 and 2015, the U advection and subsurface processes played dominant roles in the positive ocean heat content tendency. During the easterly wind anomalies period of 2014, the U advection process mainly caused a negative tendency and halted the development of the warming phase. In regard to the easterly wind anomalies of 2015, the U advection and subsurface processes were weaker negatively when compared with that in 2014. However, the V advection processes were

consistently positive, taking a leading role in the positive trends observed in the middle of 2015.

300. Making the Case for High-Resolution Regional Ocean Reanalyses: An Example with the Red Sea

Authors: Sivareddy Sanikommu (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia;); Sabique Langodan (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology, Thuwal, and Environment Protection Department, Saudi Aramco, Dhahran, Saudi Arabia;); Hari Prasad Dasari (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia;); Peng Zhan (Southern University of Science and Technology); George Krokos (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia;); Yasser O. Abualnaja (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia;); Khaled Asfahani (Environment Protection Department, Saudi Aramco, Dhahran, Saudi Arabia); Ibrahim Hoteit (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia;)

Journal: Bulletin of the American Meteorological Society

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Abstract: Coastal oceans host 40% of the world population and amount to \$1.5 trillion of the global economy. Studying, managing, and developing the coastal regions require decades-long information about their environment. Long-term ocean measurements are, however, lacking for most coastal regions and often global reanalyses are used instead. These are, however, coarse in nature and tuned for the global circulations. The Red Sea (RS) is a narrow basin connected to the Indian Ocean through the Bab-al-Mandab Strait. Despite being the busiest commercial crossroad and hosting the world's third largest coral reef system, the RS lacks long-term observations. A recent increase in population and an unprecedented acceleration in governmental and industrial developments further emphasized the need for long-term datasets to support its development and the sustainability of its habitats, and to understand its response to a changing climate. Toward this end, we have generated a 20-yr high-resolution reanalysis for the RS (RSRA) using a state-of-the-art ensemble data assimilation system incorporating available observations. Compared to global reanalyses, RSRA provides a markedly better description of the RS general and mesoscale circulation features, their variability, and trends. In particular, RSRA accurately captures the three-layer summer transport through the

Bab-al-Mandab, simulated as two-layer transport by some global reanalyses. It further reproduces the seasonal anomalies, whereas global reanalyses misidentify some seasons as anomalous. Global reanalyses further overestimate the interannual variations in salinity, misrepresent the trend in temperature, and underestimate the trend in sea level. Our study clearly emphasizes the importance of generating dedicated high-resolution regional ocean reanalyses.

301. Validation and Evaluation of GRACE-FO Estimates with In Situ Bottom Pressure Array Measurements in the South China Sea

Authors: Xuecheng Wang (Zhejiang University, Second Institute of Oceanography, Ministry of Natural Resources); Hua Zheng (Second Institute of Oceanography, Shanghai Jiao Tong University, Ministry of Natural Resources); Xiao-Hua Zhu (Zhejiang University, Ministry of Natural Resources, Second Institute of Oceanography, Southern Marine Science and Engineering Guangdong Laboratory, Shanghai Jiao Tong University); Ruixiang Zhao (Second Institute of Oceanography, Ministry of Natural Resources); Min Wang (Second Institute of Oceanography, Shanghai Jiao Tong University, Ministry of Natural Resources); Juntian Chen (Zhejiang University, Second Institute of Oceanography, Ministry of Natural Resources); Yunlong Ma (Second Institute of Oceanography, Ministry of Natural Resources); Feng Nan (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Fei Yu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology)

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Open Access: N/A

Abstract: The Gravity Recovery and Climate Experiment (GRACE), and its follow-on mission (GRACE-FO), provides a novel measurement of the variations in ocean bottom pressure (OBP) at global and basin scales, including those in marginal seas. However, these measurements have not yet been validated rigorously for the South China Sea (SCS). In this study, the accuracy in the monthly GRACE-FO mascon solutions in the SCS from the Jet Propulsion Laboratory (JPL), Center for Space Research (CSR), and Goddard Space Flight Center (GSFC) was validated with the results of the comparison with the in situ OBP records from an array of 25 pressure-recording inverted echo sounders (PIESs) that are located west of the Luzon Strait (LS). The correlation coefficient (Cor) and root mean square difference (RMSD) between the 10-month period of GSFC and PIES, spanning from July 2018 to June 2019 (with missing satellite data for August and September 2018), were 0.77 (p-value = 0.005) and 0.41 mbar (1 mbar = 100 Pa), respectively. These values suggest that the accuracy of GSFC in the SCS in this period was substantially better than that of JPL (Cor = 0.35, p-value = 0.16; RMSD = 0.74 mbar) and CSR (Cor = 0.25, p-value = 0.24; RMSD = 0.89 mbar). Moreover, the volume transport anomaly of the SCS abyssal circulation was estimated and compared based on the OBP records from GSFC and PIES observations, indicating that the GRACE-FO OBP

(GSFC) can be used to monitor seasonal or longer-period variations in the SCS abyssal volume transport. Additionally, the variations in OBP from GRACE-FO were significantly overestimated on the continental shelf of the SCS, which may be attributed to signal leakage. Our findings provide reliable evidence for the application of long-term, fully covered OBP records from GRACE-FO in the SCS, and also offer a valuable reference for the application of GRACE-FO in other regions.

302. Zonal Distribution of Circumpolar Deep Water Transformation Rates and Its Relation to Heat Content on Antarctic Shelves

Authors: Aditya Narayanan (University of Gothenburg); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Marcel D. du Plessis (University of Gothenburg); K. Murali (Indian Institute of Technology Madras Chennai India); Fabien Roquet (University of Gothenburg)

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Publisher: American Geophysical Union (AGU)

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Keywords: Southern Ocean, Circumpolar Deep Water, ice shelves, Antarctica

Funding: National Centre for Polar and Ocean Research, Ministry of Earth Sciences, Horizon 2020 Framework Programme, National Aeronautics and Space Administration (and 1 more)

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Open Access: N/A

Abstract: We analyze 15-year of observational data and a 5-year Southern Ocean model simulation to quantify the transformation rates of Circumpolar Deep Water (CDW) and the associated heat loss to the surface. This study finds that over the continental shelves of East Antarctica and the Weddell and Ross Seas, surface buoyancy fluxes transform ~ 4.4 Sv of surface waters into CDW, providing a path for CDW to lose heat to the surface. In addition, ~ 6.6 Sv of CDW are mixed with surface waters in the Weddell and Ross subpolar gyres. In contrast, enhanced stratification inhibits the outcropping of CDW isopycnals, reducing their transformation rates by a factor of ~ 8 over the continental shelf and by a factor of ~ 3 over the deeper ocean in the Amundsen and Bellingshausen Seas. The CDW retains its offshore warm properties as it intrudes over the continental shelves, resulting in elevated bottom temperatures there. This analysis demonstrates the importance of processes in subpolar gyres to erode CDW and to facilitate further transformation on the continental shelves, significantly reducing the heat able to access ice shelf fronts. This sheltering effect is strongest in the western Weddell Sea and tends to diminish toward the east, which helps explain the large zonal differences in continental-shelf bottom temperatures and the melt rates of Antarctic ice shelves.

303. Gateway to the arctic: Defining the eastern channel of the Bering Strait

Authors: Mark Zimmermann; Rebecca A. Woodgate; Megan M. Prescott

Journal: Progress in Oceanography

Publisher: Elsevier BV

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Open Access: N/A

Abstract: N/A

304. Interference of ocean and land mass changes in seasonal crustal deformation of coastal stations: A case study in northern Australia

Authors: Shuo Zheng; Kosuke Heki; Zizhan Zhang; Yuta Tokui; Haoming Yan

Journal: Earth and Planetary Science Letters

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Open Access: N/A

Abstract: N/A

305. Varying Relationship between La Niña and SST Anomalies in the Kuroshio and Adjacent Regions during Boreal Winter: Role of the East Asian Winter Monsoon

Authors: Sheng Chen (Chinese Academy of Sciences); Jiepeng Chen (Chinese Academy of Sciences); Xin Wang (Chinese Academy of Sciences); Ziniu Xiao (Chinese Academy of Sciences)

Journal: Journal of Climate

Publisher: American Meteorological Society

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Keywords: Anomaly (physics)|Siberian High

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Abstract: The winter sea surface temperature (SST) anomalies in the Kuroshio and adjacent regions (KAR), which greatly influence the East Asian-North Pacific-North American climate, are closely related to El Niño-Southern Oscillation (ENSO). This SST relationship between the KAR and the equatorial eastern-central Pacific is widely assumed to be symmetric between El Niño and La Niña. Compared to previous studies indicating the significant and strong KAR warming during El Niño winters, this study indicates weakly negative KAR SST anomalies in the composite analysis for all La Niña events. Positive winter KAR SST anomalies unexpectedly appear in approximately half of La Niña events, which counteract negative SST anomalies in the rest of La Niña events. Further analysis suggests that the impact of La Niña on KAR SST anomalies is modulated by the East Asian winter monsoon (EAWM) during early winter. The weaker-than-normal EAWM offsets the anomalous northeasterly winds in the KAR induced by La Niña and then reinforces the KAR warming through warm oceanic advection. As for strong EAWM, it enhances the northeasterly winds to the west of an anomalous Philippine Sea cyclone associated with La Niña, leading to KAR cooling with more latent heat flux loss from the ocean and anomalous cold oceanic advection. Additionally, when the EAWM is independent of ENSO and is associated with the western Pacific pattern, it also can exhibit a pronounced influence on the KAR SST anomalies via the major processes of surface latent flux and horizontal heat advection in the ocean, accompanied by a change in Kuroshio transport. Significance Statement The interannual variability of SST from Philippine Sea to the Kuroshio and adjacent regions is considered an important forecast factor of the eastern China climate. Previous studies have reported a dipole SST pattern with Philippine Sea cooling and warming in the Kuroshio and adjacent regions during El Niño winter is dominant, and simply regarded that the effect of La Niña on western North Pacific SST anomalies was a mirror image of that of El Niño events. Here, we have found two distinctive SST patterns in the western North Pacific

during La Niña winter. One type is the dipole SST pattern, characterized by Philippine Sea warming and by cooling in the Kuroshio and adjacent regions. The other type is the monopole SST pattern with uniform warming expanding from the Philippine Sea to the Kuroshio and adjacent regions. The dipole (monopole) SST pattern in western North Pacific is modulated by La Niña and a strong (weak) EAWM. Analysis shows that the above two SST patterns during La Niña winter are equally important during 1950-2021. This study has identified the influence of the EAWM independent of ENSO on the SST anomalies in the Kuroshio and adjacent regions.

306. The South Atlantic Circulation Between 34.5 degreesS, 24 degreesS and Above the Mid-Atlantic Ridge From an Inverse Box Model

Authors: Cristina Arumí-Planas (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global IOCAG Universidad de Las Palmas de Gran Canaria ULPGC Unidad Asociada ULPGC-CSIC Las Palmas de Gran Canaria Spain); María Dolores Pérez-Hernández (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global IOCAG Universidad de Las Palmas de Gran Canaria ULPGC Unidad Asociada ULPGC-CSIC Las Palmas de Gran Canaria Spain); Josep L. Pelegrí (Departament d'Oceanografia Física i Tecnològica Institut de Ciències del Mar CSIC Unidad Asociada ULPGC-CSIC Barcelona Spain); Pedro Vélez-Belchí (Centro Oceanográfico de Canarias Instituto Español de Oceanografía CSIC Santa Cruz de Tenerife Spain); Mikhail Emelianov (Departament d'Oceanografia Física i Tecnològica Institut de Ciències del Mar CSIC Unidad Asociada ULPGC-CSIC Barcelona Spain); Verónica Caínzos (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global IOCAG Universidad de Las Palmas de Gran Canaria ULPGC Unidad Asociada ULPGC-CSIC Las Palmas de Gran Canaria Spain); Luis Cana (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global IOCAG Universidad de Las Palmas de Gran Canaria ULPGC Unidad Asociada ULPGC-CSIC Las Palmas de Gran Canaria Spain); Yvonne L. Firing (National Oceanography Centre); Luis García-Weil (Instituto Universitario de Estudios Ambientales y Recursos Naturales (i-UNAT) Universidad de Las Palmas de Gran Canaria ULPGC Las Palmas de Gran Canaria Spain); Daniel Santana-Toscano (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global IOCAG Universidad de Las Palmas de Gran Canaria ULPGC Unidad Asociada ULPGC-CSIC Las Palmas de Gran Canaria Spain); Alonso Hernández-Guerra (Unidad Océano y Clima Instituto de Oceanografía y Cambio Global IOCAG Universidad de Las Palmas de Gran Canaria ULPGC Unidad Asociada ULPGC-CSIC Las Palmas de Gran Canaria Spain)

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Abstract: The South Atlantic Ocean plays a key role in the heat exchange of the climate system, as it hosts the returning flow of the Atlantic Meridional Overturning Circulation (AMOC). To gain insights on this role, using data from three hydrographic cruises conducted in the South Atlantic Subtropical gyre at 34.5 degreesS, 24 degreesS, and 10 degreesW, we identify water masses and compute absolute geostrophic circulation using inverse modeling. In the upper layers, the currents describe the South

Atlantic anticyclonic gyre with the northwest flowing Benguela Current (26.3 ± 2.0 Sv at 34.5°S , and 21.2 ± 1.8 Sv at 24°S) flowing above the Mid-Atlantic Ridge (MAR) between 22.4°S and 28.4°S (-19.2 ± 1.4 Sv), and the southward flowing Brazil Current (-16.5 ± 1.3 Sv at 34.5°S , and -7.3 ± 0.9 Sv at 24°S); the deep layers feature the southward transports of Deep Western Boundary Current (-13.9 ± 3.0 Sv at 34.5°S , and -8.7 ± 3.8 Sv at 24°S) and Deep Eastern Boundary Current (-15.1 ± 3.5 Sv at 34.5°S , and -16.3 ± 4.7 Sv at 24°S), with the interbasin west-to-east flow close to 24°S (7.5 ± 4.4 Sv); the abyssal waters present northward mass transports through the Argentina Basin (5.6 ± 1.1 Sv at 34.5°S , and 5.8 ± 1.5 Sv at 24°S) and Cape Basin (8.6 ± 3.5 Sv at 34.5°S , -3.0 ± 0.8 Sv at 24°S) before returning southward (-2.2 ± 0.7 Sv at 24°S to -7.9 ± 3.6 Sv at 34.5°S), without any interbasin exchange across the MAR. In addition, we compute the upper AMOC strength (14.8 ± 1.0 and 17.5 ± 0.9 Sv), the equatorward heat transport (0.30 ± 0.05 and 0.80 ± 0.05 PW), and the freshwater flux (0.18 ± 0.02 and -0.07 ± 0.02 Sv) at 34.5°S and 24°S , respectively.

307. The Atlantic Meridional Overturning Circulation at 35 degreesN From Deep Moorings, Floats, and Satellite Altimeter

Authors: Isabela Alexander-Astiz Le Bras (Woods Hole Oceanographic Institution); Josh Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: From 2004 to 2014, the Line W moorings measured a 0.7 Sv yr⁻¹ slowing of the deep western boundary current (DWBC) offshore of Cape Cod. Here, we combine these deep mooring observations with float and satellite altimeter data and find that this DWBC change corresponded to a slowing of the cross-basin Atlantic Meridional Overturning Circulation (AMOC) of about 0.3 Sv yr⁻¹. Our AMOC transport time series corresponds well with the Estimating the Circulation and Climate of the Ocean state estimate, particularly when the Line W mooring data influences our volume closure. We compare our 35 degreesN time series with a similar time series at 41 degreesN as well as the 26 degreesN RAPID AMOC, and find AMOC declines across datasets from 2004 to 2014. However, when we extend our analysis to 2004-2019, there are no significant trends at any latitude. These observations suggest that AMOC decadal variability is meridionally coherent from 26 degreesN to 41 degreesN and that the DWBC may reflect this variability.

308. Toward a consensus framework to evaluate air-sea CO₂ equilibration for marine CO₂ removal

Authors: Lennart T. Bach (University of Tasmania); David T. Ho (Department of Oceanography University of Hawai'i at Mānoa Honolulu Hawaii USA, École Normale Supérieure (ENS)); Philip W. Boyd (University of Tasmania); Michael D. Tyka (Google Inc. Seattle Washington USA)

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Open Access: N/A

Abstract: N/A

309. Modeling the Recent Changes of Phytoplankton Blooms Dynamics in the Arctic Ocean

Authors: Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: Arctic Ocean, phytoplankton, polar marine ecosystems, climate change, sea ice (and 1 more)

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Open Access: N/A

Abstract: In recent decades, Arctic Ocean (AO) sea ice has experienced intense climate-related modifications, which coincide with earlier melting in spring and delayed formation in the fall. In this study, we use the ECCO2-Darwin ocean biogeochemistry model to gain a mechanistic understanding of how changes in sea-ice seasonality impact the phenology of AO phytoplankton blooms. We use a 2006-2013 simulation to show that early sea-ice melt in 2007 and 2012 triggers earlier phytoplankton blooms (~1 month earlier compared to other years) and that the blooms dissipate earlier due to more rapid nutrient depletion in the surface AO. Our simulation also confirms that delayed formation of sea ice during fall triggers second fall blooms, in agreement with observations. These results suggest that in a warmer and changing climate, further changes to sea-ice extent could drastically impact AO phytoplankton, with important consequences for Arctic marine ecosystems.

310. Global ocean reanalysis CORA2 and its inter comparison with a set of other reanalysis products

Authors: Hongli Fu; Bo Dan; Zhigang Gao; Xinrong Wu; Guofang Chao; Lianxin Zhang; Yinquan Zhang; Kexiu Liu; Xiaoshuang Zhang; Wei Li

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1084186

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References: 54

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China

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Open Access: N/A

Abstract: We present the China Ocean ReAnalysis version 2 (CORA2) in this paper. We compare CORA2 with its predecessor, CORA1, and with other ocean reanalysis products created between 2004 and 2019 [GLORYS12v1 (Global Ocean reanalysis and Simulation), HYCOM (HYbrid Coordinate Ocean Model), GREP (Global ocean Reanalysis Ensemble Product), SODA3 (Simple Ocean Data Assimilation, version 3), and ECCO4 (Estimating the Circulation and Climate of the Ocean, version 4)], to demonstrate its improvements and reliability. In addition to providing tide and sea ice signals, the accuracy and eddy kinetic energy (EKE) of CORA2 are also improved owing to an enhanced resolution of 9 km and updated data assimilation scheme compared with CORA1. Error analysis shows that the root-mean-square error (RMSE) of CORA2 sea-surface temperature (SST) remains around 0.3 degreesC, which is comparable to that of GREP and smaller than those of the other products studied. The subsurface temperature (salinity) RMSE of CORA2, at 0.87 degreesC (0.15 psu), is comparable to that of SODA3, smaller than that of ECCO4, and larger than those of GLORYS12v1, HYCOM, and GREP. CORA2 and GLORYS12v1 can better represent sub-monthly-scale variations in subsurface temperature and salinity than the other products. Although the correlation coefficient of sea-level anomaly (SLA) in CORA2 does not exceed 0.8 in the whole region, as those of GREP and GLORYS12v1 do, it is more effective than ECCO4 and SODA3 in the Indian Ocean and Pacific Ocean. CORA2 can reproduce the variations in steric sea level and ocean heat content (OHC) on the multiple timescales as the other products. The linear trend of the steric sea level of CORA2 is closer to that of GREP than that of the other products, and the long-term warming trends of global OHC in the high-resolution CORA2 and GLORYS12v1 are greater than those in the low-resolution EN4 and GREP. Although CORA2 shows overall poorer performance in the Atlantic Ocean, it still achieves good results from 2009 onward. We plan to further improve CORA2 by assimilating the best available observation data using the incremental analysis update (IAU) procedure and improving the SLA assimilation method.

311. Distributional Changes in Fishery Resource Diversity Caused by Typhoon Pathways in the East/Japan Sea

Authors: Yong Woo Jung (Chonnam National University); Beom Sik Kim (Gangneung-Wonju National University); Hae Kun Jung (Fisheries Resources and Environment Research Division, East Sea Fisheries Research Institute, National Institute of Fisheries Science, Gangneung 25435, Republic of Korea); Chung Il Lee (Gangneung-Wonju National University)

Journal: Fishes

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Volume/Issue: Volume 8, Issue 5

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Keywords: N/A

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Open Access: N/A

Abstract: Typhoons disturb the upper ocean, weaken the physical stratification, and induce temporal and spatial changes in primary production, which rapidly alter the distribution and diversity of fishery resources. This study analyzed the response of oceanic conditions and fishery resources on the sea area of the typhoon pathway in the East/Japan Sea (Type A: typhoon passed from southwest to northeast; Type B: typhoon dissipated in the southwest; Type C: typhoon passed from southeast to northeast; and Type D: typhoons passed from southwest to northwest). For Types A and B, the sea surface temperature (SST) decreased in all areas, and Chl-a showed the largest fluctuations in the southwest. For Type C, the SST variation was reduced in the eastern part, stratification was strengthened, and Chl-a did not differ significantly in each area. For Type D, SST and Chl-a showed significant variations in the western part. The biomass of fishery resources increased along the typhoon path for each type, and the diversity increased for Types A and D but decreased for Type B; however, the diversity and catch of fishery resources increased in the northeast for Type C. This study contributes to understanding the impact of typhoon pathway changes on the marine environment and ecosystem.

312. Finale: impact of the ORCHESTRA/ENCORE programmes on Southern Ocean heat and carbon understanding

Authors: Andrew J. S. Meijers (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Michael P. Meredith (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Emily F. Shuckburgh (University of Cambridge); Elizabeth C. Kent (National Oceanography Centre); David R. Munday (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Yvonne L. Firing (National Oceanography Centre); Brian King (National Oceanography Centre); Tim J. Smyth (Plymouth Marine Laboratory); Melanie J. Leng (British Geological Survey, Keyworth, Nottingham NG12 5GG, UK); A. J. George Nurser (National Oceanography Centre); Helene T. Hewitt (Met Office Hadley Centre, Met Office); E. Povl Abrahamsen (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Alexandra Weiss (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Mingxi Yang (Plymouth Marine Laboratory); Thomas G. Bell (Plymouth Marine Laboratory); J. Alexander Brearley (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Emma J. D. Boland (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Daniel C. Jones (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); Simon A. Josey (National Oceanography Centre); Robyn P. Owen (National Oceanography Centre); Jeremy P. Grist (National Oceanography Centre); Adam T. Blaker (National Oceanography Centre); Stavroula Biri (National Oceanography Centre); Margaret J. Yelland (National Oceanography Centre); Ciara Pimm (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK, University of Liverpool); Shenjie Zhou (British Antarctic Survey, High Cross, Madingley Road, CB3 0ET Cambridge, UK); James Harle (National Oceanography Centre); Richard C. Cornes (National Oceanography Centre)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020 Framework Programme, European Space Agency, Natural Environment Research Council (and 2 more)

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Open Access: N/A

Abstract: The 5-year Ocean Regulation of Climate by Heat and Carbon Sequestration and Transports (ORCHESTRA) programme and its 1-year extension ENCORE (ENCORE is the National Capability ORCHESTRA Extension) was an approximately 11-million-pound programme involving seven UK research centres that finished in March 2022. The project sought to radically improve our ability to

measure, understand and predict the exchange, storage and export of heat and carbon by the Southern Ocean. It achieved this through a series of milestone observational campaigns in combination with model development and analysis. Twelve cruises in the Weddell Sea and South Atlantic were undertaken, along with mooring, glider and profiler deployments and aircraft missions, all contributing to measurements of internal ocean and air-sea heat and carbon fluxes. Numerous forward and adjoint numerical experiments were developed and supported by the analysis of coupled climate models. The programme has resulted in over 100 peer-reviewed publications to date as well as significant impacts on climate assessments and policy and science coordination groups. Here, we summarize the research highlights of the programme and assess the progress achieved by ORCHESTRA/ENCORE and the questions it raises for the future. This article is part of a discussion meeting issue 'Heat and carbon uptake in the Southern Ocean: the state of the art and future priorities'.

313. A Southern Ocean supergyre as a unifying dynamical framework identified by physics-informed machine learning

Authors: Maike Sonnewald; Krissy Anne Reeve; Redouane Lguensat

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

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Cited by: 4 publications

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Keywords: N/A

Funding: United States Department of Commerce | National Oceanic and Atmospheric Administration, Deutsche Forschungsgemeinschaft, Agence Nationale de la Recherche

License: CC BY 4.0

Open Access: N/A

Abstract: The Southern Ocean closes the global overturning circulation and is key to the regulation of carbon, heat, biological production, and sea level. However, the dynamics of the general circulation and upwelling pathways remain poorly understood. Here, a physics-informed unsupervised machine learning framework using principled constraints is used. A unifying framework is proposed invoking a semi-circumpolar supergyre south of the Antarctic circumpolar current: a massive series of leaking sub-gyres spanning the Weddell and Ross seas that are connected and maintained via rough topography that acts as scaffolding. The supergyre framework challenges the conventional view of having separate circulation structures in the Weddell and Ross seas and suggests that idealized models and zonally-averaged frameworks may be of limited utility for climate applications. Machine learning was used to reveal areas of coherent driving forces within a vorticity-based analysis. Predictions from the supergyre framework are supported by available observations and could aid observational and modelling efforts to study this climatologically key region undergoing rapid change.

314. Nonlinear response of Equatorial Western Pacific phytoplankton blooms to 'double-dip' La Niña events

Authors: Ruyan Chen; Yan Du; Ying Zhang; Jianwei Chi

Journal: Environmental Research Communications

Publisher: IOP Publishing

DOI: 10.1088/2515-7620/acd1e7

URL: <https://doi.org/10.1088/2515-7620/acd1e7>

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Keywords: N/A

Funding: National Natural Science Foundation of China, Chinese Academy of Sciences

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Open Access: N/A

Abstract: Phytoplankton in the equatorial western Pacific tends to bloom during consecutive ('double-dip') La Niña events with nonlinear characteristics: extremely high chlorophyll-a (Chl-a) concentrations typically occur during the second-year La Niña events even when the associated SST anomalies are significantly weakened. Photosynthetically available radiation is found to have the strongest correlation with the equatorial western Pacific Chl-a fluctuations. However, barrier layer variation is critical in driving the strong bloom events seen in the second-year La Niña, which can be further explained by the nonlinear heat advection within the isothermal layer. To improve the current climate models' performance in simulating the western Pacific phytoplankton bloom events, it is recommended that the influence of barrier layer should be better considered.

315. Reconstructing ocean carbon storage with CMIP6 Earth system models and synthetic Argo observations

Authors: Katherine E. Turner; Doug M. Smith; Anna Katavouta; Richard G. Williams

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-20-1671-2023

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Keywords: N/A

Funding: Department for Business, Energy and Industrial Strategy, UK Government, Department for Environment, Food and Rural Affairs, UK Government, Natural Environment Research Council (and 1 more)

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Open Access: N/A

Abstract: . The ocean carbon store plays a vital role in setting the carbon response to emissions and variability in the carbon cycle. However, due to the ocean's strong regional and temporal variability, sparse carbon observations limit our understanding of historical carbon changes. Ocean temperature and salinity profiles are more widespread and rapidly expanding due to autonomous programmes, and so we explore how temperature and salinity profiles can provide information to reconstruct ocean carbon inventories with ensemble optimal interpolation. Here, ensemble optimal interpolation is used to reconstruct ocean carbon using synthetic Argo temperature and salinity observations, with examples for both the top 100 m and top 2000 m carbon inventories. When considering reconstructions of the top 100 m carbon inventory, coherent relationships between upper-ocean carbon, temperature, salinity, and atmospheric CO₂ result in optimal solutions that reflect the controls of undersaturation, solubility, and alkalinity. Out-of-sample reconstructions of the top 100 m show that, in most regions, the trend in ocean carbon and over 60 % of detrended variability can be reconstructed using local temperature and salinity measurements, with only small changes when considering synthetic profiles consistent with irregular Argo sampling. Extending the method to reconstruct the upper 2000 m reveals that model uncertainties at depth limit the reconstruction skill. The impact of these uncertainties on reconstructing the carbon inventory over the upper 2000 m is small, and full reconstructions with historical Argo locations show that the method can reconstruct regional inter-annual and decadal variability. Hence, optimal interpolation based on model relationships combined with hydrographic measurements can provide valuable information about global ocean carbon inventory changes.

316. Electrolytic Seawater Mineralization and the Mass Balances That Demonstrate Carbon Dioxide Removal

Authors: Erika Callagon La Plante (Center for Advanced Construction Materials, University of Texas at Arlington, Arlington, Texas 76019, United States, Equatic Inc., Los Angeles, California 90024, United States, University of California Los Angeles, Department of Materials Science and Engineering, University of Texas at Arlington, Arlington, Texas 76019, United States); Xin Chen (Equatic Inc., Los Angeles, California 90024, United States, University of California Los Angeles); Steven Bustillos (University of California Los Angeles); Arnaud Bouissonnie (University of California Los Angeles); Thomas Traynor (Equatic Inc., Los Angeles, California 90024, United States, University of California Los Angeles); David Jassby (Equatic Inc., Los Angeles, California 90024, United States, University of California Los Angeles); Lorenzo Corsini (Equatic Inc., Los Angeles, California 90024, United States); Dante A. Simonetti (Equatic Inc., Los Angeles, California 90024, United States, University of California Los Angeles); Gaurav N. Sant (Equatic Inc., Los Angeles, California 90024, United States, University of California Los Angeles)

Journal: ACS ES&T; Engineering

Publisher: American Chemical Society (ACS)

DOI: 10.1021/acsestengg.3c00004

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Keywords: N/A

Funding: Public Utilities Board - Singapore, University of California, Los Angeles, Advanced Research Projects Agency - Energy (and 4 more)

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Open Access: N/A

Abstract: N/A

317. A shallow slow slip event in 2018 in the Semidi segment of the Alaska subduction zone detected by machine learning

Authors: Bing He; XiaoZhuo Wei; Meng Wei; Yang Shen; Marco Alvarez; Susan Y. Schwartz

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2023.118154

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Volume/Issue: Volume 612

References: 51

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

318. Elevated methylmercury in Antarctic surface seawater: The role of phytoplankton mass and sea ice

Authors: Fange Yue; Yanbin Li; Yanxu Zhang; Longquan Wang; Dan Li; Peipei Wu; Hongwei Liu; Lijin Lin; Dong Li; Ji Hu; Zhouqing Xie

Journal: Science of The Total Environment

Publisher: Elsevier BV

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References: 58

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

319. Bias Correction and Statistical Modeling of Variable Oceanic Forcing of Greenland Outlet Glaciers

Authors: Vincent Verjans (Georgia Institute of Technology); Alexander Robel (Georgia Institute of Technology); Andrew F. Thompson (California Institute of Technology); Helene Seroussi (Dartmouth College)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023ms003610

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References: 73

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: statistical modeling, ocean forcing, ice sheet

Funding: Heising-Simons Foundation

License: Creative Commons

Open Access: N/A

Abstract: Variability in oceanic conditions directly impacts ice loss from marine outlet glaciers in Greenland, influencing the ice sheet mass balance. Oceanic conditions are available from Atmosphere-Ocean Global Climate Model (AOGCM) output, but these models require extensive computational resources and lack the fine resolution needed to simulate ocean dynamics on the Greenland continental shelf and close to glacier marine termini. Here, we develop a statistical approach to generate ocean forcing for ice sheet model simulations, which incorporates natural spatiotemporal variability and anthropogenic changes. Starting from raw AOGCM ocean heat content, we apply: (a) a bias-correction using ocean reanalysis, (b) an extrapolation accounting for on-shelf ocean dynamics, and (c) stochastic time series models to generate realizations of natural variability. The bias-correction reduces model errors by ~25% when compared to independent in-situ measurements. The bias-corrected time series are subsequently extrapolated to fjord mouth locations using relations constrained from available high-resolution regional ocean model results. The stochastic time series models reproduce the spatial correlation, characteristic timescales, and the amplitude of natural variability of bias-corrected AOGCMs, but at negligible computational expense. We demonstrate the efficiency of this method by generating >6,000 time series of ocean forcing for >200 Greenland marine-terminating glacier locations until 2100. As our method is computationally efficient and adaptable to any ocean model output and reanalysis product, it provides flexibility in exploring sensitivity to ocean conditions in Greenland ice sheet model simulations. We provide the output and workflow in an open-source repository, and discuss advantages and future developments for our method.

320. Relative Contribution of Atmospheric Forcing, Oceanic Preconditioning and Sea Ice to Deep Convection in the Labrador Sea

Authors: Yang Wu (School of Information Engineering, Nanjing Xiaozhuang University, Nanjing 211171, China); Xiangjun Zhao (School of Information Engineering, Nanjing Xiaozhuang University, Nanjing 211171, China); Zhengdong Qi (School of Information Engineering, Nanjing Xiaozhuang University, Nanjing 211171, China); Kai Zhou (School of Information Engineering, Nanjing Xiaozhuang University, Nanjing 211171, China); Dalei Qiao (School of Information Engineering, Nanjing Xiaozhuang University, Nanjing 211171, China)

Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

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Publication Date: 2023-04-20

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References: 54

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Natural Science Foundation of Jiangsu Province, China Postdoctoral Science Foundation (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: The relative contribution of atmospheric forcing, oceanic preconditioning, and sea ice to Labrador Sea Deep Convection (LSDC) is investigated by conducting three ensemble experiments using a global coupled sea ice-ocean model for the first time. Simulated results show that the atmospheric activities dominate the interannual and decadal variability, accounting for 70% of LSDC. Oceanic preconditioning is more significant in the shallow LSDC years that the water column is stable, accounting for 21%, especially in the central Labrador Sea and Irminger Sea. Moreover, the sea ice contribution is negligible over the whole Labrador Sea, while its contribution is significant in the sea ice-covered slope regions, accounting for 20%. The increasingly importance of sea ice on LSDC and the water mass transformation will be found in the North Atlantic Ocean, if the Arctic sea ice declines continuously. Additionally, there is a 10 Sv increase (85%) in atmospheric forcing to the subpolar gyre in the North Atlantic Ocean, while oceanic preconditioning contributes a 7 Sv decrease (12%). These findings highlight the importance of summer oceanic preconditioning to LSDC and the subpolar gyre, and therefore it should be appropriately accounted for in future studies.

321. Parameterizing Submesoscale Vertical Buoyancy Flux by Simultaneously Considering Baroclinic Instability and Strain-Induced Frontogenesis

Authors: Jinchao Zhang (Ocean University of China, Laoshan Laboratory); Zhiwei Zhang (Ocean University of China, Laoshan Laboratory); Bo Qiu (University of Hawaii at Manoa)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China

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Open Access: N/A

Abstract: Oceanic submesoscale processes (submesoscales) with O(1-10) km horizontal scale can generate strong vertical buoyancy flux (VBF) that significantly modulate upper-ocean stratification. Because submesoscales cannot be resolved by the prevailing ocean models, their VBFs have to be properly parameterized in order to improve model performance. Here, based on theoretical scaling analysis, we propose a new parameterization of submesoscale VBF by simultaneously considering mixed-layer baroclinic instability (MLI) and strain-induced frontogenesis, which are two leading generation mechanisms of submesoscales that typically co-occur in open ocean. Compared with the parameterization of Fox-Kemper et al. (2008, <https://doi.org/10.1175/2007jpo3792.1>; F08) that only considers the MLI, the new parameterization includes mesoscale strain rate and improves vertical structure function. Diagnostic validations based on submesoscale permitting simulation outputs suggest that the newly parameterized VBFs are more realistic than F08 in regard to three-dimension distributions. How to incorporate this new parameterization into coarser-grid ocean models, however, needs further studies.

322. Heat stored in the Earth system 1960-2020: where does the energy go?

Authors: Karina von Schuckmann; Audrey Minière; Flora Gues; Francisco José Cuesta-Valero; Gottfried Kirchengast; Susheel Adusumilli; Fiammetta Straneo; Michaël Ablain; Richard P. Allan; Paul M. Barker; Hugo Beltrami; Alejandro Blazquez; Tim Boyer; Lijing Cheng; John Church; Damien Desbruyeres; Han Dolman; Catia M. Domingues; Almudena García-García; Donata Giglio; John E. Gilson; Maximilian Gorfer; Leopold Haimberger; Maria Z. Hakuba; Stefan Hendricks; Shigeki Hosoda; Gregory C. Johnson; Rachel Killick; Brian King; Nicolas Kolodziejczyk; Anton Korosov; Gerhard Krinner; Mikael Kuusela; Felix W. Landerer; Moritz Langer; Thomas Lavergne; Isobel Lawrence; Yuehua Li; John Lyman; Florence Marti; Ben Marzeion; Michael Mayer; Andrew H. MacDougall; Trevor McDougall; Didier Paolo Monselesan; Jan Nitzbon; Inès Otosaka; Jian Peng; Sarah Purkey; Dean Roemmich; Kanako Sato; Katsunari Sato; Abhishek Savita; Axel Schweiger; Andrew Shepherd; Sonia I. Seneviratne; Leon Simons; Donald A. Slater; Thomas Slater; Andrea K. Steiner; Toshio Suga; Tanguy Szekely; Wim Thiery; Mary-Louise Timmermans; Inne Vanderkelen; Susan E. Wijffels; Tonghua Wu; Michael Zemp

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-15-1675-2023

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Cited by: 73 publications

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Open Access: N/A

Abstract: . The Earth climate system is out of energy balance, and heat has accumulated continuously over the past decades, warming the ocean, the land, the cryosphere, and the atmosphere. According to the Sixth Assessment Report by Working Group I of the Intergovernmental Panel on Climate Change, this planetary warming over multiple decades is human-driven and results in unprecedented and committed changes to the Earth system, with adverse impacts for ecosystems and human systems. The Earth heat inventory provides a measure of the Earth energy imbalance (EEI) and allows for quantifying how much heat has accumulated in the Earth system, as well as where the heat is stored. Here we show that the Earth system has continued to accumulate heat, with 381 ± 61 ZJ accumulated from 1971 to 2020. This is equivalent to a heating rate (i.e., the EEI) of 0.48 ± 0.1 W m⁻². The majority, about 89 %, of this heat is stored in the ocean, followed by about 6 % on land, 1 % in the atmosphere, and about 4 % available for melting the cryosphere. Over the most recent period (2006-2020), the EEI amounts to 0.76 ± 0.2 W m⁻². The Earth energy imbalance is the most fundamental global climate indicator that the scientific community and the public can use as the measure of how well the world is doing in the task of bringing anthropogenic climate change under control. Moreover, this indicator is highly complementary to other established ones like global

meansurface temperature as it represents a robust measure of the rate of climatechange and its future commitment. We call for an implementation of theEarth energy imbalance into the Paris Agreement's Global Stocktake based onbest available science. The Earth heat inventory in this study, updated fromvon Schuckmann et al. (2020), is underpinned by worldwide multidisciplinarycollaboration and demonstrates the critical importance of concertedinternational efforts for climate change monitoring and community-basedrecommendations and we also call for urgently needed actions for enablingcontinuity, archiving, rescuing, and calibrating efforts to assure improvedand long-term monitoring capacity of the global climate observing system. The data for the Earth heat inventory are publicly available, and more details are provided in Table 4.

323. Biogeochemical River Runoff Drives Intense Coastal Arctic Ocean CO₂ Outgassing

Authors: C. Bertin (Littoral ENvironnement et Sociétés (LIENSs) - UMR 7266 Bâtiment ILE La Rochelle France); D. Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); S. Dutkiewicz (Massachusetts Institute of Technology); H. Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Matsuoka (University of New Hampshire); S. Tank (University of Alberta); M. Manizza (Scripps Institution of Oceanography / University of California San Diego); C. E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Babin (Université Laval); A. Mangin (ACRI-ST); V. Le Fouest (Littoral ENvironnement et Sociétés (LIENSs) - UMR 7266 Bâtiment ILE La Rochelle France)

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Open Access: N/A

Abstract: Arctic warming alters land-to-sea fluxes of nutrients and organic matter, which impact air-sea carbon exchange. Here we use an ocean-biogeochemical model of the southeastern Beaufort Sea (SBS) to investigate the role of Mackenzie River biogeochemical discharge in modulating air-sea CO₂ fluxes during 2000-2019. The contribution of six biogeochemical discharge constituents leads to a net CO₂ outgassing of 0.13 TgC yr⁻¹, with a decrease in the coastal SBS carbon sink of 0.23 and 0.4 TgC yr⁻¹ due to riverine dissolved organic and inorganic carbon, respectively. Years with high (low) discharge promote more CO₂ outgassing (uptake) from the river plume. These results demonstrate that the Mackenzie River modulates the capacity of the SBS to act as a sink or source of atmospheric CO₂. Our work suggests that accurate model representation of land-to-sea biogeochemical coupling can be critical for assessing present-day Arctic coastal ocean response to the rapidly changing environment.

324. Vertical-Slice Ocean Tomography With Seismic Waves

Authors: Jörn Callies (California Institute of Technology); Wenbo Wu (California Institute of Technology, Wood Hole Oceanographic Institution Woods Hole MA USA); Shirui Peng (California Institute of Technology); Zhongwen Zhan (California Institute of Technology)

Journal: Geophysical Research Letters

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Subjects: N/A

Keywords: ocean acoustic tomography, ocean heat uptake, T waves

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License: CC BY 4.0

Open Access: N/A

Abstract: Seismically generated sound waves that propagate through the ocean are used to infer temperature anomalies and their vertical structure in the deep East Indian Ocean. These T waves are generated by earthquakes off Sumatra and received by hydrophone stations off Diego Garcia and Cape Leeuwin. Between repeating earthquakes, a T wave's travel time changes in response to temperature anomalies along the wave's path. What part of the water column the travel time is sensitive to depends on the frequency of the wave, so measuring travel time changes at a few low frequencies constrains the vertical structure of the inferred temperature anomalies. These measurements reveal anomalies due to equatorial waves, mesoscale eddies, and decadal warming trends. By providing direct constraints on basin-scale averages with dense sampling in time, these data complement previous point measurements that alias local and transient temperature anomalies.

325. Widespread global disparities between modelled and observed mid-depth ocean currents

Authors: Fenzhen Su; Rong Fan; Fengqin Yan; Michael Meadows; Vincent Lyne; Po Hu; Xiangzhou Song; Tianyu Zhang; Zenghong Liu; Chenghu Zhou; Tao Pei; Xiaomei Yang; Yunyan Du; Zexun Wei; Fan Wang; Yiquan Qi; Fei Chai

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

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Open Access: N/A

Abstract: The mid-depth ocean circulation is critically linked to actual changes in the long-term global climate system. However, in the past few decades, predictions based on ocean circulation models highlight the lack of data, knowledge, and long-term implications in climate change assessment. Here, using 842,421 observations produced by Argo floats from 2001-2020, and Lagrangian simulations, we show that only 3.8% of the mid-depth oceans, including part of the equatorial Pacific Ocean and the Antarctic Circumpolar Current, can be regarded as accurately modelled, while other regions exhibit significant underestimations in mean current velocity. Knowledge of ocean circulation is generally more complete in the low-latitude oceans but is especially poor in high latitude regions. Accordingly, we propose improvements in forecasting, model representation of stochasticity, and enhancement of observations of ocean currents. The study demonstrates that knowledge and model representations of global circulation are substantially compromised by inaccuracies of significant magnitude and direction, with important implications for modelled predictions of currents, temperature, carbon dioxide sequestration, and sea-level rise trends.

326. Machine Learning-Derived Inference of the Meridional Overturning Circulation From Satellite-Observable Variables in an Ocean State Estimate

Authors: Aviv Solodoch (Department of Atmospheric and Oceanic Sciences University of California in Los Angeles Los Angeles CA USA); Andrew L. Stewart (Department of Atmospheric and Oceanic Sciences University of California in Los Angeles Los Angeles CA USA); Andrew McC. Hogg (ARC Centre of Excellence for Climate Extremes Sydney NSW Australia, Australian National University); Georgy E. Manucharyan (University of Washington)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: climate variability, satellite sensing, observing systems, Meridional Overturning Circulation, ocean circulation (and 1 more)

Funding: National Science Foundation

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Open Access: N/A

Abstract: The oceanic Meridional Overturning Circulation (MOC) plays a key role in the climate system, and monitoring its evolution is a scientific priority. Monitoring arrays have been established at several latitudes in the Atlantic Ocean, but other latitudes and oceans remain unmonitored for logistical reasons. This study explores the possibility of inferring the MOC from globally-available satellite measurements via machine learning (ML) techniques, using the ECCOV4 state estimate as a test bed. The methodological advantages of the present approach include the use purely of available satellite data, its applicability to multiple basins within a single ML framework, and the ML model simplicity (a feed-forward fully connected neural network (NN) with small number of neurons). The ML model exhibits high skill in MOC reconstruction in the Atlantic, Indo-Pacific, and Southern Oceans. The approach achieves a higher skill in predicting the model Southern Ocean abyssal MOC than has previously been achieved via a dynamically-based approach. The skill of the model is quantified as a function of latitude in each ocean basin, and of the time scale of MOC variability. We find that ocean bottom pressure generally has the highest reconstruction skill potential, followed by zonal wind stress. We additionally test which combinations of variables are optimal. Furthermore, ML interpretability techniques are used to show that high reconstruction skill in the Southern Ocean is mainly due to (NN processing of) bottom pressure variability at a few prominent bathymetric ridges. Finally, the potential for reconstructing MOC strength estimates from real satellite measurements is discussed.

327. Sensitivity of simulated water mass transformation on the Antarctic shelf to tides, topography and model resolution

Authors: Fabio Boeira Dias; Stephen R. Rintoul; Ole Richter; Benjamin Keith Galton-Fenzi; Jan D. Zika; Violaine Pellichero; Petteri Uotila

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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Open Access: N/A

Abstract: Water mass transformation (WMT) around the Antarctic margin controls Antarctica Bottom Water formation and the abyssal limb of the global meridional overturning circulation, besides mediating ocean-ice shelf exchange, ice sheet stability and its contribution to sea level rise. However, the mechanisms controlling the rate of WMT in the Antarctic shelf are poorly understood due to the lack of observations and the inability of climate models to simulate those mechanisms, in particular beneath the floating ice shelves. We used a circum-Antarctic ocean-ice shelf model to assess the contribution of surface fluxes, mixing, and ocean-ice shelf interaction to the WMT on the continental shelf. The salt budget dominates the WMT rates, with only a secondary contribution from the heat budget. Basal melt of ice shelves drives buoyancy gain at lighter density classes ($27.2 < \sigma_\theta < 27.6 \text{ kg m}^{-3}$), while salt input associated with sea-ice growth in coastal polynyas drives buoyancy loss at heavier densities ($\sigma_\theta > 27.6$). We found a large sensitivity of the WMT rates to model horizontal resolution, tides and topography within the Filchner-Ronne, East and West Antarctica ice shelf cavities. In the Filchner-Ronne Ice Shelf, an anticyclonic circulation in front of the Ronne Depression regulates the rates of basal melting/refreezing and WMT and is substantially affected by tides and model resolution. Model resolution is also found to affect the Antarctic Slope Current in both East and West Antarctica, impacting the on-shelf heat delivery, basal melt and WMT. Moreover, the representation of the ice shelf draft associated with model resolution impacts the freezing temperature and thus basal melt and WMT rates in the East Antarctica. These results highlight the importance of resolving small-scale features of the flow and topography, and to include the effects of tidal forcing, to adequately represent water mass transformations on the shelf that directly influence the abyssal global overturning circulation.

328. Mixing and Geometry in the North Atlantic Meridional Overturning Circulation

Authors: Renzo Bruera (Departament de Matemàtiques Universitat Politècnica de Catalunya Barcelona Spain); Jezabel Curbelo (Departament de Matemàtiques Universitat Politècnica de Catalunya Barcelona Spain); Guillermo García-Sánchez (Instituto de Ciencias Matemáticas Consejo Superior de Investigaciones Científicas C/ Nicolás Cabrera 13-15 Campus Cantoblanco Madrid Spain, Escuela Técnica Superior de Ingenieros de Telecomunicación Universidad Politécnica de Madrid Madrid Spain); Ana M. Mancho (Instituto de Ciencias Matemáticas Consejo Superior de Investigaciones Científicas C/ Nicolás Cabrera 13-15 Campus Cantoblanco Madrid Spain)

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Subjects: N/A

Keywords: ocean vertical motion, Atlantic Meridional Overturning Circulation, Lagrangian Coherent Structures

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License: CC BY 4.0

Open Access: N/A

Abstract: Vertical motions across the ocean are central to processes, like CO₂ fixation, heat removal or pollutant transport, which are essential to the Earth's climate. This work explores 3D conveyor routes associated with the Atlantic Meridional Overturning Circulation (AMOC). Our findings show the geometry of mixing structures in the upper and deep ocean layers by means of Lagrangian Coherent Structures. This tool identifies among others, zones linked to vertical transport and characterizes vertical transport time scales. We focus the study in two regions. The first one is the Flemish Cap region, a zone of interaction between the major AMOC components, where our analysis identifies a domain of deep waters that ascend very rapidly to the ocean surface. The second one is the Irminger Sea, where our analysis confirms the existence of a downwelling zone, and reveals a previously unreported upwelling connection between very deep waters and the ocean surface.

329. Pan-Arctic plankton community structure and its global connectivity

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Open Access: N/A

Abstract: The Arctic Ocean (AO) is being rapidly transformed by global warming, but its biodiversity remains understudied for many planktonic organisms, in particular for unicellular eukaryotes that play pivotal roles in marine food webs and biogeochemical cycles. The aim of this study was to characterize the biogeographic ranges of species that comprise the contemporary pool of unicellular eukaryotes in the AO as a first step toward understanding mechanisms that structure these communities and identifying potential target species for monitoring. Leveraging the Tara Oceans DNA metabarcoding data, we mapped the global distributions of operational taxonomic units (OTUs) found on Arctic shelves into five biogeographic categories, identified biogeographic indicators, and inferred the degree to which AO communities of unicellular eukaryotes share members with assemblages from lower latitudes. Arctic/Polar indicator OTUs, as well as some globally ubiquitous OTUs, dominated the detection and abundance of DNA reads in the Arctic samples. OTUs detected only in Arctic samples (Arctic-exclusives) showed restricted distribution with relatively low abundances, accounting for 10-16% of the total Arctic OTU pool. OTUs with high abundances in tropical and/or temperate latitudes (non-Polar indicators) were also found in the AO but mainly at its periphery. We observed a large change in community taxonomic composition across the Atlantic-Arctic continuum, supporting the idea that advection and environmental filtering are important processes that shape plankton assemblages in the AO. Altogether, this study highlights the connectivity between the AO and other oceans, and provides a framework for monitoring and assessing future changes in this vulnerable ecosystem.

330. Eddy and tidal driven basal melting of the Totten and Moscow University ice shelves

Authors: Yuwei Xia; David E. Gwyther; Ben Galton-Fenzi; Eva A. Cougnon; Alexander D. Fraser; John C. Moore

Journal: Frontiers in Marine Science

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Language: N/A

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Open Access: N/A

Abstract: The mass loss from the neighboring Totten and Moscow University ice shelves is accelerating and may raise global sea levels in coming centuries. Totten Glacier is mostly based on bedrock below sea level, and so is vulnerable to warm water intrusion reducing its ice shelf buttressing. The mechanisms driving the ocean forced sub-ice-shelf melting remains to be further explored. In this study, we simulate oceanic-driven ice shelf melting of the Totten (TIS) and Moscow University ice shelves (MUIS) using a high spatiotemporal resolution model that resolves both eddy and tidal processes. We selected the year 2014 as representative of the period 1992 to 2017 to investigate how basal melting varies on spatial and temporal scales. We apply the wavelet coherence method to investigate the interactions between the two ice shelves in time-frequency space and hence estimate the contributions from tidal (<1.5 days) and eddy (2-35 days) components of the ocean heat transport to the basal melting of each ice shelf. In our simulation, the 2014 mean basal melt rate for TIS is 6.7 m yr⁻¹ (42 Gt yr⁻¹) and 9.7 m yr⁻¹ (52 Gt yr⁻¹) for MUIS. We find high wavelet coherence in the eddy dominated frequency band between the two ice shelves over almost the whole year. The wavelet coherence along five transects across the ice shelves suggests that TIS basal melting is dominated by eddy processes, while MUIS basal melting is dominated by tidal processes. The eddy-dominated basal melt for TIS is probably due to the large and convoluted bathymetric gradients beneath the ice shelf, weakening higher frequency tidal mode transport. This illustrates the key role of accurate bathymetric data plays in simulating on-going and future evolution of these important ice shelves.

331. Water mass transformation variability in the Weddell Sea in ocean reanalyses

Authors: Shanice T. Bailey; C. Spencer Jones; Ryan P. Abernathey; Arnold L. Gordon; Xiaojun Yuan

Journal: Ocean Science

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . This study investigates the variability of water mass transformation (WMT) within the Weddell Gyre (WG). The WG serves as a pivotal site for the Meridional Overturning Circulation (MOC) and ocean ventilation because it is the primary origin of the largest volume of water mass in the global ocean: Antarctic Bottom Water (AABW). Recent mooring data suggest substantial seasonal and interannual variability of AABW properties exiting the WG, and studies have linked the variability to the large-scale climate forcings affecting wind stress in the WG region. However, the specific thermodynamic mechanisms that link variability in surface forcings to variability in water mass transformations and AABW export remain unclear. This study explores how current state-of-the-art data-assimilating ocean reanalyses can help fill the gaps in our understanding of the thermodynamic drivers of AABW variability in the WG via WMT volume budgets derived from Walin's classic WMT framework. The three ocean reanalyses used are the following: Estimating the Circulation and Climate of the Ocean state estimate (ECCOv4), Southern Ocean State Estimate (SOSE) and Simple Ocean Data Assimilation (SODA). From the model outputs, we diagnose a closed form of the water mass budget for AABW that explicitly accounts for transport across the WG boundary, surface forcing, interior mixing and numerical mixing. We examine the annual mean climatology of the WMT budget terms, the seasonal climatology and finally the interannual variability. Our finding suggests that the relatively coarse resolution of these models did not realistically capture AABW formation, export and variability. In ECCO and SOSE, we see strong interannual variability in AABW volume budget. In SOSE, we find an accelerating loss of AABW during 2005-2010, driven largely by interior mixing and changes in surface salt fluxes. ECCO shows a similar trend during a 4-year time period starting in late 2007 but also reveals such trends to be part of interannual variability over a much longer time period. Overall, ECCO provides the most useful time series for understanding the processes and mechanisms that drive WMT and export variability in the WG. SODA, in contrast, displays unphysically large variability in AABW volume, which we attribute to its data assimilation scheme. We also examine correlations between the WMT budgets and large-scale climate indices, including El Niño-Southern Oscillation (ENSO) and Southern Annular Mode (SAM), and find no strong relationships.

332. Middepth Recipes

Authors: Mason Rogers (Massachusetts Institute of Technology, Woods Hole Oceanographic Institution); Raffaele Ferrari (Massachusetts Institute of Technology); Louis-Philippe Nadeau (c Institut des Sciences de la Mer de Rimouski, Rimouski, Quebec)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Subjects: N/A

Keywords: Large-scale motions, Diapycnal mixing, Meridional overturning circulation, Budgets, General circulation models

Funding: National Science Foundation, National Science Foundation, National Science Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: The Indo-Pacific Ocean appears exponentially stratified between 1- and 3-km depth with a decay scale on the order of 1 km. In his celebrated paper "Abyssal recipes," W. Munk proposed a theoretical explanation of these observations by suggesting a pointwise buoyancy balance between the upwelling of cold water and the downward diffusion of heat. Assuming a constant upwelling velocity w and turbulent diffusivity κ , the model yields an exponential stratification whose decay scale is consistent with observations if $\kappa \sim 10^{-4} \text{ m}^2 \text{ s}^{-1}$. Over time, much effort has been made to reconcile Munk's ideas with evidence of vertical variability in κ , but comparably little emphasis has been placed on the even stronger evidence that w decays toward the surface. In particular, the basin-averaged w nearly vanishes at 1-km depth in the Indo-Pacific. In light of this evidence, we consider a variable-coefficient, basin-averaged analog of Munk's budget, which we verify against a hierarchy of numerical models ranging from an idealized basin-and-channel configuration to a coarse global ocean simulation. Study of the budget reveals that the decay of basin-averaged w requires a concurrent decay in basin-averaged κ to produce an exponential-like stratification. As such, the frequently cited value of $10^{-4} \text{ m}^2 \text{ s}^{-1}$ is representative only of the bottom of the middepths, whereas κ must be much smaller above. The decay of mixing in the vertical is as important to the stratification as its magnitude. Significance Statement Using a combination of theory and numerical simulations, it is argued that the observed magnitude and shape of the global ocean stratification and overturning circulation appear to demand that turbulent mixing increases quasi-exponentially toward the ocean bottom. Climate models must therefore prescribe such a vertical profile of turbulent mixing in order to properly represent the heat and carbon uptake accomplished by the global overturning circulation on centennial and longer time scales.

333. Ocean-Ice Sheet Coupling in the Totten Glacier Area, East Antarctica: Analysis of the Feedbacks and Their Response to a Sudden Ocean Warming

Authors: Guillian Van Achter (Earth and Life Institute, UCLouvain, 1348 Louvain-la-Neuve, Belgium); Thierry Fichefet (Earth and Life Institute, UCLouvain, 1348 Louvain-la-Neuve, Belgium); Hugues Goosse (Earth and Life Institute, UCLouvain, 1348 Louvain-la-Neuve, Belgium); Charles Pelletier (European Centre for Medium-Range Weather Forecasts (ECMWF)); Konstanze Haubner (University of Bergen, Laboratoire de Glaciologie, Université libre de Bruxelles, 1050 Brussels, Belgium, Bjerknes Centre for Climate Research); Frank Pattyn (Laboratoire de Glaciologie, Université libre de Bruxelles, 1050 Brussels, Belgium)

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Keywords: N/A

Funding: Fonds De La Recherche Scientifique—FNRS

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Open Access: N/A

Abstract: We coupled together high-resolution versions of the ocean-sea ice model NEMO and the ice sheet model BISICLES configured to the Totten Glacier area and ran a series of simulations over the recent past (1995-2014) and under warming conditions (2081-2100; SSP4-4.5) with NEMO in stand-alone mode and with the coupled model to assess the effects of the coupling. During the recent past, the ocean-ice sheet coupling has increased the time-averaged value of the basal melt rate in both the Totten and Moscow University ice shelf cavities by 6.7% and 14.2%, respectively. The relationship between the changes in ice shelf thickness and ice shelf basal melt rate suggests that the effect of the coupling is not a linear response to the melt rate but rather a more complex response, driven partly by the dynamical component of the ice sheet model. The response of the ice sheet-ocean coupling due to the ocean warming is a 10% and 3% basal melt rate decrease in the Totten and Moscow University ice shelf cavities, respectively. This indicates that the ocean-ice sheet coupling under climate warming conditions dampens the basal melt rates. Our study highlights the importance of incorporating ocean-ice sheet coupling in climate simulations, even over short time periods.

334. Ocean drifter velocity data assimilation, Part 1: Formulation and diagnostic results

Authors: Robert W. Helber; Scott R. Smith; Gregg A. Jacobs; Charlie N. Barron; Matt J. Carrier; Clark D. Rowley; Hans E. Ngodock; Max Yaremchuk; Ivo Pasmans; Brent P. Bartels; Chris J. DeHaan

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Open Access: N/A

Abstract: N/A

335. Sources and Sinks of Interannual Steric Sea Level Variability

Authors: Antoine Hochet (Institut de Recherche pour le Développement (IRD)); William Llovel (Institut de Recherche pour le Développement (IRD)); Florian Sévellec (Institut de Recherche pour le Développement (IRD)); Thierry Huck (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

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Abstract: Understanding the mechanisms of regional steric sea level variability is fundamental to understand the regional sea level variability recorded by satellite altimetry for years and to insure that future projections made by climate models are realistic. Here, we first develop a novel method based on steric sea level variance budget that allows to detect the sources and sinks of the variability. Using the "Estimating the Circulation and Climate of the Ocean" (ECCO V4) state estimate, we then show that interannual steric sea level variability is mainly sustained by interannual fluctuating winds via Ekman transport almost everywhere. The damping of the variability is made by both the interannual fluctuating net heat flux from the atmosphere, that largely dominates the atmospheric freshwater fluxes, and the parametrized effect of eddies. It is also found that the parametrized effect of diffusion on the variability is weak in most regions and that, although globally weak, the fluctuations of atmospheric freshwater fluxes are a source of variance close to the Equator in the Pacific Ocean.

336. On the Multiscale Oceanic Heat Transports Toward the Bases of the Antarctic Ice Shelves

Authors: Zhaomin Wang (Southern Marine Science and Engineering Guangdong Laboratory); Chengyan Liu (Southern Marine Science and Engineering Guangdong Laboratory); Chen Cheng (Southern Marine Science and Engineering Guangdong Laboratory); Qing Qin (Hohai University, Ministry of Natural Resources); Liangjun Yan (Hohai University, Ministry of Natural Resources); Jiangchao Qian (Hohai University, Ministry of Natural Resources); Chong Sun (Hohai University, Ministry of Natural Resources); Li Zhang (Sun Yat-Sen University)

Journal: Ocean-Land-Atmosphere Research

Publisher: American Association for the Advancement of Science (AAAS)

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Abstract: The mass balance of the Antarctic Ice Sheet (AIS) is important to global sea-level change. The AIS loses mass mainly through basal melting and subsequent calving of the Antarctic ice shelves. However, the simulated basal melting rates are very uncertain in ice sheet models, partially resulting from the poor understanding of oceanic heat transports. In this article, we review the recent progress in understanding and simulating such heat transports. Regulated by major circulation features, Circumpolar Deep Water (CDW) is much closer to the Bellingshausen-Amundsen Seas and the Cooperation Sea (60 degreesE to 90 degreesE) and the sector further east to 160 degreesE. The ice shelves within these sectors are experiencing enhanced basal melting resulting from tropical forcing and intensified westerlies. Around West Antarctica, the isopycnal structure favors the delivery of CDW across slopes and shelves, while around East Antarctica, the persistent and strong westward Antarctic Slope Current (Front) acts to prevent warm-water intrusion. Both eddies and troughs favor heat transport to the fronts of the ice shelves and even into the cavities. The sharp contrast between the water column thicknesses on both sides of ice shelf fronts blocks the barotropic inflows and can excite topographic Rossby waves. Inside the cavities, the heat fluxes to the bases of the ice shelves are controlled by the cavity geometry, the circulations in the cavities, and the properties of the water masses beneath the ice shelves. Limited direct observations of cavities have promoted the development of various models. To improve basal melting simulations, meltwater plume models have been developed to study meltwater-laden mixed layer dynamics by increasing the vertical resolution, with recent advanced studies considering the vertical structures of frazil ice concentration and velocity. To reduce the uncertainties in the simulated and projected basal mass loss of the Antarctic ice shelves, future efforts should be devoted to improving the bathymetry and cavity geometry, investigating

small-scale processes and parameterizing these processes in coupled climate-ice sheet models, and quantifying the feedback from the mass loss of the AIS.

337. Submesoscale inverse energy cascade enhances Southern Ocean eddy heat transport

Authors: Zhiwei Zhang; Yuelin Liu; Bo Qiu; Yiyong Luo; Wenju Cai; Qingguo Yuan; Yinxing Liu; Hong Zhang; Hailong Liu; Mingfang Miao; Jinchao Zhang; Wei Zhao; Jiwei Tian

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

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Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: Oceanic eddy-induced meridional heat transport (EHT) is an important process in the Southern Ocean heat budget, the variability of which significantly modulates global meridional overturning circulation (MOC) and Antarctic sea-ice extent. Although it is recognized that mesoscale eddies with scales of ~40-300 km greatly contribute to the EHT, the role of submesoscale eddies with scales of ~1-40 km remains unclear. Here, using two state-of-the-art high-resolution simulations (resolutions of 1/48 degrees and 1/24 degrees), we find that submesoscale eddies significantly enhance the total poleward EHT in the Southern Ocean with an enhancement percentage reaching 19-48% in the Antarctic Circumpolar Current band. By comparing the eddy energy budgets between the two simulations, we detect that the primary role of submesoscale eddies is to strengthen mesoscale eddies (and thus their heat transport capability) through inverse energy cascade rather than directly through submesoscale heat fluxes. Due to the submesoscale-mediated enhancement of mesoscale eddies in the 1/48 degrees simulation, the clockwise upper cell and anti-clockwise lower cell of the residual-mean MOC in the Southern Ocean are weakened and strengthened, respectively. This finding identifies a potential route to improve the mesoscale parameterization in climate models for more accurate simulations of the MOC and sea ice variability in the Southern Ocean.

338. What drives the decadal variability in sea surface salinity and stratification over the tropical Indian Ocean?

Authors: Sandeep Mohapatra; C. Gnanaseelan; T. S. Fousiya; Sumit Dandapat

Journal: Theoretical and Applied Climatology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00704-023-04429-w

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Language: en

Subjects: N/A

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Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

339. North Atlantic Oscillation impact on the Atlantic Meridional Overturning Circulation shaped by the mean state

Authors: Hyo-Jeong Kim; Soon-Il An; Jae-Heung Park; Mi-Kyung Sung; Daehyun Kim; Yeonju Choi; Jin-Soo Kim

Journal: npj Climate and Atmospheric Science

Publisher: Springer Science and Business Media LLC

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Cited by: 13 publications

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Keywords: N/A

Funding: National Research Foundation of Korea

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Open Access: N/A

Abstract: Accurate representation of the Atlantic Meridional Overturning Circulation (AMOC) in global climate models is crucial for reliable future climate predictions and projections. In this study, we used 42 coupled atmosphere-ocean global climate models to analyze low-frequency variability of the AMOC driven by the North Atlantic Oscillation (NAO). Our results showed that the influence of the simulated NAO on the AMOC differs significantly between the models. We showed that the large intermodel diversity originates from the diverse oceanic mean state, especially over the subpolar North Atlantic (SPNA), where deep water formation of the AMOC occurs. For some models, the climatological sea ice extent covers a wide area of the SPNA and restrains efficient air-sea interactions, making the AMOC less sensitive to the NAO. In the models without the sea-ice-covered SPNA, the upper-ocean mean stratification critically affects the relationship between the NAO and AMOC by regulating the AMOC sensitivity to surface buoyancy forcing. Our results pinpoint the oceanic mean state as an aspect of climate model simulations that must be improved for an accurate understanding of the AMOC.

340. Parameterizing the nonlinear feedback on ENSO from tropical instability waves (TIWs) by nonlinear eddy thermal diffusivity

Authors: Aoyun Xue; Fei-Fei Jin; Wenjun Zhang; Julien Boucharel; Jong-Seong Kug

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

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Funding: National Science Foundation, United States, U.S. Department of Energy, National Natural Science Foundation of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

341. Sea level anomalies in the southeastern tropical Indian Ocean as a potential predictor of La Niña beyond one-year lead

Authors: Xia Zhao; Dongliang Yuan; Jing Wang

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1141961

URL: <https://doi.org/10.3389/fmars.2023.1141961>

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Volume/Issue: Volume 10

References: 81

Cited by: 1 publications

Language: N/A

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Open Access: N/A

Abstract: Most climate forecast agencies failed to make successful predictions of the strong 2020/2021 La Niña event before May 2020. The western equatorial Pacific warm water volume (WWV) before the 2020 spring failed to predict this La Niña event because of the near neutral state of the equatorial Pacific Ocean in the year before. A strong Indian Ocean Dipole (IOD) event took place in the fall of 2019, which is used as a precursor for the La Niña prediction in this study. We used observational data to construct the precursory relationship between negative sea level anomalies (SLA) in the southeastern tropical Indian Ocean (SETIO) in boreal fall and negative Niño 3.4 sea surface temperature anomalies index one year later. The application of the above relation to the prediction of the 2020/2021 La Niña was a great success. The dynamics behind are the Indo-Pacific "oceanic channel" connection via the Indian Ocean Kelvin wave propagation through the Indonesian seas, with the atmospheric bridge playing a secondary role. The high predictability of La Niña across the spring barrier if a positive IOD should occur in the previous year suggests that the negative SETIO SLA in fall is a much better and longer predictor for this type of La Niña prediction than the WWV. In comparison, positive SETIO SLA lead either El Niño or La Niña by one year, suggesting uncertainty of El Niño predictions.

342. Physical Oceanography: The Shift to a Global View and Its Changing Culture

Authors: Carl Wunsch (Harvard University, Massachusetts Institute of Technology)

Journal: Perspectives of Earth and Space Scientists

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022cn000204

URL: <https://doi.org/10.1029/2022cn000204>

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Language: en

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Abstract: Many changes took place in physical oceanography over the last 60 years, as the science encountered the global-scale problems of climate and in the availability of numerous new technologies. Changes have been in the known science, and in the intellectual culture in which it is pursued.

343. The Salinity Budget of the Ross Sea Continental Shelf, Antarctica

Authors: Liangjun Yan (Hohai University); Zhaomin Wang (Southern Marine Science and Engineering Guangdong Laboratory); Chengyan Liu (Southern Marine Science and Engineering Guangdong Laboratory); Yang Wu (School of Information Engineering Nanjing Xiaozhuang University Nanjing China); Qing Qin (Hohai University); Chong Sun (Hohai University); Jiangchao Qian (Hohai University); Li Zhang (Sun Yat-Sen University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jc018979

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Cited by: 5 publications

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Keywords: N/A

Funding: National Natural Science Foundation of China, Natural Science Foundation of Jiangsu Province

License: N/A

Open Access: N/A

Abstract: Toward understanding the trend and variability in the bottom water salinity in the Ross Sea, this study focuses on the seasonality of the salinity budget over the Ross Sea continental shelf, using a new coupled regional ocean-sea ice-ice shelf model. Owing to the sea ice production, the upstream advection, and the glacial melting, the Ross Sea continental shelf is characterized by the relatively saltier water on the western shelf than on the eastern shelf, with the saltiest water in the Terra Nova Bay Polynya. During the early freezing season (March-April), there is remarkable sea ice production over the broad continental shelf, which makes a significant contribution to the production of High Salinity Shelf Water (HSSW). The brine rejection in the polynyas leads to the salinification and deepening of the mixed layer, yet the upstream advection acts to decrease the salinity below the mixed layer until the deep convection is fully developed throughout the water column. The seasonal cycle of the salinity budget in the Ross Sea continental shelf is largely determined by the formation/melting of sea ice, the intrusion of modified Circumpolar Deep Water, and the outflow of HSSW, while the contribution of the Ross Ice Shelf melting is relatively small.

344. UKESM1.1: development and evaluation of an updated configuration of the UK Earth System Model

Authors: Jane P. Mulcahy; Colin G. Jones; Steven T. Rumbold; Till Kuhlbrodt; Andrea J. Dittus; Edward W. Blockley; Andrew Yool; Jeremy Walton; Catherine Hardacre; Timothy Andrews; Alejandro Bodas-Salcedo; Marc Stringer; Lee de Mora; Phil Harris; Richard Hill; Doug Kelley; Eddy Robertson; Yongming Tang

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-1569-2023

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Publication Date: 2023-03-20

Volume/Issue: Volume 16, Issue 6

References: 119

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Language: en

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Keywords: N/A

Funding: Department for Business, Energy and Industrial Strategy, UK Government, Natural Environment Research Council, Horizon 2020

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Open Access: N/A

Abstract: . Many Coupled Model Intercomparison Project phase 6 (CMIP6) models have exhibited a substantial cold bias in the global mean surface temperature (GMST) in the latter part of the 20th century. An overly strong negative aerosol forcing has been suggested as a leading contributor to this bias. An updated configuration of UK Earth System Model (UKESM) version 1, UKESM1.1, has been developed with the aim of reducing the historical cold bias in this model. Changes implemented include an improved representation of SO₂ dry deposition, along with several other smaller modifications to the aerosol scheme and a retuning of some uncertain parameters of the fully coupled Earth system model. The Diagnostic, Evaluation and Characterization of Klima (DECK) experiments, a six-member historical ensemble and a subset of future scenario simulations are completed. In addition, the total anthropogenic effective radiative forcing (ERF), its components and the effective and transient climate sensitivities are also computed. The UKESM1.1 preindustrial climate is warmer than UKESM1 by up to 0.75 K, and a significant improvement in the historical GMST record is simulated, with the magnitude of the cold bias reduced by over 50 %. The warmer climate increases ocean heat uptake in the Northern Hemisphere oceans and reduces Arctic sea ice, which is in better agreement with observations. Changes to the aerosol and related cloud properties are a driver of the improved GMST simulation despite only a modest reduction in the magnitude of the negative aerosol ERF (which increases by +0.08 W m⁻²). The total anthropogenic ERF increases from 1.76 W m⁻² in UKESM1 to 1.84 W m⁻² in UKESM1.1. The effective climate sensitivity (5.27 K) and transient climate response (2.64 K) remain largely unchanged from UKESM1 (5.36 and 2.76 K respectively).

345. Ocean carbon from space: Current status and priorities for the next decade

Authors: Robert J.W. Brewin; Shubha Sathyendranath; Gemma Kulk; Marie-Hélène Rio; Javier A. Concha; Thomas G. Bell; Astrid Bracher; Cédric Fichot; Thomas L. Frölicher; Martí Galí; Dennis Arthur Hansell; Tihomir S. Kostadinov; Catherine Mitchell; Aimee Renee Neeley; Emanuele Organelli; Katherine Richardson; Cécile Rousseaux; Fang Shen; Dariusz Stramski; Maria Tzortziou; Andrew J. Watson; Charles Izuma Addey; Marco Bellacicco; Heather Bouman; Dustin Carroll; Ivona Cetinić; Giorgio Dall'Olmo; Robert Frouin; Judith Hauck; Martin Hieronymi; Chuanmin Hu; Valeria Ibello; Bror Jönsson; Christina Eunjin Kong; Žarko Kovačević; Marko Laine; Jonathan Lauderdale; Samantha Lavender; Eleni Livanou; Joan Llort; Larisa Lorinczi; Michael Nowicki; Novia Arinda Pradisty; Stella Psarra; Dionysios E. Raitsos; Ana Belén Ruescas; Joellen L. Russell; Joe Salisbury; Richard Sanders; Jamie D. Shutler; Xuerong Sun; Fernando González Taboada; Gavin H. Tilstone; Xinyuan Wei; David K. Woolf

Journal: Earth-Science Reviews

Publisher: Elsevier BV

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References: 541

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

346. Effects of including the adjoint sea ice rheology on estimating Arctic Ocean-sea ice state

Authors: Guokun Lyu; Armin Koehl; Xinrong Wu; Meng Zhou; Detlef Stammer

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-19-305-2023

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Volume/Issue: Volume 19, Issue 2

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Keywords: N/A

Funding: Ministry of Natural Resources of the People's Republic of China

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Abstract: . The adjoint assimilation method has been applied to coupled ocean and sea ice models for sensitivity studies and Arctic state estimations. However, the accuracy of the adjoint model is degraded by simplifications of the adjoint of the sea ice model, especially the adjoint sea ice rheologies. As part of ongoing developments in coupled ocean and sea ice estimation systems, we incorporate and approximate the adjoint of viscous-plastic sea ice dynamics (adjoint-VP) and compare it with the adjoint of free-drift sea ice dynamics (adjoint-FD) through assimilation experiments. Using the adjoint-VP results in a further cost reduction of 7.9 % in comparison to adjoint-FD, with noticeable improvements in the ocean temperature over the open water and the intermediate layers of the Arctic Ocean. Adjoint-VP adjusts the model input more efficiently than adjoint-FD does by involving different sea ice retreat processes. For instance, adjoint-FD melts the sea ice up to 1.0 m in the marginal seas from May to June by overadjusting air temperature (>8 K); adjoint-VP reproduces the sea ice retreat with smaller adjustments to the atmospheric state within their prior uncertainty range. These developments of the adjoint model here lay the foundation for further improving Arctic Ocean and sea ice estimations by comprehensively adjusting the initial conditions, atmospheric forcings, and parameters of the model.

347. An updated global mercury budget from a coupled atmosphere-land-ocean model: 40% more re-emissions buffer the effect of primary emission reductions

Authors: Yanxu Zhang; Peng Zhang; Zhengcheng Song; Shaojian Huang; Tengfei Yuan; Peipei Wu; Viral Shah; Maodian Liu; Long Chen; Xuejun Wang; Jun Zhou; Yannick Agnan

Journal: One Earth

Publisher: Elsevier BV

DOI: 10.1016/j.oneear.2023.02.004

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References: 66

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: N/A

348. On Dynamical Decomposition of Multiscale Oceanic Motions

Authors: Chuanyin Wang (Xiamen University); Zhiyu Liu (Xiamen University); Hongyang Lin (Xiamen University)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022ms003556

URL: <https://doi.org/10.1029/2022ms003556>

Publication Date: 2023-03-17

Volume/Issue: Volume 15, Issue 3

References: 75

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: internal gravity waves, submesoscale flows, mesoscale flows, dynamical decomposition

Funding: National Natural Science Foundation of China, Natural Science Foundation of Fujian Province

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Open Access: N/A

Abstract: The ocean hosts a variety of fluid motions characterized by contrasting dynamical regimes and spatiotemporal scales. A practical decomposition of multiscale motions in realistic oceanic settings is crucial to advancing dynamical interpretation and prediction of oceanic processes, but remains a major challenge. To this end, methodology is developed in this study for decomposing multiscale oceanic motions based on their respective dynamical characteristics. Specifically, large-scale currents and barotropic tides have the largest horizontal scales but contrasting frequencies; low-mode internal gravity waves (IGWs) are well constrained by linear dispersion relations, whereas mesoscale flows are of relatively low frequency and with horizontal scales above the first baroclinic deformation radius; the intrinsic frequency of high-mode IGWs (submesoscale flows) is above (well below) the inertial frequency. The validity and usefulness of the proposed methodology are demonstrated with a proof-of-concept application to simulated flows in the central basin of the South China Sea.

349. A global gridded ocean salinity dataset with 0.5 degrees horizontal resolution since 1960 for the upper 2000 m

Authors: Guancheng Li; Lijing Cheng; Yuying Pan; Gongjie Wang; Hailong Liu; Jiang Zhu; Bin Zhang; Huanping Ren; Xutao Wang

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1108919

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Language: N/A

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Keywords: N/A

Funding: Institute of Atmospheric Physics, Chinese Academy of Sciences

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Open Access: N/A

Abstract: A gridded salinity dataset with high resolution is essential for investigating global ocean salinity variability and understanding its role in climate and the ocean ecosystem. In this study, a new version of the Institute of Atmospheric Physics gridded salinity dataset with a higher resolution (0.5 degrees by 0.5 degrees) is provided by using a revised ensemble optimal interpolation scheme with a dynamic ensemble. The performance of this dataset is evaluated using "subsample test" and the high-resolution satellite-based data. Compared with the previous 1 degrees by 1 degrees resolution IAP product, the new dataset is more capable of representing regional salinity changes with the meso-scale and small-scale signals (i.e., in the coastal and boundary currents regions), meanwhile, maintains the large-scale structure and variability. Therefore, the new dataset complements the previous data product. Besides, the new dataset is compared with in situ observations and several international salinity products for the salinity multiscale variabilities and patterns. The comparison shows the smaller magnitude of mean difference and Root-mean-square deviation (RMSD) in basin scale for the new dataset, some differences in strength and fine structure of the "fresh gets fresher, salty gets saltier" surface and subsurface salinity pattern amplification trends from 1980 to 2017, a broad similarity for the salinity changes associated with El Niño-Southern Oscillation (ENSO) and a consistent salinity dipole mode in the tropical Indian Ocean (S-IOD). These results support the future use of gridded salinity data.

350. Role of Ocean Advections during the Evolution of the Pacific Meridional Modes

Authors: Qi Shu (Ocean University of China, Laoshan Laboratory); Yu Zhang (Ocean University of China, Laoshan Laboratory); Dillon J. Amaya (c NOAA/Earth System Research Laboratory, Physical Science Laboratory, Boulder, Colorado); Sarah M. Larson (North Carolina State University); Yu Kosaka (University of Tokyo); Jun-Chao Yang (Ocean University of China, Laoshan Laboratory); Xiaopei Lin (Ocean University of China, Laoshan Laboratory)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-22-0296.1

URL: <https://doi.org/10.1175/jcli-d-22-0296.1>

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Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Mixed layer|Ocean observations

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, National Natural Science Foundation of China (and 11 more)

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Open Access: N/A

Abstract: The Pacific meridional modes (PMMs) are the leading ocean-atmosphere coupled modes in the subtropical northeastern (NPMM) and southeastern (SPMM) Pacific, respectively, and have been suggested to be key precursors to equatorial Pacific variability. Previous studies pointed out that both NPMM- and SPMM-related sea surface temperature (SST) anomalies are primarily driven by net surface heat flux variations during their equatorward evolution. However, whether oceanic heat advective processes would play a role during the evolution remains unclear. To address this issue, we perform an ocean mixed layer heat budget analysis based on observations and three ocean reanalysis datasets, and then reveal the effect of ocean advections on the evolution by comparing a fully coupled dynamic ocean model (DOM) to a slab ocean model (SOM). Our results suggest that for the NPMM evolution, ocean advections--primarily by anomalous meridional Ekman heat advections driven by mean and anomalous zonal wind stresses--play a damping role in the south of the NPMM. Thus, the NPMM SST anomalies appear to instead exhibit a poleward shift, although still freely propagating westward from the preceding boreal winter to the following summer. This finding challenges the traditional view that the NPMM propagates equatorward through the wind-evaporation-SST feedback. For the SPMM evolution, ocean advections play a damping role in the center of the SPMM from boreal spring to summer, as well as an intensification role in the southwest Pacific during summer. However, the effect of the intensification on the SPMM evolution is hard to determine due to the strong simulation bias of the SPMM evolution in the DOM. Significance Statement While it is known that both NPMM- and SPMM-associated SST anomalies are primarily driven by net surface heat flux variations during their evolution, whether ocean advections would play a role remains unknown. Here, we show that ocean advections play a role in the evolution of both PMMs. In particular, for the NPMM evolution, ocean

advections play a damping role in the south of the NPMM center, causing a tendency for the NPMM to be displaced northward. The role of ocean advection challenges the prevailing notion that the NPMM simply evolves equatorward through the wind-evaporation-SST feedback.

351. Using satellite observations of ocean variables to improve estimates of water mass (trans)formation

Authors: Aqeel Piracha; Estrella Olmedo; Antonio Turiel; Marcos Portabella; Cristina González-Haro

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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References: 44

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Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: For the first time, an accurate and complete picture of Mixed Layer (ML) water mass dynamics can be inferred at high spatio-temporal resolution via the material derivative derived from Sea Surface Salinity/Temperature (SSS/T) and Currents (SSC). The product between this satellite derived material derivative and in-situ derived Mixed Layer Depth (MLD) provides a satellite based kinematic approach to the water mass (trans)formation framework (WMT/F) above ML. We compare this approach to the standard thermodynamic approach based on air-sea fluxes provided by satellites, an ocean state estimate and in-situ observations. Southern Hemisphere surface density flux and water mass (trans)formation framework (WMT/F) were analysed in geographic and potential density space for the year 2014. Surface density flux differences between the satellite derived thermodynamic and kinematic approaches and ECCO (an ocean state estimate) underline: 1) air-sea heat fluxes dominate variability in the thermodynamic approach; and 2) fine scale structures from the satellite derived kinematic approach are most likely geophysical and not artefacts from noise in SSS/T or SSC--as suggested by a series of smoothing experiments. Additionally, ECCO revealed surface density flux integrated over ML are positively biased as compared to similar estimates assuming that surface conditions are homogeneous over ML--in part owing to the e-folding nature of shortwave solar radiation. Major differences between the satellite derived kinematic and thermodynamic approaches are associated to: 1) lateral mixing and mesoscale dynamics in the kinematic framework; 2) vertical excursions of, and vertical velocities through the ML base; and 3) interactions between ML horizontal velocities and ML base spatial gradients.

352. Data initiatives for ocean-driven melt of Antarctic ice shelves

Authors: Sue Cook; Keith W. Nicholls; Irena Vanková; Sarah S. Thompson; Benjamin K. Galton-Fenzi

Journal: Annals of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/aog.2023.6

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Abstract: Ocean-driven melt of Antarctic ice shelves is an important control on mass loss from the ice sheet, but is complex to study due to significant variability in melt rates both spatially and temporally. Here we assess the strengths and weakness of satellite and field-based observations as tools for testing models of ice-shelf melt. We discuss how the complementary use of field, satellite and model data can be a powerful but underutilised tool for studying melt processes. Finally, we identify some community initiatives working to collate and publish coordinated melt rate datasets, which can be used in future for validating satellite-derived maps of melt and evaluating processes in numerical simulations.

353. Contemporary oceanic radiocarbon response to ocean circulation changes

Authors: Byju Pookkandy; Heather Graven; Adrian Martin

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-023-06737-3

URL: <https://doi.org/10.1007/s00382-023-06737-3>

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License: CC BY 4.0

Open Access: N/A

Abstract: Radiocarbon (^{14}C) is a valuable tracer of ocean circulation, owing to its natural decay over thousands of years and to its perturbation by nuclear weapons testing in the 1950s and 1960s. Previous studies have used ^{14}C to evaluate models or to investigate past climate change. However, the relationship between ocean ^{14}C and ocean circulation changes over the past few decades has not been explored. Here we use an Ocean-Sea-ice model (NEMO) forced with transient or fixed atmospheric reanalysis (JRA-55-do) and atmospheric ^{14}C and CO_2 boundary conditions to investigate the effect of ocean circulation trends and variability on ^{14}C . We find that $^{14}\text{C}/\text{C}$ ($\Delta^{14}\text{C}$) variability is generally anti-correlated with potential density variability. The areas where the largest variability occurs varies by depth: in upwelling regions at the surface, at the edges of the subtropical gyres at 300 m depth, and in Antarctic Intermediate Water and North Atlantic Deep Water at 1000 m depth. We find that trends in the Atlantic Meridional Overturning Circulation may influence trends in $\Delta^{14}\text{C}$ in the North Atlantic. In the high-variability regions the simulated variations are larger than typical ocean $\Delta^{14}\text{C}$ measurement uncertainty of 2-5‰ suggesting that $\Delta^{14}\text{C}$ data could provide a useful tracer of circulation changes.

354. Significance of Diapycnal Mixing Within the Atlantic Meridional Overturning Circulation

Authors: Laura Cimoli (Scripps Institution of Oceanography / University of California San Diego, University of Cambridge, University of Oxford); Ali Mashayek (University of Cambridge); Helen L. Johnson (University of Oxford); David P. Marshall (University of Oxford); Alberto C. Naveira Garabato (University of Southampton); Caitlin B. Whalen (University of Washington); Clément Vic (Institut de Recherche pour le Développement (IRD)); Casimir de Lavergne (Institut de Recherche pour le Développement (IRD)); Matthew H. Alford (Scripps Institution of Oceanography / University of California San Diego); Jennifer A. MacKinnon (Scripps Institution of Oceanography / University of California San Diego); Lynne D. Talley (Scripps Institution of Oceanography / University of California San Diego)

Journal: AGU Advances

Publisher: American Geophysical Union (AGU)

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Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: tracer ventilation, Atlantic Ocean, diapycnal mixing

Funding: Natural Environment Research Council, Division of Ocean Sciences

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Open Access: N/A

Abstract: Diapycnal mixing shapes the distribution of climatically important tracers, such as heat and carbon, as these are carried by dense water masses in the ocean interior. Here, we analyze a suite of observation-based estimates of diapycnal mixing to assess its role within the Atlantic Meridional Overturning Circulation (AMOC). The rate of water mass transformation in the Atlantic Ocean's interior shows that there is a robust buoyancy increase in the North Atlantic Deep Water (NADW, neutral density σ_{θ} 27.6-28.15), with a diapycnal circulation of 0.5-8 Sv between 48 degreesN and 32 degreesS in the Atlantic Ocean. Moreover, tracers within the southward-flowing NADW may undergo a substantial diapycnal transfer, equivalent to a vertical displacement of hundreds of meters in the vertical. This result, confirmed with a zonally averaged numerical model of the AMOC, indicates that mixing can alter where tracers upwell in the Southern Ocean, ultimately affecting their global pathways and ventilation timescales. These results point to the need for a realistic mixing representation in climate models in order to understand and credibly project the ongoing climate change.

355. Likely accelerated weakening of Atlantic overturning circulation emerges in optimal salinity fingerprint

Authors: Chenyu Zhu; Zhengyu Liu; Shaoqing Zhang; Lixin Wu

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-023-36288-4

URL: <https://doi.org/10.1038/s41467-023-36288-4>

Publication Date: 2023-03-04

Volume/Issue: Volume 14, Issue 1

References: 68

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: The long-term response of the Atlantic meridional overturning circulation (AMOC) to anthropogenic forcing has been difficult to detect from the short direct measurements available due to strong interdecadal variability. Here, we present observational and modeling evidence for a likely accelerated weakening of the AMOC since the 1980s under the combined forcing of anthropogenic greenhouse gases and aerosols. This likely accelerated AMOC weakening signal can be detected in the AMOC fingerprint of salinity pileup remotely in the South Atlantic, but not in the classic warming hole fingerprint locally in the North Atlantic, because the latter is contaminated by the "noise" of interdecadal variability. Our optimal salinity fingerprint retains much of the signal of the long-term AMOC trend response to anthropogenic forcing, while dynamically filtering out shorter climate variability. Given the ongoing anthropogenic forcing, our study indicates a potential further acceleration of AMOC weakening with associated climate impacts in the coming decades.

356. The dynamic mechanism of sea level variations in the Bohai Sea and Yellow Sea

Authors: Heng Liu; Xuhua Cheng; Jianhuang Qin; Guidi Zhou; Long Jiang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-023-06724-8

URL: <https://doi.org/10.1007/s00382-023-06724-8>

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Cited by: 4 publications

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Keywords: N/A

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Abstract: N/A

357. Optimized sea ice simulation in MITgcm-ECCO2 forced by ERA5

Authors: Estanislao Gavilan Pascual-Ahuir; Zhaomin Wang

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2023.102183

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

358. Estimating C_{30} coefficients for GRACE/GRACE-FO time-variable gravity field models using the GRACE-OBP approach

Authors: Yu Sun; Yang Li; Xiang Guo; Jinyun Guo

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-023-01707-3

URL: <https://doi.org/10.1007/s00190-023-01707-3>

Publication Date: 2023-03-02

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References: 36

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

359. A Simplified Ocean Physics? Revisiting Abyssal Recipes

Authors: Carl Wunsch (Harvard University, Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0229.1

URL: <https://doi.org/10.1175/jpo-d-22-0229.1>

Publication Date: 2023-03-02

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References: 37

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Large-scale motions, Abyssal zone|Temperature salinity diagrams|Potential temperature, Diapycnal mixing, Ocean circulation, Ocean dynamics

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Simplified descriptions of the ocean are useful both for formulating explanatory theories and for conveying meaningful global attributes. Here, using a 26-yr average of a global state estimate from ECCO, the basis for Munk's "abyssal recipes" is evaluated on a global scale between 1000- and 3000-m depth. The two specific hydrographic stations he used prove untypical, with potential temperature and salinity more generally displaying different vertical scale heights, and thus differing in one-dimensional (in the vertical) values of mixing coefficients and/or vertical velocities. The simplest explanation is that the circulation is fully three-dimensional with temperature and salinity fields not describable with a one-dimensional steady balance. In contrast, the potential density and buoyancy are quantitatively describable through a one-dimensional exponential balance, and which calls for an explanation in terms of turbulent mixing processes.

360. Are Ocean Reanalyses Useful for Earth Rotation Research?

Authors: L. Börger (University of Bonn); M. Schindelegger (University of Bonn); H. Dobslaw (Helmholtz Centre Potsdam); D. Salstein (Atmospheric and Environmental Research Inc.)

Journal: Earth and Space Science

Publisher: American Geophysical Union (AGU)

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Publication Date: 2023-02-28

Volume/Issue: Volume 10, Issue 3

References: 67

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: Earth rotation, data assimilation, ocean angular momentum, ocean reanalysis

Funding: Deutsche Forschungsgemeinschaft

License: CC BY 4.0

Open Access: N/A

Abstract: Oceanic circulation and mass-field variability play important roles in exciting Earth's wobbles and length-of-day changes ($\Delta\Lambda$), on time scales from days to several years. Modern descriptions of these effects employ oceanic angular momentum (OAM) series from numerical forward models or ocean state estimates, but nothing is known about how ocean reanalyses with sequential data assimilation (DA) would fare in that context. Here, we compute daily OAM series from three $1/4$ degrees global ocean reanalyses that are based on the same hydrodynamic core and input data (e.g., altimetry, Argo) but different DA schemes. Comparisons are carried out (a) among the reanalyses, (b) with an established ocean state estimate, and (c) with Earth rotation data, all focusing on the period 2006-2015. The reanalyses generally provide credible OAM estimates across a range of frequencies, although differences in amplitude spectra indicate a sensitivity to the adopted DA scheme. For periods less than 120 days, the reanalysis-based OAM series explain ~40%-50% and ~30%-40% of the atmosphere-corrected equatorial and axial geodetic excitation, similar to what is achieved with the state estimate. We find mixed performance of the reanalyses in seasonal excitation budgets, with some questionable mean ocean mass changes affecting the annual cycle in $\Delta\Lambda$. Modeled excitations at interannual frequencies are more uncertain compared to OAM series from the state estimate and show hints of DA artifacts in one case. If users are to choose any of the tested reanalyses for rotation research, our study points to the Ocean Reanalysis System 5 as the most sensible choice.

361. Equatorial Submesoscale Eddies Contribute to the Asymmetry in ENSO Amplitude

Authors: Chengcheng Wang (Ocean University of China); Shengpeng Wang (Ocean University of China, Laoshan Laboratory); Zhao Jing (Ocean University of China, Laoshan Laboratory); Tao Geng (Laoshan Laboratory); Hong Wang (Ocean University of China, Laoshan Laboratory); Lixin Wu (Ocean University of China, Laoshan Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: ENSO asymmetry, vertical heat flux, submesoscale eddy

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: El Niño/Southern Oscillation (ENSO) exhibits evident amplitude asymmetry with stronger El Niño than La Niña events. Equatorial submesoscale eddies act as an important damper of ENSO via their induced upward heat flux ($\bar{Q}$) from the subsurface to the surface ocean. Yet their effect on ENSO amplitude asymmetry remains unexplored. Using a high-resolution global climate simulation, we found that strengthening of $\bar{Q}$ in the Niño3.4 region during La Niña is more evident than weakening of $\bar{Q}$ during El Niño, resulting in stronger damping of La Niña than El Niño events. This asymmetric response of $\bar{Q}$ to ENSO is primarily ascribed to that of the temperature fronts associated with larger-scale processes. Using a recharge oscillator model modified to include the effects of $\bar{Q}$, we show that the asymmetric damping of ENSO by $\bar{Q}$ contributes importantly to the ENSO amplitude asymmetry.

362. Geodetic residual time series: A combined series by minimization of their internal noise level

Authors: Małgorzata Wińska

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2023.1138410

URL: <https://doi.org/10.3389/feart.2023.1138410>

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Volume/Issue: Volume 11

References: 73

Cited by: N/A publications

Language: N/A

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: This study aims to assess the hydrological effects of polar motion calculated from different combinations of geophysical excitations at decadal, seasonal, and non-seasonal periods. The geodetic residuals GAO, being a difference between observed geodetic excitation function of polar motion Geodetic Angular Momentum (GAM) and atmospheric (Atmospheric Angular Momentum--AAM) plus oceanic excitation functions (Oceanic Angular Momentum--OAM), are compared. Estimating hydrological effects on Earth's rotation differs significantly, especially when using various oceanic models. Up to now, studies of geophysical excitations of polar motion containing AAM, OAM, and hydrological angular momentum (HAM) have not achieved entire agreement between geophysical (sum of AAM, OAM, and HAM obtained from the models) and geodetic excitation. Many geophysical models of the atmosphere, oceans, and land hydrology can be used to compute polar motion excitation. However, these models are very complex and still have uncertainties in the process descriptions, parametrization, and forcing. This work aims to show differences between various GAO solutions calculated from different mass and motion terms of various AAM and OAM models. Justifying to use one combination of GAO to estimate geodetic residuals is comparing those time series to hydrological excitations computed from Gravity Recovery and Climate Experiment (GRACE) data and the Land Surface Discharge Model (LSDM) model. Especially the quality of each geodetic residual time series is determined by estimating their noise level using a generalized formulation of the "three-cornered hat method" (3CH). This study presents a combined series of geodetic residuals GAO in polar motion (PM), wherein the internal noise level is shortened to a minimum by using the 3CH method. The combined GAO time series are compared with results obtained from the GRACE/GRACE Follow-On (GRACE-FO) solution provided by International Combination Service for Time variable Gravity Fields (COST-G) and the single solution elaborated by the Center for Space Research (CSR) and from the HAM LSDM hydrological model. The results show that higher consistency between GAO and HAM excitations can be achieved by minimizing the internal noise level in the GAO combined excitation series using the 3CH method, especially for the overall broadband and seasonal oscillations. For seasonal spectral bands, an agreement between combined GAO and the best-correlated series of

GRACE CSR achieve correlations as high as 0.97 and 0.83 for the χ_1 and χ_2 equatorial components of PM excitation, respectively. This study's combined geodetic residual time series slightly improved consistency between observed geodetic polar motion excitations and geophysical ones.

363. Ecological niche modelling of a critically endangered species *Commiphora wightii* (Arn.) Bhandari using bioclimatic and non-bioclimatic variables

Authors: Manish Mathur; Preet Mathur; Harshit Purohit

Journal: Ecological Processes

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s13717-023-00423-2

URL: <https://doi.org/10.1186/s13717-023-00423-2>

Publication Date: 2023-02-27

Volume/Issue: Volume 12, Issue 1

References: 136

Cited by: 23 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: BackgroundThe aim of this study is to examine the effects of four different bioclimatic predictors (current, 2050, 2070, and 2090 under Shared Socioeconomic Pathways SSP2-4.5) and non-bioclimatic variables (soil, habitat heterogeneity index, land use, slope, and aspect) on the habitat suitability and niche dimensions of the critically endangered plant species *Commiphora wightii* in India. We also evaluate how niche modelling affects its extent of occurrence (EOO) and area of occupancy (AOO). ResultsThe area under the receiver operating curve (AUC) values produced by the maximum entropy (Maxent) under various bioclimatic time frames were more than 0.94, indicating excellent model accuracy. Non-bioclimatic characteristics, with the exception of terrain slope and aspect, decreased the accuracy of our model. Additionally, Maxent accuracy was the lowest across all combinations of bioclimatic and non-bioclimatic variables (AUC = 0.75 to 0.78). With current, 2050, and 2070 bioclimatic projections, our modelling revealed the significance of water availability parameters (BC-12 to BC-19, i.e. annual and seasonal precipitation as well as precipitation of wettest, driest, and coldest months and quarters) on habitat suitability for this species. However, with 2090 projection, energy variables such as mean temperature of wettest quarter (BC-8) and isothermality (BC-3) were identified as governing factors. Excessive salt, rooting conditions, land use type (grassland), characteristics of the plant community, and slope were also noticed to have an impact on this species. Through distribution modelling of this species in both its native (western India) and exotic (North-east, Central Part of India, as well as northern and eastern Ghat) habitats, we were also able to simulate both its fundamental niche and its realized niche. Our EOO and AOO analysis reflects the possibility of many new areas in India where this species can be planted and grown. ConclusionAccording to the calculated area under the various suitability classes, we can conclude that *C. wightii*'s potentially suitable bioclimatic distribution under the optimum and moderate classes would increase under all future bioclimatic scenarios (2090 > 2050 ≈ current), with the exception of 2070, demonstrating that there are more suitable habitats available for *C. wightii* artificial cultivation and will be available for future bioclimatic projections of 2050 and 2090. Predictive sites indicated that this species also favours various types of landforms outside rocky environments, such as sand dunes, sandy plains, young

alluvial plains, saline areas, and so on. Our research also revealed crucial information regarding the community dispersion variable, notably the coefficient of variation that, when bioclimatic + non-bioclimatic variables were coupled, disguised the effects of bioclimatic factors across all time frames.

364. A Deep Seamount Effect Enhanced the Vertical Connectivity of the Planktonic Community Across 1,000 m Above Summit

Authors: Rongjie Zhao (Chinese Academy of Sciences); Feng Zhao (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Ling Feng (Chinese Academy of Sciences); James Kar-Hei Fang (State Key Laboratory of Marine Pollution City University of Hong Kong Kowloon Tong China, Department of Applied Biology and Chemical Technology The Hong Kong Polytechnic University Hung Hom China); Chuanyu Liu (Chinese Academy of Sciences); Kuidong Xu (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: cooccurrence networks, protists, genetic diversity, seamount effects, community connectivity (and 1 more)

Funding: National Natural Science Foundation of China, Youth Innovation Promotion Association

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Open Access: N/A

Abstract: Seamount effects, which are generally defined as hydrographic disturbances caused by topography and nutrient enrichment and biological aggregations around seamounts, are normally observed in shallow seamounts due to limited sampling efforts in deep seamounts. However, it remains unclear how and to what extent do deep seamounts leave their imprint on planktonic communities. Herein ciliates, a representative protist group, were chosen to explore the effect of deep seamount on planktonic community. By investigating the vertical and horizontal distribution of ciliate communities around the Kocebu Guyot (summit at -1,198 m) and in nonseamount area, we revealed an obvious deep seamount effect, which enhanced the vertical mixing of ciliate communities to an extent of over 1,000 m above the summit. The vertical mixing was manifested by a strong uplift of bottom dwellers from waters deeper than 500 m and a weak uplift from the 300 m layer to the deep chlorophyll maximum (about 150 m) layer. Network analysis showed that the ciliate cooccurrence relationship around the seamount was much more complex than that in nonseamount area. Statistical analysis indicated that seamount significantly weakened the limitation that water depth posed on vertical ciliate distribution. Overall, the ciliate communities presented a much higher-resolution record of deep seamount effects than physico-chemical data. Deep seamount could enhance the vertical mixing of waters and cooccurrence complexity of planktonic community to the euphotic layer. Considering the wide existence of deep seamounts, such an effect may have ecological significance and enhance the cycles of matter and energy of global oceans.

365. Evidence of earthquake seasonality in the Azores Triple Junction

Authors: Ana L. Lordi; Maria C. Neves; Susana Custódio; Stéphanie Dumont

Journal: Geo-Marine Letters

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00367-023-00744-3

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References: 34

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fundação para a Ciência e a Tecnologia, Universidade de Lisboa

License: CC BY 4.0

Open Access: N/A

Abstract: This work presents evidence of seasonal and inter-annual variations of the earthquake occurrence rate in the Azores Triple Junction, on the Mid-Atlantic Ridge (MAR). Annual cycles in microearthquakes are relatively common in intraplate continental regions affected by large hydrological loads, but this is the first time that earthquake seasonality is recognized near mid-ocean ridges. First, we benchmark the methodology by matching the published results of earthquake seasonality in the intraplate New Madrid Seismic Zone (USA). Next, we analyze the Azores earthquake catalogue, from 2008 to 2018, separately for oceanic and island regions. The results demonstrate that the seasonal modulation of the seismicity rate is only observed in the ocean, especially in the vicinity of the triple junction, with more earthquakes occurring during the summer months from May to August. Monte Carlo simulations show that the probability of observing such seasonality by chance is less than 1% for the magnitude band from 3.3 to 4.5, well above the detection threshold and magnitude of completeness of the seismic catalogue. The methodology includes a Jack-Knife approach, which shows that the oceanic seasonality is not the consequence of abnormal or extreme events. Although we speculate about possible earthquake triggering processes, it remains a challenge to definitely establish the mechanism responsible for the observed earthquake seasonal modulation in the Azores.

366. Matchup Strategies for Satellite Sea Surface Salinity Validation

Authors: Elizabeth E. Westbrook (University of North Carolina Wilmington); Frederick M. Bingham (University of North Carolina Wilmington); Severine Fournier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Akiko Hayashi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs15051242

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References: 30

Cited by: 3 publications

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Open Access: N/A

Abstract: Satellite validation is the process of comparing satellite measurements with in-situ measurements to ensure their accuracy. Satellite and in-situ sea surface salinity (SSS) measurements are different due to instrumental errors (IE), retrieval errors (RE), and representation differences (RD). In real-world data, IE, RE, and RD are inseparable, but validations seek to quantify only instrumental and retrieval error. Our goal is to determine which of four methods comparing in-situ and satellite measurements minimizes RD most effectively, which includes differences due to mismatches in the location and timing of the measurement, as well as representation error caused by the averaging of satellite measurements over a footprint. IE and RE were obviated by using simulated Argo float, and L2 NASA/SAC-D Aquarius, NASA-SMAP, and ESA-SMOS data generated from the high-resolution ECCO (Estimating the Climate and Circulation of the Oceans) model SSS data. The methods tested include the all-salinity difference averaging method (ASD), the N closest method (NCLO), which is an averaging method that is optimized for different satellites and regions of the ocean, and two single salinity difference methods--closest in space (SSDS) and closest in time (SSDT). The root mean square differences (RMSD) between the simulated in-situ and satellite measurements in seven regions of the ocean are used as a measure of the effectiveness of each method. The optimization of NCLO is examined to determine how the optimum matchup strategy changes depending on satellite track and region. We find that the NCLO method marginally produces the lowest RMSD in all regions but invoking a regionally optimized method is far more computationally expensive than the other methods. We find that averaging methods smooth IE, thus perhaps misleadingly lowering the detected instrumental error in the L2 product by as much as 0.15 PSU. It is apparent from our results that the dynamics of a particular region have more of an effect on matchup success than the method used. We recommend the SSDT validation strategy because it is more computationally efficient than NCLO, considers the proximity of in-situ and satellite measurements in both time and space, does not smooth instrumental

errors with averaging, and generally produces RMSD values only slightly higher than the optimized NCLO method.

367. A Comprehensive Evaluation of GRACE-Like Terrestrial Water Storage (TWS) Reconstruction Products at an Interannual Scale During 1981-2019

Authors: Shanshan Deng (Guangxi Laboratory on the Study of Coral Reefs in the South China Sea Coral Reef Research Center of China School of Marine Sciences Guangxi University Nanning China); Yuxin Liu (Guangxi Laboratory on the Study of Coral Reefs in the South China Sea Coral Reef Research Center of China School of Marine Sciences Guangxi University Nanning China); Wenxi Zhang (Guangxi Laboratory on the Study of Coral Reefs in the South China Sea Coral Reef Research Center of China School of Marine Sciences Guangxi University Nanning China)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

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References: 59

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Given the success of the Gravity Recovery and Climate Experiment (GRACE) mission in mapping terrestrial water storage (TWS) since 2002, recent reconstructions of long-term TWS rely on the use of statistical machine learning to apply GRACE-derived information to past decades. Evaluating the interannual accuracy during nonobservational periods is a key challenge. This study develops a comprehensive framework to discuss the interannual accuracy of three different TWS reconstructions during 1981-2019, including (a) global-scale accuracy assessment using GRACE and satellite laser ranging data; (b) regional-scale accuracy testing across various underlying surfaces (i.e., rivers, lakes, and glaciers); and (c) investigation of relevant evidence from other Earth subsystems (i.e., historic climate events, sea level budget, and polar motion). Among the three reconstructions, the one that additionally corrects glacial TWS changes (REC2) detects a breaking point in the 1990s and further closes the interannual sea level budget with an absolute difference reduction to 5.13 mm; the reconstruction that is forced by local meteorological conditions (REC1), accounting for 54% of the GRACE-derived signal energy, underestimates glacial TWS variability but outperforms the other reconstructions in reproducing lake levels, basin-scale water balances, and climate events at the interannual scale, while the others consider 95%-99% of the GRACE-derived signal energy. The relatively high accuracy of REC1 (and REC2) in reflecting interannual changes in nonglacial (and glacial) regions is further confirmed by explaining the χ^2 - (and χ^1 -) component polar motion. Ten to 20% of the interannual polar motion remains unexplained, indicating room for improvement in interannual TWS reconstruction.

368. The heat storage variability in the Brazil Current

Authors: Catherine B. Bou-Haya (University of São Paulo); Olga T. Sato (University of São Paulo)

Journal: Ocean and Coastal Research

Publisher: FapUNIFESP (SciELO)

DOI: 10.1590/2675-2824070.22006cbbh

URL: <https://doi.org/10.1590/2675-2824070.22006cbbh>

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Language: N/A

Subjects: N/A

Keywords: Brazil Current, Volume Transport, ECCO model, Interannual Variability, Oceanic Heat Storage

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

369. Steric Changes Associated With the Fast Sea Level Rise in the Upper South Indian Ocean

Authors: Tangdong Qu (Joint Institute for Regional Earth System Science and Engineering University of California CA Los Angeles USA); Oleg Melnichenko (Earth and Space Research Seattle WA USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl100635

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References: 57

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Subjects: N/A

Keywords: sea level rise, South Indian Ocean, upper-ocean freshening, upper-ocean warming

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: This study provides a quantitative assessment of steric changes associated with sea level rise in the upper (0-500 m) South Indian Ocean (SIO) during 1993-2017, using the latest ocean state estimate of Estimating the Circulation and Climate of the Ocean (ECCO) combined with in-situ observations from Argo. Both the observations and ECCO estimate show a sea level rise in the low-latitude (0 degrees-30 degreesS) SIO that is faster than its South Pacific and South Atlantic counterparts by a factor at least two. Much of this fast sea level rise is due to warming and freshening of the upper ocean, with no significant contribution from the deeper layers (>500 m). We emphasize the importance of halosteric effect, whose contribution to sea level rise is comparable with that of thermosteric effect in the eastern basin. On interannual time scale, up to 90% of the region's sea level variability can be explained by steric changes that in turn are dominated by upper ocean convergence.

370. Energetics during eddy shedding in the Gulf of Mexico

Authors: Haiyuan Yang; Chen Yang; Yongzheng Liu; Zhaohui Chen

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-023-01538-y

URL: <https://doi.org/10.1007/s10236-023-01538-y>

Publication Date: 2023-02-14

Volume/Issue: Volume 73, Issue 2

References: 65

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: Using the Estimating Circulation and Climate of the Ocean (ECCO) Phase II product, this study investigates the energetic characteristics during eddy shedding in the Gulf of Mexico. Based on the sea level anomaly data between 1992 and 2016, a total of 34 eddy shedding events are identified. Drawing on multiscale energy and vorticity analysis method, the eddy kinetic energy (EKE) budgets are diagnosed based on the ensemble of 34 eddy shedding events. During the stage of eddy shedding, barotropic instability (BT) dominates the energy budget. Meanwhile, energy transfers from upper layer to the deep layer by vertical pressure work (PW), which is the main source of abyssal EKE. Before eddy detachment, cyclonic eddy appears at the southeastern side of the Loop Current. Even though buoyancy forcing (BF) dominates the energy budget, BT makes considerable contribution to the generation of cyclonic eddy. Baroclinic instability (BC) shares the similar horizontal distribution with BF which accounts for 32% of the value of BC.

371. The carbon sink of the Coral Sea, the world's second largest marginal sea, weakened during 2006-2018

Authors: Siqi Zhang; Yan Bai; Xianqiang He; Shujie Yu; Zigeng Song; Fang Gong; Qiankun Zhu; Delu Pan

Journal: Science of The Total Environment

Publisher: Elsevier BV

DOI: 10.1016/j.scitotenv.2023.162219

URL: <https://doi.org/10.1016/j.scitotenv.2023.162219>

Publication Date: 2023-02-13

Volume/Issue: Volume 872

References: 88

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Key Technologies Research and Development Program

License: N/A

Open Access: N/A

Abstract: N/A

372. Towards a fully unstructured ocean model for ice shelf cavity environments: Model development and verification using the Firedrake finite element framework

Authors: William I. Scott; Stephan C. Kramer; Paul R. Holland; Keith W. Nicholls; Martin J. Siebert; Matthew D. Piggott

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2023.102178

URL: <https://doi.org/10.1016/j.ocemod.2023.102178>

Publication Date: 2023-02-12

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References: 83

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Engineering and Physical Sciences Research Council, Natural Environment Research Council

License: N/A

Open Access: N/A

Abstract: N/A

373. Fusion of ocean data from multiple sources using deep learning: Utilizing sea temperature as an example

Authors: Mingqing Wang; Danni Wang; Yanfei Xiang; Yishuang Liang; Ruixue Xia; Jinkun Yang; Fanghua Xu; Xiaomeng Huang

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1112065

URL: <https://doi.org/10.3389/fmars.2023.1112065>

Publication Date: 2023-02-06

Volume/Issue: Volume 10

References: 55

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: For investigating ocean activities and comprehending the role of the oceans in global climate change, it is essential to gather high-quality ocean data. However, existing ocean observation data have deficiencies such as inconsistent spatial and temporal distribution, severe fragmentation, and restricted observation depth layers. Data assimilation is computationally intensive, and other conventional data fusion techniques offer poor fusion precision. This research proposes a novel multi-source ocean data fusion network (ODF-Net) based on deep learning as a solution for these issues. The ODF-Net comprises a number of one-dimensional residual blocks that can rapidly fuse conventional observations, satellite observations, and three-dimensional model output and reanalysis data. The model utilizes vertical ocean profile data as target constraints, integrating physics-based prior knowledge to improve the precision of the fusion. The network structure contains channel and spatial attention mechanisms that guide the network model's attention to the most crucial features, hence enhancing model performance and interpretability. Comparing multiple global sea temperature datasets reveals that the ODF-Net achieves the highest accuracy and correlation with observations. To evaluate the feasibility of the proposed method, a global monthly three-dimensional sea temperature dataset with a spatial resolution of 0.25 degreesx0.25 degrees is produced by fusing ocean data from multiple sources from 1994 to 2017. The rationality tests on the fusion dataset show that ODF-Net is reliable for integrating ocean data from various sources.

374. Statistical Characteristics of the Multiscale SST Fractal Structure over the Kuroshio Extension Region Using VIIRS Data

Authors: Kai Yu (Hohai University, Second Institute of Oceanography, Ministry of Natural Resources); Changming Dong (Nanjing University); Jin Wang (Nanjing University); Xuhua Cheng (Hohai University, Ministry of Natural Resources); Yi Yu (Second Institute of Oceanography, Ministry of Natural Resources)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs15040881

URL: <https://doi.org/10.3390/rs15040881>

Publication Date: 2023-02-06

Volume/Issue: Volume 15, Issue 4

References: 41

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: structure function, sea surface temperature, Kuroshio Extension, energy cascade, submesoscale (and 1 more)

Funding: National Key Research Program of China, Fundamental Research Funds for the Central Universities, National Science Foundation of China (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: The ocean behaves as a typical multiscale fractal structure, whose dynamic and thermal variabilities extend over a wide range of spatial scales, r , spanning from 10^{-3} to 107 m. Studying the statistical characteristics of multiscale fractal structures is crucial to understanding the interactions and energy cascade processes between different spatial scales. Remote sensing data are one of the best choices for revealing these statistical characteristics. This work analyzes the multiscale (1-1000 km) fractal structures of sea surface temperature (SST) from the Level-2+ (L2P) satellite orbit Visible Infrared Imaging Radiometer Suite (VIIRS) products over the Kuroshio Extension (KE) region (145 degreesE-160 degreesW, 20 degreesN-50 degreesN), using a conventional method (second-order structure function, $D(r)$) and a newly developed statistical method (spatial variance, $V(r)$). The results show that both the power-law distribution slopes of $D(r)$ and $V(r)$ are close to $2/3$, which is equivalent to the $-5/3$ wavenumber spectrum. $V(r)$ is found to be more robust when depicting the fractal structure and variance density, $V'(r)$, compared with $D(r)$. $V'(r)$ is slightly larger at the mesoscale (50-150 km) than at the large scale (higher than 150 km) and is much smaller than that at the submesoscale (smaller than 50 km). Additionally, $V'(r)$ has an indiscernible diurnal variation but remarkable seasonal and latitudinal variations. For the seasonal variability, the maximum $V'(r)$ appears in winter at the large scale and mesoscale, and gradually shifts towards spring at the submesoscale, which implies that a forward energy cascade process may occur during this period. The maximum of the latitude-dependent $V'(r)$ appears around 40 degreesN for all the scales. It is consistent with the latitude of the strongest background SST gradient, indicating that the background SST front is the main source of the strong SST multiscale spatial variabilities over the KE region. This work benefits the application of other high-resolution remote sensing data in this research field, including the forthcoming Surface Water

Ocean Topography (SWOT) satellite product.

375. Seasonality of the Sub-Mesoscale to Mesoscale Sea Surface Variability From Multi-Year Satellite Altimetry

Authors: Y. Yu (Scripps Institution of Oceanography / University of California San Diego); D. T. Sandwell (Scripps Institution of Oceanography / University of California San Diego); S. T. Gille (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jc019486

URL: <https://doi.org/10.1029/2022jc019486>

Publication Date: 2023-02-03

Volume/Issue: Volume 128, Issue 2

References: 48

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory, Office of Naval Research Global

License: N/A

Open Access: N/A

Abstract: Sea surface slope (SSS) varies in response to a range of physical processes: tides, geostrophic flows, surface and internal waves, etc. We present the sea surface variation in the form of the SSS variability using 30 years of heterogeneous satellite altimetry measurements. We apply band-pass filters to the along-track SSS, and derive the mean and seasonal (annual and semi-annual) components of SSS variability in multiple wavelength sub-bands from 10 to 1,000 km. We show that the seasonal components are generally small (<10% in amplitude) compared to the mean variability. Through correlation analysis, we show evidence that SSS variability with wavelengths less than 30 km is dominated by wave height noise. At sub-mesoscale to mesoscale (30-100 km) wavelengths, we identify high variability over western boundary currents and regions of rough topography. In this band, the high-latitude Northern Hemisphere and the south Indian Ocean are associated with large annual cycles. The variability is higher in local wintertime except for a few regions, for example, the Bay of Bengal, which shows high variability in the boreal spring and fall. Through power spectral density analysis of the seasonal SSS variability, we find that the energy differences between local winter and summer are stronger at smaller scales (<100 km). The Ka-band radar interferometry instrument on the Surface Water and Ocean Topography satellite mission will allow observation of ocean surface activities down to ~20 km at submonthly time scales, but wave-related errors (sea state bias, aliasing, wind-driven activities, etc.) will still be a major challenge.

376. One-degree resolution mascon solution over Antarctic derived from GRACE Level-2 data

Authors: Wei Wang; Yunzhong Shen; Qiujie Chen; Tianyi Chen

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2023.1129628

URL: <https://doi.org/10.3389/feart.2023.1129628>

Publication Date: 2023-02-03

Volume/Issue: Volume 11

References: 79

Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: The mass loss of the Antarctic Ice Sheet (AIS) is an important contributor to global sea-level rise in response to the warming ocean and atmospheric temperatures as well as the changes in current systems and precipitation patterns. In this study, a regional mascon method is developed to squeeze more mass change signals, in which the pseudo-observations of the geopotential are generated from the unfiltered GRACE Level-2 data whereas the regularization matrix is constructed with the prior information derived from filtered GRACE Level-2 data. A series of mascon solutions with 1 degreesx1 degrees equal-area resolution over AIS is derived from the updated Tongji-Grace2018 model spanning April 2002 to December 2016. Compared to the filtering results from P4M6 decorrelation and 100 km Gaussian filtering, our mascon solutions can effectively suppress the strips, improve the spatial resolution over AIS, and get a stronger signal with an improvement of 116.86% in the Antarctic Peninsula Ice Sheet (APIS), and more coincide with the features of glaciers and ice streams, such as the most striking ice mass loss in Totten, Getz, Thwaites and Pine Island, and the ice mass gain in Kamb Ice Stream. During the period from 2002 to 2016, the mass change rates from our mascon solution are -103.6 +/- 5.6 Gt/yr, 63.0 +/- 4.3 Gt/yr, -143.3 +/- 4.9 Gt/yr and -23.29 +/- 1.2 Gt/yr in AIS, East AIS, West AIS, and APIS, respectively. The mass change signals at the basin scale are with even more distinguishing features, with the highest mass gain rates of 18.03 +/- 1.88 Gt/yr and 14.55 +/- 0.60 Gt/yr at Basin 7 and Basin 18, and the highest mass loss rates of -58.57 +/- 2.48 Gt/yr and -44.12 +/- 2.27 Gt/yr at Basin 21 and Basin 22. Relative to the cumulated surface mass balance from the regional atmospheric climate model, the correlation coefficients of our mascon solutions are 0.91, 0.94, and 0.96 in East AIS, West AIS, and APIS.

377. Surface cross-equatorial pathways of seawater from the Bay of Bengal

Authors: Zhengbei Zhang; Jing Wang; Jiajia Hao; Dongliang Yuan; Kunxiang Wang

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1101716

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Volume/Issue: Volume 10

References: 59

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Cross-equatorial pathways of heat and salt in the Eastern Indian Ocean are analyzed using Lagrangian trajectories based on the ECCO2 model simulation and drifter-derived currents combined with Argo data. The results show that most of the particles released in the Bay of Bengal move across the equator near the eastern boundary of the upper Indian Ocean ($< 50\text{m}$) during the southwest monsoon season and join the South Equatorial Current of the wind-driven circulation. Meridional salt and heat transports associated with the cross-equatorial currents feature significant seasonal variations, with stronger southward transports during the southwest monsoon than northward transports during the northeast monsoon, which are correlated significantly with the surface Ekman transports. The estimated annual mean salt and heat transports, based on the model (observational) data, are -0.06×10^9 (-0.11×10^9) $\text{kg} \cdot \text{s}^{-1}$ and -0.20 (-0.38) PW (1PW = 1015 W), respectively, southward across the equator.

378. Seasonal Features and Potential Mechanisms of Submesoscale Processes in the Southern Bay of Bengal during 2011/12

Authors: Xuhua Cheng (Hohai University, Southern Marine Science and Engineering Guangdong Laboratory); Lanman Li (Hohai University); Zhiyou Jing (Chinese Academy of Sciences); Haijin Cao (Hohai University); Guidi Zhou (Hohai University); Wei Duan (Hohai University); Yifei Zhou (Hohai University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0078.1

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Publication Date: 2023-01-31

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References: 81

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Ocean, Indian Ocean, Frontogenesis/frontolysis

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: This study investigates the seasonal features and generation mechanisms of submesoscale processes (SMPs) in the southern Bay of Bengal (BoB) during 2011/12, based on the output of a high-resolution model, LLC4320 (latitude-longitude-polar cap). The results show that the southern BoB exhibits the most energetic SMPs, with significant seasonal variations. The SMPs are more active during the summer and winter monsoon periods. During the monsoon periods, the sharpening horizontal buoyancy gradients associated with strong straining effects favor the frontogenesis and mixed layer instability (MLI), which are responsible for the SMPs generation. The symmetric instability (SI) scale is about 3-10 km in the southern BoB, which can be partially resolved by LLC4320. The SI is more active during summer and winter, with a proportion of 40%-80% during the study period when the necessary conditions for SI are satisfied. Energetics analysis suggests that the energy source of SMPs is mainly from the local large-scale and mesoscale processes. Baroclinic instability at submesoscales plays a significant role, further confirming the importance of frontogenesis and MLI. Barotropic instability also has considerable contribution to the submesoscale kinetic energy, especially during summer. Submesoscale processes (SMPs) are ubiquitous in the Bay of Bengal (BoB). Affected by the seasonally reversing monsoon, abundant rainfall and runoff, and equatorial remote forcing, the upper circulation in the BoB is complex, featuring active mesoscale eddies and rich submesoscale phenomena, making the BoB a "natural test ground" for submesoscale studies. It is found in this work that characteristics of SMPs in the BoB are quite different from other regions. In the southern bay, SMPs are most active during the summer and winter monsoons due to the frontogenesis, enhanced mixed layer instability (MLI), and symmetric instability. These findings could deepen our understanding on multiscale dynamic processes and energy cascade in the BoB and have implications for the study of marine ecology and biogeochemical processes.

379. Computing marine plankton connectivity under thermal constraints

Authors: Darshika Manral; Doroteaciro Iovino; Olivier Jaillon; Simona Masina; Hugo Sarmento; Daniele Iudicone; Linda Amaral-Zettler; Erik van Sebille

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2023.1066050

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References: 53

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Ocean currents are a key driver of plankton dispersal across the oceanic basins. However, species specific temperature constraints may limit the plankton dispersal. We propose a methodology to estimate the connectivity pathways and timescales for plankton species with given constraints on temperature tolerances, by combining Lagrangian modeling with network theory. We demonstrate application of two types of temperature constraints: thermal niche and adaptation potential and compare it to the surface water connectivity between sample stations in the Atlantic Ocean. We find that non-constrained passive particles representative of a plankton species can connect all the stations within three years at the surface with pathways mostly along the major ocean currents. However, under thermal constraints, only a subset of stations can establish connectivity. Connectivity time increases marginally under these constraints, suggesting that plankton can keep within their favorable thermal conditions by advecting via slightly longer paths. Effect of advection depth on connectivity is observed to be sensitive to the width of the thermal constraints, along with decreasing flow speeds with depth and possible changes in pathways.

380. Evaluation of an Ocean Reanalysis System in the Indian and Pacific Oceans

Authors: Changxiang Yan (Chinese Academy of Sciences); Jiang Zhu (Chinese Academy of Sciences)

Journal: Atmosphere

Publisher: MDPI AG

DOI: 10.3390/atmos14020220

URL: <https://doi.org/10.3390/atmos14020220>

Publication Date: 2023-01-23

Volume/Issue: Volume 14, Issue 2

References: 51

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: This paper describes an ocean reanalysis system in the Indian and Pacific oceans (IPORA) and evaluates its quality in detail. The assimilation schemes based on ensemble optimal interpolation are employed in the hybrid coordinate ocean model to conduct a long-time reanalysis experiment during the period of 1993-2020. Different metrics including comparisons with satellite sea surface temperature, altimetry data, observed currents, as well as other reanalyses such as ECCO and SODA are used to validate the performance of IPORA. Compared with the control experiment without assimilation, IPORA greatly reduces the errors of temperature, salinity, sea level anomaly, and current fields, and improves the interannual variability. In contrast to ECCO and SODA products, IPORA captures the strong signals of SLA variability and reproduces the linear trend of SLA very well. Meanwhile, IPORA also shows a good consistence with observed currents, as indicated by an improved correlation and a reduced error.

381. Addendum: Impact of downward longwave radiative deficits on Antarctic sea-ice extent predictability during the sea ice growth period (2022 Environ. Res. Lett. 17 084008)

Authors: Ivana Cerovečki; Rui Sun; David H Bromwich; Xun Zou; Matthew R Mazloff; Sheng-Hung Wang

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/acb162

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References: 8

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Division of Polar Programs, King Abdullah University of Science and Technology, Office of Polar Programs (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

382. Fast Spin-Up of Geochemical Tracers in Ocean Circulation and Climate Models

Authors: Samar Khatiwala (University of Oxford)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022ms003447

URL: <https://doi.org/10.1029/2022ms003447>

Publication Date: 2023-01-19

Volume/Issue: Volume 15, Issue 2

References: 102

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Ocean geochemical tracers such as radiocarbon, protactinium and thorium isotopes, and noble gases are widely used to constrain a range of physical and biogeochemical processes in the ocean. However, their routine simulation in global ocean circulation and climate models is hindered by the computational expense of integrating them to a steady state. Here, a new approach to this long-standing "spin-up" problem is introduced to efficiently compute equilibrium distributions of such tracers in seasonally-forced models. Based on "Anderson Acceleration," a sequence acceleration technique developed in the 1960s to solve nonlinear integral equations, the new method is entirely "black box" and offers significant speed-up over conventional direct time integration. Moreover, it requires no preconditioning, ensures tracer conservation and is fully consistent with the numerical time-stepping scheme of the underlying model. It thus circumvents some of the drawbacks of other schemes such as matrix-free Newton Krylov that have been proposed to address this problem. An implementation specifically tailored for the batch HPC systems on which ocean and climate models are typically run is described, and the method illustrated by applying it to a variety of geochemical tracer problems. The new method, which provides speed-ups by over an order of magnitude, should make simulations of such tracers more feasible and enable their inclusion in climate change assessments such as IPCC.

383. Energetics of Multiscale Interactions in the Agulhas Retroflection Current System

Authors: Mengmeng Li (Chinese Academy of Sciences); Chongguang Pang (d Center for Ocean Mega-Science, Chinese Academy of Science, Qingdao, China, Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Xiaomei Yan (d Center for Ocean Mega-Science, Chinese Academy of Science, Qingdao, China, Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Linlin Zhang (d Center for Ocean Mega-Science, Chinese Academy of Science, Qingdao, China, Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Zhiliang Liu (e Research Center for Marine Science, Hebei Normal University of Science and Technology, Qinhuangdao, China, f Hebei Key Laboratory of Ocean Dynamics, Resources and Environments, Hebei Normal University of Science and Technology, Qinhuangdao, China)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-21-0275.1

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References: 83

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Mesoscale processes, Eddies, Indian Ocean, Instability

Funding: Natural Science Foundation of Shandong Province, National Natural Science Foundation of China, National Natural Science Foundation of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: Using the recently developed multiscale window transform and multiscale energy and vorticity analysis methods, this study diagnoses the climatological characteristics of the nonlinear mutual interactions among mesoscale eddies, low-frequency (seasonal to interannual) fluctuations, and the decadal modulating mean flow in the Agulhas Retroflection Current System (ARCS). It is found that mesoscale eddies are generated primarily in the retroflection region by mixed barotropic and baroclinic instabilities. The barotropic instability dominates the generation of eddy kinetic energy (EKE) here, contributing power roughly 10 times larger than the baroclinic one. These locally generated eddies are transported away. In the rings drift and meanders regions, the nonlocal transport serves as an important energy source for the eddy field, making a contribution comparable to that of the baroclinic instability for the EKE production. Contrarily, in the stable region, the EKE is generated mainly due to the baroclinic instability. In most of the ARCS area, the kinetic energy (KE) is further transferred inversely from mesoscale eddies to other lower-frequency motions. In particular, in the retroflection, rings drift, and stable regions, the inverse KE cascade plays a leading role in generating seasonal-interannual fluctuations, providing roughly 3-5 times as much power as the forward KE cascade from the mean flow and the advection effect. In the meanders region, however, the forward cascade contributes 4 times more KE to the low-frequency variabilities than the inverse one. All the

results provide a model-based benchmark for future studies on physical processes and dynamics at different scales in the ARCS.

384. Observed Opposite Fall-to-Winter Variations in the Air-Sea Latent Heat Flux Between the Western Boundary Currents and Coastal Seas

Authors: Xiangzhou Song (Hohai University, Ministry of Natural Resources)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl100875

URL: <https://doi.org/10.1029/2022gl100875>

Publication Date: 2023-01-18

Volume/Issue: Volume 50, Issue 2

References: 47

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY-NC 4.0

Open Access: N/A

Abstract: This study presents observations of opposite fall-to-winter variations in the air-sea latent heat fluxes (LHs) between the western boundary currents (WBCs) and coastal seas obtained using eight air-sea buoys in the Northern Hemisphere. Different from the WBCs, the wintertime LH along the coast is strangely reduced by approximately 60 Wm^{-2} compared to that in fall. The reduced sea-air humidity differences account for these opposite seasonal variations, although the sea-air temperature differences are enhanced, while the roles played by winds and the nonlinear effects are minor. The boundary layer stability also varies in terms of the air-sea parameters from fall to winter. However, the conditions of the WBCs meet the traditional expectation: that is, the LH in winter is 60 Wm^{-2} higher than that in fall with near-neutral/unstable boundary layer stability. Enhanced winds in winter determine the seasonal enhancement of the LH over the WBCs, indicating a specialized regional character.

385. Resonant Diurnal Internal Tides in the North Atlantic: 2. Modeling

Authors: B. D. Dushaw (University of Washington); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl101193

URL: <https://doi.org/10.1029/2022gl101193>

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References: 34

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: An unconstrained global ocean simulation for 2020 supports past observations of diurnal internal tides by acoustic tomography during the 1991-1992 Acoustic Mid-Ocean Dynamics Experiment in the Western North Atlantic. Explicitly representing the tides, the simulation reproduces the functional form and resonant state of K1 and O1 internal-tide standing waves, while providing a more realistic physical picture of them. The tomographic data were used to predict the tides in 2020. Not surprisingly, the characteristics of the barotropic and internal tides of the unconstrained simulation deviate from observations. The simulated barotropic tidal currents have excessive, irregular amplitude and lead the acoustic tidal predictions by about 2 hr. While internal-tide phase coherence is apparent, the simulated internal-tide variations were irregular in amplitude and phase, unlike the observations. The tomographic tidal measurements therefore provide a quantitative benchmark for improved model representation of tides, internal tides, and dissipation.

386. Towards the Upper-Ocean Unbalanced Submesoscale Motions in the Oleander Observations

Authors: Haijin Cao (Hohai University, Ministry of Natural Resources); Baylor Fox-Kemper (Brown University); Zhiyou Jing (Chinese Academy of Sciences); Xiangzhou Song (Hohai University, Ministry of Natural Resources); Yuyi Liu (Chinese Academy of Sciences)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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References: 60

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Eddies, Ageostrophic circulations, Small scale processes, Ocean dynamics

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License: N/A

Open Access: N/A

Abstract: Oceanic submesoscale dynamics with horizontal scales < 20 km have similar temporal and spatial scales as internal gravity waves (IGWs), but they differ dynamically and have distinct impacts on the ocean. Separating unbalanced submesoscale motions (USMs), quasi-balanced submesoscale motions (QBMs), and IGWs in observations remains a great challenge. Based on the wave-vortex decomposition and the vertical scale separation approach for distinguishing IGWs and USMs, the long-term repeat Oleander observations in the Gulf Stream region provide an opportunity to quantify these processes separately. Here in this study, the role of USMs in the divergence is emphasized, which has confounded the wave-vortex decomposition of wintertime data in previous analyses. We also adopt the vertical filtering approach to identify the USMs by applying a high-pass filter to the vertical scales, as USMs are characterized by smaller vertical scales. This approach is tested with submesoscale-permitting model data to confirm its effectiveness in filtering the submesoscale velocity perturbations, before being applied to the compiled velocity data of the Oleander dataset (years 2005-18). The results show that the averaged submesoscale eddy kinetic energy by USMs can reach $\sim 1 \times 10^{-3} \text{ m}^2 \text{ s}^{-2}$ at $z = -30$ m in winter, much stronger than found in other seasons. Importantly, this study exemplifies the possibility of obtaining USMs from existing ADCP observations and reveals the seasonal dynamical regimes for the submesoscales.

387. Identification of Sea Surface Temperature and Sea Surface Salinity Fronts along the California Coast: Application Using Saildrone and Satellite Derived Products

Authors: Jorge Vazquez-Cuervo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Marisol García-Reyes (Farallon Institute, Petaluma, CA 94952, USA); José Gómez-Valdés (Physical Oceanography Department, Center for Scientific Research and Higher Education at Ensenada, Ensenada 22860, Baja California, Mexico)

Journal: Remote Sensing

Publisher: MDPI AG

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References: 27

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Keywords: N/A

Funding: NASA

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Open Access: N/A

Abstract: Coastal upwelling regions are one of the most dynamic areas of the world's oceans. The California and Baja California Coasts are impacted by both coastal upwelling and the California Current, leading to frontal activity that is captured by gradients in both Sea Surface Temperature (SST) and Sea Surface Salinity (SSS). Satellite data are a great source of spatial data to study fronts. However, biases near coastal areas and coarse resolutions can impair its usefulness in upwelling areas. In this work gradients in SST from NASA Multi-Scale Ultra-High Resolution (MUR) and in two SSS products derived from the Soil Moisture Active Passive (SMAP) NASA mission are compared directly with gradients derived from the Saildrone uncrewed vehicles to validate the gradients as well as to assess their ability to detect known frontal features. The three remotely sensed data sets (MURSST/JPL, SMAP SSS/RSS, SMAP SSS) were co-located with the Saildrone data prior to the calculation of the gradients. Wavelet analysis is used to determine how well the satellite derived SST and SSS products are reproducing the Saildrone derived gradients. Overall results indicate the remote sensing products are reproducing features of known areas of coastal upwelling. Differences between the SST and SSS gradients are mainly associated with the limitations of the microwave derived SSS coverage near land and its reduced spatial resolution. The results are promising for using remote sensing data sets to monitor frontal structure along the California Coast and the application to long term changes in coastal upwelling and dynamics.

388. Seasonal Eddy Variability in the Northwestern Tropical Atlantic Ocean

Authors: Minghai Huang (University of Delaware); Yang Yang (Nanjing University, Xiamen University); Xinfeng Liang (University of Delaware)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0200.1

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Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Mesoscale processes, Eddies, Instability

Funding: National Science Foundation

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Open Access: N/A

Abstract: Eddies in the northwestern tropical Atlantic Ocean play a crucial role in transporting the South Atlantic Upper Ocean Water to the North Atlantic and connect the Atlantic and the Caribbean Sea. Although surface characteristics of those eddies have been well studied, their vertical structures and governing mechanisms are much less known. Here, using a time-dependent energetics framework based on the multiscale window transform, we examine the seasonal variability of the eddy kinetic energy (EKE) in the northwestern tropical Atlantic. Both altimeter-based data and ocean reanalyses show a substantial EKE seasonal cycle in the North Brazil Current Retroflexion (NBCR) region that is mostly trapped in the upper 200 m. In the most energetic NBCR region, the EKE reaches its minimum in April-June and maximum in July-September. By analyzing six ocean reanalysis products, we find that barotropic instability is the controlling mechanism for the seasonal eddy variability in the NBCR region. Nonlocal processes, including advection and pressure work, play opposite roles in the EKE seasonal cycle. In the eastern part of the NBCR region, the EKE seasonal evolution is similar to the NBCR region. However, it is the nonlocal processes that control the EKE seasonality. In the western part of the NBCR region, the EKE magnitude is one order of magnitude smaller than in the NBCR region and shows a different seasonal cycle, which peaks in March and reaches its minimum in October-November. Our results highlight the complex mechanisms governing eddy variability in the northwestern tropical Atlantic and provide insights into their potential changes with changing background conditions.

389. Quantification of Aquarius, SMAP, SMOS and Argo-Based Gridded Sea Surface Salinity Product Sampling Errors

Authors: Séverine Fournier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Frederick M. Bingham (University of North Carolina Wilmington); Cristina González-Haro (Institute of Marine Sciences); Akiko Hayashi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Karly M. Ulfax Carlin (Catlin Engineers & Scientists); Susannah K. Brodnitz (University of North Carolina Wilmington); Verónica González-Gambau (Institute of Marine Sciences); Mikael Kuusela (Carnegie Mellon University)

Journal: Remote Sensing

Publisher: MDPI AG

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Keywords: N/A

Funding: NASA Ocean Salinity Science Team, INTERACT, CSIC PTI Teledetect (and 2 more)

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Open Access: N/A

Abstract: Evaluating and validating satellite sea surface salinity (SSS) measurements is fundamental. There are two types of errors in satellite SSS: measurement error due to the instrument's inaccuracy and problems in retrieval, and sampling error due to unrepresentativeness in the way that the sea surface is sampled in time and space by the instrument. In this study, we focus on sampling errors, which impact both satellite and in situ products. We estimate the sampling errors of Level 3 satellite SSS products from Aquarius, SMOS and SMAP, and in situ gridded products. To do that, we use simulated L2 and L3 Aquarius, SMAP and SMOS SSS data, individual Argo observations and gridded Argo products derived from a 12-month high-resolution 1/48 degrees ocean model. The use of the simulated data allows us to quantify the sampling error and eliminate the measurement error. We found that the sampling errors are high in regions of high SSS variability and are globally about 0.02/0.03 psu at weekly time scales and 0.01/0.02 psu at monthly time scales for satellite products. The in situ-based product sampling error is significantly higher than that of the three satellite products at monthly scales (0.085 psu) indicating the need to be cautious when using in situ-based gridded products to validate satellite products. Similar results are found using a Correlated Triple Collocation method that quantifies the standard deviation of products' errors acquired with different instruments. By improving our understanding and quantifying the effect of sampling errors on satellite-in situ SSS consistency over various spatial and temporal scales, this study will help to improve the validation of SSS, the robustness of scientific applications and the design of future salinity missions.

390. Impact of Yucatan Channel Subsurface Velocity Observations on the Gulf of Mexico State Estimates

Authors: Heriberto J. Vazquez (Scripps Institution of Oceanography / University of California San Diego); Ganesh Gopalakrishnan (Scripps Institution of Oceanography / University of California San Diego); Julio Sheinbaum (b Physical Oceanography, Centro de Investigación Científica y de Educación Superior de Ensenada, Ensenada, Baja California, Mexico)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-21-0213.1

URL: <https://doi.org/10.1175/jpo-d-21-0213.1>

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Language: N/A

Subjects: N/A

Keywords: Eddies, Currents, Model evaluation/performance, Trajectories, Numerical analysis/modeling (and 1 more)

Funding: Gulf Research Program, Bureau of Ocean Energy Management

License: N/A

Open Access: N/A

Abstract: The Gulf of Mexico (GoM) surface circulation variability is dominated by the Loop Current (LC) and the episodically released anticyclonic Loop Current eddies (LCEs). The Yucatan Current feeds the LC through the Yucatan Channel (YC), and its flow structure at the YC is hypothesized to affect the LC evolution critically. This study examines the impact of assimilating YC subsurface velocity observations from a tall mooring array across the YC on the GoM circulation. State estimates and forecasts of the LC circulation were produced using a regional implementation of the Massachusetts Institute of Technology general circulation model (MITgcm) and its adjoint-based four-dimensional variational (4DVAR) assimilation system. The estimates were constrained by combinations of the YC observations and satellite-derived sea surface height (SSH) and sea surface temperature (SST). The results show that assimilation of both moored and satellite data improves the model hindcasts and forecasts for all LC phases. Additionally, one realization of the state estimate that assimilates only moored data matches the LCE detachment timing with that of AVISO SSH. Observations from the moorings close to the Yucatan Peninsula significantly impact the LCE detachment. A finite-time Lyapunov exponent analysis reveals the differences among the assimilation experiments, such as eddylike structures intruding into the GoM through the YC, and its relation to the typical LC sudden growth. Finally, an adjoint sensitivity analysis is used to verify the dynamic link between the LC extension and the intrusion of eddylike structures into the GoM.

391. Evaluation of a global ocean reanalysis generated by a global ocean data assimilation system based on a four-dimensional variational (4DVAR) method

Authors: Yosuke Fujii; Takuma Yoshida; Hiroyuki Sugimoto; Ichiro Ishikawa; Shogo Urakawa

Journal: Frontiers in Climate

Publisher: Frontiers Media SA

DOI: 10.3389/fclim.2022.1019673

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Abstract: Japan Meteorological Agency (JMA) started to use a new global ocean data assimilation system for the operational seasonal predictions in February 2022. The system is composed of two subsystems with non-eddy-permitting (lower) and eddy-permitting (higher) resolutions. The lower-resolution subsystem adopts a four-dimensional variational (4DVAR) method to optimize the temperature and salinity fields, and the data-assimilated fields are downscaled into the higher-resolution subsystem using incremental analysis updates. The impact of introducing the 4DVAR method in the new ocean data assimilation system is investigated through the comparison of a regular reanalysis run of the system using the 4DVAR method with another run using a three-dimensional variational (3DVAR) method. A comparison of the temperature fields before the downscaling between the two reanalysis runs indicates that the 4DVAR method can more effectively reduce the misfits between the model field and assimilated observation data. However, the increase of the temperature root mean square difference (RMSD) relative to independent Argo float data, along with the larger variance, for the run with the 4DVAR method reveals that the 4DVAR method adjusts the temperature field more significantly but the adjustments are inconsistent with the independent data due to insufficient model physics and resolution. The increase of the RMSD is mitigated after the assimilated fields are downscaled into the higher-resolution subsystem. The 4DVAR method reduces the bias and RMSD of temperature relative to the independent data along the thermocline, as well as near the surface, in the equatorial vertical section, which is expected to affect the prediction of El Niño-Southern Oscillation (ENSO).

392. Global Sea Surface Cyclogeostrophic Currents Derived From Satellite Altimetry Data

Authors: Yuhan Cao (Nanjing University, Jiangsu Key Laboratory of Marine Bioresources and Environment Jiangsu Key Laboratory of Marine Biotechnology Jiangsu Ocean University Lianyungang China, School of Marine Technology and Geomatics Jiangsu Ocean University Lianyungang China); Changming Dong (Nanjing University, University of California Los Angeles, Southern Marine Science and Engineering Guangdong Laboratory); Alexandre Stegner (École Polytechnique); Brandon J. Bethel (Small Island Sustainability University of The Bahamas Nassau The Bahamas, Southern Marine Science and Engineering Guangdong Laboratory); Chunyan Li (Louisiana State University); Jihai Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory); Haibin Lü (School of Marine Technology and Geomatics Jiangsu Ocean University Lianyungang China); Jingsong Yang (Second Institute of Oceanography, Ministry of Natural Resources)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

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Open Access: N/A

Abstract: Sea surface currents (SSC) derived from the sea surface height anomalies (SSHA) as measured by multi-satellite altimeters are widely used for various applications including studies on ocean dynamics, marine ecology, and climate change. However, present SSC products estimated on the assumption of an idealized geostrophic balance is biased. To overcome this idealization, this study considers flow curvature in the estimation of sea surface velocities in the global ocean, with the computation scheme validated using numerical model results. It is demonstrated that the inclusion of curvature into SSC estimations significantly changes SSC dynamic features in terms of kinetic energy, enstrophy (the maximum spatial difference is about 15%), and strain rate (the maximum spatial difference is about 10%). Such correction is of importance to the application studies relying on the SSC products from the satellite measured SSHA.

393. Upper-Oceanic Warming in the Gulf of Mexico between 1950 and 2020

Authors: Zhankun Wang (Mississippi State University, Northern Gulf Institute, a NOAA/National Centers for Environmental Information, Stennis Space Center, Mississippi); Tim Boyer (c NOAA/National Centers for Environmental Information, Silver Spring, Maryland); James Reagan (c NOAA/National Centers for Environmental Information, Silver Spring, Maryland); Patrick Hogan (Mississippi State University, Northern Gulf Institute)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-22-0409.1

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Language: N/A

Subjects: N/A

Keywords: Climate change, Temperature, Heat budgets/fluxes, Energy budget/balance, Advection

Funding: National Centers for Environmental Information

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Open Access: N/A

Abstract: We estimate ocean heat content (OHC) change in the upper 2000 m in the Gulf of Mexico (GOM) from 1950 to 2020 to improve understanding of regional warming. Our estimates are based on 192 890 temperature profiles from the World Ocean Database. Warming occurs at all depths and in most regions except for a small region at northeastern GOM between 200 and 600 m. GOM OHC in the upper 2000 m increases at a rate of 0.38 ± 0.13 ZJ decade⁻¹ between 1970 and 2020, which is equivalent to 1.21 ± 0.41 terawatts (TW). The GOM sea surface temperature (SST) increased ~ 1.0 degrees ± 0.25 degreesC between 1970 and 2020, equivalent to a warming rate of 0.19 degrees ± 0.05 degreesC decade⁻¹. Although SST in the GOM increases at a rate approximately twice that for the global ocean, the full-depth ocean heat storage rate in the GOM (0.86 ± 0.26 W m⁻²) applied to the entire GOM surface is comparable to that for the global ocean (0.82 - 1.11 W m⁻²). The upper-1000-m layer accounts for approximately 80%-90% of the total warming and variations in the upper 2000 m in the GOM. The Loop Current advective net heat flux is estimated to be 40.7 ± 6.3 TW through the GOM. A heat budget analysis shows the difference between the advective heat flux and the ocean heat storage rate (1.76 ± 1.36 TW, 1992-2017) can be roughly balanced with the annual net surface heat flux from ECCO (-37.9 TW).

394. Limits and CO₂ equilibration of near-coast alkalinity enhancement

Authors: Jing He; Michael D. Tyka

Journal: Biogeosciences

Publisher: Copernicus GmbH

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Abstract: . Ocean alkalinity enhancement (OAE) has recently gained attention as a potential method for carbon dioxide removal (CDR) at gigatonne (Gt) scale, with near-coast OAE operations being economically favorable due to proximity to mineral and energy sources. In this paper we study critical questions which determine the scale and viability of OAE. Which coastal locations are able to sustain a large flux of alkalinity at minimal pH and Ω_{Arag} (aragonite saturation) changes? What is the interference distance between adjacent OAE projects? How much CO₂ is absorbed per unit of alkalinity added? How quickly does the induced CO₂ deficiency equilibrate with the atmosphere? Choosing relatively conservative constraints on ΔpH or $\Delta\Omega$, we examine the limits of OAE using the ECCO LLC270 (0.3■) global circulation model. We find that the sustainable OAE rate varies over 1-2 orders of magnitude between different coasts and exhibits complex patterns and non-local dependencies which vary from region to region. In general, OAE in areas of strong coastal currents enables the largest fluxes and depending on the direction of these currents, neighboring OAE sites can exhibit dependencies as far as 400 km or more. At these steady state fluxes most regional stretches of coastline are able to accommodate on the order of 10s to 100s of megatonnes of negative emissions within 300 km of the coast. We conclude that near-coastal OAE has the potential to scale globally to several Gt CO₂ yr⁻¹ of drawdown with conservative pH constraints, if the effort is spread over the majority of available coastlines. Depending on the location, we find a diverse set of equilibration kinetics, determined by the interplay of gas exchange and surface residence time. Most locations reach an uptake efficiency plateau of 0.6-0.8 mol CO₂ per mol of alkalinity after 3-4 years, after which there is only slow additional CO₂ uptake. Regions of significant downwelling (e.g., around Iceland) should be avoided by OAE deployments, as in such locations up to half of the CDR potential of OAE can be lost to bottom waters. The most ideal locations, reaching a molar uptake ratio of around 0.8, include North Madagascar, California, Brazil, Peru and locations close to the Southern Ocean such as Tasmania, Kerguelen and Patagonia, where the gas exchange appears to occur faster than the surface residence time. However, some locations (e.g., Hawaii) take significantly longer to equilibrate (up to 8-10 years) but can still eventually achieve high uptake ratios.

395. Quality Assessment of Sea Surface Salinity from Multiple Ocean Reanalysis Products

Authors: Haodi Wang (National University of Defense Technology); Ziqi You (National University of Defense Technology); Hailong Guo (National University of Defense Technology); Wen Zhang (National University of Defense Technology); Peng Xu (Nanjing University); Kaijun Ren (National University of Defense Technology)

Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

DOI: 10.3390/jmse11010054

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, the science and technology innovation program of Hunan Province

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Open Access: N/A

Abstract: Sea surface salinity (SSS) is one of the Essential Climate Variables (ECVs) as defined by the Global Climate Observing System (GCOS). Acquiring high-quality SSS datasets with high spatial-temporal resolution is crucial for research on the hydrological cycle and the earth climate. This study assessed the quality of SSS data provided by five high-resolution ocean reanalysis products, including the Hybrid Coordinate Ocean Model (HYCOM) 1/12 degrees global reanalysis, the Copernicus Global 1/12 degrees Oceanic and Sea Ice GLORYS12 Reanalysis, the Simple Ocean Data Assimilation (SODA) reanalysis, the ECMWF Oceanic Reanalysis System 5 (ORAS5) product and the Estimating the Circulation and Climate of the Ocean Phase II (ECCO2) reanalysis. Regional comparison in the Mediterranean Sea shows that reanalysis largely depicts the accurate spatial SSS structure away from river mouths and coastal areas but slightly underestimates the mean SSS values. Better SSS reanalysis performance is found in the Levantine Sea while larger SSS uncertainties are found in the Adriatic Sea and the Aegean Sea. The global comparison with CMEMS level-4 (L4) SSS shows generally consistent large-scale structures. The mean Δ SSS between monthly gridded reanalysis data and in situ analyzed data is -0.1 PSU in the open seas between 40 degrees S and 40 degrees N with the mean Root Mean Square Deviation (RMSD) generally smaller than 0.3 PSU and the majority of correlation coefficients higher than 0.5. A comparison with collocated buoy salinity shows that reanalysis products well capture the SSS variations at the locations of tropical moored buoy arrays at weekly scale. Among all of the five products, the data quality of HYCOM reanalysis SSS is highest in marginal sea, GLORYS12 has the best performance in the global ocean especially in tropical regions. Comparatively, ECCO2 has the overall worst performance to reproduce SSS states and variations by showing the largest discrepancies with CMEMS L4 SSS.

396. A high-resolution ocean bottom temperature product for the northeast U.S. continental shelf marine ecosystem

Authors: Hubert du Pontavice; Zhuomin Chen; Vincent S. Saba

Journal: Progress in Oceanography

Publisher: Elsevier BV

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Language: en

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Keywords: N/A

Funding: National Oceanic and Atmospheric Administration, Northeast Fisheries Science Center

License: N/A

Open Access: N/A

Abstract: N/A

397. Ocean variability beneath Thwaites Eastern Ice Shelf driven by the Pine Island Bay Gyre strength

Authors: Tiago S. Dotto; Karen J. Heywood; Rob A. Hall; Ted A. Scambos; Yixi Zheng; Yoshihiro Nakayama; Shuntaro Hyogo; Tasha Snow; Anna K. Wåhlin; Christian Wild; Martin Truffer; Atsuhiko Muto; Karen E. Alley; Lars Boehme; Guilherme A. Bortolotto; Scott W. Tyler; Erin Pettit

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

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Funding: N/A

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Open Access: N/A

Abstract: West Antarctic ice-shelf thinning is primarily caused by ocean-driven basal melting. Here we assess ocean variability below Thwaites Eastern Ice Shelf (TEIS) and reveal the importance of local ocean circulation and sea-ice. Measurements obtained from two sub-ice-shelf moorings, spanning January 2020 to March 2021, show warming of the ice-shelf cavity and an increase in meltwater fraction of the upper sub-ice layer. Combined with ocean modelling results, our observations suggest that meltwater from Pine Island Ice Shelf feeds into the TEIS cavity, adding to horizontal heat transport there. We propose that a weakening of the Pine Island Bay gyre caused by prolonged sea-ice cover from April 2020 to March 2021 allowed meltwater-enriched waters to enter the TEIS cavity, which increased the temperature of the upper layer. Our study highlights the sensitivity of ocean circulation beneath ice shelves to local atmosphere-sea-ice-ocean forcing in neighbouring open oceans.

398. Opposite-Sign Sea Surface Salinity Anomalies Over the Northeastern and Southwestern South Atlantic Ocean From 2010 to 2017

Authors: Hao Liu (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Institute of Oceanographic Instrumentation Qilu University of Technology (Shandong Academy of Science) Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Xunwei Nie (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Zexun Wei (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Ingo Richter (Application Laboratory Research Institute for Value-Added-Information Generation Japan Agency for Marine-Earth Science and Technology Yokohama Japan)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2022jc019351>

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: South Atlantic Ocean, interannual variability, salinity budget, ensemble mean, signal-to-noise ratio (and 1 more)

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Open Access: N/A

Abstract: Using seven sea surface salinity (SSS) data sets, the dominant modes of interannual SSS variability over the South Atlantic Ocean are examined for the period 2010-2017, which features strong anomalies (>0.1 g/kg). The empirical orthogonal function analysis and composite analysis are used, and the results show that the South Atlantic Ocean features opposite-sign SSS anomalies over the southwestern (45 degrees-20 degreesS/40 degrees-60 degreesW) and northeastern (0 degrees-15 degreesE/0 degrees-15 degreesS) regions. SSS budget analyses based on the Estimating the Circulation and Climate of the Ocean model estimate suggest that horizontal advection promotes the opposite-sign SSS anomalies over the two regions. The interannual changes in the horizontal advection can be attributed to changes in the advection of mean salinity by the anomalous horizontal flow, in which the Ekman velocities play the dominant role. The anomalous wind stress and sea level pressure patterns suggest that the opposite SSS variations in poles A and B are probably associated with atmospheric forcings under alternating Atlantic Nino and Benguela Nina conditions. This study offers a connection in interannual SSS variations over two remote regions within the South Atlantic Ocean and suggests the possible role of the climate mode in setting up interannual salinity anomalies over the South Atlantic Ocean.

399. Causes for Atlantic Freshwater Content Variability in the GECCO3 Ocean Synthesis

Authors: Xin Liu (Institute of Coastal Systems--Analysis and Modeling Helmholtz-Zentrum Hereon Geesthacht Germany, University of Hamburg); Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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References: 59

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Regional freshwater content (FWC) changes are studied over the period 1961-2018 using the GECCO3 ocean synthesis. In four dynamically distinct regions of the Atlantic, the study identifies causes for FWC variability with a focus on interannual and decadal time-scale changes. Results show that in each region, it is a combination of the surface freshwater flux and the net freshwater transport across the region's boundaries that act jointly in changing the respective FWC. Surface flux mainly contributes to the FWC variability on multi-decadal time scales. The impact of surface flux also increases toward the tropics. On shorter time scales, it is especially horizontal transport fluctuations, leading to FWC changes in mid and high latitudes. Going from north to the south, the transport across a single meridional boundary becomes less correlated with the FWC changes but the net transport across both boundaries plays an increasingly important role. Moreover, the subpolar box is mainly gyre driven, which differs from the other two, essentially overturning driven, North Atlantic boxes. In the tropical Atlantic, the shallow overturning cell and the deep overturning contribute about equal amounts to the freshwater variations.

400. Ground deformation monitoring of the eruption offshore Mayotte

Authors: Aline Peltier; Sébastien Saur; Valérie Ballu; François Beauducel; Pierre Briole; Kristel Chanard; Denis Dausse; Jean-Bernard De Chabaliér; Raphael Grandin; Perrine Rouffiac; Yann-Treden Tranchant; Maxime Bès de Berc; Simon Besançon; Patrice Boissier; Céleste Broucke; Christophe Brunet; Kevin Canjamalé; Erwan Carme; Philippe Catherine; Alison Colombain; Wayne Crawford; Romuald Daniel; Grégoire Dectot; Nicolas Desfete; Cécile Doubre; Tom Dumouch; Cyprien Griot; Marc Grunberg; Hélène Jund; Philippe Kowalski; Frédéric Lauret; Jacques Lebreton; Frédérick Pesqueira; Frédéric Tronel; Pierre Valtý; Jérôme van der Woerd

Journal: Comptes Rendus. Géoscience

Publisher: Cellule MathDoc/Centre Mersenne

DOI: 10.5802/crgeos.176

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Abstract: N/A

401. Role of local and remote forcing on the decadal variability of Mixed Layer Depth in the Bay of Bengal

Authors: Biplab Sadhukhan; Arun Chakraborty

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2022.101349

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Publication Date: 2022-12-16

Volume/Issue: Volume 102

References: 73

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

402. Topography Effects on the Seasonal Variability of Ocean Bottom Pressure in the North Pacific Ocean

Authors: Lei Chen (Ocean University of China, Laoshan Laboratory); Jiayan Yang (Woods Hole Oceanographic Institution); Lixin Wu (Ocean University of China, Laoshan Laboratory)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0140.1

URL: <https://doi.org/10.1175/jpo-d-22-0140.1>

Publication Date: 2022-12-14

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References: 45

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Topographic effects, Rossby waves, Potential vorticity, Seasonal variability

Funding: N/A

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Open Access: N/A

Abstract: Ocean bottom pressure pB is an important oceanic variable that is dynamically related to the abyssal ocean circulation through geostrophy. In this study we examine the seasonal pB variability in the North Pacific Ocean by analyzing satellite gravimetric observations from the GRACE program and a data-assimilated ocean-state estimate from ECCOv4. The seasonal pB variability is characterized by alternations of low and high anomalies among three regions, the subpolar and subtropical basins as well as the equatorial region. A linear two-layer wind-driven model is used to examine forcing mechanisms and topographic effects on seasonal pB variations. The model control run, which uses a realistic topography, is able to simulate a basinwide seasonal pB variability that is remarkably similar to that from GRACE and ECCOv4. Since the model is driven by wind stress alone, the good model-data agreement indicates that wind stress is the leading forcing for seasonal changes in pB. An additional model simulation was conducted by setting the water depth uniformly at 5000 m. The magnitude of seasonal pB anomaly is amplified significantly in the flat-bottom simulation as compared with that in the control run. The difference can be explained in terms of the topographic Sverdrup balance. In addition, the spatial pattern of the seasonal pB variability is also profoundly affected by topography especially on continental margins, ridges, and trenches. Such differences are due to topographic effects on the propagation pathways of Rossby waves.

403. Solar Energy Production Planning in Antikythera: Adequacy Scenarios and the Effect of the Atmospheric Parameters

Authors: Panagiotis Kosmopoulos (Institute for Environmental Research and Sustainable Development, National Observatory of Athens (IERSD/NOA), 15236 Athens, Greece); Marios Mechilis (National and Kapodistrian University of Athens); Panagiota Kaoura (National and Kapodistrian University of Athens)

Journal: Energies

Publisher: MDPI AG

DOI: 10.3390/en15249406

URL: <https://doi.org/10.3390/en15249406>

Publication Date: 2022-12-13

Volume/Issue: Volume 15, Issue 24

References: 49

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The National Observatory of Athens intends to operate a European Climate Change Observatory (ECCO) on the island of Antikythera, which meets the criteria to become a first-class research infrastructure. This project requires electricity that is unprofitable to get from the thermal units of this small island (20 km²). Solar energy is the subject that was examined in case it can give an environmentally and economically viable solution, both for the observatory and for the whole island. Specifically, observational and modeled data were utilized relevant to solar dynamic and atmospheric parameters in order to simulate the solar energy production by photovoltaics (PV) and Concentrated Solar Power (CSP) plant technologies. To this direction, a synergy of aerosol and cloud optical properties from the Copernicus Atmosphere Monitoring Service (CAMS) and the Eumetsat's support to nowcasting and very short range forecasting (NWC SAF) with Radiative Transfer Model (RTM) techniques was used in order to quantify the solar radiation and energy production as well as the effect of the atmospheric parameters and to demonstrate energy adequacy scenarios and financial analysis. The ultimate goal is to highlight the opportunity for energy transition and autonomy for both the island itself and the rest of the community with the operation of ECCO, and hence to tackle climate change.

404. Testing JENDL-5 data for U-235, Pu-239 and U-238 above the resolved resonance region

Authors: Sandro Pelloni; Dimitri Rochman

Journal: Annals of Nuclear Energy

Publisher: Elsevier BV

DOI: 10.1016/j.anucene.2022.109603

URL: <https://doi.org/10.1016/j.anucene.2022.109603>

Publication Date: 2022-12-12

Volume/Issue: Volume 182

References: 33

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

405. On the detectability of the magnetic fields induced by ocean circulation in geomagnetic satellite observations

Authors: Aaron Hornschild; Julien Baerenzung; Jan Saynisch-Wagner; Christopher Irrgang; Maik Thomas

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-022-01741-z

URL: <https://doi.org/10.1186/s40623-022-01741-z>

Publication Date: 2022-12-09

Volume/Issue: Volume 74, Issue 1

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Helmholtz-Zentrum Potsdam Deutsches GeoForschungsZentrum - GFZ

License: CC BY 4.0

Open Access: N/A

Abstract: Due to their sensitivity to conductivity and oceanic transport, magnetic signals caused by the movement of the ocean are a beneficial source of information. Satellite observed tidal-induced magnetic fields have already proven to be helpful to derive Earth's conductivity or ocean heat content. However, magnetic signals caused by ocean circulation are still unobserved in satellite magnetometer data. We present a novel method to detect these magnetic signals from ocean circulation using an observing system simulation experiment. The introduced approach relies on the assimilation of satellite magnetometer data based on a Kalman filter algorithm. The separation from other magnetic contributions is attained by predicting the temporal behavior of the ocean-induced magnetic field through presumed proxies. We evaluate the proposed method in different test case scenarios. The results demonstrate a possible detectability of the magnetic signal in large parts of the ocean. Furthermore, we point out the crucial dependence on the magnetic signal's variability and show that our approach is robust to slight spatial and temporal deviations of the presumed proxies. Additionally, we showed that including simple prior spatial constraints could further improve the assimilation results. Our findings indicate an appropriate sensitivity of the detection method for an application outside the presented observing system simulation experiment. Therefore, we finally discussed potential issues and required advances toward the method's application on original geomagnetic satellite observations. Graphical Abstract

406. Seasonal modulation of oceanic seismicity in the azores

Authors: Ana L. Lordi; Maria C. Neves; Susana Custódio; Stéphanie Dumont

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2022.995401

URL: <https://doi.org/10.3389/feart.2022.995401>

Publication Date: 2022-12-08

Volume/Issue: Volume 10

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Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Faculdade de Ciências e Tecnologia, Universidade Nova de Lisboa

License: CC BY 4.0

Open Access: N/A

Abstract: The analysis of an 11-year (2008-2018) seismic catalogue of the Azores suggests the existence of secondary cyclic influences on the seismicity rate of the oceanic region, with more earthquakes observed during the summer months, from May to August, than in winter. Statistical testing based on Monte-Carlo simulations and a Jack-Knife methodology indicate that the seasonal modulation affects earthquakes with magnitudes M3.3-4.5, well above the magnitude of completeness. Here, we investigate the seasonal variations of earthquake rate considering both the whole Azores oceanic domain and four separate sub-regions, corresponding to four regional clusters identified by previous authors. The analysis shows that the seasonal modulation is particularly observed near the Triple Junction region between the Faial Island and the Mid-Atlantic Ridge. To identify possible mechanisms driving the seasonal modulation, we apply Singular Spectral Analysis to the seismicity rate and to time-series of plausible external triggers, in order to investigate possible common periodicities. We find significant correlations between the earthquake rate, sea level anomaly rate, GRACE satellite anomalies and ocean bottom pressure, suggesting that water load may modulate the Azores oceanic seismicity.

407. Editorial for the Special Issue "Remote Sensing of the Polar Oceans"

Authors: Giuseppe Aulicino (Department of Science and Technology, Università degli Studi di Napoli Parthenope, 80143 Napoli, Italy); Peter Wadhams (Università Politecnica delle Marche)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs14246195

URL: <https://doi.org/10.3390/rs14246195>

Publication Date: 2022-12-07

Volume/Issue: Volume 14, Issue 24

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Keywords: N/A

Funding: Italian National Programme for Research in Antarctica (PNRA)

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Open Access: N/A

Abstract: This Special Issue gathers papers reporting research on various aspects of the use of satellites for monitoring polar oceans. It includes contributions presenting improvements in the retrieval of sea ice concentration, extent and area, and concerning error information; the interannual and decadal variability of sea surface temperature and sea ice concentration in the Barents Sea; validation and comparison of Arctic salinity products; melt pond retrieval applying a Linear Polar algorithm to Landsat data; the characterization of surface layer freshening from sea surface salinity and coloured detrital matter in the Kara and Laptev Seas; multi-sensor estimations of chlorophyll-a concentrations in the Western Antarctic Peninsula; and enhanced techniques for detection and monitoring of glacier dynamics and iceberg paths.

408. A modeling study of hydrographic and flow variability along the river-influenced coastal ocean off central Chile

Authors: Cristian M. Rojas; Gonzalo S. Saldías; Raúl P. Flores; Sebastián I. Vásquez; Cristian Salas; Cristian A. Vargas

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2022.102155

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Publication Date: 2022-12-05

Volume/Issue: Volume 181

References: 85

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

409. Do Submesoscales Affect the Large-Scale Structure of the Upper Ocean?

Authors: Anirban Sinha (California Institute of Technology); Jörn Callies (California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0129.1

URL: <https://doi.org/10.1175/jpo-d-22-0129.1>

Publication Date: 2022-12-05

Volume/Issue: Volume 53, Issue 4

References: 60

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Eddies, Thermocline circulation, Boundary currents, Mesoscale processes, Ocean dynamics

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, California Institute of Technology

License: N/A

Open Access: N/A

Abstract: Submesoscale baroclinic instabilities have been shown to restratify the surface mixed layer and to seasonally energize submesoscale turbulence in the upper ocean. But do these instabilities also affect the large-scale circulation and stratification of the upper thermocline? This question is addressed for the North Atlantic Subtropical Mode Water region with a series of numerical simulations at varying horizontal grid spacings (16, 8, 4, and 2 km). These simulations are realistically forced and integrated long enough for the thermocline to adjust to the presence or absence of submesoscales. Linear stability analysis indicates that a 2-km grid spacing is sufficient to resolve the most unstable mode of the wintertime mixed layer instability. As the resolution is increased, spectral slopes of horizontal kinetic energy flatten and vertical velocities increase in magnitude, consistent with previous regional and short-time simulations. The equilibrium stratification of the thermocline changes drastically as the grid spacing is refined from 16 to 8 km and mesoscale eddies are fully resolved. The thermocline stratification remains largely unchanged, however, between the 8-, 4-, and 2-km runs. This robustness is argued to arise from a mesoscale constraint on the buoyancy variance budget. Once mesoscale processes are resolved, the rate of mesoscale variance production is largely fixed. This constrains the variance destruction by submesoscale vertical buoyancy fluxes, which thus remain invariant across resolutions. The bulk impact of mixed layer instabilities on upper-ocean stratification in the Subtropical Mode Water region through an enhanced vertical buoyancy flux is therefore captured at 8-km grid spacing, even though the instabilities are severely underresolved.

410. The increasing big gap of carbon sink between the western and eastern Pacific in the last three decades

Authors: Guorong Zhong; Xuegang Li; Jinming Song; Baoxiao Qu; Fan Wang; Yanjun Wang; Bin Zhang; Detong Tian; Jun Ma; Huamao Yuan; Liqin Duan; Ning Li; Qidong Wang; Jianwei Xing

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2022.1088181

URL: <https://doi.org/10.3389/fmars.2022.1088181>

Publication Date: 2022-12-02

Volume/Issue: Volume 9

References: 44

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Key Research and Development Program of China

License: CC BY 4.0

Open Access: N/A

Abstract: The Pacific Ocean is one of the important carbon sink regions, and there is a significant west-east difference in sea-air CO₂ flux. However, the influence of the long-standing greater CO₂ uptakes in the western Pacific than in the east and the dynamic change of this west-east difference remain unclear. In this paper, using the gridded surface ocean pCO₂ product constructed by the stepwise FFNN algorithm, we reported an increasing west-east CO₂ flux difference from 0.41 PgC yr⁻¹ in 1992 to 0.73 PgC yr⁻¹ in 2020. This increase was mainly attributed to the strengthening western Pacific carbon sink and relatively stable eastern Pacific carbon source. During El Nino events, the west-east CO₂ flux difference decreased significantly in a few years, and it then rose back rapidly when El Nino events ended. In addition, the increasing west-east difference in CO₂ uptakes during the last three decades did not lead to a higher acidification speed in the western surface temperate Pacific than the east. The greater CO₂ absorbed in the west was mainly transported to the deeper waters and caused a more significant carbon inventory change at 200-600 m than the eastern Pacific.

411. Physical and biogeochemical responses in the Southern Ocean to a simple parameterization of Langmuir circulation

Authors: Yong-Jin Tak; Hajoon Song; Yign Noh; Yeonju Choi

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2022.102152

URL: <https://doi.org/10.1016/j.ocemod.2022.102152>

Publication Date: 2022-12-01

Volume/Issue: Volume 181

References: 90

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Science, ICT and Future Planning, National Research Foundation of Korea

License: N/A

Open Access: N/A

Abstract: N/A

412. Standing Eddies in Glacial Fjords and Their Role in Fjord Circulation and Melt

Authors: Ken X. Zhao (University of California Los Angeles); Andrew L. Stewart (University of California Los Angeles); James C. McWilliams (University of California Los Angeles); Ian G. Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric J. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0085.1

URL: <https://doi.org/10.1175/jpo-d-22-0085.1>

Publication Date: 2022-11-30

Volume/Issue: Volume 53, Issue 3

References: 60

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Vortices, Ice sheets, Eddies, Snowmelt/icemelt

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Glacial fjord circulation modulates the connection between marine-terminating glaciers and the ocean currents offshore. These fjords exhibit a complex 3D circulation with overturning and horizontal recirculation components, which are both primarily driven by water mass transformation at the head of the fjord via subglacial discharge plumes and distributed meltwater plumes. However, little is known about the 3D circulation in realistic fjord geometries. In this study, we present high-resolution numerical simulations of three glacial fjords (Ilulissat, Sermilik, and Kangerdlugssuaq), which exhibit along-fjord overturning circulations similar to previous studies. However, one important new phenomenon that deviates from previous results is the emergence of multiple standing eddies in each of the simulated fjords, as a result of realistic fjord geometries. These standing eddies are long-lived, take months to spin up, and prefer locations over the widest regions of deep-water fjords, with some that periodically merge with other eddies. The residence time of Lagrangian particles within these eddies are significantly larger than waters outside of the eddies. These eddies are most significant for two reasons: 1) they account for a majority of the vorticity dissipation required to balance the vorticity generated by discharge and meltwater plume entrainment and act to spin down the overall recirculation and 2) if the eddies prefer locations near the ice face, their azimuthal velocities can significantly increase melt rates. Therefore, the existence of standing eddies is an important factor to consider in glacial fjord circulation and melt rates and should be taken into account in models and observations.

413. Global patterns in marine organic matter stoichiometry driven by phytoplankton ecophysiology

Authors: Keisuke Inomura; Curtis Deutsch; Oliver Jahn; Stephanie Dutkiewicz; Michael J. Follows

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-022-01066-2

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Publication Date: 2022-11-21

Volume/Issue: Volume 15, Issue 12

References: 69

Cited by: 24 publications

Language: en

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Keywords: N/A

Funding: Simons Foundation, Gordon and Betty Moore Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: The proportion of major elements in marine organic matter links cellular processes to global nutrient, oxygen and carbon cycles. Differences in the C:N:P ratios of organic matter have been observed between ocean biomes, but these patterns have yet to be quantified from the underlying small-scale physiological and ecological processes. Here we use an ecosystem model that includes adaptive resource allocation within and between ecologically distinct plankton size classes to attribute the causes of global patterns in the C:N:P ratios. We find that patterns of N:C variation are largely driven by common physiological adjustment strategies across all phytoplankton, while patterns of N:P are driven by ecological selection for taxonomic groups with different phosphorus storage capacities. Although N:C varies widely due to cellular adjustment to light and nutrients, its latitudinal gradient is modest because of depth-dependent trade-offs between nutrient and light availability. Strong latitudinal variation in N:P reflects an ecological balance favouring small plankton with lower P storage capacity in the subtropics, and larger eukaryotes with a higher cellular P storage capacity in nutrient-rich high latitudes. A weaker N:P difference between southern and northern hemispheres, and between the Atlantic and Pacific oceans, reflects differences in phosphate available for cellular storage. Despite simulating only two phytoplankton size classes, the emergent global variability of elemental ratios resembles that of all measured species, suggesting that the range of growth conditions and ecological selection sustain the observed diversity of stoichiometry among phytoplankton.

414. A regime shift in the Southeast Greenland marine ecosystem

Authors: Mads Peter Heide-Jørgensen (Greenland Institute of Natural Resources); Philippine Chambault (University of California Santa Cruz, Greenland Institute of Natural Resources); Teunis Jansen (Greenland Institute of Natural Resources, DTU Aqua Institute of Aquatic Resources Lyngby Denmark); Caroline V. B. Gjelstrup (DTU Aqua Institute of Aquatic Resources Lyngby Denmark); Aqqalu Rosing-Asvid (Greenland Institute of Natural Resources); Andreas Macrander (Marine and Freshwater Research Institute Hafnarfjörður Iceland); Gísli Víkingsson (Marine and Freshwater Research Institute Hafnarfjörður Iceland); Xiangdong Zhang (University of Alaska Fairbanks); Camilla S. Andresen (GEUS København K Denmark); Brian R. MacKenzie (DTU Aqua Institute of Aquatic Resources Lyngby Denmark)

Journal: Global Change Biology

Publisher: Wiley

DOI: 10.1111/gcb.16494

URL: <https://doi.org/10.1111/gcb.16494>

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Volume/Issue: Volume 29, Issue 3

References: 102

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Two major oceanographic changes have recently propagated through several trophic levels in coastal areas of Southeast Greenland (SEG). Firstly, the amount of drift-ice exported from the Fram Strait and transported with the East Greenland Current (EGC) has decreased significantly over the past two decades, and a main tipping element (summer sea ice) has virtually disappeared since 2003 leading to a regime shift in oceanographic and ecological conditions in the region. The following 20-year period with low or no coastal sea ice is unique in the 200-year history of ice observations in the region, and the regime shift is also obvious in the volume of ice export through the Fram Strait after 2013. In the same period, the temperature of the EGC south of 73.5 N has increased significantly (>2 degreesC) since 1980. Secondly, the warm Irminger Current, which advects warm, saline Atlantic Water into the region, has become warmer since 1990. The lack of pack ice in summer together with a warming ocean generated cascading effects on the ecosystem in SEG that are manifested in a changed fish fauna with an influx of boreal species in the south and the subarctic capelin further north. At higher trophic levels there has been an increase in the abundance of several boreal cetaceans (humpback, fin, killer, and pilot whales and dolphins) that are either new to this area or occur in historically large numbers. It is estimated that the new cetacean species in SEG are responsible for an annual predation level of 700,000 tons of fish. In addition, predation on krill species is estimated at >1,500,000 tons mainly consumed by fin whales. Simultaneously, there has been a reduction in the abundance and catches of narwhals and walrus in SEG and it is suggested that these species have been impacted by the habitat changes.

415. Mesoscale and submesoscale turbulence in the Northwest Pacific Ocean revealed by numerical simulations

Authors: Shihong Wang; Zhenya Song; Weidong Ma; Qi Shu; Fangli Qiao

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2022.105221

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Volume/Issue: Volume 206

References: 46

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Ames Research Center, National Natural Science Foundation of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

416. The reemergence of the winter sea surface temperature tripole in the North Atlantic from ocean reanalysis data

Authors: Pavel A. Sukhonos; Michael A. Alexander

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-022-06581-x

URL: <https://doi.org/10.1007/s00382-022-06581-x>

Publication Date: 2022-11-21

Volume/Issue: Volume 61, Issue 1-2

References: 37

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Russian Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

417. Using Seafloor Geodesy to Detect Vertical Deformation at the Hikurangi Subduction Zone: Insights From Self-Calibrating Pressure Sensors and Ocean General Circulation Models

Authors: K. Woods (Victoria University of Wellington); S. C. Webb (Columbia University); L. M. Wallace (Institute for Geophysics University of Texas Austin TX USA, GNS Science Lower Hutt New Zealand); Y. Ito (Kyoto University); C. Collins (National Institute of Water and Atmospheric Research Wellington New Zealand); N. Palmer (GNS Science Lower Hutt New Zealand); R. Hino (Tohoku University); M. K. Savage (Victoria University of Wellington); D. M. Saffer (Institute for Geophysics University of Texas Austin TX USA); E. E. Davis (Pacific Geoscience Centre Geological Survey of Canada Sidney BC Canada); D. H. N. Barker (GNS Science Lower Hutt New Zealand)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jb023989

URL: <https://doi.org/10.1029/2022jb023989>

Publication Date: 2022-11-18

Volume/Issue: Volume 127, Issue 12

References: 61

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry for Business Innovation and Employment, Directorate for Geosciences, Ministry of Education, Culture, Sports, Science and Technology

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Open Access: N/A

Abstract: Seafloor pressure sensor data is emerging as a promising approach to resolve vertical displacement of the seafloor in the offshore reaches of subduction zones, particularly in response to slow slip events (SSEs), although such signals are challenging to resolve due to sensor drift and oceanographic signals. Constraining offshore SSE slip distribution is of key importance to understanding earthquake and tsunami hazards posed by subduction zones. We processed seafloor pressure data from January to October 2019 acquired at the Hikurangi subduction zone, offshore New Zealand, to estimate vertical displacement associated with a large SSE that occurred beneath the seafloor array. The experiment included three self-calibrating sensors designed to remove sensor drift, which, together with ocean general circulation models, were essential to the identification and correction of long-period ocean variability remaining in the data after applying traditional processing techniques. We estimate that long-period oceanographic signals that were not synchronous between pressure sensors and reference sites influenced our inferred displacements by 0.3-2.6 cm, suggesting that regionally deployed reference sites alone may not provide sufficient ocean noise correction. After incorporating long-period ocean variability corrections into the processing, we calculate 1.0-3.3 cm of uplift during the SSE offshore Gisborne at northern Hikurangi, and 1.1-2.7 cm of uplift offshore the Hawke's Bay area at central Hikurangi. Some Hawke Bay displacements detected by pressure sensors near the trench were delayed by 6 weeks compared to the timing of slip onset detected by onshore

Global Navigation Satellite System sites, suggesting updip migration of the SSE.

418. Seasonal Variation of Intra-Seasonal Eddy Kinetic Energy along the East Australian Current

Authors: Zhipeng Xu (Hohai University, Ministry of Natural Resources); Chengcheng Yang (Hohai University, Ministry of Natural Resources); Xiao Chen (Hohai University, Southern Marine Science and Engineering Guangdong Laboratory, Ministry of Natural Resources); Yiquan Qi (Hohai University, Ministry of Natural Resources)

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w14223725

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References: 24

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key R&D Program of China, Postgraduate Research & Practice Innovation Program of Jiangsu Province

License: CC BY 4.0

Open Access: N/A

Abstract: By using satellite altimeter observations and the eddy-permitting Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2), the seasonal variation of eddy kinetic energy (EKE) along the East Australian Current (EAC) is investigated. Both observations and ECCO2 outputs indicate active intra-seasonal EKE along the EAC path. The ECCO2 result reveals that the intra-seasonal EKE is mainly concentrated in the upper 500 m layer, and shows a prominent seasonal cycle, strong in austral summer and weak in austral winter. Eddy energy budget diagnosis reveals that the evolution of EKE is controlled by barotropic instability of the mean EAC. The seasonal variation of baroclinic instability is opposite to the barotropic instability variation, but of a much smaller magnitude. Further analysis indicates that the seasonal cycle of mesoscale signals in this region is related to the transport variability of the EAC.

419. Reconstructing ocean subsurface salinity at high resolution using a machine learning approach

Authors: Tian Tian; Lijing Cheng; Gongjie Wang; John Abraham; Wangxu Wei; Shihe Ren; Jiang Zhu; Junqiang Song; Hongze Leng

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-14-5037-2022

URL: <https://doi.org/10.5194/essd-14-5037-2022>

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Cited by: 17 publications

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Keywords: N/A

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Open Access: N/A

Abstract: . A gridded ocean subsurface salinity dataset with global coverage is useful for research on climate change and its variability. Here, we explore the feed-forward neural network (FFNN) approach to reconstruct a high-resolution ($0.25^\circ \times 0.25^\circ$) ocean subsurface (1–2000 m) salinity dataset for the period 1993–2018 by merging in situ salinity profile observations with high-resolution ($0.25^\circ \times 0.25^\circ$) satellite remote-sensing altimetry absolute dynamic topography (ADT), sea surface temperature (SST), sea surface wind (SSW) field data, and a coarse-resolution ($1^\circ \times 1^\circ$) gridded salinity product. We show that the FFNN can effectively transfer small-scale spatial variations in ADT, SST, and SSW fields into the $0.25^\circ \times 0.25^\circ$ salinity field. The root-mean-square error (RMSE) can be reduced by ~11 % on a global-average basis compared with the $1^\circ \times 1^\circ$ salinity gridded field. The reduction in RMSE is much larger in the upper ocean than the deep ocean because of stronger mesoscale variations in the upper layers. In addition, the new $0.25^\circ \times 0.25^\circ$ reconstruction shows more realistic spatial signals in the regions with strong mesoscale variations, e.g., the Gulf Stream, Kuroshio, and Antarctic Circumpolar Current regions, than the $1^\circ \times 1^\circ$ resolution product, indicating the efficiency of the machine learning approach in bringing satellite observations together with in situ observations. The large-scale salinity patterns from $0.25^\circ \times 0.25^\circ$ data are consistent with the $1^\circ \times 1^\circ$ gridded salinity field, suggesting the persistence of the large-scale signals in the high-resolution reconstruction. The successful application of machine learning in this study provides an alternative approach for ocean and climate data reconstruction that can complement the existing data assimilation and objective analysis methods. The reconstructed IAP0.25 dataset is freely available at <https://doi.org/10.57760/sciencedb.o00122.00001> (Tian et al., 2022).

420. Showcasing model performance across space and time using single diagrams

Authors: Arya Paul; Afroosa M.; Balaji Baduru; Biswamoy Paul

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2022.102150

URL: <https://doi.org/10.1016/j.ocemod.2022.102150>

Publication Date: 2022-11-17

Volume/Issue: Volume 181

References: 20

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

421. Unique thermal expansion properties of water key to the formation of sea ice on Earth

Authors: Fabien Roquet (University of Gothenburg); David Ferreira (University of Reading); Romain Caneill (University of Gothenburg); Daniel Schlesinger (Stockholm University); Gurvan Madec (LOCEAN, Sorbonne Universités, UPMC, Paris, France.)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.abq0793

URL: <https://doi.org/10.1126/sciadv.abq0793>

Publication Date: 2022-11-16

Volume/Issue: Volume 8, Issue 46

References: 66

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

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License: N/A

Open Access: N/A

Abstract: The formation of sea ice in polar regions is possible because a salinity gradient or halocline keeps the water column stable despite intense cooling. Here, we demonstrate that a unique water property is central to the maintenance of the polar halocline, namely, that the thermal expansion coefficient (TEC) of seawater increases by one order of magnitude between polar and tropical regions. Using a fully coupled climate model, it is shown that, even with excess precipitations, sea ice would not form at all if the near-freezing temperature TEC was not well below its ocean average value. The leading order dependence of the TEC on temperature is essential to the coexistence of the mid/low-latitude thermally stratified and the high-latitude sea ice-covered oceans that characterize our planet. A key implication is that nonlinearities of water properties have a first-order impact on the global climate of Earth and possibly exoplanets.

422. Modulation of the Eastern Equatorial Pacific Seasonal Cycle by Tropical Instability Waves

Authors: L. Maillard (Institut de Recherche pour le Développement (IRD), University of Toulouse); J. Boucharel (Institut de Recherche pour le Développement (IRD), Department of Atmospheric Sciences School of Ocean and Earth Science and Technology (SOEST) University of Hawai'i at Mānoa Honolulu HI USA, University of Toulouse); M. F. Stuecker (Department of Oceanography and International Pacific Research Center (IPRC) School of Ocean and Earth Science and Technology (SOEST) University of Hawai'i at Mānoa Honolulu HI USA); F.-F. Jin (Department of Atmospheric Sciences School of Ocean and Earth Science and Technology (SOEST) University of Hawai'i at Mānoa Honolulu HI USA); L. Renault (Institut de Recherche pour le Développement (IRD), University of Toulouse)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl100991

URL: <https://doi.org/10.1029/2022gl100991>

Publication Date: 2022-11-14

Volume/Issue: Volume 49, Issue 23

References: 59

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Agence Nationale de la Recherche, Division of Atmospheric and Geospace Sciences, Climate Program Office (and 2 more)

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Open Access: N/A

Abstract: Feedbacks from tropical instability waves (TIWs) on the seasonal cycle of the eastern Pacific Ocean are studied using two eddy-rich ocean simulations, with and without TIWs. By warming the equatorial waters by up to 0.4 degreesC through nonlinear advection in boreal summer and fall, TIWs reduce the amplitude of the seasonal cycle in upper ocean temperatures. In addition, TIWs stabilize the upper part of the Equatorial Undercurrent (EUC) through enhanced barotropic energy conversion, leading to a year-round weakening by -0.15 m s⁻¹ and preventing an unrealistic re-intensification in boreal fall usually found in non-eddy resolving models. A coarser simulation at 1-degree horizontal resolution fails to reproduce the TIW-induced nonlinear warming of equatorial waters, but succeeds in inhibiting the EUC re-intensification. This suggests a threshold effect in TIW strength, associated with the model's ability to simulate eddies, which may be responsible for long-standing biases displayed by global climate models in this region.

423. Dispersal and coastal geomorphology limit potential for mangrove range expansion under climate change

Authors: Jacqueline L. Raw (Nelson Mandela University); Tom Van der Stocken (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Vrije Universiteit Brussel); Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); Linda R. Harris (Nelson Mandela University); Anusha Rajkaran (University of the Western Cape); Lara Van Niekerk (Nelson Mandela University, Council for Scientific and Industrial Research (CSIR)); Janine B. Adams (Nelson Mandela University)

Journal: Journal of Ecology

Publisher: Wiley

DOI: 10.1111/1365-2745.14020

URL: <https://doi.org/10.1111/1365-2745.14020>

Publication Date: 2022-11-14

Volume/Issue: Volume 111, Issue 1

References: 76

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Research Foundation, Nelson Mandela University, Water Research Commission

License: Creative Commons

Open Access: N/A

Abstract: Latitudinal range limits for mangroves on high-energy, wave-dominated coasts are controlled by geomorphological features and estuarine dynamics. Mangroves reach a southern global range limit along the South African coastline, but the distribution is patchy, with stands occurring in only 16% of the estuaries in the region. Yet, the persistence of forests planted >50 years ago beyond the natural distribution limit suggests that additional estuaries could support mangroves. Understanding regional drivers is necessary to inform global-scale estimates for how this important ecosystem is predicted to respond to climate change. Here, we combine species distribution modelling (MaxEnt), Lagrangian particle tracking using an eddy- and tide-resolving numerical ocean model, and connectivity matrices, to identify suitable mangrove habitats along the South African coastline at present, as well as under the IPCC RCP4.5 and RCP8.5 climate scenarios. Within the current South African distribution range (± 900 km), eight more estuaries were identified to be suitable under contemporary conditions. When considering potential range extension (± 110 km), an additional 14 suitable estuaries were identified. Connectivity matrices suggest limited long-distance dispersal, stranding mostly at or near the release location, and a decreased probability of connectivity towards the range limit. Under both future climate scenarios, 30% of estuaries currently supporting mangroves are predicted to become unsuitable, while an additional six estuaries beyond the current distribution are predicted to become suitable. However, there is limited connectivity between these new sites and established forests. **Synthesis.** This study shows that dispersal substantially limits mangrove distribution at the southern African range limit and highlights the importance of including this process in species distribution models. Ultimately, our results provide new insight into mangrove conservation and management at range limits that are not controlled

predominantly by temperature, as it has been assumed that mangroves will largely expand to higher latitudes under climate change.

424. Diagnosing the upper ocean variability in the Northern Bay of Bengal during the super cyclone Phailin using a high-resolution regional ocean model

Authors: Lokesh Kumar Pandey; Suneet Dwivedi; Alok Kumar Mishra

Journal: Theoretical and Applied Climatology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00704-022-04275-2

URL: <https://doi.org/10.1007/s00704-022-04275-2>

Publication Date: 2022-11-11

Volume/Issue: Volume 151, Issue 1-2

References: 66

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

425. Indonesian Throughflow Slowdown under Global Warming: Remote AMOC Effect versus Regional Surface Forcing

Authors: Qihua Peng (Scripps Institution of Oceanography / University of California San Diego); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego); Rui Xin Huang (Woods Hole Oceanographic Institution); Weiqiang Wang (Chinese Academy of Sciences); Tingting Zu (Chinese Academy of Sciences); Dongxiao Wang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-22-0331.1

URL: <https://doi.org/10.1175/jcli-d-22-0331.1>

Publication Date: 2022-11-08

Volume/Issue: Volume 36, Issue 5

References: 50

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: Climate change, Ocean, Transport, Ocean circulation, General circulation models

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, National Natural Science Foundation of China (and 3 more)

License: N/A

Open Access: N/A

Abstract: The Indonesian Throughflow (ITF) is projected to slow down under anthropogenic warming. Several mechanisms--some mutually conflicting--have been proposed but the detailed processes causing this slowdown remain unclear. By turning on/off buoyancy and wind forcings globally and in key regions, this study investigates the dynamical adjustments underlying the centennial ITF slowdown in the global oceans and climate models. Our results show that the projected weakened ITF transport in the top 1500 m is dominated by remote anomalous buoyancy forcing in the North Atlantic Ocean. Specifically, surface freshening and warming over the North Atlantic Ocean slow the Atlantic meridional overturning circulation (AMOC), and the resultant dynamic signals propagate through the coastal-equatorial waveguide into the southeastern Indian Ocean and western Pacific Ocean, causing the reduction of ITF transport over a deep layer. In contrast, the anomalous surface buoyancy flux in the Indo-Pacific affects the ocean temperature and salinity in a shallow upper layer, resulting in ITF changes in forms of high baroclinic mode structure with negligible impacts on the net ITF transport. A vertical partitioning index is proposed to distinguish the remote forcing via the AMOC and regional forcing in the Indo-Pacific Ocean, which could be useful for monitoring, attributing, and predicting the changing ITF transport under global warming.

426. Influence of GEOTRACES data distribution and misfit function choice on objective parameter retrieval in a marine zinc cycle model

Authors: Claudia Eisenring; Sophy E. Oliver; Samar Khatiwala; Gregory F. de Souza

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-19-5079-2022

URL: <https://doi.org/10.5194/bg-19-5079-2022>

Publication Date: 2022-11-07

Volume/Issue: Volume 19, Issue 21

References: 103

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Schweizerischer Nationalfonds zur Förderung der Wissenschaftlichen Forschung, Natural Environment Research Council

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Open Access: N/A

Abstract: . Biogeochemical model behaviour for micronutrients is typically hard to constrain because of the sparsity of observational data, the difficulty of determining parameters in situ, and uncertainties in observations and models. Here, we assess the influence of data distribution, model uncertainty, and the misfit function on objective parameter optimisation in a model of the oceanic cycle of zinc (Zn), an essential micronutrient for marine phytoplankton with a long whole-ocean residence time. We aim to investigate whether observational constraints are sufficient for reconstruction of biogeochemical model behaviour, given that the Zn data coverage provided by the GEOTRACES Intermediate Data Product 2017 is sparse. Furthermore, we aim to assess how optimisation results are affected by the choice of the misfit function and by confounding factors such as analytical uncertainty in the data or biases in the model related to either seasonal variability or the larger-scale circulation. The model framework applied herein combines a marine Zn cycling model with a state-of-the-art estimation of distribution algorithm (Covariance Matrix Adaption Evolution Strategy, CMA-ES) to optimise the model towards synthetic data in an ensemble of 260 optimisations. Provided with a target field that can be perfectly reproduced by the model, optimisation retrieves parameter values perfectly regardless of data coverage. As differences between the model and the system underlying the target field increase, the choice of the misfit function can greatly impact optimisation results, while limitation of data coverage is in most cases of subordinate significance. In cases where optimisation to full or limited data coverage produces relatively distinct model behaviours, we find that applying a misfit metric that compensates for differences in data coverage between ocean basins considerably improves agreement between optimisation results obtained with the two data situations.

427. The atmospheric hinder for intraseasonal sea-air interaction over the Bay of Bengal during Indian summer monsoon in CMIP6

Authors: Ze Meng; Lei Zhou; Baosheng Li; Jianhuang Qin; Juncheng Xie

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-022-2023-3

URL: <https://doi.org/10.1007/s13131-022-2023-3>

Publication Date: 2022-11-04

Volume/Issue: Volume 41, Issue 10

References: 51

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

428. Intra-seasonal variability of the abyssal currents in COMRA's contract area in the Clarion-Clipperton Zone

Authors: Fangfang Kuang; Jing Cha; Junpeng Zhang; Aijun Pan; Hangyu Chen; Xiwu Zhou; Chunsheng Jing; Xiaogang Guo

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-021-1945-5

URL: <https://doi.org/10.1007/s13131-021-1945-5>

Publication Date: 2022-11-04

Volume/Issue: Volume 41, Issue 11

References: 25

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

429. Optimally growing initial error for predicting the sudden shift in the Antarctic Circumpolar Current transport and its application to targeted observation

Authors: Li Zhou; Kun Zhang; Qiang Wang; Mu Mu

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-022-01531-x

URL: <https://doi.org/10.1007/s10236-022-01531-x>

Publication Date: 2022-11-01

Volume/Issue: Volume 72, Issue 11-12

References: 66

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fundamental Research Funds for the Central Universities, National Natural Science Foundation of China, Key Technologies Research and Development Program

License: N/A

Open Access: N/A

Abstract: N/A

430. On Constraining the Mesoscale Eddy Energy Dissipation Time-Scale

Authors: J. Mak (Department of Ocean Science Hong Kong University of Science and Technology Kowloon Hong Kong, Center for Ocean Research in Hong Kong and Macau Hong Kong University of Science and Technology Kowloon Hong Kong); A. Avdis (Imperial College London, HPC Department Boston Limited St. Albans UK); T. David (Laboratoire de Physique, École Normale Supérieure de Lyon Lyon France); H. S. Lee (Department of Ocean Science Hong Kong University of Science and Technology Kowloon Hong Kong); Y. Na (Department of Ocean Science Hong Kong University of Science and Technology Kowloon Hong Kong); Y. Wang (Department of Ocean Science Hong Kong University of Science and Technology Kowloon Hong Kong, Center for Ocean Research in Hong Kong and Macau Hong Kong University of Science and Technology Kowloon Hong Kong); F. E. Yan (Department of Ocean Science Hong Kong University of Science and Technology Kowloon Hong Kong)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022ms003223

URL: <https://doi.org/10.1029/2022ms003223>

Publication Date: 2022-10-30

Volume/Issue: Volume 14, Issue 11

References: 96

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: eddy parameterization, inverse methods, ocean energetics, parameter inference, ocean modeling

Funding: Research Grants Council, University Grants Committee

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Open Access: N/A

Abstract: A physically plausible lower bound on the spatially varying geostrophic mesoscale eddy energy dissipation time-scale within the ocean, related to the geographical energy transfer rate out of the geostrophic mesoscales, is provided by means of a simple and computational inexpensive inverse calculation. Data diagnosed from a high resolution global configuration ocean simulation is supplied to a parameterized model of the geostrophic mesoscale eddy energy, from which the dissipation time-scale results as a solution to an optimization calculation. We find that the dissipation time-scale is shortest in the Southern Ocean, in the Western Boundary Currents, and on the western boundaries, consistent with the expectation that these regions are notable sites of baroclinic activity with processes leading to energy transfer out of the geostrophic mesoscales. Although our solution should be interpreted as a lower bound given the assumptions going into the calculation, it serves as an important physically consistent base line reference for further investigations into ocean energetics, as well as for an intended inference calculation that is more complete but also much more complex.

431. Introduction to Marine Data Source Analysis and Sharing

Authors: Shanshan Geng; Zhenmin Liu; Xiaoxiao Shi

Journal: E3S Web of Conferences

Publisher: EDP Sciences

DOI: 10.1051/e3sconf/202235801046

URL: <https://doi.org/10.1051/e3sconf/202235801046>

Publication Date: 2022-10-27

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References: 18

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: You should leave 8 mm of space above the abstract and 10 mm after the abstract. The heading Abstract should be typed in bold 9-point Times. The body of the abstract should be typed in normal 9-point Arial in a single paragraph, immediately following the heading. The text should be set to 1 line spacing. The abstract should be centred across the page, indented 17 mm from the left and right page margins and justified. It should not normally exceed 200 words.

432. Wind-Induced Variability of Warm Water on the Southern Bellingshausen Sea Continental Shelf

Authors: Ria Oelerich (University of East Anglia); Karen J. Heywood (University of East Anglia); Gillian M. Damerell (University of East Anglia); Andrew F. Thompson (California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jc018636

URL: <https://doi.org/10.1029/2022jc018636>

Publication Date: 2022-10-26

Volume/Issue: Volume 127, Issue 11

References: 62

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: H2020 European Research Council, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: The Bellingshausen Sea hosts heat transport onto the continental shelf, potentially enhancing ice shelf basal melt. Here, we use the GLORYS12V1 1993-2018 reanalysis to identify physical processes that set seasonal and interannual variability of water mass properties in the Eltanin and Latady Bays on the southern Bellingshausen Sea continental shelf. Annual means of potential temperature from 300 m to the seabed reveal interannual variability and allow separation into warm and cold regimes. The Amundsen Sea Low (ASL) is more intense and extends further east during the warm regime than the cold regime. In the warm regime, a wind-induced reduction of sea ice concentration near the coast increases surface heat loss, convection, and formation of cold dense water in winter, associated with a decrease in heat content of the southern Bellingshausen Sea over time and a net northward heat transport. In contrast, in the cold regime, increased sea ice concentration reduces surface heat loss and thus formation of cold, dense water. Combined with an increase in heat content over time and a net southward heat transport, this results in a warming of the southern Bellingshausen Sea. This suggests that variability in the deep water temperature in the southern Bellingshausen Sea is primarily due to local surface heat fluxes above the shelf. The variability of surface heat fluxes is related to the variability of the ASL and its influence on sea ice extent and local formation of cold, dense water in winter.

433. Asymmetric Response of Sea Surface Salinity to Extreme Positive and Negative Indian Ocean Dipole in the Southern Tropical Indian Ocean

Authors: Qiwei Sun (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Yuhong Zhang (Chinese Academy of Sciences); Yan Du (Chinese Academy of Sciences); Xingwei Jiang (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory, National Satellite Ocean Application Service Beijing China)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jc018986

URL: <https://doi.org/10.1029/2022jc018986>

Publication Date: 2022-10-25

Volume/Issue: Volume 127, Issue 11

References: 74

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Chinese Academy of Sciences

License: N/A

Open Access: N/A

Abstract: Based on the Argo and multi-satellite observations, this study investigates the asymmetric response of sea surface salinity (SSS) to two extreme positive and negative Indian Ocean Dipole (IOD) events in the southern tropical Indian Ocean (TIO). During positive IOD events, the easterly wind anomalies trigger zonal freshwater advection along the equator, decreasing SSS, while cold sea surface temperature (SST) depress the convective precipitation in the southeastern TIO (SETIO), increasing the SSS. Contrast SSS anomalies along the equator and in the SETIO show opposite phases during positive and negative IOD events. These SSS anomalies have enhanced in recent years, reaching the strongest in the extreme negative IOD in 2016 and positive IOD in 2019. The case study shows that positive SSS anomalies occurred in the southern TIO during both positive and negative IOD events in 2016 and 2019, suggesting significant asymmetry in SSS. Analysis of the mixed layer salinity budget shows that the local precipitation and the entrainment associated with the upwelling Rossby waves played an essential role in the SSS increase in 2016, while the decreased precipitation and horizontal advection caused by the westward South Equatorial Current contributed to the SSS increase in 2019. Further analyses suggest that SSS response is more sensitive to thermocline shoaling than deepening, and SSS response to precipitation is greater at shallower mixed layer than at deeper ones, which is the main reason for the asymmetric response of the SSS to negative and positive IOD events.

434. Sensitivity of Chlorophyll Variability to Specific Growth Rate of Phytoplankton Equation over the Yangtze River Estuary in a Physical-Biogeochemical Model

Authors: Qiong Wu (Nanjing University, Division of Environment Science and Energy, Pohang University of Science and Technology, Pohang 37673, Korea, Zhejiang Ocean University, Ministry of Natural Resources); Xiaochun Wang (Nanjing University); Peng Xiu (Chinese Academy of Sciences); Fei Chai (University of Maine); Zhongxiao Chen (Nanjing University)

Journal: Atmosphere

Publisher: MDPI AG

DOI: 10.3390/atmos13111748

URL: <https://doi.org/10.3390/atmos13111748>

Publication Date: 2022-10-24

Volume/Issue: Volume 13, Issue 11

References: 53

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research Project of China, China Scholarship Council, Public Science and Technology Research Fund Projects for Ocean Research

License: CC BY 4.0

Open Access: N/A

Abstract: In addition to nutrients and light, temperature plays a crucial role in marine biogeochemical processes. In this study, the sensitivity of the growth rate of phytoplankton to temperature was systematically studied by using a two-level nested physical-biogeochemical coupled model for the Yangtze River estuary of the East China Sea. The physical component of the coupled model is configured from the Regional Ocean Modeling System (ROMS) with the highest horizontal resolution of 3 km. The biogeochemical component of the coupled model is based on the carbon, silicon and nitrogen ecosystem model (CoSiNE). Five specific growth rate of phytoplankton equations with different relation to temperature were tested with the objective of reproducing the temporal evolution of chlorophyll concentration as observed by SeaWiFS. Our results indicate that the specific growth rate of phytoplankton equation which is from Geider's work, reaches a maximum at 22 degreesC and remains constant with higher temperature, can reproduce the seasonal variation of chlorophyll very well, and may be suitable for application in the physical-biogeochemical coupled model (ROMS-CoSiNE) of the Yangtze River estuary.

435. Gravimetry-based terrigenous freshwater extension in the southwestern South China Sea and its response to monsoon under ENSO

Authors: Zhongtian Ma; Hok Sum Fok

Journal: Science of The Total Environment

Publisher: Elsevier BV

DOI: 10.1016/j.scitotenv.2022.159583

URL: <https://doi.org/10.1016/j.scitotenv.2022.159583>

Publication Date: 2022-10-19

Volume/Issue: Volume 857

References: 81

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Center for Atmospheric Research

License: N/A

Open Access: N/A

Abstract: N/A

436. Decadal Variability of the Pacific Shallow Overturning Circulation and the Role of Local Wind Forcing

Authors: Antonietta Capotondi (University of Colorado Boulder, b NOAA/Physical Sciences Laboratory, Boulder, Colorado); Bo Qiu (c Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-22-0408.1

URL: <https://doi.org/10.1175/jcli-d-22-0408.1>

Publication Date: 2022-10-19

Volume/Issue: Volume 36, Issue 3

References: 73

Cited by: 17 publications

Language: N/A

Subjects: N/A

Keywords: ENSO, Ekman pumping, Oceanic variability, Rossby waves, Decadal variability

Funding: Climate Program Office, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: The Pacific shallow meridional overturning circulations, known as subtropical cells (STCs), link subduction in the subtropical regions to equatorial upwelling, suggesting the possibility for subtropical winds to influence equatorial sea surface temperatures (SSTs) by altering the STCs' strength. Indeed, STC variability provides the basis for one of the mechanisms proposed to explain the origin of tropical Pacific decadal variability (TPDV). While the relationship between STC strength, as measured by their subsurface transport convergence, and equatorial SST variations is well documented, the location of the wind forcing most influential on STC variability is still being debated. In this study, we use the output of an ocean reanalysis to examine tropical Pacific Ocean surface and subsurface decadal changes during recent decades and relate them to STC variability and surface wind forcing. Our results indicate that the STC interior transport at each latitude is largely controlled by the wind forcing at that latitude rather than induced by remote subtropical wind variations. We also show that the establishment of the anomalous transport at each latitude is associated with the westward propagation of oceanic wind-forced Rossby waves, as part of the ocean adjustment process that also leads to a zonal redistribution of upper-ocean heat content at both interannual and decadal time scales. These results provide guidance for understanding the origin of TPDV by elucidating the underlying dynamics of STC variability and can have practical implications for monitoring STC variability in the tropical Pacific. Significance Statement Slow variations of the surface ocean temperature in the tropical Pacific Ocean have been shown to affect the global climate. Our study aims at better understanding the origin of these temperature anomalies by taking a closer look at the upper ocean circulation variability and its relationship with surface wind forcing. Unlike previous studies, which have related the upper ocean circulation changes to wind variations outside the tropical Pacific, we show here that the variations in upper-ocean circulation are primarily driven by local winds. This result not only clarifies which winds are most important, but also suggests a practical approach for monitoring circulation changes from surface observations.

437. Seasonal Structure and Interannual Variation of the South Equatorial Current in the Indian Ocean

Authors: Gengxin Chen (Chinese Academy of Sciences); Weiqing Han (University of Colorado Boulder); Dongxiao Wang (Sun Yat-Sen University); Lei Zhang (Chinese Academy of Sciences); Xiaoqing Chu (Chinese Academy of Sciences); Yunkai He (Chinese Academy of Sciences); Ju Chen (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jc018969

URL: <https://doi.org/10.1029/2022jc018969>

Publication Date: 2022-10-18

Volume/Issue: Volume 127, Issue 11

References: 46

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: SEC, freshening event, Indian ocean

Funding: Chinese Academy of Sciences, National Natural Science Foundation of China, Youth Innovation Promotion Association of the Chinese Academy of Sciences

License: N/A

Open Access: N/A

Abstract: The South Equatorial Current (SEC) in the south Indian Ocean (SIO) contributes to mass and heat exchanges among the Pacific, Indian, and Atlantic Oceans. By analyzing satellite and in situ observations, this study examines the seasonal structure and the interannual variability of the SEC. The SEC is mainly part of the Seychelles-Chagos thermocline ridge (SCTR) circulation during December to April, and is composed of the internal SCTR circulation and the intrusive Indonesian Throughflow during May to November. The SEC thus presents straight and meandering routes in these two periods, respectively. Furthermore, observations and model experiments show that the SEC's meandering route has obvious interannual variability, with exceptional northwestward extension in 1994, 1997, and 2019 during the past three decades. These exceptional routes are caused primarily by extreme positive Indian Ocean Dipole (IOD) events, which induce a strengthened westward transport of the fresh water. These exceptional routes thus reflect the abnormal freshening in the SIO. Mooring observations reveal that a significant long-term freshening event occurred in the SIO from August 2019 to at least July 2020. The persistent low salinity confirms that there is substantial water exchange in the SIO during the extreme positive IOD years.

438. Effect of tides on river water behavior over the eastern shelf seas of China

Authors: Lei Lin; Hao Liu; Xiaomeng Huang; Qingjun Fu; Xinyu Guo

Journal: Hydrology and Earth System Sciences

Publisher: Copernicus GmbH

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Keywords: N/A

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Abstract: . Rivers carry large amounts of freshwater and terrestrial material into shelf seas, which is an important part of the global water and biogeochemical cycles. The earth system model or climate model is an important instrument for simulating and projecting the global water cycle and climate change, in which tides however are commonly removed. For a better understanding of the potential effect of the absence of tides in the simulation of the water cycle, this study compared the results of a regional model with and without considering tides, and evaluated the effect of tides on the behavior of three major rivers (i.e., the Yellow, Yalujiang, and Changjiang rivers) water in the eastern shelf seas of China from the perspectives of transport pathways, timescales, and water concentration. The results showed that the tides induced more dispersed transport for the water of the Yellow and Yalujiang rivers, but more concentrated transport for the Changjiang River water. The effect of tides on the transit areas of the Yellow, Yalujiang, and Changjiang rivers was 13 %, 40 %, and 21 %, respectively. The annual mean water age and transit time of the three rivers in the model with tides were several (~ 2-10) times higher than those in the no-tide model, suggesting that tides dramatically slow the river water transport and export rate over the shelf. By slowing the river water export, tides induced a three-fold increase in river water concentration and a decrease in shelf seawater salinity by > 1. Moreover, the effect of tides on river behavior was stronger in relatively enclosed seas (i.e., the Bohai and Yellow seas) than in relatively open seas (i.e., the East China Sea). The change in the shelf currents induced by tides is the main cause of the difference in the river water behavior between the two model runs. Tides can increase bottom stress and thus weaken shelf currents and decrease the water transport timescales. The improvement in tidal parameterization in the no-tide model in the simulation of river water behavior was very limited. Given the important role of river runoff on the global water cycle and the effect of changes in river water behavior on ocean carbon cycling, it is important to include the tidal effect in earth system models to improve their projection accuracy.

439. Automated identification of dominant physical processes

Authors: Bryan E. Kaiser; Juan A. Saenz; Maike Sonnewald; Daniel Livescu

Journal: Engineering Applications of Artificial Intelligence

Publisher: Elsevier BV

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Open Access: N/A

Abstract: N/A

440. Decadal Weakening of Abyssal South China Sea Circulation

Authors: Yaohua Zhu (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Jingxin Yao (Tianjin Sub-bureau of North China Regional Air Traffic Management Bureau CAAC Tianjin China); Shujiang Li (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Tengfei Xu (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Rui Xin Huang (Woods Hole Oceanographic Institution); Xunwei Nie (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Haidong Pan (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Yonggang Wang (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Yue Fang (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Zexun Wei (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: South China Sea, weakening trend, abyssal circulation, ocean bottom pressure, ECCO (and 1 more)

Funding: N/A

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Open Access: N/A

Abstract: The global ocean mass has significantly changed during recent decades, which may induce changes in the abyssal ocean circulation. Due to the lack of in situ observations, the response of the abyssal South China Sea (SCS) circulation to ocean mass changes remains unclear. The ocean state estimate, Estimating the Circulation and Climate of the Ocean (ECCO), provides the long-term ocean mass changes in terms of ocean bottom pressure (OBP). Here we use the ECCO OBP data to quantify decadal changes in the abyssal SCS circulation and reveal a weakening trend. The OBP gradient trend manifests in forms of an anomalous anticyclonic circulation that is intensified by topographic effects along the continental slopes to reduce the existing abyssal circulation. The weakening abyssal circulation corresponds to the weakening Luzon Strait overflow that yields a decreasing vorticity budget

in the abyssal SCS.

441. Estimating Ocean Heat Uptake Using Boundary Green's Functions: A Perfect-Model Test of the Method

Authors: Quran Wu (University of Reading); Jonathan M. Gregory (Met Office Hadley Centre, Met Office, University of Reading)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: passive tracer, maximum entropy, ocean heat uptake, Green's function

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License: CC BY 4.0

Open Access: N/A

Abstract: Ocean heat uptake is caused by "excess heat" being added to the ocean surface by air-sea fluxes and then carried to depths by ocean transports. One way to estimate excess heat in the ocean is to propagate observed sea surface temperature (SST) anomalies downward using a Green's function (GF) representation of ocean transports. Taking a "perfect-model" approach, we test this GF method using a historical simulation, in which the true excess heat is diagnosed. We derive GFs from two approaches: (a) simulating GFs using idealized tracers, and (b) inferring GFs from simulated CFCs and climatological tracers. In the model world, we find that combining simulated GFs with SST anomalies reconstructs the Indo-Pacific excess heat with a root-mean-square error of 26% for depth-integrated changes; the corresponding number is 34% for inferred GFs. Simulated GFs are inaccurate because they are coarse grained in space and time to reduce computational cost. Inferred GFs are inaccurate because observations are insufficient constraints. Both kinds of GFs neglect the slowdown of the North Atlantic heat uptake as the ocean warms up. SST boundary conditions contain redistributive cooling in the Southern Ocean, which causes an underestimate of heat uptake there. All these errors are of comparable magnitude, and tend to compensate each other partially. Inferred excess heat is not sensitive to: (a) small changes in the shape of prior GFs, or (b) additional constraints from SF6 and bomb 14C.

442. Explainable Artificial Intelligence for Bayesian Neural Networks: Toward Trustworthy Predictions of Ocean Dynamics

Authors: Mariana C. A. Clare (Imperial College London, European Centre for Medium-Range Weather Forecasts (ECMWF)); Maike Sonnewald (Ocean and Cryosphere Division NOAA/Geophysical Fluid Dynamics Laboratory Princeton NJ USA, University of Washington, Princeton University); Redouane Lguensat (Institut de Recherche pour le Développement (IRD)); Julie Deshayes (LOCEAN-IPSL CNRS Sorbonne Université Paris France); V. Balaji (Laboratoire des Sciences du Climat et de l'Environnement CEA Saclay Gif Sur Yvette France, Ocean and Cryosphere Division NOAA/Geophysical Fluid Dynamics Laboratory Princeton NJ USA, Princeton University)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: changing climate, explainable AI, interpretability, dynamical ocean regimes, Bayesian Neural Networks (and 1 more)

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Open Access: N/A

Abstract: The trustworthiness of neural networks is often challenged because they lack the ability to express uncertainty and explain their skill. This can be problematic given the increasing use of neural networks in high stakes decision-making such as in climate change applications. We address both issues by successfully implementing a Bayesian Neural Network (BNN), where parameters are distributions rather than deterministic, and applying novel implementations of explainable AI (XAI) techniques. The uncertainty analysis from the BNN provides a comprehensive overview of the prediction more suited to practitioners' needs than predictions from a classical neural network. Using a BNN means we can calculate the entropy (i.e., uncertainty) of the predictions and determine if the probability of an outcome is statistically significant. To enhance trustworthiness, we also spatially apply the two XAI techniques of Layer-wise Relevance Propagation (LRP) and SHapley Additive exPlanation (SHAP) values. These XAI methods reveal the extent to which the BNN is suitable and/or trustworthy. Using two techniques gives a more holistic view of BNN skill and its uncertainty, as LRP considers neural network parameters, whereas SHAP considers changes to outputs. We verify these techniques using comparison with intuition from physical theory. The differences in explanation identify potential areas where new physical theory guided studies are needed.

443. Modelling the growth of atmospheric nitrous oxide using a global hierarchical inversion

Authors: Angharad C. Stell; Michael Bertolacci; Andrew Zammit-Mangion; Matthew Rigby; Paul J. Fraser; Christina M. Harth; Paul B. Krummel; Xin Lan; Manfredi Manizza; Jens Mühle; Simon O'Doherty; Ronald G. Prinn; Ray F. Weiss; Dickon Young; Anita L. Ganesan

Journal: Atmospheric Chemistry and Physics

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Language: en

Subjects: N/A

Keywords: Nitrous oxide

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Open Access: N/A

Abstract: . Nitrous oxide is a potent greenhouse gas (GHG) and ozone-depleting substance, whose atmospheric abundance has risen throughout the contemporary record. In this work, we carry out the first global hierarchical Bayesian inversion to solve for nitrous oxide emissions, which includes prior emissions with truncated Gaussian distributions and Gaussian model errors, in order to examine the drivers of the atmospheric surface growth rate. We show that both emissions and climatic variability are key drivers of variations in the surface nitrous oxide growth rate between 2011 and 2020. We derive increasing global nitrous oxide emissions, which are mainly driven by emissions between 0 and 30° N, with the highest emissions recorded in 2020. Our mean global total emissions for 2011–2020 of 17.2 (16.7–17.7 at the 95 % credible intervals) Tg N yr⁻¹, comprising of 12.0 (11.2–12.8) Tg N yr⁻¹ from land and 5.2 (4.5–5.9) Tg N yr⁻¹ from ocean, agrees well with previous studies, but we find that emissions are poorly constrained for some regions of the world, particularly for the oceans. The prior emissions used in this and other previous work exhibit a seasonal cycle in the extra-tropical Northern Hemisphere that is out of phase with the posterior solution, and there is a substantial zonal redistribution of emissions from the prior to the posterior. Correctly characterizing the uncertainties in the system, for example in the prior emission fields, is crucial for deriving posterior fluxes that are consistent with observations. In this hierarchical inversion, the model-measurement discrepancy and the prior flux uncertainty are informed by the data, rather than solely through "expert judgement". We show cases where this framework provides different plausible adjustments to the prior fluxes compared to inversions using widely adopted, fixed uncertainty constraints.

444. Recent Changes of Phytoplankton Bloom Phenology in the Northern High-Latitude Oceans (2003-2020)

Authors: Haipeng Zhao (Department of Earth and Environmental Systems Indiana State University Terre Haute IN USA); Atsushi Matsuoka (University of New Hampshire); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Amos Winter (Department of Earth and Environmental Systems Indiana State University Terre Haute IN USA)

Journal: Journal of Geophysical Research: Oceans

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Keywords: phytoplankton bloom, high latitude, remote sensing, phenology

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Abstract: Eighteen years (2003-2020) of data on remotely-sensed ocean color in high northern latitudes show a significant increase (0.56% yr⁻¹) in double blooms of phytoplankton. Starting in 2015 the rate of increase rose to 1.10% yr⁻¹. Double blooms have become especially prevalent in regions north of 50 degreesN in the North Atlantic, Gulf of Alaska and southern Chukchi Sea. Our correlation analysis suggests that double bloom occurrences are dependent on latitude and seasonal changes in both sea surface temperature (SST) and mixed layer depth (MLD), particularly in the subpolar North Pacific (SNP) and polar oceans (65 degrees-75 degreesN), although we cannot attribute the increased double blooms to specific causes. Single blooms indicated by Chlorophyll a (Chl a) dominate in subpolar-polar waters and occur close to the timing of the Particulate Inorganic Carbon (PIC) peak (i.e., June-August). In the SNP single bloom, Chl a peaks occur earlier than PIC peaks. This interval was shorter during 2016-2020, which is associated with warming anomalies. In the southern part of the Chukchi Sea, however, we found that PIC peaks occur earlier than Chl a peaks, suggesting a potentially different succession pattern between calcifiers and other functional groups. We speculate that in the near future double blooms of phytoplankton will become a prevalent feature in high-latitude oceans with concurrent changes in the timing of blooms and seasonal phytoplankton succession in the surface ocean due to climate change.

445. What Controls the Partition between the Cold and Warm Routes in the Meridional Overturning Circulation?

Authors: Louise Rousselet (Scripps Institution of Oceanography / University of California San Diego); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

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Keywords: Lagrangian circulation/transport, Mass fluxes/transport, Eddy diffusion|Eddy|Boundary current, Trajectories, Streamfunction (and 2 more)

Funding: National Science Foundation, National Aeronautics and Space Administration, National Science Foundation (and 3 more)

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Open Access: N/A

Abstract: The origins of the upper limb of the Atlantic meridional overturning circulation and the partition among different routes has been quantified with models at eddy-permitting and one eddy-resolving model or with low-resolution models assimilating observations. Here, a step toward bridging this gap is taken by using the Southern Ocean State Estimate (SOSE) at the eddy-permitting 1/6 degrees horizontal resolution to compute Lagrangian diagnostics from virtual particle trajectories advected between 6.7 degreesS and two meridional sections: one at Drake Passage (cold route) and the other from South Africa to Antarctica (warm route). Our results agree with the prevailing concept attributing the largest transport contribution to the warm route with 12.3 Sv (88%) (1 Sv $\equiv$ 106 m³ s⁻¹) compared with 1.7 Sv (12%) for the cold route. These results are compared with a similar Lagrangian experiment performed with the lower-resolution state estimate from Estimating the Circulation and Climate of the Ocean. Eulerian and Lagrangian means highlight an overall increase in the transport of the major South Atlantic currents with finer resolution, resulting in a relatively larger contribution from the cold route. In particular, the Malvinas Current to Antarctic Circumpolar Current (MC/ACC) ratio plays a more important role on the routes partition than the increased Agulhas Leakage. The relative influence of the mean flow versus the eddy flow on the routes partition is investigated by computing the mean and eddy kinetic energies and the Lagrangian-based eddy diffusivity. Lagrangian diffusivity estimates are largest in the Agulhas and Malvinas regions but advection by the mean flow dominates everywhere.

446. A nutrient relay sustains subtropical ocean productivity

Authors: Mukund Gupta (California Institute of Technology, Massachusetts Institute of Technology); Richard G. Williams (University of Liverpool); Jonathan M. Lauderdale (Massachusetts Institute of Technology); Oliver Jahn (Massachusetts Institute of Technology); Christopher Hill (Massachusetts Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Michael J. Follows (Massachusetts Institute of Technology)

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Publisher: Proceedings of the National Academy of Sciences

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Open Access: N/A

Abstract: The expansive gyres of the subtropical ocean account for a significant fraction of global organic carbon export from the upper ocean. In the gyre interior, vertical mixing and the heaving of nutrient-rich waters into the euphotic layer sustain local productivity, in turn depleting the layers below. However, the nutrient pathways by which these subeuphotic layers are themselves replenished remain unclear. Using a global, eddy-permitting simulation of ocean physics and biogeochemistry, we quantify nutrient resupply mechanisms along and across density surfaces, including the contribution of eddy-scale motions that are challenging to observe. We find that mesoscale eddies (10 to 100 km) flux nutrients from the shallow flanks of the gyre into the recirculating interior, through time-varying motions along density surfaces. The subeuphotic layers are ultimately replenished in approximately equal contributions by this mesoscale eddy transport and the remineralization of sinking particles. The mesoscale eddy resupply is most important in the lower thermocline for the whole subtropical region but is dominant at all depths within the gyre interior. Subtropical gyre productivity may therefore be sustained by a nutrient relay, where the lateral transport resupplies nutrients to the thermocline and allows vertical exchanges to maintain surface biological production and carbon export.

447. Submarine melting of glaciers in Greenland amplified by atmospheric warming

Authors: D. A. Slater; F. Straneo

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

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Keywords: N/A

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Open Access: N/A

Abstract: Rapid ice loss from the Greenland ice sheet since 1992 is due in equal parts to increased surface melting and accelerated ice flow. The latter is conventionally attributed to ocean warming, which has enhanced submarine melting of the fronts of Greenland's marine-terminating glaciers. Yet, through the release of ice sheet surface meltwater into the ocean, which excites near-glacier ocean circulation and in turn the transfer of heat from ocean to ice, a warming atmosphere can increase submarine melting even in the absence of ocean warming. The relative importance of atmospheric and oceanic warming in driving increased submarine melting has, however, not been quantified. Here, we reconstruct the rate of submarine melting at Greenland's marine-terminating glaciers from 1979 to 2018 and estimate the resulting dynamic mass loss. We show that in south Greenland, variability in submarine melting was indeed governed by the ocean, but, in contrast, the atmosphere dominated in the northwest. At the ice sheet scale, the atmosphere plays a first-order role in controlling submarine melting and the subsequent dynamic mass loss. Our results challenge the attribution of dynamic mass loss to ocean warming alone and show that a warming atmosphere has amplified the impact of the ocean on the Greenland ice sheet.

448. Analysis of Global Sea Level Change Based on Multi-Source Data

Authors: Yongjun Jia (Key Laboratory of Space Ocean Remote Sensing and Application (MNR), Beijing 100081, China, Southern Marine Science and Engineering Guangdong Laboratory, National Satellite Ocean Application Service, Beijing 100081, China); Kailin Xiao (School of Marine Technology and Geomatics, Jiangsu Ocean University, Lianyungang 222005, China); Mingsen Lin (National Satellite Ocean Application Service, Beijing 100081, China); Xi Zhang (National Satellite Ocean Application Service, Beijing 100081, China)

Journal: Remote Sensing

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Keywords: N/A

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Open Access: N/A

Abstract: Global sea level rise is both a major indicator and consequence of global warming. At present, global warming is causing sea level rise in two main ways: one is the thermal expansion of sea water, and the other is the injection of large amounts of fresh water into the ocean after glaciers and ice sheets melt. In this paper, satellite altimeter data are used to study the total changes of global sea level from 2002 to 2020. Different from most previous studies, this study proposes a calculation method of sea level anomaly using only the along track altimetry data, which is similar to considering the satellite points as tide gauges, in order to avoid the error caused by interpolation in the map data. In addition, GRACE satellite data are used to calculate the changes of global sea level caused by water increase; temperature and salinity data are used to calculate the changes from ocean thermal expansion. Next, using satellite altimetry data, the calculation results show that the global sea level rise rate in the period of 2002-2020 is 3.3 mm/a. During this period, the sea level change caused by the increase of sea water calculated with GRACE satellite data is 2.07 mm/a, and that caused by the thermal expansion of seawater is 0.62 mm/a. The sea level rise caused by the increase of water volume accounts for 62.7% of the total sea level rise.

449. Uncertainties in ocean biogeochemical simulations: Application of ensemble data assimilation to a one-dimensional model

Authors: Nabir Mamnun; Christoph Völker; Mihalís Vrekoussis; Lars Nerger

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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Open Access: N/A

Abstract: Marine biogeochemical (BGC) models are highly uncertain in their parameterization. The value of the BGC parameters are poorly known and lead to large uncertainties in the model outputs. This study focuses on the uncertainty quantification of model fields and parameters within a one-dimensional (1-D) ocean BGC model applying ensemble data assimilation. We applied an ensemble Kalman filter provided by the Parallel Data Assimilation Framework (PDAF) into a 1-D vertical configuration of the BGC model Regulated Ecosystem Model 2 (REcoM2) at two BGC time-series stations: the Bermuda Atlantic Time-series Study (BATS) and the Dynamique des Flux Atmosphériques en Méditerranée (DYFAMED). We assimilated 5-day satellite chlorophyll-a (chl-a) concentration and monthly in situ net primary production (NPP) data for 3 years to jointly estimate 10 preselected key BGC parameters and the model state. The estimated set of parameters resulted in improvements in the model prediction up to 66% for the surface chl-a and 56% for NPP. Results show that assimilating satellite chl-a concentration data alone degraded the prediction of NPP. Simultaneous assimilation of the satellite chl-a data and in situ NPP data improved both surface chl-a and NPP simulations. We found that correlations between parameters preclude estimating parameters independently. Co-dependencies between parameters also indicate that there is not a unique set of optimal parameters. Incorporation of proper uncertainty estimation in BGC predictions, therefore, requires ensemble simulations with varying parameter values.

450. The inhibition of warm advection on the southward expansion of sea ice during early winter in the Bering Sea

Authors: Weibo Wang; Jie Su; Chunsheng Jing; Xiaogang Guo

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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Open Access: N/A

Abstract: Recent observations demonstrate that the Bering Sea exhibits a substantial positive trend of sea ice area increment (Δ SIA, difference in SIA between the current and preceding months) in January contrasted to the considerable negative sea ice area (SIA) trend from 1979 to 2020, and the Δ SIA is unrelated to the local wind field anomaly. To better understand the January Δ SIA variability and its physical characteristics, we explore two distinct empirical orthogonal function (EOF) modes of sea ice concentration increments. EOF1 features a reduction in sea ice concentration (SIC) in the south of St. Lawrence Island. EOF2 is characterized by the rise of SIC surrounding St. Lawrence Island. EOF1 is related to the well-known physical process of December strong poleward heat transport in mixed layer depth. During the southward expansion of sea ice, the multiyear variation of the December SST tendency mostly relies on warm advection in the Bering Sea shelf rather than net air-sea heat flux, and the abnormal northeast wind in December no longer plays the role of a dynamic process dominating the ice area expansion, but generates a stronger poleward heat transport in the Bering Sea shelf to inhibit the southward development of sea ice in the later stage. The two physical processes together result in oceanic poleward heat transport regulating the Bering Sea SIA in competition with atmospheric forcing in early winter. Since PC1 (principal component (PC) time series for EOF1) has a high correlation of -0.76 with the maximum SIA in the Bering Sea, it can be used as the prediction index of the Bering Sea maximum SIA.

451. Near-Surface Oceanic Kinetic Energy Distributions From Drifter Observations and Numerical Models

Authors: Brian K. Arbic (University of Michigan); Shane Elipot (University of Miami); Jonathan M. Brasch (University of Michigan); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Aurélien L. Ponte (Institut de Recherche pour le Développement (IRD)); Jay F. Shriver (Ocean Sciences Division, Naval Research Laboratory Stennis Space Center Hancock County MS USA); Xiaolong Yu (Sun Yat-Sen University, Institut de Recherche pour le Développement (IRD)); Edward D. Zaron (Oregon State University); Matthew H. Alford (Scripps Institution of Oceanography / University of California San Diego); Maarten C. Buijsman (Division of Marine Science University of Southern Mississippi Stennis Space Center, Hancock County MS USA); Ryan Abernathey (Columbia University); Daniel Garcia (University of Michigan); Lingxiao Guan (University of Michigan); Paige E. Martin (University of Michigan, Columbia University); Arin D. Nelson (University of Rhode Island, University of Michigan)

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Open Access: N/A

Abstract: The geographical variability, frequency content, and vertical structure of near-surface oceanic kinetic energy (KE) are important for air-sea interaction, marine ecosystems, operational oceanography, pollutant tracking, and interpreting remotely sensed velocity measurements. Here, KE in high-resolution global simulations (HYbrid Coordinate Ocean Model; HYCOM, and Massachusetts Institute of Technology general circulation model; MITgcm), at the sea surface (0 m) and at 15 m, are compared with KE from undrogued and drogued surface drifters, respectively. Global maps and zonal averages are computed for low-frequency (<0.5 cpd), near-inertial, diurnal, and semidiurnal bands. Both models exhibit low-frequency equatorial KE that is low relative to drifter values. HYCOM near-inertial KE is higher than in MITgcm, and closer to drifter values, probably due to more frequently updated atmospheric forcing. HYCOM semidiurnal KE is lower than in MITgcm, and closer to drifter values, likely due to inclusion of a parameterized topographic internal wave drag. A concurrent tidal harmonic analysis in the diurnal band demonstrates that much of the diurnal flow is nontidal. We compute simple proxies of near-surface vertical structure--the ratio 0 m KE/(0 m KE + 15 m KE) in model outputs, and the ratio undrogued KE/(undrogued KE + drogued KE) in drifter observations. Over most latitudes and frequency bands, model ratios track the drifter ratios to within error bars. Values of

this ratio demonstrate significant vertical structure in all frequency bands except the semidiurnal band. Latitudinal dependence in the ratio is greatest in diurnal and low-frequency bands.

452. Sensitivity of the Ross Ice Shelf to environmental and glaciological controls

Authors: Francesca Baldacchino; Mathieu Morlighem; Nicholas R. Golledge; Huw Horgan; Alena Malyarenko

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Cited by: 3 publications

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Keywords: N/A

Funding: Ministry of Science and Innovation, New Zealand, Royal Society Te Apārangi

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Open Access: N/A

Abstract: . The Ross Ice Shelf (RIS) is currently stable but recent observations have indicated that basal melt rates beneath the ice shelf are expected to increase. It is important to know which areas of the RIS are more sensitive to enhanced basal melting as well as other external forcings or internal material properties of the ice to understand how climate change will influence RIS mass balance. In this paper, we use automatic differentiation and the Ice Sheet and Sea-level System Model to quantify the sensitivity of the RIS to changes in basal friction, ice rigidity, surface mass balance, and basal melting. Using volume above flotation (VAF) as our quantity of interest, we find that the RIS is most sensitive to changes in basal friction and ice rigidity close to grounding lines and along shear margins of the Siple Coast Ice Streams and Transantarctic Mountains Outlet Glaciers. The RIS sensitivity to surface mass balance is uniform over grounded ice, while the sensitivity to basal melting is more spatially variable. Changes in basal melting close to the grounding lines of the Siple Coast Ice Streams and Transantarctic Mountains outlet glaciers have a larger impact on the final VAF compared to elsewhere. Additionally, the pinning points and ice shelf shear margins are highly sensitive to changes in basal melt. Our sensitivity maps allow areas of greatest future vulnerability to be identified.

453. Decadal Cooling Events in the South Indian Ocean During the Argo Era

Authors: Lei Huang (Xiamen University, University of Delaware, Southern Marine Science and Engineering Guangdong Laboratory); Wei Zhuang (Xiamen University, Southern Marine Science and Engineering Guangdong Laboratory); Zelun Wu (Xiamen University, University of Delaware); Lingsheng Meng (Xiamen University, University of Delaware); Deanna Edwing (Xiamen University, University of Delaware); Kelsea Edwing (Xiamen University, University of Delaware); Lina Wang (Xiamen University, University of Delaware); Xiao-Hai Yan (Xiamen University, University of Delaware)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China

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Open Access: N/A

Abstract: Despite the rapid warming in the south Indian Ocean (SIO) during the recent two decades, the upper 300 m of the tropical southwest Indian Ocean (SWIO) experienced significant cooling during 2005-2011. The decreasing rate of the observed temperature reached -0.50 ± 0.1 degreesC/decade. Heat budget analysis reveals that the cooling trend over SWIO results from the competing effect between local air-sea heat flux and horizontal heat advection. Moreover, all the three observational datasets indicate that in the subtropical SIO, a prolonged cooling event occurred beneath the thermocline during 2010-2016. Although there exist some discrepancies between the observations and estimating the circulation and climate of the ocean (ECCO) reanalysis before 2010, the observations and ECCO reanalysis agree in terms of the post-2010 temperature variability in the intermediate layer of the subtropical region, probably because of the better constraint of the ECCO assimilation product by the increased sampling of Argo floats in the deeper ocean. The heat budget analysis indicates that during the period 2010-2016, the decadal cooling in the 400-1,000 m layer is primarily modulated by vertical ocean advection, and the horizontal ocean advection makes a secondary contribution. This study expands our knowledge of the decadal temperature variability and heat balance in the SIO during the Argo era.

454. Remote-Sensing Derived Trends in Gross Primary Production Explain Increases in the CO₂ Seasonal Cycle Amplitude

Authors: Liyin He (California Institute of Technology); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yi Yin (California Institute of Technology); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christian Frankenberg (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: satellite remote sensing, net ecosystem exchange, peak season photosynthesis, seasonal cycle of atmospheric CO₂, climate change

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License: N/A

Open Access: N/A

Abstract: An increase in the seasonal cycle amplitude (SCA) of atmospheric CO₂ since the 1960s has been observed in the Northern Hemisphere (NH). However, the underlying dominant drivers are still debated. The peak season CO₂ uptake by vegetation is critical in shaping the CO₂ seasonality. Using satellite-upscaled gross primary production (GPP) from FLUXCOM and near-infrared reflectance of vegetation (NIRV), we demonstrate that peak GPP has increased across the NH over the last two decades. We relate this productivity increase to changes in the CO₂ SCA using an atmospheric transport model. The increased photosynthesis has strongly contributed to CO₂ SCA trends, but with substantial latitudinal and longitudinal variations. Despite a general increase in the CO₂ SCA, there are distinct regional differences. These differences are mainly controlled by regional biosphere carbon fluxes, with the remainder explained by non-biome factors, including large-scale atmospheric transport, changes in fossil fuel combustion, biomass burning and oceanic fluxes. Using the global flask and in situ CO₂ measurement sites, we find that SCA trends at high latitude are mainly driven by increasingly productive natural ecosystems, whereas mid latitude sites around the Midwest United States are mainly impacted by intensified agriculture and atmospheric transport. Averaging across the 15 long-term surface sites, forests contribute 26% (7%) to the SCA trends, while crops contribute 17% (24%) and the combined shrubland, grassland and wetland regions contribute 23% (37%) for simulations driven by FLUXCOM (NIRv) ecosystem fluxes. Our findings demonstrate that satellite inferred trends of ecosystem fluxes can capture the observed CO₂ SCA trend.

455. Variability of Near-Surface Salinity in the Nordic Seas Over the Past Three Decades (1991-2019)

Authors: Ji-Eun Park; Hyun-Cheol Kim; Kyoung-Ho Cho

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2022.969159

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Publication Date: 2022-09-14

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References: 100

Cited by: N/A publications

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Keywords: N/A

Funding: Korea Polar Research Institute

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Open Access: N/A

Abstract: The Nordic Seas have been widely implicated by deep water formation as a part of Atlantic Meridional Overturning Circulation. This study explores the spatiotemporal variations in the near-surface salinity over the Nordic Seas associated with surface freshening factors by using monthly TOPAZ4 reanalysis data from 1991 to 2019. We first show that reliability of TOPAZ4 data compared to the salinity products of other reanalysis data, satellite data, and in-situ measurements in the Nordic Seas. The salinity variability was larger in the Greenland Sea (GS) than in the Norwegian Sea (NS) on both time scales of seasonal and interannual. The seasonal change of GS salinity was coincident with the seasonality of sea ice extent. The longer-time variations are decomposed by empirical orthogonal function (EOF) analysis. The GS salinity is mainly affected by current advection (29%) and sea ice extent (11%). The interannual response of salinity to the sea ice extent over the GS differs by season. NS salinity variability responds to the strength of the Subpolar Gyre associated with a large-scale atmospheric system that caused the freshening event in the mid-1990s. The propagation of the northward Atlantic Water core is observed over the period of about 3 years from the Faroe Shetland Channel to the Fram Strait at a speed of 2.6-6.5 degrees year⁻¹. Other freshening factors such as sea ice export from the Arctic, freshwater flux at the Fram Strait, and net precipitation are also discussed. For the past three decades, the continuous trend appeared only in the sea ice extent, which might be a signal of climate changes over high latitude. However, there was no significant trend other than the periodic change in a few years to the decadal time scale in the salinity of GS and NS. As preconditioning for deep convection, near-surface salinity within Greenland Sea Gyre was influenced by salinity fluctuation in both GS and NS.

456. Hydrography, circulation, and response to atmospheric forcing in the vicinity of the central Getz Ice Shelf, Amundsen Sea, Antarctica

Authors: Vår Dundas; Elin Darelius; Kjersti Daae; Nadine Steiger; Yoshihiro Nakayama; Tae-Wan Kim

Journal: Ocean Science

Publisher: Copernicus GmbH

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Language: en

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Funding: Norges Forskningsråd, Korea Polar Research Institute

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Open Access: N/A

Abstract: . Ice shelves in the Amundsen Sea are thinning rapidly as ocean currents bring warm water into the cavities beneath the floating ice. Although the reported melt rates for the Getz Ice Shelf are comparatively low for the region, its size makes it one of the largest freshwater sources around Antarctica, with potential consequences for, bottom water formation downstream, for example. Here, we use a 2-year-long novel mooring record (2016-2018) and 16-year-long regional model simulations to describe, for the first time, the hydrography and circulation in the vicinity of the ice front between Siple and Carney Island. We find that, throughout the mooring record, temperatures in the trough remain below 0.15 °C, more than 1 °C lower than in the neighboring Siple and Dotson Trough, and we observe a mean current (0.03 m s⁻¹) directed toward the ice shelf front. The variability in the heat transport toward the ice shelf appears to be governed by nonlocal ocean surface stress over the Amundsen Sea Polynya region, and northward to the continental shelf break, where strengthened westward ocean surface stress leads to increased southward flow at the mooring site. The model simulations suggest that the heat content in the trough during the observed period was lower than normal, possibly owing to anomalously low summertime sea ice concentration and weak winds.

457. Oceanic Electrical Conductivity Variability From Observations and Its Budget From an Ocean State Estimate

Authors: D. S. Trossman (Louisiana State University, University of Maryland); R. H. Tyler (University of Maryland Baltimore County, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl100453

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: ocean heat content, observations, conductivity, ECCO, variability (and 1 more)

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Open Access: N/A

Abstract: Because spatio-temporal variations in ocean heat content (OHC) are strongly predicted by ocean conductivity content (OCC) over most of the global ocean, we analyze the dynamical budget and behavior of the electrical conductivity of seawater. To perform these analyses, we use an ocean-model state estimate designed to accurately represent long-term variations in ocean properties in a dynamically and kinematically consistent way. We show that this model accurately reproduces the spatio-temporal variations in electrical conductivity seen in satellite-derived data and in a seasonal climatology product derived from in-situ data, justifying use of the model data to perform further analyses. An empirical orthogonal function analysis suggests that the vast majority of the variance in OHC and OCC can be explained by similar mechanisms. The electrical conductivity budget's most important term is the temperature forcing tendency term, suggesting that ocean heat uptake is the mechanism responsible for the strong relationship between OCC and OHC.

458. A global ocean state estimation using tidally induced vertical-mixing schemes

Authors: Satoshi Osafune; Shinya Kouketsu; Toshimasa Doi; Nozomi Sugiura; Shuhei Masuda

Journal: Ocean Modelling

Publisher: Elsevier BV

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Keywords: N/A

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Abstract: N/A

459. Multi-time scale control of Southern Ocean diapycnal mixing over Atlantic tracer budgets

Authors: Elizabeth Ellison; Laura Cimoli; Ali Mashayek

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-022-06428-5

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Oceanic cross-density (diapycnal) mixing helps sustain the ocean density stratification and its Meridional Overturning Circulation (MOC) and is key to global tracer distributions. The Southern Ocean (SO) is a key region where different overturning cells connect, allowing nutrient and carbon rich Indian and Pacific deep waters, and oxygen rich Atlantic deep waters to resurface. The SO is also rife with intense diapycnal mixing due to the interaction of energetic eddies and currents with rough topography. SO diapycnal mixing is believed to be of secondary importance for the MOC. Here we show that changes to SO mixing can cause significant alterations to biogeochemical tracer distributions over short and long time scales in an idealized model of the AMOC (Atlantic MOC). While such alterations are dominated by the direct impact of changes in diapycnal mixing on tracer fluxes on annual to decadal time scales, on centennial time scales they are dominated by the mixing-induced variations in the advective transport of the tracers by the AMOC. This work suggests that an accurate representation of spatio-temporally variable local and non-local mixing processes in the SO is essential for climate models' ability to (i) simulate the global biogeochemical cycles and air sea carbon fluxes on decadal time scales, (ii) represent the indirect impact of mixing-induced changes to AMOC on biogeochemical cycles on longer time scales.

460. Contributions of Climate Changes in Temperature and Salinity to the Formation of North Atlantic Thermohaline Circulation Trends in 1951-2017

Authors: V. A. Bagatinsky; N. A. Diansky

Journal: Moscow University Physics Bulletin

Publisher: Allerton Press

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References: 82

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

461. Quantitative Evaluation of Environmental Loading Products and Thermal Expansion Effect for Correcting GNSS Vertical Coordinate Time Series in Taiwan

Authors: Bin Liu (School of Traffic and Transportation Engineering, Changsha University of Science & Technology, Changsha 410114, China, Engineering Laboratory of Spatial Information Technology of Highway Geological Disaster Early Warning in Hunan Province, Changsha University of Science & Technology, Changsha 410114, China); Xiaojun Ma (School of Traffic and Transportation Engineering, Changsha University of Science & Technology, Changsha 410114, China, Engineering Laboratory of Spatial Information Technology of Highway Geological Disaster Early Warning in Hunan Province, Changsha University of Science & Technology, Changsha 410114, China); Xuemin Xing (School of Traffic and Transportation Engineering, Changsha University of Science & Technology, Changsha 410114, China, Engineering Laboratory of Spatial Information Technology of Highway Geological Disaster Early Warning in Hunan Province, Changsha University of Science & Technology, Changsha 410114, China); Jianbo Tan (School of Traffic and Transportation Engineering, Changsha University of Science & Technology, Changsha 410114, China, Engineering Laboratory of Spatial Information Technology of Highway Geological Disaster Early Warning in Hunan Province, Changsha University of Science & Technology, Changsha 410114, China); Wei Peng (School of Traffic and Transportation Engineering, Changsha University of Science & Technology, Changsha 410114, China, Engineering Laboratory of Spatial Information Technology of Highway Geological Disaster Early Warning in Hunan Province, Changsha University of Science & Technology, Changsha 410114, China); Liqun Zhang (School of Traffic and Transportation Engineering, Changsha University of Science & Technology, Changsha 410114, China, Engineering Laboratory of Spatial Information Technology of Highway Geological Disaster Early Warning in Hunan Province, Changsha University of Science & Technology, Changsha 410114, China)

Journal: Remote Sensing

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Language: en

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Keywords: N/A

Funding: National Natural Science Foundation of China, Natural Science Foundation of Hunan Province, China, Open Fund of Engineering Laboratory of Spatial Information Technology of Highway Geological Disaster Early Warning in Hunan Province (and 2 more)

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Open Access: N/A

Abstract: We explored the driving factors of nonlinear signals in vertical coordinate sequences of stations in a Taiwan global navigation satellite system (GNSS) network, including atmospheric loading

(ATML), hydrological loading (HYDL), and non-tidal ocean loading (NTOL) effects. At the same time, we used the finite element analysis software MIDAS to quantify the vertical displacements of different types of monuments due to the thermal expansion effect, including deep drilled braced (DDB) and short drilled braced (SDB). By quantitatively comparing the correction results of GNSS time series with different single mass loading models, we found that there was little difference in the correction of different environmental loading products. We compared different combinations of each loading product to correct the GNSS time series, and finally selected the best combination suitable for Taiwan GNSS network, that is, ATML (GFZ_ECMWF IB) + HYDL (IMLS_MERRA2) + NTOL (IMLS_MPIOM06). We found that the spatial and temporal models of ATML and NTOL are very similar, with non-tidal atmospheric loading and non-tidal ocean loading working together, a pattern that may be related to tropical cyclones. Both models also showed good correction effect on GNSS stations in the western plain of Taiwan, but with limited correction effect in the eastern part of Taiwan. This may be due to the influence of the subtropical monsoon climate in Taiwan and the barrier of the central mountain range, resulting in obvious differences between eastern and western Taiwan. The hydrological loading was found to act in the opposite way to the thermal expansion effect in the temporal domain, indicating that some displacements in hydrological loading may cancel out displacements caused by the thermal expansion effect. This aspect of displacement is not included in the hydrological loading model but should be considered when accurately estimating the temporal and spatial variation of water storage capacity in Taiwan using GNSS observed displacements.

462. Quantifying the Ocean's Biological Pump and Its Carbon Cycle Impacts on Global Scales

Authors: David A. Siegel (University of California Santa Barbara); Timothy DeVries (University of California Santa Barbara); Ivona Cetinić (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Goddard Earth Sciences Technology and Research (GESTAR) II, Morgan State University, Baltimore, Maryland, USA); Kelsey M. Bisson (Oregon State University)

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

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Abstract: The biological pump transports organic matter, created by phytoplankton productivity in the well-lit surface ocean, to the ocean's dark interior, where it is consumed by animals and heterotrophic microbes and remineralized back to inorganic forms. This downward transport of organic matter sequesters carbon dioxide from exchange with the atmosphere on timescales of months to millennia, depending on where in the water column the respiration occurs. There are three primary export pathways that link the upper ocean to the interior: the gravitational, migrant, and mixing pumps. These pathways are regulated by vastly different mechanisms, making it challenging to quantify the impacts of the biological pump on the global carbon cycle. In this review, we assess progress toward creating a global accounting of carbon export and sequestration via the biological pump and suggest a path toward achieving this goal.

463. Intra-decadal variability of the Indian Ocean shallow meridional overturning circulation during boreal winter

Authors: Rahul U. Pai; Anant Parekh; Jasti S. Chowdary; C. Gnanaseelan

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-022-06475-y

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Cited by: 5 publications

Language: en

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

464. Diabatic Transformations along the Global Routes of the Middepth Meridional Overturning Circulation

Authors: Louise Rousselet (Scripps Institution of Oceanography / University of California San Diego); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-21-0256.1

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Keywords: Abyssal circulation, Water masses/storage, Diabatic|Ekman transport|Antarctic Bottom Water|Antarctic Intermediate Water|Boundary current|Circumpolar deep water, Thermohaline circulation

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The diabatic transformations of the middepth meridional overturning circulation (MOC) as it exits and reenters the South Atlantic to close the AMOC are studied using a state estimate assimilating data into a dynamically consistent ocean model. Virtual Lagrangian parcels in the lower branch of the MOC are followed in their global tour as they return to the upper branch of the MOC. Three return pathways are identified. The first pathway enters the abyssal Indo-Pacific as Circumpolar Deep Water, directly from the northern Antarctic Circumpolar Current (ACC), and before sampling the Antarctic margin. The second pathway sinks to abyssal densities exclusively in the Southern Ocean, then upwells while circulating within the ACC and eventually enters the Indo-Pacific or Atlantic at mid- to upper depths. The third pathway never reaches densities in the abyssal range. Parcels sinking in the Antarctic Bottom Water range upwell to mid- to upper depths south of 55 degreesS. Parcels in all three pathways experience additional diabatic transformations after upwelling in the Southern Ocean, with more diabatic changes north of about 30 degreesS than elsewhere. Diabatic changes are predominantly in the mixed layer of the tropical and subpolar gyres, enabled by Ekman suction. A simple model of the wind-driven flow illustrates that parcels always reach the surface in the tropical and subpolar gyres, regardless of their initial condition, because of coupling among gyres, the Ekman transport, and its return.

465. Characteristics and Formation Conditions of Thin Phytoplankton Layers in the Northern Gulf of Mexico Revealed by Airborne Lidar

Authors: Yichen Yang (Zhejiang University); Hangkai Pan (Zhejiang University); Dekang Zheng (Zhejiang University); Hongkai Zhao (Zhejiang University); Yudi Zhou (Zhejiang University); Dong Liu (Zhejiang University)

Journal: Remote Sensing

Publisher: MDPI AG

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Keywords: N/A

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Open Access: N/A

Abstract: The thin layers in the ocean are temporally-coherent aggregations of phytoplankton with high concentrations at small vertical scales, presenting important hotspots of ecological activity. Lidar could identify thin phytoplankton layers at a large spatial scale due to its capabilities of profile detection with a high efficiency. However, studies that linked thin layers to environmental factors are few, which limits our understanding of the layer formation mechanism. This paper investigates the characteristics and formation conditions of thin phytoplankton layers in the northern Gulf of Mexico using airborne lidar. The results depict that the chlorophyll concentration determines the formation probability of the phytoplankton layer. The layer is mainly formed at concentrations less than 6 mg m⁻³ and mostly distributed at 2 mg m⁻³. In addition, layer thicknesses were within 5 m and layer depths were mainly in the range of 10-15 m. Layer depths in the nearshore region were shallower than those in the offshore region. We conclude that the characteristics and formation conditions of the thin phytoplankton layers depend on the nutrients and light that are related to the seabed topography, turbidity, eddies and upwelling. The findings of this paper will enhance the understanding of layer formation mechanisms.

466. Fast mechanisms linking the Labrador Sea with subtropical Atlantic overturning

Authors: Yavor Kostov; Marie-José Messias; Herlé Mercier; Helen L. Johnson; David P. Marshall

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-022-06459-y

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Language: en

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Keywords: N/A

Funding: Natural Environment Research Council

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Open Access: N/A

Abstract: We use an ocean general circulation model and its adjoint to analyze the causal chain linking sea surface buoyancy anomalies in the Labrador Sea to variability in the deep branch of the Atlantic meridional overturning circulation (AMOC) on inter-annual timescales. Our study highlights the importance of the North Atlantic Current (NAC) for the north-to-south connectivity in the AMOC and for the meridional transport of Lower North Atlantic Deep Water (LNADW). We identify two mechanisms that allow the Labrador Sea to impact velocities in the LNADW layer. The first mechanism involves a passive advection of surface buoyancy anomalies from the Labrador Sea towards the eastern subpolar gyre by the background NAC. The second mechanism plays a dominant role and involves a dynamical response of the NAC to surface density anomalies originating in the Labrador Sea; the NAC adjustment modifies the northward transport of salt and heat and exerts a strong positive feedback, amplifying the upper ocean buoyancy anomalies. The two mechanisms spin up/down the subpolar gyre on a timescale of years, while boundary trapped waves rapidly communicate this signal to the subtropics and trigger an adjustment of LNADW transport on a timescale of months. The NAC and the eastern subpolar gyre play an essential role in both mechanisms linking the Labrador Sea with LNADW transport variability and the subtropical AMOC. We thus reconcile two apparently contradictory paradigms about AMOC connectivity: (1) Labrador Sea buoyancy anomalies drive AMOC variability; (2) water mass transformation is largest in the eastern subpolar gyre.

467. Four-dimensional temperature, salinity and mixed-layer depth in the Gulf Stream, reconstructed from remote-sensing and in situ observations with neural networks

Authors: Etienne Pauthenet; Loïc Bachelot; Kevin Balem; Guillaume Maze; Anne-Marie Tréguier; Fabien Roquet; Ronan Fablet; Pierre Tandeo

Journal: Ocean Science

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . Despite the ever-growing number of ocean data, the interior of the ocean remains undersampled in regions of high variability such as the Gulf Stream. In this context, neural networks have been shown to be effective for interpolating properties and understanding ocean processes. We introduce OSnet (Ocean Stratification network), a new ocean reconstruction system aimed at providing a physically consistent analysis of the upper ocean stratification. The proposed scheme is a bootstrapped multilayer perceptron trained to predict simultaneously temperature and salinity (T-S) profiles down to 1000 m and the mixed-layer depth (MLD) from surface data covering 1993 to 2019. OSnet is trained to fit sea surface temperature and sea level anomalies onto all historical in situ profiles in the Gulf Stream region. To achieve vertical coherence of the profiles, the MLD prediction is used to adjust a posteriori the vertical gradients of predicted T-S profiles, thus increasing the accuracy of the solution and removing vertical density inversions. The prediction is generalized on a $1/4^\circ$ daily grid, producing four-dimensional fields of temperature and salinity, with their associated confidence interval issued from the bootstrap. OSnet profiles have root mean square error comparable with the observation-based Armor3D weekly product and the physics-based ocean reanalysis Glorys12. The lowest confidence in the prediction is located north of the Gulf Stream, between the shelf and the current, where the thermohaline variability is large. The OSnet reconstructed field is coherent even in the pre-Argo years, demonstrating the good generalization properties of the network. It reproduces the warming trend of surface temperature, the seasonal cycle of surface salinity and mesoscale structures of temperature, salinity and MLD. While OSnet delivers an accurate interpolation of the ocean stratification, it is also a tool to study how the ocean stratification relates to surface data. We can compute the relative importance of each input for each T-S prediction and analyse how the network learns which surface feature influences most which property and at which depth. Our results demonstrate the potential of machine learning methods to improve predictions of ocean interior

properties from observations of the ocean surface.

468. Spectral Kinetic-Energy Fluxes in the North Pacific: Definition Comparison and Normal- and Shear-Strain Decomposition

Authors: Yi Yang (Ministry of Natural Resources); Ru Chen (Ministry of Natural Resources)

Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

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Abstract: The spectral kinetic-energy flux is an effective tool to analyze the kinetic-energy transfer across a range of length scales, also known as the kinetic-energy cascade. Three methods to calculate spectral energy fluxes have been widely used, hereafter the Π_A , Π_F , and Π_Q definitions. However, the relations among these three definitions have not been examined in detail. Moreover, the respective contribution of the normal strain and shear strain of the flow field to kinetic-energy cascade has not been estimated before. Here, we use the kinetic energy equations to rigorously compare these definitions. Then, we evaluate the spectral energy fluxes, as well as its decomposition into the normal-strain and shear-strain components for the North Pacific, using a dynamically consistent global eddying state estimate. We find that the data must be preprocessed first to obtain stable results from the Π_F and Π_Q definitions, but not for the Π_A definition. For the upper 500 m of the North Pacific, in the wavenumber ranges with inverse kinetic-energy cascade, both the normal and shear-strain flow components contribute significantly to the spectral energy fluxes. However, at high wavenumbers, the dominant contributor to forward kinetic-energy cascade is the normal-strain component. These results should help shed light on the underlying mechanism of inverse and forward energy cascades.

469. A Comparative Study of the Argo-Era Ocean Heat Content Among Four Different Types of Data Sets

Authors: Fanglou Liao (Physical Science and Engineering Division King Abdullah University of Science and Technology Thuwal Saudi Arabia); Ibrahim Hoteit (Physical Science and Engineering Division King Abdullah University of Science and Technology Thuwal Saudi Arabia)

Journal: Earth's Future

Publisher: American Geophysical Union (AGU)

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Abstract: We conducted a comparative study of ocean heat content (OHC) in the top 2,000 m during the Argo-era using 12 latest and representative global ocean data sets. The differences in the global and basins-wide OHC trends were minor among the observation-based data sets, and remarkable among the ocean reanalyzes (RAs). Some RAs might exhibit much higher or lower basins-wide warming rates than the observation-based data sets. In the top 700 m, RAs suggested similar large-scale warming and cooling patterns, in agreement with the observation-based data sets. Below 700 m, the major warming and cooling features were significantly different between RAs and the observation-based data sets. All data sets suffered from relatively larger uncertainties in the highly dynamic regions. Special caution is suggested when estimating the OHC using only a single data set, especially a RA. Differences of RAs' OHC from observation-based data sets were significantly reduced when considering their ensemble mean, to be further confirmed with a larger sample of data sets.

470. Hindrance effect of tides on water exchanges between the Red Sea and the Gulf of Aden

Authors: Daquan Guo; Fengchao Yao; Peng Zhan; George Krokos; Ibrahim Hoteit

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2022.964097

URL: <https://doi.org/10.3389/fmars.2022.964097>

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Keywords: N/A

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Open Access: N/A

Abstract: As a semienclosed marginal sea, the Red Sea connects with the open ocean through a narrow strait at its southern end, known as the Bab-al-Mandeb (BAM) strait. The water exchange between the Red Sea and the Gulf of Aden at the BAM strait is crucial for the water mass transformations and thermohaline circulation in the Red Sea as well as for nutrient supply to the open ocean. In this study, a three-dimensional high-resolution nonhydrostatic MIT general circulation model (MITgcm) was used to investigate the tidal influence on the water exchange in the BAM strait through simulations with and without tidal forcing. We found that the tidal effects on the water exchange in winter were insignificant; however, the summer intrusion of the Gulf of Aden Intermediate Water (GAIW) was strongly affected. When the simulation includes tidal forcing, the along-axis northern extension of the GAIW intrusion is reduced by up to 100 km and the monthly mean volume transport is decreased by 20% on average. Two actors that possibly contribute to the hindrance effects of tides in summer are (i) the tidal residual currents that propagate in a direction opposite to the pathway of the GAIW intrusion currents and (ii) the enhanced vertical mixing at the pycnocline and near the benthic topography of the BAM strait, which triggers more instabilities along the pathway of the intrusion.

471. Can We Estimate Air-Sea Flux of Biological O₂ From Total Dissolved Oxygen?

Authors: Yibin Huang (Xiamen University, Duke University); Rachel Eveleth (Duke University, Now at Department of Geosciences Oberlin College Oberlin OH USA); David Nicholson (Woods Hole Oceanographic Institution); Nicolas Cassar (Institut de Recherche pour le Développement (IRD), Duke University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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References: 77

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Open Access: N/A

Abstract: In this study, we compare mechanistic and empirical approaches to reconstruct the air-sea flux of biological oxygen (ΔO_2) by parameterizing the physical oxygen saturation anomaly ($\Delta O_2[\text{phy}]$) in order to separate the biological contribution from total oxygen. The first approach matches $\Delta O_2[\text{phy}]$ to the monthly climatology of the argon saturation anomaly from a global ocean circulation model's output. The second approach derives $\Delta O_2[\text{phy}]$ from an iterative mass balance model forced by satellite-based physical drivers of $\Delta O_2[\text{phy}]$ prior to the sampling day by assuming that air-sea interactions are the dominant factors driving the surface $\Delta O_2[\text{phy}]$. The final approach leverages the machine-learning technique of Genetic Programming (GP) to search for the functional relationship between $\Delta O_2[\text{phy}]$ and biophysicochemical parameters. We compile simultaneous measurements of O_2/Ar and O_2 concentration from 14 cruises to train the GP algorithm and test the validity and applicability of our modeled $\Delta O_2[\text{phy}]$ and ΔO_2 . Among the approaches, the GP approach, which incorporates ship-based measurements and historical records of physical parameters from the reanalysis products, provides the most robust predictions ($R^2 = 0.74$ for $\Delta O_2[\text{phy}]$ and 0.72 for ΔO_2 ; $\text{RMSE} = 1.4\%$ for $\Delta O_2[\text{phy}]$ and $7.1 \text{ mmol } O_2 \text{ m}^{-2} \text{ d}^{-1}$ for ΔO_2). We use the empirical formulation derived from GP approach to reconstruct regional, inter-annual, and decadal variability of ΔO_2 based on historical oxygen records. Overall, our study represents a first attempt at deriving ΔO_2 from snapshot measurements of oxygen, thereby paving the way toward using historical O_2 data and a rapidly growing number of O_2 measurements on autonomous platforms for independent insight into the biological pump.

472. Antarctic Peninsula warming triggers enhanced basal melt rates throughout West Antarctica

Authors: M. Mar Flexas (California Institute of Technology); Andrew F. Thompson (California Institute of Technology); Michael P. Schodlok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Speer (Florida State University)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

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Publication Date: 2022-08-12

Volume/Issue: Volume 8, Issue 32

References: 63

Cited by: 29 publications

Language: en

Subjects: N/A

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Open Access: N/A

Abstract: The observed acceleration of ice shelf basal melt rates throughout West Antarctica could destabilize continental ice sheets and markedly increase global sea level. Explanations for decadal-scale melt intensification have focused on processes local to shelf seas surrounding the ice shelves. A suite of process-based model experiments, guided by CMIP6 forcing scenarios, show that freshwater forcing from the Antarctic Peninsula, propagated between marginal seas by a coastal boundary current, causes enhanced melting throughout West Antarctica. The freshwater anomaly stratifies the ocean in front of the ice shelves and modifies vertical and lateral heat fluxes, enhancing heat transport into ice shelf cavities and increasing basal melt. Increased glacial runoff at the Antarctic Peninsula, one of the first signatures of a warming climate in Antarctica, emerges as a key trigger for increased ice shelf melt rates in the Amundsen and Bellingshausen Seas.

473. Impact of Vertical Mixing Parameterizations on Internal Gravity Wave Spectra in Regional Ocean Models

Authors: Ritabrata Thakur (University of Michigan); Brian K. Arbic (University of Michigan); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kayhan Momeni (University of Toronto); Yulin Pan (University of Michigan); W. R. Peltier (University of Toronto); Joseph Skitka (University of Michigan); Matthew H. Alford (Scripps Institution of Oceanography / University of California San Diego); Yuchen Ma (University of Toronto)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl099614

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Subjects: N/A

Keywords: internal gravity waves, KPP mixing parameterization, MITgcm, vertical wavenumber spectra

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Open Access: N/A

Abstract: We present improvements in the modeling of the vertical wavenumber spectrum of the internal gravity wave continuum in high-resolution regional ocean simulations. We focus on model sensitivities to mixing parameters and comparisons to McLane moored profiler observations in a Pacific region near the Hawaiian Ridge, which features strong semidiurnal tidal beams. In these simulations, the modeled continuum exhibits high sensitivity to the background mixing components of the K-Profile Parameterization (KPP) vertical mixing scheme. Without the KPP background mixing, stronger vertical gradients in velocity are sustained in the simulations and the modeled kinetic energy and shear spectral slopes are significantly closer to the observations. The improved representation of internal wave dynamics in these simulations makes them suitable for improving ocean mixing estimates and for the interpretation of satellite missions such as the Surface Water and Ocean Topography mission.

474. Women and adaptive capacity to climate change in East African seascapes - Zanzibar as an example

Authors: Maricela de la Torre-Castro; Lars Lindström; Narriman S. Jiddawi; Felicity Pike; Astrid Max

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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Open Access: N/A

Abstract: As the climate crisis persists, there is a crucial need to increase knowledge on adaptive capacity and the underlying factors building it. This is particularly important for disadvantaged groups, such as coastal women in East Africa. Women's livelihoods in these seascapes are and will be more severely affected by climate change and the capacity of East African states to deal with these challenges is limited in terms of financial and human capital. In this research, we investigated the underlying factors building the adaptive capacity of coastal women in Zanzibar (Unguja Island), Tanzania. Coastal women (N=117) were interviewed in villages around the island to gather information about potential factors supporting adaptive capacity. This was analysed applying Cinner et al (2018) five domains typology for adaptive capacity, i.e. assets, flexibility, organizations, learning and agency. The results show that women had relatively low adaptive capacity, extended poverty and very high dependence on seaweed farming of red algae, a livelihood providing low income and already being seriously affected by climate variability and change. Women's observations of key variables related to environmental changes corresponded to most scientific findings. It was, however, unclear how that knowledge is useful and enhances adaptive capacity. Adaptive capacity was generally low but individual differences were found in which ten women had a high income. The results show that the factors underlying adaptive capacity are complex and interact with each other, being positive, negative and unclear. Many of the identified factors deserve future research. This study adds to the pool of knowledge by addressing women (not only men); coastal ecosystems (as land and freshwater systems are more studied) and the individual level (since most studies focus on national and community levels). The study illustrates that institutional renewal, bridging and cooperation is possible in Zanzibar bringing good news to the region.

475. Comparative analysis of the correction effect of different environmental loading products on global GNSS coordinate time series

Authors: Yuefan He; Guigen Nie; Shuguang Wu; Haiyang Li

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2022.08.009

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Cited by: 6 publications

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

476. Dynamical Filtering Highlights the Seasonality of Surface-Balanced Motions at Diurnal Scales in the Eastern Boundary Currents

Authors: Antonio Quintana (Physical Oceanography Department, Center for Scientific Research and Higher Education at Ensenada, Baja California, Ensenada 22860, Mexico); Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jose Gomez-Valdes (Physical Oceanography Department, Center for Scientific Research and Higher Education at Ensenada, Baja California, Ensenada 22860, Mexico)

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Volume/Issue: Volume 7, Issue 8

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Center for Scientific Research and Higher Education at Ensenada (CICESE), Consejo Nacional de Ciencia y Tecnología (CONACyT), S-MODE project, an EVS-3 Investigation awarded under NASA Research Announcement (and 1 more)

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Open Access: N/A

Abstract: Balanced motions (BM) and internal gravity waves (IGW) account for most of the kinetic energy budget and capture most of the vertical velocity in the ocean. However, estimating the contribution of BM to both issues at time scales of less than a day is a challenge because BM are obscured by IGW. To study the BM regime, we outlined the implementation of a dynamical filter that separates both classes of motion. This study used a high-resolution global simulation to analyze the Eastern Boundary Currents during the winter and summer months. Our results confirm the feasibility of recovering BM dynamics at short time scales, emphasizing the diurnal cycle in winter and its dampening in summer due to local stratification that prevents large vertical excursion of the surface boundary layer. Our filter opens up new possibilities for more accurate estimation of the vertical exchanges of any tracers at any vertical level in the water column. Moreover, it could be a valuable tool for studies focused on wave-turbulence interactions in ocean simulations.

477. A Hybrid Dynamical Approach for Seasonal Prediction of Sea-Level Anomalies: A Pilot Study for Charleston, South Carolina

Authors: Thomas Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ben Kirtman (University of Miami); Emily Becker (University of Miami); Ben Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Daniel Limonadi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Duane Waliser (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Publication Date: 2022-08-08

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Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: coastal hazards, ocean dynamics, sea-level prediction, adjoint models, climate predictability

Funding: Earth Sciences Division

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Open Access: N/A

Abstract: Using Earth system models for seasonal sea-level prediction remains challenging due to model biases and initialization shocks. Here we present a hybrid dynamical approach for seasonal sea-level prediction to alleviate some of these issues. The approach is based on convolving atmospheric forcings with sea-level sensitivities to these forcings. The sensitivities are pre-computed by the adjoint model of the Estimating Circulation and Climate of the Ocean (ECCO) system. The forcings are a concatenation of ECCO forcings before prediction initialization and a 10-member predicted atmospheric forcing ensemble from the Community Climate System Model version 4 (CCSM4) after initialization, with offline forcing bias corrections applied using the observationally-constrained ECCO seasonal forcing climatology. As a pilot study, we conducted 12-month hindcasts from 1995 to 2016 in Charleston (United States East Coast). Our approach avoids drifts in CCSM4 sea-level predictions and beats seasonal climatology and damped persistence as predictors up to a 6-month lead time. The prediction skill comes from two factors: (a) ECCO forcings prior to prediction initialization influence sea level after initialization through delayed oceanic adjustments (e.g., coastally-trapped waves, open-ocean Rossby waves, and advection of steric anomalies) leading to skillful predictions beyond 2 months after initialization, and (b) the 10-member CCSM4 ensemble forcing predictions have relatively good skill at 1-2 months lead times. Our method is computationally efficient for operational sea-level prediction at specific locations and can attribute sea-level prediction skill and uncertainty to specific

forcings or forcing from particular regions, thereby providing useful information to seasonal prediction centers for improving their prediction systems.

478. The future projection of cyclones in Bay of Bengal: a study using coupled ocean atmosphere model

Authors: P V Kiran; C Balaji

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Department of Science and Technology under SPLICE - Climate Change Programme

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Abstract: N/A

479. Pacific Equatorial Undercurrent: Mean state, sources, and future changes across models

Authors: Annette Stellema; Alex Sen Gupta; Andréa S. Taschetto; Ming Feng

Journal: Frontiers in Climate

Publisher: Frontiers Media SA

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Abstract: The Equatorial Undercurrent (EUC) stretches across the Pacific, transporting cool waters rich in oxygen and nutrients eastward to one of the most productive regions in the ocean. As an intricate part of the global climate system, EUC dynamics are essential to understanding future climate change but are poorly represented in global coupled climate models. This study examines EUC representation and future changes in the latest generations of the Coupled Model Intercomparison Project (CMIP6 and CMIP5) and an eddy-permitting ocean model. We also examine historical and projected changes in EUC source waters, including the Mindanao Current (MC), New Guinea Coastal Undercurrent (NGCU), and interior thermocline convergence. The circulation features in the models are broadly consistent with observations and ocean reanalyses, but improvements from CMIP5 to CMIP6 are limited. In the future projections, the EUC is enhanced in the western Pacific, with less prominent changes in CMIP6, but more so in the eddy-permitting model. The western Pacific EUC enhancement is likely associated with a wind-driven redirection of waters south of the equator, in which the NGCU boundary flow increases while the interior thermocline convergence decreases. This is superimposed on an overall weakening of the North Pacific subtropical overturning cell, including the MC, interior thermocline convergence, and Ekman divergence. As EUC heat and nutrient composition is linked to its sources, these projected changes have implications for the EUC's role in air-sea feedbacks, nutrient replenishment, and oxygen minimum zone ventilation in the eastern Pacific.

480. Genomic evidence for global ocean plankton biogeography shaped by large-scale current systems

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Abstract: Biogeographical studies have traditionally focused on readily visible organisms, but recent technological advances are enabling analyses of the large-scale distribution of microscopic organisms, whose biogeographical patterns have long been debated. Here we assessed the global structure of

plankton geography and its relation to the biological, chemical, and physical context of the ocean (the 'seascape') by analyzing metagenomes of plankton communities sampled across oceans during the Tara Oceans expedition, in light of environmental data and ocean current transport. Using a consistent approach across organismal sizes that provides unprecedented resolution to measure changes in genomic composition between communities, we report a pan-ocean, size-dependent plankton biogeography overlying regional heterogeneity. We found robust evidence for a basin-scale impact of transport by ocean currents on plankton biogeography, and on a characteristic timescale of community dynamics going beyond simple seasonality or life history transitions of plankton.

481. Surface Quasigeostrophic Turbulence in Variable Stratification

Authors: Houssam Yassin (Princeton University); Stephen M. Griffies (b NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey, Princeton University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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URL: <https://doi.org/10.1175/jpo-d-22-0040.1>

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Keywords: Quasigeostrophic models, Eddies, Buoyancy, Turbulence, Stratification (seeds)|Barotropic fluid|Surface layer

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Abstract: Numerical and observational evidence indicates that, in regions where mixed layer instability is active, the surface geostrophic velocity is largely induced by surface buoyancy anomalies. Yet, in these regions, the observed surface kinetic energy spectrum is steeper than predicted by uniformly stratified surface quasigeostrophic theory. By generalizing surface quasigeostrophic theory to account for variable stratification, we show that surface buoyancy anomalies can generate a variety of dynamical regimes depending on the stratification's vertical structure. Buoyancy anomalies generate longer-range velocity fields over decreasing stratification and shorter-range velocity fields over increasing stratification. As a result, the surface kinetic energy spectrum is steeper over decreasing stratification than over increasing stratification. An exception occurs if the near-surface stratification is much larger than the deep-ocean stratification. In this case, we find an extremely local turbulent regime with surface buoyancy homogenization and a steep surface kinetic energy spectrum, similar to equivalent barotropic turbulence. By applying the variable stratification theory to the wintertime North Atlantic, and assuming that mixed layer instability acts as a narrowband small-scale surface buoyancy forcing, we obtain a predicted surface kinetic energy spectrum between $k^{-4/3}$ and $k^{-7/3}$, which is consistent with the observed wintertime k^{-2} spectrum. We conclude by suggesting a method of measuring the buoyancy frequency's vertical structure using satellite observations.

482. FOCI-MOPS v1 - integration of marine biogeochemistry within the Flexible Ocean and Climate Infrastructure version 1 (FOCI 1) Earth system model

Authors: Chia-Te Chien; Jonathan V. Durgadoo; Dana Ehlert; Ivy Frenger; David P. Keller; Wolfgang Koeve; Iris Kriest; Angela Landolfi; Lavinia Patara; Sebastian Wahl; Andreas Oschlies

Journal: Geoscientific Model Development

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Abstract: . The consideration of marine biogeochemistry is essential for simulating the carbon cycle in an Earth system model. Here we present the implementation and evaluation of a marine biogeochemical model, the Model of Oceanic Pelagic Stoichiometry (MOPS) in the Flexible Ocean and Climate Infrastructure (FOCI) climate model. FOCI-MOPS enables the simulation of marine biological processes, i.e. the marine carbon, nitrogen, and oxygen cycles with prescribed or prognostic atmospheric CO₂ concentration. A series of experiments covering the historical period (1850-2014) were performed following the DECK (Diagnostic, Evaluation and Characterization of Klima) and CMIP6 (Coupled Model Intercomparison Project 6) protocols. Overall, modelled biogeochemical tracer distributions and fluxes, transient evolution in surface air temperature, air-sea CO₂ fluxes, and changes in ocean carbon and heat contents are in good agreement with observations. Modelled inorganic and organic tracer distributions are quantitatively evaluated by statistically derived metrics. Results of the FOCI-MOPS model, including sea surface temperature, surface pH, oxygen (100-600 m), nitrate (0-100 m), and primary production, are within the range of other CMIP6 model results. Overall, the evaluation of FOCI-MOPS indicates its suitability for Earth climate system simulations.

483. A global atlas of potential thermal refugia for coral reefs generated by internal gravity waves

Authors: Scott D. Bachman; Joan A. Kleypas; Mark Erdmann; Edy Setyawan

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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Abstract: Coral reefs are highly threatened by ocean warming and the majority are likely to be lost in less than three decades. A first step in maximizing reef conservation through this period is to identify where coral reefs are more likely to survive rising ocean temperatures, such as locations that experience lower temperatures than surrounding regions, high temperature variability, and high food supply. Such conditions are often the result of naturally occurring internal gravity waves (IGWs), oscillatory subsurface disturbances that can entrain cooler and/or nutrient-rich subsurface waters and cause high frequency temperature fluctuations. These features usually remain undetected because they occur subsurface and at spatial scales of $O(1 \text{ km})$ and smaller. To shed light on where IGWs are likely to impact temperature conditions within coral reef regions, we present an analysis of data from the LLC4320, a massive high resolution ($1/48^\circ$; $< 2.5 \text{ km}$) numerical global ocean simulation. The results highlight strong regional differences in the incidence of IGW-induced temperature variability. The analysis also reveals that thermal refugia are limited to depths where high temperature variability coincides with the actual reef depth and may not persist year-round. Assuming 10-m depth as the nominal reef depth, reef regions likely to benefit from IGW-induced cooling occur in SE Asia and the Coral Triangle, the Galápagos, along the Pacific shelf of Central America, and isolated locations worldwide. Such refugia are rare within the Atlantic reef sector. An interactive global atlas showing the results of this study has been made freely available online at <https://ncar.github.io/coral-viz/>.

484. The Earth system model CLIMBER-X v1.0 - Part 1: Climate model description and validation

Authors: Matteo Willeit; Andrey Ganopolski; Alexander Robinson; Neil R. Edwards

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Abstract: . The newly developed fast Earth system model CLIMBER-X is presented. The climate component of CLIMBER-X consists of a 2.5-D semi-empirical statistical-dynamical atmosphere model, a 3-D frictional-geostrophic ocean model, a dynamic-thermodynamic sea ice model and a land surface model. All the model components are discretized on a regular lat-long grid with a horizontal resolution of $5^\circ \times 5^\circ$. The model has a throughput of $\sim 10\,000$ simulation years per day on a single node with 16 CPUs on a high-performance computer and is designed to simulate the evolution of the Earth system on temporal scales ranging from decades to $>100\,000$ years. A comprehensive evaluation of the model performance for the present day and the historical period shows that CLIMBER-X is capable of realistically reproducing many observed climate characteristics, with results that generally lie within the range of state-of-the-art general circulation models. The analysis of model performance is complemented by a thorough assessment of climate feedbacks and model sensitivities to changes in external forcings and boundary conditions. Limitations and applicability of the model are critically discussed. CLIMBER-X also includes a detailed representation of the global carbon cycle and is coupled to an ice sheet model, which will be described in separate papers. CLIMBER-X is available as open-source code and is expected to be a useful tool for studying past climate changes and for the investigation of the long-term future evolution of the climate.

485. Cloud-based framework for inter-comparing submesoscale-permitting realistic ocean models

Authors: Takaya Uchida; Julien Le Sommer; Charles Stern; Ryan P. Abernathey; Chris Holdgraf; Aurélie Albert; Laurent Brodeau; Eric P. Chassignet; Xiaobiao Xu; Jonathan Gula; Guillaume Roullet; Nikolay Koldunov; Sergey Danilov; Qiang Wang; Dimitris Menemenlis; Clément Bricaud; Brian K. Arbic; Jay F. Shriver; Fangli Qiao; Bin Xiao; Arne Biastoch; René Schubert; Baylor Fox-Kemper; William K. Dewar; Alan Wallcraft

Journal: Geoscientific Model Development

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Open Access: N/A

Abstract: . With the increase in computational power, ocean models with kilometer-scale resolution have emerged over the last decade. These models have been used for quantifying the energetic exchanges between spatial scales, informing the design of eddy parametrizations, and preparing observing networks. The increase in resolution, however, has drastically increased the size of model outputs, making it difficult to transfer and analyze the data. It remains, nonetheless, of primary importance to assess more systematically the realism of these models. Here, we showcase a cloud-based analysis framework proposed by the Pangeo project that aims to tackle such distribution and analysis challenges. We analyze the output of eight submesoscale-permitting simulations, all on the cloud, for a crossover region of the upcoming Surface Water and Ocean Topography (SWOT) altimeter mission near the Gulf Stream separation. The cloud-based analysis framework (i) minimizes the cost of duplicating and storing ghost copies of data and (ii) allows for seamless sharing of analysis results amongst collaborators. We describe the framework and provide example analyses (e.g., sea-surface height variability, submesoscale vertical buoyancy fluxes, and comparison to predictions from the mixed-layer instability parametrization). Basin- to global-scale, submesoscale-permitting models are still at their early stage of development; their cost and carbon footprints are also rather large. It would, therefore, benefit the community to document the different model configurations for future best practices. We also argue that an emphasis on data analysis strategies would be crucial for improving the models themselves.

486. Impacts of the long-term atmospheric trend on the seasonality of Antarctic sea ice

Authors: Fu Zhao; Xi Liang; Zhongxiang Tian; Chengyan Liu; Xichen Li; Yun Yang; Ming Li; Na Liu

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

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Abstract: N/A

487. Excitations of Seasonal Polar Motions Derived from Satellite Gravimetry and General Circulation Models: Comparisons of Harmonic and Inharmonic Analyses

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Journal: Remote Sensing

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Abstract: Due to the conservation of global angular momentum, polar motion (PM) is dominated by global mass redistributions and relative motions in the atmosphere, oceans and land water at seasonal time scales. Thus, accurately measured PM data can be used to validate the general circulation models (GCMs) for the atmosphere, oceans and land water. This study aims to analyze geophysical excitations and observed excitations obtained from PM observations from both the harmonic and wavelet analysis perspectives, in order to refine our understanding of the geophysical excitation of PM. The geophysical excitations are derived from two sets of GCMs and a monthly gravity model combining satellite gravity data and some GCM outputs using the PM theory for an Earth model with frequency-dependent responses, while the observed excitation is obtained from the PM data using the frequency-domain Liouville's equation. Our results show that wavelet analysis can reveal the time-varying nature of all excitations and identify when changes happen and how strong they are, while harmonic analysis can only show the average amplitudes and phases. In particular, the monthly gravity model can correct the mismodeled GCM outputs, while the Earth's frequency-dependent responses provide us with a better understanding of atmosphere-ocean-land water-solid Earth interactions.

488. Deep Ocean Learning of Small Scale Turbulence

Authors: Ali Mashayek (Imperial College London); Nick Reynard (Imperial College London); Fangming Zhai (Imperial College London); Kaushik Srinivasan (University of California Los Angeles Los Angeles CA USA); Adam Jelley (University of Edinburgh); Alberto Naveira Garabato (University of Southampton); Colm-cille P. Caulfield (University of Cambridge)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Keywords: ocean turbulence, diapycnal mixing, machine learning

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Abstract: Turbulent mixing at the sub-meter scale is an essential component of the ocean's meridional overturning circulation and its associated global redistribution of heat, carbon, nutrients, pollutants, and other tracers. Whereas direct turbulence observations in the ocean interior are limited to a modest collection of field programs, basic information such as temperature, salinity, and depth is available globally. Here, we show that supervised machine learning algorithms can be trained on the existing turbulence data to develop skillful predictions of the key properties of turbulence from T, S, Z, and topographic data. This constitutes a promising first step toward a hybrid physics-artificial intelligence approach to parameterization of turbulent mixing in ocean and climate models.

489. Using a Lagrangian particle tracking model to evaluate impacts of El Niño-related advection on euphausiids in the southern California Current System

Authors: Laura E. Lilly; Bruce D. Cornuelle; Mark D. Ohman

Journal: Deep Sea Research Part I: Oceanographic Research Papers

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Abstract: N/A

490. Achievements and Prospects of Global Broadband Seismographic Networks After 30 Years of Continuous Geophysical Observations

Authors: A. T. Ringler (Albuquerque Seismological Laboratory U.S. Geological Survey Albuquerque NM USA); R. E. Anthony (Albuquerque Seismological Laboratory U.S. Geological Survey Albuquerque NM USA); R. C. Aster (Colorado State University); C. J. Ammon (Pennsylvania State University); S. Arrowsmith (Roy M. Huffington Department of Earth Sciences Southern Methodist University Dallas TX USA); H. Benz (National Earthquake Information Center U.S. Geological Survey Denver CO USA); C. Ebeling (Scripps Institution of Oceanography / University of California San Diego); A. Frassetto (Incorporated Research Institutions for Seismology Consortium Washington DC USA); W.-Y. Kim (Columbia University); P. Koelemeijer (Department of Earth Sciences Royal Holloway University of London Egham UK, University of Oxford); H. C. P. Lau (University of California Berkeley); V. Lekić (University of Maryland); J. P. Montagner (Institut de Physique du Globe); P. G. Richards (Columbia University); D. P. Schaff (Columbia University); M. Vallée (Institut de Physique du Globe); W. Yeck (National Earthquake Information Center U.S. Geological Survey Denver CO USA)

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References: 766

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Keywords: N/A

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Abstract: Global seismographic networks (GSNs) emerged during the late nineteenth and early twentieth centuries, facilitated by seminal international developments in theory, technology, instrumentation, and data exchange. The mid- to late-twentieth century saw the creation of the World-Wide Standardized Seismographic Network (1961) and International Deployment of Accelerometers (1976), which advanced global geographic coverage as seismometer bandwidth increased greatly allowing for the recording of the Earth's principal seismic spectrum. The modern era of global observations and rapid data access began during the 1980s, and notably included the inception of the GEOSCOPE initiative (1982) and GSN (1988). Through continual improvements, GEOSCOPE and the GSN have realized near-real time recording of ground motion with state-of-art data quality, dynamic range, and timing precision to encompass 180 seismic stations, many in very remote locations. Data from GSNs are increasingly integrated with other geophysical data (e.g., space geodesy, infrasound and Interferometric Synthetic Aperture Radar). Globally distributed seismic data are critical to resolving crust, mantle, and core structure; illuminating features of the plate tectonic and mantle convection system; rapid characterization of earthquakes; identification of potential tsunamis; global nuclear test verification; and provide sensitive proxies for environmental changes. As the global

geosciences community continues to advance our understanding of Earth structure and processes controlling elastic wave propagation, GSN infrastructure offers a springboard to realize increasingly multi-instrument geophysical observatories. Here, we review the historical, scientific, and monitoring heritage of GSNs, summarize key discoveries, and discuss future associated opportunities for Earth Science.

491. Long-Term Monitoring and Change Analysis of Pine Island Ice Shelf Based on Multi-Source Satellite Observations during 1973-2020

Authors: Shijie Liu (Tongji University); Shu Su (Tongji University); Yuan Cheng (Tongji University); Xiaohua Tong (Tongji University); Rongxing Li (Tongji University)

Journal: Journal of Marine Science and Engineering

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Open Access: N/A

Abstract: Pine Island Glacier (PIG) is one of the largest contributors to sea level rise in Antarctica. Continuous thinning and frequent calving imply significant destabilization of Pine Island Glacier Ice Shelf (PIGIS). To understand the mechanism of its accelerated disintegration and its future development, we conducted a long-term monitoring and comprehensive analysis of PIGIS, including ice flow velocity, ice shelf fronts, ocean water temperature, rifts, and surface strain rates, based on multi-source satellite observations during 1973-2020. The results reveal that: (1) ice flow velocities of PIGIS increased from 2.3 km/yr in 1973 to 4.5 km/yr in 2020, with two rapid acceleration periods of 1995-2009 and 2017-2020, and its change was highly correlated to the ocean water temperature variation. (2) At least 13 calving events occurred during 1973-2020, with four unprecedented successive retreats in 2015, 2017, 2018, and 2020. (3) The acceleration of ice shelf rifting and calving may correlate to the destruction of shear margins, while this damage was likely a response to the warming of bottom seawater. The weakening southern shear margin may continue to recede, indicating that the instability of PIGIS will continue.

492. Effects of solar irradiance noise on a complex marine trophic web

Authors: Roberto Grimaudo; Paolo Lazzari; Cosimo Solidoro; Davide Valenti

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

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Funding: Ministero dell'Istruzione, dell'Università e della Ricerca

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Open Access: N/A

Abstract: The analysis of experimental data of the solar irradiance, collected on the marine surface, clearly highlights the intrinsic stochasticity of such an environmental parameter. Given this result, effects of randomly fluctuating irradiance on the population dynamics of a marine ecosystem are studied on the basis of the stochastic 0-dimensional biogeochemical flux model. The noisy fluctuations of the irradiance are formally described as a multiplicative Ornstein-Uhlenbeck process, that is a self-correlated Gaussian noise. Nonmonotonic behaviours of the variance of the marine populations' biomass are found with respect to the intensity and the autocorrelation time of the noise source, manifesting a noise-induced transition of the ecosystem to an out-of-equilibrium steady state. Moreover, evidence of noise-induced effects on the organic carbon cycling processes underlying the food web dynamics are highlighted. The reported results clearly show the profound impact the stochastic environmental variables can have on both the populations and the biogeochemistry at the basis of a marine trophic network.

493. Comparison of physical and biological responses to tropical cyclones between the low and middle latitude zones of the western North Pacific

Authors: Teng Wang; Haofei Zhang; Lei Gao; Lixin Zhu

Journal: Regional Studies in Marine Science

Publisher: Elsevier BV

DOI: 10.1016/j.rsma.2022.102535

URL: <https://doi.org/10.1016/j.rsma.2022.102535>

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References: 55

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Natural Resources of the People's Republic of China, Ministry of Science and Technology of the People's Republic of China, China Meteorological Administration (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

494. Low-Frequency Dynamic Ocean Response to Barometric-Pressure Loading

Authors: Christopher G. Piecuch (Woods Hole Oceanographic Institution); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Michael Schindelegger (Universität Bonn, Bonn, Germany); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mengnan Zhao (Atmospheric and Environmental Research Inc.)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-22-0090.1

URL: <https://doi.org/10.1175/jpo-d-22-0090.1>

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References: 97

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Large-scale motions, Potential vorticity, Barotropic fluid|Extratropical cyclone|Oceanic basin|Wind Stress|Ocean surface topography, Sea level, Ocean circulation (and 2 more)

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Changes in dynamic manometric sea level ζ_m represent mass-related sea level changes associated with ocean circulation and climate. We use twin model experiments to quantify magnitudes and spatiotemporal scales of ζ_m variability caused by barometric pressure p_a loading at long periods (month) and large scales () relevant to Gravity Recovery and Climate Experiment (GRACE) ocean data. Loading by p_a drives basin-scale monthly ζ_m variability with magnitudes as large as a few centimeters. Largest ζ_m signals occur over abyssal plains, on the shelf, and in marginal seas. Correlation patterns of modeled ζ_m are determined by continental coasts and H/f contours (H is ocean depth and f is Coriolis parameter). On average, ζ_m signals forced by p_a represent departures of and from the inverted-barometer effect ζ_{ib} on monthly and annual periods, respectively. Basic magnitudes, spatial patterns, and spectral behaviors of ζ_m from the model are consistent with scaling arguments from barotropic potential vorticity conservation. We also compare ζ_m from the model driven by p_a to ζ_m from GRACE observations. Modeled and observed ζ_m are significantly correlated across parts of the tropical and extratropical oceans, on shelf and slope regions, and in marginal seas. Ratios of modeled to observed ζ_m magnitudes are as large as ~ 0.2 (largest in the Arctic Ocean) and qualitatively agree with analytical theory for the gain of the transfer function between ζ_m forced by p_a and wind stress. Results demonstrate that p_a loading is a secondary but nevertheless important contributor to monthly mass variability from GRACE over the ocean.

495. Causes of the 2015 North Atlantic cold anomaly in a global state estimate

Authors: Rachael N. C. Sanders; Daniel C. Jones; Simon A. Josey; Bablu Sinha; Gael Forget

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-18-953-2022

URL: <https://doi.org/10.5194/os-18-953-2022>

Publication Date: 2022-07-06

Volume/Issue: Volume 18, Issue 4

References: 49

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: UK Research and Innovation, Natural Environment Research Council, National Aeronautics and Space Administration (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . The subpolar North Atlantic is an important part of the global ocean and climate system, with SST variability in the region influencing the climate of Europe and North America. While the majority of the global ocean exhibited higher than average surface temperatures in 2015, the subpolar North Atlantic experienced record low temperatures. This interannual cold anomaly is thought to have been driven by surface forcing, but detailed questions remain about how the anomaly was created and maintained. To better quantify and understand the processes responsible for the cold anomaly, we computed mixed-layer temperature budgets in the Estimating the Circulation and Climate of the Ocean (ECCO) Version 4 global ocean state estimate. State estimates have been brought into consistency with a large suite of observations without using artificial sources or sinks of heat, making them ideal for temperature budget studies. We found that strong surface forcing drove approximately 75 % of the initial anomalies in the cooling of the mixed layer in December 2013, while horizontal advection drove the remaining 25 %. The cold anomaly was then sequestered beneath the mixed layer. Re-emergence of the cold anomaly during the summer and autumn of 2014 was primarily the result of a strong temperature gradient across the base of the mixed layer, with vertical diffusion accounting for approximately 70 % of the re-emergence. Weaker surface warming of the mixed layer during the summer of 2015 enhanced the anomaly, causing a temperature minimum. Spatial patterns in the budgets also show large differences between the north and south of the anomaly region, with particularly strong initial surface cooling in the south related to the positive phase of the East Atlantic Pattern. It is important to note that this interannual cold anomaly, which is thought to be primarily driven by surface forcing, is distinct from the multi-decadal North Atlantic "warming hole", which has been associated with changes in advection.

496. Diel light cycles affect phytoplankton competition in the global ocean

Authors: Ioannis Tsakalakis (Ecosystems Center Marine Biological Laboratory Woods Hole Massachusetts USA, Massachusetts Institute of Technology); Michael J. Follows (Massachusetts Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Christopher L. Follett (Massachusetts Institute of Technology); Joseph J. Vallino (Ecosystems Center Marine Biological Laboratory Woods Hole Massachusetts USA)

Journal: Global Ecology and Biogeography

Publisher: Wiley

DOI: 10.1111/geb.13562

URL: <https://doi.org/10.1111/geb.13562>

Publication Date: 2022-07-03

Volume/Issue: Volume 31, Issue 9

References: 78

Cited by: 8 publications

Language: en

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Keywords: N/A

Funding: National Science Foundation, Simons Foundation

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Open Access: N/A

Abstract: Light, essential for photosynthesis, is present in two periodic cycles in nature: seasonal and diel. Although seasonality of light is typically resolved in ocean biogeochemical-ecosystem models because of its significance for seasonal succession and biogeography of phytoplankton, the diel light cycle is generally not resolved. The goal of this study is to demonstrate the impact of diel light cycles on phytoplankton competition and biogeography in the global ocean. **Location** Global ocean. **Major taxa studied** Phytoplankton. **Methods** We use a three-dimensional global ocean model and compare simulations of high temporal resolution with and without diel light cycles. The model simulates 15 phytoplankton types with different cell sizes, encompassing two broad ecological strategies: small cells with high nutrient affinity (gleaners) and larger cells with high maximal growth rate (opportunists). Both are grazed by zooplankton and limited by nitrogen, phosphorus and iron. **Results** Simulations show that diel cycles of light induce diel cycles in limiting nutrients in the global ocean. Diel nutrient cycles are associated with higher concentrations of limiting nutrients, by 100% at low latitudes (-40 degrees to 40 degrees), a process that increases the relative abundance of opportunists over gleaners. Size classes with the highest maximal growth rates from both gleaner and opportunist groups are favoured by diel light cycles. This mechanism weakens as latitude increases, because the effects of the seasonal cycle dominate over those of the diel cycle. **Main conclusions** Understanding the mechanisms that govern phytoplankton biogeography is crucial for predicting ocean ecosystem functioning and biogeochemical cycles. We show that the diel light cycle has a significant impact on phytoplankton competition and biogeography, indicating the need for understanding the role of diel processes in shaping macroecological patterns in the global ocean.

497. Investigating the robustness of the intraseasonal see-saw in the Indo-Pacific barotropic sea level across models

Authors: M. Afroosa; B. Rohith; Arya Paul; Fabien Durand; Romain Bourdallé-Badie; Sudheer Joseph; S. Prerna; S. S. C. Shenoi

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-022-01518-8

URL: <https://doi.org/10.1007/s10236-022-01518-8>

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Volume/Issue: Volume 72, Issue 7

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Cited by: 2 publications

Language: en

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Keywords: N/A

Funding: GMMC, Council of Scientific and Industrial Research, India

License: N/A

Open Access: N/A

Abstract: N/A

498. Adjustment of JEFF-3.3 data for U-235 and Pu-239 in the fast, non-resonant energy range

Authors: Sandro Pelloni; Dimitri Rochman

Journal: Annals of Nuclear Energy

Publisher: Elsevier BV

DOI: 10.1016/j.anucene.2022.109296

URL: <https://doi.org/10.1016/j.anucene.2022.109296>

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References: 31

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

499. Role of Ocean and Atmosphere Variability in Scale-Dependent Thermodynamic Air-Sea Interactions

Authors: Lucas C. Laurindo (International Laboratory for High-Resolution Earth System Prediction (iHESP) Texas A&M; University College Station TX USA, National Center for Atmospheric Research); R. Justin Small (International Laboratory for High-Resolution Earth System Prediction (iHESP) Texas A&M; University College Station TX USA, National Center for Atmospheric Research); LuAnne Thompson (University of Washington); Leo Siqueira (University of Miami); Frank O. Bryan (National Center for Atmospheric Research); Ping Chang (International Laboratory for High-Resolution Earth System Prediction (iHESP) Texas A&M; University College Station TX USA, Department of Atmospheric Sciences Texas A&M; University College Station TX USA, Department of Oceanography Texas A&M; University College Station TX USA); Gokhan Danabasoglu (International Laboratory for High-Resolution Earth System Prediction (iHESP) Texas A&M; University College Station TX USA, National Center for Atmospheric Research); Igor V. Kamenkovich (University of Miami); Ben P. Kirtman (University of Miami); Hong Wang (International Laboratory for High-Resolution Earth System Prediction (iHESP) Texas A&M; University College Station TX USA, Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Shaoqing Zhang (International Laboratory for High-Resolution Earth System Prediction (iHESP) Texas A&M; University College Station TX USA, Ocean University of China, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Language: en

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Keywords: N/A

Funding: National Center for Atmospheric Research, National Science Foundation, Climate Program Office (and 1 more)

License: N/A

Open Access: N/A

Abstract: This study investigates the influence of oceanic and atmospheric processes in extratropical thermodynamic air-sea interactions resolved by satellite observations (OBS) and by two climate model simulations run with eddy-resolving high-resolution (HR) and eddy-parameterized low-resolution (LR) ocean components. Here, spectral methods are used to characterize the sea surface temperature (SST) and turbulent heat flux (THF) variability and co-variability over scales between 50 and 10,000 km and 60 days to 80 years in the Pacific Ocean. The relative roles of the ocean and atmosphere are interpreted using a stochastic upper-ocean temperature evolution model forced by noise terms representing intrinsic variability in each medium, defined using climate model data to produce realistic rather than white spectral power density distributions. The analysis of all datasets shows that the atmosphere dominates the SST and THF variability over zonal wavelengths larger than ~2,000-2,500

km. In HR and OBS, ocean processes dominate the variability of both quantities at scales smaller than the atmospheric first internal Rossby radius of deformation (R_1 , ~600-2,000 km) due to a substantial ocean forcing coinciding with a weaker atmospheric modulation of THF (and consequently of SST) than at larger scales. The ocean forcing also induces oscillations in SST and THF with periods ranging from intraseasonal to multidecadal, reflecting a red spectrum response to ocean forcing similar to that driven by atmospheric forcing. Such features are virtually absent in LR due to a weaker ocean forcing relative to HR.

500. The role of winter net heat fluxes on the modulation of the upper mixed layer temperature and depth in the North Atlantic by the reanalysis data

Authors: P A Sukhonos; N A Diansky

Journal: IOP Conference Series: Earth and Environmental Science

Publisher: IOP Publishing

DOI: 10.1088/1755-1315/1040/1/012032

URL: <https://doi.org/10.1088/1755-1315/1040/1/012032>

Publication Date: 2022-06-24

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References: 18

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: The role of winter net heat fluxes (NHF) on the upper mixed layer (UML) temperature and depth variability over the North Atlantic is studied using ocean reanalysis data products. The monthly data from the ORA-S3 (1959-2011), GODAS (1980-2020), GECCO3 (1948-2018) datasets were used. The correlation coefficients were calculated between the anomalies of the NHF on the ocean surface and the UML temperature and depth with a shift of 1 month when the former were leading. In most of the North Atlantic, negative correlations between the UML temperature and the NHF preceding 1 month were found. Here, the UML temperature anomalies are probably caused by changes in heat exchange with the atmosphere. Areas of significant positive correlations across all data sets are concentrated in the area of the transition of the Gulf Stream to the North Atlantic Current. Almost the entire water area of the North Atlantic is occupied by positive correlations between the mixed layer depth in winter and the NHF on the ocean surface, with the latter leading for 1 month.

501. Synchronous Retreat of Southeast Greenland's Peripheral Glaciers

Authors: Julia Liu (University of Maine, Boise State University); Ellyn Enderlin (Boise State University); Hans-Peter Marshall (Boise State University); Andre Khalil (University of Maine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl097756

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Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: Earth Sciences Division

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Open Access: N/A

Abstract: Recently, scientific attention has focused on estimating Greenland's dynamic mass loss through changes to flow speeds, thickness, and length on its marine outlet glaciers. For the ice sheet outlet glaciers, dynamic mass loss has been found to be highly sensitive to changes in climate and individual glacier geometry. For the ice-sheet-independent marine glaciers around Greenland's periphery, dynamic mass loss is presently overlooked. Here, we apply an open-source, automated method of measuring glacier length changes using satellite imagery, to produce highly detailed records of length changes for 135 peripheral marine glaciers in southeast Greenland. We find evidence for anomalous retreat across 56 glaciers coincident with elevated surface melt in 2016, with melt 22% above the 2013-2019 average. Our detailed observations resolve the widespread, rapid, and synchronous response of these independent marine glaciers to increased meltwater input in 2016, indicating that their dynamics may be more sensitive to atmospheric warming than currently thought.

502. Submesoscale processes-induced vertical heat transport modulated by oceanic mesoscale eddies

Authors: Qingyue Wang; Changming Dong; Jihai Dong; Hong Zhang; Jingsong Yang

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2022.105138

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References: 47

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

503. Ocean mover's distance: using optimal transport for analysing oceanographic data

Authors: Sangwon Hyun (University of Southern California); Aditya Mishra (Flatiron Institute); Christopher L. Follett (Massachusetts Institute of Technology); Bror Jonsson (Plymouth Marine Laboratory); Gemma Kulk (Plymouth Marine Laboratory); Gael Forget (Massachusetts Institute of Technology); Marie-Fanny Racault (University of East Anglia); Thomas Jackson (Plymouth Marine Laboratory); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Christian L. Müller (Ludwig Maximilian University of Munich (LMU), Institute of Computational Biology, Helmholtz Zentrum München, Neuherberg, Germany, Flatiron Institute); Jacob Bien (University of Southern California)

Journal: Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rspa.2021.0875

URL: <https://doi.org/10.1098/rspa.2021.0875>

Publication Date: 2022-06-22

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References: 62

Cited by: 4 publications

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Keywords: N/A

Funding: NIH, Simons Collaboration on Computational Biogeochemical Modeling of Marine Ecosystems/CBIOMES, Frontiers of instability in marine ecosystems and carbon export (and 1 more)

License: N/A

Open Access: N/A

Abstract: Remote sensing observations from satellites and global biogeochemical models have combined to revolutionize the study of ocean biogeochemical cycling, but comparing the two data streams to each other and across time remains challenging due to the strong spatial-temporal structuring of the ocean. Here, we show that the Wasserstein distance provides a powerful metric for harnessing these structured datasets for better marine ecosystem and climate predictions. The Wasserstein distance complements commonly used point-wise difference methods such as the root-mean-squared error, by quantifying differences in terms of spatial displacement in addition to magnitude. As a test case, we consider chlorophyll (a key indicator of phytoplankton biomass) in the northeast Pacific Ocean, obtained from model simulations, in situ measurements, and satellite observations. We focus on two main applications: (i) comparing model predictions with satellite observations, and (ii) temporal evolution of chlorophyll both seasonally and over longer time frames. The Wasserstein distance successfully isolates temporal and depth variability and quantifies shifts in biogeochemical province boundaries. It also exposes relevant temporal trends in satellite chlorophyll consistent with climate change predictions. Our study shows that optimal transport vectors underlying the Wasserstein distance provide a novel visualization tool for testing models and better understanding temporal dynamics in the ocean.

504. Mixed Layer Salinity Balance in the Eastern Tropical Indian Ocean

Authors: Zhengbei Zhang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Jing Wang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Dongliang Yuan (Chinese Academy of Sciences, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc018229

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References: 60

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: mixed layer, salinity balance, Indian Ocean

Funding: National Natural Science Foundation of China, Chinese Academy of Sciences, Ministry of Natural Resources of the People's Republic of China

License: N/A

Open Access: N/A

Abstract: The mixed layer salinity (MLS) balance in the eastern tropical Indian Ocean is investigated using observations and output from the Estimating the Circulation and Climate of the Ocean model. The results suggest that the annual mean MLS is determined by the balance between the surface freshwater forcing and ocean advection. The seasonal MLS variation in the eastern equatorial Indian Ocean is driven strongly by salt advection associated with the Wyrtki Jet in spring and fall despite weak seasonal cycles of precipitation and evaporation over this region. Off the south Java coast, the MLS is the lowest in summer, due to the advection of freshwater southward associated with the meridional currents, whereas it reaches the maximum in fall, due to strong evaporation. During the negative Indian Ocean Dipole (IOD), the anomalous eastward currents transported high-salinity water to the east, generating positive MLS anomalies in the eastern tropical Indian Ocean. The regional evaporation minus precipitation anomalies were negative, which reduced the MLS. The MLS balance during the positive IOD is found to be nearly a mirror reflection of that during the negative IOD.

505. Noise Analysis and Combination of Hydrology Loading-Induced Displacements

Authors: Chang Xu (School of Road and Bridge, Zhejiang Institute of Communications, Hangzhou 311112, China); Xin Yao (School of Road and Bridge, Zhejiang Institute of Communications, Hangzhou 311112, China); Xiaoxing He (School of Civil and Surveying & Mapping Engineering, Jiangxi University of Science and Technology, Ganzhou 341000, China)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs14122840

URL: <https://doi.org/10.3390/rs14122840>

Publication Date: 2022-06-15

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References: 45

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Jiangxi University of Science and Technology High-level Talent Research Startup Project

License: CC BY 4.0

Open Access: N/A

Abstract: Large uncertainties exist in the available hydrology loading prediction models, and currently no consensus is reached on which loading model is superior or appears to represent nature in a more satisfactory way. This study discusses the noise characterization and combination of the vertical loadings predicted by different hydrology reanalysis (e.g., MERRA, GLDAS/Noah, GEOS-FPIT, and ERA interim). We focused on the hydrology loading predictions in the time span from 2011 to 2014 for the 70 Global Positioning System (GPS) sites, which are located close to the great rivers, lakes, and reservoirs. The maximum likelihood estimate with Akaike information criteria (AIC) showed that the auto-regressive (AR) model with an order from 2 to 5 is a good description of the temporal correlation that exists in the hydrology loading predictions. Moreover, significant discrepancy exists in the root mean square (RMS) of different hydrology loading predictions, and none of them have the lowest noise level for the all-time domain. Principal component analysis (PCA) was therefore used to create a combined loading-induced time series. Statistical indices (e.g., mean overlapping Hadamard variance, Nash-Sutcliffe efficiency, and variance reduction) showed that our proposed algorithm had an overall good performance and seemed to be potentially feasible for performing corrections on geodetic GPS heights.

506. A northeast United States Atlantis marine ecosystem model with ocean reanalysis and ocean color forcing

Authors: Joseph C. Caracappa; Andrew Beet; Sarah Gaichas; Robert J. Gamble; Kimberly J.W. Hyde; Scott I. Large; Ryan E. Morse; Charles A. Stock; Vincent S. Saba

Journal: Ecological Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ecolmodel.2022.110038

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Publication Date: 2022-06-14

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References: 53

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Northeast Fisheries Science Center

License: N/A

Open Access: N/A

Abstract: N/A

507. Wind-Driven Mechanisms for the Variations of the Pacific Equatorial Undercurrent Based on Adjoint Sensitivity Analysis

Authors: Jin Wang; Chuanyu Liu; Xiaowei Wang; Armin Köhl; Yilong Lyu; Yuanlong Li; Fan Wang

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2022.908939

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References: 52

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Key Research Program of Frontier Science, Chinese Academy of Sciences

License: CC BY 4.0

Open Access: N/A

Abstract: The maintenance and variation of the Pacific Equatorial Undercurrent (EUC) are thought to be controlled by the zonal pressure gradient force (ZPGF). However, a recent study found that a large-scale circulation associated with Rossby waves can also lead to EUC variation, implying that the structures and timing of the influential winds and the underlying wind-driven mechanisms need to be revisited. Here, we use the adjoint-sensitivity method to obtain the crucial winds that can most efficiently influence EUC variations. The obtained adjoint sensitivities (denoting the sensitive winds) are confined to 15 degreesS-15 degreesN and exhibit a funnel-shaped pattern with high symmetry about the equator. The remote winds, which occur 4 to 11 months prior, can lead to EUC variations at the basin scale; in contrast, the near-term winds (occurring not earlier than 4 months) lead only to local EUC variations. Accordingly, we find that wind-initiated equatorial Kelvin waves, equatorial Rossby waves, and the reflected waves at both the western and eastern boundaries superimpose onto each other to result in EUC variations. Specifically, when the travel time is longer than 4 months, the waves can form a negative-positive-negative sea surface height anomaly (SSHA) pattern between 15 degreesS and 15 degreesN in the central-eastern tropical Pacific, indicating the joint effect of both types of waves; they also form a positive SSHA in the western equatorial Pacific, indicating the dominance of the Kelvin wave therein. These mechanisms are complementary to the canonical ZPGF mechanism, which provide a clear theoretical basis for EUC variation studies.

508. Local and Remote Forcing of Interannual Sea-Level Variability at Nantucket Island

Authors: Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christopher G. Piecuch (Woods Hole Oceanographic Institution); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Thomas Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc018275

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References: 40

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: sea level, adjoint sensitivity, forcing mechanism

Funding: National Aeronautics and Space Administration

License: CC BY-NC 4.0

Open Access: N/A

Abstract: The relative contributions of local and remote wind stress and air-sea buoyancy forcing to sea-level variations along the East Coast of the United States are not well quantified, hindering the understanding of sea-level predictability there. Here, we use an adjoint sensitivity analysis together with an Estimating the Circulation and Climate of the Ocean (ECCO) ocean state estimate to establish the causality of interannual variations in Nantucket dynamic sea level. Wind forcing explains 67% of the Nantucket interannual sea-level variance, while wind and buoyancy forcing together explain 97% of the variance. Wind stress contribution is near-local, primarily from the New England shelf northeast of Nantucket. We disprove a previous hypothesis about Labrador Sea wind stress being an important driver of Nantucket sea-level variations. Buoyancy forcing, as important as wind stress in some years, includes local contributions as well as remote contributions from the subpolar North Atlantic that influence Nantucket sea level a few years later. Our rigorous adjoint-based analysis corroborates previous correlation-based studies indicating that sea-level variations in the subpolar gyre and along the United States northeast coast can both be influenced by subpolar buoyancy forcing. Forward perturbation experiments further indicate remote buoyancy forcing affects Nantucket sea level mostly through slow advective processes, although coastally trapped waves can cause rapid Nantucket sea

level response within a few weeks.

509. Seasonal Variability of the Pacific South Equatorial Current during the Argo Era

Authors: Lina Yang (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Raghu Murtugudde (University of Maryland); Shaojun Zheng (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Peng Liang (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Wei Tan (e College of Ocean Science and Engineering, Shandong University of Science and Technology, Qingdao, China); Lei Wang (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Baoxin Feng (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China); Tianyu Zhang (Ministry of Natural Resources, Guangdong Ocean University, b Key Laboratory of Climate, Resources and Environment in Continental Shelf Sea and Deep Sea of Department of Education of Guangdong Province, Zhanjiang, China)

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Keywords: Kelvin waves, Wind stress curl, Ocean circulation, Rossby waves, Ocean dynamics

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Abstract: The tropical Pacific currents from January 2004 to December 2018 are computed based on the gridded Argo temperatures and salinities using the P-vector method on an f plane and the geostrophic approximation on a beta plane. Three branches of the South Equatorial Current (SEC) are identified, i.e., SEC(N) (2 degreesS-5 degreesN), SEC(M) (7 degrees-3 degreesS), and SEC(S) (20 degrees-8 degreesS), with the maximum zonal velocity of -55 cm s⁻¹ and total volume transport of -49.8 Sv (1 Sv $\equiv$ 106 m³ s⁻¹) occurring in the central-east Pacific. The seasonal variability of each branch shows a distinct and different westward propagation of zonal current anomalies, which are well mirrored by the SLA differences between 2 degreesS and 5 degreesN, between 3 degreesS and 6 degreesS, and between 8 degreesS and 15 degreesS, respectively. Most of the seasonal variations are successfully simulated by a simple analytical Rossby wave model, highlighting the significance of the

first-mode baroclinic, linear Rossby waves, particularly those driven by the wind stress curl in the central-east Pacific. However, the linear theory fails to explain the SEC(M) variations in certain months in the central-east Pacific, where the first baroclinic mode contributes only around 50% of the explained variance to the equatorial surface currents. A nonlinear model involving higher baroclinic modes is suggested for a further diagnosis. Considering the crucial role played by the tropical Pacific in the natural climate variability via the El Niño-Southern Ocean dynamics and the ocean response to anthropogenic forcing via the ocean heat uptake in the eastern tropical Pacific, advancing the process understanding of the SEC from observations is critical.

510. Preliminary study on hydrological angular momentum determined from CMIP6 historical simulations

Authors: Jolanta Nastula; Justyna ■liwi■ska; Tomasz Kur; Ma■gorzata Wi■ska; Aleksander Partyka

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-022-01636-z

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Abstract: Polar motion (PM) is an essential parameter needed to transform coordinates between celestial and terrestrial reference frames, thus playing a crucial role in precise positioning and navigation. The role of hydrological signals in PM excitation is not yet fully understood, which is largely because of the lack of agreement between estimates of hydrological angular momentum (HAM) computed from different data sources. In this study, we used data obtained from the latest, sixth phase of the Coupled Model Intercomparison Project (CMIP6) to assess the impact of the continental hydrosphere on PM excitation. To do so, we exploited soil moisture and snow water variables obtained from historical simulations of CMIP6 to estimate climate-based HAM. The HAM series were computed, then we analysed their variability in terms of trends, seasonal and non-seasonal oscillations. An important part of this study is the validation of HAM estimates based on comparison with the hydrological signal in geodetically observed PM excitation (geodetic residuals, GAO). In addition, HAM series based on climate models were compared with those determined from global gravimetric data provided by the Gravity Recovery and Climate Experiment (GRACE) mission, and from the Land Surface Discharge Model (LSDM). This study also aimed to identify the most appropriate CMIP6 models for interpretation of PM variations. Overall, the correspondence between GAO and HAM received from CMIP6 was lower than the previously obtained consistency with GRACE results, and the level of agreement was dependent on the oscillation considered and the model used. However, it may be possible to identify several CMIP6 models from among the almost 100 available that provides a HAM series more compatible with GAO than HAM from GRACE or LSDM, especially in annual oscillations. The GISS-E2-1-G_historical_r10i1p1f1 model was found to provide the highest consistency with GAO for annual prograde amplitudes, GFDL-CM4_historical_r1i1p1f1 for annual retrograde amplitudes, BCC-ESM1_historical_r3i1p1f1 for the annual prograde phase, and MIROC-ES2L_historical_r2i1p1f2 for the annual retrograde phase. Because of their length, the CMIP6 data allow for analysis of the past and future changes in HAM from 1850 to 2100, which is of particular importance in the exploration of the impact of climate change on PM excitation. Graphical Abstract

511. Region of freshwater influence (ROFI) and its impact on sediment transport in the lower Mekong Delta coastal zone of Vietnam

Authors: Nguyet-Minh Nguyen; Dinh Cong San; Kim Dan Nguyen; Quoc Bao Pham; Alexandre S. Gagnon; Son T. Mai; Duong Tran Anh

Journal: Environmental Monitoring and Assessment

Publisher: Springer Science and Business Media LLC

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Cited by: 8 publications

Language: en

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Funding: N/A

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Open Access: N/A

Abstract: N/A

512. Tracer and observationally derived constraints on diapycnal diffusivities in an ocean state estimate

Authors: David S. Trossman; Caitlin B. Whalen; Thomas W. N. Haine; Amy F. Waterhouse; An T. Nguyen; Arash Bigdeli; Matthew Mazloff; Patrick Heimbach

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-18-729-2022

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Volume/Issue: Volume 18, Issue 3

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Open Access: N/A

Abstract: . Use of an ocean parameter and state estimation framework - such as the Estimating the Circulation and Climate of the Ocean (ECCO) framework - could provide an opportunity to learn about the spatial distribution of the diapycnal diffusivity parameter (κ_p) that observations alone cannot due to gaps in coverage. However, we show that the inclusion of misfits to observed physical variables - such as in situ temperature, salinity, and pressure - currently accounted for in ECCO is not sufficient, as κ_p from ECCO does not agree closely with any observationally derived product. These observationally derived κ_p products were inferred from microstructure measurements, derived from Argo and conductivity-temperature-depth (CTD) data using a strain-based parameterization of fine-scale hydrographic structure, or calculated from climatological and seafloor data using a parameterization of tidal mixing. The κ_p products are in close agreement with one another but have both measurement and structural uncertainties, whereas tracers can have relatively small measurement uncertainties. With the ultimate goal being to jointly improve the ECCO state estimate and representation of κ_p in ECCO, we investigate whether adjustments in κ_p due to inclusion of misfits to a tracer - dissolved oxygen concentrations from an annual climatology - would be similar to those due to inclusion of misfits to observationally derived κ_p products. We do this by performing sensitivity analyses with ECCO. We compare multiple adjoint sensitivity calculations: one configuration uses misfits to observationally derived κ_p , and the other uses misfits to observed dissolved oxygen concentrations. We show that adjoint sensitivities of dissolved oxygen concentration misfits to the state estimate's control space typically direct κ_p to improve relative to the observationally derived values. These results suggest that the inclusion of oxygen in ECCO's misfits will improve κ_p in ECCO, particularly in (sub)tropical regions.

513. Seasonality and Spatial Dependence of Mesoscale and Submesoscale Ocean Currents from Along-Track Satellite Altimetry

Authors: Albion Lawrence (Brandeis University); Jörn Callies (California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Language: N/A

Subjects: N/A

Keywords: Oceanic variability, Seasonal variability, Altimetry, Turbulence, Oceanic mixed layer (and 3 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Along-track wavenumber spectral densities of sea surface height (SSH) are estimated from Jason-2 altimetry data as a function of spatial location and calendar month to understand the seasonality of meso- and submesoscale balanced dynamics across the global ocean. Regions with significant mode-1 and mode-2 baroclinic tides are rejected, restricting the analysis to the extratropics. Where balanced motion dominates, the SSH spectral density is averaged over all pass segments in a region for each calendar month and is fit to a four-parameter model consisting of a flat plateau at low wavenumbers, a transition at wavenumber k_0 to a red power law spectrum k^{-s} , and a white spectrum at high wavenumbers that models the altimeter noise. The monthly time series of the model parameters are compared to the evolution of the mixed layer. The annual mode of the spectral slope s reaches a minimum after the mixed layer deepens, and the annual mode of the bandpassed kinetic energy in the ranges $[2k_0, 4k_0]$ and $[k_0, 2k_0]$ peak ~ 2 and ~ 4 months, respectively, after the maximum of the annual mode of the mixed layer depth. This analysis is consistent with an energization of the submesoscale by a winter mixed layer instability followed by an inverse cascade of kinetic energy to the mesoscale, in agreement with prior modeling studies and in situ measurements. These results are compared to prior modeling, in situ, and satellite investigations of specific regions and are broadly consistent with them within measurement uncertainties. **Significance Statement** This paper uses satellite observations to understand the source of ocean dynamics at the 1-100-km scales at which vertical motion becomes important and which are thus relevant for biology and for the exchange of heat and carbon with the atmosphere. The observations are consistent with a seasonal variation of dynamics at these scales, predicted by a specific theory of upper-ocean turbulence and confirmed by modeling studies and regional observations. We update prior satellite-based studies by excluding regions with competing effects, by our treatment of the noise, and by our characterization of the seasonality. This work provides a template for analyzing data from the upcoming Surface Water and Ocean Topography (SWOT) satellite.

514. Data-based estimates of interannual sea-air CO₂ flux variations 1957-2020 and their relation to environmental drivers

Authors: Christian Rödenbeck; Tim DeVries; Judith Hauck; Corinne Le Quéré; Ralph F. Keeling

Journal: Biogeosciences

Publisher: Copernicus GmbH

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URL: <https://doi.org/10.5194/bg-19-2627-2022>

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References: 69

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Open Access: N/A

Abstract: . This study considers year-to-year and decadal variations in as well as secular trends of the sea-air CO₂ flux over the 1957-2020 period, as constrained by the pCO₂ measurements from the SOCATv2021 database. In a first step, we relate interannual anomalies in ocean-internal carbon sources and sinks to local interannual anomalies in sea surface temperature (SST), the temporal changes in SST ($dSST/dt$), and squared wind speed (u^2), employing a multi-linear regression. In the tropical Pacific, we find interannual variability to be dominated by $dSST/dt$, as arising from variations in the upwelling of colder and more carbon-rich waters into the mixed layer. In the eastern upwelling zones as well as in circumpolar bands in the high latitudes of both hemispheres, we find sensitivity to wind speed, compatible with the entrainment of carbon-rich water during wind-driven deepening of the mixed layer and wind-driven upwelling. In the Southern Ocean, the secular increase in wind speed leads to a secular increase in the carbon source into the mixed layer, with an estimated reduction in the sink trend in the range of 17 % to 42 %. In a second step, we combined the result of the multi-linear regression and an explicitly interannual pCO₂-based additive correction into a "hybrid" estimate of the sea-air CO₂ flux over the period 1957-2020. As a pCO₂ mapping method, it combines (a) the ability of a regression to bridge data gaps and extrapolate into the early decades almost void of pCO₂ data based on process-related observables and (b) the ability of an auto-regressive interpolation to follow signals even if not represented in the chosen set of explanatory variables. The "hybrid" estimate can be applied as an ocean flux prior for atmospheric CO₂ inversions covering the whole period of atmospheric CO₂ data since 1957.

515. A Semi-Empirical Framework for ice sheet response analysis under Oceanic forcing in Antarctica and Greenland

Authors: Xiao Luo; Ting Lin

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-022-06317-x

URL: <https://doi.org/10.1007/s00382-022-06317-x>

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Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: Texas Tech University

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Open Access: N/A

Abstract: N/A

516. Temporal Variability of Co-Occurring *Calanus finmarchicus* and *C. helgolandicus* in Skagerrak

Authors: Tone Falkenhaug; Cecilie Broms; Espen Bagøien; Nikolaos Nikolioudakis

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

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References: 100

Cited by: 3 publications

Language: N/A

Subjects: N/A

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Open Access: N/A

Abstract: The congeneric copepods *Calanus finmarchicus* and *C. helgolandicus* co-occur in the northern North Sea and Skagerrak where they play important roles as prey for higher trophic levels. This study analyses a 26-year time series (1994-2019) with a ~ two-week temporal resolution for *Calanus* spp. at a fixed monitoring station in Skagerrak, off southern Norway. Seasonal variation, inter-annual variability and long-term trends for the two species were examined. Strong differences in the species-specific seasonality were revealed, with *C. finmarchicus* dominating in spring and *C. helgolandicus* in autumn. The seasonal peak of *C. finmarchicus* was associated with relatively low temperatures (6-8 degreesC) and high chlorophyll a concentration, while *C. helgolandicus* displayed its seasonal maximum at higher temperatures (11-16 degreesC). *C. finmarchicus* was found to produce one dominant annual generation (in spring), but two or more generations are considered likely. Contrasting long-term trends in abundances were found for the two species, suggesting that their population sizes were affected by different mechanisms. The abundance of *C. helgolandicus* showed an increasing trend over the time period studied, apart from the last years. The abundance of *C. finmarchicus* was more variable and displayed no unidirectional long-term trends over the time series. The study revealed a shift in the phenology of *Calanus* spp. over the last 25 years at this site. Since 1994 the timing of the annual peaks in both *C. finmarchicus* and *C. helgolandicus* have advanced by about one month. Moreover, the seasonal pattern of *C. helgolandicus*, switched from a unimodal to a bimodal pattern around 2002, with a small additional peak also appearing in spring. The results suggest that the proximity to the Norwegian Trench influences the demography and abundance of *C. finmarchicus* in this coastal area, both as a gateway for the advective supply, as well as a habitat for local overwintering.

517. Historical Reconstruction of Subpolar North Atlantic Overturning and Its Relationship to Density

Authors: Vassil M. Roussenov (University of Liverpool); Richard G. Williams (University of Liverpool); M. Susan Lozier (Georgia Institute of Technology); N. Penny Holliday (National Oceanography Centre); Doug M. Smith (Met Office Hadley Centre, Met Office)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Abstract: The connections between the overturning of the subpolar North Atlantic and regional density changes are assessed on interannual and decadal timescales using historical, data-based reconstructions of the overturning over the last 60 years and forward model integrations with buoyancy and wind forcing. The data-based reconstructions reveal a dominant eastern basin contribution to the subpolar overturning in density space and changes in the overturning reaching ± 2.5 Sv, which are both in accord with the Overturning in the Subpolar North Atlantic Program (OSNAP). The zonally integrated geostrophic velocity across the basin is connected to boundary contrasts in Montgomery potential in density space. The overturning for the eastern side of the basin is strongly correlated with density changes in the Irminger and Labrador Seas, while the overturning for the western side is correlated with boundary density changes in the Labrador Sea. These boundary density signals are a consequence of local atmospheric forcing and transport of upstream density changes. In forward model experiments, a localized density increase over the Irminger Sea increases the overturning over both sides of the basin due to dense waters spreading to the Labrador Sea. Conversely, a localized density increase over the Labrador Sea only increases the overturning for the western basin and instead eventually decreases the overturning for the eastern basin. Labrador Sea density provides a useful overturning metric by its direct control of the overturning over the western side and lower latitudes of the subpolar basin.

518. Equatorial Origin of the Observed Tropical Pacific Quasi-Decadal Variability From ENSO Nonlinearity

Authors: Chao Liu (Nanjing University); Wenjun Zhang (Nanjing University); Fei-Fei Jin (Department of Atmospheric Sciences School of Ocean and Earth Science and Technology (SOEST) University of Hawai'i at Mānoa Honolulu HI USA); Malte F. Stuecker (Department of Oceanography & International Pacific Research Center (IPRC) School of Ocean and Earth Science and Technology (SOEST) University of Hawai'i at Mānoa Honolulu HI USA); Licheng Geng (Department of Atmospheric Sciences School of Ocean and Earth Science and Technology (SOEST) University of Hawai'i at Mānoa Honolulu HI USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 9 publications

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Funding: China Scholarship Council

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Abstract: Quasi-decadal (QD) climate variability is detected in the tropical Pacific based on the recent 70 years of observations. This QD variability is identified in equatorial sea surface temperatures (SSTs), the pattern of which resembles the Central Pacific (CP) El Niño-Southern Oscillation (ENSO) but extends further meridionally to the northeastern subtropical Pacific. Whereas equatorial upper-ocean heat content and SSTs are in quadrature for ENSO, these two quantities are almost in phase on the QD timescale. Further analysis shows that nonlinear dynamical heating, primarily originating from strong El Niño events, tends to lead QD SSTs by a quarter of its dominant period (approximately 30 months) and shapes the dominant QD periodicity in observations. Our results suggest that the observed QD variability largely originates from ENSO nonlinearity and thus is fundamentally different from ENSO's oscillatory nature.

**519. Limnology and Oceanography Bulletin Volume 31 Number 2 May 2022
31-55**

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Journal: Limnology and Oceanography Bulletin

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Open Access: N/A

Abstract: N/A

520. Mean, Annual, and Interannual Circulation and Volume Transport in the Western Tropical North Pacific From the Western Pacific Ocean State Estimates (WPOSE)

Authors: Martha C. Schönau (University of California San Diego, Applied Ocean Sciences Fairfax Station VA USA, Scripps Institution of Oceanography); Daniel L. Rudnick (University of California San Diego, Scripps Institution of Oceanography); Ganesh Gopalakrishnan (University of California San Diego, Scripps Institution of Oceanography); Bruce D. Cornuelle (University of California San Diego, Scripps Institution of Oceanography); Bo Qiu (University of Hawaii at Manoa)

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Cited by: 2 publications

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Subjects: N/A

Keywords: volume transport, tropical pacific, El Niño, MITgcm, ECCO (and 1 more)

Funding: Office of Naval Research, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Interannual volume transport anomalies in the far western tropical North Pacific were calculated using the Western Pacific Ocean State Estimates (WPOSE; 2009-2017; 115 degreesE-170 degreesE, 15 degreesS-27 degreesN) to assess the magnitude, phase, and vertical structure of volume transport, and the relationship of interannual transport to the El Niño-Southern Oscillation phenomena. The mean velocity and thermohaline structure for the WPOSE were verified using Argo climatology and autonomous glider observations. Transport was calculated for both an upper layer (surface-26 kg m⁻³) and a lower layer (26-27.3 kg m⁻³). Annual volume transport anomalies were small north of 8 degreesN but varied by 50% of the mean volume transport between the equator and 8 degreesN. Interannual anomalies exceeded annual anomalies throughout the region. Volume transports of the North Equatorial Current, the Mindanao Current, and the South Equatorial Current/New Guinea Coastal Undercurrent increased beginning in September 2014, leading to a large increase in transport in the North Equatorial Counter Current before the mature phase of El Niño in 2015/2016. The increase in transport was related to the meridional gradient in thermocline depth centered at 5 degreesN at the southern part of the Mindanao Dome, where extreme shoaling of the thermocline took place. Lower-layer volume transports were not always in phase with those of the upper-layer and made considerable contributions to total transport variability.

521. Transition from Balanced to Unbalanced Motion in the Eastern Tropical Pacific

Authors: Saulo M. Soares (Scripps Institution of Oceanography / University of California San Diego); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Teresa K. Chereskin (Scripps Institution of Oceanography / University of California San Diego); Eric Firing (b Department of Oceanography, School of Ocean and Earth Science and Technology, University of Hawai'i at Mānoa, Honolulu, Hawaii); Jules Hummon (b Department of Oceanography, School of Ocean and Earth Science and Technology, University of Hawai'i at Mānoa, Honolulu, Hawaii); Cesar B. Rocha (c Department of Marine Sciences, University of Connecticut, Groton, Connecticut)

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Language: N/A

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Keywords: Pacific Ocean, Eddies, Ship observations, Currents, Acoustic measurements/effects (and 8 more)

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Open Access: N/A

Abstract: Kinetic energy associated with inertia-gravity waves (IGWs) and other ageostrophic phenomena often overwhelms kinetic energy due to geostrophic motions for wavelengths on the order of tens of kilometers. Understanding the dependencies of the wavelength at which balanced (geostrophic) variability ceases to be larger than unbalanced variability is important for interpreting high-resolution altimetric data. This wavelength has been termed the transition scale. This study uses acoustic Doppler current profiler (ADCP) data along with auxiliary observations and a numerical model to investigate the transition scale in the eastern tropical Pacific and the mechanisms responsible for its regional and seasonal variations. One-dimensional kinetic energy wavenumber spectra are separated into rotational and divergent components, and subsequently into vortex and wave components. The divergent motions, most likely predominantly IGWs, account for most of the energy at wavelengths less than 100 km. The observed regional and seasonal patterns in the transition scale are consistent with those from a high-resolution global simulation. Observations, however, show weaker seasonality, with only modest wintertime increases in vortex energy. The ADCP-inferred IGW wavenumber spectra suggest that waves with near-inertial frequency dominate the unbalanced variability, while in model output, internal tides strongly influence the wavenumber spectrum. The ADCP-derived transition scales from the eastern tropical Pacific are typically in the 100-200-km range.

522. Upper-Ocean Lateral Heat Transports in the Niño3.4 Region and Their Connection with ENSO

Authors: He Zhao; Tongwen Wu; Laurent Z. X. Li; Fanghua Wu; Weihua Jie; Xiangwen Liu; Ronghua Zhang; Chengjun Xie

Journal: Journal of Meteorological Research

Publisher: Springer Science and Business Media LLC

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

523. Hydroacoustic Travel Time Variations as a Proxy for Passive Deep-Ocean Thermometry--A Cookbook

Authors: P. S. M. Smets (Delft University of Technology, Royal Netherlands Meteorological Institute); C. Weemstra (Delft University of Technology, Royal Netherlands Meteorological Institute); L. G. Evers (Delft University of Technology, Royal Netherlands Meteorological Institute)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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References: 55

Cited by: N/A publications

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Keywords: N/A

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Abstract: We report on the extraction of deep ocean travel time variations from time-lapse cross-correlations between a hydrophone station and a three-component broadband seismometer. The signals we cross-correlate in this study result from repeated activity by the Monowai seamount, one of the most active submarine volcanoes of the Tonga-Kermadec ridge. In particular, we introduce a specific workflow to exploit repetitive hydroacoustic underwater source activity, which we detail to such an extent that it serves as an example (or "cookbook"). For this reason, we have made the source code publicly available. The workflow proposed in this study (a) overcomes differences in instrument sensitivity and sample rates, (b) involves the selection of eligible cross-correlations based on a source activity criterium as well as slowness analysis, and (c) extracts the travel time variations in distinct frequency bands. In our case, the two frequency bands are 3-6 and 6-12 Hz. We find that the estimated travel time variations in both frequency bands consist of a complex periodic pattern superimposed on a robust linear trend. This linear trend is decreasing, which we attribute to increasing water temperatures along the propagation path of the hydroacoustic signals.

524. Inter-annual variability of biogeography-based phytoplankton seasonality in the Arabian Sea during 1998-2017

Authors: Ke Huang; Huijie Xue; Fei Chai; Dongxiao Wang; Peng Xiu; Qiang Xie; Ying Wu; Xiaomei Liao; Wentao Ma; Jinglong Yao; Wei Zhou

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

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Abstract: N/A

525. A monthly surface pCO₂ product for the California Current Large Marine Ecosystem

Authors: Jonathan D. Sharp; Andrea J. Fassbender; Brendan R. Carter; Paige D. Lavin; Adrienne J. Sutton

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-14-2081-2022

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References: 111

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Longitude

Funding: National Oceanic and Atmospheric Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . A common strategy for calculating the direction and rate of carbon dioxide gas (CO₂) exchange between the ocean and atmosphere relies on knowledge of the partial pressure of CO₂ in surface seawater (pCO₂(sw)), a quantity that is frequently observed by autonomous sensors on ships and moored buoys, albeit with significant spatial and temporal gaps. Here we present a monthly gridded data product of pCO₂(sw) at 0.25° latitude by 0.25° longitude resolution in the northeastern Pacific Ocean, centered on the California Current System (CCS) and spanning all months from January 1998 to December 2020. The data product (RFR-CCS; Sharp et al., 2022; <https://doi.org/10.5281/zenodo.5523389>) was created using observations from the most recent (2021) version of the Surface Ocean CO₂ Atlas (Bakker et al., 2016). These observations were fit against a variety of collocated and contemporaneous satellite- and model-derived surface variables using a random forest regression (RFR) model. We validate RFR-CCS in multiple ways, including direct comparisons with observations from sensors on moored buoys, and find that the data product effectively captures seasonal pCO₂(sw) cycles at nearshore sites. This result is notable because global gridded pCO₂(sw) products do not capture local variability effectively in this region, suggesting that RFR-CCS is a better option than regional extractions from global products to represent pCO₂(sw) in the CCS over the last 2 decades. Lessons learned from the construction of RFR-CCS provide insight into how global pCO₂(sw) products could effectively characterize seasonal variability in nearshore coastal environments. We briefly review the physical and biological processes - acting across a variety of spatial and temporal scales - that are responsible for the latitudinal and nearshore-to-offshore pCO₂(sw) gradients seen in the RFR-CCS reconstruction of pCO₂(sw). RFR-CCS will be valuable for the validation of high-resolution models, the attribution of spatiotemporal carbonate system variability to physical and biological drivers, and the quantification of multiyear trends and interannual variability of ocean acidification.

526. Second Baroclinic Mode Rossby Waves in the South Indian Ocean

Authors: Motoki Nagura (a Japan Agency for Marine-Earth Science and Technology, Yokosuka, Kanagawa, Japan); Satoshi Osafune (a Japan Agency for Marine-Earth Science and Technology, Yokosuka, Kanagawa, Japan)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-21-0290.1

URL: <https://doi.org/10.1175/jpo-d-21-0290.1>

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References: 79

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Interannual variability, Numerical analysis/modeling, Data assimilation, Indian Ocean, Rossby waves

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Many previous studies of midlatitude Rossby waves have examined satellite altimetry data, which reflect variability near the surface above the pycnocline. Argo float observations provide hydrographic data in the upper 2000 m, which likely monitor subsurface variability below the pycnocline. This study examines the variability in meridional velocity at midlatitudes and investigates Rossby waves in the southern Indian Ocean using an ocean reanalysis generated by a 4DVAR method. The results show two modes of variability. One is trapped near the surface and propagates to the west at a phase speed close to that of first baroclinic mode Rossby waves. This mode is representative of variability detected by satellite altimetry. The other mode has a local peak in amplitude at ~600-m depth and propagates to the west at a phase speed 3 times slower than the first baroclinic mode. Such slowly propagating signals are observed globally, but they are largest in amplitude in the southern Indian Ocean and consistent in phase speed with the second baroclinic mode. Results from numerical experiments using an OGCM show that zonal winds in the tropical Pacific Ocean related to ENSO are the primary driver of slowly propagating signals in the southern Indian Ocean. Wind forcing in the tropical Pacific Ocean drives a surface trapped jet that propagates via the Indonesian Archipelago and excites subsurface variability in meridional velocity in the southern Indian Ocean. In addition, surface heat flux and meridional winds near the west coast of Australia can drive subsurface variability. **Significance Statement** Many previous studies of midlatitude Rossby waves have used satellite altimetry measurements, which reflect variability in the upper few hundred meters of the ocean. Argo float observations have provided in situ hydrographic observations in the upper 2000 m, and these enable us to examine subsurface variability with high reliability. In this study, we used output from an ocean reanalysis, which assimilates in situ observations, and found that the meridional velocity below the surface (~600-m depth) of the southern Indian Ocean propagates at a phase speed 3 times slower than that of surface variability. These slowly propagating signals can be of climatic importance because of their possible impact on meridional heat transport. We also discuss the driving

force of these slowly propagating signals.

527. Quantifying discrepancies in the three-dimensional seasonal variations between IGS station positions and load models

Authors: Yujiao Niu; Na Wei; Min Li; Paul Rebischung; Chuang Shi; Guo Chen

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-022-01618-9

URL: <https://doi.org/10.1007/s00190-022-01618-9>

Publication Date: 2022-04-25

Volume/Issue: Volume 96, Issue 4

References: 85

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Draconitic errors, Regional/local-scale loading, GNSS, Inter-AC disagreement, 3D seasonal deformation (and 1 more)

Funding: national natural science foundation of china, wuhan municipal science and technology bureau

License: N/A

Open Access: N/A

Abstract: N/A

528. Monitoring Groundwater Storage Changes in a Karst Aquifer using Superconducting Gravimeter OSG-066 at the Lijiang Station in China

Authors: Lelin Xing; Xiaowei Niu; Lei Bai; Yaowen Yang

Journal: Pure and Applied Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00024-022-03024-w

URL: <https://doi.org/10.1007/s00024-022-03024-w>

Publication Date: 2022-04-23

Volume/Issue: Volume 179, Issue 5

References: 46

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

529. Uncertainty in GRACE/GRACE-follow on global ocean mass change estimates due to mis-modeled glacial isostatic adjustment and geocenter motion

Authors: Jae-Seung Kim; Ki-Weon Seo; Jianli Chen; Clark Wilson

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-022-10628-8

URL: <https://doi.org/10.1038/s41598-022-10628-8>

Publication Date: 2022-04-22

Volume/Issue: Volume 12, Issue 1

References: 38

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Korea Institute of Marine Science and Technology promotion, National Research Foundation of Korea, NASA GRACE and GRACE Follow-On Projects (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Global mean sea level has increased about 3 mm/yr over several decades due to increases in ocean mass and changes in sea water density. Ocean mass, accounting for about two-thirds of the increase, can be directly measured by the Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GFO) satellites. An independent measure is obtained by combining satellite altimetry (measuring total sea level change) and Argo float data (measuring steric changes associated with sea water density). Many previous studies have reported that the two estimates of global mean ocean mass (GMOM) change are in good agreement within stated confidence intervals. Recently, particularly since 2016, estimates by the two methods have diverged. A partial explanation appears to be a spurious variation in steric sea level data. An additional contributor may be deficiencies in Glacial Isostatic Adjustment (GIA) corrections and degree-1 spherical harmonic (SH) coefficients. We found that erroneous corrections for GIA contaminate GRACE/GFO estimates as time goes forward. Errors in GIA corrections affect degree-1 SH coefficients, and degree-1 errors may also be associated with ocean dynamics. Poor estimates of degree-1 SH coefficients are likely an important source of discrepancies in the two methods of estimating GMOM change.

530. 3DVar and SC-4DVar for the Lorenz 63 Model

Authors: Geir Evensen; Femke C. Vossepoel; Peter Jan van Leeuwen

Journal: Springer Textbooks in Earth Sciences, Geography and Environment

Publisher: Springer International Publishing

DOI: 10.1007/978-3-030-96709-3_16

URL: https://doi.org/10.1007/978-3-030-96709-3_16

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Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: In this chapter, we study the workings of 3DVar and SC-4DVar on the same chaotic Lorenz 1963 system as used with ensemble methods in Chap. 10.1007/978-3-030-96709-3_15. We will apply both 3DVar and SC-4DVar sequentially over multiple data-assimilation windows, and we will demonstrate the difference between the filter solution obtained by 3DVar and the recursive SC-4DVar smoother solution. We will also dive deeper into the behavior of the SC-4DVar with highly nonlinear- and chaotic dynamics and try to understand more of the method's properties and possible limitations in these cases. After studying the 3DVar and 4DVar methods, we compare them with the ensemble methods used in Chap. 10.1007/978-3-030-96709-3_15.

531. Seasonality and trend of the global upper-ocean vertical velocity over 1998-2017

Authors: Fanglou Liao; Guandong Gao; Peng Zhan; Yan Wang

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2022.102804

URL: <https://doi.org/10.1016/j.pocean.2022.102804>

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Volume/Issue: Volume 204

References: 71

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Chinese Academy of Sciences, Research Grants Council, University Grants Committee (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

532. The impact of tides on Antarctic ice shelf melting

Authors: Ole Richter; David E. Gwyther; Matt A. King; Benjamin K. Galton-Fenzi

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-1409-2022

URL: <https://doi.org/10.5194/tc-16-1409-2022>

Publication Date: 2022-04-21

Volume/Issue: Volume 16, Issue 4

References: 73

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: . Tides influence basal melting of individual Antarctic ice shelves, but their net impact on Antarctic-wide ice-ocean interaction has yet to be constrained. Here we quantify the impact of tides on ice shelf melting and the continental shelf seas using a 4 km resolution circum-Antarctic ocean model. Activating tides in the model increases the total basal mass loss by 57 Gt yr⁻¹ (4 %) while decreasing continental shelf temperatures by 0.04 °C. The Ronne Ice Shelf features the highest increase in mass loss (44 Gt yr⁻¹, 128 %), coinciding with strong residual currents and increasing temperatures on the adjacent continental shelf. In some large ice shelves tides strongly affect melting in regions where the ice thickness is of dynamic importance to grounded ice flow. Further, to explore the processes that cause variations in melting we apply dynamical-thermodynamical decomposition to the melt drivers in the boundary layer. In most regions, the impact of tidal currents on the turbulent exchange of heat and salt across the ice-ocean boundary layer has a strong contribution. In some regions, however, mechanisms driven by thermodynamic effects are equally or more important, including under the frontal parts of Ronne Ice Shelf. Our results support the importance of capturing tides for robust modelling of glacier systems and shelf seas, as well as motivate future studies to directly assess friction-based parameterizations for the pan-Antarctic domain.

533. Surface warming-induced global acceleration of upper ocean currents

Authors: Qihua Peng (Scripps Institution of Oceanography / University of California San Diego, Chinese Academy of Sciences); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego); Dongxiao Wang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Rui Xin Huang (Woods Hole Oceanographic Institution); Gengxin Chen (Chinese Academy of Sciences); Yeqiang Shu (Chinese Academy of Sciences); Jia-Rui Shi (Scripps Institution of Oceanography / University of California San Diego, Woods Hole Oceanographic Institution); Wei Liu (University of California Riverside)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.abj8394

URL: <https://doi.org/10.1126/sciadv.abj8394>

Publication Date: 2022-04-20

Volume/Issue: Volume 8, Issue 16

References: 69

Cited by: 74 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: How the ocean circulation changes in a warming climate is an important but poorly understood problem. Using a global ocean model, we decompose the problem into distinct responses to changes in sea surface temperature, salinity, and wind. Our results show that the surface warming effect, a robust feature of anthropogenic climate change, dominates and accelerates the upper ocean currents in 77% of the global ocean. Specifically, the increased vertical stratification intensifies the upper subtropical gyres and equatorial currents by shoaling these systems, while the differential warming between the Southern Ocean upwelling zone and the region to the north accelerates surface zonal currents in the Southern Ocean. In comparison, the wind stress and surface salinity changes affect regional current systems. Our study points a way forward for investigating ocean circulation change and evaluating the uncertainty.

534. Heavy footprints of upper-ocean eddies on weakened Arctic sea ice in marginal ice zones

Authors: Georgy E. Manucharyan; Andrew F. Thompson

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-022-29663-0

URL: <https://doi.org/10.1038/s41467-022-29663-0>

Publication Date: 2022-04-20

Volume/Issue: Volume 13, Issue 1

References: 72

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: United States Department of Defense | United States Navy | Office of Naval Research, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Arctic sea ice extent continues to decline at an unprecedented rate that is commonly underestimated by climate projection models. This disagreement may imply biases in the representation of processes that bring heat to the sea ice in these models. Here we reveal interactions between ocean-ice heat fluxes, sea ice cover, and upper-ocean eddies that constitute a positive feedback missing in climate models. Using an eddy-resolving global ocean model, we demonstrate that ocean-ice heat fluxes are predominantly induced by localized and intermittent ocean eddies, filaments, and internal waves that episodically advect warm subsurface waters into the mixed layer where they are in direct contact with sea ice. The energetics of near-surface eddies interacting with sea ice are modulated by frictional dissipation in ice-ocean boundary layers, being dominant under consolidated winter ice but substantially reduced under low-concentrated weak sea ice in marginal ice zones. Our results indicate that Arctic sea ice loss will reduce upper-ocean dissipation, which will produce more energetic eddies and amplified ocean-ice heat exchange. We thus emphasize the need for sea ice-aware parameterizations of eddy-induced ice-ocean heat fluxes in climate models.

535. The Global Patterns of Interannual and Intraseasonal Mass Variations in the Oceans from GRACE and GRACE Follow-On Records

Authors: Damien Delforge (Royal Observatory of Belgium, Littoral, Environnement et Sociétés, Université de La Rochelle and CNRS (UMR7266), 17000 La Rochelle, France); Olivier de Viron (Littoral, Environnement et Sociétés, Université de La Rochelle and CNRS (UMR7266), 17000 La Rochelle, France); Fabien Durand (Institut de Recherche pour le Développement (IRD), University of Brasilia); Véronique Dehant (Royal Observatory of Belgium)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs14081861

URL: <https://doi.org/10.3390/rs14081861>

Publication Date: 2022-04-13

Volume/Issue: Volume 14, Issue 8

References: 84

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Centre National d'Études Spatiales, European Research Council (ERC), H2020 research and innovation programme, Synergy grant

License: CC BY 4.0

Open Access: N/A

Abstract: We decompose the monthly global ocean bottom pressure (OBP) from GRACE(-FO) mass concentration solutions, with trends and seasonal harmonics removed from the signal, to extract 23 significant regional modes of variability. The 23 modes are analyzed and discussed considering sea-level anomalies (SLA), wind stress curl (WSC), and major climate indices. A total of two-thirds of the patterns correspond to extratropical regions and are substantially documented in other global or regional studies. Over the equatorial band, the identified modes are unprecedented, with an amplitude ranging between 0.5 and 1 cm. With smaller amplitude than extratropical patterns, they appear to be less correlated with the local SLA or WSC; yet they present significantly coherent dynamics. The Pacific Ocean modes show significant correlations with the Pacific decadal oscillation (PDO) and El Niño southern oscillation (ENSO).

536. Simulated sea levels during 1948-2009 in a global ocean-sea ice model for OMIP

Authors: Yaqi Wang; Hailong Liu; Zipeng Yu; Pengfei Lin; Mengrong Ding

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2022.105082

URL: <https://doi.org/10.1016/j.dsr2.2022.105082>

Publication Date: 2022-04-09

Volume/Issue: Volume 199

References: 43

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

537. Downscaling Satellite-Based Estimates of Ocean Bottom Pressure for Tracking Deep Ocean Mass Transport

Authors: Andrew Delman (Joint Institute for Regional Earth System Science and Engineering, University of California-Los Angeles, Los Angeles, CA 90095, USA); Felix Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs14071764

URL: <https://doi.org/10.3390/rs14071764>

Publication Date: 2022-04-08

Volume/Issue: Volume 14, Issue 7

References: 43

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: Gravimetry measurements from the GRACE and GRACE-Follow-On satellites provide observations of ocean bottom pressure (OBP), which can be differenced between basin boundaries to infer mass transport variability at a given level in the deep ocean. However, GRACE data products are limited in spatial resolution, and conflate signals from many depth levels along steep continental slopes. To improve estimates of OBP variability near steep bathymetry, ocean bottom pressure observations from a JPL GRACE mascon product are downscaled using an objective analysis procedure, with OBP covariance information from an ocean model with horizontal grid spacing of ~18 km. In addition, a depth-based adjustment was applied to enhance correlations at similar depths. Downscaled GRACE OBP shows realistic representations of sharp OBP gradients across bathymetry contours and strong currents, albeit with biases in the shallow ocean. In validations at intraannual (3-12 month) timescales, correlations of downscaled GRACE data (with depth adjustment) and in situ bottom pressure recorder time series were improved in ~79% of sites, compared to correlations that did not involve downscaled GRACE. Correlations tend to be higher at sites where the amplitude of the OBP signal is larger, while locations where surface eddy kinetic energy is high (e.g., Gulf Stream extension) are more likely to have no improvement from the downscaling procedure. The downscaling procedure also increases the amplitude (standard deviation) of OBP variability compared to the non-downscaled GRACE at most sites, resulting in standard deviations that are closer to in situ values. A comparison of hydrography-based transport from RAPID with estimates based on downscaled GRACE data suggests substantial improvement from the downscaling at intraannual timescales, though this improvement does not extend to longer interannual timescales. Possible efforts to improve the downscaling technique through process studies and analysis of alongtrack GRACE/GRACE-FO observations are discussed.

538. Seasonal and spatial variations in spice generation in the South Indian Ocean salinity maxima

Authors: Madhu Kaundal; Nadimpalli Jithendra Raju; Dhrubajyoti Samanta; Mihir Kumar Dash

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-022-01502-2

URL: <https://doi.org/10.1007/s10236-022-01502-2>

Publication Date: 2022-04-08

Volume/Issue: Volume 72, Issue 5

References: 48

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

539. How Well Do We Understand the Land-Ocean-Atmosphere Carbon Cycle?

Authors: David Crisp (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Han Dolman (Vrije Universiteit Amsterdam, Royal NIOZ Texel The Netherlands); Toste Tanhua (GEOMAR Helmholtz Centre for Ocean Research); Galen A. McKinley (Columbia University); Judith Hauck (Alfred Wegener Institute); Ana Bastos (Max Planck Institute for Biogeochemistry); Stephen Sitch (University of Exeter); Simon Eggleston (Global Climate Observing System World Meteorological Organization Geneva Switzerland); Valentin Aich (Global Climate Observing System World Meteorological Organization Geneva Switzerland)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021rg000736

URL: <https://doi.org/10.1029/2021rg000736>

Publication Date: 2022-04-08

Volume/Issue: Volume 60, Issue 2

References: 462

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: carbon stocks, anthropogenic emissions, carbon cycle, carbon fluxes

Funding: Science Mission Directorate, National Science Foundation

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Fossil fuel combustion, land use change and other human activities have increased the atmospheric carbon dioxide (CO₂) abundance by about 50% since the beginning of the industrial age. The atmospheric CO₂ growth rates would have been much larger if natural sinks in the land biosphere and ocean had not removed over half of this anthropogenic CO₂. As these CO₂ emissions grew, uptake by the ocean increased in response to increases in atmospheric CO₂ partial pressure (pCO₂). On land, gross primary production also increased, but the dynamics of other key aspects of the land carbon cycle varied regionally. Over the past three decades, CO₂ uptake by intact tropical humid forests declined, but these changes are offset by increased uptake across mid- and high-latitudes. While there have been substantial improvements in our ability to study the carbon cycle, measurement and modeling gaps still limit our understanding of the processes driving its evolution. Continued ship-based observations combined with expanded deployments of autonomous platforms are needed to quantify ocean-atmosphere fluxes and interior ocean carbon storage on policy-relevant spatial and temporal scales. There is also an urgent need for more comprehensive measurements of stocks, fluxes and atmospheric CO₂ in humid tropical forests and across the Arctic and boreal regions, which are experiencing rapid change. Here, we review our understanding of the atmosphere, ocean, and land carbon cycles and their interactions, identify emerging measurement and modeling capabilities and gaps and the need for a sustainable, operational framework to ensure a scientific basis for carbon management.

540. Observed Linkages Between the Atmospheric Circulation and Oceanic-Forced Sea-Surface Temperature Variability in the Western North Pacific

Authors: Casey R. Patrizio (Colorado State University); David W. J. Thompson (Colorado State University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl095172

URL: <https://doi.org/10.1029/2021gl095172>

Publication Date: 2022-04-02

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References: 58

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Earth Sciences Division, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Previous research suggests the extratropical atmospheric circulation responds to that sea-surface temperature (SST) variability in the western North Pacific. However, the relative roles of oceanic and atmospheric processes in driving the SST anomalies that, in turn, seemingly influence the atmospheric circulation are unclear. Here, we exploit a simple stochastic climate model to subdivide the SST variability in the Kuroshio-Oyashio Extension region into components forced by oceanic and atmospheric processes. We then probe the lead/lag relationships between the atmospheric circulation and both components of the SST variability. Importantly, only the oceanic-forced SST variability is associated with robust atmospheric anomalies that lag the SSTs by 1 month. The results are consistent with the surface heat fluxes associated with atmospheric and oceanic-forced components of the SST variability. Overall, the findings suggest that ocean dynamical processes in the western North Pacific play an important role in influencing the variability of the extratropical circulation.

541. Weakening Trend of Luzon Strait Overflow Transport in the Past Two Decades

Authors: Yaohua Zhu (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Jingxin Yao (Nanjing University, Ministry of Natural Resources); Tengfei Xu (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Shujiang Li (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Yonggang Wang (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Zexun Wei (Shandong Key Laboratory of Marine Science and Numerical Modeling Qingdao China, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl097395

URL: <https://doi.org/10.1029/2021gl097395>

Publication Date: 2022-04-02

Volume/Issue: Volume 49, Issue 7

References: 23

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: The Luzon Strait overflow is a deep current of dense water flowing persistently into the South China Sea (SCS) from the Pacific. It is the primary driving force of the deep SCS circulation and potentially influenced by ocean mass changes. Limited by the lengths of in situ observations, long-term changes in the Luzon Strait overflow transport (LSOT) have not been ascertained. Based on 14 years of satellite-based ocean bottom pressure (OBP) time series, here we provide an evidence of a weakening of the LSOT by $-0.43 \text{ Sv decade}^{-1}$ between 2003 and 2016. Moreover, both OBP data and mooring time series reveal a large annual cycle in the LSOT. The weakening trend in the LSOT may profoundly influence the deep SCS circulation, as well as heat and mass fluxes between the SCS and surrounding oceans.

542. Modulation of the Intraseasonal Chlorophyll-a Concentration in the Tropical Indian Ocean by the Central Indian Ocean Mode

Authors: Jianhuang Qin (Hohai University); Ze Meng (Shanghai Jiao Tong University); Wenlong Xu (Hohai University); Baosheng Li (Second Institute of Oceanography, Ministry of Natural Resources); Xuhua Cheng (Hohai University); Raghu Murtugudde (University of Maryland)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl097802

URL: <https://doi.org/10.1029/2022gl097802>

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Volume/Issue: Volume 49, Issue 7

References: 47

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: This study diagnoses the drivers of intraseasonal variability of mixed layer chlorophyll-a (Chl) concentration in the tropical Indian Ocean during boreal summer (June-September) using a well-tested coupled ocean-ecosystem model. Results show that wind forcing is the primary source for energetic intraseasonal Chl concentration variability in the eastern Arabian Sea and western Bay of Bengal (BoB) modulated by the Central Indian Ocean (CIO) mode. The atmospheric anomalous anticyclone associated with the positive phase of CIO mode drives southeastward currents in the Arabian Sea, which transport nutrient and Chl towards the east. Simultaneously, the wind-induced offshore currents strengthen upwelling in the western boundary of BoB, resulting in enhanced Chl in the upper ocean. Conversely, because of the formation of a barrier layer, the nutrient supply is suppressed along the equator, which causes a negative correlation between the CIO mode and Chl. This study complements the Madden-Julian Oscillation forcing on the intraseasonal Chl concentration anomaly in the tropical Indian Ocean.

543. Response to Reviewer Comments for essd-2021-328

Authors: Jonathan Sharp

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/essd-2021-328-ac1

URL: <https://doi.org/10.5194/essd-2021-328-ac1>

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References: N/A

Cited by: N/A publications

Language: N/A

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Keywords: Life span|Longitude

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Open Access: N/A

Abstract: N/A

544. What Made the Sustained Intensification of Tropical Cyclone Fani in the Bay of Bengal? An Investigation Using Coupled Atmosphere-Ocean Model

Authors: Kumar Ravi Prakash (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Vimlesh Pant (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); T. V. S. Udaya Bhaskar (Indian National Centre for Ocean Information Services, Hyderabad 500090, India); Navin Chandra (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences)

Journal: Atmosphere

Publisher: MDPI AG

DOI: 10.3390/atmos13040535

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References: 49

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The extremely severe tropical cyclone Fani (25 April-5 May 2019) unusually sustained high intensity for a prolonged duration over the Bay of Bengal (BoB). A regional coupled atmosphere-ocean model was used to investigate the atmospheric and oceanic conditions and processes responsible for the sustained intensification of the tropical cyclone (TC) Fani. The coupled model simulated the track and intensification/weakening stages of the cyclone reasonably well. A reduction in sea surface temperature (by -2 degreesC) and an increase in sea surface salinity due to cyclone-induced upwelling and inertial mixing was noticed in both observations and model. The passage of TC Fani over two geostrophic mesoscale warm-core eddies along the cyclone track was found to supply the necessary energy for the intensification of TC Fani. The sea surface height anomaly and tropical cyclone heat potential (TCHP) were higher during TC Fani than other pre-monsoon cyclones in the BoB. The anomalous TCHP in the warm-core eddy zones (i.e., in excess of >160 kJ cm⁻²) maintained the warm surface temperature and high air-sea heat fluxes. The air-sea latent heat flux and atmospheric wind shear were favourable for the intensification of the cyclone. The atmospheric moist static energy enhanced up to 360 kJ kg⁻¹ with a deep vertical extension in the atmospheric column supporting the further intensification of TC Fani. Therefore, the unusual oceanic TCHP associated with mesoscale eddies, higher latent heat flux, and enhanced moist static energy in the atmosphere contributed to the sustained intensification of TC Fani for a prolonged period in the BoB.

545. Effects of grid spacing on high-frequency precipitation variance in coupled high-resolution global ocean-atmosphere models

Authors: Charles X. Light; Brian K. Arbic; Paige E. Martin; Laurent Brodeau; J. Thomas Farrar; Stephen M. Griffies; Ben P. Kirtman; Lucas C. Laurindo; Dimitris Menemenlis; Andrea Molod; Arin D. Nelson; Ebenezer Nyadjro; Amanda K. O'Rourke; Jay F. Shriver; Leo Siqueira; R. Justin Small; Ehud Strobach

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-022-06257-6

URL: <https://doi.org/10.1007/s00382-022-06257-6>

Publication Date: 2022-03-29

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References: 64

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Office of Naval Research, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: High-frequency precipitation variance is calculated in 12 different free-running (non-data-assimilative) coupled high resolution atmosphere-ocean model simulations, an assimilative coupled atmosphere-ocean weather forecast model, and an assimilative reanalysis. The results are compared with results from satellite estimates of precipitation and rain gauge observations. An analysis of irregular sub-daily fluctuations, which was applied by Covey et al. (Geophys Res Lett 45:12514-12522, 2018. <https://doi.org/10.1029/2018GL078926>) to satellite products and low-resolution climate models, is applied here to rain gauges and higher-resolution models. In contrast to lower-resolution climate simulations, which Covey et al. (2018) found to be lacking with respect to variance in irregular sub-daily fluctuations, the highest-resolution simulations examined here display an irregular sub-daily fluctuation variance that lies closer to that found in satellite products. Most of the simulations used here cannot be analyzed via the Covey et al. (2018) technique, because they do not output precipitation at sub-daily intervals. Thus the remainder of the paper focuses on frequency power spectral density of precipitation and on cumulative distribution functions over time scales (2-100 days) that are still relatively "high-frequency" in the context of climate modeling. Refined atmospheric or oceanic model grid spacing is generally found to increase high-frequency precipitation variance in simulations, approaching the values derived from observations. Mesoscale-eddy-rich ocean simulations significantly increase precipitation variance only when the atmosphere grid spacing is sufficiently fine (< 0.5 degrees). Despite the improvements noted above, all of the simulations examined here suffer from the "drizzle effect", in which precipitation is not temporally intermittent to the extent found in observations.

546. Surprises in Physical Oceanography: Contributions from Ocean Acoustic Tomography

Authors: Brian D. Dushaw

Journal: Tellus A: Dynamic Meteorology and Oceanography

Publisher: Stockholm University Press

DOI: 10.16993/tellusa.39

URL: <https://doi.org/10.16993/tellusa.39>

Publication Date: 2022-03-29

Volume/Issue: Volume 74, Issue 2022

References: 134

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

547. Forced vs. Intrinsic Wintertime Submonthly Variability of Sea Surface Temperature in the Midlatitude Western North Pacific

Authors: Yusheng Wu; Guidi Zhou; Guifen Wang; Xuhua Cheng

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2022.847144

URL: <https://doi.org/10.3389/fmars.2022.847144>

Publication Date: 2022-03-28

Volume/Issue: Volume 9

References: 64

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Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Natural Science Foundation of Jiangsu Province, Fundamental Research Funds for the Central Universities

License: CC BY 4.0

Open Access: N/A

Abstract: The relative importance of wintertime forced and intrinsic SST variability in the Kuroshio-Oyashio Extension (KOE) region on submonthly timescales (2-10 and 10-30 days) is evaluated based on theoretical, observational, and modeling analysis. It is shown that the theoretical framework extended from the stochastic climate model has difficulties in representing observed SST variability on such short scales. We then employ the single-column General Ocean Turbulence Model (GOTM) to explicitly evaluate the SST variability forced by atmospheric disturbances. Results show that in the KOE region forced SST variability is responsible for a very small fraction of the total variability (<10%) on the submonthly scales, indicating the dominance of intrinsic oceanic processes. Outside the KOE forced variability dominates. By means of sensitivity experiments, the key physical factors are identified: upper ocean vertical mixing, wind stress forcing (mainly for outside KOE), and latent heat flux, the former two of which are not considered in the theoretical framework. The above results are robust against different levels of submonthly SST variability.

548. Seasonal variation of deep vertical velocity in the South China Sea

Authors: Jiaqian Li; Jian Lan

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2022.101990

URL: <https://doi.org/10.1016/j.ocemod.2022.101990>

Publication Date: 2022-03-23

Volume/Issue: Volume 172

References: 45

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

549. Interdecadal Modulation of ENSO-Related Anomalous Equatorial Intermediate Currents in the Western Pacific by the PDO

Authors: Qiang Ma (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Jianing Wang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Fan Wang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Yilong Lyu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Zhixiang Zhang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl098409

URL: <https://doi.org/10.1029/2022gl098409>

Publication Date: 2022-03-22

Volume/Issue: Volume 49, Issue 6

References: 34

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: As the most prominent decadal variability in the Pacific Ocean, the Pacific Decadal Oscillation (PDO) can influence the El Niño-Southern Oscillation-related atmospheric and upper oceanic motions. Based on results from ocean reanalysis products and linear continuously stratified model experiments, we reveal that the PDO can further extend its interdecadal modulation effect to the equatorial intermediate currents below the thermocline. For the maximum El Niño-related intermediate eastward transport anomaly in the western Pacific, its magnitude during the warm PDO is nearly twice as large as that during the cold PDO, and its occurrence time in the warm PDO leads that in the cold PDO by about 2 months. Such modulation is achieved through westward and downward propagation of the reflected Kelvin-to-Rossby waves, triggered by enhanced and eastward-displaced anomalous westerly wind. The decisive wind forcing region for the modulation is located over the equatorial central Pacific of 5 degreesS-5 degreesN and 180 degrees-140 degreesW.

550. Tracking the source direction of surface mass loads using vertical and horizontal displacements from satellite geodesy: A case study of the inter-annual fluctuations in the water level in the Great Lakes

Authors: Linsong Wang; Michael Bevis; Zhenran Peng; Mikhail K. Kaban; Maik Thomas; Chao Chen

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2022.113001

URL: <https://doi.org/10.1016/j.rse.2022.113001>

Publication Date: 2022-03-21

Volume/Issue: Volume 274

References: 74

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

551. Carbon Dynamics in the Open Indian Ocean

Authors: Abhra Chanda

Journal: Blue Carbon Dynamics of the Indian Ocean

Publisher: Springer International Publishing

DOI: 10.1007/978-3-030-96558-7_2

URL: https://doi.org/10.1007/978-3-030-96558-7_2

Publication Date: 2022-03-21

Volume/Issue: N/A

References: 143

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Deep ocean burial, Dissolved organic carbon, Air-sea CO₂ exchange, Partial pressure of CO₂, Upwelling (and 6 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

552. Local Air-Sea Interactions at Ocean Mesoscale and Submesoscale in a Western Boundary Current

Authors: Ehud Strobach (Agricultural Research Organization); Patrice Klein (California Institute of Technology, PSL Research University); Andrea Molod (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Abdullah A. Fahad (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Atanas Trayanov (Science Systems and Applications Inc. Lanham MD USA, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hector Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl097003

URL: <https://doi.org/10.1029/2021gl097003>

Publication Date: 2022-03-18

Volume/Issue: Volume 49, Issue 7

References: 42

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: SST fronts, high-resolution, air-sea interactions, latent heat, wind stress (and 1 more)

Funding: Goddard Space Flight Center

License: CC BY 4.0

Open Access: N/A

Abstract: We present results from a new, global, high-resolution (~3-km for ocean and ~6-km for atmosphere) realistic earth system simulation. This simulation allows us to examine aspects of small-scale air-sea interaction beyond what previous studies have reported. Our study focuses on recurring intermittent wind events in the Gulf Stream region. These events induce local air-sea heat fluxes above Sea Surface Temperature (SST) anomalies with horizontal scales smaller than 500-km. In particular, strong latent heat bursts above warm SST anomalies are observed during these wind events. We show that such wind events are associated with a secondary circulation that acts to fuel the latent heat bursts by transferring dry air and momentum down to the surface. The intensity of this secondary circulation is related to the strength of small-scale SST fronts that border SST anomalies. The study of such phenomena requires high-resolution in both the atmospheric and oceanic components of the model.

553. Earth system model parameter adjustment using a Green's functions approach

Authors: Ehud Strobach; Andrea Molod; Donifan Barahona; Atanas Trayanov; Dimitris Menemenlis; Gael Forget

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-2309-2022

URL: <https://doi.org/10.5194/gmd-15-2309-2022>

Publication Date: 2022-03-17

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References: 32

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . We demonstrate the practicality and effectiveness of using a Green's functions estimation approach for adjusting uncertain parameters in an Earth system model (ESM). This estimation approach has previously been applied to an intermediate-complexity climate model and to individual ESM components, e.g., ocean, sea ice, or carbon cycle components. Here, the Green's functions approach is applied to a state-of-the-art ESM that comprises a global atmosphere/land configuration of the Goddard Earth Observing System (GEOS) coupled to an ocean and sea ice configuration of the Massachusetts Institute of Technology general circulation model (MITgcm). Horizontal grid spacing is approximately 110 km for GEOS and 37-110 km for MITgcm. In addition to the reference GEOS-MITgcm simulation, we carried out a series of model sensitivity experiments, in which 20 uncertain parameters are perturbed. These "control" parameters can be used to adjust sea ice, microphysics, turbulence, radiation, and surface schemes in the coupled simulation. We defined eight observational targets: sea ice fraction, net surface shortwave radiation, downward longwave radiation, near-surface temperature, sea surface temperature, sea surface salinity, and ocean temperature and salinity at 300 m. We applied the Green's functions approach to optimize the values of the 20 control parameters so as to minimize a weighted least-squares distance between the model and the eight observational targets. The new experiment with the optimized parameters resulted in a total cost reduction of 9 % relative to a simulation that had already been adjusted using other methods. The optimized experiment attained a balanced cost reduction over most of the observational targets. We also report on results from a set of sensitivity experiments that are not used in the final optimized simulation but helped explore options and guided the optimization process. These experiments include an assessment of sensitivity to the number of control parameters and to the selection of observational targets and weights in the cost function. Based on these sensitivity experiments, we selected a specific definition for the cost function. The sensitivity experiments also revealed a decreasing overall cost as the number of control variables was increased. In summary, we recommend using the Green's functions estimation approach as an additional fine-tuning step in the model development process. The

method is not a replacement for modelers' experience in choosing and adjusting sensitive model parameters. Instead, it is an additional practical and effective tool for carrying out final adjustments of uncertain ESM parameters.

554. Offshore Freshwater Pathways in the Northern Gulf of Mexico: Impacts of Modeling Choices

Authors: Guangpeng Liu; Annalisa Bracco; Daoxun Sun

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2022.841900

URL: <https://doi.org/10.3389/fmars.2022.841900>

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Volume/Issue: Volume 9

References: 62

Cited by: 6 publications

Language: N/A

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Keywords: N/A

Funding: National Academies of Sciences, Engineering, and Medicine

License: CC BY 4.0

Open Access: N/A

Abstract: The Gulf of Mexico is a very productive and economically important system where riverine runoff acts as a linkage between the continental shelf and the open ocean, providing nutrients in addition to freshwater. This work investigates the three-dimensional transport and pathway structure of this river runoff offshore the continental shelf using ensembles of numerical simulations with different configurations regarding grid resolution (mesoscale resolving and submesoscale permitting) and river setup using suites of 5-months long integrations covering nearly 3 years. The riverine forcing is applied only at the surface over an area around the river mouth, a strategy often adopted in numerical studies, or as a meridional flux with a vertical extension. The simulated flow captures the southward offshore transport of river runoff driven by its interaction with the largest mesoscale circulations in the basin, the Loop Current and Loop Current eddies. This pathway is strong and well-documented during summer but also active and relevant in winter, despite a less obvious surface signature. The most intense transport occurs primarily at the peripheries of the Loop Current and the detached eddies, and the freshwater is subducted as deep as 600 m around the mesoscale anticyclonic eddies. Submesoscale motions strengthen slightly the spread of freshwater plumes in summer but their contribution is negligible, if not negative, in winter. Differences in the freshwater distribution and transport volume among runs are small and generally less than 10% among ensembles, with overall slightly higher volume of freshwater transported off-shore and at depth in submesoscale permitting runs that include a velocity flux in their riverine input representation.

555. Attribution of Space-Time Variability in Global-Ocean Dissolved Inorganic Carbon

Authors: Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, San José State University); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Jonathan M. Lauderdale (Massachusetts Institute of Technology); Jess F. Adkins (California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Holger Brix (University of California Los Angeles, Institute of Coastal Ocean Dynamics Helmholtz-Zentrum Hereon Geesthacht Germany); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michelle M. Gierach (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Chris Hill (Massachusetts Institute of Technology); Oliver Jahn (Massachusetts Institute of Technology); Peter Landschützer (Max Planck Institute for Meteorology); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Matt R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ariane Verdy (Scripps Institution of Oceanography / University of California San Diego); Daniel B. Whitt (National Aeronautics and Space Administration (NASA) Ames Research Center); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gb007162

URL: <https://doi.org/10.1029/2021gb007162>

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Volume/Issue: Volume 36, Issue 3

References: 102

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: equatorial, sink, ocean, budget, carbon (and 1 more)

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: The inventory and variability of oceanic dissolved inorganic carbon (DIC) is driven by the interplay of physical, chemical, and biological processes. Quantifying the spatiotemporal variability of these drivers is crucial for a mechanistic understanding of the ocean carbon sink and its future trajectory. Here, we use the Estimating the Circulation and Climate of the Ocean-Darwin ocean biogeochemistry state estimate to generate a global-ocean, data-constrained DIC budget and investigate how spatial and seasonal-to-interannual variability in three-dimensional circulation, air-sea

CO₂ flux, and biological processes have modulated the ocean sink for 1995-2018. Our results demonstrate substantial compensation between budget terms, resulting in distinct upper-ocean carbon regimes. For example, boundary current regions have strong contributions from vertical diffusion while equatorial regions exhibit compensation between upwelling and biological processes. When integrated across the full ocean depth, the 24-year DIC mass increase of 64 Pg C (2.7 Pg C year⁻¹) primarily tracks the anthropogenic CO₂ growth rate, with biological processes providing a small contribution of 2% (1.4 Pg C). In the upper 100 m, which stores roughly 13% (8.1 Pg C) of the global increase, we find that circulation provides the largest DIC gain (6.3 Pg C year⁻¹) and biological processes are the largest loss (8.6 Pg C year⁻¹). Interannual variability is dominated by vertical advection in equatorial regions, with the 1997-1998 El Niño-Southern Oscillation causing the largest year-to-year change in upper-ocean DIC (2.1 Pg C). Our results provide a novel, data-constrained framework for an improved mechanistic understanding of natural and anthropogenic perturbations to the ocean sink.

556. Vertical Variations of Submesoscale Motions Between New Jersey Shelf and Bermuda

Authors: Jianing Li; Xin Wang

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2022.848916

URL: <https://doi.org/10.3389/feart.2022.848916>

Publication Date: 2022-03-10

Volume/Issue: Volume 10

References: 31

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Based on the measurements from the Oleander Project, the behaviors of submesoscale motions are examined in the area between New Jersey Shelf and Bermuda. The vertical variation of Rossby number, the kinetic energy in the submesoscale range, and the power law of kinetic energy spectra suggest that submesoscale motions are mainly confined within the surface mixed layer with seasonality that is strong in winter and weak in summer. Besides, submesoscale motions with no significant seasonality were also found beneath the surface mixed layer, which could reach 500 m depth. A possible explanation is that the drastically varying flows in the Gulf Stream and mesoscale eddy periphery could generate strong lateral shear throughout their influence depth, which is favorable for breaking the geostrophic balance and causing submesoscale motions beneath the surface mixed layer.

557. Variability in the meridional overturning circulation at 32 degreesS in the Pacific Ocean diagnosed by inverse box models

Authors: Cristina Arumí-Planas; Alonso Hernández-Guerra; Verónica Caínzos; Pedro Vélez-Belchí; Riccardo Farneti; Matthew R. Mazloff; Sabine Mecking; Isabella Rosso; Lena M. Schulze Chretien; Kevin G. Speer; Lynne D. Talley

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2022.102780

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Publication Date: 2022-03-10

Volume/Issue: Volume 203

References: 103

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

558. Multi-decadal retreat of marine-terminating outlet glaciers in northwest and central-west Greenland

Authors: Taryn E. Black; Ian Joughin

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-807-2022

URL: <https://doi.org/10.5194/tc-16-807-2022>

Publication Date: 2022-03-10

Volume/Issue: Volume 16, Issue 3

References: 84

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . The retreat and acceleration of marine-terminating outlet glaciers in Greenland over the past 2 decades have been widely attributed to climate change. Here we present a comprehensive annual record of glacier terminus positions in northwest and central-west Greenland and compare it against local and regional climatology to assess the regional sensitivity of glacier termini to different climatic factors. This record is derived from optical and radar satellite imagery and spans 87 marine-terminating outlet glaciers from 1972 through 2021. We find that in this region, most glaciers have retreated over the observation period and widespread regional retreat accelerated from around 1996. The acceleration of glacier retreat coincides with the timing of sharp shifts in ocean surface temperatures, the duration of the sea-ice season, ice-sheet surface mass balance, and meltwater and runoff production. Regression analysis indicates that terminus retreat is most sensitive to increases in runoff and ocean temperatures, while the effect of offshore sea ice is weak. Because runoff and ocean temperatures can influence terminus positions through several mechanisms, our findings suggest that a variety of processes - such as ocean-interface melting, mélange presence and rigidity, and hydrofracture-induced calving - may contribute to, but do not conclusively dominate, the observed regional retreat.

559. Seasonal Surface Eddy Mixing in the Kuroshio Extension: Estimation and Machine Learning Prediction

Authors: Wenting Guan (School of Marine Science and Technology Tianjin University Tianjin China); Ru Chen (School of Marine Science and Technology Tianjin University Tianjin China); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yi Yang (School of Marine Science and Technology Tianjin University Tianjin China); Hao Wei (School of Marine Science and Technology Tianjin University Tianjin China)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017967

URL: <https://doi.org/10.1029/2021jc017967>

Publication Date: 2022-03-08

Volume/Issue: Volume 127, Issue 3

References: 85

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Results of coarse-resolution climate models are sensitive to the specification of ocean eddy mixing coefficients. Therefore, it is important to estimate, rationalize and predict eddy diffusivities. Here, we estimate the seasonal variability of surface eddy diffusivities in the Kuroshio Extension region using numerical particles advected by a submesoscale-permitting model solution. We find that both the spatial structure and the domain-averaged value of the particle-based eddy diffusivities have a significant seasonal cycle. We also assess the predictability of cross-stream mixing lengths in this region using the methods of machine learning, suppressed mixing length theory (SMLT), and multiple linear regression (LR). The predictors we choose are all variables from SMLT that represent eddy- and mean-flow properties, and these predictors correlate well with the particle-based cross-stream mixing lengths. We demonstrate that, compared to SMLT and LR, machine learning methods, in particular the random forest (RF) and convolutional neural network (CNN), can better represent both the spatial structure and the domain-averaged value of cross-stream mixing lengths. The skill in predicting the mixing lengths with CNN has much less seasonal variability than that with RF. Our results indicate that the machine learning approach may be useful in future development of eddy parameterization schemes.

560. Derivation of bedrock topography measurement requirements for the reduction of uncertainty in ice-sheet model projections of Thwaites Glacier

Authors: Blake A. Castleman; Nicole-Jeanne Schlegel; Lambert Caron; Eric Larour; Ala Khazendar

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-761-2022

URL: <https://doi.org/10.5194/tc-16-761-2022>

Publication Date: 2022-03-08

Volume/Issue: Volume 16, Issue 3

References: 71

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Determining the future evolution of the Antarctic IceSheet is critical for understanding and narrowing the large existing uncertainties in century-scale global mean sea-level-rise (SLR) projections. One of the most significant glaciers and ice streams in Antarctica, Thwaites Glacier, is at risk of destabilization and, if destabilized, has the potential to be the largest regional-scale contributor of SLR on Earth. This is because Thwaites Glacier is vulnerable to the marine ice-sheet instability as its grounding line is significantly influenced by ocean-driven basal melting rates, and its bedrock topography retrogrades into kilometer-deep troughs. In this study, we investigate how bedrock topography features influence the grounding line migration beneath Thwaites Glacier when extreme ocean-driven basal melt rates are applied. Specifically, we design experiments using the Ice-sheet and Sea-level System Model (ISSM) to quantify the SLR projection uncertainty due to reported errors in the current bedrock topography maps that are often used by ice-sheet models. We find that spread in model estimates of sea-level-rise contribution from Thwaites Glacier due to the reported bedrock topography error could be as large as 21.9 cm after 200 years of extreme ocean warming. Next, we perturb the bedrock topography beneath Thwaites Glacier using wavelet decomposition techniques to introduce realistic noise (within error). We explore the model space with multiple realizations of noise to quantify what spatial and vertical resolutions in bedrock topography are required to minimize the uncertainty in our 200-year experiment. We conclude that at least a 2 km spatial and 8 m vertical resolution would independently constrain possible SLR to ± 2 cm over 200 years, fulfilling requirements outlined by the 2017 Decadal Survey for Earth Science. Lastly, we perform an ensemble of simulations to determine in which regions our model of Thwaites Glacier is most sensitive to perturbations in bedrock topography. Our results suggest that the retreat of the grounding line is most sensitive to bedrock topography in proximity to the grounding line's initial position. Additionally, we find that the location and amplitude of the bedrock perturbation is more significant than its sharpness and shape. Overall, these findings inform and benchmark observational requirements for future missions that will measure ice-sheet bedrock topography, not only in the case of Thwaites Glacier but for Antarctica on the continental scale.

561. Hidden Upwelling Systems Associated With Major Western Boundary Currents

Authors: Fanglou Liao (University of Delaware); Xinfeng Liang (University of Delaware); Yun Li (University of Delaware); Michael Spall (Woods Hole Oceanographic Institution)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: upwelling, ocean synthesis products, ocean vertical velocity, overturning circulation, ocean vertical transport (and 1 more)

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Open Access: N/A

Abstract: Western boundary currents (WBCs) play an essential role in regulating global climate. In contrast to their widely examined horizontal motions, less attention has been paid to vertical motions associated with WBCs. Here, we examine vertical motions associated with the major WBCs by analyzing vertical velocity from five ocean synthesis products and one eddy-resolving ocean simulation. These data reveal robust and intense subsurface upwelling systems, which are primarily along isopycnal surfaces, in five major subtropical WBC systems. These upwelling systems are part of basin-scale overturning circulations and are likely driven by meridional pressure gradients along the western boundary. Globally, the WBC upwelling contributes significantly to the vertical transport of water mass and ocean properties and is an essential yet overlooked branch of the global ocean circulation. In addition, the WBC upwelling intersects the oceanic euphotic and mixed layers, and thus likely plays an important role in ocean biological and chemical processes by transporting nutrients, carbon and other tracers vertically inside the ocean. This study calls for more research into the dynamics of the WBC upwelling and their role in the ocean and climate systems.

562. Separating Energetic Internal Gravity Waves and Small-Scale Frontal Dynamics

Authors: Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Patrice Klein (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Laboratoire de Météorologie Dynamique Ecole Normale Supérieure CNRS Paris France); Eric D'Asaro (University of Washington); Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Andrew F. Thompson (California Institute of Technology); Lia Siegelman (Scripps Institution of Oceanography); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ernesto Rodriguez (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alexander Wineteer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dragana Perkovic-Martin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Abstract: Oceanic fronts with lateral scales less than 20 km are now known to be one of the major contributors to vertical heat fluxes in the global ocean, which highlights their potential impact on Earth's climate. However, frontal dynamics with time scales less than 1 day, whose contribution to vertical heat fluxes is thought to be significant, are obscured by energetic internal gravity waves (IGWs). In this study, we address this critical issue by separating IGWs and frontal dynamics using an approach based on their respective vertical scales of variability. Results using a numerical model with a horizontal grid spacing of 500 m confirm that it is possible to recover frontal dynamics at short time scales as well as associated intense vertical velocities and vertical heat fluxes. This opens up new possibilities for a more accurate estimation of the vertical exchanges of any tracers between the surface and the ocean interior.

563. Analysis of the Periodic Component of Vertical Land Motion in the Po Delta (Northern Italy) by GNSS and Hydrological Data

Authors: Eleonora Vitagliano (Department of Earth, Environment and Resources Sciences, University of Naples Federico II, 80126 Naples, Italy, Istituto Nazionale di Geofisica e Vulcanologia (INGV), 00143 Rome, Italy); Enza Vitale (Department of Earth, Environment and Resources Sciences, University of Naples Federico II, 80126 Naples, Italy); Giacomo Russo (Department of Earth, Environment and Resources Sciences, University of Naples Federico II, 80126 Naples, Italy); Leonardo Piccinini (Department of Geosciences, University of Padua, 35131 Padua, Italy); Massimo Fabris (Department of Civil, Environmental and Architectural Engineering, University of Padua, 35131 Padua, Italy); Domenico Calcaterra (Department of Earth, Environment and Resources Sciences, University of Naples Federico II, 80126 Naples, Italy); Rosa Di Maio (Department of Earth, Environment and Resources Sciences, University of Naples Federico II, 80126 Naples, Italy)

Journal: Remote Sensing

Publisher: MDPI AG

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References: 84

Cited by: 6 publications

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Open Access: N/A

Abstract: Nowadays, several methodologies, implemented for satellite or terrestrial surveys, reveal that daily and weekly site-positioning time series can exhibit linear trends plus seasonal oscillations. Such periodic components affect the evaluation of subsidence rates and, thus, they must be recognized and properly modelled. In this work, the periodic component of vertical land motion in Po Delta (Northern Italy) is estimated by a multi-component and multi-source procedure recently proposed by some of the authors for studying land subsidence in delta areas. First, land vertical motion data, acquired in the central part of the Po Delta over a six-year time interval, were compared with hydro-meteorological and climate datasets collected from nineteen stations distributed over the entire Delta. Then, four physically based models of the test site were implemented to verify the water pressure- and water mass-dependent processes inferred from the analytical phase. Modelling results show that the annual ground oscillation is better explained by soil moisture change, although river water mass variation gives a relevant contribution to land deformation, especially in the wet periods. Finally, to account for intra-annual processes, the joint contributions of all the inferred sources were treated as a nonlinear problem and solved applying the generalized reduced gradient method. The obtained combination is well supported by statistical parameters and provides the best agreement with the geodetic observations.

564. An investigation of the Helmholtz and wave-vortex decompositions on surface currents in a coastal region

Authors: Eun Ae Lee; Sung Yong Kim

Journal: Continental Shelf Research

Publisher: Elsevier BV

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References: 93

Cited by: 2 publications

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: N/A

565. A Prototype for Remote Monitoring of Ocean Heat Content Anomalies

Authors: David S. Trossman (Louisiana State University, Global Science and Technology); Robert H. Tyler (University of Maryland Baltimore County, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

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Language: N/A

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Keywords: Argo|Ocean observations

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: To overcome challenges with observing ocean heat content (OHC) over the entire ocean, we propose a novel approach that exploits the abundance of satellite data, including data from modern satellite geomagnetic surveys such as Swarm. The method considers a novel combination of conventional in situ (temperature and pressure) as well as satellite (altimetry and gravimetry) data with estimates of ocean electrical conductance (depth-integrated conductivity), which can potentially be obtained from magnetic observations (by satellite, land, seafloor, ocean, and airborne magnetometers). To demonstrate the potential benefit of the proposed method, we sample model output of an ocean state estimate to reflect existing observations and train a machine learning algorithm [Generalized Additive Model (GAM)] on these samples. We then calculate OHC everywhere using information potentially derivable from various global satellite coverage--including magnetic observations--to gauge the GAM's goodness of fit on a global scale. Inclusion of in situ observations of OHC in the upper 2000 m from Argo-like floats and conductance data each reduce the root-mean-square error by an order of magnitude. Retraining the GAM with recent ship-based hydrographic data attains a smaller RMSE in polar oceans than training the GAM only once on all available historical ship-based hydrographic data; the opposite is true elsewhere. The GAM more accurately calculates OHC anomalies throughout the water column than below 2000 m and can detect global OHC anomalies over multiyear time scales, even when considering hypothetical measurement errors. Our method could complement existing methods and its accuracy could be improved through careful ship-based campaign planning. **Significance Statement** The purpose of this manuscript is to demonstrate the potential for practical implementation of a remote monitoring method for ocean heat content (OHC) anomalies. To do this, we sample data from a reanalysis product primarily because of the dearth of observations below 2000 m depth that can be used for validation and the fact that full-depth-integrated electrical seawater conductivity data products derived from satellite magnetometry are not yet available. We evaluate multiple factors related to the accuracy of OHC anomaly estimation and find that, even with

hypothetical measurement errors, our method can be used to monitor OHC anomalies on multiyear time scales.

566. Deviation of the Lagrangian particle tracing method in the evaluation of the Southern Hemisphere annual subduction rate

Authors: Kai Liu; Shan Gao; Fan Wang

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-021-1097-7

URL: <https://doi.org/10.1007/s00343-021-1097-7>

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References: 45

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Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

567. A new state-dependent parameterization for the free drift of sea ice

Authors: Charles Brunette; L. Bruno Tremblay; Robert Newton

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-533-2022

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Volume/Issue: Volume 16, Issue 2

References: 92

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . Free-drift estimates of sea ice motion are necessary to produce a seamless observational record combining buoy and satellite-derived sea ice motion vectors. We develop a new parameterization for the free drift of sea ice based on wind forcing, wind turning angle, sea ice state variables (thickness and concentration), and estimates of the ocean currents. Given the fact that the spatial distribution of the wind-ice-ocean transfer coefficient has a similar structure to that of the spatial distribution of sea ice thickness, we take the standard free-drift equation and introduce a wind-ice-ocean transfer coefficient that scales linearly with ice thickness. Results show a mean bias error of -0.5 cm s^{-1} (low-speed bias) and a root-mean-square error of 5.1 cm s^{-1} , considering daily buoy drift data as truth. This represents a 35 % reduction of the error on drift speed compared to the free-drift estimates used in the Polar Pathfinder dataset (Tschudi et al., 2019b). The thickness-dependent transfer coefficient provides an improved seasonality and long-term trend of the sea ice drift speed, with a minimum (maximum) drift speed in May (October), compared to July (January) for the constant transfer coefficient parameterizations which simply follow the peak in mean surface wind stresses. Over the 1979–2019 period, the trend in sea ice drift in this new model is $+0.45 \text{ cm s}^{-1}$ per decade compared with $+0.39 \text{ cm s}^{-1}$ per decade from the buoy observations, whereas there is essentially no trend in a free-drift parameterization with a constant transfer coefficient (-0.09 cm s^{-1} per decade) or the Polar Pathfinder free-drift input data (-0.01 cm s^{-1} per decade). The optimal wind turning angle obtained from a least-squares fitting is 25° , resulting in a mean error and a root-mean-square error of $+3^\circ$ and 42° on the direction of the drift, respectively. The ocean current estimates obtained from the minimization procedure resolve key large-scale features such as the Beaufort Gyre and Transpolar Drift Stream and are in good agreement with ocean state estimates from the ECCO, GLORYS, and PIOMAS ice-ocean reanalyses, as well as geostrophic currents from dynamical ocean topography, with a root-mean-square difference of 2.4, 2.9, 2.6, and 3.8 cm s^{-1} , respectively. Finally, a repeat of the analysis on two sub-sections of the time series (pre- and post-2000) clearly shows the acceleration of the Beaufort Gyre (particularly along the Alaskan coastline) and an expansion of the gyre in the post-2000s, concurrent with a thinning of the sea ice cover and the observed acceleration of the ice drift speed and

oceancurrents. This new dataset is publicly available for complementing merged observation-based sea ice drift datasets that include satellite and buoydrift records.

568. Interannual variability of sea level in the southern Indian Ocean: local vs. remote forcing mechanisms

Authors: Marion Kersalé; Denis L. Volkov; Kandaga Pujiana; Hong Zhang

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-18-193-2022

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Language: en

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Keywords: N/A

Funding: National Aeronautics and Space Administration, Cooperative Institute for Marine and Atmospheric Studies, University of Miami

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Open Access: N/A

Abstract: . The subtropical southern Indian Ocean (SIO) has been described as one of the world's largest heat accumulators due to its remarkable warming during the past 2 decades. However, the relative contributions of remote (of Pacific origin) forcing and local wind forcing to the variability of heat content and sea level in the SIO have not been fully attributed. Here, we combine a general circulation model, an analytic linear reduced-gravity model, and observations to disentangle the spatial and temporal inputs of each forcing component on interannual to decadal timescales. A sensitivity experiment is conducted with artificially closed Indonesian straits to physically isolate the Indian Ocean and Pacific Ocean, intentionally removing the Indonesian Throughflow (ITF) influence on the Indian Ocean heat content and sea level variability. We show that the relative contribution of the signals originating in the equatorial Pacific vs. signals caused by local wind forcing to the interannual variability of sea level and heat content in the SIO is dependent on location within the basin (low latitude vs. midlatitude and western side vs. eastern side of the basin). The closure of the ITF in the numerical experiment reduces the amplitude of interannual-to-decadal sea level changes compared to the simulation with realistic ITF. However, the spatial and temporal evolution of sea level patterns in the two simulations remain similar and correlated with El Niño-Southern Oscillation (ENSO). This suggests that these patterns are mostly determined by local wind forcing and oceanic processes, linked to ENSO via the "atmospheric bridge" effect. We conclude that local wind forcing is an important driver for the interannual changes of sea level, heat content, and meridional transports in the SIO subtropical gyre, while oceanic signals originating in the Pacific amplify locally forced signals.

569. El Niño/Southern Oscillation inhibited by submesoscale ocean eddies

Authors: Shengpeng Wang; Zhao Jing; Lixin Wu; Wenju Cai; Ping Chang; Hong Wang; Tao Geng; Gohkan Danabasoglu; Zhaohui Chen; Xiaohui Ma; Bolan Gan; Haiyuan Yang

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

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References: 60

Cited by: 24 publications

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Keywords: N/A

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Open Access: N/A

Abstract: The El Niño/Southern Oscillation is characterized by irregular warm (El Niño) and cold (La Niña) events in the tropical Pacific Ocean, which have substantial global environmental and socioeconomic impacts. These events are generally attributed to the instability of basin-scale air-sea interactions in the equatorial Pacific. However, the role of sub-basin-scale processes in the El Niño/Southern Oscillation life cycle remains unknown due to the scarcity of observations and coarse resolution of climate models. Here, using a long-term high-resolution global climate simulation, we find that equatorial ocean eddies with horizontal wavelengths less than several hundred kilometres substantially inhibit the growth of La Niña and El Niño events. These submesoscale eddies are regulated by the intensity of Pacific cold-tongue temperature fronts. The eddies generate an anomalous surface cooling tendency during El Niño by inducing a reduced upward heat flux from the subsurface to the surface in the central-eastern equatorial Pacific; the opposite occurs during La Niña. This dampening effect is missing in the majority of state-of-the-art climate models. Our findings identify a pathway to resolve the long-standing overestimation of El Niño and La Niña amplitudes in climate simulations.

570. Reconstruction of global surface ocean pCO₂ using region-specific predictors based on a stepwise FFNN regression algorithm

Authors: Guorong Zhong; Xuegang Li; Jinming Song; Baoxiao Qu; Fan Wang; Yanjun Wang; Bin Zhang; Xiaoxia Sun; Wuchang Zhang; Zhenyan Wang; Jun Ma; Huamao Yuan; Liqin Duan

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-19-845-2022

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Cited by: 8 publications

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Keywords: N/A

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Open Access: N/A

Abstract: . Various machine learning methods were attempted in the global mapping of surface ocean partial pressure of CO₂ (pCO₂) to reduce the uncertainty of the global ocean CO₂ sink estimate due to undersampling of pCO₂. In previous research, the predictors of pCO₂ were usually selected empirically based on theoretic drivers of surface ocean pCO₂, and the same combination of predictors was applied in all areas except where there was a lack of coverage. However, the differences between the drivers of surface ocean pCO₂ in different regions were not considered. In this work, we combined the stepwise regression algorithm and a feed-forward neural network (FFNN) to select predictors of pCO₂ based on the mean absolute error in each of the 11 biogeochemical provinces defined by the self-organizing map (SOM) method. Based on the predictors selected, a monthly global 1° x 1° surface ocean pCO₂ product from January 1992 to August 2019 was constructed. Validation of different combinations of predictors based on the Surface Ocean CO₂ Atlas (SOCAT) dataset version 2020 and independent observations from time series stations was carried out. The prediction of pCO₂ based on region-specific predictors selected by the stepwise FFNN algorithm was more precise than that based on predictors from previous research. Applying the FFNN size-improving algorithm in each province decreased the mean absolute error (MAE) of the global estimate to 11.32 μatm and the root mean square error (RMSE) to 17.99 μatm . The script file of the stepwise FFNN algorithm and pCO₂ product are distributed through the Institute of Oceanology of the Chinese Academy of Sciences Marine Science Data Center (IOCAS, <https://doi.org/10.12157/iocas.2021.0022>, Zhong, 2021).

571. Comparison of ocean heat content estimated using two eddy-resolving hindcast simulations based on OFES1 and OFES2

Authors: Fanglou Liao; Xiao Hua Wang; Zhiqiang Liu

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-1129-2022

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References: 58

Cited by: 4 publications

Language: en

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Keywords: N/A

Funding: Southern Marine Science and Engineering Guangdong Laboratory

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Open Access: N/A

Abstract: . In this study, we have compared the ocean heat content(OHC), estimated using two eddy-resolving hindcast simulations based on Ocean General Circulation Model for the Earth Simulator version 1 (OFES1) and version 2 (OFES2). Results from a global objective analysis of subsurface temperature (EN4) were taken as a reference. Both EN4 and OFES1 suggest that OHC has increased in most regions of the top 2000 m during 1960-2016, which is mainly associated with the deepening of neutral density surfaces and variations along the neutral density surfaces of regional importance. Upon comparing the results obtained from the two OFES hindcasts, we found substantial differences in the temporal and spatial distributions of the OHC, especially in the Atlantic Ocean. A basin-wide heat budget analysis showed that there was less surface heating for the major basins in OFES2. The horizontal heat advection was mostly similar; however, OFES2 had a significantly stronger meridional heat advection associated with the Indonesian Throughflow (ITF) above 300 m. Additionally, large discrepancies in the vertical heat advection were also evinced when the two OFES results were compared, especially at a depth of 300 m in the Indian Ocean. We inferred that there are large discrepancies in the vertical heat diffusion (those that cannot be directly evaluated in this study due to data unavailability), which, along with the different magnitudes of sea surface heat flux and vertical heat advection, were the major factors responsible for the examined differences in OHC. This work suggests that OFES1 provides a reasonable multi-decadal estimate of global and basin-integrated warming trends above 700 m, except for the top 300 m for the Pacific Ocean and between 300-700 m for the Indian Ocean. Although the estimates of the global OHC during 1960-2016 are consistent with observations between 700-2000 m, caution is warranted while examining the basin-wide multi-decadal OHC variations using OFES1. The seemingly suboptimal OHC estimate based on OFES2 suggests that any conclusions on long-term climate variations derived from OFES2 might suffer from large drifts, necessitating audits.

572. Possible Thermal Effect of Tibetan Plateau on the Atlantic Meridional Overturning Circulation

Authors: Qin Wen (Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China, School of Geography Nanjing Normal University Nanjing China, Key Laboratory of Virtual Geographic Environment (Nanjing Normal University) Ministry of Education Nanjing China); Haijun Yang (Fudan University); Kai Yang (Lanzhou University); Gen Li (Hohai University, Southern Marine Science and Engineering Guangdong Laboratory); Zhengyu Liu (Ohio State University); Jian Liu (Key Laboratory of Virtual Geographic Environment (Nanjing Normal University) Ministry of Education Nanjing China, Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application Nanjing China, School of Geography Nanjing Normal University Nanjing China, Jiangsu Provincial Key Laboratory for Numerical Simulation of Large Scale Complex Systems, School of Mathematical Science Nanjing Normal University Nanjing China, Pilot National Laboratory for Marine Science and Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 61

Cited by: 5 publications

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Open Access: N/A

Abstract: The Tibetan Plateau (TP) is important in shaping the global climate. To understand how its thermal condition may affect the Atlantic meridional overturning circulation (AMOC), we conduct a series of albedo forcing experiments to simulate the TP warming and cooling conditions. The results show that the TP warming can immediately generate anticyclone in the subtropical North Atlantic, leading to more water vapor transporting from the North Atlantic to eastern tropical Pacific. This results in less precipitation over the North Atlantic and increases the sea surface salinity. The latter enhances the deep-water formation in the subpolar North Atlantic and thus the AMOC. In contrast, the TP cooling weakens the AMOC via opposite processes. These findings imply that the AMOC change can be affected not only by local factors but also remote forcing from Asia.

573. The Equatorial Pacific Cold Tongue Bias in CESM1 and Its Influence on ENSO Forecasts

Authors: Xian Wu (National Center for Atmospheric Research); Yuko M. Okumura (University of Texas at Austin); Pedro N. DiNezio (University of Colorado Boulder); Stephen G. Yeager (National Center for Atmospheric Research); Clara Deser (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

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References: 166

Cited by: 14 publications

Language: N/A

Subjects: N/A

Keywords: Pacific Ocean, Seasonal forecasting, ENSO, Model errors, Initialization|Teleconnection|Forcing (mathematics)|Hindcast|Atmospheric models|Multivariate ENSO index

Funding: climate program office, climate program office, national science foundation

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Open Access: N/A

Abstract: The mean-state bias and the associated forecast errors of the El Niño-Southern Oscillation (ENSO) are investigated in a suite of 2-yr-lead retrospective forecasts conducted with the Community Earth System Model, version 1, for 1954-2015. The equatorial Pacific cold tongue in the forecasts is too strong and extends excessively westward due to a combination of the model's inherent climatological bias, initialization imbalance, and errors in initial ocean data. The forecasts show a stronger cold tongue bias in the first year than that inherent to the model due to the imbalance between initial subsurface oceanic states and model dynamics. The cold tongue bias affects not only the pattern and amplitude but also the duration of ENSO in the forecasts by altering ocean-atmosphere feedbacks. The predicted sea surface temperature anomalies related to ENSO extend to the far western equatorial Pacific during boreal summer when the cold tongue bias is strong, and the predicted ENSO anomalies are too weak in the central-eastern equatorial Pacific. The forecast errors of pattern and amplitude subsequently lead to errors in ENSO phase transition by affecting the amplitude of the negative thermocline feedback in the equatorial Pacific and tropical interbasin adjustments during the mature phase of ENSO. These ENSO forecast errors further degrade the predictions of wintertime atmospheric teleconnections, land surface air temperature, and rainfall anomalies over the Northern Hemisphere. These mean-state and ENSO forecast biases are more pronounced in forecasts initialized in boreal spring-summer than other seasons due to the seasonal intensification of the Bjerknes feedback.

574. Assessment of GRACE/GRACE Follow-On Terrestrial Water Storage Estimates Using an Improved Forward Modeling Method: A Case Study in Africa

Authors: Hao Zhou; Min Dai; Penghui Wang; Min Wei; Lu Tang; Siyou Xu; Zhicai Luo

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2021.796723

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References: 58

Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: Leakage (economics)|Jet propulsion

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License: CC BY 4.0

Open Access: N/A

Abstract: Leakage errors derived from spatial filters are the major limitation for estimating terrestrial water storage via the Gravity Recovery and Climate Experiment (GRACE) mission and the recently launched GRACE Follow-On mission. Here we develop an improved forward modeling method and assess its performance of reducing leakage errors over Africa. In noise-free condition, the forward modeling method shows its outperformance in restoring signals, and the improved forward modeling method can further reduce the leakage errors along the coastline of Africa. In noise-contaminated condition, the simulated environment is set as real as possible to GRACE mission and GRACE Follow-On mission. The results based on the simulated GRACE and GRACE Follow-On solutions demonstrate the capacity of improved forward modeling method in reducing leakage errors. In the case of simulated GRACE data, the average improvements of 24 basins over Africa are respectively 37% for annual amplitudes and 36% for trends. When compared with these simulated GRACE data, the improvements via simulated GRACE Follow-On solutions are minor over large and medium size river basins, but they are significant over small size river basins. In the case of simulated GRACE Follow-On solutions, the average improvements over Africa are 39% for annual amplitudes and 41% for trends. Eventually, the improved forward modeling method is used to process GRACE spherical harmonic datasets from the Center for Space Research (CSR). The results present better agreement with those derived from the newly released mascon solutions from Jet Propulsion Laboratory (JPL) and CSR, when compared with those derived from CSR Tellus grids with scale factors. The better consistency between these model-independent approaches indicates the good performance of our improved forward modeling method and the further necessity of careful evaluation of model-dependent approaches when using different prior hydrological models. Overall, experiments based on noise-free observations, noise-contaminated observations, and GRACE datasets indicate that improved forward modeling method is capable of restoring temporal signals.

575. Deep Lagrangian Connectivity in the Global Ocean Inferred from Argo Floats

Authors: Ryan Abernathey (Columbia University); Christopher Bladwell (University of New South Wales); Gary Froyland (University of New South Wales); Konstantinos Sakellariou (University of New South Wales)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Cited by: 1 publications

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Subjects: N/A

Keywords: Lagrangian circulation/transport, Spectral analysis/models/distribution, Ocean dynamics

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License: N/A

Open Access: N/A

Abstract: The connectivity between ocean basins and subbasin regions strongly influences the transport of ocean tracers and thus plays a role in regulating climate and ocean ecosystems. We describe the application of a new technique from nonlinear dynamical systems to infer the Lagrangian connectivity of the deep global ocean. We approximate the dynamic Laplacian using Argo trajectories from January 2011 to January 2017 and extract the eight dominant coherent (or dynamically self-connected) regions at 1500 m depth. Our approach overcomes issues such as sparsity of observed data and floats continually leaving and entering the dataset; only 10% of floats record continuously for the full six years. The identified coherent regions maximally trap water within them over the six-year time frame, providing a distinct analysis of the deep global ocean and relevant information for planning future float deployment. A key result is that the coherent regions are highly stationary, showing minimal displacement over the six-year period. Although our study is concerned with ocean circulation at a multiyear, global scale, the dynamic Laplacian approach may be applied at any temporal or spatial scale to identify coherent structures in ocean flow from positional time series information arising from observations or models.

576. Admittance of the Earth Rotational Response to Zonal Tide Potential

Authors: C. Bizouard (Observatoire de Paris PSL Paris France); L. I. Fernández (MAGGIA Lab. Fac. de Cs. Astronómicas y Geofísicas University Nac. de La Plata La Plata Argentina, Consejo Nacional de Investigaciones Científicas y Técnicas); L. Zotov (National Research University Higher School of Economics Moscow Russia, Lomonosov Moscow State University Moscow Russia)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jb022962

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Publication Date: 2022-02-01

Volume/Issue: Volume 127, Issue 3

References: 31

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Consejo Nacional de Investigaciones Científicas y Técnicas

License: N/A

Open Access: N/A

Abstract: This study is devoted to the determination of the admittance parameters describing the Earth rotational response to the components of the zonal tide potential. First, in order to better grasp the physical content of those admittance coefficients, we revisit the theoretical description of the length of day (LOD) changes at sub-decadal time scale, where forcing is dominated by zonal tides and hydro-atmospheric mass transports. This theoretical reminder specifies the rheological coefficients permitting to apply the hydro-atmospheric corrections to isolate the tidal part of the LOD. Then, the admittances are determined from the LOD series corrected from hydro-atmospheric contributions at the frequencies of the dominant zonal tidal terms between 7 and 365 days. In contrast of the former kindred studies, we both address the discrepancy of the results brought by various EOP series and the hydro-atmospheric corrections on the LOD. Our study forwards the complementary corrections brought by the ocean, the land water and sea level changes. Below 32 days, removing the atmospheric-oceanic excitation from LOD allows to much better constraint the admittance complex coefficients κ than applying the atmospheric correction only: the discrepancy with respect to modeled values is reduced up to 70%, and the frequency dependence of the imaginary part brought by the ocean dynamical response is confirmed. A systematic effect with respect to the values modeled by Ray and Erofeeva (2014), <https://doi.org/10.1002/2013jb010830> has been detected and hints a defect of this model. Moreover, the role of land water and associated sea level variation is notable at the semi-annual period.

577. The Whole Antarctic Ocean Model (WAOM v1.0): development and evaluation

Authors: Ole Richter; David E. Gwyther; Benjamin K. Galton-Fenzi; Kaitlin A. Naughten

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-617-2022

URL: <https://doi.org/10.5194/gmd-15-617-2022>

Publication Date: 2022-01-26

Volume/Issue: Volume 15, Issue 2

References: 115

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Government

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Open Access: N/A

Abstract: . The Regional Ocean Modeling System (ROMS), including an ice shelf component, has been applied on a circum-Antarctic domain to derive estimates of ice shelf basal melting. Significant improvements made compared to previous models of this scale are the inclusion of tides and a horizontal spatial resolution of 2 km, which is sufficient to resolve on-shelf heat transport by bathymetric troughs and eddy-scale circulation. We run the model with ocean-atmosphere-sea ice conditions from the year 2007 to represent nominal present-day climate. We force the ocean surface with buoyancy fluxes derived from sea ice concentration observations and wind stress from ERA-Interim atmospheric reanalysis. Boundary conditions are derived from the ECCO2 ocean state estimate; tides are incorporated as sea surface height and barotropic currents at the open boundary. We evaluate model results using satellite-derived estimates of ice shelf melting and established compilations of ocean hydrography. The Whole Antarctic Ocean Model (WAOM v1.0) qualitatively captures the broad scale difference between warm and cold regimes as well as many of the known characteristics of regional ice-ocean interaction. We identify a cold bias for some warm-water ice shelves and a lack of high-salinity shelf water (HSSW) formation. We conclude that further calibration and development of our approach are justified. At its current state, the model is ideal for addressing specific, process-oriented questions, e.g. related to tide-driven ice shelf melting at large scales.

578. On the contribution of Rossby waves driven by surface buoyancy fluxes to low-frequency North Atlantic steric sea surface height variations

Authors: Peter Kowalski

Journal: Atmospheric and Oceanic Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.aosl.2022.100153

URL: <https://doi.org/10.1016/j.aosl.2022.100153>

Publication Date: 2022-01-25

Volume/Issue: Volume 15, Issue 3

References: 24

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

579. Dynamics of ENSO Phase-Locking and Its Biases in Climate Models

Authors: Han-Ching Chen (Nanjing University, Department of Atmospheric Sciences University of Hawaii¹ at M²noa Honolulu HI USA); Fei-Fei Jin (Department of Atmospheric Sciences University of Hawaii¹ at M²noa Honolulu HI USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2021gl097603>

Publication Date: 2022-01-25

Volume/Issue: Volume 49, Issue 3

References: 40

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, U.S. Department of Energy

License: N/A

Open Access: N/A

Abstract: The contributions of different oceanic feedbacks to the El Niño-Southern Oscillation (ENSO) phase-locking are examined by deriving ENSO dynamics based on the recharge-discharge framework. In observations, the significant winter preference of the ENSO peak is determined by a strong seasonal modulation of SST growth rate, which is controlled by the zonal advective and thermodynamic feedbacks. However, the majority of climate models fail to simulate ENSO phase-locking because the contribution of zonal advective feedback to the seasonal modulation of the SST growth rate is much smaller compared to observations. The weak annual cycle of the SST-current coupling coefficient and small annual mean of the negative climatological zonal SST gradient are two factors contributing to the weak-biased seasonality of zonal advective feedback. Further analysis shows that the Niño3.4 SSTA has better phase-locking performance than Niño3 SSTA in the climate models due to the better simulation of zonal advection feedback in the Niño3.4 region.

580. Components of 21 years (1995-2015) of absolute sea level trends in the Arctic

Authors: Carsten Bjerre Ludwigsen; Ole Baltazar Andersen; Stine Kildegaard Rose

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-18-109-2022

URL: <https://doi.org/10.5194/os-18-109-2022>

Publication Date: 2022-01-21

Volume/Issue: Volume 18, Issue 1

References: 78

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The Arctic Ocean is at the frontier of the fast-changing climate in the northern latitudes, and sea level trends are a bulk measure of ongoing processes related to climate change. Observations of sea level in the Arctic Ocean are nonetheless difficult to validate with independent measurements, and this is globally the region where the sea level trend (SLT) is most uncertain. The aim of this study is to create a satellite-independent reconstruction of Arctic SLT, as it is observed by altimetry and tide gauges (TGs). Previous studies use Gravity Recovery and Climate Experiment (GRACE) observations to estimate the manometric (mass component of) SLT. GRACE estimates, however, are challenged by large mass changes on land, which are difficult to separate from much smaller ocean mass changes. Furthermore, GRACE is not available before 2003, which significantly limits the period and makes the trend more vulnerable to short-term changes. As an alternative approach, this study estimates the climate-change-driven Arctic manometric SLT from the Arctic sea level fingerprints of glaciers, Greenland, Antarctica and glacial isostatic adjustment (GIA) with the addition of the long-term inverse barometer (IB) effect. The halosteric and thermosteric components complete the reconstructed Arctic SLT and are estimated by interpolating 300 000 temperature (T) and salinity (S) in situ observations. The SLT from 1995-2015 is compared to the observed SLT from altimetry and 12 selected tide gauges (TGs) corrected for vertical land movement (VLM). The reconstructed estimate manifests the salinity-driven halosteric component as dominating the spatial SLT pattern with variations between -7 and 10 mm yr⁻¹. The manometric SLT in comparison is estimated to be 1-2 mm yr⁻¹ for most of the Arctic Ocean. The reconstructed SLT shows a larger sea level rise in the Beaufort Sea compared to altimetry, an issue that is also identified by previous studies. There is a TG-observed sea level rise in the Siberian Arctic in contrast to the sea level fall from the reconstructed and altimetric estimate. From 1995-2015 the reconstructed SLT agrees within the 68 % confidence interval with the SLT from observed altimetry in 87 % of the Arctic between 65° N and 82° N ($R=0.50$) and with 5 of 12 TG-derived (VLM-corrected) SLT estimates. The residuals are seemingly smaller than results from previous studies using GRACE estimates and modeled T-S data. The spatial correlation of the reconstructed SLT to altimetric SLT during the GRACE period (2003-2015) is $R=0.38$ and

$R=0.34/R=0.37$ if GRACE estimates are used instead of the constructed manometric component. Thus, the reconstructed manometric component is suggested as a legitimate alternative to GRACE that can be projected into the past and future.

581. Mixed layer heat budget in the eastern equatorial Indian Ocean during the two consecutive positive Indian Ocean dipole events in 2018 and 2019

Authors: A. R. Aparna; M. S. Girishkumar

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-06099-8

URL: <https://doi.org/10.1007/s00382-021-06099-8>

Publication Date: 2022-01-19

Volume/Issue: Volume 58, Issue 11-12

References: 63

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

582. Biogeochemical Dynamics in Adjacent Mesoscale Eddies of Opposite Polarity

Authors: Benedetto Barone (Daniel K. Inouye Center for Microbial Oceanography: Research and Education University of Hawai'i at Mānoa Honolulu HI USA, Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Matthew J. Church (University of Montana); Mathilde Dugenne (Daniel K. Inouye Center for Microbial Oceanography: Research and Education University of Hawai'i at Mānoa Honolulu HI USA, Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Nicholas J. Hawco (University of Southern California, Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Oliver Jahn (Massachusetts Institute of Technology); Angelique E. White (Daniel K. Inouye Center for Microbial Oceanography: Research and Education University of Hawai'i at Mānoa Honolulu HI USA, Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Seth G. John (University of Southern California); Michael J. Follows (Massachusetts Institute of Technology); Edward F. DeLong (Daniel K. Inouye Center for Microbial Oceanography: Research and Education University of Hawai'i at Mānoa Honolulu HI USA, Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); David M. Karl (Daniel K. Inouye Center for Microbial Oceanography: Research and Education University of Hawai'i at Mānoa Honolulu HI USA, Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gb007115

URL: <https://doi.org/10.1029/2021gb007115>

Publication Date: 2022-01-19

Volume/Issue: Volume 36, Issue 2

References: 85

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: nutrient injection, deep chlorophyll maximum, mesoscale eddy, chemical wake, diapycnal mixing

Funding: Simons Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: We examined the biogeochemical impact of paired mesoscale cyclones and anticyclones in spatial proximity (<200 km apart) in the North Pacific Subtropical Gyre. While previous studies have demonstrated that upwelling associated with the intensification of cyclonic eddies can supply nutrients supporting plankton productivity, we observed that steeper vertical gradients in inorganic nutrients increased nutrient fluxes due to diapycnal mixing during the mature stage of cyclonic eddies. The increased diapycnal nutrient supply was linked with expansion of eukaryotic phytoplankton biomass and intensification of the deep chlorophyll maximum (DCM) layer. This perturbation in the plankton community was associated with increased fluxes of biominerals (specifically particulate inorganic carbon and particulate silica) and isotopically enriched organic nitrogen in particles exported in the cyclone. The time-integrated effects of thermocline vertical displacements on the lower euphotic zone were predictable deficits and surpluses of inorganic nutrients and dissolved oxygen, respectively.

However, the stoichiometry of oxygen and inorganic nutrients differed from that predicted for production and consumption of phytoplankton biomass, requiring additional biological processes that decouple changes in oxygen and nutrient concentrations. The dynamics revealed by this study may be a common feature of oligotrophic ecosystems, where mesoscale biogeochemical perturbations are buffered by the DCM layer, which limits the ecological impact of eddies in the well-lit, near-surface ocean.

583. Influence of Nonseasonal River Discharge on Sea Surface Salinity and Height

Authors: Hrishikesh A. Chandanpurkar (FLAME University Pune India, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of Saskatchewan Saskatoon SK Canada); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Xiaochun Wang (University of California Los Angeles); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Severine Fournier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christopher G. Piecuch (Woods Hole Oceanographic Institution); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021ms002715

URL: <https://doi.org/10.1029/2021ms002715>

Publication Date: 2022-01-18

Volume/Issue: Volume 14, Issue 2

References: 68

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: sea surface salinity, river discharge, sea surface height

Funding: Earth Sciences Division

License: CC BY-NC 4.0

Open Access: N/A

Abstract: River discharge influences ocean dynamics and biogeochemistry. Due to the lack of a systematic, up-to-date global measurement network for river discharge, global ocean models typically use seasonal discharge climatology as forcing. This compromises the simulated nonseasonal variation (the deviation from seasonal climatology) of the ocean near river plumes and undermines their usefulness for interdisciplinary research. Recently, a reanalysis-based daily varying global discharge data set was developed, providing the first opportunity to quantify nonseasonal discharge effects on global ocean models. Here we use this data set to force a global ocean model for the 1992-2017 period. We contrast this experiment with another experiment (with identical atmospheric forcings) forced by seasonal climatology from the same discharge data set to isolate nonseasonal discharge effects, focusing on sea surface salinity (SSS) and sea surface height (SSH). Near major river mouths, nonseasonal discharge causes standard deviations in SSS (SSH) of 1.3-3 practical salinity unit (1-2.7

cm). The inclusion of nonseasonal discharge results in notable improvement of model SSS against satellite SSS near most of the tropical-to-midlatitude river mouths and minor improvement of model SSH against satellite or in-situ SSH near some of the river mouths. SSH changes associated with nonseasonal discharge can be explained by salinity effects on halosteric height and estimated accurately through the associated SSS changes. A recent theory predicting river discharge impact on SSH is found to perform reasonably well overall but underestimates the impact on SSH around the global ocean and has limited skill when applied to rivers near the equator and in the Arctic Ocean.

584. Seasonality in the relationship between equatorial-mean heat content and interannual eastern equatorial Atlantic sea surface temperature variability

Authors: Kyle J. Turner; Natalie J. Burls; Anna von Brandis; Joke Lübbecke; Martin Claus

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-06116-w

URL: <https://doi.org/10.1007/s00382-021-06116-w>

Publication Date: 2022-01-14

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References: 45

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Directorate for Geosciences, National Oceanic and Atmospheric Administration, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: Interannual sea surface temperature (SST) variations in the tropical Atlantic Ocean lead to anomalous atmospheric circulation and precipitation patterns with important ecological and socioeconomic consequences for the semiarid regions of sub-Saharan Africa and northeast Brazil. This interannual SST variability is characterized by three modes: an Atlantic meridional mode featuring an anomalous cross-equatorial SST gradient that peaks in boreal spring; an Atlantic zonal mode (Atlantic Niño mode) with SST anomalies in the eastern equatorial Atlantic cold tongue region that peaks in boreal summer; and a second zonal mode of variability with eastern equatorial SST anomalies peaking in boreal winter. Here we investigate the extent to which there is any seasonality in the relationship between equatorial warm water recharge and the development of eastern equatorial Atlantic SST anomalies. Seasonally stratified cross-correlation analysis between eastern equatorial Atlantic SST anomalies and equatorial heat content anomalies (evaluated using warm water volume and sea surface height) indicate that while equatorial heat content changes do occasionally play a role in the development of boreal summer Atlantic zonal mode events, they contribute more consistently to Atlantic Niño II, boreal winter events. Event and composite analysis of ocean adjustment with a shallow water model suggest that the warm water volume anomalies originate mainly from the off-equatorial northwestern Atlantic, in agreement with previous studies linking them to anomalous wind stress curl associated with the Atlantic meridional mode.

585. Validation of satellite-based sea surface temperature products against in situ observations off the western coast of Sumatra

Authors: Qoosaku Moteki

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-021-04156-0

URL: <https://doi.org/10.1038/s41598-021-04156-0>

Publication Date: 2022-01-07

Volume/Issue: Volume 12, Issue 1

References: 26

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: This study validated the sea surface temperature (SST) datasets from the Group for High-Resolution SST Multi Product Ensemble (GMPE), National Oceanic and Atmospheric Administration (NOAA) Optimal Interpolation (OI) SST version 2 and 2.1 (Olv2 and Olv2.1), and Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2) in the area off the western coast of Sumatra against in situ observations. Furthermore, the root mean square differences (RMSDs) of Olv2, Olv2.1, and ECCO2 were investigated with respect to GMPE, whose small RMSD < 0.2 K against in situ observations confirmed its suitability as a reference. Although Olv2 showed a large RMSD (1-1.5 K) with a significant negative bias, Olv2.1 (RMSD < 0.4 K) improved remarkably. In the average SST distributions for December 2017, the differences among the 4 datasets were significant in the areas off the western coast of Sumatra, along the southern coast of Java, and in the Indonesian inland sea. These results were consistent with the ensemble spread distribution obtained with GMPE. The large RMSDs of Olv2 corresponded to high clouds, and it was suggested that the change in the satellites used for SST estimation contributed to the improvement in Olv2.1.

586. Arctic sea level variability from high-resolution model simulations and implications for the Arctic observing system

Authors: Guokun Lyu; Nuno Serra; Meng Zhou; Detlef Stammer

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-18-51-2022

URL: <https://doi.org/10.5194/os-18-51-2022>

Publication Date: 2022-01-06

Volume/Issue: Volume 18, Issue 1

References: 65

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020, National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: . Two high-resolution model simulations are used to investigate the spatiotemporal variability of the Arctic Ocean sea level. The model simulations reveal barotropic sea level variability at periods of < 30 d, which is strongly captured by bottom pressure observations. The seasonal sea level variability is driven by volume exchanges with the Pacific and Atlantic oceans and the redistribution of the water by the wind. Halosteric effects due to river runoff and evaporation minus precipitation ice melting/formation also contribute in the marginal seas and seasonal sea ice extent regions. In the central Arctic Ocean, especially the Canadian Basin, the decadal halosteric effect dominates sea level variability. The study confirms that satellite altimetric observations and Gravity Recovery and Climate Experiment (GRACE) could infer the total freshwater content changes in the Canadian Basin at periods longer than 1 year, but they are unable to depict the seasonal and subseasonal freshwater content changes. The increasing number of profiles seems to capture freshwater content changes since 2007, encouraging further data synthesis work with a more complicated interpolation method. Further, in situ hydrographic observations should be enhanced to reveal the freshwater budget and close the gaps between satellite altimetry and GRACE, especially in the marginal seas.

587. Export of Ice Sheet Meltwater from Upernavik Fjord, West Greenland

Authors: Morven Muilwijk (University of Bergen, Bjerknes Centre for Climate Research); Fiamma Straneo (Scripps Institution of Oceanography / University of California San Diego, University of Bergen); Donald A. Slater (University of St Andrews, Scripps Institution of Oceanography / University of California San Diego, University of Edinburgh); Lars H. Smedsrud (University of Bergen, Bjerknes Centre for Climate Research); James Holte (Scripps Institution of Oceanography / University of California San Diego); Michael Wood (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Camilla S. Andresen (Geological Survey of Denmark and Greenland); Ben Harden (h Sea Education Association, Woods Hole, Massachusetts)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-21-0084.1

URL: <https://doi.org/10.1175/jpo-d-21-0084.1>

Publication Date: 2022-01-04

Volume/Issue: Volume 52, Issue 3

References: 69

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: Arctic, Buoyancy, Atlantic Ocean, Annual variations, Glaciers (and 5 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Meltwater from Greenland is an important freshwater source for the North Atlantic Ocean, released into the ocean at the head of fjords in the form of runoff, submarine melt, and icebergs. The meltwater release gives rise to complex in-fjord transformations that result in its dilution through mixing with other water masses. The transformed waters, which contain the meltwater, are exported from the fjords as a new water mass Glacially Modified Water (GMW). Here we use summer hydrographic data collected from 2013 to 2019 in Upernavik, a major glacial fjord in northwest Greenland, to describe the water masses that flow into the fjord from the shelf and the exported GMWs. Using an optimum multi-parameter technique across multiple years we then show that GMW is composed of 57.8% $\pm$ 8.1% Atlantic Water (AW), 41.0% $\pm$ 8.3% Polar Water (PW), 1.0% $\pm$ 0.1% subglacial discharge, and 0.2% $\pm$ 0.2% submarine meltwater. We show that the GMW fractional composition cannot be described by buoyant plume theory alone since it includes lateral mixing within the upper layers of the fjord not accounted for by buoyant plume dynamics. Consistent with its composition, we find that changes in GMW properties reflect changes in the AW and PW source waters. Using the obtained dilution ratios, this study suggests that the exchange across the fjord mouth during summer is on the order of 50 mSv (1 Sv $\equiv$ 106 m³ s⁻¹) (compared to a freshwater input of 0.5 mSv). This study provides a first-order parameterization for the exchange at the mouth of glacial fjords for large-scale ocean models.

588. Reduction in Meridional Heat Export Contributes to Recent Indian Ocean Warming

Authors: K. McMonigal (University of Miami); Kathryn L. Gunn (University of Miami); Lisa M. Beal (University of Miami); Shane Elipot (University of Miami); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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URL: <https://doi.org/10.1175/jpo-d-21-0085.1>

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Volume/Issue: Volume 52, Issue 3

References: 76

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: Seasonal cycle, In situ oceanic observations, Ocean circulation, Indian Ocean, Boundary current|Throughflow|Oceanic basin (and 1 more)

Funding: National Science Foundation, National Science Foundation, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: Since 2000, the Indian Ocean has warmed more rapidly than the Atlantic or Pacific Oceans. Air-sea fluxes alone cannot explain the rapid Indian Ocean warming, which has so far been linked to an increase in temperature transport into the basin through the Indonesian Throughflow (ITF). Here, we investigate the role that the heat transport out of the basin at 36 degreesS plays in the warming. Adding the heat transport out of the basin to the ITF temperature transport into the basin, we calculate the decadal mean Indian Ocean heat budget over the 2010s. We find that heat convergence increased within the Indian Ocean over 2000-19. The heat convergence over the 2010s is of the same order as the warming rate, and thus the net air-sea fluxes are near zero. This is a significant change from previous analyses using transbasin hydrographic sections from 1987, 2002, and 2009, which all found divergences of heat. A 2-yr time series shows that seasonal aliasing is not responsible for the decadal change. The anomalous ocean heat convergence over the 2010s in comparison with previous estimates is due to changes in ocean currents at both the southern boundary (33%) and the ITF (67%). We hypothesize that the changes at the southern boundary are linked to an observed broadening of the Agulhas Current, implying that temperature and velocity data at the western boundary are crucial to constrain heat budget changes.

589. A Comparative Study of Interannual Oscillation Models for Determining Geophysical Polar Motion Excitations

Authors: Małgorzata Wiśka (Warsaw University of Technology)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs14010147

URL: <https://doi.org/10.3390/rs14010147>

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References: 60

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Similar to seasonal and intraseasonal variations in polar motion (PM), interannual variations are also largely caused by changes in the angular momentum of the Earth's geophysical fluid layers composed of the atmosphere, the oceans, and in-land hydrologic flows (AOH). Not only are inland freshwater systems crucial for interannual PM fluctuations, but so are atmospheric surface pressures and winds, oceanic currents, and ocean bottom pressures. However, the relationship between observed geodetic PM excitations and hydro-atmospheric models has not yet been determined. This is due to defects in geophysical models and the partial knowledge of atmosphere-ocean coupling and hydrological processes. Therefore, this study provides an analysis of the fluctuations of PM excitations for equatorial geophysical components χ_1 and χ_2 at interannual time scales. The geophysical excitations were determined from different sources, including atmospheric, ocean models, Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On data, as well as from the Land Surface Discharge Model. The Multi Singular Spectrum Analysis method was applied to retain interannual variations in χ_1 and χ_2 components. None of the considered mass and motion terms studied for the different atmospheric and ocean models were found to have a negligible effect on interannual PM. These variables, derived from different Atmospheric Angular Momentum (AAM) and Oceanic Angular Momentum (OAM) models, differ from each other. Adding hydrologic considerations to the coupling of AAM and OAM excitations was found to provide benefits for achieving more consistent interannual geodetic budgets, but none of the AOH combinations fully explained the total observed PM excitations.

590. Examination of seasonal variation of the equatorial undercurrent termination in the Eastern Pacific diagnosed by ECCO2

Authors: Jie Wang; Xuezhi Bai; Hengling Leng

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-021-0308-6

URL: <https://doi.org/10.1007/s00343-021-0308-6>

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References: 45

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License: N/A

Open Access: N/A

Abstract: N/A

591. Intercomparison of Salinity Products in the Beaufort Gyre and Arctic Ocean

Authors: Sarah B. Hall (School of the Earth, Ocean and Environment, University of South Carolina, Columbia, SC 29208, USA); Bulusu Subrahmanyam (School of the Earth, Ocean and Environment, University of South Carolina, Columbia, SC 29208, USA); James H. Morison (University of Washington)

Journal: Remote Sensing

Publisher: MDPI AG

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URL: <https://doi.org/10.3390/rs14010071>

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Volume/Issue: Volume 14, Issue 1

References: 47

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research

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Open Access: N/A

Abstract: Salinity is the primary determinant of the Arctic Ocean's density structure. Freshwater accumulation and distribution in the Arctic Ocean have varied significantly in recent decades and certainly in the Beaufort Gyre (BG). In this study, we analyze salinity variations in the BG region between 2012 and 2017. We use in situ salinity observations from the Seasonal Ice Zone Reconnaissance Surveys (SIZRS), CTD casts from the Beaufort Gyre Exploration Project (BGP), and the EN4 data to validate and compare with satellite observations from Soil Moisture Active Passive (SMAP), Soil Moisture and Ocean Salinity (SMOS), and Aquarius Optimally Interpolated Sea Surface Salinity (OISSS), and Arctic Ocean models: ECCO, MIZMAS, HYCOM, ORAS5, and GLORYS12. Overall, satellite observations are restricted to ice-free regions in the BG area, and models tend to overestimate sea surface salinity (SSS). Freshwater Content (FWC), an important component of the BG, is computed for EN4 and most models. ORAS5 provides the strongest positive SSS correlation coefficient (0.612) and lowest bias to in situ observations compared to the other products. ORAS5 subsurface salinity and FWC compare well with the EN4 data. Discrepancies between models and SIZRS data are highest in GLORYS12 and ECCO. These comparisons identify dissimilarities between salinity products and extend challenges to observations applicable to other areas of the Arctic Ocean.

592. Geomorphological imprint of opposing ocean bottom currents, a case study from the southeastern Brazilian Atlantic margin

Authors: M.M. de Mahiques; F.J. Lobo; U. Schattner; A. López-Quirós; C.B. Rocha; R.J.S. Dias; I. Montoya-Montes; A.C.B. Vieira

Journal: Marine Geology

Publisher: Elsevier BV

DOI: 10.1016/j.margeo.2021.106715

URL: <https://doi.org/10.1016/j.margeo.2021.106715>

Publication Date: 2021-12-24

Volume/Issue: Volume 444

References: 80

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: Contourite|Seafloor Spreading|Salt tectonics|Continental Margin|Nepheloid layer

Funding: FAPESP, CNPq

License: N/A

Open Access: N/A

Abstract: N/A

593. An Objective Method with a Continuity Constraint for Improving Surface Velocity Estimates from the Geostationary Ocean Color Imager

Authors: Zifeng Hu (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Lan Li (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Jun Zhao (Sun Yat-Sen University, Pearl River Estuary Marine Ecosystem Research Station, Ministry of Education, Zhuhai 519000, China, Guangdong Provincial Key Laboratory of Marine Resources and Coastal Engineering, Guangzhou 510275, China, Southern Marine Science and Engineering Guangdong Laboratory); Dongxiao Wang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs14010014

URL: <https://doi.org/10.3390/rs14010014>

Publication Date: 2021-12-21

Volume/Issue: Volume 14, Issue 1

References: 32

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the Innovation Group Project of Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: Mapping surface currents with high spatiotemporal resolution over a wide coverage is crucial for understanding ocean dynamics and associated biogeochemical processes. The most widely used algorithm for estimating surface velocities from sequential satellite observations is the maximum cross-correlation (MCC) method. However, many unrealistic vectors still exist, despite the utilization of various filtering techniques. In this study, an objective method has been developed through the combination of MCC and multivariate optimum interpolation (MOI) analysis under a continuity constraint. The MCC method, with and without MOI, is applied to sequences of simulated sea surface temperature (SST) fields with a 1/48 degrees spatial resolution over the East China Sea continental shelf. Integration of MOI into MCC reduces the average absolute differences between the model's 'actual' velocity and the SST-derived velocity by 19% in relative magnitude and 22% in direction, respectively. Application of the proposed method to Geostationary Ocean Color Imager (GOCI) satellite observations produces good agreement between derived surface velocities and the Oregon State University (OSU) regional tidal model outputs. Our results demonstrate that the incorporation of MOI into MCC can provide a significant improvement in the reliability and accuracy of satellite-derived velocity fields.

594. A Deep Learning Model for Forecasting Velocity Structures of the Loop Current System in the Gulf of Mexico

Authors: Ali Muhamed Ali (EECS Department, Florida Atlantic University, Boca Raton, FL 33431, USA); Hanqi Zhuang (EECS Department, Florida Atlantic University, Boca Raton, FL 33431, USA); James VanZwieten (EECS Department, Florida Atlantic University, Boca Raton, FL 33431, USA); Ali K. Ibrahim (Harbor Branch Oceanographic Institute, Florida Atlantic University, Boca Raton, FL 33431, USA); Laurent Chérubin (Harbor Branch Oceanographic Institute, Florida Atlantic University, Boca Raton, FL 33431, USA)

Journal: Forecasting

Publisher: MDPI AG

DOI: 10.3390/forecast3040056

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Publication Date: 2021-12-16

Volume/Issue: Volume 3, Issue 4

References: 48

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Academics of science

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Open Access: N/A

Abstract: Despite the large efforts made by the ocean modeling community, such as the GODAE (Global Ocean Data Assimilation Experiment), which started in 1997 and was renamed as OceanPredict in 2019, the prediction of ocean currents has remained a challenge until the present day--particularly in ocean regions that are characterized by rapid changes in their circulation due to changes in atmospheric forcing or due to the release of available potential energy through the development of instabilities. Ocean numerical models' useful forecast window is no longer than two days over a given area with the best initialization possible. Predictions quickly diverge from the observational field throughout the water and become unreliable, despite the fact that they can simulate the observed dynamics through other variables such as temperature, salinity and sea surface height. Numerical methods such as harmonic analysis are used to predict both short- and long-term tidal currents with significant accuracy. However, they are limited to the areas where the tide was measured. In this study, a new approach to ocean current prediction based on deep learning is proposed. This method is evaluated on the measured energetic currents of the Gulf of Mexico circulation dominated by the Loop Current (LC) at multiple spatial and temporal scales. The approach taken herein consists of dividing the velocity tensor into planes perpendicular to each of the three Cartesian coordinate system directions. A Long Short-Term Memory Recurrent Neural Network, which is best suited to handling long-term dependencies in the data, was thus used to predict the evolution of the velocity field in each plane, along each of the three directions. The predicted tensors, made of the planes perpendicular to each Cartesian direction, revealed that the model's prediction skills were best for the flow field in the planes perpendicular to the direction of prediction. Furthermore, the fusion of all three predicted tensors significantly increased the overall skills of the flow prediction over the individual model's predictions.

The useful forecast period of this new model was greater than 4 days with a root mean square error less than $0.05 \text{ cm}\cdot\text{s}^{-1}$ and a correlation coefficient of 0.6.

595. Stochastic Guidance of Buoyancy Controlled Vehicles under Ice Shelves using Ocean Currents

Authors: Federico Rossi; Andrew Branch; Michael P. Schodlok; Timothy Stanton; Ian G. Fenty; Joshua Vander Hook; Evan B. Clark

Journal: 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Publisher: IEEE

DOI: 10.1109/iros51168.2021.9635987

URL: <https://doi.org/10.1109/iros51168.2021.9635987>

Publication Date: 2021-12-16

Volume/Issue: N/A

References: 37

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Traverse|Lead (geology)

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: N/A

596. Decadal Variability of the Upper-Ocean Salinity in the Southeast Indian Ocean: Role of Local Ocean-Atmosphere Dynamics

Authors: Yue Wu (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Xiao-Tong Zheng (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Qi-Wei Sun (Chinese Academy of Sciences); Yu Zhang (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Yan Du (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Lin Liu (h Shandong Key Laboratory of Marine Science and Numerical Modeling, Qingdao, China, Southern Marine Science and Engineering Guangdong Laboratory, Ministry of Natural Resources)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-21-0122.1

URL: <https://doi.org/10.1175/jcli-d-21-0122.1>

Publication Date: 2021-12-15

Volume/Issue: Volume 34, Issue 19

References: N/A

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: Salinity, Indian Ocean, Atmosphere-ocean interaction, Decadal variability, Ocean dynamics

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean salinity plays a crucial role in the upper-ocean stratification and local marine ecosystem. This study reveals that ocean salinity presents notable decadal variability in upper 200 m over the southeast Indian Ocean (SEIO). Previous studies linked this salinity variability with precipitation anomalies over the Indo-Pacific region modulated by the tropical Pacific decadal variability. Here we conduct a quantitative salinity budget analysis and show that, in contrast, oceanic advection, especially the anomalous meridional advection, plays a dominant role in modulating the SEIO salinity on the decadal time scale. The anomalous meridional advection is mainly associated with a zonal dipole pattern of sea level anomaly (SLA) in the south Indian Ocean (SIO). Specifically, positive and negative SLAs in the east and west of the SIO correspond to anomalous southward oceanic current, which transports much fresher seawater from the warm pool into the SEIO and thereby decreases the local upper-ocean salinity, and vice versa. Further investigation reveals that the local anomalous wind stress curl associated with tropical Pacific forcing is responsible for generating the sea level dipole pattern via oceanic Rossby wave adjustment on decadal time scale. This study highlights that the local ocean-atmosphere dynamical adjustment is critical for the decadal salinity variability in the SEIO.

597. Response to referee comments, tc-2021-249

Authors: Charles Brunette

Journal: Unknown Journal

Publisher: Copernicus GmbH

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URL: <https://doi.org/10.5194/tc-2021-249-ac1>

Publication Date: 2021-12-14

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Buoy|Forcing (mathematics)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

598. The Eastern Atlantic Basin Pathway for the Export of the North Atlantic Deep Waters

Authors: Yujia Zhai (Ocean University of China, Woods Hole Oceanographic Institution); Jiayan Yang (Woods Hole Oceanographic Institution); Xiuquan Wan (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Sijia Zou (Xiamen University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl095615

URL: <https://doi.org/10.1029/2021gl095615>

Publication Date: 2021-12-13

Volume/Issue: Volume 48, Issue 24

References: 56

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The North Atlantic deep water (NADW), according to the classic ocean circulation theory, moves southward as a deep western boundary current (DWBC) even though it may veer into interior and then rejoin DWBC when encountering regional circulation features, such as eddy-driven recirculation. In potential vorticity dynamics, the eastern side of the Mid-Atlantic Ridge (MAR) may provide a similar topographic support as the continental slope off the western boundary for a southward transport of NADW. In this article, we quantify the mean meridional NADW transports on both sides of the MAR using a data-assimilated product and find that the flow in the eastern basin contributes about of the net southward transport of NADW from 50 degrees to 35 degreesN. Our study points to the importance of observing NADW transport variations on the eastern side of the MAR in order to monitor the transport strength of Atlantic Meridional Overturning Circulation.

599. Influence of river discharge on circulation and tidal process in the Java Sea, Indonesia

Authors: H Ramadhan; D Nugroho; I W Nurjaya; A S Atmadipoera

Journal: IOP Conference Series: Earth and Environmental Science

Publisher: IOP Publishing

DOI: 10.1088/1755-1315/944/1/012068

URL: <https://doi.org/10.1088/1755-1315/944/1/012068>

Publication Date: 2021-12-09

Volume/Issue: Volume 944, Issue 1

References: 16

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: This study investigates the effect of river discharge in transport and tidal processes in the Java Sea using the Coastal and Regional Ocean Community (CROCO) hydrodynamic model. The model has 20 vertical layers and a horizontal resolution of 1/18 degrees. The oceanic and atmospheric forcing of this model is taken from the global Copernicus Marine Environment Monitoring Service (CMEMS) model and the fifth generation ECMWF atmospheric reanalysis (ERA5) hourly data. Daily Global Flood Awareness System (GloFAS) data has been successfully implemented as river flow data for this study. Two scenarios have been applied, namely, with and without river discharge. This study shows that the two scenarios and the satellite observational data agree in terms of water level with Root Mean Square Difference (RMSD) about 4 cm, Sea Surface Temperature with RMSD about 0.29 degreesC, and Sea Surface Salinity with RMSD about 0.39 psu. The model was also validated using seven tide gauges and produced a good agreement. River discharge increase eastward transport in the eastern part of the Java Sea up to 0.1 Sv (1 Sv= 10⁶ m³s⁻¹). Both scenarios produce similar tidal amplitude and phase and agree well with previous studies and other tidal data sources.

600. Impact of Local Atmospheric Intraseasonal Variability on Mean Sea Ice State in the Arctic Ocean

Authors: Xi Liang (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Chengyan Liu (Southern Marine Science and Engineering Guangdong Laboratory); Lejiang Yu (Polar Research Institute of China, Southern Marine Science and Engineering Guangdong Laboratory); Martin Losch (Alfred Wegener Institute); Lujun Zhang (Nanjing University); Xichen Li (Chinese Academy of Sciences); Fu Zhao (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Zhongxiang Tian (National Marine Environmental Forecasting Center, Ministry of Natural Resources)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-21-0376.1

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Publication Date: 2021-12-06

Volume/Issue: Volume 35, Issue 5

References: 74

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Arctic, Sea ice, Intraseasonal variability, Sea ice concentration|Arctic geoengineering|Arctic dipole anomaly

Funding: National Key R&D; Program of China, National Key R&D; Program of China, National Key R&D; Program of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: The Arctic atmosphere shows significant variability on intraseasonal time scales of 10-90 days. The intraseasonal variability in the Arctic sea ice is clearly related to that in the Arctic atmosphere. It is well known that the Arctic mean sea ice state is governed by the local mean atmospheric state. However, the response of the Arctic mean sea ice state to the local atmospheric intraseasonal variability is unclear. The Arctic atmospheric intraseasonal variability exists in both the thermodynamical and dynamical variables. Based on a sea ice-ocean coupled simulation with a quantitative sea ice budget analysis, this study finds that 1) the intraseasonal atmospheric thermodynamical variability tends to reduce sea ice melting through changing the downward heat flux on the open water area in the marginal sea ice zone, and the intraseasonal atmospheric dynamical variability tends to increase sea ice melting by a combination of modified air-ocean heat fluxes, ice-ocean heat fluxes, and sea ice deformation; 2) the intraseasonal atmospheric dynamical variability increases summertime sea ice concentration in the Beaufort Sea and the Greenland Sea but decreases summertime sea ice concentration along the Eurasian continent in the East Siberia-Laptev-Kara Seas, resulting from the joint effects of the modified air-ocean heat fluxes, ice-ocean heat fluxes, and the sea ice deformation, as well as the mean sea ice advection due to the changes of sea ice drift. The large spread in sea ice in the CMIP models may be partly attributed to the different model performances in representing the observed atmospheric intraseasonal variability. Reliable modeling of atmospheric intraseasonal variability is an essential condition in correctly

projecting future sea ice evolution.

601. An assessment of the North Atlantic (25-75 degreesN) air-sea CO₂ flux in 12 CMIP6 models

Authors: Yujie Jing; Yangchun Li; Yongfu Xu

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2021.103682

URL: <https://doi.org/10.1016/j.dsr.2021.103682>

Publication Date: 2021-12-05

Volume/Issue: Volume 180

References: 47

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

602. Simulation and future projection of the mixed layer depth and subduction process in the subtropical Southeast Pacific

Authors: Ruibin Xia; Yijun He; Tingting Yang

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-021-1877-0

URL: <https://doi.org/10.1007/s13131-021-1877-0>

Publication Date: 2021-12-04

Volume/Issue: Volume 40, Issue 12

References: 30

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

603. A Comparison of Factors That Led to the Extreme Sea Ice Minima in the Twenty-First Century in the Arctic Ocean

Authors: Xi Liang (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Xichen Li (Chinese Academy of Sciences); Haibo Bi (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Martin Losch (Alfred Wegener Institute); Yongqi Gao (Nansen Environmental and Remote Sensing Center); Fu Zhao (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Zhongxiang Tian (National Marine Environmental Forecasting Center, Ministry of Natural Resources); Chengyan Liu (Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-21-0199.1

URL: <https://doi.org/10.1175/jcli-d-21-0199.1>

Publication Date: 2021-12-03

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References: 78

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: Anomalies, Sea ice concentration, Ice loss/growth, Arctic

Funding: National Key R&D; Program of China, National Key R&D; Program of China, national key r&d; program of china

License: N/A

Open Access: N/A

Abstract: The extreme Arctic sea ice minima in the twenty-first century have been attributed to multiple factors, such as anomalous atmospheric circulation, excess solar radiation absorbed by open ocean, and thinning sea ice in a warming world. Most likely it is the combination of these factors that drives the extreme sea ice minima, but how the factors rank in setting the conditions for these events has not been quantified. To address this question, the sea ice budget of an Arctic regional sea ice-ocean model forced by atmospheric reanalysis data is analyzed to assess the development of the observed sea ice minima. Results show that the ice area difference in the years 2012, 2019, and 2007 is driven to over 60% by the difference in summertime sea ice area loss due to air-ocean heat flux over open water. Other contributions are small. For the years 2012 and 2020 the situation is different and more complex. The air-ice heat flux causes more sea ice area loss in summer 2020 than in 2012 due to warmer air temperatures, but this difference in sea ice area loss is compensated by reduced advective sea ice loss out of the Arctic Ocean mainly caused by the relaxation of the Arctic dipole. The difference in open water area in early August leads to different air-ocean heat fluxes, which distinguishes the sea ice minima in 2012 and 2020. Further, sensitivity experiments indicate that both the atmospheric circulation associated with the Arctic dipole and extreme storms are essential conditions for a new low record of sea ice extent.

604. Ocean warming and accelerating Southern Ocean zonal flow

Authors: Jia-Rui Shi; Lynne D. Talley; Shang-Ping Xie; Qihua Peng; Wei Liu

Journal: Nature Climate Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41558-021-01212-5

URL: <https://doi.org/10.1038/s41558-021-01212-5>

Publication Date: 2021-11-29

Volume/Issue: Volume 11, Issue 12

References: 61

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, the Southern Ocean Carbon and Climate Observations and Modeling project

License: N/A

Open Access: N/A

Abstract: N/A

605. Sensitivity of Steady State, Deep Ocean Dissolved Organic Carbon to Surface Boundary Conditions

Authors: Katsumi Matsumoto (University of Minnesota); Tatsuro Tanioka (University of California Irvine, University of Minnesota); Maya Gilchrist (University of Minnesota)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gb007102

URL: <https://doi.org/10.1029/2021gb007102>

Publication Date: 2021-11-26

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References: 41

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: We use the transport matrices of a data-constrained circulation model to efficiently compute the steady state distribution of the deep ocean dissolved organic carbon (DOC) at a 1 degrees horizontal resolution by propagating the surface DOC boundary conditions into the ocean interior. An equivalent simulation in the traditional forward modeling approach would be prohibitively computationally expensive. Our model simulates the total DOC as the sum of two DOC pools, the refractory and the semi-labile. The model is able to simulate the large-scale features of the deep ocean DOC without local sources or sinks of DOC in the ocean interior. The deep ocean DOC in the model is sensitive to the preformed DOC concentrations in the formation sites of deep and bottom waters, where observations are lacking. Furthermore, our model experiments indicate that the deep Atlantic DOC gradient is sensitive to the mixing of deep waters with different concentrations of preformed refractory DOC, the transport of semi-labile DOC from the surface North Atlantic, and the decay rate of semi-labile DOC. These, combined with the observation that much of the deep ocean DOC gradient is in the Atlantic, suggests that the semi-labile DOC may be an important component of the deep Atlantic DOC. Finally, we show that DOC export depends substantially on the depth level where it is evaluated.

606. Progress in understanding of Indian Ocean circulation, variability, air-sea exchange, and impacts on biogeochemistry

Authors: Helen E. Phillips; Amit Tandon; Ryo Furue; Raleigh Hood; Caroline C. Ummenhofer; Jessica A. Benthuyssen; Viviane Menezes; Shijian Hu; Ben Webber; Alejandra Sanchez-Franks; Deepak Cherian; Emily Shroyer; Ming Feng; Hemantha Wijesekera; Abhisek Chatterjee; Lisan Yu; Juliet Hermes; Raghu Murtugudde; Tomoki Tozuka; Danielle Su; Arvind Singh; Luca Centurioni; Satya Prakash; Jerry Wiggert

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-17-1677-2021

URL: <https://doi.org/10.5194/os-17-1677-2021>

Publication Date: 2021-11-26

Volume/Issue: Volume 17, Issue 6

References: 728

Cited by: 66 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Over the past decade, our understanding of the Indian Ocean has advanced through concerted efforts toward measuring the ocean circulation and air-sea exchanges, detecting changes in water masses, and linking physical processes to ecologically important variables. New circulation pathways and mechanisms have been discovered that control atmospheric and oceanic mean state and variability. This review brings together new understanding of the ocean-atmosphere system in the Indian Ocean since the last comprehensive review, describing the Indian Ocean circulation patterns, air-sea interactions, and climate variability. Coordinated international focus on the Indian Ocean has motivated the application of new technologies to deliver higher-resolution observations and models of Indian Ocean processes. As a result we are discovering the importance of small-scale processes in setting the large-scale gradients and circulation, interactions between physical and biogeochemical processes, interactions between boundary currents and the interior, and interactions between the surface and the deep ocean. A newly discovered regional climate mode in the southeast Indian Ocean, the Ningaloo Niño, has instigated more regional air-sea coupling and marine heatwave research in the global oceans. In the last decade, we have seen rapid warming of the Indian Ocean overlaid with extremes in the form of marine heatwaves. These events have motivated studies that have delivered new insight into the variability in ocean heat content and exchanges in the Indian Ocean and have highlighted the critical role of the Indian Ocean as a clearing house for anthropogenic heat. This synthesis paper reviews the advances in these areas in the last decade.

607. Mapping the Sensitivity of the Amundsen Sea Embayment to Changes in External Forcings Using Automatic Differentiation

Authors: Mathieu Morlighem (University of California Irvine, Dartmouth College); Daniel Goldberg (University of Edinburgh); Thiago Dias dos Santos (University of California Irvine, University of São Paulo, Centro Polar e Climático Universidade Federal do Rio Grande do Sul Porto Alegre Brazil); Jane Lee (University of California Irvine); Max Sagebaum (TU Kaiserslautern Kaiserslautern Germany)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl095440

URL: <https://doi.org/10.1029/2021gl095440>

Publication Date: 2021-11-25

Volume/Issue: Volume 48, Issue 23

References: 31

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Natural Environment Research Council

License: N/A

Open Access: N/A

Abstract: Thwaites and Pine Island Glaciers as well as other ice streams in West Antarctica have been changing dramatically over the past decades. Although changes in ocean conditions are likely the primary driver of these changes, it remains unclear where other processes could cause more mass loss. By employing Automatic Differentiation and two independent ice-sheet models, we construct maps of the sensitivity of the volume above floatation to changes in ocean-induced melt rates, ice rigidity, basal friction, and surface mass balance. We find that changes in basal melt close to the grounding lines and along shear margins have a larger impact on the glaciers' final volume. The glaciers are sensitive to changes in basal friction on regions close to the grounding lines, while changes in ice rigidity has a larger impact along the shear margins of Pine Island. The sensitivity to surface mass balance is uniform over grounded ice.

608. Kinetic Energy Transfers between Mesoscale and Submesoscale Motions in the Open Ocean's Upper Layers

Authors: Alberto C. Naveira Garabato (University of Southampton); Xiaolong Yu (Sun Yat-Sen University, c Laboratoire d'Océanographie Physique et Spatiale, Ifremer, Université de Bretagne Occidentale, Brest, France); Jörn Callies (California Institute of Technology); Roy Barkan (University of California Los Angeles, Tel Aviv University); Kurt L. Polzin (Woods Hole Oceanographic Institution); Eleanor E. Frajka-Williams (National Oceanography Centre); Christian E. Buckingham (c Laboratoire d'Océanographie Physique et Spatiale, Ifremer, Université de Bretagne Occidentale, Brest, France, i British Antarctic Survey, Cambridge, United Kingdom); Stephen M. Griffies (Princeton University, j NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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References: 184

Cited by: 30 publications

Language: N/A

Subjects: N/A

Keywords: Eddies, Ageostrophic circulations, Dynamics, Instability, Ocean circulation (and 7 more)

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License: N/A

Open Access: N/A

Abstract: Mesoscale eddies contain the bulk of the ocean's kinetic energy (KE), but fundamental questions remain on the cross-scale KE transfers linking eddy generation and dissipation. The role of submesoscale flows represents the key point of discussion, with contrasting views of submesoscales as either a source or a sink of mesoscale KE. Here, the first observational assessment of the annual cycle of the KE transfer between mesoscale and submesoscale motions is performed in the upper layers of a typical open-ocean region. Although these diagnostics have marginal statistical significance and should be regarded cautiously, they are physically plausible and can provide a valuable benchmark for model evaluation. The cross-scale KE transfer exhibits two distinct stages, whereby submesoscales energize mesoscales in winter and drain mesoscales in spring. Despite this seasonal reversal, an inverse KE cascade operates throughout the year across much of the mesoscale range. Our results are not incompatible with recent modeling investigations that place the headwaters of the inverse KE cascade at the submesoscale, and that rationalize the seasonality of mesoscale KE as an inverse cascade-mediated response to the generation of submesoscales in winter. However, our findings may challenge those investigations by suggesting that, in spring, a downscale KE transfer could dampen the inverse KE cascade. An exploratory appraisal of the dynamics governing mesoscale-submesoscale KE exchanges suggests that the upscale KE transfer in winter is underpinned by mixed layer baroclinic instabilities, and that the downscale KE transfer in spring is associated with frontogenesis. Current submesoscale-permitting ocean models may substantially understate this downscale KE transfer, due to the models' muted representation of frontogenesis.

609. Different mechanisms for the seasonal variations of the mesoscale eddy energy in the South China Sea

Authors: Yuhui Zhao; Yang Yang; X. San Liang; Yuanzhi Zhang

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2021.103677

URL: <https://doi.org/10.1016/j.dsr.2021.103677>

Publication Date: 2021-11-20

Volume/Issue: Volume 179

References: 41

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

610. The Salinity Pilot-Mission Exploitation Platform (Pi-MEP): A Hub for Validation and Exploitation of Satellite Sea Surface Salinity Data

Authors: Sébastien Guimbard (OceanScope); Nicolas Reul (Institut de Recherche pour le Développement (IRD)); Roberto Sabia (Telespazio-UK); Sylvain Herlédan (OceanDataLab); Ziad El Khoury Hanna (OceanDataLab); Jean-Francois Piollé (Institut de Recherche pour le Développement (IRD)); Frédéric Paul (Institut de Recherche pour le Développement (IRD)); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Julian J. Schanze (Earth & Space Research); Frederick M. Bingham (University of North Carolina Wilmington); David Le Vine (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Nadya Vinogradova-Shiffer (NASA Headquarters, Washington, DC 20001, USA); Susanne Mecklenburg (European Space Agency); Klaus Scipal (European Space Agency); Henri Laur (European Space Agency)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs13224600

URL: <https://doi.org/10.3390/rs13224600>

Publication Date: 2021-11-17

Volume/Issue: Volume 13, Issue 22

References: 40

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency

License: CC BY 4.0

Open Access: N/A

Abstract: The Pilot-Mission Exploitation Platform (Pi-MEP) for salinity is an ESA initiative originally meant to support and widen the uptake of Soil Moisture and Ocean Salinity (SMOS) mission data over the ocean. Starting in 2017, the project aims at setting up a computational web-based platform focusing on satellite sea surface salinity data, supporting studies on enhanced validation and scientific process over the ocean. It has been designed in close collaboration with a dedicated science advisory group in order to achieve three main objectives: gathering all the data required to exploit satellite sea surface salinity data, systematically producing a wide range of metrics for comparing and monitoring sea surface salinity products' quality, and providing user-friendly tools to explore, visualize and exploit both the collected products and the results of the automated analyses. The Salinity Pi-MEP is becoming a reference hub for the validation of satellite sea surface salinity missions by providing valuable information on satellite products (SMOS, Aquarius, SMAP), an extensive in situ database (e.g., Argo, thermosalinographs, moorings, drifters) and additional thematic datasets (precipitation, evaporation, currents, sea level anomalies, sea surface temperature, etc.). Co-localized databases between satellite products and in situ datasets are systematically generated together with validation analysis reports for 30 predefined regions. The data and reports are made fully accessible through the web interface of the platform. The datasets, validation metrics and tools (automatic, user-driven) of the platform are described in detail in this paper. Several dedicated scientific case studies involving satellite SSS data

are also systematically monitored by the platform, including major river plumes, mesoscale signatures in boundary currents, high latitudes, semi-enclosed seas, and the high-precipitation region of the eastern tropical Pacific. Since 2019, a partnership in the Salinity Pi-MEP project has been agreed between ESA and NASA to enlarge focus to encompass the entire set of satellite salinity sensors. The two agencies are now working together to widen the platform features on several technical aspects, such as triple-collocation software implementation, additional match-up collocation criteria and sustained exploitation of data from the SPURS campaigns.

611. Analysis of signal propagation in the UNDER-ICE experiment

Authors: Espen Storheim; Hanne Sagen; Florian Geyer

Journal: Proceedings of Meetings on Acoustics

Publisher: ASA

DOI: 10.1121/2.0001500

URL: <https://doi.org/10.1121/2.0001500>

Publication Date: 2021-11-17

Volume/Issue: Volume 44

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Mooring|SIGNAL (programming language)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

612. Optimal Precursors Triggering Sudden Shifts in the Antarctic Circumpolar Current Transport Through Drake Passage

Authors: Li Zhou (Chinese Academy of Sciences); Qiang Wang (Hohai University, Ministry of Natural Resources); Mu Mu (Fudan University, Chinese Academy of Sciences); Kun Zhang (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017899

URL: <https://doi.org/10.1029/2021jc017899>

Publication Date: 2021-11-16

Volume/Issue: Volume 126, Issue 12

References: 70

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: Antarctic Circumpolar Current, baroclinic instability, optimal precursors, CNOP, short-range prediction

Funding: Fundamental Research Funds for the Central Universities, National Outstanding Youth Science Fund Project of National Natural Science Foundation of China, Chinese Academy of Sciences

License: N/A

Open Access: N/A

Abstract: Due to the high nonlinearity, the accurate prediction of the Antarctic circumpolar current (ACC) transport is still challenging. Using the eddy-permitting regional ocean modeling system and the conditional nonlinear optimal perturbation approach, the sudden shift in the ACC transport through Drake Passage (DP) is investigated by exploring its optimal precursor (OPR). Here, the sudden shift in the ACC transport is defined as a fluctuation exceeding double STD (~ 16 Sv; $1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$) within 30 days. The results indicate that the OPRs for all three cases exhibit specific structures in the middle DP (58 degrees-62 degreesS, 72 degrees-64 degreesW) at the depth of 1,000-3,000 m, implying that the ACC transport is most sensitive to the initial perturbations there. The OPRs' evolutions show similar features: the OPR for each case triggers an eddy-like dipole perturbation with a northern cyclone and a southern anticyclone, leading to a sudden reduction in the ACC transport of ~ 40 Sv. It is verified that the density components in the OPRs determine such evolution processes. Furthermore, baroclinic instability is diagnosed to play a dominant role in the development of OPRs by analyzing the perturbed kinetic energy budget. This study suggests that the deep-layer density perturbations in the Southern Ocean should be carefully monitored when considering the short-range prediction of the ACC transport.

613. Review of Sharp et al „A monthly surface pCO₂ product for the California Current Large Marine Ecosystem”

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/essd-2021-328-rc1

URL: <https://doi.org/10.5194/essd-2021-328-rc1>

Publication Date: 2021-11-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

614. Mesoscale Eddy-Induced Ocean Dynamic and Thermodynamic Anomalies in the North Pacific

Authors: Jiang Zhou; Guidi Zhou; Hailong Liu; Zhuhua Li; Xuhua Cheng

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2021.756918

URL: <https://doi.org/10.3389/fmars.2021.756918>

Publication Date: 2021-11-12

Volume/Issue: Volume 8

References: 123

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Oceanic mesoscale eddies are associated with large thermodynamic anomalies, yet so far they are most commonly studied in terms of surface temperature and in the sense of composite mean. Here we employ an objective eddy identification and tracking algorithm together with a novel matching and filling procedure to more thoroughly examine eddy-induced thermodynamic anomalies in the North Pacific, their relationship with eddy amplitude (SSH), and the percentage of variability they explain on various timescales from submonthly to interannual. The thermodynamic anomalies are investigated in terms of sea surface temperature (SST), isothermal layer depth (ITD), and upper ocean heat content (HCT). Most eddies are weak in amplitude and are associated with small thermodynamic anomalies. In the sense of composite mean, anticyclonic eddies are generally warm eddies with deeper isothermal layer and larger heat content, and the reverse is true for cyclonic eddies. A small fraction of eddies, most probably subsurface eddies, exhibits the opposite polarities. Linear relationships with eddy amplitude are found for each of the thermodynamic parameters but with different level of scatter and seasonality. HCT-amplitude relation scatters the least and has the smallest seasonal difference, ITD-amplitude relation has the largest scatter and seasonality, while SST-amplitude relation is in between. For the Kuroshio and Oyashio Extension region, the most eddy-rich region in the North Pacific, eddies are responsible for over 50% of the total SSH variability up to the intra-seasonal scale, and ITD and HCT variability up to interannual. Eddy-induced SST variability is the highest along the Oyashio Extension Front on the order of 40-60% on submonthly scales. These results highlight the role of mesoscale eddies in ocean thermodynamic variability and in air-sea interaction.

615. Drawdown of Atmospheric pCO₂ Via Variable Particle Flux Stoichiometry in the Ocean Twilight Zone

Authors: Tatsuro Tanioka (University of California Irvine, University of Minnesota); Katsumi Matsumoto (University of Minnesota); Michael W. Lomas (Bigelow Laboratory for Ocean Sciences East Boothbay ME USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094924

URL: <https://doi.org/10.1029/2021gl094924>

Publication Date: 2021-11-11

Volume/Issue: Volume 48, Issue 22

References: 68

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Simons Foundation

License: N/A

Open Access: N/A

Abstract: The strength of the biological soft tissue pump in the ocean critically depends on how much organic carbon is produced via photosynthesis and how efficiently the carbon is transferred to the ocean interior. For a given amount of limiting nutrient, phosphate, soft tissue pump would be strengthened if the carbon (C) to phosphorus (P) ratio of sinking organic matter increases as the remineralization length scale of C increases. Here, we present a new data compilation of particle flux stoichiometry and show that C:P of sinking particulate organic matter (POM) in the ocean twilight zone on average is likely to be higher than the C:P ratio of surface suspended POM. We further demonstrate using a physics-biology coupled global ocean model combined with a theory from first principles that an increase in C:P export flux ratio in the ocean's twilight zone can lead to a considerable drawdown of atmospheric pCO₂.

616. Ocean mass, sterodynamic effects, and vertical land motion largely explain US coast relative sea level rise

Authors: T. C. Harvey; B. D. Hamlington; T. Frederikse; R. S. Nerem; C. G. Piecuch; W. C. Hammond; G. Blewitt; P. R. Thompson; D. P. S. Bekaert; F. W. Landerer; J. T. Reager; R. E. Kopp; H. Chandanpurkar; I. Fenty; D. Trossman; J. S. Walker; C. Boening

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-021-00300-w

URL: <https://doi.org/10.1038/s43247-021-00300-w>

Publication Date: 2021-11-09

Volume/Issue: Volume 2, Issue 1

References: 88

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: Regional sea-level changes are caused by several physical processes that vary both in space and time. As a result of these processes, large regional departures from the long-term rate of global mean sea-level rise can occur. Identifying and understanding these processes at particular locations is the first step toward generating reliable projections and assisting in improved decision making. Here we quantify to what degree contemporary ocean mass change, sterodynamic effects, and vertical land motion influence sea-level rise observed by tide-gauge locations around the contiguous U.S. from 1993 to 2018. We are able to explain tide gauge-observed relative sea-level trends at 47 of 55 sampled locations. Locations where we cannot explain observed trends are potentially indicative of shortcomings in our coastal sea-level observational network or estimates of uncertainty.

617. Improved High Resolution Ocean Reanalyses Using a Simple Smoother Algorithm

Authors: Bo Dong (University of Reading); Keith Haines (University of Reading); Matthew Martin (Met Office)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021ms002626

URL: <https://doi.org/10.1029/2021ms002626>

Publication Date: 2021-11-08

Volume/Issue: Volume 13, Issue 12

References: 58

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: ocean heat content, data assimilation, Kalman smoother, AMOC, salinity (and 1 more)

Funding: National Centre for Earth Observation, UK Research and Innovation

License: CC BY 4.0

Open Access: N/A

Abstract: We present a simple smoother designed for smooth data adjustments in sequentially generated reanalysis products by utilizing knowledge of future assimilation increments. A decay time parameter is applied to the smoother increments to account for memory decay timescales in the ocean. The result is different from simply time smoothing the reanalysis itself as only the increments are smoothed so the reanalysis product can retain high frequency variability that is being internally generated by the model and the atmospheric forcing. The smoother is applied first to the Lorenz 1963 model and then to the daily Met Office GloSea5 Global 1/4 degrees ocean reanalysis during 2016. Results show significant improvement over the original reanalysis in the 3D temperature and salinity state and variability, as well as in the sea surface height (SSH) and ocean currents. Comparisons are made directly against temperature, salinity and SSH observations, as well as independent 15 m drifter velocities. The impact on the time variability of conservative quantities, particularly ocean heat and salt content, as well as kinetic energy and the Atlantic Meridional Overturning Circulation (AMOC), is also demonstrated.

618. Overlooked Current Estimation Biases Arising from the Lagrangian Argo Trajectory Derivation Method

Authors: Tianyu Wang (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Yan Du (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Minyang Wang (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0287.1

URL: <https://doi.org/10.1175/jpo-d-20-0287.1>

Publication Date: 2021-11-04

Volume/Issue: Volume 52, Issue 1

References: 70

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Eddies, Bias, Numerical analysis/modeling, Ocean circulation, Buoy observations

Funding: national key r&d; program of china, chinese academy of sciences, chinese academy of sciences (and 5 more)

License: N/A

Open Access: N/A

Abstract: An Argo simulation system is used to provide synthetic Lagrangian trajectories based on the Estimating the Circulation and Climate of the Ocean Model, phase II (ECCO2). In combination with ambient Eulerian velocity at the reference layer (1000 m) from the model, quantitative metrics of the Lagrangian trajectory-derived velocities are computed. The result indicates that the biases induced by the derivation algorithm are strongly linked with ocean dynamics. In low latitudes, Ekman currents and vertically sheared geostrophic currents influence both the magnitude and the direction of the derivation velocity vectors. The maximal shear-induced biases exist near the equator with the amplitudes reaching up to about 1.2 cm s⁻¹. The angles of the shear biases are pronounced in the low-latitude oceans, ranging from -8 degrees to 8 degrees. Specifically, the study shows an overlooked bias from the float drifting motions that mainly occurs in the western boundary current and Antarctic Circumpolar Current (ACC) regions. In these regions, a recently reported horizontal acceleration measured via Lagrangian floats is significantly associated with the strong eddy-jet interactions. The acceleration could induce an overestimation of Eulerian current velocity magnitudes. For the common Argo floats with a 9-day float parking period, the derivation speed biases induced by velocity acceleration would be as large as 3 cm s⁻¹, approximately 12% of the ambient velocity. It might have implications to map the mean middepth ocean currents from Argo trajectories, as well as to understand the dynamics of eddy-jet interactions in the ocean.

619. Indo-Pacific Warming Induced by a Weakening of the Atlantic Meridional Overturning Circulation

Authors: Shantong Sun (California Institute of Technology); Andrew F. Thompson (California Institute of Technology); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego); Shang-Min Long (Hohai University, Ministry of Natural Resources)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-21-0346.1

URL: <https://doi.org/10.1175/jcli-d-21-0346.1>

Publication Date: 2021-11-04

Volume/Issue: Volume 35, Issue 2

References: 134

Cited by: 15 publications

Language: N/A

Subjects: N/A

Keywords: Climate change, General circulation models, Meridional overturning circulation

Funding: National Science Foundation, National Natural Science Foundation of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: The reorganization of the Atlantic meridional overturning circulation (AMOC) is often associated with changes in Earth's climate. These AMOC changes are communicated to the Indo-Pacific basins via wave processes and induce an overturning circulation anomaly that opposes the Atlantic changes on decadal to centennial time scales. We examine the role of this transient, interbasin overturning response, driven by an AMOC weakening, both in an ocean-only model with idealized geometry and in a coupled CO₂ quadrupling experiment, in which the ocean warms on two distinct time scales: a fast decadal surface warming and a slow centennial subsurface warming. We show that the transient interbasin overturning produces a zonal heat redistribution between the Atlantic and Indo-Pacific basins. Following a weakened AMOC, an anomalous northward heat transport emerges in the Indo-Pacific, which substantially compensates for the Atlantic southward heat transport anomaly. This zonal heat redistribution manifests as a thermal interbasin seesaw between the high-latitude North Atlantic and the subsurface Indo-Pacific and helps to explain why Antarctic temperature records generally show more gradual changes than the Northern Hemisphere during the last glacial period. In the coupled CO₂ quadrupling experiment, we find that the interbasin heat transport due to a weakened AMOC contributes substantially to the slow centennial subsurface warming in the Indo-Pacific, accounting for more than half of the heat content increase and sea level rise. Thus, our results suggest that the transient interbasin overturning circulation is a key component of the global ocean heat budget in a changing climate.

620. Sea Surface Salinity Subfootprint Variability from a Global High-Resolution Model

Authors: Frederick M. Bingham (University of North Carolina Wilmington); Susannah Brodnitz (University of North Carolina Wilmington); Severine Fournier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Karly Ulfsax (Catlin Engineers & Scientists); Akiko Hayashi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs13214410

URL: <https://doi.org/10.3390/rs13214410>

Publication Date: 2021-11-03

Volume/Issue: Volume 13, Issue 21

References: 44

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: Subfootprint variability (SFV) is variability at a spatial scale smaller than the footprint of a satellite, and it cannot be resolved by satellite observations. It is important to quantify and understand, as it contributes to the error budget for satellite data. The purpose of this study was to estimate the SFV for sea surface salinity (SSS) satellite observations. This was performed by using a high-resolution numerical model, a 1/48 degrees version of the MITgcm simulation, from which one year of output has recently become available. SFV, defined as the weighted standard deviation of SSS within the satellite footprint, was computed from the model for a 2 degrees x 2 degrees grid of points for the one model year. We present maps of median SFV for 40 and 100 km footprint size, display histograms of its distribution for a range of footprint sizes and quantify its seasonality. At a 100 km (40 km) footprint size, SFV has a mode of 0.06 (0.04). It is found to vary strongly by location and season. It has larger values in western-boundary and eastern-equatorial regions, as well as in a few other areas. SFV has strong variability throughout the year, with the largest values generally being in the fall season. We also quantified the representation error, the degree of mismatch between random samples within a footprint and the footprint average. Our estimates of SFV and representation error can be used in understanding errors in the satellite observation of SSS.

621. Defining Southern Ocean fronts using unsupervised classification

Authors: Simon D. A. Thomas; Daniel C. Jones; Anita Faul; Erik Mackie; Etienne Pauthenet

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-17-1545-2021

URL: <https://doi.org/10.5194/os-17-1545-2021>

Publication Date: 2021-11-02

Volume/Issue: Volume 17, Issue 6

References: 70

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, UK Research and Innovation, H2020 European Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: . Oceanographic fronts are transitions between thermohaline structures with different characteristics. Such transitions are ubiquitous, and their locations and properties affect how the ocean operates as part of the global climate system. In the Southern Ocean, fronts have classically been defined using a small number of continuous, circumpolar features in sea surface height or dynamic height. Modern observational and theoretical developments are challenging and expanding this traditional framework to accommodate a more complex view of fronts. Here, we present a complementary new approach for calculating fronts using an unsupervised classification method called Gaussian mixture modelling (GMM) and a novel inter-class parameter called the I-metric. The I-metric approach produces a probabilistic view of front location, emphasising the fact that the boundaries between water masses are not uniformly sharp across the entire Southern Ocean. The I-metric approach uses thermohaline information from a range of depth levels, making it more general than approaches that only use near-surface properties. We train the GMM using an observationally constrained state estimate in order to have more uniform spatial and temporal data coverage. The probabilistic boundaries defined by the I-metric roughly coincide with several classically defined fronts, offering a novel view of this structure. The I-metric fronts appear to be relatively sharp in the open ocean and somewhat diffuse near large topographic features, possibly highlighting the importance of topographically induced mixing. For comparison with a more localised method, we also use an edge detection approach for identifying fronts. We find a strong correlation between the edge field of the leading principal component and the zonal velocity; the edge detection method highlights the presence of jets, which are supported by thermal wind balance. This more localised method highlights the complex, multiscale structure of Southern Ocean fronts, complementing and contrasting with the more domain-wide view offered by the I-metric. The Sobel edge detection method may be useful for defining and tracking smaller-scale fronts and jets in model or reanalysis data. The I-metric approach may prove to be a useful method for inter-model comparison, as it uses the thermohaline structure of those models instead of tracking somewhat ad hoc values of sea surface height and/or dynamic height, which can vary considerably between models. In addition, the general I-metric approach allows front definitions to

shift with changing temperature and salinity structures, which may be useful for characterising fronts in a changing climate.

622. Characterization of far-field Mekong freshwater mass transport in the southern South China Sea using satellite gravimetry

Authors: Hok Sum Fok; Zhongtian Ma

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2021.103686

URL: <https://doi.org/10.1016/j.gloplacha.2021.103686>

Publication Date: 2021-10-30

Volume/Issue: Volume 207

References: 68

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

623. Modeling the mixed layer depth in Southern Ocean using high resolution regional coupled ocean sea ice model

Authors: Anurag Kumar; R. Bhatla

Journal: Modeling Earth Systems and Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s40808-021-01321-2

URL: <https://doi.org/10.1007/s40808-021-01321-2>

Publication Date: 2021-10-29

Volume/Issue: Volume 8, Issue 3

References: 59

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Science and Engineering Research Board

License: N/A

Open Access: N/A

Abstract: N/A

624. Formation Mechanism of Warm SST Anomalies in 2010s Around Hawaii

Authors: Naoya Takahashi (International Pacific Research Center School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu Hawaii); Kelvin J. Richards (International Pacific Research Center School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu Hawaii); Niklas Schneider (International Pacific Research Center School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu Hawaii); H. Annamalai (International Pacific Research Center School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu Hawaii); Wei-Ching Hsu (International Pacific Research Center School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu Hawaii); Masami Nonaka (Japan Agency for Marine-Earth Science and Technology Yokohama Japan)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017763

URL: <https://doi.org/10.1029/2021jc017763>

Publication Date: 2021-10-29

Volume/Issue: Volume 126, Issue 11

References: 56

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Agency for Marine-Earth Science and Technology

License: N/A

Open Access: N/A

Abstract: Warm sea surface temperature (SST) anomalies have been observed in the subtropical North Pacific around Hawaii in the recent decade, appearing from 2013. We examined the formation mechanisms of the warm SST anomalies in terms of relative contribution of atmospheric surface forcing and oceanic dynamics, using the latest reanalysis products from ECMWF (ERA5 for atmosphere and ORAS5 for ocean). Results of the mixed layer temperature budget diagnosis in the target area (10-20 degreesN and 180 degrees-160 degreesW) indicates that contributions from anomalous latent heat fluxes to the subtropical SST anomalies are dominant. Oceanic advective contributions play a secondary role, dampen the SST anomalies, and are negatively correlated ($r = -0.38$) with the latent heat fluxes. For example, the +1.0 K SST increased from 2011 to 2015 results from +1.5 K contributions from sum of surface heat flux and -0.5 K from meridional oceanic advection. The anti-correlation between atmospheric forcing and oceanic meridional advection reflects co-variations of wind-driven latent heat flux and meridional Ekman advection due to the weakening of the zonal component of the surface winds.

625. Modeling Untreated Wastewater Evolution and Swimmer Illness for Four Wastewater Infrastructure Scenarios in the San Diego-Tijuana (US/MX) Border Region

Authors: Falk Feddersen (Scripps Institution of Oceanography); Alexandria B. Boehm (Dept Civil and Environmental Engineering Stanford University Stanford CA USA); Sarah N. Giddings (Scripps Institution of Oceanography); Xiaodong Wu (Scripps Institution of Oceanography); Doug Liden (Environmental Protection Agency San Diego CA USA)

Journal: GeoHealth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gh000490

URL: <https://doi.org/10.1029/2021gh000490>

Publication Date: 2021-10-28

Volume/Issue: Volume 5, Issue 11

References: 61

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

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License: CC BY-NC 4.0

Open Access: N/A

Abstract: The popular beaches of the San Diego-Tijuana (US/MX) border region are often impacted by untreated wastewater sourced from Mexico--via the Tijuana River Estuary (TJRE) and San Antonio de los Buenos outfall at the Pt. Bandera (SAB/PTB) shoreline, leading to impacted beaches and human illness. The US-Mexico-Canada trade agreement will fund border infrastructure projects reducing untreated wastewater discharges. However, estimating project benefits such as reduced human illness and beach impacts is challenging. We develop a coupled hydrodynamic, norovirus (NoV) pathogen, and swimmer illness risk model with the wastewater sources for the year 2017. The model is used to evaluate the reduction in human illness and beach impacts under baseline conditions and three infrastructure diversion scenarios which (Scenario A) reduce SAB/PTB discharges and moderately reduce TJRE inflows or (Scenarios B, C) strongly reduce TJRE in inflows only. The model estimates shoreline untreated wastewater and NoV concentrations, and the number of NoV ill swimmers at Imperial Beach CA. In the Baseline, the percentage of swimmers becoming ill is 3.8% over 2017, increasing to 4.5% during the tourist season (Memorial to Labor Day) due to south-swell driven SAB/PTB plumes. Overall, Scenario A provides the largest reduction in ill swimmers and beach impacts for the tourist season and full year. The 2017 tourist season TJRE inflows were not representative of those in 2020, yet, Scenario A likely still provides the greatest benefit in other years. This methodology can be applied to other coastal regions with wastewater inputs.

626. Advective pathways and transit times of the Red Sea Overflow Water in the Arabian Sea from Lagrangian simulations

Authors: Viviane V. Menezes

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2021.102697

URL: <https://doi.org/10.1016/j.pocean.2021.102697>

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Volume/Issue: Volume 199

References: 102

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

627. Energy Transfers between Multidecadal and Turbulent Variability

Authors: Antoine Hochet (Institut de Recherche pour le Développement (IRD)); Thierry Huck (Institut de Recherche pour le Développement (IRD)); Olivier Arzel (Institut de Recherche pour le Développement (IRD)); Florian Sévellec (Institut de Recherche pour le Développement (IRD)); Alain Colin de Verdière (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-21-0136.1

URL: <https://doi.org/10.1175/jcli-d-21-0136.1>

Publication Date: 2021-10-27

Volume/Issue: Volume 35, Issue 4

References: 68

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Energy budget|Energy flux|Mode (computer interface)

Funding: H2020 Marie Skłodowska-Curie Actions

License: N/A

Open Access: N/A

Abstract: One of the proposed mechanisms to explain the multidecadal variability observed in sea surface temperature of the North Atlantic Ocean consists of a large-scale low-frequency internal mode spontaneously developing because of the large-scale baroclinic instability of the time-mean circulation. Even though this mode has been extensively studied in terms of the buoyancy variance budget, its energetic properties remain poorly known. Here we perform the full mechanical energy budget including available potential energy (APE) and kinetic energy (KE) of this internal mode and decompose the budget into three frequency bands: mean, low frequency (LF) associated with the large-scale mode, and high frequency (HF) associated with mesoscale eddy turbulence. This decomposition allows us to diagnose the energy fluxes between the different reservoirs and to understand the sources and sinks. Because of the large scale of the mode, most of its energy is contained in the APE. In our configuration, the only source of LF APE is the transfer from mean APE to LF APE that is attributed to the large-scale baroclinic instability. In return the sinks of LF APE are the parameterized diffusion, the flux toward HF APE, and, to a much lesser extent, the flux toward LF KE. The presence of an additional wind stress component weakens multidecadal oscillations and modifies the energy fluxes between the different energy reservoirs. The KE transfer appears to only have a minor influence on the multidecadal mode relative to the other energy sources involving APE, in all experiments. These results highlight the utility of the full APE-KE budget.

628. A Surface pCO₂ Increasing Hiatus in the Equatorial Pacific Ocean Since 2010

Authors: Shuang Qiu (Chinese Academy of Sciences); Yang Feng (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Yuhong Zhang (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Di Qi (Polar and Marine Research Institute, Jimei University Xiamen China); Yingxu Wu (Polar and Marine Research Institute, Jimei University Xiamen China); Yan Du (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl093612

URL: <https://doi.org/10.1029/2021gl093612>

Publication Date: 2021-10-26

Volume/Issue: Volume 48, Issue 21

References: 75

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Southern Marine Science and Engineering Guangdong Laboratory, National Outstanding Youth Science Fund Project of National Natural Science Foundation of China, Chinese Academy of Sciences

License: N/A

Open Access: N/A

Abstract: We used multiple data sets to investigate the decadal and longer time scale variability of sea surface partial pressure of CO₂ (pCO₂) over the equatorial Pacific Ocean from 1990 to 2019. Unlike the increasing trend in the global oceanic pCO₂, both the Niño 3.4 and warm pool regions featured distinct decadal variabilities. A pCO₂ increasing stagnation, a hiatus stage, was identified in the Niño 3.4 region from 2010 to 2019. Further analysis demonstrated that low-frequency pCO₂ was negatively correlated with the Interdecadal Pacific Oscillation (IPO) index. Both the 2009/10 and 2015/16 El Niño events and an increasing phase of IPO contributed to this long pCO₂ stagnation. Correspondingly, a weaker upwelling was induced in the central and eastern tropical Pacific as the trade winds weakened and thus stagnated the increase in pCO₂ in the Niño 3.4 region. Our results imply a significant impact of climate variabilities on sea surface pCO₂ in the equatorial Pacific.

629. Sea surface salinity short-term variability in the tropics

Authors: Frederick M. Bingham; Susannah Brodnitz

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-17-1437-2021

URL: <https://doi.org/10.5194/os-17-1437-2021>

Publication Date: 2021-10-22

Volume/Issue: Volume 17, Issue 5

References: 48

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Using data from the Global Tropical Moored Buoy Array, we study the validation process for satellite measurement of sea surface salinity (SSS). We compute short-term variability (STV) of SSS, variability on timescales of 2-17 d. It is a proxy for subfootprint variability over a 100 km footprint as seen by a satellite measuring SSS. We also compute representation error, which is meant to mimic the SSS satellite validation process where footprint averages are compared to pointwise in situ values. We present maps of these quantities over the tropical array. We also look at seasonality in the variability of SSS and find which months have maximum and minimum amounts. STV is driven at least partly by rainfall. Moorings exhibit larger STV during rainy periods than during non-rainy ones. The same computations are also done using output from a high-resolution global ocean model to see how it might be used to study the validation process. The model gives good estimates of STV, in line with the moorings, although tending to have smaller values.

630. Evaluation of the Local Sea-Level Budget at Tide Gauges Since 1958

Authors: Jinping Wang (Ocean University of China, Centre for Southern Hemisphere Oceans Research (CSHOR) CSIRO Oceans and Atmosphere Hobart TAS Australia); John A. Church (University of New South Wales); Xuebin Zhang (Centre for Southern Hemisphere Oceans Research (CSHOR) CSIRO Oceans and Atmosphere Hobart TAS Australia); Jonathan M. Gregory (Met Office Hadley Centre, Met Office, University of Reading); Laure Zanna (Courant Institute of Mathematical Sciences New York University New York NY USA); Xian Yao Chen (Ocean University of China)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094502

URL: <https://doi.org/10.1029/2021gl094502>

Publication Date: 2021-10-15

Volume/Issue: Volume 48, Issue 20

References: 76

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: sterodynamic sea level, tide gauge, sea-level budget, sea-level fingerprint, vertical land motion (and 1 more)

Funding: N/A

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Although global mean sea-level rise since 1900 and regional mean sea-level change since the 1960s have been accounted for in terms of the sum of contributions, the same budget closure has not been achieved for local relative sea-level change from a global network of tide gauges. To address this, we combine new estimates of sterodynamic sea-level change (SDSL; including ocean dynamics), glacial isostatic adjustment (GIA), change in land ice mass and terrestrial water storage, and other local vertical land motion. We find that the observed trends over 1958-2015 at all 272 tide gauges distributed worldwide agree with the sum of contributions (within 90% confidence estimates), with similar mean trend (1.1 mm yr⁻¹) and comparable spatial variability (standard deviation of 2.0 and 1.9 mm yr⁻¹ respectively). SDSL is the dominant contribution to both local observed mean trend and spatial variability, except at locations close to former ice-sheets, where GIA dominates.

631. Interannual variability of sea level in the South Indian Ocean: Local versus remote forcing mechanisms

Authors: Marion Kersalé; Denis L. Volkov; Kandaga Pujiana; Hong Zhang

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/os-2021-97

URL: <https://doi.org/10.5194/os-2021-97>

Publication Date: 2021-10-11

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Forcing (mathematics)|Throughflow|Oceanic basin

Funding: National Aeronautics and Space Administration, Cooperative Institute for Marine and Atmospheric Studies, University of Miami

License: CC BY 4.0

Open Access: N/A

Abstract: . The subtropical South Indian Ocean (SIO) has been described as one of the world's largest heat accumulators due to its remarkable warming during the past two decades. However, the relative contributions of the remote (of Pacific origin) forcing and local wind forcing to the variability of heat content and sea level in the SIO have not been fully attributed. Here, we combine a general circulation model, an analytic linear reduced gravity model, and observations to disentangle the spatial and temporal inputs of each forcing component on interannual to decadal timescales. A sensitivity experiment is conducted with artificially closed Indonesian straits to physically isolate the Indian and Pacific Oceans, thus, intentionally removing the Indonesian throughflow (ITF) influence on the Indian Ocean heat content and sea level variability. We show that the relative contribution of the signals originating in the equatorial Pacific versus signals caused by local wind forcing to the interannual variability of sea level and heat content in the SIO is dependent on location within the basin (low vs. mid latitude; western vs. eastern side of the basin). The closure of the ITF in the numerical experiment reduces the amplitude of interannual-to-decadal sea level changes compared to the simulation with a realistic ITF. However, the spatial and temporal evolution of sea level patterns in the two simulations remain similar and correlated with El Nino Southern Oscillation (ENSO). This suggests that these patterns are mostly determined by local wind forcing and oceanic processes, linked to ENSO via the 'atmospheric bridge' effect. We conclude that local wind forcing is an important driver for the interannual changes of sea level, heat content, and meridional transports in the SIO subtropical gyre, while oceanic signals originating in the Pacific amplify locally-forced signals.

632. Volume Budget of Subantarctic Mode Water in the Southern Ocean From an Ocean General Circulation Model

Authors: Wandi Jing (Ocean University of China); Yiyong Luo (Ocean University of China, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc017040

URL: <https://doi.org/10.1029/2020jc017040>

Publication Date: 2021-10-09

Volume/Issue: Volume 126, Issue 10

References: 50

Cited by: 4 publications

Language: en

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Keywords: N/A

Funding: National Natural Science Foundation of China, National Key Research and Development Program of China

License: N/A

Open Access: N/A

Abstract: The volume budget of Subantarctic Mode Water (SAMW) in the Southern Ocean (SO) is examined using a 0.1 degrees ocean model output from 1950 to 2017. The results show that air-sea buoyancy fluxes, advective exports, and diapycnal mixing are all important to the SAMW volume budget. A climatological mean analysis finds that, for the SO as a whole, water in the SAMW density range (kg m^{-3}) is balanced between the following: formation due to air-sea buoyancy fluxes (Sv), removal due to advective export northward out of the SO (Sv), destruction due to diapycnal mixing (Sv), and storage (Sv). Decadal changes in the SAMW volume in the Indian and Pacific sectors display a two-layer density structure, in which the volumes of the upper and lower layers vary predominantly out of phase with each other. In addition, the SAMW volume changes in the Indian and Pacific sectors appear to compensate each other on decadal time scales. The volume of SAMW in the SO has increased by $\sim 11\%$ since the 1950s, and this increase tends to emerge gradually from denser to lighter layers. The increased SAMW volume mainly results from a consistent increase in oceanic buoyancy gain over the outcropping area at approximately 27.1° .

633. Derivation of bedrock topography measurement requirements for the reduction of uncertainty in ice sheet model projections of Thwaites Glacier

Authors: Blake A. Castleman; Nicole-Jeanne Schlegel; Lambert Caron; Eric Larour; Ala Khazendar

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Publication Date: 2021-10-06

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Bedrock|Glacier morphology|Antarctic ice sheet|Future sea level

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Determining the future evolution of the Antarctic Ice Sheet is critical for understanding and narrowing the large existing uncertainties in century-scale global mean sea level rise (SLR) projections. One of the most significant glaciers/ice streams in Antarctica, Thwaites Glacier, is at risk of destabilization and, if destabilized, has the potential to be the largest regional-scale contributor of SLR on Earth. This is because Thwaites Glacier is vulnerable to the marine ice sheet instability, as its grounding line is significantly influenced by ocean-driven basal melting rates, and its bedrock topography retrogrades into kilometer deep troughs. In this study, we investigate how bedrock topography features influence the grounding line migration beneath Thwaites Glacier when extreme ocean-driven basal melt rates are applied. Specifically, we design experiments using the Ice-Sheet and Sea-level System Model (ISSM) to quantify the SLR projection uncertainty due to reported errors in the current bedrock topography maps that are often used by ice sheet models. We find that spread in model estimates of sea level rise contribution from Thwaites glacier due to the reported bedrock topography error could be as large as 21.9 cm after 200 years of extreme ocean warming. Next, we perturb the bedrock topography beneath Thwaites Glacier using wavelet decomposition techniques to introduce realistic noise (within error). We explore the model space with multiple realizations of noise to quantify what spatial and vertical resolutions in bedrock topography are required to minimize the uncertainty in our 200-year experiment. We conclude that at least a 2 km spatial and 8 m vertical resolution would independently constrain possible SLR to ± 2 cm over 200 years, fulfilling requirements outlined by the 2017 Decadal Survey for Earth Science. Lastly, we perform an ensemble of simulations to determine in which regions our model of Thwaites Glacier is most sensitive to perturbations in bedrock topography. Our results suggest that the retreat of the grounding line is most sensitive to bedrock topography in proximity to the grounding line's initial position. Additionally, we find that the location and amplitude of the bedrock perturbation is more significant than its sharpness and shape. Overall, these findings inform and benchmark observational requirements for future missions that will measure ice sheet bedrock topography, not only in the case of Thwaites Glacier but for Antarctica on the continental scale.

634. Relative vs. absolute wind stress in a circumpolar model of the Southern Ocean

Authors: David R. Munday; Xiaoming Zhai; James Harle; Andrew C. Coward; A.J. George Nurser

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2021.101891

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Publication Date: 2021-10-01

Volume/Issue: Volume 168

References: 115

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

635. Water Mass Transformation and Overturning Circulation in the Arabian Gulf

Authors: Maryam R. Al-Shehhi (a Civil and Environmental Engineering, Khalifa University, Abu Dhabi, United Arab Emirates); Hajoon Song (Yonsei University); Jeffery Scott (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0249.1

URL: <https://doi.org/10.1175/jpo-d-20-0249.1>

Publication Date: 2021-09-30

Volume/Issue: Volume 51, Issue 11

References: 56

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: Mixing, Upwelling/downwelling, Buoyancy, Thermocline circulation, Boundary current|Mass flux (and 2 more)

Funding: National Research Foundation of Korea, National Research Foundation of Korea, Khalifa University of Science, Technology and Research

License: N/A

Open Access: N/A

Abstract: We diagnose the ocean's residual overturning circulation of the Arabian Gulf in a high-resolution model and interpret it in terms of water-mass transformation processes mediated by air-sea buoyancy fluxes and interior mixing. We attempt to rationalize the complex three-dimensional flow in terms of the superposition of a zonal (roughly along axis) and meridional (transverse) overturning pattern. Rates of overturning and the seasonal cycle of air-sea fluxes sustaining them are quantified and ranked in order of importance. Air-sea fluxes dominate the budget so that, at zero order, the magnitude and sense of the overturning circulation can be inferred from air-sea fluxes, with interior mixing playing a lesser role. We find that wintertime latent heat fluxes dominate the water-mass transformation rate in the interior waters of the Gulf leading to a diapycnal volume flux directed toward higher densities. In the zonal overturning cell, fluid is drawn in from the Sea of Oman through the Strait of Hormuz, transformed, and exits the Strait near the southern and bottom boundaries. Along the southern margin of the Gulf, evaporation plays an important role in the meridional overturning pattern inducing sinking there.

636. Tracer and observationally-derived constraints on diapycnal diffusivities in an ocean state estimate

Authors: David S. Trossman; Caitlin B. Whalen; Thomas W. N. Haine; Amy F. Waterhouse; An T. Nguyen; Arash Bigdeli; Matthew Mazloff; Patrick Heimbach

Journal: Unknown Journal

Publisher: Copernicus GmbH

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References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Argo|Temperature salinity diagrams|TRACER|Optode

Funding: Earth Sciences Division, NOAA Research, Directorate for Geosciences

License: CC BY 4.0

Open Access: N/A

Abstract: . Use of an ocean parameter and state estimation framework-such as the Estimating the Circulation & Climate of the Ocean (ECCO) framework-could provide an opportunity to learn about the spatial distribution of the diapycnal diffusivity parameter (κ_p) that observations alone cannot due to gaps in coverage. However, we show that the assimilation of existing in situ temperature, salinity, and pressure observations is not sufficient to constrain κ_p estimated with ECCO, as κ_p from ECCO does not agree closely with observations-specifically, κ_p inferred from microstructure measurements. We investigate whether there are observations with more global coverage and well-understood measurement uncertainties that can be assimilated by ECCO to improve its representation of κ_p . Argo-derived κ_p using a strain-based parameterization of finescale hydrographic structure is one potential source of information. Argo-derived κ_p agrees well with microstructure. However, because Argo-derived κ_p has both measurement and structural uncertainties, we propose dissolved oxygen concentrations as a candidate for future data assimilation with ECCO. We perform sensitivity analyses with ECCO to test whether oxygen concentrations provide information about κ_p . We compare two adjoint sensitivity calculations: one that uses misfits to Argo-derived κ_p and the other uses misfits to dissolved oxygen concentrations. We show that adjoint sensitivities of dissolved oxygen concentration misfits to the state estimate's control space typically direct κ_p to improve relative to the Argo-derived and microstructure-inferred values. However, assimilation of dissolved oxygen concentrations would likely not serve as a substitute for assimilating accurately measured κ_p .

637. Earth System Model Parameter Adjustment Using a Green's Functions Approach

Authors: Ehud Strobach; Andrea Molod; Donifan Barahona; Atanas Trayanov; Dimitris Menemenlis; Gael Forget

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2021-251

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Publication Date: 2021-09-27

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Shortwave|Longwave|Earth system science|Shortwave radiation

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . We demonstrate the practicality and effectiveness of using a Green's functions estimation approach for adjusting uncertain parameters in an Earth System Model (ESM). This estimation approach had previously been applied to an intermediate-complexity climate model and to individual ESM components, e.g., ocean, sea-ice, or carbon-cycle components. Here, the Green's functions approach is applied to a state-of-the-art ESM that comprises a global atmosphere-land configuration of the Goddard Earth Observing System (GEOS) coupled to an ocean and sea-ice configuration of the Massachusetts Institute of Technology general circulation model (MITgcm). Horizontal grid spacing is approximately 110 km for GEOS and 37-110 km for MITgcm. In addition to the reference GEOS-MITgcm simulation, we carry out a series of model sensitivity experiments, in which 20 uncertain parameters are perturbed. These control parameters can be used to adjust sea-ice, microphysics, turbulence, radiation, and surface schemes in the coupled simulation. We define eight observational targets: sea-ice fraction, net surface shortwave radiation, downward longwave radiation, near-surface temperature, sea surface temperature, sea surface salinity, and ocean temperature and salinity at 300 m. We applied the Green's functions approach to optimize the values of the 20 control parameters so as to minimize a weighted least-squares distance between the model and the eight observational targets. The new experiment with the optimized parameters resulted in a total cost reduction of 9 % relative to a simulation that had already been adjusted using other methods. The optimized experiment attained a balanced cost reduction over most of the observational targets. We also report on results from a set of sensitivity experiments that are not used in the final optimized simulation but helped explore options and guided the optimization process. These experiments include an assessment of sensitivity to the number of control parameters and to the selection of observational targets and weights in the cost function. Based on these sensitivity experiments, we selected a specific definition for the cost function. The sensitivity experiments also revealed a decreasing overall cost as the number of control variables was increased. In summary, we recommend using the Green's functions estimation approach as an additional fine-tuning step in the model development process. The method is not a

replacement for modelers' experience in choosing and adjusting sensitive model parameters. Instead, it is an additional practical and effective tool for carrying out final adjustments of uncertain ESM parameters.

638. Tropical oceanic intraseasonal variabilities associated with central Indian Ocean mode

Authors: Ze Meng; Lei Zhou; Raghu Murtugudde; Qingxuan Yang; Kandaga Pujiana; Jingyuan Xi

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-05951-1

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Publication Date: 2021-09-22

Volume/Issue: Volume 58, Issue 3-4

References: 74

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, the China Ocean Mineral Resources Research and Development Association Program

License: N/A

Open Access: N/A

Abstract: N/A

639. Analysis of Earth's polar motion and length of day trends in comparison with estimates using second degree stokes coefficients from satellite gravimetry

Authors: Leonid Zotov; Christian Bizouard; C.K. Shum; Chaoyang Zhang; Nikolay Sidorenkov; Victor Yushkin

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2021.09.010

URL: <https://doi.org/10.1016/j.asr.2021.09.010>

Publication Date: 2021-09-22

Volume/Issue: Volume 69, Issue 1

References: 69

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

640. Understanding Lead Times of Warm Water Volumes to ENSO Sea Surface Temperature Anomalies

Authors: Sen Zhao (Nanjing University, Department of Atmospheric Sciences School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu HI USA); Fei-Fei Jin (Department of Atmospheric Sciences School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu HI USA); Malte F. Stuecker (Department of Oceanography and International Pacific Research Center School of Ocean and Earth Science and Technology University of Hawai'i at Mānoa Honolulu HI USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094366

URL: <https://doi.org/10.1029/2021gl094366>

Publication Date: 2021-09-21

Volume/Issue: Volume 48, Issue 19

References: 58

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The equatorial Pacific warm water volume (WWV) and western equatorial Pacific WWV (WWVw) are two important long-lead time El Niño-Southern Oscillation (ENSO) event precursors. Here, we utilize the recharge-discharge oscillator framework to obtain the analytical solution of WWV and WWVw lead times to ENSO sea surface temperatures (SSTs). Both WWV and WWVw lead times show a strong dependence on the inherent ENSO periodicity, whereas WWV lead time is further modulated by the air-sea coupling strength and ENSO SST spatial pattern. The derived solution is able to explain observed decadal variations in lead times of WWV to SST in terms of variations in ENSO periodicity and air-sea coupling strength.

641. Geostrophy Assessment and Momentum Balance of the Global Oceans in a Tide- and Eddy-Resolving Model

Authors: Xiaolong Yu (Institut de Recherche pour le Développement (IRD)); Aurélien L. Ponte (Institut de Recherche pour le Développement (IRD)); Noé Lahaye (Inria & IRMAR Campus Universitaire de Beaulieu Rennes France); Zoé Caspar-Cohen (Institut de Recherche pour le Développement (IRD)); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017422

URL: <https://doi.org/10.1029/2021jc017422>

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References: 62

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Agence Nationale de la Recherche

License: N/A

Open Access: N/A

Abstract: The future wide-swath satellite altimeters, such as the upcoming Surface Water Ocean Topography (SWOT) mission, will provide instantaneous 2D measurements of sea level down to the spatial scale of $O(10\text{ km})$ for the first time. However, the validity of the geostrophic assumption for estimating surface currents from these instantaneous maps is not known a priori. In this study, we quantify the accuracy of geostrophy for the estimation of surface currents from a knowledge of instantaneous sea level using the hourly snapshots from a tide- and eddy-resolving global numerical simulation. Geostrophic balance is found to be the leading-order balance in frontal regions characterized by large kinetic energy, such as the western boundary currents and the Antarctic Circumpolar Current. Everywhere else, geostrophic approximation ceases to be a useful predictor of ocean velocity, which may result in significant high-frequency contamination of geostrophically computed velocities by fast variability (e.g., inertial and higher). As expected, the validity of geostrophy is shown to improve at low frequencies (typically 0.5 cpd). Global estimates of the horizontal momentum budget reveal that the tropical and mid-latitude regions where geostrophic balance fails are dominated by fast variability and turbulent stress divergence terms rather than higher-order geostrophic terms. These findings indicate that the estimation of velocity from geostrophy applied on SWOT instantaneous sea level maps may be challenging away from energetic areas.

642. Mechanism of Interannual Cross-Equatorial Overturning Anomalies in the Pacific Ocean

Authors: Devanarayana R. M. Rao (Department of Earth and Space Science and Engineering York University Toronto ON Canada); Neil F. Tandon (Department of Earth and Space Science and Engineering York University Toronto ON Canada)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017509

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References: 37

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Pacific Ocean, interannual variability, meridional overturning circulation, SODA, ECCO

Funding: Natural Sciences and Engineering Research Council of Canada

License: N/A

Open Access: N/A

Abstract: The meridional overturning circulation (MOC) transports heat and mass between the tropics and the extratropics. Recent research has shown that the variability of the Indo-Pacific MOC dominates the variability of the global MOC on interannual timescales, and this variability is characterized by a prominent cross-equatorial cell (CEC) spanning the tropics. This CEC is a potentially important influence on interannual climate variability, but the mechanism responsible for the CEC is not understood. This study seeks to elucidate the mechanism of the CEC using two observational estimates of the ocean. Our analysis shows that the CEC can be explained by the following mechanistic chain: (a) Anomalies in the atmospheric circulation and hydrological cycle produce equatorially antisymmetric density anomalies in the upper Pacific Ocean (above approximately 500 m); (b) these density anomalies generate equatorially antisymmetric anomalies of sea surface height (SSH); (c) these SSH anomalies generate a cross-equatorial flow above approximately 1,000 m; and (d) this anomalous cross-equatorial flow drives compensating flow below approximately 1,000 m. This mechanism contrasts with that responsible for anomalous cross-equatorial overturning on seasonal timescales, which is primarily the Ekman response to equatorially antisymmetric anomalies of zonal wind stress. On interannual timescales, the zonal wind stress anomalies associated with the CEC are equatorially symmetric, and steric SSH variations are the dominant driver of the CEC. These insights may lead to improved understanding and prediction of interannual climate variability.

643. Dynamics of counter wind current along the south Sri Lanka coast during the Southwest Monsoon

Authors: C. H. Wickramage; Weiqiang Wang; K. Arulanathan; Ruchira Jayathilake

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-021-01477-6

URL: <https://doi.org/10.1007/s10236-021-01477-6>

Publication Date: 2021-09-15

Volume/Issue: Volume 71, Issue 11-12

References: 50

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

644. Control of oceanic circulation on sediment distribution in the southwestern Atlantic margin (23 to 55 degrees S)

Authors: Michel Michaelovitch de Mahiques; Roberto Violante; Paula Franco-Fraguas; Leticia Burone; Cesar Barbedo Rocha; Leonardo Ortega; Rosangela Felicio dos Santos; Bianca Sung Mi Kim; Rubens Cesar Lopes Figueira; Marcia Caruso Bicego

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-17-1213-2021

URL: <https://doi.org/10.5194/os-17-1213-2021>

Publication Date: 2021-09-15

Volume/Issue: Volume 17, Issue 5

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Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fundação de Amparo à Pesquisa do Estado de São Paulo, Conselho Nacional de Desenvolvimento Científico e Tecnológico

License: CC BY 4.0

Open Access: N/A

Abstract: . In this study, we interpret the role played by ocean circulation in sediment distribution on the southwestern Atlantic margin using radiogenic Nd and Pb isotopes. The latitudinal trends for Pb and Nd isotopes reflect the different current systems acting on the margin. The utilization of the sediment fingerprinting method allowed us to associate the isotopic signatures with the main oceanographic features in the area. We recognized differences between Nd and Pb sources to the Argentinean shelf (carried by the flow of Subantarctic Shelf Water) and slopes (transported by deeper flows). Sediments from Antarctica extend up to the Uruguayan margin, carried by the Upper and Lower Circumpolar Deep Water. Our data confirm that, for shelf and intermediate areas (the upper 1200 m), the transfer of sediments from the Argentinean margin to the north of 35° S is limited by the Subtropical Shelf Front and the basin-wide recirculated Antarctic Intermediate Water. On the southern Brazilian inner and middle shelf, it is possible to recognize the northward influence of the Río de la Plata sediments carried by the Plata Plume Water. Another flow responsible for sediment transport and deposition on the outer shelf and slope is the southward flow of the Brazil Current. Finally, we propose that the Brazil-Malvinas Confluence and the Santos Bifurcation act as boundaries of geochemical provinces in the area. A conceptual model of sediment sources and transport is provided for the southwestern Atlantic margin.

645. An analysis of Atlantic water in the Arctic Ocean using the Arctic subpolar gyre state estimate and observations

Authors: Jeffrey S. Grabon; John M. Toole; An T. Nguyen; Richard A. Krishfield

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2021.102685

URL: <https://doi.org/10.1016/j.pocean.2021.102685>

Publication Date: 2021-09-13

Volume/Issue: Volume 198

References: 67

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

646. Covariation of Airborne Biogenic Tracers (CO₂, COS, and CO) Supports Stronger Than Expected Growing Season Photosynthetic Uptake in the Southeastern US

Authors: Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bianca C. Baier (NOAA Earth System Research Laboratory Global Monitoring Division Boulder CO USA, Cooperative Institute for Research in Environmental Sciences University of Colorado-Boulder Boulder CO USA); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Le Kuai (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yoichi Shiga (Universities Space Research Association); Ian Baker (Colorado State University); Mary E. Whelan (Department of Environmental Sciences Rutgers, The State University of New Jersey New Brunswick NJ USA); Sha Feng (Pennsylvania State University); Maarten Krol (Wageningen University, Utrecht University); Colm Sweeney (NOAA Earth System Research Laboratory Global Monitoring Division Boulder CO USA); Benjamin R. Runkle (University of Arkansas); Elahe Tajfar (University of Arkansas); Kenneth J. Davis (Pennsylvania State University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gb006956

URL: <https://doi.org/10.1029/2021gb006956>

Publication Date: 2021-09-12

Volume/Issue: Volume 35, Issue 10

References: 98

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: carbon cycle, carbonyl sulfide, act-America, ecosystem, carbon monoxide (and 1 more)

Funding: NASA

License: N/A

Open Access: N/A

Abstract: The Atmospheric Carbon Transport (ACT)-America Earth Venture mission conducted five airborne campaigns across four seasons from 2016 to 2019, to study the transport and fluxes of Greenhouse gases across the eastern United States. Unprecedented spatial sampling of atmospheric tracers (CO₂, carbon monoxide [CO], carbonyl sulfide [COS]) related to biospheric processes offers opportunities to improve our qualitative and quantitative understanding of seasonal and spatial patterns of biospheric carbon uptake. Here, we examine co-variation of boundary layer enhancements of CO₂, CO, and COS across three diverse regions: the crop-dominated Midwest, evergreen-dominated South, and deciduous broadleaf-dominated Northeast. To understand the biogeochemical processes controlling these tracers, we compare the observed co-variation to simulated co-variation resulting from model- and satellite- constrained surface carbon fluxes. We found indication of a common terrestrial

biogenic sink of CO₂ and COS and secondary production of CO from biogenic sources in summer throughout the eastern US, driven by stomatal conductance. Upper Midwest crops drive CO₂ and COS depletion from early to late summer. Northeastern temperate forests drive CO₂ and COS depletion in late summer. The unprecedented ACT-America flask samples uncovered evidence that southern humid temperate forests photosynthesize and absorb CO₂ and COS, and emit CO precursors, deep into the growing season. Satellite- constrained carbon fluxes capture much of the observed seasonal and spatial variability, but underestimate the magnitude of net CO₂ and COS depletion in the South, indicating a stronger than expected net sink of CO₂ in late summer. Additional sampling of the South will more accurately constrain underlying biological processes and climate sensitivities governing southern carbon dynamics.

647. Modeling Photosynthesis and Exudation in Subtropical Oceans

Authors: Zhen Wu (Massachusetts Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Oliver Jahn (Massachusetts Institute of Technology); Daniel Sher (Department of Marine Biology Leon H. Charney School of Marine Sciences University of Haifa Haifa Israel); Angelicque White (University of Hawaii at Manoa); Michael J. Follows (Massachusetts Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gb006941

URL: <https://doi.org/10.1029/2021gb006941>

Publication Date: 2021-09-12

Volume/Issue: Volume 35, Issue 9

References: 63

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Simons Foundation, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Marine phytoplankton contributes nearly half of the total primary production on Earth through photosynthesis. Parameterizations of algal photosynthesis commonly employed in global biogeochemical simulations generally fail to capture the observed vertical structure of primary production. Here we examined the consequences of decoupling photosynthesis (carbon fixation) and biosynthesis (biomass building) with accumulation or exudation of excess photosynthate under energy rich conditions in both regional and global models. The results show that the decoupling of these two processes improved the simulated vertical profile of primary production, increased modeled primary production over 30% globally and over 40% in subtropical oceans, improved simulated meridional patterns of particulate C:N:P and increased modeled surface pool of labile/semi-labile dissolved organic carbon. More generally, these results highlight the importance of exudation, which results from the decoupling of photosynthesis and biosynthesis, as a major physiological process affecting ocean biogeochemistry.

648. The adjoint-based Two Oceans One Sea State Estimate (TOOSSE)

Authors: Xiaowei Wang; Chuanyu Liu; Armin Köhl; Wu Geng; Fan Wang; Detlef Stammer

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-021-0439-9

URL: <https://doi.org/10.1007/s00343-021-0439-9>

Publication Date: 2021-09-09

Volume/Issue: Volume 40, Issue 1

References: 80

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

649. Modeling ocean-induced rapid Earth rotation variations: an update

Authors: Alexander A. Harker; Michael Schindelegger; Rui M. Ponte; David A. Salstein

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-021-01555-z

URL: <https://doi.org/10.1007/s00190-021-01555-z>

Publication Date: 2021-09-07

Volume/Issue: Volume 95, Issue 9

References: 73

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Austrian Science Fund, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: We revisit the problem of modeling the ocean's contribution to rapid, non-tidal Earth rotation variations at periods of 2-120 days. Estimates of oceanic angular momentum (OAM, 2007-2011) are drawn from a suite of established circulation models and new numerical simulations, whose finest configuration is on a 1° grid. We show that the OAM product by the Earth System Modeling Group at GeoForschungsZentrum Potsdam has spurious short period variance in its equatorial motion terms, rendering the series a poor choice for describing oceanic signals in polar motion on time scales of less than ~ 2 weeks. Accounting for OAM in rotation budgets from other models typically reduces the variance of atmosphere-corrected geodetic excitation by $\sim 54\%$ for deconvolved polar motion and by $\sim 60\%$ for length-of-day. Use of OAM from the 1° model does provide for an additional reduction in residual variance such that the combined oceanic-atmospheric effect explains as much as 84% of the polar motion excitation at periods < 120 days. Employing statistical analysis and bottom pressure changes from daily Gravity Recovery and Climate Experiment solutions, we highlight the tendency of ocean models run at a 1° grid spacing to misrepresent topographically constrained dynamics in some deep basins of the Southern Ocean, which has adverse effects on OAM estimates taken along the 90° meridian. Higher model resolution thus emerges as a sensible target for improving the oceanic component in broader efforts of Earth system modeling for geodetic purposes.

650. Terrestrial Water Storage Anomalies Emphasize Interannual Variations in Global Mean Sea Level During 1997-1998 and 2015-2016 El Niño Events

Authors: Yan-Ning Kuo (National Taiwan University); Min-Hui Lo (National Taiwan University); Yu-Chiao Liang (National Taiwan University, Columbia University, Woods Hole Oceanographic Institution); Yu-Heng Tseng (National Taiwan University); Chia-Wei Hsu (Colorado State University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094104

URL: <https://doi.org/10.1029/2021gl094104>

Publication Date: 2021-09-07

Volume/Issue: Volume 48, Issue 18

References: 62

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Forcing (mathematics)|Anomaly (physics)

Funding: Ministry of Science and Technology, Taiwan

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Interannual variations in global mean sea level (GMSL) closely correlate with the evolution of El Niño-Southern Oscillation. However, GMSL differences occur in extreme El Niños; for example, in the 2015-2016 and 1997-1998 El Niños, the peak GMSL during the mature stage of the former (9.00 mm) is almost 2.5 times higher than the latter (3.72 mm). Analyses from satellite and reanalysis data sets show that the disparity in GMSL is primarily due to barystatic (ocean mass) changes. We find that the 2015-2016 event developed not purely as an Eastern Pacific El Niño event but with Central Pacific (CP) El Niño forcing. CP El Niños contribute to a stronger negative anomaly of global terrestrial water storage and subsequent higher barystatic heights. Our results suggest that the mechanism of hydrology-related interannual variations of GMSL should be further emphasized, as more CP El Niño events are projected to occur.

651. Subpolar Southern Ocean Response to Changes in the Surface Momentum, Heat, and Freshwater Fluxes under 2xCO₂

Authors: Fabio Boeira Dias (University of Tasmania, b CSIRO Oceans and Atmosphere, Aspendale, Victoria, Australia); Catia M. Domingues (d National Oceanographic Centre, Southampton, United Kingdom, University of Tasmania); Simon J. Marsland (University of Tasmania, b CSIRO Oceans and Atmosphere, Aspendale, Victoria, Australia); Stephen R. Rintoul (f Centre for Southern Hemisphere Oceans Research, Hobart, Tasmania, Australia, e CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia, g Australian Antarctic Program Partnership, Hobart, Tasmania, Australia); Petteri Uotila (h Institute for Atmospheric and Earth Science Research, University of Helsinki, Helsinki, Finland); Russell Fiedler (e CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Mauricio M. Mata (i Instituto de Oceanografia, Universidade Federal do Rio Grande, Rio Grande, Brazil); Nathaniel L. Bindoff (University of Tasmania); Abhishek Savita (University of Tasmania, b CSIRO Oceans and Atmosphere, Aspendale, Victoria, Australia)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-21-0161.1

URL: <https://doi.org/10.1175/jcli-d-21-0161.1>

Publication Date: 2021-09-02

Volume/Issue: Volume 34, Issue 21

References: 107

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: Deep convection, Climate variability, Abyssal circulation, Ocean circulation, Meridional overturning circulation (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Antarctic subpolar Southern Ocean (sSO) has fundamental climate importance. Antarctic Bottom Water (AABW) originates in the sSO and supplies the lower limb of the meridional overturning circulation (MOC), occupying 36% of ocean volume. Climate models struggle to represent continental shelf processes that form AABW. We explore sources of persistent model biases by examining response of the sSO to perturbations in surface forcing in a global ocean-sea ice model (ACCESS-OM2) that forms AABW both on shelf and in open ocean. The sSO response to individual and combined perturbations of surface heat, freshwater, and momentum fluxes follows the WCRP CMIP6 FAFMIP-protocol. Wind perturbation (i.e., a poleward shift and intensification of the westerlies) is dominant, enhancing AABW formation and accelerating the global MOC. This occurs through upwelling of warm waters and inhibition of sea ice growth during winter, which triggers large open water polynya (OWP) events with associated deep convection. These events occur in the Weddell and Ross Seas and their variability is associated with availability of heat at midocean depths. These OWPs cease when the heat reservoir is depleted. Effects of surface warming and freshening only partially compensate changes from increasing winds on ocean stratification and depletion of AABW formation. These results indicate that overly convective models, such ACCESS-OM2, can respond to

CO₂-perturbed scenarios by forming too much AABW in OWP, which might not hold in models without OWPs. This might contribute to the large intermodel spread thermosteric sea level projections, being relevant to the interpretation of future projections by current climate models.

652. Evaluation of intraseasonal wind rectification on recent Indian Ocean dipole events using LICOM

Authors: Xia Zhao; Dongliang Yuan

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-05943-1

URL: <https://doi.org/10.1007/s00382-021-05943-1>

Publication Date: 2021-08-28

Volume/Issue: Volume 58, Issue 3-4

References: 52

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Project, cas key laboratory of receptor research, strategic priority research program of chinese academy of sciences (and 4 more)

License: N/A

Open Access: N/A

Abstract: N/A

653. Mechanisms linking multi-year La Niña with preceding strong El Niño

Authors: Tomoki Iwakiri; Masahiro Watanabe

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-021-96056-6

URL: <https://doi.org/10.1038/s41598-021-96056-6>

Publication Date: 2021-08-25

Volume/Issue: Volume 11, Issue 1

References: 53

Cited by: 51 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science, Ministry of Education, Culture, Sports, Science and Technology, Integrated Research Program for Advancing Climate Models

License: CC BY 4.0

Open Access: N/A

Abstract: El Niño-Southern Oscillation (ENSO), characterized by anomalous sea surface temperature in the central-eastern equatorial Pacific, is a dominant interannual variability, impacting worldwide weather and socioeconomics. The ENSO cycle contains irregularity, in which La Niña often persists for more than two years, called "multi-year La Niña". Observational records show that multi-year La Niña tends to accompany strong El Niño in the preceding year, but their physical linkage remains unclear. Here we show using reanalysis data that a strong El Niño excites atmospheric conditions that favor the generation of multi-year La Niña in subsequent years. Easterly wind anomalies along the northern off-equatorial Pacific during the decay phase of the strong El Niño are found crucial as they act to discharge ocean heat content (OHC) via an anomalous northward Ekman transport. The negative OHC anomaly is large enough to be restored by a single La Niña and, therefore, causes another La Niña to occur in the second year. Furthermore, analyses of the Coupled Model Intercomparison Project Phase 6 (CMIP6) climate models support the abovementioned mechanisms and indicate that the occurrence frequencies of multi-year La Niña and strong El Niño are highly correlated.

654. High-Frequency Fluctuations in Antarctic Bottom Water Transport Driven by Southern Ocean Winds

Authors: Andrew L. Stewart (University of California Los Angeles); Xiaoyang Chi (University of California Los Angeles); Aviv Solodoch (University of California Los Angeles); Andrew McC. Hogg (Australian National University, Australian Research Council Centre of Excellence for Climate Extremes)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094569

URL: <https://doi.org/10.1029/2021gl094569>

Publication Date: 2021-08-25

Volume/Issue: Volume 48, Issue 17

References: 53

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: overturning circulation, wind variability, momentum balance, Southern Ocean

Funding: Earth Sciences Division, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Northward flow of Antarctic Bottom Water (AABW) across the Southern Ocean comprises a key component of the global overturning circulation. Yet AABW transport remains poorly constrained by observations and state estimates, and there is presently no means of directly monitoring any component of the Southern Ocean overturning. However, AABW flow is dynamically linked to Southern Ocean surface circulation via the zonal momentum balance, offering potential routes to indirect monitoring of the transport. Exploiting this dynamical link, this study shows that wind stress (WS) fluctuations drive large AABW transport fluctuations on time scales shorter than 2 years, which comprise almost all of the transport variance. This connection occurs due to differing time scales on which topographic and interfacial form stresses respond to wind variability, likely associated with differences in barotropic versus baroclinic Rossby wave propagation. These findings imply that AABW transport variability can largely be reconstructed from the surface WS alone.

655. Arctic sea level variability from high-resolution model simulations and implications for the Arctic observing system

Authors: Guokun Lyu; Nuno Serra; Meng Zhou; Detlef Stammer

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/os-2021-79

URL: <https://doi.org/10.5194/os-2021-79>

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Arctic geoengineering

Funding: Horizon 2020, National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: . Two high-resolution model simulations are used to investigate the spatio-temporal variability of the Arctic Ocean sea level. The model simulations reveal barotropic sea level variability at periods < 30 days, which is strongly captured by bottom pressure observations. The seasonal sea level variability is driven by volume ex-changes with the Pacific and Atlantic Oceans and the redistribution of the water by the wind. Halosteric effects due to river runoff and evaporation minus precipitation (EmPmR), ice melting/formation also contribute in the marginal seas and seasonal sea ice extent regions. In the central Arctic Ocean, especially the Canadian Basin, the decadal halosteric effect dominates sea level variability. Satellite altimetric observations and Gravity Re-covery and Climate Experiment (GRACE) measurements could be used to infer freshwater content changes in the Canadian Basin at periods longer than one year. The increasing number of profiles seems to capture fresh-water content changes since 2007, encouraging further data synthesis work with a more complicated interpola-tion method. Further, in-situ hydrographic observations should be enhanced to reveal the freshwater budget and close the gaps between satellite altimetry and GRACE, especially in the marginal seas.

656. Error Assessment of GRACE and GRACE Follow-On Mass Change

Authors: Jianli Chen (University of Texas at Austin); Byron Tapley (University of Texas at Austin); Mark E. Tamisiea (University of Texas at Austin); Himanshu Save (University of Texas at Austin); Clark Wilson (University of Texas at Austin); Srinivas Bettadpur (University of Texas at Austin); Ki-Weon Seo (Seoul National University)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jb022124

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Publication Date: 2021-08-19

Volume/Issue: Volume 126, Issue 9

References: 52

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Earth Sciences Division, Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: We carry out a comprehensive error assessment of Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GFO) Release-6 (RL06) solutions from the Center for Space Research (CSR) at the University of Texas at Austin, NASA Jet Propulsion Laboratory (JPL), and Geoforschungszentrum (GFZ). The study covers the period April 2002 to August 2020 and uses two different methods, one based upon open ocean residuals (OOR) and the other a Three-Cornered Hat (TCH) calculation. General results from the two methods are similar. With 300 km Gaussian smoothing OOR RMS errors for CSR, JPL, and GFZ solutions are ~2.01, 3.19, and 3.67 cm, respectively. With additional decorrelation filtering OOR RMS values are reduced to ~1.24, 1.53, and 1.69 cm, respectively. TCH analysis also shows that CSR has the lowest noise levels with similar RMS values, and additional decorrelation filtering reduces error levels. TCH may underestimate errors if there are common errors among geophysical background models. Errors in GFO's first two years (25 solutions for 2018.06 to 2020.08) are comparable to those of GRACE when zonal degree 2 and 3 coefficients are replaced by Satellite Laser Ranging estimates. The OOR method reveals mismodeled intra-seasonal dynamic ocean signals associated with the Argentine Gyre during de-aliasing, while the TCH method shows differences between ocean tide models near Australia and Antarctica. Both OOR and TCH RMS analysis offer a means to assess the noise level of GRACE/GFO estimated mass change. The actual uncertainty of GRACE/GFO estimate averaged (or totaled) over a given region is also affected by other error sources.

657. Causal Mechanisms of Sea-level and Freshwater Content Change in the Beaufort Sea

Authors: Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-21-0069.1

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Publication Date: 2021-08-17

Volume/Issue: N/A

References: 74

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: Arctic, Seasonal variability, Ocean, Numerical analysis/modeling, Climate variability (and 3 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In the Arctic's Beaufort Sea, the rate of sea-level rise over the last two decades has been an order of magnitude greater than that of its global mean. This rapid regional sea-level rise is mainly a halosteric change, reflecting an increase in Beaufort Sea's freshwater content comparable to that associated with the Great Salinity Anomaly of the 1970s in the North Atlantic Ocean. Here we provide a new perspective of these Beaufort Sea variations by quantifying their causal mechanisms from 1992 to 2017 using a global, data-constrained ocean and sea-ice estimate of the Estimating the Circulation and Climate of the Ocean (ECCO) consortium. Our analysis reveals wind and sea-ice jointly driving the variations. Seasonal variation mainly reflects near-surface change due to annual melting and freezing of sea-ice, while interannual change extends deeper and mostly relates to wind-driven Ekman transport. Increasing wind stress and sea-ice melt are, however, equally important for decadal change that dominates the overall variation. Strengthening anticyclonic wind stress surrounding the Beaufort Sea intensifies the ocean's lateral Ekman convergence of relatively fresh near-surface waters. The strengthening stress also enhances convergence of sea-ice and ocean heat that increase the amount of Beaufort Sea's net sea-ice melt. The enhanced significance at longer time-scales of sea-ice melt relative to direct wind forcing can be attributed to ocean's advection and mixing of melt-water being slower than its dynamic adjustment to mechanical perturbations. The adjustments' difference implies that the sea-ice-melt-driven diabatic change will persist longer than the direct wind-driven kinematic anomaly.

658. Antarctic Slope Current Modulates Ocean Heat Intrusions Towards Totten Glacier

Authors: Yoshihiro Nakayama (Hokkaido University); Chad A. Greene (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Fernando S. Paolo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Vigan Mensah (Hokkaido University); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Haruhiko Kashiwase (National Institute of Polar Research Tachikawa Japan); Daisuke Simizu (National Institute of Polar Research Tachikawa Japan); Jamin S. Greenbaum (Scripps Institution of Oceanography / University of California San Diego); Donald D. Blankenship (University of Texas at Austin); Ayako Abe-Ouchi (University of Tokyo); Shigeru Aoki (Hokkaido University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094149

URL: <https://doi.org/10.1029/2021gl094149>

Publication Date: 2021-08-17

Volume/Issue: Volume 48, Issue 17

References: 66

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: ice shelf, Southern Ocean, ice shelf-ocean interaction, ocean heat intrusions, ocean modeling (and 1 more)

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: The Totten ice shelf (TIS) in East Antarctica has received increasing attention in recent years due to high basal melt rates, which have been linked to a presence of warm modified Circumpolar Deep Water (mCDW) observed at the ice front. We show that mCDW on-shelf intrusions towards the TIS strengthen when the Antarctic Slope Current (ASC) weakens. This demonstrates that the ASC has a blocking effect and ASC weakening leads to on-shelf intrusions, as proposed by previous observational studies. The interannual variability of the ASC is controlled primarily by atmospheric and oceanic conditions beyond our regional model domain. We further show that heat intrusions onto the continental shelf off the TIS are not influenced by off-shelf warming but are enhanced with coastal freshening, suggesting positive feedback whereby ice melt and freshening upstream could start a chain reaction, leading to increased melt, and further coastal freshening.

659. An iterative process for efficient optimisation of parameters in geoscientific models: a demonstration using the Parallel Ice Sheet Model (PISM) version 0.7.3

Authors: Steven J. Phipps; Jason L. Roberts; Matt A. King

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-5107-2021

URL: <https://doi.org/10.5194/gmd-14-5107-2021>

Publication Date: 2021-08-17

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References: 69

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council

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Abstract: . Physical processes within geoscientific models are sometimes described by simplified schemes known as parameterisations. The values of the parameters within these schemes can be poorly constrained by theory or observation. Uncertainty in the parameter values translates into uncertainty in the outputs of the models. Proper quantification of the uncertainty in model predictions therefore requires a systematic approach for sampling parameter space. In this study, we develop a simple and efficient approach to identify regions of multi-dimensional parameter space that are consistent with observations. Using the Parallel Ice Sheet Model to simulate the present-day state of the Antarctic Ice Sheet, we find that co-dependencies between parameters preclude any simple identification of a single optimal set of parameter values. Approaches such as large ensemble modelling are therefore required in order to generate model predictions that incorporate proper quantification of the uncertainty arising from the parameterisation of physical processes.

660. Widespread Grounding Line Retreat of Totten Glacier, East Antarctica, Over the 21st Century

Authors: Tyler Pelle (University of California Irvine); Mathieu Morlighem (University of California Irvine, Dartmouth College); Yoshihiro Nakayama (Hokkaido University); H  l  ne Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Dartmouth College)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: ice-ocean interactions, model coupling, Totten Glacier, ice sheet modeling, climate change (and 1 more)

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Open Access: N/A

Abstract: Totten Glacier (TG), the primary ice discharger of East Antarctica, contains 3.85 m sea level rise equivalent (SLRe) ice mass and has displayed ocean-driven dynamic change since at least the early 2000s. We project TG's evolution through 2100 in an asynchronously coupled ice-ocean model, forced at the ocean boundaries with anomalies in CMIP6 projected temperature, salinity, and velocity. Consistent with previous studies, the Antarctic Slope Current continues to modulate warm water inflow toward TG in future simulations. Warm water (-0.5 - $^{\circ}\text{C}$) accesses TG's sub-ice shelf cavity through depressions along the eastern ice front, driving sustained retreat of TG's eastern grounding zone that cannot be captured in uncoupled models. In high emission scenarios, warm water overcomes topographic barriers and dislodges TG's southern grounding zone around 2070, increasing the rate of grounded ice loss 3.5-fold (10-35 Gt/yr) and resulting in a total 4.20 mm SLRe loss by 2100.

661. A new sea ice state dependent parameterization for the free drift of sea ice

Authors: Charles Brunette; L. Bruno Tremblay; Robert Newton

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2021-249

URL: <https://doi.org/10.5194/tc-2021-249>

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Buoy|Sea ice concentration|Wind Stress

Funding: Marine Environmental Observation Prediction and Response Network, Natural Sciences and Engineering Research Council of Canada, Environment and Climate Change Canada

License: CC BY 4.0

Open Access: N/A

Abstract: . Free drift estimates of sea ice motion are necessary to produce a seamless observational record combining buoy and satellite-derived sea ice motion vectors. We develop a new parameterization for the free drift of sea ice based on wind forcing, wind turning angle, sea ice state variables (concentration and thickness) and ocean current (as a residual). Building on the fact that the spatially varying standard wind-ice transfer coefficient (considering only surface wind stress) has a structure as the spatial distribution of sea ice thickness, we introduce a wind-ice transfer coefficient that scales linearly with thickness. Results show a mean error of -0.5 cm/s (low-speed bias) and a root-mean-square error of 5.1 cm/s, considering daily buoy drift data as truth. This represents a 31 % reduction of the error on drift speed compared to the free drift estimates used in the Polar Pathfinder dataset. The thickness-dependent wind transfer coefficient provides an improved seasonality and long-term trend of the sea ice drift speed, with a minimum (maximum) drift speed in May (October), compared to July (January) for the constant wind transfer coefficient parameterizations which simply follow the peak in mean surface wind stresses. The trend in sea ice drift in this new model is +0.45 cm/s decade⁻¹ compared with +0.39 cm/s decade⁻¹ from the buoy observations, whereas there is essentially no trend in the standard free drift parameterization (-0.01 cm/s decade⁻¹) or the Polar Pathfinder free drifts (-0.03 cm/s decade⁻¹). The wind turning angle that minimize the cost function is equal of 25 degrees, with a mean and root-mean square error of +2.6 degrees and 51 degrees on the direction of the drift, respectively. The residual from the minimization procedure (i.e. the ocean currents) resolves key large scale features such as the Beaufort Gyre and Transpolar Drift Stream, and is in good agreement with ocean state estimates from the ECCO, GLORYS and PIOMAS ice-ocean reanalyses, and geostrophic currents from dynamical ocean topography, with a root-mean-square difference of 2.4, 2.9, 2.6 and 3.8 cm/s, respectively. Finally, a repeat of the analysis on a two sub-section of the time series (pre- and post-2000) clearly shows the acceleration of the Beaufort Gyre (particularly along the Alaskan coastline) and an expansion of the gyre in the post-2000 concurrent with a thinning of the sea ice cover and observations acceleration of the ice drift speed and ocean current. This new dataset is publicly available for complementing merged observations-based sea ice drift

datasets that includes satellite and buoy drift records.

662. Untangling local and remote influences in two major petrel habitats in the oligotrophic Southern Ocean

Authors: Daniel C. Jones (British Antarctic Survey Natural Environment Research Council Cambridge UK); Filipe R. Ceia (University of Coimbra); Eugene Murphy (British Antarctic Survey Natural Environment Research Council Cambridge UK); Karine Delord (Centre d'Etudes Biologiques de Chizé UMR 7372 du CNRS-La Rochelle Université Villiers-en-Bois France); Robert W. Furness (University of Glasgow); Ariane Verdy (Scripps Institution of Oceanography); Matthew Mazloff (Scripps Institution of Oceanography); Richard A. Phillips (British Antarctic Survey Natural Environment Research Council Cambridge UK); Paul M. Sagar (National Institute of Water and Atmospheric Research Ltd Christchurch New Zealand); Jean-Baptiste Sallée (L'Océan CNRS UPMC Paris France); Ben Schreiber (British Antarctic Survey Natural Environment Research Council Cambridge UK, University of Cambridge); David R. Thompson (National Institute of Water and Atmospheric Research Ltd Wellington New Zealand); Leigh G. Torres (Oregon State University); Philip J. Underwood (British Antarctic Survey Natural Environment Research Council Cambridge UK); Henri Weimerskirch (Centre d'Etudes Biologiques de Chizé UMR 7372 du CNRS-La Rochelle Université Villiers-en-Bois France); José C. Xavier (British Antarctic Survey Natural Environment Research Council Cambridge UK, University of Coimbra)

Journal: Global Change Biology

Publisher: Wiley

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Funding: Natural Environment Research Council, Institut Polaire Français Paul Emile Victor, Fundação para a Ciência e a Tecnologia

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Abstract: Ocean circulation connects geographically distinct ecosystems across a wide range of spatial and temporal scales via exchanges of physical and biogeochemical properties. Remote oceanographic processes can be especially important for ecosystems in the Southern Ocean, where the Antarctic Circumpolar Current transports properties across ocean basins through both advection and mixing. Recent tracking studies have indicated the existence of two large-scale, open ocean habitats in the Southern Ocean used by grey petrels (*Procellaria cinerea*) from two populations (i.e., Kerguelen and Antipodes islands) during their nonbreeding season for extended periods during austral summer (i.e., October to February). In this work, we use a novel combination of large-scale oceanographic observations, surface drifter data, satellite-derived primary productivity, numerical adjoint sensitivity experiments, and output from a biogeochemical state estimate to examine local and remote influences on these grey petrel habitats. Our aim is to understand the oceanographic features

that control these isolated foraging areas and to evaluate their ecological value as oligotrophic open ocean habitats. We estimate the minimum local primary productivity required to support these populations to be much $<1\%$ of the estimated local primary productivity. The region in the southeast Indian Ocean used by the birds from Kerguelen is connected by circulation to the productive Kerguelen shelf. In contrast, the region in the south-central Pacific Ocean used by seabirds from the Antipodes is relatively isolated suggesting it is more influenced by local factors or the cumulative effects of many seasonal cycles. This work exemplifies the potential use of predator distributions and oceanographic data to highlight areas of the open ocean that may be more dynamic and productive than previously thought. Our results highlight the need to consider advective connections between ecosystems in the Southern Ocean and to re-evaluate the ecological relevance of oligotrophic Southern Ocean regions from a conservation perspective.

663. Imminent calving accelerated by increased instability of the Brunt Ice Shelf, in response to climate warming

Authors: Yuan Cheng; Menglian Xia; Gang Qiao; Da Lv; Yanjun Li; Gang Hai

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2021.117132

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Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Key Research and Development Program of China

License: N/A

Open Access: N/A

Abstract: N/A

664. Parallel-Dynamic Interpolation Algorithm of Sea Surface Height for Future 2D Altimetry Mapping of Sea Surface Height

Authors: Jiankai Di; Chunyong Ma; Ge Chen

Journal: Journal of Ocean University of China

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11802-021-4664-9

URL: <https://doi.org/10.1007/s11802-021-4664-9>

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Volume/Issue: Volume 20, Issue 5

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: N/A

665. Ocean salinity indices of interannual modes in the tropical Pacific

Authors: Jianwei Chi; Tangdong Qu; Yan Du; Jifeng Qi; Ping Shi

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-05911-9

URL: <https://doi.org/10.1007/s00382-021-05911-9>

Publication Date: 2021-08-07

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References: 74

Cited by: 5 publications

Language: en

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Keywords: N/A

Funding: National Natural Science Foundation of China, Chinese Academy of Sciences, National Basic Research Program of China (and 3 more)

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Open Access: N/A

Abstract: This study investigates the interannual modes of the tropical Pacific using salinity from observations, ocean reanalysis output and CMIP6 products. Here we propose two indices of sea surface salinity (SSS), a monopole mode and a dipole mode, to identify the El Niño--South Oscillation (ENSO) and its diversity, respectively. The monopole mode is primarily controlled by atmospheric forcing, namely, the enhanced precipitation that induces negative SSS anomalies across nearly the entire tropical Pacific. The dipole mode is mainly forced by oceanic dynamics, with zonal current transporting fresh water from the western fresh pool into the western-central and salty water from the subtropics into the eastern tropical Pacific. Under a global warming condition, an increase in the monopole and dipole mode variance indicates an increase in both the central and eastern Pacific El Niño variability. The increase in central Pacific El Niño variability is largely due to enhanced vertical stratification during global warming in the upper layer, with intensified zonal advection. An eastern Pacific El Niño-like warming pattern contributes to the increase in eastern Pacific El Niño, with enhanced precipitation over the central-eastern tropical Pacific.

666. Development of adjoint-based ocean state estimation for the Amundsen and Bellingshausen seas and ice shelf cavities using MITgcm-ECCO (66j)

Authors: Yoshihiro Nakayama; Dimitris Menemenlis; Ou Wang; Hong Zhang; Ian Fenty; An T. Nguyen

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-4909-2021

URL: <https://doi.org/10.5194/gmd-14-4909-2021>

Publication Date: 2021-08-06

Volume/Issue: Volume 14, Issue 8

References: 72

Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: Japan Society for the Promotion of Science, Directorate for Geosciences

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Open Access: N/A

Abstract: . The Antarctic coastal ocean impacts sea level rise, deep-ocean circulation, marine ecosystems, and the global carbon cycle. To better describe and understand these processes and their variability, it is necessary to combine the sparse available observations with the best-possible numerical descriptions of ocean circulation. In particular, high ice shelf melting rates in the Amundsen Sea have attracted many observational campaigns, and we now have some limited oceanographic data that capture seasonal and interannual variability during the past decade. One method to combine observations with numerical models that can maximize the information extracted from the sparse observations is the adjoint method, a.k.a. 4D-Var (4-dimensional variational assimilation), as developed and implemented for global ocean state estimation by the Estimating the Circulation and Climate of the Ocean (ECCO) project. Here, for the first time, we apply the adjoint-model estimation method to a regional configuration of the Amundsen and Bellingshausen seas, Antarctica, including explicit representation of sub-ice-shelf cavities. We utilize observations available during 2010-2014, including ship-based and seal-tagged CTD measurements, moorings, and satellite sea-ice concentration estimates. After 20 iterations of the adjoint-method minimization algorithm, the cost function, here defined as a sum of the weighted model-data difference, is reduced by 65 % relative to the baseline simulation by adjusting initial conditions, atmospheric forcing, and vertical diffusivity. The sea-ice and ocean components of the cost function are reduced by 59 % and 70 %, respectively. Major improvements include better representations of (1) Winter Water (WW) characteristics and (2) intrusions of modified Circumpolar Deep Water (mCDW) towards the Pine Island Glacier. Sensitivity experiments show that ~40 % and ~10 % of improvements in sea ice and ocean state, respectively, can be attributed to the adjustment of air temperature and wind. This study is a preliminary demonstration of adjoint-method optimization with explicit representation of ice shelf cavity circulation. Despite the 65 % cost reduction, substantial model-data discrepancies remain, in particular with annual and interannual variability observed by moorings in front of the Pine Island Ice Shelf. We list a series of possible causes for these residuals, including limitations of the model, the optimization methodology, and observational sampling. In particular, we hypothesize that residuals could be further reduced if

the model could more accurately represent sea-ice concentration and coastal polynyas.

667. Matchup Characteristics of Sea Surface Salinity Using a High-Resolution Ocean Model

Authors: Frederick M. Bingham (University of North Carolina Wilmington); Severine Fournier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Susannah Brodnitz (University of North Carolina Wilmington); Karly Ulfsax (University of North Carolina Wilmington); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs13152995

URL: <https://doi.org/10.3390/rs13152995>

Publication Date: 2021-08-02

Volume/Issue: Volume 13, Issue 15

References: 41

Cited by: 7 publications

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Keywords: N/A

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Open Access: N/A

Abstract: Sea surface salinity (SSS) satellite measurements are validated using in situ observations usually made by surfacing Argo floats. Validation statistics are computed using matched values of SSS from satellites and floats. This study explores how the matchup process is done using a high-resolution numerical ocean model, the MITgcm. One year of model output is sampled as if the Aquarius and Soil Moisture Active Passive (SMAP) satellites flew over it and Argo floats popped up into it. Statistical measures of mismatch between satellite and float are computed, RMS difference (RMSD) and bias. The bias is small, less than 0.002 in absolute value, but negative with float values being greater than satellites. RMSD is computed using an "all salinity difference" method that averages level 2 satellite observations within a given time and space window for comparison with Argo floats. RMSD values range from 0.08 to 0.18 depending on the space-time window and the satellite. This range gives an estimate of the representation error inherent in comparing single point Argo floats to area-average satellite values. The study has implications for future SSS satellite missions and the need to specify how errors are computed to gauge the total accuracy of retrieved SSS values.

668. Reanalysis in Earth System Science: Toward Terrestrial Ecosystem

Reanalysis

Authors: R. Baatz (International Soil Modelling Consortium, Forschungszentrum Jülich); H. J. Hendricks Franssen (Forschungszentrum Jülich); E. Euskirchen (University of Alaska Fairbanks); D. Sihi (Emory University); M. Dietze (Boston University); S. Ciavatta (Plymouth Marine Laboratory); K. Fennel (Dalhousie University); H. Beck (Princeton University); G. De Lannoy (KU Leuven); V. R. N. Pauwels (Monash University); A. Raiho (Colorado State University); C. Montzka (Forschungszentrum Jülich); M. Williams (University of Edinburgh); U. Mishra (Sandia National Laboratory); C. Poppe (Forschungszentrum Jülich); S. Zacharias (Department of Monitoring and Exploration Technologies UFZ Helmholtz Centre for Environmental Research Leipzig Germany); A. Lausch (Department Computational Landscape Ecology Helmholtz Centre for Environmental Research-UFZ Leipzig Germany, Geography Department Humboldt University Berlin Berlin Germany); L. Samaniego (Department Computational Hydrosystems Helmholtz Centre for Environmental Research - UFZ Leipzig Germany); K. Van Looy (OVAM, International Policy Unit Mechelen Belgium); H. Bogaen (Forschungszentrum Jülich); M. Adamescu (University of Bucharest); M. Mirtl (Department of Monitoring and Exploration Technologies UFZ Helmholtz Centre for Environmental Research Leipzig Germany); A. Fox (Joint Center for Satellite Data Assimilation UCAR Boulder CO USA); K. Goergen (Centre for High-Performance Scientific Computing in Terrestrial Systems Geoverbund ABC/J Jülich Germany, Forschungszentrum Jülich); B. S. Naz (Centre for High-Performance Scientific Computing in Terrestrial Systems Geoverbund ABC/J Jülich Germany, Forschungszentrum Jülich); Y. Zeng (Faculty of Geo-information Science and Earth Observation (ITC) University of Twente Enschede The Netherlands); H. Vereecken (International Soil Modelling Consortium, Forschungszentrum Jülich)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

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Cited by: 44 publications

Language: en

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Keywords: N/A

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Abstract: A reanalysis is a physically consistent set of optimally merged simulated model states and historical observational data, using data assimilation. High computational costs for modeled processes and assimilation algorithms has led to Earth system specific reanalysis products for the atmosphere, the ocean and the land separately. Recent developments include the advanced uncertainty quantification and the generation of biogeochemical reanalysis for land and ocean. Here, we review atmospheric and oceanic reanalyses, and more in detail biogeochemical ocean and terrestrial reanalyses. In particular, we identify land surface, hydrologic and carbon cycle reanalyses which are

nowadays produced in targeted projects for very specific purposes. Although a future joint reanalysis of land surface, hydrologic, and carbon processes represents an analysis of important ecosystem variables, biotic ecosystem variables are assimilated only to a very limited extent. Continuous data sets of ecosystem variables are needed to explore biotic-abiotic interactions and the response of ecosystems to global change. Based on the review of existing achievements, we identify five major steps required to develop terrestrial ecosystem reanalysis to deliver continuous data streams on ecosystem dynamics.

669. Analysis of the Seasonal and Interannual Variations of Air-Sea CO₂ Flux in the Chukchi Sea Using a Coupled Ocean-Sea Ice-Biogeochemical Model

Authors: Zijia Zheng (School of Marine Science and Technology Tianjin University Tianjin China); Xiaofan Luo (School of Marine Science and Technology Tianjin University Tianjin China); Hao Wei (School of Marine Science and Technology Tianjin University Tianjin China); Wei Zhao (School of Marine Science and Technology Tianjin University Tianjin China); Di Qi (Ministry of Natural Resources)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017550

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References: 67

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Natural Science Foundation of Tianjin Municipal Science and Technology Commission

License: N/A

Open Access: N/A

Abstract: Understanding the responses of air-sea carbon dioxide (CO₂) flux to rapid environmental changes in the highly productive Chukchi Sea is of great significance for the climate prediction in the Arctic Ocean. Through analyzing the 1998-2015 hindcast simulation of a coupled ocean-sea ice-biogeochemical model, this study identifies that the key factors influencing oceanic carbon uptake in the Chukchi Sea vary seasonally. The effects of these seasonal carbon uptakes on the total annual carbon sink vary at different timescales. The model solution suggests that the Chukchi Sea acts as a perennial carbon sink of atmospheric CO₂ with the annual mean influx of -3.73 ± 3.90 mmol m⁻² d⁻¹. Quantitative analyses reveal that the sea ice concentration predominates the annual cycle of air-sea CO₂ flux, while primary production and wind speed strengthen the capacity for carbon uptake during June-July (melt season) and October-December (freeze season), respectively. The total annual carbon sink shows an increasing trend during 1998-2015 with a multi-year average of 9.22 ± 1.93 Tg C y⁻¹. The carbon uptakes during June-July and August-September (open water season), which are respectively regulated by primary production and open water area at interannual scale, present significant long-term increases and jointly determine the secular increase trend of the total annual carbon sink. Interannual CO₂ absorption during the freeze season is sensitive to the sea ice concentration and wind conditions, presenting the greatest year-to-year variation but insignificant long-term trend, and modulates the interannual amplitude of the total annual carbon sink.

670. Turbulence theories and statistical closure approaches

Authors: Ye Zhou

Journal: Physics Reports

Publisher: Elsevier BV

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URL: <https://doi.org/10.1016/j.physrep.2021.07.001>

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Cited by: 80 publications

Language: en

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Keywords: N/A

Funding: US Department of Energy

License: N/A

Open Access: N/A

Abstract: N/A

671. The relation of wind-driven coastal and offshore upwelling in the Benguela Upwelling System

Authors: Mohammad Hadi Bordbar (Leibniz Institute for Baltic Sea Research); Volker Mohrholz (Leibniz Institute for Baltic Sea Research); Martin Schmidt (Leibniz Institute for Baltic Sea Research)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0297.1

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References: 138

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: Wind Stress|Downwelling

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Open Access: N/A

Abstract: Spatial and temporal variations of nutrient-rich upwelled water across the major eastern boundary upwelling systems are primarily controlled by the surface wind with different, and sometimes contrasting, impacts on coastal upwelling systems driven by alongshore wind and offshore upwelling systems driven by the local wind-stress-curl. Here, concurrently measured wind-fields, satellite-derived Chlorophyll-a concentration along with a state-of-the-art ocean model simulation spanning 2008-2018 are used to investigate the connection between coastal and offshore physical drivers of the Benguela Upwelling System (BUS). Our results indicate that the spatial structure of long-term mean upwelling derived from Ekman theory and the numerical model are fairly consistent across the entire BUS and closely followed by the Chlorophyll-a pattern. The variability of the upwelling from the Ekman theory is proportionally diminished with offshore distance, whereas different and sometimes opposite structures are revealed in the model-derived upwelling. Our result suggests the presence of sub-mesoscale activity (i.e., filaments and eddies) across the entire BUS with a large modulating effect on the wind-stress-curl-driven upwelling off Lüderitz and Walvis Bay. In Kunene and Cape Frio upwelling cells, located in the northern sector of the BUS, the coastal upwelling and open-ocean upwelling frequently alternate each other, whereas they are modulated by the annual cycle and mostly in phase off Walvis Bay. Such a phase relationship appears to be strongly seasonally dependent off Lüderitz and across the southern BUS. Thus, our findings suggest this relationship is far more complex than currently thought and seems to be sensitive to climate changes with short- and far-reaching consequences for this vulnerable marine ecosystem.

672. Icelandic Low and Azores High Migrations Impact Florida Current Transport in Winter

Authors: Sultan Hameed (a School of Marine and Atmospheric Sciences, Stony Brook University, State University of New York, Stony Brook, New York); Christopher L. P. Wolfe (a School of Marine and Atmospheric Sciences, Stony Brook University, State University of New York, Stony Brook, New York); Lequan Chi (b Skidaway Institute of Oceanography, The University of Georgia, Savannah, Georgia)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: Boundary currents, Atmosphere-ocean interaction

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Previous work by Meinen and coworkers to find an association between variations of annually averaged Florida Current transport (FCT) and the North Atlantic Oscillation (NAO) has yielded negative results. Here we show that the Florida Current in winter is impacted by displacements in the positions of the Azores high and the Icelandic low, the constituent pressure centers of the NAO. As a one-dimensional representation of North Atlantic atmospheric circulation, the NAO index does not distinguish displacements of the pressure centers from fluctuations in their intensity. FCT is significantly correlated with Icelandic low longitude with a lag of less than one season. We carried out perturbation experiments in the ECCOV4 model to investigate these correlations. These experiments reveal that east-west shifts of the Icelandic low perturb the wind stress in midlatitudes adjacent to the American coast, driving downwelling (through longshore winds) and offshore sea level anomalies (through wind stress curl) that travel to the Straits of Florida within the same season. FCT is also correlated with the latitude variations of both the Icelandic low and the Azores high with a lag of 4 years. Regression analysis shows that latitude variations of the Icelandic low and the Azores high are associated with positive wind stress curl anomalies over extended regions in the ocean east of Florida. Rossby wave propagation from this region to the Straits of Florida has been suggested as a mechanism for perturbing FCT in several previous studies by various researchers, as detailed in sections 4b and 5.

673. An Atmospheric Constraint on the Seasonal Air-Sea Exchange of Oxygen and Heat in the Extratropics

Authors: Eric J. Morgan (Scripps Institution of Oceanography / University of California San Diego); Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Ralph F. Keeling (Scripps Institution of Oceanography / University of California San Diego); Laure Resplandy (Princeton University); Sara E. Mikaloff-Fletcher (National Institute of Water and Atmospheric Research Wellington New Zealand); Cynthia D. Nevison (University of Colorado Boulder); Yuming Jin (Scripps Institution of Oceanography / University of California San Diego); Jonathan D. Bent (Scripps Institution of Oceanography / University of California San Diego, Now at Picarro, Inc. Santa Clara CA USA, National Center of Atmospheric Research Boulder CO USA); Olivier Aumont (LOCEAN Paris France); Scott C. Doney (University of Virginia); John P. Dunne (NOAA/OAR Geophysical Fluid Dynamics Laboratory Princeton NJ USA); Jasmin John (NOAA/OAR Geophysical Fluid Dynamics Laboratory Princeton NJ USA); Ivan D. Lima (Woods Hole Oceanographic Institution); Matthew C. Long (National Center of Atmospheric Research Boulder CO USA); Keith B. Rodgers (Center for Climate Physics, Institute for Basic Science, Pusan National University Busan South Korea)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

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References: 74

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, NOAA Research

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Open Access: N/A

Abstract: The air-sea exchange of oxygen (O_2) is driven by changes in solubility, biological activity, and circulation. The total air-sea exchange of O_2 has been shown to be closely related to the air-sea exchange of heat on seasonal timescales, with the ratio of the seasonal flux of O_2 to heat varying with latitude, being higher in the extratropics and lower in the subtropics. This O_2 /heat ratio is both a fundamental biogeochemical property of air-sea exchange and a convenient metric for testing earth system models. Current estimates of the O_2 /heat flux ratio rely on sparse observations of dissolved O_2 , leaving it fairly unconstrained. From a model ensemble we show that the ratio of the seasonal amplitude of two atmospheric tracers, atmospheric potential oxygen (APO) and the argon-to-nitrogen ratio (Ar/O_2), exhibits a close relationship to the O_2 /heat ratio of the extratropics (40-). The amplitude ratio, I , is relatively constant within the extratropics of each hemisphere due to the zonal mixing of the atmosphere. I is not sensitive to atmospheric transport, as most of the observed spatial variability in the seasonal amplitude of APO is compensated by similar variations in (Ar/O_2). From the relationship between I /heat and I in the model ensemble, we determine that the atmospheric observations suggest hemispherically distinct I /heat flux ratios of 3.3 0.3 and 4.7 0.8 nmol between 40 and in the Northern and

Southern Hemispheres respectively, providing a useful constraint for and heat air-sea fluxes in earth system models and observation-based data products.

674. Period-Lengthening of the Mindanao Current Variability From the Long-Term Tide Gauge Sea Level Measurements

Authors: Xiao Chen (Hohai University, Southern Marine Science and Engineering Guangdong Laboratory); Bo Qiu (University of Hawaii at Manoa); Shuiming Chen (University of Hawaii at Manoa); Yiquan Qi (Hohai University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016932

URL: <https://doi.org/10.1029/2020jc016932>

Publication Date: 2021-07-15

Volume/Issue: Volume 126, Issue 8

References: 50

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Natural Science Foundation of Jiangsu Province

License: N/A

Open Access: N/A

Abstract: Long-term tide gauge sea level data from 1969 to 2014 at Davao (7.08 degreesN, 125.63 degreesE) and Malakal (7.33 degreesN, 134.46 degreesE) are analyzed to examine the decadal frequency modulations embedded in the Mindanao Current (MC) variability. The MC variability inferred from the Davao-Malakal sea level was predominantly biennial in the 1970s. This prevailing period switched to interannual in the 1980s and lengthened to decadal during the last two decades. With the aid of the basin-scale sea level information from satellite altimeter measurements, it is found that the sea level-inferred MC variability represents the coherent changes of the wind-driven tropical gyre in the western North Pacific. An investigation into the long-term wind stress curl data reveals that its prevailing period underwent similar biennial→interannual→decadal transitions in the western tropical Pacific, implying the forced nature of the period-lengthening of the MC variability during the past half-a-century. While the sign of the MC variability is largely determined by the Malakal sea level signals on the interannual and decadal time scales, the Davao sea level change becomes important when the time scale extends to multi-decades.

675. Time-Varying Upper Ocean Circulation and Control of Coral Bleaching in the Western Tropical Pacific

Authors: Bo Qiu (University of Hawaii at Manoa); Patrick L. Colin (Coral Reef Research Foundation); Shuiming Chen (University of Hawaii at Manoa)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl093632

URL: <https://doi.org/10.1029/2021gl093632>

Publication Date: 2021-07-15

Volume/Issue: Volume 48, Issue 14

References: 47

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: The western tropical Pacific Ocean (WTPO) features complicated ocean circulation systems and has the warmest world open-ocean waters. Small upper ocean temperature change there can exert significant impact on the regional coral reef ecosystems. In the past three decades, moderate to severe coral bleaching events have been observed in the WTPO surrounding Palau in 1998, 2010, 2016, 2017, and 2020. Reflecting the diversity of El Niño-Southern Oscillation (ENSO) variability, the observed coral bleaching severity does not correspond simply to the amplitude of an ENSO index, such as Niño-3.4. By conducting an upper ocean temperature budget, we found the time-varying upper ocean circulation advection acted to damp the anomalous surface heat flux forcing and played critical roles in controlling the surface ocean thermal conditions around Palau. This happened either directly via the advective temperature flux convergence, or indirectly through the pre-conditioning of upper ocean thermal structures.

676. Dense water downslope flow and AABW production in a numerical model: Sensitivity to horizontal and vertical resolution in the region off Cape Darnley polynya

Authors: Vigan Mensah; Yoshihiro Nakayama; Masakazu Fujii; Yoshifumi Nogi; Kay I. Ohshima

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2021.101843

URL: <https://doi.org/10.1016/j.ocemod.2021.101843>

Publication Date: 2021-07-07

Volume/Issue: Volume 165

References: 90

Cited by: 19 publications

Language: en

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Keywords: N/A

Funding: University of Tokyo, Hokkaido University, Ministry of Education, Culture, Sports, Science and Technology

License: N/A

Open Access: N/A

Abstract: N/A

677. Revealing the Impact of Global Heating on North Atlantic Circulation Using Transparent Machine Learning

Authors: Maïke Sonnewald (University of Washington, Princeton University, NOAA/OAR Geophysical Fluid Dynamics Laboratory Ocean and Cryosphere Division Princeton NJ USA); Redouane Lguensat (Laboratoire des Sciences du Climat et de l'Environnement CEA Saclay Institut Pierre Simon Laplace Gif-Sur-Yvette France, LOCEAN-IPSL Sorbonne Université Institut Pierre Simon Laplace Paris France)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021ms002496

URL: <https://doi.org/10.1029/2021ms002496>

Publication Date: 2021-07-03

Volume/Issue: Volume 13, Issue 8

References: 84

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: explainable and interpretable AI, climate modeling, transparent machine learning, oceanography, North Atlantic Ocean (and 1 more)

Funding: NOAA Research, Agence Nationale de la Recherche

License: CC BY 4.0

Open Access: N/A

Abstract: The North Atlantic ocean is key to climate through its role in heat transport and storage. Climate models suggest that the circulation is weakening but the physical drivers of this change are poorly constrained. Here, the root mechanisms are revealed with the explicitly transparent machine learning (ML) method Tracking global Heating with Ocean Regimes (THOR). Addressing the fundamental question of the existence of dynamical coherent regions, THOR identifies these and their link to distinct currents and mechanisms such as the formation regions of deep water masses, and the location of the Gulf Stream and North Atlantic Current. Beyond a black box approach, THOR is engineered to elucidate its source of predictive skill rooted in physical understanding. A labeled data set is engineered using an explicitly interpretable equation transform and k-means application to model data, allowing theoretical inference. A multilayer perceptron is then trained, explaining its skill using a combination of layerwise relevance propagation and theory. With abrupt CO₂ quadrupling, the circulation weakens due to a shift in deep water formation regions, a northward shift of the Gulf Stream and an eastward shift in the North Atlantic Current. If CO₂ is increased 1% yearly, similar but weaker patterns emerge influenced by natural variability. THOR is scalable and applicable to a range of models using only the ocean depth, dynamic sea level and wind stress, and could accelerate the analysis and dissemination of climate model data. THOR constitutes a step toward trustworthy ML called for within oceanography and beyond, as its predictions are physically tractable.

678. The ENEA-REG system (v1.0), a multi-component regional Earth system model: sensitivity to different atmospheric components over the Med-CORDEX (Coordinated Regional Climate Downscaling Experiment) region

Authors: Alessandro Anav; Adriana Carillo; Massimiliano Palma; Maria Vittoria Struglia; Ufuk Utku Turuncoglu; Gianmaria Sannino

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-4159-2021

URL: <https://doi.org/10.5194/gmd-14-4159-2021>

Publication Date: 2021-07-02

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References: 105

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Center for Atmospheric Research

License: CC BY 4.0

Open Access: N/A

Abstract: . In this study, a new regional Earth system model is developed and applied to the Med-CORDEX (Coordinated Regional Climate Downscaling Experiment) region. The ENEA-REG system is made up of two interchangeable regional climate models as atmospheric components (RegCM, REGIONal Climate Model, and WRF, Weather Research and Forecasting), a river model (Hydrological Discharge, HD), and an ocean model (Massachusetts Institute of Technology General Circulation Model, MITgcm); processes taking place at the land surface are represented within the atmospheric models with the possibility to use several land surface schemes of different complexity. The coupling between these components is performed through the RegESM driver. Here, we present and describe our regional Earth system model and evaluate its components using a multidecadal hindcast simulation over the period 1980-2013 driven by ERA-Interim reanalysis. We show that the atmospheric components correctly reproduce both large-scale and local features of the Euro-Mediterranean climate, although we found some remarkable biases: in particular, WRF has a significant cold bias during winter over the northeastern bound of the domain and a warm bias in the whole continental Europe during summer, while RegCM overestimates the wind speed over the Mediterranean Sea. Similarly, the ocean component correctly reproduces the analyzed ocean properties with performances comparable to the state-of-art coupled regional models contributing to the Med-CORDEX initiative. Our regional Earth system model allows studying the Euro-Mediterranean climate system and can be applied to both hindcast and scenario simulations.

679. Bridging observations, theory and numerical simulation of the ocean using machine learning

Authors: Maike Sonnewald; Redouane Lguensat; Daniel C Jones; Peter D Dueben; Julien Brajard; V Balaji

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/ac0eb0

URL: <https://doi.org/10.1088/1748-9326/ac0eb0>

Publication Date: 2021-06-25

Volume/Issue: Volume 16, Issue 7

References: 271

Cited by: 78 publications

Language: N/A

Subjects: N/A

Keywords: Sophistication

Funding: Norges Forskningsråd, Agence National de Recherche, UK Research and Innovation (and 2 more)

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Open Access: N/A

Abstract: Progress within physical oceanography has been concurrent with the increasing sophistication of tools available for its study. The incorporation of machine learning (ML) techniques offers exciting possibilities for advancing the capacity and speed of established methods and for making substantial and serendipitous discoveries. Beyond vast amounts of complex data ubiquitous in many modern scientific fields, the study of the ocean poses a combination of unique challenges that ML can help address. The observational data available is largely spatially sparse, limited to the surface, and with few time series spanning more than a handful of decades. Important timescales span seconds to millennia, with strong scale interactions and numerical modelling efforts complicated by details such as coastlines. This review covers the current scientific insight offered by applying ML and points to where there is imminent potential. We cover the main three branches of the field: observations, theory, and numerical modelling. Highlighting both challenges and opportunities, we discuss both the historical context and salient ML tools. We focus on the use of ML in situ sampling and satellite observations, and the extent to which ML applications can advance theoretical oceanographic exploration, as well as aid numerical simulations. Applications that are also covered include model error and bias correction and current and potential use within data assimilation. While not without risk, there is great interest in the potential benefits of oceanographic ML applications; this review caters to this interest within the research community.

680. A diagnosis of surface currents and sea surface heights in a coastal region

Authors: Eun Ae Lee; Sung Yong Kim

Journal: Continental Shelf Research

Publisher: Elsevier BV

DOI: 10.1016/j.csr.2021.104486

URL: <https://doi.org/10.1016/j.csr.2021.104486>

Publication Date: 2021-06-23

Volume/Issue: Volume 226

References: 106

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

681. Intercomparison of Arctic sea ice simulation in ROMS-CICE and ROMS-Budgell

Authors: Rajesh Kumar; Junde Li; Kate Hedstrom; Alexander V. Babanin; David M. Holland; Petra Heil; Youmin Tang

Journal: Polar Science

Publisher: Elsevier BV

DOI: 10.1016/j.polar.2021.100716

URL: <https://doi.org/10.1016/j.polar.2021.100716>

Publication Date: 2021-06-21

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References: 79

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

682. Multi-decadal retreat of marine-terminating outlet glaciers in northwest and central-west Greenland

Authors: Taryn E. Black; Ian Joughin

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2021-164

URL: <https://doi.org/10.5194/tc-2021-164>

Publication Date: 2021-06-18

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Glacier morphology|Glacier mass balance|Meltwater|Greenland ice sheet|Future sea level|Tidewater glacier cycle

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . The retreat and acceleration of marine-terminating outlet glaciers in Greenland over the past two decades has been widely attributed to climate change. Here we present a comprehensive annual record of glacier terminus positions in northwest and central-west Greenland and compare it against local and regional climatology to assess the regional sensitivity of glacier termini to different climatic factors. This record is derived from optical and radar satellite imagery and spans 87 marine-terminating outlet glaciers from 1972 through 2021. We find that in this region, most glaciers have retreated over the observation period, and widespread regional retreat accelerated around 1996. The acceleration of glacier retreat coincides with the timing of sharp shifts in ocean surface temperatures, duration of sea-ice season, ice-sheet surface mass balance, and meltwater and runoff production. Our findings suggest that a variety of processes - such as ocean-interface melting, mélange presence and rigidity, and hydrofracture-induced calving - contribute to, but do not conclusively dominate, the observed regional retreat.

683. The Effect of Resolution on Vertical Heat and Carbon Transports in a Regional Ocean Circulation Model of the Argentine Basin

Authors: Stan Swierczek (University of Arizona); Matthew R. Mazloff (Scripps Institution of Oceanography); Matthias Morzfeld (Scripps Institution of Oceanography); Joellen L. Russell (University of Arizona)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017235

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Publication Date: 2021-06-17

Volume/Issue: Volume 126, Issue 7

References: 56

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: air-sea carbon flux, model resolution, air-sea heat flux, biogeochemistry, Argentine Basin

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Simulations of the Argentine Basin have large uncertainties associated with quantities such as air-sea exchanges of heat and carbon in current generation climate models and ocean reanalysis products. This is due to the complex topography, profound undersampling until recent years, and strong currents and mixing of subpolar and subtropical water masses in the basin. Because mixing of water masses is important here, model resolution is hypothesized to play an important role in estimating ocean quantities and determining overall budgets. We construct three regional ocean models with biogeochemistry at 1/3 degrees, 1/6 degrees, and 1/12 degrees resolutions for the year 2017 to investigate heat and carbon dynamics in the region and determine the effect of model resolution on these dynamics. Initial conditions and boundary forcing from BSOSE (the Biogeochemical Southern Ocean State Estimate (Verdy & Mazloff, 2017), <https://doi.org/10.1002/2016JC012650>) and atmospheric forcing from ERA5 are used. The models are evaluated for accuracy by comparing output to Argo and BGC-Argo float profiles, BSOSE, and other reanalyses and mapped products. We then quantify the effect of resolution on model upper ocean heat and carbon transport and the associated air-sea exchanges. We determine that increasing the resolution from 1/3 degrees to 1/12 degrees enhances the upward vertical transport and surface exchanges of heat but causes no significant effect on surface carbon fluxes despite enhancing downward transport of anomalous DIC.

684. Sinking Diatom Assemblages as a Key Driver for Deep Carbon and Silicon Export in the Scotia Sea (Southern Ocean)

Authors: D. Zúñiga; A. Sanchez-Vidal; M. M. Flexas; D. Carroll; M. M. Rufino; G. Spreen; A. Calafat; F. Abrantes

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2021.579198

URL: <https://doi.org/10.3389/feart.2021.579198>

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Volume/Issue: Volume 9

References: 129

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Ministerio de Ciencia y Tecnología, Fundação para a Ciência e a Tecnologia, Generalitat de Catalunya

License: CC BY 4.0

Open Access: N/A

Abstract: Physical and biogeochemical processes in the Southern Ocean are fundamental for modulating global climate. In this context, a process-based understanding of how Antarctic diatoms control primary production and carbon export, and hence global-ocean carbon sequestration, has been identified as a scientific priority. Here we use novel sediment trap observations in combination with a data-assimilative ocean biogeochemistry model (ECCO-Darwin) to understand how environmental conditions trigger diatom ecology in the iron-fertilized southern Scotia Sea. We unravel the role of diatoms assemblage in controlling the biogeochemistry of sinking material escaping from the euphotic zone, and discuss the link between changes in upper-ocean environmental conditions and the composition of settling material exported from the surface to 1,000 m depth from March 2012 to January 2013. The combined analysis of in situ observations and model simulation suggests that an anomalous sea-ice episode in early summer 2012-2013 favored (via restratification due to sea-ice melt) an early massive bloom of *Corethron pennatum* that rapidly sank to depth. This event drove high biogenic silicon to organic carbon export ratios, while modulating the carbon and nitrogen isotopic signals of sinking organic matter reaching the deep ocean. Our findings highlight the role of diatom ecology in modulating silicon vs. carbon sequestration efficiency, a critical factor for determining the stoichiometric relationship of limiting nutrients in the Southern Ocean.

685. Combining Argo and Satellite Data Using Model-Derived Covariances: Blue Maps

Authors: Peter R. Oke; Matthew A. Chamberlain; Russell A. S. Fiedler; Hugo Bastos de Oliveira; Helen M. Beggs; Gary B. Brassington

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2021.696985

URL: <https://doi.org/10.3389/feart.2021.696985>

Publication Date: 2021-06-17

Volume/Issue: Volume 9

References: 75

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Blue Maps aims to exploit the versatility of an ensemble data assimilation system to deliver gridded estimates of ocean temperature, salinity, and sea-level with the accuracy of an observation-based product. Weekly maps of ocean properties are produced on a 1/10 degrees, near-global grid by combining Argo profiles and satellite observations using ensemble optimal interpolation (EnOI). EnOI is traditionally applied to ocean models for ocean forecasting or reanalysis, and usually uses an ensemble comprised of anomalies for only one spatiotemporal scale (e.g., mesoscale). Here, we implement EnOI using an ensemble that includes anomalies for multiple space- and time-scales: mesoscale, intraseasonal, seasonal, and interannual. The system produces high-quality analyses that produce mis-fits to observations that compare well to other observation-based products and ocean reanalyses. The accuracy of Blue Maps analyses is assessed by comparing background fields and analyses to observations, before and after each analysis is calculated. Blue Maps produces analyses of sea-level with accuracy of about 4 cm; and analyses of upper-ocean (deep) temperature and salinity with accuracy of about 0.45 (0.15) degrees and 0.1 (0.015) practical salinity units, respectively. We show that the system benefits from a diversity of ensemble members with multiple scales, with different types of ensemble members weighted accordingly in different dynamical regions.

686. Energetics of Eddy-Mean Flow Interactions in the Amery Ice Shelf Cavity

Authors: Yang Wu; Zhaomin Wang; Chengyan Liu; Liangjun Yan

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2021.638741

URL: <https://doi.org/10.3389/fmars.2021.638741>

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References: 106

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, China Postdoctoral Science Foundation

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Open Access: N/A

Abstract: Previous studies demonstrated that eddy processes play an important role in ice shelf basal melting and the water mass properties of ice shelf cavities. However, the eddy energy generation and dissipation mechanisms in ice shelf cavities have not been studied systematically. The dynamic processes of the ocean circulation in the Amery Ice Shelf cavity are studied quantitatively through a Lorenz energy cycle approach for the first time by using the outputs of a high-resolution coupled regional ocean-sea ice-ice shelf model. Over the entire sub-ice-shelf cavity, mean available potential energy (MAPE) is the largest energy reservoir (112 TJ), followed by the mean kinetic energy (MKE, 70 TJ) and eddy available potential energy (EAPE, 10 TJ). The eddy kinetic energy (EKE) is the smallest pool (5.5 TJ), which is roughly 8% of the MKE, indicating significantly suppressed eddy activities by the drag stresses at ice shelf base and bottom topography. The total generation rate of available potential energy is about 1.0 GW, almost all of which is generated by basal melting and seawater refreezing, i.e., the so-called "ice pump." The energy generated by ice pump is mainly dissipated by the ocean-ice shelf and ocean-bottom drag stresses, amounting to 0.3 GW and 0.2 GW, respectively. The EKE is generated through two pathways: the barotropic pathway MAPE→MKE→EKE (0.03 GW) and the baroclinic pathway MAPE→EAPE→EKE (0.2 GW). In addition to directly supplying the EAPE through baroclinic pathway (0.2 GW), MAPE also provides 0.5 GW of power to MKE to facilitate the barotropic pathway.

687. Estimating Ocean Surface Currents With Machine Learning

Authors: Anirban Sinha; Ryan Abernathey

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2021.672477

URL: <https://doi.org/10.3389/fmars.2021.672477>

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Volume/Issue: Volume 8

References: 31

Cited by: 29 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Global surface currents are usually inferred from directly observed quantities like sea-surface height, wind stress by applying diagnostic balance relations (like geostrophy and Ekman flow), which provide a good approximation of the dynamics of slow, large-scale currents at large scales and low Rossby numbers. However, newer generation satellite altimeters (like the upcoming SWOT mission) will capture more of the high wavenumber variability associated with the unbalanced components, but the low temporal sampling can potentially lead to aliasing. Applying these balances directly may lead to an incorrect un-physical estimate of the surface flow. In this study we explore Machine Learning (ML) algorithms as an alternate route to infer surface currents from satellite observable quantities. We train our ML models with SSH, SST, and wind stress from available primitive equation ocean GCM simulation outputs as the inputs and make predictions of surface currents (u,v), which are then compared against the true GCM output. As a baseline example, we demonstrate that a linear regression model is ineffective at predicting velocities accurately beyond localized regions. In comparison, a relatively simple neural network (NN) can predict surface currents accurately over most of the global ocean, with lower mean squared errors than geostrophy + Ekman. Using a local stencil of neighboring grid points as additional input features, we can train the deep learning models to effectively "learn" spatial gradients and the physics of surface currents. By passing the stenciled variables through convolutional filters we can help the model learn spatial gradients much faster. Various training strategies are explored using systematic feature hold out and multiple combinations of point and stenciled input data fed through convolutional filters (2D/3D), to understand the effect of each input feature on the NN's ability to accurately represent surface flow. A model sensitivity analysis reveals that besides SSH, geographic information in some form is an essential ingredient required for making accurate predictions of surface currents with deep learning models.

688. Argo--Two Decades: Global Oceanography, Revolutionized

Authors: Gregory C. Johnson (Pacific Marine Environmental Laboratory, National Oceanic and Atmospheric Administration, Seattle, Washington 98115, USA;); Shigeki Hosoda (Japan Agency for Marine-Earth Science and Technology, Kanagawa 237-0061, Japan;); Steven R. Jayne (Woods Hole Oceanographic Institution); Peter R. Oke (Division of Marine and Atmospheric Research, Commonwealth Scientific and Industrial Research Organisation, Hobart, Tasmania 7001, Australia;); Stephen C. Riser (University of Washington); Dean Roemmich (Scripps Institution of Oceanography / University of California San Diego); Tohsio Suga (Tohoku University); Virginie Thierry (Institut de Recherche pour le Développement (IRD)); Susan E. Wijffels (Woods Hole Oceanographic Institution); Jianping Xu (Second Institute of Oceanography, Ministry of Natural Resources)

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

DOI: 10.1146/annurev-marine-022521-102008

URL: <https://doi.org/10.1146/annurev-marine-022521-102008>

Publication Date: 2021-06-08

Volume/Issue: Volume 14, Issue 1

References: 151

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Argo, an international, global observational array of nearly 4,000 autonomous robotic profiling floats, each measuring ocean temperature and salinity from 0 to 2,000 m on nominal 10-day cycles, has revolutionized physical oceanography. Argo started at the turn of the millennium, growing out of advances in float technology over the previous several decades. After two decades, with well over 2 million profiles made publicly available in real time, Argo data have underpinned more than 4,000 scientific publications and improved countless nowcasts, forecasts, and projections. We review a small subset of those accomplishments, such as elucidating remarkable zonal jets spanning the deep tropical Pacific; increasing understanding of ocean eddies and the roles of mixing in shaping water masses and circulation; illuminating interannual to decadal ocean variability; quantifying, in concert with satellite data, contributions of ocean warming and ice melting to sea level rise; improving coupled numerical weather predictions; and underpinning decadal climate forecasts.

689. Evaluating the physical and biogeochemical state of the global ocean component of UKESM1 in CMIP6 historical simulations

Authors: Andrew Yool; Julien Palmiéri; Colin G. Jones; Lee de Mora; Till Kuhlbrodt; Ekatarina E. Popova; A. J. George Nurser; Joel Hirschi; Adam T. Blaker; Andrew C. Coward; Edward W. Blockley; Alistair A. Sellar

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-3437-2021

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References: 142

Cited by: 26 publications

Language: en

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Keywords: N/A

Funding: Natural Environment Research Council, European Commission

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Open Access: N/A

Abstract: . The ocean plays a key role in modulating the climate of the Earth system (ES). At the present time it is also a major sink both for the carbon dioxide (CO₂) released by human activities and for the excess heat driven by the resulting atmospheric greenhouse effect. Understanding the ocean's role in these processes is critical for model projections of future change and its potential impacts on human societies. A necessary first step in assessing the credibility of such future projections is an evaluation of their performance against the present state of the ocean. Here we use a range of observational fields to validate the physical and biogeochemical performance of the ocean component of UKESM1, a new Earth system model (ESM) for CMIP6 built upon the HadGEM3-GC3.1 physical climate model. Analysis focuses on the realism of the ocean's physical state and circulation, its key elemental cycles, and its marine productivity. UKESM1 generally performs well across a broad spectrum of properties, but it exhibits a number of notable biases. Physically, these include a global warm bias inherited from model spin-up, excess northern sea ice but insufficient southern sea ice and sluggish interior circulation. Biogeochemical biases found include shallow remineralization of sinking organic matter, excessive iron stress in regions such as the equatorial Pacific, and generally lower surface alkalinity that results in decreased surface and interior dissolved inorganic carbon (DIC) concentrations. The mechanisms driving these biases are explored to identify consequences for the behaviour of UKESM1 under future climate change scenarios and avenues for model improvement. Finally, across key biogeochemical properties, UKESM1 improves in performance relative to its CMIP5 precursor and performs well alongside its fellow members of the CMIP6 ensemble.

690. Distinct sources of interannual subtropical and subpolar Atlantic overturning variability

Authors: Yavor Kostov; Helen L. Johnson; David P. Marshall; Patrick Heimbach; Gael Forget; N. Penny Holliday; M. Susan Lozier; Feili Li; Helen R. Pillar; Timothy Smith

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-021-00759-4

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Cited by: 26 publications

Language: en

Subjects: N/A

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Funding: RCUK | Natural Environment Research Council, United States Department of Commerce | National Oceanic and Atmospheric Administration, National Science Foundation (and 4 more)

License: N/A

Open Access: N/A

Abstract: N/A

691. Water-Mass Properties and Circulation in the Deep and Abyssal Philippine Sea

Authors: Zichen Tian (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Chun Zhou (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Xin Xiao (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Tinghao Wang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Tangdong Qu (University of California Los Angeles); Qingxuan Yang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Wei Zhao (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Jiwei Tian (Ocean University of China, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016994

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Publication Date: 2021-05-31

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References: 71

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: The National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Water-mass properties and circulation in the deep and abyssal Philippine Sea are investigated using repeatedly occupied full-depth hydrographic data. The abyssal water-mass, the Lower Circumpolar Deep Water (LCDW), is captured. After crossing the Yap Mariana Junction, the LCDW spreads to the Parece Vela Basin, the southern part of the Shikoku Basin, the Minami-Daito Basin, and the northern Philippine Basin, with its upper interface sinking from ~4,000 m to ~5,000 m. Blocked by the Central Basin Fault, the LCDW cannot penetrate further southward into the southern Philippine Basin. An abyssal current across the Kyushu-Palau Ridge connecting the eastern and western parts of the Philippine Sea is identified at 22.5 degreesN-29.5 degreesN, which is suggested to be the main branch transporting LCDW from the Yap Mariana Junction westward into the northern Philippine Basin. Synoptic analysis of hydrographic observations suggests the existence of a cyclonic circulation in the abyssal Philippine Sea. During 1990-2020, a warming trend of $0.33 \pm 0.19 \times 10^{-3}$ degreesC yr⁻¹ is identified at depths below 2,000 m, while a SiO₂ maximum core at 30 degreesN implies a possible cooling of the Mixed Deep Water (more influenced by North Pacific Deep Water) in the Philippine Sea. Under 4,000 m, freshening together with warming south of 20 degreesN indicate contraction of LCDW after entering the Yap Mariana Junction.

692. The effects of historical ozone changes on Southern Ocean heat uptake and storage

Authors: Shouwei Li; Wei Liu; Kewei Lyu; Xuebin Zhang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-05803-y

URL: <https://doi.org/10.1007/s00382-021-05803-y>

Publication Date: 2021-05-22

Volume/Issue: Volume 57, Issue 7-8

References: 121

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Alfred P. Sloan Foundation, Commonwealth Scientific and Industrial Research Organisation

License: N/A

Open Access: N/A

Abstract: N/A

693. Coupling of the mid-depth and abyssal components of the global overturning circulation according to a state estimate

Authors: Louise Rousselet (Scripps Institution of Oceanography / University of California San Diego); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego); Gael Forget (Massachusetts Institute of Technology)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.abf5478

URL: <https://doi.org/10.1126/sciadv.abf5478>

Publication Date: 2021-05-21

Volume/Issue: Volume 7, Issue 21

References: 74

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation, National Aeronautic Space Administration (and 2 more)

License: CC BY-NC 4.0

Open Access: N/A

Abstract: The global routes of mid-depth overturning circulation are mapped with Lagrangian analysis using velocities from a state estimate.

694. How Salty Is the Global Ocean: Weighing It All or Tasting It a Sip at a Time?

Authors: R. M. Ponte (Atmospheric and Environmental Research (AER) Lexington MA USA); Q. Sun (Atmospheric and Environmental Research (AER) Lexington MA USA); C. Liu (University of Delaware); X. Liang (University of Delaware)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl092935

URL: <https://doi.org/10.1029/2021gl092935>

Publication Date: 2021-05-21

Volume/Issue: Volume 48, Issue 11

References: 25

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Earth Sciences Division, National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Global ocean mean salinity is a key indicator of the Earth's hydrological cycle and the exchanges of freshwater between land and ocean, but its determination remains a challenge. Aside from traditional methods based on gridded salinity fields derived from in situ measurements, we explore estimates of based on liquid freshwater changes derived from space gravimetry data corrected for sea ice effects. For the 2005-2019 period analyzed, the different series show little consistency in seasonal, interannual, and long-term variability. In situ estimates show sensitivity to choice of product and unrealistic variations. A suspiciously large rise in since ~2015 is enough to measurably affect halosteric sea level estimates and can explain recent discrepancies in the global mean sea level budget. Gravimetry-based estimates are more realistic, inherently consistent with estimated freshwater contributions to global mean sea level, and provide a way to calibrate the in situ estimates.

695. The Impact of Southern Ocean Ekman Pumping, Heat and Freshwater Flux Variability on Intermediate and Mode Water Export in CMIP Models: Present and Future Scenarios

Authors: Lucas Almeida (Laboratório de Estudos dos Oceanos e Clima (LEOC) Instituto de Oceanografia Universidade Federal do Rio Grande (FURG) Rio Grande RS Brazil); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Mauricio M. Mata (Laboratório de Estudos dos Oceanos e Clima (LEOC) Instituto de Oceanografia Universidade Federal do Rio Grande (FURG) Rio Grande RS Brazil)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017173

URL: <https://doi.org/10.1029/2021jc017173>

Publication Date: 2021-05-17

Volume/Issue: Volume 126, Issue 6

References: 61

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, climate, climate models, physical ocean, water masses (and 1 more)

Funding: Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Conselho Nacional de Desenvolvimento Científico e Tecnológico, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: Subduction in the Antarctic circumpolar region of the Southern Ocean (SO) results in the formation of Antarctic Intermediate Water (AAIW) and Subantarctic Mode Water (SAMW). Theoretical understanding predicts that subduction rates of these waters masses is driven by wind stress curl and buoyancy fluxes. The objective of this work is to evaluate the extent to which AAIW and SAMW variability are correlated to SO air-sea fluxes and how potential changes to those forcings would impact the future water mass export rates. We correlate the water mass volume transport at 30 degreesS with Ekman pumping, freshwater and heat fluxes in the Coupled Model Intercomparison Project. The export of these water masses varies across models, with most overestimating the total transport. Correlation coefficients between the air-sea fluxes and exports are consistent with theoretical expectations. In the picontrol/historical scenarios, the highest correlations with AAIW export variability are heat flux, while Ekman pumping best explains SAMW. However, multivariate regressions show that both AAIW and SAMW export variability are better explained using the combination of all three fluxes. In future scenario simulations air-sea fluxes trend significantly in the catastrophic scenario (RCP8.5 and SSP8.5). Both AAIW and SAMW are still highly correlated to the fluxes, but with different correlation coefficients. The dominant forcing components even change from the present simulations to the future scenario runs. Thus, correlations between AAIW and SAMW transports and air-sea fluxes are not stationary in time, limiting the predictive skill of statistical models and highlighting the importance of using complex climate models.

696. A multi-model architecture based on Long Short-Term Memory neural networks for multi-step sea level forecasting

Authors: Gabriele Accarino; Marco Chiarelli; Sandro Fiore; Ivan Federico; Salvatore Causio; Giovanni Coppini; Giovanni Aloisio

Journal: Future Generation Computer Systems

Publisher: Elsevier BV

DOI: 10.1016/j.future.2021.05.008

URL: <https://doi.org/10.1016/j.future.2021.05.008>

Publication Date: 2021-05-11

Volume/Issue: Volume 124

References: 51

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

697. State Estimates and Forecasts of the Northern Philippine Sea Circulation including Ocean Acoustic Travel Times

Authors: No authors listed

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-20-0178.1

URL: <https://doi.org/10.1175/jtech-d-20-0178.1>

Publication Date: 2021-05-07

Volume/Issue: Volume 38, Issue 11

References: 136

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: Data assimilation, Ocean models, Mesoscale models, Optimization

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The 2010-11 North Pacific Acoustic Laboratory (NPAL) Philippine Sea experiment measured travel times between six acoustic transceiver moorings in a 660-km diameter ocean acoustic tomography array in the northern Philippine Sea (NPS). The travel-time series compare favorably with travel times computed for a yearlong series of state estimates produced for this region using the Massachusetts Institute of Technology General Circulation Model-Estimating the Circulation and Climate of the Ocean four-dimensional variational (MITgcm-ECCO 4DVAR) assimilation system constrained by satellite sea surface height and sea surface temperature observations and by Argo temperature and salinity profiles. Fluctuations in the computed travel times largely match the fluctuations in the measurements caused by the intense mesoscale eddy field in the NPS, providing a powerful test of the observations and state estimates. The computed travel times tend to be shorter than the measured travel times, however, reflecting a warm bias in the state estimates. After processing the travel times to remove tidal signals and extract the low-frequency variability, the differences between the measured and computed travel times were used in addition to SSH, SST, and Argo temperature and salinity observations to further constrain the model and generate improved state estimates. The assimilation of the travel times reduced the misfit between the measured and computed travel times, while not increasing the misfits with the other assimilated observations. The state estimates that used the travel times are more consistent with temperature measurements from an independent oceanographic mooring than the state estimates that did not incorporate the travel times.

698. Variability of the North Atlantic Thermohaline Circulation in Different Phases of the Atlantic Multidecadal Oscillation from Ocean Objective Analyses and Reanalyses

Authors: V. A. Bagatinsky; N. A. Diansky

Journal: Izvestiya, Atmospheric and Oceanic Physics

Publisher: Pleiades Publishing Ltd

DOI: 10.1134/s000143382102002x

URL: <https://doi.org/10.1134/s000143382102002x>

Publication Date: 2021-05-04

Volume/Issue: Volume 57, Issue 2

References: 40

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

699. Linking global terrestrial CO₂ fluxes and environmental drivers: inferences from the Orbiting Carbon Observatory 2 satellite and terrestrial biospheric models

Authors: Zichong Chen; Junjie Liu; Daven K. Henze; Deborah N. Huntzinger; Kelley C. Wells; Stephen Sitch; Pierre Friedlingstein; Emilie Joetzjer; Vladislav Bastrikov; Daniel S. Goll; Vanessa Haverd; Atul K. Jain; Etsushi Kato; Sebastian Lienert; Danica L. Lombardozzi; Patrick C. McGuire; Joe R. Melton; Julia E. M. S. Nabel; Benjamin Poulter; Hanqin Tian; Andrew J. Wiltshire; Sönke Zaehle; Scot M. Miller

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-21-6663-2021

URL: <https://doi.org/10.5194/acp-21-6663-2021>

Publication Date: 2021-05-04

Volume/Issue: Volume 21, Issue 9

References: 80

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Observations from the Orbiting Carbon Observatory 2 (OCO-2) satellite have been used to estimate CO₂ fluxes in many regions of the globe and provide new insight into the global carbon cycle. The objective of this study is to infer the relationships between patterns in OCO-2 observations and environmental drivers (e.g., temperature, precipitation) and therefore inform a process understanding of carbon fluxes using OCO-2. We use a multiple regression and inverse model, and the regression coefficients quantify the relationships between observations from OCO-2 and environmental driver datasets within individual years for 2015-2018 and within seven global biomes. We subsequently compare these inferences to the relationships estimated from 15 terrestrial biosphere models (TBMs) that participated in the TRENDY model inter-comparison. Using OCO-2, we are able to quantify only a limited number of relationships between patterns in atmospheric CO₂ observations and patterns in environmental driver datasets (i.e., 10 out of the 42 relationships examined). We further find that the ensemble of TBMs exhibits a large spread in the relationships with these key environmental driver datasets. The largest uncertainty in the models is in the relationship with precipitation, particularly in the tropics, with smaller uncertainties for temperature and photosynthetically active radiation (PAR). Using observations from OCO-2, we find that precipitation is associated with increased CO₂ uptake in all tropical biomes, a result that agrees with half of the TBMs. By contrast, the relationships that we infer from OCO-2 for temperature and PAR are similar to the ensemble mean of the TBMs, though the results differ from many individual TBMs. These results point to the limitations of current space-based observations for inferring environmental relationships but also indicate the potential to help inform key relationships that are very uncertain in state-of-the-art TBMs.

700. Interannual Variability of Air-Sea Exchange of Mercury in the Global Ocean: The "Seesaw Effect" in the Equatorial Pacific and Contributions to the Atmosphere

Authors: Shaojian Huang (Nanjing University); Yanxu Zhang (Nanjing University)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

DOI: 10.1021/acs.est.1c00691

URL: <https://doi.org/10.1021/acs.est.1c00691>

Publication Date: 2021-04-30

Volume/Issue: Volume 55, Issue 10

References: 99

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Education of the People's Republic of China, Jiangsu Province Collaborative Innovation Center of Climate Change, Jiangsu Innovative and Entrepreneurial Talents Plan

License: N/A

Open Access: N/A

Abstract: N/A

701. Origin and Fate of the Chukchi Slope Current Using a Numerical Model and In-Situ Data

Authors: Hengling Leng (Hohai University, Ministry of Natural Resources, Woods Hole Oceanographic Institution); Michael A. Spall (Woods Hole Oceanographic Institution); Robert S. Pickart (Woods Hole Oceanographic Institution); Peigen Lin (Woods Hole Oceanographic Institution); Xuezhi Bai (Hohai University, Ministry of Natural Resources)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017291

URL: <https://doi.org/10.1029/2021jc017291>

Publication Date: 2021-04-29

Volume/Issue: Volume 126, Issue 5

References: 62

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China Stem Cell and Translational Research, National Natural Science Foundation of China, National Science Foundation

License: N/A

Open Access: N/A

Abstract: A regional coupled sea ice-ocean model and mooring/shipboard measurements are used to investigate the origins, seasonality, and downstream fate of the Chukchi Slope Current (CSC). Three years (2013-2015) of model integration indicates that, in the mean, the model slope current transports ~0.45 Sv of Pacific water northwestward along the Chukchi continental slope. Only 62% of this water emanates from Barrow Canyon, while the rest (38%) is fed by a westward jet extending from the southern Beaufort Sea. The jet merges with the outflow from the canyon, forming the CSC. Due to these two distinct origins, the slope current in the model has a double velocity core at times. This is consistent with the double-core structure of the slope current seen in ship-based observations. Seasonal changes in the volume, heat, and freshwater transports by the slope current appear to be related to the changes in the upstream flows. A tracer diagnostic in the model suggests that the part of the slope current over the upper continental slope continues westward toward the East Siberian Sea, while the portion of the current overlying deeper isobaths flows northward into the Chukchi Borderland, where it ultimately gets entrained into the Beaufort Gyre. Our study provides a detailed and complete picture of the slope current.

702. Assessing the current and future Arctic Ocean observing system with observing system simulating experiments

Authors: Guokun Lyu (Shanghai Jiao Tong University, University of Hamburg); Armin Koehl (University of Hamburg); Nuno Serra (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.4044

URL: <https://doi.org/10.1002/qj.4044>

Publication Date: 2021-04-29

Volume/Issue: Volume 147, Issue 738

References: 68

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission, National Natural Science Foundation of China

License: Creative Commons

Open Access: N/A

Abstract: Arctic Ocean Observing System Simulation Experiments (OSSEs) were performed with a pan-Arctic coupled ocean-sea ice data assimilation system to assess the impacts of assimilating available observations on the Arctic ocean-sea ice state. To this end, the adjoint method with a 3-year assimilation window was used to assimilate synthetic observations sampled from a 4 km model simulation at the spatio-temporal distribution of the existing observing system. After data assimilation, the sea ice state, including sea ice concentration (SIC), sea ice thickness (SIT), and sea ice volume (SIV), were significantly improved, benefiting from the high spatio-temporal coverage of SIC observations and the wintertime SIT observations. In contrast, the ocean state is not very well constrained with the existing hydrographic observing system. An additional 1 degrees x 1 degrees ocean profiling array with a 10-day sampling frequency was seen to substantially improve the estimated ocean temperature and freshwater content. Data from additional moorings deployed in the Fram Strait and continental slope of the Laptev Sea could also improve the pathway of Atlantic inflow to the Arctic Ocean and the temperature of the Atlantic inflow but degrade the mean volume transport through the Fram Strait. Moreover, estimated exchanges between the Arctic Ocean and the Atlantic Ocean through the Fram Strait, Davis Strait and the St. Anna Trough were found to benefit from the high-density profiling array.

703. Oceans

Authors: Guy Jacques; Paul Tréguer; Herlé Mercier

Journal: Unknown Journal

Publisher: Wiley

DOI: 10.1002/9781119818038

URL: <https://doi.org/10.1002/9781119818038>

Publication Date: 2021-04-29

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

704. Numerical study of the seasonal salinity budget of the upper ocean in the Bay of Bengal in 2014

Authors: Qingyue Wang; Changming Dong; Junde Li; Jingsong Yang; Qing Tian

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-020-0285-1

URL: <https://doi.org/10.1007/s00343-020-0285-1>

Publication Date: 2021-04-28

Volume/Issue: Volume 39, Issue 4

References: 54

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

705. Strong Solvability of a Variational Data Assimilation Problem for the Primitive Equations of Large-Scale Atmosphere and Ocean Dynamics

Authors: Peter Korn

Journal: Journal of Nonlinear Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00332-021-09707-3

URL: <https://doi.org/10.1007/s00332-021-09707-3>

Publication Date: 2021-04-26

Volume/Issue: Volume 31, Issue 3

References: 46

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Max Planck Institute for Meteorology

License: CC BY 4.0

Open Access: N/A

Abstract: For the primitive equations of large-scale atmosphere and ocean dynamics, we study the problem of determining by means of a variational data assimilation algorithm initial conditions that generate strong solutions which minimize the distance to a given set of time-distributed observations. We suggest a modification of the adjoint algorithm whose novel elements is to use norms in the variational cost functional that reflects the H^1 -regularity of strong solutions of the primitive equations. For such a cost functional, we prove the existence of minima and a first-order adjoint condition for strong solutions that provides the basis for computing these minima. We prove the local convergence of a gradient-based descent algorithm to optimal initial conditions using the second-order adjoint primitive equations. The algorithmic modifications due to the H^1 -norms are straightforwardly to implement into a variational algorithm that employs the standard L^2 -metrics.

706. IndividualDisplacements.jl: a Julia package to simulate and study particle displacements within the climate system

Authors: Gael Forget

Journal: Journal of Open Source Software

Publisher: The Open Journal

DOI: 10.21105/joss.02813

URL: <https://doi.org/10.21105/joss.02813>

Publication Date: 2021-04-26

Volume/Issue: Volume 6, Issue 60

References: 16

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Software package|R package|Particle system|Particle (ecology)

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

707. Cross-Equatorial Anti-Symmetry in the Seasonal Transport of the Western Boundary Current in the Atlantic Ocean

Authors: Yujia Zhai (Ocean University of China, Woods Hole Oceanographic Institution); Jiayan Yang (Woods Hole Oceanographic Institution); Xiuquan Wan (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017184

URL: <https://doi.org/10.1029/2021jc017184>

Publication Date: 2021-04-23

Volume/Issue: Volume 126, Issue 5

References: 32

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Ocean University of China, China Scholarship Council (and 2 more)

License: N/A

Open Access: N/A

Abstract: The western boundary current in the equatorial Atlantic Ocean is a main conduit for water-mass exchanges across the equator and thus a major pathway for the interhemispheric transports in the Atlantic Meridional Overturning Circulation (AMOC) system. In this study we quantify and examine the mean and seasonal variability of the equatorial western boundary current (EWBC) in the upper ocean layer using two data-assimilated products, the Estimating the Circulation and Climate of the Ocean (ECCO4r3) and the Simple Ocean Data Assimilation (SODA3). It is found that the EWBC between 10 degreesS and 10 degreesN exhibits two pronounced features in its seasonal variability: (1) the transport varies anti-symmetrically across the equator, that is, the northward EWBC strengthens to the north of the equator when it weakens to the south of the equator, and vice versa; and (2) the amplitude of seasonal variations is much greater in the northern hemisphere than in the south. We hypothesize that the cross-equatorial anti-symmetry in EWBC transport variability is attributable to the impingement of equatorial Rossby waves at the western boundary and the shape of the western boundary is the main cause for the amplified seasonal variability in the northern hemisphere. A simple 1 and 1/2-layer model is used to test and validate this hypothesis and to elucidate the role of wind forcing and topography plays in the seasonal variability in the EWBC transport.

708. What Role Does the Barrier Layer Play During Extreme El Niño Events?

Authors: Xiaolin Zhang (University of Hamburg); Janet Sprintall (Scripps Institution of Oceanography); Lili Zeng (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc017001

URL: <https://doi.org/10.1029/2020jc017001>

Publication Date: 2021-04-23

Volume/Issue: Volume 126, Issue 5

References: 38

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: extreme El Niño, salinity budget analysis, zonal current, barrier layer thickness, heat flux

Funding: Academic Medical Organization of Southwestern Ontario, National Aeronautics and Space Administration, National Oceanic and Atmospheric Administration (and 1 more)

License: Creative Commons

Open Access: N/A

Abstract: Intensive air-sea interaction and the formation of the salinity barrier layer (BL) in the Pacific has fundamental importance to the El Niño evolution. The structure and formation of the BL in the equatorial Pacific Ocean during moderate and extreme El Niños over the past 30 years are investigated using in situ temperature and salinity data measured by the TAO/TRITON array and the data-assimilating ECCO2 product. In the western and central Pacific Ocean, the BL is thicker during moderate El Niños compared to extreme El Niños due to a deeper isothermal layer depth compared to the density defined mixed layer depth. Moreover, in the western and central Pacific Ocean, the anomalous zonal eastward current related to the westerly wind event that initiates El Niños is found to be stronger during extreme El Niños, advecting the thicker BLs from west to east. A salinity budget suggests that during both moderate and extreme El Niño events, surface freshwater flux dominates at the equator. During extreme El Niños, the change in the freshwater flux drives a strong surface jet in the far western Pacific at 1 degreesS, 156 degreesE. North of the equator, the surface freshwater flux largely dampens this advective impact. Thus during the different El Niño strengths, the BL distribution, evolution and impact are also different. This suggests that climate models need to better distinguish different types of El Niño events in order to simulate the ENSO dynamics correctly.

709. The transferability of adjoint inversion products between different ice flow models

Authors: Jowan M. Barnes; Thiago Dias dos Santos; Daniel Goldberg; G. Hilmar Gudmundsson; Mathieu Morlighem; Jan De Rydt

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-1975-2021

URL: <https://doi.org/10.5194/tc-15-1975-2021>

Publication Date: 2021-04-23

Volume/Issue: Volume 15, Issue 4

References: 64

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: . Among the most important challenges faced by ice flow models is how to represent basal and rheological conditions, which are challenging to obtain from direct observations. A common practice is to use numerical inversions to calculate estimates for the unknown properties, but there are many possible methods and not one standardised approach. As such, every ice flow model has a unique initialisation procedure. Here we compare the outputs of inversions from three different ice flow models, each employing a variant of adjoint-based optimisation to calculate basal sliding coefficients and flow rate factors using the same observed surface velocities and ice thickness distribution. The region we focus on is the Amundsen Sea Embayment in West Antarctica, the subject of much investigation due to rapid changes in the area over recent decades. We find that our inversions produce similar distributions of basal sliding across all models, despite using different techniques, implying that the methods used are highly robust and represent the physical equations without much influence by individual model behaviours. Transferring the products of inversions between models results in time-dependent simulations displaying variability on the order of or lower than existing model intercomparisons. Focusing on contributions to sea level, the highest variability we find in simulations run in the same model with different inversion products is 32 %, over a 40-year period, a difference of 3.67 mm. There is potential for this to be improved with further standardisation of modelling processes, and the lowest variability within a single model is 13 %, or 1.82 mm over 40 years. While the successful transfer of inversion outputs from one model to another requires some extra effort and technical knowledge of the particular models involved, it is certainly possible and could indeed be useful for future intercomparison projects.

710. Five years of variability in the global carbon cycle: comparing an estimate from the Orbiting Carbon Observatory-2 and process-based models

Authors: Zichong Chen; Deborah N Huntzinger; Junjie Liu; Shilong Piao; Xuhui Wang; Stephen Sitch; Pierre Friedlingstein; Peter Anthoni; Almut Arneth; Vladislav Bastrikov; Daniel S Goll; Vanessa Haverd; Atul K Jain; Emilie Joetzjer; Etsushi Kato; Sebastian Lienert; Danica L Lombardozzi; Patrick C McGuire; Joe R Melton; Julia E M S Nabel; Julia Pongratz; Benjamin Poulter; Hanqin Tian; Andrew J Wiltshire; Sönke Zaehle; Scot M Miller

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/abfac1

URL: <https://doi.org/10.1088/1748-9326/abfac1>

Publication Date: 2021-04-22

Volume/Issue: Volume 16, Issue 5

References: 81

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Year-to-year variability in CO₂ fluxes can yield insight into climate-carbon cycle relationships, a fundamental yet uncertain aspect of the terrestrial carbon cycle. In this study, we use global observations from NASA's Orbiting Carbon Observatory-2 (OCO-2) satellite for years 2015-2019 and a geostatistical inverse model to evaluate 5 years of interannual variability (IAV) in CO₂ fluxes and its relationships with environmental drivers. OCO-2 launched in late 2014, and we specifically evaluate IAV during the time period when OCO-2 observations are available. We then compare inferences from OCO-2 with state-of-the-art process-based models (terrestrial biosphere model, TBMs). Results from OCO-2 suggest that the tropical grasslands biome (including grasslands, savanna, and agricultural lands within the tropics) makes contributions to global IAV during the 5 year study period that are comparable to tropical forests, a result that differs from a majority of TBMs. Furthermore, existing studies disagree on the environmental variables that drive IAV during this time period, and the analysis using OCO-2 suggests that both temperature and precipitation make comparable contributions. TBMs, by contrast, tend to estimate larger IAV during this time and usually estimate larger relative contributions from the extra-tropics. With that said, TBMs show little consensus on both the magnitude and the contributions of different regions to IAV. We further find that TBMs show a wide range of responses on the relationships of CO₂ fluxes with annual anomalies in temperature and precipitation, and these relationships across most of the TBMs have a larger magnitude than inferred from OCO-2. Overall, the findings of this study highlight large uncertainties in process-based estimates of IAV during recent years and provide an avenue for evaluating these processes against inferences from OCO-2.

711. Assessing Global Present-Day Surface Mass Transport and Glacial Isostatic Adjustment From Inversion of Geodetic Observations

Authors: Yan Jiang (Natural Resources Canada Geological Survey of Canada Sidney Canada, School of Earth and Ocean Science University of Victoria Victoria Canada); Xiaoping Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michiel R. van den Broeke (Utrecht University); Peter Kuipers Munneke (Utrecht University); Sebastian B. Simonsen (Technical University of Denmark); Wouter van der Wal (Delft University of Technology); Bert L. Vermeersen (Delft University of Technology, Royal Netherlands Institute for Sea Research)

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Keywords: N/A

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Abstract: Long-term monitoring of global mass transport within the Earth system improves our ability to mitigate natural hazards and better understand their relations to climate change. Satellite gravity is widely used to monitor surface mass variations for its unprecedented spatial and temporal coverage. However, the gravity data contain signals from visco-elastic deformation in response to past ice sheet melting, preventing us from extracting signals of present-day surface mass trend (PDMT) directly. Here we present a global inversion scheme that separates PDMT and visco-elastic glacial isostatic adjustment (GIA) signatures by combining satellite gravimetry with satellite altimetry and ground observations. Our inversion provides global dual data coverage that enables a robust separation of PDMT and GIA spherical harmonic coefficients. It has the advantage of providing estimates of Earth's long wavelength deformation signatures and their uncertainties. Our GIA result, along with its uncertainty estimates, can be used in future GRACE processing to better assess the impact of GIA on surface mass change. Our GIA estimates include a rapid GIA uplift in the Southeast Alaska and the Amundsen Sea Embayment, due to the visco-elastic response to recent glacial unloading. We estimate the average surface mass change rate from 2002-2010 to be $-203 \pm 3 \text{ GT} \cdot \text{a}^{-1}$ in Greenland, $-126 \pm 18 \text{ GT} \cdot \text{a}^{-1}$ in Antarctica and, $-62 \pm 5 \text{ GT} \cdot \text{a}^{-1}$ in Alaska. The GIA low degree spherical harmonic coefficients are sensitive to rheological properties in Earth's deep interior. Our low-degree GIA estimates include geocenter motion and which provide unique constraints to understand Earth's lower mantle and ice history.

712. Counteracting Contributions of the Upper and Lower Meridional Overturning Limbs to the North Atlantic Nutrient Budgets: Enhanced Imbalance in 2010

Authors: L. I. Carracedo (Institut de Recherche pour le Développement (IRD), National Oceanography Center (NOC) Southampton UK); H. Mercier (Institut de Recherche pour le Développement (IRD)); E. McDonagh (NORCE Norwegian Research Centre, National Oceanography Center (NOC) Southampton UK, Bjerknes Centre for Climate Research); G. Rosón (Faculty of Marine Sciences University of Vigo Campus Lagoas-Marcosende Vigo Spain); R. Sanders (NORCE Norwegian Research Centre, National Oceanography Center (NOC) Southampton UK, Bjerknes Centre for Climate Research); C. M. Moore (University of Southampton, National Oceanography Centre); S. Torres-Valdés (Alfred Wegener Institute); P. Brown (National Oceanography Center (NOC) Southampton UK); P. Lherminier (Institut de Recherche pour le Développement (IRD)); F. F. Pérez (Instituto de Investigaciones Marinas CSIC Vigo Spain)

Journal: Global Biogeochemical Cycles

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Subjects: N/A

Keywords: Nutrient budgets, North Atlantic, Carbon cycle, Nutrient and oxygen transports, Nutrient cycles (and 1 more)

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Open Access: N/A

Abstract: The North Atlantic Basin is a major sink for atmospheric carbon dioxide (CO₂) due in part to the extensive plankton blooms which form there supported by nutrients supplied by the three-dimensional ocean circulation. Hence, changes in ocean circulation and/or stratification may influence primary production and biological carbon export. In this study, we assess this possibility by evaluating inorganic nutrient budgets for 2004 and 2010 in the North Atlantic based on observations from the transatlantic A05-24.5 degreesN and the Greenland-Portugal OVIDE hydrographic sections, to which we applied a box inverse model to solve the circulation and estimate the across-section nutrient transports. Full water column nutrient budgets were split into upper and lower meridional overturning circulation (MOC) limbs. According to our results, anomalous circulation in early 2010, linked to extreme negative NAO conditions, led to an enhanced northward advection of more nutrient-rich waters by the upper overturning limb, which resulted in a significant nitrate and phosphate convergence north of 24.5 degreesN. Combined with heaving of the isopycnals, this anomalous circulation event in 2010 favored an enhancement of the nutrient consumption ($5.7 \pm 4.1 \text{ kmol-P s}^{-1}$) and associated biological CO₂ uptake ($0.25 \pm 0.18 \text{ Pg-C yr}^{-1}$, upper-bound estimate), which represents a 50% of the mean

annual sea-air CO₂ flux in the region. Our results also suggest a transient state of deep silicate divergence in both years. Both results are indicative of a MOC-driven modulation of the biological carbon uptake (by the upper MOC limb) and nutrient inventories (by the lower MOC limb) in the North Atlantic.

713. The observed North Equatorial Counter Current in the far western Pacific Ocean during the 2014-2016 El Niño

Authors: Hui Zhou (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Hengchang Liu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Shuwen Tan (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Wenlong Yang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Yao Li (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Xueqi Liu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Qiang Ren (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); William K. Dewar (Florida State University, Institut de Recherche pour le Développement (IRD))

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: Argo|Barotropic fluid|Boundary current

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Open Access: N/A

Abstract: The structure and variations of the North Equatorial Counter Current (NECC) in the far western Pacific Ocean during 2014-2016 are investigated using repeated in-situ hydrographic data, altimeter data, Argo data, and reanalysis data. The NECC shifted ~1 degree southward and intensified significantly with its transport exceeding 40 Sv (1 Sv = $10^6 \text{ m}^3 \text{ s}^{-1}$), nearly double its climatology value, during the developing phase of the 2015/16 El Niño event. Observations show that the 2015/16 El Niño exerted a comparable impact on the NECC with that of the extreme 1997/98 El Niño in the far western Pacific Ocean. Baroclinic instability provided the primary energy source for the eddy kinetic energy (EKE) in the 2015/16 El Niño, which differs from the traditional understanding of the energy source of EKE as barotropic instability in low latitude ocean. The enhanced vertical shear and the reduced density jump between the NECC layer and the subsurface North Equatorial Subsurface Current (NESC) layer renders the NECC-NESC system baroclinically unstable in the western Pacific Ocean during El Niño developing phase. The eddy-mean flow interactions here are diverse associated with various states of the El Niño Southern Oscillation (ENSO).

714. A Strong Sub-Thermocline Intrusion of the North Equatorial Subsurface Current Into the Makassar Strait in 2016-2017

Authors: Mingting Li (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Dongliang Yuan (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Arnold L. Gordon (Columbia University); Laura K. Gruenbug (Columbia University); Xiang Li (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Rui Li (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Xueli Yin (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Ya Yang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Corry Corvianatie (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Jun Wei (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Song Yang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China

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Open Access: N/A

Abstract: The poorly resolved pathway of the sub-thermocline waters (>300 m) within the Makassar Strait, the primary inflow path of the Indonesian Throughflow, is investigated using in situ mooring measurements and Argo profiles at the entrance to the Indonesian Seas. We focus on the strong sub-thermocline intrusion in the summer of 2016, when significant changes of sub-thermocline transport thrice as large as the interannual standard deviation occurred. Analysis suggests that the intrusion was drawn from the North Equatorial Subsurface Current (NESC) flowing westward below the North Equatorial Countercurrent, which was composed of a mixture of intermediate-depth waters from both the North and South Pacific. The anomalously strong NESC to the Makassar sub-thermocline in the summer 2016 is suggested to be in response to the 2015/2016 extreme El Niño event, forced by the trade wind anomalies over the western-central Pacific Ocean through westward and downward propagating baroclinic Rossby waves.

715. Buoyancy and Wind Driven Changes in Subantarctic Mode Water During 2004-2019

Authors: Lixiao Xu (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Yang Ding (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 18 publications

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Keywords: N/A

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Open Access: N/A

Abstract: The Subantarctic Mode Water (SAMW) is a thick low-stratified subsurface water mass located north of the Antarctic Circumpolar Current, and it is important for heat uptake. How the SAMW has changed and what drives that change remain to be quantified. By synthesizing observations, eddy-resolving ocean model hindcasts and climate model simulations, we show that the SAMW has become warmer, fresher, lighter, and weaker in recent decades. As revealed by partially coupled model experiments, the SAMW core has remained at a constant depth since 2004 due to a balance between the enhanced wind stress curl, which deepens the SAMW core, and the buoyancy gain, which shallows the core. The lightening and weakening of the SAMW mainly result from the increased surface buoyancy gain due to climate warming, while the enhanced westerlies play a secondary role in slowing this trend. As greenhouse warming continues, the increased stratification will keep weakening the SAMW.

716. The Arctic Subpolar Gyre sTate Estimate: Description and Assessment of a Data-Constrained, Dynamically Consistent Ocean-Sea Ice Estimate for 2002-2017

Authors: An T. Nguyen (University of Texas at Austin); Helen Pillar (University of Texas at Austin); Victor Ocaña (University of Texas at Austin); Arash Bigdeli (University of Texas at Austin); Timothy A. Smith (University of Texas at Austin); Patrick Heimbach (University of Texas at Austin)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: ocean-sea ice, subpolar north Atlantic, data assimilation, Arctic Mediterranean, state estimate (and 1 more)

Funding: National Science Foundation, National Science Foundation, National Science Foundation (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: A description and assessment of the first release of the Arctic Subpolar gyre sTate Estimate (ASTE_R1), a data-constrained ocean-sea ice model-data synthesis, is presented. ASTE_R1 has a nominal resolution of 1/3 degrees and spans the period 2002-2017. The fit of the model to an extensive (O(109)) set of satellite and in situ observations was achieved through adjoint-based nonlinear least squares optimization. The improvement of the solution compared to an unconstrained simulation is reflected in misfit reductions of 77% for Argo, 50% for satellite sea surface height, 58% for the Fram Strait mooring, 65% for Ice Tethered Profilers, and 83% for sea ice extent. Exact dynamical and kinematic consistency is a key advantage of ASTE_R1, distinguishing the state estimate from existing ocean reanalyses. Through strict adherence to conservation laws, all sources and sinks within ASTE_R1 can be accounted for, permitting meaningful analysis of closed budgets at the grid-scale, such as contributions of horizontal and vertical convergence to the tendencies of heat and salt. ASTE_R1 thus serves as the biggest effort undertaken to date of producing a specialized Arctic ocean-ice estimate over the 21st century. Transports of volume, heat, and freshwater are consistent with published observation-based estimates across important Arctic Mediterranean gateways. Interannual variability and low frequency trends of freshwater and heat content are well represented in the Barents Sea, western Arctic halocline, and east subpolar North Atlantic. Systematic biases remain in ASTE_R1, including a warm bias in the Atlantic Water layer in the Arctic and deficient freshwater inputs from rivers and Greenland discharge.

717. Interannual variability in the subduction of the South Atlantic subtropical underwater

Authors: Hao Liu; Shuijiang Li; Zexun Wei

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-05758-0

URL: <https://doi.org/10.1007/s00382-021-05758-0>

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Cited by: 2 publications

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Keywords: N/A

Funding: National key Research and Development Program of China, the National Natural Science Foundation of China, The Postdoctoral Innovation Research Program of Shandong Province (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: The South Atlantic subtropical underwater (STUW) is a high-salinity water mass formed by subduction within the subtropical gyre. It is a major component of the subtropical cell and affects stratification in the downstream direction due to its high salinity characteristics. Understanding the interannual variability in STUW subduction is essential for quantifying the impact of subtropical variability on the tropical Atlantic. Using the output from the ocean state estimate of the Consortium for Estimating the Circulation and Climate of the Ocean (ECCO), this study investigates the interannual variability in STUW subduction from 1992 to 2016. We find that heat fluxes, wind stress, and wind stress curl cause interannual variability in the subduction rate. Heat fluxes over the subduction area modulate the sea surface buoyancy and regulate the mixed layer depth (MLD) during its deepening and shoaling phases. Additionally, the wind stress curl and zonal wind stress can modulate the size of the subduction area by regulating the probability of particles entrained into the mixed layer within 1 year of tracing. This analysis evaluates the influence of subtropical wind patterns on the South Atlantic subsurface high-salinity water mass, highlighting the impact of heat and wind on the interannual changes in the oceanic component of the hydrological cycle.

718. Brazil Current Volume Transport Variability During 2009-2015 From a Long-Term Moored Array at 34.5 degreesS

Authors: M. P. Chidichimo (Consejo Nacional de Investigaciones Científicas y Técnicas, Institut de Recherche pour le Développement (IRD), Servicio de Hidrografía Naval); A. R. Piola (Consejo Nacional de Investigaciones Científicas y Técnicas, Institut de Recherche pour le Développement (IRD), Universidad de Buenos Aires Buenos Aires Argentina, Servicio de Hidrografía Naval); C. S. Meinen (National Oceanic and Atmospheric Administration Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); R. C. Perez (National Oceanic and Atmospheric Administration Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); E. J. D. Campos (American University of Sharjah, Oceanographic Institute of the University of Sao Paulo São Paulo Brazil); S. Dong (National Oceanic and Atmospheric Administration Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); R. Lumpkin (National Oceanic and Atmospheric Administration Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); S. L. Garzoli (National Oceanic and Atmospheric Administration Atlantic Oceanographic and Meteorological Laboratory Miami FL USA)

Journal: Journal of Geophysical Research: Oceans

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Keywords: N/A

Funding: Consejo Nacional de Investigaciones Científicas y Técnicas, Inter-American Institute for Global Change Research, Conselho Nacional de Desenvolvimento Científico e Tecnológico (and 3 more)

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Open Access: N/A

Abstract: The Brazil Current, the western limb of the subtropical gyre of the South Atlantic Ocean, is one of the major Western Boundary Currents of the global ocean. Here, we present the first multiyear continuous daily time series of Brazil Current absolute volume transport obtained using 6+ years of observations from a line of four pressure-recording inverted echo sounders (PIES) deployed at 34.5 degreesS. The array was augmented in December 2012 with two current meter-equipped PIES and in December 2013 with a moored Acoustic Doppler Current Profiler on the upper continental slope. The Brazil Current is bounded by the sea surface and the neutral density interface separating South Atlantic Central Water and Antarctic Intermediate Water, which is on average at a reference pressure of 628 +/- 46 dbar, and it is confined west of 49.5 degreesW. The Brazil Current has a mean strength of -14.0 +/- 2.8 Sv (1 Sv $\equiv$ 10⁶ m³ s⁻¹; negative indicates southward flow) with a temporal standard deviation of 8.8 Sv and peak-to-peak range from -41.7 to +20 Sv. About 80% of the absolute transport variance is concentrated at periods shorter than 150 days with a prominent peak at 100 days. The baroclinic component accounts for 85% of the absolute transport variance, but the barotropic variance is not

negligible. The baroclinic and barotropic transports are uncorrelated, demonstrating the need to measure both transport components independently. Given the energetic high frequency transport variations, statistically significant seasonal to interannual variability and trends have yet to be detected.

719. On the Breakdown of ENSO's Relationship With Thermocline Depth in the Central-Equatorial Pacific

Authors: Sen Zhao (Nanjing University, Department of Atmospheric Sciences University of Hawaiⁱⁱ at M^{noa} Honolulu HI USA); Fei-Fei Jin (Department of Atmospheric Sciences University of Hawaiⁱⁱ at M^{noa} Honolulu HI USA); Xiaoyu Long (Joint Institute for Marine and Atmospheric Research School of Ocean and Earth Science and Technology University of Hawaiⁱⁱ at M^{noa} Honolulu HI USA); Mark A. Cane (Columbia University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 51

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. Department of Energy

License: N/A

Open Access: N/A

Abstract: Recent classification of El Nino-Southern Oscillation (ENSO) into two types, Eastern (EP) and Central (CP) events, has highlighted the importance of the central Pacific. We show here that the local correlation between ENSO subsurface temperatures (T_{sub}) in the upper 100-m and thermocline depth anomalies breaks down in the central equatorial Pacific, whereas T_{sub} remains well correlated with sea surface height anomalies. This observed difference in the central equatorial Pacific is simulated by almost all climate models. It arises from a vertically slanted T_{sub} anomaly structure that is unique to the central equatorial Pacific. We show that this feature is an adiabatic response to wind-driving that is even present in a linear dynamic model, as long as the model has enough baroclinic modes to adequately represent the observed vertical complexity. Our findings have implications for better understanding the different importance of thermocline feedback in CP and EP events.

720. Evolving patterns of sterodynamic sea-level rise under mitigation scenarios and insights from linear system theory

Authors: Quran Wu; Xuebin Zhang; John A. Church; Jianyu Hu; Jonathan M. Gregory

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-05727-7

URL: <https://doi.org/10.1007/s00382-021-05727-7>

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: UK Natural Environment Research Council, Australian Research Council, National Natural Science Foundation of China (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Long-term behaviour of sea-level rise is an important factor in assessing the impact of climate change on multi-century timescales. Under the stabilisation scenario RCP4.5, Sterodynamic Sea-Level (SdynSL) and ocean density change in the CMIP5 models exhibit distinct patterns over the periods before and after Radiative Forcing (RF) stabilisation (2000-2070 vs. 2100-2300). The stabilisation pattern is more geographically uniform and involves deeper penetration of density change than the transient pattern. In RCP2.6, 4.5 and 8.5, the spatiotemporal evolution of SdynSL change can be approximated as a linear combination of the transient and stabilisation patterns. Specifically, SdynSL change is dominated by the transient pattern when RF increases rapidly, but it is increasingly affected by the stabilisation pattern once RF starts to stabilise. The growth of the stabilisation pattern could persist for centuries after RF ceases increasing. The evolving patterns of SdynSL change can also be approximated as a linear system's responses (characterised by its Green's function) to time-dependent boundary conditions. By examining SdynSL change simulated in linear system models with different estimates of Green's functions, we find that both the climatological ocean circulation and the ocean's dynamical response to RF play a role in shaping the patterns of SdynSL change. The linear system model is more accurate than the univariate pattern scaling in emulating the CMIP5 SdynSL change beyond 2100. The emergence of the stabilisation pattern leads to a 1-10% decrease in the ocean's expansion efficiency of heat over 2000-2300 in RCP2.6 and 4.5.

721. State Estimates and Forecasts of the Eddy Field in the Subtropical Countercurrent in the Northern Philippine Sea

Authors: No authors listed

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-20-0083.1

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References: 112

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Ocean models, Eddies, Forecasting, Countercurrent exchange|Eddy, Hindcasts

Funding: Office of Naval Research

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Open Access: N/A

Abstract: A strongly nonlinear eddy field is present in and around the subtropical countercurrent in the northern Philippine Sea (NPS). A regional implementation of the Massachusetts Institute of Technology General Circulation Model-Estimating the Circulation and Climate of the Ocean four-dimensional variational assimilation (MITgcm-ECCO 4DVAR) system is found to be able to produce a series of 2-month-long dynamically consistent optimized state estimates between April 2010 and April 2011 for the eddy-rich NPS region. The assimilation provides a stringent dynamical test of the model, showing that a free run of the model forced using adjusted controls remains consistent with the observations for 2 months. The 4DVAR iterative optimization reduced the total cost function for the observations and controls by 40%-50% from the reference solution, initialized using the Hybrid Coordinate Ocean Model 1/12 degrees global daily analysis, achieving residuals approximately equal to the assumed uncertainties for the assimilated observations. The state estimates are assessed by comparing with assimilated and withheld observations and also by comparing 1-month model forecasts with future data. The state estimates and forecasts were more skillful than model persistence and the reference solutions. Finally, the continuous state estimates were used to detect and track the eddies, analyze their structure, and quantify their vertically integrated meridional heat and salt transports.

722. Multi-Year Estimates of Daily Heat Transport by the Atlantic Meridional Overturning Circulation at 34.5 degreesS

Authors: M. Kersalé (NOAA Atlantic Oceanographic and Meteorological Laboratory Miami FL USA, University of Miami); C. S. Meinen (NOAA Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); R. C. Perez (NOAA Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); A. R. Piola (Consejo Nacional de Investigaciones Científicas y Técnicas, Universidad de Buenos Aires Buenos Aires Argentina, Servicio de Hidrografía Naval); S. Speich (Laboratoire de Météorologie Dynamique-IPSL); E. J. D. Campos (American University of Sharjah, University of São Paulo); S. L. Garzoli (NOAA Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); I. Ansgore (University of Cape Town); D. L. Volkov (NOAA Atlantic Oceanographic and Meteorological Laboratory Miami FL USA, University of Miami); M. Le Hénaff (NOAA Atlantic Oceanographic and Meteorological Laboratory Miami FL USA, University of Miami); S. Dong (NOAA Atlantic Oceanographic and Meteorological Laboratory Miami FL USA); T. Lamont (Department of Environment, Forestry and Fisheries Oceans and Coasts Research Branch Cape Town South Africa, University of Cape Town, Bayworld Centre for Research and Education Cape Town South Africa); O. T. Sato (University of São Paulo); M. van den Berg (Department of Environment, Forestry and Fisheries Oceans and Coasts Research Branch Cape Town South Africa)

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References: 75

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Keywords: N/A

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Open Access: N/A

Abstract: Variations in the mass and heat transported by the meridional overturning circulation (MOC) have important, well-documented, influences on global and regional climate, weather, ecosystems, and coastal sea levels. However, continuous, high-frequency, observations of these quantities have been limited to date. Multiple years of full-depth daily observations from moored instruments in the South Atlantic at 34.5 degreesS are combined with satellite observations to resolve the volume transports in both the upper and abyssal MOC cells, and the associated full-depth meridional heat transport (MHT), on daily to interannual timescales. A newly developed method for combining satellite sea level observations with historical hydrographic measurements was used to estimate daily full-depth ocean profiles of temperature in the ocean interior where mooring coverage is sparse. The average MHT during 2013-2017 is 0.5 PW, with a daily standard deviation of 0.8 PW. The MHT variability is most strongly driven by the geostrophic relative velocity contributions (horizontal density-gradient changes). This variability is highly correlated with the volume transport variability of the MOC upper cell ($r = 0.96$)

and modestly anti-correlated ($r = -0.52$) with the abyssal cell variations. An empirical relationship between the MHT and MOC values was developed allowing the reconstruction of a longer MHT time series including the pilot array period (2009-2010). Seasonal variation of the MHT is significant, and results from strong variations of all terms (Ekman, barotropic, and baroclinic). Although the 2013-2017 shows an increasing MHT trend (0.14 PW/year), the longer time period record suggests that the apparent trend may simply be interannual modulation of MHT at 34.5 degreesS.

723. Improved representation of river runoff in Estimating the Circulation and Climate of the Ocean Version 4 (ECCOV4) simulations: implementation, evaluation, and impacts to coastal plume regions

Authors: Yang Feng; Dimitris Menemenlis; Huijie Xue; Hong Zhang; Dustin Carroll; Yan Du; Hui Wu

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . In this study, we improve the representation of global river runoff in the Estimating the Circulation and Climate of the Ocean Version 4 (ECCOV4) framework, allowing for a more realistic treatment of coastal plume dynamics. We use a suite of experiments to explore the sensitivity of coastal plume regions to runoff forcing, model grid resolution, and grid type. The results show that simulated sea surface salinity (SSS) is reduced as the model grid resolution increases. Compared to Soil Moisture Active Passive (SMAP) observations, simulated SSS is closest to SMAP when using daily, point-source runoff (DPR) and the intermediate-resolution LLC270 grid. The Willmott skill score, which quantifies agreement between models and SMAP, yields up to 0.92 for large rivers such as the Amazon. There was no major difference in SSS for tropical and temperate coastal rivers when the model grid type was changed from the ECCO v4 latitude-longitude-polar-cap grid to the ECCO2 cube-sphere grid. We also found that using DPR forcing and increasing model resolution from the coarse-resolution LLC90 grid to the intermediate-resolution LLC270 grid elevated the river plume area, volume, stabilized the stratification and shoaled the mixed layer depth (MLD). Additionally, we find that the impacts of increasing model resolution from the intermediate-resolution LLC270 grid to the high-resolution LLC540 grid are regionally dependent. The Mississippi River Plume is more sensitive than other regions, possibly because the wider and shallower Texas-Louisiana shelf drives a stronger baroclinic effect, as well as relatively weak sub-grid vertical mixing and adjustment in this region. Since rivers deliver large amounts of freshwater and anthropogenic materials to coastal regions, improving the representation of river runoff in global, high-resolution models will advance studies of coastal hypoxia, carbon cycling, and regional weather and climate and will ultimately help to predict land-ocean-atmospheric feedbacks seamlessly in the next generation of Earth system models.

724. Impacts of Changed Ice-Ocean Stress on the North Atlantic Ocean: Role of Ocean Surface Currents

Authors: Yang Wu; Zhaomin Wang; Chengyan Liu

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2021.628892

URL: <https://doi.org/10.3389/fmars.2021.628892>

Publication Date: 2021-04-01

Volume/Issue: Volume 8

References: 81

Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, China Postdoctoral Science Foundation

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Open Access: N/A

Abstract: The importance of considering ocean surface currents in ice-ocean stress calculation in the North Atlantic Ocean and Arctic sea ice is investigated for the first time using a global coupled ocean-sea ice model. Considering ocean surface currents in ice-ocean stress calculation weakens the ocean surface stress and Ekman pumping by about 7.7 and 15% over the North Atlantic Ocean, respectively. It also significantly reduces the mechanical energy input to ageostrophic and geostrophic currents, and weakens the mean and eddy kinetic energy by reducing the energy conversion rates of baroclinic and barotropic pathways. Furthermore, the strength of the Atlantic Meridional Overturning Circulation (AMOC), the Nordic Seas MOC, and the North Atlantic subpolar gyre are found to be reduced considerably (by 14.3, 31.0, and 18.1%, respectively). The weakened AMOC leads to a 0.12 PW reduction in maximum northward ocean heat transport, resulting in a reduced surface heat loss and lower sea surface temperature over the North Atlantic Ocean. This reduction also leads to a shrink in sea ice extent and an attenuation of sea ice thickness. These findings highlight the importance of properly considering both the geostrophic and ageostrophic components of ocean surface currents in ice-ocean stress calculation on ocean circulation and climate studies.

725. Sea-Level Change: Emergent Hazard in a Warming World

Authors: John J. Clague

Journal: Treatise on Geomorphology

Publisher: Elsevier

DOI: 10.1016/b978-0-12-818234-5.00065-1

URL: <https://doi.org/10.1016/b978-0-12-818234-5.00065-1>

Publication Date: 2021-04-01

Volume/Issue: N/A

References: 211

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: Risk, Inundation, Hazard, Coastline, Shorelines (and 2 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

726. Observational evidence of stratification control of upwelling and pelagic fishery in the eastern Arabian Sea

Authors: Jayu Narvekar; Riyanka Roy Chowdhury; Diksha Gaonkar; P. K. Dinesh Kumar; S. Prasanna Kumar

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-021-86594-4

URL: <https://doi.org/10.1038/s41598-021-86594-4>

Publication Date: 2021-03-31

Volume/Issue: Volume 11, Issue 1

References: 57

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Upwelling is a physical phenomenon that occurs globally along the eastern boundary of the ocean and supports pelagic fishery which is an important source of protein for the coastal population. Though upwelling and associated small pelagic fishery along the eastern Arabian Sea (EAS) is known to exist at least for the past six decades, our understanding of the factors controlling them are still elusive. Based on observation and data analysis we hypothesize that upwelling in the EAS during 2017 was modulated by freshwater-induced stratification. To validate this hypothesis, we examined 17 years of data from 2001 and show that inter-annual variability of freshwater influx indeed controls the upwelling in the EAS through stratification, a mechanism hitherto unexplored. The upper ocean stratification in turn is regulated by the fresh water influx through a combination of precipitation and river runoff. We further show that the oil sardine which is one of the dominant fish of the small pelagic fishery of the EAS varied inversely with stratification. Our study for the first time underscored the role of freshwater influx in regulating the coastal upwelling and upper ocean stratification controlling the regional pelagic fishery of the EAS.

727. Local and Remote Influences on the Heat Content of Southern Ocean Mode Water Formation Regions

Authors: Emma J. D. Boland (British Antarctic Survey Cambridge UK); Daniel C. Jones (British Antarctic Survey Cambridge UK); Andrew J. S. Meijers (British Antarctic Survey Cambridge UK); Gael Forget (Massachusetts Institute of Technology); Simon A. Josey (National Oceanography Centre)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016585

URL: <https://doi.org/10.1029/2020jc016585>

Publication Date: 2021-03-31

Volume/Issue: Volume 126, Issue 4

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Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, feat flux, heat content, wind stress, sensitivity (and 1 more)

Funding: Natural Environment Research Council, National Aeronautics and Space Administration, Simons Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: The Southern Ocean (SO) is a crucial region for the global ocean uptake of heat and carbon. There are large uncertainties in the observations of fluxes of heat and carbon between the atmosphere and the ocean mixed layer, which lead to large uncertainties in the amount entering into the global overturning circulation. In order to better understand where and when fluxes of heat and momentum have the largest impact on near-surface heat content, we use an adjoint model to calculate the linear sensitivities of heat content in SO mode water formation regions (MWFRs) to surface fluxes. We find that the heat content of these regions is, in all three basins, most sensitive to same-winter, local heat fluxes, and to local and remote wind one to eight years (the maximum lead-time of our simulations) previously. This is supported by sensitivities to potential temperature changes, which reveal the sources of the MWFRs as well as dynamic links with boundary current regions and the Antarctic Circumpolar Current. We use the adjoint sensitivity fields to design a set of targeted perturbation experiments, allowing us to examine the linear and non-linear responses of the heat content to changes in surface forcing. In these targeted experiments, the heat content is sensitive to both temperature changes and mixed layer volume changes in roughly equal magnitude.

728. Sea Surface Height Variability in the 30-120 km Wavelength Band From Altimetry Along-Track Observations

Authors: Shuiming Chen (University of Hawaii at Manoa); Bo Qiu (University of Hawaii at Manoa)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jc017284

URL: <https://doi.org/10.1029/2021jc017284>

Publication Date: 2021-03-31

Volume/Issue: Volume 126, Issue 4

References: 35

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: satellite altimetry along-track data, sea surface height variability, small mesoscale eddies

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Reprocessed altimetry along-track sea surface height (SSH) observations from multiple missions are analyzed for the SSH variability in the 30-120 km wavelength band. This band of variability is missing in the widely used global multi-mission mapped SSH data set, known as the Archiving, Validation and Interpretation of Satellite Oceanographic Data SSH product. The SSH variability is derived from along-track SSH variance spectra after subtracting noise floors, and its global spatial patterns are consistent among the missions and the output of a high-resolution ocean general circulation model simulation. High SSH variability in the 30-120 km wavelength band is found in the intratropics of all ocean basins (25 degreesS-25 degreesN), the western boundary current outflow regions, and in five regions along the Antarctic Circumpolar Current: Agulhas Retroflexion zone, Kerguelen Plateau, Macquarie Ridge, Pacific Antarctic Ridge, and Drake Passage. The average slope of the along-track SSH spectra in the 30-120 km wavelength band in the intratropics falls between -2.14 and -1.56, consistent with the Garrett-Munk spectral slope. The spatial patterns of the SSH variability could be used as a metric to evaluate high-resolution numerical models and to shed light on what to expect from future high-resolution altimetry missions.

729. Algorithmic Improvements to Finding Approximately Neutral Surfaces

Authors: Geoffrey J. Stanley (University of New South Wales); Trevor J. McDougall (University of New South Wales); Paul M. Barker (University of New South Wales)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020ms002436

URL: <https://doi.org/10.1029/2020ms002436>

Publication Date: 2021-03-30

Volume/Issue: Volume 13, Issue 5

References: 32

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Interior oceanic motions occur predominantly along, rather than across, the neutral tangent plane. These planes do not link together to form well-defined surfaces, so oceanographers use approximately neutral surfaces. To date, the most accurate such surface is the omega-surface, but its practical utility was limited because its numerical implementation was slow and sometimes unstable. This work upgrades the speed, robustness, and utility of omega-surfaces. First, we switch from solving an overdetermined matrix problem by minimal least squares, to solving an exactly determined matrix problem, obtained either by the normal equations (multiplication by the matrix's transpose) or by discretizing Poisson's equation derived from the original optimization problem by the calculus of variations. This reduces the computational complexity, roughly from $O(N^3)$ to $O(N^2)$, where N is the number of grid points in the surface. Second, we update the surface's vertical position by solving a nonlinear equation in each water column, rather than assuming the stratification is vertically uniform. This reduces the number of iterations required for convergence by an order of magnitude and eliminates the need for a damping factor that stabilized the original software. Additionally, we add "wetting" capacity, whereby incrops and outcrops are reincorporated into the surface should they become neutrally linked as iterations proceed. The new algorithm computes an omega-surface in a 1,440 by 720 gridded ocean in roughly 15 s, down from roughly 11 h for the original software. We also provide two simple methods to label an omega-surface with a (neutral) density value.

730. Seasonal and Interannual Variability of EAPE in the South China Sea Derived from ECCO2 Data from 1997 to 2019

Authors: Qiang Li (Guangdong Ocean University); Lei Zhou (Shanghai Jiao Tong University); Lingling Xie (Guangdong Ocean University)

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w13070926

URL: <https://doi.org/10.3390/w13070926>

Publication Date: 2021-03-29

Volume/Issue: Volume 13, Issue 7

References: 65

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: IPOVAR Project (GASI-IPOVAI-01-02), the National Natural Science Foundation of China, the Guangdong Province First-Class Discipline Plan,

License: CC BY 4.0

Open Access: N/A

Abstract: Using Estimating the Circulation and Climate of the Ocean (phase 2, ECCO2) reanalysis products from 1997 to 2019, this study analyzes the spatiotemporal features of the eddy available gravitational potential energy (EAPE) in the South China Sea (SCS). The results indicate that the EAPE accounts for 64% of the total APE in the SCS with the climatological mean. The 2D EAPE distribution images manifest show high-value regions which are generally consistent with the eddy distributions. One region is located around 21 degrees N and west of the Luzon Strait, the second around 17 degrees N and near Luzon Island, and the third off the Vietnam coast. In the region around 21 degrees N and 17 degrees N, both the seasonal variability and the interannual variability associated with the El Niño-Southern Oscillation (ENSO) are significant. Off the Vietnam coast, the EAPE is closely associated with coastal processes which heavily depend on the seasonal monsoon, the El Niño/La Niña events, and the Indian Ocean Dipole (IOD). The results provide new insights into SCS dynamics from the point of view of ocean energy sources.

731. Effects of Wave-Induced Sea Ice Break-Up and Mixing in a High-Resolution Coupled Ice-Ocean Model

Authors: Junde Li (University of Melbourne); Alexander V. Babanin (Qingdao National Laboratory for Marine Science and Technology, University of Melbourne); Qingxiang Liu (University of Melbourne); Joey J. Voermans (University of Melbourne); Petra Heil (University of Tasmania); Youmin Tang (University of Northern British Columbia)

Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

DOI: 10.3390/jmse9040365

URL: <https://doi.org/10.3390/jmse9040365>

Publication Date: 2021-03-29

Volume/Issue: Volume 9, Issue 4

References: 93

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: DISI Australia-China Centre, Office of Naval Research Global, Australian Antarctic Program

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Open Access: N/A

Abstract: Arctic sea ice plays a vital role in modulating the global climate. In the most recent decades, the rapid decline of the Arctic summer sea ice cover has exposed increasing areas of ice-free ocean, with sufficient fetch for waves to develop. This has highlighted the complex and not well-understood nature of wave-ice interactions, requiring modeling effort. Here, we introduce two independent parameterizations in a high-resolution coupled ice-ocean model to investigate the effects of wave-induced sea ice break-up (through albedo change) and mixing on the Arctic sea ice simulation. Our results show that wave-induced sea ice break-up leads to increases in sea ice concentration and thickness in the Bering Sea, the Baffin Sea and the Barents Sea during the ice growth season, but accelerates the sea ice melt in the Chukchi Sea and the East Siberian Sea in summer. Further, wave-induced mixing can decelerate the sea ice formation in winter and the sea ice melt in summer by exchanging the heat fluxes between the surface and subsurface layer. As our baseline model underestimates sea ice cover in winter and produces more sea ice in summer, wave-induced sea ice break-up plays a positive role in improving the sea ice simulation. This study provides two independent parameterizations to directly include the wave effects into the sea ice models, with important implications for the future sea ice model development.

732. Progress in understanding of Indian Ocean circulation, variability, air-sea exchange and impacts on biogeochemistry

Authors: Helen E. Phillips; Amit Tandon; Ryo Furue; Raleigh Hood; Caroline Ummenhofer; Jessica Benthuyssen; Viviane Menezes; Shijian Hu; Ben Webber; Alejandra Sanchez-Franks; Deepak Cherian; Emily Shroyer; Ming Feng; Hemantha Wijeskera; Abhishek Chatterjee; Lisan Yu; Juliet Hermes; Raghu Murtugudde; Tomoki Tozuka; Danielle Su; Arvind Singh; Luca Centurioni; Satya Prakash; Jerry Wiggert

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/os-2021-1

URL: <https://doi.org/10.5194/os-2021-1>

Publication Date: 2021-03-29

Volume/Issue: N/A

References: N/A

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Biogeochemistry|Biogeochemical Cycle|Ocean observations|Physical oceanography|Boundary current|Atmospheric Circulation

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Over the past decade, our understanding of the Indian Ocean has advanced through concerted efforts toward measuring the ocean circulation and its water properties, detecting changes in water masses, and linking physical processes to ecologically important variables. New circulation pathways and mechanisms have been discovered, which control atmospheric and oceanic mean state and variability. This review brings together new understanding of the ocean-atmosphere system in the Indian Ocean since the last comprehensive review, describing the Indian Ocean circulation patterns, air-sea interactions and climate variability. The second International Indian Ocean Expedition (IIOE-2) and related efforts have motivated the application of new technologies to deliver higher-resolution observations and models of Indian Ocean processes. As a result we are discovering the importance of small scale processes in setting the large-scale gradients and circulation, interactions between physical and biogeochemical processes, interactions between boundary currents and the interior, and between the surface and the deep ocean. In the last decade we have seen rapid warming of the Indian Ocean overlaid with extremes in the form of marine heatwaves. These events have motivated studies that have delivered new insight into the variability in ocean heat content and exchanges in the Indian Ocean, and climate variability on interannual to decadal timescales. This synthesis paper reviews the advances in these areas in the last decade.

733. Water Depth Dependence of Long-Range Correlation in Nontidal Variations in Seafloor Pressure

Authors: Tomohiro Inoue (Kyoto University); Yoshihiro Ito (Kyoto University); Laura M. Wallace (GNS Science Lower Hutt New Zealand, University of Texas at Austin); Yutaka Yoshikawa (Kyoto University); Daisuke Inazu (Department of Marine Resources and Energy Tokyo University of Marine Science and Technology Tokyo Japan); Emmanuel Soliman M. Garcia (Kyoto University); Tomoya Muramoto (National Metrology Institute of Japan National Institute of Advanced Industrial Science and Technology Tsukuba Japan); Spahr C. Webb (Columbia University); Kazuaki Ohta (National Research Institute for Earth Science and Disaster Resilience Tsukuba Japan); Syuichi Suzuki (Tohoku University); Ryota Hino (Tohoku University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl092173

URL: <https://doi.org/10.1029/2020gl092173>

Publication Date: 2021-03-27

Volume/Issue: Volume 48, Issue 8

References: 36

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: ocean bottom pressure, correlation, water depth dependence, nontidal variation

Funding: Japan Society for the Promotion of Science, Japan Science and Technology Agency, National Science Foundation (and 4 more)

License: Creative Commons

Open Access: N/A

Abstract: Isolating the source of nontidal oceanographic noise in seafloor pressure data is critical for improving the use of these data for seafloor geodetic applications. Residuals between nearby bottom pressure records have typically been used to remove the nontidal components, as these are largely common-mode. To evaluate the similarities between pairs of observed bottom pressure records at a range of water depths, we calculate the standard deviations of the time series of residuals between data from all site pairs, installed in the Hikurangi subduction zone offshore New Zealand. We find that the magnitude of the standard deviation depends more on relative water depth than the distance between sites. This confirms the result of previous studies from Cascadia that nontidal components are more similar along isobaths even if the distance between sites is large. Furthermore, in order to reduce noises, the required depth difference between site pairs also varies with site depths.

734. The imperative of physics-based modeling and inverse theory in computational science

Authors: Karen E. Willcox; Omar Ghattas; Patrick Heimbach

Journal: Nature Computational Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43588-021-00040-z

URL: <https://doi.org/10.1038/s43588-021-00040-z>

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Volume/Issue: Volume 1, Issue 3

References: 15

Cited by: 69 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. Department of Energy, National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: N/A

735. Analysis of Ocean Bottom Pressure Anomalies and Seismic Activities in the MedRidge Zone

Authors: Hakan S. Kutoglu (Zonguldak Bulent Ecevit University); Kazimierz Becek (Wroclaw University of Science and Technology)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs13071242

URL: <https://doi.org/10.3390/rs13071242>

Publication Date: 2021-03-25

Volume/Issue: Volume 13, Issue 7

References: 39

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The Mediterranean Ridge accretionary complex (MAC) is a product of the convergence of Africa-Europe-Aegean plates. As a result, the region exhibits a continuous mass change (horizontal/vertical movements) that generates earthquakes. Over the last 50 years, approximately 430 earthquakes with $M \geq 5$, including 36 $M \geq 6$ earthquakes, have been recorded in the region. This study aims to link the ocean bottom deformations manifested through ocean bottom pressure variations with the earthquakes' time series. To this end, we investigated the time series of the ocean bottom pressure (OBP) anomalies derived from the Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GRACE-FO) satellite missions. The OBP time series comprises a decreasing trend in addition to 1.02, 1.52, 4.27, and 10.66-year periodic components, which can be explained by atmosphere, oceans, and hydrosphere (AOH) processes, the Earth's pole movement, solar activity, and core-mantle coupling. It can be inferred from the results that the OBP anomalies time series/mass change is linked to a rising trend and periods in the earthquakes' energy time series. Based on this preliminary work, ocean-bottom pressure variation appears to be a promising lead for further research.

736. CAFE60v1: A 60-year large ensemble climate reanalysis. Part I: System design, model configuration and data assimilation.

Authors: Terence J. O'Kane (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Paul A. Sandery (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Vassili Kitsios (Monash University); Pavel Sakov (Bureau of Meteorology, Docklands, Victoria, Australia); Matthew A. Chamberlain (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Mark A. Collier (CSIRO Oceans and Atmosphere, Aspendale, Victoria, Australia); Russell Fiedler (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Thomas S. Moore (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Christopher C. Chapman (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Bernadette M. Sloyan (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Richard J. Matear (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-20-0974.1

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Publication Date: 2021-03-22

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Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: Atmosphere, Ocean, Data assimilation, Ensembles, Climate prediction

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We detail the system design, model configuration and data assimilation evaluation for the CSIRO Climate retrospective Analysis and Forecast Ensemble system: version 1. CAFE60v1 has been designed with the intention of simultaneously generating both initial conditions for multi-year climate forecasts and a large ensemble retrospective analysis of the global climate system from 1960 to present. Strongly coupled data assimilation (SCDA) is implemented via an ensemble transform Kalman filter in order to constrain a general circulation climate model to observations. Satellite (altimetry, sea surface temperature, sea ice concentration) and in-situ ocean temperature and salinity profiles are directly assimilated each month, whereas atmospheric observations are sub-sampled from the JRA-55 atmospheric reanalysis. Strong coupling is implemented via explicit cross domain covariances between ocean, atmosphere, sea ice and ocean biogeochemistry. Atmospheric and surface ocean fields are available at daily resolution and monthly resolution for the land, subsurface ocean and sea ice. The system produces 96 climate trajectories (state estimates) over the most recent six decades as well as a complete data archive of initial conditions potentially enabling individual forecasts for all members each month over the 60 year period. The size of the ensemble and application of strongly coupled data assimilation lead to new insights for future reanalyses.

737. CAFE60v1: A 60-year large ensemble climate reanalysis. Part II: Evaluation

Authors: Terence J. O'Kane (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Paul A. Sandery (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Vassili Kitsios (Monash University); Pavel Sakov (Bureau of Meteorology, Docklands, Victoria, Australia); Matthew A. Chamberlain (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Dougal T. Squire (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Mark A. Collier (CSIRO Oceans and Atmosphere, Aspendale, Victoria, Australia); Christopher C. Chapman (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Russell Fiedler (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Dylan Harries (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Thomas S. Moore (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Doug Richardson (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); James S. Risbey (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Benjamin J. E. Schroeter (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Serena Schroeter (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Bernadette M. Sloyan (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Carly Tozer (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Ian G. Watterson (CSIRO Oceans and Atmosphere, Aspendale, Victoria, Australia); Amanda Black (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Courtney Quinn (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Richard J. Matear (CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-20-0518.1

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Publication Date: 2021-03-22

Volume/Issue: N/A

References: N/A

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Reanalysis data, Data assimilation, Model comparison, Ensembles

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The CSIRO Climate retrospective Analysis and Forecast Ensemble system: version 1 (CAFE60v1) provides a large (96 member) ensemble retrospective analysis of the global climate system from 1960 to present with sufficiently many realizations and at spatio-temporal resolutions suitable to enable probabilistic climate studies. Using a variant of the ensemble Kalman filter, 96 climate state estimates are generated over the most recent six decades. These state estimates are constrained by monthly mean ocean, atmosphere and sea ice observations such that their trajectories track the observed state while enabling estimation of the uncertainties in the approximations to the retrospective mean climate over recent decades. For the atmosphere, we evaluate CAFE60v1 in comparison to empirical indices of the major climate teleconnections and blocking with various reanalysis products. Estimates of the large scale ocean structure, transports and biogeochemistry are

compared to those derived from gridded observational products and climate model projections (CMIP). Sea ice (extent, concentration and variability) and land surface (precipitation and surface air temperatures) are also compared to a variety of model and observational products. Our results show that CAFE60v1 is a useful, comprehensive and unique data resource for studying internal climate variability and predictability, including the recent climate response to anthropogenic forcing on multi-year to decadal time scales.

738. Interannual variability of the spring Wyrski Jet

Authors: Kangping Deng; Xuhua Cheng; Tao Feng; Tian Ma; Wei Duan; Jiajia Chen

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-020-9330-3

URL: <https://doi.org/10.1007/s00343-020-9330-3>

Publication Date: 2021-03-11

Volume/Issue: Volume 39, Issue 1

References: 67

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

739. Using Drifter Velocity Measurements to Assess and Constrain Coarse-Resolution Ocean Models

Authors: Mengnan Zhao (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rick Lumpkin (c Physical Oceanography Division, NOAA/Atlantic Oceanographic and Meteorological Laboratory, Miami, Florida)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-20-0159.1

URL: <https://doi.org/10.1175/jtech-d-20-0159.1>

Publication Date: 2021-03-10

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References: N/A

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Data assimilation, Error analysis, Ocean circulation, Climate models

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License: N/A

Open Access: N/A

Abstract: Properly fitting ocean models to observations is crucial for improving model performance and understanding ocean dynamics. Near-surface velocity measurements from the Global Drifter Program (GDP) contain valuable information about upper-ocean circulation and air-sea fluxes on various space and time scales. This study explores whether GDP measurements can be used for usefully constraining the surface circulation from coarse-resolution ocean models, using global solutions produced by the consortium for Estimating the Circulation and Climate of the Ocean (ECCO) as an example. To address this problem, a careful examination of velocity data errors is required. Comparisons between an ECCO model simulation, performed without any data constraints, and GDP and Ocean Surface Current Analyses Real-Time (OSCAR) velocity data, over the period 1992-2017, reveal considerable differences in magnitude and pattern. These comparisons are used to estimate GDP data errors in the context of the time-mean and time-variable surface circulations. Both instrumental errors and errors associated with limitations in model physics and resolution (representation errors) are considered. Given the estimated model-data differences, errors, and signal-to-noise ratios, our results indicate that constraining ocean-state estimates to GDP can have a substantial impact on the ECCO large-scale time-mean surface circulation over extensive areas. Impact of GDP data constraints on the ECCO time-variable circulation would be weaker and mainly limited to low latitudes. Representation errors contribute substantially to degrading the data impacts.

740. The Scale and Activity of Symmetric Instability Estimated from a Global Submesoscale-Permitting Ocean Model

Authors: Jihai Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory); Baylor Fox-Kemper (Brown University); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Changming Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0159.1

URL: <https://doi.org/10.1175/jpo-d-20-0159.1>

Publication Date: 2021-03-10

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References: 89

Cited by: 31 publications

Language: N/A

Subjects: N/A

Keywords: Ocean models, Eddies, Potential vorticity, Instability, Oceanic mixed layer (and 1 more)

Funding: National Key Research Program of China, National Natural Science Foundation of China, China Scholarship Council (and 2 more)

License: N/A

Open Access: N/A

Abstract: Symmetric instability (SI) extracts kinetic energy from fronts in the surface mixed layer (SML), potentially affecting the SML structure and dynamics. Here, a global submesoscale-permitting ocean model named MITgcm LLC4320 simulation is used to examine the Stone linear prediction of the maximum SI scale to estimate grid spacings needed to begin resolving SI. Furthermore, potential effects of SI on the usable wind work are estimated roughly: this estimate of SI "activity" is useful for assessing if these modes should be resolved or parameterized. The maximum SI scale varies by latitude with median values from 568 to 23 m. Strong seasonality is observed in the SI scale and activity. The median scale in winter is 188 m globally, 2.5 times of that of summer (75 m). SI is more active in winter: 15% of the time compared with 6% in summer. The strongest SI activity is found in the western Pacific, western Atlantic, and Southern Oceans. The required grid spacings for a global model to begin resolving SI eddies in the SML are 24 m (50% of regions resolved) and 7.9 m (90%) in winter, decreasing to 9.4 m (50%) and 3.6 m (90%) in summer. It is also estimated that SI may reduce usable wind work by an upper bound of 0.83 mW m⁻² globally, or 5% of the global magnitude. The sensitivity of these estimates to empirical thresholds is provided in the text.

741. Assessment of degree-2 order-1 gravitational changes from GRACE and GRACE Follow-on, Earth rotation, satellite laser ranging, and models

Authors: Jianli Chen; John C. Ries; Byron D. Tapley

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-021-01492-x

URL: <https://doi.org/10.1007/s00190-021-01492-x>

Publication Date: 2021-03-06

Volume/Issue: Volume 95, Issue 4

References: 43

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Jet Propulsion Laboratory

License: N/A

Open Access: N/A

Abstract: N/A

742. Meridional Asymmetry in Recent Decadal Sea-Level Trends in the Subtropical Pacific Ocean

Authors: F. Schloesser (Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); P. R. Thompson (Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); C. G. Piecuch (Woods Hole Oceanographic Institution)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl091959

URL: <https://doi.org/10.1029/2020gl091959>

Publication Date: 2021-03-03

Volume/Issue: Volume 48, Issue 6

References: 30

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: wind driven circulation, Pacific heat content trends, Pacific sea level trends, thermosteric sea level

Funding: National Aeronautics and Space Administration, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Recent sea surface height (SSH) trends in the South Pacific are substantially greater than trends in the North Pacific. Here, we use the Estimating the Climate and Circulation of the Ocean Version 4 Release 4 ocean state estimate and the Ocean Reanalysis System 5 to identify the forcing and mechanisms underlying that meridional asymmetry during 2005-2015. Thermosteric contributions dominate the spatial structure in Pacific SSH trends, but contributions from local surface heat fluxes are small. Wind stress trends drive a spin-up of the South Pacific subtropical gyre and a northward shift of the North Pacific subtropical gyre. A reduced gravity model forced with reanalysis winds qualitatively reproduces the meridional seesaw in sea level, suggesting that asymmetric trends in subtropical wind stress drive a cross-equatorial heat transport. A reversal in forcing associated with this process could impact near-term rates of coastal sea-level change, particularly in Pacific Island communities.

743. Correlation between subsurface salinity anomalies in the Bay of Bengal and the Indian Ocean Dipole and governing mechanisms

Authors: Zheen Zhang; Thomas Pohlmann; Xueen Chen

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-17-393-2021

URL: <https://doi.org/10.5194/os-17-393-2021>

Publication Date: 2021-03-03

Volume/Issue: Volume 17, Issue 1

References: 75

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Lead-lag correlations between the subsurface temperature and salinity anomalies in the Bay of Bengal (BoB) and the Indian Ocean Dipole (IOD) are revealed in model results, ocean synthesis, and observations. Mechanisms for such correlations are further investigated using the Hamburg Shelf Ocean Model (HAMSOM), mainly relating to the salinity variability. It is found that the subsurface salinity anomaly of the BoB positively correlates to the IOD, with a lag of 3 months on average, while the subsurface temperature anomaly correlates negatively. The model results suggest the remote forcing from the equatorial Indian Ocean dominates the interannual subsurface salinity variability in the BoB. The coastal Kelvin waves carry signals of positive (negative) salinity anomalies from the eastern equatorial Indian Ocean and propagate counterclockwise along the coasts of the BoB during positive (negative) IOD events. Subsequently, westward Rossby waves propagate these signals to the basin at a relatively slow speed, which causes a considerable delay of the subsurface salinity anomalies in the correlation. By analyzing the salinity budget of the BoB, it is found that diffusion dominates the salinity changes near the surface, while advection dominates the subsurface; the vertical advection of salinity contributes positively to this correlation, while the horizontal advection contributes negatively. These results suggest that the IOD plays a crucial role in the interannual subsurface salinity variability in the BoB.

744. Modeling rapid Earth rotation variations – where are we going next?

Authors: Michael Schindelegger

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-egu21-1616

URL: <https://doi.org/10.5194/egusphere-egu21-1616>

Publication Date: 2021-03-03

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Momentum (technical analysis)

Funding: N/A

License: N/A

Open Access: N/A

Abstract:

Non-tidal Earth rotation variations at intraseasonal periods are almost exclusively driven by mass redistributions within the atmosphere and ocean. Our capacity to model these signals has advanced over the past decades, but differences between the observed and modeled portions of the planetary angular momentum budget are still as large as 1.5 cm when expressed as axis displacement at the Earth's surface. A likely source for these significant errors is the ocean, poorly sampled with observations and thus not amenable to the sequential data assimilation machinery developed for the atmosphere. Moreover, the recent delineation of basin-wide ocean mass exchanges associated with the Madden-Julian Oscillation (MJO) in a high-resolution baroclinic model emphasizes that a revisit of standard forward modeling choices (e.g., grid spacings of 8764;100 km) may be in order to better describe rapid, large-scale oceanic mass motions. In this contribution, I provide a brief overview of recent progress in the field and suggest that dynamically consistent model-data syntheses, as practiced by the consortium on Estimating the Circulation and Climate of the Ocean (ECCO), are a viable route to mitigate deficiencies in present oceanic angular momentum (OAM) series on intraseasonal (but also longer) time scales. As ocean state estimates continue to be refined by the central ECCO production, I assess the benefits of higher model resolution with OAM series from an eddy-permitting (1/6°) forward simulation, descending from the current ECCO release in its discrete setup. The resulting mass and motion terms indeed provide smaller Earth rotation residuals than other available OAM estimates, possibly due to the model resolving important topographic interactions and the dynamic response to MJO in the 30–80-day band. However, these improvements come at disproportionately large computational costs, and iteratively fitting an eddy-permitting general circulation model to oceanographic observations may still be prohibitive in the near future. Instead, efforts should be devoted to extending the present coarser-resolution ECCO framework to new data constraints and shorter adjustment intervals. Of particular interest in the context of Earth rotation are non-standard daily GRACE gravity field solutions, which contain realistic information on oceanic mass-field variability below the nominal GRACE Nyquist period of 60 days.

745. Selective constraints on global plankton dispersal

Authors: Ben A. Ward (University of Southampton); B. B. Cael (National Oceanography Centre); Sinead Collins (University of Edinburgh); C. Robert Young (National Oceanography Centre)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.2007388118

URL: <https://doi.org/10.1073/pnas.2007388118>

Publication Date: 2021-03-01

Volume/Issue: Volume 118, Issue 10

References: 29

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Royal Society

License: Creative Commons

Open Access: N/A

Abstract: Significance Microscopic plankton form the ecological and biogeochemical foundation of almost all marine ecosystems. In the fluid ocean environment, biodiversity and community structure are determined by the poorly constrained balance of local selection and global dispersal. While ocean currents have the capacity to rapidly connect distant locations, we use numerical simulations to show that extremely high rates of adaptation are required for populations to traverse large-scale gradients in environmental variables such as temperature. Changing the assumed balance of selection, adaptation, and dispersal in our simulations has pronounced effects on the simulated community structure, accounting for emergent patterns in the global ocean microbiome and emphasizing the importance of evolutionary history in global marine biodiversity and biogeography.

746. A shift in the ocean circulation has warmed the subpolar North Atlantic Ocean since 2016

Authors: Damien Desbruyères; Léon Chafik; Guillaume Maze

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-021-00120-y

URL: <https://doi.org/10.1038/s43247-021-00120-y>

Publication Date: 2021-02-26

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References: 63

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The Subpolar North Atlantic is known for rapid reversals of decadal temperature trends, with ramifications encompassing the large-scale meridional overturning and gyre circulations, Arctic heat and mass balances, or extreme continental weather. Here, we combine datasets derived from sustained ocean observing systems (satellite and in situ), idealized observation-based modelling (advection-diffusion of a passive tracer), and a machine learning technique (ocean profile clustering) to document and explain the most-recent and ongoing cooling-to-warming transition of the Subpolar North Atlantic. Following a gradual cooling of the region that was persisting since 2006, a surface-intensified and large-scale warming sharply emerged in 2016 following an ocean circulation shift that enhanced the northeastward penetration of warm and saline waters from the western subtropics. The long ocean memory of the Subpolar North Atlantic implies that this advection-driven warming is likely to persist in the near-future with possible implications for the Atlantic multidecadal variability and its global impacts.

747. A novel hydrographic gridded data set for the northern Antarctic Peninsula

Authors: Tiago S. Dotto; Mauricio M. Mata; Rodrigo Kerr; Carlos A. E. Garcia

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-13-671-2021

URL: <https://doi.org/10.5194/essd-13-671-2021>

Publication Date: 2021-02-26

Volume/Issue: Volume 13, Issue 2

References: 97

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . The northern Antarctic Peninsula (NAP) is a highly dynamic transitional zone between the subpolar-polar and oceanic-coastal environments, and it is located in an area affected by intense climate change, including intensification and spatial shifts of the westerlies as well as atmospheric and oceanic warming. In the NAP area, the water masses originate mainly from the Bellingshausen and Weddell seas, which create a marked regional dichotomy thermohaline characteristic. Although the NAP area has relatively easy access when compared to other Southern Ocean environments, our understanding of the water masses' distribution and the dynamical processes affecting the variability of the region is still limited. That limitation is closely linked to the sparse data coverage, as is commonly the case in most Southern Ocean environments. This work provides a novel seasonal three-dimensional high-resolution hydrographic gridded data set for the NAP (version 1), namely the NAPv1.0. Hydrographic measurements from 1990 to 2019 comprising data collected by conductivity, temperature, depth (CTD) casts; sensors from the Marine Mammals Exploring the Oceans Pole to Pole (MEOP) consortium; and Argo floats have been optimally interpolated to produce maps of in situ temperature, practical salinity, and dissolved oxygen at ~ 10 km spatial resolution and 90 depth levels. The water masses and oceanographic features in this regional gridded product are more accurate than other climatologies and state estimate products currently available. The data sets are available in netCDF format at <https://doi.org/10.5281/zenodo.4420006> (Dotto et al., 2021). The novel and comprehensive data sets presented here for the NAPv1.0 product are a valuable tool to be used in studies addressing climatological changes in the unique NAP region since they provide accurate initial conditions for ocean models and improve the end of the 20th- and early 21st-century ocean mean-state representation for that area.

748. Subseasonal Predictability of Arctic Ocean Sea Ice Conditions: Bering Strait and Ekman-Driven Ocean Heat Transport

Authors: Jed E. Lenetsky (McGill University, University of Colorado Boulder); Bruno Tremblay (McGill University); Charles Brunette (McGill University); Gianluca Meneghello (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-20-0544.1

URL: <https://doi.org/10.1175/jcli-d-20-0544.1>

Publication Date: 2021-02-26

Volume/Issue: Volume 34, Issue 11

References: 134

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: Statistical forecasting, Seasonal forecasting, Ekman transport|Hindcast|Predictability, Heat budgets/fluxes, Sea ice

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We use ocean observations and reanalyses to investigate the subseasonal predictability of summer and fall sea ice area (SIA) in the western Arctic Ocean associated with lateral ocean heat transport (OHT) through Bering Strait and vertical OHT along the Alaskan coastline from Ekman divergence and upwelling. Results show predictive skill of spring Bering Strait OHT anomalies in the Chukchi Sea and eastern East Siberian Sea for June and July SIA, followed by a sharp drop in predictive skill in August, September, and October and a resurgence of the correlation in November during freeze-up. Fall upwelling of Pacific Water along the Alaskan coastline--a mechanism that was proposed as a preconditioner for lower sea ice concentration (SIC) in the Beaufort Sea the following summer--shows minimal predictive strength on both local and regional scales for any months of the melt season. A statistical hindcast based on May Bering Strait OHT anomalies explains 77% of July Chukchi Sea SIA variance. Using OHT as a predictor of SIA anomalies in the Chukchi Sea improves hindcasts from the simple linear trend by 35% and predictions from spring sea ice thickness anomalies by 24%. This work highlights the importance of ocean heat anomalies for melt season sea ice prediction and provides observational evidence of subseasonal changes in forecast skill observed in model-based forecasts of the Chukchi Sea.

749. Using Saildrones to Validate Arctic Sea-Surface Salinity from the SMAP Satellite and from Ocean Models

Authors: Jorge Vazquez-Cuervo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Chelle Gentemann (Farallon Institute, Petaluma, CA 94952, USA); Wenqing Tang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dustin Carroll (San José State University); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jose Gomez-Valdes (Physical Oceanography Department, Center for Scientific Research and Higher Education at Ensenada, Ensenada 22860, Mexico); Marouan Bouali (Institute of Oceanography, University of Sao Paulo, São Paulo 05508-120, Brazil); Michael Steele (University of Washington)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs13050831

URL: <https://doi.org/10.3390/rs13050831>

Publication Date: 2021-02-24

Volume/Issue: Volume 13, Issue 5

References: 40

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: The Arctic Ocean is one of the most important and challenging regions to observe--it experiences the largest changes from climate warming, and at the same time is one of the most difficult to sample because of sea ice and extreme cold temperatures. Two NASA-sponsored deployments of the Saildrone vehicle provided a unique opportunity for validating sea-surface salinity (SSS) derived from three separate products that use data from the Soil Moisture Active Passive (SMAP) satellite. To examine possible issues in resolving mesoscale-to-submesoscale variability, comparisons were also made with two versions of the Estimating the Circulation and Climate of the Ocean (ECCO) model (Carroll, D; Menmenlis, D; Zhang, H.). The results indicate that the three SMAP products resolve the runoff signal associated with the Yukon River, with high correlation between SMAP products and Saildrone SSS. Spectral slopes, overall, replicate the -2.0 slopes associated with mesoscale-submesoscale variability. Statistically significant spatial coherences exist for all products, with peaks close to 100 km. Based on these encouraging results, future research should focus on improving derivations of satellite-derived SSS in the Arctic Ocean and integrating model results to complement remote sensing observations.

750. A fully coupled Arctic sea-ice-ocean-atmosphere model (ArcIOAM v1.0) based on C-Coupler2: model description and preliminary results

Authors: Shihe Ren; Xi Liang; Qizhen Sun; Hao Yu; L. Bruno Tremblay; Bo Lin; Xiaoping Mai; Fu Zhao; Ming Li; Na Liu; Zhikun Chen; Yunfei Zhang

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-1101-2021

URL: <https://doi.org/10.5194/gmd-14-1101-2021>

Publication Date: 2021-02-24

Volume/Issue: Volume 14, Issue 2

References: 76

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The Arctic regional coupled sea-ice-ocean-atmosphere model (ArcIOAM) has been developed to provide reliable Arctic sea ice prediction on seasonal timescales. The description and implementation of ArcIOAM and its preliminary results for the year of 2012 are presented in this paper. In the ArcIOAM configuration, the Community Coupler 2 (C-Coupler2) is used to couple the Arctic sea-ice-oceanic configuration of the MITgcm (Massachusetts Institute of Technology general circulation model) with the Arctic atmospheric configuration of the Polar WRF (Weather Research and Forecasting) model. A scalability test is performed to investigate the parallelization of the coupled model. As the first step toward reliable Arctic seasonal sea ice prediction, ArcIOAM, implemented with two-way coupling strategy along with one-way coupling strategy, is evaluated with respect to available observational data and reanalysis products for the year of 2012. A stand-alone MITgcm run with prescribed atmospheric forcing is performed for reference. From the comparison, all the experiments simulate reasonable evolution of sea ice and ocean states in the Arctic region over a 1-year simulation period. The two-way coupling has better performance in terms of sea ice extent, concentration, thickness and sea surface temperature (SST), especially in summer. This result indicates that sea-ice-ocean-atmosphere interaction plays a crucial role in controlling Arctic summertime sea ice distribution.

751. Short-Term Predictability of the Bay of Bengal Region Using a High-Resolution Indian Ocean Model

Authors: Lokesh Kumar Pandey (University of Allahabad); Suneet Dwivedi (University of Allahabad); Matthew Martin (Met Office)

Journal: Marine Geodesy

Publisher: Informa UK Limited

DOI: 10.1080/01490419.2021.1894273

URL: <https://doi.org/10.1080/01490419.2021.1894273>

Publication Date: 2021-02-23

Volume/Issue: Volume 44, Issue 3

References: 68

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

752. Can seafloor voltage cables be used to study large-scale circulation? An investigation in the Pacific Ocean

Authors: Jakub Velímský; Neesha R. Schnepf; Manoj C. Nair; Natalie P. Thomas

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-17-383-2021

URL: <https://doi.org/10.5194/os-17-383-2021>

Publication Date: 2021-02-22

Volume/Issue: Volume 17, Issue 1

References: 47

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Marine electromagnetic (EM) signals largely depend on three factors: flow velocity, Earth's main magnetic field, and seawater's electrical conductivity (which depends on the local temperature and salinity). Because of this, there has been recent interest in using marine EM signals to monitor and study ocean circulation. Our study utilizes voltage data from retired seafloor telecommunication cables in the Pacific Ocean to examine whether such cables could be used to monitor circulation velocity or transport on large oceanic scales. We process the cable data to isolate the seasonal and monthly variations and then evaluate the correlation between the processed data and numerical predictions of the electric field induced by an estimate of ocean circulation. We find that the correlation between cable voltage data and numerical predictions strongly depends on both the strength and coherence of the model velocities flowing across the cable, the local EM environment, as well as the length of the cable. The cable within the Kuroshio Current had good correlation between data and predictions, whereas two of the cables in the Eastern Pacific Gyre - a region with both low flow speeds and interfering velocity directions across the cable - did not have any clear correlation between data and predictions. Meanwhile, a third cable also located in the Eastern Pacific Gyre showed good correlation between data and predictions - although the cable is very long and the speeds were low, it was located in a region of coherent flow velocity across the cable. While much improvement is needed before utilizing seafloor voltage cables to study and monitor oceanic circulation across wide regions, we believe that with additional work, the answer to the question of whether or not seafloor voltage cables can be used to study large-scale circulation may eventually be yes.

753. Inverse estimation of diffusivity coefficients from salinity distributions on isopycnal surfaces using Argo float array data

Authors: Shinya Kouketsu

Journal: Journal of Oceanography

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10872-021-00595-5

URL: <https://doi.org/10.1007/s10872-021-00595-5>

Publication Date: 2021-02-20

Volume/Issue: Volume 77, Issue 4

References: 32

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science

License: N/A

Open Access: N/A

Abstract: N/A

754. Arctic ocean-sea ice reanalysis for the period 2007-2016 using the adjoint method

Authors: Guokun Lyu (Shanghai Jiao Tong University, University of Hamburg); Armin Koehl (University of Hamburg); Nuno Serra (University of Hamburg); Detlef Stammer (University of Hamburg); Jiping Xie (Nansen Environmental and Remote Sensing Center)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.4002

URL: <https://doi.org/10.1002/qj.4002>

Publication Date: 2021-02-16

Volume/Issue: Volume 147, Issue 736

References: 55

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: We present an Arctic ocean-sea ice reanalysis covering the period 2007-2016 based on the adjoint approach of the Estimating the Circulation and Climate of the Ocean (ECCO) consortium. The spatiotemporal variation of Arctic sea surface temperature (SST), sea ice concentration (SIC), and sea ice thickness (SIT) is substantially improved after the assimilation of ocean and sea ice observations. By assimilating additional World Ocean Atlas 2018 (WOA18) hydrographic data, the freshwater content of the Canadian Basin becomes closer to the observations and translates into changes of the ocean circulation and of transports through the Fram and Davis straits. This new reanalysis compares well with previous filter-based (TOPAZ4) and nudging-based (PIOMAS) reanalyses regarding SIC and SST. Benefiting from using the adjoint of the sea ice model, our reanalysis is superior to the ECCOv4r4 product considering sea ice parameters. However, the mean state and variability of the freshwater content and the transport properties of our reanalysis remain different from TOPAZ4 and ECCOv4r4, likely because of a lack of hydrographic observations.

755. Role of Extratropical Wintertime Cyclones in Regulating the Variations of Baffin Bay Sea Ice Export

Authors: Yu Liang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Haibo Bi (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Yunhe Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Haijun Huang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Zehua Zhang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Jue Huang (College of Geodesy and Geomatics Shandong University of Science and Technology Qingdao China); Yanxia Liu (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jd033616

URL: <https://doi.org/10.1029/2020jd033616>

Publication Date: 2021-02-13

Volume/Issue: Volume 126, Issue 5

References: 71

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, NSAF Joint Fund

License: N/A

Open Access: N/A

Abstract: A cyclone is an intensive synoptic activity that occurs frequently over Baffin Bay. By modifying the large-scale distribution pattern of sea level pressure, a passing cyclone can serve as an important regulator of sea ice outflow via the Davis Strait. We obtain a nearly 40-year-long record (1979/1980-2017/2018) of the sea ice area flux (SIAF) through the Davis Strait and Arctic cyclone activities in winter. A case study and statistical results indicate that the sea ice concentration and motion fields can be greatly altered by the occurrence of cyclones, thereby contributing to changes in sea ice export. Moreover, the effects of cyclones on sea ice export in Baffin Bay are dependent on the spatial distribution pattern of the storms. In terms of the cyclone center count and intensity, the key regions with significant impacts on sea ice export out of Baffin Bay are identified, one around Baffin Island (80 degreesW-60 degreesW, 60 degreesN-70 degreesN) and the other over the southern Labrador Peninsula (70 degreesW-50 degreesW, 40 degreesN-60 degreesN). A robust correlation exists between the winter-accumulated SIAF via the Davis Strait and the average winter cyclone intensity (center count) in the critical regions with $R = -0.57 (+0.49)$, affirming the vital role of cyclone activity in modulating the interannual variability of sea ice export in Baffin Bay.

756. Synchronization of Geophysically Driven Oscillators With Short-Range Interaction

Authors: Cong Wei (University of Delaware); Herbert G. Tanner (University of Delaware)

Journal: IEEE Transactions on Automatic Control

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tac.2021.3058960

URL: <https://doi.org/10.1109/tac.2021.3058960>

Publication Date: 2021-02-13

Volume/Issue: Volume 67, Issue 3

References: 48

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research

License: N/A

Open Access: N/A

Abstract: N/A

757. Spiciness Anomalies of Subantarctic Mode Water in the South Indian Ocean

Authors: Motoki Nagura (a Japan Agency for Marine-Earth Science and Technology, Yokosuka, Japan)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-20-0482.1

URL: <https://doi.org/10.1175/jcli-d-20-0482.1>

Publication Date: 2021-02-12

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References: N/A

Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: Sea surface temperature, Water masses/storage, Interannual variability, Heat budgets/fluxes, Indian Ocean (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study investigates spreading and generation of spiciness anomalies of the Subantarctic Mode Water (SAMW) located on 26.6 to 26.8 σ_{θ} in the south Indian Ocean, using in situ hydrographic observations, satellite measurements, reanalysis datasets, and numerical model output. The amplitude of spiciness anomalies is about 0.03 psu or 0.13 degreesC and tends to be large along the streamline of the subtropical gyre, whose upstream end is the outcrop region south of Australia. The speed of spreading is comparable to that of the mean current, and it takes about a decade for a spiciness anomaly in the outcrop region to spread into the interior up to Madagascar. In the outcrop region, interannual variability in mixed layer temperature and salinity tends to be density compensating, which indicates that Eulerian temperature or salinity changes account for the generation of isopycnal spiciness anomalies. It is known that wintertime temperature and salinity in the surface mixed layer determine the temperature and salinity relationship of a subducted water mass. Considering this, the mixed layer heat budget in the outcrop region is estimated based on the concept of effective mixed layer depth, the result of which shows the primary contribution from horizontal advection. The contributions from Ekman and geostrophic currents are comparable. Ekman flow advection is caused by zonal wind stress anomalies and the resulting meridional Ekman current anomalies, as is pointed out by a previous study. Geostrophic velocity is decomposed into large-scale and mesoscale variability, both of which significantly contribute to horizontal advection.

758. Carbon Monitoring System Flux Net Biosphere Exchange 2020 (CMS-Flux NBE 2020)

Authors: Junjie Liu; Latha Baskaran; Kevin Bowman; David Schimel; A. Anthony Bloom; Nicholas C. Parazoo; Tomohiro Oda; Dustin Carroll; Dimitris Menemenlis; Joanna Joiner; Roisin Commane; Bruce Daube; Lucianna V. Gatti; Kathryn McKain; John Miller; Britton B. Stephens; Colm Sweeney; Steven Wofsy

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-13-299-2021

URL: <https://doi.org/10.5194/essd-13-299-2021>

Publication Date: 2021-02-10

Volume/Issue: Volume 13, Issue 2

References: 81

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Here we present a global and regionally resolved terrestrial netbiosphere exchange (NBE) dataset with corresponding uncertainties between 2010–2018: Carbon Monitoring System Flux Net Biosphere Exchange 2020 (CMS-Flux NBE 2020). It is estimated using the NASA Carbon Monitoring System Flux (CMS-Flux) top-down flux inversion system that assimilates column CO₂ observations from the Greenhouse Gases Observing Satellite (GOSAT) and NASA's Observing Carbon Observatory 2 (OCO-2). The regional monthly fluxes are readily accessible as tabular files, and the gridded fluxes are available in NetCDF format. The fluxes and their uncertainties are evaluated by extensively comparing the posterior CO₂ mole fractions with CO₂ observations from aircraft and the NOAA marine boundary layer reference sites. We describe the characteristics of the dataset as the global total, regional climatological mean, and regional annual fluxes and seasonal cycles. We find that the global total fluxes of the dataset agree with atmospheric CO₂ growth observed by the surface-observation network within uncertainty. Averaged between 2010 and 2018, the tropical regions range from close to neutral in tropical South America to a net source in Africa; these contrast with the extra-tropics, which are a net sink of 2.5±0.3 Gt C/year. The regional satellite-constrained NBE estimates provide a unique perspective for understanding the terrestrial biosphere carbon dynamics and monitoring changes in regional contributions to the changes of atmospheric CO₂ growth rate. The gridded and regional aggregated dataset can be accessed at <https://doi.org/10.25966/4v02-c391> (Liu et al., 2020).

759. Convergence of Daily GRACE Solutions and Models of Submonthly Ocean Bottom Pressure Variability

Authors: Michael Schindelegger (University of Bonn); Alexander A. Harker (University of Bonn); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Henryk Dobslaw (Department 1: Geodesy GFZ German Research Centre for Geosciences Potsdam Germany); David A. Salstein (Atmospheric and Environmental Research Inc.)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc017031

URL: <https://doi.org/10.1029/2020jc017031>

Publication Date: 2021-02-08

Volume/Issue: Volume 126, Issue 2

References: 66

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: barotropic, GRACE, ocean bottom pressure, time-variable gravity

Funding: Austrian Science Fund, National Aeronautics and Space Administration, Deutsche Forschungsgemeinschaft

License: CC BY 4.0

Open Access: N/A

Abstract: Knowledge of submonthly variability in ocean bottom pressure (pb) is an essential element in space-geodetic analyses and global gravity field research. Estimates of these mass changes are typically drawn from numerical ocean models and, more recently, GRACE (Gravity Recovery and Climate Experiment) series at daily sampling. However, the quality of pb fields from either source has been difficult to assess and reservations persist as to the dependence of regularized GRACE solutions on their oceanographic priors. Here, we make headway on the subject by comparing two daily satellite gravimetry products (years 2007-2009) both with each other and with pb output from a diverse mix of ocean models, complemented by insights from bottom pressure gauges. Emphasis is given to large spatial scales and periods <60 days. Satellite-based mass changes are in good agreement over basin interiors and point to excess pb signals (~2 cm root-mean-square error) over Southern Ocean abyssal plains in the present GRACE de-aliasing model. These and other imperfections in baroclinic models are especially apparent at periods <10 days, although none of the GRACE series presents a realistic ground truth on time scales of a few days. A barotropic model simulation with parameterized topographic wave drag is most commensurate with the GRACE fields over the entire submonthly band, allowing for first-order inferences about error and noise in the gravimetric mass changes. Estimated pb errors vary with signal magnitude and location but are generally low enough (0.5-1.5 cm) to judge model skill in dynamically active regions.

760. Advances in the Estimation of Global Surface Net Heat Flux Based on Satellite Observation: J-OFURO3 V1.1

Authors: Hiroyuki Tomita; Kunio Kutsuwada; Masahisa Kubota; Tsutomu Hihara

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2021.612361

URL: <https://doi.org/10.3389/fmars.2021.612361>

Publication Date: 2021-02-08

Volume/Issue: Volume 8

References: 58

Cited by: 18 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science, Japan Aerospace Exploration Agency

License: CC BY 4.0

Open Access: N/A

Abstract: The reliability of surface net heat flux data obtained from the latest satellite-based estimation [the third-generation Japanese Ocean Flux Data Sets with Use of Remote Sensing Observations (J-OFURO3, V1.1)] was investigated. Three metrics were utilized: (1) the global long-term (30 years) mean for 1988-2017, (2) the local accuracy evaluation based on comparison with observations recorded at buoys located at 11 global oceanic points with varying climatological characteristics, and (3) the physical consistency with the freshwater balance related to the global water cycle. The globally averaged value of the surface net heat flux of J-OFURO3 was -22.2 W m^{-2} , which is largely imbalanced to heat the ocean surface. This imbalance was due to the turbulent heat flux being smaller than the net downward surface radiation. On the other hand, compared with the local buoy observations, the average difference was -5.8 W m^{-2} , indicating good agreement. These results indicate a paradox of the global surface net heat flux. In relation to the global water cycle, the balance between surface latent heat flux (ocean evaporation) and precipitation was estimated to be almost 0 when river runoff from the land was taken into consideration. The reliability of the estimation of the latent heat flux was reconciled by two different methods. Systematic ocean-heating biases by surface sensible heat flux (SHF) and long wave radiation were identified. The bias in the SHF was globally persistent and especially large in the mid- and high latitudes. The correction of the bias has an impact on improving the global mean net heat flux by $+5.5 \text{ W m}^{-2}$. Furthermore, since J-OFURO3 SHF has low data coverage in high-latitudes areas containing sea ice, its impact on global net heat flux was assessed using the latest atmospheric reanalysis product. When including the sea ice region, the globally averaged value of SHF was approximately 1.4 times larger. In addition to the bias correction mentioned above, when assuming that the global ocean average of J3 SHF is 1.4 times larger, the net heat flux value changes to the improved value (-11.3 W m^{-2}), which is approximately half the original value (-22.2 W m^{-2}).

761. Impacts of climate change on methylmercury formation and bioaccumulation in the 21st century ocean

Authors: Yanxu Zhang; Stephanie Dutkiewicz; Elsie M. Sunderland

Journal: One Earth

Publisher: Elsevier BV

DOI: 10.1016/j.oneear.2021.01.005

URL: <https://doi.org/10.1016/j.oneear.2021.01.005>

Publication Date: 2021-02-07

Volume/Issue: Volume 4, Issue 2

References: 49

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

762. Intraseasonal Variations in Atmospheric and Oceanic Excitation of Length-of-Day

Authors: Nan Yu (College of Marine Science and Technology China University of Geosciences Wuhan China); Jim Ray (National Oceanic and Atmospheric Administration Silver Spring MD USA); Jiancheng Li (Wuhan University); Gang Chen (College of Marine Science and Technology China University of Geosciences Wuhan China); Nengfang Chao (College of Marine Science and Technology China University of Geosciences Wuhan China); Wei Chen (Wuhan University)

Journal: Earth and Space Science

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020ea001563

URL: <https://doi.org/10.1029/2020ea001563>

Publication Date: 2021-02-06

Volume/Issue: Volume 8, Issue 3

References: 49

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: atmospheric and oceanic excitations, intraseasonal length-of-day variations, sliding window average filtering, geodetic excitations

Funding: National Natural Science Foundation of China

License: Creative Commons

Open Access: N/A

Abstract: Like the seasonal and interannual variations in length-of-day (LOD), variations on intraseasonal timescales (periods shorter than 183 days) are also predominantly caused by changes in the angular momentum of the zonal winds. But the smaller intraseasonal LOD excitations from atmospheric surface pressure, oceanic currents, and the ocean bottom pressure are less clear. Here sliding window average filtering is applied to isolate intraseasonal signals from both geodetically observed LOD excitation (shorten to geodetic excitation for convenience) and atmospheric and oceanic LOD excitations. Based on careful comparison between these two-excitation series, we find that during 1993-2018, atmospheric winds reduce the RMS of observed intraseasonal LOD series from 276.1 to 55.6 μ s with a correlation coefficient of 0.9795. Including the effect of surface pressure changes with that of the winds reduced the RMS from 55.6 to 43.3 μ s, and increased the correlation coefficient with the observations from 0.9795 to 0.9884. Additionally, including the effects of changes in oceanic currents and bottom pressure further reduced the RMS from 43.3 to 33.9 μ s, and further increased the correlation coefficient from 0.9884 to 0.9931. Thus, although the impact of oceans is relatively minor, closer agreement with the observations in the intraseasonal frequency band is obtained when the effects of oceanic processes are added to that of atmospheric. The RMS values and correlation coefficients of ESMGFZ and MPIOM residual series are better than NCEP and ECCO in every component, indicating that NCEP model set is generally poorer than ESMGFZ in the intraseasonal band.

763. Deep learning to infer eddy heat fluxes from sea surface height patterns of mesoscale turbulence

Authors: Tom M. George; Georgy E. Manucharyan; Andrew F. Thompson

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-020-20779-9

URL: <https://doi.org/10.1038/s41467-020-20779-9>

Publication Date: 2021-02-05

Volume/Issue: Volume 12, Issue 1

References: 74

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Mesoscale eddies have strong signatures in sea surface height (SSH) anomalies that are measured globally through satellite altimetry. However, monitoring the transport of heat associated with these eddies and its impact on the global ocean circulation remains difficult as it requires simultaneous observations of upper-ocean velocity fields and interior temperature and density properties. Here we demonstrate that for quasigeostrophic baroclinic turbulence the eddy patterns in SSH snapshots alone contain sufficient information to estimate the eddy heat fluxes. We use simulations of baroclinic turbulence for the supervised learning of a deep Convolutional Neural Network (CNN) to predict up to 64% of eddy heat flux variance. CNNs also significantly outperform other conventional data-driven techniques. Our results suggest that deep CNNs could provide an effective pathway towards an operational monitoring of eddy heat fluxes using satellite altimetry and other remote sensing products.

764. Leveraging Uncertainty Quantification to Design Ocean Climate Observing Systems

Authors: Nora Loose (University of Texas at Austin); Patrick Heimbach (University of Texas at Austin)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020ms002386

URL: <https://doi.org/10.1029/2020ms002386>

Publication Date: 2021-02-05

Volume/Issue: Volume 13, Issue 4

References: 47

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: observing system design, data assimilation, adjoint model, uncertainty quantification, North Atlantic

Funding: National Science Foundation, National Science Foundation, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Ocean observations are expensive and difficult to collect. Designing effective ocean observing systems therefore warrants deliberate, quantitative strategies. We leverage adjoint modeling and Hessian uncertainty quantification (UQ) within the ECCO (Estimating the Circulation and Climate of the Ocean) framework to explore a new design strategy for ocean climate observing systems. Within this context, an observing system is optimal if it minimizes uncertainty in a set of investigator-defined quantities of interest (QoIs), such as oceanic transports or other key climate indices. We show that Hessian UQ unifies three design concepts. (1) An observing system reduces uncertainty in a target QoI most effectively when it is sensitive to the same dynamical controls as the QoI. The dynamical controls are exposed by the Hessian eigenvector patterns of the model-data misfit function. (2) Orthogonality of the Hessian eigenvectors rigorously accounts for redundancy between distinct members of the observing system. (3) The Hessian eigenvalues determine the overall effectiveness of the observing system, and are controlled by the sensitivity-to-noise ratio of the observational assets (analogous to the statistical signal-to-noise ratio). We illustrate Hessian UQ and its three underlying concepts in a North Atlantic case study. Sea surface temperature observations inform mainly local air-sea fluxes. In contrast, subsurface temperature observations reduce uncertainty over basin-wide scales, and can therefore inform transport QoIs at great distances. This research provides insight into the design of effective observing systems that maximally inform the target QoIs, while being complementary to the existing observational database.

765. Absolute instabilities in the spatially developing Kuroshio Extension

Authors: X. San Liang; Jianyu Hu

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2021.101205

URL: <https://doi.org/10.1016/j.dynatmoce.2021.101205>

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References: 54

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

766. Low-frequency variability of the Pacific Subtropical Cells as reproduced by coupled models and ocean reanalyses

Authors: Giorgio Graffino; Riccardo Farneti; Fred Kucharski

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-021-05639-6

URL: <https://doi.org/10.1007/s00382-021-05639-6>

Publication Date: 2021-01-26

Volume/Issue: Volume 56, Issue 9-10

References: 95

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Low-frequency variability of the Pacific Subtropical Cells (STCs) is investigated using outputs from several models included in the two latest phases of Coupled Model Intercomparison Project (CMIP), CMIP5 and CMIP6, as well as ocean reanalysis products. Our analysis focuses on historical simulations and an idealised future scenario integration. Mass and heat transport diagnostics are employed to assess how coupled models and ocean reanalyses reproduce Pacific STCs total and interior transport convergence at the equator and their relationship with equatorial Pacific sea surface temperature (SST). Trends of mass and heat transport are also evaluated, in order to study how the STCs are expected to change in a warming climate. A large spread is obtained across models in simulated mass transports, confirming that coupled models do not agree on reproducing observed Pacific STCs dynamics, with very limited improvement by CMIP6 models. Compared to ocean reanalysis products, coupled models tend to underestimate the STCs interior transport convergence, and are less efficient on propagating the signal generated by the subtropical wind stress towards the equator. Also, mass transport obtained from ocean reanalyses exhibit larger variability, and these products also better reproduce the STCs-SST relationship. Future scenario simulations suggest a weakening (strengthening) of the heat transport by the North (South) Pacific cell under warmer conditions, with a general agreement across models. Equatorward mass transport trends do not confirm this for total and interior components, but they do for the western boundary component.

767. Drivers of Local Ocean Heat Content Variability in ECCOv4

Authors: Jan-Erik Tesdal (Columbia University); Ryan P. Abernathey (Columbia University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-20-0058.1

URL: <https://doi.org/10.1175/jcli-d-20-0058.1>

Publication Date: 2021-01-26

Volume/Issue: Volume 34, Issue 8

References: 102

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Reanalysis data, Ocean, Interannual variability, Clustering, Oceanic variability (and 2 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Variation in upper-ocean heat content is a critical factor in understanding global climate variability. Using temperature anomaly budgets in a two-decade-long physically consistent ocean state estimate (ECCOv4r3; 1992-2015), we describe the balance between atmospheric forcing and ocean transport mechanisms for different depth horizons and at varying temporal and spatial resolutions. Advection dominates in the tropics, while forcing is most relevant at higher latitudes and in parts of the subtropics, but the balance of dominant processes changes when integrating over greater depths and considering longer time scales. While forcing is shown to increase with coarser resolution, overall the heat budget balance between it and advection is remarkably insensitive to spatial scale. A novel perspective on global ocean heat content variability was made possible by combining unsupervised classification with a measure of temporal variability in heat budget terms to identify coherent dynamical regimes with similar underlying mechanisms, which are consistent with prior research. The vast majority of the ocean includes significant contributions by both forcing and advection. However advection-driven regions were identified that coincide with strong currents, such as western boundary currents, the Antarctic Circumpolar Current, and the tropics, while forcing-driven regions were defined by shallower wintertime mixed layers and weak velocity fields. This identification of comprehensive dynamical regimes and the sensitivity of the ocean heat budget analysis to exact resolution (for different depth horizons and at varying temporal and spatial resolutions) should provide a useful orientation for future studies of ocean heat content variability in specific ocean regions.

768. CYCLOCIM: A 4-D variational assimilation system for the climatological mean seasonal cycle of the ocean circulation

Authors: Qian Huang; François Primeau; Tim DeVries

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2021.101762

URL: <https://doi.org/10.1016/j.ocemod.2021.101762>

Publication Date: 2021-01-23

Volume/Issue: Volume 159

References: 73

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

769. Improving Detectability of Seafloor Deformation From Bottom Pressure Observations Using Numerical Ocean Models

Authors: Yoichiro Dobashi; Daisuke Inazu

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2020.598270

URL: <https://doi.org/10.3389/feart.2020.598270>

Publication Date: 2021-01-15

Volume/Issue: Volume 8

References: 95

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science

License: CC BY 4.0

Open Access: N/A

Abstract: We investigated ocean bottom pressure (OBP) observation data at six plate subduction zones around the Pacific Ocean. The six regions included the Hikurangi Trough, the Nankai Trough, the Japan Trench, the Aleutian Trench, the Cascadia Subduction Zone, and the Chile Trench. For the sake of improving the detectability of seafloor deformation using OBP observations, we used numerical ocean models to represent realistic oceanic variations, and subtracted them from the observed OBP data. The numerical ocean models included four ocean general circulation models (OGCMs) of HYCOM, GLORYS, ECCO2, and JCOPE2M, and a single-layer ocean model (SOM). The OGCMs are mainly driven by the wind forcing. The SOM is driven by wind and/or atmospheric pressure loading. The modeled OBP was subtracted from the observed OBP data, and root-mean-square (RMS) amplitudes of the residual OBP variations at a period of 3-90 days were evaluated by the respective regions and by the respective numerical ocean models. The OGCMs and SOM driven by wind alone (SOMw) contributed to 5-27% RMS reduction in the residual OBP. When SOM driven by atmospheric pressure alone (SOMp) was added to the modeled OBP, residual RMS amplitudes were additionally reduced by 2-15%. This indicates that the atmospheric pressure is necessary to explain substantial amounts of observed OBP variations at the period. The residual RMS amplitudes were 1.0-1.7 hPa when SOMp was added. The RMS reduction was relatively effective as 16-42% at the Hikurangi Trough, the Nankai Trough, and the Japan Trench. The residual RMS amplitudes were relatively small as 1.0-1.1 hPa at the Nankai Trough and the Chile Trench. These results were discussed with previous studies that had identified slow slips using OBP observations. We discussed on further accurate OBP modeling, and on improving detectability of seafloor deformation using OBP observation arrays.

770. Evaluation of global high-resolution reanalysis products based on the Chinese global oceanography forecasting system

Authors: Zhijie Li; Zhaoyi Wang; Yun Li; Yu Zhang; Jingjing Zheng; Shan Gao

Journal: Atmospheric and Oceanic Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.aosl.2021.100032

URL: <https://doi.org/10.1016/j.aosl.2021.100032>

Publication Date: 2021-01-14

Volume/Issue: Volume 14, Issue 5

References: 25

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

771. Analysis of Hydroacoustic Signals Associated to the Loss of the Argentinian ARA San Juan Submarine

Authors: Julien Vergoz; Yves Cansi; Yoann Cano; Pierre Gaillard

Journal: Pure and Applied Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00024-020-02625-7

URL: <https://doi.org/10.1007/s00024-020-02625-7>

Publication Date: 2021-01-08

Volume/Issue: Volume 178, Issue 7

References: 31

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: On November 15, 2017, an event related to the disappearance of the Argentine military submarine ARA San Juan was detected by two hydrophone triplet stations of the IMS network, established to enforce the nuclear test ban treaty (CTBT). From the two direct hydroacoustic arrivals recorded at 6000 and 8000 km from the localized wreckage, calculated location based on hydroacoustic data only is poorly constrained, and the associated uncertainties are large. In an attempt to interpret the recorded signals, an air dropped calibration grenade was conducted by the Argentine Navy two weeks later, on December 1, 2017, near the last known position of the submarine. From the comparison of temporal and spectral features of both events, we confirm the impulsive nature of the San Juan event. Array processing was performed with a progressive multi-channel correlation method (PMCC). Fine propagation details of direct arrivals are very well resolved in time-frequency space and thirteen secondary arrivals are revealed for the San Juan event, within the fifteen minutes following direct arrivals. The detections presented in this paper were calculated with DTK-PMCC software embedded in the NDC-In-A-Box virtual machine, and can be reproduced by any CTBTO principal user (Member State user which can access raw waveform data and data bulletins). All the identified late arrivals are associated to reflections or refractions from seamounts, islands and the South American continental Slope. The accurate identification of all the reflectors allows to significantly improve the source location accuracy: 95% confidence ellipse area has been reduced by a factor of 100 compared to location obtained from direct arrivals only, and the estimated location is 3.5 km from the known location of the wreckage. The originality of the relocation method is that it is based on the joint inversion of both San Juan and calibration events unknown parameters, and from the selection of only a well-chosen subset of secondary arrivals. Its calculation did not require either the need of advanced oceanographic specifications, or sophisticated methods requiring heavy computational means. Finally, a detailed cepstral analysis of the direct and secondary arrivals has allowed to detect the existence of a second impulse (doublet) in the signals associated to both San Juan and calibration events. Unlike the calibration event, the anisotropic character of the delays measured from the San Juan cepstra suggests that the 15 November signal was generated by two impulsive acoustic sources closely separated in

space and time over scales comparable to the size of the submarine. This study demonstrates the capability of the hydroacoustic component of the IMS network to accomplish its mission of Comprehensive Nuclear-Test-Ban Treaty monitoring.

772. The ENEA-REG system (v1.0), a multi-component regional earth system model. Sensitivity to different atmospheric component over Med-CORDEX region

Authors: Alessandro Anav; Adriana Carillo; Massimiliano Palma; Maria Vittoria Struglia; Ufuk Utku Turuncoglu; Gianmaria Sannino

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2020-248

URL: <https://doi.org/10.5194/gmd-2020-248>

Publication Date: 2021-01-08

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Hindcast|Earth system science|Component (thermodynamics)|Atmospheric models

Funding: National Center for Atmospheric Research

License: CC BY 4.0

Open Access: N/A

Abstract: . In this study, a new regional Earth system model is developed and applied to the Med-CORDEX region. The ENEA-REG system is made up of two interchangeable regional climate models as atmospheric components (RegCM and WRF), a river model (HD), and an ocean model (MITgcm); processes taking place at the land surface are represented within the atmospheric models with the possibility to use several land surface schemes of different complexity. The coupling between these components is performed through the RegESM driver. Here, we present and describe our regional Earth system model and evaluate its components using a multidecadal hindcast simulation over the period 1980-2013 driven by ERA-INTERIM reanalysis. We show how the atmospheric components are able to correctly reproduce both large-scale and local features of the Euro-Mediterranean climate, although some remarkable biases are relevant for some variables. In particular, WRF has a significant cold bias during winter over North-Eastern bound of the domain, while RegCM systematically overestimates the wind speed over the Mediterranean Sea. This latter bias has severe consequences on the ocean component: we show that when WRF is used as the atmospheric component of the Earth system, the performances of the ocean model are remarkably better compared with the RegCM version. Our regional Earth system model allows studying the Euro-Mediterranean climate system and can be applied to both hindcast and scenario simulations.

773. Ocean state estimations for synthesis of ocean-mixing observations

Authors: Shuhei Masuda; Satoshi Osafune

Journal: Journal of Oceanography

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10872-020-00587-x

URL: <https://doi.org/10.1007/s10872-020-00587-x>

Publication Date: 2021-01-07

Volume/Issue: Volume 77, Issue 3

References: 59

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science

License: CC BY 4.0

Open Access: N/A

Abstract: Vertical mixing in oceans is an essential component of the dynamics of ocean circulation, including meridional circulation. Nevertheless, various aspects of mixing, particularly in conjunction with global ocean energetics, remain debatable. One of the biggest reasons is the lack of observational facts. With the recent expansion of global vertical-mixing observations, attempts have been made to estimate the ocean state using vertical-mixing observation data to better understand the role of mixing in oceanography. In this review, we discuss the current status of the ocean state estimation and future synthesis of vertically mixing observation data into the oceanic basin-scale state estimation, including progress of data assimilation studies using numerical models. These will contribute to the construction of the future line of observation, model, and data synthesis studies along which the issues on ocean mixing can be consistently resolved.

774. Quantifying the Role of Ocean Dynamics in Ocean Mixed Layer Temperature Variability

Authors: Casey R. Patrizio (Colorado State University); David W. J. Thompson (Colorado State University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-20-0476.1

URL: <https://doi.org/10.1175/jcli-d-20-0476.1>

Publication Date: 2021-01-07

Volume/Issue: Volume 34, Issue 7

References: 170

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: Mixed layer|Ocean dynamics, Interdecadal variability, Interannual variability, North pacific ocean, Climate variability (and 3 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Understanding the role of the ocean in climate variability requires first understanding the role of ocean dynamics in the ocean mixed layer and thus sea surface temperature variability. However, key aspects of the spatially and temporally varying contributions of ocean dynamics to such variability remain unclear. Here, the authors quantify the contributions of ocean dynamical processes to mixed layer temperature variability on monthly to multiannual time scales across the globe. To do so, they use two complementary but distinct methods: 1) a method in which ocean heat transport is estimated directly from a state-of-the-art ocean state estimate spanning 1992-2015 and 2) a method in which it is estimated indirectly from observations between 1980-2017 and the energy budget of the mixed layer. The results extend previous studies by providing quantitative estimates of the role of ocean dynamics in mixed layer temperature variability throughout the globe, across a range of time scales, in a range of available measurements, and using two different methods. Consistent with previous studies, both methods indicate that the ocean-dynamical contribution to mixed layer temperature variance is largest over western boundary currents, their eastward extensions, and regions of equatorial upwelling. In contrast to previous studies, the results suggest that ocean dynamics reduce the variance of Northern Hemisphere mixed layer temperatures on time scales longer than a few years. Hence, in the global mean, the fractional contribution of ocean dynamics to mixed layer temperature variability decreases at increasingly low frequencies. Differences in the magnitude of the ocean dynamical contribution based on the two methods highlight the critical need for improved and continuous observations of the ocean mixed layer.

775. Evaluation of sea-ice thickness from four reanalyses in the Antarctic Weddell Sea

Authors: Qian Shi; Qinghua Yang; Longjiang Mu; Jinfei Wang; François Massonnet; Matthew R. Mazloff

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-31-2021

URL: <https://doi.org/10.5194/tc-15-31-2021>

Publication Date: 2021-01-05

Volume/Issue: Volume 15, Issue 1

References: 68

Cited by: 14 publications

Language: en

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Keywords: N/A

Funding: National Natural Science Foundation of China, Fonds De La Recherche Scientifique - FNRS, National Science Foundation (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . Ocean-sea-ice coupled models constrained by various observations provide different ice thickness estimates in the Antarctic. We evaluate contemporary monthly ice thickness from four reanalyses in the Weddell Sea: the German contribution of the project Estimating the Circulation and Climate of the Ocean Version 2 (GECCO2), the Southern Ocean State Estimate (SOSE), the Ensemble Kalman Filter system based on the Nucleus for European Modelling of the Ocean (NEMO-EnKF) and the Global Ice-Ocean Modeling and Assimilation System (GIOMAS). The evaluation is performed against reference satellite and in situ observations from ICESat-1, Envisat, upward-looking sonars and visual ship-based sea-ice observations. Compared with ICESat-1, NEMO-EnKF has the highest correlation coefficient (CC) of 0.54 and lowest root mean square error (RMSE) of 0.44 m. Compared with in situ observations, SOSE has the highest CC of 0.77 and lowest RMSE of 0.72 m. All reanalyses underestimate ice thickness near the coast of the western Weddell Sea with respect to ICESat-1 and in situ observations even though these observational estimates may be biased low. GECCO2 and NEMO-EnKF reproduce the seasonal variation in first-year ice thickness reasonably well in the eastern Weddell Sea. In contrast, GIOMAS ice thickness performs best in the central Weddell Sea, while SOSE ice thickness agrees most with the observations from the southern coast of the Weddell Sea. In addition, only NEMO-EnKF can reproduce the seasonal evolution of the large-scale spatial distribution of ice thickness, characterized by the thick ice shifting from the southwestern and western Weddell Sea in summer to the western and northwestern Weddell Sea in spring. We infer that the thick ice distribution is correlated with its better simulation of northward ice motion in the western Weddell Sea. These results demonstrate the possibilities and limitations of using current sea-ice reanalysis for understanding the recent variability of sea-ice volume in the Antarctic.

776. Barotropic and Baroclinic Inverse Kinetic Energy Cascade in the Antarctic Circumpolar Current

Authors: Hongjie Li (Chinese Academy of Sciences, c Department of mathematics and statistics, Ludong University, Yantai, China); Yongsheng Xu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0053.1

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Language: N/A

Subjects: N/A

Keywords: Baroclinic flows, Antarctica, Jets, Fluxes, Barotropic flows (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Stratified geostrophic turbulence theory predicts an inverse energy cascade for the barotropic (BT) mode. Satellite altimetry has revealed a net inverse cascade in the baroclinic (BC) mode. Here the spatial variabilities of BT and BC kinetic energy fluxes in the Antarctic Circumpolar Current (ACC) were investigated using ECCO2 data, which synthesize satellite data and in situ measurements with an eddy-permitting general circulation model containing realistic bathymetry and wind forcing. The BT and BC inverse kinetic energy cascades both reveal complex spatial variations that could not be explained fully by classical arguments. For example, the BC injection scales match better with most unstable scales than with the first-mode deformation scales, but the opposite is true for the BT mode. In addition, the BT and BC arrest scales do not follow the Rhines scale well in terms of spatial variation, but show better consistency with their own energy-containing scales. The reverse cascade of the BT and BC modes was found related to their EKE, and better correlation was found between the BT inverse cascade and barotropization. Speculations of the findings were proposed; however, further observations and modeling experiments are needed to test these interpretations. Spectral flux anisotropy exhibits a feature associated with oceanic jets that is consistent with classical expectations. Specifically, the spectral flux along the along-stream direction remains negative at scales up to that of the studied domain (~2000 km), while that in the perpendicular direction becomes positive close to the scale of the width of a typical jet.

777. Effects of eddies on the subduction and movement of water masses reaching the 137°E section using Lagrangian particles in an eddy-resolving OGCM

Authors: Hideyuki Nakano; Yoshimasa Matsumura; Hiroyuki Tsujino; Shogo Urakawa; Kei Sakamoto; Takahiro Toyoda; Goro Yamanaka

Journal: Journal of Oceanography

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10872-020-00573-3

URL: <https://doi.org/10.1007/s10872-020-00573-3>

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Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science, Japan Society for the Promotion of Science

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Open Access: N/A

Abstract: N/A

778. Ocean forcing drives glacier retreat in Greenland

Authors: Michael Wood (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lu An (University of California Irvine); Anders Bjørk (University of Copenhagen); Michiel van den Broeke (Utrecht University); Cilan Cai (University of California Irvine); Emily Kane (University of California Irvine); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Romain Millan (University of Grenoble Alpes, Institut de Recherche pour le Développement (IRD)); Mathieu Morlighem (University of California Irvine); Jeremie Mouginot (University of Grenoble Alpes, University of California Irvine, Institut de Recherche pour le Développement (IRD)); Brice Noël (Utrecht University); Bernd Scheuchl (University of California Irvine); Isabella Velicogna (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

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URL: <https://doi.org/10.1126/sciadv.aba7282>

Publication Date: 2021-01-01

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Language: en

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Keywords: N/A

Funding: National Science Foundation, Jet Propulsion Laboratory, Netherlands Earth System Science Centre (and 6 more)

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Open Access: N/A

Abstract: Many Greenland glaciers have been attacked by the ocean from below and, in turn, contributed to sea-level rise from the ice sheet.

779. Ocean melting of the Zachariae Isstrøm and Nioghalvfjærdsfjorden glaciers, northeast Greenland

Authors: Lu An (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael Wood (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jérémie Mouginot (University of Grenoble Alpes, University of California Irvine, Institut de Recherche pour le Développement (IRD)); Shfaqat A. Khan (Technical University of Denmark)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

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Cited by: 35 publications

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Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: Significance Oceanography and gravity data of northeastern Greenland reveal ocean temperature and bathymetry in front of Zachariae Isstrøm (ZI) and Nioghalvfjærdsfjorden (79N), which hold a 1.1-m sea level rise ice volume equivalent but underwent different evolutions. Subsurface, warm, salty water of Atlantic origin has easier access to ZI than to 79N because of bathymetric barriers. We reconstruct ice removal by the ocean at the grounding line and flotation retreat from thinning to explain the observed grounding line retreat since 1979. The agreement between observed and calculated retreat demonstrates that ocean thermal forcing is a major control on the glacier evolution, and undercutting of grounded ice must be included in numerical ice sheet models to reproduce the high rates of retreat.

780. The Sloshing and Diapycnal Meridional Overturning Circulations in the Indian Ocean

Authors: Lei Han (b Laboratory for Regional Oceanography and Numerical Modeling, National Laboratory for Marine Science and Technology, Qingdao, China, a Institute of Marine Science and Technology, Shandong University, Qingdao, China)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0211.1

URL: <https://doi.org/10.1175/jpo-d-20-0211.1>

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Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: Seasonal variability, Ekman pumping/transport, Diapycnal mixing, Indian Ocean, Vertical motion (and 1 more)

Funding: National Basic Research Program of China, Fundamental Research Funds of Shandong University

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Open Access: N/A

Abstract: The meridional overturning circulation (MOC) seasonality in the Indian Ocean is investigated with the ocean state estimate product ECCO v4r3. The vertical movements of water parcels are predominantly due to the heaving of the isopycnals all over the basin except off the western coast. Aided by the linear propagation equation of long baroclinic Rossby waves, the driving factor determining the strength of the seasonal MOC in the Indian Ocean is identified as the zonally integrated Ekman pumping anomaly, rather than the Ekman transport concluded in earlier studies. A new concept of sloshing MOC is proposed, and its difference with the classic Eulerian MOC leads to the so-called diapycnal MOC. The striking resemblance of the Eulerian and sloshing MOCs implies the seasonal variation of the Eulerian MOC in the Indian Ocean is a sloshing mode. The shallow overturning cells manifest themselves in the diapycnal MOC as the most remarkable structure. New perspectives on the upwelling branch of the shallow overturn in the Indian Ocean are offered based on diapycnal vertical velocity. The discrepancy among the observation-based estimates on the bottom inflow across 32 degreesS of the basin is interpreted with the seasonal sloshing mode. Consequently, the "missing mixing" in the deep Indian Ocean is attributed to the overestimated diapycnal volume fluxes. Decomposition of meridional heat transport (MHT) into sloshing and diapycnal components clearly shows the dominant mechanism of MHT in the Indian Ocean in various seasons.

781. The surface salinity maximum of the South Atlantic

Authors: N. Aubone; E.D. Palma; A.R. Piola

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2020.102499

URL: <https://doi.org/10.1016/j.pocean.2020.102499>

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Volume/Issue: Volume 191

References: 55

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Ministerio de Ciencia, Tecnología e Innovación Productiva, Agencia Nacional de Promoción Científica y Tecnológica (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

782. Past, Present, and Future Pacific Sea-Level Change

Authors: B. D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); T. Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); P. R. Thompson (University of Hawaii at Manoa); J. K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); R. S. Nerem (Cooperative Institute for Research in Environmental Sciences University of Colorado at Boulder Boulder CO USA); J. T. Fasullo (Cooperative Institute for Research in Environmental Sciences University of Colorado at Boulder Boulder CO USA)

Journal: Earth's Future

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020ef001839

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Publication Date: 2020-12-24

Volume/Issue: Volume 9, Issue 4

References: 45

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: trends, sea-level, satellite altimetry

Funding: N/A

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Open Access: N/A

Abstract: Satellite altimeters have provided near-global coverage of the ocean with a continuous record now approaching 3 decades. These observations have led to definitive evidence of an increase in global mean sea-level, while offering a depiction of the spatial variability in sea-level change. As a result of the increasing length of the altimeter record, studies have sought to understand and detect the emergence of a forced or anthropogenic response in sea-level. The central idea behind these studies is that the altimeter record is now long enough that the influence of interannual to decadal variability is reduced and does not obscure underlying regional trends in the way that it did for the early part of the altimeter record. Two locations--the western tropical Pacific and the northeastern Pacific off the west coast of the United States--are noteworthy for their sea-level variations during the altimeter era, exhibiting large departures from the global average albeit in opposite directions. In this study, we examine satellite altimeter-measured sea-level in these two regions. The goals are to understand the evolution of the sea-level trend, and determine what can be inferred about future sea-level change from the satellite altimeter record. We find that in the two locations considered here the altimetry record is still too short to rule out natural variability contributions to the trend on the order of 1 mm/yr. Using longer records from tide-gauge observations and models, an assessment of the potential future changes in both regions is made for both regions.

783. Evaluation of Arctic Sea-ice Cover and Thickness Simulated by MITgcm

Authors: Fei Zheng; Yue Sun; Qinghua Yang; Longjiang Mu

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-020-9223-6

URL: <https://doi.org/10.1007/s00376-020-9223-6>

Publication Date: 2020-12-23

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References: 59

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

784. New Circulation Features in the Okhotsk Sea from a Numerical Model

Authors: P. A. Fayman; S. V. Prants; M. V. Budyansky; M. Yu. Uleysky

Journal: Izvestiya, Atmospheric and Oceanic Physics

Publisher: Pleiades Publishing Ltd

DOI: 10.1134/s0001433820060043

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Publication Date: 2020-12-23

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References: 44

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: Okhotsk Sea, Numerical circulation model RIAMOM, monthly-averaged volume transport of the main currents, seasonal variability of circulation

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

785. Is the Surface Salinity Difference between the Atlantic and Indo-Pacific a Signature of the Atlantic Meridional Overturning Circulation?

Authors: Johan Nilsson (Stockholm University); David Ferreira (University of Reading); Tapio Schneider (California Institute of Technology); Robert C. J. Wills (University of Washington)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0126.1

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Cited by: 3 publications

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Subjects: N/A

Keywords: Temperature salinity diagrams

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License: N/A

Open Access: N/A

Abstract: The high Atlantic surface salinity has sometimes been interpreted as a signature of the Atlantic meridional overturning circulation and an associated salt advection feedback. Here, the role of oceanic and atmospheric processes for creating the surface salinity difference between the Atlantic and Indo-Pacific is examined using observations and a conceptual model. In each basin, zonally averaged data are represented in diagrams relating net evaporation and surface salinity S . The data-pair curves in the $-S$ plane share common features in both basins. However, the slopes of the curves are generally smaller in the Atlantic than in the Indo-Pacific, indicating a weaker sensitivity of the Atlantic surface salinity to net evaporation variations. To interpret these observations, a conceptual advective-diffusive model of the upper-ocean salinity is constructed. Notably, the $-S$ relations can be qualitatively reproduced with only meridional diffusive salt transport. In this limit, the interbasin difference in salinity is caused by the spatial structure of net evaporation, which in the Indo-Pacific oceans contains lower meridional wavenumbers that are weakly damped by the diffusive transport. The observed Atlantic $-S$ relationship at the surface reveals no clear influence of northward advection associated with the meridional overturning circulation; however, a signature of northward advection emerges in the relationship when the salinity is vertically averaged over the upper kilometer. The results indicate that the zonal-mean near-surface salinity is shaped primarily by the spatial pattern of net evaporation and the diffusive meridional salt transport due to wind-driven gyres and mesoscale ocean eddies, rather than by salt advection within the meridional overturning circulation.

786. Modeling the breakup of tabular icebergs

Authors: Mark R. England (Scripps Institution of Oceanography / University of California San Diego, University of North Carolina Wilmington); Till J. W. Wagner (University of North Carolina Wilmington); Ian Eisenman (Scripps Institution of Oceanography / University of California San Diego)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

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Publication Date: 2020-12-16

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Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation

License: CC BY-NC 4.0

Open Access: N/A

Abstract: New model of giant Antarctic icebergs produces simulations that match satellite observations, with implications for climate.

787. Interannual variations of degree 2 from geodetic observations and surface processes

Authors: S Rosat (Université de Strasbourg, CNRS, EOST, IPGS UMR 7516, F-67000 Strasbourg, France); N Gillet (Institut de Recherche pour le Développement (IRD)); J-P Boy (Université de Strasbourg, CNRS, EOST, IPGS UMR 7516, F-67000 Strasbourg, France); A Couhert (Centre National d'Etudes Spatiales, 18 avenue Edouard Belin, 31401 Toulouse, France); M Dumberry (University of Alberta)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Summary Geodetic observations from space continuously record surface deformation and global mass redistribution with an increasing accuracy. In parallel, surficial processes (oceanic, atmospheric, and hydrological loading) are more and more precisely modeled. We propose a confrontation of the geodetic Global Positioning System (GPS) and gravity-field satellite laser ranging (SLR) observations at decadal and interannual time scales, in terms of resolution, correlation and comparison with surficial loading models. We focus on the largest global scale signals of degree 2. At interannual periods, surface deformations retrieved from GPS time-series do not exceed 0.8 mm. Our analysis does not reveal the presence of a dominant signal at a specific period, except perhaps for a signal of approximately 3 yr likely connected to the loading response to El Nio / Southern Oscillations. Contrary to the results of previous studies, we do not find in GPS time-series a clear 6-yr oscillation associated with a degree-2 order-2 pattern. Interannual variations in the degree-2 Stokes coefficients of the gravity field do not exceed 2×10^{-11} . We do not detect a dominant gravity signal at one specific period but instead a broad spectrum of frequencies. The comparison between the degree 2 deformations built from GPS time-series with a prediction from SLR derived gravity variations reveals some correlations, though their differences remain important. This highlights the present day limitations of these techniques in their ability to characterize global scale interannual variations. Hydrological loading models show some correlations with both GPS and SLR signals, but we cannot firmly establish that continental hydrology is dominantly responsible for the observed variations. Given the current limits in the resolution of both gravity and surface deformation and in the modelling of surface processes, we conclude that it will be a challenge to retrieve a geodetic signal of sub-decadal period originating in the Earth's core.

788. Spatial and Temporal Characteristics of the Submesoscale Energetics in the Gulf of Mexico

Authors: Yang Yang (Nanjing University, University of South Florida); James C. McWilliams (Nanjing University, University of California Los Angeles); X. San Liang (Nanjing University); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Robert H. Weisberg (University of South Florida); Yonggang Liu (University of South Florida); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Cited by: 18 publications

Language: N/A

Subjects: N/A

Keywords: Small scale processes, Eddies, Instability, Frontogenesis/frontolysis

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License: N/A

Open Access: N/A

Abstract: The submesoscale energetics of the eastern Gulf of Mexico (GoM) are diagnosed using outputs from a 1/48 degrees MITgcm simulation. Employed is a recently developed, localized multiscale energetics formalism with three temporal-scale ranges (or scale windows), namely, a background flow window, a mesoscale window, and a submesoscale window. It is found that the energy cascades are highly inhomogeneous in space. Over the eastern continental slope of the Campeche Bank, the submesoscale eddies are generated via barotropic instability, with forward cascades of kinetic energy (KE) following a weak seasonal variation. In the deep basin of the eastern GoM, the submesoscale KE exhibits a seasonal cycle, peaking in winter, maintained via baroclinic instability, with forward available potential energy (APE) cascades in the mixed layer, followed by a strong buoyancy conversion. A spatially coherent pool of inverse KE cascade is found to extract energy from the submesoscale KE reservoir in this region to replenish the background flow. The northern GoM features the strongest submesoscale signals with a similar seasonality as seen in the deep basin. The dominant source for the submesoscale KE during winter is from buoyancy conversion and also from the forward KE cascades from mesoscale processes. To maintain the balance, the excess submesoscale KE must be dissipated by smaller-scale processes via a forward cascade, implying a direct route to finescale dissipation. Our results highlight that the role of submesoscale turbulence in the ocean energy cycle is region and time dependent.

789. Truncated Singular Value Decomposition Regularization for Estimating Terrestrial Water Storage Changes Using GPS: A Case Study over Taiwan

Authors: Yen Ru Lai (Ohio State University); Lei Wang (Ohio State University); Michael Bevis (Ohio State University); Hok Sum Fok (Wuhan University); Abdullah Alanazi (Ohio State University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs12233861

URL: <https://doi.org/10.3390/rs12233861>

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Volume/Issue: Volume 12, Issue 23

References: 47

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: It is a typical ill-conditioned problem to invert GPS-measured loading deformations for terrestrial water storage (TWS) changes. While previous studies commonly applied the 2nd-order Tikhonov regularization, we demonstrate the truncated singular value decomposition (TSVD) regularization can also be applied to solve the inversion problem. Given the fact that a regularized estimate is always biased, it is valuable to obtain estimates with different methods for better assessing the uncertainty in the solution. We also show the general cross validation (GCV) can be applied to select the truncation term for the TSVD regularization, producing a solution that minimizes predictive mean-square errors. Analyzing decade-long GPS position time series over Taiwan, we apply the TSVD regularization to estimate mean annual TWS variations for Taiwan. Our results show that the TSVD estimates can sufficiently fit the GPS-measured annual displacements, resulting in randomly distributed displacement residuals with a zero mean and small standard deviation (around 0.1 cm). On the island-wide scale, the GPS-inferred annual TWS variation is consistent with the general seasonal cycle of precipitations. However, on smaller spatial scales, we observe significant differences between the TWS changes estimated by GPS and simulated by GLDAS land surface models in terms of spatiotemporal pattern and magnitude. Based on the results, we discuss some challenges in the characterization of TWS variations using GPS observations over Taiwan.

790. Observed Transport Variability of the Atlantic Subtropical Cells and Their Connection to Tropical Sea Surface Temperature Variability

Authors: Franz Philip Tuchen (GEOMAR Helmholtz Centre for Ocean Research); Joke F. Lübbecke (Faculty of Mathematics and Natural Sciences Kiel University Kiel Germany, GEOMAR Helmholtz Centre for Ocean Research); Peter Brandt (Faculty of Mathematics and Natural Sciences Kiel University Kiel Germany, GEOMAR Helmholtz Centre for Ocean Research); Yao Fu (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016592

URL: <https://doi.org/10.1029/2020jc016592>

Publication Date: 2020-11-22

Volume/Issue: Volume 125, Issue 12

References: 45

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The shallow meridional overturning cells of the Atlantic Ocean, the subtropical cells (STCs), consist of poleward Ekman transport at the surface, subduction in the subtropics, equatorward flow at thermocline level and upwelling along the equator and at the eastern boundary. In this study, we provide the first observational estimate of transport variability associated with the horizontal branches of the Atlantic STCs in both hemispheres based on Argo float data and supplemented by reanalysis products. Thermocline layer transport convergence and surface layer transport divergence between 10 degreesN and 10 degreesS are dominated by seasonal variability. Meridional thermocline layer transport anomalies at the western boundary and in the interior basin are anti-correlated and partially compensate each other at all resolved time scales. It is suggested that the seesaw-like relation is forced by the large-scale off-equatorial wind stress changes through low-baroclinic-mode Rossby wave adjustment. We further show that anomalies of the thermocline layer interior transport convergence modulate sea surface temperature (SST) variability in the upwelling regions along the equator and at the eastern boundary at time scales longer than 5 years. Phases of weaker (stronger) interior transport are associated with phases of higher (lower) equatorial SST. At these time scales, STC transport variability is forced by off-equatorial wind stress changes, especially by those in the southern hemisphere. At shorter time scales, equatorial SST anomalies are, instead, mainly forced by local changes of zonal wind stress.

791. The impact of vertical resolution in reducing biases in sea surface temperature in a tropical Pacific Ocean model

Authors: Yanli Jia; Kelvin J. Richards; H. Annamalai

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2020.101722

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References: 41

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration

License: N/A

Open Access: N/A

Abstract: N/A

792. Optimal 3D time-energy trajectory planning for AUVs using ocean general circulation models

Authors: Sultan Albarakati; Ricardo M. Lima; Thomas Theußl; Ibrahim Hoteit; Omar M. Knio

Journal: Ocean Engineering

Publisher: Elsevier BV

DOI: 10.1016/j.oceaneng.2020.108057

URL: <https://doi.org/10.1016/j.oceaneng.2020.108057>

Publication Date: 2020-11-01

Volume/Issue: Volume 218

References: 53

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: King Abdullah University of Science and Technology

License: N/A

Open Access: N/A

Abstract: N/A

793. Bathymetric Influences on Antarctic Ice-Shelf Melt Rates

Authors: D. N. Goldberg (University of Edinburgh); T. A. Smith (University of Texas at Austin); S. H. K. Narayanan (Argonne National Laboratory); P. Heimbach (University of Texas at Austin); M. Morlighem (University of California Irvine)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016370

URL: <https://doi.org/10.1029/2020jc016370>

Publication Date: 2020-10-31

Volume/Issue: Volume 125, Issue 11

References: 88

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: ice-ocean interaction, Ocean model, Ice shelf, Adjoint, Bathymetry

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean bathymetry exerts a strong control on ice sheet-ocean interactions within Antarctic ice-shelf cavities, where it can limit the access of warm, dense water at depth to the underside of floating ice shelves. However, ocean bathymetry is challenging to measure within or close to ice-shelf cavities. It remains unclear how uncertainty in existing bathymetry datasets affect simulated sub-ice-shelf melt rates. Here we infer linear sensitivities of ice-shelf melt rates to bathymetric shape with grid-scale detail by means of the adjoint of an ocean general circulation model. Both idealized and realistic-geometry experiments of sub-ice-shelf cavities in West Antarctica reveal that bathymetry has a strong impact on melt in localized regions such as topographic obstacles to flow. Moreover, response of melt to bathymetric perturbation is found to be non-monotonic, with deepening leading to either increased or decreased melt depending on location. Our computational approach provides a comprehensive way of identifying regions, where refined knowledge of bathymetry is most impactful, and also where bathymetric errors have relatively little effect on modeled ice sheet-ocean interactions.

794. A Comparison between Coastal Altimetry Data and Tidal Gauge Measurements in the Gulf of Genoa (NW Mediterranean Sea)

Authors: Paola Picco (Istituto Idrografico della Marina, passo dell'Osservatorio, 4, 16135 Genova, Italy); Stefano Vignudelli (Institute of Biophysics, National Research Council, CNR, via Moruzzi 1, 56124 Pisa, Italy); Luca Repetti (Istituto Idrografico della Marina, passo dell'Osservatorio, 4, 16135 Genova, Italy)

Journal: Journal of Marine Science and Engineering

Publisher: MDPI AG

DOI: 10.3390/jmse8110862

URL: <https://doi.org/10.3390/jmse8110862>

Publication Date: 2020-10-31

Volume/Issue: Volume 8, Issue 11

References: 53

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Satellite altimetry data from X-TRACK products were analyzed for an overall assessment of their capability to detect coastal sea level variability in the Ligurian Sea. Near-coastal altimetry data, collected from 2009 to 2016 along track n.044, were compared with simultaneous high frequency sampled data at the tidal station in Genoa (NW Mediterranean Sea). The two time series show a very good agreement: correlation between total sea level elevation from the altimeter and sea level variation from the tidal gauge is 0.92 and root mean square difference is 4.5 cm. Some relevant mismatches can be ascribed to the local high frequency coastal variability due to shelf and harbor oscillation detected at the tidal station, which might not be observed at the location of the altimetry points of measurement. The analysis evidences discrepancies (root mean square difference of 4.7 cm) between model results for open sea tides and harmonic analysis at the tidal station, mainly occurring at the annual and semiannual period. On the contrary, the important part of dynamic atmospheric correction due to the inverse barometer effect, well agrees with that computed at the tidal station.

795. Hydrodynamic Modeling of Vertical Velocities in the Lofoten Vortex

Authors: A. V. Koldunov; T. V. Belonenko

Journal: Izvestiya, Atmospheric and Oceanic Physics

Publisher: Pleiades Publishing Ltd

DOI: 10.1134/s0001433820040040

URL: <https://doi.org/10.1134/s0001433820040040>

Publication Date: 2020-10-27

Volume/Issue: Volume 56, Issue 5

References: 34

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

796. Multi-year ocean thermal variability

Authors: Carl Wunsch

Journal: Tellus A: Dynamic Meteorology and Oceanography

Publisher: Stockholm University Press

DOI: 10.1080/16000870.2020.1824485

URL: <https://doi.org/10.1080/16000870.2020.1824485>

Publication Date: 2020-10-24

Volume/Issue: Volume 72, Issue 1

References: 36

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

License: CC BY-NC 4.0

Open Access: N/A

Abstract: N/A

797. Evaluation of multi-model current data in the East/Japan Sea

Authors: Haodi Wang; Shiyao Chen; Ning Wang; Peilong Yu; Xiao Yang; Yang Wang; Yongchui Zhang

Journal: 2020 IEEE 3rd International Conference on Information Communication and Signal Processing (ICICSP)

Publisher: IEEE

DOI: 10.1109/icicsp50920.2020.9232090

URL: <https://doi.org/10.1109/icicsp50920.2020.9232090>

Publication Date: 2020-10-21

Volume/Issue: N/A

References: 14

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Hindcast|Drifter|Buoy|Mooring|Interpolation|Tide gauge

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

798. On the Surface Current Measurement Capabilities of Spaceborne Doppler Scatterometry

Authors: Alexander Wineteer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ernesto Rodriguez (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl090116

URL: <https://doi.org/10.1029/2020gl090116>

Publication Date: 2020-10-15

Volume/Issue: Volume 47, Issue 21

References: 44

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: Wide-swath spaceborne Doppler scatterometry is a promising technique for the simultaneous measurement of global ocean surface winds and currents. The technique has been proven from airborne platforms, and here we use the lessons learned to examine a range of implications for a spaceborne system. We use a Doppler scatterometer simulator and a state-of-the-art global circulation model to generate surface current measurements and their random errors. We find that a feasible instrument could measure 5 km gridded surface currents with typical random errors between 10 and 25 cm/s. For higher wind speeds, the random error in surface current decreases logarithmically. This level of accuracy allows for the computation of surface current relative vorticity and horizontal divergence with typical wavelength resolutions of 15-30 and 25-60 km, respectively. Unlike previous studies, we find that these measurements do not require multiday averaging, opening up new avenues for monitoring global ocean circulation.

799. Atmospheric Warming Drives Growth in Arctic Sea Ice: A Key Role for Snow

Authors: A. Bigdeli (University of Texas at Austin); A. T. Nguyen (University of Texas at Austin); H. R. Pillar (University of Texas at Austin); V. Ocaña (University of Texas at Austin); P. Heimbach (University of Texas at Austin)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl090236

URL: <https://doi.org/10.1029/2020gl090236>

Publication Date: 2020-10-14

Volume/Issue: Volume 47, Issue 20

References: 58

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: sea-ice growth, Arctic sea-ice sensitivity to atmospheric warming, numerical adjoint, conductivity feedback

Funding: National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: A number of feedbacks regulate the response of Arctic sea ice to local atmospheric warming. Using a realistic coupled ocean-sea ice model and its adjoint, we isolate a mechanism by which significant ice growth at the end of the melt season may occur as a lagged response to Arctic atmospheric warming. A series of perturbation simulations informed by adjoint model-derived sensitivity patterns reveal the enhanced ice growth to be accompanied by a reduction of snow thickness on the ice pack. Detailed analysis of ocean-ice-snow heat budgets confirms the essential role of the reduced snow thickness for persistence and delayed overshoot of ice growth. The underlying mechanism is a snow-melt-conductivity feedback, wherein atmosphere-driven snow melt leads to a larger conductive ocean heat loss through the overlying ice layer. Our results highlight the need for accurate observations of snow thickness to constrain climate models and to initialize sea ice forecasts.

800. Comparisons of Net Heat Flux Data Sets Over the Western North Pacific

Authors: Gyundo Pak; Jae-Hyoung Park; Seok-Joon Lee; Young-Gyu Park; You-Soon Chang

Journal: Ocean Science Journal

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12601-020-0036-4

URL: <https://doi.org/10.1007/s12601-020-0036-4>

Publication Date: 2020-10-12

Volume/Issue: Volume 55, Issue 4

References: 43

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

801. Eddy-Induced Acceleration of Argo Floats

Authors: Tianyu Wang (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Nathalie V. Zilberman (Scripps Institution of Oceanography / University of California San Diego); Yan Du (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc016042

URL: <https://doi.org/10.1029/2019jc016042>

Publication Date: 2020-10-03

Volume/Issue: Volume 125, Issue 10

References: 34

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: eddy-mean flow interaction, circulation, acceleration, Lagrangian motion, Argo floats (and 1 more)

Funding: DOC Office of the Secretary, National Natural Science Foundation of China, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: Float trajectories are simulated using Lagrangian particle tracking software and eddy-permitting ocean model output from the Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2) project. We find that Argo-like particles near strong mean flows tend to accelerate while at their parking depth. This effect is pronounced in western boundary current regions and in the Antarctic Circumpolar Current system. The acceleration is associated with eddy-mean flow interactions: Eddies converge particles toward regions with stronger mean currents. Particles do not accelerate when they are advected by the eddy or mean flow alone. During a 9-day parking period, speed increases induced by the eddy-mean flow interactions can be as large as 2 cm s⁻¹, representing roughly 10% of the mean velocity. If unaccounted for, this acceleration could bias velocities inferred from observed Argo float trajectories.

802. Combined Effects of Tidal Mixing in Narrow Straits and the Ekman Transport on the Sea Surface Temperature Cooling in the Southern Indonesian Seas

Authors: T. Nagai (University of Tokyo); T. Hibiya (University of Tokyo)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016314

URL: <https://doi.org/10.1029/2020jc016314>

Publication Date: 2020-10-03

Volume/Issue: Volume 125, Issue 11

References: 20

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Although the SST cooling events around the Lesser Sunda Islands (LSIs), which contribute to the seasonal variation of SST in the southern Indonesian Seas, are thought to be controlled by tidal mixing, the distribution of tidal mixing was highly idealized in previous studies. This study reinvestigates the mechanisms of such SST cooling events using MITgcm, which incorporates the realistic distribution of tidal mixing. The observed SST variability is well reproduced in the numerical experiment; SST is largely reduced along the northern (or southern) coast of the LSIs during austral summer (or winter). Particle tracking suggests that the narrow tidal straits between the LSIs, where well-mixed waters are produced by vigorous tidal mixing, play a key role in controlling the SST cooling. During austral summer, the cold SST signal of well-mixed waters propagates both northward and southward from the narrow straits as baroclinic Kelvin waves. On the northern side of the LSIs, in particular, coastal upwelling caused by the northwest monsoon raises the background thermocline, which affects the propagation of baroclinic Kelvin waves such that the accompanying cold SST signal becomes much enhanced compared with that on the southern side of the LSIs. During austral winter, in contrast, the Ekman transport enhances southward through-strait flow so that baroclinic Kelvin waves can propagate only southward affecting the SST on the southern side of the LSIs. This demonstrates the importance of the combined effects of tidal mixing in narrow straits and the Ekman transport on the SST cooling in the southern Indonesian Seas.

803. Dominant Terms in the Freshwater and Heat Budgets of the Subpolar North Atlantic Ocean and Nordic Seas From 1992 to 2015

Authors: Jan-Erik Tesdal (Columbia University); Thomas W. N. Haine (Johns Hopkins University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016435

URL: <https://doi.org/10.1029/2020jc016435>

Publication Date: 2020-10-02

Volume/Issue: Volume 125, Issue 10

References: 108

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: ocean heat content, decadal variability, Nordic Seas, Atlantic Meridional Overturning Circulation, liquid freshwater content (and 1 more)

Funding: Connecticut Space Grant College Consortium, Connecticut Space Grant College Consortium

License: N/A

Open Access: N/A

Abstract: The Arctic and subarctic oceans exhibit distinct decadal variations in freshwater and heat content. We describe freshwater and heat budgets with the ECCOv4 reanalysis product and compare budget variability and mechanisms within the subpolar North Atlantic Ocean, Nordic Seas, and Labrador Sea from 1992 to 2015. For all regions, changes in freshwater content are largely anticorrelated with changes in heat content. Since 1995, the subpolar North Atlantic Ocean has undergone a decade of warming and salinification followed by ongoing cooling and freshening. The recent increase in freshwater content and the reduction in heat in the subpolar North Atlantic can largely be attributed to anomalous circulation of mean salinity and temperature, respectively. Interannual variability in heat and freshwater mostly corresponds to boundary fluxes from the subtropics. Meanwhile, the Nordic Seas have undergone an overall warming and salinification from the mid-1990s to 2015. Salinification is primarily driven by reduced sea ice flux through Fram Strait, while warming is due to changes in both sea surface heating and advective flux. In the last 5 years, Labrador Sea freshwater convergence remained unchanged, as increased inflow via the Baffin Island Current is balanced by increased outflow via the Labrador Current. Hence, the observed freshening of the Arctic Ocean is expected to be an increasingly important source of future freshwater increases in the subpolar North Atlantic. This stands in contrast to variability in freshwater flux from the subtropical North Atlantic, which is associated with variability in the Atlantic Meridional Overturning Circulation.

804. Production and Fate of the South Atlantic Subtropical Underwater

Authors: Hao Liu (Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Tangdong Qu (University of California Los Angeles)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016309

URL: <https://doi.org/10.1029/2020jc016309>

Publication Date: 2020-09-29

Volume/Issue: Volume 125, Issue 10

References: 46

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: subtropical underwater, subduction

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: The South Atlantic Subtropical Underwater (STUW) was identified from Argo floats between 2001 and 2019. Expressed as a vertical salinity maximum at the subsurface, the STUW spreads over the western subtropical and tropical South Atlantic. The mean salinity, conservative temperature, and potential density of the STUW are 36.66 ± 0.43 g/kg, 22.62 ± 2.27 degreesC, and 25.14 ± 0.49 kg/m³, respectively. Based on the criteria derived from the STUW properties described above and the results from the Estimating the Circulation and Climate of the Ocean (ECCO) model, the production and fate of the STUW are calculated. The annual mean subduction rate is 3.26 Sv, and effective subduction occurs from September to November. Approximately 73% (2.38 Sv) of the subducted STUW is transported toward the tropical Atlantic, and 23% (0.75 Sv) of the subducted STUW is dissipated near the STUW formation region; 99% of the subducted STUW particles are dissipated within 2 years. Water particles can enter/exit the STUW along the advection route of the STUW. A total of 3.92 Sv of the STUW volume dissipates within the tropical region between 1.5 degreesS and 1 degreesN, implying that STUW contributes to 98% of the total transport of the subtropical cell. This analysis tracks the South Atlantic STUW from formation to dissipation and highlights the importance of the STUW in governing the subtropical cell.

805. Vertical Land Motion From Present-Day Deglaciation in the Wider Arctic

Authors: Carsten Ankjær Ludwigsen (Technical University of Denmark); Shfaqat Abbas Khan (Technical University of Denmark); Ole Baltazar Andersen (Technical University of Denmark); Ben Marzeion (University of Bremen)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl088144

URL: <https://doi.org/10.1029/2020gl088144>

Publication Date: 2020-09-29

Volume/Issue: Volume 47, Issue 19

References: 76

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: vertical land motion, sea level, Arctic, Greenland, coastal sea level (and 1 more)

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Vertical land motion (VLM) from past and ongoing glacial changes can amplify or mitigate ongoing relative sea level change. We present a high-resolution VLM model for the wider Arctic, that includes both present-day ice loading (PDIL) and glacial isostatic adjustment (GIA). The study shows that the nonlinear elastic uplift from PDIL is significant (0.5-1 mm yr⁻¹) in most of the wider Arctic and exceeds GIA at 15 of 54 Arctic GNSS sites, including sites in nonglaciaded areas of the North Sea region and the east coast of North America. Thereby the sea level change from PDIL (1.85 mm yr⁻¹) is significantly mitigated from VLM caused by PDIL. The combined VLM model was consistent with measured VLM at 85% of the GNSS sites () and outperformed a GIA-only model (). Deviations from GNSS-measured VLM can be attributed to local circumstances causing VLM.

806. Deep circulation in the South China Sea simulated in a regional model

Authors: Xiaolong Zhao; Chun Zhou; Xiaobiao Xu; Ruijie Ye; Wei Zhao

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-020-01411-2

URL: <https://doi.org/10.1007/s10236-020-01411-2>

Publication Date: 2020-09-28

Volume/Issue: Volume 70, Issue 11

References: 74

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, National Natural Science Foundation of China, Global Change and Air–Sea Interaction Project (and 5 more)

License: N/A

Open Access: N/A

Abstract: N/A

807. A Deep Learning Network to Retrieve Ocean Hydrographic Profiles from Combined Satellite and In Situ Measurements

Authors: Bruno Buongiorno Nardelli (Consiglio Nazionale delle Ricerche, Istituto di Scienze Marine (CNR-ISMAR), 80133 Naples, Italy)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs12193151

URL: <https://doi.org/10.3390/rs12193151>

Publication Date: 2020-09-25

Volume/Issue: Volume 12, Issue 19

References: 58

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency

License: CC BY 4.0

Open Access: N/A

Abstract: An efficient combination of remotely-sensed data and in situ measurements is needed to obtain accurate 3D ocean state estimates, representing a fundamental step to describe ocean dynamics and its role in the Earth climate system and marine ecosystems. Observations can either be assimilated in ocean general circulation models or used to feed data-driven reconstructions and diagnostic models. Here we describe an innovative deep learning algorithm that projects sea surface satellite data at depth after training with sparse co-located in situ vertical profiles. The technique is based on a stacked Long Short-Term Memory neural network, coupled to a Monte-Carlo dropout approach, and is applied here to the measurements collected between 2010 and 2018 over the North Atlantic Ocean. The model provides hydrographic vertical profiles and associated uncertainties from corresponding remotely sensed surface estimates, outperforming similar reconstructions from simpler statistical algorithms and feed-forward networks.

808. Towards an End-to-End Analysis and Prediction System for Weather, Climate, and Marine Applications in the Red Sea

Authors: Ibrahim Hoteit (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Yasser Abualnaja (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Shehzad Afzal (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Boujemaa Ait-El-Fquih (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Triantaphyllos Akyias (Massachusetts Institute of Technology); Charls Antony (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Clint Dawson (University of Texas at Austin); Khaled Asfahani (Saudi Aramco, Damam, Saudi Arabia); Robert J. Brewin (University of Exeter); Luigi Cavaleri (Institute of Marine Sciences); Ivana Cerovecki (Scripps Institution of Oceanography); Bruce Cornuelle (Scripps Institution of Oceanography); Srinivas Desamsetti (National Center for Medium Range Weather Forecasting, Noida, India); Raju Attada (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Hari Dasari (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Jose Sanchez-Garrido (University of Malaga, Malaga, Spain); Lily Genevier (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Mohamad El Gharamti (National Center of Atmospheric Research, Boulder, Colorado); John A. Gittings (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Elamurugu Gokul (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Ganesh Gopalakrishnan (Scripps Institution of Oceanography); Daquan Guo (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Bilel Hadri (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Markus Hadwiger (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Mohammed Abed Hammoud (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Myrl Hendershott (Scripps Institution of Oceanography); Mohamad Hittawe (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Ashok Karumuri (University of Hyderabad, Hyderabad, India); Omar Knio (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Armin Köhl (University of Hamburg); Samuel Kortas (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); George Krokos (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Ravi Kunchala (Indian Institute of Technology Delhi); Leila Issa (Lebanese American University, Beirut, Lebanon); Issam Lakkis (American University of Beirut); Sabique Langodan (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Pierre Lermusiaux (Massachusetts Institute of Technology); Thang Luong (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Jingyi Ma (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Olivier Le Maitre (École Polytechnique); Matthew Mazloff (Scripps Institution of Oceanography); Samah El Mohtar (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Vassilis P. Papadopoulos (Hellenic Centre for Marine Research); Trevor Platt (Plymouth Marine Laboratory); Larry Pratt (Woods Hole Oceanographic Institution); Naila Raboudi (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Marie-Fanny Racault (Plymouth Marine Laboratory); Dionysios E. Raitsos (National and Kapodistrian University of Athens); Shanas Razak (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Sivareddy Sanikommu (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Shubha Sathyendranath (Plymouth Marine Laboratory); Sarantis Sofianos (National and Kapodistrian University of Athens); Aneesh Subramanian (University of Colorado Boulder); Rui Sun (Scripps Institution of Oceanography); Edriss Titi (University of Cambridge); Habib Toye (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); George Triantafyllou (Hellenic Centre for Marine Research); Kostas Tsiaras (Hellenic Centre for Marine Research); Panagiotis Vasou (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Yesubabu Viswanadhapalli (National Atmospheric Research Laboratories, Gadanki, India); Yixin Wang (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Fengchao Yao (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Peng Zhan (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); George Zodiatis (Coastal and Marine Research Laboratory, Crete, Greece)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/bams-d-19-0005.1

URL: <https://doi.org/10.1175/bams-d-19-0005.1>

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Volume/Issue: Volume 102, Issue 1

References: 118

Cited by: 33 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Red Sea, home to the second-longest coral reef system in the world, is a vital resource for the Kingdom of Saudi Arabia. The Red Sea provides 90% of the Kingdom's potable water by desalinization, supporting tourism, shipping, aquaculture, and fishing industries, which together contribute about 10%-20% of the country's GDP. All these activities, and those elsewhere in the Red Sea region, critically depend on oceanic and atmospheric conditions. At a time of mega-development projects along the Red Sea coast, and global warming, authorities are working on optimizing the harnessing of environmental resources, including renewable energy and rainwater harvesting. All these require high-resolution weather and climate information. Toward this end, we have undertaken a multipronged research and development activity in which we are developing an integrated data-driven regional coupled modeling system. The telescopically nested components include 5-km- to 600-m-resolution atmospheric models to address weather and climate challenges, 4-km- to 50-m-resolution ocean models with regional and coastal configurations to simulate and predict the general and mesoscale circulation, 4-km- to 100-m-resolution ecosystem models to simulate the biogeochemistry, and 1-km- to 50-m-resolution wave models. In addition, a complementary probabilistic transport modeling system predicts dispersion of contaminant plumes, oil spill, and marine ecosystem connectivity. Advanced ensemble data assimilation capabilities have also been implemented for accurate forecasting. Resulting achievements include significant advancement in our understanding of the regional circulation and its connection to the global climate, development, and validation of long-term Red Sea regional atmospheric-oceanic-wave reanalyses and forecasting capacities. These products are being extensively used by academia, government, and industry in various weather and marine studies and operations, environmental policies, renewable energy applications, impact assessment, flood forecasting, and more.

809. Seismic ocean thermometry

Authors: Wenbo Wu (California Institute of Technology); Zhongwen Zhan (California Institute of Technology); Shirui Peng (California Institute of Technology); Sidao Ni (Chinese Academy of Sciences); Jörn Callies (California Institute of Technology)

Journal: Science

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/science.abb9519

URL: <https://doi.org/10.1126/science.abb9519>

Publication Date: 2020-09-17

Volume/Issue: Volume 369, Issue 6510

References: 56

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Resnick Sustainability Institute for Science, Energy and Sustainability, California Institute of Technology, National Natural Science Foundation of China, Chinese Academy of Sciences (and 1 more)

License: N/A

Open Access: N/A

Abstract: Hearing the heat Most of the excess heat that causes global warming is absorbed by the oceans. Quantifying that heat increase is challenging because it requires many different temperature measurements over both the vertical and horizontal extent of the oceans. Wu et al. report success in this effort through the use of a different method: They inferred temperature changes from sound waves generated by repeating earthquakes (see the Perspective by Wunsch). The travel time of these earthquakes from source to receiver reflects changes in the average water temperature that they encounter. This technique should substantially enhance our ability to monitor ocean warming. Science , this issue p. 1510 ; see also p. 1433

810. Is the Ocean Speeding Up? Ocean Surface Energy Trends

Authors: Carl Wunsch (Harvard University, Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0082.1

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References: 54

Cited by: 18 publications

Language: N/A

Subjects: N/A

Keywords: Climate variability, Ocean circulation, Oceanic variability, Decadal variability, Trends

Funding: Jet Propulsion Laboratory

License: N/A

Open Access: N/A

Abstract: A recent paper by Hu et al. (<https://doi.org/10.1126/sciadv.aax7727>) has raised the interesting question of whether the ocean circulation has been "speeding up" in the last decades. Their result contrasts with some estimates of the lack of major trends in oceanic surface gravity waves and wind stress. In general, both the increased energy and implied power inputs of the calculated circulation correspond to a small fraction of the very noisy background values. An example is the implied power increase of about 3×10^8 W, as compared to wind energy inputs of order 10^{12} W. Here the problem is reexamined using a state estimate that has the virtue of being energy, mass, etc. conserving. Because it is an estimate over an entire recent 26-yr interval, it is less sensitive to the strong changes in observational data density and distribution, and it does not rely upon nonconservative "reanalyses." The focus is on the energy lying in the surface layers of the ocean. A potential energy increase is found, but it is almost completely unavailable--arising from the increase in mean sea level. A weak increase in kinetic energy in the top layer (10 m) is confirmed, corresponding to an increase of order $1 \text{ cm s}^{-1} \text{ yr}^{-1}$ over 26 years. An estimate of kinetic energy in the full water column shows no monotonic trend, but the changes in the corresponding available potential energy are not calculated here.

811. Improved Estimates of Changes in Upper Ocean Salinity and the Hydrological Cycle

Authors: Lijing Cheng (Chinese Academy of Sciences); Kevin E. Trenberth (National Center for Atmospheric Research); Nicolas Gruber (ETH Zurich); John P. Abraham (University of St. Thomas, School of Engineering, St. Paul, Minnesota); John T. Fasullo (National Center for Atmospheric Research); Guancheng Li (Chinese Academy of Sciences); Michael E. Mann (Pennsylvania State University); Xuanming Zhao (Chinese Academy of Sciences); Jiang Zhu (Chinese Academy of Sciences)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-20-0366.1

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Publication Date: 2020-09-09

Volume/Issue: Volume 33, Issue 23

References: 98

Cited by: 133 publications

Language: N/A

Subjects: N/A

Keywords: Climate change, Climate records, Sampling, Salinity

Funding: National Key R&D; Program of China, Key Deployment Project of Centre for Ocean Mega-Research of Science, CAS, Horizon 2020 (and 1 more)

License: N/A

Open Access: N/A

Abstract: Ocean salinity records the hydrological cycle and its changes, but data scarcity and the large changes in sampling make the reconstructions of long-term salinity changes challenging. Here, we present a new observational estimate of changes in ocean salinity since 1960 from the surface to 2000 m. We overcome some of the inconsistencies present in existing salinity reconstructions by using an interpolation technique that uses information on the spatiotemporal covariability of salinity taken from model simulations. The interpolation technique is comprehensively evaluated using recent Argo-dominated observations through subsample tests. The new product strengthens previous findings that ocean surface and subsurface salinity contrasts have increased (i.e., the existing salinity pattern has amplified). We quantify this contrast by assessing the difference between the salinity in regions of high and low salinity averaged over the top 2000 m, a metric we refer to as SC2000. The increase in SC2000 is highly distinguishable from the sampling error and less affected by interannual variability and sampling error than if this metric was computed just for the surface. SC2000 increased by 1.9% $\pm$ 0.6% from 1960 to 1990 and by 3.3% $\pm$ 0.4% from 1991 to 2017 (5.2% $\pm$ 0.4% for 1960-2017), indicating an acceleration of the pattern amplification in recent decades. Combining this estimate with model simulations, we show that the change in SC2000 since 1960 emerges clearly as an anthropogenic signal from the natural variability. Based on the salinity-contrast metrics and model simulations, we find a water cycle amplification of 2.6% $\pm$ 4.4% K⁻¹ since 1960, with the larger error than salinity metric mainly being due to model uncertainty.

812. Heat stored in the Earth system: where does the energy go?

Authors: Karina von Schuckmann; Lijing Cheng; Matthew D. Palmer; James Hansen; Caterina Tassone; Valentin Aich; Susheel Adusumilli; Hugo Beltrami; Tim Boyer; Francisco José Cuesta-Valero; Damien Desbruyères; Catia Domingues; Almudena García-García; Pierre Gentine; John Gilson; Maximilian Gorfer; Leopold Haimberger; Masayoshi Ishii; Gregory C. Johnson; Rachel Killick; Brian A. King; Gottfried Kirchengast; Nicolas Kolodziejczyk; John Lyman; Ben Marzeion; Michael Mayer; Maeva Monier; Didier Paolo Monselesan; Sarah Purkey; Dean Roemmich; Axel Schweiger; Sonia I. Seneviratne; Andrew Shepherd; Donald A. Slater; Andrea K. Steiner; Fiammetta Straneo; Mary-Louise Timmermans; Susan E. Wijffels

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-12-2013-2020

URL: <https://doi.org/10.5194/essd-12-2013-2020>

Publication Date: 2020-09-07

Volume/Issue: Volume 12, Issue 3

References: 231

Cited by: 207 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Human-induced atmospheric composition changes cause a radiative imbalance at the top of the atmosphere which is driving global warming. This Earth energy imbalance (EEI) is the most critical number defining the prospects for continued global warming and climate change. Understanding the heat gain of the Earth system - and particularly how much and where the heat is distributed - is fundamental to understanding how this affects warming of the ocean, atmosphere and land; rising surface temperature; sea level; and loss of grounded and floating ice, which are fundamental concerns for society. This study is a Global Climate Observing System (GCOS) concerted international effort to update the Earth heat inventory and presents an updated assessment of ocean warming estimates as well as new and updated estimates of heat gain in the atmosphere, cryosphere and land over the period 1960-2018. The study obtains a consistent long-term Earth system heat gain over the period 1971-2018, with a total heat gain of 358 ± 37 ZJ, which is equivalent to a global heating rate of 0.47 ± 0.1 W m⁻². Over the period 1971-2018 (2010-2018), the majority of heat gain is reported for the global ocean with 89 % (90 %), with 52 % for both periods in the upper 700 m depth, 28 % (30 %) for the 700-2000 m depth layer and 9 % (8 %) below 2000 m depth. Heat gain over land amounts to 6 % (5 %) over these periods, 4 % (3 %) is available for the melting of grounded and floating ice, and 1 % (2 %) is available for atmospheric warming. Our results also show that EEI is not only continuing, but also increasing: the EEI amounts to 0.87 ± 0.12 W m⁻² during 2010-2018. Stabilization of climate, the goal of the universally agreed United Nations Framework Convention on Climate Change (UNFCCC) in 1992 and the Paris Agreement in 2015, requires that EEI be reduced to approximately zero to achieve Earth's system quasi-equilibrium. The amount of CO₂ in the atmosphere would need to be reduced from 410 to

353 ppm to increase heat radiation to space by 0.87 W m^{-2} , bringing Earth back towards energy balance. This simple number, EEI, is the most fundamental metric that the scientific community and public must be aware of as the measure of how well the world is doing in the task of bringing climate change under control, and we call for an implementation of the EEI into the global stocktake based on best available science. Continued quantification and reduced uncertainties in the Earth heat inventory can be best achieved through the maintenance of the current global climate observing system, its extension into areas of gaps in the sampling, and the establishment of an international framework for concerted multidisciplinary research of the Earth heat inventory as presented in this study. This Earth heat inventory is published at the German Climate Computing Centre (DKRZ, <https://www.dkrz.de/>, last access: 7 August 2020) under the DOI https://doi.org/10.26050/WDCC/GCOS_EHI_EXP_v2 (von Schuckmann et al., 2020).

813. The role of oceanic feedbacks in the 2014-2016 El Niño events as derived from ocean reanalysis data

Authors: Cong Guan; Fan Wang; Shijian Hu

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-020-0038-1

URL: <https://doi.org/10.1007/s00343-020-0038-1>

Publication Date: 2020-09-03

Volume/Issue: Volume 38, Issue 5

References: 69

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

814. A Fukushima tracer perspective on four years of North Pacific mode water evolution

Authors: Alison M. Macdonald; Sachiko Yoshida; Steven M. Pike; Ken O. Buesseler; Irina I. Rypina; Steven R. Jayne; Vincent Rossi; Jennifer Kenyon; Jessica A. Drysdale

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2020.103379

URL: <https://doi.org/10.1016/j.dsr.2020.103379>

Publication Date: 2020-09-02

Volume/Issue: Volume 166

References: 89

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission, National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

815. Amplified seasonal cycle in hydroclimate over the Amazon river basin and its plume region

Authors: Yu-Chiao Liang; Min-Hui Lo; Chia-Wei Lan; Hyodae Seo; Caroline C. Ummenhofer; Stephen Yeager; Ren-Jie Wu; John D. Steffen

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-020-18187-0

URL: <https://doi.org/10.1038/s41467-020-18187-0>

Publication Date: 2020-09-01

Volume/Issue: Volume 11, Issue 1

References: 77

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The Amazon river basin receives ~2000 mm of precipitation annually and contributes ~17% of global river freshwater input to the oceans; its hydroclimatic variations can exert profound impacts on the marine ecosystem in the Amazon plume region (APR) and have potential far-reaching influences on hydroclimate over the tropical Atlantic. Here, we show that an amplified seasonal cycle of Amazonia precipitation, represented by the annual difference between maximum and minimum values, during the period 1979-2018, leads to enhanced seasonalities in both Amazon river discharge and APR ocean salinity. An atmospheric moisture budget analysis shows that these enhanced seasonal cycles are associated with similar amplifications in the atmospheric vertical and horizontal moisture advections. Hierarchical sensitivity experiments using global climate models quantify the relationships of these enhanced seasonalities. The results suggest that an intensified hydroclimatological cycle may develop in the Amazonia atmosphere-land-ocean coupled system, favouring more extreme terrestrial and marine conditions.

816. Routes of the Upper Branch of the Atlantic Meridional Overturning Circulation according to an Ocean State Estimate

Authors: Louise Rousselet (Scripps Institution of Oceanography / University of California San Diego); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego); Gael Forget (Massachusetts Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl089137

URL: <https://doi.org/10.1029/2020gl089137>

Publication Date: 2020-08-31

Volume/Issue: Volume 47, Issue 18

References: 43

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: Lagrangian ocean analysis, meridional overturning circulation

Funding: National Aeronautics and Space Administration, Office of the Director

License: CC BY 4.0

Open Access: N/A

Abstract: The origins of the upper branch of the Atlantic meridional overturning circulation (AMOC) are traced with backward-in-time Lagrangian trajectories, quantifying the partition of volume transport between different routes of entry from the Indo-Pacific into the Atlantic. Particles are advected by the velocity field from a recent release of "Estimating the Circulation and Climate of the Ocean" (ECCOV4). This global time-variable velocity field is a dynamically consistent interpolation of over 1 billion oceanographic observations collected between 1992 and 2015. Of the 13.6 Sverdrups (1 Sv = 106 m³/s) flowing northward across 6 degreesS, 15% enters the Atlantic from Drake Passage, 35% enters from the straits between Asia and Australia (Indonesian Throughflow), and 49% comes from the region south of Australia (Tasman Leakage). Because of blending in the Agulhas region, water mass properties in the South Atlantic are not a good indicator of origin.

817. Physical Mechanisms Driving Oxygen Subduction in the Global Ocean

Authors: Esther Portela (Institut de Recherche pour le Développement (IRD)); Nicolas Kolodziejczyk (Institut de Recherche pour le Développement (IRD)); Clément Vic (Institut de Recherche pour le Développement (IRD)); Virginie Thierry (Institut de Recherche pour le Développement (IRD))

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl089040

URL: <https://doi.org/10.1029/2020gl089040>

Publication Date: 2020-08-31

Volume/Issue: Volume 47, Issue 17

References: 58

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: oxygen diffusion, ocean ventilation, oxygen subduction, mode waters

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Future changes in subduction are suspected to be critical for the ocean deoxygenation predicted by climate models over the 21st century. However, the drivers of global oxygen subduction have not been fully described or quantified. Here, we address the physical mechanisms responsible for the oxygen transport across the late-winter mixed layer base and their relation with water mass formation. Up to 70% of the global oxygen uptake takes place during Mode Water subduction mostly in the Southern Ocean and the North Atlantic. The driving mechanisms are (i) the combination of strong currents with large mixed-layer-depth gradients at localized hot spots and (ii) the wind-driven vertical velocity within the subtropical gyres. Oxygen diffusion, despite being underestimated in this study, is likely to play an important role in the global ocean oxygenation. The physical mass flux dominates the total oxygen subduction while the oxygen solubility plays a minor role in its modulation.

818. Comparative Analysis of the Effect of the Loading Series from GFZ and EOST on Long-Term GPS Height Time Series

Authors: Shuguang Wu (Wuhan University, The University of Nottingham, Nottingham Geospatial Institute); Guigen Nie (Wuhan University, Collaborative Innovation Center for Geospatial Information Technology, Wuhan 430072, China); Xiaolin Meng (The University of Nottingham, Nottingham Geospatial Institute); Jingnan Liu (Wuhan University); Yuefan He (Wuhan University); Changhu Xue (Southern University of Science and Technology); Haiyang Li (Wuhan University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs12172822

URL: <https://doi.org/10.3390/rs12172822>

Publication Date: 2020-08-31

Volume/Issue: Volume 12, Issue 17

References: 43

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: China Scholarship Council, National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: In order to investigate the effect of different loading models on the nonlinear variations in Global Positioning System (GPS) height time series, the characteristics of annual signals (amplitude and phase) of GPS time series, loading series from Deutsche GeoForschungsZentrum, Germany (GFZ) and School and Observatory of Earth Sciences, France (EOST) at 633 global GPS stations are processed and analyzed. The change characteristics of the root mean square (RMS) reduction rate, annual amplitude and phase of GPS time series after environmental loading corrections (ELCs) are then detected. Results show that ELCs have a positive effect on the reduction in the nonlinear deformation contained in most GPS stations around the world. RMS reduction rates are positive at 82.6% stations after GFZ correction and 87.4% after EOST correction, and the average reduction rates of all stations are 10.6% and 15.4%, respectively. As for the environmental loading series from GFZ and EOST, their average annual amplitudes are 2.7 and 3.1 mm, which explains ~40% annual amplitude of GPS height time series (7.2 mm). Further analysis of some specific stations indicates that the annual phase difference between GPS height time series and the environmental loading series is an important reason that affects the reduction rates of the RMS and annual amplitude. The linear relationship between the annual phase difference and the annual amplitude reduction rate is significant. The linear fitting results show that when there is no annual phase difference between GPS and loading series, the reduction rates of the RMS and annual amplitude will increase to the maximum of 15.6% and 41.6% for GFZ, and 22.0% and 46.6% for EOST.

819. Impact of Assimilation of Moored Velocity Data on Low-Frequency Current Estimation in Northwestern Tropical Pacific

Authors: Danian Liu (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Fan Wang (Chinese Academy of Sciences); Jiang Zhu (Chinese Academy of Sciences); Dongxiao Wang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Jianing Wang (Chinese Academy of Sciences); Qiang Xie (Chinese Academy of Sciences); Yeqiang Shu (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015829

URL: <https://doi.org/10.1029/2019jc015829>

Publication Date: 2020-08-28

Volume/Issue: Volume 125, Issue 9

References: 61

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: northwestern tropical Pacific, moored current, ensemble optimal interpolation

Funding: National Natural Science Foundation of China, National Basic Research Program of China

License: N/A

Open Access: N/A

Abstract: This study investigates the impact of assimilation of Northwestern Tropical Pacific Ocean (NWTPO) moored currents observation system on low-frequency (over a period of more than 90 days) current estimation. A low-frequency scale-selective ensemble optimal interpolation (EnOI) scheme is designed to assimilate the horizontally sparse moored currents into an eddy-permitting ocean model. In this assimilation scheme, low-frequency current signals are selected by removing intraseasonal signals in both observation and ensemble data, and a horizontal spatial filter is implemented to match the model forecast to the low-frequency ensemble and observation. The assimilation of currents efficiently resolves simulated low-frequency current displacements, especially for the Equatorial Undercurrent (EUC), the North Equatorial Countercurrent (NECC), and the North Equatorial Current (NEC). More realistic intensities of the NWTPO low-frequency currents are also reproduced. With respect to the moored velocity profiles, the root-mean-square error (RMSE) of the simulated low-frequency zonal velocity above a 500-m depth is reduced from 10.9 to 5.8 cm/s using this low-frequency EnOI scheme. The low-frequency variation of NWTPO currents is reproduced by correcting the structures of different baroclinic modes. The increments of zonal velocities mainly correspond to the first baroclinic mode for the NECC and NEC, while for the EUC, they correspond to the leading three and higher-order baroclinic modes. The improvements of the simulated stratification and pressure gradient remark the role of the assimilation of moored currents in adjusting available potential energy.

820. Inconsistent Atmosphere-Ocean Dynamics and Multidecadal Zonal SST Gradient Trends Across the Equatorial Pacific Ocean in Reanalysis Products

Authors: T. Jayasankar (Indian Institute of Technology Bombay); R. Murtugudde (University of Maryland, Indian Institute of Technology Bombay); T. I. Eldho (Indian Institute of Technology Bombay)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016297

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Volume/Issue: Volume 125, Issue 9

References: 60

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean reanalysis products are routinely employed as reality checks in model evaluations and for process studies. This is especially so for critical regions such as the equatorial cold tongue (ECT) in the eastern equatorial Pacific where models suffer a chronic cold bias. ECT is a major player in the Pacific equatorial zonal sea surface temperature (SST) gradient (ΔEWSST) that has a significant impact on oceanic heat uptake and thus global climate. Hence, we investigate the reliability of three ocean reanalysis products for surface flux and ocean dynamic controls on ΔEWSST and Niño3.4 SST trends. We infer that while Niño3.4 SST trends are positive in all products, the signs of ΔEWSST trends do not agree with each other because initial conditions likely play a big role in their evolution. However, for ΔEWSST trends, the effect of initial conditions gets canceled out to some extent. Mixed layer heat budget and trends in ocean dynamic features such as tropical and subtropical cells, equatorial undercurrent, and subsurface temperatures are also diagnosed. We show that two reanalysis products that show a strengthening of ΔEWSST have contradicting trends in their surface heat flux and ocean dynamic contributions. This suggests that without accurate surface heat and momentum fluxes, data assimilation techniques may produce an east-west trend that is inconsistent among each other. Reanalysis products must address these issues considering the importance of this gradient.

821. High-Frequency Submesoscale Motions Enhance the Upward Vertical Heat Transport in the Global Ocean

Authors: Zhan Su (University of Toronto); Hector Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Patrice Klein (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, IFREMER, CNRS Paris France); Andrew F. Thompson (California Institute of Technology); Lia Siegelman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christopher Hill (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016544

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Publication Date: 2020-08-28

Volume/Issue: Volume 125, Issue 9

References: 52

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: eddies, eddy transport, ocean heat transport, ocean front, high frequency

Funding: National Aeronautics and Space Administration, Norsk Sykepleierforbund

License: N/A

Open Access: N/A

Abstract: The rate of ocean heat uptake depends on the mechanisms that transport heat between the surface and the ocean interior. A recent study found that the vertical heat transport driven by motions with scales smaller than 0.5 degrees (submesoscales) and frequencies smaller than 1 day⁻¹ is upward. This transport competes with the other major components of the global heat transport, namely, the downward heat transport explained by the large-scale wind-driven vertical circulation and vertical diffusion at small scales and the upward heat transport associated with mesoscale eddies (50- to 300-km size). The contribution from motions with small spatial scales (<0.5 degrees) and frequencies larger than 1 day⁻¹, including internal gravity waves, has never been explicitly estimated. This study investigates this high-frequency (subdaily) submesoscale contribution to the global heat transport. The major result of this study, based on the analysis of a high-resolution ocean model, is that including this high-frequency contribution surprisingly doubles the upward heat transport due to submesoscales in winter in the global ocean. This contribution typically concerns depths down to 200-500 m and can have a magnitude of up to 500 W m⁻² in terms of wintertime heat fluxes at 40-m depth, which causes a significant upward heat transport of ~7 PW when integrated over the global ocean. Thus, such submesoscale heat transport, which is not resolved by climate models, impacts the heat uptake in the global ocean. The mechanisms involved in these results still need to be understood, which should be the scope of future work.

822. The Influence of Warming Patterns on Passive Ocean Heat Uptake

Authors: Emily Newsom (University of Oxford); Laure Zanna (University of Oxford, Courant Institute of Mathematical Sciences New York University New York NY USA); Samar Khatiwala (University of Oxford); Jonathan M. Gregory (Met Office Hadley Centre, Met Office, University of Reading)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl088429

URL: <https://doi.org/10.1029/2020gl088429>

Publication Date: 2020-08-26

Volume/Issue: Volume 47, Issue 18

References: 53

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: The climate's response to forcing depends on how efficiently heat is absorbed by the ocean. Much, if not most, of this ocean heat uptake results from the passive transport of warm surface waters into the ocean's interior. Here we examine how geographic patterns of surface warming influence the efficiency of this passive heat uptake process. We show that the average pattern of surface warming in CMIP5 damps passive ocean heat uptake efficiency by nearly 25%, as compared to homogeneous surface warming. This "pattern effect" occurs because strong ventilation and weak surface warming are robustly colocated, particularly in the Southern Ocean. However, variations in warming patterns across CMIP5 do not drive significant ensemble spread in passive ocean heat uptake efficiency. This spread is likely linked to intermodel differences in ocean circulation, which our idealized results suggest may be dominated by differences in Southern Ocean and subtropical ventilation processes.

823. Rossby and Yanai Modes of Tropical Instability Waves in the Equatorial Pacific Ocean and a Diagnostic Model for Surface Currents

Authors: Minyang Wang (d Department of Mathematics and Statistics, San Diego State University, San Diego, California, Scripps Institution of Oceanography / University of California San Diego, Chinese Academy of Sciences); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego); Samuel S. P. Shen (d Department of Mathematics and Statistics, San Diego State University, San Diego, California, Scripps Institution of Oceanography / University of California San Diego); Yan Du (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0063.1

URL: <https://doi.org/10.1175/jpo-d-20-0063.1>

Publication Date: 2020-08-26

Volume/Issue: Volume 50, Issue 10

References: 60

Cited by: 13 publications

Language: N/A

Subjects: N/A

Keywords: Diagnostics, Currents, Pattern detection, Instability, Altimetry (and 1 more)

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, State Oceanic Administration (and 7 more)

License: N/A

Open Access: N/A

Abstract: Mesoscale activities over the equatorial Pacific Ocean are dominated by the Rossby and Yanai modes of tropical instability waves (TIWs). The TIW-induced surface velocity has not been accurately estimated in previous diagnostic models, especially for the meridional component across the equator. This study develops a diagnostic model that retains the acceleration terms to estimate the TIW surface velocity from the satellite-observed sea surface height. Validated against moored observations, the velocity across the equator is accurately estimated for the first time, much improved from existing products. The results identify the Rossby- and Yanai-mode TIWs as the northwest-southeastward (NW-SE) velocity oscillations north of the equator and the northeast-southwestward (NE-SW) velocity oscillations on the equator, respectively. Barotropic instability is the dominant energy source of the two TIW modes. The NE-SW velocity oscillation of the Yanai mode is associated with the counterclockwise shear of the South Equatorial Current on the equator. The two TIW modes induce different sea surface temperature patterns and vertical motions. Accurate estimates of TIW velocity are important for studying equatorial ocean dynamics and climate variability in the tropical Pacific Ocean.

824. On the heat budget and water mass exchange in the Andaman Sea

Authors: Jiawen Liao; Shiqiu Peng; Xixi Wen

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-019-1627-8

URL: <https://doi.org/10.1007/s13131-019-1627-8>

Publication Date: 2020-08-24

Volume/Issue: Volume 39, Issue 7

References: 30

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

825. Intraseasonal modulation of Wyrтки jet in the eastern Indian Ocean by equatorial waves during spring 2013

Authors: Yongliang Duan; Hongwei Liu; Lin Liu; Weidong Yu

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-020-1576-2

URL: <https://doi.org/10.1007/s13131-020-1576-2>

Publication Date: 2020-08-24

Volume/Issue: Volume 39, Issue 7

References: 72

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

826. Generation and propagation of 21-day bottom pressure variability driven by wind stress curl in the East China Sea

Authors: Hua Zheng; Xiao-Hua Zhu; Hirohiko Nakamura; Jae-Hun Park; Chanhung Jeon; Ruixiang Zhao; Ayako Nishina; Chuanzheng Zhang; Hanna Na; Ze-Nan Zhu; Hong-Sik Min

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-020-1603-3

URL: <https://doi.org/10.1007/s13131-020-1603-3>

Publication Date: 2020-08-24

Volume/Issue: Volume 39, Issue 7

References: 45

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

827. Elucidating Large-Scale Atmospheric Controls on Bering Strait Throughflow Variability Using a Data-Constrained Ocean Model and Its Adjoint

Authors: An T. Nguyen (University of Texas at Austin); Rebecca A. Woodgate (University of Washington); Patrick Heimbach (University of Texas at Austin)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016213

URL: <https://doi.org/10.1029/2020jc016213>

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References: 72

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: atmospheric control, Bering Strait throughflow, transport variability, adjoint sensitivity, ocean model (and 1 more)

Funding: National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: A regional data-constrained coupled ocean-sea ice general circulation model and its adjoint are used to investigate mechanisms controlling the volume transport variability through Bering Strait during 2002 to 2013. Comprehensive time-resolved sensitivity maps of Bering Strait transport to atmospheric forcing can be accurately computed with the adjoint along the forward model trajectory to identify spatial and temporal scales most relevant to the strait's transport variability. The simulated Bering Strait transport anomaly is found to be controlled primarily by the wind stress on short time scales of order 1 month. Spatial decomposition indicates that on monthly time scales winds over the Bering and the combined Chukchi and East Siberian Seas are the most significant drivers. Continental shelf waves and coastally trapped waves are suggested as the dominant mechanisms for propagating information from the far field to the strait. In years with transport extrema, eastward wind stress anomalies in the Arctic sector are found to be the dominant control, with correlation coefficient of 0.94. This implies that atmospheric variability over the Arctic plays a substantial role in determining Bering Strait flow variability. The near-linear response of the transport anomaly to wind stress allows for predictive skill at interannual time scales, thus potentially enabling skillful prediction of changes at this important Pacific-Arctic gateway, provided that accurate measurements of surface winds in the Arctic can be obtained. The novelty of this work is the use of space and time-resolved adjoint-based sensitivity maps, which enable detailed dynamical, that is, causal attribution of the impacts of different forcings.

828. The Enhancement of the Impact of the Wintertime North Atlantic Oscillation on the Subsequent Sea Surface Temperature over the Tropical Atlantic since the Middle 1990s

Authors: Shaobo Qiao (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Meng Zou (Sun Yat-Sen University); Shankai Tang (Lanzhou University); Ho Nam Cheung (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Haijing Su (GoldWind Smart Energy Technology); Qingxiang Li (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Guolin Feng (National Climate Center, China Meteorological Administration, Yangzhou University, Southern Marine Science and Engineering Guangdong Laboratory); Wenjie Dong (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0934.1

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Publication Date: 2020-08-20

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References: 86

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: Tropical Atlantic|Atlantic Equatorial mode|Gulf Stream|Shortwave radiation|Hadley cell

Funding: National Key Research and Development Program of China, Key Program of the National Natural Science Foundation of China, General Program of the National Natural Science Foundation of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: The impact of the wintertime North Atlantic Oscillation (NAO) on the subsequent sea surface temperature (SST) anomalies over the tropical Atlantic has experienced obvious interdecadal changes during 1950-2015. During 1995-2015, the negative (positive) phase of the wintertime NAO favors positive (negative) SST anomalies over the tropical Atlantic in the subsequent spring-summer, whereas the NAO-SST connection is insignificant during 1970-94 and is confined to the northern tropical Atlantic (NTA) during 1950-69. Compared to 1970-94, the much stronger influence on the NTA SST during 1995-2015 and 1950-69 is associated with a southward shift of the southern boundary of the NAO. During 1995-2015, the inverted NAO-related warming of the tropical Atlantic consists of three stages: 1) the pronounced increase in SST over the subtropical North Atlantic (SNA) and the tropical South Atlantic (TSA) during December-January, 2) the pronounced increase in SST over the northwestern tropical Atlantic (NWTa) during February-April, and 3) the persistent warming over the tropical Atlantic during May-August. The increases in SST over the SNA and the TSA are attributed to significant positive latent heat flux anomalies via the wind-evaporation effect, which are connected by the suppressed regional Hadley circulation. Afterward, the associated anomalous downward motion over the NWTa persists into February-April, which induces more incoming shortwave radiation and results in

a significant increase in the local SST via the cloud-radiation effect. In contrast, during 1950-69, due to the decreased interannual variability of the vertical motion over the NHTA, the anomalous downward branch aloft and the low-level cross-equatorial northwesterly winds associated with the inverted NAO are not evident, and thus the regions with an increase in SST are confined to the Northern Hemisphere.

829. The Scale of Submesoscale Baroclinic Instability Globally

Authors: Jihai Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory); Baylor Fox-Kemper (Brown University); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Changming Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0043.1

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Publication Date: 2020-08-18

Volume/Issue: Volume 50, Issue 9

References: 69

Cited by: 41 publications

Language: N/A

Subjects: N/A

Keywords: Eddy|Mixed layer|Anticyclone|Seasonality|Boundary current

Funding: National Key Research Program of China, National Natural Science Foundation of China, National Natural Science Foundation of China (and 3 more)

License: N/A

Open Access: N/A

Abstract: The spatial scale of submesoscales is an important parameter for studies of submesoscale dynamics and multiscale interactions. The horizontal spatial scales of baroclinic, geostrophic-branch mixed layer instabilities (MLI) are investigated globally (without the equatorial or Arctic oceans) based on observations and simulations in the surface and bottom mixed layers away from significant topography. Three high-vertical-resolution boundary layer schemes driven with profiles from a MITgcm global submesoscale-permitting model improve robustness. The fastest-growing MLI wavelength decreases toward the poles. The zonal median surface MLI wavelength is 51-2.9 km when estimated from the observations and from 32, 25, and 27 km to 2.5, 1.2, and 1.1 km under the K-profile parameterization (KPP), Mellor-Yamada (MY), and κ -epsilon schemes, respectively. The surface MLI wavelength has a strong seasonality with a median value 1.6 times smaller in summer (10 km) than winter (16 km) globally from the observations. The median bottom MLI wavelengths estimated from simulations are 2.1, 1.4, and 0.41 km globally under the KPP, MY, and κ -epsilon schemes, respectively, with little seasonality. The estimated required ocean model grid spacings to resolve wintertime surface mixed layer eddies are 1.9 km (50% of regions resolved) and 0.92 km (90%) globally. To resolve summertime eddies or MLI seasonality requires grids finer than 1.3 km (50%) and 0.55 km (90%). To resolve bottom mixed layer eddies, grids finer than 257, 178, and 51 m (50%) and 107, 87, and 17 m (90%) are estimated under the KPP, MY, and κ -epsilon schemes.

830. Freshwater forcing control on early-Holocene South American monsoon

Authors: Wilton Aguiar; Luciana F. Prado; Ilana Wainer; Zhengyu Liu; Alvaro Montenegro; Katrin J. Meissner; Mauricio M. Mata

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2020.106498

URL: <https://doi.org/10.1016/j.quascirev.2020.106498>

Publication Date: 2020-08-17

Volume/Issue: Volume 245

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Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fundação de Amparo à Pesquisa do Estado do Rio Grande do Sul, Conselho Nacional de Desenvolvimento Científico e Tecnológico, Conselho Nacional de Desenvolvimento Científico e Tecnológico (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

831. Impact of Synthetic Arctic Argo-Type Floats in a Coupled Ocean-Sea Ice State Estimation Framework

Authors: An T. Nguyen (University of Texas at Austin); Patrick Heimbach (University of Texas at Austin); Vikram V. Garg (University of Texas at Austin); Victor Ocaña (University of Texas at Austin); Craig Lee (University of Washington); Luc Rainville (University of Washington)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-19-0159.1

URL: <https://doi.org/10.1175/jtech-d-19-0159.1>

Publication Date: 2020-08-14

Volume/Issue: Volume 37, Issue 8

References: 79

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: Argo|Float (project management)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The lack of continuous spatial and temporal sampling of hydrographic measurements in large parts of the Arctic Ocean remains a major obstacle for quantifying mean state and variability of the Arctic Ocean circulation. This shortcoming motivates an assessment of the utility of Argo-type floats, the challenges of deploying such floats due to the presence of sea ice, and the implications of extended times of no surfacing on hydrographic inferences. Within the framework of an Arctic coupled ocean-sea ice state estimate that is constrained to available satellite and in situ observations, we establish metrics for quantifying the usefulness of such floats. The likelihood of float surfacing strongly correlates with the annual sea ice minimum cover. Within the float lifetime of 4-5 years, surfacing frequency ranges from 10-100 days in seasonally sea ice-covered regions to 1-3 years in multiyear sea ice-covered regions. The longer the float drifts under ice without surfacing, the larger the uncertainty in its position, which translates into larger uncertainties in hydrographic measurements. Below the mixed layer, especially in the western Arctic, normalized errors remain below 1, suggesting that measurements along a path whose only known positions are the beginning and end points can help constrain numerical models and reduce hydrographic uncertainties. The error assessment presented is a first step in the development of quantitative methods for guiding the design of observing networks. These results can and should be used to inform a float network design with suggested locations of float deployment and associated expected hydrographic uncertainties.

832. An analysis of Atlantic Water in the Arctic Ocean using the Arctic Subpolar Gyre State Estimate and observations

Authors: Jeffrey S. Grabon

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/26071

URL: <https://doi.org/10.1575/1912/26071>

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Volume/Issue: N/A

References: N/A

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Physical oceanography

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

833. Quantifying Dynamical Proxy Potential Through Shared Adjustment Physics in the North Atlantic

Authors: N. Loose (University of Bergen, Bjerknes Centre for Climate Research, University of Texas at Austin); P. Heimbach (University of Texas at Austin); H. R. Pillar (University of Texas at Austin); K. H. Nisancioglu (University of Oslo, University of Bergen, Bjerknes Centre for Climate Research)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016112

URL: <https://doi.org/10.1029/2020jc016112>

Publication Date: 2020-08-13

Volume/Issue: Volume 125, Issue 9

References: 83

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: observing system design, proxy, adjoint model, North Atlantic, uncertainty quantification (and 1 more)

Funding: H2020 European Research Council, National Science Foundation

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Open Access: N/A

Abstract: Oceanic quantities of interest (QoIs), for example, ocean heat content or transports, are often inaccessible to direct observation, due to the high cost of instrument deployment and logistical challenges. Therefore, oceanographers seek proxies for undersampled or unobserved QoIs. Conventionally, proxy potential is assessed via statistical correlations, which measure covariability without establishing causality. This paper introduces an alternative method: quantifying dynamical proxy potential. Using an adjoint model, this method unambiguously identifies the physical origins of covariability. A North Atlantic case study illustrates our method within the ECCO (Estimating the Circulation and Climate of the Ocean) state estimation framework. We find that wind forcing along the eastern and northern boundaries of the Atlantic drives a basin-wide response in North Atlantic circulation and temperature. Due to these large-scale teleconnections, a single subsurface temperature observation in the Irminger Sea informs heat transport across the remote Iceland-Scotland ridge (ISR), with a dynamical proxy potential of 19%. Dynamical proxy potential allows two equivalent interpretations: Irminger Sea subsurface temperature (i) shares 19% of its adjustment physics with ISR heat transport and (ii) reduces the uncertainty in ISR heat transport by 19% (independent of the measured temperature value), if the Irminger Sea observation is added without noise to the ECCO state estimate. With its two interpretations, dynamical proxy potential is simultaneously rooted in (i) ocean dynamics and (ii) uncertainty quantification and optimal observing system design, the latter being an emerging branch in computational science. The new method may therefore foster dynamics-based, quantitative ocean observing system design in the coming years.

834. The parameterization of mean dynamic topography based on the Lagrange basis functions

Authors: Hongkai Shi; Xiufeng He; Yihao Wu; Jia Huang

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2020.07.042

URL: <https://doi.org/10.1016/j.asr.2020.07.042>

Publication Date: 2020-08-11

Volume/Issue: Volume 66, Issue 9

References: 81

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

835. Evaluation of Global Ocean Models on Simulating the Deep Western Boundary Current in the Pacific

Authors: Huichang Jiang (Chinese Academy of Sciences); Liu Yu (Chinese Academy of Sciences); Hongzhou Xu (Chinese Academy of Sciences); Philip A. Vetter (Chinese Academy of Sciences)

Journal: Atmosphere-Ocean

Publisher: Informa UK Limited

DOI: 10.1080/07055900.2020.1789547

URL: <https://doi.org/10.1080/07055900.2020.1789547>

Publication Date: 2020-08-06

Volume/Issue: Volume 58, Issue 3

References: 46

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Chinese National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

836. Highly variable upper and abyssal overturning cells in the South Atlantic

Authors: M. Kersalé (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA., University of Miami); C. S. Meinen (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA.); R. C. Perez (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA.); M. Le Hénaff (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA., University of Miami); D. Valla (Servicio de Hidrografía Naval); T. Lamont (University of Cape Town, Oceans and Coasts Research Branch, Department of Environmental Affairs, Cape Town, South Africa.); O. T. Sato (University of São Paulo); S. Dong (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA.); T. Terre (Institut de Recherche pour le Développement (IRD)); M. van Caspel (University of São Paulo); M. P. Chidichimo (Consejo Nacional de Investigaciones Científicas y Técnicas, Servicio de Hidrografía Naval); M. van den Berg (Oceans and Coasts Research Branch, Department of Environmental Affairs, Cape Town, South Africa.); S. Speich (Laboratoire de Météorologie Dynamique-IPSL); A. R. Piola (Consejo Nacional de Investigaciones Científicas y Técnicas, Universidad de Buenos Aires, Buenos Aires, Argentina., Servicio de Hidrografía Naval); E. J. D. Campos (American University of Sharjah, University of São Paulo); I. Ansorge (University of Cape Town); D. L. Volkov (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA., University of Miami); R. Lumpkin (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA.); S. L. Garzoli (NOAA Atlantic Oceanographic and Meteorological Laboratory, Miami, FL, USA., University of Miami)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.aba7573

URL: <https://doi.org/10.1126/sciadv.aba7573>

Publication Date: 2020-08-05

Volume/Issue: Volume 6, Issue 32

References: 40

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Inter-American Institute for Global Change Research, CNPq (and 17 more)

License: CC BY-NC 4.0

Open Access: N/A

Abstract: We present the first-ever daily time series of the full-depth volume transports by the overturning circulation cells at 34.5 degreesS.

837. Observed Triple Mode of Salinity Variability in the Thermocline of Tropical Pacific Ocean

Authors: Shijian Hu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego); Cong Guan (Chinese Academy of Sciences, Institute of Oceanographic Instrumentation, Shandong Academy of Sciences Qingdao China, Pilot National Laboratory for Marine Science and Technology); Dunxin Hu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Fan Wang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Xi Lu (Chinese Academy of Sciences); Shihan Li (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016210

URL: <https://doi.org/10.1029/2020jc016210>

Publication Date: 2020-08-04

Volume/Issue: Volume 125, Issue 9

References: 47

Cited by: 6 publications

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Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China

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Open Access: N/A

Abstract: A well-defined Triple Mode of Salinity (TMOS) variability in the tropical Pacific thermocline layer (24-25.5 σ_{θ} isopycnal surfaces) is revealed from the Argo observations during 2004-2018. Thermocline salinity in the tropical northern and southwestern Pacific Ocean varies out of phase on interannual to near-decadal time scales with that in the tropical southeastern Pacific Ocean. The TMOS is attributed to anomalous advection of mean salinity as well as advection of anomalous salinity by the mean oceanic current on isopycnal surfaces. While the TMOS pattern is quite different from the thermal patterns associated with the El Niño-Southern Oscillation (ENSO) or the Pacific Decadal Oscillation (PDO), it is significantly associated with the ENSO and PDO indices with a time lag of 10 months. Isopycnal propagation of salinity anomalies and anomalous freshwater input in the outcropping region, also both associated with ENSO and PDO, are important in the formation of TMOS. The propagation speed of isopycnal salinity anomalies is close to documented phase speeds of baroclinic Rossby waves and velocities of zonal currents. The TMOS acts to shape the basin-scale Pacific water masses and potentially plays an important role in climate.

838. A multi-year time series of observation-based 3D horizontal and vertical quasi-geostrophic global ocean currents

Authors: Bruno Buongiorno Nardelli

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-12-1711-2020

URL: <https://doi.org/10.5194/essd-12-1711-2020>

Publication Date: 2020-08-03

Volume/Issue: Volume 12, Issue 3

References: 45

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Estimates of 3D ocean circulation are needed to improve our understanding of ocean dynamics and to assess their impact on marine ecosystems and Earth climate. Here we present the OMEGA3D product, an observation-based time series of (quasi-)global 3D ocean currents covering the 1993-2018 period, developed by the Italian Consiglio Nazionale delle Ricerche within the European Copernicus Marine Environment Monitoring Service (CMEMS). This dataset was obtained by applying a diabatic quasi-geostrophic (QG) diagnostic model to the data-driven CMEMS-ARMOR3D weekly reconstruction of temperature and salinity as well as ERA Interim fluxes. Outside the equatorial band, vertical velocities were retrieved in the upper 1500 m at 1°4' nominal resolution and successively used to compute the horizontal ageostrophic components. Root mean square differences between OMEGA3D total horizontal velocities and totally independent drifter observations at two different depths (15 and 1000 m) decrease with respect to corresponding estimates obtained from zero-order geostrophic balance, meaning that estimated vertical velocities can also be deemed reliable. OMEGA3D horizontal velocities are also closer to drifter observations than velocities provided by a set of reanalyses spanning a comparable time period but based on data assimilation in ocean general circulation numerical models. The full OMEGA3D product (released on 31 March 2020) is available upon free registration at https://doi.org/10.25423/cmcc/multiobs_glo_phy_w_rep_015_007 (Buongiorno Nardelli, 2020a). The reduced subset used here for validation and review purposes is openly available at <https://doi.org/10.5281/zenodo.3696885> (Buongiorno Nardelli, 2020b).

839. Non-Tidal Mass Variations in the IGS Second Reprocessing Campaign: Interpretations and Noise Analysis from GRACE and Geophysical Models

Authors: Liansheng Deng (School of Electrical and Electronic Information Engineering, Hubei Polytechnic University, Huangshi 435003, China); Hua Chen (Wuhan University); Ailong Ma (Wuhan University); Qusen Chen (Wuhan University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs12152477

URL: <https://doi.org/10.3390/rs12152477>

Publication Date: 2020-08-03

Volume/Issue: Volume 12, Issue 15

References: 60

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Scientific Research Project of Hubei Provincial Department of Education, National Natural Science Foundation of China

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Open Access: N/A

Abstract: Vertical deformations caused by non-tidal mass variations still remain in global navigation satellite system (GNSS) height time series, and can be computed from both Gravity Recovery and Climate Experiment (GRACE) and geophysical models. In this study, we provide a thorough evaluation of the relationships between these different techniques in the global scale by comparing non-tidal vertical deformations from IGS second reprocessing campaign (IG2), GRACE and Global Geophysical Fluid Center (GGFC) solutions, and investigate the noise properties of the GNSS corrected by GRACE solutions and GNSS corrected by GGFC solutions for global stations using optimal noise models. Our results demonstrate that the consistency between seasonal vertical deformations derived from GNSS, GRACE and GGFC is high. When correcting GNSS deformations with GRACE and GGFC solutions, 81% and 73% of the 186 stations have the weight root mean square (WRMS) reduction, respectively. The WRMS variations averaged over all stations are -12.3% and -5.6%, respectively for GNSS corrected by GRACE and GNSS corrected by GGFC solutions. The obvious difference occurs in the GNSS corrected by GGFC solutions WRMS increase, with the mean increase value up to 29%, mainly happening to stations located on islands or small peninsulas. In addition, noise properties of the GNSS corrected by GRACE solutions and GNSS corrected by GGFC solutions for global stations are investigated using optimal noise models. After correcting non-tidal loading effects, the solutions of GNSS corrected by GRACE solutions have the lowest noise level, and can occupy 5% of the noise behavior presenting in global stations, while the solutions of GNSS corrected by GGFC solutions can bring more than 5% of the noise into global stations, implying that GRACE correction solutions can present more favorable results when interpreting GNSS non-tidal loading deformations.

840. Transport of Legacy Perfluoroalkyl Substances and the Replacement Compound HFPO-DA through the Atlantic Gateway to the Arctic Ocean--Is the Arctic a Sink or a Source?

Authors: Hanna Joerss (Department for Environmental Chemistry, Helmholtz-Zentrum Geesthacht, Centre for Materials and Coastal Research, 21502 Geesthacht, Germany, University of Hamburg); Zhiyong Xie (Department for Environmental Chemistry, Helmholtz-Zentrum Geesthacht, Centre for Materials and Coastal Research, 21502 Geesthacht, Germany); Charlotte C. Wagner (Harvard University); Wilken-Jon von Appen (Alfred Wegener Institute); Elsie M. Sunderland (Harvard University); Ralf Ebinghaus (Department for Environmental Chemistry, Helmholtz-Zentrum Geesthacht, Centre for Materials and Coastal Research, 21502 Geesthacht, Germany)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

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Publication Date: 2020-07-29

Volume/Issue: Volume 54, Issue 16

References: 62

Cited by: 102 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Institute of Environmental Health Sciences, Bundesministerium für Bildung und Forschung, Alfred Wegener Institute Helmholtz Centre for Polar and Marine Research (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

841. Boundary-Included Enhanced Water Storage Changes Inferred by GPS in the Pacific Rim of the Western United States

Authors: Yingchun Shen (Chinese Academy of Sciences); Haoming Yan (Chinese Academy of Sciences); Peng Peng (Chinese Academy of Sciences); Wei Feng (Chinese Academy of Sciences); Zizhan Zhang (Chinese Academy of Sciences); Yunji Song (Qingdao Surveying and Mapping Institute, Qingdao 266034, China); Xixuan Bai (Wuhan Institute of Technology, Wuhan 430205, China)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs12152429

URL: <https://doi.org/10.3390/rs12152429>

Publication Date: 2020-07-29

Volume/Issue: Volume 12, Issue 15

References: 32

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: Vertical deformation, Water storage changes, GPS, Disc load empirical function (DLEF), Boundary-included inversion

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: We developed a new boundary-included inversion model to improve the terrestrial water storage (TWS) inverted from regional GPS vertical deformation data. Through defining a new disc load empirical function (DLEF) and considering the mass change effect from the near but outside region, the result shows the TWS is more reasonable than the one inverted directly. Six simulation tests further confirmed the effectiveness of the boundary-included model. Finally, our new boundary-included model was used to derive the TWS in the Pacific Rim of the western United States based on the GPS-observed vertical deformation information. The inversion results show that our boundary-included inversion model can effectively improve the inversion results by 10-20% in terms of variance reduction in the boundary regions.

842. Feedbacks Driving Interdecadal Variability in Southern Ocean Convection in Climate Models: A Coupled Oscillator Mechanism

Authors: Anand Gnanadesikan (Johns Hopkins University); Cassidy M. Speller (Johns Hopkins University); Grace Ringlein (Department of Physics and Astronomy, University of Pennsylvania, Philadelphia, Pennsylvania); John San Soucie (Massachusetts Institute of Technology); Jordan Thomas (Johns Hopkins University); Marie-Aude Pradal (Johns Hopkins University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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References: 68

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Division of Ocean Sciences

License: N/A

Open Access: N/A

Abstract: Numerous climate models display large-amplitude, long-period variability associated with quasiperiodic convection in the Southern Ocean, but the mechanisms responsible for producing such oscillatory convection are poorly understood. In this paper we identify three feedbacks that help generate such oscillations within an Earth system model with a particularly regular oscillation. The first feedback involves increased (decreased) upward mixing of warm interior water to the surface, resulting in more (less) evaporation and loss of heat to the atmosphere which produces more (less) mixing. This positive feedback helps explain why temperature anomalies are not damped out by surface forcing. A second key mechanism involves convective (nonconvective) events in the Weddell Sea causing a relaxation (intensification) of westerly winds, which at some later time results in a pattern of currents that reduces (increases) the advection of freshwater out of the Weddell Sea. This allows for the surface to become lighter (denser) which in turn can dampen (trigger) convection--so that the overall feedback is a negative one with a delay--helping to produce a multidecadal oscillation time scale. The decrease (increase) in winds associated with convective (nonconvective) states also results in a decrease (increase) in the upward mixing of salt in the Eastern Weddell Sea, creating a negative (positive) salinity anomaly that propagates into the Western Weddell Sea and dampens (triggers) convection--again producing a negative feedback with a delay. A principal oscillatory pattern analysis yields a reasonable prediction for the period of oscillation. Strengths of the feedbacks are sensitive to parameterization of mesoscale eddies.

843. The ECCO-Darwin Data-Assimilative Global Ocean Biogeochemistry Model: Estimates of Seasonal to Multidecadal Surface Ocean pCO₂ and Air-Sea CO₂ Flux

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Keywords: air-sea CO₂ flux, data assimilation, ocean carbon cycle, ecosystem model, ocean modeling (and 1 more)

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Open Access: N/A

Abstract: Quantifying variability in the ocean carbon sink remains problematic due to sparse observations and spatiotemporal variability in surface ocean pCO₂. To address this challenge, we have updated and improved ECCO-Darwin, a global ocean biogeochemistry model that assimilates both physical and biogeochemical observations. The model consists of an adjoint-based ocean circulation estimate from the Estimating the Circulation and Climate of the Ocean (ECCO) consortium and an

ecosystem model developed by the Massachusetts Institute of Technology Darwin Project. In addition to the data-constrained ECCO physics, a Green's function approach is used to optimize the biogeochemistry by adjusting initial conditions and six biogeochemical parameters. Over seasonal to multidecadal timescales (1995-2017), ECCO-Darwin exhibits broad-scale consistency with observed surface ocean pCO₂ and air-sea CO₂ flux reconstructions in most biomes, particularly in the subtropical and equatorial regions. The largest differences between CO₂ uptake occur in subpolar seasonally stratified biomes, where ECCO-Darwin results in stronger winter uptake. Compared to the Global Carbon Project OBMs, ECCO-Darwin has a time-mean global ocean CO₂ sink (2.47 +/- 0.50 Pg C year⁻¹) and interannual variability that are more consistent with interpolation-based products. Compared to interpolation-based methods, ECCO-Darwin is less sensitive to sparse and irregularly sampled observations. Thus, ECCO-Darwin provides a basis for identifying and predicting the consequences of natural and anthropogenic perturbations to the ocean carbon cycle, as well as the climate-related sensitivity of marine ecosystems. Our study further highlights the importance of physically consistent, property-conserving reconstructions, as are provided by ECCO, for ocean biogeochemistry studies.

844. Atmospheric Simulations of Total Column CO₂ Mole Fractions from Global to Mesoscale within the Carbon Monitoring System Flux Inversion Framework

Authors: Martha P. Butler (Pennsylvania State University); Thomas Lauvaux (Pennsylvania State University); Sha Feng (Pennsylvania State University); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kenneth J. Davis (Pennsylvania State University)

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Keywords: Model coupling, Mesoscale modeling, Carbon cycle, GOSAT, XCO₂

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Open Access: N/A

Abstract: Quantifying the uncertainty of inversion-derived CO₂ surface fluxes and attributing the uncertainty to errors in either flux or atmospheric transport simulations continue to be challenges in the characterization of surface sources and sinks of carbon dioxide (CO₂). Despite recent studies inferring fluxes while using higher-resolution modeling systems, the utility of regional-scale models remains unclear when compared to existing coarse-resolution global systems. Here, we present an off-line coupling of the mesoscale Weather Research and Forecasting (WRF) model to optimized biogenic CO₂ fluxes and mole fractions from the global Carbon Monitoring System inversion system (CMS-Flux). The coupling framework consists of methods to constrain the mass of CO₂ introduced into WRF, effectively nesting our regional domain covering most of North America (except the northern half of Canada) within the CMS global model. We test the coupling by simulating Greenhouse gases Observing SATellite (GOSAT) column-averaged dry-air mole fractions (XCO₂) over North America for 2010. We find mean model-model differences in summer of ~0.12 ppm, significantly lower than the original coupling scheme (from 0.5 to 1.5 ppm, depending on the boundary). While 85% of the XCO₂ values are due to long-range transport from outside our North American domain, most of the model-model differences appear to be due to transport differences in the fraction of the troposphere below 850 hPa. Satellite data from GOSAT and tower and aircraft data are used to show that vertical transport above the Planetary Boundary Layer is responsible for significant model-model differences in the horizontal distribution of column XCO₂ across North America.

845. Tidal Modulation of Antarctic Ice Shelf Melting

Authors: Ole Richter; David E. Gwyther; Matt A. King; Benjamin K. Galton-Fenzi

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2020-169

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Keywords: Iceberg|Antarctic ice sheet

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Open Access: N/A

Abstract: . Tides influence basal melting of individual Antarctic ice shelves, but their net impact on Antarctic-wide ice-ocean interaction has yet to be constrained. Here we quantify the impact of tides on ice shelf melting and the continental shelf seas by means of a 4 km resolution circum-Antarctic ocean model. Activating tides in the model increases the total basal mass loss by 57 Gt/yr (4 %), while decreasing continental shelf temperatures by 0.04 degreesC, indicating a slightly more efficient conversion of ocean heat into ice shelf melting. Regional variations can be larger, with melt rate modulations exceeding 500 % and temperatures changing by more than 0.5 degreesC, highlighting the importance of capturing tides for robust modelling of glacier systems and coastal oceans. Tide-induced changes around the Antarctic Peninsula have a dipolar distribution with decreased ocean temperatures and reduced melting towards the Bellingshausen Sea and warming along the continental shelf break on the Weddell Sea side. This warming extends under the Ronne Ice Shelf, which also features one of the highest increases in area-averaged basal melting (150 %) when tides are included. Further, by means of a singular spectrum analysis, we explore the processes that cause variations in melting and its drivers in the boundary layer over periods of up to one month. At most places friction velocity varies at tidal timescales (one day or faster), while thermal driving changes at slower rates (longer than one day). In some key regions under the large cold-water ice shelves, however, thermal driving varies faster than friction velocity and this can not be explained by tidal modulations in boundary layer exchange rates alone. Our results suggest that large scale ocean models aiming to predict accurate ice shelf melt rates will need to explicitly resolve tides.

846. Impacts of Oceanic and Atmospheric Heat Transports on Sea Ice Extent

Authors: Jake Aylmer (University of Reading); David Ferreira (University of Reading); Daniel Feltham (University of Reading)

Journal: Journal of Climate

Publisher: American Meteorological Society

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Abstract: Climate-model biases in ocean heat transport (OHT) have been proposed as a major contributor to uncertainties in projections of sea ice extent. To better understand the impact of OHT on sea ice extent and compare it to that of atmospheric heat transport (AHT), an idealized, zonally averaged energy balance model (EBM) is developed. This is distinguished from previous EBM work by coupling a diffusive mixed layer OHT and a prescribed OHT contribution, with an atmospheric EBM and a reduced-complexity sea ice model. The ice-edge latitude is roughly linearly related to the convergence of each heat transport component, with different sensitivities depending on whether the ice cover is perennial or seasonal. In both regimes, Bjerknes compensation (BC) occurs such that the response of AHT partially offsets the impact of changing OHT. As a result, the effective sensitivity of ice-edge retreat to increasing OHT is only $\sim 2/3$ of the actual sensitivity (i.e., eliminating the BC effect). In the perennial regime, the sensitivity of the ice edge to OHT is about twice that to AHT, while in the seasonal regime they are similar. The ratio of sensitivities is, to leading order, determined by atmospheric longwave feedback parameters in the perennial regime. Here, there is no parameter range in which the ice edge is more sensitive to AHT than OHT.

847. Comparing Arctic Sea Ice Model Simulations to Satellite Observations by Multiscale Directional Analysis of Linear Kinematic Features

Authors: Mahdi Mohammadi-Aragh (Alfred Wegener Institute, Leibniz Institute for Baltic Sea Research); Martin Losch (Alfred Wegener Institute); Helge F. Goessling (Alfred Wegener Institute)

Journal: Monthly Weather Review

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Abstract: Sea ice models have become essential components of weather, climate, and ocean models. A realistic representation of sea ice affects the reliability of process representation, environmental forecast, and climate projections. Realistic simulations of sea ice kinematics require the consideration of both large-scale and finescale geomorphological structures such as linear kinematic features (LKF). We propose a multiscale directional analysis (MDA) that diagnoses the spatial characteristics of LKFs. The MDA is different from previous analyses in that it (i) does not detect LKFs as objects, (ii) takes into account the width of LKFs, and (iii) estimates scale-dependent orientation and intersection angles. The MDA is applied to pairs of deformation fields derived from satellite remote sensing data and from a numerical model simulation with a horizontal grid spacing of ~4.5 km. The orientation and intersection angles of LKFs agree with the observations and confirm the visual impression that the intersection angles tend to be smaller in the satellite data compared to the model data. The MDA distributions can be used to compare satellite data and numerical model fields using conventional metrics such as a Euclidean distance, the Bhattacharyya coefficient, or the Earth mover's distance. The latter is found to be the most meaningful metric to compare distributions of LKF orientations and intersection angles. The MDA proposed here provides a tool to diagnose if modified sea ice rheologies lead to more realistic simulations of LKFs.

848. A Dynamical Framework for Interpreting Ancient Sea Surface Temperatures

Authors: Emily J. Judd (Department of Earth Sciences Syracuse University Syracuse NY USA); Tripti Bhattacharya (Department of Earth Sciences Syracuse University Syracuse NY USA); Linda C. Ivany (Department of Earth Sciences Syracuse University Syracuse NY USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Keywords: paleoceanography, paleoclimate, proxy data, sea surface temperature, Paleozoic (and 1 more)

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Open Access: N/A

Abstract: Efforts to estimate past global mean temperature and latitudinal gradients must contend with spatial heterogeneity in sea surface temperatures (SSTs). Here, we use modern SSTs to show that the environments from which most paleoclimatic data are drawn, shallow epeiric seas and continental margins, are systematically offset from zonal mean temperatures. Epeiric seas are warmer and more seasonal than open-ocean values from the same latitudes, while continental margins exhibit consistent and predictable deviations related to gyre circulation. Warm temperatures inferred from Paleozoic proxy data may largely reflect that these data derive almost entirely from epeiric seas. Moreover, pseudoproxy analysis using Paleogene sampling localities demonstrates how undersampling of the full range of dynamical environments associated with gyre circulation can generate spurious estimates of latitudinal temperature gradients. Recognition of these global patterns permits a predictive framework within which to more robustly interpret proxy data, improve Earth system models, and reconstruct ancient dynamic regimes.

849. Sensitivity of the Atlantic Meridional Overturning Circulation to Model Resolution in CMIP6 HighResMIP Simulations and Implications for Future Changes

Authors: Malcolm J. Roberts (Met Office); Laura C. Jackson (Met Office); Christopher D. Roberts (European Centre for Medium-Range Weather Forecasts (ECMWF)); Virna Meccia (Istituto di Scienze dell'Atmosfera e del Clima (CNR-ISAC), Bologna Bologna Italy); David Docquier (Rossby Centre Swedish Meteorological and Hydrological Institute (SMHI) Norrköping Sweden); Torben Koenigk (Rossby Centre Swedish Meteorological and Hydrological Institute (SMHI) Norrköping Sweden); Pablo Ortega (Barcelona Supercomputing Center); Eduardo Moreno-Chamorro (Barcelona Supercomputing Center); Alessio Bellucci (Fondazione Centro Euro-Mediterraneo sui Cambiamenti Climatici (CMCC) Bologna Italy); Andrew Coward (National Oceanography Centre); Sybren Drijfhout (Koninklijk Nederlands Meteorologisch Instituut (KNMI) De Bilt The Netherlands); Eleftheria Exarchou (Barcelona Supercomputing Center); Oliver Gutjahr (Max Planck Institute for Meteorology); Helene Hewitt (Met Office); Doroteaciro Iovino (Fondazione Centro Euro-Mediterraneo sui Cambiamenti Climatici (CMCC) Bologna Italy); Katja Lohmann (Max Planck Institute for Meteorology); Dian Putrasahan (Max Planck Institute for Meteorology); Reinhard Schiemann (University of Reading); Jon Seddon (Met Office); Laurent Terray (CECI, Université de Toulouse, CERFACS/CNRS Toulouse France); Xiaobiao Xu (Florida State University); Qiuying Zhang (Department of Oceanography Texas A&M; University College Station USA, International Laboratory for High Resolution Earth System Prediction (iHESP) College Station TX USA); Ping Chang (Department of Oceanography Texas A&M; University College Station USA, International Laboratory for High Resolution Earth System Prediction (iHESP) College Station TX USA); Stephen G. Yeager (National Center for Atmospheric Research, International Laboratory for High Resolution Earth System Prediction (iHESP) College Station TX USA); Frederic S. Castruccio (National Center for Atmospheric Research, International Laboratory for High Resolution Earth System Prediction (iHESP) College Station TX USA); Shaoqing Zhang (Qingdao National Laboratory for Marine Science (QNLN) Qingdao China, International Laboratory for High Resolution Earth System Prediction (iHESP) College Station TX USA); Lixin Wu (Qingdao National Laboratory for Marine Science (QNLN) Qingdao China, International Laboratory for High Resolution Earth System Prediction (iHESP) College Station TX USA)

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Funding: Horizon 2020 Framework Programme, National Science Foundation, Natural Environment Research Council (and 1 more)

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Abstract: A multimodel, multiresolution ensemble using Coupled Model Intercomparison Project Phase 6 (CMIP6) High Resolution Model Intercomparison Project (HighResMIP) coupled experiments is used to assess the performance of key aspects of the North Atlantic circulation. The Atlantic Meridional Overturning Circulation (AMOC), and related heat transport, tends to become stronger as ocean model resolution is enhanced, better agreeing with observations at 26.5 degreesN. However, for most models the circulation remains too shallow compared to observations and has a smaller temperature contrast between the northward and southward limbs of the AMOC. These biases cause the northward heat transport to be systematically too low for a given overturning strength. The higher-resolution models also tend to have too much deep mixing in the subpolar gyre. In the period 2015-2050 the overturning circulation tends to decline more rapidly in the higher-resolution models, which is related to both the mean state and to the subpolar gyre contribution to deep water formation. The main part of the decline comes from the Florida Current component of the circulation. Such large declines in AMOC are not seen in the models with resolutions more typically used for climate studies, suggesting an enhanced risk for Northern Hemisphere climate change. However, only a small number of different ocean models are included in the study.

850. On the seasonality of waters below the seasonal thermocline in the Gulf of Cádiz

Authors: R.F. Sánchez-Leal; M.J. Bellanco; C. Naranjo; J. García-Lafuente; C. González-Pola

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851. GISS-E2.1: Configurations and Climatology

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Abstract: This paper describes the GISS-E2.1 contribution to the Coupled Model Intercomparison Project, Phase 6 (CMIP6). This model version differs from the predecessor model (GISS-E2) chiefly due to parameterization improvements to the atmospheric and ocean model components, while keeping atmospheric resolution the same. Model skill when compared to modern era climatologies is significantly higher than in previous versions. Additionally, updates in forcings have a material impact on the results. In particular, there have been specific improvements in representations of modes of variability (such as the Madden-Julian Oscillation and other modes in the Pacific) and significant improvements in the simulation of the climate of the Southern Oceans, including sea ice. The effective climate sensitivity to 2 x CO₂ is slightly higher than previously at 2.7-3.1 degreesC (depending on version) and is a result of lower CO₂ radiative forcing and stronger positive feedbacks.

852. Contributions of Greenland GPS Observed Deformation From Multisource Mass Loading Induced Seasonal and Transient Signals

Authors: Wenhao Li (Wuhan University); C. K. Shum (Ohio State University, Chinese Academy of Sciences); Fei Li (Wuhan University); Shengkai Zhang (Wuhan University); Feng Ming (State Key Laboratory of Geo-information Engineering, Xi'an Institute of Surveying and Mapping Xi'an China); Wei Chen (Chinese Academy of Sciences); Bao Zhang (Wuhan University); Jintao Lei (Department of Land Surveying and Geo-Informatics Hong Kong Polytechnic University Hong Kong); Qingchuan Zhang (Wuhan University)

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Abstract: Global Positioning System (GPS) provides critical geodetic constraints on Earth's elastic response induced by ice and other mass loadings in Greenland. Previous studies focused on long-term ice mass changes, with relatively fewer studies on transient signals and had used fewer GPS stations. Here we reconstructed 44 Greenland-wide coastal decadal GPS time series using multichannel singular spectral analysis and quantified the origins of spatiotemporal patterns of transient to seasonal signals. We used an exhaustive list of geophysical processes, including surface mass balance (SMB), atmospheric pressure, continental hydrology, nontidal ocean loading, bedrock thermal expansion, precipitation, runoff, and ice discharge, and identified that most of Greenland's GPS sites exhibit SMB-induced transient signals during 2012-2015, with maximum displacement reaching 20.45 mm. Analyses of differenced time series between adjacent GPS stations near peripheral glaciers reveal that locally varying seasonal signals are primarily attributable to SMB, followed by atmosphere.

853. Basin-Width Dependence of Northern Deep Convection

Authors: Madeleine K. Youngs (Massachusetts Institute of Technology, Woods Hole Oceanographic Institution); Raffaele Ferrari (Massachusetts Institute of Technology); Glenn R. Flierl (Massachusetts Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 40

Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: American Meteorological Society, National Science Foundation

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Open Access: N/A

Abstract: Convection penetrates to the ocean bottom in the North Atlantic but not in the North Pacific. This study examines the role of basin width in shutting down high-latitude ocean convection. Deep convection is triggered by polar cooling, but it is opposed by precipitation. A two-layer analytical model illustrates that the overturning circulation acts to mitigate the effect of precipitation by advecting salty, dense water from subtropical latitudes to polar latitudes. The nonlinear dependence of the overturning strength on basin width makes it more efficient in a narrow basin, resulting in a convection shutdown at a stronger freshwater forcing. These predictions are confirmed by simulations with a general circulation model configured with a single closed basin to the north and a reentrant channel to the south. This suggests that basin width may play a role in suppressing convection in the North Pacific but not in the North Atlantic.

854. Why Does the CP El Niño less Frequently Evolve Into La Niña than the EP El Niño?

Authors: Shan He (Sun Yat-Sen University, Department of Earth System Science University of California at Irvine Irvine CA USA); Jin-Yi Yu (Department of Earth System Science University of California at Irvine Irvine CA USA); Song Yang (Sun Yat-Sen University, Southern Marine Science and Engineering Guangdong Laboratory); Shih-Wei Fang (Department of Earth System Science University of California at Irvine Irvine CA USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 50

Cited by: 4 publications

Language: en

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Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, China Postdoctoral Science Foundation

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Open Access: N/A

Abstract: During 1958-2017, the Central-Pacific (CP) El Niño evolved into La Niña less frequently (38%) than the Eastern-Pacific (EP) El Niño (75%). Composite analyses reveal that the reversal of zonal wind anomalies in the tropical western Pacific is the key mechanism for the EP El Niño's transitions to La Niña. This reversal induces oceanic Kelvin waves to promote the La Niña onset. This reversal mechanism is often triggered by the Indo-Pacific teleconnection produced by the EP El Niño. Consequently, the EP El Niño often experiences the transitional evolution. For the CP events, the southward shift of westerly anomalies in the tropical central Pacific is the key mechanism for transitions from El Niño to La Niña via local processes. However, the subtropical teleconnection of the CP El Niño prevents this mechanism from occurring and often causes the nontransitional evolution for the CP events.

855. Advective Controls on the North Atlantic Anthropogenic Carbon Sink

Authors: S. M. Ridge (Columbia University); G. A. McKinley (Columbia University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gb006457

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References: 56

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Columbia University

License: CC BY 4.0

Open Access: N/A

Abstract: Though it is clear that the North Atlantic is the site of the highest storage of anthropogenic carbon (Cant) per area, it is uncertain whether the air-sea Cant fluxes contributing to North Atlantic Cant storage occur in the subpolar gyre or upstream in the subtropical gyre. Using data and models, we show that air-sea Cant uptake capacity is advected into the subpolar gyre along the same subsurface pathway as nutrients. This pathway is known as the nutrient stream. On the A22 section between Woods Hole and Bermuda, nutrient stream waters are the oldest in the upper 2,000 m and contain low Cant. These northward moving waters are sufficiently depleted in Cant such that they could sustain a subpolar air-sea Cant flux of $-0.19 \text{ Pg Cant year}^{-1}$. The ocean hindcast model used here indicates that despite some subtropical re-circulation, uptake capacity is transported into the subpolar gyre where it sustains subpolar air-sea Cant uptake. With this model, we show that high- and low-end estimates of subpolar air-sea Cant flux are reconciled by accounting for a factor of two difference in their respective study areas. If half of the observed air-sea Cant uptake capacity transport at A22 is ventilated in the subpolar region, it can fully support high-end estimates of the subpolar air-sea Cant sink ($-0.09 \pm 0.01 \text{ Pg Cant year}^{-1}$).

856. Importance of Boundary Processes for Heat Uptake in the Subpolar North Atlantic

Authors: D. G. Desbruyères (Institut de Recherche pour le Développement (IRD)); B. Sinha (National Oceanography Centre); E. L. McDonagh (NORCE Norwegian Research Centre, Bjerknes Centre for Climate Research, National Oceanography Centre); S. A. Josey (National Oceanography Centre); N. P. Holliday (National Oceanography Centre); D. A. Smeed (National Oceanography Centre); A. L. New (National Oceanography Centre); A. Megann (National Oceanography Centre); B. I. Moat (National Oceanography Centre)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016366

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References: 56

Cited by: 8 publications

Language: en

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Keywords: N/A

Funding: Natural Environment Research Council

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Open Access: N/A

Abstract: The decadal to multidecadal temperature variability of the intermediate (700-2,000 m) North Atlantic Subpolar Gyre (SPG) significantly imprints the global pattern of ocean heat uptake. Here, the origins and dominant pathways of this variability are investigated with an ocean analysis product (EN4), an ocean state estimate (ECCOV4), and idealized modeling approaches. Sustained increases and decreases of intermediate temperature in the SPG correlate with long-lasting warm and cold states of the upper ocean with the largest anomalous vertical heat exchanges confined to the vicinity of continental boundaries and strong ocean currents. In particular, vertical diffusive processes along the boundaries of the Labrador, Irminger, and Newfoundland basins are important drivers of the recent intermediate depth warming trend observed during 1996-2014. The overall effect of those processes is captured by a one-dimensional diffusive model with appropriate boundary-like parametrization and demonstrated through the boundary-focused downward propagation of a passive tracer in a 3-D numerical simulation. Our results imply that the slow and quasi-periodic ventilation of intermediate thermohaline properties and associated heat uptake in the SPG are not strictly driven by convection-restratification events in the open seas but also receives a key contribution from boundary sinking and mixing. Increased skill for modeling and predicting intermediate-depth ocean properties in the North Atlantic will hence require the appropriate representation of surface-deep dynamical connections within the boundary currents encircling Greenland and Newfoundland.

857. Drivers of Marine Heatwaves in the East China Sea and the South Yellow Sea in Three Consecutive Summers During 2016-2018

Authors: Guandong Gao (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences, Center for Mega Science Chinese Academy of Science Qingdao China); Maxime Marin (CSIRO Oceans and Atmosphere Crawley Western Australia Australia, University of Tasmania); Ming Feng (CSIRO Oceans and Atmosphere Crawley Western Australia Australia); Baoshu Yin (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences, Center for Mega Science Chinese Academy of Science Qingdao China); Dezhou Yang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences, Center for Mega Science Chinese Academy of Science Qingdao China); Xingru Feng (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences, Center for Mega Science Chinese Academy of Science Qingdao China); Yang Ding (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Dehai Song (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016518

URL: <https://doi.org/10.1029/2020jc016518>

Publication Date: 2020-07-01

Volume/Issue: Volume 125, Issue 8

References: 77

Cited by: 80 publications

Language: en

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Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: In 3 consecutive years from 2016 to 2018, extreme ocean warming events, or marine heatwaves (MHWs), occurred during boreal summers in the East China Sea (ECS) and South Yellow Sea (SYS), which is unprecedented in the past four decades based on the satellite record. In this study, we used a high-resolution hydrodynamic model based on Finite Volume Community Ocean Model (FVCOM) to simulate the evolution of these warming events. An upper ocean temperature budget (0-20 m) analysis based on the model results shows that the shortwave radiation and the ocean advection anomalies jointly contributed to the anomalous warming in the three successive summers (June-August) in the SYS and the north part of the ECS. In addition, the reduction of surface wind speeds during the 2016 and 2017 summers further weakened the vertical mixing, thereby enhancing the anomalous warming in the north part of the ECS adjacent to the SYS. During the three summers, the increases of shortwave radiation were closely related to the East Asian Summer Monsoon (EASM) variability, which reduced the cloud cover in the ECS and SYS, whereas the advection anomalies were mostly associated with regional wind anomalies. In summer 2018, upper ocean heat was transported into the central trough of the SYS, accumulated in an anticyclonic eddy generated by the anomalous wind stress curls. Therefore, despite the primary driver of the MHWs is the EASM variation, regional

processes are critical to driving the spatial pattern of the MHW intensity in the ECS and SYS.

858. Evaluation of heat and salt transports by mesoscale eddies in the Lofoten Basin

Authors: Tatyana Belonenko; Vadim Zinchenko; Svetlana Gordeeva; Roshin P. Raj

Journal: Russian Journal of Earth Sciences

Publisher: Geophysical Center of the Russian Academy of Sciences

DOI: 10.2205/2020es000720

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Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: N/A

859. Improved Constraints on Northern Extratropical CO₂ Fluxes Obtained by Combining Surface-Based and Space-Based Atmospheric CO₂ Measurements

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Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

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References: 104

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: carbon cycle, data assimilation, CO₂ flux, TCCON, OCO-2 (and 1 more)

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Open Access: N/A

Abstract: Top-down estimates of CO₂ fluxes are typically constrained by either surface-based or space-based CO₂ observations. Both of these measurement types have spatial and temporal gaps in observational coverage that can lead to differences in inferred fluxes. Assimilating both surface-based and space-based measurements concurrently in a flux inversion framework improves observational

coverage and reduces sampling related artifacts. This study examines the consistency of flux constraints provided by these different observations and the potential to combine them by performing a series of 6-year (2010-2015) CO₂ flux inversions. Flux inversions are performed assimilating surface-based measurements from the in situ and flask network, measurements from the Total Carbon Column Observing Network (TCCON), and space-based measurements from the Greenhouse Gases Observing Satellite (GOSAT), or all three data sets combined. Combining the data sets results in more precise flux estimates for subcontinental regions relative to any of the data sets alone. Combining the data sets also improves the accuracy of the posterior fluxes, based on reduced root-mean-square differences between posterior flux-simulated CO₂ and aircraft-based CO₂ over midlatitude regions (0.33-0.56 ppm) in comparison to GOSAT (0.37-0.61 ppm), TCCON (0.50-0.68 ppm), or in situ and flask measurements (0.46-0.56 ppm) alone. These results suggest that surface-based and GOSAT measurements give complementary constraints on CO₂ fluxes in the northern extratropics and can be combined in flux inversions to improve constraints on regional fluxes. This stands in contrast with many earlier attempts to combine these data sets and suggests that improvements in the NASA Atmospheric CO₂ Observations from Space (ACOS) retrieval algorithm have significantly improved the consistency of space-based and surface-based flux constraints.

860. Mechanisms Underlying Recent Arctic Atlantification

Authors: Helene Asbjørnsen (University of Bergen, Bjerknes Centre for Climate Research); Marius Årthun (University of Bergen, Bjerknes Centre for Climate Research); Øystein Skagseth (Institute of Marine Research, Bjerknes Centre for Climate Research); Tor Eldevik (University of Bergen, Bjerknes Centre for Climate Research)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl088036

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Publication Date: 2020-06-20

Volume/Issue: Volume 47, Issue 15

References: 42

Cited by: 76 publications

Language: en

Subjects: N/A

Keywords: Arctic climate change, Barents Sea, ECCOv4-r3, Atlantification

Funding: Horizon 2020, Norges Forskningsråd

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Open Access: N/A

Abstract: Recent warming and reduced sea ice concentrations in the Atlantic sector of the Arctic Ocean are the main signatures of ongoing Arctic "Atlantification." The mechanisms driving the warming trends are nevertheless still debated, particularly regarding the relative importance of oceanic and atmospheric heat fluxes. Here, heat budgets along main Atlantic water pathways through the Barents Sea and Fram Strait are constructed to investigate the mechanisms of Atlantification during 1993-2014. The largest warming trends occur south of the winter ice edge, with ocean advection as the main driver. Warming in the marginal ice zone is mainly due to low surface heat loss from the 1990s to the mid-2000s. In the ice-covered northwestern Barents Sea, ocean advection and air-sea heat fluxes act in concert to drive a gradual warming of the upper ocean. Despite a weakened stratification, no evidence is found of vertical oceanic temperature fluxes driving this upper-ocean warming.

861. Marine Nitrous Oxide Emissions From Three Eastern Boundary Upwelling Systems Inferred From Atmospheric Observations

Authors: A. L. Ganesan (University of Bristol); M. Manizza (Scripps Institution of Oceanography / University of California San Diego); E. J. Morgan (Scripps Institution of Oceanography / University of California San Diego); C. M. Harth (Scripps Institution of Oceanography / University of California San Diego); E. Kozlova (University of Exeter); T. Lueker (Scripps Institution of Oceanography / University of California San Diego); A. J. Manning (Met Office); M. F. Lunt (University of Edinburgh); J. Mühle (Scripps Institution of Oceanography / University of California San Diego); J. V. Lavric (Max Planck Institute for Biogeochemistry); M. Heimann (Institute for Atmospheric and Earth System Research (INAR)/Physics University of Helsinki Helsinki Finland, Max Planck Institute for Biogeochemistry); R. F. Weiss (Scripps Institution of Oceanography / University of California San Diego); M. Rigby (University of Bristol)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Publication Date: 2020-06-20

Volume/Issue: Volume 47, Issue 14

References: 48

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: Nitrous oxide

Funding: Natural Environment Research Council, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Eastern Boundary Upwelling Systems (EBUSs) are coastal hotspots of the potent greenhouse gas nitrous oxide (N₂O). However, estimates of their emissions suffer from large uncertainties due to their significant spatial and temporal heterogeneity. Here, we derive the first multiyear, monthly resolution N₂O emissions from three of the four major EBUSs using high-frequency coastal atmospheric measurements and an inverse method. We find average combined N₂O emissions from the northern California, Benguela, and southern Canary upwelling systems to be 57.7 (51.4-63.9) Gg-N yr⁻¹. We also find an offshore region near the Benguela EBUS that exhibits large pulses of emissions with emissions that reach 677 Gg-N yr⁻¹ in 1 month. Our findings highlight that atmospheric measurements coupled with inverse modeling can capture the large variability in EBUS emissions by quantifying emissions over large spatial distances and over long time periods compared to previous methods using traditional oceanographic measurements.

862. One size fits all? Calibrating an ocean biogeochemistry model for different circulations

Authors: Iris Kriest; Paul Kähler; Wolfgang Koeve; Karin Kvale; Volkmar Sauerland; Andreas Oschlies

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-17-3057-2020

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Publication Date: 2020-06-18

Volume/Issue: Volume 17, Issue 12

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Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . Global biogeochemical ocean models are often tuned to match the observed distributions and fluxes of inorganic and organic quantities. This tuning is typically carried out "by hand". However, this rather subjective approach might not yield the best fit to observations, is closely linked to the circulation employed and is thus influenced by its specific features and even its faults. We here investigate the effect of model tuning, via objective optimisation, of one biogeochemical model of intermediate complexity when simulated in five different offline circulations. For each circulation, three of six model parameters have been adjusted to characteristic features of the respective circulation. The values of these three parameters - namely, the oxygen utilisation of remineralisation, the particle flux parameter and potential nitrogen fixation rate - correlate significantly with deep mixing and ideal age of North Atlantic Deep Water (NADW) and the outcrop area of Antarctic Intermediate Waters (AAIW) and Subantarctic Mode Water (SAMW) in the Southern Ocean. The clear relationship between these parameters and circulation characteristics, which can be easily diagnosed from global models, can provide guidance when tuning global biogeochemistry within any new circulation model. The results from 20 global cross-validation experiments show that parameter sets optimised for a specific circulation can be transferred between similar circulations without losing too much of the model's fit to observed quantities. When compared to model intercomparisons of subjectively tuned, global coupled biogeochemistry-circulation models, each with different circulation and/or biogeochemistry, our results show a much lower range of oxygen inventory, oxygen minimum zone (OMZ) volume and global biogeochemical fluxes. Export production depends to a large extent on the circulation applied, while deep particle flux is mostly determined by the particle flux parameter. Oxygen inventory, OMZ volume, primary production and fixed-nitrogen turnover depend more or less equally on both factors, with OMZ volume showing the highest sensitivity, and residual variability. These results show a beneficial effect of optimisation, even when a biogeochemical model is first optimised in a relatively coarse circulation and then transferred to a different finer-resolution circulation model.

863. The Regional Earth System Model RegCM-ES: Evaluation of the Mediterranean Climate and Marine Biogeochemistry

Authors: Marco Reale (National Institute of Oceanography and Applied Geophysics--OGS Trieste Italy, Earth System Physics Section The Abdus Salam International Centre for Theoretical Physics Trieste Italy); Filippo Giorgi (Earth System Physics Section The Abdus Salam International Centre for Theoretical Physics Trieste Italy); Cosimo Solidoro (National Institute of Oceanography and Applied Geophysics--OGS Trieste Italy); Valeria Di Biagio (National Institute of Oceanography and Applied Geophysics--OGS Trieste Italy); Fabio Di Sante (National Institute of Oceanography and Applied Geophysics--OGS Trieste Italy, Earth System Physics Section The Abdus Salam International Centre for Theoretical Physics Trieste Italy); Laura Mariotti (National Institute of Oceanography and Applied Geophysics--OGS Trieste Italy); Riccardo Farneti (Earth System Physics Section The Abdus Salam International Centre for Theoretical Physics Trieste Italy); Gianmaria Sannino (Climate Modelling Laboratory Italian National Agency for New Technologies-Energy-Sustainable Economic Development (ENEA) Rome Italy)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019ms001812

URL: <https://doi.org/10.1029/2019ms001812>

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References: 136

Cited by: 24 publications

Language: en

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Keywords: N/A

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License: Creative Commons

Open Access: N/A

Abstract: We introduce a new version of the Earth System Regional Climate model RegCM-ES and evaluate its performances for the first time over the Mediterranean region. The novel aspect of this coupled system is the possibility to simulate the dynamics of the marine ecosystem through a biogeochemical model, BFM (Biogeochemical Flux Model), coupled online with the ocean circulation model MITgcm (MIT general circulation model). The validation of atmosphere and ocean components has shown that the model is able to capture interannual and intermonthly variabilities of the atmospheric heat fluxes and spatial patterns of land surface temperature, precipitation, evaporation, and sea surface temperature with a general improvement compared to previous versions. At the same time, we diagnosed some prominent deficiencies as a warm and dry bias associated in summer with the resolution of the atmospheric module and the tuning of the boundary layer and convective precipitation scheme. On the biogeochemical side, RegCM-ES shows good skills in reproducing mean values and spatial patterns of net primary production, phosphate, and horizontal/vertical patterns of chlorophyll-a. Limitations in this case include deficiencies mainly in the simulation of mean values of nitrate and dissolved oxygen in the basin which have been associated with too large vertical mixing throughout the water column, deficiencies in the boundary conditions, and solubility computations.

Overall, RegCM-ES has the potential to become a suitable tool for the analysis of the impacts of climate change on the ocean and marine biogeochemistry in the Mediterranean region and many other domains.

864. The Whole Antarctic Ocean Model (WAOM v1.0): Development and Evaluation

Authors: Ole Richter; David E. Gwyther; Benjamin K. Galton-Fenzi; Kaitlin A. Naughten

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2020-164

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Barotropic fluid|Ocean surface topography|Wind Stress|Iceberg|Sea-surface height

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Open Access: N/A

Abstract: . The Regional Ocean Modeling System (ROMS), including an ice shelf component, has been applied on a circum-Antarctic domain to derive estimates of ice shelf basal melting. Significant improvements made compared to previous models of this scale are the inclusion of tides and a horizontal spatial resolution of 2 km, which is sufficient to resolve onshelf heat transport by bathymetric troughs and eddy scale circulation. We run the model with ocean-atmosphere-sea ice conditions from the year 2007, to represent nominal present day climate. We force the ocean surface with buoyancy fluxes derived from sea ice concentration observations and wind stress from ERA-Interim atmospheric reanalysis. At the northern boundaries ocean conditions are derived from the ECCO2 reanalysis and tides are incorporated as sea surface height and barotropic currents. The accuracy of tidal height signals close to the coast is comparable to those simulated from widely-used barotropic tide models, while off-shelf hydrography agrees well with the Southern Ocean State Estimate (SOSE) model. On the shelf, most details of ice shelf-ocean interaction are consistent with results from regional modelling and observational studies, although a paucity of observational data (particularly taken during 2007) prohibits a full verification. We conclude that our improved model is well suited to derive a new estimate of present day Antarctic ice shelf melting at high resolution and is able to quantify its sensitivity to tides.

865. Overturning Circulation Pathways in a Two-Basin Ocean Model

Authors: Louis-Philippe Nadeau (Institut des Sciences de la Mer de Rimouski, Université du Québec à Rimouski, Rimouski, Quebec, Canada); Malte F. Jansen (University of Chicago)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0034.1

URL: <https://doi.org/10.1175/jpo-d-20-0034.1>

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References: 39

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Division of Ocean Sciences, NSERC Discovery Individual

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Open Access: N/A

Abstract: A toy model for the deep ocean overturning circulation in multiple basins is presented and applied to study the role of buoyancy forcing and basin geometry in the ocean's global overturning. The model reproduces the results from idealized general circulation model simulations and provides theoretical insights into the mechanisms that govern the structure of the overturning circulation. The results highlight the importance of the diabatic component of the meridional overturning circulation (MOC) for the depth of North Atlantic Deep Water (NADW) and for the interbasin exchange of deep ocean water masses. This diabatic component, which extends the upper cell in the Atlantic below the depth of adiabatic upwelling in the Southern Ocean, is shown to be sensitive to the global area-integrated diapycnal mixing rate and the density contrast between NADW and Antarctic Bottom Water (AABW). The model also shows that the zonally averaged global overturning circulation is to zeroth-order independent of whether the ocean consists of one or multiple connected basins, but depends on the total length of the southern reentrant channel region (representing the Southern Ocean) and the global ocean area integrated diapycnal mixing. Common biases in single-basin simulations can thus be understood as a direct result of the reduced domain size.

866. Harmonizing models and observations: Data assimilation in Earth system science

Authors: Xin Li; Feng Liu; Miao Fang

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-019-9620-x

URL: <https://doi.org/10.1007/s11430-019-9620-x>

Publication Date: 2020-06-02

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References: 63

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

867. Elucidating ecological complexity: Unsupervised learning determines global marine eco-provinces

Authors: Maike Sonnewald (Harvard University, Massachusetts Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Christopher Hill (Massachusetts Institute of Technology); Gael Forget (Massachusetts Institute of Technology)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.aay4740

URL: <https://doi.org/10.1126/sciadv.aay4740>

Publication Date: 2020-05-29

Volume/Issue: Volume 6, Issue 22

References: 56

Cited by: 52 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory, NASA

License: N/A

Open Access: N/A

Abstract: Global marine eco-provinces are determined toward tackling complexity and to explore community structure using unsupervised ML.

868. Zooplankton-Mediated Fluxes in the Eastern Tropical North Atlantic

Authors: Rainer Kiko; Peter Brandt; Svenja Christiansen; Jannik Faustmann; Iris Kriest; Elizandro Rodrigues; Florian Schütte; Helena Hauss

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2020.00358

URL: <https://doi.org/10.3389/fmars.2020.00358>

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Volume/Issue: Volume 7

References: 82

Cited by: 31 publications

Language: N/A

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Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Agence Nationale de la Recherche, Horizon 2020

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Open Access: N/A

Abstract: N/A

869. Sea Surface Salinity as a Proxy for Arctic Ocean Freshwater Changes

Authors: Séverine Fournier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Xiaochun Wang (University of California Los Angeles); Thomas W. K. Armitage (University of Maryland); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ron Kwok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Publisher: American Geophysical Union (AGU)

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Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: sea surface height, Arctic, freshwater, gravimetry, sea surface salinity

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The changing Arctic freshwater content and distribution have a significant implications for ocean circulation, climate, and water and biogeochemical cycles. The paucity of in situ salinity measurements in the Arctic Ocean has limited our ability to study Arctic-Ocean freshwater variability. Although satellite-derived sea surface height (SSH) and ocean bottom pressure (OBP) have been used together to infer depth-integrated freshwater content changes, these measurements are limited in sampling and resolution. Motivated by the recent development of sea surface salinity (SSS) remote sensing, we explore the use of SSS as a proxy for Arctic freshwater changes. As a first step, here we conduct a proof-of-concept study by analyzing the output of an ocean-ice state estimation product. We find that SSS variations are coherent with those of SSH-minus-OBP across the Arctic basin (with $R \sim -0.8$) except for those in the center of the Beaufort Gyre and the region affected by the subpolar North Atlantic inflow. On Arctic shelves, the linear regression coefficient between SSS and SSH-minus-OBP is -0.3 pss/cm or larger in magnitude. The results suggest that SSS is a good proxy for Arctic freshwater changes, and satellite SSH-minus-OBP can be used to evaluate satellite SSS quality.

870. Contributions of Indonesian Throughflow to eastern Indian Ocean surface variability during ENSO events

Authors: Xiaolin Jin (Civil Aviation Flight University of China Guanghan China); Jonathon S. Wright (Tsinghua University)

Journal: Atmospheric Science Letters

Publisher: Wiley

DOI: 10.1002/asl.979

URL: <https://doi.org/10.1002/asl.979>

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References: 63

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Science and Technology of the People's Republic of China

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Open Access: N/A

Abstract: The contribution of ocean transport to the transition between the Indian Ocean Dipole mode (IOD) and the Indian Ocean Basin mode (IOB) during El Niño-Southern Oscillation (ENSO) years is investigated using reanalysis products. Composite analysis suggests that the IOD-IOB transition is robust among ENSO years, with variations in the eastern Indian Ocean mixed layer playing a key role. Although this transition has typically been attributed to changes in surface heat flux in the eastern Indian Ocean, our results suggest that approximately one third of the mixed layer temperature tendency during the transition results from anomalous ocean heat transport. Enhanced surface heat flux anomalies and oceanic advection starting from October during El Niño developing years combine to warm the eastern Indian Ocean, promoting the decay of the positive IOD cold tongue and the emergence of a positive IOB pattern. The contribution of ocean advection to the IOD-IOB transition is dominated by the Indonesian Throughflow (ITF), which accounts for around 70-80% of total ocean advection contribution in the southeastern Indian Ocean (20-25% of the total warming tendency). These variations in ITF heat transport arise in large part from local wind anomalies. Southeasterly wind anomalies along the northeastern edge of an anomalous Indian Ocean anticyclone associated with El Niño intensify surface-layer heat transport into the Indian Ocean in the ITF outflow region. This change in surface transport contrasts with total ITF transport anomalies during El Niño years, as total transport is dominated by negative subsurface transport anomalies.

871. Control of Bering Strait Transport by the Meridional Overturning Circulation

Authors: Paola Cessi (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-20-0026.1

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Publication Date: 2020-05-19

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References: 42

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation

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Open Access: N/A

Abstract: It is well established that the mean transport through Bering Strait is balanced by a sea level difference between the North Pacific and the Arctic Ocean, but no mechanism has been proposed to explain this sea level difference. It is argued that the sea level difference across Bering Strait, which geostrophically balances the northward throughflow, is associated with the sea level difference between the North Pacific and the North Atlantic/Arctic. In turn, the latter difference is caused by deeper middepth isopycnals in the Indo-Pacific than in the Atlantic, especially in the northern high latitudes because there is deep water formation in the Atlantic, but not in the Pacific. Because the depth of the middepth isopycnals is associated with the dynamics of the upper branch of the meridional overturning circulation (MOC), a model is formulated that quantitatively relates the sea level difference between the North Pacific and the Arctic/North Atlantic with the wind stress in the Antarctic Circumpolar region, since this forcing powers the MOC, and with the outcropping isopycnals shared between the Northern Hemisphere and the Antarctic circumpolar region, since this controls the location of deep water formation. This implies that if the sinking associated with the MOC were to occur in the North Pacific, rather than the North Atlantic, then the Bering Strait flow would reverse. These predictions, formalized in a theoretical box model, are confirmed by a series of numerical experiments in a simplified geometry of the World Ocean, forced by steady surface wind stress, temperature, and freshwater flux.

872. Impacts of the upper-ocean salinity variations on the decadal sea level change in the Southeast Indian Ocean during the Argo era

Authors: Jiamei Huang; Wei Zhuang; Xiao-Hai Yan; Zelun Wu

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-020-1574-4

URL: <https://doi.org/10.1007/s13131-020-1574-4>

Publication Date: 2020-05-14

Volume/Issue: Volume 39, Issue 7

References: 43

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

873. Intensification of the global water cycle and evidence from ocean salinity: a synthesis review

Authors: Lisan Yu (Woods Hole Oceanographic Institution); Simon A. Josey (National Oceanography Centre); Frederick M. Bingham (University of North Carolina Wilmington); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Annals of the New York Academy of Sciences

Publisher: Wiley

DOI: 10.1111/nyas.14354

URL: <https://doi.org/10.1111/nyas.14354>

Publication Date: 2020-05-09

Volume/Issue: Volume 1472, Issue 1

References: 126

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The ocean plays a dominant role in the global water cycle. It is the center of action for global evaporation and precipitation and supplies the moisture that falls as continental precipitation. It also acts to some extent as nature's rain gauge, as it tells us about the long-term changes in the global water cycle through monitoring of the changes in ocean surface salinity. As climate warms, the global water cycle is expected to intensify as a result of the strong nonlinear dependence of water vapor pressure (moisture-holding capacity) on temperature. Such change is of great concern, as it has profound socioeconomic impacts throughout the globe. Despite the evidence of an intensified global water cycle, two important questions remain: What is the pattern of the warming-induced intensification of the water cycle? and What is the rate of intensification? Our article provides a synthesis review of recent progress in diagnosing and understanding the changes in both the global water cycle and ocean salinity in recent decades. Targeted numerical ocean model experiments are also reviewed to provide insights into the response of salinity to the changes in evaporation-minus-precipitation flux, meltwater runoff, and ocean warming.

874. A comprehensive model-based index for identification of larval retention areas: A case study for Japanese anchovy *Engraulis japonicus* in the Yellow Sea

Authors: Qinwang Xing; Huaming Yu; Haiqing Yu; Peng Sun; Yang Liu; Zhenjiang Ye; Jianchao Li; Yongjun Tian

Journal: Ecological Indicators

Publisher: Elsevier BV

DOI: 10.1016/j.ecolind.2020.106479

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Keywords: N/A

Funding: Department of Science and Technology of Shandong Province, Ocean University of China, Second Institute of Oceanography, State Oceanic Administration (and 2 more)

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Open Access: N/A

Abstract: N/A

875. Understanding the Role of Nutrient Limitation on Plankton Biomass Over Arabian Sea Via 1-D Coupled Biogeochemical Model and Bio-Argo Observations

Authors: M. Anju (Indian Institute of Tropical Meteorology, Department of Meteorology and Oceanography, College of Science and Technology Andhra University Visakhapatnam India); M. G. Sreeush (Indian Institute of Tropical Meteorology); V. Valsala (Indian Institute of Tropical Meteorology); B. R. Smitha (Centre for Marine Living Resources and Ecology (CMLRE) Kerala India); Faseela Hamza (Indian Institute of Tropical Meteorology); G. Bharathi (Department of Meteorology and Oceanography, College of Science and Technology Andhra University Visakhapatnam India); C. V. Naidu (Department of Meteorology and Oceanography, College of Science and Technology Andhra University Visakhapatnam India)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015502

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References: 80

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Arabian Sea (AS) is known to have seasonal phytoplankton blooms during winter and summer driven by dynamically distinct physical forcing mechanisms and associated nutrient dynamics. A 1-D coupled physical-biogeochemical model based on North Pacific Ecosystem Model for Understanding Regional Oceanography (NEMURO) with nitrogen and silicon cycles is adapted for the AS environment. The model is implemented to investigate the role of nitrogen (nitrate + ammonium) versus silicate limitation on plankton biomass. The seasonal cycle of plankton biomass is well simulated by the model along bio-Argo (during a period from 2013 to 2016) and ship cruise tracks (for the single year 2009). Further, three sensitivity simulations are conducted by suppressing (1) nitrate availability (representing new production), (2) ammonium availability (representing regenerated production), and (3) silicate availability (for diatom production). The new production represents 80% of the total primary production in the AS and implicitly controls 70% of total zooplankton production annually. The regenerated production augments small phytoplankton (by ~50%; e.g., flagellates) and small zooplankton (by ~20%; e.g., ciliates) growth with negligible effects on large phytoplankton (e.g., diatom) and predatory zooplankton (e.g., copepods). The diatom production remains within the observed range due to silicate limitation which is fundamental in the model for realistic simulation of chlorophyll. Silicate is the primary limiting nutrient in diatom bloom in the subsurface chlorophyll maxima with maximum limitation occurring during the winter season, due to the deeper silicicline as compared to shallower nitricline. At the surface, both nitrogen and silicate co-limit the total production; however, nitrogen is a stronger limiter than silicate from March to June. The study highlights the relative role of silicate versus

total nitrogen (nitrate + ammonium) in limiting primary production in the AS.

876. Explaining the Zonal Asymmetry in the Air-Sea Net Heat Flux Climatology Over the Antarctic Circumpolar Current

Authors: Xiangzhou Song (Hohai University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jc016215

URL: <https://doi.org/10.1029/2020jc016215>

Publication Date: 2020-04-29

Volume/Issue: Volume 125, Issue 6

References: 63

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: Air-sea turbulent heat fluxes, Surface heat budget balance, Air-sea heat flux, Antarctic Circumpolar Current

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Surface heating occurs over the streamline of the Antarctic Circumpolar Current (ACC) and balances the heat transport associated with the global meridional overturning circulation. With a combination of ocean assimilation model output, objectively analyzed products, and atmospheric reanalysis, this paper investigates the mechanism of climatological surface cooling in the Pacific sector with zonal asymmetry. The poleward shift of the ACC path in the Pacific with barotropic potential vorticity conservation was found to account for this cooling. The southward transport of warm/wet surface waters encounters relatively cold/dry air at high latitudes, causing strong turbulent heat fluxes. Sensible heat flux related to the sea-air heat convection plays an equivalent role alongside the sea surface evaporative latent heat flux, both of which dominate the surface cooling in the Pacific sector. Solar radiation decreases with a poleward meandering of the ACC, which aggravates the cooling process. A significant seasonal cycle of the ensemble mean net surface heat flux (Q_{net}) is demonstrated, that is, approximately half-year heating (cooling) during austral warm (cool) seasons. The estimated magnitude of Q_{net} over the ACC is approximately 20 W m^{-2} by averaging 10 heat flux climatologies and 4 W m^{-2} in an ocean state estimate model. However, efforts are still needed to reduce the excess heat flux over the sea surface in the Southern Ocean for the heat flux products.

877. Hydrodynamic effect of biofouling in fish cage aquaculture netting

Authors: Pablo Cornejo; Nicolás M. Guerrero; Rodrigo M. Montes; Renato A. Quiñones; Hector H. Sepúlveda

Journal: Aquaculture

Publisher: Elsevier BV

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Publication Date: 2020-04-28

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References: 98

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Comisión Nacional de Investigación Científica y Tecnológica

License: N/A

Open Access: N/A

Abstract: N/A

878. Resource Availability and Entrainment Are Driven by Offsets Between Nutriclines and Winter Mixed-Layer Depth

Authors: S. J. Rigby (University of Liverpool); R. G. Williams (University of Liverpool); E. P. Achterberg (GEOMAR Helmholtz Centre for Ocean Research); A. Tagliabue (University of Liverpool)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gb006497

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References: 101

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: Atlantic, cycling, nutrient, mixed-layer, availability

Funding: Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: While phytoplankton play a key role in ocean biogeochemical cycles, the availability and supply pathways of resources that support their growth remain poorly constrained. Here, we show that the availability of various resources varies over several orders of magnitude throughout the Atlantic Ocean, causing regional contrasts in resource deficiency. Regional variations in the relative availability of nitrogen, phosphorous, silicon, iron, zinc, manganese, cobalt, and cadmium are important and result from the contrasts between winter mixing depths and differences in vertical profiles of the different resources. The winter-time thickening of the mixed layer may replenish or deplete resources via entrainment, depending on the vertical nutrient profile. For nutrients like nitrate, phosphate, and cadmium, entrainment is a consistent source term. While for others, such as manganese and iron, entrainment can reduce ocean resource availability, particularly in subtropical regions. Any future change to the depth of winter-time mixing will cause region-specific changes in relative availability of different resources that may have important ecological consequences.

879. A Comparative Analysis of the Bay of Bengal Ocean State Using Standalone and Coupled Numerical Models

Authors: T. S Anandh; Bijan Kumar Das; J. Kuttippurath; Arun Chakraborty

Journal: Asia-Pacific Journal of Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13143-020-00197-z

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References: 46

Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: MHRD, SPLICE MDRP

License: N/A

Open Access: N/A

Abstract: N/A

880. Barotropic vorticity balance of the North Atlantic subpolar gyre in an eddy-resolving model

Authors: Mathieu Le Corre; Jonathan Gula; Anne-Marie Tréguier

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-16-451-2020

URL: <https://doi.org/10.5194/os-16-451-2020>

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References: 77

Cited by: 23 publications

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Subjects: N/A

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Funding: N/A

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Open Access: N/A

Abstract: . The circulation in the North Atlantic subpolar gyre is complex and strongly influenced by the topography. The gyre dynamics are traditionally understood as the result of a topographic Sverdrup balance, which corresponds to a first-order balance between the planetary vorticity advection, the bottom pressure torque, and the wind stress curl. However, these dynamics have been studied mostly with non-eddy-resolving models and a crude representation of the bottom topography. Here we revisit the barotropic vorticity balance of the North Atlantic subpolar gyre using a new eddy-resolving simulation (with a grid space of ≈ 2 km) with topography-following vertical coordinates to better represent the mesoscale turbulence and flow-topography interactions. Our findings highlight that, locally, there is a first-order balance between the bottom pressure torque and the nonlinear terms, albeit with a high degree of cancellation between them. However, balances integrated over different regions of the gyre - shelf, slope, and interior - still highlight the important role played by nonlinearities and bottom drag curls. In particular, the Sverdrup balance cannot describe the dynamics in the interior of the gyre. The main sources of cyclonic vorticity are nonlinear terms due to eddies generated along eastern boundary currents and time-mean nonlinear terms in the northwest corner. Our results suggest that a good representation of the mesoscale activity and a good positioning of mean currents are two important conditions for a better representation of the circulation in the North Atlantic subpolar gyre.

881. Spin-up of UK Earth System Model 1 (UKESM1) for CMIP6

Authors: A. Yool (National Oceanography Centre); J. Palmiéri (National Oceanography Centre); C. G. Jones (University of Leeds); A. A. Sellar (Met Office); L. de Mora (Plymouth Marine Laboratory); T. Kuhlbrodt (University of Reading); E. E. Popova (National Oceanography Centre); J. P. Mulcahy (Met Office); A. Wiltshire (Met Office); S. T. Rumbold (University of Reading); M. Stringer (University of Reading); R. S. R. Hill (Met Office); Y. Tang (Met Office); J. Walton (Met Office); A. Blaker (National Oceanography Centre); A. J. G. Nurser (National Oceanography Centre); A. C. Coward (National Oceanography Centre); J. Hirschi (National Oceanography Centre); S. Woodward (Met Office); D. I. Kelley (Centre for Ecology and Hydrology Wallingford UK); R. Ellis (Centre for Ecology and Hydrology Wallingford UK); S. Rumbold-Jones (Met Office, University of Reading)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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References: 103

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: carbon cycle, spin-up, CMIP6, equilibrium, Earth system model

Funding: Horizon 2020 Framework Programme, Natural Environment Research Council

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Open Access: N/A

Abstract: For simulations intended to study the influence of anthropogenic forcing on climate, temporal stability of the Earth's natural heat, freshwater, and biogeochemical budgets is critical. Achieving such coupled model equilibration is scientifically and computationally challenging. We describe the protocol used to spin-up the UK Earth system model (UKESM1) with respect to preindustrial forcing for use in the sixth Coupled Model Intercomparison Project (CMIP6). Due to the high computational cost of UKESM1's atmospheric model, especially when running with interactive full chemistry and aerosols, spin-up primarily used parallel configurations using only ocean/land components. For the ocean, the resulting spin-up permitted the carbon and heat contents of the ocean's full volume to approach equilibrium over 5,000 years. On land, a spin-up of 1,000 years brought UKESM1's dynamic vegetation and soil carbon reservoirs toward near-equilibrium. The end-states of these parallel ocean- and land-only phases then initialized a multicentennial period of spin-up with the full Earth system model, prior to this simulation continuing as the UKESM1 CMIP6 preindustrial control (piControl). The realism of the fully coupled spin-up was assessed for a range of ocean and land properties, as was the degree of equilibration for key variables. Lessons drawn include the importance of consistent interface physics across ocean- and land-only models and the coupled (parent) model, the extreme simulation duration required to approach equilibration targets, and the occurrence of significant regional land carbon drifts despite global-scale equilibration. Overall, the UKESM1 spin-up underscores the expense involved and argues in favor of future development of more efficient spin-up techniques.

882. Impacts of High-Frequency Atmospheric Forcing on Southern Ocean Circulation and Antarctic Sea Ice

Authors: Yang Wu; Zhaomin Wang; Chengyan Liu; Xia Lin

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-020-9203-x

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References: 68

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

883. Understanding of Contemporary Regional Sea-Level Change and the Implications for the Future

Authors: Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alex S. Gardner (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Erik Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jan T. M. Lenaerts (University of Colorado Boulder); J. T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Trossman (University of Texas at Austin); Edward D. Zaron (Portland State University); Surendra Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anthony Arendt (University of Washington); Andy Aschwanden (University of Alaska Fairbanks); Brian D. Beckley (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, KBR Inc.); David P. S. Bekaert (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Geoffrey Blewitt (University of Nevada); Lambert Caron (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Don P. Chambers (University of South Florida); Hrishikesh A. Chandanpurkar (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Knut Christianson (University of Washington); Beata Csatho (University of Buffalo); Richard I. Cullather (University of Maryland); Robert M. DeConto (University of Massachusetts Amherst); John T. Fasullo (National Center for Atmospheric Research); Thomas Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jeffrey T. Freymueller (Michigan State University); Daniel M. Gilford (Rutgers University); Manuela Girotto (University of California Berkeley); William C. Hammond (University of Nevada); Regine Hock (University of Alaska Fairbanks); Nicholas Holschuh (University of Washington); Robert E. Kopp (University of California Berkeley); Felix Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mark Merrifield (Scripps Institution of Oceanography / University of California San Diego); Jerry X. Mitrovica (Harvard University); R. Steven Nerem (University of Colorado Boulder); Isabel J. Nias (University of Maryland, University of Colorado Boulder); Veronica Nieves (Institute of Marine Sciences); Sophie Nowicki (University of Colorado Boulder); Kishore Pangaluru (University of California Irvine); Christopher G. Piecuch (Woods Hole Oceanographic Institution); Richard D. Ray (University of Colorado Boulder); David R. Rounce (University of Alaska Fairbanks); Nicole-Jeanne Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H  l  ne Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Manoochehr Shirzaei (Arizona State University); William V. Sweet (Center for Operational Oceanographic Products and Services NOAA Silver Spring MD USA); Isabella Velicogna (Institute of Marine Sciences); Nadya Vinogradova (Science Mission Directorate NASA Headquarters Washington DC USA); Thomas Wahl (Civil, Environmental and Construction Engineering and National Center for Integrated Coastal Research University of Central Florida Orlando FL USA); David N. Wiese (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael J. Willis (University of Colorado Boulder)

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Subjects: N/A

Keywords: satellite observations, remote sensing, sea level

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Global sea level provides an important indicator of the state of the warming climate, but changes in regional sea level are most relevant for coastal communities around the world. With improvements to the sea-level observing system, the knowledge of regional sea-level change has advanced dramatically in recent years. Satellite measurements coupled with in situ observations have allowed for comprehensive study and improved understanding of the diverse set of drivers that lead to variations in sea level in space and time. Despite the advances, gaps in the understanding of contemporary sea-level change remain and inhibit the ability to predict how the relevant processes may lead to future change. These gaps arise in part due to the complexity of the linkages between the drivers of sea-level change. Here we review the individual processes which lead to sea-level change and then describe how they combine and vary regionally. The intent of the paper is to provide an overview of the current state of understanding of the processes that cause regional sea-level change and to identify and discuss limitations and uncertainty in our understanding of these processes. Areas where the lack of understanding or gaps in knowledge inhibit the ability to provide the needed information for comprehensive planning efforts are of particular focus. Finally, a goal of this paper is to highlight the role of the expanded sea-level observation network--particularly as related to satellite observations--in the improved scientific understanding of the contributors to regional sea-level change.

884. Phase synchronisation in the Kuroshio Current System

Authors: Ann Kristin Klose; René M. van Westen; Henk A. Dijkstra

Journal: Ocean Science

Publisher: Copernicus GmbH

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URL: <https://doi.org/10.5194/os-16-435-2020>

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Cited by: 1 publications

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Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . The Kuroshio Current System in the North Pacific displays path transitions on a decadal timescale. It is known that both internal variability involving barotropic and baroclinic instabilities and remote Rossby waves induced by North Pacific wind stress anomalies are involved in these path transitions. However, the precise coupling of both processes and its consequences for the dominant decadal transition timescale are still under discussion. Here, we analyse the output of a multi-centennial high-resolution global climate model simulation and study phase synchronisation between Pacific zonal wind stress anomalies and Kuroshio Current System path variability. We apply the Hilbert transform technique to determine the phase and find epochs where such phase synchronisation appears. The physics of this synchronisation are shown to occur through the effect of the vertical motion of isopycnals, as induced by the propagating Rossby waves, on the instabilities of the Kuroshio Current System.

885. Statistical Comparisons of Temperature Variance and Kinetic Energy in Global Ocean Models and Observations: Results From Mesoscale to Internal Wave Frequencies

Authors: Conrad A. Luecke (Scripps Institution of Oceanography / University of California San Diego, University of Michigan); Brian K. Arbic (University of Michigan); James G. Richman (Florida State University); Jay F. Shriver (Oceanographic Division, Naval Research Laboratory Stennis Space Center Hancock County MS USA); Matthew H. Alford (Scripps Institution of Oceanography / University of California San Diego); Joseph K. Ansong (University of Michigan, Department of Mathematics University of Ghana Accra Ghana); Steven L. Bassette (University of Michigan); Maarten C. Buijsman (Department of Marine Sciences University of Southern Mississippi, Stennis Space Center Hancock County MS USA); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Robert B. Scott (Univ Brest, UMR CNRS 6205, Laboratoire de Mathématiques de Bretagne Atlantique Brest France); Patrick G. Timko (University of Michigan); Gunnar Voet (Scripps Institution of Oceanography / University of California San Diego); Alan J. Wallcraft (Florida State University); Luis Zamudio (Florida State University)

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Keywords: embedded tides, kinetic energy, model-data comparison, model resolution, temperature variance (and 1 more)

Funding: N/A

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Open Access: N/A

Abstract: Temperature variance and kinetic energy (KE) from two global simulations of the HYbrid Coordinate Ocean Model (HYCOM; 1/12 degrees and 1/25 degrees) and three global simulations of the Massachusetts Institute of Technology general circulation model (MITgcm; 1/12 degrees, 1/24 degrees, and 1/48 degrees), all of which are forced by atmospheric fields and the astronomical tidal potential, are compared with temperature variance and KE from a database of about 2,000 moored historical observations (MHOs). The variances are computed across frequencies ranging from supertidal, dominated by the internal gravity wave continuum, to subtidal, dominated by currents and mesoscale eddies. The most important qualitative difference between HYCOM and MITgcm, and between simulations of different resolutions, is in the supertidal band, where the 1/48 degrees MITgcm lies closest to observations. Across all frequency bands examined, the HYCOM simulations display higher spatial correlation with the MHO than do the MITgcm simulations. The supertidal, semidiurnal, and diurnal velocities in the HYCOM simulations also compare more closely with observations than do the

MITgcm simulations in a number of specific continental margin/marginal sea regions. To complement the model-MHO comparisons, this paper also compares the surface ocean geostrophic eddy KE in HYCOM, MITgcm, and a gridded satellite altimeter product. Consistent with the model-MHO comparisons, the HYCOM simulations have a higher spatial correlation with the altimeter product than the MITgcm simulations do. On the other hand, the surface ocean geostrophic eddy KE is too large, relative to the altimeter product, in the HYCOM simulations.

886. SICOPOLIS-AD v1: an open-source adjoint modeling framework for ice sheet simulation enabled by the algorithmic differentiation tool OpenAD

Authors: Liz C. Logan; Sri Hari Krishna Narayanan; Ralf Greve; Patrick Heimbach

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-13-1845-2020

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Cited by: 4 publications

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Keywords: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: . We present a new capability of the ice sheet model SICOPOLIS that enables flexible adjoint code generation via source transformation using the open-source algorithmic differentiation (AD) tool OpenAD. The adjoint code enables efficient calculation of the sensitivities of a scalar-valued objective function or quantity of interest (QoI) to a range of important, often spatially varying and uncertain model input variables, including initial and boundary conditions, as well as model parameters. Compared to earlier work on the adjoint code generation of SICOPOLIS, our work makes several important advances: (i) it is embedded within the up-to-date trunk of the SICOPOLIS repository - accounting for 1.5 decades of code development and improvements - and is readily available to the wider community; (ii) the AD tool used, OpenAD, is an open-source tool; (iii) the adjoint code developed is applicable to both Greenland and Antarctica, including grounded ice as well as floating ice shelves, with an extended choice of thermodynamical representations. A number of code refactorization steps were required. They are discussed in detail in an Appendix as they hold lessons for the application of AD to legacy codes at large. As an example application, we examine the sensitivity of the total Antarctic Ice Sheet volume to changes in initial ice thickness, austral summer precipitation, and basal and surface temperatures across the ice sheet. Simulations of Antarctica with floating ice shelves show that over 100 years of simulation the sensitivity of total ice sheet volume to the initial ice thickness and precipitation is almost uniformly positive, while the sensitivities to surface and basal temperature are almost uniformly negative. Sensitivity to austral summer precipitation is largest on floating ice shelves from Queen Maud to Queen Mary Land. The largest sensitivity to initial ice thickness is at outlet glaciers around Antarctica. Comparison between total ice sheet volume sensitivities to surface and basal temperature shows that surface temperature sensitivities are higher broadly across the floating ice shelves, while basal temperature sensitivities are highest at the grounding lines of floating ice shelves and outlet glaciers. A uniformly perturbed region of East Antarctica reveals that, among the four control variables tested here, total ice sheet volume is the most sensitive to variations in austral summer precipitation as formulated in SICOPOLIS. Comparison between adjoint- and finite-difference-derived sensitivities shows good agreement, lending confidence that the AD tool is

producing correct adjoint code. The new modeling infrastructure is freely available at <http://www.sicopolis.net> (last access: 2 April 2020) under the development trunk.

887. Locally and Remotely Forced Subtropical AMOC Variability: A Matter of Time Scales

Authors: Quentin Jamet (Institut de Recherche pour le Développement (IRD)); William K. Dewar (Florida State University, Institut de Recherche pour le Développement (IRD)); Nicolas Wienders (Florida State University); Bruno Deremble (c Laboratoire de Météorologie Dynamique, Paris, France); Sally Close (d Laboratoire d'Océanographie Physique et Spatiale, Univ. de Bretagne Occidentale, Brest, France); Thierry Penduff (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Climate

Publisher: American Meteorological Society

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References: 70

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Mechanisms driving the North Atlantic meridional overturning circulation (AMOC) variability at low frequency are of central interest for accurate climate predictions. Although the subpolar gyre region has been identified as a preferred place for generating climate time-scale signals, their southward propagation remains under consideration, complicating the interpretation of the observed time series provided by the Rapid Climate Change-Meridional Overturning Circulation and Heatflux Array-Western Boundary Time Series (RAPID-MOCHA-WBTS) program. In this study, we aim at disentangling the respective contribution of the local atmospheric forcing from signals of remote origin for the subtropical low-frequency AMOC variability. We analyze for this a set of four ensembles of a regional (20 degreesS-55 degreesN), eddy-resolving (1/12 degrees) North Atlantic oceanic configuration, where surface forcing and open boundary conditions are alternatively permuted from fully varying (realistic) to yearly repeating signals. Their analysis reveals the predominance of local, atmospherically forced signal at interannual time scales (2-10 years), whereas signals imposed by the boundaries are responsible for the decadal (10-30 years) part of the spectrum. Due to this marked time-scale separation, we show that, although the intergyre region exhibits peculiarities, most of the subtropical AMOC variability can be understood as a linear superposition of these two signals. Finally, we find that the decadal-scale, boundary-forced AMOC variability has both northern and southern origins, although the former dominates over the latter, including at the site of the RAPID array (26.5 degreesN).

888. GRACE constraints on Earth rheology of the Barents Sea and Fennoscandia

Authors: Marc Rovira-Navarro; Wouter van der Wal; Valentina R. Barletta; Bart C. Root; Louise Sandberg Sørensen

Journal: Solid Earth

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . The Barents Sea is situated on a continental margin and was home to a large ice sheet at the Last Glacial Maximum. Studying the solid Earth response to the removal of this ice sheet (glacial isostatic adjustment; GIA) can give insight into the subsurface rheology of this region. However, because the region is currently covered by ocean, uplift measurements from the center of the former ice sheet are not available. The Gravity Recovery and Climate Experiment (GRACE) gravity data have been shown to be able to constrain GIA. Here we analyze GRACE data for the period 2003-2015 in the Barents Sea and use the data to constrain GIA models for the region. We study the effect of uncertainty in non-tidal ocean mass models that are used to correct GRACE data and find that it should be taken into account when studying solid Earth signals in oceanic areas from GRACE. We compare GRACE-derived gravity disturbance rates with GIA model predictions for different ice deglaciation chronologies of the last glacial cycle and find that best-fitting models have an upper mantle viscosity equal or higher than 3×10^{20} Pa s. Following a similar procedure for Fennoscandia we find that the preferred upper mantle viscosity there is a factor 2 larger than in the Barents Sea for a range of lithospheric thickness values. This factor is shown to be consistent with the ratio of viscosities derived for both regions from global seismic models. The viscosity difference can serve as constraint for geodynamic models of the area.

889. Evaluating the GECCO3 1948-2018 ocean synthesis - a configuration for initializing the MPI-ESM climate model

Authors: Armin Köhl (University of Hamburg)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

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URL: <https://doi.org/10.1002/qj.3790>

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Keywords: N/A

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Open Access: N/A

Abstract: The paper describes an update of the GECCO (German contribution to the Estimating the Circulation and Climate of the Ocean project) ocean synthesis, now in its version 3, and provides an evaluation of the results with assimilated and independent data. GECCO3 covers the 71-year period 1948-2018 and differs from its predecessor by returning to a single assimilation window instead of partitioning the period in 5-year-long overlapping windows which was previously necessary to yield convergence. A solution to the convergence problem is presented. GECCO3 is intended to be used for the initialization of coupled climate models and is configured for the higher-resolution version of the earth system model (MPI-ESM) developed at the Max Planck Institute for Meteorology. It uses the bathymetry and grid of the MPI-ESM with quasi-uniform resolution of 0.4 degrees, thereby providing the first global eddy-permitting synthesis based on the adjoint method. The synthesis additionally features the estimation of various mixing parameters and can regionally choose between explicit or parametrized eddy fluxes. Except for the altimeter data in tropical regions, GECCO3 is in better agreement with the assimilated data than GECCO2. The improvements relative to the in situ data partly result from the much larger amount of Argo data, which show lower model-data differences. Global heat content changes are in good agreement with recent estimates, but show uptake almost exclusively in the top 700 m. An alternative version of GECCO3, created by starting from different first-guess control parameters, was used to evaluate the uncertainty of the estimated parameters and state due to lack of convergence. This estimate suggests a large uncertainty related to the uptake of heat into the lower layers, while estimates of mean meridional transport of heat and freshwater are not affected.

890. Cropland Carbon Uptake Delayed and Reduced by 2019 Midwest Floods

Authors: Yi Yin (California Institute of Technology); Brendan Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Paul O. Wennberg (California Institute of Technology); Kenneth J. Davis (Pennsylvania State University); Troy Magney (California Institute of Technology, Department of Plant Sciences Davis CA USA); Philipp Köhler (California Institute of Technology); Liyin He (California Institute of Technology); Rupesh Jeyaram (California Institute of Technology); Vincent Humphrey (California Institute of Technology); Tobias Gerken (Pennsylvania State University); Sha Feng (Pennsylvania State University); Joshua P. Digangi (National Aeronautics and Space Administration (NASA) Langley Research Center); Christian Frankenberg (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: AGU Advances

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019av000140

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Cited by: 44 publications

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Keywords: N/A

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Open Access: N/A

Abstract: While large-scale floods directly impact human lives and infrastructures, they also profoundly impact agricultural productivity. New satellite observations of vegetation activity and atmospheric CO₂ offer the opportunity to quantify the effects of such extreme events on cropland carbon sequestration. Widespread flooding during spring and early summer 2019 induced conditions that delayed crop planting across the U.S. Midwest. As a result, satellite observations of solar-induced chlorophyll fluorescence from TROPospheric Monitoring Instrument and Orbiting Carbon Observatory reveal a 16-day shift in the seasonal cycle of photosynthesis relative to 2018, along with a 15% lower peak value. We estimate a reduction of 0.21 PgC in cropland gross primary productivity in June and July, partially compensated in August and September (+0.14 PgC). The extension of the 2019 growing season into late September is likely to have benefited from increased water availability and late-season temperature. Ultimately, this change is predicted to reduce the crop productivity in the Midwest Corn/Soy belt by ~15% compared to 2018. Using an atmospheric transport model, we show that a decline of ~0.1 PgC in the net carbon uptake during June and July is consistent with observed CO₂ enhancements of up to 10 ppm in the midday boundary layer from Atmospheric Carbon and Transport-America aircraft and over 3 ppm in column-averaged dry-air mole fractions from Orbiting Carbon Observatory. This study quantifies the impact of floods on cropland productivity and

demonstrates the potential of combining solar-induced chlorophyll fluorescence with atmospheric CO₂ observations to monitor regional carbon flux anomalies.

891. Three-to-Six-Day Air-Sea Oscillation in Models and Observations

Authors: Ehud Strobach (University of Maryland, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Andrea Molod (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Atanas Trayanov (Science Systems and Applications Inc. Greenbelt MD USA, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Gael Forget (Massachusetts Institute of Technology); Jean-Michel Campin (Massachusetts Institute of Technology); Chris Hill (Massachusetts Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 13 publications

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Analysis of coupled ocean-atmosphere numerical simulations revealed a widespread oscillation of sea surface temperature (SST) and surface wind anomalies with a cyclic nature characterized by a period of 3 to 6 days. Similar oscillations were also found in buoy data at different locations around the globe and in an atmospheric reanalysis, albeit the oscillations were noisier and stronger in the observations, and they were weaker and longer period in the atmospheric reanalysis. These oscillations did not, however, develop in an atmosphere-only simulation, suggesting the importance of ocean-atmosphere coupling for simulating this phenomenon. We propose a mechanism for this interaction involving a feedback between SST and surface wind.

892. Impacts of ocean current-atmosphere interactions on mesoscale eddy energetics in the Kuroshio extension region

Authors: Xuan Shan; Zhao Jing; Bingrong Sun; Lixin Wu

Journal: Geoscience Letters

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40562-020-00152-w

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Cited by: 4 publications

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Keywords: N/A

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Open Access: N/A

Abstract: The surface ocean current's imprint on the wind stress (known as the current feedback) has a significant influence on the wind power input to the ocean. In this study, we investigate the effect of the current feedback on mesoscale eddy energetics in the Kuroshio extension region using a high-resolution (9 km) coupled regional climate model. We perform three sets of simulations: one calculates the wind stress without the surface current, one includes only the mesoscale eddy's current and another includes the entire current in the computation. In this way, the mesoscale eddy's current feedback can be isolated and its contribution to the entire current feedback can be assessed. The simulation results show that the mesoscale eddy's current feedback results in negative wind power input to mesoscale eddies and reduces the surface eddy kinetic energy (EKE) by ~ 20% in the Kuroshio extension region. This negative wind power is compensated primarily by the enhanced conversion from mean flow kinetic energy (MKE) and eddy potential energy (EPE) to EKE and secondarily by the reduced pressure flux divergence and horizontal dissipation. Compared to the mesoscale eddy's current feedback, the entire current feedback has a similar impact on EKE and the wind power on mesoscale eddies, while it affects the EKE budget differently. It weakens the MKE to EKE conversion partly due to the reduced kinetic energy input to background flows by wind. Correspondingly, the negative wind power on mesoscale eddies is primarily compensated by the enhanced EPE to EKE conversion.

893. Heat stored in the Earth system: Where does the energy go? The GCOS Earth heat inventory team

Authors: Karina von Schuckmann; Lijing Cheng; Matthew D. Palmer; Caterina Tassone; Valentin Aich; Susheel Adusumilli; Hugo Beltrami; Tim Boyer; Francisco José Cuesta-Valero; Damien Desbruyères; Catia Domingues; Almudena García-García; Pierre Gentine; John Gilson; Maximilian Gorfer; Leopold Haimberger; Masayoshi Ishii; Gregory C. Johnson; Rachel Killik; Brian A. King; Gottfried Kirchengast; Nicolas Kolodziejczyk; John Lyman; Ben Marzeion; Michael Mayer; Maeva Monier; Didier Paolo Monselesan; Sarah Purkey; Dean Roemmich; Axel Schweiger; Sonia I. Seneviratne; Andrew Shepherd; Donald A. Slater; Andrea K. Steiner; Fiammetta Straneo; Mary-Louise Timmermans; Susan E. Wijffels

Journal: Unknown Journal

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Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Earth system science|Radiant heat

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Open Access: N/A

Abstract: . Human-induced atmospheric composition changes cause a radiative imbalance at the top-of-atmosphere which is driving global warming. This Earth Energy Imbalance (EEI) is a fundamental metric of climate change. Understanding the heat gain of the Earth system from this accumulated heat - and particularly how much and where the heat is distributed in the Earth system - is fundamental to understanding how this affects warming oceans, atmosphere and land, rising temperatures and sea level, and loss of grounded and floating ice, which are fundamental concerns for society. This study is a Global Climate Observing System (GCOS) concerted international effort to update the Earth heat inventory, and presents an updated international assessment of ocean warming estimates, and new and updated estimates of heat gain in the atmosphere, cryosphere and land over the period 1960-2018. The study obtains a consistent long-term Earth system heat gain over the past 58 years, with a total heat gain of 393 ± 40 ZJ, which is equivalent to a heating rate of 0.42 ± 0.04 W m⁻². The majority of the heat gain (89 %) takes place in the global ocean (0-700 m: 53 %; 700-2000 m: 28 %; > 2000 m: 8 %), while it amounts to 6 % for the land heat gain, to 4 % available for the melting of grounded and floating ice, and to 1 % for atmospheric warming. These new estimates indicate a larger contribution of land and ice heat gain (10 % in total) compared to previous estimates (7 %). There is a regime shift of the Earth heat inventory over the past 2 decades, which appears to be predominantly driven by heat sequestration into the deeper layers of the global ocean, and a doubling of heat gain in the atmosphere. However, a major challenge is to reduce uncertainties in the Earth heat inventory, which can be best achieved through the maintenance of the current global climate observing

system, its extension into areas of gaps in the sampling, as well as to establish an international framework for concerted multi-disciplinary research of the Earth heat inventory. Earth heat inventory is published at DKRZ (<https://www.dkrz.de/>) under the doi: https://doi.org/10.26050/WDCC/GCOS_EHI_EXP (von Schuckmann et al., 2020).

894. Volume and Heat Budgets in the Coastal California Current System: Means, Annual Cycles, and Interannual Anomalies of 2014-16

Authors: Katherine D. Zaba (Scripps Institution of Oceanography); Daniel L. Rudnick (Scripps Institution of Oceanography); Bruce D. Cornuelle (Scripps Institution of Oceanography); Ganesh Gopalakrishnan (Scripps Institution of Oceanography); Matthew R. Mazloff (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Cited by: 14 publications

Language: N/A

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Keywords: N/A

Funding: National Oceanic and Atmospheric Administration, National Oceanic and Atmospheric Administration, National Science Foundation (and 3 more)

License: N/A

Open Access: N/A

Abstract: The data-assimilating California State Estimate (CASE) enables the explicit evaluation of spatiotemporally varying volume and heat budgets in the coastal California Current System (CCS). An analysis of over 10 years of CASE model output (2007-17) diagnoses the physical drivers of the CCS mean state, annual cycles, and the 2014-16 temperature anomalies associated with a marine heat wave and an El Niño event. The largest terms in the mean mixed layer (from -50 to 0 m) volume budgets are upward vertical transport at the coast and offshore-flowing ageostrophic Ekman transport at the surface, the two branches of the coastal upwelling overturning cell. Contributions from onshore geostrophic flow in the Southern California Bight and alongshore geostrophic convergence in the central CCS balance the mean volume budgets. The depth-dependent annual cycle of vertical velocity exhibits the strongest upward velocity between -40- and -30-m depth in April. Interannual volume budgets show that over 50% of the 2013.5-16.5 time period experienced downwelling anomalies, which were balanced predominantly by alongshore transport convergence and, less often, by onshore transport anomalies. Mixed layer temperature anomalies persisted for the entirety of 2014-16, reaching a maximum of +3 degrees in October 2015. The mixed layer heat budget shows that intermittent high air-sea heat flux anomalies and alongshore and vertical heat advection anomalies all contributed to warming during 2014-16. A subsurface (from -210 to -100 m) heat budget reveals that in September 2015 anomalous poleward heat advection into the Southern California Bight by the California Undercurrent caused deeper warming during the 2015/16 El Niño.

895. Half-precessional cycle of thermocline temperature in the western equatorial Pacific and its bihemispheric dynamics

Authors: Zhimin Jian (Tongji University); Yue Wang (Tongji University); Haowen Dang (Tongji University); David W. Lea (University of California Santa Barbara); Zhengyu Liu (Ohio State University); Haiyan Jin (Tongji University); Yaqian Yin (Tongji University)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

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Publication Date: 2020-03-17

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References: 73

Cited by: 51 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Significance The precessional insolation plays a critical role in modulating the low-latitude oceanographic changes through ENSO-like dynamics. While previous studies focused on the precessional insolation-induced low-latitude surface water changes, our well-dated, centennial-resolution proxy data and numerical simulation reveal the half-precession cycle of thermocline water temperature (TWT) in the western equatorial Pacific that directly linked to ENSO-like zonal temperature seesaw over the equatorial Pacific. These half-precessional TWT changes are proposed to originate from the interplay of subtropical-to-tropical thermocline anomalies forced by the antiphased meridional insolation gradients in the two hemispheres at the precessional band. This work illustrates that the tropical thermocline water could bridge the meridional climatic processes and the east-west ENSO-like dynamics at orbital timescales.

896. Assessment of Numerical Simulations of Deep Circulation and Variability in the Gulf of Mexico Using Recent Observations

Authors: Steven L. Morey (a Florida Agricultural and Mechanical University, Tallahassee, Florida, Florida State University); Ganesh Gopalakrishnan (Scripps Institution of Oceanography / University of California San Diego); Enric Pallás Sanz (d Centro de Investigacion Cientifica y de Educacion Superior de Ensenada, Ensenada, Baja California, Mexico); Joao Marcos Azevedo Correia De Souza (e MetOcean Solutions, New Plymouth, New Zealand, d Centro de Investigacion Cientifica y de Educacion Superior de Ensenada, Ensenada, Baja California, Mexico); Kathleen Donohue (University of Rhode Island); Paula Pérez-Brunius (d Centro de Investigacion Cientifica y de Educacion Superior de Ensenada, Ensenada, Baja California, Mexico); Dmitry Dukhovskoy (Florida State University); Eric Chassignet (Florida State University); Bruce Cornuelle (Scripps Institution of Oceanography / University of California San Diego); Amy Bower (Woods Hole Oceanographic Institution); Heather Furey (Woods Hole Oceanographic Institution); Peter Hamilton (North Carolina State University); Julio Candela (d Centro de Investigacion Cientifica y de Educacion Superior de Ensenada, Ensenada, Baja California, Mexico)

Journal: Journal of Physical Oceanography

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Keywords: N/A

Funding: Gulf Research Program, Gulf Research Program, CICESE-PetroleosMexicanos

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Open Access: N/A

Abstract: Three simulations of the circulation in the Gulf of Mexico (the "Gulf") using different numerical general circulation models are compared with results of recent large-scale observational campaigns conducted throughout the deep (>1500 m) Gulf. Analyses of these observations have provided new understanding of large-scale mean circulation features and variability throughout the deep Gulf. Important features include cyclonic flow along the continental slope, deep cyclonic circulation in the western Gulf, a counterrotating pair of cells under the Loop Current region, and a cyclonic cell to the south of this pair. These dominant circulation features are represented in each of the ocean model simulations, although with some obvious differences. A striking difference between all the models and the observations is that the simulated deep eddy kinetic energy under the Loop Current region is generally less than one-half of that computed from observations. A multidecadal integration of one of these numerical simulations is used to evaluate the uncertainty of estimates of velocity statistics in the deep Gulf computed from limited-length (4 years) observational or model records. This analysis shows that the main deep circulation features identified from the observational studies appear to be robust and are not substantially impacted by variability on time scales longer than the observational

records. Differences in strengths and structures of the circulation features are identified, however, and quantified through standard error analysis of the statistical estimates using the model solutions.

897. The Seasonality of Submesoscale Energy Production, Content, and Cascade

Authors: Jihai Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory); Baylor Fox-Kemper (Brown University); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Changming Dong (Nanjing University, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Keywords: Kuroshio extension, mixed layer instability, nonlinear interaction, energy cascade, Submesoscale

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License: CC BY 4.0

Open Access: N/A

Abstract: Submesoscale processes in the upper ocean vary seasonally, in tight correspondence with mixed layer thickness variability. Based on a global high-resolution MITgcm simulation, seasonal evaluation of strong vorticity and spectral analysis of the kinetic energy in the Kuroshio Extension System show the strongest submesoscales occur in March, implying a lag of about a month behind mixed layer thickness maximum in February. An analysis of spectral energy sources and transfers indicates that the seasonality of the submesoscale energy content is a result of the competition between the conversion of available potential energy into submesoscale kinetic energy via a buoyancy production/vertical buoyancy flux associated with mixed layer instability and nonlinear energy transfers to other scales associated with an energy cascade. The buoyancy production is seasonally in phase with the mixed layer depth, but the transfers of energy across scales makes energizing the reservoir of submesoscale kinetic energy lag behind by a month.

898. Self-Consistent Ice Mass Balance and Regional Sea Level From Time-Variable Gravity

Authors: Tyler C. Sutterley (University of Washington, University of California Irvine, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Isabella Velicogna (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Chia-Wei Hsu (University of California Irvine, University of Arizona)

Journal: Earth and Space Science

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019ea000860

URL: <https://doi.org/10.1029/2019ea000860>

Publication Date: 2020-03-11

Volume/Issue: Volume 7, Issue 3

References: 77

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Measurements of time-variable gravity from the Gravity Recovery and Climate Experiment (GRACE) and the GRACE Follow-on (GRACE-FO) missions are an invaluable tool for monitoring changes in the mass of the Earth's glaciated regions. We improve upon estimates of glacier and ice sheet mass balance from time-variable gravity by including instantaneous spatiotemporal variations in sea level. Here, a least squares mascon technique is combined with solutions to the sea level equation to iteratively correct the GRACE/GRACE-FO data for the induced sea level response on a monthly basis. We find that variations in regional sea level affect ice sheet mass balance estimates in Greenland by approximately 4% and in Antarctic by approximately 5%. Since 2002, the Greenland ice sheet has been losing mass at an average rate of 263 ± 23 Gt/yr, and the Antarctic ice sheet has been losing mass at average rates between 90 ± 52 and 122 ± 53 Gt/yr depending on the rate of glacial isostatic adjustment. The mass losses from both ice sheets represent an increase of 15.6 ± 2.0 mm to global mean sea levels since 2002.

899. Instabilities and Multiscale Interactions Underlying the Loop Current Eddy Shedding in the Gulf of Mexico

Authors: Yang Yang (Nanjing University, University of South Florida); Robert H. Weisberg (University of South Florida); Yonggang Liu (University of South Florida); X. San Liang (Nanjing University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0202.1

URL: <https://doi.org/10.1175/jpo-d-19-0202.1>

Publication Date: 2020-03-11

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References: 82

Cited by: 39 publications

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Subjects: N/A

Keywords: N/A

Funding: National Science Foundation of China, National Science Foundation of China, Jiangsu Program of Entrepreneurship and Innovation Group (and 6 more)

License: N/A

Open Access: N/A

Abstract: A recently developed tool, the multiscale window transform, along with the theory of canonical energy transfer is used to investigate the roles of multiscale interactions and instabilities in the Gulf of Mexico Loop Current (LC) eddy shedding. A three-scale energetics framework is employed, in which the LC system is reconstructed onto a background flow window, a mesoscale eddy window, and a high-frequency eddy window. The canonical energy transfer between the background flow and the mesoscale windows plays an important role in LC eddy shedding. Barotropic instability contributes to the generation/intensification of the mesoscale eddies over the eastern continental slope of the Campeche Bank. Baroclinic instability favors the growth of the mesoscale eddies that propagate downstream to the northeastern portion of the well-extended LC, eventually causing the shedding by cutting through the neck of the LC. These upper-layer mesoscale eddies lose their kinetic energy back to the background LC through inverse cascade processes in the neck region. The deep eddies obtain energy primarily from the upper layer through vertical pressure work and secondarily from baroclinic instability in the deep layer. In contrast, the canonical energy transfer between the mesoscale and the high-frequency frontal eddy windows accounts for only a small fraction in the mesoscale eddy energy balance, and this generally acts as a damping mechanism for the mesoscale eddies. A budget analysis reveals that the mesoscale eddy energy gained through the instabilities is balanced by horizontal advection, pressure work, and dissipation.

900. Some Statistical Issues in Climate Science

Authors: Michael L. Stein

Journal: Statistical Science

Publisher: Institute of Mathematical Statistics

DOI: 10.1214/19-sts730

URL: <https://doi.org/10.1214/19-sts730>

Publication Date: 2020-03-03

Volume/Issue: Volume 35, Issue 1

References: 90

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

901. A preliminary evaluation of some elements for designation of preservation and impact reference zones in deep sea in the Clarion-Clipperton Zone: A case study of the China ocean mineral resources association contract area

Authors: Huang Hao; Wang Lei; Ou Danyun; Li Weiwen; Kuang Fangfang; Lin Cai; He Xuebao; An Lina; Wang Weobo

Journal: Ocean & Coastal Management

Publisher: Elsevier BV

DOI: 10.1016/j.ocecoaman.2020.105135

URL: <https://doi.org/10.1016/j.ocecoaman.2020.105135>

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Volume/Issue: Volume 188

References: 9

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the Ocean 13.5 Environmental Project, China, the Basic Research Foundation of the Third Institute of Oceanography of MNR, China, the Natural Sciences Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

902. Ocean Bottom Pressure Variability: Can It Be Reliably Modeled?

Authors: Alexey Androsov (Alfred Wegener Institute, Shirshov Institute of Oceanology Moscow Russia); Olaf Boebel (Alfred Wegener Institute); Jens Schröter (Alfred Wegener Institute); Sergey Danilov (Alfred Wegener Institute, Department of Mathematics and Logistics Jacobs University Bremen Germany, A. M. Obukhov Institute of Atmospheric Physics); Andreas Macrander (Marine and Freshwater Research Institute Reykjavik Iceland); Ioana Ivanciu (GEOMAR Helmholtz Centre for Ocean Research)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015469

URL: <https://doi.org/10.1029/2019jc015469>

Publication Date: 2020-02-20

Volume/Issue: Volume 125, Issue 3

References: 43

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Ocean bottom pressure (OBP) variability serves as a proxy of ocean mass variability, the knowledge of which is needed in geophysical applications. The question of how well it can be modeled by the present general ocean circulation models on time scales in excess of 1 day is addressed here by comparing the simulated OBP variability with the observed one. To this end, a new multiyear data set is used, obtained with an array of bottom pressure gauges deployed deeply along a transect across the Southern Ocean. We present a brief description of OBP data and show large-scale correlations over several thousand kilometers at all time scales using daily and monthly averaged data. Annual and semiannual cycles are weak. Close to the Agulhas Retroflection, signals of up to 30 cm equivalent water height are detected. Further south, signals are mostly intermittent and noisy. It is shown that the models simulate consistent patterns of bottom pressure variability on monthly and longer scales except for areas with high mesoscale eddy activity, where high resolution is needed to capture the variability due to eddies. Furthermore, despite good agreement in the amplitude of variability, the in situ and simulated OBP show only modest correlation.

903. A Quasi-Decadal Oscillation of Sea-Level Variation in the South China Sea

Authors: Hui Xi (Chinese Academy of Sciences); Zizhan Zhang (Chinese Academy of Sciences); Yang Lu (Chinese Academy of Sciences)

Journal: Journal of Coastal Research

Publisher: Coastal Education and Research Foundation

DOI: 10.2112/jcoastres-d-19-00078.1

URL: <https://doi.org/10.2112/jcoastres-d-19-00078.1>

Publication Date: 2020-02-19

Volume/Issue: Volume 36, Issue 2

References: 64

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: altimetry, sunspot, Pacific decadal oscillation (PDO), Ishii, GRACE

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

904. Patterns and Dynamics of SST Fronts in the California Current System

Authors: Y. L. Mauzole (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H. S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); L.-L. Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015499

URL: <https://doi.org/10.1029/2019jc015499>

Publication Date: 2020-02-15

Volume/Issue: Volume 125, Issue 2

References: 34

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: compensation, California, simulation, fronts, patterns (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The California Current System (CCS) is one of the four major Eastern Boundary Current Systems, where major coastal upwelling is observed in the ocean. Oceanic fronts are abundant in the CCS, and are known to be associated with coastal upwelling. Analyses of satellite observations (altimetry and sea surface temperature, SST) of the CCS highlight the presence of SST fronts in the southern part of the CCS outside of the coastal upwelling season, which will be investigated. The submesoscale-permitting and tidal-resolving LLC4320-simulation was used to analyze the nature of these features, through the study of the surface density gradients, the vertical velocities and the horizontal strain. An algorithm was also developed to track SST fronts in the model. The classical scenario of coastal upwelling is confirmed in the central CCS (off California), with a strain field correlated to vertical velocities associated with the SST fronts, and collocated density fronts. In contrast, in the southern CCS (off Baja California), vertical velocities were much weaker and density gradients were not always observed where SST fronts were detected, leading to the assumption that the fronts there may be affected by thermohaline compensation occurring in the region. These findings confirm the existence of SST fronts in the CCS outside of the traditional coastal upwelling framework, with part of the fronts in the southern CCS being passive and advected by the background flow.

905. Correcting GPS measurements for non-tidal loading

Authors: Anthony Mémin; Jean-Paul Boy; Alvaro Santamaría-Gómez

Journal: GPS Solutions

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10291-020-0959-3

URL: <https://doi.org/10.1007/s10291-020-0959-3>

Publication Date: 2020-02-14

Volume/Issue: Volume 24, Issue 2

References: 48

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: CNES, Australian Research Council, FP7 People: Marie-Curie Actions

License: N/A

Open Access: N/A

Abstract: N/A

906. Response of Air-Sea Fluxes and Oceanic Features to the Coupling of Ocean-Atmosphere-Wave During the Passage of a Tropical Cyclone

Authors: Vimlesh Pant; Kumar Ravi Prakash

Journal: Pure and Applied Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00024-020-02441-z

URL: <https://doi.org/10.1007/s00024-020-02441-z>

Publication Date: 2020-02-13

Volume/Issue: Volume 177, Issue 8

References: 68

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

907. The Observation-Based Application of a Regional Thermohaline Inverse Method to Diagnose the Formation and Transformation of Water Masses North of the OSNAP Array from 2013 to 2015

Authors: Neill Mackay (National Oceanography Centre); Chris Wilson (National Oceanography Centre); N. Penny Holliday (National Oceanography Centre); Jan D. Zika (University of New South Wales)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0188.1

URL: <https://doi.org/10.1175/jpo-d-19-0188.1>

Publication Date: 2020-02-13

Volume/Issue: Volume 50, Issue 6

References: 42

Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Natural Environment Research Council, Australian Research Council (and 4 more)

License: CC BY 4.0

Open Access: N/A

Abstract: The strength of the meridional overturning circulation (MOC) in the North Atlantic is dependent upon the formation of dense waters that occurs at high northern latitudes. Wintertime deep convection in the Labrador and Irminger Seas forms the intermediate water mass known as Labrador Sea Water (LSW). Changes in the rate of formation and subsequent export of LSW are thought to play a role in MOC variability, but formation rates are uncertain and the link between formation and export is complex. We present the first observation-based application of a recently developed regional thermohaline inverse method (RTHIM) to a region encompassing the Arctic and part of the North Atlantic subpolar gyre for the years 2013, 2014, and 2015. RTHIM is a novel method that can diagnose the formation and export rates of water masses such as the LSW identified by their temperature and salinity, apportioning the formation rates into contributions from surface fluxes and interior mixing. We find LSW formation rates of up to 12 Sv ($1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) during 2014-15, a period of strong wintertime convection, and around half that value during 2013 when convection was weak. We also show that the newly convected water is not exported directly, but instead is mixed isopycnally with warm, salty waters that have been advected into the region, before the products are then exported. RTHIM solutions for 2015 volume, heat, and freshwater transports are compared with observations from a mooring array deployed for the Overturning in the Subpolar North Atlantic Program (OSNAP) and show good agreement, lending validity to our results.

908. Using a regional ocean model to understand the structure and variability of acoustic arrivals in Fram Strait

Authors: Florian Geyer (Nansen Environmental and Remote Sensing Center); Hanne Sagen (Nansen Environmental and Remote Sensing Center); Bruce Cornuelle (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Heriberto J. Vazquez (Scripps Institution of Oceanography / University of California San Diego)

Journal: The Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

DOI: 10.1121/10.0000513

URL: <https://doi.org/10.1121/10.0000513>

Publication Date: 2020-02-11

Volume/Issue: Volume 147, Issue 2

References: 24

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research Global, National Aeronautics and Space Administration, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: A regional ocean model for Fram Strait provides a framework for interpretation of the variability and structure of acoustic tomography arrivals. The eddy-permitting model (52 vertical levels and 4.5 km horizontal resolution) was evaluated using long-term moored hydrography data and time series of depth-range averaged temperature obtained from the inversion of acoustic tomography measurements. Geometric ray modeling using the ocean model fields reproduces the measured arrival structure of the acoustic tomography experiment. The combination of ocean and acoustic models gives insights into acoustic propagation during winter and spring. Moreover, overlapping arrivals coming from different vertical angles can be resolved and explained. The overlapping arrival of purely refracted rays and surface-reflected/bottom-reflected (SRBR) rays has implications for the inversion of tomography data in Fram Strait. The increased knowledge about the ray-length variations of SRBR rays is valuable for choosing appropriate observation kernels for the data assimilation of acoustic tomography data in Fram Strait.

909. Quantitative Evaluation of Environmental Loading Induced Displacement Products for Correcting GNSS Time Series in CMONOC

Authors: Chenfeng Li (Wuhan University, University of Luxembourg); Shengxiang Huang (Wuhan University); Qiang Chen (University of Luxembourg); Tonie van Dam (University of Luxembourg); Hok Sum Fok (Wuhan University); Qian Zhao (China Earthquake Administration); Weiwei Wu (China Earthquake Administration); Xinpeng Wang (Wuhan University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs12040594

URL: <https://doi.org/10.3390/rs12040594>

Publication Date: 2020-02-11

Volume/Issue: Volume 12, Issue 4

References: 62

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: Mass redistribution within the Earth system deforms the surface elastically. Loading theory allows us to predict loading induced displacement anywhere on the Earth's surface using environmental loading models, e.g., Global Land Data Assimilation System. In addition, different publicly available loading products are available. However, there are differences among those products and the differences among the combinations of loading models cannot be ignored when precisions of better than 1 cm are required. Many scholars have applied these loading corrections to Global Navigation Satellite System (GNSS) time series from mainland China without considering or discussing the differences between the available models. Evaluating the effects of different loading products over this region is of paramount importance for accurately removing the loading signal. In this study, we investigate the performance of these different publicly available loading products on the scatter of GNSS time series from the Crustal Movement Observation Network of China. We concentrate on five different continental water storage loading models, six different non-tidal atmospheric loading models, and five different non-tidal oceanic loading models. We also investigate all the different combinations of loading products. The results show that the difference in RMS reduction can reach 20% in the vertical component depending on the loading correction applied. We then discuss the performance of different loading combinations and their effects on the noise characteristics of GNSS height time series and horizontal velocities. The results show that the loading products from NASA may be the best choice for corrections in mainland China. This conclusion could serve as an important reference for loading products users in this region.

910. Variability of spice injection in the upper ocean of the southeastern Pacific during 1992-2016

Authors: Yingying Wang; Yiyong Luo

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-020-05164-y

URL: <https://doi.org/10.1007/s00382-020-05164-y>

Publication Date: 2020-02-11

Volume/Issue: Volume 54, Issue 5-6

References: 37

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

911. Energetics of a Rotating Wind-forced Horizontal Convection Model of a Reentrant Channel

Authors: Varvara E. Zemskova (University of North Carolina at Chapel Hill, Chapel Hill, North Carolina); Brian L. White (University of North Carolina at Chapel Hill, Chapel Hill, North Carolina); Alberto Scotti (University of North Carolina at Chapel Hill, Chapel Hill, North Carolina)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0169.1

URL: <https://doi.org/10.1175/jpo-d-19-0169.1>

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We present numerical results for an idealized rotating, buoyancy- and windforced channel as a simple model for the Southern Ocean branch of the Meridional Overturning Circulation (MOC). Differential buoyancy forcing is applied along the top horizontal surface, with surface cooling at one end (to represent the pole) and surface warming at the other (to represent the equatorial region) and a zonally re-entrant channel to represent the Antarctic Circumpolar Current (ACC). Zonally-uniform surface wind forcing is applied with a similar pattern to the westerlies and easterlies with varying magnitude relative to the buoyancy forcing. The problem is solved numerically using a 3D DNS model based on a finite-volume solver for the Boussinesq Navier-Stokes equations with rotation. The overall dynamics, including large-scale overturning, baroclinic eddying, turbulent mixing, and resulting energy cascades are studied by calculating terms in the energy budget using the local Available Potential Energy framework. The basic physics of the overturning in the Southern Ocean are investigated at multiple scales and the output from the fully-resolved DNS simulations is compared with the results from previous studies of the global (ECCO2) and Southern Ocean eddy-permitting state estimates. We find that both the magnitude and shape of the zonal wind stress profile are important to the spatial pattern of the overturning circulation. However, the available potential energy budget and the diapycnal mixing are not significantly affected by the surface wind stress and are primarily set by the buoyancy forcing at the surface.

912. Deep-reaching acceleration of global mean ocean circulation over the past two decades

Authors: Shijian Hu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego); Cong Guan (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Michael J. McPhaden (NOAA/Pacific Marine Environmental Laboratory, Seattle, WA 98115, USA.); Fan Wang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Dunxin Hu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Wenju Cai (CSIRO Oceans and Atmosphere Flagship, Aspendale, Victoria 3195, Australia., Centre for Southern Hemisphere Oceans Research (CSHOR), CSIRO Oceans and Atmosphere, Hobart, Tasmania 7004, Australia., Pilot National Laboratory for Marine Science and Technology)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.aax7727

URL: <https://doi.org/10.1126/sciadv.aax7727>

Publication Date: 2020-02-06

Volume/Issue: Volume 6, Issue 6

References: 50

Cited by: 107 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, National Natural Science Foundation of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: The global mean ocean circulation from surface to deep oceans has accelerated over the past two decades.

913. Dynamic Sea Level Variability Due to Seasonal River Discharge: A Preliminary Global Ocean Model Study

Authors: Christopher G. Piecuch (Woods Hole Oceanographic Institution); Riley Wadehra (Colorado College Colorado Springs CO USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl086984

URL: <https://doi.org/10.1029/2020gl086984>

Publication Date: 2020-02-04

Volume/Issue: Volume 47, Issue 4

References: 58

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: ocean modeling, seasonal cycle, river discharge, sea level

Funding: National Science Foundation, Woods Hole Oceanographic Institution

License: N/A

Open Access: N/A

Abstract: River discharge is an important driver of the coastal ocean circulation. However, the influence of variable effects on large-scale sea level related to ocean dynamics is not well understood. We perform numerical experiments using a coarse (degrees) global ocean state estimate to quantify effects on the seasonal cycle in dynamic over the global ocean. Forcing by drives substantial dynamic signals (cm) on regional scales around major rivers (e.g., Amazon, Ganges, Brahmaputra, Irrawaddy, Ob, Lena, and Yenisei). Modeled changes largely reflect halosteric (salinity) effects, which are partly affected by transport processes involving planetary waves and advection. The modeled response to is informative for interpreting observations from satellite altimetry and tide gauges, explaining 10-20% of the regional-scale seasonal data variance around some major rivers. Future studies should consider finer-resolution models and forcing on nonseasonal time scales of relevance to climate.

914. The Sensitivity of Southeast Pacific Heat Distribution to Local and Remote Changes in Ocean Properties

Authors: Daniel C. Jones (British Antarctic Survey, NERC, UKRI, Cambridge, United Kingdom); Emma Boland (British Antarctic Survey, NERC, UKRI, Cambridge, United Kingdom); Andrew J. S. Meijers (British Antarctic Survey, NERC, UKRI, Cambridge, United Kingdom); Gael Forget (Massachusetts Institute of Technology); Simon Josey (National Oceanography Centre); Jean-Baptiste Sallée (Sorbonne Universités, UPMC Université, LOCEAN-IPSL, Paris, France); Emily Shuckburgh (University of Cambridge)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0155.1

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Publication Date: 2020-01-28

Volume/Issue: Volume 50, Issue 3

References: 77

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Natural Environment Research Council, Natural Environment Research Council (and 2 more)

License: N/A

Open Access: N/A

Abstract: The Southern Ocean features ventilation pathways that transport surface waters into the subsurface thermocline on time scales from decades to centuries, sequestering anomalies of heat and carbon away from the atmosphere and thereby regulating the rate of surface warming. Despite its importance for climate sensitivity, the factors that control the distribution of heat along these pathways are not well understood. In this study, we use an observationally constrained, physically consistent global ocean model to examine the sensitivity of heat distribution in the recently ventilated subsurface Pacific (RVP) sector of the Southern Ocean to changes in ocean temperature and salinity. First, we define the RVP using numerical passive tracer release experiments that highlight the ventilation pathways. Next, we use an ensemble of adjoint sensitivity experiments to quantify the sensitivity of the RVP heat content to changes in ocean temperature and salinity. In terms of sensitivities to surface ocean properties, we find that RVP heat content is most sensitive to anomalies along the Antarctic Circumpolar Current (ACC), upstream of the subduction hotspots. In terms of sensitivities to subsurface ocean properties, we find that RVP heat content is most sensitive to basin-scale changes in the subtropical Pacific Ocean, around the same latitudes as the RVP. Despite the localized nature of mode water subduction hotspots, changes in basin-scale density gradients are an important controlling factor on heat distribution in the southeast Pacific.

915. The Atlantic Meridional Overturning Circulation in High-Resolution Models

Authors: Joël J.-M. Hirschi (National Oceanography Centre); Bernard Barnier (Institut de Recherche pour le Développement (IRD)); Claus Böning (GEOMAR Helmholtz Centre for Ocean Research); Arne Biastoch (GEOMAR Helmholtz Centre for Ocean Research); Adam T. Blaker (National Oceanography Centre); Andrew Coward (National Oceanography Centre); Sergey Danilov (Alfred Wegener Institute); Sybren Drijfhout (University of Southampton); Klaus Getzlaff (GEOMAR Helmholtz Centre for Ocean Research); Stephen M. Griffies (Princeton University); Hiroyasu Hasumi (University of Tokyo); Helene Hewitt (Met Office); Dorotea Iovino (Ocean Modeling and Data Assimilation Division Centro Euro-Mediterraneo sui Cambiamenti Climatici Italy); Takao Kawasaki (University of Tokyo); Andrew E. Kiss (Australian National University); Nikolay Koldunov (Alfred Wegener Institute); Alice Marzocchi (National Oceanography Centre); Jennifer V. Mecking (National Oceanography Centre); Ben Moat (National Oceanography Centre); Jean-Marc Molines (Institut de Recherche pour le Développement (IRD)); Paul G. Myers (University of Alberta); Thierry Penduff (Institut de Recherche pour le Développement (IRD)); Malcolm Roberts (Met Office); Anne-Marie Treguier (Laboratoire d'océanographie Physique et Spatiale, CNRS Brest France); Dmitry V. Sein (Alfred Wegener Institute); Dmitry Sidorenko (Alfred Wegener Institute); Justin Small (NCAR/UCAR Boulder CO USA); Paul Spence (University of New South Wales); LuAnne Thompson (University of Washington); Wilbert Weijer (Los Alamos National Laboratory); Xiaobiao Xu (Florida State University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015522

URL: <https://doi.org/10.1029/2019jc015522>

Publication Date: 2020-01-27

Volume/Issue: Volume 125, Issue 4

References: 284

Cited by: 84 publications

Language: en

Subjects: N/A

Keywords: mesoscale, high-resolution modeling, Atlantic Meridional Overturning

Funding: National Aeronautics and Space Administration, Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: The Atlantic meridional overturning circulation (AMOC) represents the zonally integrated stream function of meridional volume transport in the Atlantic Basin. The AMOC plays an important role in transporting heat meridionally in the climate system. Observations suggest a heat transport by the AMOC of 1.3 PW at 26 degreesN--a latitude which is close to where the Atlantic northward heat transport is thought to reach its maximum. This shapes the climate of the North Atlantic region as we know it today. In recent years there has been significant progress both in our ability to observe the AMOC in nature and to simulate it in numerical models. Most previous modeling investigations of the AMOC and its impact on climate have relied on models with horizontal resolution that does not resolve ocean mesoscale eddies and the dynamics of the Gulf Stream/North Atlantic Current system. As a result of recent increases in computing power, models are now being run that are able to represent mesoscale ocean dynamics and the circulation features that rely on them. The aim of this review is to

describe new insights into the AMOC provided by high-resolution models. Furthermore, we will describe how high-resolution model simulations can help resolve outstanding challenges in our understanding of the AMOC.

916. Observed variability of the West India Coastal Current on the continental slope from 2009-2018

Authors: Anya Chaudhuri; D Shankar; S G Aparna; P Amol; V Fernando; A Kankonkar; G S Michael; N P Satelkar; S T Khalap; A P Tari; M G Gaonkar; S Ghatkar; R R Khedekar

Journal: Journal of Earth System Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12040-019-1322-3

URL: <https://doi.org/10.1007/s12040-019-1322-3>

Publication Date: 2020-01-22

Volume/Issue: Volume 129, Issue 1

References: 33

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Department of Science and Technology, SERB, Ministry of Science and Technology, Council of Scientific and Industrial Research, India, Indian National Centre for Ocean Information Services

License: N/A

Open Access: N/A

Abstract: N/A

917. Eastern Pacific Wind Effect on the Evolution of El Niño: Implications for ENSO Diversity

Authors: Qihua Peng (Scripps Institution of Oceanography / University of California San Diego, Chinese Academy of Sciences); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego); Dongxiao Wang (Sun Yat-Sen University, Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory); Youichi Kamae (Faculty of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Japan); Hong Zhang (University of California Los Angeles); Shineng Hu (Columbia University); Xiao-Tong Zheng (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Weiqiang Wang (Chinese Academy of Sciences, Southern Marine Science and Engineering Guangdong Laboratory)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0435.1

URL: <https://doi.org/10.1175/jcli-d-19-0435.1>

Publication Date: 2020-01-22

Volume/Issue: Volume 33, Issue 8

References: 59

Cited by: 22 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Key Research and Development Program of China, Key Special Project for Introduced Talents Team of Southern Marine Science and Engineering Guangdong Laboratory (and 4 more)

License: N/A

Open Access: N/A

Abstract: The influence of eastern tropical Pacific (EPAC; 10 degreesS-10 degreesN, 140 degrees-80 degreesW) wind anomalies on El Niño is investigated using observations and model experiments. Extreme and moderate El Niños exhibit contrasting anomalous wind patterns in the EPAC during the peak and decay phases: westerly wind anomalies during extreme El Niño and southeasterly (southwesterly) wind anomalies south (north) of the equator during moderate El Niño. Experiments with an ocean general circulation model indicate that for extreme El Niño, the eastward intrusion of westerly wind anomalies contributes to the prolonged positive sea surface temperature (SST) anomalies in the eastern equatorial Pacific throughout boreal spring by weakened upwelling and horizontal advection. For moderate El Niño, by contrast, both the meridional and zonal anomalous winds over the EPAC are important in the rapid (slow) SST cooling south (north) of the equator through advection and wind-evaporation-SST feedback. Atmospheric model experiments confirm that these EPAC anomalous winds are primarily forced by tropical SST anomalies. The interplay between wind and SST anomalies suggests positive air-sea feedbacks over EPAC during the decay phase of El Niño. Ocean model results show that the frequency of extreme El Niño increases when EPAC wind anomalies are removed, suggesting the importance of EPAC winds for El Niño diversity.

918. Using the Helmholtz Decomposition to Define the Indian Ocean Meridional Overturning Streamfunction

Authors: Lei Han (Qingdao National Laboratory for Marine Science and Technology); Rui Xin Huang (Chinese Academy of Sciences, Woods Hole Oceanographic Institution)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0218.1

URL: <https://doi.org/10.1175/jpo-d-19-0218.1>

Publication Date: 2020-01-17

Volume/Issue: Volume 50, Issue 3

References: 33

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: The Fundamental Research Funds of Shandong University, National Basic Research Program of China

License: N/A

Open Access: N/A

Abstract: The zonally integrated flow in a basin can be separated into the divergent/nondivergent parts, and a uniquely defined meridional overturning circulation (MOC) can be calculated. For a basin with significant volume exchange at zonal open boundaries, this method is competent in removing the components associated with the nonzero source terms due to zonal transports at open boundaries. This method was applied to the zonally integrated flow in the Indian Ocean basin extended all the way to the Antarctic by virtue of the ECCO dataset. The contributions due to two major zonal flow systems at open boundaries, the Indonesian Throughflow (ITF) and the Antarctic Circumpolar Current (ACC), were well separated from the rotational flow component, and a nondivergent overturning circulation pattern was identified. Comparisons with previous studies on the MOC of the Indian Ocean in different seasons showed overall consistency but with refinements in details to the south of the entry of the ITF, reflecting the influence of ITF on the MOC pattern in the domain. Other options of decomposition are also examined.

919. Preserving the coupled atmosphere-ocean feedback in initializations of decadal climate predictions

Authors: Sebastian Brune (University of Hamburg); Johanna Baehr (University of Hamburg)

Journal: WIREs Climate Change

Publisher: Wiley

DOI: 10.1002/wcc.637

URL: <https://doi.org/10.1002/wcc.637>

Publication Date: 2020-01-17

Volume/Issue: Volume 11, Issue 3

References: 95

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung, Deutsche Forschungsgemeinschaft

License: CC BY-NC 4.0

Open Access: N/A

Abstract: On interannual to decadal time scales, memory in the Earth's climate system resides to a large extent in the slowly varying heat content of the ocean, which responds to fast atmospheric variability and in turn sets the frame for large-scale atmospheric circulation patterns. This large-scale coupled atmosphere-ocean feedback is generally well represented in today's Earth system models. This may fundamentally change when data assimilation is used to bring such models close to an observed state to initialize interannual to decadal climate predictions. Here, we review how the large-scale coupled atmosphere-ocean feedback is preserved in common approaches to construct such initial conditions, with the focus on the initialized ocean state. In a set of decadal prediction experiments, ranging from an initialization of atmospheric variability only to full-field nudging of both atmosphere and ocean, we evaluate the variability and predictability of the Atlantic meridional overturning circulation, of the Atlantic multidecadal variability and North Atlantic subpolar gyre sea surface temperatures. We argue that the quality of initial conditions for decadal predictions should not purely be assessed by their closeness to observations, but also by the closeness of their respective predictions to observations. This prediction quality may depend on the representation of the simulated large-scale atmosphere-ocean feedback. This article is categorized under: Climate Models and Modeling > Knowledge Generation with Models

920. The establishment of a pelagic Sargassum population in the tropical Atlantic: Biological consequences of a basin-scale long distance dispersal event

Authors: Elizabeth M. Johns; Rick Lumpkin; Nathan F. Putman; Ryan H. Smith; Frank E. Muller-Karger; Digna T. Rueda-Roa; Chuanmin Hu; Mengqiu Wang; Maureen T. Brooks; Lewis J. Gramer; Francisco E. Werner

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2020.102269

URL: <https://doi.org/10.1016/j.pocean.2020.102269>

Publication Date: 2020-01-16

Volume/Issue: Volume 182

References: 137

Cited by: 151 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Oceanic and Atmospheric Administration

License: N/A

Open Access: N/A

Abstract: N/A

921. Feature-based comparison of sea ice deformation in lead-permitting sea ice simulations

Authors: Nils Hutter; Martin Losch

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-93-2020

URL: <https://doi.org/10.5194/tc-14-93-2020>

Publication Date: 2020-01-16

Volume/Issue: Volume 14, Issue 1

References: 86

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The sea ice modeling community is progressing towards pan-Arctic simulations that explicitly resolve leads in the simulated sea ice cover. Evaluating these simulations against observations poses new challenges. A new feature-based evaluation of simulated deformation fields is introduced, and the results are compared to a scaling analysis of sea ice deformation. Leads and pressure ridges - here combined into linear kinematic features (LKFs) - are detected and tracked automatically from deformation and drift data. LKFs in two pan-Arctic sea ice simulations with a horizontal grid spacing of 2 km are compared with an LKF dataset derived from the RADARSAT Geophysical Processor System (RGPS). One simulation uses a five-class ice thickness distribution (ITD). The simulated sea ice deformation follows a multi-fractal spatial and temporal scaling, as observed from RGPS. The heavy-tailed distribution of LKF lengths and the scale invariance of LKF curvature, which points to the self-similar nature of sea ice deformation fields, are reproduced by the model. Interannual and seasonal variations in the number of LKFs, LKF densities, and LKF orientations in the ITD simulation are found to be consistent with RGPS observations. The lifetimes and growth rates follow a distribution with an exponential tail. The model overestimates the intersection angle of LKFs, which is attributed to the model's viscous-plastic rheology with an elliptical yield curve. In conclusion, the new feature-based analysis of LKF statistics is found to be useful for a comprehensive evaluation of simulated deformation features, which is required before the simulated features can be used with confidence in the context of climate studies. As such, it complements the commonly used scaling analysis and provides new useful information for comparing deformation statistics. The ITD simulation is shown to reproduce LKFs sufficiently well for it to be used for studying the effect of directly resolved leads in climate simulations. The feature-based analysis of LKFs also identifies specific model deficits that may be addressed by specific parameterizations, for example, a damage parameter, a grounding scheme, and a Mohr-Coulombic yield curve.

922. Oceanographic characteristics at fish aggregating device sites for tuna pole-and-line fishery in eastern Indonesia

Authors: Alexander M.A. Khan; Anta M. Nasution; Noir P. Purba; Achmad Rizal; Zahidah; Herman Hamdani; Lantun P. Dewanti; Junianto; Isnurruhwati; Asep Sahidin; Dedi Supriyadi; Heti Herawati; Izza M. Apriliani; Monika Ridwan; Tim S. Gray; Mingguo Jiang; Hazmi Arief; Aileen C. Mill; Nicholas V.C. Polunin

Journal: Fisheries Research

Publisher: Elsevier BV

DOI: 10.1016/j.fishres.2019.105471

URL: <https://doi.org/10.1016/j.fishres.2019.105471>

Publication Date: 2020-01-11

Volume/Issue: Volume 225

References: 54

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Direktorat Jenderal Pendidikan Tinggi

License: N/A

Open Access: N/A

Abstract: N/A

923. Can We Resolve the Basin-Scale Sea Level Trend Budget From GRACE Ocean Mass?

Authors: Sam Royston (University of Bristol); Bramha Dutt Vishwakarma (University of Bristol); Richard Westaway (University of Bristol); Jonathan Rougier (Second Institute of Oceanography, Ministry of Natural Resources); Zhe Sha (Second Institute of Oceanography, University of Bristol, Ministry of Natural Resources); Jonathan Bamber (University of Bristol)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015535

URL: <https://doi.org/10.1029/2019jc015535>

Publication Date: 2020-01-09

Volume/Issue: Volume 125, Issue 1

References: 89

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: regional sea level, sea level budget, sea level

Funding: Seventh Framework Programme, Leverhulme Trust, Royal Society

License: N/A

Open Access: N/A

Abstract: Understanding sea level changes at a regional scale is important for improving local sea level projections and coastal management planning. Sea level budget (SLB) estimates derived from the sum of observation of each component close for the global mean. The sum of steric and Gravity Recovery and Climate Experiment (GRACE) ocean mass contributions to sea level calculated from measurements does not match the spatial patterns of sea surface height trends from satellite altimetry at 1 degrees grid resolution over the period 2005-2015. We investigate potential drivers of this mismatch aggregating to subbasin regions and find that the steric plus GRACE ocean mass observations do not represent the small-scale features seen in the satellite altimetry. In addition, there are discrepancies with large variance apparent at the global and hemispheric scale. Thus, the SLB closure on the global scale to some extent represents a cancelation of errors. The SLB is also sensitive to the glacial isostatic adjustment correction for GRACE and to altimetry orbital altitude. Discrepancies in the SLB are largest for the Indian-South Pacific Ocean region. Taking the spread of plausible sea level trends, the SLB closes at the ocean-basin scale () but with large spread of magnitude, one third or more of the trend signal. Using the most up-to-date observation products, our ocean-region SLB does not close everywhere, and consideration of systematic uncertainties diminishes what information can be gained from the SLB about sea level processes, quantifying contributions, and validating Earth observation systems.

924. Energetic Submesoscale Dynamics in the Ocean Interior

Authors: Lia Siegelman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0253.1

URL: <https://doi.org/10.1175/jpo-d-19-0253.1>

Publication Date: 2020-01-09

Volume/Issue: Volume 50, Issue 3

References: 59

Cited by: 26 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Centre National d'Etudes Spatiales, Région Bretagne

License: N/A

Open Access: N/A

Abstract: Submesoscale ocean processes, characterized by order-1 Rossby and Richardson numbers, are currently thought to be mainly confined to the ocean surface mixed layer, whereas the ocean interior is commonly assumed to be in quasigeostrophic equilibrium. Here, a realistic numerical simulation in the Antarctic Circumpolar Current, with a $1/48$ degrees horizontal resolution and tidal forcing, is used to demonstrate that the ocean interior departs from the quasigeostrophic regime down to depths of 900 m, that is, well below the mixed layer. Results highlight that, contrary to the classical paradigm, the ocean interior is strongly ageostrophic, with a pronounced cyclone-anticyclone asymmetry and a dominance of frontogenesis over frontolysis. Numerous vortices and filaments, from the surface down to 900 m, are characterized by large Rossby and low Richardson numbers, strong lateral gradients of buoyancy, and vigorous ageostrophic frontogenesis. These deep submesoscale fronts are only weakly affected by internal gravity waves and drive intense upward vertical heat fluxes, consistent with recent observations in the Antarctic Circumpolar Current and the Gulf Stream. As such, deep submesoscale fronts are an efficient pathway for the transport of heat from the ocean interior to the surface, suggesting the presence of an intensified oceanic restratification at depth.

925. Estimation and prediction of the upper ocean circulation in the Bay of Bengal

Authors: Ganesh Gopalakrishnan; Aneesh C. Subramanian; Arthur J. Miller; Hyodae Seo; Debasis Sengupta

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2019.104721

URL: <https://doi.org/10.1016/j.dsr2.2019.104721>

Publication Date: 2020-01-07

Volume/Issue: Volume 172

References: 56

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research

License: N/A

Open Access: N/A

Abstract: N/A

926. Contributions to Arctic sea level from 2003 to 2015

Authors: Carsten A. Ludwigsen; Ole B. Andersen

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2019.12.027

URL: <https://doi.org/10.1016/j.asr.2019.12.027>

Publication Date: 2020-01-07

Volume/Issue: Volume 68, Issue 2

References: 31

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission, European Space Agency

License: N/A

Open Access: N/A

Abstract: N/A

927. Improved Global Nonlinear Surface Mass Variation Estimates From Geodetic Displacements and Reconciliation With GRACE Data

Authors: Xiaoping Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bruce J. Haines (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael B. Heflin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Felix W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jb018355

URL: <https://doi.org/10.1029/2019jb018355>

Publication Date: 2020-01-03

Volume/Issue: Volume 125, Issue 2

References: 37

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: geocenter motion, Global Navigation Satellite System, terrestrial reference frame, surface mass variations, GRACE

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Spatiotemporal coverage gaps and the nature of distant signature sources in Global Navigation Satellite System (GNSS) and The Gravity Recovery and Climate Experiment (GRACE) data require global inversions of data combinations for cross validation and complete and continuous monitoring of surface water mass variations. For spectral completeness, accurate data-based degree-1 surface mass variation coefficient or equivalent geocenter motion estimates are also desired but very difficult to obtain. To improve the reliability of such estimates, we carry out a new effort to combine different space geodetic technique data in a Kalman filter and time series approach to Terrestrial Reference Frame realization using an advanced formulation. With origin at the center of mass of the Earth system, the new parameterization now includes site displacements in the state vector explicitly for realistic assessment of their covariance matrices. More robust geocenter motion results are achieved when the displacement information is combined further with GRACE gravity data in a unified inversion. Previously, various systematic effects in GNSS data resulted in rather large discrepancies in global surface mass change estimates when compared with those from GRACE gravity data. Here, we show that significantly improved GNSS data after reprocessing and the refined Estimating the Circulation and Climate of the Ocean (ECCO) bottom pressure model reveal a global surface mass variation pattern that has been largely reconciled with that from GRACE data.

928. A coupled model analyses on the interaction between oceanic eddies and tropical cyclones over the Bay of Bengal

Authors: Thanka Swamy Anandh; Bijan Kumar Das; J. Kuttippurath; Arun Chakraborty

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-019-01330-x

URL: <https://doi.org/10.1007/s10236-019-01330-x>

Publication Date: 2020-01-02

Volume/Issue: Volume 70, Issue 3

References: 48

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Human Resource Development, Government of India.

License: N/A

Open Access: N/A

Abstract: N/A

929. An assessment of the East Australian Current as a renewable energy resource

Authors: Amandine Schaeffer; Matthew R. Archer; Quentin Baumard; Moninya Roughan; Colette Kerry

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2019.103285

URL: <https://doi.org/10.1016/j.jmarsys.2019.103285>

Publication Date: 2019-12-23

Volume/Issue: Volume 204

References: 43

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

930. Nonlinear Short-Term Upper Ocean Circulation Variability in the Tropical Western Pacific

Authors: Unknown Author; Bo Qiu; Shuiming Chen; Brian Powell; Patrick Colin; Daniel Rudnick; Martha Schönau

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2019.408

URL: <https://doi.org/10.5670/oceanog.2019.408>

Publication Date: 2019-12-19

Volume/Issue: Volume 32, Issue 4

References: 32

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

931. The End of an El Niño: A View from Palau

Authors: Unknown Author; Martha Schönau; Hemantha Wijesekera; William Teague; Patrick Colin; Ganesh Gopalakrishnan; Daniel Rudnick; Bruce Cornuelle; Zachariah Hallock; David Wang

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2019.409

URL: <https://doi.org/10.5670/oceanog.2019.409>

Publication Date: 2019-12-19

Volume/Issue: Volume 32, Issue 4

References: 52

Cited by: 15 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

932. Palau's Effects on Regional-Scale Ocean Circulation

Authors: Unknown Author; Ganesh Gopalakrishnan; Bruce Cornuelle

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2019.418

URL: <https://doi.org/10.5670/oceanog.2019.418>

Publication Date: 2019-12-19

Volume/Issue: Volume 32, Issue 4

References: 46

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

933. Effects of the Sea Surface Roughness and Sea Spray-Induced Flux Parameterization on the Simulations of a Tropical Cyclone

Authors: Kumar Ravi Prakash (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Vimlesh Pant (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Tanuja Nigam (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jd029760

URL: <https://doi.org/10.1029/2018jd029760>

Publication Date: 2019-12-19

Volume/Issue: Volume 124, Issue 24

References: 85

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A tropical cyclone (TC) interacts with the ocean through the interchange of surface fluxes of heat and momentum at the air-sea interface. The sustainability and intensification of TCs are maintained by the enhancement in the latent heat flux in the high wind speed regime. An accurate calculation of heat fluxes is vital to understand the dynamics of the TC. The sea surface roughness is a crucial parameter that modulates the air-sea fluxes. A coupled ocean-atmosphere-wave model is used to simulate TC Vardah over the Bay of Bengal during 10-15 December 2016. The study focuses on the influence of sea surface roughness on the heat fluxes, track, and intensity of TC Vardah. Four numerical experiments are performed with different parameterization schemes for surface roughness on the basis of (i) frictional velocity (Charnock), (ii) wave age, (iii) wave steepness, and (iv) a combination of wave steepness and wave age. A change in ocean surface roughness results in the modification of the cyclone track and heat fluxes. However, the effect of sea spray on the simulation of wind intensity of TC Vardah is only marginal. Out of the four experiments, the wave age-based roughness parameterization was found to improve cyclone track, wind, and, hence, the generation of wind waves on the sea surface. In the case study of TC Vardah, the dynamic sea spray reduces surface wind speed up to 1.5 m/s which is within the uncertainty range of the model. An intercomparison of model experiments reveals the impact of surface heat fluxes on the TC Vardha characteristics.

934. Tropical Western Pacific Thermal Structure and its Relationship to Ocean Surface Variables: A Numerical State Estimate and Forereef Temperature Records

Authors: Unknown Author; Travis Schramek; Bruce Cornuelle; Ganesh Gopalakrishnan; Patrick Colin; Sonia Rowley; Mark Merrifield; Eric Terrill

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2019.421

URL: <https://doi.org/10.5670/oceanog.2019.421>

Publication Date: 2019-12-19

Volume/Issue: Volume 32, Issue 4

References: 33

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

935. Interannual Variability of the Global Meridional Overturning Circulation Dominated by Pacific Variability

Authors: Neil F. Tandon (Department of Earth and Space Science and Engineering, York University, Toronto, Ontario, Canada); Oleg A. Saenko (Canadian Centre for Climate Modelling and Analysis); Mark A. Cane (Columbia University); Paul J. Kushner (University of Toronto)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0129.1

URL: <https://doi.org/10.1175/jpo-d-19-0129.1>

Publication Date: 2019-12-18

Volume/Issue: Volume 50, Issue 3

References: 39

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The most prominent feature of the time-mean global meridional overturning circulation (MOC) is the Atlantic MOC (AMOC). However, interannual variability of the global MOC is shown here to be dominated by Pacific MOC (PMOC) variability over the full depth of the ocean at most latitudes. This dominance of interannual PMOC variability is robust across modern climate models and an observational state estimate. PMOC interannual variability has large-scale organization, its most prominent feature being a cross-equatorial cell spanning the tropics. Idealized experiments show that this variability is almost entirely wind driven. Interannual anomalies of zonal mean zonal wind stress produce zonally integrated Ekman transport anomalies that are larger in the Pacific Ocean than in the Atlantic Ocean, simply because the Pacific is wider than the Atlantic at most latitudes. This contrast in Ekman transport variability implies greater variability in the near-surface branch of the PMOC when compared with the near-surface branch of the AMOC. These near-surface variations in turn drive compensating flow anomalies below the Ekman layer. Because the baroclinic adjustment time is longer than a year at most latitudes, these compensating flow anomalies have baroclinic structure spanning the full depth of the ocean. Additional analysis reveals that interannual PMOC variations are the dominant contribution to interannual variations of the global meridional heat transport. There is also evidence of interaction between interannual PMOC variability and El Niño-Southern Oscillation.

936. Arctic climate change and oil spill risk analysis

Authors: William B. Samuels; David E. Amstutz; Heather A. Crowley

Journal: Frontiers of Earth Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11707-011-0198-6

URL: <https://doi.org/10.1007/s11707-011-0198-6>

Publication Date: 2019-12-15

Volume/Issue: Volume 5, Issue 4

References: 17

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

937. Scale Transition From Geostrophic Motions to Internal Waves in the Northern South China Sea

Authors: Haijin Cao (Hohai University); Zhiyou Jing (Chinese Academy of Sciences); Baylor Fox-Kemper (Brown University); Tong Yan (Chinese Academy of Sciences); Yiquan Qi (Hohai University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015575

URL: <https://doi.org/10.1029/2019jc015575>

Publication Date: 2019-12-11

Volume/Issue: Volume 124, Issue 12

References: 68

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: Balanced to unbalanced transition, Wave-vortex decomposition, Spectral analysis, Northern South China Sea

Funding: National Basic Research Program of China, National Basic Research Program of China, National Natural Science Foundation of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: The eddy-abundant circulation in the northern South China Sea (NSCS) tends to be dynamically complex due to monsoon forcing, Kuroshio intrusion, and emitted internal waves from the Luzon Strait. This study uses 13-year shipboard acoustic Doppler current profiler measurements (2004-2016), orbital altimeter data, and high-resolution model output to perform wave-vortex decompositions and investigate the scale of transition from dominantly geostrophic flows to internal wave motions in the northern South China Sea. The upper ocean kinetic energy spectra transition on scales exceeds 200 km. This large scale of transition is attributed to the energetic low-mode internal waves (e.g., internal tides and inertia-gravity waves). However, inconsistencies in the decomposition reveal that the assumptions of homogeneity and isotropy required for the 1-D decomposition (Bühler et al., 2014) are sufficiently violated at smaller scales to affect the subdominant member of the decomposition on scales below 100 km. A method for direct quantification of the consequences by degree of violation using bootstrapping of the 2-D model data is developed and illustrated. Observed and modeled sea surface height spectra flatten at scales smaller than 125 km, which is found in the model to be due to the coherent, semidiurnal internal waves. The large scale of transition between geostrophic and wave motions in the South China Sea is an irreducible uncertainty for altimeter velocities (e.g., the Surface Water and Ocean Topography mission).

938. Mass balance of the Greenland Ice Sheet from 1992 to 2018

Authors: Unknown Author

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-019-1855-2

URL: <https://doi.org/10.1038/s41586-019-1855-2>

Publication Date: 2019-12-10

Volume/Issue: Volume 579, Issue 7798

References: 119

Cited by: 469 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

939. Temporal and Spatial Scales of Correlation in Marine Phytoplankton Communities

Authors: A. M. Kuhn (Scripps Institution of Oceanography / University of California San Diego); S. Dutkiewicz (Massachusetts Institute of Technology); O. Jahn (Massachusetts Institute of Technology); S. Clayton (Old Dominion University); T. A. Ryneerson (University of Rhode Island); M. R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); A. D. Barton (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015331

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Publication Date: 2019-12-10

Volume/Issue: Volume 124, Issue 12

References: 61

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: phytoplankton traits, phytoplankton dispersal, global plankton model, phytoplankton, phytoplankton correlation scales (and 1 more)

Funding: Simons Foundation, National Aeronautics and Space Administration, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Ocean circulation shapes marine phytoplankton communities by setting environmental conditions and dispersing organisms. In addition, processes acting on the water column (e.g., heat fluxes and mixing) affect the community structure by modulating environmental variables that determine in situ growth and loss rates. Understanding the scales over which phytoplankton communities vary in time and space is key to elucidate the relative contributions of local processes and ocean circulation on phytoplankton distributions. Using a global ocean ecosystem model, we quantify temporal and spatial correlation scales for phytoplankton phenotypes with diverse functional traits and cell sizes. Through this analysis, we address these questions: (1) Over what timescales do perturbations in phytoplankton populations persist? and (2) over what distances are variations in phytoplankton populations synchronous? We find that correlation timescales are short in regions of strong currents, such as the Gulf Stream and Antarctic Circumpolar Current. Conversely, in the subtropical gyres, phytoplankton population anomalies persist for relatively long periods. Spatial correlation length scales are elongated near ocean fronts and narrow boundary currents, reflecting flow paths and frontal patterns. In contrast, we find nearly isotropic spatial correlation fields where current speeds are small, or where mixing acts roughly equally in all directions. Phytoplankton timescales and length scales also vary coherently with phytoplankton body size. In addition to aiding understanding of phytoplankton population dynamics, our results provide global insights to guide the design of biological ocean observing networks and to better interpret data collected at long-term monitoring stations.

940. Evaluation of the China Ocean Reanalysis (CORA) in the South China Sea

Authors: Maoting Fan; Huizan Wang; Weimin Zhang; Guijun Han; Pinqiang Wang

Journal: Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-019-9146-1

URL: <https://doi.org/10.1007/s00343-019-9146-1>

Publication Date: 2019-12-05

Volume/Issue: Volume 38, Issue 6

References: 22

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

941. Interior Water-Mass Variability in the Southern Hemisphere Oceans during the Last Decade

Authors: Esther Portela (Institut de Recherche pour le Développement (IRD)); Nicolas Kolodziejczyk (Institut de Recherche pour le Développement (IRD)); Christophe Maes (Institut de Recherche pour le Développement (IRD)); Virginie Thierry (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0128.1

URL: <https://doi.org/10.1175/jpo-d-19-0128.1>

Publication Date: 2019-11-25

Volume/Issue: Volume 50, Issue 2

References: 69

Cited by: 23 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: CNES/INSU/Ifremer

License: N/A

Open Access: N/A

Abstract: Using an Argo dataset and the ECCOv4 reanalysis, a volume budget was performed to address the main mechanisms driving the volume change of the interior water masses in the Southern Hemisphere oceans between 2006 and 2015. The subduction rates and the isopycnal and diapycnal water-mass transformation were estimated in a density-spiciness (σ - τ) framework. Spiciness, defined as thermohaline variations along isopycnals, was added to the potential density coordinates to discriminate between water masses spreading on isopycnal layers. The main positive volume trends were found to be associated with the Subantarctic Mode Waters (SAMW) in the South Pacific and South Indian Ocean basins, revealing a lightening of the upper waters in the Southern Hemisphere. The SAMW exhibits a two-layer density structure in which subduction and diapycnal transformation from the lower to the upper layers accounted for most of the upper-layer volume gain and lower-layer volume loss, respectively. The Antarctic Intermediate Waters, defined here between the 27.2 and 27.5 kg m⁻³ isopycnals, showed the strongest negative volume trends. This volume loss can be explained by their negative isopycnal transformation southward of the Antarctic Circumpolar Current into the fresher and colder Antarctic Winter Waters (AAWW) and northward into spicier tropical/subtropical Intermediate Waters. The AAWW is destroyed by obduction back into the mixed layer so that its net volume change remains nearly zero. The proposed mechanisms to explain the transformation within the Intermediate Waters are discussed in the context of Southern Ocean dynamics. The σ - τ decomposition provided new insight on the spatial and temporal water-mass variability and driving mechanisms over the last decade.

942. Pathways of ocean heat towards Pine Island and Thwaites grounding lines

Authors: Yoshihiro Nakayama; Georgy Manucharyan; Hong Zhang; Pierre Dutrieux; Hector S. Torres; Patrice Klein; Helene Seroussi; Michael Schodlok; Eric Rignot; Dimitris Menemenlis

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-019-53190-6

URL: <https://doi.org/10.1038/s41598-019-53190-6>

Publication Date: 2019-11-22

Volume/Issue: Volume 9, Issue 1

References: 53

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: In the Amundsen Sea, modified Circumpolar Deep Water (mCDW) intrudes into ice shelf cavities, causing high ice shelf melting near the ice sheet grounding lines, accelerating ice flow, and controlling the pace of future Antarctic contributions to global sea level. The pathways of mCDW towards grounding lines are crucial as they directly control the heat reaching the ice. A realistic representation of mCDW circulation, however, remains challenging due to the sparsity of in-situ observations and the difficulty of ocean models to reproduce the available observations. In this study, we use an unprecedentedly high-resolution (200 m horizontal and 10 m vertical grid spacing) ocean model that resolves shelf-sea and sub-ice-shelf environments in qualitative agreement with existing observations during austral summer conditions. We demonstrate that the waters reaching the Pine Island and Thwaites grounding lines follow specific, topographically-constrained routes, all passing through a relatively small area located around 104 degreesW and 74.3 degreesS. The temporal and spatial variabilities of ice shelf melt rates are dominantly controlled by the sub-ice shelf ocean current. Our findings highlight the importance of accurate and high-resolution ocean bathymetry and subglacial topography for determining mCDW pathways and ice shelf melt rates.

943. AWESOME OCIM: A simple, flexible, and powerful tool for modeling elemental cycling in the oceans

Authors: Seth G. John; Hengdi Liang; Tom Weber; Tim DeVries; Francois Primeau; Keith Moore; Mark Holzer; Natalie Mahowald; Wilford Gardner; Alexey Mishonov; Mary Jo Richardson; Yannice Faugere; Guillaume Taburet

Journal: Chemical Geology

Publisher: Elsevier BV

DOI: 10.1016/j.chemgeo.2019.119403

URL: <https://doi.org/10.1016/j.chemgeo.2019.119403>

Publication Date: 2019-11-20

Volume/Issue: Volume 533

References: 60

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Simons Foundation, National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

944. Evaluation of GRACE mascon solutions using in-situ geodetic data: The case of hydrologic-induced crust displacement in the Yangtze River Basin

Authors: Linsong Wang; Chao Chen; Xian Ma; Zhengyan Fu; Yuhao Zheng; Zhenran Peng

Journal: Science of The Total Environment

Publisher: Elsevier BV

DOI: 10.1016/j.scitotenv.2019.135606

URL: <https://doi.org/10.1016/j.scitotenv.2019.135606>

Publication Date: 2019-11-19

Volume/Issue: Volume 707

References: 69

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Fundamental Research Funds for the Central Universities, China University of Geosciences, Wuhan

License: N/A

Open Access: N/A

Abstract: N/A

945. Mapping near surface global marine ecosystems through cluster analysis of environmental data

Authors: Qianshuo Zhao (Institute of Marine Science University of Auckland Auckland New Zealand); Zeenatul Basher (Institute of Marine Science University of Auckland Auckland New Zealand); Mark J. Costello (Institute of Marine Science University of Auckland Auckland New Zealand)

Journal: Ecological Research

Publisher: Wiley

DOI: 10.1111/1440-1703.12060

URL: <https://doi.org/10.1111/1440-1703.12060>

Publication Date: 2019-11-19

Volume/Issue: Volume 35, Issue 2

References: 70

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Almost all classifications of the world ocean are based on expert opinion or ad hoc management areas. A quantitative analysis of environmental variables may provide a more objective basis for mapping and classifying the oceans to support data management, reporting, and conservation efforts. Here, we used long-term averages of 20 ocean variables to classify the ocean surface waters using PCA and k-means clustering. We identified seven distinct areas that fit the definition of "ecosystems," that is, enduring regions demarcated by environmental characteristics. Of all the variables, temperature had the greatest importance and correlated with many other variables. However, some variables had uniquely significant effects on the classification, namely slope, surface current, pH, and wave height. Thus, while the present classification is robust for available data, future analyses with variables not presently available may improve it. How the ecosystems correlate with species richness, endemism, or abundance will inform on the factors that most influence species' abundance, and thus support global modeling of the effects of climate change, for example, with regard to biological carbon fluxes.

946. Global Trends in Marine Plankton Diversity across Kingdoms of Life

Authors: Federico M. Ibarbalz; Nicolas Henry; Manoela C. Brandão; Séverine Martini; Greta Busseni; Hannah Byrne; Luis Pedro Coelho; Hisashi Endo; Josep M. Gasol; Ann C. Gregory; Frédéric Mahé; Janaina Rigonato; Marta Royo-Llonch; Guillem Salazar; Isabel Sanz-Sáez; Eleonora Scalco; Dodji Soviadan; Ahmed A. Zayed; Adriana Zingone; Karine Labadie; Joannie Ferland; Claudie Marec; Stefanie Kandels; Marc Picheral; Céline Dimier; Julie Poulain; Sergey Pisarev; Margaux Carmichael; Stéphane Pesant; Marcel Babin; Emmanuel Boss; Daniele Iudicone; Olivier Jaillon; Silvia G. Acinas; Hiroyuki Ogata; Eric Pelletier; Lars Stemmann; Matthew B. Sullivan; Shinichi Sunagawa; Laurent Bopp; Colombar de Vargas; Lee Karp-Boss; Patrick Wincker; Fabien Lombard; Chris Bowler; Lucie Zinger; Silvia G. Acinas; Marcel Babin; Peer Bork; Emmanuel Boss; Chris Bowler; Guy Cochrane; Colombar de Vargas; Mick Follows; Gabriel Gorsky; Nigel Grimsley; Lionel Guidi; Pascal Hingamp; Daniele Iudicone; Olivier Jaillon; Stefanie Kandels; Lee Karp-Boss; Eric Karsenti; Fabrice Not; Hiroyuki Ogata; Stéphane Pesant; Nicole Poulton; Jeroen Raes; Christian Sardet; Sabrina Speich; Lars Stemmann; Matthew B. Sullivan; Shinichi Sunagawa; Patrick Wincker

Journal: Cell

Publisher: Elsevier BV

DOI: 10.1016/j.cell.2019.10.008

URL: <https://doi.org/10.1016/j.cell.2019.10.008>

Publication Date: 2019-11-14

Volume/Issue: Volume 179, Issue 5

References: 137

Cited by: 340 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: CNRS, Groupement de Recherche GDR3280, Research Federation for the Study of Global Ocean Systems Ecology and Evolution (and 28 more)

License: N/A

Open Access: N/A

Abstract: N/A

947. The Vertical Structure of Open-Ocean Submesoscale Variability during a Full Seasonal Cycle

Authors: Zachary K. Erickson (California Institute of Technology); Andrew F. Thompson (California Institute of Technology); Jörn Callies (California Institute of Technology); Xiaolong Yu (Institut de Recherche pour le Développement (IRD)); Alberto Naveira Garabato (University of Southampton); Patrice Klein (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0030.1

URL: <https://doi.org/10.1175/jpo-d-19-0030.1>

Publication Date: 2019-11-14

Volume/Issue: Volume 50, Issue 1

References: 63

Cited by: 18 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Division of Ocean Sciences, David and Lucile Packard Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: Submesoscale dynamics are typically intensified at boundaries and assumed to weaken below the mixed layer in the open ocean. Here, we assess both the seasonality and the vertical distribution of submesoscale motions in an open-ocean region of the northeast Atlantic. Second-order structure functions, or variance in properties separated by distance, are calculated from submesoscale-resolving ocean glider and mooring observations, as well as a 1/48 degrees numerical ocean model. This dataset combines a temporal coverage that extends through a full seasonal cycle, a horizontal resolution that captures spatial scales as small as 1 km, and vertical sampling that provides near-continuous coverage over the upper 1000 m. While kinetic and potential energies undergo a seasonal cycle, being largest during the winter, structure function slopes, influenced by dynamical characteristics, do not exhibit a strong seasonality. Furthermore, structure function slopes show weak vertical variations; there is not a strong change in properties across the base of the mixed layer. Additionally, we compare the observations to output from a high-resolution numerical model. The model does not represent variability associated with superinertial motions and does not capture an observed reduction in submesoscale kinetic energy that occurs throughout the water column in spring. Overall, these results suggest that the transfer of mixed layer submesoscale variability down to depths below the traditionally defined mixed layer is important throughout the weakly stratified subpolar mode waters.

948. A New Multigrid 3D-VAR Optimization Method for Bottom Friction Using HF Radar Current Observation

Authors: Xiaoshuang Zhang; Xinrong Wu; Wei Li; Bin Wang; Xiande Zhu; Kexiu Liu; Caixia Shao; Chunjian Sun; Guofang Chao; Tiecheng Zhang

Journal: Journal of Ocean University of China

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11802-019-4020-5

URL: <https://doi.org/10.1007/s11802-019-4020-5>

Publication Date: 2019-11-12

Volume/Issue: Volume 18, Issue 6

References: 25

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

949. Heat Distribution in the Southeast Pacific Is Only Weakly Sensitive to High-Latitude Heat Flux and Wind Stress

Authors: Daniel C. Jones (British Antarctic Survey Natural Environment Research Council Cambridge UK); Emma Boland (British Antarctic Survey Natural Environment Research Council Cambridge UK); Andrew J.S. Meijers (British Antarctic Survey Natural Environment Research Council Cambridge UK); Gael Forget (Massachusetts Institute of Technology); Simon A. Josey (National Oceanography Centre); Jean-Baptiste Sallee (Sorbonne Universités, UPMC Université, LOCEAN-IPSL Paris France); Emily Shuckburgh (British Antarctic Survey Natural Environment Research Council Cambridge UK, University of Cambridge)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015460

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Publication Date: 2019-11-11

Volume/Issue: Volume 124, Issue 12

References: 45

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Natural Environment Research Council, Simons Foundation

License: N/A

Open Access: N/A

Abstract: The Southern Ocean features regionally varying ventilation pathways that transport heat and carbon from the surface ocean to the interior thermocline on timescales of decades to centuries, but the factors that control the distribution of heat along these pathways are not well understood. In this study, we use a global ocean state estimate (ECCOv4) to (1) define the recently ventilated interior Pacific (RVP) using numerical passive tracer experiments over a 10-year period and (2) use an adjoint approach to calculate the sensitivities of the RVP heat content (RVPh) to changes in net heat flux and wind stress. We find that RVPh is most sensitive to local heat flux and wind stress anomalies north of the sea surface height contours that delineate the Antarctic Circumpolar Current, with especially high sensitivities over the South Pacific Gyre. Surprisingly, RVPh is not especially sensitive to changes at higher latitudes. We perform a set of step response experiments over the South Pacific Gyre, the subduction region, and the high-latitude Southern Ocean. In consistency with the adjoint sensitivity fields, RVPh is most sensitive to wind stress curl over the subtropical gyre, which alter isopycnal heave, and it is only weakly sensitive to changes at higher latitudes. Our results suggest that despite the localized nature of mode water subduction hot spots, changes in basin-scale pressure gradients are an important controlling factor on RVPh. Because basin-scale wind stress is expected to change in the coming decades to centuries, our results may have implications for climate, via the atmosphere/ocean partitioning of heat.

950. Reducing filter effects in GRACE-derived polar motion excitations

Authors: Franziska Göttl; Michael Murböck; Michael Schmidt; Florian Seitz

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-019-1101-z

URL: <https://doi.org/10.1186/s40623-019-1101-z>

Publication Date: 2019-11-08

Volume/Issue: Volume 71, Issue 1

References: 52

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft

License: CC BY 4.0

Open Access: N/A

Abstract: Polar motion is caused by mass redistribution and motion within the Earth system. The GRACE (Gravity Recovery and Climate Experiment) satellite mission observed variations of the Earth's gravity field which are caused by mass redistribution. Therefore GRACE time variable gravity field models are a valuable source to estimate individual geophysical mass-related excitations of polar motion. Since GRACE observations contain erroneous meridional stripes, filtering is essential to retrieve meaningful information about mass redistribution within the Earth system. However filtering reduces not only the noise but also smoothes the signal and induces leakage of neighboring subsystems into each other. We present a novel approach to reduce these filter effects in GRACE-derived equivalent water heights and polar motion excitation functions which is based on once- and twice-filtered gravity field solutions. The advantages of this method are that it is independent from geophysical model information, works on global grid point scale and can therefore be used for mass variation estimations of several subsystems of the Earth. A closed-loop simulation reveals that due to application of the new filter effect reduction approach the uncertainties in GRACE-derived polar motion excitations can be decreased from 12-48% to 5-29%, especially for the oceanic excitations. Comparisons of real GRACE data with model-based oceanic excitations show that the agreement can be improved by up to 15 percentage points.

951. The Mean State and Variability of the North Atlantic Circulation: A Perspective From Ocean Reanalyses

Authors: L. C. Jackson (Met Office Hadley Centre, Met Office); C. Dubois (Mercator Ocean International Ramonville-Saint-Agne France, Météo France Toulouse France); G. Forget (Massachusetts Institute of Technology); K. Haines (University of Reading); M. Harrison (Geophysical Fluid Dynamics Laboratory Princeton NJ USA); D. Iovino (Ocean Modelling and Data Assimilation Division, Centro Euro-Mediterraneo sui Cambiamenti Climatici Bologna Italy); A. Köhl (University of Hamburg); D. Mignac (University of Reading); S. Masina (Ocean Modelling and Data Assimilation Division, Centro Euro-Mediterraneo sui Cambiamenti Climatici Bologna Italy); K. A. Peterson (Met Office Hadley Centre, Environment and Climate Change Canada, Met Office); C. G. Piecuch (Woods Hole Oceanographic Institution); C. D. Roberts (European Centre for Medium-Range Weather Forecasts (ECMWF)); J. Robson (University of Reading); A. Storto (NATO STO Centre for Maritime Research and Experimentation La Spezia Italy, Ocean Modelling and Data Assimilation Division, Centro Euro-Mediterraneo sui Cambiamenti Climatici Bologna Italy); T. Toyoda (Meteorological Research Institute); M. Valdivieso (University of Reading); C. Wilson (National Oceanography Centre); Y. Wang (Nansen Environmental and Remote Sensing Centre/Bjerknes Center for Climate Research Bergen Norway); H. Zuo (European Centre for Medium-Range Weather Forecasts (ECMWF))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015210

URL: <https://doi.org/10.1029/2019jc015210>

Publication Date: 2019-11-06

Volume/Issue: Volume 124, Issue 12

References: 108

Cited by: 56 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission, Ministério da Educação, NordForsk (and 2 more)

License: N/A

Open Access: N/A

Abstract: The observational network around the North Atlantic has improved significantly over the last few decades with subsurface profiling floats and satellite observations and the recent efforts to monitor the Atlantic Meridional Overturning Circulation (AMOC). These have shown decadal time scale changes across the North Atlantic including in heat content, heat transport, and the circulation. However, there are still significant gaps in the observational coverage. Ocean reanalyses integrate the observations with a dynamically consistent ocean model and can be used to understand the observed changes. However, the ability of the reanalyses to represent the dynamics must also be assessed. We use an ensemble of global ocean reanalyses to examine the time mean state and interannual-decadal variability of the North Atlantic ocean since 1993. We assess how well the reanalyses are able to capture processes and whether any understanding can be gained. In particular, we examine aspects of the circulation including convection, AMOC and gyre strengths, and transports. We find that reanalyses show some consistency, in particular showing a weakening of the subpolar gyre and AMOC at 50

degreesN from the mid-1990s until at least 2009 (related to decadal variability in previous studies), a strengthening and then weakening of the AMOC at 26.5 degreesN since 2000, and impacts of circulation changes on transports. These results agree with model studies and the AMOC observations at 26.5 degreesN since 2005. We also see less spread across the ensemble in AMOC strength and mixed layer depth, suggesting improvements as the observational coverage has improved.

952. Separation of Sources of Seasonal Uplift in China Using Independent Component Analysis of GNSS Time Series

Authors: Jun Yan (Key Laboratory of Geographic Information Science, Ministry of Education East China Normal University Shanghai China, University of California Berkeley, Engineering Center of SHMEC for Space Information and GNSS East China Normal University Shanghai China); Danan Dong (Key Laboratory of Geographic Information Science, Ministry of Education East China Normal University Shanghai China, Engineering Center of SHMEC for Space Information and GNSS East China Normal University Shanghai China); Roland Bürgmann (University of California Berkeley); Kathryn Materna (University of California Berkeley); Weijie Tan (Chinese Academy of Sciences); Yu Peng (Key Laboratory of Geographic Information Science, Ministry of Education East China Normal University Shanghai China, Engineering Center of SHMEC for Space Information and GNSS East China Normal University Shanghai China); Junping Chen (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jb018139

URL: <https://doi.org/10.1029/2019jb018139>

Publication Date: 2019-11-01

Volume/Issue: Volume 124, Issue 11

References: 61

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, East China Normal University, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: With the improvement of Global Navigation Satellite System (GNSS) observation accuracy and the establishment of large continuously operating networks, long GNSS time series are now widely used to understand a range of Earth deformation processes. The continuously operating stations of the Crustal Movement Observation Network of China capture deformation signals due to time-dependent tectonic, nontectonic mass loading, and potentially unknown geophysical processes. In order to separate and recover these underlying sources accurately and effectively, we apply the independent component analysis (ICA) to decompose the observed time series of vertical displacements. Then, we compare these signals with those predicted from independently developed geophysical process models of atmospheric, nontidal ocean, snow, soil moisture mass loading, and the Land Surface Discharge Model, as well as with Gravity Recovery and Climate Experiment observations. For comparison, we also perform the principal component analysis decomposition of time series and find that the ICA achieves a more consistent representation of multiple geophysical contributors to annual vertical GNSS displacements. ICA can decompose the long-term trend and different seasonal and multiannual signals that closely correspond to the independently derived mass loading models. We find that independent contributions from atmospheric, soil moisture, and snow mass loading can be resolved from the GNSS data. Discrepancies are likely due to the correlated nature of some of the loading processes and

unmodeled contributions from groundwater and surface water changes in South Central China and the Ganges Basin.

953. What Caused Recent Shifts in Tropical Pacific Decadal Sea-Level Trends?

Authors: Christopher G. Piecuch (Woods Hole Oceanographic Institution); Philip R. Thompson (Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Mark A. Merrifield (Scripps Institution of Oceanography); Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015339

URL: <https://doi.org/10.1029/2019jc015339>

Publication Date: 2019-10-31

Volume/Issue: Volume 124, Issue 11

References: 61

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: state estimation, sea-level change, tropical Pacific, decadal variability, sea-level variability

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Satellite altimetry reveals substantial decadal variability in sea level ζ across the tropical Pacific during 1993-2015. An ocean state estimate that faithfully reproduces the observations is used to elucidate the origin of these low-frequency tropical Pacific ζ variations. Analysis of the hydrostatic equation reveals that recent decadal ζ changes in the tropical Pacific are mainly thermosteric in nature, related to changes in upper-ocean heat content. A forcing experiment performed with the numerical model suggests that anomalous wind stress was an important driver of the relevant heat storage and thermosteric variation. Closed budget diagnostics further clarify that the wind-stress-related thermosteric ζ variation resulted from the joint actions of large-scale ocean advection and local surface heat flux, such that advection controlled the budget over shorter, intraseasonal to interannual time scales, and local surface heat flux became increasingly influential at longer decadal periods. In particular, local surface heat flux was important in contributing to a recent reversal of decadal ζ trends in the tropical Pacific. Contributions from local surface heat flux partly reflect damping latent heat flux tied to wind-stress-driven sea-surface-temperature variations.

954. Intergyre Salt Transport in the Climate Warming Response

Authors: Samuel J. Levang (Woods Hole Oceanographic Institution); Raymond W. Schmitt (Woods Hole Oceanographic Institution)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0166.1

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References: 37

Cited by: 2 publications

Language: N/A

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Keywords: N/A

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License: N/A

Open Access: N/A

Abstract: Regional connectivity is important to the global climate salinity response, particularly because salinity anomalies do not have a damping feedback with atmospheric freshwater fluxes and may therefore be advected over long distances by ocean circulation, resulting in nonlocal influences. Climate model intercomparison experiments such as CMIP5 exhibit large uncertainty in some aspects of the salinity response, hypothesized here to be a result of ocean dynamics. We use two types of Lagrangian particle tracking experiments to investigate pathways of exchange for salinity anomalies. The first uses forward trajectories to estimate average transport time scales between water cycle regimes. The second uses reverse trajectories and a freshwater accumulation method to quantitatively identify remote influences in the salinity response. Additionally, we compare velocity fields with both resolved and parameterized eddies to understand the impact of eddy stirring on intergyre exchange. These experiments show that surface anomalies are readily exchanged within the ocean gyres by the mean circulation, but intergyre exchange is slower and largely eddy driven. These dynamics are used to analyze the North Atlantic salinity response to climate warming and water cycle intensification, where the system is broadly forced with fresh surface anomalies in the subpolar gyre and salty surface anomalies in the subtropical gyres. Under these competing forcings, strong intergyre eddy fluxes carry anomalously salty subtropical water into the subpolar gyre which balances out much of the local freshwater input.

955. Reconstructing Upper-Ocean Vertical Velocity Field from Sea Surface Height in the Presence of Unbalanced Motion

Authors: Bo Qiu (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Shuiming Chen (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Patrice Klein (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hector Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0172.1

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Publication Date: 2019-10-22

Volume/Issue: Volume 50, Issue 1

References: 58

Cited by: 41 publications

Language: N/A

Subjects: N/A

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Open Access: N/A

Abstract: Reconstructability of upper-ocean vertical velocity w and vorticity ζ fields from high-resolution sea surface height (SSH) data is explored using the global $1/48$ degrees horizontal-resolution MITgcm output in the context of the forthcoming Surface Water and Ocean Topography (SWOT) mission. By decomposing w with an omega equation of the primitive equation system and by taking into account the measurement design of the SWOT mission, this study seeks to reconstruct the subinertial, balanced w and ζ signals. By adopting the effective surface quasigeostrophic (eSQG) framework and applying to the Kuroshio Extension region of the North Pacific, we find that the target and reconstructed fields have a spatial correlation of ~ 0.7 below the mixed layer for w and 0.7 - 0.9 throughout the 1000 -m upper ocean for ζ in the error-free scenario. By taking the SWOT sampling and measurement errors into account, the spatial correlation is found to decrease to 0.4 - 0.6 below the mixed layer for w and 0.6 - 0.7 for ζ , respectively. For both w and ζ reconstruction, the degradation due to the SWOT errors is more significant in the surface layer and for smaller-scale signals. The impact of errors lessens with the increasing depth and lengthening horizontal scales.

956. The Indian Ocean Deep Meridional Overturning Circulation in Three Ocean Reanalysis Products

Authors: T. Jayasankar (Indian Institute of Technology Bombay); R. Murtugudde (University of Maryland, Indian Institute of Technology Bombay); T.I. Eldho (Indian Institute of Technology Bombay)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl084244

URL: <https://doi.org/10.1029/2019gl084244>

Publication Date: 2019-10-18

Volume/Issue: Volume 46, Issue 21

References: 46

Cited by: 9 publications

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Keywords: N/A

Funding: Indian Institute of Technology Bombay, Indian Institute of Technology Bombay

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Open Access: N/A

Abstract: The time mean Indian Ocean (IO) deep meridional overturning circulation (MOC) is compared across three ocean reanalysis products (ORAS4, GECCO2, and GFDL). The MOC stream functions obtained by vertically integrating the mass flux across a latitude-depth section in three products are found to be significantly different from each other. Detailed analysis suggests that ORAS4 delivers the best depiction of IO MOC. The inferred IO deep MOC consists of two deep and strong counterclockwise cells located south of 30 degreesS and around 10 degreesS, respectively. The geostrophic component along with the barotropic or external mode dominates the former, and a combination of Ekman and geostrophic components dominates the latter. GECCO2 depicts a steady decline in the northward meridional transport in the bottom layer and a consequent reduction in the MOC strength. The tropical thermocline in GECCO2 responds to this MOC variability leading to rapid and monotonic warming of the tropical IO.

957. What Drives Upper-Ocean Temperature Variability in Coupled Climate Models and Observations?

Authors: R. Justin Small (National Center for Atmospheric Research); Frank O. Bryan (National Center for Atmospheric Research); Stuart P. Bishop (North Carolina State University); Sarah Larson (North Carolina State University); Robert A. Tomas (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0295.1

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Publication Date: 2019-10-10

Volume/Issue: Volume 33, Issue 2

References: 59

Cited by: 41 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, Biological and Environmental Research

License: N/A

Open Access: N/A

Abstract: A key question in climate modeling is to what extent sea surface temperature and upper-ocean heat content are driven passively by air-sea heat fluxes, as opposed to forcing by ocean dynamics. This paper investigates the question using a climate model at different resolutions, and observations, for monthly variability. At the grid scale in a high-resolution climate model with resolved mesoscale ocean eddies, ocean dynamics (i.e., ocean heat flux convergence) dominates upper 50 m heat content variability over most of the globe. For deeper depths of integration to 400 m, the heat content variability at the grid scale is almost totally controlled by ocean heat flux convergence. However, a strong dependence on spatial scale is found--for the upper 50 m of ocean, after smoothing the data to around 7 degrees, air-sea heat fluxes, augmented by Ekman heat transports, dominate. For deeper depths of integration to 400 m, the transition scale becomes larger and is above 10 degrees in western boundary currents. Comparison of climate model results with observations show that the small-scale influence of ocean intrinsic variability is well captured by the high-resolution model but is missing from a comparable model with parameterized ocean-eddy effects. In the deep tropics, ocean dynamics dominates in all cases and all scales. In the subtropical gyres at large scales, air-sea heat fluxes play the biggest role. In the midlatitudes, at large scales >10 degrees, atmosphere-driven air-sea heat fluxes and Ekman heat transport variability are the dominant processes except in the western boundary currents for the 400 m heat content.

958. On the wave-current interaction during the passage of a tropical cyclone in the Bay of Bengal

Authors: Kumar Ravi Prakash; Vimlesh Pant

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2019.104658

URL: <https://doi.org/10.1016/j.dsr2.2019.104658>

Publication Date: 2019-10-07

Volume/Issue: Volume 172

References: 91

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Space Applications Centre, ISRO, Ocean Mixing and Monsoon (OMM) programme of the National Monsoon Mission, Govt. of India, UGC-CSIR (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

959. Hydrometeorological and gravity signals at the Argentine-German Geodetic Observatory (AGGO) in La Plata

Authors: Michal Mikolaj; Andreas Güntner; Claudio Brunini; Hartmut Wziontek; Mauricio Gende; Stephan Schröder; Augusto M. Cassino; Alfredo Pasquaré; Marvin Reich; Anne Hartmann; Fernando A. Oreiro; Jonathan Pendiuk; Luis Guarracino; Ezequiel D. Antokoletz

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-11-1501-2019

URL: <https://doi.org/10.5194/essd-11-1501-2019>

Publication Date: 2019-10-02

Volume/Issue: Volume 11, Issue 4

References: 51

Cited by: 7 publications

Language: en

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Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung

License: CC BY 4.0

Open Access: N/A

Abstract: . The Argentine-German Geodetic Observatory (AGGO) is one of the very few sites in the Southern Hemisphere equipped with comprehensive cutting-edge geodetic instrumentation. The employed observation techniques are used for a wide range of geophysical applications. The data set provides gravity time series and selected gravity models together with the hydrometeorological monitoring data of the observatory. These parameters are of great interest to the scientific community, e.g. for achieving accurate realization of terrestrial and celestial reference frames. Moreover, the availability of the hydrometeorological products is beneficial to inhabitants of the region as they allow for monitoring of environmental changes and natural hazards including extreme events. The hydrological data set is composed of time series of groundwater level, modelled and observed soil moisture content, soil temperature, and physical soil properties and aquifer properties. The meteorological time series include air temperature, humidity, pressure, wind speed, solar radiation, precipitation, and derived reference evapotranspiration. These data products are extended by gravity models of hydrological, oceanic, La Plata estuary, and atmospheric effects. The quality of the provided meteorological time series is tested via comparison to the two closest WMO (World Meteorological Organization) sites where data are available only in an inferior temporal resolution. The hydrological series are validated by comparing the respective forward-modelled gravity effects to independent gravity observations reduced up to a signal corresponding to local water storage variation. Most of the time series cover the time span between April 2016 and November 2018 with either no or only few missing data points. The data set is available at <https://doi.org/10.5880/GFZ.5.4.2018.001> (Mikolaj et al., 2018).

960. Interannual Variability of the Overturning and Energy Transport in the Atmosphere and Ocean During the Late Twentieth Century with Implications for Precipitation and Sea Level

Authors: Magnus Hieronymus (Stockholm University); Jonas Nycander (Stockholm University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0204.1

URL: <https://doi.org/10.1175/jcli-d-19-0204.1>

Publication Date: 2019-10-01

Volume/Issue: Volume 33, Issue 1

References: 66

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The overturning circulations in the atmosphere and ocean transport energy from the tropics to higher latitudes and thereby modulate Earth's climate. The interannual variability in the overturning over the last 40 years is found to be dominated by two coupled atmosphere-ocean modes. The first is related to the meridional motion of the intertropical convergence zone and the second to El Niño. Both modes have a strong influence on the sea level variability in the tropical Indo-Pacific Ocean. The interannual variability of the cross-equatorial energy transport is dominated by the first mode, and the variability is larger in the Indo-Pacific Ocean than in the Atlantic Ocean or the atmosphere. Our results suggest an important role of oceanic energy transport in setting precipitation patterns in the tropics and a key role of the Indo-Pacific Ocean as a climate modulator.

961. Dipole Structure of Mixed Layer Salinity in Response to El Niño-La Niña Asymmetry in the Tropical Pacific

Authors: Cong Guan (Institute of Oceanographic Instrumentation Shandong Academy of Sciences Qingdao China, Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Shijian Hu (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Michael J. McPhaden (NOAA/Pacific Marine Environmental Laboratory Seattle WA USA); Fan Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Shan Gao (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Yinglin Hou (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl084817

URL: <https://doi.org/10.1029/2019gl084817>

Publication Date: 2019-10-01

Volume/Issue: Volume 46, Issue 21

References: 41

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: tropical Pacific, ENSO asymmetry, El Niño±o, salinity, nonlinear (and 1 more)

Funding: National Natural Science Foundation of China, China Postdoctoral Science Foundation

License: Creative Commons

Open Access: N/A

Abstract: The asymmetry of mixed layer salinity (MLS) anomalies in response to El Niño and La Niña events in the tropical Pacific is examined for the first time based on Argo observations and Estimating the Circulation and Climate of the Ocean simulation. The difference of MLS anomalies between El Niño and La Niña shows a dipole structure, with El Niño featuring strong negative salinity anomalies east of 160 degreesE while La Niña shows remarkable positive salinity anomalies west of 160 degreesE. A salinity budget analysis suggests that nonlinear zonal advection plays a dominant role in generating the asymmetric MLS structure. This dipole MLS structure acts to generate a dipole structure of barrier layer thickness and thus likely plays an important role in the development of El Niño-Southern Oscillation asymmetries in sea surface temperature.

962. Poly- and Perfluoroalkyl Substances in Seawater and Plankton from the Northwestern Atlantic Margin

Authors: Xianming Zhang (Harvard University); Rainer Lohmann (University of Rhode Island); Elsie M. Sunderland (Harvard University)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

DOI: 10.1021/acs.est.9b03230

URL: <https://doi.org/10.1021/acs.est.9b03230>

Publication Date: 2019-09-29

Volume/Issue: Volume 53, Issue 21

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Cited by: 103 publications

Language: en

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Keywords: N/A

Funding: National Institute of Environmental Health Sciences, Office of Polar Programs, Richard and Susan Smith Family Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

963. Steric Sea Level Changes from Ocean Reanalyses at Global and Regional Scales

Authors: Andrea Storto (Centre for Maritime Research and Experimentation, 19126 La Spezia SP, Italy); Antonio Bonaduce (Helmholtz-Zentrum Geesthacht Centre for Materials and Coastal Research (HZG), 21502 Geesthacht, Germany); Xiangbo Feng (Hohai University, University of Reading); Chunxue Yang (Institute of Marine Sciences)

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w11101987

URL: <https://doi.org/10.3390/w11101987>

Publication Date: 2019-09-25

Volume/Issue: Volume 11, Issue 10

References: 122

Cited by: 24 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Sea level has risen significantly in the recent decades and is expected to rise further based on recent climate projections. Ocean reanalyses that synthesize information from observing networks, dynamical ocean general circulation models, and atmospheric forcing data offer an attractive way to evaluate sea level trend and variability and partition the causes of such sea level changes at both global and regional scales. Here, we review recent utilization of reanalyses for steric sea level trend investigations. State-of-the-science ocean reanalysis products are then used to further infer steric sea level changes. In particular, we used an ensemble of centennial reanalyses at moderate spatial resolution (between 0.5 x 0.5 and 1 x 1 degree) and an ensemble of eddy-permitting reanalyses to quantify the trends and their uncertainty over the last century and the last two decades, respectively. All the datasets showed good performance in reproducing sea level changes. Centennial reanalyses reveal a 1900-2010 trend of steric sea level equal to 0.47 +/- 0.04 mm year⁻¹, in agreement with previous studies, with unprecedented rise since the mid-1990s. During the altimetry era, the latest vintage of reanalyses is shown to outperform the previous ones in terms of skill scores against the independent satellite data. They consistently reproduce global and regional upper ocean steric expansion and the association with climate variability, such as ENSO. However, the mass contribution to the global mean sea level rise is varying with products and its representability needs to be improved, as well as the contribution of deep and abyssal waters to the steric sea level rise. Similarly, high-resolution regional reanalyses for the European seas provide valuable information on sea level trends, their patterns, and their causes.

964. Portraying the Impact of the Tibetan Plateau on Global Climate

Authors: Haijun Yang (Peking University); Xingchen Shen (Peking University); Jie Yao (Peking University); Qin Wen (Peking University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0734.1

URL: <https://doi.org/10.1175/jcli-d-18-0734.1>

Publication Date: 2019-09-25

Volume/Issue: Volume 33, Issue 9

References: 83

Cited by: 28 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: As the most extensive highland in the world, the Tibetan Plateau (TP) plays an important role in shaping the global climate. Quantifying the effect of the TP on global climate is the first step for a full understanding of the TP's standing on planet Earth. Through coupled model sensitivity experiments, we draw a panorama of the TP's global impact in this paper. Our model results show that the absence of the TP would result in a 4 degreesC colder and 10% drier climate in the Northern Hemisphere (NH). The TP has a striking remote effect on the North Atlantic. Removing the TP would enhance the westerlies in the mid- to high latitudes of the NH and weaken the easterlies over the tropical Pacific. More moisture would be relocated from the tropical Pacific to the North Atlantic, shutting down the Atlantic thermohaline circulation, which would eventually result in more than 15 degreesC colder and 20% drier climate over the North Atlantic. Our model results suggest that the presence of the TP may have contributed greatly to the hospitable modern climate in the NH, by promoting the establishment of the thermohaline circulation in the Atlantic, and therefore enhancing the northward ocean heat transport and atmosphere moisture transport across the equator.

965. Mesoscale Air-Sea Interaction and Its Role in Eddy Energy Dissipation in the Kuroshio Extension

Authors: Haiyuan Yang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Ping Chang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Bo Qiu (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Qiuying Zhang (Department of Oceanography, and Department of Atmospheric Sciences, Texas A&M University, College Station, Texas); Lixin Wu (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Zhaohui Chen (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Hong Wang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0155.1

URL: <https://doi.org/10.1175/jcli-d-19-0155.1>

Publication Date: 2019-09-18

Volume/Issue: Volume 32, Issue 24

References: 54

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National key research and development program of China

License: N/A

Open Access: N/A

Abstract: Using the high-resolution Community Earth System Model (CESM) output, this study investigates air-sea interaction and its role in eddy energy dissipation in the Kuroshio Extension (KE) region. Based on an eddy energetics analysis, it is found that the baroclinic pathway associated with temperature variability is the main eddy energy source in this region. Both the air-sea heat flux and wind stress act as eddy killers that remove energy from oceanic eddies. Heat exchange between atmosphere and oceanic eddies dominates the dissipation of eddy temperature variance within the surface layer and accounts for 36% of the total dissipation in the upper 350-m layer. Compared to the heat exchange, the role of wind power in damping the eddy kinetic energy (EKE) is relatively small. Only 18% of EKE dissipation in the upper 350 m is attributed to eddy wind power. Misrepresentation of the damping role of mesoscale ocean-atmosphere interaction can result in an incorrect vertical structure of eddy energy dissipation, leading to an erroneous representation of vertical mixing in the interior ocean.

966. A global vertical datum defined by the conventional geoid potential and the Earth ellipsoid parameters

Authors: Hadi Amin; Lars E. Sjöberg; Mohammad Bagherbandi

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-019-01293-3

URL: <https://doi.org/10.1007/s00190-019-01293-3>

Publication Date: 2019-09-12

Volume/Issue: Volume 93, Issue 10

References: 68

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The geoid, according to the classical Gauss-Listing definition, is, among infinite equipotential surfaces of the Earth's gravity field, the equipotential surface that in a least squares sense best fits the undisturbed mean sea level. This equipotential surface, except for its zero-degree harmonic, can be characterized using the Earth's global gravity models (GGM). Although, nowadays, satellite altimetry technique provides the absolute geoid height over oceans that can be used to calibrate the unknown zero-degree harmonic of the gravimetric geoid models, this technique cannot be utilized to estimate the geometric parameters of the mean Earth ellipsoid (MEE). The main objective of this study is to perform a joint estimation of W_0 , which defines the zero datum of vertical coordinates, and the MEE parameters relying on a new approach and on the newest gravity field, mean sea surface and mean dynamic topography models. As our approach utilizes both satellite altimetry observations and a GGM model, we consider different aspects of the input data to evaluate the sensitivity of our estimations to the input data. Unlike previous studies, our results show that it is not sufficient to use only the satellite-component of a quasi-stationary GGM to estimate W_0 . In addition, our results confirm a high sensitivity of the applied approach to the altimetry-based geoid heights, i.e., mean sea surface and mean dynamic topography models. Moreover, as W_0 should be considered a quasi-stationary parameter, we quantify the effect of time-dependent Earth's gravity field changes as well as the time-dependent sea level changes on the estimation of W_0 . Our computations resulted in the geoid potential $W_0 = 62636848.102 \pm 0.004 \text{ m}^2 \text{ s}^{-2}$ and the semi-major and minor axes of the MEE, $a = 6378137.678 \pm 0.0003 \text{ m}$ and $b = 6356752.964 \pm 0.0005 \text{ m}$, which are 0.678 and 0.650 m larger than those axes of GRS80 reference ellipsoid, respectively. Moreover, a new estimation for the geocentric gravitational constant was obtained as $GM = (398600460.55 \pm 0.03) \times 10^6 \text{ m}^3 \text{ s}^{-2}$.

967. Improved Estimates of Geocenter Variability from Time-Variable Gravity and Ocean Model Outputs

Authors: Tyler C. Sutterley (University of California Irvine, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Isabella Velicogna (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs11182108

URL: <https://doi.org/10.3390/rs11182108>

Publication Date: 2019-09-10

Volume/Issue: Volume 11, Issue 18

References: 69

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Geocenter variations relate the motion of the Earth's center of mass with respect to its center of figure, and represent global-scale redistributions of the Earth's mass. We investigate different techniques for estimating of geocenter motion from combinations of time-variable gravity measurements from the Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On missions, and bottom pressure outputs from ocean models. Here, we provide self-consistent estimates of geocenter variability incorporating the effects of self-attraction and loading, and investigate the effect of uncertainties in atmospheric and oceanic variation. The effects of self-attraction and loading from changes in land water storage and ice mass change affect both the seasonality and long-term trend in geocenter position. Omitting the redistribution of sea level affects the average annual amplitudes of the x, y, and z components by 0.2, 0.1, and 0.3 mm, respectively, and affects geocenter trend estimates by 0.02, 0.04 and 0.05 mm/yr for the the x, y, and z components, respectively. Geocenter estimates from the GRACE Follow-On mission are consistent with estimates from the original GRACE mission.

968. Synchronous Rendezvous for Networks of Marine Robots in Large Scale Ocean Monitoring

Authors: Xi Yu; M. Ani Hsieh; Cong Wei; Hebert G. Tanner

Journal: Frontiers in Robotics and AI

Publisher: Frontiers Media SA

DOI: 10.3389/frobt.2019.00076

URL: <https://doi.org/10.3389/frobt.2019.00076>

Publication Date: 2019-09-04

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References: 26

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

969. Upper ocean high resolution regional modeling of the Arabian Sea and Bay of Bengal

Authors: Dwivedi Suneet; Mishra Alok Kumar; Srivastava Atul

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-019-1439-x

URL: <https://doi.org/10.1007/s13131-019-1439-x>

Publication Date: 2019-09-04

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References: 78

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

970. Extratropical-Tropical Interaction Model Intercomparison Project (Etin-Mip): Protocol and Initial Results

Authors: Sarah M. Kang (School of Urban and Environmental Engineering, Ulsan National Institute of Science and Technology, Ulsan, South Korea); Matt Hawcroft (University of Exeter); Baoqiang Xiang (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey, and University Corporation for Atmospheric Research, Boulder, Colorado); Yen-Ting Hwang (National Taiwan University); Gabriel Cazes (School of Engineering, Universidad de la República, Montevideo, Uruguay); Francis Codron (Institut de Recherche pour le Développement (IRD)); Traute Crueger (Max Planck Institute for Meteorology); Clara Deser (National Center for Atmospheric Research); Øivind Hodnebrog (Center for International Climate Research (CICERO), Oslo, Norway); Hanjun Kim (School of Urban and Environmental Engineering, Ulsan National Institute of Science and Technology, Ulsan, South Korea); Jiyeong Kim (School of Urban and Environmental Engineering, Ulsan National Institute of Science and Technology, Ulsan, South Korea); Yu Kosaka (University of Tokyo); Teresa Losada (Department of Earth Physics and Astrophysics, Faculty of Physics, Universidad Complutense de Madrid, Madrid, Spain); Carlos R. Mechoso (University of California Los Angeles); Gunnar Myhre (Center for International Climate Research (CICERO), Oslo, Norway); Øyvind Seland (Norwegian Meteorological Institute); Bjorn Stevens (Max Planck Institute for Meteorology); Masahiro Watanabe (University of Tokyo); Sungduk Yu (Department of Geology and Geophysics, Yale University, New Haven, Connecticut)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/bams-d-18-0301.1

URL: <https://doi.org/10.1175/bams-d-18-0301.1>

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Volume/Issue: Volume 100, Issue 12

References: 74

Cited by: 30 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This article introduces the Extratropical-Tropical Interaction Model Intercomparison Project (ETIN-MIP), where a set of fully coupled model experiments are designed to examine the sources of longstanding tropical precipitation biases in climate models. In particular, we reduce insolation over three targeted latitudinal bands of persistent model biases: the southern extratropics, the southern tropics, and the northern extratropics. To address the effect of regional energy bias corrections on the mean distribution of tropical precipitation, such as the double intertropical convergence zone problem, we evaluate the quasi-equilibrium response of the climate system corresponding to a 50-yr period after the 100 years of prescribed energy perturbation. Initial results show that, despite a large intermodel spread in each perturbation experiment due to differences in ocean heat uptake response and climate feedbacks across models, the southern tropics is most efficient at driving a meridional shift of tropical

precipitation. In contrast, the extratropical energy perturbations are effectively damped by anomalous heat uptake over the subpolar oceans, thereby inducing a smaller meridional shift of tropical precipitation compared with the tropical energy perturbations. The ETIN-MIP experiments allow us to investigate the global implications of regional energy bias corrections, providing a route to guide the practice of model development, with implications for understanding dynamical responses to anthropogenic climate change and geoengineering.

971. An assessment of the impact of oceanic initial conditions on the interaction of upper ocean with the tropical cyclones in the Arabian Sea

Authors: Tanuja Nigam (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Kumar Ravi Prakash (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences); Vimlesh Pant (Indian Institute of Technology Delhi, Centre for Atmospheric Sciences)

Journal: Journal of Operational Oceanography

Publisher: Informa UK Limited

DOI: 10.1080/1755876x.2019.1658567

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References: 78

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License: N/A

Open Access: N/A

Abstract: N/A

972. New insight into 3-D mesoscale eddy properties from CMEMS operational models in the western Mediterranean

Authors: Evan Mason; Simón Ruiz; Romain Bourdalle-Badie; Guillaume Reffray; Marcos García-Sotillo; Ananda Pascual

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-15-1111-2019

URL: <https://doi.org/10.5194/os-15-1111-2019>

Publication Date: 2019-08-21

Volume/Issue: Volume 15, Issue 4

References: 71

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Rapid evolution of operational ocean forecasting systems is driven by advances in numerics and data assimilation schemes, and increase of in situ and satellite observations. The Copernicus Marine Service (CMEMS) is a major provider of operational products that are made available through an online catalogue. The service includes global and regional forecasts in near-real-time and reanalysis modes. Here, we apply an eddy tracker to daily sea surface height (SSH) fields from three such reanalysis products from the CMEMS catalogue, with the objective to evaluate their performance in terms of their eddy properties and three-dimensional composite structures over the 2013-2016 period. The products are (i) the Global Analysis Forecast, (ii) the Mediterranean Analysis Forecast and (iii) the Iberia-Biscay-Ireland Analysis Forecast. The common domain between these reanalyses is the western Mediterranean Sea (WMED) between the Strait of Gibraltar and Sardinia. This is a complex region with strong density gradients, especially in the Alboran Sea in the west where Atlantic and Mediterranean waters compete. Surface eddy property maps over the WMED of eddy radii, amplitudes and nonlinearity are consistent between the models, as well as with gridded altimetric data that serve as a reference. Mean 3-D eddy composites are shown only for three subregions in the Alboran Sea. These are mostly consistent between the models, with minor differences being attributed to details of the respective model configurations. This information can be informative for the ongoing development of these CMEMS operational modeling systems. The mesoscale data provided here may be of interest to CMEMS users and in the future could be a useful addition to a more diverse CMEMS catalogue.

973. Measuring Global Ocean Heat Content to Estimate the Earth Energy Imbalance

Authors: Benoit Meyssignac; Tim Boyer; Zhongxiang Zhao; Maria Z. Hakuba; Felix W. Landerer; Detlef Stammer; Armin Köhl; Seiji Kato; Tristan L'Ecuyer; Michael Ablain; John Patrick Abraham; Alejandro Blazquez; Anny Cazenave; John A. Church; Rebecca Cowley; Lijing Cheng; Catia M. Domingues; Donata Giglio; Viktor Gouretski; Masayoshi Ishii; Gregory C. Johnson; Rachel E. Killick; David Legler; William Llovel; John Lyman; Matthew Dudley Palmer; Steve Piotrowicz; Sarah G. Purkey; Dean Roemmich; Rémy Roca; Abhishek Savita; Karina von Schuckmann; Sabrina Speich; Graeme Stephens; Gongjie Wang; Susan Elisabeth Wijffels; Nathalie Zilberman

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00432

URL: <https://doi.org/10.3389/fmars.2019.00432>

Publication Date: 2019-08-20

Volume/Issue: Volume 6

References: 177

Cited by: 132 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

974. Impacts of Basin-Scale Climate Modes on Coastal Sea Level: a Review

Authors: Weiqing Han; Detlef Stammer; Philip Thompson; Tal Ezer; Hindu Palanisamy; Xuebin Zhang; Catia M. Domingues; Lei Zhang; Dongliang Yuan

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-019-09562-8

URL: <https://doi.org/10.1007/s10712-019-09562-8>

Publication Date: 2019-08-19

Volume/Issue: Volume 40, Issue 6

References: 309

Cited by: 55 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration, DFG (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

975. A Barotropic Vorticity Budget for the Subtropical North Atlantic Based on Observations

Authors: Isabela Alexander-Astiz Le Bras (Scripps Institution of Oceanography / University of California San Diego, Massachusetts Institute of Technology); Maike Sonnewald (Harvard University, Massachusetts Institute of Technology); John M. Toole (Woods Hole Oceanographic Institution)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0111.1

URL: <https://doi.org/10.1175/jpo-d-19-0111.1>

Publication Date: 2019-08-19

Volume/Issue: Volume 49, Issue 11

References: 60

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: US National Science Foundation, US National Science Foundation, US National Science Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: To ground truth the large-scale dynamical balance of the North Atlantic subtropical gyre with observations, a barotropic vorticity budget is constructed in the ECCO state estimate and compared with hydrographic observations and wind stress data products. The hydrographic dataset at the center of this work is the A22 WOCE section, which lies along 66 degreesW and creates a closed volume with the North and South American coasts to its west. The planetary vorticity flux across A22 is quantified, providing a metric for the net meridional flow in the western subtropical gyre. The wind stress forcing over the subtropical gyre to the west and east of the A22 section is calculated from several wind stress data products. These observational budget terms are found to be consistent with an approximate barotropic Sverdrup balance in the eastern subtropical gyre and are on the same order as budget terms in the ECCO state estimate. The ECCO vorticity budget is closed by bottom pressure torques in the western subtropical gyre, which is consistent with previous studies. In sum, the analysis provides observational ground truth for the North Atlantic subtropical vorticity balance and explores the seasonal variability of this balance for the first time using the ECCO state estimate. This balance is found to hold on monthly time scales in ECCO, suggesting that the integrated subtropical gyre responds to forcing through fast barotropic adjustment.

976. Observed Interhemispheric Meridional Heat Transports and the Role of the Indonesian Throughflow in the Pacific Ocean

Authors: Kevin E. Trenberth (National Center for Atmospheric Research); Yongxin Zhang (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0465.1

URL: <https://doi.org/10.1175/jcli-d-19-0465.1>

Publication Date: 2019-08-19

Volume/Issue: Volume 32, Issue 24

References: 49

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The net surface energy flux is computed as a residual of the energy budget using top-of-atmosphere radiation combined with the divergence of the column-integrated atmospheric energy transports, and then used with the vertically integrated ocean heat content tendencies to compute the ocean meridional heat transports (MHTs). The mean annual cycles and 12-month running mean MHTs as a function of latitude are presented for 2000-16. Effects of the Indonesian Throughflow (ITF), associated with a net volume flow around Australia accompanied by a heat transport, are fully included. Because the ITF-related flow necessitates a return current northward in the Tasman Sea that relaxes during El Niño, the reduced ITF during El Niño may contribute to warming in the south Tasman Sea by allowing the East Australian Current to push farther south even as it gains volume from the tropical waters not flowing through the ITF. Although evident in 2015/16, when a major marine heat wave occurred, these effects can be overwhelmed by changes in the atmospheric circulation. Large interannual MHT variability in the Pacific is 4 times that of the Atlantic. Strong relationships reveal influences from the southern subtropics on ENSO for this period. At the equator, northward ocean MHT arises mainly in the Atlantic (0.75 PW), offset by the Pacific (-0.33 PW) and Indian Oceans (-0.20 PW) while the atmosphere transports energy southward (-0.35 PW). The net equatorial MHT southward (-0.18 PW) is enhanced by -0.1 PW that contributes to the greater warming of the southern (vs northern) oceans.

977. Reconciling Observation and Model Trends in North Atlantic Surface CO₂

Authors: Alice D. Lebehent (Southern Ocean Carbon-Climate Observatory (SOCCO) CSIR Cape Town South Africa, University of Cape Town, University of Exeter); Paul R. Halloran (University of Exeter); Andrew J. Watson (University of Exeter); Doug McNeall (Met Office); David A. Ford (Met Office); Peter Landschützer (Max Planck Institute for Meteorology); Siv K. Lauvset (University of Bergen, Bjerknes Centre for Climate Research); Ute Schuster (University of Exeter)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gb006186

URL: <https://doi.org/10.1029/2019gb006186>

Publication Date: 2019-08-19

Volume/Issue: Volume 33, Issue 10

References: 89

Cited by: 14 publications

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Keywords: N/A

Funding: Department for Business, Energy and Industrial Strategy, Department for Environment, Food and Rural Affairs, Natural Environment Research Council (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: The North Atlantic Ocean is a region of intense uptake of atmospheric CO₂. To assess how this CO₂ sink has evolved over recent decades, various approaches have been used to estimate basin-wide uptake from the irregularly sampled in situ CO₂ observations. Until now, the lack of robust uncertainties associated with observation-based gap-filling methods required to produce these estimates has limited the capacity to validate climate model simulated surface ocean CO₂ concentrations. After robustly quantifying basin-wide and annually varying interpolation uncertainties using both observational and model data, we show that the North Atlantic surface ocean fugacity of CO₂ (fCO₂-ocean) increased at a significantly slower rate than that simulated by the latest generation of Earth System Models during the period 1992-2014. We further show, with initialized model simulations, that the inability of these models to capture the observed trend in surface fCO₂-ocean is primarily due to biases in the models' ocean biogeochemistry. Our results imply that current projections may underestimate the contribution of the North Atlantic to mitigating increasing future atmospheric CO₂ concentrations.

978. The $^{210}\text{Po}/^{210}\text{Pb}$ method to calculate particle export: Lessons learned from the results of three GEOTRACES transects

Authors: Yi Tang; Gillian Stewart

Journal: Marine Chemistry

Publisher: Elsevier BV

DOI: 10.1016/j.marchem.2019.103692

URL: <https://doi.org/10.1016/j.marchem.2019.103692>

Publication Date: 2019-08-12

Volume/Issue: Volume 217

References: 77

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: Geotraces

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

979. Surface Kinetic Energy Distributions in the Global Oceans From a High-Resolution Numerical Model and Surface Drifter Observations

Authors: Xiaolong Yu (Institut de Recherche pour le Développement (IRD)); Aurélien L. Ponte (Institut de Recherche pour le Développement (IRD)); Shane Elipot (University of Miami); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Edward D. Zaron (Portland State University); Ryan Abernathey (Columbia University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl083074

URL: <https://doi.org/10.1029/2019gl083074>

Publication Date: 2019-08-09

Volume/Issue: Volume 46, Issue 16

References: 54

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: LLC4320, rotary spectrum, surface drifter, SWOT

Funding: Agence Nationale de la Recherche, National Science Foundation

License: N/A

Open Access: N/A

Abstract: The surface kinetic energy of a 1/48 degrees global ocean simulation and its distribution as a function of frequency and location are compared with the one estimated from 15,329 globally distributed surface drifter observations at hourly resolution. These distributions follow similar patterns with a dominant low-frequency component and well-defined tidal and near-inertial peaks globally. Quantitative differences are identified with deficits of low-frequency energy near the equator (factor 2) and at near-inertial frequencies (factor 3) and an excess of energy at semidiurnal frequencies (factor 4) for the model. Owing to its hourly resolution and its near-global spatial coverage, the array of surface drifters is an invaluable tool to evaluate the realism of tide-resolving high-resolution ocean simulations used in observing system simulation experiments. Sources of bias between model and drifter data are discussed, and associated leads for future work highlighted.

980. Toward an Energetically Consistent, Resolution Aware Parameterization of Ocean Mesoscale Eddies

Authors: Malte F. Jansen (University of Chicago); Alistair Adcroft (Princeton University, NOAA Geophysical Fluid Dynamics Laboratory Princeton NJ USA); Sina Khani (University of Chicago, Princeton University); Hailu Kong (University of Chicago)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019ms001750

URL: <https://doi.org/10.1029/2019ms001750>

Publication Date: 2019-08-07

Volume/Issue: Volume 11, Issue 8

References: 65

Cited by: 64 publications

Language: en

Subjects: N/A

Keywords: eddy parameterization, backscatter, mesoscale, energy budget

Funding: Office of the Director

License: CC BY 4.0

Open Access: N/A

Abstract: A subgrid-scale eddy parameterization is developed, which makes use of an explicit eddy kinetic energy budget and can be applied at both "non-eddy" and "eddy-permitting" resolutions. The subgrid-scale eddies exchange energy with the resolved flow in both directions via a parameterization of baroclinic instability (based on the established formulation of Gent and McWilliams) and biharmonic and negative Laplacian viscosity terms. This formulation represents the turbulent cascades of energy and enstrophy consistent with our current understanding of the turbulent eddy energy cycle. At the same time, the approach is simple and general enough to be readily implemented in ocean climate models, without adding significant computational cost. The closure has been implemented in the Modular Ocean Model Version 6 and tested in the "Neverworld" configuration, which employs an idealized analytically defined topography designed as a testbed for mesoscale eddy parameterizations. The parameterization performs well over a range of resolutions, seamlessly connecting non-eddy and eddy-resolving regimes.

981. Recent Contributions of Theory to Our Understanding of the Atlantic Meridional Overturning Circulation

Authors: Helen L. Johnson (University of Oxford); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego); David P. Marshall (University of Oxford); Fabian Schloesser (International Pacific Research Center Honolulu HI USA); Michael A. Spall (Woods Hole Oceanographic Institution)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015330

URL: <https://doi.org/10.1029/2019jc015330>

Publication Date: 2019-08-06

Volume/Issue: Volume 124, Issue 8

References: 181

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Revolutionary observational arrays, together with a new generation of ocean and climate models, have provided new and intriguing insights into the Atlantic Meridional Overturning Circulation (AMOC) over the last two decades. Theoretical models have also changed our view of the AMOC, providing a dynamical framework for understanding the new observations and the results of complex models. In this paper we review recent advances in conceptual understanding of the processes maintaining the AMOC. We discuss recent theoretical models that address issues such as the interplay between surface buoyancy and wind forcing, the extent to which the AMOC is adiabatic, the importance of mesoscale eddies, the interaction between the middepth North Atlantic Deep Water cell and the abyssal Antarctic Bottom Water cell, the role of basin geometry and bathymetry, and the importance of a three-dimensional multiple-basin perspective. We review new paradigms for deep water formation in the high-latitude North Atlantic and the impact of diapycnal mixing on vertical motion in the ocean interior. And we discuss advances in our understanding of the AMOC's stability and its scaling with large-scale meridional density gradients. Along with reviewing theories for the mean AMOC, we consider models of AMOC variability and discuss what we have learned from theory about the detection and meridional propagation of AMOC anomalies. Simple theoretical models remain a vital and powerful tool for articulating our understanding of the AMOC and identifying the processes that are most critical to represent accurately in the next generation of numerical ocean and climate models.

982. Variability of Thermohaline Characteristics at 26.5 degrees N in Reanalyses and Oceanographic Section Data

Authors: A. R. Gorbushkin; A. N. Demidov

Journal: Russian Meteorology and Hydrology

Publisher: Allerton Press

DOI: 10.3103/s1068373919070069

URL: <https://doi.org/10.3103/s1068373919070069>

Publication Date: 2019-08-06

Volume/Issue: Volume 44, Issue 7

References: 15

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

983. Examining the salinity change in the upper Pacific Ocean during the Argo period

Authors: Guancheng Li; Yuhong Zhang; Jingen Xiao; Xiangzhou Song; John Abraham; Lijing Cheng; Jiang Zhu

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-019-04912-z

URL: <https://doi.org/10.1007/s00382-019-04912-z>

Publication Date: 2019-08-02

Volume/Issue: Volume 53, Issue 9-10

References: 106

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key R&D; Program of China, National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

984. The 2015-2016 carbon cycle as seen from OCO-2 and the global in situ network

Authors: Sean Crowell; David Baker; Andrew Schuh; Sourish Basu; Andrew R. Jacobson; Frederic Chevallier; Junjie Liu; Feng Deng; Liang Feng; Kathryn McKain; Abhishek Chatterjee; John B. Miller; Britton B. Stephens; Annmarie Eldering; David Crisp; David Schimel; Ray Nassar; Christopher W. O'Dell; Tomohiro Oda; Colm Sweeney; Paul I. Palmer; Dylan B. A. Jones

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-19-9797-2019

URL: <https://doi.org/10.5194/acp-19-9797-2019>

Publication Date: 2019-08-02

Volume/Issue: Volume 19, Issue 15

References: 112

Cited by: 130 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . The Orbiting Carbon Observatory-2 has been on orbit since 2014, and its global coverage holds the potential to reveal new information about the carbon cycle through the use of top-down atmospheric inversion methods combined with column average CO₂ retrievals. We employ a large ensemble of atmospheric inversions utilizing different transport models, data assimilation techniques, and prior flux distributions in order to quantify the satellite-informed fluxes from OCO-2 Version 7r land observations and their uncertainties at continental scales. Additionally, we use in situ measurements to provide a baseline against which to compare the satellite-constrained results. We find that within the ensemble spread, in situ observations, and satellite retrievals constrain a similar global total carbon sink of 3.7 $\pm$ 0.5 PgC yr⁻¹, and 1.5 $\pm$ 0.6 PgC yr⁻¹ for global land, for the 2015-2016 annual mean. This agreement breaks down in smaller regions, and we discuss the differences between the experiments. Of particular interest is the difference between the different assimilation constraints in the tropics, with the largest differences occurring in tropical Africa, which could be an indication of the global perturbation from the 2015-2016 El Niño. Evaluation of posterior concentrations using TCCON and aircraft observations gives some limited insight into the quality of the different assimilation constraints, but the lack of such data in the tropics inhibits our ability to make strong conclusions there.

985. Vertical redistribution of salt and layered changes in global ocean salinity

Authors: Chao Liu; Xinfeng Liang; Rui M. Ponte; Nadya Vinogradova; Ou Wang

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-019-11436-x

URL: <https://doi.org/10.1038/s41467-019-11436-x>

Publication Date: 2019-08-01

Volume/Issue: Volume 10, Issue 1

References: 48

Cited by: 20 publications

Language: en

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Keywords: N/A

Funding: National Aeronautics and Space Administration, National Science Foundation, Alfred P. Sloan Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Salinity is an essential proxy for estimating the global net freshwater input into the ocean. Due to the limited spatial and temporal coverage of the existing salinity measurements, previous studies of global salinity changes focused mostly on the surface and upper oceans. Here, we examine global ocean salinity changes and ocean vertical salt fluxes over the full depth in a dynamically consistent and data-constrained ocean state estimate. The changes of the horizontally averaged salinity display a vertically layered structure, consistent with the profiles of the ocean vertical salt fluxes. For salinity changes in the relatively well-observed upper ocean, the contribution of vertical exchange of salt can be on the same order of the net surface freshwater input. The vertical redistribution of salt thus should be considered in inferring changes in global ocean salinity and the hydrological cycle from the surface and upper ocean measurements.

986. Ocean Reanalyses: Recent Advances and Unsolved Challenges

Authors: Andrea Storto; Aida Alvera-Azcárate; Magdalena A. Balmaseda; Alexander Barth; Matthieu Chevallier; Francois Counillon; Catia M. Domingues; Marie Drevillon; Yann Drillet; Gaël Forget; Gilles Garric; Keith Haines; Fabrice Hernandez; Doroteaciro Iovino; Laura C. Jackson; Jean-Michel Lellouche; Simona Masina; Michael Mayer; Peter R. Oke; Stephen G. Penny; K. Andrew Peterson; Chunxue Yang; Hao Zuo

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00418

URL: <https://doi.org/10.3389/fmars.2019.00418>

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Volume/Issue: Volume 6

References: 114

Cited by: 85 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: European Cooperation in Science and Technology

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

987. Strong North Pacific Subtropical Mode Water Volume and Density Decrease in Year 1999

Authors: I. Cerovečki (Scripps Institution of Oceanography / University of California San Diego); M. C. Hendershott (Scripps Institution of Oceanography / University of California San Diego); E. Yulaeva (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc014956

URL: <https://doi.org/10.1029/2019jc014956>

Publication Date: 2019-07-31

Volume/Issue: Volume 124, Issue 9

References: 51

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: Kuroshio Extension, quasi-decadal variability, North Pacific Subtropical Mode Water

Funding: U.S. Department of Energy, National Science Foundation

License: N/A

Open Access: N/A

Abstract: A strong decrease of volume and density of North Pacific Subtropical Mode Water (NPSTMW) in 1999 was analyzed in a regional high-resolution (0.1 degrees) numerical ocean circulation model simulation. Both shoaling of the bottom and deepening of the top of the NPSTMW layer contributed equally to volume decrease. They were locally governed by different physical processes, but both seem to be associated with basin-wide changes in wind. A westward propagating negative thermocline depth anomaly that developed in the Central Pacific when the Pacific Decadal Oscillation index changed from a positive to a negative phase in 1998 caused shoaling of the bottom of the NPSTMW layer. Deepening of the top of the NPSTMW layer was due to an increase in the near surface stratification, caused by an increase in wind-driven lateral heat transport convergence by the Kuroshio Extension jet starting in 1997. Both processes increased the potential vorticity in the NPSTMW region, decreasing the volume of water in the NPSTMW density range that satisfied the low potential vorticity constraint that is part of the definition of "mode water." The strong near-surface density decrease provided preconditioning for preferential surface formation of a lighter variety of NPSTMW, further decreasing its density. It also resulted in decrease of the outcrop window in the NPSTMW density range, strongly reducing its formation rate in 1998 and 1999 despite strong surface heat loss.

988. Diagnosing Ocean-Wave-Turbulence Interactions From Space

Authors: H. S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); P. Klein (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, LOPS/IFREMER Plouzane France); L. Siegelman (LEMAR Plouzane France, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); B. Qiu (University of Hawaii at Manoa); S. Chen (University of Hawaii at Manoa); C. Uebelmann (Collecte Localisation Satellites Ramonville St-Agne France); J. Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); L.-L. Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl083675

URL: <https://doi.org/10.1029/2019gl083675>

Publication Date: 2019-07-30

Volume/Issue: Volume 46, Issue 15

References: 29

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Numerical studies indicate that interactions between ocean internal gravity waves (especially those <100 km) and geostrophic (or balanced) motions associated with mesoscale eddy turbulence (involving eddies of 100-300 km) impact the ocean's kinetic energy budget and therefore its circulation. Results from these studies have never been confirmed by observations in regional or basin-scale domains. Here we show that internal gravity waves have a spectral signature on sea surface height during summer that significantly differs from that of balanced motions. These spectral differences lead us to propose a new dynamical framework that quantifies the interactions between internal gravity waves and balanced motions in physical space from sea surface height snapshots, and in particular the energy exchanges between them. Our results, using this dynamical framework, highlight the strong potential of future satellite altimeter missions to make critical advances in assessing the ocean's kinetic energy budget from observations in large domains.

989. Observing System Evaluation Based on Ocean Data Assimilation and Prediction Systems: On-Going Challenges and a Future Vision for Designing and Supporting Ocean Observational Networks

Authors: Yosuke Fujii; Elisabeth Rémy; Hao Zuo; Peter Oke; George Halliwell; Florent Gasparin; Mounir Benkiran; Nora Loose; James Cummings; Jiping Xie; Yan Xue; Shuhei Masuda; Gregory C. Smith; Magdalena Balmaseda; Cyril Germaineaud; Daniel J. Lea; Gilles Larnicol; Laurent Bertino; Antonio Bonaduce; Pierre Brasseur; Craig Donlon; Patrick Heimbach; YoungHo Kim; Villy Kourafalou; Pierre-Yves Le Traon; Matthew Martin; Shastri Paturi; Benoit Tranchant; Norihisa Usui

Journal: Frontiers in Marine Science

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Volume/Issue: Volume 6

References: 108

Cited by: 61 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

990. Differences in Response of Terrestrial Water Storage Components to Precipitation over 168 Global River Basins

Authors: Yafeng Zhang (Beijing Normal University); Bin He (Beijing Normal University); Lanlan Guo (Beijing Normal University); Daochen Liu (School of Environment and Planning, Liaocheng University, Shandong, China)

Journal: Journal of Hydrometeorology

Publisher: American Meteorological Society

DOI: 10.1175/jhm-d-18-0253.1

URL: <https://doi.org/10.1175/jhm-d-18-0253.1>

Publication Date: 2019-07-29

Volume/Issue: Volume 20, Issue 9

References: 85

Cited by: 43 publications

Language: en

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License: N/A

Open Access: N/A

Abstract: A time lag exists between precipitation P falling and being converted into terrestrial water. The responses of terrestrial water storage (TWS) and its individual components to P over the global scale, which are vital for understanding the interactions and mechanisms between climatic variables and hydrological components, are not well constrained. In this study, relying on land surface models, we isolate five component storage anomalies from TWS anomalies (TWSA) derived from the Gravity Recovery and Climate Experiment mission (GRACE): canopy water storage anomalies (CWSA), surface water storage anomalies (SWSA), snow water equivalent anomalies (SWEA), soil moisture storage anomalies (SMSA), and groundwater storage anomalies (GWSA). The responses of TWSA and of the individual components of TWSA to P are then evaluated over 168 global basins. The lag between TWSA and P is quantified by calculating the correlation coefficients between GRACE-based TWSA and P for different time lags, then identifying the lag (measured in months) corresponding to the maximum correlation coefficient. A multivariate regression model is used to explore the relationship between climatic and basin characteristics and the lag between TWSA and P . Results show that the spatial distribution of TWSA trend presents a similar global pattern to that of P for the period January 2004–December 2013. TWSA is positively related to P over basins but with lags of variable duration. The lags are shorter in the low- and midlatitude basins (1–2 months) than those in the high-latitude basins (6–9 months). The spatial patterns of the maximum correlations and the corresponding lags between individual components of the TWSA and P are consistent with those of the GRACE-based analysis, except for SWEA (3–8 months) and CWSA (0 months). The lags between GWSA, SMSA, and SWSA to P can be arranged as $GWSA > SMSA \geq SWSA$. Regression analysis results show that the lags between TWSA and P are related to the mean temperature, mean precipitation, mean latitude, mean longitude, mean elevation, and mean slope.

991. Observing the Oceans Acoustically

Authors: Bruce M. Howe; Jennifer Miksis-Olds; Eric Rehm; Hanne Sagen; Peter F. Worcester; Georgios Haralabus

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00426

URL: <https://doi.org/10.3389/fmars.2019.00426>

Publication Date: 2019-07-26

Volume/Issue: Volume 6

References: 117

Cited by: 84 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

992. Revisiting the circulation of the East Australian Current: Its path, separation, and eddy field

Authors: Peter R. Oke; Moninya Roughan; Paulina Cetina-Heredia; Gabriela S. Pilo; Kenneth R. Ridgway; Tatiana Rykova; Matthew R. Archer; Richard C. Coleman; Colette G. Kerry; Carlos Rocha; Amandine Schaeffer; Eduardo Vitarelli

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2019.102139

URL: <https://doi.org/10.1016/j.pocean.2019.102139>

Publication Date: 2019-07-24

Volume/Issue: Volume 176

References: 133

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council

License: N/A

Open Access: N/A

Abstract: N/A

993. More Than 50 Years of Successful Continuous Temperature Section Measurements by the Global Expendable Bathythermograph Network, Its Integrability, Societal Benefits, and Future

Authors: Gustavo J. Goni; Janet Sprintall; Francis Bringas; Lijing Cheng; Mauro Cirano; Shenfu Dong; Ricardo Domingues; Marlos Goes; Hosmay Lopez; Rosemary Morrow; Ulises Rivero; Thomas Rossby; Robert E. Todd; Joaquin Trinanes; Nathalie Zilberman; Molly Baringer; Tim Boyer; Rebecca Cowley; Catia M. Domingues; Katherine Hutchinson; Martin Kramp; Mauricio M. Mata; Franco Reseghetti; Charles Sun; Udaya Bhaskar TVS; Denis Volkov

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00452

URL: <https://doi.org/10.3389/fmars.2019.00452>

Publication Date: 2019-07-24

Volume/Issue: Volume 6

References: 161

Cited by: 40 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration, Commonwealth Scientific and Industrial Research Organisation

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

994. Hemispheric Asymmetry of the Pacific Shallow Meridional Overturning Circulation

Authors: Mathias Zeller (ARC Centre of Excellence for Climate System Science Sydney New South Wales Australia, Monash University); Shayne McGregor (ARC Centre of Excellence for Climate Extremes Sydney New South Wales Australia, Monash University); Paul Spence (ARC Centre of Excellence for Climate Extremes Sydney New South Wales Australia, University of New South Wales)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014840

URL: <https://doi.org/10.1029/2018jc014840>

Publication Date: 2019-07-20

Volume/Issue: Volume 124, Issue 8

References: 75

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: decadal variability, tropical Pacific, subtropical cells, ocean GCM

Funding: Australian Research Council

License: N/A

Open Access: N/A

Abstract: The shallow subtropical cells (STCs) in the Pacific Ocean are thought to modulate the background state that the El Niño-Southern Oscillation (ENSO) operates in. This modulation is proposed to impact the frequency and intensity of ENSO events and their teleconnections. We use a high-resolution ocean model to investigate the volume transports associated with the STC branches along 5 degrees N and 5 degrees S. We find three prominent differences between the Southern hemisphere (SH) STC and the Northern hemisphere (NH) STC: (i) the NH STC varies 26% stronger than the SH STC; (ii) the NH STC appears to lead the SH STC by 3 months which causes the NH and SH STCs to play different roles during the course of El Niño and La Niña events; and (iii) in spite of the relative symmetry of the wind stress trends, the STCs have differing decadal trends, with the SH STC clearly dominating the changes in the post-1993 period. To investigate the mechanisms driving the STC variability, we identify winds that are linearly and nonlinearly related to ENSO to force the ocean model. The hemispheric difference in interannual variance as well as the phase difference between the STCs can be explained with ENSO forcing. Our results suggest ENSO to be an important factor in modulating its own background state, with a prominent role for the winds that are nonlinearly related to ENSO. The decadal trends and their interhemispheric disparity, however, cannot be reproduced by our targeted ENSO experiments.

995. Near surface ocean temperature uncertainty related to initial condition uncertainty

Authors: Robin Tokmakian; Peter Challenor

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-019-04872-4

URL: <https://doi.org/10.1007/s00382-019-04872-4>

Publication Date: 2019-07-16

Volume/Issue: Volume 53, Issue 7-8

References: 40

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

996. Toward Optimization of Rheology in Sea Ice Models through Data Assimilation

Authors: J. N. Stroh (University of Alaska Fairbanks); Gleb Panteleev (Naval Research Laboratory, Stennis Space Center, Mississippi); Max Yaremchuk (Naval Research Laboratory, Stennis Space Center, Mississippi); Oceana Francis (Department of Civil and Environmental Engineering, University of Hawai'i at Mānoa, Honolulu, Hawaii); Richard Allard (Naval Research Laboratory, Stennis Space Center, Mississippi)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-18-0239.1

URL: <https://doi.org/10.1175/jtech-d-18-0239.1>

Publication Date: 2019-07-12

Volume/Issue: Volume 36, Issue 12

References: 68

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research Global, Office of Naval Research Global, Hawai'i Sea Grant, University of Hawai'i

License: N/A

Open Access: N/A

Abstract: Sea ice models that allow for deformation are primarily based on rheological formulations originally developed in the 1970s. In both the original viscoplastic (VP) and elastic-VP schemes, the internal pressure term is modeled as a function of variable sea ice thickness and concentration with spatially and temporally constant empirical parameters for ice strength. This work considers a spatially variable extension of the rheology parameters as well as wind stress in a one-dimensional VP sea ice data assimilation system. In regions of total ice cover, experiments that assimilate synthetic ice-state observations using variable rheological parameters show larger improvements than equivalent experiments using homogeneous parameters. For partially ice-covered regions where internal ice stresses are relatively unimportant, experiments assimilating synthetic sea ice velocity observations demonstrate reasonable reconstruction of spatially variable wind stresses. These results suggest practical benefits for sea ice-state reconstruction and forecasts by using sea ice velocity, thickness, and concentration observations to optimize spatially varying rheological parameters and to improve wind stress forcing.

997. Observational Needs for Improving Ocean and Coupled Reanalysis, S2S Prediction, and Decadal Prediction

Authors: Stephen G. Penny; Santha Akella; Magdalena A. Balmaseda; Philip Browne; James A. Carton; Matthieu Chevallier; Francois Counillon; Catia Domingues; Sergey Frolov; Patrick Heimbach; Patrick Hogan; Ibrahim Hoteit; Doroteaciro Iovino; Patrick Laloyaux; Matthew J. Martin; Simona Masina; Andrew M. Moore; Patricia de Rosnay; Dinand Schepers; Bernadette M. Sloyan; Andrea Storto; Aneesh Subramanian; SungHyun Nam; Frederic Vitart; Chunxue Yang; Yosuke Fujii; Hao Zuo; Terry O'Kane; Paul Sandery; Thomas Moore; Christopher C. Chapman

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00391

URL: <https://doi.org/10.3389/fmars.2019.00391>

Publication Date: 2019-07-11

Volume/Issue: Volume 6

References: 180

Cited by: 25 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

998. Interannual to Decadal Variability of Upper-Ocean Salinity in the Southern Indian Ocean and the Role of the Indonesian Throughflow

Authors: Shijian Hu (Second Institute of Oceanography, Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Ying Zhang (CSIRO Oceans and Atmosphere, Crawley, Western Australia, Australia, Chinese Academy of Sciences); Ming Feng (CSIRO Oceans and Atmosphere, Crawley, Western Australia, Australia, Centre for Southern Hemisphere Oceans Research, CSIRO Oceans and Atmosphere, Hobart, Tasmania, Australia); Yan Du (Chinese Academy of Sciences); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego); Fan Wang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Dunxin Hu (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Qiang Xie (Chinese Academy of Sciences); Fei Chai (Second Institute of Oceanography)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0056.1

URL: <https://doi.org/10.1175/jcli-d-19-0056.1>

Publication Date: 2019-07-08

Volume/Issue: Volume 32, Issue 19

References: 67

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, Climate Program Office

License: N/A

Open Access: N/A

Abstract: Variability of oceanic salinity, an indicator of the global hydrological cycle, plays an important role in the basin-scale ocean circulation. In this study, interannual to decadal variability of salinity in the upper layer of the Indian Ocean is investigated using Argo observations since 2004 and data assimilating model outputs (1992-2015). The southeastern Indian Ocean shows the strongest interannual to decadal variability of upper-ocean salinity in the Indian Ocean. Westward propagation of salinity anomalies along isopycnal surfaces is detected in the southern Indian Ocean and attributed to zonal salinity advection anomalies associated with the Indonesian Throughflow and the South Equatorial Current. Composite and salinity budget analyses show that horizontal advection is a major contributor to the interannual to decadal salinity variability of the southern Indian Ocean, and the local air-sea freshwater flux plays a secondary role. The Pacific decadal oscillation (PDO) and El Niño-Southern Oscillation (ENSO) modulate the salinity variability in the southeastern Indian Ocean, with low salinity anomalies occurring during the negative phases of the PDO and ENSO and high salinity anomalies during their positive phases. The Indonesian Throughflow plays an essential role in transmitting the PDO- and ENSO-related salinity signals into the Indian Ocean. A statistical model is proposed based on the PDO index, which successfully predicts the southeastern Indian Ocean salinity variability with a lead time of 10 months.

999. Meridional Overturning and Heat Transport From Argo Floats Displacements and the Planetary Geostrophic Method (PGM): Application to the subpolar North Atlantic

Authors: Alain Colin de Verdière (Institut de Recherche pour le Développement (IRD)); Thomas Meunier (Department of Physical Oceanography Ensenada Center for Scientific Research and Higher Education, CICESE Ensenada Mexico); Michel Ollitrault (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014565

URL: <https://doi.org/10.1029/2018jc014565>

Publication Date: 2019-07-06

Volume/Issue: Volume 124, Issue 8

References: 51

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A direct method is presented to obtain the meridional overturning and heat transport in oceanic basins from observations under the sole assumptions of geostrophy and hydrostatics. The method is made possible because of the rising Argo float displacements database that can provide a reference level at 1000 dbar for the time mean circulation at 1 degrees x 1 degrees resolution. To achieve the overturning and heat transport objectives, the absolute geostrophic time mean circulation must have nondivergent barotropic transports and this requires the solutions of two Poisson equations with suitable boundary conditions, one for the geopotential at 1000 dbar and one for the barotropic stream function. Applied to the subpolar Atlantic for the period 2000-2009, an overturning of 16-18 Sv is found around 40-50 degreesN, and a meridional heat transport of 0.59 petawatt (PW, 1 PW = 10¹⁵ W) is found at 40 degreesN (0.23 PW at 60 degreesN) so that on average ~50 W/m² is exported from ocean to atmosphere to feed the atmospheric storm track. The zonally averaged flow (the overturning) falls short of explaining the observed heat transport, and the barotropic component of the circulation accounts for up to 50% of the heat transport poleward of 55 degreesN. With the rising Argo float database, the method offers high potential to reconstruct the World Ocean time mean circulation and its heat transport away from the equator at higher resolution. The drawback is that it requires in some critical places additional current observations on the shallow shelves that are not sampled by the Argo floats.

1000. GPS-derived geocenter motion from the IGS second reprocessing campaign

Authors: Liansheng Deng; Zhao Li; Na Wei; Yifang Ma; Hua Chen

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-019-1054-2

URL: <https://doi.org/10.1186/s40623-019-1054-2>

Publication Date: 2019-07-05

Volume/Issue: Volume 71, Issue 1

References: 45

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1001. Surface chlorophyll-a variations in the Southeastern Tropical Indian Ocean during various types of the positive Indian Ocean Dipole events

Authors: Q.W. Sari (Sriwijaya University); P.A. Utari (Sriwijaya University); D. Setiabudidaya (Sriwijaya University); I. Yustian (Sriwijaya University); E. Siswanto (Department of Environmental Geochemical Cycle Research, JAMSTEC Yokohama Institute for Earth Sciences (YES), Yokohama, Japan); I. Iskandar (Sriwijaya University)

Journal: International Journal of Remote Sensing

Publisher: Informa UK Limited

DOI: 10.1080/01431161.2019.1637962

URL: <https://doi.org/10.1080/01431161.2019.1637962>

Publication Date: 2019-07-04

Volume/Issue: Volume 41, Issue 1

References: 27

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1002. Global Estimates of the Energy Transfer From the Wind to the Ocean, With Emphasis on Near-Inertial Oscillations

Authors: M. Mar Flexas (California Institute of Technology); Andrew F. Thompson (California Institute of Technology); Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Patrice Klein (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); J. Thomas Farrar (Woods Hole Oceanographic Institution); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, University of California Los Angeles); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014453

URL: <https://doi.org/10.1029/2018jc014453>

Publication Date: 2019-07-03

Volume/Issue: Volume 124, Issue 8

References: 78

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: wind power, kinetic energy budget, global ocean model, surface fluxes, inertial oscillations (and 1 more)

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: Estimates of the kinetic energy transfer from the wind to the ocean are often limited by the spatial and temporal resolution of surface currents and surface winds. Here we examine the wind work in a pair of global, very high-resolution (1/48 degrees and 1/24 degrees) MIT general circulation model simulations in Latitude-Longitude-polar Cap (LLC) configuration that provide hourly output at spatial resolutions of a few kilometers and include tidal forcing. A cospectrum analysis of wind stress and ocean surface currents shows positive contribution at large scales (>300 km) and near-inertial frequency and negative contribution from mesoscales, tidal frequencies, and internal gravity waves. Larger surface kinetic energy fluxes are in the Kuroshio in winter at large scales (40 mW/m²) and mesoscales (-30 mW/m²). The Kerguelen region is dominated by large scale (~20 mW/m²), followed by inertial oscillations in summer (13 mW/m²) and mesoscale in winter (-12 mW/m²). Kinetic energy fluxes from internal gravity waves (-0.1 to -9.9 mW/m²) are generally stronger in summer. Surface kinetic energy fluxes in the LLC simulations are 4.71 TW, which is 25-85% higher than previous global estimates from coarser (1/6-1/10 degrees) general ocean circulation models; this is likely due to improved representation of wind variability (6-hourly, 0.14 degrees, operational European Center for Medium-Range Weather Forecasts). However, the low wind power input to the near-inertial frequency band obtained with LLC (0.16 TW) compared to global slab models suggests that wind variability on time scales less than 6 hr and spatial scales less than 15 km are critical to better representing the wind power input in ocean circulation models.

1003. Using Sea Surface Temperature Observations to Constrain Upper Ocean Properties in an Arctic Sea Ice-Ocean Data Assimilation System

Authors: Xi Liang (National Marine Environmental Forecasting Center); Martin Losch (Alfred Wegener Institute); Lars Nerger (Alfred Wegener Institute); Longjiang Mu (Alfred Wegener Institute); Qinghua Yang (Sun Yat-Sen University, Chinese Academy of Sciences, Southern Laboratory of Ocean Science and Engineering (Guangdong, Zhuhai) Zhuhai China); Chengyan Liu (Hohai University, Nanjing University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015073

URL: <https://doi.org/10.1029/2019jc015073>

Publication Date: 2019-07-01

Volume/Issue: Volume 124, Issue 7

References: 52

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: sea ice edge, data assimilation, sea ice thickness, upper ocean temperature, sea ice concentration (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sea ice data assimilation can greatly improve forecasts of Arctic sea ice evolution. Many previous sea ice data assimilation studies were conducted without assimilating ocean state variables, even though the sea ice evolution is closely linked to the oceanic conditions, both dynamically and thermodynamically. Based on the method of a localized ensemble error subspace transform Kalman filter, satellite-retrieved sea ice concentration and sea ice thickness are assimilated into an Arctic sea ice-ocean model. As a new addition, sea surface temperature (SST) data are also assimilated. The additional assimilation of SST improves not only the simulated ocean temperature in the mixed layer of the ocean substantially but also the accuracy of sea ice edge position, sea ice extent, and sea ice thickness in the marginal sea ice zone. The improvement in the simulated potential temperature in the upper 1,000 m can be attributed to the enhanced vertical convection processes in the regions where the assimilated observational SST is colder than the simulated SST without assimilation. The improvements in the sea ice edge position and sea ice thickness simulations are primarily caused by the SST data assimilation reducing biases in the simulated SST and the associated coupled ocean-sea ice processes. Our investigation suggests that, due to the complex interaction between the sea ice and ocean, assimilating ocean data should be an indispensable component of numerical polar sea ice forecasting systems.

1004. A Sustained Ocean Observing System in the Indian Ocean for Climate Related Scientific Knowledge and Societal Needs

Authors: J. C. Hermes; Y. Masumoto; L. M. Beal; M. K. Roxy; J. Vialard; M. Andres; H. Annamalai; S. Behera; N. D'Adamo; T. Doi; M. Feng; W. Han; N. Hardman-Mountford; H. Hendon; R. Hood; S. Kido; C. Lee; T. Lee; M. Lengaigne; J. Li; R. Lumpkin; K. N. Navaneeth; B. Milligan; M. J. McPhaden; M. Ravichandran; T. Shinoda; A. Singh; B. Sloyan; P. G. Strutton; A. C. Subramanian; S. Thurston; T. Tozuka; C. C. Ummerhofer; A. S. Unnikrishnan; R. Venkatesan; D. Wang; J. Wiggert; L. Yu; W. Yu

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00355

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Publication Date: 2019-06-28

Volume/Issue: Volume 6

References: 160

Cited by: 51 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1005. Some Expectations for Submesoscale Sea Surface Height Variance Spectra

Authors: Jörn Callies (California Institute of Technology); Weiguang Wu (University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-18-0272.1

URL: <https://doi.org/10.1175/jpo-d-18-0272.1>

Publication Date: 2019-06-28

Volume/Issue: Volume 49, Issue 9

References: 89

Cited by: 15 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In anticipation of the Surface Water and Ocean Topography (SWOT) wide-swath altimetry mission, this study reviews expectations for sea surface height (SSH) variance spectra at wavelengths of 10-100 km. Kinetic energy spectra from in situ observations and numerical simulations indicate that SSH variance spectra associated with balanced flow drop off steeply with wavenumber, with at least the negative fourth power of the wavenumber. Such a steep drop-off implies that even drastic reductions in altimetry noise yield only a modest improvement in the resolution of balanced flow. This general expectation is made concrete by extrapolating SSH variance spectra from existing altimetry to submesoscales, the results of which suggest that in the extratropics (poleward of 20 degrees latitude) SWOT will improve the resolution from currently about 100 km to a median of 51 or 74 km, depending on whether or not submesoscale balanced flows are energetic. Internal waves, in contrast to balanced flow, give rise to SSH variance spectra that drop off relatively gently with wavenumber, so SSH variance should become strongly dominated by internal waves in the submesoscale range. In situ observations of the internal-wave field suggest that the internal-wave signal accessible by SWOT will be largely dominated by internal tides. The internal-wave continuum is estimated to have a spectral level close to but somewhat lower than SWOT's expected noise level.

1006. Interannual Sea Level Variability Along the Southeastern Seaboard of the United States in Relation to the Gyre-Scale Heat Divergence in the North Atlantic

Authors: Denis L. Volkov (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami FL USA, University of Miami); Sang-Ki Lee (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami FL USA); Ricardo Domingues (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami FL USA, University of Miami); Hong Zhang (University of California Los Angeles); Marlos Goes (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami FL USA, University of Miami)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl083596

URL: <https://doi.org/10.1029/2019gl083596>

Publication Date: 2019-06-20

Volume/Issue: Volume 46, Issue 13

References: 61

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: The low-amplitude, large-scale, interannual, and longer-term sea level changes are linked to the variations of ocean heat and freshwater content and strongly controlled by ocean dynamics. Near the coast, especially in low-lying and flood-vulnerable regions, these changes can provide background conditions favorable for the occurrence of extreme sea levels that represent a threat for coastal communities and ecosystems. In this study, we identify a tripole mode of the ocean gyre-scale sea surface height variability in the North Atlantic and show that this mode is responsible for most of the interannual-to-decadal sea surface height changes along the southeast coast of the United States, including the Gulf of Mexico. We also show that these changes are largely driven by the large-scale heat divergence related to the Atlantic Meridional Overturning Circulation and linked to the low-frequency North Atlantic Oscillation.

1007. Poleward Shift of the Pacific North Equatorial Current Bifurcation

Authors: Haihong Guo (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Zhaohui Chen (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Haiyuan Yang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015019

URL: <https://doi.org/10.1029/2019jc015019>

Publication Date: 2019-06-18

Volume/Issue: Volume 124, Issue 7

References: 43

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: poleward, bifurcation, shift

Funding: National Natural Science Foundation of China

License: Creative Commons

Open Access: N/A

Abstract: The dynamics of the poleward shift of the Pacific North Equatorial Current bifurcation latitude (NBL) is studied using a 5.5-layer reduced gravity model. It is found that the poleward shift of the NBL is associated with the asymmetric intensity of the wind stress curl input to the Pacific tropical and subtropical gyres. Stronger wind stress curl in the subtropical gyre leads to equatorward transport in the interior upper ocean across the boundary between the two gyres, causing a poleward transport compensation at the western boundary. In the lower layer ocean, in turn, there is poleward (equatorward) transport at the interior (western boundary) due to Sverdrup balance which requires zero transport at the gyre boundary where zonally integrated wind stress curl is zero. Therefore, the NBL exhibits a tilting feature, with its position being more equatorward in the upper layer and more poleward in the lower layer. The equatorial currents bifurcations in other basins are also characterized by the poleward tilting vertical structure. The wind stress curl over the subtropical gyre is generally stronger than that over the tropical gyre, resulting in the bifurcations shifting poleward with increasing depth.

1008. Characterizing the Seasonal Hydrological Loading Over the Asian Continent Using GPS, GRACE, and Hydrological Model

Authors: Yunfei Xiang; Jianping Yue; Kanglin Cong; Yin Xing; Dongjian Cai

Journal: Pure and Applied Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00024-019-02251-y

URL: <https://doi.org/10.1007/s00024-019-02251-y>

Publication Date: 2019-06-18

Volume/Issue: Volume 176, Issue 11

References: 42

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fundamental Research Funds for the Central Universities, Postgraduate Research & Practice Innovation Program of Jiangsu Province, National Key R&D; Program of China

License: N/A

Open Access: N/A

Abstract: N/A

1009. Variability in the Northern North Atlantic and Arctic Oceans Across the Last Two Millennia: A Review

Authors: P. Moffa-Sánchez (Department of Geography Durham University Durham UK, Cardiff University); E. Moreno-Chamarro (Barcelona Supercomputing Center); D. J. Reynolds (University of Arizona, Cardiff University); P. Ortega (Barcelona Supercomputing Center); L. Cunningham (Department of Geography University of Portsmouth Portsmouth UK); D. Swingedouw (Environnements et Paléoenvironnements Océaniques et Continentaux (EPOC), CNRS-Université de Bordeaux Pessac France); D. E. Amrhein (University of Washington); J. Halfar (University of Toronto); L. Jonkers (University of Bremen); J. H. Jungclaus (Max Planck Institute for Meteorology); K. Perner (Leibniz Institute for Baltic Sea Research); A. Wanamaker (Iowa State University); S. Yeager (National Center for Atmospheric Research)

Journal: Paleoceanography and Paleoclimatology

Publisher: American Geophysical Union (AGU)

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References: 269

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020 Framework Programme, Natural Environment Research Council, National Science Foundation (and 1 more)

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Open Access: N/A

Abstract: The climate of the last two millennia was characterized by decadal to multicentennial variations, which were recorded in terrestrial records and had important societal impacts. The cause of these climatic events is still under debate, but changes in the North Atlantic circulation have often been proposed to play an important role. In this review we compile available high-resolution paleoceanographic data sets from the northern North Atlantic and Nordic Seas. The records are grouped into regions related to modern ocean conditions, and their variability is discussed. We additionally discuss our current knowledge from modeling studies, with a specific focus on the dynamical changes that are not well inferred from the proxy records. An illustration is provided through the analysis of two climate model ensembles and an individual simulation of the last millennium. This review thereby provides an up-to-date paleoperspective on the North Atlantic multidecadal to multicentennial ocean variability across the last two millennia.

1010. Insights into Decadal North Atlantic Sea Surface Temperature and Ocean Heat Content Variability from an Eddy-Permitting Coupled Climate Model

Authors: B. I. Moat (University of Southampton, National Oceanography Centre); B. Sinha (University of Southampton, National Oceanography Centre); S. A. Josey (University of Southampton, National Oceanography Centre); J. Robson (University of Reading); P. Ortega (Barcelona Supercomputing Center); F. Sévellec (Institut de Recherche pour le Développement (IRD)); N. P. Holliday (University of Southampton, National Oceanography Centre); G. D. McCarthy (National University of Ireland, Maynooth, Maynooth, Ireland); A. L. New (University of Southampton, National Oceanography Centre); J. J.-M. Hirschi (University of Southampton, National Oceanography Centre)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0709.1

URL: <https://doi.org/10.1175/jcli-d-18-0709.1>

Publication Date: 2019-06-14

Volume/Issue: Volume 32, Issue 18

References: 59

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Natural Environment Research Council, Natural Environment Research Council (and 3 more)

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Open Access: N/A

Abstract: An ocean mixed layer heat budget methodology is used to investigate the physical processes determining subpolar North Atlantic (SPNA) sea surface temperature (SST) and ocean heat content (OHC) variability on decadal to multidecadal time scales using the state-of-the-art climate model HadGEM3-GC2. New elements include development of an equation for evolution of anomalous SST for interannual and longer time scales in a form analogous to that for OHC, parameterization of the diffusive heat flux at the base of the mixed layer, and analysis of a composite Atlantic meridional overturning circulation (AMOC) event. Contributions to OHC and SST variability from two sources are evaluated: 1) net ocean-atmosphere heat flux and 2) all other processes, including advection, diffusion, and entrainment for SST. Anomalies in OHC tendency propagate anticlockwise around the SPNA on multidecadal time scales with a clear relationship to the phase of the AMOC. AMOC anomalies lead SST tendencies, which in turn lead OHC tendencies in both the eastern and western SPNA. OHC and SST variations in the SPNA on decadal time scales are dominated by AMOC variability because it controls variability of advection, which is shown to be the dominant term in the OHC budget. Lags between OHC and SST are traced to differences between the advection term for OHC and the advection-entrainment term for SST. The new results have implications for interpretation of variations in Atlantic heat uptake in the CMIP6 climate model assessment.

1011. Ligand Binding Strength Explains the Distribution of Iron in the North Atlantic Ocean

Authors: Anh L. D. Pham (Georgia Institute of Technology); Takamitsu Ito (Georgia Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl083319

URL: <https://doi.org/10.1029/2019gl083319>

Publication Date: 2019-06-10

Volume/Issue: Volume 46, Issue 13

References: 46

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Observations of dissolved iron (dFe) in the subtropical North Atlantic revealed remarkable features: While the near-surface dFe concentration is low despite receiving high dust deposition, the subsurface dFe concentration is high. We test several hypotheses that might explain this feature in an ocean biogeochemistry model with a refined Fe cycling scheme. These hypotheses invoke a stronger lithogenic scavenging rate, rapid biological uptake, and a weaker binding between Fe and a ubiquitous, refractory ligand. While the standard model overestimates the surface dFe concentration, a 10-time stronger biological uptake run causes a slight reduction in the model surface dFe. A tenfold decrease in the binding strength of the refractory ligand, suggested by recent observations, starts reproducing the observed dFe pattern, with a potential impact for the global nutrient distribution. An extreme value for the lithogenic scavenging rate can also match the model dFe with observations, but this process is still poorly constrained.

1012. Atlantic Meridional Overturning Circulation: Observed Transport and Variability

Authors: Eleanor Frajka-Williams; Isabelle J. Ansorge; Johanna Baehr; Harry L. Bryden; Maria Paz Chidichimo; Stuart A. Cunningham; Gokhan Danabasoglu; Shenfu Dong; Kathleen A. Donohue; Shane Elipot; Patrick Heimbach; N. Penny Holliday; Rebecca Hummels; Laura C. Jackson; Johannes Karstensen; Matthias Lankhorst; Isabela A. Le Bras; M. Susan Lozier; Elaine L. McDonagh; Christopher S. Meinen; Herlé Mercier; Bengamin I. Moat; Renellys C. Perez; Christopher G. Piecuch; Monika Rhein; Meric A. Srokosz; Kevin E. Trenberth; Sheldon Bacon; Gael Forget; Gustavo Goni; Dagmar Kieke; Jannes Koelling; Tarron Lamont; Gerard D. McCarthy; Christian Mertens; Uwe Send; David A. Smeed; Sabrina Speich; Marcel van den Berg; Denis Volkov; Chris Wilson

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00260

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Publication Date: 2019-06-07

Volume/Issue: Volume 6

References: 122

Cited by: 146 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, National Natural Science Foundation of China, Bundesministerium für Bildung und Forschung (and 4 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1013. Rapid Decline of Total Antarctic Sea Ice Extent during 2014-16 Controlled by Wind-Driven Sea Ice Drift

Authors: Zhaomin Wang (Hohai University); John Turner (British Antarctic Survey, Cambridge, United Kingdom); Yang Wu (Hohai University); Chengyan Liu (Hohai University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0635.1

URL: <https://doi.org/10.1175/jcli-d-18-0635.1>

Publication Date: 2019-06-07

Volume/Issue: Volume 32, Issue 17

References: 54

Cited by: 40 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Ministry of Science and Technology of the People's Republic of China, Fundamental Research Funds for the Central Universities, Fundamental Research Funds for the Central Universities (and 2 more)

License: N/A

Open Access: N/A

Abstract: Between 2014 and 2016 the annual mean total extent of Antarctic sea ice decreased by a record, unprecedented amount of 1.6×10^6 km², the largest in a record starting in the late 1970s. The mechanisms behind such a rapid decrease remain unknown. Using the outputs of a high-resolution, global ocean-sea ice model we show that the change was predominantly a result of record atmospheric low pressure systems over sectors of the Southern Ocean in 2016, with the associated winds inducing strong sea ice drift. Regions of large positive and negative sea ice extent anomaly were generated by both thermal and dynamic effects of the wind anomalies. Although the strong wind forcing also generated the warmest ocean surface state from April to December 2016, we show that enhanced northward sea ice drift and hence increased melting at lower latitudes driven by strong winds made the dominant contribution to the large decrease in total Antarctic sea ice extent between 2014 and 2016.

1014. New Perspectives on the Generation and Maintenance of the Kuroshio Large Meander

Authors: Yang Yang (Nanjing University); X. San Liang (Nanjing University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-18-0276.1

URL: <https://doi.org/10.1175/jpo-d-18-0276.1>

Publication Date: 2019-06-05

Volume/Issue: Volume 49, Issue 8

References: 70

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, National Program on Global Change and Air-Sea Interaction (and 4 more)

License: N/A

Open Access: N/A

Abstract: The internal dynamical processes underlying the Kuroshio large meander are investigated using a recently developed analysis tool, multiscale window transform (MWT), and the MWT-based canonical transfer theory. Oceanic fields are reconstructed on a low-frequency mean flow window, a mesoscale eddy window, and a high-frequency synoptic window with reference to the three typical path states south of Japan, that is, the typical large meander (tLM), nearshore non-large meander (nNLM), and offshore non-large meander (oNLM) path states. The interactions between the scale windows are quantitatively evaluated in terms of canonical transfer, which bears a Lie bracket form and conserves energy in the space of scale. In general, baroclinic (barotropic) instability is strengthened (weakened) during the tLM state. For the first time we found a spatially coherent inverse cascade of kinetic energy (KE) from the synoptic eddies to the slowly varying mean flow; it occupies the whole large meander region but exists only in the tLM state. By the time-varying multiscale energetics, a typical large meander is preceded by a strong influx of mesoscale eddy energy from upstream with a cyclonic eddy, which subsequently triggers a strong inverse KE cascade from the mesoscale window to the mean flow window to build up the KE reservoir for the meander. Synoptic frontal eddies are episodically intensified due to the baroclinic instability of the meander, but they immediately feed back to the mean flow window through inverse KE cascade. These results highlight the important role played by inverse KE cascades in generating and maintaining the Kuroshio large meander.

1015. Acceleration and Overturning of the Antarctic Slope Current by Winds, Eddies, and Tides

Authors: Andrew L. Stewart (University of California Los Angeles); Andreas Klocker (Antarctic Climate and Ecosystems Cooperative Research Centre, University of Tasmania); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-18-0221.1

URL: <https://doi.org/10.1175/jpo-d-18-0221.1>

Publication Date: 2019-06-03

Volume/Issue: Volume 49, Issue 8

References: 91

Cited by: 40 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Directorate for Geosciences, Directorate for Geosciences, Directorate for Geosciences (and 1 more)

License: N/A

Open Access: N/A

Abstract: All exchanges between the open ocean and the Antarctic continental shelf must cross the Antarctic Slope Current (ASC). Previous studies indicate that these exchanges are strongly influenced by mesoscale and tidal variability, yet the mechanisms responsible for setting the ASC's transport and structure have received relatively little attention. In this study the roles of winds, eddies, and tides in accelerating the ASC are investigated using a global ocean-sea ice simulation with very high resolution (1/48 degrees grid spacing). It is found that the circulation along the continental slope is accelerated both by surface stresses, ultimately sourced from the easterly winds, and by mesoscale eddy vorticity fluxes. At the continental shelf break, the ASC exhibits a narrow (~30-50 km), swift ($>0.2 \text{ m s}^{-1}$) jet, consistent with in situ observations. In this jet the surface stress is substantially reduced, and may even vanish or be directed eastward, because the ocean surface speed matches or exceeds that of the sea ice. The shelfbreak jet is shown to be accelerated by tidal momentum advection, consistent with the phenomenon of tidal rectification. Consequently, the shoreward Ekman transport vanishes and thus the mean overturning circulation that steepens the Antarctic Slope Front (ASF) is primarily due to tidal acceleration. These findings imply that the circulation and mean overturning of the ASC are not only determined by near-Antarctic winds, but also depend crucially on sea ice cover, regionally-dependent mesoscale eddy activity over the continental slope, and the amplitude of tidal flows across the continental shelf break.

1016. Assessment of the interannual variability of local atmospheric and ITF contribution to the subsurface heat content of southern tropical Indian Ocean in GECCO2 and ORAS4 using ROMS

Authors: T. Jayasankar; T.I. Eldho; S. Ghosh; R. Murtugudde

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2019.05.014

URL: <https://doi.org/10.1016/j.gloplacha.2019.05.014>

Publication Date: 2019-05-31

Volume/Issue: Volume 181

References: 63

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Indian Institute of Technology, Bombay

License: N/A

Open Access: N/A

Abstract: N/A

1017. GOCE-Derived Coseismic Gravity Gradient Changes Caused by the 2011 Tohoku-Oki Earthquake

Authors: Xinyu Xu (Wuhan University); Hao Ding (Wuhan University); Yongqi Zhao (Wuhan University); Jin Li (Chinese Academy of Sciences); Minzhang Hu (China Earthquake Administration)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs11111295

URL: <https://doi.org/10.3390/rs11111295>

Publication Date: 2019-05-30

Volume/Issue: Volume 11, Issue 11

References: 62

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, DAAD Thematic Network Project, The Major Project of High resolution Earth Observation System

License: CC BY 4.0

Open Access: N/A

Abstract: In contrast to most of the coseismic gravity change studies, which are generally based on data from the Gravity field Recovery and Climate Experiment (GRACE) satellite mission, we use observations from the Gravity field and steady-state Ocean Circulation Explorer (GOCE) Satellite Gravity Gradient (SGG) mission to estimate the coseismic gravity and gravity gradient changes caused by the 2011 Tohoku-Oki Mw 9.0 earthquake. We first construct two global gravity field models up to degree and order 220, before and after the earthquake, based on the least-squares method, with a bandpass Auto Regression Moving Average (ARMA) filter applied to the SGG data along the orbit. In addition, to reduce the influences of colored noise in the SGG data and the polar gap problem on the recovered model, we propose a tailored spherical harmonic (TSH) approach, which only uses the spherical harmonic (SH) coefficients with the degree range 30-95 to compute the coseismic gravity changes in the spatial domain. Then, both the results from the GOCE observations and the GRACE temporal gravity field models (with the same TSH degrees and orders) are simultaneously compared with the forward-modeled signals that are estimated based on the fault slip model of the earthquake event. Although there are considerable misfits between GOCE-derived and modeled gravity gradient changes (ΔV_{xx} , ΔV_{yy} , ΔV_{zz} , and ΔV_{xz}), we find analogous spatial patterns and a significant change (greater than 3sigma) in gravity gradients before and after the earthquake. Moreover, we estimate the radial gravity gradient changes from the GOCE-derived monthly time-variable gravity field models before and after the earthquake, whose amplitudes are at a level over three times that of their corresponding uncertainties, and are thus significant. Additionally, the results show that the recovered coseismic gravity signals in the west-to-east direction from GOCE are closer to the modeled signals than those from GRACE in the TSH degree range 30-95. This indicates that the GOCE-derived gravity models might be used as additional observations to infer/explain some time-variable geophysical signals of interest.

1018. LSCE-FFNN-v1: a two-step neural network model for the reconstruction of surface ocean pCO₂ over the global ocean

Authors: Anna Denvil-Sommer; Marion Gehlen; Mathieu Vrac; Carlos Mejia

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-12-2091-2019

URL: <https://doi.org/10.5194/gmd-12-2091-2019>

Publication Date: 2019-05-29

Volume/Issue: Volume 12, Issue 5

References: 47

Cited by: 82 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . A new feed-forward neural network (FFNN) model is presented to reconstruct surface ocean partial pressure of carbon dioxide (pCO₂) over the global ocean. The model consists of two steps: (1) the reconstruction of pCO₂ climatology, and (2) the reconstruction of pCO₂ anomalies with respect to the climatology. For the first step, a gridded climatology was used as the target, along with sea surface salinity (SSS), sea surface temperature (SST), sea surface height (SSH), chlorophyll a (Chl a), mixed layer depth (MLD), as well as latitude and longitude as predictors. For the second step, data from the Surface Ocean CO₂ Atlas (SOCAT) provided the target. The same set of predictors was used during step (2) augmented by their anomalies. During each step, the FFNN model reconstructs the nonlinear relationships between pCO₂ and the ocean predictors. It provides monthly surface ocean pCO₂ distributions on a 1°x1° grid for the period from 2001 to 2016. Global ocean pCO₂ was reconstructed with satisfying accuracy compared with independent observational data from SOCAT. However, errors were larger in regions with poor data coverage (e.g., the Indian Ocean, the Southern Ocean and the subpolar Pacific). The model captured the strong interannual variability of surface ocean pCO₂ with reasonable skill over the equatorial Pacific associated with ENSO (the El Niño-Southern Oscillation). Our model was compared to three pCO₂ mapping methods that participated in the Surface Ocean pCO₂ Mapping intercomparison (SOCOM) initiative. We found a good agreement in seasonal and interannual variability between the models over the global ocean. However, important differences still exist at the regional scale, especially in the Southern Hemisphere and, in particular, in the southern Pacific and the Indian Ocean, as these regions suffer from poor data coverage. Large regional uncertainties in reconstructed surface ocean pCO₂ and sea-air CO₂ fluxes have a strong influence on global estimates of CO₂ fluxes and trends.

1019. Subtropical-Tropical Transfer in the South Atlantic Ocean

Authors: A. Cabré (Institut de Ciències del Mar, CSIC Barcelona Spain); J. L. Pelegrí (Institut de Ciències del Mar, CSIC Barcelona Spain); I. Vallès-Casanova (Institut de Ciències del Mar, CSIC Barcelona Spain)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc015160

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Publication Date: 2019-05-25

Volume/Issue: Volume 124, Issue 7

References: 39

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Here we explore the water transfer between the subtropical and tropical gyres of the South Atlantic Ocean to better understand its unique equatorward heat delivery. A Lagrangian technique is applied to the reanalysis product GLORYS2V4 in order to trace back the western boundary flow in the tropical (North Brazil Undercurrent, NBUC) and subtropical (Brazil Current) gyres. Most of the northward NBUC core transport (14.9 Sv at 8 degreesS) arrives from the eastern boundary subtropical current (Benguela Current) via the zonal South Equatorial Current. This subtropical-tropical transfer represents the core of the returning limb of the Atlantic meridional overturning circulation and accounts for most of the observed increase in heat and salt-volume transports (0.18 PW and 0.19 Sv from 30 degreesS to 8 degreesS, respectively) across the South Atlantic. The NBUC also includes Antarctic Intermediate Water below 400 m (7.4 Sv at 8 degreesS) coming from the interior subtropical gyre, as well as water from the current's surface and peripheral components coming from the tropical gyre (13.3 Sv at 8 degreesS). The Brazil Current (9.9 Sv at 29 degreesS) is mostly composed of subtropical water originating in the upper 800 m west of the eastern boundary current at 30 degreesS (8.5 Sv), with a minor contribution of surface tropical water that transfers to the subtropics (1.4 Sv).

1020. Interannual Variability of Antarctic Intermediate Water in the Tropical North Atlantic

Authors: Yao Fu (Chinese Academy of Sciences); Chunzai Wang (Chinese Academy of Sciences); Peter Brandt (GEOMAR Helmholtz Centre for Ocean Research, Faculty of Mathematics and Natural Sciences Christian-Albrechts-Universität zu Kiel Kiel Germany); Richard J. Greatbatch (GEOMAR Helmholtz Centre for Ocean Research, Faculty of Mathematics and Natural Sciences Christian-Albrechts-Universität zu Kiel Kiel Germany)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014878

URL: <https://doi.org/10.1029/2018jc014878>

Publication Date: 2019-05-22

Volume/Issue: Volume 124, Issue 6

References: 54

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: zonal propagating AAIW anomalies related to baroclinic Rossby wave propagation, interannual variability of Antarctic Intermediate Water in the tropical North Atlantic, meridional propagating AAIW salinity anomalies associated with the western boundary transport

Funding: Deutsche Forschungsgemeinschaft, Chinese Academy of Sciences, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Interannual variability of Antarctic Intermediate Water (AAIW) in the tropical North Atlantic is investigated using the GECCO2 ocean state estimate and Argo data. AAIW salinity variability near the western boundary is highly correlated with the transport along the western boundary on interannual timescales. Northward propagating anomalies are associated with the western boundary transport variability that, to some extent, is related to the large-scale wind stress curl forcing by means of the Sverdrup balance. AAIW anomalies also propagate westward with the speed of baroclinic Rossby waves, indicating that the displacement of the meridional salinity gradient by westward propagation of baroclinic Rossby waves plays a role in the variability of AAIW characteristics. Slower eastward spreading of AAIW anomalies is identified on decadal timescales likely associated with the advection of salinity anomalies by weak eastward current bands. Understanding the observed interannual and decadal variability of AAIW salinity is important to properly interpret salinity changes reported in response to changes in the hydrological cycle.

1021. Satellite Salinity Observing System: Recent Discoveries and the Way Forward

Authors: Nadya Vinogradova; Tong Lee; Jacqueline Boutin; Kyla Drushka; Severine Fournier; Roberto Sabia; Detlef Stammer; Eric Bayler; Nicolas Reul; Arnold Gordon; Oleg Melnichenko; Laifang Li; Eric Hackert; Matthew Martin; Nicolas Kolodziejczyk; Audrey Hasson; Shannon Brown; Sidharth Misra; Eric Lindstrom

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00243

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Publication Date: 2019-05-22

Volume/Issue: Volume 6

References: 217

Cited by: 144 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1022. Modelling of electromagnetic signatures of global ocean circulation: physical approximations and numerical issues

Authors: Libor Šachl; Zdeněk Martinec; Jakub Velímský; Christopher Irrgang; Johannes Petereit; Jan Saynisch; David Einšpigel; Neesha Regmi Schnepf

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-019-1033-7

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References: 45

Cited by: 14 publications

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Subjects: N/A

Keywords: N/A

Funding: Ministerstvo Školství, Mládeže a Tělovýchovy, Science Foundation Ireland, Grantová Agentura České Republiky (and 3 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1023. The Atlantic inflow across the Greenland-Scotland ridge in global climate models (CMIP5)

Authors: Céline Heuzé (University of Gothenburg); Marius Årthun (University of Bergen, Bjerknes Centre for Climate Research)

Journal: Elementa: Science of the Anthropocene

Publisher: University of California Press

DOI: 10.1525/elementa.354

URL: <https://doi.org/10.1525/elementa.354>

Publication Date: 2019-05-21

Volume/Issue: Volume 7

References: 65

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Oceanic heat transport from the North Atlantic to the Arctic through the Nordic Seas is a key component of the climate system that has to be modelled accurately in order to predict, for example, future Arctic sea ice changes or European climate. Here we quantify biases in the climatological state and dynamics of the transport of oceanic heat into the Nordic Seas across the Greenland-Scotland ridge in 23 state-of-the-art global climate models that participated in the Climate Model Intercomparison Project phase 5. The mean poleward heat transport, its seasonal cycle and interannual variability are inconsistently represented across these models, with a vast majority underestimating them and a few models greatly overestimating them. The main predictor for these biases is the resolution of the model via its representation of the Greenland-Scotland ridge bathymetry: the higher the resolution, the larger the heat transport through the section. The second predictor is the large-scale ocean circulation, which is also connected to the bathymetry: models with the largest heat transport import water from the European slope current into all three straits of the Greenland-Scotland ridge, whereas those with a weak transport import water from the Labrador Sea. The third predictor is the spatial pattern of their main atmospheric modes of variability (North Atlantic Oscillation, East Atlantic and Scandinavian patterns), where the models with a weak inflow have their atmospheric low-pressure centre shifted south towards the central Atlantic. We argue that the key to a better representation of the large-scale oceanic heat transport from the North Atlantic to the Arctic in global models resides not only in higher resolution, but also in a better bathymetry and representation of the complex ocean-ice-atmosphere interactions.

1024. AMOC sensitivity to surface buoyancy fluxes: the role of air-sea feedback mechanisms

Authors: Yavor Kostov; Helen L. Johnson; David P. Marshall

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-019-04802-4

URL: <https://doi.org/10.1007/s00382-019-04802-4>

Publication Date: 2019-05-21

Volume/Issue: Volume 53, Issue 7-8

References: 48

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1025. Annihilation of the Somali upwelling system during summer monsoon

Authors: Abhisek Chatterjee; B. Praveen Kumar; Satya Prakash; Purna Singh

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-019-44099-1

URL: <https://doi.org/10.1038/s41598-019-44099-1>

Publication Date: 2019-05-20

Volume/Issue: Volume 9, Issue 1

References: 51

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Somali upwelling system during northern summer is believed to be the largest upwelling region in the Indian Ocean and has motivated some of the early studies on the Indian Ocean. Here we present results from observations and ocean model to show that the upwelling along the Somali coast is limited to the early phase of the summer monsoon and later primarily limited to the eddy dominated flows in the northern and some extent in the southern part of the coast. Major part of the Somali coast (~60% of the entire coastal length) shows prominent downwelling features driven by offshore negative windstress curl and subsurface entrainment mixing. Further, we show that the surface cooling of coastal waters are dominantly driven by subsurface entrainment and surface heat fluxes. These findings not only augment the existing knowledge of the Somali upwelling system, but also have serious implications on the regional climate. Most importantly, our analysis underscores the use of alongshore winds only to project future (climate driven) changes in the upwelling intensity along this coast.

1026. Pacific modulation of accelerated south Indian Ocean sea level rise during the early 21st Century

Authors: J. Jyoti; P. Swapna; R. Krishnan; C. V. Naidu

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-019-04795-0

URL: <https://doi.org/10.1007/s00382-019-04795-0>

Publication Date: 2019-05-17

Volume/Issue: Volume 53, Issue 7-8

References: 80

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1027. Seasonal and Interannual Variability of Bering Strait Throughflow from AO-FVCOM and Observation

Authors: Yongfei Deng; Guoping Gao; Yu Zhang; Changsheng Chen

Journal: Journal of Ocean University of China

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11802-019-3906-6

URL: <https://doi.org/10.1007/s11802-019-3906-6>

Publication Date: 2019-05-14

Volume/Issue: Volume 18, Issue 3

References: 47

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1028. Erosion of the coastal Mekong delta: Assessing natural against man induced processes

Authors: Patrick Marchesiello; Nguyet Minh Nguyen; Nicolas Gratiot; Hubert Loisel; Edward J. Anthony; Cong San Dinh; Thong Nguyen; Rafael Almar; Elodie Kestenare

Journal: Continental Shelf Research

Publisher: Elsevier BV

DOI: 10.1016/j.csr.2019.05.004

URL: <https://doi.org/10.1016/j.csr.2019.05.004>

Publication Date: 2019-05-14

Volume/Issue: Volume 181

References: 53

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: EU, AFD, SIWRR

License: N/A

Open Access: N/A

Abstract: N/A

1029. The Ability of Barotropic Models to Simulate Historical Mean Sea Level Changes from Coastal Tide Gauge Data

Authors: C. G. Piecuch; F. M. Calafat; S. Dangendorf; G. Jordà

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-019-09537-9

URL: <https://doi.org/10.1007/s10712-019-09537-9>

Publication Date: 2019-05-06

Volume/Issue: Volume 40, Issue 6

References: 102

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Woods Hole Oceanographic Institution, Woods Hole Oceanographic Institution

License: N/A

Open Access: N/A

Abstract: N/A

1030. Polar climate system modeling in China: Recent progress and future challenges

Authors: Zhaomin Wang; Dake Chen

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-018-9355-2

URL: <https://doi.org/10.1007/s11430-018-9355-2>

Publication Date: 2019-05-03

Volume/Issue: Volume 62, Issue 7

References: 140

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1031. Evolution of Sea Surface Salinity Anomalies in the Southwestern Tropical Indian Ocean During 2010-2011 Influenced by a Negative IOD Event

Authors: Qiwei Sun (Chinese Academy of Sciences); Yan Du (Chinese Academy of Sciences); Yuhong Zhang (Chinese Academy of Sciences); Ming Feng (CSIRO Oceans and Atmosphere Crawley Western Australia Australia, Centre for Southern Hemisphere Oceans Research Hobart Tasmania Australia); Jasti S. Chowdary (Indian Institute of Tropical Meteorology); Jianwei Chi (Chinese Academy of Sciences); Shuang Qiu (Chinese Academy of Sciences); Weidong Yu (Marine Environmental Forecasting Center, State Oceanic Administration Beijing China, Center for Ocean and Climate Research First Institute of Oceanography, State Oceanic Administration Qingdao China, Qingdao National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014580

URL: <https://doi.org/10.1029/2018jc014580>

Publication Date: 2019-05-01

Volume/Issue: Volume 124, Issue 5

References: 89

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: southwestern tropical Indian Ocean, horizontal advection, Indian Ocean dipole, Rossby waves, sea surface salinity

Funding: State Key Laboratory of Tropical Oceanography, South China Sea Institute of Oceanology, Chinese Academy of Sciences, State Oceanic Administration, National Natural Science Foundation of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: Using Argo observations and Estimating the Circulation and Climate of the Ocean data sets, this study investigates the sea surface salinity (SSS) variability and its relationship with ocean dynamics in the southwestern tropical Indian Ocean (SWTIO) associated with the 2010 negative Indian Ocean dipole (IOD) event. The results show that the ocean circulation in the tropical southern Indian Ocean (IO) significantly contributes to the SSS anomalies during the evolution of the negative IOD event. The anomalous eastward current advected the high-salinity water into the eastern equatorial IO during July-October 2010. Then the positive SSS anomalies expanded southward to 6-8 degreesS due to the southeastward mean seasonal currents in the eastern equatorial IO during November-December 2010. A salinity budget analysis has been used to identify key processes in contributing to the SSS variations in the SWTIO. The decreased precipitation in the western and central equatorial IO leads to high SSS anomalies from January to April 2011. Then the anomalous southwestward currents transport the higher SSS water from the eastern equatorial IO to the SWTIO until May-June 2011. At the same time, the upwelling Rossby waves shoal the thermocline and mixed layer depth, bringing high-salinity subsurface water to the surface layer and cooling the sea surface temperature, which further depresses local precipitation to provide a positive feedback for the SSS increase.

1032. The exact geostrophic streamfunction for neutral surfaces

Authors: Geoffrey J. Stanley

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2019.04.002

URL: <https://doi.org/10.1016/j.ocemod.2019.04.002>

Publication Date: 2019-05-01

Volume/Issue: Volume 138

References: 41

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Clarendon Scholarship, Linacre College, University of Oxford

License: N/A

Open Access: N/A

Abstract: N/A

1033. Long-Term and Interannual Variation of the Steric Sea Level in the South China Sea and the Connection with ENSO

Authors: Hui Xi (Chinese Academy of Sciences); Zizhan Zhang (Chinese Academy of Sciences); Yang Lu (Chinese Academy of Sciences); Yan Li (Chinese Academy of Sciences)

Journal: Journal of Coastal Research

Publisher: Coastal Education and Research Foundation

DOI: 10.2112/jcoastres-d-18-00080.1

URL: <https://doi.org/10.2112/jcoastres-d-18-00080.1>

Publication Date: 2019-04-29

Volume/Issue: Volume 35, Issue 3

References: 56

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1034. Seasonal variations in global mean sea level and consequences on the excitation of length-of-day changes

Authors: R Dill (Helmholtz Centre Potsdam); H Dobslaw (Helmholtz Centre Potsdam)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggz201

URL: <https://doi.org/10.1093/gji/ggz201>

Publication Date: 2019-04-27

Volume/Issue: Volume 218, Issue 2

References: 40

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1035. Improving Arctic sea ice seasonal outlook by ensemble prediction using an ice-ocean model

Authors: Qinghua Yang; Longjiang Mu; Xingren Wu; Jiping Liu; Fei Zheng; Jinlun Zhang; Chuanjin Li

Journal: Atmospheric Research

Publisher: Elsevier BV

DOI: 10.1016/j.atmosres.2019.04.021

URL: <https://doi.org/10.1016/j.atmosres.2019.04.021>

Publication Date: 2019-04-26

Volume/Issue: Volume 227

References: 39

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key R&D; Program of China, State Key Laboratory of Cryospheric Science, Federal Ministry of Education and Research of Germany (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

1036. Global Patterns of Submesoscale Surface Salinity Variability

Authors: Kyla Drushka (University of Washington); William E. Asher (University of Washington); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Clifford Hoang (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-19-0018.1

URL: <https://doi.org/10.1175/jpo-d-19-0018.1>

Publication Date: 2019-04-25

Volume/Issue: Volume 49, Issue 7

References: 61

Cited by: 20 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Surface salinity variability on O(1-10) km lateral scales (the submesoscale) generates density variability and thus has implications for submesoscale dynamics. Satellite salinity measurements represent a spatial average over horizontal scales of approximately 40-100 km but are compared to point measurements for validation, so submesoscale salinity variability also complicates validation of satellite salinities. Here, we combine several databases of historical thermosalinograph (TSG) measurements made from ships to globally characterize surface submesoscale salinity, temperature, and density variability. In river plumes; regions affected by ice melt or upwelling; and the Gulf Stream, South Atlantic, and Agulhas Currents, submesoscale surface salinity variability is large. In these regions, horizontal salinity variability appears to explain some of the differences between surface salinities from the Aquarius and SMOS satellites and salinities measured with Argo floats. In other words, apparent satellite errors in highly variable regions in fact arise because Argo point measurements do not represent spatially averaged satellite data. Salinity dominates over temperature in generating submesoscale surface density variability throughout the tropical rainbands, in river plumes, and in polar regions. Horizontal density fronts on 10-km scales tend to be compensated (salinity and temperature have opposing effects on density) throughout most of the global oceans, with the exception of the south Indian and southwest Pacific Oceans between 20 degrees and 30 degreesS, where fronts tend to be anticompensated.

1037. Radiocarbon and Helium Isotope Constraints on Deep Ocean Ventilation and Mantle-³He Sources

Authors: Tim DeVries (University of California Santa Barbara); Mark Holzer (University of New South Wales)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014716

URL: <https://doi.org/10.1029/2018jc014716>

Publication Date: 2019-04-22

Volume/Issue: Volume 124, Issue 5

References: 91

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Radiocarbon ($\Delta^{14}\text{C}$) and helium isotopes ($\delta^3\text{He}$) have long been used to constrain the ocean's ventilation rates and to trace regional deep ocean circulation pathways, but they have not been fully exploited together to constrain the deep circulation in global models. Here we assimilate $\Delta^{14}\text{C}$ and $\delta^3\text{He}$ measurements into a global ocean circulation inverse model (OCIM) to jointly constrain the deep ocean circulation and the rate of mantle-helium injection at seafloor spreading ridges. We find that the new version of the inverse model (OCIM2) matches the observed $\Delta^{14}\text{C}$ and $\delta^3\text{He}$ distributions much better than a previous version (OCIM1) that assimilated objectively mapped $\Delta^{14}\text{C}$ but not $\delta^3\text{He}$. OCIM2 features faster-ventilated bottom waters and slower-ventilated intermediate-depth waters in the Pacific and Indian Oceans. The mean time since last ventilation (ideal mean age) in Pacific bottom waters is up to 150 years younger, while middepth Pacific waters are up to several hundred years older. The $\delta^3\text{He}$ constraints are shown to be important for estimates of the mean time to next ventilation in the Pacific Ocean. The $\delta^3\text{He}$ constraints also favor jet-like currents in the deep equatorial Pacific to capture realistic westward propagating helium plumes emanating from the East Pacific Rise. The globally integrated mantle-helium source is 585-672 mol/year, compared to 400-1,000 mol/year from previous estimates. The major regional difference occurs in the Southern Ocean, where the OCIM2 mantle-helium source is up to threefold smaller than estimates based on ridge spreading rates.

1038. Pacific Abyssal Transport and Mixing: Through the Samoan Passage versus around the Manihiki Plateau

Authors: Larry J. Pratt (Woods Hole Oceanographic Institution); Gunnar Voet (Scripps Institution of Oceanography); Astrid Pacini (Woods Hole Oceanographic Institution); Shuwen Tan (Woods Hole Oceanographic Institution); Matthew H. Alford (Scripps Institution of Oceanography); Glenn S. Carter (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); James B. Girtton (University of Washington); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-18-0124.1

URL: <https://doi.org/10.1175/jpo-d-18-0124.1>

Publication Date: 2019-04-22

Volume/Issue: Volume 49, Issue 6

References: 38

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: Abyssal zone|Isopycnal|Samoa|Sill|Hydraulic jump

Funding: National Science Foundation, National Science Foundation, National Science Foundation

License: N/A

Open Access: N/A

Abstract: The main source feeding the abyssal circulation of the North Pacific is the deep, northward flow of 5-6 Sverdrups (Sv; $1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) through the Samoan Passage. A recent field campaign has shown that this flow is hydraulically controlled and that it experiences hydraulic jumps accompanied by strong mixing and dissipation concentrated near several deep sills. By our estimates, the diapycnal density flux associated with this mixing is considerably larger than the diapycnal flux across a typical isopycnal surface extending over the abyssal North Pacific. According to historical hydrographic observations, a second source of abyssal water for the North Pacific is 2.3-2.8 Sv of the dense flow that is diverted around the Manihiki Plateau to the east, bypassing the Samoan Passage. This bypass flow is not confined to a channel and is therefore less likely to experience the strong mixing that is associated with hydraulic transitions. The partitioning of flux between the two branches of the deep flow could therefore be relevant to the distribution of Pacific abyssal mixing. To gain insight into the factors that control the partitioning between these two branches, we develop an abyssal and equator-proximal extension of the "island rule." Novel features include provisions for the presence of hydraulic jumps as well as identification of an appropriate integration circuit for an abyssal layer to the east of the island. Evaluation of the corresponding circulation integral leads to a prediction of 0.4-2.4 Sv of bypass flow. The circulation integral clearly identifies dissipation and frictional drag effects within the Samoan Passage as crucial elements in partitioning the flow.

1039. Mechanisms of Ocean Heat Anomalies in the Norwegian Sea

Authors: Helene Asbjørnsen (University of Bergen, Bjerknes Centre for Climate Research); Marius Årthun (University of Bergen, Bjerknes Centre for Climate Research); Øystein Skagseth (Institute of Marine Research, Bjerknes Centre for Climate Research); Tor Eldevik (University of Bergen, Bjerknes Centre for Climate Research)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014649

URL: <https://doi.org/10.1029/2018jc014649>

Publication Date: 2019-04-10

Volume/Issue: Volume 124, Issue 4

References: 73

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Ocean heat content in the Norwegian Sea exhibits pronounced variability on interannual to decadal time scales. These ocean heat anomalies are known to influence Arctic sea ice extent, marine ecosystems, and continental climate. It nevertheless remains unknown to what extent such heat anomalies are produced locally within the Norwegian Sea, and to what extent the region is more of a passive receiver of anomalies formed elsewhere. A main practical challenge has been the lack of closed heat budget diagnostics. In order to address this issue, a regional heat budget is calculated for the Norwegian Sea using the ECCOv4 ocean state estimate--a dynamically and kinematically consistent model framework fitted to ocean observations for the period 1992-2015. The depth-integrated Norwegian Sea heat budget shows that both ocean advection and air-sea heat fluxes play an active role in the formation of interannual heat content anomalies. A spatial analysis of the individual heat budget terms shows that ocean advection is the primary contributor to heat content variability in the Atlantic domain of the Norwegian Sea. Anomalous heat advection furthermore depends on the strength of the Atlantic water inflow, which is related to large-scale circulation changes in the subpolar North Atlantic. This result suggests a potential for predicting Norwegian Sea heat content based on upstream conditions. However, local surface forcing (air-sea heat fluxes and Ekman forcing) within the Norwegian Sea substantially modifies the phase and amplitude of ocean heat anomalies along their poleward pathway, and, hence, acts to limit predictability.

1040. Ocean-Scale Interactions From Space

Authors: Patrice Klein (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, LOPS-Ifrermer/CNRS Brest France); Guillaume Lapeyre (LMD/IPSL,CNES,ENS,PSLRU Paris France); Lia Siegelman (California Institute of Technology, LEMAR/IUEM/CNRS Brest France, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bo Qiu (Department of Oceanography University of Hawai'i Honolulu HI USA); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hector Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zhan Su (California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Sylvie Le Gentil (LOPS-Ifrermer/CNRS Brest France)

Journal: Earth and Space Science

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018ea000492

URL: <https://doi.org/10.1029/2018ea000492>

Publication Date: 2019-04-10

Volume/Issue: Volume 6, Issue 5

References: 155

Cited by: 99 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Satellite observations of the last two decades have led to a major breakthrough emphasizing the existence of a strongly energetic mesoscale turbulent eddy field in all the oceans. This ocean mesoscale turbulence is characterized by cyclonic and anticyclonic eddies (with a 100- to 300-km size and depth scales of ~500-1,000 m) that capture approximatively 80% of the total kinetic energy and is now known to significantly impact the large-scale ocean circulation, the ocean's carbon storage, the air-sea interactions, and therefore the Earth climate as a whole. However, ocean mesoscale turbulence revealed by satellite observations has properties that differ from those related to classical geostrophic turbulence theories. In the last decade, a large number of theoretical and numerical studies has pointed to submesoscale surface fronts (1-50 km), not resolved by satellite altimeters, as the key suspect explaining these discrepancies. Submesoscale surface fronts have been shown to impact mesoscale eddies and the large-scale ocean circulation in counterintuitive ways, leading in particular to up-gradient fluxes. The ocean engine is now known to involve energetic scale interactions, over a much broader range of scales than expected one decade ago, from 1 to 5,000 km. New space observations with higher spatial resolution are however needed to validate and improve these recent theoretical and numerical results.

1041. Overview of the multi-layer circulation in the South China Sea

Authors: Yaohua Zhu; Junchuan Sun; Yonggang Wang; Shujiang Li; Tengfei Xu; Zexun Wei; Tangdong Qu

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2019.04.001

URL: <https://doi.org/10.1016/j.pocean.2019.04.001>

Publication Date: 2019-04-10

Volume/Issue: Volume 175

References: 95

Cited by: 80 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China, State Oceanic Administration Program on Global Change and Air-Sea interactions of China, National High Technology Research and Development Program of China (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

1042. Sensitivity of submarine melting on North East Greenland towards ocean forcing

Authors: Philipp Anhaus; Lars H. Smedsrud; Marius Årthun; Fiammetta Straneo

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2019-35

URL: <https://doi.org/10.5194/tc-2019-35>

Publication Date: 2019-04-10

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Meltwater|Greenland ice sheet|Forcing (mathematics)

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The Nioghalvfjærdsbræ (79NG) is a floating ice tongue on Northeast Greenland draining a large part of the Greenland Ice Sheet. A CTD profile from a rift on the ice tongue close to the northern front shows that Atlantic Water (AW) is present in the cavity below, with maximum temperature of approximately 1 degreesC at 610 m depth. The AW present in the cavity thus has the potential to drive submarine melting along the ice base. Here, we simulate melt rates from the 79NG with a 1D numerical Ice Shelf Water (ISW) plume model. A meltwater plume is initiated at the grounding line depth (600 m) and rises along the ice base as a result of buoyancy contrast to the underlying AW. Ice melts as the plume entrains the warm AW. Maximum simulated melt rates are 50-76 m yr⁻¹ within 10 km of the grounding line. Within a zone of rapid decay between 10 km and 20 km melt rates drop to roughly 6 m yr⁻¹. Further downstream, melt rates are between 15 m yr⁻¹ and 6 m yr⁻¹. The melt-rate sensitivity to variations in AW temperatures is assessed by forcing the model with AW temperatures between 0.1-1.4 degreesC, as identified from the ECCOV4 ocean state estimate. The melt rates increase linearly with rising AW temperature, ranging from 10 m yr⁻¹ to 21 m yr⁻¹ along the centerline. The corresponding freshwater flux ranges between 11 km³ yr⁻¹ (0.4 mSv) and 30 km³ yr⁻¹ (1.0 mSv), which is 5 % and 12 % of the total freshwater flux from the Greenland Ice Sheet since 1995, respectively. Our results improve the understanding of processes driving submarine melting of marine-terminating glaciers around Greenland, and its sensitivity to changing ocean conditions.

1043. A Coupled Global Atmosphere-Ocean Model for Air-Sea Exchange of Mercury: Insights into Wet Deposition and Atmospheric Redox Chemistry

Authors: Yanxu Zhang (Nanjing University); Hannah Horowitz (University of Washington); Jiancheng Wang (University of Science and Technology of China); Zhouqing Xie (University of Science and Technology of China); Joachim Kuss (Leibniz Institute for Baltic Sea Research); Anne L. Soerensen (Stockholm University)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

DOI: 10.1021/acs.est.8b06205

URL: <https://doi.org/10.1021/acs.est.8b06205>

Publication Date: 2019-04-04

Volume/Issue: Volume 53, Issue 9

References: 56

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Svenska Forskningsrådet Formas, Government of Jiangsu Province (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1044. Global ocean heat transport dominated by heat export from the tropical Pacific

Authors: Gaël Forget; David Ferreira

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-019-0333-7

URL: <https://doi.org/10.1038/s41561-019-0333-7>

Publication Date: 2019-04-01

Volume/Issue: Volume 12, Issue 5

References: 41

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1045. A Three-Way Balance in the Beaufort Gyre: The Ice-Ocean Governor, Wind Stress, and Eddy Diffusivity

Authors: Edward W. Doddridge (Massachusetts Institute of Technology); Gianluca Meneghello (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology); Jeffery Scott (Massachusetts Institute of Technology); Camille Lique (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014897

URL: <https://doi.org/10.1029/2018jc014897>

Publication Date: 2019-04-01

Volume/Issue: Volume 124, Issue 5

References: 65

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: The Beaufort Gyre (BG) is a large anticyclonic circulation in the Arctic Ocean. Its strength is directly related to the halocline depth, and therefore also to the storage of freshwater. It has recently been proposed that the equilibrium state of the BG is set by the Ice-Ocean Governor, a negative feedback between surface currents and ice-ocean stress, rather than a balance between lateral mesoscale eddy fluxes and surface Ekman pumping. However, mesoscale eddies are present in the Arctic Ocean; it is therefore important to extend the Ice-Ocean Governor theory to include lateral fluxes due to mesoscale eddies. Here, a nonlinear ordinary differential equation is derived that represents the effects of wind stress, the Ice-Ocean Governor, and eddy fluxes. Equilibrium and time-varying solutions to this three-way balance equation are obtained and shown to closely match the output from a hierarchy of numerical simulations, indicating that the analytical model represents the processes controlling BG equilibration. The equilibration timescale derived from this three-way balance is faster than the eddy equilibration timescale and slower than the Ice-Ocean Governor equilibration timescales for most values of eddy diffusivity. The sensitivity of the BG equilibrium depth to changes in eddy diffusivity and the presence of the Ice-Ocean Governor is also explored. These results show that predicting the response of the BG to changing surface forcing and sea ice conditions requires faithfully capturing the three-way balance between the Ice-Ocean Governor, wind stress, and eddy fluxes.

1046. Reexamination of Fram Strait sea ice export and its role in recently accelerated Arctic sea ice retreat

Authors: Jianfen Wei; Xiangdong Zhang; Zhaomin Wang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-019-04741-0

URL: <https://doi.org/10.1007/s00382-019-04741-0>

Publication Date: 2019-03-29

Volume/Issue: Volume 53, Issue 3-4

References: 66

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key R&D; Program of China, Ministry of Science and Technology, the Fundamental Research Funds for the Central Universities (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1047. Decadal Variations of the Mindanao Current During 1960-2010

Authors: Jing Duan (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Yuanlong Li (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Fan Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Zhaohui Chen (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc014975

URL: <https://doi.org/10.1029/2019jc014975>

Publication Date: 2019-03-26

Volume/Issue: Volume 124, Issue 4

References: 78

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Qingdao National Laboratory for Marine Science and Technology, National Natural Science Foundation of China

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Open Access: N/A

Abstract: Results from an eddy-resolving ocean model simulation, an ocean reanalysis data, and reduced-gravity ocean (RGO) model simulations reach consensus in suggesting prominent decadal fluctuations of the Mindanao Current (MC) during 1960-2010. The MC was evidently enhanced during 1974-1982 and 2002-2008 and weakened during 1983-1990 and 1994-2001, with typical amplitude of ~ 2.0 Sv ($1 \text{ Sv} \equiv 10^6 \text{ m}^3/\text{s}$) for decadal volume transport anomaly. These variations are closely associated with the strength of the westward flowing North Equatorial Current at 130°E and not evidently related with its bifurcation latitude. Experiments using a 1.5-layer RGO model are performed to understand the mechanisms. Decadal variations of the MC are primarily caused by off-equatorial wind changes in the tropical North Pacific, and winds in the latitude range of the MC ($3\text{--}13^\circ\text{N}$) play the dominant role. Equatorial winds between 3°S and 3°N generally act to attenuate the MC variability, and subtropical winds have little contribution. For zonal distribution, local wind forcing between 120 and 160°E explains 67.8% of the MC decadal variability, while those in the central and eastern Pacific have weaker effects (25.0% and 7.2%, respectively). RGO model runs forced by nine wind data sets confirm the tight relationship between the MC and the western Pacific winds since the 1980s. The anomalous westerly winds under El Niño-like condition drive upwelling in the Philippine Sea through off-equatorial Ekman pumping that strengthens the MC. The relationship prior to 1980 is sensitive to the choice of wind data set, with four data sets yielding low or negative correlations.

1048. Interruption of two decades of Jakobshavn Isbrae acceleration and thinning as regional ocean cools

Authors: Ala Khazendar; Ian G. Fenty; Dustin Carroll; Alex Gardner; Craig M. Lee; Ichiro Fukumori; Ou Wang; Hong Zhang; Hélène Seroussi; Delwyn Moller; Brice P. Y. Noël; Michiel R. van den Broeke; Steven Dinardo; Josh Willis

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-019-0329-3

URL: <https://doi.org/10.1038/s41561-019-0329-3>

Publication Date: 2019-03-25

Volume/Issue: Volume 12, Issue 4

References: 65

Cited by: 90 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1049. Mass sea level variation in the South China Sea from GRACE, altimetry and model and the connection with ENSO

Authors: Hui Xi; Zizhan Zhang; Yang Lu; Yan Li

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2019.03.027

URL: <https://doi.org/10.1016/j.asr.2019.03.027>

Publication Date: 2019-03-23

Volume/Issue: Volume 64, Issue 1

References: 52

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Key Basic Research Program of China

License: N/A

Open Access: N/A

Abstract: N/A

1050. Revisiting the Circulation of Hudson Bay: Evidence for a Seasonal Pattern

Authors: Natasha A. Ridenour (University of Alberta); Xianmin Hu (University of Alberta); Kevin Sydor (Manitoba Hydro Winnipeg Manitoba Canada); Paul G. Myers (University of Alberta); David G. Barber (University of Manitoba)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl082344

URL: <https://doi.org/10.1029/2019gl082344>

Publication Date: 2019-03-19

Volume/Issue: Volume 46, Issue 7

References: 40

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: anticyclonic circulation, freshwater, Hudson Bay, atmospheric forcing, numerical modeling (and 1 more)

Funding: Canadian Network for Research and Innovation in Machining Technology, Natural Sciences and Engineering Research Council of Canada

License: N/A

Open Access: N/A

Abstract: The Hudson Bay Complex is the outlet for many Canadian rivers, receiving roughly 900 km³/year of river runoff. Historically, studies found a consistent cyclonic flow year-round in Hudson Bay, due to the geostrophic boundary current induced by river discharge and cyclonic wind forcing that was supported by available observations at that time. Using a high-resolution ocean general circulation model, we show that in summer, the mean circulation is not cyclonic but consists of multiple small cyclonic and anticyclonic features, with the mean flow directed through the center of the bay. Absolute Dynamic Topography and velocity observations also show this seasonal flow pattern. We find that this summer circulation is driven by geostrophic currents, generated by steric height gradients, which are induced by increased river discharge during the spring freshet, and reinforced by anticyclonic seasonal wind patterns.

1051. Origins and Pathways of the Pacific Equatorial Undercurrent Identified by a Simulated Adjoint Tracer

Authors: Xunwei Nie (Chinese Academy of Sciences, University of California Los Angeles, Ministry of Natural Resources, Pilot National Laboratory for Marine Science and Technology); Shan Gao (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Fan Wang (Chinese Academy of Sciences, Pilot National Laboratory for Marine Science and Technology); Jianwei Chi (University of California Los Angeles, Chinese Academy of Sciences); Tangdong Qu (University of California Los Angeles, Ministry of Natural Resources)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014212

URL: <https://doi.org/10.1029/2018jc014212>

Publication Date: 2019-03-14

Volume/Issue: Volume 124, Issue 4

References: 64

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: transit time, Pacific Equatorial Undercurrent, source water, pathway, cells (and 1 more)

Funding: National Natural Science Foundation of China, State Key Laboratory of Tropical Oceanography, South China Sea Institute of Oceanology, Chinese Academy of Sciences

License: Creative Commons

Open Access: N/A

Abstract: The origins and pathways of the Pacific Equatorial Undercurrent (EUC) are investigated using a simulated adjoint tracer of the consortium Estimating the Circulation and Climate of the Ocean. The main source waters of the EUC, as well as their transit time and contributions, are identified. The zonal variability of the EUC water composition and the contributions from different pathways are also estimated. Results show that the ratio of the EUC water coming from the tropics to that from extratropics is relatively stable (1 vs. 4) along the equator, except in the western Pacific where extratropical water is more dominant. The main body of extratropical water within the EUC is transported through the western boundary pathways, while the percent transported via the interior pathways gradually increases toward the east. Tropical water merges into the EUC mainly through the tropical cells with a larger portion through the northern side of the equator.

1052. Unsupervised Learning Reveals Geography of Global Ocean Dynamical Regions

Authors: Maike Sonnewald (Harvard University, Massachusetts Institute of Technology); Carl Wunsch (Harvard University, Massachusetts Institute of Technology); Patrick Heimbach (University of Texas at Austin)

Journal: Earth and Space Science

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018ea000519

URL: <https://doi.org/10.1029/2018ea000519>

Publication Date: 2019-03-13

Volume/Issue: Volume 6, Issue 5

References: 36

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: big data, ocean dynamics, physical oceanography, machine learning, ocean modeling (and 1 more)

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: Dynamically similar regions of the global ocean are identified using a barotropic vorticity (BV) framework from a 20-year mean of the Estimating the Circulation and Climate of the Ocean state estimate at 1 degrees resolution. An unsupervised machine learning algorithm, K-means, objectively clusters the standardized BV equation, identifying five unambiguous regimes. Cluster 1 covers 43 +/- 3.3% of the ocean area. Surface and bottom stress torque are balanced by the bottom pressure torque and the nonlinear torque. Cluster 2 covers 24.8 +/- 1.2%, where the beta effect balances the bottom pressure torque. Cluster 3 covers 14.6 +/- 1.0%, characterized by a "Quasi-Sverdrupian" regime where the beta effect is balanced by the wind and bottom stress term. The small region of Cluster 4 has baroclinic dynamics covering 6.9 +/- 2.9% of the ocean. Cluster 5 occurs primarily in the Southern Ocean. Residual "dominantly nonlinear" regions highlight where the BV approach is inadequate, found in areas of rough topography in the Southern Ocean and along western boundaries.

1053. Decomposing Mean Sea Level Rise in a Semi-Enclosed Basin, the Baltic Sea

Authors: Ulf Gräwe (Leibniz Institute for Baltic Sea Research); Knut Klingbeil (Leibniz Institute for Baltic Sea Research); Jessica Kelln (University of Siegen, Siegen, Germany); Sönke Dangendorf (University of Siegen, Siegen, Germany)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0174.1

URL: <https://doi.org/10.1175/jcli-d-18-0174.1>

Publication Date: 2019-03-12

Volume/Issue: Volume 32, Issue 11

References: 97

Cited by: 32 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Bundesministerium für Bildung und Forschung, Bundesministerium für Bildung und Forschung

License: N/A

Open Access: N/A

Abstract: We analyzed changes in mean sea level (MSL) for the period 1950-2015 using a regional ocean model for the Baltic Sea. Sensitivity experiments allowed us to separate external from local drivers and to investigate individual forcing agents triggering basin-internal spatial variations. The model reveals a basin-average MSL rise (MSLR) of 2.08 ± 0.49 mm yr⁻¹, a value that is slightly larger than the simultaneous global average of 1.63 ± 0.32 mm yr⁻¹. This MSLR is, however, spatially highly nonuniform with lower than average increases in the southwestern part (1.71 ± 0.51 mm yr⁻¹) and higher than average rates in the northeastern parts (2.34 ± 1.05 mm yr⁻¹). While 75% of the basin-average MSL externally enters the Baltic basin as a mass signal from the adjacent North Sea, intensified westerly winds and a poleward shift of low pressure systems explain the majority of the spatial variations in the rates. Minor contributions stem from local changes in baroclinicity leading to a basin-internal redistribution of water masses. An observed increase in local ocean temperature further adds to the total basinwide MSLR through thermal expansion but has little effect on the spatial pattern. To test the robustness of these results, we further assessed the sensitivity to six different atmospheric surface forcing reanalysis products over their common period from 1980 to 2005. The ensemble runs indicated that there are significant differences between individual ensemble members increasing the total trend uncertainty for the basin average by 0.22 mm yr⁻¹ (95% confidence intervals). Locally the uncertainty varies from 0.05 mm yr⁻¹ in the central part to up to 0.4 mm yr⁻¹ along the coasts.

1054. Influence of initial ocean conditions on temperature and precipitation in a coupled climate model's solution

Authors: Robin Tokmakian; Peter Challenor

Journal: Advances in Statistical Climatology, Meteorology and Oceanography

Publisher: Copernicus GmbH

DOI: 10.5194/ascmo-5-17-2019

URL: <https://doi.org/10.5194/ascmo-5-17-2019>

Publication Date: 2019-03-12

Volume/Issue: Volume 5, Issue 1

References: 44

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Division of Ocean Sciences

License: CC BY 4.0

Open Access: N/A

Abstract: . This paper describes results of an experiment that perturbed the initial conditions for the ocean's temperature field of the Community Earth System Model (CESM) with a well defined design. The resulting 30-member ensemble of CESM simulations, each of 10 years in length, is used to create an emulator (a nonlinear regression relating the initial conditions to various outcomes) from the simulations. Through the use of the emulator to expand the output distribution space, we estimate the spatial uncertainties at 10 years for surface air temperature, 25 m ocean temperature, precipitation, and rain. Outside the tropics, basin averages for the uncertainty in the ocean temperature field range between 0.48 °C (Indian Ocean) and 0.87 °C (North Pacific) (2 standard deviation). The tropical Pacific uncertainty is the largest due to different phasings of the ENSO signal. Over land areas, the regional temperature uncertainty varies from 1.03 °C (South America) to 10.82 °C (Europe) (2 standard deviation). Similarly, the regional average uncertainty in precipitation varies from 0.001 cm day⁻¹ over Antarctica to 0.163 cm day⁻¹ over Australia with a global average of 0.075 cm day⁻¹. In general, both temperature and precipitation uncertainties are larger over land than over the ocean. A maximum covariance analysis is used to examine how ocean temperatures affect both surface air temperatures and precipitation over land. The analysis shows that the tropical Pacific influences the temperature over North America, but the North America surface temperature is also moderated by the state of the North Pacific outside the tropics. It also indicates which regions show a high degree of variance between the simulations in the ensemble and are, therefore, less predictable. The calculated uncertainties are also compared to an estimate of internal variability within CESM. Finally, the importance of feedback processes on the solution of the simulation over the 10 years of the experiment is quantified. These estimates of uncertainty do not take into consideration the anthropogenic effect on warming of the atmosphere and ocean.

1055. On the Long-Wavelength Validation of the SWOT KaRIn Measurement

Authors: Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-18-0148.1

URL: <https://doi.org/10.1175/jtech-d-18-0148.1>

Publication Date: 2019-03-11

Volume/Issue: Volume 36, Issue 5

References: 15

Cited by: 14 publications

Language: N/A

Subjects: N/A

Keywords: Nadir|Ocean surface topography|Sea-surface height

Funding: Jet Propulsion Laboratory

License: N/A

Open Access: N/A

Abstract: The Surface Water and Ocean Topography (SWOT) mission will measure the sea surface height (SSH) using a Ka-band radar interferometer (KaRIn) over a swath off the nadir of the satellite tracks. The mission requires calibration and validation (CalVal) of the SSH wavenumber spectrum at wavelengths between 15 and 1000 km. The CalVal in the short-wavelength range (15-150 km) requires in situ observations. In the long-wavelength range (150-1000 km), the CalVal will use the onboard Jason-class nadir altimeter. Using a high-resolution global ocean simulation, this study identifies the spatial scales beyond which the nadir and off-nadir observations can be considered comparable. Our results suggest that the ocean signals at nadir can represent off-nadir ocean signals at wavelengths longer than 120 and 70 km along the midswath and the inner edge of the KaRIn grid, respectively, indicating that the nadir altimeter is able to fulfill its goal to validate the long-wavelength KaRIn measurement. The wavelength along the inner edge is limited around 70 km because the onboard nadir altimeter cannot resolve spatial scales longer than ~70 km. These wavelengths provide a reference point for the required spatial coverage of the SWOT SSH in situ CalVal.

1056. Neutral surface topology

Authors: Geoffrey J. Stanley

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2019.01.008

URL: <https://doi.org/10.1016/j.ocemod.2019.01.008>

Publication Date: 2019-03-05

Volume/Issue: Volume 138

References: 45

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Clarendon Scholarship, University of Oxford

License: N/A

Open Access: N/A

Abstract: N/A

1057. Shelf Waves in the Beaufort Sea in a High-Resolution Ocean Model

Authors: T. V. Belonenko; D. L. Volkov; A. V. Koldunov

Journal: Oceanology

Publisher: Pleiades Publishing Ltd

DOI: 10.1134/s0001437018060024

URL: <https://doi.org/10.1134/s0001437018060024>

Publication Date: 2019-03-05

Volume/Issue: Volume 58, Issue 6

References: 23

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1058. Putting It All Together: Adding Value to the Global Ocean and Climate Observing Systems With Complete Self-Consistent Ocean State and Parameter Estimates

Authors: Patrick Heimbach; Ichiro Fukumori; Christopher N. Hill; Rui M. Ponte; Detlef Stammer; Carl Wunsch; Jean-Michel Campin; Bruce Cornuelle; Ian Fenty; Gaël Forget; Armin Köhl; Matthew Mazloff; Dimitris Menemenlis; An T. Nguyen; Christopher Piecuch; David Trossman; Ariane Verdy; Ou Wang; Hong Zhang

Journal: Frontiers in Marine Science

Publisher: Frontiers Media SA

DOI: 10.3389/fmars.2019.00055

URL: <https://doi.org/10.3389/fmars.2019.00055>

Publication Date: 2019-03-04

Volume/Issue: Volume 6

References: 104

Cited by: 24 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1059. A process study of interactions between a warm eddy and the Kuroshio Current in Luzon Strait: The fate of eddies

Authors: Shengmu Yang; Jiuxing Xing; Jinyu Sheng; Shengli Chen; Daoyi Chen

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2019.02.009

URL: <https://doi.org/10.1016/j.jmarsys.2019.02.009>

Publication Date: 2019-03-02

Volume/Issue: Volume 194

References: 44

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Basic Research Program of ChinaNational Basic Research Program of China (973 Program), Environmental Protection Special Funds for Public Welfare, Shenzhen Government

License: N/A

Open Access: N/A

Abstract: N/A

1060. Modeling the response of northwest Greenland to enhanced ocean thermal forcing and subglacial discharge

Authors: Mathieu Morlighem; Michael Wood; H  l  ne Seroussi; Youngmin Choi; Eric Rignot

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-13-723-2019

URL: <https://doi.org/10.5194/tc-13-723-2019>

Publication Date: 2019-03-01

Volume/Issue: Volume 13, Issue 2

References: 52

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Division of Arctic Sciences

License: CC BY 4.0

Open Access: N/A

Abstract: Calving-front dynamics is an important control on Greenland's ice mass balance. Ice front retreat of marine-terminating glaciers may, for example, lead to a loss in resistive stress, which ultimately results in glacier acceleration and thinning. Over the past decade, it has been suggested that such retreats may be triggered by warm and salty Atlantic Water, which is typically found at a depth below 200–300 m. An increase in subglacial water discharge at glacier ice fronts due to enhanced surface runoff may also be responsible for an intensification of undercutting and calving. An increase in ocean thermal forcing or subglacial discharge therefore has the potential to destabilize marine-terminating glaciers along the coast of Greenland. It remains unclear which glaciers are currently stable but may retreat in the future and how far inland and how fast they will retreat. Here, we quantify the sensitivity and vulnerability of marine-terminating glaciers along the northwest coast of Greenland (from 72.5 to 76° N) to ocean forcing and subglacial discharge using the Ice Sheet System Model (ISSM). We rely on a parameterization of undercutting based on ocean thermal forcing and subglacial discharge and use ocean temperature and salinity from high-resolution ECCO2 (Estimating the Circulation and Climate of the Ocean, Phase II) simulations at the fjord mouth to constrain the ocean thermal forcing. The ice flow model includes a calving law based on a tensile von Mises criterion. We find that some glaciers, such as Dietrichson Gletscher or Alison Glacier, are sensitive to small increases in ocean thermal forcing, while others, such as Illullip Sermia or Cornell Gletscher, are remarkably stable, even in a +3 °C ocean warming scenario. Under the most intense experiment, we find that Hayes Gletscher retreats by more than 50 km inland by 2100 into a deep trough, and its velocity increases by a factor of 3 over only 23 years. The model confirms that ice-ocean interactions can trigger extensive and rapid glacier retreat, but the bed controls the rate and magnitude of the retreat. Under current oceanic and atmospheric conditions, we find that this sector of the Greenland ice sheet alone will contribute more than 1 cm to sea level rise and up to 3 cm by 2100 under the most extreme scenario.

1061. Temperature and Salinity Variability in the SODA3, ECCO4r3, and ORAS5 Ocean Reanalyses, 1993-2015

Authors: James A. Carton (University of Maryland); Stephen G. Penny (University of Maryland); Eugenia Kalnay (University of Maryland)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0605.1

URL: <https://doi.org/10.1175/jcli-d-18-0605.1>

Publication Date: 2019-02-27

Volume/Issue: Volume 32, Issue 8

References: 53

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Directorate for Geosciences

License: N/A

Open Access: N/A

Abstract: This study extends recent ocean reanalysis comparisons to explore improvements to several next-generation products, the Simple Ocean Data Assimilation, version 3 (SODA3); the Estimating the Circulation and Climate of the Ocean, version 4, release 3 (ECCO4r3); and the Ocean Reanalysis System 5 (ORAS5), during their 23-yr period of overlap (1993-2015). The three reanalyses share similar historical hydrographic data, but the forcings, forward models, estimation algorithms, and bias correction methods are different. The study begins by comparing the reanalyses to independent analyses of historical SST, heat, and salt content, as well as examining the analysis-minus-observation misfits. While the misfits are generally small, they still reveal some systematic biases that are not present in the reference Hadley Center EN4 objective analysis. We next explore global trends in temperature averaged into three depth intervals: 0-300, 300-1000, and 1000-2000 m. We find considerable similarity in the spatial structure of the trends and their distribution among different ocean basins; however, the trends in global averages do differ by 30%-40%, which implies an equivalent level of disagreement in net surface heating rates. ECCO4r3 is distinct in having quite weak warming trends while ORAS5 has stronger trends that are noticeable in the deeper layers. To examine the performance of the reanalyses in the Arctic we explore representation of Atlantic Water variability on the Atlantic side of the Arctic and upper-halocline freshwater storage on the Pacific side of the Arctic. These comparisons are encouraging for the application of ocean reanalyses to track ocean climate variability and change at high northern latitudes.

1062. Striations in Marginal Seas and the Mediterranean Sea

Authors: Y. Zhang (Chinese Academy of Sciences); Y. P. Guan (Chinese Academy of Sciences, Zhuhai Joint Innovative Center for Climate-Environment-Ecosystem Zhuhai China)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl081050

URL: <https://doi.org/10.1029/2018gl081050>

Publication Date: 2019-02-26

Volume/Issue: Volume 46, Issue 5

References: 39

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: marginal seas, filter parameters, spatial distribution, striations, the Mediterranean sea

Funding: State Key Laboratory of Marine Environmental Science, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Striations, which are jet-like structures comprising alternating ridges and furrows within large-scale circulations, are widely studied in the open ocean, although it is unclear whether they exist in marginal or semiclosed seas. Based on an appropriate choice of averaging time scale and cutoff wavelength, our analysis reveals striations in marginal seas, including the South China Sea (SCS), Sea of Japan, and Gulf of Mexico, and in the Mediterranean Sea (MS). Compared with the open ocean, these striations have a narrower bandwidth (about 60 km in marginal seas and 100 km in the MS), fewer zonal orientations, stronger baroclinic structure, and deeper intensification. Striations in the open ocean are usually barotropic; however, within the South China Sea and MS, they are baroclinic in the upper 1,000 m and intensified at depths greater than 3,000 m.

1063. A new perspective of the 2014/15 failed El Niño as seen from ocean salinity

Authors: J. Chi; Y. Du; Y. Zhang; X. Nie; P. Shi; T. Qu

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-019-38743-z

URL: <https://doi.org/10.1038/s41598-019-38743-z>

Publication Date: 2019-02-25

Volume/Issue: Volume 9, Issue 1

References: 46

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Ministry of Land and Resources | State Oceanic Administration, Chinese Academy of Sciences (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: This study investigates the 2014/15 failed El Niño using salinity from an ocean general circulation model. The results indicate that subsurface processes were especially strong in the summer of 2014 and they led to positive sea surface salinity anomalies in the central equatorial Pacific. The positive sea surface salinity anomalies induced a westward displacement of the sea surface salinity front that represents the eastern boundary of the western Pacific warm pool, preventing the warm surface water from shifting eastward as seen in a typical El Niño event. In the meantime, more salty water was transported equatorward by a strengthening subtropical cell in the South Pacific. The enhanced subsurface processes in the central equatorial Pacific conveyed the salinity anomalies of subtropical origin to the sea surface and were largely responsible for the sea surface salinity variability but had less impacts on sea surface temperature during the 2014/15 failed El Niño, suggesting some potential advantage of ocean salinity in the El Niño-Southern Oscillation prediction.

1064. Characterizing the Transition From Balanced to Unbalanced Motions in the Southern California Current

Authors: Teresa K. Chereskin (Scripps Institution of Oceanography / University of California San Diego); Cesar B. Rocha (Scripps Institution of Oceanography / University of California San Diego, Woods Hole Oceanographic Institution); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Marcello Passaro (Deutsches Geodätisches Forschungsinstitut der Technischen Universität München Munich Germany)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014583

URL: <https://doi.org/10.1029/2018jc014583>

Publication Date: 2019-02-22

Volume/Issue: Volume 124, Issue 3

References: 63

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: internal gravity waves, mesoscale, submesoscale

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: As observations and models improve their resolution of oceanic motions at ever finer horizontal scales, interest has grown in characterizing the transition from the geostrophically balanced flows that dominate at large-scale to submesoscale turbulence and waves that dominate at small scales. In this study we examine the mesoscale-to-submesoscale (100 to 10 km) transition in an eastern boundary current, the southern California Current System (CCS), using repeated acoustic Doppler current profiler transects, sea surface height from high-resolution nadir altimetry and output from a (1/48) degrees global model simulation. In the CCS, the submesoscale is as energetic as in western boundary current regions, but the mesoscale is much weaker, and as a result the transition lacks the change in kinetic energy (KE) spectral slope observed for western boundary currents. Helmholtz and vortex-wave decompositions of the KE spectra are used to identify balanced and unbalanced contributions. At horizontal scales greater than 70 km, we find that observed KE is dominated by balanced geostrophic motions. At scales from 40 to 10 km, unbalanced contributions such as inertia-gravity waves contribute as much as balanced motions. The model KE transition occurs at longer scales, around 125 km. The altimeter spectra are consistent with acoustic Doppler current profiler/model spectra at scales longer than 70/125 km, respectively. Observed seasonality is weak. Taken together, our results suggest that geostrophic velocities can be diagnosed from sea surface height on scales larger than about 70 km in the southern CCS.

1065. Relative roles of dynamic and thermodynamic processes in causing positive and negative global mean SST trends during the past 100 years

Authors: Zhibo Li; Jia Liu; Tim Li; Ying Sun

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2019.02.002

URL: <https://doi.org/10.1016/j.dynatmoce.2019.02.002>

Publication Date: 2019-02-16

Volume/Issue: Volume 86

References: 36

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: China National Key R&D; Program, NSFC, NSF (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1066. Resolving Geophysical Signals by Terrestrial Gravimetry: A Time Domain Assessment of the Correction-Induced Uncertainty

Authors: M. Mikolaj (Section Hydrology GFZ German Research Centre for Geosciences Potsdam Germany); M. Reich (Section Hydrology GFZ German Research Centre for Geosciences Potsdam Germany); A. Güntner (Institute of Environmental Science and Geography University of Potsdam Potsdam Germany, Section Hydrology GFZ German Research Centre for Geosciences Potsdam Germany)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jb016682

URL: <https://doi.org/10.1029/2018jb016682>

Publication Date: 2019-02-15

Volume/Issue: Volume 124, Issue 2

References: 66

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: Earth tides, atmosphere, hydrology, gravity observations, nontidal ocean loading

Funding: Bundesministerium für Bildung und Forschung

License: N/A

Open Access: N/A

Abstract: Terrestrial gravimetry is increasingly used to monitor mass transport processes in geophysics boosted by the ongoing technological development of instruments. Resolving a particular phenomenon of interest, however, requires a set of gravity corrections of which the uncertainties have not been addressed up to now. In this study, we quantify the time domain uncertainty of tide, global atmospheric, large-scale hydrological, and nontidal ocean loading corrections. The uncertainty is assessed by comparing the majority of available global models for a suite of sites worldwide. The average uncertainty expressed as root-mean-square error equals 5.1 nm/s², discounting local hydrology or air pressure. The correction-induced uncertainty of gravity changes over various time periods of interest ranges from 0.6 nm/s² for hours up to a maximum of 6.7 nm/s² for 6 months. The corrections are shown to be significant and should be applied for most geophysical applications of terrestrial gravimetry. From a statistical point of view, however, resolving subtle gravity effects in the order of few nanometers per square second is challenged by the uncertainty of the corrections.

1067. Ice flow variations at Polar Record Glacier, East Antarctica

Authors: QI LIANG; CHUNXIA ZHOU; IAN M. HOWAT; SEONGSU JEONG; RUIXI LIU; YIMING CHEN

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2019.6

URL: <https://doi.org/10.1017/jog.2019.6>

Publication Date: 2019-02-15

Volume/Issue: Volume 65, Issue 250

References: 48

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: Meltwater|Fast ice|Ice divide

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Relatively little is known about the physical mechanisms that drive the dynamics of the East Antarctic outlet glaciers. Here we conduct a remote-sensing investigation of the Polar Record Glacier (PRG), East Antarctica to analyze its ice flow acceleration, ice front variations and ice surface melting. Ice flow speeds at PRG increased by up to 15% from 2005 to 2015, with substantial interannual fluctuations. The ice velocities also showed seasonal variations, accelerating by up to 9% between September and January. Multiple mechanisms contribute to the observed seasonal variations: the initial acceleration may result from the lost back-stress provided by the sea ice in the austral spring and the later speedup relate to the surface meltwater that leads to weakened ice shelf and shear margins. The sensitivity of the PRG to oceanic forcing is confirmed by comparing the secular ice velocity increases with ocean temperatures. These measurements suggest that the dynamics of East Antarctic ice shelves are sensitive to melt at both the surface and base, at a range of timescales.

1068. On the timescales and length scales of the Arctic sea ice thickness anomalies: a study based on 14 reanalyses

Authors: Leandro Ponsoni; François Massonnet; Thierry Fichefet; Matthieu Chevallier; David Docquier

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-13-521-2019

URL: <https://doi.org/10.5194/tc-13-521-2019>

Publication Date: 2019-02-14

Volume/Issue: Volume 13, Issue 2

References: 77

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . The ocean-sea ice reanalyses are one of the main sources of Arctic sea ice thickness data both in terms of spatial and temporal resolution, since observations are still sparse in time and space. In this work, we first aim at comparing how the sea ice thickness from an ensemble of 14 reanalyses compares with different sources of observations, such as moored upward-looking sonars, submarines, airborne, satellites, and ice boreholes. Second, based on the same reanalyses, we intend to characterize the timescales (persistence) and length scales of sea ice thickness anomalies. We investigate whether data assimilation of sea ice concentration by the reanalyses impacts the realism of sea ice thickness as well as its respective timescales and length scales. The results suggest that reanalyses with sea ice data assimilation do not necessarily perform better in terms of sea ice thickness compared with the reanalyses which do not assimilate sea ice concentration. However, data assimilation has a clear impact on the timescales and length scales: reanalyses built with sea ice data assimilation present shorter timescales and length scales. The mean timescales and length scales for reanalyses with data assimilation vary from 2.5 to 5.0 months and 337.0 to 732.5 km, respectively, while reanalyses with no data assimilation are characterized by values from 4.9 to 7.8 months and 846.7 to 935.7 km, respectively.

1069. Three-Dimensional Signature of the Red Sea Eddies and Eddy-Induced Transport

Authors: Peng Zhan (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia); George Krokos (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia); Daquan Guo (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia); Ibrahim Hoteit (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl081387

URL: <https://doi.org/10.1029/2018gl081387>

Publication Date: 2019-02-14

Volume/Issue: Volume 46, Issue 4

References: 42

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: Eddy

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Mesoscale eddies are a dominant feature of the Red Sea circulation, yet their three-dimensional characteristics remain largely unexplored. This hinders our understanding about eddy-induced transport in the basin. This study analyzes 14-year outputs from a high-resolution eddy-resolving model to investigate the three-dimensional signature of the Red Sea eddies, their contribution to the air-sea flux, and the eddy-induced transport of heat and salt. Eddies are mostly active and energetic in the central and northern Red Sea. Their variability explains ~8% of the total variance in the surface heat flux and, particularly, ~39% in the salt flux. The asymmetric eddy structure and meridional gradient drive significant transport of heat and salt across the basin. A negative feedback mechanism is identified that relates the eddy intensity and the meridional steepness of the mixed layer depth in the basin.

1070. 50 Years of Satellite Remote Sensing of the Ocean

Authors: Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); W. Timothy Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ronald Kwok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Meteorological Monographs

Publisher: American Meteorological Society

DOI: 10.1175/amsmonographs-d-18-0010.1

URL: <https://doi.org/10.1175/amsmonographs-d-18-0010.1>

Publication Date: 2019-02-08

Volume/Issue: Volume 59

References: 514

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: The development of the technologies of remote sensing of the ocean was initiated in the 1970s, while the ideas of observing the ocean from space were conceived in the late 1960s. The first global view from space revealed the expanse and complexity of the state of the ocean that had perplexed and inspired oceanographers ever since. This paper presents a glimpse of the vast progress made from ocean remote sensing in the past 50 years that has a profound impact on the ways we study the ocean in relation to weather and climate. The new view from space in conjunction with the deployment of an unprecedented amount of in situ observations of the ocean has led to a revolution in physical oceanography. The highlights of the achievement include the description and understanding of the global ocean circulation, the air-sea fluxes driving the coupled ocean-atmosphere system that is most prominently illustrated in the tropical oceans. The polar oceans are most sensitive to climate change with significant consequences, but owing to remoteness they were not accessible until the space age. Fundamental discoveries have been made on the evolution of the state of sea ice as well as the circulation of the ice-covered ocean. Many surprises emerged from the extraordinary accuracy and expanse of the space observations. Notable examples include the determination of the global mean sea level rise as well as the role of the deep ocean in tidal mixing and dissipation.

1071. A Global 3-D Ocean Model for PCBs: Benchmark Compounds for Understanding the Impacts of Global Change on Neutral Persistent Organic Pollutants

Authors: Charlotte C. Wagner (Harvard University); Helen M. Amos (Harvard University); Colin P. Thackray (Harvard University); Yanxu Zhang (Harvard University); Elizabeth W. Lundgren (Harvard University); Gael Forget (Massachusetts Institute of Technology); Carey L. Friedman (Corning School of Ocean Studies Maine Maritime Academy Castine ME USA, Massachusetts Institute of Technology); Noelle E. Selin (Massachusetts Institute of Technology); Rainer Lohmann (University of Rhode Island); Elsie M. Sunderland (Harvard University, Department of Environmental Health Harvard T.H. Chan School of Public Health Boston MA USA)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gb006018

URL: <https://doi.org/10.1029/2018gb006018>

Publication Date: 2019-02-07

Volume/Issue: Volume 33, Issue 3

References: 91

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: persistent pollutants, biogeochemistry, ocean, fate and transport modeling

Funding: National Aeronautics and Space Administration, Simons Foundation, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Human activities have released large quantities of neutral persistent organic pollutants (POPs) that may be biomagnified in food webs and pose health risks to wildlife, particularly top predators. Here we develop a global 3-D ocean simulation for four polychlorinated biphenyls (PCBs) spanning a range of molecular weights and volatilities to better understand effects of climate-driven changes in ocean biogeochemistry on the lifetime and distribution of POPs. Observations are most abundant in the Arctic Ocean. There, model results reproduce spatial patterns and magnitudes of measured PCB concentrations. Sorption of PCBs to suspended particles and subsequent burial in benthic marine sediment is the dominant oceanic loss process globally. Results suggest benthic sediment burial has removed 75% of cumulative PCB releases since the onset of production in 1930. Wind speed, light penetration, and ocean circulation exert a stronger and more variable influence on volatile PCB congeners with lower particle affinity such as chlorinated biphenyl-28 and chlorinated biphenyl-101. In the Arctic Ocean between 1992 and 2015, modeled evasion (losses) of the more volatile PCB congeners from the surface ocean increased due to declines in sea ice and changes in ocean circulation. By contrast, net deposition increased slightly for higher molecular weight congeners with stronger partitioning to particles. Our results suggest future climate changes will have the greatest impacts on the chemical lifetimes and distributions of volatile POPs with lower molecular weights.

1072. The Effects of Enhanced Sea Ice Export from the Ross Sea on Recent Cooling and Freshening of the Southeast Pacific

Authors: Ivana Cerovečki (Scripps Institution of Oceanography); Andrew J. S. Meijers (British Antarctic Survey, Cambridge, United Kingdom); Matthew R. Mazloff (Scripps Institution of Oceanography); Sarah T. Gille (Scripps Institution of Oceanography); Veronica M. Tamsitt (Scripps Institution of Oceanography); Paul R. Holland (British Antarctic Survey, Cambridge, United Kingdom)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0205.1

URL: <https://doi.org/10.1175/jcli-d-18-0205.1>

Publication Date: 2019-02-05

Volume/Issue: Volume 32, Issue 7

References: 78

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation, National Science Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: The top 2000 m of the Southern Ocean has freshened and warmed over recent decades. However, the high-latitude (south of 50 degreesS) southeast Pacific was observed to be cooler and fresher in the years 2008-10 compared to 2005-07 over a wide depth range including surface, mode, and intermediate waters. The causes and impacts of this event are analyzed using the ocean-sea ice data-assimilating Southern Ocean State Estimate (SOSE) and observationally based products. In 2008-10, a strong positive southern annular mode coincided with a negative El Niño-Southern Oscillation and a deep Amundsen Sea low. Enhanced meridional winds drove strong sea ice export from the eastern Ross Sea, bringing large amounts of ice to the Amundsen Sea ice edge. In 2008, together with increased precipitation, this introduced a strong freshwater anomaly that was advected eastward by the Antarctic Circumpolar Current (ACC), mixing along the way. This anomaly entered the ocean interior not only as Antarctic Intermediate Water, but also as lighter Southeast Pacific Subantarctic Mode Water (SEPSAMW). A numerical particle release experiment carried out in SOSE showed that the Ross Sea sector was the dominant source of particles reaching the SEPSAMW formation region. This suggests that large-scale climate fluctuations can induce strong interannual variability of volume and properties of SEPSAMW. These fluctuations act at different time scales: instantaneously via direct forcing and also lagged over advective time scales of several years from upstream regions.

1073. Comparison of 4DVAR and EnKF state estimates and forecasts in the Gulf of Mexico

Authors: Ganesh Gopalakrishnan (Scripps Institution of Oceanography); Ibrahim Hoteit (King Abdullah University of Science and Technology Jeddah Kingdom of Saudi Arabia); Bruce D. Cornuelle (Scripps Institution of Oceanography); Daniel L. Rudnick (Scripps Institution of Oceanography)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.3493

URL: <https://doi.org/10.1002/qj.3493>

Publication Date: 2019-02-04

Volume/Issue: Volume 145, Issue 721

References: 69

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An experiment is conducted to compare four-dimensional variational (4DVAR) and ensemble Kalman filter (EnKF) assimilation systems and their predictability in the Gulf of Mexico (GoM) using the Massachusetts Institute of Technology general circulation model (MITgcm). The quality of the ocean-state estimates, forecasts, and the contribution of ensemble prediction are evaluated. The MITgcm-Estimating the Circulation and Climate of the Ocean (ECCO) 4DVAR (MITgcm-ECCO) and the MITgcm-Data Assimilation Research Testbed (DART) EnKF (MITgcm-DART) systems were used to compute two-month hindcasts (March-April, 2010) by assimilating satellite-derived along-track sea-surface height (SSH) and gridded sea-surface temperature (SST) observations. The estimates from both methods at the end of the hindcast period were then used to initialize forecasts for two months (May-June, 2010). This period was selected because a loop current (LC) eddy (Eddy Franklin: Eddy-F) detachment event occurred at the end of May 2010, immediately after the Deepwater Horizon (DwH) oil spill. Despite some differences between the setups, both systems produce analyses and forecasts of comparable quality and both solutions significantly outperformed model persistence. A reference forecast initialized from the 1/12 degrees Hybrid Coordinate Ocean Model (HYCOM)/NCODA global analysis also performed well. The EnKF experiments for sensitivity to filter parameters showed enhanced predictability when using more ensemble members and stronger covariance localization, but not for larger inflation. The EnKF experiments varying the number of assimilation cycles showed enhanced short-term (long-term) predictability with fewer (more) assimilation cycles. Additional hindcast and forecast experiments at other times of significant LC evolution showed mixed performance of both systems, which depends strongly on the background state of the GoM circulation. The present work demonstrates a practical application of both assimilation methods for the GoM and compares them in a limited number of realizations. The overall conclusion showing improved short-term (long-term) predictability for EnKF (4DVAR) carries an important caveat that the results from this study are specific to a few 4DVAR and EnKF LC eddy separation experiments in the GoM and cannot be generalized to

conclude the relative performance of both methods, especially in other applications. However, some of the concepts and methods should carry over to other applications.

1074. Observational and model studies of synoptic current fluctuations in the Bohai Strait on the Chinese continental shelf

Authors: Yang Ding; Xianwen Bao; Zhigang Yao; Congcong Bi; Kai Wan; Min Bao; Zhipeng Jiang; Jun Song; Jia Gao

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-019-01247-5

URL: <https://doi.org/10.1007/s10236-019-01247-5>

Publication Date: 2019-02-01

Volume/Issue: Volume 69, Issue 3

References: 42

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

1075. Interannual Salinity Variability in the Tropical Pacific in CMIP5 Simulations

Authors: Hai Zhi; Rong-Hua Zhang; Pengfei Lin; Peng Yu

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-018-7309-1

URL: <https://doi.org/10.1007/s00376-018-7309-1>

Publication Date: 2019-01-31

Volume/Issue: Volume 36, Issue 4

References: 73

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1076. A Strong Role for the AMOC in Partitioning Global Energy Transport and Shifting ITCZ Position in Response to Latitudinally Discrete Solar Forcing in CESM1.2

Authors: Sungduk Yu (University of California Irvine); Michael S. Pritchard (University of California Irvine)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0360.1

URL: <https://doi.org/10.1175/jcli-d-18-0360.1>

Publication Date: 2019-01-31

Volume/Issue: Volume 32, Issue 8

References: 88

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Ocean circulation responses to interhemispheric radiative imbalance can damp north-south migrations of the intertropical convergence zone (ITCZ) by reducing the burden on atmospheric energy transport. The role of the Atlantic meridional overturning circulation (AMOC) in such dynamics has not received much attention. Here, we present coupled climate modeling results that suggest AMOC responses are of first-order importance to muting ITCZ shift magnitudes as a pair of hemispherically asymmetric solar forcing bands is moved from equatorial to polar latitudes. The cross-equatorial energy transport response to the same amount of interhemispheric forcing becomes systematically more ocean-centric when higher latitudes are perturbed in association with strengthening AMOC responses. In contrast, the responses of the Pacific subtropical cell are not monotonic and cannot predict this variance in the ITCZ's equilibrium position. Overall, these results highlight the importance of the meridional distribution of interhemispheric radiative imbalance and the rich buffering of internal feedbacks that occurs in dynamic versus thermodynamic (slab) ocean modeling experiments. Mostly, the results imply that the problem of developing a theory of ITCZ migration is entangled with that of understanding the AMOC's response to hemispherically asymmetric radiative forcing--a difficult topic deserving of focused analysis across more climate models.

1077. The intrinsic nonlinear multiscale interactions among the mean flow, low frequency variability and mesoscale eddies in the Kuroshio region

Authors: Yang Yang; X. San Liang

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-018-9289-4

URL: <https://doi.org/10.1007/s11430-018-9289-4>

Publication Date: 2019-01-26

Volume/Issue: Volume 62, Issue 3

References: 67

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1078. The Trend and Seasonal Change of Sediment in the East China Sea Detected by GRACE

Authors: Le Chang (Chinese Academy of Sciences); He Tang (Chinese Academy of Sciences); Shuang Yi (Chinese Academy of Sciences); Wenke Sun (Chinese Academy of Sciences)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl081652

URL: <https://doi.org/10.1029/2018gl081652>

Publication Date: 2019-01-24

Volume/Issue: Volume 46, Issue 3

References: 34

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: The deposition of land sediments onto the broad continental shelf of the East China Sea (ECS) is controlled by complex, dynamic hydrological mechanisms. In this study, we apply the data from Gravity Recovery and Climate Experiment (GRACE) satellite to detect the gravity signal caused by changes in sedimentation. The GRACE-observed gravity signal is separated into the individual contributions using multisource data. After correction of each known individual signal, the sediment-caused gravity change signal is used to invert for the sediment mass change in the ECS using a space domain inverse method. The results indicate that the total GRACE-inverted sediment deposition rate was 0.82 ± 0.3 Gt/year from 2005 to 2015, which is in agreement with in situ sediment observations. The seasonal sediment change is also calculated and discussed. It is caused by sediment transport between the ECS and the Yellow Sea.

1079. The export flux of particulate organic carbon derived from ^{210}Po ■ ^{210}Pb disequilibria along the North Atlantic GEOTRACES GA01 transect: GEOVIDE cruise

Authors: Yi Tang; Nolwenn Lemaitre; Maxi Castrillejo; Montserrat Roca-Martí; Pere Masqué; Gillian Stewart

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-16-309-2019

URL: <https://doi.org/10.5194/bg-16-309-2019>

Publication Date: 2019-01-24

Volume/Issue: Volume 16, Issue 2

References: 65

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Division of Ocean Sciences, Agence Nationale de la Recherche, Centre National de la Recherche Scientifique

License: CC BY 4.0

Open Access: N/A

Abstract: . The disequilibrium between ^{210}Po activity and ^{210}Pb activity in seawater samples was determined along the GEOTRACES GA01 transect in the North Atlantic during the GEOVIDE cruise (May-June 2014). A steady-state model was used to quantify vertical export of particulate ^{210}Po . Vertical advection was incorporated into one version of the model using time-averaged vertical velocity, which had substantial variance. This resulted in large uncertainties for the ^{210}Po export flux in this model, suggesting that those calculations of ^{210}Po export fluxes should be used with great care. Despite the large uncertainties, there is no question that the deficits of ^{210}Po in the Iberian Basin and at the Greenland Shelf have been strongly affected by vertical advection. Using the export flux of ^{210}Po and the particulate organic carbon (POC) to ^{210}Po ratio of total ($> 1 \mu\text{m}$) particles, we determined the POC export fluxes along the transect. Both the magnitude and efficiency of the estimated POC export flux from the surface ocean varied spatially within our study region. Export fluxes of POC ranged from negligible to $10 \text{ mmol C m}^{-2} \text{ d}^{-1}$, with enhanced POC export in the Labrador Sea. The cruise track was characterized by overall low POC export relative to net primary production (export efficiency $< 1\%$ -15%), but relatively high export efficiencies were seen in the basins where diatoms dominated the phytoplankton community. The particularly low export efficiencies in the Iberian Basin, on the other hand, were explained by the dominance of smaller phytoplankton, such as cyanobacteria or coccolithophores. POC fluxes estimated from the ^{210}Po ■ ^{210}Pb and ^{234}Th ■ ^{238}U disequilibria agreed within a factor of 3 along the transect, with higher POC estimates generally derived from ^{234}Th . The differences were attributed to integration timescales and the history of bloom events.

1080. The tropical-subtropical coupling in the Southeast Atlantic from the perspective of the northern Benguela upwelling system

Authors: Lydia Siegfried; Martin Schmidt; Volker Mohrholz; Hans Pogrzeba; Pascal Nardini; Michael Böttinger; Gerik Scheuermann

Journal: PLOS ONE

Publisher: Public Library of Science (PLoS)

DOI: 10.1371/journal.pone.0210083

URL: <https://doi.org/10.1371/journal.pone.0210083>

Publication Date: 2019-01-22

Volume/Issue: Volume 14, Issue 1

References: 116

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1081. Modeling the Recent Changes in the Arctic Ocean CO₂Sink (2006-2013)

Authors: Manfredi Manizza (Scripps Institution of Oceanography / University of California San Diego); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gb006070

URL: <https://doi.org/10.1029/2018gb006070>

Publication Date: 2019-01-19

Volume/Issue: Volume 33, Issue 3

References: 68

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: sea-ice, carbon cycle, Arctic Ocean, polar oceans, climate change (and 1 more)

Funding: Connecticut Space Grant College Consortium

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Open Access: N/A

Abstract: The Arctic Ocean (AO) and its associated marginal seas have recently experienced rapid climate and environmental changes, most notably sea-ice area (SIA) loss and warming potentially impacting its uptake of carbon dioxide (CO₂). We used the state-of-the-art ECCO2-Darwin coupled ocean-biogeochemistry model to simulate the 2006-2013 period and investigate the impact of changing SIA on the CO₂uptake of the AO. We find that the mean annual CO₂sink of the AO is 153 +/- 14 TgC a⁻¹and the CO₂sink decreased at a rate of 3.6 TgC a⁻¹even though SIA decreased by 8 x 10⁴ km² a⁻¹over the same period. Extreme SIA loss in 2007 resulted in a 185.4 TgC CO₂sink, an increase ~20% over the 2006-2013 mean. In contrast, extreme SIA loss of 2012 resulted in a CO₂sink of the AO of only 146.3 TgC due to two main factors: (1) increased both wind speed and stratification in the Eastern Siberian Sea absorbing less CO₂and (2) decreased primary production and area of air-sea gas exchange in the Chukchi and Nordic Seas. Our model captures a trend of decreasing CO₂sink in most of the Chukchi Sea during fall but does not show the changes in winter CO₂sink in the Nordic and Barents Seas as previous independent studies have suggested. Our results indicate that future AO-atmosphere CO₂exchange will be determined by complex interplay of SIA and other environmental drivers.

1082. Toward modular in situ visualization in Earth system models: the regional modeling system RegESM 1.1

Authors: Ufuk Utku Turuncoglu

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-12-233-2019

URL: <https://doi.org/10.5194/gmd-12-233-2019>

Publication Date: 2019-01-16

Volume/Issue: Volume 12, Issue 1

References: 63

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The data volume produced by regional and global multicomponent Earth system models is rapidly increasing because of the improved spatial and temporal resolution of the model components and the sophistication of the numerical models regarding represented physical processes and their complex non-linear interactions. In particular, very small time steps need to be defined in non-hydrostatic high-resolution modeling applications to represent the evolution of the fast-moving processes such as turbulence, extratropical cyclones, convective lines, jet streams, internal waves, vertical turbulent mixing and surface gravity waves. Consequently, the employed small time steps cause extra computation and disk input-output overhead in the modeling system even if today's most powerful high-performance computing and data storage systems are considered. Analysis of the high volume of data from multiple Earth system model components at different temporal and spatial resolutions also poses a challenging problem to efficiently perform integrated data analysis of the massive amounts of data when relying on the traditional postprocessing methods today. This study mainly aims to explore the feasibility and added value of integrating existing in situ visualization and data analysis methods within the model coupling framework. The objective is to increase interoperability between Earth system multicomponent code and data-processing systems by providing an easy-to-use, efficient, generic and standardized modeling environment. The new data analysis approach enables simultaneous analysis of the vast amount of data produced by multicomponent regional Earth system models during the runtime. The presented methodology also aims to create an integrated modeling environment for analyzing fast-moving processes and their evolution both in time and space to support a better understanding of the underlying physical mechanisms. The state-of-the-art approach can also be employed to solve common problems in the model development cycle, e.g., designing a new subgrid-scale parameterization that requires inspecting the integrated model behavior at a higher temporal and spatial scale simultaneously and supporting visual debugging of the multicomponent modeling systems, which usually are not facilitated by existing model coupling libraries and modeling systems.

1083. Quantifying the Role of Oceanic Feedbacks on ENSO Asymmetry

Authors: Cong Guan (Institute of Oceanographic Instrumentation Shandong Academy of Sciences Qingdao China, Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Michael J. McPhaden (NOAA/Pacific Marine Environmental Laboratory Seattle WA USA); Fan Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Shijian Hu (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl081332

URL: <https://doi.org/10.1029/2018gl081332>

Publication Date: 2019-01-15

Volume/Issue: Volume 46, Issue 4

References: 46

Cited by: 27 publications

Language: en

Subjects: N/A

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Open Access: N/A

Abstract: The role of oceanic feedbacks in determining the asymmetry of El Niño-Southern Oscillation (ENSO) magnitude, spatial structure, and duration is quantified on the basis of a novel temperature variance budget. Results confirm previous studies that in the eastern Pacific, El Niño warm temperature anomalies are larger in magnitude than La Niña cold temperature anomalies mainly due to stronger positive oceanic feedbacks for El Niño. We find that La Niña cold anomalies are typically stronger than El Niño warm anomalies in the central Pacific with a faster growth rate for cold anomalies, due to a stronger positive thermocline feedback and weaker nonlinear damping. The thermocline feedback related to recharge oscillator dynamics plays a dominate role and leads to asymmetry in the duration of El Niño and La Niña events. In particular, the thermocline feedback becomes significantly negative during the late decaying phase of El Niño and speeds up its demise.

1084. Predictability of Ocean Heat Content From Electrical Conductance

Authors: D. S. Trossman (University of Texas at Austin); R. H. Tyler (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014740

URL: <https://doi.org/10.1029/2018jc014740>

Publication Date: 2019-01-14

Volume/Issue: Volume 124, Issue 1

References: 64

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: conductance, prediction, conductivity, ocean, state estimate (and 1 more)

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Ocean heat content (OHC) is a key climate variable that needs to be monitored to know how Earth's energy imbalance is changing, yet observing OHC remains a challenge. The present study examines whether a depth integral of the ocean's electrical conductivity ("conductance"), which may be inferred from both in situ methods and satellite magnetometers over the global ocean, could help monitor OHC. The ocean's electrical conductivity locally depends on temperature, salinity, and pressure, but it is not as well known how the conductance depends on OHC and ocean salt content. By examining the output of an ocean state estimate shown to agree well with observations that have not been assimilated, this study evaluates the fundamental limitations of using perfectly known ocean conductance to predict OHC, rather than the challenges associated with accounting for observational error. It is found that the ocean's conductance and OHC fields are nonlinearly related but nevertheless highly correlated. A statistical framework tends to predict OHC more accurately than ocean salt content from ocean conductance in regions where conductivity is more sensitive to salinity than temperature. The annually (bidecadally) averaged OHC can be predicted from a combination of conductance and depth-averaged conductivity ocean fields to within nearly 0.1% (1%) error globally and even more accurately in many poorly observed (e.g., ice-covered) regions. Practical application of this statistical framework to monitor OHC requires examination of the effect of uncertainties in the observed bathymetry and ocean conductance, which vary with application.

1085. Coupled ocean-atmosphere dynamics of the 2017 extreme coastal El Niño

Authors: Qihua Peng; Shang-Ping Xie; Dongxiao Wang; Xiao-Tong Zheng; Hong Zhang

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-018-08258-8

URL: <https://doi.org/10.1038/s41467-018-08258-8>

Publication Date: 2019-01-11

Volume/Issue: Volume 10, Issue 1

References: 41

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: In March 2017, sea surface temperatures off Peru rose above 28 degreesC, causing torrential rains that affected the lives of millions of people. This coastal warming is highly unusual in that it took place with a weak La Niña state. Observations and ocean model experiments show that the downwelling Kelvin waves caused by strong westerly wind events over the equatorial Pacific, together with anomalous northerly coastal winds, are important. Atmospheric model experiments further show the anomalous coastal winds are forced by the coastal warming. Taken together, these results indicate a positive feedback off Peru between the coastal warming, atmospheric deep convection, and the coastal winds. These coupled processes provide predictability. Indeed, initialized on as early as 1 February 2017, seasonal prediction models captured the extreme rainfall event. Climate model projections indicate that the frequency of extreme coastal El Niño will increase under global warming.

1086. On the Spatial Scales to be Resolved by the Surface Water and Ocean Topography Ka-Band Radar Interferometer

Authors: Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Shuiming Chen (University of Hawai'i at Mānoa, Honolulu, Hawaii); Bo Qiu (University of Hawai'i at Mānoa, Honolulu, Hawaii); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-18-0119.1

URL: <https://doi.org/10.1175/jtech-d-18-0119.1>

Publication Date: 2019-01-11

Volume/Issue: Volume 36, Issue 1

References: 37

Cited by: 55 publications

Language: N/A

Subjects: N/A

Keywords: Altimetry, Sea state, Satellite observations, Seasonal variability

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration, National Aeronautics and Space Administration (and 2 more)

License: N/A

Open Access: N/A

Abstract: The Surface Water and Ocean Topography (SWOT) mission aims to measure the sea surface height (SSH) at a high spatial resolution using a Ka-band radar interferometer (KaRIn). The primary oceanographic objective is to characterize the ocean eddies at a spatial resolution of 15 km for 68% of the ocean surface. This resolution is derived from the ratio between the wavenumber spectrum of the conventional altimeter (projected to submesoscale) and the SWOT SSH errors. While the 15-km threshold is useful as a global approximation of the spatial scales resolved by SWOT (SWOT scale), it can be misleading for regional studies. Here we revisit the problem using a high-resolution (~2-km horizontal grid spacing) tide-resolving global ocean simulation and map the SWOT scale as a function of location and season. The results show that the SWOT scale increases, in general, from about 15 km at low latitudes to ~30-45 km at mid- and high latitudes but with a large geographical dependence. A SWOT scale smaller than 30 km is expected in the high-latitude energetic regions. The SWOT scale varies seasonally as a result of the seasonality in both the noise and ocean signals. The seasonality also has a geographical dependence. Both eddies and internal gravity waves/tides contribute significantly to the SWOT scale variation. Our analysis provides model predictions for interpreting the anticipated observations from SWOT and guidance for the development of analysis methodologies.

1087. Hydrological signals in polar motion excitation - Evidence after fifteen years of the GRACE mission

Authors: Jolanta Nastula; Małgorzata Wiśka; Justyna Śliwińska; David Salstein

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2019.01.014

URL: <https://doi.org/10.1016/j.jog.2019.01.014>

Publication Date: 2019-01-10

Volume/Issue: Volume 124

References: 46

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Polish National Science Centre NCN

License: N/A

Open Access: N/A

Abstract: N/A

1088. Global reconstruction of historical ocean heat storage and transport

Authors: Laure Zanna (University of Oxford); Samar Khatiwala (University of Oxford); Jonathan M. Gregory (Met Office Hadley Centre, Met Office, University of Reading); Jonathan Ison (University of Oxford); Patrick Heimbach (University of Texas at Austin)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.1808838115

URL: <https://doi.org/10.1073/pnas.1808838115>

Publication Date: 2019-01-08

Volume/Issue: Volume 116, Issue 4

References: 50

Cited by: 185 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: RCUK | Natural Environment Research Council, NASA | Jet Propulsion Laboratory

License: CC BY 4.0

Open Access: N/A

Abstract: SignificanceSince the 19th century, rising greenhouse gas concentrations have caused the ocean to absorb most of the Earth's excess heat and warm up. Before the 1990s, most ocean temperature measurements were above 700 m and therefore, insufficient for an accurate global estimate of ocean warming. We present a method to reconstruct ocean temperature changes with global, full-depth ocean coverage, revealing warming of $436 \times 10^{21} \text{ J}$ since 1871. Our reconstruction, which agrees with other estimates for the well-observed period, demonstrates that the ocean absorbed as much heat during 1921-1946 as during 1990-2015. Since the 1950s, up to one-half of excess heat in the Atlantic Ocean at midlatitudes has come from other regions via circulation-related changes in heat transport.

1089. The global toroidal magnetic field generated in the Earth's oceans

Authors: Jakub Velímský; Libor Šachl; Zdeněk Martinec

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2018.12.026

URL: <https://doi.org/10.1016/j.epsl.2018.12.026>

Publication Date: 2019-01-08

Volume/Issue: Volume 509

References: 33

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency, Ministerstvo Školství, Mládeže a Tělovýchovy, Grantová Agentura České Republiky (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1090. The Global Overturning Circulation

Authors: Paola Cessi (Scripps Institution of Oceanography / University of California San Diego)

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

DOI: 10.1146/annurev-marine-010318-095241

URL: <https://doi.org/10.1146/annurev-marine-010318-095241>

Publication Date: 2019-01-04

Volume/Issue: Volume 11, Issue 1

References: 67

Cited by: 65 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In this article, I use the Estimating the Circulation and Climate of the Ocean version 4 (ECCO4) reanalysis to estimate the residual meridional overturning circulation, zonally averaged, over the separate Atlantic and Indo-Pacific sectors. The abyssal component of this estimate differs quantitatively from previously published estimates that use comparable observations, indicating that this component is still undersampled. I also review recent conceptual models of the oceanic meridional overturning circulation and of the mid-depth and abyssal stratification. These theories show that dynamics in the Antarctic circumpolar region are essential in determining the deep and abyssal stratification. In addition, they show that a mid-depth cell consistent with observational estimates is powered by the wind stress in the Antarctic circumpolar region, while the abyssal cell relies on interior diapycnal mixing, which is bottom intensified.

1091. Global-scale dispersal and connectivity in mangroves

Authors: Tom Van der Stocken (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dustin Carroll (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Marc Simard (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nico Koedam (Vrije Universiteit Brussel)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.1812470116

URL: <https://doi.org/10.1073/pnas.1812470116>

Publication Date: 2019-01-01

Volume/Issue: Volume 116, Issue 3

References: 81

Cited by: 91 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Significance Mangroves are of considerable ecological and socioeconomical importance; however, substantial areal losses have been recorded in many regions, driven primarily by anthropogenic disturbances and sea level rise. Oceanic dispersal of mangrove propagules provides a key mechanism for shifting distributions in response to environmental change. Here we provide a model framework for describing global dispersal and connectivity in mangroves. We identify important dispersal routes, barriers, and stepping-stones and quantify the influence of minimum and maximum floating periods on simulated connectivity patterns. Our study provides a baseline to improve our understanding of observed mangrove species distributions and, in combination with climate data, the expected range shifts under climate change.

1092. Assessing hydrological signal in polar motion from observations and geophysical models

Authors: Małgorzata Wiśka; Justyna Śliwińska

Journal: Studia Geophysica et Geodaetica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11200-018-1028-z

URL: <https://doi.org/10.1007/s11200-018-1028-z>

Publication Date: 2018-12-19

Volume/Issue: Volume 63, Issue 1

References: 47

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1093. Spin-Up of the Southern Hemisphere Super Gyre

Authors: Tangdong Qu (University of California Los Angeles); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rana A. Fine (University of Miami)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014391

URL: <https://doi.org/10.1029/2018jc014391>

Publication Date: 2018-12-15

Volume/Issue: Volume 124, Issue 1

References: 80

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: Tasman leakage, Southern Hemisphere super gyre, southern annular mode, Agulhas leakage, westerly winds

Funding: National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: This study investigates the variability of the Southern Hemisphere super gyre (SHSG), using remotely sensed altimeter measurements, in situ Argo observations, and results from an ocean state estimate of the Consortium for Estimating the Circulation and Climate of the Ocean. Analyses of altimeter data show large trends of sea surface height, and their positive-negative contrast suggests a strengthening of subtropical gyres in all the three Southern Hemisphere oceans since 1993. Analyses of Argo data and the Estimating the Circulation and Climate of the Ocean estimate indicate that these dynamic signals of southern subtropical gyres extend to at least 2,000 m. The three southern subtropical gyres are interconnected through the Tasman and Agulhas leakages and vary consistently during the period 1993-2016. The Tasman and Agulhas leakages also show an increasing trend of inter-ocean water exchange with a typical increase of ~ 2 Sv (1 Sv = 10^6 m³/s) per decade, indicative of a two-decade-long spin-up of the SHSG. The strengthening and poleward shift of westerly winds are associated with an increasing southern annular mode, which affect the midlatitude and high-latitude Southern Hemisphere oceans and contribute to the spin-up of the SHSG.

1094. Atmospheric Origins of Variability in the South Atlantic Meridional Overturning Circulation

Authors: Timothy Smith (University of Texas at Austin); Patrick Heimbach (University of Texas at Austin)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0311.1

URL: <https://doi.org/10.1175/jcli-d-18-0311.1>

Publication Date: 2018-12-12

Volume/Issue: Volume 32, Issue 5

References: 51

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration, Jet Propulsion Laboratory

License: N/A

Open Access: N/A

Abstract: Insights from the RAPID-MOCHA observation network in the North Atlantic have motivated a recent focus on the South Atlantic, where water masses are exchanged with neighboring ocean basins. In this study, variability in the South Atlantic meridional overturning circulation (SAMOC) at 34 degreesS is attributed to global atmospheric forcing using an inverse modeling approach. The sensitivity of the SAMOC to atmospheric state variables is computed with the adjoint of the Massachusetts Institute of Technology general circulation model, which is fit to 20 years of observational data in a dynamically consistent framework. The dynamical pathways highlighted by these sensitivity patterns show that the domain of influence for the SAMOC is broad, covering neighboring ocean basins even on short time scales. This result differs from what has previously been shown in the North Atlantic, where Atlantic meridional overturning circulation (AMOC) variability is largely governed by dynamics confined to that basin. The computed sensitivities are convolved with surface atmospheric state variability from ERA-Interim to attribute the influence of each external forcing variable (e.g., wind stress, precipitation) on the SAMOC from 1992 to 2011. Here, local wind stress perturbations are shown to dominate variability on seasonal time scales while buoyancy forcing plays a minor role, confirming results from past forward perturbation experiments. Interannual variability, however, is shown to have originated from remote locations across the globe, including a nontrivial component originating from the tropical Pacific. The influence of atmospheric forcing emphasizes the importance of continuous widespread observations of the global atmospheric state for attributing observed AMOC variability.

1095. Mass-related excitation of polar motion: an assessment of the new RL06 GRACE gravity field models

Authors: Franziska Göttl; Michael Schmidt; Florian Seitz

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-018-0968-4

URL: <https://doi.org/10.1186/s40623-018-0968-4>

Publication Date: 2018-12-12

Volume/Issue: Volume 70, Issue 1

References: 35

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: Polar motion

Funding: Deutsche Forschungsgemeinschaft

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1096. Ocean Acoustic Tomography in the North Atlantic

Authors: Brian D. Dushaw (Nansen Environmental and Remote Sensing Center)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-18-0082.1

URL: <https://doi.org/10.1175/jtech-d-18-0082.1>

Publication Date: 2018-12-07

Volume/Issue: Volume 36, Issue 2

References: 78

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Acoustic measurements/effects, Optimization, Principal components analysis, In situ oceanic observations (and 1 more)

Funding: Office of Naval Research

License: N/A

Open Access: N/A

Abstract: An objective mapping exercise simulating observations of temperature in the North Atlantic Ocean was used to assess the resolution capabilities of ocean acoustic tomography in combination with Argo floats. A set of basis functions for a basinwide area was obtained from a singular value decomposition of a covariance derived from an ocean state estimate. As demonstrated by the formal uncertainty estimates from the objective maps, Argo and tomography are complementary measurements. In several examples, each separately obtained uncertainty for determining large-scale monthly average temperature of about 50% of prior (resolved 75% of variance), while when both data were employed, uncertainties were reduced to about 25% of prior (resolved 94% of variance). Possible tomography configurations range from arrays that span specific regions to line arrays that supplement existing observations to arrays that span the Atlantic basin. A basinwide array consisting of two acoustic sources and seven receivers can be used to significantly reduce the uncertainties of estimated broad-scale temperature. An optimal observing system study would comprise simulated measurements in combination with data assimilation techniques and numerical ocean modeling. This objective map study, however, showed that the addition of tomography to the existing observing system could substantially reduce the uncertainties for estimated large-scale temperature. To the extent that tomography offers a 50% reduction in uncertainty at a fraction of the cost of the Argo program, it is a cost-effective contribution to the ocean observing system.

1097. Biogeography of zooplankton feeding strategy

Authors: A. E. Friederike Prowe (GEOMAR Helmholtz Centre for Ocean Research, Technical University of Denmark); André W. Visser (Technical University of Denmark); Ken H. Andersen (Technical University of Denmark); Sanae Chiba (JAMSTEC, Yokohama Institute for Earth Sciences Yokohama City Kanagawa Japan, UNEP-WCMC Cambridge UK); Thomas Kiørboe (Technical University of Denmark)

Journal: Limnology and Oceanography

Publisher: Wiley

DOI: 10.1002/lno.11067

URL: <https://doi.org/10.1002/lno.11067>

Publication Date: 2018-12-07

Volume/Issue: Volume 64, Issue 2

References: 90

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, V. Kann Rasmussen Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: The trait-based approach is increasingly used in plankton ecology to understand diversity, community dynamics, and biogeography. While on the global scale phytoplankton traits are fairly well established, zooplankton traits are only beginning to be understood. One taxa-transcending aspect of zooplankton diversity is the distinction between ambush and active feeding strategies. We present a global-scale empirical estimate of feeding strategy derived from copepod abundance observations, which for the first time suggests a distinct trait biogeography with ambush feeding as the dominant feeding strategy at higher, but not at lower latitudes. To explain this trait biogeography, we develop a minimalist trade-off based model of feeding strategies based on encounter rates between zooplankton predators and their phyto- and zooplankton prey. Encounter rates are governed by the two traits, size and motility, that trade off against predation risk. Coupled to a three-dimensional dynamic green ocean model, our idealized encounter model captures the observed feeding strategy biogeography. In the model, this pattern arises from competing dominant food chains within the food web and is shaped by a trophic trait cascade of active vs. passive feeding in adjacent trophic levels. The dominant feeding strategy structures the pathways and efficiency of energy and biomass transfer through the model food web, with consequences for primary production, export and higher trophic levels. Understanding feeding strategies is therefore important for fisheries, biogeochemical cycling, and long-term predictions of ecosystem dynamics and functioning by global dynamic green ocean models.

1098. Terrestrial water storage variations and their effect on polar motion

Authors: Justyna ■liwi■ska; Ma■gorzata Wi■ska; Jolanta Nastula

Journal: Acta Geophysica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11600-018-0227-x

URL: <https://doi.org/10.1007/s11600-018-0227-x>

Publication Date: 2018-12-06

Volume/Issue: Volume 67, Issue 1

References: 85

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Narodowe Centrum Nauki

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1099. On the Effects of Increased Vertical Mixing on the Arctic Ocean and Sea Ice

Authors: Xi Liang (National Marine Environmental Forecasting Center); Martin Losch (Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014303

URL: <https://doi.org/10.1029/2018jc014303>

Publication Date: 2018-12-06

Volume/Issue: Volume 123, Issue 12

References: 54

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: ocean circulation, sea ice, vertical mixing, Arctic

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Against the backdrop of Arctic sea ice decline, vertical mixing in the interior Arctic Ocean will most likely change, but it is still unclear how the Arctic Ocean and sea ice will respond. In this paper, a sea ice-ocean model with a simple parameterization for interior background mixing is used to investigate the Arctic Ocean and sea ice response to a scenario of increased vertical mixing. It is found that more vertical mixing reduces sea ice thickness all year round and decreases summertime sea ice concentration. More vertical mixing leads to a cooling of the Arctic halocline layer and Atlantic Water layer below. The increased vertical mixing speeds up vertical heat and salinity exchange, brings the underlying warm and saline water into the surface layer, and contributes to the sea ice decline. Vertical salinity gradient of the Arctic halocline layer reduces together with a much fresher Atlantic Water layer, and more volume of saline water enters the deep ocean below the Atlantic Water layer. As a result, the reduced Arctic Ocean stratification leads to an adjustment of the circulation pattern. Cyclonic circulation anomalies occur in the surface layer shallower than 20-m depth and in the interior ocean deeper than 700-m depth, while anticyclonic circulation anomalies occur between these depths. Our study suggests that the extra heat and salinity exchange induced by more vertical mixing will have a noticeable impact on the upper ocean structure, ocean circulation, and sea ice in a changing Arctic Ocean.

1100. Locally stationary spatio-temporal interpolation of Argo profiling float data

Authors: Mikael Kuusela (Carnegie Mellon University); Michael L. Stein (University of Chicago)

Journal: Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rspa.2018.0400

URL: <https://doi.org/10.1098/rspa.2018.0400>

Publication Date: 2018-12-05

Volume/Issue: Volume 474, Issue 2220

References: 50

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF, US Department of Energy

License: N/A

Open Access: N/A

Abstract: Argo floats measure seawater temperature and salinity in the upper 2000 m of the global ocean. Statistical analysis of the resulting spatio-temporal dataset is challenging owing to its non-stationary structure and large size. We propose mapping these data using locally stationary Gaussian process regression where covariance parameter estimation and spatio-temporal prediction are carried out in a moving-window fashion. This yields computationally tractable non-stationary anomaly fields without the need to explicitly model the non-stationary covariance structure. We also investigate Student-t-distributed fine-scale variation as a means to account for non-Gaussian heavy tails in ocean temperature data. Cross-validation studies comparing the proposed approach with the existing state of the art demonstrate clear improvements in point predictions and show that accounting for the non-stationarity and non-Gaussianity is crucial for obtaining well-calibrated uncertainties. This approach also provides data-driven local estimates of the spatial and temporal dependence scales for the global ocean, which are of scientific interest in their own right.

1101. The Cold Transit of Southern Ocean Upwelling

Authors: Dafydd Gwyn Evans (University of Southampton, National Oceanography Centre); Jan D. Zika (University of New South Wales); Alberto C. Naveira Garabato (University of Southampton, National Oceanography Centre); A. J. George Nurser (National Oceanography Centre)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl079986

URL: <https://doi.org/10.1029/2018gl079986>

Publication Date: 2018-12-05

Volume/Issue: Volume 45, Issue 24

References: 42

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Centre of Excellence for Environmental Decisions, Australian Research Council, Natural Environment Research Council

License: Creative Commons

Open Access: N/A

Abstract: The upwelling of deep waters in the Southern Ocean is a critical component of the climate system. The time and zonal mean dynamics of this circulation describe the upwelling of Circumpolar Deep Water and the downwelling of Antarctic Intermediate Water. The thermodynamic drivers of the circulation and their seasonal cycle play a potentially key regulatory role. Here an observationally constrained ocean model and an observation-based seasonal climatology are analyzed from a thermodynamic perspective, to assess the diabatic processes controlling overturning in the Southern Ocean. This reveals a seasonal two-stage cold transit in the formation of intermediate water from upwelled deep water. First, relatively warm and saline deep water is transformed into colder and fresher near-surface winter water via wintertime mixing. Second, winter water warms to form intermediate water through summertime surface heat fluxes. The mixing-driven pathway from deep water to winter water follows mixing lines in thermohaline coordinates indicative of nonlinear processes.

1102. GPS Time Series Analysis from Aboa the Finnish Antarctic Research Station

Authors: Constantin-Octavian Andrei (Department of Geodesy and Geodynamics, Finnish Geospatial Research Institute FGI, Geodeetinrinne 2, FI-02430 Masala, Finland); Sonja Lahtinen (Department of Geodesy and Geodynamics, Finnish Geospatial Research Institute FGI, Geodeetinrinne 2, FI-02430 Masala, Finland); Maaria Nordman (Department of Geodesy and Geodynamics, Finnish Geospatial Research Institute FGI, Geodeetinrinne 2, FI-02430 Masala, Finland, Stockholm University); Jyri Näränen (Department of Geodesy and Geodynamics, Finnish Geospatial Research Institute FGI, Geodeetinrinne 2, FI-02430 Masala, Finland); Hannu Koivula (Department of Geodesy and Geodynamics, Finnish Geospatial Research Institute FGI, Geodeetinrinne 2, FI-02430 Masala, Finland); Markku Poutanen (Department of Geodesy and Geodynamics, Finnish Geospatial Research Institute FGI, Geodeetinrinne 2, FI-02430 Masala, Finland); Juha Hyypä (Department of Remote Sensing and Photogrammetry, Finnish Geospatial Research Institute FGI, Geodeetinrinne 2, FI-02430 Masala, Finland)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs10121937

URL: <https://doi.org/10.3390/rs10121937>

Publication Date: 2018-12-03

Volume/Issue: Volume 10, Issue 12

References: 89

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: Uncertainty, Antarctica, Trend, Noise, Coordinates (and 4 more)

Funding: Academy of Finland

License: CC BY 4.0

Open Access: N/A

Abstract: Continuous Global Positioning System (GPS) observations have been logged at the Finnish Antarctic research station (Aboa) since February 2003. The station is located in Dronning Maud Land, East Antarctica. Almost 5000 daily observation files have been archived based on yearly scientific expeditions. These files have not been fully analysed until now. This study reports for the first time on the consistent and homogeneous data processing and analysis of the 15-year long time series. Daily coordinates are obtained using Precise Point Positioning (PPP) processing based on two approaches. The first approach is based on the Kalman filter and uses the RTKLIB open source library to produce daily solutions by unconventionally running the filter in the forward and backward direction. The second approach uses APPS web service and is based on GIPSY scientific processing engine. The two approaches show an excellent agreement with less than 3 mm rms error horizontally and 6 mm rms error vertically. The derived position time series is analysed in terms of trend, periodicity and noise characteristics. The noise of the time series was found to be power-law noise model with spectral index closer to flicker noise. In addition, several periodic signals were found at 5, 14, 183 and 362 days. Furthermore, most of the horizontal movement was found to be in the North direction at a rate of 11.23 +/- 0.09 mm/y, whereas the rate in the East direction was estimated to be 1.46 +/- 0.05 mm/y. Lastly,

the 15-year long time series revealed a movement upwards at a rate of 0.79 ± 0.35 mm/y. Despite being an unattended station, Aboa provides one of the most continuous and longest GPS time series in Antarctica. Therefore, we believe that this research increases the awareness of local geophysical phenomena in a less reported area of the Antarctic continent.

1103. The Effect of Wind Stress Anomalies and Location in Driving Pacific Subtropical Cells and Tropical Climate

Authors: Giorgio Graffino (ESFM Doctorate School, Università degli Studi di Trieste, and Earth System Physics Section, International Centre for Theoretical Physics, Trieste, Italy); Riccardo Farneti (Earth System Physics Section, International Centre for Theoretical Physics, Trieste, Italy); Fred Kucharski (Earth System Physics Section, International Centre for Theoretical Physics, Trieste, Italy, and Center of Excellence for Climate Change Research, Department of Meteorology, King Abdulaziz University, Jeddah, Saudi Arabia); Franco Molteni (European Centre for Medium-Range Weather Forecasts (ECMWF))

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Cited by: 15 publications

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Subjects: N/A

Keywords: Ocean models, Pacific Ocean, Meridional overturning circulation, Tropics, Atmosphere-ocean interaction (and 1 more)

Funding: Abdus Salam International Centre for Theoretical Physics

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Abstract: The importance of subtropical and extratropical zonal wind stress anomalies on Pacific subtropical cell (STC) strength is assessed through several idealized and realistic numerical experiments with a global ocean model. Different zonal wind stress anomalies are employed, and their intensity is strengthened or weakened with respect to the climatological value throughout a suite of simulations. Subtropical strengthened (weakened) zonal wind stress anomalies result in increased (decreased) STC meridional mass and energy transport. When upwelling of subsurface water into the tropics is intensified (reduced), a distinct cold (warm) anomaly appears in the equatorial thermocline and up to the surface, resulting in significant tropical sea surface temperature (SST) anomalies. The use of realistic wind stress anomalies also suggests a potential impact of midlatitude atmospheric modes of variability on tropical climate through STC dynamics. The remotely driven response is compared with a set of simulations where an equatorial zonal wind stress anomaly is imposed. A dynamically distinct response is achieved, whereby the equatorial thermocline adjusts to the wind stress anomaly, resulting in significant equatorial SST anomalies as in the remotely forced simulations but with no role for STCs. Significant anomalies in Indonesian Throughflow transport are generated only when equatorial wind stress anomalies are applied, leading to remarkable heat content anomalies in the Indian Ocean. Equatorial wind stress anomalies do not involve modifications of STC transport but could set up the appropriate initial conditions for a tropical-extratropical teleconnection involving Hadley cells, exciting an STC anomalous transport, which ultimately feeds back on the tropics.

1104. A Deep Eastern Boundary Current in the Chile Basin

Authors: Lena M. Schulze Chretien (Marine Science Research Institute, Department of Biology and Marine Science Jacksonville University Jacksonville FL USA); Kevin Speer (Florida State University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014400

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Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: Hydrographic data and tracers from several ship-based sections show the location and structure of a deep eastern boundary current in the Chile Basin. The current is centered above the Peru-Chile Trench at 2,500-3,400 m and transports up to 6 Sv of low-potential vorticity, high-silicate water south toward Drake Passage. Deep current velocities from direct Lowered Acoustic Doppler Current Profiler measurements are up to about 15 cm/s southward. The hydrographic data, as well as potential vorticity and silicate distributions, show that the current is comprised to a large extent of flow from the west moving along the southern flank of the Sala y Gomez Ridge and Nazca Ridge, and to a lesser extent from a flow along the eastern boundary entering directly from the Panama Basin. At the southern edge of the Chile Trench, the current weakens and partly turns offshore to cross the Chile Ridge through a complex region of passages. Above the southern flank of the Chile Rise the flow joins a broader eastward flow; together, these waters return to the eastern boundary before entering Drake Passage.

1105. What historical landfast ice observations tell us about projected ice conditions in Arctic archipelagoes and marginal seas under anthropogenic forcing

Authors: Frédéric Laliberté; Stephen E. L. Howell; Jean-François Lemieux; Frédéric Dupont; Ji Lei

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Language: en

Subjects: N/A

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Funding: Natural Sciences and Engineering Research Council of Canada

License: CC BY 4.0

Open Access: N/A

Abstract: . Arctic landfast ice extent and duration are examined from observations, ice assimilations, ocean reanalyses and coupled models. From observations and assimilations, it is shown that in areas where landfast ice conditions last more than 5 months the first-year ice typically grows to more than 2 m and is rarely less than 1 m. The observed spatial distribution of landfast ice closely matches assimilation products but less so for ocean reanalyses and coupled models. Although models generally struggle to represent the landfast ice necessary to emulate the observed import/export of sea ice in regions favourable to landfast ice conditions, some do exhibit both a realistic climatology and a realistic decline of landfast ice extent under an anthropogenic forcing scenario. In these more realistic simulations, projections show that an extensive landfast ice cover should remain for at least 5 months of the year, well into the end of the 21st century. This is in stark contrast with the simulations that have an unrealistic emulation of landfast ice conditions. In these simulations, slow and packed ice conditions shrink markedly over the same period. In all simulations and in areas with landfast ice that lasts more than 5 months, the end-of-winter sea ice thickness remains between 1 and 2 m, well beyond the second half of the century. It is concluded that in the current generation of climate models, projections of winter sea ice conditions in the Canadian Arctic Archipelago and the Laptev Sea are overly sensitive to the representation of landfast ice conditions and that ongoing development in landfast ice parameterization will likely better constrain these projections.

1106. Deep Transient Slow Slip Detected by Survey GPS in the Region of Atacama, Chile

Authors: E. Klein (PSL Research University, Institut de Physique du Globe, Scripps Institution of Oceanography / University of California San Diego); Z. Duputel (Institut de Physique du Globe); D. Zigone (Institut de Physique du Globe); C. Vigny (PSL Research University); J.-P. Boy (Institut de Physique du Globe); C. Doubre (Institut de Physique du Globe); G. Meneses (PSL Research University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 75

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: Chile, GPS, subduction zone, SSE

Funding: Agence Nationale de la Recherche, H2020 European Research Council

License: Creative Commons

Open Access: N/A

Abstract: We detected a long-term transient deformation signal between 2014 and 2016 in the Atacama region (Chile) using survey Global Positioning System (GPS) observations. Over an ~150 km along-strike region, survey GPS measurements in 2014 and 2016 deviate significantly from the interseismic trend estimated using previous observations. This deviation from steady state deformation is spatially coherent and reveals a horizontal westward diverging motion of several centimeters, along with a significant uplift. It is confirmed by continuous measurements of recently installed GPS stations. We discard instrumental, hydrological, oceanic, or atmospheric loading effects and show that the transient is likely due to deep slow slip in the transition zone of the subduction interface (~40- to 60-km depth). In addition, daily observations recorded by a continuous GPS station operating between 2002 and 2015 highlight similar transient signals in 2005 and 2009, suggesting a recurrent pattern.

1107. Control of Ocean Temperature on Jakobshavn Isbræ's Present and Future Mass Loss

Authors: Johannes H. Bondzio (University of California Irvine); Mathieu Morlighem (University of California Irvine); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael H. Wood (University of California Irvine); Jérémie Mouginot (University of California Irvine, Institut des Géosciences de l'Environnement Grenoble France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Publication Date: 2018-11-16

Volume/Issue: Volume 45, Issue 23

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Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: sea level projection, glaciology, ice flow, Jakobshavn Isbræ, statistical approach (and 1 more)

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Large uncertainties in model parameterizations and input data sets make projections of future sea level rise contributions of outlet glaciers challenging. Here we introduce a novel technique for weighing large ensemble model simulations that uses information of key observables. The approach is robust to input errors and yields calibrated means and error estimates of a glacier's mass balance. We apply the technique to Jakobshavn Isbræ, using a model that includes a dynamic calving law, and closely reproduce the observed behavior from 1985 to 2018 by forcing the model with ocean temperatures only. Our calibrated projection suggests that the glacier will continue to retreat and contribute about 5.1 mm to eustatic sea level rise by 2100 under present-day climatic forcing. Our analysis shows that the glacier's future evolution will strongly depend on the ambient oceanic setting.

1108. Use of integral data assimilation and differential measurements as a contribution to improve ^{235}U and ^{238}U cross sections evaluations in the fast and epithermal energy range

Authors: Virginie Huy; Gilles Noguère; Gérard Rimpault

Journal: EPJ Nuclear Sciences & Technologies

Publisher: EDP Sciences

DOI: 10.1051/epjn/2018035

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References: 19

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Critical mass calculations of various HEU-fueled fast reactors result in large discrepancies in C/E values, depending on the nuclear data library used and the configuration modeled. Thus, it seems relevant to use integral experiments to try to reassess cross sections that might be responsible for such a dispersion in critical mass results. This work makes use of the Generalized Least Square method to solve Bayes equation, as implemented in the CONRAD code. Experimental database used includes ICSBEP Uranium based critical experiments and benefits from recent re-analyses of MASURCA and FCA-IX criticality experiments (with Monte-Carlo calculations) and of PROFIL irradiation experiments. These last ones provide very specific information on ^{235}U and ^{238}U capture cross sections. Due to high experimental uncertainties associated to fission spectra, we chose to consider either fitting these data or set them to JEFF-3.1.1 evaluations. The work focused on JEFF-3.1.1 ^{235}U and ^{238}U evaluations and results presented in this paper for ^{235}U capture and ^{238}U capture, and inelastic cross sections are compared to recent differential experiment or recent evaluations. Our integral experiment assimilation work notably suggests a 30% decrease for ^{235}U capture around 1-2.25 keV, a 10% increase in the unresolved resonance range when using JEFF-3.1.1 as "a priori" data. These results are in agreement with recent microscopic measurements from Danon et al. [Nucl. Sci. Eng. 187, 291 (2017)] and Jandel et al. [Phys. Rev. Lett. 109, 202506 (2012)]. For ^{238}U cross sections, results are highly dependent on fission spectra.

1109. Improving the Coastal Mean Dynamic Topography by Geodetic Combination of Tide Gauge and Satellite Altimetry

Authors: Ole Baltazar Andersen (Technical University of Denmark); Karina Nielsen (Technical University of Denmark); Per Knudsen (Technical University of Denmark); Chris W. Hughes (University of Liverpool, National Oceanography Centre); Rory Bingham (University of Bristol); Luciana Fenoglio-Marc (University of Bonn); Médéric Gravelle (Institut du Littoral et de l'Environnement, LIENSs, Université de La Rochelle-CNRS, La Rochelle, France;); Michael Kern (Science, Applications and Climate Department (EOP-SME), ESA ESTEC, Noordwijk, The Netherlands;); Sara Padilla Polo (Institut de Recherche pour le Développement (IRD))

Journal: Marine Geodesy

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References: 39

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1110. Exploration of Antarctic Ice Sheet 100-year contribution to sea level rise and associated model uncertainties using the ISSM framework

Authors: Nicole-Jeanne Schlegel; Helene Seroussi; Michael P. Schodlok; Eric Y. Larour; Carmen Boening; Daniel Limonadi; Michael M. Watkins; Mathieu Morlighem; Michiel R. van den Broeke

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3511-2018

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References: 85

Cited by: 53 publications

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Funding: N/A

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Open Access: N/A

Abstract: . Estimating the future evolution of the Antarctic Ice Sheet (AIS) is critical for improving future sea level rise (SLR) projections. Numerical ice sheet models are invaluable tools for bounding Antarctic vulnerability; yet, few continental-scale projections of century-scale AIS SLR contribution exist, and those that do vary by up to an order of magnitude. This is partly because model projections of future sea level are inherently uncertain and depend largely on the model's boundary conditions and climate forcing, which themselves are unknown due to the uncertainty in the projections of future anthropogenic emissions and subsequent climate response. Here, we aim to improve the understanding of how uncertainties in model forcing and boundary conditions affect ice sheet model simulations. With use of sampling techniques embedded within the Ice Sheet System Model (ISSM) framework, we assess how uncertainties in snow accumulation, ocean-induced melting, ice viscosity, basal friction, bedrock elevation, and the presence of ice shelves impact continental-scale 100-year model simulations of AIS future sea level contribution. Overall, we find that AIS sea level contribution is strongly affected by grounding line retreat, which is driven by the magnitude of ice shelf basal melt rates and by variations in bedrock topography. In addition, we find that over 1.2 m of AIS global mean sea level contribution over the next century is achievable, but not likely, as it is tenable only in response to unrealistically large melt rates and continental ice shelf collapse. Regionally, we find that under our most extreme 100-year warming experiment generalized for the entire ice sheet, the Amundsen Sea sector is the most significant source of model uncertainty (1032 mm 6sigma spread) and the region with the largest potential for future sea level contribution (297 mm). In contrast, under a more plausible forcing informed regionally by literature and model sensitivity studies, the Ronne basin has a greater potential for local increases in ice shelf basal melt rates. As a result, under this more likely realization, where warm waters reach the continental shelf under the Ronne ice shelf, it is the Ronne basin, particularly the Evans and Rutford ice streams, that are the greatest contributors to potential SLR (161 mm) and to simulation uncertainty (420 mm 6sigma spread).

1111. Heat, salt, and volume transports in the eastern Eurasian Basin of the Arctic Ocean from 2 years of mooring observations

Authors: Andrey V. Pnyushkov; Igor V. Polyakov; Robert Rember; Vladimir V. Ivanov; Matthew B. Alkire; Igor M. Ashik; Till M. Baumann; Genrikh V. Alekseev; Arild Sundfjord

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-14-1349-2018

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References: 59

Cited by: 22 publications

Language: en

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Keywords: N/A

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Open Access: N/A

Abstract: . This study discusses along-slope volume, heat, and salt transports derived from observations collected in 2013-2015 using a cross-slope array of six moorings ranging from 250 to 3900 m in the eastern Eurasian Basin (EB) of the Arctic Ocean. These observations demonstrate that in the upper 780 m layer, the along-slope boundary current advected, on average, 5.1 ± 0.1 Sv of water, predominantly in the eastward (shallow-to-right) direction. Monthly net volume transports across the Laptev Sea slope vary widely, from -0.3 ± 0.8 in April 2014 to -9.9 ± 0.8 Sv in June 2014; 3.1 ± 0.1 Sv (or 60 %) of the net transport was associated with warm and salty intermediate-depth Atlantic Water (AW). Calculated heat transport for 2013-2015 (relative to -1.8°C) was 46.0 ± 1.7 TW, and net salt transport (relative to zero salinity) was 172 ± 6 Mkg s⁻¹. Estimates for AW heat and salt transports were 32.7 ± 1.3 TW (71 % of net heat transport) and 112 ± 4 Mkg s⁻¹ (65 % of net salt transport). The variability of currents explains ~90 % of the variability in the heat and salt transports. The remaining ~10 % is controlled by temperature and salinity anomalies together with the temporal variability of the AW layer thickness. The annual mean volume transports decreased by 25 % from 5.8 ± 0.2 Sv in 2013-2014 to 4.4 ± 0.2 Sv in 2014-2015, suggesting that changes in the transports at interannual and longer timescales in the eastern EB may be significant.

1112. Sensitivity Studies of the Red Sea Eddies Using Adjoint Method

Authors: Peng Zhan (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia); Ganesh Gopalakrishnan (Scripps Institution of Oceanography / University of California San Diego); Aneesh C. Subramanian (Scripps Institution of Oceanography / University of California San Diego); Daquan Guo (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia); Ibrahim Hoteit (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia)

Journal: Journal of Geophysical Research: Oceans

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References: 50

Cited by: 21 publications

Language: en

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Keywords: N/A

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Abstract: Adjoint sensitivity analysis is applied to a set of eddies in the Red Sea using a high-resolution Massachusetts Institute of Technology general circulation model and its adjoint model. Previous studies have reported several eddy events in the Red Sea, namely, a dipole captured on 17 August 2001 in the southern Red Sea, a cyclonic eddy in November 2011 in the northern Red Sea, and an anticyclonic eddy in April 2010 in the central Red Sea. Sensitivity analysis is applied here to investigate the governing factors that control the intensity and evolution of these eddies. The eddies are first reproduced by running the Massachusetts Institute of Technology general circulation model forward and their sensitivities to external atmospheric forcing and previous model states are then computed using the adjoint model. In the experiments, (relative) surface vorticity (curl of horizontal velocity) is defined as the objective function. The contributions of forcings and model states are quantified and investigated. The sensitivities to external forcings are distinct in different eddy events. The dipole in the central Red Sea is dominantly sensitive to the cross-basin eastward wind jet. The anticyclonic eddy in the central Red Sea is most sensitive to the along-basin wind stress. The cyclonic eddy in the northern Red Sea is sensitive to the net heat flux and to surface elevation perturbations even from the remote southern Red Sea, which is attributed to the propagation of baroclinic Kelvin waves along the coast. Analysis of the sensitivity to model state variables suggests that these eddies are also modulated by the boundary currents and the temperature profile distributions.

1113. Interannual to Multidecadal Forcing of Mesoscale Eddy Kinetic Energy in the Subtropical Southern Indian Ocean

Authors: Andrew S. Delman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bo Qiu (Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA)

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Cited by: 14 publications

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Subjects: N/A

Keywords: eddies, decadal, mesoscale, interannual, Indian Ocean (and 1 more)

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: A region of elevated mesoscale eddy activity spans the subtropical southern Indian Ocean (SSIO) between Madagascar and Australia. The interannual and decadal changes in eddy activity in the SSIO eddy band, as represented by the variability of eddy kinetic energy (EKE), have implications for the large-scale circulation, mixed-layer budgets, and biological activity. An analysis of nearly two and a half decades of sea level anomaly (SLA) data from merged satellite altimetry shows that, in the southeast Indian Ocean east of 90 degreesE, the variations of EKE and SLA are positively correlated on interannual and decadal time scales. Moreover, EKE exhibits a multidecadal increasing linear trend that corresponds to an increasing trend of SLA in the region. The EKE-SLA covariability in the southeast Indian Ocean does not appear to be associated with a preference for anticyclonic over cyclonic eddy activity; rather, it can be attributed to the common remote forcing from the tropical Pacific associated with the El Niño-Southern Oscillation and the Pacific Decadal Oscillation. In the west central SSIO, wind stress curl just south of the eddy band forces potential vorticity anomalies that affect conditions for instability in the west central SSIO; potential density and potential vorticity gradient anomalies also suggest a remote forcing mechanism originating in the region southwest of Australia. The interannual to multidecadal variability of EKE in the SSIO and its relationship with large-scale SLA has implications for mixed-layer dynamics and biogeochemistry and provides a basis for assessment of model simulations of eddy activity in the region.

1114. An Assessment of Representation of Oceanic Mesoscale Eddy-Atmosphere Interaction in the Current Generation of General Circulation Models and Reanalyses

Authors: Peiran Yang (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Zhao Jing (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Lixin Wu (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology)

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Publisher: American Geophysical Union (AGU)

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References: 24

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

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Abstract: Oceanic mesoscale eddies interact strongly with the atmosphere, inducing heat flux that acts to dissipate the eddy potential energy. So far it remains unknown how well this oceanic mesoscale eddy-atmosphere (OMEA) interaction is represented in the current generation of general circulation models. Here we evaluate the intensity of OMEA interaction in numerical models widely used by the community. It is found that the intensity of OMEA interaction differs significantly among models in its overall magnitude and spatial distribution. In eddy-rich regions such as Kuroshio Extension and Antarctic Circumpolar Current, the intermodel difference can reach 40%. Surface wind strength and marine atmospheric boundary layer adjustment to mesoscale heat flux anomaly are two important factors accounting for the intermodel difference. Models with stronger surface wind tend to have higher OMEA interaction. Moreover, neglecting the marine atmospheric boundary layer adjustment, ocean-alone model simulations overestimate OMEA interaction especially at middle and high latitudes by 20%-50%.

1115. ENSO-Related Global Ocean Heat Content Variations

Authors: Quran Wu (Xiamen University); Xuebin Zhang (University of Tasmania); John A. Church (University of New South Wales); Jianyu Hu (Xiamen University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0861.1

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Cited by: 15 publications

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Keywords: N/A

Funding: National Natural Science Foundation of China, National Basic Research Program of China

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Abstract: The modulation of the full-depth global integrated ocean heat content (GOHC) by El Niño-Southern Oscillation (ENSO) has been estimated in various studies. However, the quantitative results and the mechanisms at work remain uncertain. Here, a dynamically consistent ocean state estimate is utilized to study the large-scale integrated heat content variations during ENSO events for the global ocean. The full-depth GOHC exhibits a cooling tendency during the peak and decaying phases of El Niño, which is a result of the negative surface heat flux (SHF) anomaly in the tropics (30 degreesS-30 degreesN), partially offset by the positive SHF anomaly at higher latitudes. The tropical SHF anomaly acts as a lagged response to damp the convergence of oceanic heat transport, which redistributes heat from the extratropics and the subsurface layers (100-440 m) into the upper tropical oceans (0-100 m) during the onset and peak of El Niño. These results highlight the global nature of the oceanic heat redistribution during ENSO events, as well as how the redistribution process affects the full-depth GOHC. The meridional heat exchange across 30 degreesS and 30 degreesN is driven by ocean current anomalies, while multiple processes contribute to the vertical heat exchange across 100 m simultaneously. Heat advection due to unbalanced mass transport is distinguished from the mass balanced one, with significant contributions from the meridional and zonal overturning cells being identified for the latter in the vertical direction. Results presented here have implications for monitoring the planetary energy budget and evaluating ENSO's global imprints on ocean heat content in different estimates.

1116. Ocean angular momentum from a recent global state estimate, with assessment of uncertainties

Authors: K J Quinn (Atmospheric and Environmental Research Inc.); R M Ponte (Atmospheric and Environmental Research Inc.); P Heimbach (University of Texas at Austin); I Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J-M Campin (Massachusetts Institute of Technology)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggy452

URL: <https://doi.org/10.1093/gji/ggy452>

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References: 40

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration GRACE Science

License: N/A

Open Access: N/A

Abstract: N/A

1117. Partitioning Ocean Motions Into Balanced Motions and Internal Gravity Waves: A Modeling Study in Anticipation of Future Space Missions

Authors: Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Patrice Klein (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Lops Ifremer/CNRS Plouzané France); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bo Qiu (University of Hawaii at Manoa); Zhan Su (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Shuiming Chen (University of Hawaii at Manoa); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Keywords: oceanic surface motions, internal gravity waves, satellite observations, balanced motions

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License: N/A

Open Access: N/A

Abstract: Internal gravity waves (IGWs) and balanced motions (BM) with scales <100-km capture most of the vertical velocity field in the upper ocean. They have, however, different impacts on the ocean energy budget, which explains the need to partition motions into BMs and IGWs. One way is to exploit the synergy of using different satellite observations, the only observations with global coverage, and a reasonable spatial and temporal resolution. But we need first to characterize and understand their signatures on the different surface oceanic fields. This study addresses this issue by using an ocean global numerical simulation with high-resolution (1/48 degrees). Our methodology is based on the analysis of the 12,000 frequency-wave number spectra to discriminate these two classes of motions in the surface kinetic energy, sea surface height, sea surface temperature, sea surface salinity, relative vorticity, and divergence fields and for two seasons. Results reveal a complex picture worldwide of the partition of motions between IGWs and BMs in the different surface fields, depending on the season, the hemisphere, and low and high eddy kinetic energy regions. But they also highlight some generic properties on the impact of these two classes of motions on the different fields. This points to the synergy of using present and future satellite observations to assess the ocean kinetic energy on a global scale. The 12,000 frequency-wave number spectra represent a World Ocean Atlas of the surface ocean dynamics not fully exploited in the present study. We hope the use of this World Ocean Atlas by other studies will lead to extend much these results.

1118. On the Modified Circumpolar Deep Water Upwelling Over the Four Ladies Bank in Prydz Bay, East Antarctica

Authors: Chengyan Liu (Hohai University, Nanjing University); Zhaomin Wang (Hohai University); Chen Cheng (Nanjing University); Yang Wu (Hohai University); Ruibin Xia (Nanjing University); Bingrui Li (Polar Research Institute of China); Xiang Li (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014026

URL: <https://doi.org/10.1029/2018jc014026>

Publication Date: 2018-10-26

Volume/Issue: Volume 123, Issue 11

References: 63

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: Prdyz Bay, potential vorticity, onshore upwelling, Antarctic Slope Current, topography dynamic effect (and 1 more)

Funding: China Scholarship Council

License: N/A

Open Access: N/A

Abstract: Analyses of hydrographic observation data sets revealed that modified Circumpolar Deep Water (mCDW) tends to flood the western flank (73-78 degreesE, 67-68 degreesS) of the Four Ladies Bank (FLB) in Prydz Bay. In this study, we investigated the mechanism responsible for mCDW upwelling over the FLB based on an eddy-resolving coupled ocean-sea ice-ice shelf model in conjunction with the latest high-accuracy bathymetry. It was found that zonal step-like declines in the seabed over the FLB are crucial for the mCDW onshore upwelling, through topographic dynamic effects on the alongshore Antarctic Slope Current. In the presence of meridional cross-shelf σ_θ isopycnals, warm mCDW at ~500-m depth in the deep sea can make its approach to the shallower continental shelf (~200- to 500-m depths), characterized as a warm layer (> -1.8 degreesC) bounded by the cross-shelf σ_θ isopycnal surfaces of 27.5 and 27.7 kg/m³. The simulated outflow of the Prydz Bay gyre is concentrated to the eastern flank of the Prydz Channel. These results suggest that, in addition to the depressions and troughs around the Antarctic, smaller topographic features such as step-like declines in the seabed near the Antarctic Slope Current are also favorable for onshore mCDW intrusion. This study demonstrates that mCDW onshore transport around the Antarctic continental shelf might be significantly underestimated by numerical models with coarse spatial representations of the topography.

1119. Modeling the response of Northwest Greenland to enhanced ocean thermal forcing and subglacial discharge

Authors: Mathieu Morlighem; Michael Wood; H            ; Youngmin Choi; Eric Rignot

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2018-214

URL: <https://doi.org/10.5194/tc-2018-214>

Publication Date: 2018-10-26

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Forcing (mathematics)|Glacier ice accumulation|Future sea level|Fjord|Greenland ice sheet|Tidewater glacier cycle|Glacier morphology

Funding: National Aeronautics and Space Administration, Division of Arctic Sciences

License: CC BY 4.0

Open Access: N/A

Abstract: . Calving front dynamics is an important control on Greenland's ice mass balance. Ice front retreat of marine-terminating glaciers may, for example, lead to a loss in resistive stress, which ultimately results in glacier acceleration and thinning. Over the past decade, it has been suggested that such retreats may be triggered by warm and salty Atlantic water, which is typically found at a depth below 200-300 m. An increase in subglacial water discharge at glacier ice fronts due to enhanced surface runoff may also be responsible for an intensification of undercutting and calving. An increase in ocean thermal forcing or subglacial discharge therefore has the potential to destabilize marine terminating glaciers along the coast of Greenland. It remains unclear which glaciers are currently stable but may retreat in the future, and how far inland and how fast they will retreat. Here, we quantify the sensitivity and vulnerability of marine-terminating glaciers along the Northwest coast of Greenland (from 72.5   to 76   N) to the ocean forcing and subglacial discharge using the Ice Sheet System Model (ISSM). We rely on the undercutting parameterization based on ocean thermal forcing and subglacial discharge, and use ocean temperature and salinity from high-resolution ECCO2 (Estimating the Circulation & Climate of the Ocean, Phase II) simulations at the fjords mouth to constrain the ocean thermal forcing. The ice flow model includes a calving law based on a tensile Von Mises criterion. While these parameterizations remain approximations and do not include all the physical processes at play, they have been shown to provide reliable estimates of undercutting and calving rates, respectively, on a number of glaciers along the coast of Greenland. We find that some glaciers, such as Dietrichson Gletscher or Alison Gletscher, are sensitive to small increases in ocean thermal forcing, while others, such as Illullip Sermia or Cornell Gletscher, are remarkably stable and remain stable, even in a 3-degree ocean warming scenario. Under the most intense experiment, we find that Hayes Gletscher retreats by more than 50 km inland into a deep trough and its velocity increases by a factor of 10 over only 15 years. The model confirms that ice-ocean interactions can trigger extensive and rapid glacier retreat, but the bed controls the rate and magnitude of the retreat. Under current oceanic and atmospheric condition, we find that this sector alone will contribute more than 1 cm to sea level, and up

to 3 cm under the most extreme scenario.

1120. Island Wakes

Authors: R.M.A. Caldeira

Journal: Encyclopedia of Ocean Sciences

Publisher: Elsevier

DOI: 10.1016/b978-0-12-409548-9.11614-8

URL: <https://doi.org/10.1016/b978-0-12-409548-9.11614-8>

Publication Date: 2018-10-26

Volume/Issue: N/A

References: 22

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Eddy|Biogeochemical Cycle

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1121. A variational approach for parameter estimation based on balanced proper orthogonal decomposition

Authors: M.U. Altaf; M.F. McCabe

Journal: Computer Methods in Applied Mechanics and Engineering

Publisher: Elsevier BV

DOI: 10.1016/j.cma.2018.10.013

URL: <https://doi.org/10.1016/j.cma.2018.10.013>

Publication Date: 2018-10-25

Volume/Issue: Volume 344

References: 51

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: King Abdullah University of Science and Technology

License: N/A

Open Access: N/A

Abstract: N/A

1122. Quantifying the paleogeographic driver of Cretaceous carbonate platform development using paleoecological niche modeling

Authors: Alexandre Pohl; Marie Laugié; Jean Borgomano; Julien Michel; Cyprien Lanteaume; Christopher R. Scotese; Camille Frau; Emmanuelle Poli; Yannick Donnadieu

Journal: Palaeogeography, Palaeoclimatology, Palaeoecology

Publisher: Elsevier BV

DOI: 10.1016/j.palaeo.2018.10.017

URL: <https://doi.org/10.1016/j.palaeo.2018.10.017>

Publication Date: 2018-10-23

Volume/Issue: Volume 514

References: 77

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Total

License: N/A

Open Access: N/A

Abstract: N/A

1123. Scavenging and transport of thorium radioisotopes in the North Atlantic Ocean

Authors: Paul Lerner

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/10652

URL: <https://doi.org/10.1575/1912/10652>

Publication Date: 2018-10-22

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Geotraces|Scavenging|Isotopes of thorium|Particle (ecology)|Settling

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1124. An Observationally Based Evaluation of Subgrid Scale Ice Thickness Distributions Simulated in a Large-Scale Sea Ice-Ocean Model of the Arctic Ocean

Authors: Mischa Ungermann (Alfred Wegener Institute); Martin Losch (Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014022

URL: <https://doi.org/10.1029/2018jc014022>

Publication Date: 2018-10-20

Volume/Issue: Volume 123, Issue 11

References: 68

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: airborne electromagnetic sounding, decadal variability, MITgcm, regional, upward-looking sonar (and 2 more)

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: A key parameterization in sea ice models describes the subgrid scale ice thickness distribution. Based on only a few observations, the ice thickness distribution model was shown to be consistent with field data and to improve the simulation's large-scale properties. The available submarine and airborne observations enable to evaluate in greater detail the ability of a pan-Arctic sea ice-ocean model with an ice thickness distribution parameterization to reproduce observed thickness distributions in different regions and seasons. Many observations are reproduced accurately. Some cases of poorly simulated modes and tails of the distributions are tentatively attributed to simplified thermodynamics and inaccurate deformation fields. Variability on decadal timescales, however, is generally underestimated. Thickness distributions in individual grid cells of the model show similar differences between regions and seasons as observed regional mean distributions, but the modeled grid-scale variability is lower than observed. Simulated modal thicknesses of first-year ice are only insufficiently different from those of multiyear ice. The modal thickness proves to be a useful metric for quantifying model biases in both dynamics and thermodynamics. In addition to improving basin-wide mean variables, the ice thickness distribution parameterization provides reliable and valuable additional subgrid scale data. At the same time the low climate sensitivity of the parameterization may affect longer simulations with strong climate change aspects.

1125. Consequences of different air-sea feedbacks on ocean using MITgcm and MERRA-2 forcing: Implications for coupled data assimilation systems

Authors: Ehud Strobach; Andrea Molod; Gael Forget; Jean-Michel Campin; Chris Hill; Dimitris Menemenlis; Patrick Heimbach

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2018.10.006

URL: <https://doi.org/10.1016/j.ocemod.2018.10.006>

Publication Date: 2018-10-18

Volume/Issue: Volume 132

References: 66

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA

License: N/A

Open Access: N/A

Abstract: N/A

1126. Tomography

Authors: Peter F. Worcester

Journal: Encyclopedia of Ocean Sciences

Publisher: Elsevier

DOI: 10.1016/b978-0-12-409548-9.11591-x

URL: <https://doi.org/10.1016/b978-0-12-409548-9.11591-x>

Publication Date: 2018-10-18

Volume/Issue: N/A

References: 28

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Speed of Sound

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1127. Annual and Interannual Variability in the California Current System: Comparison of an Ocean State Estimate with a Network of Underwater Gliders

Authors: Katherine D. Zaba (Scripps Institution of Oceanography); Daniel L. Rudnick (Scripps Institution of Oceanography); Bruce D. Cornuelle (Scripps Institution of Oceanography); Ganesh Gopalakrishnan (Scripps Institution of Oceanography); Matthew R. Mazloff (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-18-0037.1

URL: <https://doi.org/10.1175/jpo-d-18-0037.1>

Publication Date: 2018-10-16

Volume/Issue: Volume 48, Issue 12

References: 44

Cited by: 22 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration, National Oceanic and Atmospheric Administration, National Science Foundation

License: N/A

Open Access: N/A

Abstract: A data-constrained state estimate of the southern California Current System (CCS) is presented and compared with withheld California Cooperative Oceanic Fisheries Investigations (CalCOFI) data and assimilated glider data over 2007-17. The objective of this comparison is to assess the ability of the California State Estimate (CASE) to reproduce the key physical features of the CCS mean state, annual cycles, and interannual variability along the three sections of the California Underwater Glider Network (CUGN). The assessment focuses on several oceanic metrics deemed most important for characterizing physical variability in the CCS: 50-m potential temperature, 80-m salinity, and 26 kg m⁻³ isopycnal depth and salinity. In the time mean, the CASE reproduces large-scale thermohaline and circulation structures, including observed temperature gradients, shoaling isopycnals, and the locations and magnitudes of the equatorward California Current and poleward California Undercurrent. With respect to the annual cycle, the CASE captures the phase and, to a lesser extent, the magnitude of upper-ocean warming and stratification from late summer to early fall and of isopycnal heave during springtime upwelling. The CASE also realistically captures near-surface diapycnal mixing during upwelling season and the semiannual cycle of the California Undercurrent. In terms of interannual variability, the most pronounced signals are the persistent warming and downwelling anomalies of 2014-16 and a positive isopycnal salinity anomaly that peaked with the 2015-16 El Niño.

1128. 100 Years of the Ocean General Circulation

Authors: Carl Wunsch (Harvard University, Massachusetts Institute of Technology); Raffaele Ferrari (Massachusetts Institute of Technology)

Journal: Meteorological Monographs

Publisher: American Meteorological Society

DOI: 10.1175/amsmonographs-d-18-0002.1

URL: <https://doi.org/10.1175/amsmonographs-d-18-0002.1>

Publication Date: 2018-10-12

Volume/Issue: Volume 59

References: 390

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: The central change in understanding of the ocean circulation during the past 100 years has been its emergence as an intensely time-dependent, effectively turbulent and wave-dominated, flow. Early technologies for making the difficult observations were adequate only to depict large-scale, quasi-steady flows. With the electronic revolution of the past 50+ years, the emergence of geophysical fluid dynamics, the strongly inhomogeneous time-dependent nature of oceanic circulation physics finally emerged. Mesoscale (balanced), submesoscale oceanic eddies at 100-km horizontal scales and shorter, and internal waves are now known to be central to much of the behavior of the system. Ocean circulation is now recognized to involve both eddies and larger-scale flows with dominant elements and their interactions varying among the classical gyres, the boundary current regions, the Southern Ocean, and the tropics.

1129. Arctic-Wide Sea Ice Thickness Estimates From Combining Satellite Remote Sensing Data and a Dynamic Ice-Ocean Model with Data Assimilation During the CryoSat-2 Period

Authors: Longjiang Mu (Sun Yat-Sen University, Alfred Wegener Institute); Martin Losch (Alfred Wegener Institute); Qinghua Yang (Sun Yat-Sen University); Robert Ricker (Alfred Wegener Institute); Svetlana N. Losa (Alfred Wegener Institute, Russian Academy of Sciences); Lars Nerger (Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014316

URL: <https://doi.org/10.1029/2018jc014316>

Publication Date: 2018-10-10

Volume/Issue: Volume 123, Issue 11

References: 64

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: Arctic, data assimilation, sea ice thickness, CryoSat-2, CS2SMOS

Funding: Bundesministerium für Bildung und Forschung, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Exploiting the complementary character of CryoSat-2 and Soil Moisture and Ocean Salinity satellite sea ice thickness products, daily Arctic sea ice thickness estimates from October 2010 to December 2016 are generated by an Arctic regional ice-ocean model with satellite thickness assimilated. The assimilation is performed by a Local Error Subspace Transform Kalman filter coded in the Parallel Data Assimilation Framework. The new estimates can be generally thought of as combined model and satellite thickness (CMST). It combines the skill of satellite thickness assimilation in the freezing season with the model skill in the melting season, when neither CryoSat-2 nor Soil Moisture and Ocean Salinity sea ice thickness is available. Comparisons with in situ observations from the Beaufort Gyre Exploration Project, Ice Mass Balance Buoys, and the NASA Operation IceBridge demonstrate that CMST reproduces most of the observed temporal and spatial variations. Results also show that CMST compares favorably to the Pan-Arctic Ice-Ocean Modeling and Assimilation System product and even appears to correct known thickness biases in the Pan-Arctic Ice-Ocean Modeling and Assimilation System. Due to imperfect parameterizations in the sea ice model and satellite thickness retrievals, CMST does not reproduce the heavily deformed and ridged sea ice along the northern coast of the Canadian Arctic Archipelago and Greenland. With the new Arctic sea ice thickness estimates sea ice volume changes in recent years can be further assessed.

1130. Examining the Origins of Ocean Heat Content Variability in the Eastern North Atlantic Subpolar Gyre

Authors: Nicholas P. Foukal (Duke University, Woods Hole Oceanographic Institution); M. Susan Lozier (Duke University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl079122

URL: <https://doi.org/10.1029/2018gl079122>

Publication Date: 2018-10-10

Volume/Issue: Volume 45, Issue 20

References: 53

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: ocean heat content, Subtropical gyre, heat budget, Subpolar gyre, Lagrangian trajectories (and 1 more)

Funding: National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: We analyze sources of ocean heat content (OHC) variability in the eastern North Atlantic subpolar gyre from both Eulerian and Lagrangian perspectives within two ocean simulations from 1990 to 2015. Heat budgets reveal that while the OHC seasonal cycle is driven by air-sea fluxes, interannual OHC variability is driven by both air-sea fluxes and the divergence of ocean heat transport, the latter of which is dominated by the oceanic flux through the southern face of the study area. Lagrangian trajectories initialized along the southern face and run backward in time indicate that interannual variability in the subtropical-origin volume flux (i.e., the upper limb of the overturning circulation) drives variability in the temperature flux through the southern face. As such, the heat carried by the imported subtropical waters is an important component of the eastern subpolar gyre heat budget on interannual time scales.

1131. Variability of Southern Ocean Transports

Authors: Brady S. Ferster (School of the Earth, Ocean, and Environment, University of South Carolina, Columbia, South Carolina); Bulusu Subrahmanyam (School of the Earth, Ocean, and Environment, University of South Carolina, Columbia, South Carolina); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ebenezer S. Nyadjro (Department of Physics, University of New Orleans, New Orleans, Louisiana)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-18-0055.1

URL: <https://doi.org/10.1175/jpo-d-18-0055.1>

Publication Date: 2018-10-05

Volume/Issue: Volume 48, Issue 11

References: 64

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Ocean models, Mass fluxes/transport, Dynamics, Ocean, Ocean circulation (and 1 more)

Funding: NASA/South Carolina Space Grant Consortium

License: N/A

Open Access: N/A

Abstract: The Southern Ocean (SO) is capable of transporting vast amounts of salt, heat, and nutrients, which allows it to influence and regulate global climate. The variability of depth- and density-integrated volume transports in the SO is studied using the Estimating the Circulation and Climate of the Ocean (ECCO), version 4, release 3 (1992-2015), ocean state estimate. The estimate has a net eastward transport of 150.6 $\pm$ 5.5, 162.6 $\pm$ 7.4, and 148.2 $\pm$ 5.4 Sv (1 Sv $\equiv$ 10⁶ m³ s⁻¹) between the Atlantic-Indian, Indian-Pacific, and Pacific-Atlantic basins, respectively. The time-mean meridional volume transport across 30 degreesS in the Atlantic is estimated to be -1.4 $\pm$ 0.6 Sv, -14.4 $\pm$ 3.5 Sv in the Indian basin, and 15.5 $\pm$ 4.1 Sv in the Pacific, where negative values are southward. Trends in net volume transport between the basins are statistically insignificant. Within the water column, however, the middle and lower branches of the meridional overturning circulation have trends of -0.289 and 0.248 Sv decade⁻¹ in the Atlantic basin. The Indian and Pacific basins have decreasing trends in their lower overturning cells. These results indicate increased overturning circulation within the lower branch in the South Atlantic and decreased lower branch circulation within the Indian and Pacific basins and have implications on the thermohaline-driven circulation. Using ECCO, we estimate a southward potential temperature transport of -176.2 degrees $\pm$ 197.2 degreesC Sv and salinity transport of -1.71 $\pm$ 22.4 psu Sv into the SO and indicate potential temperature transport is increasing by -15.0 degrees $\pm$ 13.2 degreesC Sv decade⁻¹.

1132. Mass Loss of Totten and Moscow University Glaciers, East Antarctica, Using Regionally Optimized GRACE Mascons

Authors: Yara Mohajerani (University of California Irvine); Isabella Velicogna (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl078173

URL: <https://doi.org/10.1029/2018gl078173>

Publication Date: 2018-10-05

Volume/Issue: Volume 45, Issue 14

References: 44

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: Glacier mass balance|Ice caps|Post-glacial rebound

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Totten and Moscow University glaciers, in the marine-based sector of East Antarctica, contain enough ice to raise sea level by 5 m. Obtaining precise measurements of their mass balance is challenging owing to large area of the basins and the small mass balance signal compared to West Antarctic glaciers. Here we employ a locally optimized processing of Gravity Recovery and Climate Experiment (GRACE) harmonics to evaluate their mass balance at the sub-basin scale and compare the results with mass budget method (MBM) estimates using regional atmospheric climate model version 2.3 (RACMO2.3) or Modèle Atmosphérique Régional version 3.6.4 (MAR3.6.4). The sub-basin mass loss estimate for April 2002 to November 2015 is 14.8 ± 4.3 Gt/yr, which is weakly affected by glacial isostatic adjustment uncertainties (± 1.4 Gt/yr). This result agrees with MBM/RACMO2.3 (15.8 ± 2.0 Gt/yr), whereas MBM/MAR3.6.4 underestimates the loss (6.6 ± 1.6 Gt/yr). For the entire drainage, the mass loss for April 2002 to August 2016 is 18.5 ± 6.6 Gt/yr, or $15 \pm 4\%$ of its ice flux. These results provide unequivocal evidence for mass loss in this East Antarctic sector.

1133. 100 Years of Progress in Tropical Cyclone Research

Authors: Kerry Emanuel (Massachusetts Institute of Technology)

Journal: Meteorological Monographs

Publisher: American Meteorological Society

DOI: 10.1175/amsmonographs-d-18-0016.1

URL: <https://doi.org/10.1175/amsmonographs-d-18-0016.1>

Publication Date: 2018-10-02

Volume/Issue: Volume 59

References: 1504

Cited by: 151 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A century ago, meteorologists regarded tropical cyclones as shallow vortices, extending upward only a few kilometers into the troposphere, and nothing was known about their physics save that convection was somehow involved. As recently as 1938, a major hurricane struck the densely populated northeastern United States with no warning whatsoever, killing hundreds. In the time since the American Meteorological Society was founded, however, tropical cyclone research blossomed into an endeavor of great breadth and depth, encompassing fields ranging from atmospheric and oceanic dynamics to biogeochemistry, and the precision and scope of forecasts and warnings have achieved a level of success that would have been regarded as impossible only a few decades ago. This chapter attempts to document the extraordinary progress in tropical cyclone research over the last century and to suggest some avenues for productive research over the next one.

1134. World Ocean Circulation Experiment - Argo Global Hydrographic Climatology

Authors: Viktor Gouretski

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-14-1127-2018

URL: <https://doi.org/10.5194/os-14-1127-2018>

Publication Date: 2018-09-27

Volume/Issue: Volume 14, Issue 5

References: 27

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The paper describes the new gridded World Ocean Circulation Experiment-Argo GlobalHydrographic Climatology (WAGHC). The climatology has a 1°4' spatial resolution resolving the annual cycle of temperature and salinity on a monthly basis. Two versions of the climatology were produced and differ with respect to whether the spatial interpolation was performed on isobaric or isopycnal surfaces, respectively. The WAGHC climatology is based on the quality controlled temperature and salinity profiles obtained before January 2016, and the average climatological year is in the range from 2008 to 2012. To avoid biases due to the significant step-like decrease of the data below 2 km, the profile extrapolation procedure is implemented. We compare the WAGHC climatology to the 1°4' resolution isobarically averaged WOA13 climatology, produced by the NOAA Ocean Climate Laboratory (Locarnini et al., 2013) and diagnose a generally good agreement between these two gridded products. The differences between the two climatologies are basically attributed to the interpolation method and the considerably extended data basis. Specifically, the WAGHC climatology improved the representation of the thermohaline structure, in both the data poor polar regions and several data abundant regions like the Baltic Sea, the Caspian sea, the Gulf of California, the Caribbean Sea, and the Weddell Sea. Further, the dependence of the ocean heat content anomaly (OHCA) time series on the baseline climatology was tested. Since the 1950s, both of the baseline climatologies produce almost identical OHCA time series. The gridded dataset can be found at https://doi.org/10.1594/WDCC/WAGHC_V1.0 (Gouretski, 2018).

1135. Satellite SST and SSS Observations and Their Role to Constrain Ocean Models

Authors: Tong Lee; Chelle Gentemann

Journal: New Frontiers in Operational Oceanography

Publisher: GODAE OceanView

DOI: 10.17125/gov2018.ch11

URL: <https://doi.org/10.17125/gov2018.ch11>

Publication Date: 2018-09-19

Volume/Issue: N/A

References: N/A

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Ocean observations|Earth system science

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1136. Using Noble Gases to Assess the Ocean's Carbon Pumps

Authors: Roberta C. Hamme (School of Earth and Ocean Sciences, University of Victoria, Victoria, British Columbia V8W 2Y2, Canada;); David P. Nicholson (Woods Hole Oceanographic Institution); William J. Jenkins (Woods Hole Oceanographic Institution); Steven R. Emerson (University of Washington)

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

DOI: 10.1146/annurev-marine-121916-063604

URL: <https://doi.org/10.1146/annurev-marine-121916-063604>

Publication Date: 2018-09-16

Volume/Issue: Volume 11, Issue 1

References: 132

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Natural mechanisms in the ocean, both physical and biological, concentrate carbon in the deep ocean, resulting in lower atmospheric carbon dioxide. The signals of these carbon pumps overlap to create the observed carbon distribution in the ocean, making the individual impact of each pump difficult to disentangle. Noble gases have the potential to directly quantify the physical carbon solubility pump and to indirectly improve estimates of the biological organic carbon pump. Noble gases are biologically inert, can be precisely measured, and span a range of physical properties. We present dissolved neon, argon, and krypton data spanning the Atlantic, Southern, Pacific, and Arctic Oceans. Comparisons between deep-ocean observations and models of varying complexity enable the rates of processes that control the carbon solubility pump to be quantified and thus provide an important metric for ocean model skill. Noble gases also provide a powerful means of assessing air-sea gas exchange parameterizations.

1137. State of the Climate in 2017

Authors: Gail Hartfield; Jessica Blunden; Derek S. Arndt

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/2018bamsstateoftheclimate.1

URL: <https://doi.org/10.1175/2018bamsstateoftheclimate.1>

Publication Date: 2018-09-14

Volume/Issue: Volume 99, Issue 8

References: N/A

Cited by: 155 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Editor's note: For easy download the posted pdf of the State of the Climate for 2018 is a low-resolution file. A high-resolution copy of the report is available by clicking [here](#). Please be patient as it may take a few minutes for the high-resolution file to download.

1138. A Dynamical Reconstruction of the Global Monthly Mean Oxygen Isotopic Composition of Seawater

Authors: Charlotte Breitzkreuz (University of Bremen); André Paul (University of Bremen); Takasumi Kurahashi-Nakamura (University of Bremen); Martin Losch (Alfred Wegener Institute); Michael Schulz (University of Bremen)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014300

URL: <https://doi.org/10.1029/2018jc014300>

Publication Date: 2018-09-07

Volume/Issue: Volume 123, Issue 10

References: 57

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung

License: N/A

Open Access: N/A

Abstract: We present a dynamically consistent gridded data set of the global, monthly mean oxygen isotope ratio of seawater ($\delta^{18}\text{O}$). The data set was created from an optimized simulation of an ocean general circulation model constrained by global monthly data collected from 1950 to 2011 and climatological salinity and temperature data collected from 1951 to 1980. The optimization was obtained using the adjoint method for variational data assimilation, which yields a simulation that is consistent with the observational data and the physical laws embedded in the model. Our data set performs equally well as a previous data set in terms of model-data misfit but brings an improvement in terms of the seasonal cycle and physical consistency. As a result the data set does not show any sharp transitions between water masses or in areas where the data coverage is low. The data assimilation method shows high potential for interpolating sparse data sets in a physically meaningful way. Comparatively big errors, however, are found in our data set in the surface levels in the Arctic Ocean mainly because the influence of isotopically highly depleted precipitation is not preserved in the sea ice model, and the low model resolution of about 285 km horizontally. The data set is publicly available, and it is anticipated to be useful for a large range of applications in (paleo-) oceanographic studies.

1139. Seasonal dynamics of Totten Ice Shelf controlled by sea ice buttressing

Authors: Chad A. Greene; Duncan A. Young; David E. Gwyther; Benjamin K. Galton-Fenzi; Donald D. Blankenship

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-2869-2018

URL: <https://doi.org/10.5194/tc-12-2869-2018>

Publication Date: 2018-09-06

Volume/Issue: Volume 12, Issue 9

References: 76

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Division of Polar Programs, G. Unger Vetlesen Foundation, Australian Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: . Previous studies of Totten Ice Shelf have employed surface velocity measurements to estimate its mass balance and understand its sensitivities to interannual changes in climate forcing. However, displacement measurements acquired over timescales of days to weeks may not accurately characterize long-term flow rates wherein ice velocity fluctuates with the seasons. Quantifying annual mass budgets or analyzing interannual changes in ice velocity requires knowing when and where observations of glacier velocity could be aliased by subannual variability. Here, we analyze 16 years of velocity data for Totten Ice Shelf, which we generate at subannual resolution by applying feature-tracking algorithms to several hundred satellite image pairs. We identify a seasonal cycle characterized by a spring to autumn speedup of more than 100 m yr⁻¹ close to the ice front. The amplitude of the seasonal cycle diminishes with distance from the open ocean, suggesting the presence of a resistive back stress at the ice front that is strongest in winter. Springtime acceleration precedes summer surface melt and is not attributable to thinning from basal melt. We attribute the onset of ice shelf acceleration each spring to the loss of buttressing from the breakup of seasonal landfast sea ice.

1140. Decadal Variability of Eddy Characteristics and Energetics in the Kuroshio Extension: Unstable Versus Stable States

Authors: Haiyuan Yang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Bo Qiu (University of Hawaii at Manoa, Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Ping Chang (Ocean University of China, Department of Oceanography and Department of Atmospheric Sciences Texas A&M; University, College Station College Station TX USA, Pilot National Laboratory for Marine Science and Technology); Lixin Wu (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Shengpeng Wang (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Zhaohui Chen (Ocean University of China, Pilot National Laboratory for Marine Science and Technology); Yun Yang (Beijing Normal University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014081

URL: <https://doi.org/10.1029/2018jc014081>

Publication Date: 2018-09-04

Volume/Issue: Volume 123, Issue 9

References: 63

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: eddy energetics, decadal variability, Kuroshio Extension, eddy characteristics

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Using the Estimating Circulation and Climate of the Ocean Phase II product, this study investigates the eddy characteristics and energetics in two dynamical regimes within the Kuroshio Extension (KE) region. Based on the empirical orthogonal function analysis, it is found that the decadal evolution of eddy kinetic energy in the KE region is characterized by a delayed negative correlation between the upstream and downstream. Besides the out-of-phase change in eddy activity levels, eddy characteristics and energetics in the upstream KE are also different from those in the downstream. In the upstream region, eddies are stretched in the zonal direction and the unstable state is characterized by strong eddy-shedding processes. During these processes, cyclonic eddies are found to be formed in the first quasi-stationary meander and finally pinched off from the KE jet. Energy budget illustrates that the eddy shedding processes are triggered by baroclinic instability, while barotropic instability becomes the dominated energy source after the meander is fully developed. Accompanied by the generation of cyclonic eddies, significant inverse energy cascade is detected. When the upstream KE is stable, the eddy activity is dominated by baroclinic instability and the inverse energy cascade occurs similarly. Distinct from the upstream, eddies in the downstream KE tend to be stretched in the meridional direction and the decadal variability of eddy kinetic energy in this region is mainly regulated by the upstream through lateral advection.

1141. Ocean Circulation in the Toarcian (Early Jurassic): A Key Control on Deoxygenation and Carbon Burial on the European Shelf

Authors: Itzel Ruvalcaba Baroni (Utrecht University); Alexandre Pohl (Institut de Recherche pour le Développement (IRD)); Niels A. G. M. van Helmond (Utrecht University); Nina M. Papadomanolaki (Utrecht University); Angela L. Coe (School of Environment, Earth and Ecosystem Sciences The Open University, Walton Hall Milton Keynes UK); Anthony S. Cohen (School of Environment, Earth and Ecosystem Sciences The Open University, Walton Hall Milton Keynes UK); Bas van de Schootbrugge (Utrecht University); Yannick Donnadieu (Institut de Recherche pour le Développement (IRD)); Caroline P. Slomp (Utrecht University)

Journal: Paleoceanography and Paleoclimatology

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018pa003394

URL: <https://doi.org/10.1029/2018pa003394>

Publication Date: 2018-08-25

Volume/Issue: Volume 33, Issue 9

References: 137

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: oxygenation, Toarcian Oceanic Anoxic Event, organic carbon burial, European Shelf, Early Jurassic (and 1 more)

Funding: FP7 Ideas: European Research Council

License: N/A

Open Access: N/A

Abstract: The Toarcian Oceanic Anoxic Event (T-OAE, ~183 Myr) was a long-lasting episode of ocean deoxygenation during the Early Jurassic. The event is related to a period of global warming and characterized by major perturbations to the hydrological and carbon cycles with high rates of organic matter burial in shelf seas. Ocean circulation during the Toarcian and its influence on marine biogeochemical cycles are still not fully understood. Here we assess the spatial extent of anoxia in the NW Tethys Ocean during the T-OAE, the relationship with ocean circulation and the impact on organic carbon burial, using new and existing sedimentary records from the European Epicontinental Shelf in combination with general circulation model results. We demonstrate that bottom waters on the southwestern part of the shelf were mainly oxic during the T-OAE, while those in the northeastern basins were mostly anoxic or even sulfidic. Results for two ocean-atmosphere models (Fast Ocean-Atmosphere Model and Massachusetts Institute of Technology general circulation model) suggest the presence of a strong clockwise gyre over the European Epicontinental Shelf, which brought oxygenated equatorial waters from the Tethys Ocean to the southern shelf. The northward limb of the gyre was significantly weakened due to the rough bathymetry of the northern shelf, making this relative small region highly sensitive to local ocean stratification. These sluggish ocean dynamics promoted bottom water anoxia and enhanced burial of organic carbon in the northeastern basins, which accounted for 3-5% of the total carbon extracted from the ocean-atmosphere system as recorded by the positive carbon isotope shift.

1142. Impact of the Atlantic Meridional Mode on Gulf Stream North Wall Position

Authors: Sultan Hameed (School of Marine and Atmospheric Sciences, Stony Brook University, State University of New York, Stony Brook, New York); Christopher L. P. Wolfe (School of Marine and Atmospheric Sciences, Stony Brook University, State University of New York, Stony Brook, New York); Lequan Chi (School of Marine and Atmospheric Sciences, Stony Brook University, State University of New York, Stony Brook, New York)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0098.1

URL: <https://doi.org/10.1175/jcli-d-18-0098.1>

Publication Date: 2018-08-24

Volume/Issue: Volume 31, Issue 21

References: 68

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: North Atlantic Oscillation, Boundary currents, Tropical variability, Ocean circulation, North Atlantic Ocean (and 1 more)

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: The path of the Gulf Stream as it leaves the continental shelf near Cape Hatteras is marked by a sharp gradient in ocean temperature known as the North Wall. Previous work in the literature has considered processes related to the North Atlantic Oscillation (NAO) in triggering latitudinal displacements of the North Wall position. This paper presents evidence that the Atlantic meridional mode (AMM) also impacts interannual variations of the North Wall position. The AMM signal from the tropics propagates to the Gulf Stream near the 200-m depth, and there are two time scales for this interaction. Anomalous Ekman suction induced by AMM cools the tropical Atlantic. The cold water in the Caribbean Sea is entrained into the currents feeding the Gulf Stream, and this cooling signal reaches the North Wall within a year. A second mechanism involves cold anomalies in the western tropical Atlantic, which initially propagate westward as baroclinic planetary waves, reaching the Gulf Stream and resulting in a southward shift in the North Wall position after a delay of about one year. In an analysis for the period 1961-2015, AMM's signal dominates North Wall fluctuations in the upper 300 m, while NAO is the major influence below ~500 m; the influence of both the teleconnections is seen between 300 and 500 m. The relationship between the Atlantic meridional overturning circulation (AMOC) and the North Wall is investigated for the 2005-15 period and found to be statistically significant only at the sea surface in one of the three North Wall indices used.

1143. Interannual Eddy Kinetic Energy Modulations in the Agulhas Return Current

Authors: Yanan Zhu (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology, Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Bo Qiu (Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Xiaopei Lin (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Fan Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014333

URL: <https://doi.org/10.1029/2018jc014333>

Publication Date: 2018-08-23

Volume/Issue: Volume 123, Issue 9

References: 48

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: eddy kinetic energy, Agulhas Return Current, interannual variability

Funding: China Scholarship Council, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Interannual variability in the mesoscale eddy field in the Agulhas Return Current (ARC) of 32-42 degreesS and 15-35 degreesE is investigated based on satellite altimeter observations and state estimate from the Estimating the Circulation and Climate of the Ocean, Phase II from 1993 to 2016. It is found that the interannual modulation of eddy kinetic energy in the ARC region is externally mediated by the wind stress forcing that generates the westward propagating sea surface height anomalies across the South Indian Ocean subtropical gyre. The wind-forced sea surface height anomalies influence the upstream Agulhas Current volume transports. By modulating the intensity of barotropic instability of the ARC mean flow centered around the retroflection region, the Agulhas Current inflow variability leads to the downstream interannual eddy kinetic energy fluctuations in the ARC region.

1144. Contrasting Intraseasonal Variations of the Equatorial Pacific Ocean Between the 1997-1998 and 2015-2016 El Niño Events

Authors: Yilong Lyu (Chinese Academy of Sciences); Yuanlong Li (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Xiaohui Tang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Fan Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Jianing Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl078915

URL: <https://doi.org/10.1029/2018gl078915>

Publication Date: 2018-08-23

Volume/Issue: Volume 45, Issue 18

References: 52

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: intraseasonal variability, MJO, WWE, El Niño, ocean current (and 1 more)

Funding: National Natural Science Foundation of China, Chinese Academy of Sciences

License: Creative Commons

Open Access: N/A

Abstract: Oceanic intraseasonal variability (ISV) plays a vital role in the initiation and development of El Niño events. Mooring observations reveal contrasting characteristics of upper-ocean zonal current ISV during different El Niño events. Compared to the 1997-1998 event, ISV of the 2015-2016 event was significantly weaker (by around 30-50%) over the equatorial Pacific basin, and the variability maximum was shifted to the east (~160 degreesE). These differences can be largely explained by wind forcing features. ISV of surface winds in 2015 is overall weaker than in 1997. Intraseasonal westerly wind bursts in 1997 occurred near the western boundary and originated from the Indian Ocean, whereas in 2015, perturbations were mainly propagated from the northeast Pacific and achieved the peak power at ~160 degreesE of the equator. These changes are likely associated with the decadal warming of the northeast subtropical Pacific and weakening of the Indian Ocean intraseasonal oscillation.

1145. Pattern of vertical velocity in the Lofoten vortex (the Norwegian Sea)

Authors: Igor Bashmachnikov; Tatyana Belonenko; Pavel Kuibin; Denis Volkov; Victor Foux

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-018-1213-1

URL: <https://doi.org/10.1007/s10236-018-1213-1>

Publication Date: 2018-08-23

Volume/Issue: Volume 68, Issue 12

References: 63

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Russian Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: N/A

1146. A Filtering of Incomplete GNSS Position Time Series with Probabilistic Principal Component Analysis

Authors: Maciej Gruszczynski; Anna Klos; Janusz Bogusz

Journal: Pageoph Topical Volumes

Publisher: Springer International Publishing

DOI: 10.1007/978-3-319-96277-1_19

URL: https://doi.org/10.1007/978-3-319-96277-1_19

Publication Date: 2018-08-23

Volume/Issue: N/A

References: N/A

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Position (finance)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: For the first time, we introduced the probabilistic principal component analysis (pPCA) regarding the spatio-temporal filtering of Global Navigation Satellite System (GNSS) position time series to estimate and remove Common Mode Error (CME) without the interpolation of missing values. We used data from the International GNSS Service (IGS) stations which contributed to the latest International Terrestrial Reference Frame (ITRF2014). The efficiency of the proposed algorithm was tested on the simulated incomplete time series, then CME was estimated for a set of 25 stations located in Central Europe. The newly applied pPCA was compared with previously used algorithms, which showed that this method is capable of resolving the problem of proper spatio-temporal filtering of GNSS time series characterized by different observation time span. We showed, that filtering can be carried out with pPCA method when there exist two time series in the dataset having less than 100 common epoch of observations. The 1st Principal Component (PC) explained more than 36% of the total variance represented by time series residuals' (series with deterministic model removed), what compared to the other PCs variances (less than 8%) means that common signals are significant in GNSS residuals. A clear improvement in the spectral indices of the power-law noise was noticed for the Up component, which is reflected by an average shift towards white noise from - 0.98 to - 0.67 (30%). We observed a significant average reduction in the accuracy of stations' velocity estimated for filtered residuals by 35, 28 and 69% for the North, East, and Up components, respectively. CME series were also subjected to analysis in the context of environmental mass loading influences of the filtering results. Subtraction of the environmental loading models from GNSS residuals provides to reduction of the estimated CME variance by 20 and 65% for horizontal and vertical components, respectively.

1147. Lagrangian tracking in stochastic fields with application to an ensemble of velocity fields in the Red Sea

Authors: Samah El Mohtar; Ibrahim Hoteit; Omar Knio; Leila Issa; Issam Lakkis

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2018.08.008

URL: <https://doi.org/10.1016/j.ocemod.2018.08.008>

Publication Date: 2018-08-21

Volume/Issue: Volume 131

References: 24

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: King Abdullah University of Science and Technology

License: N/A

Open Access: N/A

Abstract: N/A

1148. Origin of Circumpolar Deep Water intruding onto the Amundsen and Bellingshausen Sea continental shelves

Authors: Yoshihiro Nakayama; Dimitris Menemenlis; Hong Zhang; Michael Schodlok; Eric Rignot

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-018-05813-1

URL: <https://doi.org/10.1038/s41467-018-05813-1>

Publication Date: 2018-08-20

Volume/Issue: Volume 9, Issue 1

References: 52

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Melting of West Antarctic ice shelves is enhanced by Circumpolar Deep Water (CDW) intruding onto the Amundsen and Bellingshausen Seas (ABS) continental shelves. Despite existing studies of cross-shelf and on-shelf CDW transports, CDW pathways onto the ABS originating from further offshore have never been investigated. Here, we investigate CDW pathways onto the ABS using a regional ocean model. Simulated CDW tracers from a zonal section across 67 degreesS (S04P) circulate along the Antarctic Circumpolar Current (ACC) and Ross Gyre (RG) and travel into ABS continental shelf after 3-5 years, but source locations are shifted westward by ~900 km along S04P in 2001-2006 compared to 2009-2014. We find that simulated on- and off-shelf CDW is ~0.1-0.2 degreesC warmer in the 2009-2014 case than in the 2001-2006 case together with changes in simulated ocean circulation. These differences are primarily caused by lateral, rather than surface, boundary conditions, implying that large-scale atmospheric and ocean circulations are able to control CDW pathways and thus off- and on-shelf CDW properties.

1149. Visual Analysis of Spatia-temporal Relations of Pairwise Attributes in Unsteady Flow

Authors: Marzieh Berenjkoub; Rodolfo Ostilla Monico; Robert S. Laramée; Guoning Chen

Journal: IEEE Transactions on Visualization and Computer Graphics

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tvcg.2018.2864817

URL: <https://doi.org/10.1109/tvcg.2018.2864817>

Publication Date: 2018-08-20

Volume/Issue: Volume 25, Issue 1

References: 63

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: NSF

License: N/A

Open Access: N/A

Abstract: N/A

1150. Decadal Variability of North Pacific Eastern Subtropical Mode Water

Authors: Yongqing Guo (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Xiaopei Lin (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Meng Wei (Qingdao National Laboratory for Marine Science and Technology, First Institute of Oceanography, State Oceanic Administration Qingdao China); Cong Liu (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology); Guang Men (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc013890

URL: <https://doi.org/10.1029/2018jc013890>

Publication Date: 2018-08-20

Volume/Issue: Volume 123, Issue 9

References: 56

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Science and Technology of the People's Republic of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: A significant decadal variability of the North Pacific Eastern Subtropical Mode Water (NPESTMW), related to the Pacific Decadal Oscillation, is identified from the gridded Argo and EN4 observational data sets. To investigate the decadal variability of the NPESTMW volume, the high-resolution Estimating the Circulation and Climate of the Ocean, Phase II simulation is used to diagnose the volume budget of the NPESTMW density layer in the NPESTMW formation region (20 degrees-36 degreesN, 160 degrees-120 degreesW) between 1992 and 2016. Results show that NPESTMW is formed through subduction of low potential vorticity (PV) water. This low-PV water subduction is mainly determined by mixed layer depth change and lateral induction, which account for approximately 58% and 32%, respectively. The decadal NPESTMW volume anomaly is originated from the Pacific Decadal Oscillation-induced decadal anomaly of air-sea buoyancy flux. This decadal buoyancy flux anomaly drives the decadal anomaly of mixed layer depth change, then causes anomalous subduction, and leads to anomalous low-PV water subduction. The decadal NPESTMW volume anomaly is finally generated by this anomalous low-PV water subduction.

1151. Ocean modelling and altimeter data reveal the possible occurrence of intrinsic low-frequency variability of the Kuroshio Extension

Authors: Vittorio Gentile; Stefano Pierini; Paola de Ruggiero; Luca Pietranera

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2018.08.006

URL: <https://doi.org/10.1016/j.ocemod.2018.08.006>

Publication Date: 2018-08-20

Volume/Issue: Volume 131

References: 67

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1152. Projections of Future Sea Level Contributions from the Greenland and Antarctic Ice Sheets: Challenges Beyond Dynamical Ice Sheet Modeling

Authors: Unknown Author; Sophie Nowicki; Helene Seroussi

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2018.216

URL: <https://doi.org/10.5670/oceanog.2018.216>

Publication Date: 2018-08-16

Volume/Issue: Volume 31, Issue 2

References: 94

Cited by: 22 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1153. Detailed investigation of the three-dimensional structure of a mesoscale cold eddy in the Kuroshio extension region

Authors: Chen Xi (National University of Defense Technology); Hu Dong (The 31010 Army of PLA, Beijing, People's Republic of China); Mao Ke-feng (National University of Defense Technology); Li Yan (National University of Defense Technology)

Journal: Journal of Operational Oceanography

Publisher: Informa UK Limited

DOI: 10.1080/1755876x.2018.1505069

URL: <https://doi.org/10.1080/1755876x.2018.1505069>

Publication Date: 2018-08-15

Volume/Issue: Volume 11, Issue 2

References: 43

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Natural Science Foundation of Jiangsu Province

License: N/A

Open Access: N/A

Abstract: N/A

1154. Interannual variability and dynamics of intraseasonal wind rectification in the equatorial Pacific Ocean

Authors: Xia Zhao; Dongliang Yuan; Guang Yang; Jing Wang; Hailong Liu; Renhe Zhang; Weiqing Han

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-018-4383-0

URL: <https://doi.org/10.1007/s00382-018-4383-0>

Publication Date: 2018-08-09

Volume/Issue: Volume 52, Issue 7-8

References: 48

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSFC, QMSNL, Shandong Provincial projects (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1155. On the Dynamics and Water Mass Transformation of a Boundary Current Connecting Alpha and Beta Oceans

Authors: Erwin Lambert (University of Bergen, Utrecht University, Bjerknes Centre for Climate Research); Tor Eldevik (University of Bergen, Bjerknes Centre for Climate Research); Michael A. Spall (Woods Hole Oceanographic Institution)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-17-0186.1

URL: <https://doi.org/10.1175/jpo-d-17-0186.1>

Publication Date: 2018-08-08

Volume/Issue: Volume 48, Issue 10

References: 45

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Buoyancy, Baroclinic flows, Thermohaline circulation, Halocline|Outflow|Boundary current|Circumpolar deep water|Stratification (seeds)|Deep ocean water|Antarctic Intermediate Water|Gravity current, Boundary currents (and 2 more)

Funding: Norges Forskningsråd, National Science Foundation

License: N/A

Open Access: N/A

Abstract: A subpolar marginal sea, like the Nordic seas, is a transition zone between the temperature-stratified subtropics (the alpha ocean) and the salinity-stratified polar regions (the beta ocean). An inflow of Atlantic Water circulates these seas as a boundary current that is cooled and freshened downstream, eventually to outflow as Deep and Polar Water. Stratification in the boundary region is dominated by a thermocline over the continental slope and a halocline over the continental shelves, separating Atlantic Water from Deep and Polar Water, respectively. A conceptual model is introduced for the circulation and water mass transformation in a subpolar marginal sea to explore the potential interaction between the alpha and beta oceans. Freshwater input into the shelf regions has a slight strengthening effect on the Atlantic inflow, but more prominently impacts the water mass composition of the outflow. This impact of freshwater, characterized by enhancing Polar Water outflow and suppressing Deep Water outflow, is strongly determined by the source location of freshwater. Concretely, perturbations in upstream freshwater sources, like the Baltic freshwater outflow into the Nordic seas, have an order of magnitude larger potential to impact water mass transports than perturbations in downstream sources like the Arctic freshwater outflow. These boundary current dynamics are directly related to the qualitative stratification in transition zones and illustrate the interaction between the alpha and beta oceans.

1156. Validation of ocean model syntheses against hydrography using a new web application

Authors: AL Gemmell (University of Reading); Gc Smith (University of Reading); K Haines (University of Reading); Jd Blower (University of Reading)

Journal: Journal of Operational Oceanography

Publisher: Informa UK Limited

DOI: 10.1080/1755876x.2009.12027739

URL: <https://doi.org/10.1080/1755876x.2009.12027739>

Publication Date: 2018-08-06

Volume/Issue: Volume 2, Issue 2

References: 18

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1157. Impacts of Atmospheric Reanalysis Uncertainty on Atlantic Overturning Estimates at 25 degreesN

Authors: Helen R. Pillar (University of Oxford); Helen L. Johnson (University of Oxford); David P. Marshall (University of Oxford); Patrick Heimbach (University of Texas at Austin); So Takao (Imperial College London)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0241.1

URL: <https://doi.org/10.1175/jcli-d-18-0241.1>

Publication Date: 2018-08-01

Volume/Issue: Volume 31, Issue 21

References: 97

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Natural Environment Research Council, National Science Foundation (and 3 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Atmospheric reanalyses are commonly used to force numerical ocean models, but despite large discrepancies reported between different products, the impact of reanalysis uncertainty on the simulated ocean state is rarely assessed. In this study, the impact of uncertainty in surface fluxes of buoyancy and momentum on the modeled Atlantic meridional overturning at 25 degreesN is quantified for the period January 1994-December 2011. By using an ocean-only climate model and its adjoint, the space and time origins of overturning uncertainty resulting from air-sea flux uncertainty are fully explored. Uncertainty in overturning induced by prior air-sea flux uncertainty can exceed 4 Sv (where 1 Sv $\equiv$ 106 m³ s⁻¹) within 15 yr, at times exceeding the amplitude of the ensemble-mean overturning anomaly. A key result is that, on average, uncertainty in the overturning at 25 degreesN is dominated by uncertainty in the zonal wind at lags of up to 6.5 yr and by uncertainty in surface heat fluxes thereafter, with winter heat flux uncertainty over the Labrador Sea appearing to play a critically important role.

1158. On the time and length scales of the Arctic sea ice thickness anomalies: a study based on fourteen reanalyses

Authors: Leandro Ponsoni; François Massonnet; Thierry Fichefet; Matthieu Chevallier; David Docquier

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2018-133

URL: <https://doi.org/10.5194/tc-2018-133>

Publication Date: 2018-07-27

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Sea ice concentration

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The ocean-sea ice reanalyses are the main source of Arctic sea ice thickness information in terms of spatio-temporal coverage, since observations are still sparse in time and space. In this work, we first aim at comparing how the sea ice thickness from an ensemble of fourteen reanalyses compares with different sources of observations, such as moored upward-looking sonars, submarines, airborne, satellites and ice boreholes. Second, based on the same reanalyses, we intend to characterize the time (persistence) and length scales of sea ice thickness anomalies. We investigate whether data assimilation of sea ice concentration by the reanalyses impacts the realism of sea ice thickness as well as its respective time and length scales. The results suggest that reanalyses with sea ice data assimilation do not necessarily perform better in terms of sea ice thickness compared with the reanalyses which do not assimilate sea ice concentration. However, data assimilation has a clear impact on the time and length scales: reanalyses built with sea ice data assimilation present shorter time and length scales. The mean time and length scales for reanalyses with data assimilation vary from 2.5-5.0 months and 337.0-732.5 km, respectively, while reanalyses with no data assimilation are characterized by values from 4.9-7.8 months and 846.7-935.7 km, respectively.

1159. Detecting drought impact on terrestrial biosphere carbon fluxes over contiguous US with satellite observations

Authors: Junjie Liu; Kevin Bowman; Nicholas C Parazoo; A Anthony Bloom; Debra Wunch; Zhe Jiang; Kevin R Gurney; Dave Schimel

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/aad5ef

URL: <https://doi.org/10.1088/1748-9326/aad5ef>

Publication Date: 2018-07-26

Volume/Issue: Volume 13, Issue 9

References: 54

Cited by: 23 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: N/A

1160. Widespread Reemergence of Sea Surface Temperature Anomalies in the Global Oceans, Including Tropical Regions Forced by Reemerging Winds

Authors: P. Byju (Monash University); D. Dommenges (Monash University); M. A. Alexander (NOAA/ESRL Boulder CO USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl079137

URL: <https://doi.org/10.1029/2018gl079137>

Publication Date: 2018-07-25

Volume/Issue: Volume 45, Issue 15

References: 39

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: ARC Climate System Science

License: N/A

Open Access: N/A

Abstract: Sea surface temperature anomalies (SSTA) in portions of the extratropics are known to recur from one winter to the next without persisting through the intervening summer. Previous studies identified only a limited number of midlatitude regions where this reemergence occurs. Here we find that most of the global oceans exhibit winter-to-winter recurrence of SSTA. Indeed, recurrence of SSTA is the default process in the global ocean. Only regions strongly linked to El Niño do not show signs of SST reemergence. In midlatitudes, the temperature anomalies that recur persist below the shallow mixed layer in summer. However, SST recurrence is also found at some tropical locations and appears to be independent of subsurface ocean heat storage. Reemergence at these locations is linked to the recurrence of atmospheric drivers of SSTA, predominantly the wind-stress forcing. Our results are supported by different ocean data sets and by state-of-the-art climate model simulations.

1161. Variations of the Argentine Gyre Observed in the GRACE Time-Variable Gravity and Ocean Altimetry Measurements

Authors: Y. Yu (Wuhan University, Academia Sinica); B. F. Chao (Academia Sinica); D. García-García (University of Alicante); Z. Luo (MOE Key Laboratory of Fundamental Physical Quantities Measurement, School of Physics Huazhong University of Science and Technology Wuhan China)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc014189

URL: <https://doi.org/10.1029/2018jc014189>

Publication Date: 2018-07-23

Volume/Issue: Volume 123, Issue 8

References: 41

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: eddies, altimetry, Argentine Gyre, time-variable gravity, GRACE

Funding: Ministry of Science and Technology, Taiwan

License: N/A

Open Access: N/A

Abstract: We investigate the nonseasonal and high-frequency variations of the Argentine Gyre in the south Atlantic Ocean by analyzing the time-variable gravity (TVG) measurements from the Gravity Recovery and Climate Experiment (GRACE) satellite mission in conjunction with the satellite ocean altimetry and two ocean general circulation model outputs (GLORYS2V4 and ECCO V4R3). We solve the empirical orthogonal functions (EOF) and complex EOF and find good agreement between TVG and altimetry observations, confirming the barotropic structure of the Argentine Gyre. In particular, the leading EOF modes of the overall up-and-down undulation in TVG and altimetry variations are found to be in pace temporally with the Antarctic Oscillation Index with correlation as high as 0.69 at zero time shift. Furthermore, the leading complex EOF mode signifies a counterclockwise dipole pattern of ~25-day periodicity within the overall gyre with multiscale amplitude modulation. The fact that GRACE does observe these signals, while the de-aliasing background ocean model fails to, ascertains that GRACE data have adequate sensitivity to allow the detection of TVG signals at spatial and temporal resolutions higher than practiced hitherto. The ~25-day oscillation is well recovered in the GLORYS2V4 ocean general circulation model, but not in ECCO V4R3. Our study demonstrates that satellite-observed TVG fields can be useful in studying oceanographic gyres, particularly the polar gyres, that are not well observed by altimetry and in situ data.

1162. Evaluation of multi-decadal UCLA-CFSv2 simulation and impact of interactive atmospheric-ocean feedback on global and regional variability

Authors: Jiwoo Lee; Yongkang Xue; Fernando De Sales; Ismaila Diallo; Larry Marx; Michael Ek; Kenneth R. Sperber; Peter J. Gleckler

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-018-4351-8

URL: <https://doi.org/10.1007/s00382-018-4351-8>

Publication Date: 2018-07-13

Volume/Issue: Volume 52, Issue 5-6

References: 118

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, U.S. Department of Energy

License: N/A

Open Access: N/A

Abstract: N/A

1163. Identifying Spatial Variability in Greenland's Outlet Glacier Response to Ocean Heat

Authors: David F. Porter; Kirsty J. Tinto; Alexandra L. Boghosian; Beata M. Csatho; Robin E. Bell; James R. Cochran

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2018.00090

URL: <https://doi.org/10.3389/feart.2018.00090>

Publication Date: 2018-07-13

Volume/Issue: Volume 6

References: 45

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1164. Ocean-Induced Melt Triggers Glacier Retreat in Northwest Greenland

Authors: M. Wood (Department of Earth System Science University of California CA Irvine USA); E. Rignot (Department of Earth System Science University of California CA Irvine USA); I. Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); R. Millan (Department of Earth System Science University of California CA Irvine USA); M. Morlighem (Department of Earth System Science University of California CA Irvine USA); J. Mouginot (Department of Earth System Science University of California CA Irvine USA, University of Grenoble Alpes, Institut de Recherche pour le Développement (IRD)); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl078024

URL: <https://doi.org/10.1029/2018gl078024>

Publication Date: 2018-07-11

Volume/Issue: Volume 45, Issue 16

References: 44

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Cryosphere Program, NASA Physical Oceanography program, NASA Ocean Melting Greenland EVS-3 mission (and 1 more)

License: N/A

Open Access: N/A

Abstract: In recent decades, tidewater glaciers in Northwest Greenland contributed significantly to sea level rise but exhibited a complex spatial pattern of retreat. Here we use novel observations of bathymetry and water temperature from NASA's Ocean Melting Greenland mission to quantify the role of warm, salty Atlantic Water in controlling the evolution of 37 glaciers. Modeled ocean-induced undercutting of calving margins compared with ice advection and ice front retreat observed by satellites from 1985 to 2015 indicate that 35 glaciers retreated when cumulative anomalies in ocean-induced undercutting rose above the range of seasonal variability of calving-front positions, while two glaciers standing on shallow sills and colder water did not retreat. Deviations in the observed timing of retreat are explained by residual uncertainties in bathymetry, inefficient mixing of waters in shallow fjords, and the presence of small floating sections. Overall, warmer ocean temperature triggered the retreat, but calving processes dominate ablation (71%).

1165. Vulnerability of flatfish and their fisheries to climate change

Authors: William W.L. Cheung; Muhammed A. Oyinlola

Journal: Journal of Sea Research

Publisher: Elsevier BV

DOI: 10.1016/j.seares.2018.06.006

URL: <https://doi.org/10.1016/j.seares.2018.06.006>

Publication Date: 2018-07-11

Volume/Issue: Volume 140

References: 38

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: Vulnerability

Funding: Natural Sciences and Engineering Research Council of Canada

License: N/A

Open Access: N/A

Abstract: N/A

1166. Distinct Mechanisms of Decadal Subsurface Heat Content Variations in the Eastern and Western Indian Ocean Modulated by Tropical Pacific SST

Authors: Xiaolin Jin (Tsinghua University); Young-Oh Kwon (Woods Hole Oceanographic Institution); Caroline C. Ummenhofer (Woods Hole Oceanographic Institution); Hyodae Seo (Woods Hole Oceanographic Institution); Yu Kosaka (University of Tokyo); Jonathon S. Wright (Tsinghua University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-18-0184.1

URL: <https://doi.org/10.1175/jcli-d-18-0184.1>

Publication Date: 2018-07-09

Volume/Issue: Volume 31, Issue 19

References: 77

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Independent Research & Development Program at WHOI, Division of Ocean Sciences, National Oceanic and Atmospheric Administration (and 1 more)

License: N/A

Open Access: N/A

Abstract: Decadal variability of the subsurface ocean heat content (OHC) in the Indian Ocean is investigated using a coupled climate model experiment, in which observed eastern tropical Pacific sea surface temperature (EPSST) anomalies are specified. This study intends to understand the contributions of external forcing relative to those of internal variability associated with EPSST, as well as the mechanisms by which the Pacific impacts Indian Ocean OHC. Internally generated variations associated with EPSST dominate decadal variations in the subsurface Indian Ocean. Consistent with ocean reanalyses, the coupled model reproduces a pronounced east-west dipole structure in the southern tropical Indian Ocean and discontinuities in westward-propagating signals in the central Indian Ocean around 100 degreesE. This implies distinct mechanisms by which the Pacific impacts the eastern and western Indian Ocean on decadal time scales. Decadal variations of OHC in the eastern Indian Ocean are attributed to 1) western Pacific surface wind anomalies, which trigger oceanic Rossby waves propagating westward through the Indonesian Seas and influence Indonesian Throughflow transport, and 2) zonal wind anomalies over the central tropical Indian Ocean, which trigger eastward-propagating Kelvin waves. Decadal variations of OHC in the western Indian Ocean are linked to conditions in the Pacific via changes in the atmospheric Walker cell, which trigger anomalous wind stress curl and Ekman pumping in the central tropical Indian Ocean. Westward-propagating oceanic Rossby waves extend the influence of this anomalous Ekman pumping to the western Indian Ocean.

1167. Evaluation of a global ocean general circulation model; The Lat-Lon-Cap (LLC90) configuration of the MITgcm

Authors: Bernawis Lamona; Judith Hauck; Christoph Völker

Journal: IOP Conference Series: Earth and Environmental Science

Publisher: IOP Publishing

DOI: 10.1088/1755-1315/162/1/012002

URL: <https://doi.org/10.1088/1755-1315/162/1/012002>

Publication Date: 2018-07-06

Volume/Issue: Volume 162

References: 21

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1168. A Global Glacial Ocean State Estimate Constrained by Upper-Ocean Temperature Proxies

Authors: Daniel E. Amrhein (Massachusetts Institute of Technology, Woods Hole Oceanographic Institution); Carl Wunsch (Harvard University); Olivier Marchal (Woods Hole Oceanographic Institution); Gael Forget (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0769.1

URL: <https://doi.org/10.1175/jcli-d-17-0769.1>

Publication Date: 2018-07-05

Volume/Issue: Volume 31, Issue 19

References: 85

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Sea surface temperature, Ocean, Inverse methods, Abyssal circulation (and 1 more)

Funding: National Science Foundation, National Science Foundation, National Science Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: We use the method of least squares with Lagrange multipliers to fit an ocean general circulation model to the Multiproxy Approach for the Reconstruction of the Glacial Ocean Surface (MARGO) estimate of near sea surface temperature (NSST) at the Last Glacial Maximum (LGM; circa 23-19 thousand years ago). Compared to a modern simulation, the resulting global, last-glacial ocean state estimate, which fits the MARGO data within uncertainties in a free-running coupled ocean-sea ice simulation, has global-mean NSSTs that are 2 degreesC lower and greater sea ice extent in all seasons in both the Northern and Southern Hemispheres. Increased brine rejection by sea ice formation in the Southern Ocean contributes to a stronger abyssal stratification set principally by salinity, qualitatively consistent with pore fluid measurements. The upper cell of the glacial Atlantic overturning circulation is deeper and stronger. Dye release experiments show similar distributions of Southern Ocean source waters in the glacial and modern western Atlantic, suggesting that LGM NSST data do not require a major reorganization of abyssal water masses. Outstanding challenges in reconstructing LGM ocean conditions include reducing effects from model biases and finding computationally efficient ways to incorporate abyssal tracers in global circulation inversions. Progress will be aided by the development of coupled ocean-atmosphere-ice inverse models, by improving high-latitude model processes that connect the upper and abyssal oceans, and by the collection of additional paleoclimate observations.

1169. Modeling the enhancement of sea surface chlorophyll concentration during the cyclonic events in the Arabian Sea

Authors: Kunal Chakraborty; Kumar Nimit; Anirban Akhand; Satya Prakash; Arya Paul; Jayashree Ghosh; T.V.S. Udaya Bhaskar; Abhra Chanda

Journal: Journal of Sea Research

Publisher: Elsevier BV

DOI: 10.1016/j.seares.2018.07.003

URL: <https://doi.org/10.1016/j.seares.2018.07.003>

Publication Date: 2018-07-04

Volume/Issue: Volume 140

References: 51

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Earth Sciences

License: N/A

Open Access: N/A

Abstract: N/A

1170. Effect of Helmert Transformation Parameters and Weight Matrix on Seasonal Signals in GNSS Coordinate Time Series

Authors: Guo Chen (Wuhan University); Qile Zhao (Wuhan University); Na Wei (Wuhan University); Min Li (Wuhan University)

Journal: Sensors

Publisher: MDPI AG

DOI: 10.3390/s18072127

URL: <https://doi.org/10.3390/s18072127>

Publication Date: 2018-07-03

Volume/Issue: Volume 18, Issue 7

References: 40

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: Seasonal signals caused by the Earth's surface mass redistribution can be detected by Global Navigation Satellite Systems (GNSS). The authors analyze the effect of Helmert transformation parameters and weight matrices, as well as the additional draconic signals on seasonal signals, in the GNSS coordinate time series. Moreover, the contribution of environmental loading models to the GNSS position series is assessed. Position time series of 647 global stations, with spans of 2–21 years are collected to generate six cumulative solutions using different parameters estimated in a deterministic model, as well as weight matrices. Comparison among the different solutions indicates that Helmert transformation parameters and weight matrices can result in a root mean square of 0.1 mm and 0.3 mm for seasonal signals, respectively. Compared to the displacements obtained from environmental loading models, seasonal signals estimated with the Helmert parameters and full weight matrices considered seems to have the best agreement with the results of the loading model. Meanwhile, the additional draconic signals are not effective to be parameterized in the deterministic model with an observation time span less than 15 years, marginally. There are 62%, 72% and 90% of 647 stations with weight root mean squares (WRMS) reduced by removing the loading-model-induced changes from the GNSS residual series for the east, north and vertical components, respectively. Finally, to obtain a velocity estimation with a bias of less than 0.1 mm/yr induced by seasonal signals, the position series with a time span greater than seven years is suggested.

1171. Targeted observation analysis of a Northwestern Tropical Pacific Ocean mooring array using an ensemble-based method

Authors: Danian Liu; Jiang Zhu; Yeqiang Shu; Dongxiao Wang; Weiqiang Wang; Changxiang Yan; Wei Zhou

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-018-1188-y

URL: <https://doi.org/10.1007/s10236-018-1188-y>

Publication Date: 2018-06-29

Volume/Issue: Volume 68, Issue 9

References: 54

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the National Key Research and Development Program of China, the “Strategic Priority Research Program” of the Chinese Academy of Sciences, the National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

1172. Rapid Red Sea Deep Water renewals caused by volcanic eruptions and the North Atlantic Oscillation

Authors: Fengchao Yao (King Abdullah University of Science and Technology, Biological and Environmental Science and Engineering Division, Thuwal 23955-6900, Saudi Arabia.); Ibrahim Hoteit (King Abdullah University of Science and Technology, Physical Science and Engineering Division, Thuwal 23955-6900, Saudi Arabia.)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.aar5637

URL: <https://doi.org/10.1126/sciadv.aar5637>

Publication Date: 2018-06-27

Volume/Issue: Volume 4, Issue 6

References: 29

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Saudi Aramco, KAUST

License: N/A

Open Access: N/A

Abstract: Large volcanic eruptions and the North Atlantic Oscillation caused rapid renewals of the Red Sea Deep Water.

1173. Inverse Problems, Inverse Methods, and Inverse Models

Authors: Carl Wunsch

Journal: Encyclopedia of Ocean Sciences

Publisher: Elsevier

DOI: 10.1016/b978-0-12-409548-9.11262-x

URL: <https://doi.org/10.1016/b978-0-12-409548-9.11262-x>

Publication Date: 2018-06-27

Volume/Issue: N/A

References: 71

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: Models, Statistics, Ill-posed problems, Data analysis

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1174. Evaluation of a global eddy-permitting hybrid coordinate ocean model

Authors: Yue-Liang CHEN (Chinese Academy of Sciences); Chang-Xiang YAN (Chinese Academy of Sciences); Jiang ZHU (Chinese Academy of Sciences); Yi-Neng LI (Chinese Academy of Sciences)

Journal: Atmospheric and Oceanic Science Letters

Publisher: Elsevier BV

DOI: 10.1080/16742834.2018.1490625

URL: <https://doi.org/10.1080/16742834.2018.1490625>

Publication Date: 2018-06-26

Volume/Issue: Volume 11, Issue 4

References: 26

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key R&D; Program of China, the National Natural Science Foundation of China, Chinese Academy Sciences Project 'Western Pacific Ocean System: Structure, Dynamics and Consequences' (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1175. Drastic thickening of the barrier layer off the western coast of Sumatra due to the Madden-Julian oscillation passage during the Pre-Years of the Maritime Continent campaign

Authors: Qoosaku Moteki; Masaki Katsumata; Kunio Yoneyama; Kentaro Ando; Takuya Hasegawa

Journal: Progress in Earth and Planetary Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40645-018-0190-9

URL: <https://doi.org/10.1186/s40645-018-0190-9>

Publication Date: 2018-06-25

Volume/Issue: Volume 5, Issue 1

References: 44

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Agency for Marine-Earth Science and Technology, Japan Society for the Promotion of Science

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1176. Inferring Connectivity Range in Submerged Aquatic Populations (*Ruppia* L.) Along European Coastal Lagoons From Genetic Imprint and Simulated Dispersal Trajectories

Authors: Ludwig Triest; Tim Sierens; Dimitris Menemenlis; Tom Van der Stocken

Journal: Frontiers in Plant Science

Publisher: Frontiers Media SA

DOI: 10.3389/fpls.2018.00806

URL: <https://doi.org/10.3389/fpls.2018.00806>

Publication Date: 2018-06-13

Volume/Issue: Volume 9

References: 92

Cited by: 15 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1177. Climatic Changes in the Dynamic Topography and Geostrophic Circulation of the Arctic Ocean

Authors: L. A. Timokhov; V. V. Ivanov; H. Kassens; N. V. Lebedev; I. Ye. Frolov; J. Hölemann

Journal: Doklady Earth Sciences

Publisher: Pleiades Publishing Ltd

DOI: 10.1134/s1028334x18050318

URL: <https://doi.org/10.1134/s1028334x18050318>

Publication Date: 2018-06-13

Volume/Issue: Volume 480, Issue 1

References: 11

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1178. Interannual Modulations of the 50-Day Oscillations in the Celebes Sea: Dynamics and Impact

Authors: Xiao Chen (Hohai University); Bo Qiu (Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Shuiming Chen (Department of Oceanography University of Hawai'i at Mānoa Honolulu HI USA); Xuhua Cheng (Hohai University); Yiquan Qi (Hohai University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc013960

URL: <https://doi.org/10.1029/2018jc013960>

Publication Date: 2018-06-12

Volume/Issue: Volume 123, Issue 7

References: 35

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: 50-day oscillation, Celebes Sea, interannual variability

Funding: Natural Science Foundation of Jiangsu Province, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Intense 50-day oscillations have been previously observed at the entrance of Celebes Sea, and their formation has been suggested to be a result of Rossby wave resonance where the frequency of cyclonic eddy shedding by the intruding Mindanao Current matches that of the gravest Rossby mode of the semiencllosed Celebes Sea basin. Using the ocean state estimate of 1993-2016 from the Estimating the Circulation and Climate of the Ocean, Phase II, we detected strong interannual modulations in the shedding of cyclonic eddies at the Celebes Sea entrance. Active eddy sheddings occurred during 1993, 2002-2003, 2006-2010, and 2013-2015. Southward shifting of the wind-driven North Pacific tropical gyre and the concurrent strengthening of the Mindanao Current southeast of the Mindanao Island in these years are found to be conducive for the generation of cyclonic eddies intruding into the Celebes Sea. Modulated by the activity of eddy sheddings, the upper ocean water mass properties in both the Celebes Sea and Makassar Strait exhibit noticeable interannual changes with less saline waters appearing in the 75- to 175-m layer during the active eddy shedding years.

1179. Strengthened Indonesian Throughflow Drives Decadal Warming in the Southern Indian Ocean

Authors: Ying Zhang (CSIRO Oceans and Atmosphere Crawley Western Australia Australia, Chinese Academy of Sciences); Ming Feng (CSIRO Oceans and Atmosphere Crawley Western Australia Australia, Centre for Southern Hemisphere Oceans Research Hobart Tasmania Australia); Yan Du (Chinese Academy of Sciences); Helen E. Phillips (University of Tasmania, ARC Centre of Excellence for Climate Systems Science Hobart Tasmania Australia); Nathaniel L. Bindoff (Antarctic Climate and Ecosystems Cooperative Research Centre, University of Tasmania, ARC Centre of Excellence for Climate Systems Science Hobart Tasmania Australia, CSIRO Oceans and Atmosphere Hobart Tasmania Australia); Michael J. McPhaden (NOAA Pacific Marine Environmental Laboratory Seattle WA USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl078265

URL: <https://doi.org/10.1029/2018gl078265>

Publication Date: 2018-06-11

Volume/Issue: Volume 45, Issue 12

References: 62

Cited by: 81 publications

Language: en

Subjects: N/A

Keywords: upper ocean, heat budget, Ekman transport, Indonesian Throughflow, ocean boundary current (and 1 more)

Funding: Department of the Environment and Energy, Australian Government, Chinese Academy of Sciences, China Scholarship Council (and 2 more)

License: N/A

Open Access: N/A

Abstract: Remarkable warming of the Southern Indian Ocean during the recent two decades is assessed using a heat budget analysis based on the Estimating the Circulation and Climate of the Ocean version 4 release 3 model results. The annual mean temperature averaged in the upper-700 m Southern Indian Ocean during 1998-2015 has experienced significant warming at a rate of 1.03×10^{-2} degreesC/year. A heat budget analysis indicates that the increase is mostly driven by decreased cooling from net air-sea heat flux and increased warming from heat advection. Increased Indonesian Throughflow advection is the largest contributor to warming the upper 700 m of the Southern Indian Ocean, while the reduction of surface turbulent heat flux is of secondary importance. These results expand our understanding of the decadal heat balance in the Indian Ocean and of Indo-Pacific decadal climate variability.

1180. A Comparison of Satellite-Derived Sea Surface Salinity and Salt Fluxes in the Southern Ocean

Authors: Brady S. Ferster; Bulusu Subrahmanyam

Journal: Remote Sensing in Earth Systems Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s41976-018-0001-5

URL: <https://doi.org/10.1007/s41976-018-0001-5>

Publication Date: 2018-06-11

Volume/Issue: Volume 1, Issue 1-2

References: 39

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA/South Carolina Space Grant Consortium

License: N/A

Open Access: N/A

Abstract: N/A

1181. Global Observations of Horizontal Mixing from Argo Float and Surface Drifter Trajectories

Authors: Christopher J. Roach (University of Tasmania); Dhruv Balwada (Courant Institute of Mathematical Sciences New York University New York NY USA); Kevin Speer (Florida State University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jc013750

URL: <https://doi.org/10.1029/2018jc013750>

Publication Date: 2018-06-08

Volume/Issue: Volume 123, Issue 7

References: 59

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Mixing by mesoscale eddies in the ocean plays a major role in setting the distribution of oceanic tracers, with important implications for physical and biochemical systems at local to global scales. Roach et al. (2016; <https://doi.org/10.1002/2015JC011440>) demonstrated that a two-particle analysis of Argo trajectories produces robust estimates of horizontal mixing in the Southern Ocean. Here we extend this analysis to produce global 1 degrees x 1 degrees maps of eddy diffusivity at the nominal Argo parking depth of 1,000 m. We also applied this methodology to estimate surface eddy diffusivities from Global Drifter Program (GDP) surface drifters. The global mean eddy diffusivity was 543 +/- 155 m²/s at 1,000 m and 2637 +/- 311 m²/s at the surface, with elevated diffusivities in regions of enhanced eddy kinetic energy, such as western boundary currents and along the Antarctic Circumpolar Current. The eddy kinetic energy at the equator is high at both the surface and depth, but the eddy diffusivity is only enhanced near the surface. At depth the eddy diffusivity is strongly suppressed due to the presence of mean flow. We used our observational estimates to test the validity of an eddy diffusivity parameterization that accounts for mixing suppression in the presence of zonal mean flows. Our results indicated that this parameterization generally agrees with the directly observed eddy diffusivities in the midlatitude and high-latitude oceans.

1182. Towards determining uncertainties in global oceanic mean values of heat, salt, and surface elevation

Authors: Carl Wunsch

Journal: Tellus A: Dynamic Meteorology and Oceanography

Publisher: Stockholm University Press

DOI: 10.1080/16000870.2018.1471911

URL: <https://doi.org/10.1080/16000870.2018.1471911>

Publication Date: 2018-06-05

Volume/Issue: Volume 70, Issue 1

References: 43

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1183. Changes to the Air-Sea Flux and Distribution of Radiocarbon in the Ocean Over the 21st Century

Authors: Samar Khatiwala (University of Oxford); Heather Graven (Imperial College London); Sarah Payne (Now at Risk Management Solutions London UK, University of Oxford); Patrick Heimbach (University of Texas at Austin)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl078172

URL: <https://doi.org/10.1029/2018gl078172>

Publication Date: 2018-05-29

Volume/Issue: Volume 45, Issue 11

References: 46

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission

License: N/A

Open Access: N/A

Abstract: We investigate the spatiotemporal evolution of radiocarbon ($\Delta^{14}\text{C}$) in the ocean over the 21st century under different scenarios for anthropogenic CO_2 emissions and atmospheric CO_2 and radiocarbon changes using a 3-D ocean carbon cycle model. Strong decreases in atmospheric $\Delta^{14}\text{C}$ in the high-emission scenario result in strong outgassing of ^{14}C over 2050-2100, causing $\Delta^{14}\text{C}$ spatial gradients in the surface ocean and vertical gradients between the surface and intermediate waters to reverse sign. Surface $\Delta^{14}\text{C}$ in the subtropical gyres is lower than $\Delta^{14}\text{C}$ in Pacific Deep Water and Southern Ocean surface water in 2100. In the low-emission scenario, ocean $\Delta^{14}\text{C}$ remains slightly higher than in 1950 and relatively constant over 2050-2100. Over the next 20 years we find decadal changes in $\Delta^{14}\text{C}$ of -30‰ to +5‰ in the upper 2 km of the ocean, which should be detectable with continued hydrographic surveys. Our simulations can help in planning future observations, and they provide a baseline for investigating natural or anthropogenic changes in ocean circulation using ocean $\Delta^{14}\text{C}$ observations and models.

1184. Monitoring Groundwater Storage Changes Using the Gravity Recovery and Climate Experiment (GRACE) Satellite Mission: A Review

Authors: Frédéric Frappart (Institut de Recherche pour le Développement (IRD)); Guillaume Ramillien (Institut de Recherche pour le Développement (IRD))

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs10060829

URL: <https://doi.org/10.3390/rs10060829>

Publication Date: 2018-05-28

Volume/Issue: Volume 10, Issue 6

References: 142

Cited by: 224 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The Gravity Recovery and Climate Experiment (GRACE) satellite mission, which was in operation from March 2002 to June 2017, was the first remote sensing mission to provide temporal variations of Terrestrial Water Storage (TWS), which is the sum of the water masses that were contained in the soil column (i.e., snow, surface water, soil moisture, and groundwater), at a spatial resolution of a few hundred kilometers. As in situ level measurements are generally not sufficiently available for monitoring groundwater changes at the regional-scale, this unique dataset, combined with external information, is widely used to quantify the interannual variations of groundwater storage in the world's major aquifers. GRACE-based groundwater changes revealed significant aquifer depletion over large regions, such as the Middle East, the northwest India aquifer, the North China Plain aquifer, the Murray-Darling Basin in Australia, the High Plains, and the California Central Valley aquifers in the United States of America (USA), but were also used to estimate groundwater-related parameters such as the specific yield, which relates groundwater level to storage, or to define the indices of groundwater depletion and stress. In this review, the approaches used for estimating groundwater storage variations are presented along with the main applications of GRACE data for groundwater monitoring. Issues that were related to the use of GRACE-based TWS are also addressed.

1185. Tide gauge records reveal improved processing of gravity recovery and climate experiment time-variable mass solutions over the coastal ocean

Authors: Christopher G Piecuch (Atmospheric and Environmental Research Inc.); Felix W Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rui M Ponte (Atmospheric and Environmental Research Inc.)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggy207

URL: <https://doi.org/10.1093/gji/ggy207>

Publication Date: 2018-05-19

Volume/Issue: Volume 214, Issue 2

References: 62

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1186. Improved geophysical excitation of length-of-day constrained by Earth orientation parameters and satellite gravimetry products

Authors: Nan Yu (Wuhan University, Helmholtz Centre Potsdam); Jiancheng Li (Wuhan University); Jim Ray (National Oceanic and Atmospheric Administration, Silver Spring, MD 20910, USA); Wei Chen (Wuhan University)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggy204

URL: <https://doi.org/10.1093/gji/ggy204>

Publication Date: 2018-05-18

Volume/Issue: Volume 214, Issue 3

References: 85

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National key research and development program of China

License: N/A

Open Access: N/A

Abstract: N/A

1187. Inverse Cascades of Kinetic Energy as a Source of Intrinsic Variability: A Global OGCM Study

Authors: Guillaume S  razin (Institut de Recherche pour le D  veloppement (IRD)); Thierry Penduff (Institut de Recherche pour le D  veloppement (IRD)); Bernard Barnier (Institut de Recherche pour le D  veloppement (IRD)); Jean-Marc Molines (Institut de Recherche pour le D  veloppement (IRD)); Brian K. Arbic (University of Michigan); Malte M  ller (Norwegian Meteorological Institute); Laurent Terray (Sciences de l'Univers au CERFACS, CERFACS/CNRS, URA1857, Toulouse, France)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-17-0136.1

URL: <https://doi.org/10.1175/jpo-d-17-0136.1>

Publication Date: 2018-05-17

Volume/Issue: Volume 48, Issue 6

References: 84

Cited by: 45 publications

Language: N/A

Subjects: N/A

Keywords: Interannual variability, Sea level, Mesoscale processes, Turbulence, Rossby waves (and 1 more)

Funding: Centre National d'Etudes Spatiales, Agence Nationale de la Recherche

License: N/A

Open Access: N/A

Abstract: A seasonally forced 1/12 degrees global ocean/sea ice simulation is used to characterize the spatiotemporal inverse cascade of kinetic energy (KE). Nonlinear scale interactions associated with relative vorticity advection are evaluated using cross-spectral analysis in the frequency-wavenumber domain from sea level anomaly (SLA) time series. This analysis is applied within four eddy-active midlatitude regions having large intrinsic variability spread over a wide range of scales. Over these four regions, mesoscale surface KE is shown to spontaneously cascade toward larger spatial scales--between the deformation scale and the Rhines scale--and longer time scales (possibly exceeding 10 years). Other nonlinear processes might have to be invoked to explain the longer time scales of intrinsic variability, which have a substantial surface imprint at midlatitudes. The analysis of a fully forced 1/12 degrees hindcast shows that low-frequency and synoptic atmospheric forcing barely affects this inverse KE cascade. The inverse cascade is also at work in a 1/4 degrees simulation, albeit with a weaker intensity, consistent with the weaker intrinsic variability found at this coarser resolution. In the midlatitude North Pacific, the spatiotemporal cascade transfers KE from high-frequency frontal Rossby waves (FRWs), probably generated by baroclinic instability, toward the lower-frequency, westward-propagating mesoscale eddy (WME) field. The WMEs provide local gradients of potential vorticity that support these short Doppler-shifted FRWs. FRWs have periods shorter than 2 months and might be subsampled by altimetric observations, perhaps explaining why the temporal inverse cascade deduced from high-resolution models and mapped altimeter products can be quite different. The nature of the nonlinear interactions between FRWs and WMEs remains unclear but might involve wave turbulence processes.

1188. An assessment of ten ocean reanalyses in the polar regions

Authors: Petteri Uotila; Hugues Goosse; Keith Haines; Matthieu Chevallier; Antoine Barthélemy; Clément Bricaud; Jim Carton; Neven Fučkar; Gilles Garric; Doroteaciro Iovino; Frank Kauker; Meri Korhonen; Vidar S. Lien; Marika Marnela; François Massonnet; Davi Mignac; K. Andrew Peterson; Remon Sadikni; Li Shi; Steffen Tietsche; Takahiro Toyoda; Jiping Xie; Zhaoru Zhang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-018-4242-z

URL: <https://doi.org/10.1007/s00382-018-4242-z>

Publication Date: 2018-05-17

Volume/Issue: Volume 52, Issue 3-4

References: 112

Cited by: 94 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: H2020 Marie Skłodowska-Curie Actions

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1189. Impacts of extratropical storm tracks on Arctic sea ice export through Fram Strait

Authors: Jianfen Wei; Xiangdong Zhang; Zhaomin Wang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-018-4254-8

URL: <https://doi.org/10.1007/s00382-018-4254-8>

Publication Date: 2018-05-16

Volume/Issue: Volume 52, Issue 3-4

References: 57

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Science and Technology of the People's Republic of China, Research Innovation Program for College Graduates of Jiangsu Province, the Fundamental Research Funds for the Central Universities

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1190. Abrupt Transitions in Submesoscale Structure in Southern Drake Passage: Glider Observations and Model Results

Authors: Giuliana A. Viglione (California Institute of Technology); Andrew F. Thompson (California Institute of Technology); M. Mar Flexas (California Institute of Technology); Janet Sprintall (Scripps Institution of Oceanography); Sebastiaan Swart (University of Cape Town, University of Gothenburg)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-17-0192.1

URL: <https://doi.org/10.1175/jpo-d-17-0192.1>

Publication Date: 2018-05-10

Volume/Issue: Volume 48, Issue 9

References: 72

Cited by: 42 publications

Language: N/A

Subjects: N/A

Keywords: Antarctica, Potential vorticity, Instability, Mixed layer, Oceanic mixed layer (and 1 more)

Funding: National Science Foundation, National Science Foundation, National Research Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: Enhanced vertical velocities associated with submesoscale motions may rapidly modify mixed layer depths and increase exchange between the mixed layer and the ocean interior. These dynamics are of particular importance in the Southern Ocean, where the ventilation of many density classes occurs. Here we present results from an observational field program in southern Drake Passage, a region preconditioned for submesoscale instability owing to its strong mesoscale eddy field, persistent fronts, strong down-front winds, and weak vertical stratification. Two gliders sampled from December 2014 through March 2015 upstream and downstream of the Shackleton Fracture Zone (SFZ). The acquired time series of mixed layer depths and buoyancy gradients enabled calculations of potential vorticity and classifications of submesoscale instabilities. The regions flanking the SFZ displayed remarkably different characteristics despite similar surface forcing. Mixed layer depths were nearly twice as deep, and horizontal buoyancy gradients were larger downstream of the SFZ. Upstream of the SFZ, submesoscale variability was confined to the edges of topographically steered fronts, whereas downstream these motions were more broadly distributed. Comparisons to a one-dimensional (1D) mixing model demonstrate the role of submesoscale instabilities in generating mixed layer variance. Numerical output from a submesoscale-resolving simulation indicates that submesoscale instabilities are crucial for correctly reproducing upper-ocean stratification. These results show that bathymetry can play a key role in generating dynamically distinct submesoscale characteristics over short spatial scales and that submesoscale motions can be locally active during summer months.

1191. Formation and Maintenance of the GEOTRACES Subsurface-Dissolved Iron Maxima in an Ocean Biogeochemistry Model

Authors: Anh L. D. Pham (Georgia Institute of Technology); Takamitsu Ito (Georgia Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2017gb005852

URL: <https://doi.org/10.1029/2017gb005852>

Publication Date: 2018-05-10

Volume/Issue: Volume 32, Issue 6

References: 87

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Recent GEOTRACES transects revealed basin-scale patterns of dissolved iron in the global oceans, providing a unique opportunity to test numerical models and to improve our understanding of the iron cycling. Subsurface maxima of dissolved iron in the upper ocean thermocline are observed in various transects, which can play an important role in regulating marine productivity due to their proximity to the surface euphotic layer. An ocean biogeochemistry model with refined parameterizations of iron cycling is used to examine the mechanisms controlling the formation and maintenance of these subsurface maxima. The model includes the representation of three iron sources including dust deposition, continental shelves, and hydrothermal vents. Two classes of organic ligands are parameterized based on the dissolved organic matter and apparent oxygen utilization. Parameterizations of particle-dependent scavenging and desorption are included. Although the model still struggles in fully capturing the observed dissolved iron distribution, it starts reproducing some major features, especially in the main thermocline. A suite of numerical sensitivity experiments suggests that the release of scavenged iron associated with sinking organic particles forms the subsurface-dissolved iron maxima in high-dust regions of the Indian and Atlantic Oceans. In low-dust regions of the Pacific basin, the subsurface-dissolved iron extrema are sustained by inputs from the continental shelves or hydrothermal vents. In all cases, subsurface ligands produced by the remineralization of organic particles retain the dissolved iron and play a central role in the maintenance of the subsurface maxima in our model. Thus, the parameterization of subsurface ligands has a far-reaching impact on the representation of global iron cycling and biological productivity in ocean biogeochemistry models.

1192. Impact of Indian ocean dipole on the coastal upwelling features off the southwest coast of India

Authors: Tanuja Nigam; Vimlesh Pant; Kumar Ravi Prakash

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-018-1152-x

URL: <https://doi.org/10.1007/s10236-018-1152-x>

Publication Date: 2018-05-09

Volume/Issue: Volume 68, Issue 6

References: 64

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1193. 3D Geostrophy and Volume Transport in the Southern Ocean

Authors: María Isabel Vigo (University of Alicante); David García-García (University of Alicante); María Dolores Sempere (University of Alicante); Ben F. Chao (Academia Sinica)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs10050715

URL: <https://doi.org/10.3390/rs10050715>

Publication Date: 2018-05-07

Volume/Issue: Volume 10, Issue 5

References: 43

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The 3D geostrophic currents and the associated volume transport (VT) can be estimated from the GOCE and Altimetry satellite data and in-situ temperature and salinity profiles measured by the Argo floats. We do so for the Southern Ocean between 20 degreesS and 65 degreesS with their time variability down to the depth of 1975 m (in 58 layers) over the 11-year period of 2004-2014. The results depict the Southern Ocean circulation where a zonal Antarctic Circumpolar Current (ACC) interacts with a meridional thermohaline circulation. The VT reproduces the polar front and the subantarctic front of the ACC, as well as the large scale and mesoscale currents in the Southern Ocean. Our estimates for the Agulhas current and the East Australia currents are also quantitatively comparable with results from other approaches in the literature based on in-situ data. For ACC, the estimated VT at the Drake Passage is 185 Sv for the norm of the time average VT, or 202 Sv for the mean of the norms of the monthly VT, which are larger than previous estimations (ranging from 134 to 175 Sv). The estimate is potentially reconciled when only the zonal transport is considered (181 Sv). The Drake Passage total VT appears to be quite stable during the studied period, unlike its (dominant) zonal and meridional components which show higher variability that mostly compensate each other. The spatially averaged ACC VT shows per 1 degrees width in the main stream a mean value of 29.6 Sv or 35.8 Sv (depending on the method used), an annual signal with an amplitude of 0.33 ± 0.06 Sv that peaks in early April, with no significant semi-annual signals nor linear trend. Water transports of barotropic and baroclinic origin have been isolated in the VT series showing that 75% of transport is barotropic and the remaining 25% baroclinic, while the variability and annual signal in the ACC is fully barotropic.

1194. Process-Specific Contributions to Anomalous Java Mixed Layer Cooling During Positive IOD Events

Authors: Andrew S. Delman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Scripps Institution of Oceanography / University of California San Diego); Julie L. McClean (Scripps Institution of Oceanography / University of California San Diego); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego); Lynne D. Talley (Scripps Institution of Oceanography / University of California San Diego); Frank O. Bryan (National Center for Atmospheric Research)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2017jc013749

URL: <https://doi.org/10.1029/2017jc013749>

Publication Date: 2018-05-04

Volume/Issue: Volume 123, Issue 6

References: 52

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, U.S. Department of Energy, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: Negative sea surface temperature (SST) anomalies associated with positive Indian Ocean Dipole (pIOD) events first appear in the seasonal upwelling zone along the southern coast of Java during May-July. The evolution of anomalous SSTs in this coastal region is analyzed by computing a temperature budget using output from a strongly eddy-active ocean general circulation model. The seasonal cooling south of Java in May-July is driven by a reduction in incoming shortwave radiation and by vertical mixing, consistent with earlier studies in the region; however, the model budget also shows an advective contribution that drives anomalous cooling at the onset of pIOD events. To identify which process(es) are responsible for the anomalous advective cooling during pIOD events, a novel process index regression method is used to estimate the contributions of wind stress, equatorial Kelvin waves, mesoscale eddies, and Lombok Strait flow to anomalous cooling south of Java. Using this method, wind stress forcing along the west coast of Sumatra is found to make the most substantial contribution to anomalous cooling south of Java, with lesser contributions from equatorially sourced Kelvin waves and local wind stress. Mesoscale eddies redistribute heat from the Lombok Strait outflow, and have an anomalous warming effect on the eastern side of the upwelling region. The process-specific temperature budget south of Java highlights the importance of wind stress forcing west of Sumatra relative to equatorial and local forcing, and explains most of the mixed layer temperature anomaly evolution associated with advection during pIOD events.

1195. A novel approach to assess the hydrodynamic effects of a salmon farm in a Patagonian channel: Coupling between regional ocean modeling and high resolution les simulation

Authors: Julio Herrera; Pablo Cornejo; Héctor Hito Sepúlveda; Osvaldo Artal; Renato A. Quiñones

Journal: Aquaculture

Publisher: Elsevier BV

DOI: 10.1016/j.aquaculture.2018.05.003

URL: <https://doi.org/10.1016/j.aquaculture.2018.05.003>

Publication Date: 2018-05-03

Volume/Issue: Volume 495

References: 55

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Interdisciplinary Center for Aquaculture Research, CONICYT

License: N/A

Open Access: N/A

Abstract: N/A

1196. Seasonal and Interannual Variations of Heat Fluxes in the Barents Sea Region

Authors: I. L. Bashmachnikov; A. Yu. Yurova; L. P. Bobylev; A. V. Vesman

Journal: Izvestiya, Atmospheric and Oceanic Physics

Publisher: Pleiades Publishing Ltd

DOI: 10.1134/s0001433818020032

URL: <https://doi.org/10.1134/s0001433818020032>

Publication Date: 2018-04-27

Volume/Issue: Volume 54, Issue 2

References: 48

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1197. Groundwater Storage Changes in China from Satellite Gravity: An Overview

Authors: Wei Feng (Chinese Academy of Sciences); C. Shum (Ohio State University, Chinese Academy of Sciences); Min Zhong (Chinese Academy of Sciences); Yun Pan (State Key Laboratory of Urban Environmental Processes and Digital Modeling, Capital Normal University, Beijing 100048, China)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs10050674

URL: <https://doi.org/10.3390/rs10050674>

Publication Date: 2018-04-27

Volume/Issue: Volume 10, Issue 5

References: 117

Cited by: 165 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Strategic Priority Research Program of the Chinese Academy of Sciences

License: CC BY 4.0

Open Access: N/A

Abstract: Groundwater plays a critical role in the global water cycle and is the drinking source for almost half of the world's population. However, exact quantification of its storage change remains elusive due primarily to limited ground observations in space and time. The Gravity Recovery and Climate Experiment (GRACE) twin-satellite data have provided global observations of water storage variations at monthly sampling for over a decade and a half, and is enable to estimate changes in groundwater storage (GWS) after removing other water storage components using auxiliary datasets and models. In this paper, we present an overview of GWS changes in three main aquifers within China using GRACE data, and conduct a comprehensive accuracy assessment using in situ ground well observations and hydrological models. GRACE detects a significant GWS depletion rate of 7.2 ± 1.1 km³/yr in the North China Plain (NCP) during 2002-2014, consistent with ground well observations and model predictions. The Liaohe River Basin (LRB) experienced a pronounced GWS decline during 2005-2009, at a depletion rate of 5.0 ± 1.2 km³/yr. Since 2010, GRACE-based GWS reveal a slow recovery in the LRB, with excellent agreement with ground well observations. For the whole study period 2002-2014, no significant long-term GWS depletion is found in the LRB nor in the Tarim Basin. A case study in the Inner Tibetan Plateau highlights there still exist large uncertainties in GRACE-based GWS change estimates.

1198. A fault-tolerant HPC scheduler extension for large and operational ensemble data assimilation: Application to the Red Sea

Authors: Habib Teye; Samuel Kortas; Peng Zhan; Ibrahim Hoteit

Journal: Journal of Computational Science

Publisher: Elsevier BV

DOI: 10.1016/j.jocs.2018.04.018

URL: <https://doi.org/10.1016/j.jocs.2018.04.018>

Publication Date: 2018-04-26

Volume/Issue: Volume 27

References: 41

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: King Abdullah University of Science and Technology, Saudi ARAMCO

License: N/A

Open Access: N/A

Abstract: N/A

1199. Twentieth Century Seawater delta18O Dynamics and Implications for Coral-Based Climate Reconstruction

Authors: S. Stevenson (University of California Santa Barbara); B. Powell (Department of Oceanography University of Hawai'i Honolulu HI USA); K. M. Cobb (Georgia Institute of Technology); J. Nusbaumer (National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); M. Merrifield (Scripps Institution of Oceanography / University of California San Diego); D. Noone (Oregon State University)

Journal: Paleoceanography and Paleoclimatology

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2017pa003304

URL: <https://doi.org/10.1029/2017pa003304>

Publication Date: 2018-04-21

Volume/Issue: Volume 33, Issue 6

References: 72

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: The oxygen isotopic composition of tropical coral skeletons (delta18O) is a crucial source of information on past El Niño/Southern Oscillation behavior. Both temperature and the delta18O of the surrounding seawater (delta18Osw) affect coral delta18O; a linear proportionality with delta18Osw is often used to infer past salinity variations, but the degree to which dynamical influences on delta18Osw may affect that relationship is still unclear. Here we use the isotope-enabled Regional Ocean Modeling System to investigate the dynamics of delta18Osw and salinity variations in different twentieth century climate regimes. The dominant modes of delta18Osw variability are the background trend and eastern/central Pacific El Niño, similar to salinity; likewise, budget analysis reveals a strong impact of ocean dynamics (both advection and vertical mixing/diffusion) on delta18Osw variations during El Niño and La Niña events. These dynamics lead to alterations in the delta18Osw:salinity relationship across the Pacific: the linear approximation is most accurate near the eastern edge of the western Pacific warm pool but with nonlinearities due to large delta18Osw excursions during El Niño. At other sites, the delta18Osw:salinity relationship has substantial scatter and explains relatively little delta18Osw variance. Strikingly, interannual variability can lead to site-specific delta18Osw:salinity regression coefficient changes of up to 0.4‰/psu on multiyear time scales. This calls the reliability of coral-based estimates of El Niño/Southern Oscillation extremes into question and highlights the need for increased dynamical understanding as well as sustained in situ observations of delta18Osw and salinity variability.

1200. Going with the flow: a graph based approach to optimal path planning in general flows

Authors: Dhanushka Kularatne; Subhrajit Bhattacharya; M. Ani Hsieh

Journal: Autonomous Robots

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10514-018-9741-6

URL: <https://doi.org/10.1007/s10514-018-9741-6>

Publication Date: 2018-04-21

Volume/Issue: Volume 42, Issue 7

References: 27

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1201. Predictability of Arctic sea ice on weather time scales

Authors: M. Mohammadi-Aragh; H. F. Goessling; M. Losch; N. Hutter; T. Jung

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-018-24660-0

URL: <https://doi.org/10.1038/s41598-018-24660-0>

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References: 43

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The field of Arctic sea ice prediction on "weather time scales" is still in its infancy with little existing understanding of the limits of predictability. This is especially true for sea ice deformation along so-called Linear Kinematic Features (LKFs) including leads that are relevant for marine operations. Here the potential predictability of the sea ice pack in the wintertime Arctic up to ten days ahead is determined, exploiting the fact that sea ice-ocean models start to show skill at representing sea ice deformation at high spatial resolutions. Results are based on ensemble simulations with a high-resolution sea ice-ocean model driven by atmospheric ensemble forecasts. The predictability of LKFs as measured by different metrics drops quickly, with predictability being almost completely lost after 4-8 days. In contrast, quantities such as sea ice concentration or the location of the ice edge retain high levels of predictability throughout the full 10-day forecast period. It is argued that the rapid error growth for LKFs is mainly due to the chaotic behaviour of the atmosphere associated with the low predictability of near surface wind divergence and vorticity; initial condition uncertainty for ice thickness is found to be of minor importance as long as LKFs are initialized at the right locations.

1202. Interannual drivers of the seasonal cycle of CO₂ in the Southern Ocean

Authors: Luke Gregor; Schalk Kok; Pedro M. S. Monteiro

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-15-2361-2018

URL: <https://doi.org/10.5194/bg-15-2361-2018>

Publication Date: 2018-04-19

Volume/Issue: Volume 15, Issue 8

References: 74

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Resolving and understanding the drivers of variability of CO₂ in the Southern Ocean and its potential climate feedback is one of the major scientific challenges of the ocean-climate community. Here we use a regional approach on empirical estimates of pCO₂ to understand the role that seasonal variability has in long-term CO₂ changes in the Southern Ocean. Machine learning has become the preferred empirical modelling tool to interpolate time- and location-restricted ship measurements of pCO₂. In this study we use an ensemble of three machine-learning products: support vector regression (SVR) and random forest regression (RFR) from Gregor et al. (2017), and the self-organising-map feed-forward neural network (SOM-FFN) method from Landschützer et al. (2016). The interpolated estimates of Δ pCO₂ are separated into nine regions in the Southern Ocean defined by basin (Indian, Pacific, and Atlantic) and biomes (as defined by Fay and McKinley, 2014a). The regional approach shows that, while there is good agreement in the overall trend of the products, there are periods and regions where the confidence in estimated Δ pCO₂ is low due to disagreement between the products. The regional breakdown of the data highlighted the seasonal decoupling of the modes for summer and winter interannual variability. Winter interannual variability had a longer mode of variability compared to summer, which varied on a 4-6-year timescale. We separate the analysis of the Δ pCO₂ and its drivers into summer and winter. We find that understanding the variability of Δ pCO₂ and its drivers on shorter timescales is critical to resolving the long-term variability of Δ pCO₂. Results show that Δ pCO₂ is rarely driven by thermodynamics during winter, but rather by mixing and stratification due to the stronger correlation of Δ pCO₂ variability with mixed layer depth. Summer pCO₂ variability is consistent with chlorophyll a variability, where higher concentrations of chlorophyll a correspond with lower pCO₂ concentrations. In regions of low chlorophyll a concentrations, wind stress and sea surface temperature emerged as stronger drivers of Δ pCO₂. In summary we propose that sub-decadal variability is explained by summer drivers, while winter variability contributes to the long-term changes associated with the SAM. This approach is a useful framework to assess the drivers of Δ pCO₂ but would greatly benefit from improved estimates of Δ pCO₂ and a longer time series.

1203. A Dynamically Consistent, Multivariable Ocean Climatology

Authors: Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Patrick Heimbach (University of Texas at Austin); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Carl Wunsch (Harvard University, Massachusetts Institute of Technology)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/bams-d-17-0213.1

URL: <https://doi.org/10.1175/bams-d-17-0213.1>

Publication Date: 2018-04-18

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References: 68

Cited by: 27 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A dynamically consistent 20-yr average ocean climatology based on monthly values during the years 1994-2013 has been produced from the most recent state estimate of the Estimating the Circulation and Climate of the Ocean (ECCO) project, globally, top to bottom. The estimate was produced from a least squares fit of a free-running ocean general circulation model to almost all available near-global data. Data coverage in space and time during this period is far more homogeneous than in any earlier interval and includes CTD, elephant seal, and Argo temperature and salinity profiles; sea ice coverage; full altimetric and gravity-field coverage; satellite sea surface temperatures; and the initializing meteorological coverage from the European Centre for Medium-Range Weather Forecasts (ECMWF) interim reanalysis (ERA-Interim). Dominant remaining data inhomogeneity arises from increasing coverage from the Argo profiles from about 2000 to the present. The state estimate exactly satisfies the primitive equations of the free-running Massachusetts Institute of Technology General Circulation Model (MITgcm) at all times and hence produces values satisfying the fundamental conservation laws of energy, freshwater, and so forth, permitting its use for climate change studies. Quantities such as calculated heat content depend upon all observations, not just temperature, for example, altimetric height and meteorological exchanges. Output files are publicly available in Network Common Data Form (netCDF) and MATLAB form and include hydrographic variables, three components of velocity, and pressure at all depths, as well as other variables, including inferred air-sea momentum and buoyancy fluxes, 3D mixing parameters, and sea ice cover.

1204. Signature of Indian Ocean Dipole on the western boundary current of the Bay of Bengal

Authors: V.R. Sherin; F. Durand; V.V. Gopalkrishna; S. Anuvinda; A.V.S. Chaitanya; R. Bourdallé-Badie; F. Papa

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2018.04.002

URL: <https://doi.org/10.1016/j.dsr.2018.04.002>

Publication Date: 2018-04-11

Volume/Issue: Volume 136

References: 64

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: INCOIS, CNES, French Space Agency (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1205. Quantifying predictability of sea ice around the Indian Antarctic stations using coupled ocean sea ice model with shelf ice

Authors: Anurag Kumar; Suneet Dwivedi; Avinash C. Pandey

Journal: Polar Science

Publisher: Elsevier BV

DOI: 10.1016/j.polar.2018.04.003

URL: <https://doi.org/10.1016/j.polar.2018.04.003>

Publication Date: 2018-04-10

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References: 82

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: ISRO/DST/NCAOR

License: N/A

Open Access: N/A

Abstract: N/A

1206. Intercomparison of Antarctic ice-shelf, ocean, and sea-ice interactions simulated by MetROMS-iceshelf and FESOM 1.4

Authors: Kaitlin A. Naughten; Katrin J. Meissner; Benjamin K. Galton-Fenzi; Matthew H. England; Ralph Timmermann; Hartmut H. Hellmer; Tore Hattermann; Jens B. Debernard

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-11-1257-2018

URL: <https://doi.org/10.5194/gmd-11-1257-2018>

Publication Date: 2018-04-10

Volume/Issue: Volume 11, Issue 4

References: 181

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . An increasing number of Southern Ocean models now include Antarctic ice-shelf cavities, and simulate thermodynamics at the ice-shelf/ocean interface. This adds another level of complexity to Southern Ocean simulations, as ice shelves interact directly with the ocean and indirectly with sea ice. Here, we present the first model intercomparison and evaluation of present-day ocean/sea-ice/ice-shelf interactions, as simulated by two models: a circumpolar Antarctic configuration of MetROMS (ROMS: Regional Ocean Modelling System coupled to CICE: Community Ice CodE) and the global model FESOM (Finite Element Sea-ice Ocean Model), where the latter is run at two different levels of horizontal resolution. From a circumpolar Antarctic perspective, we compare and evaluate simulated ice-shelf basal melting and sub-ice-shelf circulation, as well as sea-ice properties and Southern Ocean water mass characteristics as they influence the sub-ice-shelf processes. Despite their differing numerical methods, the two models produce broadly similar results and share similar biases in many cases. Both models reproduce many key features of observations but struggle to reproduce others, such as the high melt rates observed in the small warm-cavity ice shelves of the Amundsen and Bellingshausen seas. Several differences in model design show a particular influence on the simulations. For example, FESOM's greater topographic smoothing can alter the geometry of some ice-shelf cavities enough to affect their melt rates; this improves at higher resolution, since less smoothing is required. In the interior Southern Ocean, the vertical coordinate system affects the degree of water mass erosion due to spurious diapycnal mixing, with MetROMS' terrain-following coordinate leading to more erosion than FESOM's z coordinate. Finally, increased horizontal resolution in FESOM leads to higher basal melt rates for small ice shelves, through a combination of stronger circulation and small-scale intrusions of warm water from offshore.

1207. Future Projections of Antarctic Ice Shelf Melting Based on CMIP5 Scenarios

Authors: Kaitlin A. Naughten (Antarctic Climate and Ecosystems Cooperative Research Centre, University of New South Wales); Katrin J. Meissner (University of New South Wales); Benjamin K. Galton-Fenzi (Antarctic Climate and Ecosystems Cooperative Research Centre); Matthew H. England (University of New South Wales); Ralph Timmermann (Alfred Wegener Institut, Bremerhaven, Germany); Hartmut H. Hellmer (Alfred Wegener Institut, Bremerhaven, Germany)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0854.1

URL: <https://doi.org/10.1175/jcli-d-17-0854.1>

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References: 57

Cited by: 62 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Climate change, Southern Ocean, Ocean, Antarctica

Funding: Australian Postgraduate Award, Research Training Program Scheme, UNSW Research Excellence Award (and 4 more)

License: N/A

Open Access: N/A

Abstract: Basal melting of Antarctic ice shelves is expected to increase during the twenty-first century as the ocean warms, which will have consequences for ice sheet stability and global sea level rise. Here we present future projections of Antarctic ice shelf melting using the Finite Element Sea Ice/Ice-Shelf Ocean Model (FESOM) forced with atmospheric output from models from phase 5 of the Coupled Model Intercomparison Project (CMIP5). CMIP5 models are chosen based on their agreement with historical atmospheric reanalyses over the Southern Ocean; the best-performing models are ACCESS 1.0 and the CMIP5 multimodel mean. Their output is bias-corrected for the representative concentration pathway (RCP) 4.5 and 8.5 scenarios. During the twenty-first-century simulations, total ice shelf basal mass loss increases by between 41% and 129%. Every sector of Antarctica shows increased basal melting in every scenario, with the largest increases occurring in the Amundsen Sea. The main mechanism driving this melting is an increase in warm Circumpolar Deep Water on the Antarctic continental shelf. A reduction in wintertime sea ice formation simulated during the twenty-first century stratifies the water column, allowing a warm bottom layer to develop and intrude into ice shelf cavities. This effect may be overestimated in the Amundsen Sea because of a cold bias in the present-day simulation. Other consequences of weakened sea ice formation include freshening of High Salinity Shelf Water and warming of Antarctic Bottom Water. Furthermore, freshening around the Antarctic coast in our simulations causes the Antarctic Circumpolar Current to weaken and the Antarctic Coastal Current to strengthen.

1208. Local and Remote Influences on the Heat Content of the Labrador Sea: An Adjoint Sensitivity Study

Authors: Daniel C. Jones (British Antarctic Survey, Natural Environment Research Council Cambridge UK); Gael Forget (Massachusetts Institute of Technology); Bablu Sinha (National Oceanography Centre); Simon A. Josey (National Oceanography Centre); Emma J. D. Boland (British Antarctic Survey, Natural Environment Research Council Cambridge UK); Andrew J. S. Meijers (British Antarctic Survey, Natural Environment Research Council Cambridge UK); Emily Shuckburgh (British Antarctic Survey, Natural Environment Research Council Cambridge UK)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2018jc013774

URL: <https://doi.org/10.1002/2018jc013774>

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Volume/Issue: Volume 123, Issue 4

References: 43

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: Labrador Sea, adjoint sensitivity, heat content, teleconnection, heat flux (and 1 more)

Funding: NASA, Simons Foundation

License: N/A

Open Access: N/A

Abstract: The Labrador Sea is one of the few regions on the planet where the interior ocean can exchange heat directly with the atmosphere via strong, localized, wintertime convection, with possible implications for the state of North Atlantic climate and global surface warming. Using an observationally constrained ocean adjoint model, we find that annual-mean Labrador Sea heat content is sensitive to temperature/salinity changes (1) along potential source water pathways (e.g., the subpolar gyre, the North Atlantic Current, the Gulf Stream) and (2) along the West African and European shelves, which are not significant source water regions for the Labrador Sea. The West African coastal/shelf adjustment mechanism, which may be excited by changes in along-shelf wind stress, involves pressure anomalies that propagate along a coastal waveguide toward Greenland, changing the across-shelf pressure gradient in the North Atlantic and altering heat convergence in the Labrador Sea. We also find that nonlocal (in space and time) heat fluxes (e.g., in the Irminger Sea, the seas south of Iceland) can have a strong impact on Labrador Sea heat content. Understanding and predicting the state of the Labrador Sea and its potential impacts on North Atlantic climate and global surface warming will require monitoring of oceanic and atmospheric properties at remote sites in the Irminger Sea, the subpolar gyre, and along the West African and European shelf/coast system, among others.

1209. WOCE-Argo Global Hydrographic Climatology

Authors: Viktor Gouretski

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/os-2018-34

URL: <https://doi.org/10.5194/os-2018-34>

Publication Date: 2018-04-06

Volume/Issue: N/A

References: N/A

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Argo|Anomaly (physics)|Temperature salinity diagrams

Funding: N/A

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Open Access: N/A

Abstract: . The paper describes the new gridded WOCE-Argo Global Hydrographic Climatology (WAGHC) (Gouretski, 2018). The climatology has a one-fourth degree spatial resolution resolving the annual cycle of temperature and salinity on a monthly basis. Two versions of the climatology were produced differing by the spatial interpolation performed on isobaric or isopycnal surfaces respectively. The WAGHC climatology is based on the quality controlled temperature and salinity profiles obtained before January 2016 with the average climatological year being in the range 2008 to 2012. To avoid biases due to the significant step-like decrease of the data below 2 km the profile extrapolation procedure is implemented. We compare the WAGHC climatology to the one-fourth degree resolution isobarically averaged climatology WOA13, produced by the NOAA Ocean Climate Laboratory (Boyer et al., 2013) and diagnose a generally good agreement between these two gridded products. The differences between the two climatologies are attributed basically to the interpolation method and to the considerably extended data basis. Specifically, the WAGHC climatology improved the representation of the thermohaline structure both in the data poor polar regions and in several data abundant regions like the Baltic sea, Caspian sea, Gulf of California, Caribbean Sea, and the Weddell Sea. Further, the dependence of the ocean heat content anomaly (OHCA) time series on the baseline climatology was tested. Since the 1950s, the both baseline climatologies produce almost identical OHCA time series.

1210. Ecological control of nitrite in the upper ocean

Authors: Emily J. Zakem; Alia Al-Haj; Matthew J. Church; Gert L. van Dijken; Stephanie Dutkiewicz; Sarah Q. Foster; Robinson W. Fulweiler; Matthew M. Mills; Michael J. Follows

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-018-03553-w

URL: <https://doi.org/10.1038/s41467-018-03553-w>

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References: 86

Cited by: 125 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Microorganisms oxidize organic nitrogen to nitrate in a series of steps. Nitrite, an intermediate product, accumulates at the base of the sunlit layer in the subtropical ocean, forming a primary nitrite maximum, but can accumulate throughout the sunlit layer at higher latitudes. We model nitrifying chemoautotrophs in a marine ecosystem and demonstrate that microbial community interactions can explain the nitrite distributions. Our theoretical framework proposes that nitrite can accumulate to a higher concentration than ammonium because of differences in underlying redox chemistry and cell size between ammonia- and nitrite-oxidizing chemoautotrophs. Using ocean circulation models, we demonstrate that nitrifying microorganisms are excluded in the sunlit layer when phytoplankton are nitrogen-limited, but thrive at depth when phytoplankton become light-limited, resulting in nitrite accumulation there. However, nitrifying microorganisms may coexist in the sunlit layer when phytoplankton are iron- or light-limited (often in higher latitudes). These results improve understanding of the controls on nitrification, and provide a framework for representing chemoautotrophs and their biogeochemical effects in ocean models.

1211. Estimation of oceanic subsurface mixing under a severe cyclonic storm using a coupled atmosphere-ocean-wave model

Authors: Kumar Ravi Prakash; Tanuja Nigam; Vimlesh Pant

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-14-259-2018

URL: <https://doi.org/10.5194/os-14-259-2018>

Publication Date: 2018-04-03

Volume/Issue: Volume 14, Issue 2

References: 78

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . A coupled atmosphere-ocean-wave model was used to examine mixing in the upper-oceanic layers under the influence of a very severe cyclonic storm Phailin over the Bay of Bengal (BoB) during 10-14 October 2013. The coupled model was found to improve the sea surface temperature over the uncoupled model. Model simulations highlight the prominent role of cyclone-induced near-inertial oscillations in subsurface mixing up to the thermocline depth. The inertial mixing introduced by the cyclone played a central role in the deepening of the thermocline and mixed layer depth by 40 and 15 m, respectively. For the first time over the BoB, a detailed analysis of inertial oscillation kinetic energy generation, propagation, and dissipation was carried out using an atmosphere-ocean-wave coupled model during a cyclone. A quantitative estimate of kinetic energy in the oceanic water column, its propagation, and its dissipation mechanisms were explained using the coupled atmosphere-ocean-wave model. The large shear generated by the inertial oscillations was found to overcome the stratification and initiate mixing at the base of the mixed layer. Greater mixing was found at the depths where the eddy kinetic diffusivity was large. The baroclinic current, holding a larger fraction of kinetic energy than the barotropic current, weakened rapidly after the passage of the cyclone. The shear induced by inertial oscillations was found to decrease rapidly with increasing depth below the thermocline. The dampening of the mixing process below the thermocline was explained through the enhanced dissipation rate of turbulent kinetic energy upon approaching the thermocline layer. The wave-current interaction and nonlinear wave-wave interaction were found to affect the process of downward mixing and cause the dissipation of inertial oscillations.

1212. Comparison between geodetic and oceanographic approaches to estimate mean dynamic topography for vertical datum unification: evaluation at Australian tide gauges

Authors: M. S. Filmer; C. W. Hughes; P. L. Woodworth; W. E. Featherstone; R. J. Bingham

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-018-1131-5

URL: <https://doi.org/10.1007/s00190-018-1131-5>

Publication Date: 2018-04-02

Volume/Issue: Volume 92, Issue 12

References: 123

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency

License: N/A

Open Access: N/A

Abstract: N/A

1213. Representing grounding line migration in synchronous coupling between a marine ice sheet model and a z-coordinate ocean model

Authors: D.N. Goldberg; K. Snow; P. Holland; J.R. Jordan; J.-M. Campin; P. Heimbach; R. Arthern; A. Jenkins

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2018.03.005

URL: <https://doi.org/10.1016/j.ocemod.2018.03.005>

Publication Date: 2018-03-29

Volume/Issue: Volume 125

References: 58

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NERC, NSF

License: N/A

Open Access: N/A

Abstract: N/A

1214. The deforming and rotating Earth - A review of the 18th International Symposium on Geodynamics and Earth Tide, Trieste 2016

Authors: Carla Braitenberg

Journal: Geodesy and Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.geog.2018.03.003

URL: <https://doi.org/10.1016/j.geog.2018.03.003>

Publication Date: 2018-03-29

Volume/Issue: Volume 9, Issue 3

References: 119

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: IAG: Commission 3, IAG Sub-commission 3.1, International Geodynamics and Earth Tide Service (and 8 more)

License: N/A

Open Access: N/A

Abstract: N/A

1215. Uncertainty characterization of HOAPS 3.3 latent heat-flux-related parameters

Authors: Julian Liman; Marc Schröder; Karsten Fennig; Axel Andersson; Rainer Hollmann

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

DOI: 10.5194/amt-11-1793-2018

URL: <https://doi.org/10.5194/amt-11-1793-2018>

Publication Date: 2018-03-29

Volume/Issue: Volume 11, Issue 3

References: 87

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Latent heat flux (LHF) is one of the main contributors to the global energy budget. As the density of in situ LHF measurements over the global oceans is generally poor, the potential of remotely sensed LHF for meteorological applications is enormous. However, to date none of the available satellite products have included estimates of systematic, random, and sampling uncertainties, all of which are essential for assessing their quality. Here, the challenge is taken on by matching LHF-related pixel-level data of the Hamburg Ocean Atmosphere Parameters and Fluxes from Satellite (HOAPS) climatology (version 3.3) to in situ measurements originating from a high-quality data archive of buoys and selected ships. Assuming the ground reference to be bias-free, this allows for deriving instantaneous systematic uncertainties as a function of four atmospheric predictor variables. The approach is regionally independent and therefore overcomes the issue of sparse in situ data densities over large oceanic areas. Likewise, random uncertainties are derived, which include not only a retrieval component but also contributions from in situ measurement noise and the collocation procedure. A recently published random uncertainty decomposition approach is applied to isolate the random retrieval uncertainty of all LHF-related HOAPS parameters. It makes use of two combinations of independent data triplets of both satellite and in situ data, which are analysed in terms of their pairwise variances of differences. Instantaneous uncertainties are finally aggregated, allowing for uncertainty characterizations on monthly to multi-annual timescales. Results show that systematic LHF uncertainties range between 15 and 50 W m⁻² with a global mean of 25 W m⁻². Local maxima are mainly found over the subtropical ocean basins as well as along the western boundary currents. Investigations indicate that contributions from q_a (U) to the overall LHF uncertainty are on the order of 60 % (25 %). From an instantaneous point of view, random retrieval uncertainties are specifically large over the subtropics with a global average of 37 W m⁻². In a climatological sense, their magnitudes become negligible, as do respective sampling uncertainties. Regional and seasonal analyses suggest that largest total LHF uncertainties are seen over the Gulf Stream and the Indian monsoon region during boreal winter. In light of the uncertainty measures, the observed continuous global mean LHF increase up to 2009 needs to be treated with caution. The demonstrated approach can easily be transferred to

other satellite retrievals, which increases the significance of the present work.

1216. Mechanisms Controlling Global Mean Sea Surface Temperature Determined From a State Estimate

Authors: R. M. Ponte (Atmospheric and Environmental Research Inc.); C. G. Piecuch (Atmospheric and Environmental Research Inc., Woods Hole Oceanographic Institution)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl076821

URL: <https://doi.org/10.1002/2017gl076821>

Publication Date: 2018-03-25

Volume/Issue: Volume 45, Issue 7

References: 21

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: sea surface temperature

Funding: National Sleep Foundation

License: N/A

Open Access: N/A

Abstract: Global mean sea surface temperature (T_{SST}) is a variable of primary interest in studies of climate variability and change. The temporal evolution of T_{SST} can be influenced by surface heat fluxes (Q_{net}) and by diffusion (K_{diff}) and advection (U_{adv}) processes internal to the ocean, but quantifying the contribution of these different factors from data alone is prone to substantial uncertainties. Here we derive a closed budget for the period 1993-2015 based on a global ocean state estimate, which is an exact solution of a general circulation model constrained to most extant ocean observations through advanced optimization methods. The estimated average temperature of the top (10-m thick) level in the model, taken to represent T_{SST} , shows relatively small variability at most time scales compared to Q_{net} , K_{diff} , or U_{adv} , reflecting the tendency for largely balancing effects from all the latter terms. The seasonal cycle in T_{SST} is mostly determined by small imbalances between Q_{net} and K_{diff} , with negligible contributions from U_{adv} . While Q_{net} seems to simply damp at the annual period, a different dynamical role for U_{adv} at semiannual period is suggested by it being larger than K_{diff} . At periods longer than annual, U_{adv} contributes importantly to variability, pointing to the direct influence of the variable ocean circulation on T_{SST} and mean surface climate.

1217. Improved Global Net Surface Heat Flux

Authors: James A. Carton (University of Maryland); Gennady A. Chepurin (University of Maryland); Ligan Chen (University of Maryland); Semyon A. Grodsky (University of Maryland)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013137

URL: <https://doi.org/10.1002/2017jc013137>

Publication Date: 2018-03-23

Volume/Issue: Volume 123, Issue 5

References: 44

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: Surface heat flux estimates from widely used atmospheric reanalyses differ locally by 10-30 W m⁻² even in time mean. Here a method is presented to help identify the errors causing these differences and to reduce these errors by exploiting hydrographic observations and the resulting temperature increments produced by an ocean reanalysis. The method is applied to improve the climatological monthly net surface heat fluxes from three atmospheric reanalyses: MERRA-2, ERA-Interim, and JRA-55, during an 8 year test period 2007-2014. The results show that the time mean error, as evaluated by consistency with the ocean heat budget, is reduced to less than +/-5 W m⁻² over much of the subtropical and midlatitude ocean. For the global ocean, after all the corrections have been made, the 8 year mean global net surface heat imbalance has been reduced to 3.4 W m⁻². A method is also presented to quantify the uncertainty in the heat flux estimates by repeating the procedure with many different atmospheric reanalyses and then examining the resulting spread in estimates. This reevaluation of net surface flux reveals, among other results, that the Southern Ocean is a source of heat to the atmosphere.

1218. Climate-driven seasonal geocenter motion during the GRACE period

Authors: Hongyue Zhang; Yu Sun

Journal: Acta Geophysica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11600-018-0130-5

URL: <https://doi.org/10.1007/s11600-018-0130-5>

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References: 50

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key Research and Development Program of China from MOST, Hainan Provincial Department of Science and Technology under Grant

License: N/A

Open Access: N/A

Abstract: N/A

1219. Atlantic-Pacific Asymmetry in Deep Water Formation

Authors: David Ferreira (University of Reading); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego); Helen K. Coxall (Stockholm University); Agatha de Boer (Stockholm University); Henk A. Dijkstra (Utrecht University); Sybren S. Drijfhout (University of Southampton, National Oceanography Centre); Tor Eldevik (University of Bergen, Bjerknes Centre for Climate Research); Nili Harnik (Tel Aviv University); Jerry F. McManus (Columbia University); David P. Marshall (University of Oxford); Johan Nilsson (Stockholm University); Fabien Roquet (Stockholm University); Tapio Schneider (California Institute of Technology); Robert C. Wills (University of Washington)

Journal: Annual Review of Earth and Planetary Sciences

Publisher: Annual Reviews

DOI: 10.1146/annurev-earth-082517-010045

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References: 152

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: While the Atlantic Ocean is ventilated by high-latitude deep water formation and exhibits a pole-to-pole overturning circulation, the Pacific Ocean does not. This asymmetric global overturning pattern has persisted for the past 2-3 million years, with evidence for different ventilation modes in the deeper past. In the current climate, the Atlantic-Pacific asymmetry occurs because the Atlantic is more saline, enabling deep convection. To what extent the salinity contrast between the two basins is dominated by atmospheric processes (larger net evaporation over the Atlantic) or oceanic processes (salinity transport into the Atlantic) remains an outstanding question. Numerical simulations have provided support for both mechanisms; observations of the present climate support a strong role for atmospheric processes as well as some modulation by oceanic processes. A major avenue for future work is the quantification of the various processes at play to identify which mechanisms are primary in different climate states.

1220. Transformation of Deep Water Masses Along Lagrangian Upwelling Pathways in the Southern Ocean

Authors: V. Tamsitt (Scripps Institution of Oceanography / University of California San Diego); R. P. Abernathey (Columbia University); M. R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); J. Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); L. D. Talley (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013409

URL: <https://doi.org/10.1002/2017jc013409>

Publication Date: 2018-03-15

Volume/Issue: Volume 123, Issue 3

References: 64

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF

License: N/A

Open Access: N/A

Abstract: Upwelling of northern deep waters in the Southern Ocean is fundamentally important for the closure of the global meridional overturning circulation and delivers carbon and nutrient-rich deep waters to the sea surface. We quantify water mass transformation along upwelling pathways originating in the Atlantic, Indian, and Pacific and ending at the surface of the Southern Ocean using Lagrangian trajectories in an eddy-permitting ocean state estimate. Recent related work shows that upwelling in the interior below about 400 m depth is localized at hot spots associated with major topographic features in the path of the Antarctic Circumpolar Current, while upwelling through the surface layer is more broadly distributed. In the ocean interior upwelling is largely isopycnal; Atlantic and to a lesser extent Indian Deep Waters cool and freshen while Pacific deep waters are more stable, leading to a homogenization of water mass properties. As upwelling water approaches the mixed layer, there is net strong transformation toward lighter densities due to mixing of freshwater, but there is a divergence in the density distribution as Upper Circumpolar Deep Water tends become lighter and dense Lower Circumpolar Deep Water tends to become denser. The spatial distribution of transformation shows more rapid transformation at eddy hot spots associated with major topography where density gradients are enhanced; however, the majority of cumulative density change along trajectories is achieved by background mixing. We compare the Lagrangian analysis to diagnosed Eulerian water mass transformation to attribute the mechanisms leading to the observed transformation.

1221. On the Use of Satellite Altimetry to Detect Ocean Circulation's Magnetic Signals

Authors: J. Saynisch (Helmholtz Centre Potsdam); C. Irrgang (Helmholtz Centre Potsdam); M. Thomas (Institute of Meteorology, Freie Universität Berlin Berlin Germany, Helmholtz Centre Potsdam)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013742

URL: <https://doi.org/10.1002/2017jc013742>

Publication Date: 2018-03-12

Volume/Issue: Volume 123, Issue 3

References: 50

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: oceanic induction, ocean heat budget, ocean currents, remote sensing, satellite altimetry

Funding: German Research Foundation

License: N/A

Open Access: N/A

Abstract: Oceanic magnetic signals are sensitive to ocean velocity, salinity, and heat content. The detection of respective signals with global satellite magnetometers would pose a very valuable source of information. While tidal magnetic fields are already detected, electromagnetic signals of the ocean circulation still remain unobserved from space. We propose to use satellite altimetry to construct proxy magnetic signals of the ocean circulation. These proxy time series could subsequently be fitted to satellite magnetometer data. The fitted data could be removed from the observations or the fitting constants could be analyzed for physical properties of the ocean, e.g., the heat budget. To test and evaluate this approach, synthetic true and proxy magnetic signals are derived from a global circulation model of the ocean. Both data sets are compared in dependence of location and time scale. We study and report when and where the proxy data describe the true signal sufficiently well. Correlations above 0.6 and explained variances of above 80% can be reported for large parts of the Antarctic ocean, thus explaining the major part of the global, subseasonal magnetic signal.

1222. Intercomparison of the Gulf Stream in ocean reanalyses: 1993-2010

Authors: Lequan Chi; Christopher L.P. Wolfe; Sultan Hameed

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2018.02.008

URL: <https://doi.org/10.1016/j.ocemod.2018.02.008>

Publication Date: 2018-03-06

Volume/Issue: Volume 125

References: 80

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1223. Linear Predictability: A Sea Surface Height Case Study

Authors: Maïke Sonnewald (Massachusetts Institute of Technology); Carl Wunsch (Harvard University, Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology, University of Texas at Austin)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0142.1

URL: <https://doi.org/10.1175/jcli-d-17-0142.1>

Publication Date: 2018-03-05

Volume/Issue: Volume 31, Issue 7

References: 38

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: Statistical forecasting, Ocean models, Sea level, Oceanic variability, Forecast verification/skill

Funding: Jet Propulsion Laboratory

License: N/A

Open Access: N/A

Abstract: A benchmark of linear predictability of sea surface height (SSH) globally is presented, complementing more complicated studies of SSH predictability. Twenty years of the Estimating the Circulation and Climate of the Ocean (ECCOv4) state estimate (1992-2011) are used, fitting autoregressive moving average [ARMA([Formula: see text])] models where the order of the coefficients is chosen by the Akaike information criteria (AIC). Up to 50% of the ocean SSH variability is dominated by the seasonal signal. The variance accounted for by the nonseasonal SSH is particularly distinct in the Southern and Pacific Oceans, containing >95% of the total SSH variance, and the expected prediction error growth takes a few months to reach a threshold of 1 cm. Isolated regions take 12 months or more to cross an accuracy threshold of 1 cm. Including the trend significantly increases the time taken to reach the threshold, particularly in the South Pacific. Annual averaging has expected prediction error growth of a few years to reach a threshold of 1 cm. Including the trend mainly increases the time taken to reach the threshold, but the time series is short and noisy.

1224. Towards an assessment of riverine dissolved organic carbon in surface waters of the western Arctic Ocean based on remote sensing and biogeochemical modeling

Authors: Vincent Le Fouest; Atsushi Matsuoka; Manfredi Manizza; Mona Shernetsky; Bruno Tremblay; Marcel Babin

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-15-1335-2018

URL: <https://doi.org/10.5194/bg-15-1335-2018>

Publication Date: 2018-03-05

Volume/Issue: Volume 15, Issue 5

References: 74

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Future climate warming of the Arctic could potentially enhance the load of terrigenous dissolved organic carbon (tDOC) of Arctic rivers due to increased carbon mobilization within watersheds. A greater flux of tDOC might impact the biogeochemical processes of the coastal Arctic Ocean (AO) and ultimately its capacity to absorb atmospheric CO₂. In this study, we show that sea-surface tDOC concentrations simulated by a physical-biogeochemical coupled model in the Canadian Beaufort Sea for 2003-2011 compare favorably with estimates retrieved by satellite imagery. Our results suggest that, over spring-summer, tDOC of riverine origin contributes to 35 % of primary production and that an equivalent of ~ 10 % of tDOC is exported westwards with the potential of fueling the biological production of the eastern Alaskan nearshore waters. The combination of model and satellite data provides promising results to extend this work to the entire AO so as to quantify, in conjunction with in situ data, the expected changes in tDOC fluxes and their potential impact on the AO biogeochemistry at basin scale.

1225. Influences of Pacific Climate Variability on Decadal Subsurface Ocean Heat Content Variations in the Indian Ocean

Authors: Xiaolin Jin (Tsinghua University); Young-Oh Kwon (Woods Hole Oceanographic Institution); Caroline C. Ummenhofer (Woods Hole Oceanographic Institution); Hyodae Seo (Woods Hole Oceanographic Institution); Franziska U. Schwarzkopf (GEOMAR Helmholtz Centre for Ocean Research); Arne Biastoch (GEOMAR Helmholtz Centre for Ocean Research); Claus W. Böning (GEOMAR Helmholtz Centre for Ocean Research); Jonathon S. Wright (Tsinghua University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0654.1

URL: <https://doi.org/10.1175/jcli-d-17-0654.1>

Publication Date: 2018-03-03

Volume/Issue: Volume 31, Issue 10

References: 77

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: China Scholarship Council, Ministry of Science and Technology of the People's Republic of China, National Science Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: Decadal variabilities in Indian Ocean subsurface ocean heat content (OHC; 50-300 m) since the 1950s are examined using ocean reanalyses. This study elaborates on how Pacific variability modulates the Indian Ocean on decadal time scales through both oceanic and atmospheric pathways. High correlations between OHC and thermocline depth variations across the entire Indian Ocean Basin suggest that OHC variability is primarily driven by thermocline fluctuations. The spatial pattern of the leading mode of decadal Indian Ocean OHC variability closely matches the regression pattern of OHC on the interdecadal Pacific oscillation (IPO), emphasizing the role of the Pacific Ocean in determining Indian Ocean OHC decadal variability. Further analyses identify different mechanisms by which the Pacific influences the eastern and western Indian Ocean. IPO-related anomalies from the Pacific propagate mainly through oceanic pathways in the Maritime Continent to impact the eastern Indian Ocean. By contrast, in the western Indian Ocean, the IPO induces wind-driven Ekman pumping in the central Indian Ocean via the atmospheric bridge, which in turn modifies conditions in the southwestern Indian Ocean via westward-propagating Rossby waves. To confirm this, a linear Rossby wave model is forced with wind stresses and eastern boundary conditions based on reanalyses. This linear model skillfully reproduces observed sea surface height anomalies and highlights both the oceanic connection in the eastern Indian Ocean and the role of wind-driven Ekman pumping in the west. These findings are also reproduced by OGCM hindcast experiments forced by interannual atmospheric boundary conditions applied only over the Pacific and Indian Oceans, respectively.

1226. Multichannel Singular Spectrum Analysis in the Estimates of Common Environmental Effects Affecting GPS Observations

Authors: Marta Gruszczynska; Severine Rosat; Anna Klos; Maciej Gruszczynski; Janusz Bogusz

Journal: Pure and Applied Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00024-018-1814-0

URL: <https://doi.org/10.1007/s00024-018-1814-0>

Publication Date: 2018-03-03

Volume/Issue: Volume 175, Issue 5

References: 63

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Polish National Science Centre

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1227. A New Coupled Ocean-Waves-Atmosphere Model Designed for Tropical Storm Studies: Example of Tropical Cyclone Bejisa (2013-2014) in the South-West Indian Ocean

Authors: J. Pianezze (Laboratoire de l'Atmosphère et des Cyclones (LACy), Université La Réunion, CNRS, Météo-France Saint-Denis France); C. Barthe (Laboratoire de l'Atmosphère et des Cyclones (LACy), Université La Réunion, CNRS, Météo-France Saint-Denis France); S. Bielli (Laboratoire de l'Atmosphère et des Cyclones (LACy), Université La Réunion, CNRS, Météo-France Saint-Denis France); P. Tulet (Laboratoire de l'Atmosphère et des Cyclones (LACy), Université La Réunion, CNRS, Météo-France Saint-Denis France); S. Jullien (Institut de Recherche pour le Développement (IRD)); G. Cambon (Institut de Recherche pour le Développement (IRD)); O. Bousquet (Laboratoire de l'Atmosphère et des Cyclones (LACy), Université La Réunion, CNRS, Météo-France Saint-Denis France); M. Claeys (Laboratoire de l'Atmosphère et des Cyclones (LACy), Université La Réunion, CNRS, Météo-France Saint-Denis France); E. Cordier (Observatoire des Sciences de l'Univers de La Réunion (OSU-R), Université La Réunion, CNRS Saint-Denis France)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017ms001177

URL: <https://doi.org/10.1002/2017ms001177>

Publication Date: 2018-03-01

Volume/Issue: Volume 10, Issue 3

References: 116

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: stratocumulus, trade-cumulus, boundary layer, numerical methods

Funding: Agence Nationale de la Recherche

License: Creative Commons

Open Access: N/A

Abstract: Ocean-Waves-Atmosphere (OWA) exchanges are not well represented in current Numerical Weather Prediction (NWP) systems, which can lead to large uncertainties in tropical cyclone track and intensity forecasts. In order to explore and better understand the impact of OWA interactions on tropical cyclone modeling, a fully coupled OWA system based on the atmospheric model Meso-NH, the oceanic model CROCO, and the wave model WW3 and called MSWC was designed and applied to the case of tropical cyclone Bejisa (2013-2014). The fully coupled OWA simulation shows good agreement with the literature and available observations. In particular, simulated significant wave height is within 30 cm of measurements made with buoys and altimeters. Short-term (< 2 days) sensitivity experiments used to highlight the effect of oceanic waves coupling show limited impact on the track, the intensity evolution, and the turbulent surface fluxes of the tropical cyclone. However, it is also shown that using a fully coupled OWA system is essential to obtain consistent sea salt emissions. Spatial and temporal coherence of the sea state with the 10 m wind speed are necessary to produce sea salt aerosol emissions in the right place (in the eyewall of the tropical cyclone) and with the right size distribution, which is critical for cloud microphysics.

1228. Multidecadal Changes of Upper-Ocean Thermal Conditions in the Tropical Northwest Pacific Ocean versus South China Sea during 1960-2015

Authors: Tzu-Ling Chiang (Department of Earth Sciences, National Taiwan Normal University, Taipei, Taiwan); Yi-Chia Hsin (Academia Sinica); Chau-Ron Wu (Department of Earth Sciences, National Taiwan Normal University, Taipei, Taiwan)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0394.1

URL: <https://doi.org/10.1175/jcli-d-17-0394.1>

Publication Date: 2018-03-01

Volume/Issue: Volume 31, Issue 10

References: 37

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: By analyzing the upper-ocean properties of observation-based hydrographic data and validated oceanic reanalysis products, this study presents multidecadal changes of oceanic surface and subsurface thermal conditions in the tropical northwest Pacific Ocean (TNWP) and South China Sea (SCS) during 1960-2015. The analysis reveals that a transition of a 30-yr trend took place in 1980s during the analyzed period for both the surface and subsurface environment. Generally, the warming trend of sea surface temperature (SST) in the TNWP has a similar multidecadal change to that in the SCS. However, a huge accumulating rate of upper-ocean heat content above the 26 degreesC isotherm (UOHC26) showed up in the TNWP (about 3 times compared to that in the SCS) in the last 30 years. In the TNWP, the southward shift of the North Equatorial Current on the multidecadal time scale induces the vertical displacement of isotherms, leading to a strong subsurface warming around the top of the thermocline. Secondly, the Pacific decadal oscillation (PDO)-related SST regulates the thermal structure in the mixed layer. The multidecadal UOHC26 in the SCS is mainly attributed to the PDO-related SST and further modulated by the isothermal variability caused by the change of basin-scale SCS circulation.

1229. Assessment of mean world ocean meridional\vspace *{.5ex} heat transport characteristics by a high-resolution model

Authors: K. V. Ushakov; R. A. Ibrayev

Journal: Russian Journal of Earth Sciences

Publisher: Geophysical Center of the Russian Academy of Sciences

DOI: 10.2205/2018es000616

URL: <https://doi.org/10.2205/2018es000616>

Publication Date: 2018-03-01

Volume/Issue: Volume 18, Issue 1

References: 56

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1230. Ocean submesoscales as a key component of the global heat budget

Authors: Zhan Su; Jinbo Wang; Patrice Klein; Andrew F. Thompson; Dimitris Menemenlis

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-018-02983-w

URL: <https://doi.org/10.1038/s41467-018-02983-w>

Publication Date: 2018-02-16

Volume/Issue: Volume 9, Issue 1

References: 59

Cited by: 261 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Recent studies highlight that oceanic motions associated with horizontal scales smaller than 50 km, defined here as submesoscales, lead to anomalous vertical heat fluxes from colder to warmer waters. This unique transport property is not captured in climate models that have insufficient resolution to simulate these submesoscale dynamics. Here, we use an ocean model with an unprecedented resolution that, for the first time, globally resolves submesoscale heat transport. Upper-ocean submesoscale turbulence produces a systematically-upward heat transport that is five times larger than mesoscale heat transport, with winter-time averages up to 100 W/m² for mid-latitudes. Compared to a lower-resolution model, submesoscale heat transport warms the sea surface up to 0.3 degreesC and produces an upward annual-mean air-sea heat flux anomaly of 4-10 W/m² at mid-latitudes. These results indicate that submesoscale dynamics are critical to the transport of heat between the ocean interior and the atmosphere, and are thus a key component of the Earth's climate.

1231. Seasonal resource conditions favor a summertime increase in North Pacific diatom-diazotroph associations

Authors: Christopher L Follett (Massachusetts Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); David M Karl (Daniel K. Inouye Center for Microbial Oceanography: Research and Education, University of Hawai'i at Mānoa , Honolulu, HI, USA, Department of Oceanography, School of Ocean and Earth Science and Technology, University of Hawai'i at Mānoa , Honolulu, HI, USA); Keisuke Inomura (Massachusetts Institute of Technology); Michael J Follows (Massachusetts Institute of Technology)

Journal: The ISME Journal

Publisher: Oxford University Press (OUP)

DOI: 10.1038/s41396-017-0012-x

URL: <https://doi.org/10.1038/s41396-017-0012-x>

Publication Date: 2018-02-14

Volume/Issue: Volume 12, Issue 6

References: 111

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: In the North Pacific Subtropical Gyre (NPSG), an annual pulse of sinking organic carbon is observed at 4000 m between July and August, driven by large diatoms found in association with nitrogen fixing, heterocystous, cyanobacteria: Diatom-Diazotroph Associations (DDAs). Here we ask what drives the bloom of DDAs and present a simplified trait-based model of subtropical phototroph populations driven by observed, monthly averaged, environmental characteristics. The ratio of resource supply rates favors nitrogen fixation year round. The relative fitness of DDA traits is most competitive in early summer when the mixed layer is shallow, solar irradiance is high, and phosphorus and iron are relatively abundant. Later in the season, as light intensity drops and phosphorus is depleted, the traits of small unicellular diazotrophs become more competitive. The competitive transition happens in August, at the time when the DDA export event occurs. This seasonal dynamic is maintained when embedded in a more complex, global-scale, ecological model, and provides predictions for the extent of the North Pacific DDA bloom. The model provides a parsimonious and testable hypothesis for the stimulation of DDA blooms.

1232. Temperature and sea ice hindcast skill of the MiKlip decadal prediction system in the Arctic

Authors: Daniel Senftleben; Veronika Eyring; Axel Lauer; Mattia Righi

Journal: Meteorologische Zeitschrift

Publisher: Schweizerbart

DOI: 10.1127/metz/2018/0871

URL: <https://doi.org/10.1127/metz/2018/0871>

Publication Date: 2018-02-07

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Hindcast

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1233. Reconstruction of Local Sea Levels at South West Pacific Islands--A Multiple Linear Regression Approach (1988-2014)

Authors: V. Kumar (Institut de Recherche pour le Développement (IRD), PaCE-SD, University of the South Pacific Suva Fiji); A. Melet (Mercator Ocean Ramonville-Saint-Agne France); B. Meyssignac (Institut de Recherche pour le Développement (IRD)); A. Ganachaud (Institut de Recherche pour le Développement (IRD)); W. S. Kessler (NOAA/Pacific Marine Environmental Laboratory Seattle WA USA); A. Singh (PaCE-SD, University of the South Pacific Suva Fiji); J. Aucan (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013053

URL: <https://doi.org/10.1002/2017jc013053>

Publication Date: 2018-02-05

Volume/Issue: Volume 123, Issue 2

References: 78

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: statistical downscaling, local sea levels

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Rising sea levels are a critical concern in small island nations. The problem is especially serious in the western south Pacific, where the total sea level rise over the last 60 years has been up to 3 times the global average. In this study, we aim at reconstructing sea levels at selected sites in the region (Suva, Lautoka--Fiji, and Nouméa--New Caledonia) as a multilinear regression (MLR) of atmospheric and oceanic variables. We focus on sea level variability at interannual-to-interdecadal time scales, and trend over the 1988-2014 period. Local sea levels are first expressed as a sum of steric and mass changes. Then a dynamical approach is used based on wind stress curl as a proxy for the thermosteric component, as wind stress curl anomalies can modulate the thermocline depth and resultant sea levels via Rossby wave propagation. Statistically significant predictors among wind stress curl, halosteric sea level, zonal/meridional wind stress components, and sea surface temperature are used to construct a MLR model simulating local sea levels. Although we are focusing on the local scale, the global mean sea level needs to be adjusted for. Our reconstructions provide insights on key drivers of sea level variability at the selected sites, showing that while local dynamics and the global signal modulate sea level to a given extent, most of the variance is driven by regional factors. On average, the MLR model is able to reproduce 82% of the variance in island sea level, and could be used to derive local sea level projections via downscaling of climate models.

1234. Satellite Altimetry over Oceans and Land Surfaces

Authors: Detlef Stammer; Anny Cazenave

Journal: Unknown Journal

Publisher: CRC Press

DOI: 10.1201/9781315151779

URL: <https://doi.org/10.1201/9781315151779>

Publication Date: 2018-02-02

Volume/Issue: N/A

References: N/A

Cited by: 88 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1235. Modelling ocean-colour-derived chlorophyll a

Authors: Stephanie Dutkiewicz; Anna E. Hickman; Oliver Jahn

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-15-613-2018

URL: <https://doi.org/10.5194/bg-15-613-2018>

Publication Date: 2018-01-30

Volume/Issue: Volume 15, Issue 2

References: 60

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . This article provides a proof of concept for using a biogeochemical/ecosystem/optical model with a radiative transfer component as a laboratory to explore aspects of ocean colour. We focus here on the satellite ocean colour chlorophyll a (Chl a) product provided by the often-used blue/green reflectance ratio algorithm. The model produces output that can be compared directly to the real-world ocean colour remotely sensed reflectance. This model output can then be used to produce an ocean colour satellite-like Chl a product using an algorithm linking the blue versus green reflectance similar to that used for the real world. Given that the model includes complete knowledge of the (model) water constituents, optics and reflectance, we can explore uncertainties and their causes in this proxy for Chl a (called derived Chl a in this paper). We compare the derived Chl a to the actual model Chl a field. In the model we find that the mean absolute bias due to the algorithm is 22 % between derived and actual Chl a. The real-world algorithm is found using concurrent in situ measurement of Chl a and radiometry. We ask whether increased in situ measurements to train the algorithm would improve the algorithm, and find a mixed result. There is a global overall improvement, but at the expense of some regions, especially in lower latitudes where the biases increase. Not surprisingly, we find that region-specific algorithms provide a significant improvement, at least in the annual mean. However, in the model, we find that no matter how the algorithm coefficients are found there can be a temporal mismatch between the derived Chl a and the actual Chl a. These mismatches stem from temporal decoupling between Chl a and other optically important water constituents (such as coloured dissolved organic matter and detrital matter). The degree of decoupling differs regionally and over time. For example, in many highly seasonal regions, the timing of initiation and peak of the spring bloom in the derived Chl a lags the actual Chl a by days and sometimes weeks. These results indicate that care should also be taken when studying phenology through satellite-derived products of Chl a. This study also reemphasizes that ocean-colour-derived Chl a is not the same as the real in situ Chl a. In fact the model derived Chl a compares better to real-world satellite-derived Chl a than the model actual Chl a. Modellers should keep this in mind when evaluating model output with ocean colour Chl a and in particular when assimilating this product. Our goal is to illustrate the use of a numerical laboratory that (a) helps users of ocean colour, particularly modellers, gain further understanding of the products they use and (b)

helps the ocean colour community to explore other ocean colour products, their biases and uncertainties, as well as to aid in future algorithm development.

1236. Seasonality in Transition Scale from Balanced to Unbalanced Motions in the World Ocean

Authors: Bo Qiu (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Shuiming Chen (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Patrice Klein (California Institute of Technology); Jinbo Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hector Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-17-0169.1

URL: <https://doi.org/10.1175/jpo-d-17-0169.1>

Publication Date: 2018-01-26

Volume/Issue: Volume 48, Issue 3

References: 47

Cited by: 115 publications

Language: N/A

Subjects: N/A

Keywords: Seasonal variability, Inertia-gravity waves, Small scale processes, Altimetry, General circulation models (and 1 more)

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, Agence Nationale pour le Développement de la Recherche en Santé (and 1 more)

License: N/A

Open Access: N/A

Abstract: The transition scale L_t from balanced geostrophic motions to unbalanced wave motions, including near-inertial flows, internal tides, and inertia-gravity wave continuum, is explored using the output from a global 1/48 degrees horizontal resolution Massachusetts Institute of Technology general circulation model (MITgcm) simulation. Defined as the wavelength with equal balanced and unbalanced motion kinetic energy (KE) spectral density, L_t is detected to be geographically highly inhomogeneous: it falls below 40 km in the western boundary current and Antarctic Circumpolar Current regions, increases to 40-100 km in the interior subtropical and subpolar gyres, and exceeds, in general, 200 km in the tropical oceans. With the exception of the Pacific and Indian sectors of the Southern Ocean, the seasonal KE fluctuations of the surface balanced and unbalanced motions are out of phase because of the occurrence of mixed layer instability in winter and trapping of unbalanced motion KE in shallow mixed layer in summer. The combined effect of these seasonal changes renders L_t to be 20 km during winter in 80% of the Northern Hemisphere oceans between 25 degrees and 45 degreesN and all of the Southern Hemisphere oceans south of 25 degreesS. The transition scale's geographical and seasonal changes are highly relevant to the forthcoming Surface Water and Ocean Topography (SWOT) mission. To improve the detection of balanced submesoscale signals from SWOT, especially in the tropical oceans, efforts to remove stationary internal tidal signals are called for.

1237. Accounting for Gravitational Attraction and Loading Effects from Land Ice on Absolute Sea Level

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.); Katherine J. Quinn (Atmospheric and Environmental Research Inc.); Christopher G. Piecuch (Atmospheric and Environmental Research Inc.)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-17-0092.1

URL: <https://doi.org/10.1175/jtech-d-17-0092.1>

Publication Date: 2018-01-18

Volume/Issue: Volume 35, Issue 2

References: 34

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Altimetry, Ocean, Ocean circulation, Decadal variability

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Gravitational attraction and loading (GAL) effects associated with ongoing long-term changes in land ice are expected to cause spatially varying trends in absolute sea level ζ , as measured by satellite altimeters. The largest spatial gradients in ζ trends, predicted from solving the sea level equation using GRACE retrievals of mass distribution over land for the period 2005-15, occur near Greenland and West Antarctica, consistent with a strong local land ice loss. Misinterpreting the estimated static GAL trends in ζ as dynamic pressure gradients can lead to substantial errors in large-scale geostrophic transports across the Southern Ocean and the subpolar North Atlantic over the analyzed decade. South of Greenland, where altimeter sea level and hydrography (Argo) data coverage is good, the residual ζ minus steric height trends are similar in magnitude and sign to the gravitationally based predictions. In addition, estimated GAL-related trends are as large--if not larger than--other factors, such as deep steric height, dynamic bottom pressure, and glacial isostatic rebound. Thus, accounting for static GAL effects on ζ records, which are commonly neglected in oceanographic studies, seems important for a quantitative interpretation of the observed ζ trends.

1238. Shingle 2.0: generalising self-consistent and automated domain discretisation for multi-scale geophysical models

Authors: Adam S. Candy; Julie D. Pietrzak

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-11-213-2018

URL: <https://doi.org/10.5194/gmd-11-213-2018>

Publication Date: 2018-01-17

Volume/Issue: Volume 11, Issue 1

References: 57

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The approaches taken to describe and develop spatial discretisations of the domains required for geophysical simulation models are commonly ad hoc, model- or application-specific, and under-documented. This is particularly acute for simulation models that are flexible in their use of multi-scale, anisotropic, fully unstructured meshes where a relatively large number of heterogeneous parameters are required to constrain their full description. As a consequence, it can be difficult to reproduce simulations, to ensure a provenance in model data handling and initialisation, and a challenge to conduct model intercomparisons rigorously. This paper takes a novel approach to spatial discretisation, considering it much like a numerical simulation model problem of its own. It introduces a generalised, extensible, self-documenting approach to carefully describe, and necessarily fully, the constraints over the heterogeneous parameter space that determine how a domain is spatially discretised. This additionally provides a method to accurately record these constraints, using high-level natural language based abstractions that enable full accounts of provenance, sharing, and distribution. Together with this description, a generalised consistent approach to unstructured mesh generation for geophysical models is developed that is automated, robust and repeatable, quick-to-draft, rigorously verified, and consistent with the source data throughout. This interprets the description above to execute a self-consistent spatial discretisation process, which is automatically validated to expected discrete characteristics and metrics. Library code, verification tests, and examples available in the repository at <https://github.com/shingleproject/Shingle>. Further details of the project presented at <http://shingleproject.org>.

1239. Salinity Trends within the Upper Layers of the Subpolar North Atlantic

Authors: Jan-Erik Tesdal (Columbia University); Ryan P. Abernathey (Columbia University); Joaquim I. Goes (Columbia University); Arnold L. Gordon (Columbia University); Thomas W. N. Haine (Johns Hopkins University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0532.1

URL: <https://doi.org/10.1175/jcli-d-17-0532.1>

Publication Date: 2018-01-11

Volume/Issue: Volume 31, Issue 7

References: 73

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: Salinity, North Atlantic Oscillation, Ocean, Arctic Oscillation, North Atlantic Ocean (and 1 more)

Funding: Goddard Space Flight Center, Goddard Space Flight Center, Gordon and Betty Moore Foundation (and 4 more)

License: N/A

Open Access: N/A

Abstract: Examination of a range of salinity products collectively suggests widespread freshening of the North Atlantic from the mid-2000s to the present. Monthly salinity fields reveal negative trends that differ in magnitude and significance between western and eastern regions of the North Atlantic. These differences can be attributed to the large negative interannual excursions in salinity in the western subpolar gyre and the Labrador Sea, which are not apparent in the central or eastern subpolar gyre. This study demonstrates that temporal trends in salinity in the northwest (including the Labrador Sea) are subject to mechanisms that are distinct from those responsible for the salinity trends in the central and eastern North Atlantic. In the western subpolar gyre a negative correlation between near-surface salinity and the circulation strength of the subpolar gyre suggests that negative salinity anomalies are connected to an intensification of the subpolar gyre, which is causing increased flux of freshwater from the East Greenland Current and subsequent transport into the Labrador Sea during the melting season. Analyses of sea surface wind fields suggest that the strength of the subpolar gyre is linked to the North Atlantic Oscillation- and Arctic Oscillation-driven changes in wind stress curl in the eastern subpolar gyre. If this trend of decreasing salinity continues, it has the potential to enhance water column stratification, reduce vertical fluxes of nutrients, and cause a decline in biological production and carbon export in the North Atlantic Ocean.

1240. Scaling Properties of Arctic Sea Ice Deformation in a High-Resolution Viscous-Plastic Sea Ice Model and in Satellite Observations

Authors: Nils Hutter (Alfred Wegener Institute); Martin Losch (Alfred Wegener Institute); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013119

URL: <https://doi.org/10.1002/2017jc013119>

Publication Date: 2018-01-10

Volume/Issue: Volume 123, Issue 1

References: 53

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: high-resolution modeling, Arctic Ocean, scaling, sea ice rheology, sea ice deformation

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: Sea ice models with the traditional viscous-plastic (VP) rheology and very small horizontal grid spacing can resolve leads and deformation rates localized along Linear Kinematic Features (LKF). In a 1 km pan-Arctic sea ice-ocean simulation, the small-scale sea ice deformations are evaluated with a scaling analysis in relation to satellite observations of the Envisat Geophysical Processor System (EGPS) in the Central Arctic. A new coupled scaling analysis for data on Eulerian grids is used to determine the spatial and temporal scaling and the coupling between temporal and spatial scales. The spatial scaling of the modeled sea ice deformation implies multifractality. It is also coupled to temporal scales and varies realistically by region and season. The agreement of the spatial scaling with satellite observations challenges previous results with VP models at coarser resolution, which did not reproduce the observed scaling. The temporal scaling analysis shows that the VP model, as configured in this 1 km simulation, does not fully resolve the intermittency of sea ice deformation that is observed in satellite data.

1241. Correlation Lengths for Estimating the Large-Scale Carbon and Heat Content of the Southern Ocean

Authors: M. R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); B. D. Cornuelle (Scripps Institution of Oceanography / University of California San Diego); S. T. Gille (Scripps Institution of Oceanography / University of California San Diego); A. Verdy (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013408

URL: <https://doi.org/10.1002/2017jc013408>

Publication Date: 2018-01-10

Volume/Issue: Volume 123, Issue 2

References: 29

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF

License: N/A

Open Access: N/A

Abstract: The spatial correlation scales of oceanic dissolved inorganic carbon, heat content, and carbon and heat exchanges with the atmosphere are estimated from a realistic numerical simulation of the Southern Ocean. Biases in the model are assessed by comparing the simulated sea surface height and temperature scales to those derived from optimally interpolated satellite measurements. While these products do not resolve all ocean scales, they are representative of the climate scale variability we aim to estimate. Results show that constraining the carbon and heat inventory between 35 degreesS and 70 degreesS on time-scales longer than 90 days requires approximately 100 optimally spaced measurement platforms: approximately one platform every 20 degrees longitude by 6 degrees latitude. Carbon flux has slightly longer zonal scales, and requires a coverage of approximately 30 degrees by 6 degrees. Heat flux has much longer scales, and thus a platform distribution of approximately 90 degrees by 10 degrees would be sufficient. Fluxes, however, have significant subseasonal variability. For all fields, and especially fluxes, sustained measurements in time are required to prevent aliasing of the eddy signals into the longer climate scale signals. Our results imply a minimum of 100 biogeochemical-Argo floats are required to monitor the Southern Ocean carbon and heat content and air-sea exchanges on time-scales longer than 90 days. However, an estimate of formal mapping error using the current Argo array implies that in practice even an array of 600 floats (a nominal float density of about 1 every 7 degrees longitude by 3 degrees latitude) will result in nonnegligible uncertainty in estimating climate signals.

1242. Large-scale ocean connectivity and planktonic body size

Authors: Ernesto Villarino; James R. Watson; Bror Jönsson; Josep M. Gasol; Guillem Salazar; Silvia G. Acinas; Marta Estrada; Ramón Massana; Ramiro Logares; Caterina R. Giner; Massimo C. Pernice; M. Pilar Olivar; Leire Citores; Jon Corell; Naiara Rodríguez-Ezpeleta; José Luis Acuña; Axayacatl Molina-Ramírez; J. Ignacio González-Gordillo; Andrés Cózar; Elisa Martí; José A. Cuesta; Susana Agustí; Eugenio Fraile-Nuez; Carlos M. Duarte; Xabier Irigoien; Guillem Chust

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-017-02535-8

URL: <https://doi.org/10.1038/s41467-017-02535-8>

Publication Date: 2018-01-04

Volume/Issue: Volume 9, Issue 1

References: 80

Cited by: 107 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Global patterns of planktonic diversity are mainly determined by the dispersal of propagules with ocean currents. However, the role that abundance and body size play in determining spatial patterns of diversity remains unclear. Here we analyse spatial community structure - beta-diversity - for several planktonic and nektonic organisms from prokaryotes to small mesopelagic fishes collected during the Malaspina 2010 Expedition. beta-diversity was compared to surface ocean transit times derived from a global circulation model, revealing a significant negative relationship that is stronger than environmental differences. Estimated dispersal scales for different groups show a negative correlation with body size, where less abundant large-bodied communities have significantly shorter dispersal scales and larger species spatial turnover rates than more abundant small-bodied plankton. Our results confirm that the dispersal scale of planktonic and micro-nektonic organisms is determined by local abundance, which scales with body size, ultimately setting global spatial patterns of diversity.

1243. Circum-Antarctic Shoreward Heat Transport Derived From an Eddy- and Tide-Resolving Simulation

Authors: Andrew L. Stewart (University of California Los Angeles); Andreas Klocker (Antarctic Climate and Ecosystems Cooperative Research Centre, University of Tasmania); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl075677

URL: <https://doi.org/10.1002/2017gl075677>

Publication Date: 2018-01-03

Volume/Issue: Volume 45, Issue 2

References: 77

Cited by: 89 publications

Language: en

Subjects: N/A

Keywords: eddies, overturning circulation, Antarctic margins, heat flux, modeling (and 1 more)

Funding: National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Almost all heat reaching the bases of Antarctica's ice shelves originates from warm Circumpolar Deep Water in the open Southern Ocean. This study quantifies the roles of mean and transient flows in transporting heat across almost the entire Antarctic continental slope and shelf using an ocean/sea ice model run at eddy- and tide-resolving (1/48 degrees) horizontal resolution. Heat transfer by transient flows is approximately attributed to eddies and tides via a decomposition into time scales shorter than and longer than 1 day, respectively. It is shown that eddies transfer heat across the continental slope (ocean depths greater than 1,500 m), but tides produce a stronger shoreward heat flux across the shelf break (ocean depths between 500 m and 1,000 m). However, the tidal heat fluxes are approximately compensated by mean flows, leaving the eddy heat flux to balance the net shoreward heat transport. The eddy-driven cross-slope overturning circulation is too weak to account for the eddy heat flux. This suggests that isopycnal eddy stirring is the principal mechanism of shoreward heat transport around Antarctica, though likely modulated by tides and surface forcing.

1244. Mechanisms of Mixed-Layer Salinity Seasonal Variability in the Indian Ocean

Authors: Julia Köhler (University of Hamburg); Nuno Serra (University of Hamburg); Frank O. Bryan (National Center for Atmospheric Research); Benjamin K. Johnson (North Carolina State University); Detlef Stammer (University of Hamburg)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013640

URL: <https://doi.org/10.1002/2017jc013640>

Publication Date: 2017-12-28

Volume/Issue: Volume 123, Issue 1

References: 75

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Based on a joint analysis of an ensemble mean of satellite sea surface salinity retrievals and the output of a high-resolution numerical ocean circulation simulation, physical processes are identified that control seasonal variations of mixed-layer salinity (MLS) in the Indian Ocean, a basin where salinity changes dominate changes in density. In the northern and near-equatorial Indian Ocean, annual salinity changes are mainly driven by respective changes of the horizontal advection. South of the equatorial region, between 45 degreesE and 90 degreesE, where evaporation minus precipitation has a strong seasonal cycle, surface freshwater fluxes control the seasonal MLS changes. The influence of entrainment on the salinity variance is enhanced in mid-ocean upwelling regions but remains small. The model and observational results reveal that vertical diffusion plays a major role in precipitation and river runoff dominated regions balancing the surface freshwater flux. Vertical diffusion is important as well in regions where the advection of low salinity leads to strong gradients across the mixed-layer base. There, vertical diffusion explains a large percentage of annual MLS variance. The simulation further reveals that (1) high-frequency small-scale eddy processes primarily determine the salinity tendency in coastal regions (in particular in the Bay of Bengal) and (2) shear horizontal advection, brought about by changes in the vertical structure of the mixed layer, acts against mean horizontal advection in the equatorial salinity frontal regions. Observing those latter features with the existing observational components remains a future challenge.

1245. Extraordinary boulder transport by storm waves (west of Ireland, winter 2013-2014), and criteria for analysing coastal boulder deposits

Authors: Rónadh Cox; Kalle L. Jahn; Oona G. Watkins; Peter Cox

Journal: Earth-Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.earscirev.2017.12.014

URL: <https://doi.org/10.1016/j.earscirev.2017.12.014>

Publication Date: 2017-12-23

Volume/Issue: Volume 177

References: 129

Cited by: 89 publications

Language: en

Subjects: N/A

Keywords: Coastal erosion, Storm waves, Coastal boulder deposits, Coastal hazard, Megagravel (and 1 more)

Funding: NSF, Williams College Class of 1963 Sustainability Fund

License: N/A

Open Access: N/A

Abstract: N/A

1246. Interactive visualization of high-dimensional petascale ocean data

Authors: David A. Ellsworth; Christopher E. Henze; Bron C. Nelson

Journal: 2017 IEEE 7th Symposium on Large Data Analysis and Visualization (LDAV)

Publisher: IEEE

DOI: 10.1109/ldav.2017.8231849

URL: <https://doi.org/10.1109/ldav.2017.8231849>

Publication Date: 2017-12-21

Volume/Issue: N/A

References: 39

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1247. Improving sea ice thickness estimates by assimilating CryoSat-2 and SMOS sea ice thickness data simultaneously

Authors: Longjiang Mu (National Marine Environmental Forecasting Center); Qinghua Yang (National Marine Environmental Forecasting Center); Martin Losch (Alfred Wegener Institute); Svetlana N. Losa (Alfred Wegener Institute, Russian Academy of Sciences); Robert Ricker (Alfred Wegener Institute); Lars Nerger (Alfred Wegener Institute); Xi Liang (National Marine Environmental Forecasting Center)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.3225

URL: <https://doi.org/10.1002/qj.3225>

Publication Date: 2017-12-20

Volume/Issue: Volume 144, Issue 711

References: 51

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: The impact of assimilating weekly CryoSat-2 sea ice thickness data together with daily SMOS sea ice thickness and daily SSMIS sea ice concentration data on the sea ice fields of a coupled sea ice-ocean model of the Arctic Ocean is investigated. The sea-ice model is based on the Massachusetts Institute of Technology general circulation model (MITgcm) and the assimilation is performed by a localized Singular Evolutive Interpolated Kalman (LSEIK) filter coded in the Parallel Data Assimilation Framework (PDAF). A period of three months from 1 November 2011 to 30 January 2012 is selected to assess the skill of the assimilation system in the cold season. Compared to the unassimilated solution and a solution where only sea ice concentration is assimilated, the model-data misfits are substantially reduced in areas of both thick and thin ice. The sea ice thickness estimates agree significantly better with in situ observations in the central Arctic Ocean than the sea ice thickness obtained from assimilating SMOS data alone, while the sea ice concentration shows very small improvements. The sea ice fields obtained by the joint assimilation of SMOS and CryoSat-2 data also have lower errors in thickness and concentration than those obtained from directly assimilating a statistically merged SMOS and CryoSat-2 sea ice thickness product. These lower errors suggest that model dynamics play a significant role in data blending.

1248. On the vertical velocity component in the mesoscale Lofoten vortex of the Norwegian Sea

Authors: T. V. Belonenko; I. L. Bashmachnikov; A. V. Koldunov; P. A. Kuibin

Journal: Izvestiya, Atmospheric and Oceanic Physics

Publisher: Pleiades Publishing Ltd

DOI: 10.1134/s0001433817060032

URL: <https://doi.org/10.1134/s0001433817060032>

Publication Date: 2017-12-14

Volume/Issue: Volume 53, Issue 6

References: 28

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1249. Stress Models of the Annual Hydrospheric, Atmospheric, Thermal, and Tidal Loading Cycles on California Faults: Perturbation of Background Stress and Changes in Seismicity

Authors: Christopher W. Johnson (University of California Berkeley); Yuning Fu (School of Earth, Environment and Society Bowling Green State University Bowling Green OH USA); Roland Bürgmann (University of California Berkeley)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jb014778

URL: <https://doi.org/10.1002/2017jb014778>

Publication Date: 2017-12-14

Volume/Issue: Volume 122, Issue 12

References: 76

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation, Southern California Earthquake Center (and 1 more)

License: N/A

Open Access: N/A

Abstract: Stresses in the lithosphere arise from multiple natural loading sources that include both surface and body forces. The largest surface loads include near-surface water storage, snow and ice, atmosphere pressure, ocean loading, and temperature changes. The solid Earth also deforms from celestial body interactions and variations in Earth's rotation. We model the seasonal stress changes in California from 2006 through 2014 for seven different loading sources with annual periods to produce an aggregate stressing history for faults in the study area. Our modeling shows that the annual water loading, atmosphere, temperature, and Earth pole tides are the largest loading sources and should each be evaluated to fully describe seasonal stress changes. In California we find that the hydrological loads are the largest source of seasonal stresses. We explore the seasonal stresses with respect to the background principal stress orientation constrained with regional focal mechanisms and analyze the modulation of seismicity. Our results do not suggest a resolvable seasonal variation for the ambient stress orientation in the shallow crust. When projecting the seasonal stresses into the background stress orientation we find that the timing of microseismicity modestly increases from an ~8 kPa seasonal mean-normal-stress perturbation. The results suggest that faults in California are optimally oriented with the background stress field and respond to subsurface pressure changes, possibly due to processes we have not considered in this study. At any time a population of faults are near failure as evident from earthquakes triggered by these slight seasonal stress perturbations.

1250. The role of simulated small-scale ocean variability in inverse computations for ocean acoustic tomography

Authors: Brian D. Dushaw (Nansen Environmental and Remote Sensing Center); Hanne Sagen (Nansen Environmental and Remote Sensing Center)

Journal: The Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

DOI: 10.1121/1.5016816

URL: <https://doi.org/10.1121/1.5016816>

Publication Date: 2017-12-13

Volume/Issue: Volume 142, Issue 6

References: 31

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean acoustic tomography depends on a suitable reference ocean environment with which to set the basic parameters of the inverse problem. Some inverse problems may require a reference ocean that includes the small-scale variations from internal waves, small mesoscale, or spice. Tomographic inversions that employ data of stable shadow zone arrivals, such as those that have been observed in the North Pacific and Canary Basin, are an example. Estimating temperature from the unique acoustic data that have been obtained in Fram Strait is another example. The addition of small-scale variability to augment a smooth reference ocean is essential to understanding the acoustic forward problem in these cases. Rather than a hindrance, the stochastic influences of the small scale can be exploited to obtain accurate inverse estimates. Inverse solutions are readily obtained, and they give computed arrival patterns that matched the observations. The approach is not ad hoc, but universal, and it has allowed inverse estimates for ocean temperature variations in Fram Strait to be readily computed on several acoustic paths for which tomographic data were obtained.

1251. Space Observation of Carbon Dioxide Partial Pressure at Ocean Surface

Authors: W. Timothy Liu; Xiaosu Xie

Journal: IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/jstars.2017.2766138

URL: <https://doi.org/10.1109/jstars.2017.2766138>

Publication Date: 2017-12-13

Volume/Issue: Volume 10, Issue 12

References: 80

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: JPL Research and Technology Development Program, NASA Ocean Surface Salinity Science Team

License: N/A

Open Access: N/A

Abstract: N/A

1252. Characterizing the Seasonal Crustal Motion in Tianshan Area Using GPS, GRACE and Surface Loading Models

Authors: Yunlong Wu (China Earthquake Administration); Qian Zhao (China Earthquake Administration); Bao Zhang (Chinese University of Hong Kong); Weiwei Wu (Tongji University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs9121303

URL: <https://doi.org/10.3390/rs9121303>

Publication Date: 2017-12-12

Volume/Issue: Volume 9, Issue 12

References: 49

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Complex tectonic and non-tectonic movements exist in the Tianshan area. However, we have not acquired good knowledge of such movements yet. In this study, we combine Global Positioning System (GPS), the Gravity Recovery and Climate Experiment (GRACE) and Surface Loading Models (SLMs) data to study the seasonal vertical crustal displacements in the Tianshan area. The results show that all three datasets exhibit significant annual variations at all 26 local GPS stations. Correlation coefficients higher than 0.8 between the GRACE and GPS data were observed at 85% of the stations, and it became 92% when comparing GPS and SLMs. The Weighted Root Mean Squares (WRMS) reductions were 41% and 47% after removing the annual displacements of GRACE and SLMs from the GPS time series, respectively. The consistency between the GPS and SLMs data was higher than that between the GPS and GRACE data, which is mainly due to the dominant position of atmospheric loading in the study area. For the abnormal station XJYN (43 degreesN, 81 degreesE), the GPS time series showed an abnormal uplift from early 2013 to early 2015, but this not shown in the GRACE and SLMs results. We attribute this discrepancy to groundwater variations, which are not resolvable by GRACE and SLMs for small-scale regions.

1253. A Pilot Climate Sensitivity Study Using the CEN Coupled Adjoint Model (CESAM)

Authors: D. Stammer (University of Hamburg); A. Köhl (University of Hamburg); A. Vlasenko (University of Hamburg); I. Matei (University of Hamburg); F. Lunkeit (University of Hamburg); S. Schubert (University of Hamburg)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0183.1

URL: <https://doi.org/10.1175/jcli-d-17-0183.1>

Publication Date: 2017-12-11

Volume/Issue: Volume 31, Issue 5

References: 81

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: EU, EU, BMBF (and 1 more)

License: N/A

Open Access: N/A

Abstract: A pilot coupled climate sensitivity study is presented based on the newly developed adjoint coupled climate model, Centrum für Erdsystemforschung und Nachhaltigkeit (CEN) Earth System Assimilation Model (CESAM). To this end the components of the coupled forward model are summarized, and the generation of the adjoint code out of the model forward code through the application of the Transformation of Algorithms in FORTRAN (TAF) adjoint compiler is discussed. It is shown that simulations of the intermediate-complexity CESAM are comparable in quality to CMIP-type coupled climate models, justifying the usage of the model to compute adjoint sensitivities of the northern Europe near-surface temperature to anomalies in surface temperature, sea surface salinity, and sea ice over the North Atlantic and the Arctic on time scales of up to one month. Results confirm that on a time scale of up to a few days surface temperatures over northern Europe are influenced by Atlantic temperature anomalies just upstream of the target location. With increasingly longer time lapse, however, it is the influence of SSTs over the central and western North Atlantic on the overlying atmosphere and the associated changes in storm-track pattern that dominate the evolution of the surface European temperature. Influences of surface salinity and sea ice on the northern European temperature appear to have similar sensitivity mechanisms, invoked indirectly through their influence on near-surface temperature anomalies. The adjoint study thus confirms that the SST's impact on the atmospheric dynamics, notably storm tracks, is the primary cause for the influence of northern European temperature changes.

1254. An Enhanced PIRATA Dataset for Tropical Atlantic Ocean-Atmosphere Research

Authors: Gregory R. Foltz (NOAA/Atlantic Oceanographic and Meteorological Laboratory, Miami, Florida); Claudia Schmid (NOAA/Atlantic Oceanographic and Meteorological Laboratory, Miami, Florida); Rick Lumpkin (NOAA/Atlantic Oceanographic and Meteorological Laboratory, Miami, Florida)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0816.1

URL: <https://doi.org/10.1175/jcli-d-16-0816.1>

Publication Date: 2017-12-08

Volume/Issue: Volume 31, Issue 4

References: 78

Cited by: 21 publications

Language: N/A

Subjects: N/A

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License: N/A

Open Access: N/A

Abstract: The Prediction and Research Moored Array in the Tropical Atlantic (PIRATA) provides measurements of the upper ocean and near-surface atmosphere at 18 locations. Time series from many moorings are nearly 20 years in length. However, instrumental biases, data dropouts, and the coarse vertical resolutions of the oceanic measurements complicate their use for research. Here an enhanced PIRATA dataset (ePIRATA) is presented for the 17 PIRATA moorings with record lengths of at least seven years. Data in ePIRATA are corrected for instrumental biases, temporal gaps are filled using supplementary datasets, and the subsurface temperature and salinity time series are mapped to a uniform 5-m vertical grid. All original PIRATA data that pass quality control and that do not require bias correction are retained without modification, and detailed error estimates are provided. The terms in the mixed-layer heat and temperature budgets are calculated and included, with error bars. As an example of ePIRATA's application, the vertical exchange of heat at the base of the mixed layer (Q_h) is calculated at each PIRATA location as the difference between the heat storage rate and the sum of the net surface heat flux and horizontal advection. Off-equatorial locations are found to have annual mean cooling rates of 20-60 W m⁻², while cooling at equatorial locations reaches 85-110 W m⁻² between 10 degrees and 35 degreesW and decreases to 40 W m⁻² at 0 degrees. At most off-equatorial locations, the strongest seasonal cooling from Q_h occurs when winds are weak. Possible explanations are discussed, including the importance of seasonal modulations of mixed-layer depth and the diurnal cycle.

1255. Empirical methods for the estimation of Southern Ocean CO₂: support vector and random forest regression

Authors: Luke Gregor; Schalk Kok; Pedro M. S. Monteiro

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-14-5551-2017

URL: <https://doi.org/10.5194/bg-14-5551-2017>

Publication Date: 2017-12-08

Volume/Issue: Volume 14, Issue 23

References: 53

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Southern Ocean accounts for 40 % of oceanic CO₂ uptake, but the estimates are bound by large uncertainties due to a paucity in observations. Gap-filling empirical methods have been used to good effect to approximate pCO₂ from satellite observable variables in other parts of the ocean, but many of these methods are not in agreement in the Southern Ocean. In this study we propose two additional methods that perform well in the Southern Ocean: support vector regression (SVR) and random forest regression (RFR). The methods are used to estimate ΔpCO₂ in the Southern Ocean based on SOCAT v3, achieving similar trends to the SOM-FFN method by Landschützer et al. (2014). Results show that the SOM-FFN and RFR approaches have RMSEs of similar magnitude (14.84 and 16.45 μatm, where 1 atm = 101 325 Pa) where the SVR method has a larger RMSE (24.40 μatm). However, the larger errors for SVR and RFR are, in part, due to an increase in coastal observations from SOCAT v2 to v3, where the SOM-FFN method used v2 data. The success of both SOM-FFN and RFR depends on the ability to adapt to different modes of variability. The SOM-FFN achieves this by having independent regression models for each cluster, while this flexibility is intrinsic to the RFR method. Analyses of the estimates shows that the SVR and RFR's respective sensitivity and robustness to outliers define the outcome significantly. Further analyses on the methods were performed by using a synthetic dataset to assess the following: which method (RFR or SVR) has the best performance? What is the effect of using time, latitude and longitude as proxy variables on ΔpCO₂? What is the impact of the sampling bias in the SOCAT v3 dataset on the estimates? We find that while RFR is indeed better than SVR, the ensemble of the two methods outperforms either one, due to complementary strengths and weaknesses of the methods. Results also show that for the RFR and SVR implementations, it is better to include coordinates as proxy variables as RMSE scores are lowered and the phasing of the seasonal cycle is more accurate. Lastly, we show that there is only a weak bias due to undersampling. The synthetic data provide a useful framework to test methods in regions of sparse data coverage and show potential as a useful tool to evaluate methods in future studies.

1256. Three decades of deep water mass investigation in the Weddell Sea (1984-2014): Temporal variability and changes

Authors: Rodrigo Kerr; Tiago S. Dotto; Mauricio M. Mata; Hartmut H. Hellmer

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2017.12.002

URL: <https://doi.org/10.1016/j.dsr2.2017.12.002>

Publication Date: 2017-12-07

Volume/Issue: Volume 149

References: 112

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Brazilian Ministry of the Environment, Brazilian Ministry of Science, Technology, Innovation and Communication, Brazilian National Institute of Science and Technology of Cryosphere (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

1257. Enabling Reanalysis Research Using the Collaborative Reanalysis Technical Environment (CREATE)

Authors: Gerald L. Potter (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Laura Carriere (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Judy Hertz (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Michael Bosilovich (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Daniel Duffy (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Tsengdar Lee (NASA Headquarters, Washington, D.C.); Dean N. Williams (Lawrence Livermore National Laboratory)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/bams-d-17-0174.1

URL: <https://doi.org/10.1175/bams-d-17-0174.1>

Publication Date: 2017-12-04

Volume/Issue: Volume 99, Issue 4

References: 24

Cited by: 15 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper describes the repackaging and consistent distribution of the world's major atmospheric and oceanic reanalyses. It also presents examples of the usefulness of examining multiple reanalyses. This service will make it much easier for anybody using reanalysis to access multiple datasets using an approach similar to that of phase 5 of the Coupled Model Intercomparison Project (CMIP5). Experienced users as well as students will find the standardized formatted data convenient to use.

1258. Stochastic models in the DORIS position time series: estimates for IDS contribution to ITRF2014

Authors: Anna Klos; Janusz Bogusz; Guilhem Moreaux

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-017-1092-0

URL: <https://doi.org/10.1007/s00190-017-1092-0>

Publication Date: 2017-11-30

Volume/Issue: Volume 92, Issue 7

References: 55

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Doris (gastropod)|Stochastic modelling

Funding: Faculty of Civil Engineering and Geodesy MUT statutory research funds, Centre National d'Etudes Spatiales

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1259. Upper Ocean Four-Dimensional Variational Data Assimilation in the Arabian Sea and Bay of Bengal

Authors: Suneet Dwivedi (University of Allahabad); Atul Srivastava (University of Allahabad); Alok Kumar Mishra (University of Allahabad)

Journal: Marine Geodesy

Publisher: Informa UK Limited

DOI: 10.1080/01490419.2017.1405128

URL: <https://doi.org/10.1080/01490419.2017.1405128>

Publication Date: 2017-11-28

Volume/Issue: Volume 41, Issue 3

References: 53

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1260. Lagrangian ocean analysis: Fundamentals and practices

Authors: Erik van Sebille; Stephen M. Griffies; Ryan Abernathey; Thomas P. Adams; Pavel Berloff; Arne Biastoch; Bruno Blanke; Eric P. Chassignet; Yu Cheng; Colin J. Cotter; Eric Deleersnijder; Kristofer Döös; Henri F. Drake; Sybren Drijfhout; Stefan F. Gary; Arnold W. Heemink; Joakim Kjellsson; Inga Monika Koszalka; Michael Lange; Camille Lique; Graeme A. MacGilchrist; Robert Marsh; C. Gabriela Mayorga Adame; Ronan McAdam; Francesco Nencioli; Claire B. Paris; Matthew D. Piggott; Jeff A. Polton; Siren Rühls; Syed H.A.M. Shah; Matthew D. Thomas; Jinbo Wang; Phillip J. Wolfram; Laure Zanna; Jan D. Zika

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2017.11.008

URL: <https://doi.org/10.1016/j.ocemod.2017.11.008>

Publication Date: 2017-11-24

Volume/Issue: Volume 121

References: 302

Cited by: 368 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: EPSRC, European Research Council, U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research (and 4 more)

License: N/A

Open Access: N/A

Abstract: N/A

1261. Transient Variations in Glacial Mass Near Upernavik Isstrøm (West Greenland) Detected by the Combined Use of GPS and GRACE Data

Authors: Bao Zhang (Wuhan University, Chinese University of Hong Kong); Lin Liu (Chinese University of Hong Kong); Shfaqat Abbas Khan (Technical University of Denmark); Tonie van Dam (University of Luxembourg); Enze Zhang (Chinese University of Hong Kong); Yibin Yao (Wuhan University)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jb014529

URL: <https://doi.org/10.1002/2017jb014529>

Publication Date: 2017-11-23

Volume/Issue: Volume 122, Issue 12

References: 66

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Carlsbergfondet, National Aeronautics and Space Administration, National Science Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: The continuously operating Global Positioning System (GPS) sites mounted on bedrock around the coast of Greenland provide important geodetic data sets to quantify the solid Earth's response to historical and present-day ice mass variations. However, the presence of colored noise and irregular seasonal signals makes it difficult to detect transient changes in GPS time series. Here we apply the multichannel singular spectral analysis to the combination of GPS data and Gravity Recovery and Climate Experiment (GRACE) data so that we can identify and fully utilize the spatial correlations from these two independent data sets. Using the GPS and GRACE data near Upernavik Isstrøm in West Greenland as an example, we demonstrate that this method successfully detects two transient signals in ice mass variations during 2008 and 2014. Our forward modeling of loading displacements due to changes in surface mass balance (SMB) and ice dynamics suggests that the transient change starting in mid-2008 was due to the combined contributions from dynamically induced mass loss and SMB. The transient change starting in mid-2011 was mainly due to ablation. Specifically, the ice melted more in 2012 and less in 2013 with little contribution from anomalies in accumulation.

1262. Niño4 as a Key Region for the Interannual Variability of the Western Pacific Warm Pool

Authors: Fan Jia (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Dunxin Hu (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Shijian Hu (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Junqiao Feng (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013208

URL: <https://doi.org/10.1002/2017jc013208>

Publication Date: 2017-11-17

Volume/Issue: Volume 122, Issue 11

References: 38

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation of China, Chinese Academy of Sciences, CAS (and 1 more)

License: N/A

Open Access: N/A

Abstract: The Western Pacific Warm Pool (WPWP) plays an important role in the global climate through modulating deep convections, ENSO, monsoon onsets, etc. Due to the vast spatial range and huge heat storage of the WPWP, near-real-time monitoring of its three-dimensional variations remains challenging. Based on Argo observations and three reanalysis data sets, we find that the Niño4 sea surface temperature (SST) index captures the interannual variability of the WPWP well. The Niño4 SST can explain approximately half of the variance of the WPWP heat content and almost all the variance of the east-west migration of the WPWP. An assessment of 31 CMIP5 models also reveals that models with larger interannual spectral powers and amplitudes of the Niño4 SST tend to simulate larger variations in the heat content and east-west migration of the WPWP. A surface heat budget analysis further shows that the Niño4 SST and WPWP are physically connected through basin-scale horizontal advections of mean temperatures by anomalous horizontal currents, which dominate the interannual variations of both the Niño4 SST and WPWP. Our results indicate that the Niño4 SST can efficiently estimate the interannual WPWP changes and a reliable predictor of the onset time of the South China Sea summer monsoon and Bay of Bengal summer monsoon, without the need to calculate the eastern boundary location and heat content of the WPWP. Moreover, a better simulation of the SST and horizontal currents in the Niño4 region can help to reduce model bias when reproducing the WPWP's interannual variabilities.

1263. Agreement of CMIP5 Simulated and Observed Ocean Anthropogenic CO₂ Uptake

Authors: Benjamin Bronselaer (Geophysical Fluid Dynamics Laboratory Princeton NJ USA, University of Arizona); Michael Winton (Geophysical Fluid Dynamics Laboratory Princeton NJ USA); Joellen Russell (University of Arizona); Christopher L. Sabine (NOAA Pacific Marine Environmental Laboratory Seattle WA USA); Samar Khatiwala (University of Oxford)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl074435

URL: <https://doi.org/10.1002/2017gl074435>

Publication Date: 2017-11-16

Volume/Issue: Volume 44, Issue 24

References: 32

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Southern Ocean Carbon and Climate Observations and Modeling

License: N/A

Open Access: N/A

Abstract: Previous studies found large biases between individual observational and model estimates of historical ocean anthropogenic carbon uptake. We show that the largest bias between the Coupled Model Intercomparison Project phase 5 (CMIP5) ensemble mean and between two observational estimates of ocean anthropogenic carbon is due to a difference in start date. After adjusting the CMIP5 and observational estimates to the 1791-1995 period, all three carbon uptake estimates agree to within 3 Pg of C, about 4% of the total. The CMIP5 ensemble mean spatial bias compared to the observations is generally smaller than the observational error, apart from a negative bias in the Southern Ocean and a positive bias in the Southern Indian and Pacific Oceans compensating each other in the global mean. This dipole pattern is likely due to an equatorward and weak bias in the position of Southern Hemisphere westerlies and lack of mode and intermediate water ventilation.

1264. The Defining Characteristics of ENSO Extremes and the Strong 2015/2016 El Niño

Authors: Agus Santoso (University of New South Wales, Centre for Southern Hemisphere Oceans Research CSIRO Oceans and Atmosphere Hobart Tasmania Australia); Michael J. Mcphaden (Pacific Marine Environmental Laboratory National Oceanic and Atmospheric Administration Seattle WA USA); Wenju Cai (Ocean University of China, Qingdao National Laboratory for Marine Science and Technology, Centre for Southern Hemisphere Oceans Research CSIRO Oceans and Atmosphere Hobart Tasmania Australia)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017rg000560

URL: <https://doi.org/10.1002/2017rg000560>

Publication Date: 2017-11-14

Volume/Issue: Volume 55, Issue 4

References: 284

Cited by: 388 publications

Language: en

Subjects: N/A

Keywords: El Nino, ENSO, extremes, La Nina, Pacific (and 1 more)

Funding: Australian Research Council, Australian Research Council

License: N/A

Open Access: N/A

Abstract: The year 2015 was special for climate scientists, particularly for the El Niño Southern Oscillation (ENSO) research community, as a major El Niño finally materialized after a long pause since the 1997/1998 extreme El Niño. It was scientifically exciting since, due to the short observational record, our knowledge of an extreme El Niño has been based only on the 1982/1983 and 1997/1998 events. The 2015/2016 El Niño was marked by many environmental disasters that are consistent with what is expected for an extreme El Niño. Considering the dramatic impacts of extreme El Niño, and the risk of a potential increase in frequency of ENSO extremes under greenhouse warming, it is timely to evaluate how the recent event fits into our understanding of ENSO extremes. Here we provide a review of ENSO, its nature and dynamics, and through analysis of various observed key variables, we outline the processes that characterize its extremes. The 2015/2016 El Niño brings a useful perspective into the state of understanding of these events and highlights areas for future research. While the 2015/2016 El Niño is characteristically distinct from the 1982/1983 and 1997/1998 events, it still can be considered as the first extreme El Niño of the 21st century. Its extremity can be attributed in part to unusually warm condition in 2014 and to long-term background warming. In effect, this study provides a list of physically meaningful indices that are straightforward to compute for identifying and tracking extreme ENSO events in observations and climate models.

1265. Impacts of the IOD-associated temperature and salinity anomalies on the intermittent equatorial undercurrent anomalies

Authors: Junde Li; Chujin Liang; Youmin Tang; Xiaohui Liu; Tao Lian; Zheqi Shen; Xiaojing Li

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-017-3961-x

URL: <https://doi.org/10.1007/s00382-017-3961-x>

Publication Date: 2017-11-11

Volume/Issue: Volume 51, Issue 4

References: 48

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Key R&D; Program of China, Scientific Research Fund of the Second Institute of Oceanography, SOA, National Natural Science Foundation of China (and 3 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1266. On deep convection events and Antarctic Bottom Water formation in ocean reanalysis products

Authors: Wilton Aguiar; Mauricio M. Mata; Rodrigo Kerr

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-13-851-2017

URL: <https://doi.org/10.5194/os-13-851-2017>

Publication Date: 2017-11-07

Volume/Issue: Volume 13, Issue 6

References: 80

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Conselho Nacional de Desenvolvimento Científico e Tecnológico

License: Creative Commons

Open Access: N/A

Abstract: . Open ocean deep convection is a common source of error in the representation of Antarctic Bottom Water (AABW) formation in ocean general circulation models. Although those events are well described in non-assimilatory ocean simulations, the recent appearance of a massive open ocean polynya in the Estimating the Circulation and Climate of the Ocean Phase II reanalysis product (ECCO2) raises questions on which mechanisms are responsible for those spurious events and whether they are also present in other state-of-the-art assimilatory reanalysis products. To investigate this issue, we evaluate how three recently released high-resolution ocean reanalysis products form AABW in their simulations. We found that two of the products create AABW by open ocean deep convection events in the Weddell Sea that are triggered by the interaction of sea ice with the Warm Deep Water, which shows that the assimilation of sea ice is not enough to avoid the appearance of open ocean polynyas. The third reanalysis, My Ocean University Reading UR025.4, creates AABW using a rather dynamically accurate mechanism. The UR025.4 product depicts both continental shelf convection and the export of Dense Shelf Water to the open ocean. Although the accuracy of the AABW formation in this reanalysis product represents an advancement in the representation of the Southern Ocean dynamics, the differences between the real and simulated processes suggest that substantial improvements in the ocean reanalysis products are still needed to accurately represent AABW formation.

1267. SOMAR-LES: A framework for multi-scale modeling of turbulent stratified oceanic flows

Authors: Vamsi K Chalamalla; Edward Santilli; Alberto Scotti; Masoud Jalali; Sutanu Sarkar

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2017.11.003

URL: <https://doi.org/10.1016/j.ocemod.2017.11.003>

Publication Date: 2017-11-07

Volume/Issue: Volume 120

References: 71

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF, ONR

License: N/A

Open Access: N/A

Abstract: N/A

1268. Regional Changes in Icescape Impact Shelf Circulation and Basal Melting

Authors: E. A. Cougnon (Antarctic Climate and Ecosystems Cooperative Research Centre, University of Tasmania); B. K. Galton-Fenzi (Antarctic Climate and Ecosystems Cooperative Research Centre, Australian Antarctic Division Kingston Tasmania Australia); S. R. Rintoul (Ocean and Atmosphere Commonwealth Scientific and Industrial Research Organisation Hobart Tasmania Australia, Antarctic Climate and Ecosystems Cooperative Research Centre, Centre for Southern Hemisphere Oceans Research Hobart Tasmania Australia); B. Legrésy (Ocean and Atmosphere Commonwealth Scientific and Industrial Research Organisation Hobart Tasmania Australia, Antarctic Climate and Ecosystems Cooperative Research Centre); G. D. Williams (Antarctic Climate and Ecosystems Cooperative Research Centre, University of Tasmania); A. D. Fraser (Antarctic Climate and Ecosystems Cooperative Research Centre); J. R. Hunter (Antarctic Climate and Ecosystems Cooperative Research Centre)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl074943

URL: <https://doi.org/10.1002/2017gl074943>

Publication Date: 2017-11-02

Volume/Issue: Volume 44, Issue 22

References: 57

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: regional icescape changes, Antarctica, calving, iceberg, Mertz Glacier (and 1 more)

Funding: Australian Government's Cooperative Research Centre Program, Earth System and Climate Change Hub of the National Environmental Science Program, Centre for Southern Hemisphere Ocean Research (and 2 more)

License: N/A

Open Access: N/A

Abstract: Ice shelf basal melt is the dominant contribution to mass loss from Antarctic ice shelves. However, the sensitivity of basal melt to changes in icescape (grounded icebergs, ice shelves, and sea ice) and related ocean circulation is poorly understood. Here we simulate the impact of the major 2010 calving event of the Mertz Glacier Tongue (MGT), East Antarctica, and related redistribution of sea ice and icebergs on the basal melt rate of the local ice shelves. We find that the position of the grounded tabular iceberg B9B controls the water masses that reach the nearby ice shelf cavities. After the calving of the MGT and the removal of B9B, warmer water is present both within the MGT cavity and on the continental shelf driving a 57% increase of the deep MGT basal melting. Major changes in icescape influence the oceanic heat flux responsible for basal ice shelf melting.

1269. Submesoscale Sea Ice-Ocean Interactions in Marginal Ice Zones

Authors: Georgy E. Manucharyan (California Institute of Technology); Andrew F. Thompson (California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc012895

URL: <https://doi.org/10.1002/2017jc012895>

Publication Date: 2017-11-02

Volume/Issue: Volume 122, Issue 12

References: 93

Cited by: 91 publications

Language: en

Subjects: N/A

Keywords: ocean heat flux, submesoscale eddies, melt water fronts, marginal ice zone, localized upwelling (and 1 more)

Funding: NSF

License: N/A

Open Access: N/A

Abstract: Signatures of ocean eddies, fronts, and filaments are commonly observed within marginal ice zones (MIZs) from satellite images of sea ice concentration, and in situ observations via ice-tethered profilers or underice gliders. However, localized and intermittent sea ice heating and advection by ocean eddies are currently not accounted for in climate models and may contribute to their biases and errors in sea ice forecasts. Here, we explore mechanical sea ice interactions with underlying submesoscale ocean turbulence. We demonstrate that the release of potential energy stored in meltwater fronts can lead to energetic submesoscale motions along MIZs with spatial scales $O(10\text{ km})$ and Rossby numbers $O(1)$. In low-wind conditions, cyclonic eddies and filaments efficiently trap the sea ice and advect it over warmer surface ocean waters where it can effectively melt. The horizontal eddy diffusivity of sea ice mass and heat across the MIZ can reach $O(200\text{ m}^2\text{ s}^{-1})$. Submesoscale ocean variability also induces large vertical velocities (order 10 m d^{-1}) that can bring relatively warm subsurface waters into the mixed layer. The ocean-sea ice heat fluxes are localized over cyclonic eddies and filaments reaching about 100 W m^{-2} . We speculate that these submesoscale-driven intermittent fluxes of heat and sea ice can contribute to the seasonal evolution of MIZs. With the continuing global warming and sea ice thickness reduction in the Arctic Ocean, submesoscale sea ice-ocean processes are expected to become increasingly prominent.

1270. Regional Sea Level Variability and Trends, 1960-2007: A Comparison of Sea Level Reconstructions and Ocean Syntheses

Authors: M. Carson (University of Hamburg); A. Köhl (University of Hamburg); D. Stammer (University of Hamburg); B. Meyssignac (LEGOS, CNES Toulouse France); J. Church (University of New South Wales); J. Schröter (Alfred Wegener Institute); M. Wenzel (Alfred Wegener Institute); B. Hamlington (Old Dominion University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc012992

URL: <https://doi.org/10.1002/2017jc012992>

Publication Date: 2017-10-25

Volume/Issue: Volume 122, Issue 11

References: 54

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: DFG, BMBF

License: N/A

Open Access: N/A

Abstract: Several existing statistical and dynamical reconstructions of past regional sea level variability and trends are compared with each other and with tide gauges over the 48 year period 1960-2007, partially predating the satellite altimetry era. Evaluated statistical reconstructions were built from tide-gauge data (TGR), and dynamical reconstructions from ocean data assimilation (ODA) approaches. Although most of the TGRs yield global-mean time series of sea level with trends deviating within only ± 0.1 mm yr⁻¹, the spatial anomalies of the trends deviate substantially between the reconstructions over the period predating altimetry. In contrast, TGRs match observed regional trend patterns fairly well during the satellite altimetry era. TGRs match tide-gauge data better than ODA results; however, they exhibit less variability in the open ocean compared to altimetric data. Over the prealtimetry period, all reconstructed regional sea level trend patterns deviate substantially from each other. In terms of detrended correlations in this earlier period, the reconstructions match tide gauges, and each other, much better in the Pacific than in the Atlantic. An ensemble of all TGR and ODA estimates provides some improvements in correlations and trends to both tide gauges and altimetry. Nevertheless, a lack of independent open ocean sea surface height data predating altimetry makes impossible the validation of the ensemble for prealtimetry open ocean sea level trends and variability. Estimating regional sea level changes prior to altimetry therefore remains an unsolved challenge.

1271. Investigating the role of air-sea forcing on the variability of hydrography, circulation, and mixed layer depth in the Arabian Sea and Bay of Bengal

Authors: Atul Srivastava; Suneet Dwivedi; Alok Kumar Mishra

Journal: Oceanologia

Publisher: Instytut Oceanologii PAN

DOI: 10.1016/j.oceano.2017.10.001

URL: <https://doi.org/10.1016/j.oceano.2017.10.001>

Publication Date: 2017-10-23

Volume/Issue: Volume 60, Issue 2

References: 46

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Indian Space Research Organisation, Ministry of Earth Sciences, Department of Science and Technology, Ministry of Science and Technology, India

License: N/A

Open Access: N/A

Abstract: N/A

1272. Regional ocean forecasting systems and their applications: Design considerations of such a system for the South China Sea

Authors: Xiaochun Wang (Nanjing University, Joint Institute for Regional Earth System Science and Engineering, UCLA, 607 Charles E. Young Drive East, Young Hall, Room 4242, Los Angeles, California 90095, USA); Liqing Zhao (Nanjing University); Zhijin Li (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Aquatic Ecosystem Health & Management

Publisher: Michigan State University Press

DOI: 10.1080/14634988.2015.1112123

URL: <https://doi.org/10.1080/14634988.2015.1112123>

Publication Date: 2017-10-18

Volume/Issue: Volume 18, Issue 4

References: 45

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Essential components and technical details of regional ocean forecasting systems configured from the Regional Ocean Modeling System are discussed with the goal of bridging the gap between user and ocean modeling communities. Recent development of these systems and applications are also surveyed. Design considerations of such a system for the South China Sea are discussed, based on regional dynamic characteristics and potential applications.

1273. Contrasting carbon cycle responses of the tropical continents to the 2015-2016 El Niño

Authors: Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicolas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zhe Jiang (National Center for Atmospheric Research); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Debra Wunch (University of Toronto); Christian Frankenberg (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ying Sun (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christopher W. O'Dell (Colorado State University); Kevin R. Gurney (Arizona State University); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michelle Gierach (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David Crisp (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Annmarie Eldering (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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References: 96

Cited by: 331 publications

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Subjects: N/A

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Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: INTRODUCTION The influence of El Niño on climate is accompanied by large changes to the carbon cycle, and El Niño-induced variability in the carbon cycle has been attributed mainly to the tropical continents. However, owing to a dearth of observations in the tropics, tropical carbon fluxes are poorly quantified, and considerable debate exists over the dominant mechanisms (e.g., plant growth, respiration, fire) and regions (e.g., humid versus semiarid tropics) on the net carbon balance.

RATIONALE The launch of the Orbiting Carbon Observatory-2 (OCO-2) shortly before the 2015-2016 El Niño, the second strongest since the 1950s, has provided an opportunity to understand how tropical

land carbon fluxes respond to the warm and dry climate characteristics of El Niño conditions. The El Niño events may also provide a natural experiment to study the response of tropical land carbon fluxes to future climate changes, because anomalously warm and dry tropical environments typical of El Niño are expected to be more frequent under most emission scenarios. RESULTS The tropical regions of three continents (South America, Asia, and Africa) had heterogeneous responses to the 2015-2016 El Niño, in terms of both climate drivers and the carbon cycle. The annual mean precipitation over tropical South America and tropical Asia was lower by 3.0σ and 2.8σ , respectively, in 2015 relative to the 2011 La Niña year. Tropical Africa, on the other hand, had near equal precipitation and the same number of dry months between 2015 and 2011; however, surface temperatures were higher by 1.6σ , dominated by the positive anomaly over its eastern and southern regions. In response to the warmer and drier climate anomaly in 2015, the pantropical biosphere released 2.5 ± 0.34 gigatons more carbon into the atmosphere than in 2011, which accounts for 83.3% of the global total 3.0 -gigatons of carbon (gigatons C) net biosphere flux differences and 92.6% of the atmospheric CO₂ growth-rate differences between 2015 and 2011. It indicates that the tropical land biosphere flux anomaly was the driver of the highest atmospheric CO₂ growth rate in 2015. The three tropical continents had an approximately even contribution to the pantropical net carbon flux anomaly in 2015, but had diverse dominant processes: gross primary production (GPP) reduced carbon uptake (0.9 ± 0.96 gigatons C) in tropical South America, fire increased carbon release (0.4 ± 0.08 gigatons C) in tropical Asia, and respiration increased carbon release (0.6 ± 1.01 gigatons C) in Africa. We found that most of the excess carbon release in 2015 was associated with either extremely low precipitation or high temperatures, or both. CONCLUSION Our results indicate that the global El Niño effect is a superposition of regionally specific effects. The heterogeneous climate forcing and carbon response over the three tropical continents to the 2015-2016 El Niño challenges previous studies that suggested that a single dominant process determines carbon cycle interannual variability, which could also be due to previous disturbance and soil and vegetation structure. The similarity between the 2015 tropical climate anomaly and the projected climate changes imply that the role of the tropical land as a buffer for fossil fuel emissions may be reduced in the future. The heterogeneous response may reflect differences in temperature and rainfall anomalies, but intrinsic differences in vegetation species, soils, and prior disturbance may contribute as well. A synergistic use of multiple satellite observations and a long time series of spatially resolved fluxes derived from sustained satellite observations will enable tests of these hypotheses, allow for a more process-based understanding, and, ultimately, aid improved carbon-climate model projections. Diverse climate driver anomalies and carbon cycle responses to the 2015-2016 El Niño over the three tropical continents. Schematic of climate anomaly patterns over the three tropical continents and the anomalies of the net carbon flux and its dominant constituent flux (i.e., GPP, respiration, and fire) relative to the 2011 La Niña during the 2015-2016 El Niño. GtC, gigatons C.

1274. Constraining the Global Ocean Heat Content Through Assimilation of CERES-Derived TOA Energy Imbalance Estimates

Authors: Andrea Storto (Euro-Mediterranean Center on Climate Change (CMCC) Bologna Italy); Chunxue Yang (Institute of Atmospheric Sciences and Climate (CNR-ISAC), (CNR) Bologna Italy); Simona Masina (National Institute for Geophysics and Volcanology Bologna Italy, Euro-Mediterranean Center on Climate Change (CMCC) Bologna Italy)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl075396

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Volume/Issue: Volume 44, Issue 20

References: 47

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Seventh Framework Programme

License: N/A

Open Access: N/A

Abstract: The Earth's energy imbalance (EEI) is stored in the oceans for the most part. Thus, estimates of its variability can be ingested in ocean retrospective analyses to constrain the global ocean heat budget. Here we propose a scheme to assimilate top of the atmosphere global radiation imbalance estimates from Clouds and the Earth's Radiant Energy System (CERES) in a coarse-resolution variational ocean reanalysis system (2000-2014). The methodology proves able to shape the heat content tendencies according to the EEI estimates, without compromising the reanalysis accuracy. Spurious variability and underestimation (overestimation) present in experiments with in situ (no) data assimilation disappear when EEI data are assimilated. The warming hiatus present without the assimilation of EEI data is mitigated, inducing ocean warming at depths below 1,500 m and slightly larger in the Southern Hemisphere, in accordance with recent studies. Furthermore, the methodology may be applied to Earth System reanalyses and climate simulations to realistically constrain the global energy budget.

1275. Nonhydrostatic Simulations of Tide-Induced Mixing in the Halmahera Sea: A Possible Role in the Transformation of the Indonesian Throughflow Waters

Authors: Taira Nagai (University of Tokyo); Toshiyuki Hibiya (University of Tokyo); Pascale Bouruet-Aubertot (Laboratoire d'Océanographie et du Climat: Expérimentations et Approches Numériques Institut Pierre Simon Laplace (LOCEAN-IPSL), Sorbonnes Universités, UPMC Univ. Paris 06 Paris France)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013381

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Publication Date: 2017-10-10

Volume/Issue: Volume 122, Issue 11

References: 18

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Indonesian Throughflow (ITF) waters are significantly transformed within the Indonesian Archipelago and consequently influence the large-scale ocean circulation such as Agulhas and Leeuwin Currents. Existing ocean general circulation models (OGCMs) are, however, incapable of reproducing the transformation of the ITF waters, since tidal forcing is neglected in such models. In the present study, we first conduct high-resolution nonhydrostatic three-dimensional numerical experiments focusing on the transformation of the ITF waters in the Halmahera Sea which is thought to be the most important bottleneck in simulating the ITF water mass properties. It is shown that intensive vertical mixing induced by breaking of internal tides in the shallow regions in the Halmahera Sea dilutes the ITF waters, significantly reducing model biases found in the existing OGCMs. We next evaluate quantitatively the effect of tide-induced vertical mixing on the transformation of the ITF waters. It is shown that tide-induced vertical mixing dominates the transformation of the ITF waters, although some supplementary processes such as horizontal mixing associated with the submesoscale eddies resulting from tidal interaction with land configurations cannot be ignored.

1276. Marine Pollution and Climate Change

Authors: No authors listed

Journal: Unknown Journal

Publisher: CRC Press

DOI: 10.1201/9781315119243

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Publication Date: 2017-10-06

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References: N/A

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1277. Tidal Variability Related to Sea Level Variability in the Pacific Ocean

Authors: Adam T. Devlin (Chinese University of Hong Kong, Portland State University); David A. Jay (Portland State University); Edward D. Zaron (Portland State University); Stefan A. Talke (Portland State University); Jiayi Pan (Nanjing University, Chinese University of Hong Kong); Hui Lin (Chinese University of Hong Kong)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013165

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Publication Date: 2017-09-30

Volume/Issue: Volume 122, Issue 11

References: 95

Cited by: 69 publications

Language: en

Subjects: N/A

Keywords: coastal defense, nuisance flooding, sea level rise, seasonality, climate change (and 1 more)

Funding: National Science Foundation, Hong Kong University of Science and Technology, National Natural Science Foundation of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: Ocean tides are changing worldwide for reasons unrelated to astronomical forcing. Changes in tidal properties coupled with altered mean sea level (MSL) may yield higher peak water levels and increased occurrence of short-term exceedance events, such as storm surge and nuisance flooding. Here we investigate the hypothesis that changes in relative sea level are correlated with alterations in tidal amplitudes. Our approach focuses on the correlation between short-term (monthly to interannual) fluctuations in sea level with changes in tidal properties of major ocean tides (M2, and K1; S2 and O1) at 152 gauges. Results suggest that sea level variability is correlated to interannual tidal variability at most (92%) of tide gauges in the Pacific, with statistically significant rates between +/-10 and +/-500 mm per meter sea level rise observed. These tidal anomalies, while influenced by basin-scale climate processes and sea level changes, appear to be locally forced (in part) and not coherent over amphidromic or basin-wide scales. Overall, the Western Pacific shows a greater concentration of tide/sea level correlations at interannual time scales than the Eastern Pacific; 44% and 46% of gauges are significant in K1 and O1 in the west compared to 29% and 30% in the east, and 63% and 53% of gauges in the west are significant in M2 and S2 versus 47% and 32% in the east. Seasonal variation in tidal properties is less apparent in the empirical record, with statistically significant seasonal variations observed at only 35% of all gauges, with the largest concentrations in Southeast Asia.

1278. Sea ice assimilation into a coupled ocean-sea ice model using its adjoint

Authors: Nikolay V. Koldunov; Armin Köhl; Nuno Serra; Detlef Stammer

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-11-2265-2017

URL: <https://doi.org/10.5194/tc-11-2265-2017>

Publication Date: 2017-09-29

Volume/Issue: Volume 11, Issue 5

References: 56

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Seventh Framework Programme, Deutsche Forschungsgemeinschaft

License: Creative Commons

Open Access: N/A

Abstract: . Satellite sea ice concentrations (SICs), together with several ocean parameters, are assimilated into a regional Arctic coupled ocean-sea ice model covering the period of 2000-2008 using the adjoint method. There is substantial improvement in the representation of the SIC spatial distribution, in particular with respect to the position of the ice edge and to the concentrations in the central parts of the Arctic Ocean during summer months. Seasonal cycles of total Arctic sea ice area show an overall improvement. During summer months, values of sea ice extent (SIE) integrated over the model domain become underestimated compared to observations, but absolute differences of mean SIE to the data are reduced in nearly all months and years. Along with the SICs, the sea ice thickness fields also become closer to observations, providing added value by the assimilation. Very sparse ocean data in the Arctic, corresponding to a very small contribution to the cost function, prevent sizable improvements of assimilated ocean variables, with the exception of the sea surface temperature.

1279. Evaluation of the Impact of Argo Data on Ocean Reanalysis in the Pacific Region

Authors: Xuefeng Zhang (Key Laboratory of Marine Environmental Information Technology, National Marine Data and Information Service, State Oceanic Administration, Tianjin, China, College of Automation, Harbin Engineering University, Harbin, China); Chaohui Sun (Second Institute of Oceanography); Chang Liu (College of Automation, Harbin Engineering University, Harbin, China); Lianxin Zhang (Key Laboratory of Marine Environmental Information Technology, National Marine Data and Information Service, State Oceanic Administration, Tianjin, China); Caixia Shao (Key Laboratory of Marine Environmental Information Technology, National Marine Data and Information Service, State Oceanic Administration, Tianjin, China); Xiaoshuang Zhang (Key Laboratory of Marine Environmental Information Technology, National Marine Data and Information Service, State Oceanic Administration, Tianjin, China); Yuxin Zhao (College of Automation, Harbin Engineering University, Harbin, China)

Journal: Advances in Meteorology

Publisher: Wiley

DOI: 10.1155/2017/7314106

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References: 18

Cited by: 2 publications

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Keywords: N/A

Funding: National Basic Research Program of China, National Natural Science Foundation of China, National Natural Science Foundation of China (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Observing System Simulation Experiments (OSSEs) have been conducted to evaluate the effect of Argo data assimilation on ocean reanalysis in the Pacific region. The "truth" is obtained from a 5-year model integration from 2003 to 2007 based on the MIT general circulation model with the truly varying atmospheric forcing. The "observations" are the projections of the truth onto the observational network including ocean station data, CTD, and various BTs and Argo, by adding white noise to simulate observational errors. The data assimilation method employed is a sequential three-dimensional variational (3D-Var) scheme within a multigrid framework. Results show the interannual variability of temperature, salinity, and current fields can be reconstructed fairly well. The spread of temperature anomalies in the tropical Pacific region is also able to be reflected accurately when Argo data is assimilated, which may provide a reliable initial field for the forecast of temperature and currents for the subsurface in the tropical Pacific region. The adjustment of salinity by using T-S relationship is vital in the tropical Pacific region. However, the adjustment of salinity is almost meaningless in the northwest Pacific if Argo data is included during the reanalysis.

1280. Acoustic Tomography in the Canary Basin: Meddies and Tides

Authors: Brian D. Dushaw (Nansen Environmental and Remote Sensing Center); Fabienne Gaillard (Institut de Recherche pour le Développement (IRD)); Thierry Terre (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013356

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Volume/Issue: Volume 122, Issue 11

References: 57

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: An acoustic propagation experiment over 308 km range conducted in the Canary Basin in 1997-1998 was used to assess the ability of ocean acoustic tomography to measure the flux of Mediterranean water and Meddies. Instruments on a mooring adjacent to the acoustic path measured the southwestward passage of a strong Meddy in temperature, salinity, and current. Over 9 months of transmissions, the acoustic arrival pattern was an initial broad stochastic pulse varying in duration by 250-500 ms, followed eight stable, identified-ray arrivals. Small-scale sound speed fluctuations from Mediterranean water parcels littered around the sound channel axis caused acoustic scattering. Internal waves contributed more modest acoustic scattering. Based on simulations, the main effect of a Meddy passing across the acoustic path is the formation of many early-arriving, near-axis rays, but these rays are thoroughly scattered by the small-scale Mediterranean-water fluctuations. A Meddy decreases the deep-turning ray travel times by 10-30 ms. The dominant acoustic signature of a Meddy is therefore the expansion of the width of the initial stochastic pulse. While this signature appears inseparable from the other effects of Mediterranean water in this region, the acoustic time series indicates the steady passage of Mediterranean water across the acoustic path. Tidal variations caused by the mode-1 internal tides were measured by the acoustic travel times. The observed internal tides were partly predicted using a recent global model for such tides derived from satellite altimetry.

1281. The Recent Atlantic Cold Anomaly: Causes, Consequences, and Related Phenomena

Authors: Simon A. Josey (University of Southampton, National Oceanography Centre); Joel J.-M. Hirschi (University of Southampton, National Oceanography Centre); Bablu Sinha (University of Southampton, National Oceanography Centre); Aurélie Duchez (University of Southampton, National Oceanography Centre); Jeremy P. Grist (University of Southampton, National Oceanography Centre); Robert Marsh (University of Southampton)

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

DOI: 10.1146/annurev-marine-121916-063102

URL: <https://doi.org/10.1146/annurev-marine-121916-063102>

Publication Date: 2017-09-21

Volume/Issue: Volume 10, Issue 1

References: 110

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Cold ocean temperature anomalies have been observed in the mid- to high-latitude North Atlantic on interannual to centennial timescales. Most notably, a large region of persistently low surface temperatures accompanied by a sharp reduction in ocean heat content was evident in the subpolar gyre from the winter of 2013-2014 to 2016, and the presence of this feature at a time of pervasive warming elsewhere has stimulated considerable debate. Here, we review the role of air-sea interaction and ocean processes in generating this cold anomaly and place it in a longer-term context. We also discuss the potential impacts of surface temperature anomalies for the atmosphere, including the North Atlantic Oscillation and European heat waves; contrast the behavior of the Atlantic with the extreme warm surface event that occurred in the North Pacific over a similar timescale; and consider the possibility that these events represent a response to a change in atmospheric planetary wave forcing.

1282. Global Ocean Vertical Velocity From a Dynamically Consistent Ocean State Estimate

Authors: Xinfeng Liang (University of South Florida); Michael Spall (Woods Hole Oceanographic Institution); Carl Wunsch (Harvard University, Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc012985

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Volume/Issue: Volume 122, Issue 10

References: 45

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, vertical velocity, vertical transport, vertical exchange, climate change (and 1 more)

Funding: National Science Foundation, National Oceanic and Atmospheric Administration

License: N/A

Open Access: N/A

Abstract: Estimates of the global ocean vertical velocities (Eulerian, eddy-induced, and residual) from a dynamically consistent and data-constrained ocean state estimate are presented and analyzed. Conventional patterns of vertical velocity, Ekman pumping, appear in the upper ocean, with topographic dominance at depth. Intense and vertically coherent upwelling and downwelling occur in the Southern Ocean, which are likely due to the interaction of the Antarctic Circumpolar Current and large-scale topographic features and are generally canceled out in the conventional zonally averaged results. These "elevators" at high latitudes connect the upper to the deep and abyssal oceans and working together with isopycnal mixing are likely a mechanism, in addition to the formation of deep and abyssal waters, for fast responses of the deep and abyssal oceans to the changing climate. Also, Eulerian and parameterized eddy-induced components are of opposite signs in numerous regions around the global ocean, particularly in the ocean interior away from surface and bottom. Nevertheless, residual vertical velocity is primarily determined by the Eulerian component, and related to winds and large-scale topographic features. The current estimates of vertical velocities can serve as a useful reference for investigating the vertical exchange of ocean properties and tracers, and its complex spatial structure ultimately permits regional tests of basic oceanographic concepts such as Sverdrup balance and coastal upwelling/downwelling.

1283. Bio-physical characterisation of polynyas as a key foraging habitat for juvenile male southern elephant seals (*Mirounga leonina*) in Prydz Bay, East Antarctica

Authors: Veda Malpress; Sophie Bestley; Stuart Corney; Dirk Welsford; Sara Labrousse; Michael Sumner; Mark Hindell

Journal: PLOS ONE

Publisher: Public Library of Science (PLoS)

DOI: 10.1371/journal.pone.0184536

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References: 77

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1284. Dynamical Links between the Decadal Variability of the Oyashio and Kuroshio Extensions

Authors: Bo Qiu (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Shuiming Chen (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Niklas Schneider (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0397.1

URL: <https://doi.org/10.1175/jcli-d-17-0397.1>

Publication Date: 2017-09-08

Volume/Issue: Volume 30, Issue 23

References: 57

Cited by: 56 publications

Language: en

Subjects: N/A

Keywords: Boundary currents, Advection, Fronts, Oceanic variability, Atmosphere-ocean interaction (and 1 more)

Funding: National Science Foundation, National Aeronautics and Space Administration, U.S. Department of Energy (and 1 more)

License: N/A

Open Access: N/A

Abstract: Rather than a single and continuous boundary current outflow, long-term satellite observations reveal that the Oyashio Extension (OE) in the North Pacific Subarctic Gyre comprises two independent, northeast-southwest-slanted front systems. With a mean latitude along 40 degreesN, the western OE front exists primarily west of 153 degreesE and is a continuation of the subarctic gyre western boundary current. The eastern OE front, also appearing along 40 degreesN, is located between 153 degrees and 170 degreesE, whose entity is disconnected from its western counterpart. During 1982-2016, both of the OE fronts exhibit prominent decadal fluctuations, although their signals show little contemporaneous correlation. An upper-ocean temperature budget analysis based on the Estimating the Circulation and Climate of the Ocean, phase II (ECCO2), state estimate reveals that the advective temperature flux convergence plays a critical role in determining the low-frequency temperature changes relating to the OE fronts. Specifically, the western OE front variability is controlled by the decadal mesoscale eddy modulations in the upstream Kuroshio Extension (KE). An enhanced eddy activity increases the poleward heat transport and works to strengthen the western OE front. The eastern OE front variability, on the other hand, is dictated by both the meridional shift of the KE position and the circulation intensity change immediately north of the eastern OE. Different baroclinic adjustment speeds for the KE and OE are found to cause the in-phase changes between these latter two processes. Lack of contemporaneous correlation between the decadal western and eastern OE variability is found to be related to the interaction of the meridionally migrating KE jet with the Shatsky Rise near 159 degreesE.

1285. On the Benefit of Current and Future ALPS Data for Improving Arctic Coupled Ocean-Sea Ice State Estimation

Authors: Unknown Author; An Nguyen; Victor Ocaña; Vikram Garg; Patrick Heimbach; John Toole; Richard Krishfield; Craig Lee; Luc Rainville

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2017.223

URL: <https://doi.org/10.5670/oceanog.2017.223>

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Volume/Issue: Volume 30, Issue 2

References: N/A

Cited by: 17 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1286. Improved estimations of low-degree coefficients using GPS displacements with reduced non-loading errors

Authors: Na Wei (Wuhan University); Chuang Shi (Wuhan University, Beihang University); Guangxing Wang (Faculty of Information Engineering, China University of Geosciences, 430074 Wuhan, Hubei, China); Jingnan Liu (Wuhan University)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggx357

URL: <https://doi.org/10.1093/gji/ggx357>

Publication Date: 2017-09-01

Volume/Issue: Volume 212, Issue 2

References: 43

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

1287. Spectral decomposition of internal gravity wave sea surface height in global models

Authors: Anna C. Savage (University of Michigan); Brian K. Arbic (University of Michigan); Matthew H. Alford (Scripps Institution of Oceanography / University of California San Diego); Joseph K. Ansong (University of Michigan); J. Thomas Farrar (Woods Hole Oceanographic Institution); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Amanda K. O'Rourke (University of Michigan); James G. Richman (Florida State University); Jay F. Shriver (Oceanography Division, Naval Research Laboratory Stennis Space Center Mississippi USA); Gunnar Voet (Scripps Institution of Oceanography / University of California San Diego); Alan J. Wallcraft (Oceanography Division, Naval Research Laboratory Stennis Space Center Mississippi USA); Luis Zamudio (Florida State University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013009

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Publication Date: 2017-08-31

Volume/Issue: Volume 122, Issue 10

References: 66

Cited by: 87 publications

Language: en

Subjects: N/A

Keywords: internal tides, internal gravity waves, sea surface height variability, high-resolution ocean models

Funding: National Aeronautics and Space Administration, Office of Naval Research, National Science Foundation

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Open Access: N/A

Abstract: Two global ocean models ranging in horizontal resolution from 1/12 degrees to 1/48 degrees are used to study the space and time scales of sea surface height (SSH) signals associated with internal gravity waves (IGWs). Frequency-horizontal wavenumber SSH spectral densities are computed over seven regions of the world ocean from two simulations of the HYbrid Coordinate Ocean Model (HYCOM) and three simulations of the Massachusetts Institute of Technology general circulation model (MITgcm). High wavenumber, high-frequency SSH variance follows the predicted IGW linear dispersion curves. The realism of high-frequency motions ($>$) in the models is tested through comparison of the frequency spectral density of dynamic height variance computed from the highest-resolution runs of each model (1/25 degrees HYCOM and 1/48 degrees MITgcm) with dynamic height variance frequency spectral density computed from nine in situ profiling instruments. These high-frequency motions are of particular interest because of their contributions to the small-scale SSH variability that will be observed on a global scale in the upcoming Surface Water and Ocean Topography (SWOT) satellite altimetry mission. The variance at supertidal frequencies can be comparable to the tidal and low-frequency variance for high wavenumbers (length scales smaller than ~ 50 km), especially in the higher-resolution simulations. In the highest-resolution simulations, the high-frequency variance can be greater than the low-frequency variance at these scales.

1288. Moored observations of the Deep Western Boundary Current in the NWAatlantic: 2004-2014

Authors: John M. Toole (Woods Hole Oceanographic Institution); Magdalena Andres (Woods Hole Oceanographic Institution); Isabela A. Le Bras (Woods Hole Oceanographic Institution); Terrence M. Joyce (Woods Hole Oceanographic Institution); Michael S. McCartney (Woods Hole Oceanographic Institution)

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Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc012984

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References: 56

Cited by: 49 publications

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Subjects: N/A

Keywords: Deep Western Boundary Current, Atlantic Meridional Overturning Circulation

Funding: G. Unger Vetlesen Foundation, Woods Hole Oceanographic Institution, National Science Foundation

License: N/A

Open Access: N/A

Abstract: A moored array spanning the continental slope southeast of Cape Cod sampled the equatorward-flowing Deep Western Boundary Current (DWBC) for a 10 year period: May 2004 to May 2014. Daily profiles of subinertial velocity, temperature, salinity, and neutral density are constructed for each mooring site and cross-line DWBC transport time series are derived for specified water mass layers. Time-averaged transports based on daily estimates of the flow and density fields in Stream coordinates are contrasted with those derived from the Eulerian-mean flow field, modes of DWBC transport variability are investigated through compositing, and comparisons are made to transport estimates for other latitudes. Integrating the daily velocity estimates over the neutral density range of 27.8-28.125 kg/m³ (encompassing Labrador Sea and Overflow Water layers), a mean equatorward DWBC transport of $22.8 \times 10^6 \pm 1.9 \times 10^6$ m³/s is obtained. Notably, a statistically significant trend of decreasing equatorward transport is observed in several of the DWBC components as well as the current as a whole. The largest linear change (a 4% decrease per year) is seen in the layer of Labrador Sea Water that was renewed by deep convection in the early 1990s whose transport fell from 9.0×10^6 m³/s at the beginning of the field program to 5.8×10^6 m³/s at its end. The corresponding linear fit to the combined Labrador Sea and Overflow Water DWBC transport decreases from 26.4×10^6 to 19.1×10^6 m³/s. In contrast, no long-term trend is observed in upper ocean Slope Water transport. These trends are discussed in the context of decadal observations of the North Atlantic circulation, and subpolar air-sea interaction/water mass transformation.

1289. Decadal-Mean Impact of Including Ocean Surface Currents in Bulk Formulas on Surface Air-Sea Fluxes and Ocean General Circulation

Authors: Yang Wu (Hohai University); Xiaoming Zhai (University of East Anglia); Zhaomin Wang (Hohai University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-17-0001.1

URL: <https://doi.org/10.1175/jcli-d-17-0001.1>

Publication Date: 2017-08-30

Volume/Issue: Volume 30, Issue 23

References: 42

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: Surface fluxes, Atmosphere-ocean interaction, Ocean circulation, Climate models, Wind stress

Funding: Royal Society, National Natural Science Foundation of China, National Natural Science Foundation of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: The decadal-mean impact of including ocean surface currents in the bulk formulas on surface air-sea fluxes and the ocean general circulation is investigated for the first time using a global eddy-permitting coupled ocean-sea ice model. Although including ocean surface currents in air-sea flux calculations only weakens the surface wind stress by a few percent, it significantly reduces wind power input to both geostrophic and ageostrophic motions, and damps the eddy and mean kinetic energy throughout the water column. Furthermore, the strength of the horizontal gyre circulations and the Atlantic meridional overturning circulation are found to decrease considerably (by 10%-15% and ~13%, respectively). As a result of the weakened ocean general circulation, the maximum northward global ocean heat transport decreases by about 0.2 PW, resulting in a lower sea surface temperature and reduced surface heat loss in the northern North Atlantic. Additional sensitivity model experiments further demonstrate that it is including ocean surface currents in the wind stress calculation that dominates this decadal impact, with including ocean surface currents in the turbulent heat flux calculations making only a minor contribution. These results highlight the importance of properly accounting for ocean surface currents in surface air-sea fluxes in modeling the ocean circulation and climate.

1290. Evaluation of Korean-Manufactured Non-Tactical Vehicle JP-8 Conversion Demonstration Conducted by the 19th Theatre Area Army Command, Korea

Authors: Ruben A. Alvarez; Douglas M. Yost; Bernard R. Wright

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada395540

URL: <https://doi.org/10.21236/ada395540>

Publication Date: 2017-08-25

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1291. Mechanisms underlying recent decadal changes in subpolar North Atlantic Ocean heat content

Authors: Christopher G. Piecuch (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Christopher M. Little (Atmospheric and Environmental Research Inc.); Martha W. Buckley (George Mason University); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc012845

URL: <https://doi.org/10.1002/2017jc012845>

Publication Date: 2017-08-25

Volume/Issue: Volume 122, Issue 9

References: 91

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: ocean state estimation, Atlantic multidecadal variability, decadal variability, heat content, overturning circulation (and 1 more)

Funding: National Oceanic and Atmospheric Administration, National Science Foundation

License: N/A

Open Access: N/A

Abstract: The subpolar North Atlantic (SPNA) is subject to strong decadal variability, with implications for surface climate and its predictability. In 2004-2005, SPNA decadal upper ocean and sea-surface temperature trends reversed from warming during 1994-2004 to cooling over 2005-2015. This recent decadal trend reversal in SPNA ocean heat content (OHC) is studied using a physically consistent, observationally constrained global ocean state estimate covering 1992-2015. The estimate's physical consistency facilitates quantitative causal attribution of ocean variations. Closed heat budget diagnostics reveal that the SPNA OHC trend reversal is the result of heat advection by midlatitude ocean circulation. Kinematic decompositions reveal that changes in the deep and intermediate vertical overturning circulation cannot account for the trend reversal, but rather ocean heat transports by horizontal gyre circulations render the primary contributions. The shift in horizontal gyre advection reflects anomalous circulation acting on the mean temperature gradients. Maximum covariance analysis (MCA) reveals strong covariation between the anomalous horizontal gyre circulation and variations in the local wind stress curl, suggestive of a Sverdrup response. Results have implications for decadal predictability.

1292. Ocean forced variability of Totten Glacier mass loss

Authors: Jason Roberts (University of Tasmania, Australian Antarctic Division, Hobart, Tasmania, Australia); Benjamin K. Galton-Fenzi (University of Tasmania, Australian Antarctic Division, Hobart, Tasmania, Australia); Fernando S. Paolo (Scripps Institution of Oceanography / University of California San Diego); Claire Donnelly (University of Bristol); David E. Gwyther (University of Tasmania); Laurie Padman (Earth & Space Research, Corvallis, Oregon, USA); Duncan Young (University of Texas at Austin); Roland Warner (University of Tasmania); Jamin Greenbaum (University of Texas at Austin); Helen A. Fricker (Scripps Institution of Oceanography / University of California San Diego); Antony J. Payne (University of Bristol); Stephen Cornford (University of Bristol); Anne Le Brocq (University of Exeter); Tas van Ommen (University of Tasmania, Australian Antarctic Division, Hobart, Tasmania, Australia); Don Blankenship (University of Texas at Austin); Martin J. Siegert (Imperial College London)

Journal: Geological Society, London, Special Publications

Publisher: Geological Society of London

DOI: 10.1144/sp461.6

URL: <https://doi.org/10.1144/sp461.6>

Publication Date: 2017-08-24

Volume/Issue: Volume 461, Issue 1

References: 55

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: A large volume of the East Antarctic Ice Sheet drains through the Totten Glacier (TG) and is thought to be a potential source of substantial global sea-level rise over the coming centuries. We show that the surface velocity and height of the floating part of the TG, which buttresses the grounded component, have varied substantially over two decades (1989-2011), with variations in surface height strongly anti-correlated with simulated basal melt rates ($r = 0.70$, $p < 0.05$). Coupled glacier-ice shelf simulations confirm that ice flow and thickness respond to both basal melting of the ice shelf and grounding on bed obstacles. We conclude the observed variability of the TG is primarily ocean-driven. Ocean warming in this region will lead to enhanced ice-sheet dynamism and loss of upstream grounded ice.

1293. Detection of sea level fingerprints derived from GRACE gravity data

Authors: Chia-Wei Hsu (University of California Irvine); Isabella Velicogna (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl074070

URL: <https://doi.org/10.1002/2017gl074070>

Publication Date: 2017-08-22

Volume/Issue: Volume 44, Issue 17

References: 48

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: land hydrology, sea level fingerprint, barystatic sea level, glaciers and ice caps, ice sheet mass balance (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Mass changes of ice sheets, glaciers and ice caps, land water hydrology, atmosphere, and ocean cause a nonuniform sea level rise due to the self-attraction and loading effects called sea level fingerprints (SLF). SLF have been previously derived from a combination of modeled and observed mass fluxes from the continents into the ocean. Here we derive improved SLF from time series of time variable gravity data from the Gravity Recovery and Climate Experiment (GRACE) mission for April 2002 to October 2014. We evaluate the GRACE-derived SLF using ocean bottom pressure (OBP) data from stations in the tropics, where OBP errors are the lowest. We detect the annual phase of the SLF in the OBP signal and separate it unambiguously from the barystatic sea level (BSL) at two stations. At the basin scale, the SLF explain a larger fraction of the variance in steric-corrected altimetry than the BSL, which has implications for evaluating mass transport between ocean basins.

1294. Numerical simulation of the observed near-surface East India Coastal Current on the continental slope

Authors: A. Mukherjee; D. Shankar; Abhisek Chatterjee; P. N. Vinayachandran

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-017-3856-x

URL: <https://doi.org/10.1007/s00382-017-3856-x>

Publication Date: 2017-08-22

Volume/Issue: Volume 50, Issue 11-12

References: 121

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Council of Scientific and Industrial Research, Indian National Centre for Ocean Information Services

License: N/A

Open Access: N/A

Abstract: N/A

1295. JTRF2014, the JPL Kalman filter and smoother realization of the International Terrestrial Reference System

Authors: Claudio Abbondanza (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Toshio M. Chin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Richard S. Gross (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael B. Heflin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jay W. Parker (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Benedikt S. Soja (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tonie van Dam (Faculté des Sciences, de la Technologie et de la Communication Université du Luxembourg Luxembourg City Luxembourg); Xiaoping Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jb014360

URL: <https://doi.org/10.1002/2017jb014360>

Publication Date: 2017-08-17

Volume/Issue: Volume 122, Issue 10

References: 80

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: geocenter motion, GNSS VLBI SLR DORIS, time series, Earth rotation, Terrestrial Reference Frames (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We present and discuss JTRF2014, the Terrestrial Reference Frame (TRF) the Jet Propulsion Laboratory constructed by combining space-geodetic inputs from very long baseline interferometry (VLBI), satellite laser ranging (SLR), Global Navigation Satellite Systems (GNSS), and Doppler orbitography and radiopositioning integrated by satellite submitted for the realization of ITRF2014. Determined through a Kalman filter and Rauch-Tung-Striebel smoother assimilating position observations, Earth orientation parameters, and local ties, JTRF2014 is a subsecular, time series-based TRF whose origin is at the quasi-instantaneous center of mass (CM) as sensed by SLR and whose scale is determined by the quasi-instantaneous VLBI and SLR scales. The dynamical evolution of the positions accounts for a secular motion term, annual, and semiannual periodic modes. Site-dependent variances based on the analysis of loading displacements induced by mass redistributions of terrestrial fluids have been used to control the extent of random walk adopted in the combination. With differences in the amplitude of the annual signal within the range 0.5-0.8 mm, JTRF2014-derived center of network-to-center of mass (CM-CN) is in remarkable agreement with the geocenter motion obtained via spectral inversion of GNSS, Gravity Recovery and Climate Experiment

(GRACE) observations and modeled ocean bottom pressure from Estimating the Circulation and Climate of the Ocean (ECCO). Comparisons of JTRF2014 to ITRF2014 suggest high-level consistency with time derivatives of the Helmert transformation parameters connecting the two frames below 0.18 mm/yr and weighted root-mean-square differences of the polar motion (polar motion rate) in the order of 30 mas (17 mas/d).

1296. Observations and modeling of ocean-induced melt beneath Petermann Glacier Ice Shelf in northwestern Greenland

Authors: Cilan Cai (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yoshihiro Nakayama (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl073711

URL: <https://doi.org/10.1002/2017gl073711>

Publication Date: 2017-08-17

Volume/Issue: Volume 44, Issue 16

References: 37

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We update observationally based estimates of subaqueous melt, Q_m , beneath Petermann Glacier Ice Shelf (PGIS), Greenland, and model its sensitivity to oceanic thermal forcing, TF, and subglacial runoff, Q_{sg} , using the Massachusetts Institute of Technology general circulation model (MITgcm), in a two-dimensional domain, with 20 m vertical and 40 m horizontal resolution at the grounding line. We adjust the drag coefficient to match the observationally based Q_m . With the inclusion of Q_{sg} , the maximum melt rate ($\dot{m}$) is 2 times larger in summer and 1/3 larger annually than in winter. $\dot{m}$ increases above linear with TF and below linear with Q_{sg} . We estimate that $\dot{m}$ increased by 24% (+8.1 m/yr) beneath PGIS from the 1990s to the 2000s from a 0.21 degreesC warming in ocean temperature and a doubling in Q_{sg} , hence contributing to its thinning. If the PGIS is removed, we estimate that the modeled melt rate near the grounding line will increase 13-16 times.

1297. A data assimilating model for estimating Southern Ocean biogeochemistry

Authors: A. Verdy (Scripps Institution of Oceanography); M. R. Mazloff (Scripps Institution of Oceanography)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012650

URL: <https://doi.org/10.1002/2016jc012650>

Publication Date: 2017-08-15

Volume/Issue: Volume 122, Issue 9

References: 76

Cited by: 142 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, state estimation, biogeochemistry

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: A Biogeochemical Southern Ocean State Estimate (B-SOSE) is introduced that includes carbon and oxygen fields as well as nutrient cycles. The state estimate is constrained with observations while maintaining closed budgets and obeying dynamical and thermodynamic balances. Observations from profiling floats, shipboard data, underway measurements, and satellites are used for assimilation. The years 2008-2012 are chosen due to the relative abundance of oxygen observations from Argo floats during this time. The skill of the state estimate at fitting the data is assessed. The agreement is best for fields that are constrained with the most observations, such as surface pCO₂ in Drake Passage (44% of the variance captured) and oxygen profiles (over 60% of the variance captured at 200 and 1000 m). The validity of adjoint method optimization for coupled physical-biogeochemical state estimation is demonstrated with a series of gradient check experiments. The method is shown to be mature and ready to synthesize in situ biogeochemical observations as they become more available. Documenting the B-SOSE configuration and diagnosing the strengths and weaknesses of the solution informs usage of this product as both a climate baseline and as a way to test hypotheses. Transport of Intermediate Waters across 32 degreesS supplies significant amounts of nitrate to the Atlantic Ocean (5.57 +/- 2.94 Tmol yr⁻¹) and Indian Ocean (5.09 +/- 3.06 Tmol yr⁻¹), but much less nitrate reaches the Pacific Ocean (1.78 +/- 1.91 Tmol yr⁻¹). Estimates of air-sea carbon dioxide fluxes south of 50 degreesS suggest a mean uptake of 0.18 Pg C/yr for the time period analyzed.

1298. On the Relationship between Regional Ocean Heat Content and Sea Surface Height

Authors: John T. Fasullo (National Center for Atmospheric Research); Peter R. Gent (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0920.1

URL: <https://doi.org/10.1175/jcli-d-16-0920.1>

Publication Date: 2017-08-15

Volume/Issue: Volume 30, Issue 22

References: 36

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Directorate for Geosciences, National Aeronautics and Space Administration, U.S. Department of Energy

License: N/A

Open Access: N/A

Abstract: An accurate diagnosis of ocean heat content (OHC) is essential for interpreting climate variability and change, as evidenced for example by the broad range of hypotheses that exists for explaining the recent hiatus in global mean surface warming. Potential insights are explored here by examining relationships between OHC and sea surface height (SSH) in observations and two recently available large ensembles of climate model simulations from the mid-twentieth century to 2100. It is found that in decadal-length observations and a model control simulation with constant forcing, strong ties between OHC and SSH exist, with little temporal or spatial complexity. Agreement is particularly strong on monthly to interannual time scales. In contrast, in forced transient warming simulations, important dependencies in the relationship exist as a function of region and time scale. Near Antarctica, low-frequency SSH variability is driven mainly by changes in the circumpolar current associated with intensified surface winds, leading to correlations between OHC and SSH that are weak and sometimes negative. In subtropical regions, and near other coastal boundaries, negative correlations are also evident on long time scales and are associated with the accumulated effects of changes in the water cycle and ocean dynamics that underlie complexity in the OHC relationship to SSH. Low-frequency variability in observations is found to exhibit similar negative correlations. Combined with altimeter data, these results provide evidence that SSH increases in the Indian and western Pacific Oceans during the hiatus are suggestive of substantial OHC increases. Methods for developing the applicability of altimetry as a constraint on OHC more generally are also discussed.

1299. Space and time variability of the Southern Ocean carbon budget

Authors: Isabella Rosso (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Ariane Verdy (Scripps Institution of Oceanography / University of California San Diego); Lynne D. Talley (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012646

URL: <https://doi.org/10.1002/2016jc012646>

Publication Date: 2017-08-11

Volume/Issue: Volume 122, Issue 9

References: 89

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, carbon budget, state estimate

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: The upper ocean dissolved inorganic carbon (DIC) concentration is regulated by advective and diffusive transport divergence, biological processes, freshwater, and air-sea CO₂ fluxes. The relative importance of these mechanisms in the Southern Ocean is uncertain, as year-round observations in this area have been limited. We use a novel physical-biogeochemical state estimate of the Southern Ocean to construct a closed DIC budget of the top 650 m and investigate the spatial and temporal variability of the different components of the carbon system. The dominant mechanisms of variability in upper ocean DIC depend on location and time and space scales considered. Advective transport is the most influential mechanism and governs the local DIC budget across the 10 day-5 year timescales analyzed. Diffusive effects are nearly negligible. The large-scale transport structure is primarily set by upwelling and downwelling, though both the lateral ageostrophic and geostrophic transports are significant. In the Antarctic Circumpolar Current, the carbon budget components are also influenced by the presence of topography and biological hot spots. In the subtropics, evaporation and air-sea CO₂ flux primarily balances the sink due to biological production and advective transport. Finally, in the subpolar region sea ice processes, which change the seawater volume and thus the DIC concentration, compensate the large impact of the advective transport and modulate the timing of biological activity and air-sea CO₂ flux.

1300. A multi-scale high-resolution analysis of global sea surface temperature

Authors: Toshio Michael Chin; Jorge Vazquez-Cuervo; Edward M. Armstrong

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2017.07.029

URL: <https://doi.org/10.1016/j.rse.2017.07.029>

Publication Date: 2017-08-11

Volume/Issue: Volume 200

References: 61

Cited by: 273 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1301. On the vertical structure and stability of the Lofoten vortex in the Norwegian Sea

Authors: I.L. Bashmachnikov; M.A. Sokolovskiy; T.V. Belonenko; D.L. Volkov; P.E. Isachsen; X. Carton

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2017.08.001

URL: <https://doi.org/10.1016/j.dsr.2017.08.001>

Publication Date: 2017-08-10

Volume/Issue: Volume 128

References: 104

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Saint Petersburg State University, Université de Bretagne Occidentale, LabexMER (and 10 more)

License: N/A

Open Access: N/A

Abstract: N/A

1302. North Atlantic Deep Water formation inhibits high Arctic contamination by continental perfluorooctane sulfonate discharges

Authors: Xianming Zhang (Harvard University, Department of Environmental Health Harvard T. H. Chan School of Public Health Boston Massachusetts USA); Yanxu Zhang (Harvard University); Clifton Dassuncao (Harvard University, Department of Environmental Health Harvard T. H. Chan School of Public Health Boston Massachusetts USA); Rainer Lohmann (University of Rhode Island); Elsie M. Sunderland (Harvard University, Department of Environmental Health Harvard T. H. Chan School of Public Health Boston Massachusetts USA)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gb005624

URL: <https://doi.org/10.1002/2017gb005624>

Publication Date: 2017-08-04

Volume/Issue: Volume 31, Issue 8

References: 78

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Natural Sciences and Engineering Research Council of Canada, Natural Sciences and Engineering Research Council of Canada (and 1 more)

License: N/A

Open Access: N/A

Abstract: Perfluorooctane sulfonate (PFOS) is an aliphatic fluorinated compound with eight carbon atoms that is extremely persistent in the environment and can adversely affect human and ecological health. The stability, low reactivity, and high water solubility of PFOS combined with the North American phaseout in production around the year 2000 make it a potentially useful new tracer for ocean circulation. Here we characterize processes affecting the lifetime and accumulation of PFOS in the North Atlantic Ocean and transport to sensitive Arctic regions by developing a 3-D simulation within the MITgcm. The model captures variability in measurements across biogeographical provinces ($R^2 = 0.90$, $p = 0.01$). In 2015, the North Atlantic PFOS reservoir was equivalent to 60% of cumulative inputs from the North American and European continents (1400 Mg). Cumulative inputs to the Arctic accounted for 30% of continental discharges, while the remaining 10% was transported to the tropical Atlantic and other regions. PFOS concentrations declined rapidly after 2002 in the surface mixed layer (half-life: 1-2 years) but are still increasing below 1000 m depth. During peak production years (1980-2000), plumes of PFOS-enriched seawater were transported to the sub-Arctic in energetic surface ocean currents. However, Atlantic Meridional Overturning Circulation (AMOC) and deep ocean transport returned a substantial fraction of this northward transport (20%, 530 Mg) to southern latitudes and reduced cumulative inputs to the Arctic (730 Mg) by 70%. Weakened AMOC due to climate change is thus likely to increase the magnitude of persistent bioaccumulative pollutants entering the Arctic Ocean.

1303. Towards Improved Ocean State Estimation

Authors: Steven Jayne

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada455141

URL: <https://doi.org/10.21236/ada455141>

Publication Date: 2017-08-01

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1304. Getting the right wind-forcing for an ecosystem model: A case study from the eastern Arabian Sea

Authors: Kunal Chakraborty (Indian National Centre for Ocean Information Services, Hyderabad, India); Nimit Kumar (Indian National Centre for Ocean Information Services, Hyderabad, India); G. V. M. Gupta (Ministry of Earth Sciences, Centre for Marine Living Resources and Ecology, Kochi, India)

Journal: Journal of Operational Oceanography

Publisher: Informa UK Limited

DOI: 10.1080/1755876x.2017.1354686

URL: <https://doi.org/10.1080/1755876x.2017.1354686>

Publication Date: 2017-07-31

Volume/Issue: Volume 10, Issue 2

References: 61

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Indian National Centre for Ocean Information Services, Ministry of Earth Sciences

License: N/A

Open Access: N/A

Abstract: N/A

1305. Robustness of observation-based decadal sea level variability in the Indo-Pacific Ocean

Authors: A. G. Nidheesh (Institut de Recherche pour le Développement (IRD), CSIR-National Institute of Oceanography Goa India, Sorbonne University); M. Lengaigne (Institut de Recherche pour le Développement (IRD), Sorbonne University); J. Vialard (Institut de Recherche pour le Développement (IRD), Sorbonne University); T. Izumo (Institut de Recherche pour le Développement (IRD), Sorbonne University); A. S. Unnikrishnan (CSIR-National Institute of Oceanography Goa India); B. Meyssignac (Institut de Recherche pour le Développement (IRD)); B. Hamlington (Old Dominion University); C. de Boyer Montegut (Institut de Recherche pour le Développement (IRD))

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl073955

URL: <https://doi.org/10.1002/2017gl073955>

Publication Date: 2017-07-31

Volume/Issue: Volume 44, Issue 14

References: 49

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: "Mordicus" Agence Nationale pour la Recherche (ANR), "SeaLevelALK" Centre National d'Etudes Spatiales (CNES)

License: N/A

Open Access: N/A

Abstract: We examine the consistency of Indo-Pacific decadal sea level variability in 10 gridded, observation-based sea level products for the 1960-2010 period. Decadal sea level variations are robust in the Pacific, with more than 50% of variance explained by decadal modulation of two flavors of El Niño-Southern Oscillation (classical ENSO and Modoki). Amplitude of decadal sea level variability is weaker in the Indian Ocean than in the Pacific. All data sets indicate a transmission of decadal sea level signals from the western Pacific to the northwest Australian coast through the Indonesian throughflow. The southern tropical Indian Ocean sea level variability is associated with decadal modulations of ENSO in reconstructions but not in reanalyses or in situ data set. The Pacific-independent Indian Ocean decadal sea level variability is not robust but tends to be maximum in the southwestern tropical Indian Ocean. The inconsistency of Indian Ocean decadal variability across the sea level products calls for caution in making definitive conclusions on decadal sea level variability in this basin.

1306. A Consortium for Ocean Circulation and Climate Estimation

Authors: Detlef Stammer

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada441744

URL: <https://doi.org/10.21236/ada441744>

Publication Date: 2017-07-28

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1307. North Pacific Acoustic Laboratory: Analysis of Shadow Zone Arrivals and Acoustic Propagation in Numerical Ocean Models

Authors: Brian Dushaw

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada494669

URL: <https://doi.org/10.21236/ada494669>

Publication Date: 2017-07-27

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1308. Dynamical ocean response to projected changes of the global water cycle

Authors: Xin Liu (University of Hamburg); Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013061

URL: <https://doi.org/10.1002/2017jc013061>

Publication Date: 2017-07-22

Volume/Issue: Volume 122, Issue 8

References: 56

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft

License: N/A

Open Access: N/A

Abstract: Over the next century substantial changes will occur in the ocean as a consequence of an accelerated global hydrological cycle and the associated net surface freshwater flux change is projected to result from global warming. This paper is concerned with the dynamical response to the associated surface volume flux anomalies. Based on ocean model runs driven by RCP8.5 surface freshwater flux anomalies over the period 2081-2100 relative to 1986-2005, we show that the adjustment of the circulation involves a barotropic circulation response as predicted from the Goldsbrough-Stommel theory. The corresponding barotropic circulation intensifies by approximately 20% with a stronger intensification of about 50% in the Southern Ocean, comparing to the present-day Goldsbrough-Stommel Circulation. The barotropic circulation anomaly induced by intensified freshwater flux reaches to 0.6 Sv in the Antarctic Circumpolar Current region. The adjustment also involves changes in the meridional overturning circulation mirroring the basin-wide averages of changes in the convergence and divergence of the mass transport driven by the surface volume flux. The subsequent pathways of fresh water match with the spreading of volume flux in the shallow cells but diverge substantially with depth. Associated with changes of the flow field are the changes in meridional heat and freshwater transports. Changes in the circulation also lead to a redistribution of temperature and salinity from which a significant contribution results in form of regional steric sea level changes. These changes are of the order of 0.5 cm and can be largely attributed to the displacement of the isopycnals.

1309. On the representation of capsizing in iceberg models

Authors: Till J.W. Wagner; Alon A. Stern; Rebecca W. Dell; Ian Eisenman

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2017.07.003

URL: <https://doi.org/10.1016/j.ocemod.2017.07.003>

Publication Date: 2017-07-21

Volume/Issue: Volume 117

References: 40

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1310. "Climate response functions" for the Arctic Ocean: a proposed coordinated modelling experiment

Authors: John Marshall; Jeffery Scott; Andrey Proshutinsky

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-10-2833-2017

URL: <https://doi.org/10.5194/gmd-10-2833-2017>

Publication Date: 2017-07-21

Volume/Issue: Volume 10, Issue 7

References: 44

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A coordinated set of Arctic modelling experiments, which explore how the Arctic responds to changes in external forcing, is proposed. Our goal is to compute and compare climate response functions (CRFs) - the transient response of key observable indicators such as sea-ice extent, freshwater content of the Beaufort Gyre, etc. - to abrupt step changes in forcing fields across a number of Arctic models. Changes in wind, freshwater sources, and inflows to the Arctic basin are considered. Convolutions of known or postulated time series of these forcing fields with their respective CRFs then yield the (linear) response of these observables. This allows the project to inform, and interface directly with, Arctic observations and observers and the climate change community. Here we outline the rationale behind such experiments and illustrate our approach in the context of a coarse-resolution model of the Arctic based on the MITgcm. We conclude by summarizing the expected benefits of such an activity and encourage other modelling groups to compute CRFs with their own models so that we might begin to document their robustness to model formulation, resolution, and parameterization.

1311. Controllability, not chaos, key criterion for ocean state estimation

Authors: Geoffrey Gebbie; Tsung-Lin Hsieh

Journal: Nonlinear Processes in Geophysics

Publisher: Copernicus GmbH

DOI: 10.5194/npg-24-351-2017

URL: <https://doi.org/10.5194/npg-24-351-2017>

Publication Date: 2017-07-19

Volume/Issue: Volume 24, Issue 3

References: 64

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Lagrange multiplier method for combining observations and models (i.e., the adjoint method or 4D-VAR) has been avoided or approximated when the numerical model is highly nonlinear or chaotic. This approach has been adopted primarily due to difficulties in the initialization of low-dimensional chaotic models, where the search for optimal initial conditions by gradient-descent algorithms is hampered by multiple local minima. Although initialization is an important task for numerical weather prediction, ocean state estimation usually demands an additional task - a solution of the time-dependent surface boundary conditions that result from atmosphere-ocean interaction. Here, we apply the Lagrange multiplier method to an analogous boundary control problem, tracking the trajectory of the forced chaotic pendulum. Contrary to previous assertions, it is demonstrated that the Lagrange multiplier method can track multiple chaotic transitions through time, so long as the boundary conditions render the system controllable. Thus, the nonlinear timescale poses no limit to the time interval for successful Lagrange multiplier-based estimation. That the key criterion is controllability, not a pure measure of dynamical stability or chaos, illustrates the similarities between the Lagrange multiplier method and other state estimation methods. The results with the chaotic pendulum suggest that nonlinearity should not be a fundamental obstacle to ocean state estimation with eddy-resolving models, especially when using an improved first-guess trajectory.

1312. Terrorism Knowledge Base (TKB)

Authors: Douglas Lenat; Chris Deaton

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada481100

URL: <https://doi.org/10.21236/ada481100>

Publication Date: 2017-07-14

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1313. A Prototype User Interface for Coarse-Grained Desktop Access Control

Authors: A. C. Long; Courtney Moskowitz; Greg Ganger

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada488514

URL: <https://doi.org/10.21236/ada488514>

Publication Date: 2017-07-13

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1314. Investigation of the Role of Transient Growth in Roughness-Induced Transition

Authors: Edward B. White; Eli Reshotko

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada483692

URL: <https://doi.org/10.21236/ada483692>

Publication Date: 2017-07-13

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1315. Trends and interannual variability of mass and steric sea level in the Tropical Asian Seas

Authors: Marcel Kleinherenbrink (Delft University of Technology); Riccardo Riva (Delft University of Technology); Thomas Frederikse (Delft University of Technology); Mark Merrifield (Department of Oceanography University of Hawaii Honolulu Hawaii USA); Yoshihide Wada (Utrecht University, International Institute for Applied Systems Analysis Laxenburg Austria, National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Columbia University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc012792

URL: <https://doi.org/10.1002/2017jc012792>

Publication Date: 2017-07-13

Volume/Issue: Volume 122, Issue 8

References: 68

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: steric sea level, Tropical Asian Seas, reanalysis, Jason, GRACE (and 1 more)

Funding: Netherlands Organisation for Scientific Research

License: Creative Commons

Open Access: N/A

Abstract: The mass and steric components of sea level changes have been separated in the Tropical Asian Seas (TAS) using a statistically optimal combination of Jason satellite altimetry, GRACE satellite gravimetry, and ocean reanalyses. Using observational uncertainties, statistically optimally weighted time series for both components have been obtained in four regions within the TAS over the period January 2005 to December 2012. The mass and steric sea level variability is regressed with the first two principal components (PC1&2) of Pacific equatorial wind stress and the Dipole Mode Index (DMI). Sea level in the South China Sea is not affected by any of the indices. Steric variability in the TAS is largest in the deep Banda and Celebes seas and is affected by both PCs and the DMI. Mass variability is largest on the continental shelves, which is primarily controlled by PC1. We argue that a water flux from the Western Tropical Pacific Ocean is the cause for mass variability in the TAS. The steric trends are about 2 mm yr⁻¹ larger than the mass trends in the TAS. A significant part of the mass trend can be explained by the aforementioned indices and the nodal cycle. Trends obtained from fingerprints of mass redistribution are statistically equal to mass trends after subtracting the nodal cycle and the indices. Ultimately, the effect of omitting the TAS in global sea level budgets is estimated to be 0.3 mm yr⁻¹.

1316. On the meridional ageostrophic transport in the tropical Atlantic

Authors: Yao Fu; Johannes Karstensen; Peter Brandt

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-13-531-2017

URL: <https://doi.org/10.5194/os-13-531-2017>

Publication Date: 2017-07-06

Volume/Issue: Volume 13, Issue 4

References: 53

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft

License: Creative Commons

Open Access: N/A

Abstract: . The meridional Ekman volume, heat, and salt transport across two trans-Atlantic sections near 14.5 degrees N and 11 degrees S were estimated using in situ observations, wind products, and model data. A meridional ageostrophic velocity was obtained as the difference between the directly measured total velocity and the geostrophic velocity derived from observations. Interpreting the section mean ageostrophy to be the result of an Ekman balance, the meridional Ekman transport of 6.2 ± 2.3 Sv northward at 14.5 degrees N and 11.7 ± 2.1 Sv southward at 11 degrees S is estimated. The integration uses the top of the pycnocline as an approximation for the Ekman depth, which is on average about 20 m deeper than the mixed layer depth. The Ekman transport estimated based on the velocity observations agrees well with the predictions from in situ wind stress data of 6.7 ± 3.5 Sv at 14.5 degrees N and 13.6 ± 3.3 Sv at 11 degrees S. The meridional Ekman heat and salt fluxes calculated from sea surface temperature and salinity data or from high-resolution temperature and salinity profile data differ only marginally. The errors in the Ekman heat and salt flux calculation were dominated by the uncertainty of the Ekman volume transport estimates.

1317. Sea-ice deformation in a coupled ocean-sea-ice model and in satellite remote sensing data

Authors: Gunnar Spreen; Ron Kwok; Dimitris Menemenlis; An T. Nguyen

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-11-1553-2017

URL: <https://doi.org/10.5194/tc-11-1553-2017>

Publication Date: 2017-07-04

Volume/Issue: Volume 11, Issue 4

References: 56

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A realistic representation of sea-ice deformation in models is important for accurate simulation of the sea-ice mass balance. Simulated sea-ice deformation from numerical simulations with 4.5, 9, and 18 km horizontal grid spacing and a viscous-plastic (VP) sea-ice rheology are compared with synthetic aperture radar (SAR) satellite observations (RGPS, RADARSAT Geophysical Processor System) for the time period 1996-2008. All three simulations can reproduce the large-scale ice deformation patterns, but small-scale sea-ice deformations and linear kinematic features (LKFs) are not adequately reproduced. The mean sea-ice total deformation rate is about 40 % lower in all model solutions than in the satellite observations, especially in the seasonal sea-ice zone. A decrease in model grid spacing, however, produces a higher density and more localized ice deformation features. The 4.5 km simulation produces some linear kinematic features, but not with the right frequency. The dependence on length scale and probability density functions (PDFs) of absolute divergence and shear for all three model solutions show a power-law scaling behavior similar to RGPS observations, contrary to what was found in some previous studies. Overall, the 4.5 km simulation produces the most realistic divergence, vorticity, and shear when compared with RGPS data. This study provides an evaluation of high and coarse-resolution viscous-plastic sea-ice simulations based on spatial distribution, time series, and power-law scaling metrics.

1318. Impact of Geophysical and Datum Corrections on Absolute Sea-Level Trends from Tide Gauges around Taiwan, 1993-2015

Authors: Wen-Hau Lan (Department of Geomatics, National Cheng Kung University, Tainan 70101, Taiwan); Chung-Yen Kuo (Department of Geomatics, National Cheng Kung University, Tainan 70101, Taiwan); Huan-Chin Kao (Department of Geomatics, National Cheng Kung University, Tainan 70101, Taiwan); Li-Ching Lin (Academia Sinica); C. Shum (Ohio State University, Chinese Academy of Sciences); Kuo-Hsin Tseng (National Central University); Jung-Chieh Chang (Department of Soil and Water Conservation, National Chung Hsing University, Taichung 402, Taiwan)

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w9070480

URL: <https://doi.org/10.3390/w9070480>

Publication Date: 2017-07-03

Volume/Issue: Volume 9, Issue 7

References: 50

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The Taiwanese government has established a complete tide gauge network along the coastline for accurate sea-level monitoring. In this study, we analyze several factors impacting the determination of absolute or geocentric sea-level trends--including ocean tides, inverted barometer effect, datum shift, and vertical land motion--using tide gauge records near Taiwan, from 1993-2015. The results show that datum shifts and vertical land motion have a significant impact on sea-level trends with a respective average contribution of 7.3 and 8.0 mm/yr, whereas ocean tides and inverted barometer effects have a relatively minor impact, representing 9% and 14% of the observed trend, respectively. These results indicate that datum shifts and vertical land motion effects have to be removed in the tide gauge records for accurate sea-level estimates. Meanwhile, the estimated land motions show that the southwestern plain has larger subsidence rates, for example, the Bozilliao, Dongshi, and Wengang tide gauge stations exhibit a rate of 24-31 mm/yr as a result of groundwater pumping. We find that the absolute sea-level trends around Taiwan derived from tide gauges or satellite altimetry agree well with each other, and are estimated to be 2.2 mm/yr for 1993-2015, which is significantly slower than the global average sea-level rise trend of 3.2 mm/yr from satellite altimeters. Finally, a recent hiatus in sea-level rise in this region exhibits good agreement with the interannual and decadal variabilities associated with the El Niño-Southern Oscillation and Pacific Decadal Oscillation.

1319. Interannual Variability of Sea Surface Temperature in the Southwest Pacific and the Role of Ocean Dynamics

Authors: Melissa Bowen (School of Environment, University of Auckland, Auckland, New Zealand); Jordan Markham (School of Environment, University of Auckland, Auckland, New Zealand); Philip Sutton (National Institute of Water and Atmospheric Research, Wellington, New Zealand); Xuebin Zhang (CSIRO Oceans and Atmospheres, Hobart, Australia); Quran Wu (CSIRO Oceans and Atmospheres, Hobart, Australia); Nick T. Shears (Leigh Marine Laboratory, Institute of Marine Science, University of Auckland, Auckland, New Zealand); Denise Fernandez (National Institute of Water and Atmospheric Research, Wellington, New Zealand)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0852.1

URL: <https://doi.org/10.1175/jcli-d-16-0852.1>

Publication Date: 2017-06-26

Volume/Issue: Volume 30, Issue 18

References: 41

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: Interannual variability, Ekman pumping/transport, Surface temperature, Rossby waves, Atmosphere-ocean interaction (and 1 more)

Funding: New Zealand Deep South National Science Challenge, Australian National Environmental Science Programme

License: N/A

Open Access: N/A

Abstract: This paper investigates the mechanisms causing interannual variability of upper ocean heat content and sea surface temperature (SST) in the southwest Pacific. Using the ECCOv4 ocean reanalysis it is shown that air-sea heat flux and ocean heat transport convergence due to ocean dynamics both contribute to the variability of upper ocean temperatures around New Zealand. The ocean dynamics responsible for the ocean heat transport convergence are investigated. It is shown that SSTs are significantly correlated with the arrival of barotropic Rossby waves estimated from the South Pacific wind stress over the latitudes of New Zealand. Both Argo observations and the ECCOv4 reanalysis show deep isotherms fluctuate coherently around the country. The authors suggest that the depth of the thermocline around New Zealand adjusts to changes in the South Pacific winds, modifies the vertical advection of heat into the upper ocean, and contributes to the interannual variability of SST in the region.

1320. Low-Pass Filtering, Heat Flux, and Atlantic Multidecadal Variability

Authors: Mark A. Cane (Columbia University); Amy C. Clement (University of Miami); Lisa N. Murphy (University of Miami); Katinka Bellomo (Columbia University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0810.1

URL: <https://doi.org/10.1175/jcli-d-16-0810.1>

Publication Date: 2017-06-22

Volume/Issue: Volume 30, Issue 18

References: 47

Cited by: 69 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration (US) and Columbia University, Office of Naval Research, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: Abstract In this model study the authors explore the possibility that the internal component of the Atlantic multidecadal oscillation (AMO) sea surface temperature (SST) signal is indistinguishable from the response to white noise forcing from the atmosphere and ocean. Here, complex models are compared without externally varying forcing with a one-dimensional noise-driven model for SST. General analytic expressions are obtained for both unfiltered and low-pass filtered lead-lag correlations. It is shown that this simple model reproduces many of the simulated lead-lag relationships among temperature, rate of change of temperature, and surface heat flux. It is concluded that the finding that at low frequencies the ocean loses heat to the atmosphere when the temperature is warm, which has been interpreted as showing that the ocean circulation drives the AMO, is a necessary consequence of the fact that at long periods the net heat flux (ocean plus atmosphere) is zero to a good approximation. It does not distinguish between the atmosphere and ocean as the source of the AMO and is consistent with the hypothesis that the AMO is driven by white noise heat fluxes. It is shown that some results in the literature are artifacts of low-pass filtering, which creates spurious low-frequency signals when the underlying data are white or red noise. It is concluded that in the absence of external forcing the AMO in most GCMs is consistent with being driven by white noise, primarily from the atmosphere.

1321. The Changing El Niño-Southern Oscillation and Associated Climate Extremes

Authors: Jin-Yi Yu; Xin Wang; Song Yang; Houk Paek; Mengyan Chen

Journal: Geophysical Monograph Series

Publisher: Wiley

DOI: 10.1002/9781119068020.ch1

URL: <https://doi.org/10.1002/9781119068020.ch1>

Publication Date: 2017-06-19

Volume/Issue: N/A

References: 346

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1322. Biogeochemical versus ecological consequences of modeled ocean physics

Authors: Sophie Clayton; Stephanie Dutkiewicz; Oliver Jahn; Christopher Hill; Patrick Heimbach; Michael J. Follows

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-14-2877-2017

URL: <https://doi.org/10.5194/bg-14-2877-2017>

Publication Date: 2017-06-16

Volume/Issue: Volume 14, Issue 11

References: 44

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Gordon and Betty Moore Foundation, Division of Ocean Sciences, Simmons Family Foundation (and 3 more)

License: Creative Commons

Open Access: N/A

Abstract: . We present a systematic study of the differences generated by coupling the same ecological-biogeochemical model to a 1 degrees, coarse-resolution, and 1/6 degrees, eddy-permitting, global ocean circulation model to (a) biogeochemistry (e.g., primary production) and (b) phytoplankton community structure. Surprisingly, we find that the modeled phytoplankton community is largely unchanged, with the same phenotypes dominating in both cases. Conversely, there are large regional and seasonal variations in primary production, phytoplankton and zooplankton biomass. In the subtropics, mixed layer depths (MLDs) are, on average, deeper in the eddy-permitting model, resulting in higher nutrient supply driving increases in primary production and phytoplankton biomass. In the higher latitudes, differences in winter mixed layer depths, the timing of the onset of the spring bloom and vertical nutrient supply result in lower primary production in the eddy-permitting model. Counterintuitively, this does not drive a decrease in phytoplankton biomass but results in lower zooplankton biomass. We explain these similarities and differences in the model using the framework of resource competition theory, and find that they are the consequence of changes in the regional and seasonal nutrient supply and light environment, mediated by differences in the modeled mixed layer depths. Although previous work has suggested that complex models may respond chaotically and unpredictably to changes in forcing, we find that our model responds in a predictable way to different ocean circulation forcing, despite its complexity. The use of frameworks, such as resource competition theory, provides a tractable way to explore the differences and similarities that occur. As this model has many similarities to other widely used biogeochemical models that also resolve multiple phytoplankton phenotypes, this study provides important insights into how the results of running these models under different physical conditions might be more easily understood.

1323. A Consortium for Ocean Circulation and Climate Estimation

Authors: Detlef Stammer

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada626856

URL: <https://doi.org/10.21236/ada626856>

Publication Date: 2017-06-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1324. A Consortium for Ocean Circulation and Climate Estimation

Authors: Detlef Stammer

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada627102

URL: <https://doi.org/10.21236/ada627102>

Publication Date: 2017-06-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1325. A Consortium for Ocean Circulation and Climate Estimation

Authors: Detlef Stammer

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada625416

URL: <https://doi.org/10.21236/ada625416>

Publication Date: 2017-06-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1326. A Consortium for Ocean Circulation and Climate Estimation - JPL

Authors: Ichiro Fukumori; Tong Lee; Dimitris Menemenlis; Lee-Lueng Fu; Victor Zlotnicki

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada625234

URL: <https://doi.org/10.21236/ada625234>

Publication Date: 2017-06-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1327. A Consortium for Ocean Circulation and Climate Estimation - JPL

Authors: Ichiro Fukumori; Tong Lee; Dimitris Menemenlis; Lee-Lueng Fu; Victor Zlotnicki

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada627013

URL: <https://doi.org/10.21236/ada627013>

Publication Date: 2017-06-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1328. A Consortium for Ocean Circulation and Climate Estimation

Authors: John Marshall; Carl Wunsch

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada625411

URL: <https://doi.org/10.21236/ada625411>

Publication Date: 2017-06-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1329. A Consortium for Ocean Circulation and Climate Estimation

Authors: John Marshall; Carl Wunsch

Journal: Unknown Journal

Publisher: Defense Technical Information Center

DOI: 10.21236/ada627144

URL: <https://doi.org/10.21236/ada627144>

Publication Date: 2017-06-15

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1330. Detecting seasonal and long-term vertical displacement in the North China Plain using GRACE and GPS

Authors: Linsong Wang; Chao Chen; Jinsong Du; Tongqing Wang

Journal: Hydrology and Earth System Sciences

Publisher: Copernicus GmbH

DOI: 10.5194/hess-21-2905-2017

URL: <https://doi.org/10.5194/hess-21-2905-2017>

Publication Date: 2017-06-15

Volume/Issue: Volume 21, Issue 6

References: 57

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, China Postdoctoral Science Foundation

License: Creative Commons

Open Access: N/A

Abstract: . In total, 29 continuous Global Positioning System (GPS) time series data together with data from Gravity Recovery and Climate Experiment (GRACE) are analysed to determine the seasonal displacements of surface loadings in the North China Plain (NCP). Results show significant seasonal variations and a strong correlation between GPS and GRACE results in the vertical displacement component; the average correlation and weighted root-mean-squares (WRMS) reduction between GPS and GRACE are 75.6 and 28.9 % respectively, when atmospheric and non-tidal ocean effects were removed, but the annual peak-to-peak amplitude of GPS (1.2-6.3 mm) is greater than the data (1.0-2.2 mm) derived from GRACE. We also calculate the trend rate as well as the seasonal signal caused by the mass load change from GRACE data; the rate of GRACE-derived terrestrial water storage (TWS) loss (after multiplying by the scaling factor) in the NCP was 3.39 cm yr⁻¹ (equivalent to 12.42 km³ yr⁻¹) from 2003 to 2009. For a 10-year time span (2003 to 2012), the rate loss of TWS was 2.57 cm yr⁻¹ (equivalent to 9.41 km³ yr⁻¹), which is consistent with the groundwater storage (GWS) depletion rate (the rate losses of GWS were 2.49 and 2.72 cm yr⁻¹ during 2003-2009 and 2003-2012 respectively) estimated from GRACE-derived results after removing simulated soil moisture (SM) data from the Global Land Data Assimilation System (GLDAS)/Noah model. We also found that GRACE-derived GWS changes are in disagreement with the groundwater level changes from observations of shallow aquifers from 2003 to 2009, especially between 2010 and 2013. Although the shallow groundwater can be recharged from the annual climate-driven rainfall, the important facts indicate that GWS depletion is more serious in deep aquifers. The GRACE-derived result shows an overall uplift in the whole region at the 0.37-0.95 mm yr⁻¹ level from 2004 to 2009, but the rate of change direction is inconsistent in different GPS stations at the -0.40-0.51 mm yr⁻¹ level from 2010 to 2013. Then we removed the vertical rates, which are induced by TWS from GPS-derived data, to obtain the corrected vertical velocities caused by tectonic movement and human activities. The results show that there are uplift areas and subsidence areas in NCP. Almost the whole central and eastern region of NCP suffers serious ground subsidence caused by the anthropogenic-induced groundwater exploitation in the deep confined aquifers. In addition, the slight ground uplifts in the western region of NCP are mainly controlled by

tectonic movement (e.g. Moho uplifting or mantle upwelling).

1331. The contrast between Atlantic and Pacific surface water fluxes

Authors: Philip M. Craig; David Ferreira; John Methven

Journal: Tellus A: Dynamic Meteorology and Oceanography

Publisher: Stockholm University Press

DOI: 10.1080/16000870.2017.1330454

URL: <https://doi.org/10.1080/16000870.2017.1330454>

Publication Date: 2017-06-08

Volume/Issue: Volume 69, Issue 1

References: 52

Cited by: 15 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1332. The geocentre inversion based on the global climate models and GPS site displacements

Authors: Nan Yu (Wuhan University, Chinese Academy of Surveying and Mapping, Beijing 100830, China); Pengfei Cheng (Chinese Academy of Surveying and Mapping, Beijing 100830, China); Yingyan Cheng (Chinese Academy of Surveying and Mapping, Beijing 100830, China); Hanjiang Wen (Chinese Academy of Surveying and Mapping, Beijing 100830, China); Kaifa Kuang (Wuhan University); Xinyun Cao (Wuhan University)

Journal: Survey Review

Publisher: Informa UK Limited

DOI: 10.1080/00396265.2017.1329077

URL: <https://doi.org/10.1080/00396265.2017.1329077>

Publication Date: 2017-06-08

Volume/Issue: Volume 50, Issue 363

References: 31

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the key technology of millimeter level's global epoch terrestrial reference frame building, the sea ice and ice-sheet mass variations in Antarctica and its contributions to the global sea level change, the Fundamental Research Funds for the Central Universities

License: N/A

Open Access: N/A

Abstract: N/A

1333. Preconditioning of Antarctic maximum sea ice extent by upper ocean stratification on a seasonal timescale

Authors: Zhan Su (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl073236

URL: <https://doi.org/10.1002/2017gl073236>

Publication Date: 2017-06-06

Volume/Issue: Volume 44, Issue 12

References: 60

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: Ekman transport, vertical heat flux, Antarctic sea ice, stratification

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study uses an observationally constrained and dynamically consistent ocean and sea ice state estimate. The author presents a remarkable agreement between the location of the edge of Antarctic maximum sea ice extent, reached in September, and the narrow transition band for the upper ocean (0-100 m depths) stratification, as early as April to June. To the south of this edge, the upper ocean has high stratification, which forbids convective fluxes to cross through; consequently, the ocean heat loss to the atmosphere is an efficient way to cool the surface ocean to the freezing point during April to September. To the north, the upper ocean has low stratification such that the ocean heat loss to the atmosphere is not efficient to cool the upper ocean. The upper ocean is instead cooled mainly through mixing with the colder inflow carried by northward Ekman transport but cannot reach the freezing point due to the nature of mixing. Therefore, upper ocean stratification, dominated by salinity here, provides an important constraint on the northward expansion of Antarctic sea ice to its maximum.

1334. Improved geophysical excitations constrained by polar motion observations and GRACE/SLR time-dependent gravity

Authors: Wei Chen; Jiancheng Li; Jim Ray; Minkang Cheng

Journal: Geodesy and Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.geog.2017.04.006

URL: <https://doi.org/10.1016/j.geog.2017.04.006>

Publication Date: 2017-06-05

Volume/Issue: Volume 8, Issue 6

References: 52

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: Polar motion|Geopotential|Satellite laser ranging|Momentum (technical analysis)

Funding: National 973 Project of China, National Natural Science Foundation of China, R&D; Special Fund for Public Welfare Industry (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

1335. Modification of the deep salinity-maximum in the Southern Ocean by circulation in the Antarctic Circumpolar Current and the Weddell Gyre

Authors: Matthew Donnelly; Harry Leach; Volker Strass

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-017-1054-3

URL: <https://doi.org/10.1007/s10236-017-1054-3>

Publication Date: 2017-05-28

Volume/Issue: Volume 67, Issue 7

References: 61

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council (GB)

License: N/A

Open Access: N/A

Abstract: N/A

1336. Ensemble data assimilation in the Red Sea: sensitivity to ensemble selection and atmospheric forcing

Authors: Habib Toye; Peng Zhan; Ganesh Gopalakrishnan; Aditya R. Kartadikaria; Huang Huang; Omar Knio; Ibrahim Hoteit

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-017-1064-1

URL: <https://doi.org/10.1007/s10236-017-1064-1>

Publication Date: 2017-05-26

Volume/Issue: Volume 67, Issue 7

References: 65

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1337. Short-term variations of Icelandic ice cap mass inferred from cGPS coordinate time series

Authors: Kathleen Compton (University of Arizona); Richard A. Bennett (University of Arizona); Sigrún Hreinsdóttir (Now at GNS Science Lower Hutt New Zealand, Institute of Earth Sciences, University of Iceland Reykjavík Iceland); Tonie van Dam (University of Luxembourg); Andrea Bordonì (Technical University of Denmark); Valentina Barletta (Technical University of Denmark); Giorgio Spada (Dipartimento di Scienze Pure e Applicate Università degli Studi di Urbino "Carlo Bo," Urbino Italy)

Journal: Geochemistry, Geophysics, Geosystems

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gc006831

URL: <https://doi.org/10.1002/2017gc006831>

Publication Date: 2017-05-23

Volume/Issue: Volume 18, Issue 6

References: 69

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, University of Arizona, Icelandic Centre for Research

License: N/A

Open Access: N/A

Abstract: As the global climate changes, understanding short-term variations in water storage is increasingly important. Continuously operating Global Positioning System (cGPS) stations in Iceland record annual periodic motion--the elastic response to winter accumulation and spring melt seasons--with peak-to-peak vertical amplitudes over 20 mm for those sites in the Central Highlands. Here for the first time for Iceland, we demonstrate the utility of these cGPS-measured displacements for estimating seasonal and shorter-term ice cap mass changes. We calculate unit responses to each of the five largest ice caps in central Iceland at each of the 62 cGPS locations using an elastic half-space model and estimate ice mass variations from the cGPS time series using a simple least squares inversion scheme. We utilize all three components of motion, taking advantage of the seasonal motion recorded in the horizontal. We remove secular velocities and accelerations and explore the impact that seasonal motions due to atmospheric, hydrologic, and nontidal ocean loading have on our inversion results. Our results match available summer and winter mass balance measurements well, and we reproduce the seasonal stake-based observations of loading and melting within the 1 confidence bounds of the inversion. We identify nonperiodic ice mass changes associated with interannual variability in precipitation and other processes such as increased melting due to reduced ice surface albedo or decreased melting due to ice cap insulation in response to tephra deposition following volcanic eruptions, processes that are not resolved with once or twice-yearly stake measurements.

1338. Hydrological excitation of polar motion by different variables from the GLDAS models

Authors: Malgorzata Winska; Jolanta Nastula; David Salstein

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-017-1036-8

URL: <https://doi.org/10.1007/s00190-017-1036-8>

Publication Date: 2017-05-17

Volume/Issue: Volume 91, Issue 12

References: 44

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: DEKABAN Fund Program, Polish National Science Foundation NCN

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1339. Causes of accelerating sea level on the East Coast of North America

Authors: James L. Davis (Columbia University); Nadya T. Vinogradova (Cambridge Climate Institute)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl072845

URL: <https://doi.org/10.1002/2017gl072845>

Publication Date: 2017-05-16

Volume/Issue: Volume 44, Issue 10

References: 57

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: climate impacts, sea level acceleration, sea level change, ECCO, GRACE (and 1 more)

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: The tide-gauge record from the North American East Coast reveals significant accelerations in sea level starting in the late twentieth century. The estimated post-1990 accelerations range from near zero to ~ 0.3 mm yr⁻². We find that the observed sea level acceleration is well modeled using several processes: mass change in Greenland and Antarctica as measured by the Gravity Recovery and Climate Experiment satellites; ocean dynamic and steric variability provided by the GECCO2 ocean synthesis; and the inverted barometer effect. However, to achieve this fit requires estimation of an admittance for the dynamical and steric contribution, possibly due to the coarse resolution of this analysis or to simplifications associated with parameterization of bottom friction in the shallow coastal areas. The acceleration from ice loss alone is equivalent to a regional sea level rise in one century of 0.2 m in the north and 0.75 m in the south of this region.

1340. Modeling modified Circumpolar Deep Water intrusions onto the Prydz Bay continental shelf, East Antarctica

Authors: Chengyan Liu (Nanjing University, Center for Coastal Physical Oceanography Norfolk Virginia USA); Zhaomin Wang (Hohai University); Chen Cheng (Nanjing University); Ruibin Xia (Nanjing University); Bingrui Li (Polar Research Institute of China); Zelin Xie (Nanjing University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012336

URL: <https://doi.org/10.1002/2016jc012336>

Publication Date: 2017-05-16

Volume/Issue: Volume 122, Issue 7

References: 54

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: intrusion path, Circumpolar Deep Water, modified Circumpolar Deep Water, Amery Ice Shelf, Prydz Bay

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: An eddy-resolving coupled regional ocean-sea ice-ice shelf model is employed to locate the hot spots where modified Circumpolar Deep Water (mCDW) intrudes onto the continental shelf within Prydz Bay, and locate the paths through which mCDW is transported to the Amery Ice Shelf (AIS) calving front. Evaluation of the model output is with satellite, hydrographic and borehole data. Two critical windows responsible for mCDW intrusions are identified. The first is the eastern branch of the cyclonic Prydz Bay gyre (PBG) that carries mCDW to the ice front line, accounting for an annual mean heat transport of $\sim 8.7 \times 10^{11} \text{ J s}^{-1}$. The second is located to the east of the Four Ladies Bank (FLB) where mCDW is channeled through submarine troughs, accounting for an annual mean heat transport of $\sim 16.2 \times 10^{11} \text{ J s}^{-1}$. The eddy-induced heat transport accounts for $\sim 23\%$ in the path of the PBG and $\sim 52\%$ in the path of the eastern coastal current, with respect to their total onshore heat transport. The seasonal pulsing of mCDW intrusions is greatly dependent on the seasonal cycle of the Antarctic Slope Current (ASC) that peaks with a maximum of $\sim 29.3 \text{ Sv}$ at 75°E in June. In austral winter, mCDW is allowed to access the eastern flank of the AIS calving front with potential consequences for the basal mass balance of the AIS. The dynamic effects of small-scale troughs on the longshore ASC play an important role in the onshore mCDW transport.

1341. Mindanao Current and Undercurrent: Thermohaline Structure and Transport from Repeat Glider Observations

Authors: Martha C. Schönau (Scripps Institution of Oceanography / University of California San Diego); Daniel L. Rudnick (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0274.1

URL: <https://doi.org/10.1175/jpo-d-16-0274.1>

Publication Date: 2017-05-12

Volume/Issue: Volume 47, Issue 8

References: 82

Cited by: 23 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research, Office of Naval Research

License: N/A

Open Access: N/A

Abstract: Autonomous underwater Spray gliders made repeat transects of the Mindanao Current (MC), a low-latitude western boundary current in the western tropical North Pacific Ocean, from September 2009 to October 2013. In the thermocline ($<26 \text{ kg m}^{-3}$), the MC has a maximum velocity core of -0.95 m s^{-1} , weakening with distance offshore until it intersects with the intermittent Mindanao Eddy (ME) at 129.25°E . In the subthermocline ($>26 \text{ kg m}^{-3}$), a persistent Mindanao Undercurrent (MUC), with a velocity core of 0.2 m s^{-1} and mean net transport, flows poleward. Mean transport and standard deviation integrated from the coast to 130°E is $-19 \pm 3.1 \text{ Sv}$ ($1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) in the thermocline and $-3 \pm 12 \text{ Sv}$ in the subthermocline. Subthermocline transport has an inverse linear relationship with the Niño-3.4 index and is the primary influence of total transport variability. Interannual anomalies during El Niño are greater than the annual cycle for sea surface salinity and thermocline depth. Water masses transported by the MC/MUC are identified by subsurface salinity extrema and are on isopycnals that have increased finescale salinity variance (spice variance) from eddy stirring. The MC/MUC spice variance is smaller in the thermocline and greater in the subthermocline when compared to the North Equatorial Current and its undercurrents.

1342. Kinetics of thorium and particle cycling along the U.S. GEOTRACES North Atlantic Transect

Authors: Paul Lerner; Olivier Marchal; Phoebe J. Lam; Ken Buesseler; Matthew Charette

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2017.05.003

URL: <https://doi.org/10.1016/j.dsr.2017.05.003>

Publication Date: 2017-05-12

Volume/Issue: Volume 125

References: 102

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. National Science Foundation, U.S. NSF

License: N/A

Open Access: N/A

Abstract: N/A

1343. Reviews and syntheses: Flying the satellite into your model: on the role of observation operators in constraining models of the Earth system and the carbon cycle

Authors: Thomas Kaminski; Pierre-Philippe Mathieu

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-14-2343-2017

URL: <https://doi.org/10.5194/bg-14-2343-2017>

Publication Date: 2017-05-08

Volume/Issue: Volume 14, Issue 9

References: 105

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The vehicles that fly the satellite into a model of the Earth system are observation operators. They provide the link between the quantities simulated by the model and the quantities observed from space, either directly (spectral radiance) or indirectly estimated through a retrieval scheme (biogeophysical variables). By doing so, observation operators enable modellers to properly compare, evaluate, and constrain their models with the model analogue of the satellite observations. This paper provides the formalism and a few examples of how observation operators can be used in combination with data assimilation techniques to better ingest satellite products in a manner consistent with the dynamics of the Earth system expressed by models. It describes commonalities and potential synergies between assimilation and classical retrievals. This paper explains how the combination of observation operators and their derivatives (linearizations) form powerful research tools. It introduces a technique called automatic differentiation that greatly simplifies both the development and the maintenance of code for the evaluation of derivatives. Throughout this paper, a special focus lies on applications to the carbon cycle.

1344. Association between mean and interannual equatorial Indian Ocean subsurface temperature bias in a coupled model

Authors: G. Srinivas; Jasti S. Chowdary; C. Gnanaseelan; K. V. S. R. Prasad; Ananya Karmakar; Anant Parekh

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-017-3713-y

URL: <https://doi.org/10.1007/s00382-017-3713-y>

Publication Date: 2017-05-03

Volume/Issue: Volume 50, Issue 5-6

References: 59

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1345. Estimation of the Tropical Pacific Ocean State 2010-13

Authors: Ariane Verdy (Scripps Institution of Oceanography); Bruce Cornuelle (Scripps Institution of Oceanography); Matthew R. Mazloff (Scripps Institution of Oceanography); Daniel L. Rudnick (Scripps Institution of Oceanography)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-16-0223.1

URL: <https://doi.org/10.1175/jtech-d-16-0223.1>

Publication Date: 2017-05-03

Volume/Issue: Volume 34, Issue 7

References: 29

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: Ocean models, Data assimilation, Tropical variability, Regional models, Hindcasts (and 1 more)

Funding: National Oceanic and Atmospheric Administration, Office of Polar Programs

License: N/A

Open Access: N/A

Abstract: A data-assimilating 1/3 degrees regional dynamical ocean model is evaluated on its ability to synthesize components of the Tropical Pacific Ocean Observing System. The four-dimensional variational data assimilation (4DVAR) method adjusts initial conditions and atmospheric forcing for overlapping 4-month model runs, or hindcasts, that are then combined to give an ocean state estimate for the period 2010-13. Consistency within uncertainty with satellite SSH and Argo profiles is achieved. Comparison to independent observations from Tropical Atmosphere Ocean (TAO) moorings shows that for time scales shorter than 100 days, the state estimate improves estimates of TAO temperature relative to an optimally interpolated Argo product. The improvement is greater at time scales shorter than 20 days, although unpredicted variability in the TAO temperatures implies that TAO observations provide significant information in that band. Larger discrepancies between the state estimate and independent observations from Spray gliders deployed near the Galápagos, Palau, and Solomon Islands are attributed to insufficient model resolution to capture the dynamics in strong current regions and near coasts. The sea surface height forecast skill of the model is assessed. Model forecasts using climatological forcing and boundary conditions are more skillful than climatology out to 50 days compared to persistence, which is a more skillful forecast than climatology out to approximately 20 days. Hindcasts using reanalysis products for atmospheric forcing and open boundary conditions are more skillful than climatology for approximately 120 days or longer, with the exact time scale depending on the accuracy of the state estimate used for initializing and on the reanalysis forcing. Estimating the model representational error is a goal of these experiments.

1346. Causes of the Regional Variability in Observed Sea Level, Sea Surface Temperature and Ocean Colour Over the Period 1993-2011

Authors: B. Meyssignac; C. G. Piecuch; C. J. Merchant; M.-F. Racault; H. Palanisamy; C. MacIntosh; S. Sathyendranath; R. Brewin

Journal: Space Sciences Series of ISSI

Publisher: Springer International Publishing

DOI: 10.1007/978-3-319-56490-6_9

URL: https://doi.org/10.1007/978-3-319-56490-6_9

Publication Date: 2017-04-30

Volume/Issue: N/A

References: N/A

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: Sea-surface height|Forcing (mathematics)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1347. Quantifying the impact of the major driving mechanisms of inter-annual variability of salinity in the North Sea

Authors: Ismael Núñez-Riboni; Anna Akimova

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2017.04.004

URL: <https://doi.org/10.1016/j.pocean.2017.04.004>

Publication Date: 2017-04-26

Volume/Issue: Volume 154

References: 42

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1348. Amundsen and Bellingshausen Seas simulation with optimized ocean, sea ice, and thermodynamic ice shelf model parameters

Authors: Y. Nakayama (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Schodlok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012538

URL: <https://doi.org/10.1002/2016jc012538>

Publication Date: 2017-04-25

Volume/Issue: Volume 122, Issue 8

References: 68

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: glacial melt water, Bellingshausen Sea, Pine Island Glacier, Amundsen Sea, winter water (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Recent studies suggest that the thickness of Winter Water (WW), that is, water with potential temperature below ~ -1 degreesC located below Antarctic Surface Water and above Circumpolar Deep Water (CDW) is critical in determining the ice shelf melt rate, especially for the Pine Island Glacier (PIG). Existing model studies, however, misrepresent WW thickness and properties in the Amundsen Sea (AS). Here, we adjust a small number of model parameters in a regional Amundsen and Bellingshausen Seas configuration of the Massachusetts Institute of Technology general circulation model in order to reproduce properties and thickness of WW and CDW close to observations, with significant improvement for WW compared to previous studies. The cost, which is defined as weighted model-data difference squared, is reduced by 23%. Although a previous modeling study points out that the local surface heat loss upstream from Pine Island Polynya could be the reason for the observed 2012 PIG melt decline and WW thickening, they did not show WW freshening, which was observed at the same time. Model sensitivity experiments for surface heat loss, PIG melt rate, and precipitation fail to replicate WW freshening concurrent with PIG melt decline, implying that these processes cannot fully explain the observed PIG melt decrease.

1349. Modelling mangrove propagule dispersal trajectories using high-resolution estimates of ocean surface winds and currents

Authors: Tom Van der Stocken (Vrije Universiteit Brussel); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Biotropica

Publisher: Wiley

DOI: 10.1111/btp.12440

URL: <https://doi.org/10.1111/btp.12440>

Publication Date: 2017-04-21

Volume/Issue: Volume 49, Issue 4

References: 88

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory, Vlaamse Interuniversitaire Raad, Vrije Universiteit Brussel (and 2 more)

License: N/A

Open Access: N/A

Abstract: Mangrove forests are systems that provide ecosystem services and rely on floating propagules of which the dispersal trajectories are determined by ocean currents and winds. Quantitating connectivity of mangrove patches is an important conservation concern. Current estimates of connectivity, however, fail to integrate the link between ocean currents at different spatial scales and dispersal trajectories. Here, we use high-resolution estimates of ocean currents and surface winds from meteorological and oceanographic analyses, in conjunction with experimental data on propagule traits (e.g., density, size, and shape) and dispersal vector properties (e.g., strength and direction of water and wind currents). We incorporate these data in a dispersal model to illustrate the potential effect of wind on dispersal trajectories of hydrochorous propagules from different mangrove species. We focus on the Western Indian Ocean, including the Mozambique Channel, which has received much attention because of its reported oceanic complexity, to illustrate the effect of oceanic features such as eddy currents and tides. In spite of the complex pattern of ocean surface currents and winds, some propagules are able to cross the Mozambique Channel. Eddy currents and tides may delay arrival at a suitable site. Experimentally demonstrated differences in wind sensitivity among propagule types were shown to affect the probability of departure and the shape of dispersal trajectories. The model could be used to reconstruct current fluxes of mangrove propagules that may help explain past and current distributions of mangrove forests and assess the potential for natural expansion of these forests.

1350. In Search of Fingerprints of the Recent Intensification of the Ocean Water Cycle

Authors: Nadya T. Vinogradova (Cambridge Climate Institute); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0626.1

URL: <https://doi.org/10.1175/jcli-d-16-0626.1>

Publication Date: 2017-04-19

Volume/Issue: Volume 30, Issue 14

References: 104

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: Trends, Hydrologic cycle, Freshwater, Atmosphere-ocean interaction, Decadal variability (and 1 more)

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Unprecedented changes in Earth's water budget and a recent boom in salinity observations prompted the use of long-term salinity trends to fingerprint the amount of freshwater entering and leaving the oceans (the ocean water cycle). Here changes in the ocean water cycle in the past two decades are examined to evaluate whether the rain-gauge notion can be extended to shorter time scales. Using a novel framework it is demonstrated that there have been persistent changes (defined as significant trends) in both salinity and the ocean water cycle in many ocean regions, including the subtropical gyres in both hemispheres, low latitudes of the tropical Pacific, the North Atlantic Subpolar Gyre, and the Arctic Ocean. On average, the ocean water cycle has amplified by approximately 5% since 1993, but strong regional variations exist (as well as dependency on the surface freshwater flux products chosen). Despite an intensified ocean water cycle in the last two decades, changes in surface salinity do not follow expected patterns of amplified salinity contrasts, challenging the perception that if it rains more the seas always get fresher and if it evaporates more the seas always get saltier. These findings imply a time of emergence of anthropogenic hydrological signals shorter in surface freshwater fluxes than in surface salinity and point to the importance of ocean circulation, salt transports, and natural climate variability in shaping patterns of decadal change in surface salinity. Therefore, the use of salinity measurements in conjunction with ocean salt fluxes can provide a more meaningful way of fingerprinting changes in the global water cycle on decadal time scales.

1351. PMIP Ocean - Understanding changes since the Last Glacial Maximum

Authors: Andreas Schmittner; SP Harrison; AE Carlson; AC Mix; M Kageyama; Michal Kucera

Journal: Past Global Changes Magazine

Publisher: Past Global Changes (PAGES)

DOI: 10.22498/pages.22.1.45

URL: <https://doi.org/10.22498/pages.22.1.45>

Publication Date: 2017-04-19

Volume/Issue: Volume 22, Issue 1

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Last Glacial Maximum

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1352. An Analytical Model of Iceberg Drift

Authors: Till J. W. Wagner (University of California San Diego); Rebecca W. Dell (University of California San Diego); Ian Eisenman (University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0262.1

URL: <https://doi.org/10.1175/jpo-d-16-0262.1>

Publication Date: 2017-04-18

Volume/Issue: Volume 47, Issue 7

References: 53

Cited by: 39 publications

Language: N/A

Subjects: N/A

Keywords: Arctic, Southern Ocean, Trajectories, Ice loss/growth, North Atlantic Ocean (and 1 more)

Funding: National Science Foundation - Division of Ocean Sciences

License: N/A

Open Access: N/A

Abstract: The fate of icebergs in the polar oceans plays an important role in Earth's climate system, yet a detailed understanding of iceberg dynamics has remained elusive. Here, the central physical processes that determine iceberg motion are investigated. This is done through the development and analysis of an idealized model of iceberg drift. The model is forced with high-resolution surface velocity and temperature data from an observational state estimate. It retains much of the most salient physics, while remaining sufficiently simple to allow insight into the details of how icebergs drift. An analytical solution of the model is derived, which highlights how iceberg drift patterns depend on iceberg size, ocean current velocity, and wind velocity. A long-standing rule of thumb for Arctic icebergs estimates their drift velocity to be 2% of the wind velocity relative to the ocean current. Here, this relationship is derived from first principles, and it is shown that the relationship holds in the limit of small icebergs or strong winds, which applies for typical Arctic icebergs. For the opposite limit of large icebergs (length > 12 km) or weak winds, which applies for typical Antarctic tabular icebergs, it is shown that this relationship is not applicable and icebergs move with the ocean current, unaffected by the wind. The latter regime is confirmed through comparisons with observed iceberg trajectories near the Antarctic Peninsula.

1353. Decoding the origins of vertical land motions observed today at coasts

Authors: J. Pfeffer (Université Lyon 1, Ens de Lyon, CNRS, UMR 5276 LGL-TPE, F-69622 Villeurbanne, France. E-mail: pfeffer.julia@gmail.com); G. Spada (Dipartimento di Scienze Pure e Applicate (DiSPeA), Università degli Studi di Urbino "Carlo Bo", I-61029, Urbino, Italy); A. Mémin (University of Tasmania, Institut de Recherche pour le Développement (IRD)); J.-P. Boy (EOST, IPGS, UMR 7516, CNRS, Université de Strasbourg, F-67084 Strasbourg Cedex, France); P. Allemand (Université Lyon 1, Ens de Lyon, CNRS, UMR 5276 LGL-TPE, F-69622 Villeurbanne, France. E-mail: pfeffer.julia@gmail.com)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggx142

URL: <https://doi.org/10.1093/gji/ggx142>

Publication Date: 2017-04-11

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Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: ERC, CUP, CUP (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1354. Erosion of a California Undercurrent eddy by bottom topography

Authors: Hector S. Torres (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Departamento de Oceanografía Física CICESE, Carretera Tijuana-Ensenada No. 3918 Ensenada Baja California México); Jose Gomez-Valdes (Departamento de Oceanografía Física CICESE, Carretera Tijuana-Ensenada No. 3918 Ensenada Baja California México)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc011870

URL: <https://doi.org/10.1002/2016jc011870>

Publication Date: 2017-04-07

Volume/Issue: Volume 122, Issue 5

References: 68

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Centre National d'Etudes Spatiales, Consejo Nacional de Ciencia y Tecnología, Centro de Investigación Científica y de Educación Superior de Ensenada, Baja California

License: N/A

Open Access: N/A

Abstract: Subsurface eddies are ubiquitous features in eastern boundary current systems. These phenomena tend to modulate the across-shore distribution of heat and biogeochemical tracers. A California Undercurrent eddy was detected by shipboard observations in October 2009 off the northern Baja California continental slope. The spatiotemporal variation in the California Undercurrent eddy is investigated by using a mesoscale-resolving hindcast ocean simulation. A poleward coastal current that is driven by an upwelling-wind relaxation event and the coastline geometry instigated the separation of the California Undercurrent from the slope, forming a meander-like structure, which evolved as a mesoscale eddy-like structure. The latter structure evolved as a subsurface eddy with a warm anomaly core, a distinctive feature of eddies that form from the California Undercurrent. During the initial stage, the subsurface eddy presented a cone-shape form, with the maximum amplitude of the relative vorticity in the upper section. The inviscid effect of the irregular bottom topography altered both the initial direction of propagation and the initial eddy shape: the propagation direction of the eddy changed from north-south to southwest, and the shape changed from a cone shape to a subsurface lens shape. The strong eddy-topography interactions triggered a sign change in the heat flux direction from the environment toward the eddy and vice versa through the horizontally divergent component of the velocity field, which accelerated the eddy's decay. This study shows the relevance of the synoptic wind stress events and the irregularity of the bottom topography on the mesoscale eddy activity in the southern portion of the California Current.

1355. Continued retreat of Thwaites Glacier, West Antarctica, controlled by bed topography and ocean circulation

Authors: H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Nakayama (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Morlighem (University of California Irvine); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Khazendar (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl072910

URL: <https://doi.org/10.1002/2017gl072910>

Publication Date: 2017-04-05

Volume/Issue: Volume 44, Issue 12

References: 60

Cited by: 166 publications

Language: en

Subjects: N/A

Keywords: ice sheet dynamics, Antarctica, coupled model, ocean, Amundsen Sea (and 1 more)

Funding: National Science Foundation Program

License: N/A

Open Access: N/A

Abstract: The Amundsen Sea sector is experiencing the largest mass loss, glacier acceleration, and grounding line retreat in Antarctica. Enhanced intrusion of Circumpolar Deep Water onto the continental shelf has been proposed as the primary forcing mechanism for the retreat. Here we investigate the dynamics and evolution of Thwaites Glacier with a novel, fully coupled, ice-ocean numerical model. We obtain a significantly improved agreement with the observed pattern of glacial retreat using the coupled model. Coupled simulations over the coming decades indicate a continued mass loss at a sustained rate. Uncoupled simulations using a depth-dependent parameterization of sub-ice-shelf melt significantly overestimate the rate of grounding line retreat compared to the coupled model, as the parameterization does not capture the complexity of the ocean circulation associated with the formation of confined cavities during the retreat. Bed topography controls the pattern of grounding line retreat, while oceanic thermal forcing impacts the rate of grounding line retreat. The importance of oceanic forcing increases with time as Thwaites grounding line retreats farther inland.

1356. Reviews and syntheses: parameter identification in marine planktonic ecosystem modelling

Authors: Markus Schartau; Philip Wallhead; John Hemmings; Ulrike Löptien; Iris Kriest; Shubham Krishna; Ben A. Ward; Thomas Slawig; Andreas Oschlies

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-14-1647-2017

URL: <https://doi.org/10.5194/bg-14-1647-2017>

Publication Date: 2017-03-29

Volume/Issue: Volume 14, Issue 6

References: 347

Cited by: 53 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . To describe the underlying processes involved in oceanic plankton dynamics is crucial for the determination of energy and mass flux through an ecosystem and for the estimation of biogeochemical element cycling. Many planktonic ecosystem models were developed to resolve major processes so that flux estimates can be derived from numerical simulations. These results depend on the type and number of parameterizations incorporated as model equations. Furthermore, the values assigned to respective parameters specify a model's solution. Representative model results are those that can explain data; therefore, data assimilation methods are utilized to yield optimal estimates of parameter values while fitting model results to match data. Central difficulties are (1) planktonic ecosystem models are imperfect and (2) data are often too sparse to constrain all model parameters. In this review we explore how problems in parameter identification are approached in marine planktonic ecosystem modelling. We provide background information about model uncertainties and estimation methods, and how these are considered for assessing misfits between observations and model results. We explain differences in evaluating uncertainties in parameter estimation, thereby also discussing issues of parameter identifiability. Aspects of model complexity are addressed and we describe how results from cross-validation studies provide much insight in this respect. Moreover, approaches are discussed that consider time- and space-dependent parameter values. We further discuss the use of dynamical/statistical emulator approaches, and we elucidate issues of parameter identification in global biogeochemical models. Our review discloses many facets of parameter identification, as we found many commonalities between the objectives of different approaches, but scientific insight differed between studies. To learn more from results of planktonic ecosystem models we recommend finding a good balance in the level of sophistication between mechanistic modelling and statistical data assimilation treatment for parameter estimation.

1357. Change of the Global Ocean Vertical Heat Transport over 1993-2010

Authors: Xinfeng Liang (University of South Florida); Christopher G. Piecuch (Atmospheric and Environmental Research, Lexington, Massachusetts); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts); Gael Forget (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology); Patrick Heimbach (University of Texas at Austin)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0569.1

URL: <https://doi.org/10.1175/jcli-d-16-0569.1>

Publication Date: 2017-03-28

Volume/Issue: Volume 30, Issue 14

References: 40

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: Mixing, Climate variability, Data assimilation, Ocean circulation, Decadal variability

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: A dynamically and data-consistent ocean state estimate during 1993-2010 is analyzed for bidecadal changes in the mechanisms of heat exchange between the upper and lower oceans. Many patterns of change are consistent with prior studies. However, at various levels above 1800 m the global integral of the change in ocean vertical heat flux involves the summation of positive and negative regional contributions and is not statistically significant. The nonsignificance of change in the global ocean vertical heat transport from an ocean state estimate that provides global coverage and regular sampling, spatially and temporally, raises the question of whether an adequate observational database exists to assess changes in the upper ocean heat content over the past few decades. Also, whereas the advective term largely determines the spatial pattern of the change in ocean vertical heat flux, its global integral is not significantly different from zero. In contrast, the diffusive term, although regionally weak except in high-latitude oceans, produces a statistically significant extra downward heat flux during the 2000s. This result suggests that besides ocean advection, ocean mixing processes, including isopycnal and diapycnal as well as convective mixing, are important for the decadal variation of the heat exchange between upper and deep oceans as well. Furthermore, the analyses herein indicate that focusing on any particular region in explaining changes of the global ocean heat content is misleading.

1358. Resurfacing of South Pacific Tropical Water in the Equatorial Pacific and Its Variability Associated with ENSO

Authors: Tangdong Qu (University of California Los Angeles); Shan Gao (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0078.1

URL: <https://doi.org/10.1175/jpo-d-16-0078.1>

Publication Date: 2017-03-27

Volume/Issue: Volume 47, Issue 5

References: 55

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Science Foundation, National Natural Science Foundation of China (and 1 more)

License: N/A

Open Access: N/A

Abstract: Analysis of results from a simulated passive tracer confirms the resurfacing of South Pacific Tropical Water in the equatorial Pacific. Over the period of integration (1993-2011), both the volume and barycenter of the South Pacific Tropical Water that resurfaces in the equatorial Pacific are tightly linked to El Niño-Southern Oscillation (ENSO), with their correlation with the Niño-3.4 index reaching -0.79 and 0.84, respectively. Their correlation (-0.75 and 0.85) with the sea surface salinity index, Niño-S34.8, is also high. Of particular interest is that both the volume and barycenter of the resurfacing South Pacific Tropical Water peak earlier than the ENSO indices by about 3 months. On the interannual time scale, the resurfacing of South Pacific Tropical Water may modulate the sea surface salinity in the equatorial Pacific at a rate equivalent to as much as 25% of the surface freshwater flux. The results suggest that the resurfacing of South Pacific Tropical Water directly contributes to the sea surface salinity variability in the equatorial Pacific and potentially plays a role in ENSO evolution.

1359. On the Decadal Variability of the Eddy Kinetic Energy in the Kuroshio Extension

Authors: Yang Yang (Nanjing University); X. San Liang (Nanjing University); Bo Qiu (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Shuiming Chen (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0201.1

URL: <https://doi.org/10.1175/jpo-d-16-0201.1>

Publication Date: 2017-03-20

Volume/Issue: Volume 47, Issue 5

References: 90

Cited by: 46 publications

Language: N/A

Subjects: N/A

Keywords: Eddies, Kinetic energy, Instability, Mesoscale processes, Decadal variability (and 1 more)

Funding: National Natural Science Foundation of China, State Oceanic Administration

License: N/A

Open Access: N/A

Abstract: Previous studies have found that the decadal variability of eddy kinetic energy (EKE) in the upstream Kuroshio Extension is negatively correlated with the jet strength, which seems counterintuitive at first glance because linear stability analysis usually suggests that a stronger jet would favor baroclinic instability and thus lead to stronger eddy activities. Using a time-varying energetics diagnostic methodology, namely, the localized multiscale energy and vorticity analysis (MS-EVA), and the MS-EVA-based nonlinear instability theory, this study investigates the physical mechanism responsible for such variations with the state estimate from the Estimating the Circulation and Climate of the Ocean (ECCO), Phase II. For the first time, it is found that the decadal modulation of EKE is mainly controlled by the barotropic instability of the background flow. During the high-EKE state, violent meanderings efficiently induce strong barotropic energy transfer from mean kinetic energy (MKE) to EKE despite the rather weak jet strength. The reverse is true in the low-EKE state. Although the enhanced meander in the high-EKE state also transfers a significant portion of energy from mean available potential energy (MAPE) to eddy available potential energy (EAPE) through baroclinic instability, the EAPE is not efficiently converted to EKE as the two processes are not well correlated at low frequencies revealed in the time-varying energetics. The decadal modulation of barotropic instability is found to be in pace with the North Pacific Gyre Oscillation but with a time lag of approximately 2 years.

1360. Stochastic Subgrid-Scale Ocean Mixing: Impacts on Low-Frequency Variability

Authors: Stephan Juricke (University of Oxford); Tim N. Palmer (University of Oxford); Laure Zanna (University of Oxford)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0539.1

URL: <https://doi.org/10.1175/jcli-d-16-0539.1>

Publication Date: 2017-03-20

Volume/Issue: Volume 30, Issue 13

References: 53

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Stochastic models, Subgrid-scale processes, Parameterization, Model errors (and 1 more)

Funding: European Research Council, Natural Environment Research Council, Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: In global ocean models, the representation of small-scale, high-frequency processes considerably influences the large-scale oceanic circulation and its low-frequency variability. This study investigates the impact of stochastic perturbation schemes based on three different subgrid-scale parameterizations in multidecadal ocean-only simulations with the ocean model NEMO at 1 degrees resolution. The three parameterizations are an enhanced vertical diffusion scheme for unstable stratification, the Gent-McWilliams (GM) scheme, and a turbulent kinetic energy mixing scheme, all commonly used in state-of-the-art ocean models. The focus here is on changes in interannual variability caused by the comparatively high-frequency stochastic perturbations with subseasonal decorrelation time scales. These perturbations lead to significant improvements in the representation of low-frequency variability in the ocean, with the stochastic GM scheme showing the strongest impact. Interannual variability of the Southern Ocean eddy and Eulerian streamfunctions is increased by an order of magnitude and by 20%, respectively. Interannual sea surface height variability is increased by about 20%-25% as well, especially in the Southern Ocean and in the Kuroshio region, consistent with a strong underestimation of interannual variability in the model when compared to reanalysis and altimetry observations. These results suggest that enhancing subgrid-scale variability in ocean models can improve model variability and potentially its response to forcing on much longer time scales, while also providing an estimate of model uncertainty.

1361. Dynamical reconstruction of the global ocean state during the Last Glacial Maximum

Authors: Takasumi Kurahashi-Nakamura (University of Bremen); André Paul (University of Bremen); Martin Losch (University of Bremen, Alfred Wegener Institute)

Journal: Paleoceanography

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016pa003001

URL: <https://doi.org/10.1002/2016pa003001>

Publication Date: 2017-03-17

Volume/Issue: Volume 32, Issue 4

References: 101

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: ocean general circulation model, data assimilation, AMOC, Last Glacial Maximum, ocean circulation (and 1 more)

Funding: DFG-Research Center/Center of Excellence MARUM, project PalMod

License: N/A

Open Access: N/A

Abstract: The global ocean state for the modern age and for the Last Glacial Maximum (LGM) was dynamically reconstructed with a sophisticated data assimilation technique. A substantial amount of data including global seawater temperature, salinity (only for the modern estimate), and the isotopic composition of oxygen and carbon (only in the Atlantic for the LGM) were integrated into an ocean general circulation model with the help of the adjoint method, thereby the model was optimized to reconstruct plausible continuous fields of tracers, overturning circulation and water mass distribution. The adjoint-based LGM state estimation of this study represents the state of the art in terms of the length of forward model runs, the number of observations assimilated, and the model domain. Compared to the modern state, the reconstructed continuous sea-surface temperature field for the LGM shows a global-mean cooling of 2.2 K, and the reconstructed LGM ocean has a more vigorous Atlantic meridional overturning circulation, shallower North Atlantic Deep Water (NADW) equivalent, stronger stratification, and more saline deep water.

1362. A reanalysis of the Chilean ocean circulation: preliminary results for the region between 20 S to 40 S

Authors: Christopher M. Aiken

Journal: Latin American Journal of Aquatic Research

Publisher: Pontificia Universidad Catolica de Valparaiso

DOI: 10.3856/vol45-issue1-fulltext-19

URL: <https://doi.org/10.3856/vol45-issue1-fulltext-19>

Publication Date: 2017-03-15

Volume/Issue: Volume 45, Issue 1

References: N/A

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1363. Seasonal variation analysis of Greenland ice mass time-series

Authors: Roghayeh Shamshiri; Hossein Nahavandchi; Gholamreza Joodaki

Journal: Acta Geodaetica et Geophysica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s40328-017-0198-4

URL: <https://doi.org/10.1007/s40328-017-0198-4>

Publication Date: 2017-03-13

Volume/Issue: Volume 53, Issue 1

References: 30

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1364. How climate model biases skew the distribution of iceberg meltwater

Authors: Till J. W. Wagner (Scripps Institution of Oceanography / University of California San Diego); Ian Eisenman (Scripps Institution of Oceanography / University of California San Diego)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl071645

URL: <https://doi.org/10.1002/2016gl071645>

Publication Date: 2017-03-03

Volume/Issue: Volume 44, Issue 8

References: 34

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: westerlies, freshwater, North Atlantic, climate models, Greenland (and 1 more)

Funding: Office of Naval Research, National Science Foundation

License: N/A

Open Access: N/A

Abstract: The discharge of icebergs into the polar oceans is expected to increase over the coming century, which raises the importance of accurate representations of icebergs in global climate models (GCMs) used for future projections. Here we analyze the prospects for interactive icebergs in GCMs by forcing an iceberg drift and decay model with circulation and temperature fields from (i) state-of-the-art GCM output and (ii) an observational state estimate. The spread of meltwater is found to be smaller for the GCM than for the observational state estimate, despite a substantial high wind bias in the GCM--a bias that is similar to most current GCMs. We argue that this large-scale reduction in the spread of meltwater occurs primarily due to localized differences in ocean currents, which may be related to the coarseness of the horizontal resolution in the GCM. The high wind bias in the GCM is shown to have relatively little impact on the meltwater distribution, despite Arctic iceberg drift typically being dominated by the wind forcing. We find that this is due to compensating effects between faster drift under stronger winds and larger wind-driven wave erosion. These results may have implications for future changes in the Atlantic meridional overturning circulation simulated with iceberg-enabled GCMs.

1365. On the response of subduction in the South Pacific to an intensification of westerlies and heat flux in an eddy permitting ocean model

Authors: Chengyan Liu; Zhaomin Wang; Bingrui Li; Chen Cheng; Ruibin Xia

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-016-6021-2

URL: <https://doi.org/10.1007/s00376-016-6021-2>

Publication Date: 2017-02-28

Volume/Issue: Volume 34, Issue 4

References: 55

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1366. Ice and groundwater effects on long term polar motion (1979-2010)

Authors: Kookhyoun Youm; Ki-Weon Seo; Taehwan Jeon; Sung-Ho Na; Jianli Chen; Clark R. Wilson

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2017.01.008

URL: <https://doi.org/10.1016/j.jog.2017.01.008>

Publication Date: 2017-02-28

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References: 54

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Korea Polar Research Institute (KOPRI), Korea Meteorological Industry Promotion Agency (KMIPA)

License: N/A

Open Access: N/A

Abstract: N/A

1367. Contributions of thermoelastic deformation to seasonal variations in GPS station position

Authors: Xueqing Xu; Danan Dong; Ming Fang; Yonghong Zhou; Na Wei; Feng Zhou

Journal: GPS Solutions

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10291-017-0609-6

URL: <https://doi.org/10.1007/s10291-017-0609-6>

Publication Date: 2017-02-24

Volume/Issue: Volume 21, Issue 3

References: 30

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSFC grant, State Key Laboratory of Geodesy and Earth's Dynamics grant

License: N/A

Open Access: N/A

Abstract: N/A

1368. Correlations of surface ocean pCO₂ to satellite chlorophyll on monthly to interannual timescales

Authors: Amanda R. Fay (University of Wisconsin-Madison); Galen A. McKinley (University of Wisconsin-Madison)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gb005563

URL: <https://doi.org/10.1002/2016gb005563>

Publication Date: 2017-02-18

Volume/Issue: Volume 31, Issue 3

References: 92

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: chlorophyll, biological pump, carbon pump, pCO₂

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: On the mean, ocean carbon uptake is linked to biological productivity, but how biological variability impacts carbon uptake is poorly quantified. Our ability to diagnose past change, understand present variability, and predict the future state of the global carbon cycle requires improving mechanistic understanding in this area. Here we make use of colocated pCO₂ and temperature data, a merged surface ocean color product, and physical fields from an ocean state estimate to assess relationships between surface ocean biology and the carbon cycle on seasonal, monthly anomaly, and interannual timescales over the period 1998-2014. Using a correlation analysis on spatial scales from local to basin-scale biomes, we identify the timescales on which ocean productivity could be directly modifying ocean carbon uptake. On seasonal timescales outside of the equatorial Pacific, biome-scale correlations are negative between chlorophyll and pCO₂. Though this relationship is pervasive, the underlying mechanisms vary across timescales and biomes. Consistent with previous findings, biological activity is a significant driver of pCO₂ seasonality only in the subpolar biomes. For monthly anomalies acting on top of the mean seasonality, productivity and pCO₂ changes are significantly correlated in the subpolar North Pacific and Southern Ocean. Only in the Southern Ocean are correlations consistent with a dominant role for biology in the surface ocean carbon cycle on all timescales.

1369. Eastern tropical North Pacific coral radiocarbon reveals North Pacific Gyre Oscillation (NPGO) variability

Authors: Patrick A. Rafter; Sara C. Sanchez; Julie Ferguson; Jose D. Carriquiry; Ellen R.M. Druffel; J.A. Villaescusa; John R. Southon

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2017.02.002

URL: <https://doi.org/10.1016/j.quascirev.2017.02.002>

Publication Date: 2017-02-16

Volume/Issue: Volume 160

References: 44

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: UC MEXUS

License: N/A

Open Access: N/A

Abstract: N/A

1370. Estimates of Vertical Velocity Errors for IGS ITRF2014 Stations by Applying the Improved Singular Spectrum Analysis Method and Environmental Loading Models

Authors: Anna Klos; Marta Gruszczynska; Machiel Simon Bos; Jean-Paul Boy; Janusz Bogusz

Journal: Pure and Applied Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00024-017-1494-1

URL: <https://doi.org/10.1007/s00024-017-1494-1>

Publication Date: 2017-02-15

Volume/Issue: Volume 175, Issue 5

References: 71

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Polish National Science Centre, Centre National d'Etudes Spatiales, Portuguese Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1371. Variability and change of sea level and its components in the Indo-Pacific region during the altimetry era

Authors: Quran Wu (Xiamen University, CSIRO Oceans and Atmosphere Hobart Tasmania Australia); Xuebin Zhang (CSIRO Oceans and Atmosphere Hobart Tasmania Australia); John A. Church (University of New South Wales); Jianyu Hu (Xiamen University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012345

URL: <https://doi.org/10.1002/2016jc012345>

Publication Date: 2017-02-14

Volume/Issue: Volume 122, Issue 3

References: 85

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: interannual and decadal variability, steric sea level, sea level, mass-induced sea level, linear trend

Funding: China Scholarship Council

License: N/A

Open Access: N/A

Abstract: Previous studies have shown that regional sea level exhibits interannual and decadal variations associated with the modes of climate variability. A better understanding of those low-frequency sea level variations benefits the detection and attribution of climate change signals. Nonetheless, the contributions of thermosteric, halosteric, and mass sea level components to sea level variability and trend patterns remain unclear. By focusing on signals associated with dominant climate modes in the Indo-Pacific region, we estimate the interannual and decadal fingerprints and trend of each sea level component utilizing a multivariate linear regression of two adjoint-based ocean reanalyses. Sea level interannual, decadal, and trend patterns primarily come from thermosteric sea level (TSSL). Halosteric sea level (HSSL) is of regional importance in the Pacific Ocean on decadal time scale and dominates sea level trends in the northeast subtropical Pacific. The compensation between TSSL and HSSL is identified in their decadal variability and trends. The interannual and decadal variability of temperature generally peak at subsurface around 100 m but that of salinity tend to be surface-intensified. Decadal temperature and salinity signals extend deeper into the ocean in some regions than their interannual equivalents. Mass sea level (MassSL) is critical for the interannual and decadal variability of sea level over shelf seas. Inconsistencies exist in MassSL trend patterns among various estimates. This study highlights regions where multiple processes work together to control sea level variability and change. Further work is required to better understand the interaction of different processes in those regions.

1372. On the response of the Lorenz energy cycle for the Southern Ocean to intensified westerlies

Authors: Yang Wu (Nanjing University); Zhaomin Wang (Hohai University); Chengyan Liu (Nanjing University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012539

URL: <https://doi.org/10.1002/2016jc012539>

Publication Date: 2017-02-11

Volume/Issue: Volume 122, Issue 3

References: 67

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, westerlies, Lorenz energy cycle, wave-mean flow interaction

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, China Scholarship Council

License: N/A

Open Access: N/A

Abstract: The potential impact of intensified westerlies on the Lorenz energy cycle for the Southern Ocean is examined by employing a global eddy-permitting ocean-sea ice model. Two idealized sensitivity experiments are designed for this purpose: one is driven by 1992 forcing with weaker westerlies and the other driven by 1998 forcing with stronger westerlies. The intensified westerlies lead to the most significant increase of about 30% in the eddy kinetic energy (EKE) reservoir, followed by the mean kinetic energy (MKE) reservoir increase (17.9%), eddy available potential energy (EAPE) reservoir increase (8.6%), and mean available potential energy (MAPE) reservoir increase (6.5%). In contrast, the increases in the generations of kinetic energy and available potential energy are quite similar, ranging from 21% for EAPE generation to 26% for MKE generation. There are considerably increased energy transfers from MKE to MAPE (about 75%) and from MAPE to EAPE (about 78%), reflecting greatly enhanced baroclinic instability pathway. The conversion rates are strongly influenced by large topography; in particular, a relatively large energy conversion from EKE to MKE exists in the regions associated with large topography, in contrast to the energy flow from MKE to EKE over the broad Southern Ocean. Under stronger wind forcing, all energy conversions are enhanced, and the increases in the conversion rates from EAPE to EKE and from EKE to MKE are more prominent than the increases from MKE to MAPE and from MAPE to EAPE near large topography.

1373. Global negative emissions capacity of ocean macronutrient fertilization

Authors: Daniel P Harrison

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/aa5ef5

URL: <https://doi.org/10.1088/1748-9326/aa5ef5>

Publication Date: 2017-02-07

Volume/Issue: Volume 12, Issue 3

References: 80

Cited by: 23 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Department of Education, Australian Government

License: N/A

Open Access: N/A

Abstract: N/A

1374. Lateral Eddy Mixing in the Subtropical Salinity Maxima of the Global Ocean

Authors: Julius Busecke (Division of Ocean and Climate Physics, Lamont-Doherty Earth Observatory, Palisades, New York); Ryan P. Abernathey (Division of Ocean and Climate Physics, Lamont-Doherty Earth Observatory, Palisades, New York); Arnold L. Gordon (Division of Ocean and Climate Physics, Lamont-Doherty Earth Observatory, Palisades, New York)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0215.1

URL: <https://doi.org/10.1175/jpo-d-16-0215.1>

Publication Date: 2017-02-01

Volume/Issue: Volume 47, Issue 4

References: 41

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: A suite of observationally driven model experiments is used to investigate the contribution of near-surface lateral eddy mixing to the subtropical surface salinity maxima in the global ocean. Surface fields of salinity are treated as a passive tracer and stirred by surface velocities derived from altimetry, leading to irreversible water-mass transformation. In the absence of surface forcing and vertical processes, the transformation rate can be directly related to the integrated diffusion across tracer contours, which is determined by the observed velocities. The destruction rates of the salinity maxima by lateral mixing can be compared to the production rates by surface forcing, which act to strengthen the maxima. The ratio of destruction by eddy mixing in the surface layer versus the surface forcing exhibits regional differences in the mean--from 10% in the South Pacific to up to 25% in the south Indian. Furthermore, the regional basins show seasonal and interannual variability in eddy mixing. The dominant mechanism for this temporal variability varies regionally. Most notably, the North Pacific shows a large sensitivity to the background salinity fields and a weak sensitivity to the velocity fields, while the North Atlantic exhibits the opposite behavior. The different mechanism for temporal variability could have impacts on the manifestation of a changing hydrological cycle in the sea surface salinity (SSS) field specifically in the North Pacific. The authors find evidence for large-scale interannual changes of eddy diffusivity and transformation rate in several ocean basins that could be related to large-scale climate forcing.

1375. Sea Ice Assimilation into a Coupled Ocean-Sea Ice Adjoint Model of the Arctic Ocean

Authors: Nikolay V. Koldunov; Armin Köhl; Nuno Serra; Detlef Stammer

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2017-2

URL: <https://doi.org/10.5194/tc-2017-2>

Publication Date: 2017-01-31

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Sea ice concentration

Funding: Seventh Framework Programme, Deutsche Forschungsgemeinschaft

License: Creative Commons

Open Access: N/A

Abstract: . High-resolution satellite sea ice concentrations (SIC), together with several ocean parameters, are assimilated into a regional Arctic coupled ocean-sea ice model covering the period 2000-2008 using the adjoint method. There is substantial improvement in the representation of the SIC spatial distribution, in particular with respect to the position of the ice edge and to the concentrations in the central parts of the Arctic Ocean during summer months. Seasonal cycles of total Arctic sea ice area show an overall improvement. During summer months values of sea ice extent become underestimated, however, it is shown that this metric is not suitable to characterize the quality of the sea ice simulation, as the root-mean-square difference to the data is reduced in nearly all months and years. Along with the SIC the sea ice thickness fields also become closer to observations, providing added-value by the assimilation. Very sparse data in the Arctic ocean, corresponding to a very small contribution to the cost function, prevent sizable improvements of assimilated ocean variables, with the exception of the sea surface temperature. The bias between simulated and observed SIC decreases mainly due to corrections to the surface atmospheric temperature, while contributions of other control variables remain small.

1376. An Intrinsic Mode of Interannual Variability in the Indian Ocean

Authors: Christopher L. Wolfe (School of Marine and Atmospheric Sciences, Stony Brook University, Stony Brook, New York); Paola Cessi (Scripps Institution of Oceanography / University of California San Diego); Bruce D. Cornuelle (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0177.1

URL: <https://doi.org/10.1175/jpo-d-16-0177.1>

Publication Date: 2017-01-30

Volume/Issue: Volume 47, Issue 3

References: 32

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Directorate for Geosciences

License: N/A

Open Access: N/A

Abstract: An intrinsic mode of self-sustained, interannual variability is identified in a coarse-resolution ocean model forced by an annually repeating atmospheric state. The variability has maximum loading in the Indian Ocean, with a significant projection into the South Atlantic Ocean. It is argued that this intrinsic mode is caused by baroclinic instability of the model's Leeuwin Current, which radiates out to the tropical Indian and South Atlantic Oceans as long Rossby waves at a period of 4 yr. This previously undescribed mode has a remarkably narrowband time series. However, the variability is not synchronized with the annual cycle; the phase of the oscillation varies chaotically on decadal time scales. The presence of this internal mode reduces the predictability of the ocean circulation by obscuring the response to forcing or initial condition perturbations. The signature of this mode can be seen in higher-resolution global ocean models driven by high-frequency atmospheric forcing, but altimeter and assimilation analyses do not show obvious signatures of such a mode, perhaps because of insufficient duration.

1377. Excitation of the Earth's Chandler wobble by a turbulent oceanic double-gyre

Authors: S. E. Naghibi; M. A. Jalali; S. A. Karabasov; M.-R. Alam

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggx032

URL: <https://doi.org/10.1093/gji/ggx032>

Publication Date: 2017-01-26

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1378. Recent Wind-Driven Variability in Atlantic Water Mass Distribution and Meridional Overturning Circulation

Authors: Dafydd Gwyn Evans (University of Southampton, National Oceanography Centre); John Toole (Woods Hole Oceanographic Institution); Gael Forget (Massachusetts Institute of Technology); Jan D. Zika (Imperial College London); Alberto C. Naveira Garabato (University of Southampton, National Oceanography Centre); A. J. George Nurser (National Oceanography Centre); Lisan Yu (Woods Hole Oceanographic Institution)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0089.1

URL: <https://doi.org/10.1175/jpo-d-16-0089.1>

Publication Date: 2017-01-25

Volume/Issue: Volume 47, Issue 3

References: 42

Cited by: 33 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Natural Environment Research Council, National Science Foundation (and 6 more)

License: N/A

Open Access: N/A

Abstract: Interannual variability in the volumetric water mass distribution within the North Atlantic Subtropical Gyre is described in relation to variability in the Atlantic meridional overturning circulation. The relative roles of diabatic and adiabatic processes in the volume and heat budgets of the subtropical gyre are investigated by projecting data into temperature coordinates as volumes of water using an Argo-based climatology and an ocean state estimate (ECCO version 4). This highlights that variations in the subtropical gyre volume budget are predominantly set by transport divergence in the gyre. A strong correlation between the volume anomaly due to transport divergence and the variability of both thermocline depth and Ekman pumping over the gyre suggests that wind-driven heave drives transport anomalies at the gyre boundaries. This wind-driven heaving contributes significantly to variations in the heat content of the gyre, as do anomalies in the air-sea fluxes. The analysis presented suggests that wind forcing plays an important role in driving interannual variability in the Atlantic meridional overturning circulation and that this variability can be unraveled from spatially distributed hydrographic observations using the framework presented here.

1379. The pelagic habitat analysis module for ecosystem-based fisheries science and management

Authors: Daniel P. Harrison (University of Southern California); Michael G. Hinton (Inter-American Tropical Tuna Commission); Suzanne Kohin (National Oceanic and Atmospheric Administration Southwest Fisheries Science Center 8901 La Jolla Shores Dr La Jolla CA 92037 U.S.A); Edward M. Armstrong (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Stephanie Snyder (National Oceanic and Atmospheric Administration Southwest Fisheries Science Center 8901 La Jolla Shores Dr La Jolla CA 92037 U.S.A); Frank O'Brien (System Science Applications Inc.); Dale K. Kiefer (University of Southern California)

Journal: Fisheries Oceanography

Publisher: Wiley

DOI: 10.1111/fog.12194

URL: <https://doi.org/10.1111/fog.12194>

Publication Date: 2017-01-24

Volume/Issue: Volume 26, Issue 3

References: 38

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Science Foundation

License: N/A

Open Access: N/A

Abstract: We have developed a set of tools that operate within an aquatic geographic information system to improve the accessibility, and usability of remote-sensed satellite and computer-modeled oceanographic data for marine science and ecosystem-based management. The tools form the Pelagic Habitat Analysis Module (PHAM), which can be applied as a modeling platform, an investigative aid in scientific research, or utilized as a decision support system for marine ecological management. Applications include fisheries, marine biology, physical and biological oceanography, and marine spatial management. The GIS provides a home for diverse data types and automated tools for downloading remote sensed and global circulation model data. Within the GIS environment, PHAM provides a framework for seamless interactive four-dimensional visualization, for matching between disparate data types, for flexible statistic or mechanistic model development, and for dynamic application of user developed models for habitat, density, and probability predictions. Here we describe PHAM in the context of ecosystem-based fisheries management, and present results from case study projects which guided development. In the first, an analysis of the purse seine fishery for tropical tuna in the eastern Pacific Ocean revealed oceanographic drivers of the catch distribution and the influence of climate-driven circulation patterns on the location of fishing grounds. To support management of the Common Thresher Shark (*Alopias vulpinus*) in the California Current Ecosystem, a simple empirical habitat utilization model was developed and used to dynamically predict the seasonal range expansion of common thresher shark based on oceanographic conditions.

1380. Mixing Inferred from an Ocean Climatology and Surface Fluxes

Authors: Sjoerd Groeskamp (Columbia University); Bernadette M. Sloyan (Oceans and Atmosphere, CSIRO, Hobart, Tasmania, Australia); Jan D. Zika (University of New South Wales); Trevor J. McDougall (University of New South Wales)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0125.1

URL: <https://doi.org/10.1175/jpo-d-16-0125.1>

Publication Date: 2017-01-24

Volume/Issue: Volume 47, Issue 3

References: 105

Cited by: 28 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Natural Environment Research Council

License: N/A

Open Access: N/A

Abstract: This study provides observation-based estimates, determined by inverse methods, of horizontal and isopycnal eddy diffusion coefficients K_H and K_I , respectively, the small-scale mixing coefficient D , and the diathermohaline streamfunction Ψ . The inverse solution of Ψ represents the ocean circulation in Absolute Salinity SA and Conservative Temperature Θ coordinates. The authors suggest that the observation-based estimate of Ψ will be useful for comparison with equivalent diagnostics from numerical climate models. The estimates of K_H and K_I represent horizontal eddy mixing in the mixed layer and isopycnal eddy mixing in the ocean interior, respectively. This study finds that the solution for D and K_H are comparable to existing estimates. The solution for K_I is one of the first observation-based global and full-depth constrained estimates of isopycnal mixing and indicates that K_I is an order of magnitude smaller than K_H . This suggests that there is a large vertical variation in the eddy mixing coefficient, which is generally not included in ocean models. With ocean models being very sensitive to the choice of isopycnal mixing, this result suggests that further investigation of the spatial structure of isopycnal eddy mixing from observations is required.

1381. Numerical investigation of the Arctic ice-ocean boundary layer and implications for air-sea gas fluxes

Authors: Arash Bigdeli; Brice Loose; An T. Nguyen; Sylvia T. Cole

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-13-61-2017

URL: <https://doi.org/10.5194/os-13-61-2017>

Publication Date: 2017-01-23

Volume/Issue: Volume 13, Issue 1

References: 81

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Division of Polar Programs

License: Creative Commons

Open Access: N/A

Abstract: . In ice-covered regions it is challenging to determine constituent budgets - for heat and momentum, but also for biologically and climatically active gases like carbon dioxide and methane. The harsh environment and relative data scarcity make it difficult to characterize even the physical properties of the ocean surface. Here, we sought to evaluate if numerical model output helps us to better estimate the physical forcing that drives the air-sea gas exchange rate (k) in sea ice zones. We used the budget of radioactive ^{222}Rn in the mixed layer to illustrate the effect that sea ice forcing has on gas budgets and air-sea gas exchange. Appropriate constraint of the ^{222}Rn budget requires estimates of sea ice velocity, concentration, mixed-layer depth, and water velocities, as well as their evolution in time and space along the Lagrangian drift track of a mixed-layer water parcel. We used 36, 9 and 2 km horizontal resolution of regional Massachusetts Institute of Technology general circulation model (MITgcm) configuration with fine vertical spacing to evaluate the capability of the model to reproduce these parameters. We then compared the model results to existing field data including satellite, moorings and ice-tethered profilers. We found that mode sea ice coverage agrees with satellite-derived observation 88 to 98 % of the time when averaged over the Beaufort Gyre, and model sea ice speeds have 82 % correlation with observations. The model demonstrated the capacity to capture the broad trends in the mixed layer, although with a significant bias. Model water velocities showed only 29 % correlation with point-wise in situ data. This correlation remained low in all three model resolution simulations and we argued that is largely due to the quality of the input atmospheric forcing. Overall, we found that even the coarse-resolution model can make a modest contribution to gas exchange parameterization, by resolving the time variation of parameters that drive the ^{222}Rn budget, including rate of mixed-layer change and sea ice forcings.

1382. Air-sea heat flux climatologies in the Mediterranean Sea: Surface energy balance and its consistency with ocean heat storage

Authors: Xiangzhou Song (Ocean University of China, National Marine Environmental Forecasting Center); Lisan Yu (Woods Hole Oceanographic Institution)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012254

URL: <https://doi.org/10.1002/2016jc012254>

Publication Date: 2017-01-21

Volume/Issue: Volume 122, Issue 5

References: 84

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: Mediterranean Sea, heat content changes, air-sea heat flux, heat budget analysis

Funding: National Natural Science Foundation of China, China Postdoctoral Science Foundation, Natural Science Foundation of Shandong

License: N/A

Open Access: N/A

Abstract: This study provides an analysis of the Mediterranean Sea surface energy budget using nine surface heat flux climatologies. The ensemble mean estimation shows that the net downward shortwave radiation ($192 \pm 19 \text{ W m}^{-2}$) is balanced by latent heat flux ($-98 \pm 10 \text{ W m}^{-2}$), followed by net longwave radiation ($-78 \pm 13 \text{ W m}^{-2}$) and sensible heat flux ($-13 \pm 4 \text{ W m}^{-2}$). The resulting net heat budget (Q_{net}) is $2 \pm 12 \text{ W m}^{-2}$ into the ocean, which appears to be warm biased. The annual-mean Q_{net} should be $-5.6 \pm 1.6 \text{ W m}^{-2}$ when estimated from the observed net transport through the Strait of Gibraltar. To diagnose the uncertainty in nine Q_{net} climatologies, we constructed Q_{net} from the heat budget equation by using historic hydrological observations to determine the heat content changes and advective heat flux. We also used the Q_{net} from a data-assimilated global ocean state estimation as an additional reference. By comparing with the two reference Q_{net} estimates, we found that seven products (NCEP 1, NCEP 2, CFSR, ERA-Interim, MERRA, NOCSv2.0, and OAFux+ISCCP) overestimate Q_{net} , with magnitude ranging from 6 to 27 W m^{-2} , while two products underestimate Q_{net} by -6 W m^{-2} (JRA55) and -14 W m^{-2} (CORE.2). Together with the previous warm pool work of Song and Yu (2013), we show that CFSR, MERRA, NOCSv2.0, and OAFux+ISCCP are warm-biased not only in the western Pacific warm pool but also in the Mediterranean Sea, while CORE.2 is cold-biased in both regions. The NCEP 1, 2, and ERA-Interim are cold-biased over the warm pool but warm-biased in the Mediterranean Sea.

1383. The dependence of the ocean's MOC on mesoscale eddy diffusivities: A model study

Authors: John Marshall; Jeffery R. Scott; Anastasia Romanou; Maxwell Kelley; Anthony Leboissetier

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2017.01.001

URL: <https://doi.org/10.1016/j.ocemod.2017.01.001>

Publication Date: 2017-01-16

Volume/Issue: Volume 111

References: 36

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1384. Modelling the long-term evolution of worst-case Arctic oil spills

Authors: Hauke Blanken; Louis Bruno Tremblay; Susan Gaskin; Alexander Slavin

Journal: Marine Pollution Bulletin

Publisher: Elsevier BV

DOI: 10.1016/j.marpolbul.2016.12.070

URL: <https://doi.org/10.1016/j.marpolbul.2016.12.070>

Publication Date: 2017-01-15

Volume/Issue: Volume 116, Issue 1-2

References: 90

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Québec Océan, Natural Sciences and Engineering Research Council, Office of Naval Research

License: N/A

Open Access: N/A

Abstract: N/A

1385. Two decades [1992-2012] of surface wind analyses based on satellite scatterometer observations

Authors: Fabien Desbiolles; Abderrahim Bentamy; Bruno Blanke; Claude Roy; Alberto M. Mestas-Nuñez; Semyon A. Grodsky; Steven Herbette; Gildas Cambon; Christophe Maes

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2017.01.003

URL: <https://doi.org/10.1016/j.jmarsys.2017.01.003>

Publication Date: 2017-01-12

Volume/Issue: Volume 168

References: 58

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: CNES, IFREMER, CNRS (and 4 more)

License: N/A

Open Access: N/A

Abstract: N/A

1386. Annual variations in GPS-measured vertical displacements near Upernavik Isstrøm (Greenland) and contributions from surface mass loading

Authors: Lin Liu (Chinese University of Hong Kong); Shfaqat Abbas Khan (Technical University of Denmark); Tonie van Dam (University of Luxembourg); Joseph Ho Yin Ma (National University of Singapore, Chinese University of Hong Kong); Michael Bevis (Ohio State University)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jb013494

URL: <https://doi.org/10.1002/2016jb013494>

Publication Date: 2017-01-11

Volume/Issue: Volume 122, Issue 1

References: 44

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Carlsbergfondet, Det Frie Forskningsråd, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: In response to present-day ice mass loss on and near the Greenland Ice Sheet, steady crustal uplifts have been observed from the network of Global Positioning System (GPS) stations mounted on bedrock. In addition to the secular uplift trends, the GPS time series also show prominent annual variability. Here we examine the annual changes of the vertical displacements measured at two GPS stations (SRMP and UPVK) near Upernavik Isstrøm in western Greenland. We model elastic loading displacements due to various surface mass loading including three nonice components: atmospheric pressure, ocean bottom pressure, continental water storage, and one ice component, i.e., surface mass balance (SMB). We find that the contribution from atmospheric pressure changes can explain 46% and 78% of the annual amplitude observed in the GPS verticals at SRMP and UPVK, respectively. We also show that removing the predicted loading displacements due to SMB adversely increases the annual variance of the GPS residuals. However, using the GPS data alone, we cannot identify the exact cause(s) of this discrepancy because the annual loading displacements are sensitive to the SMB changes from over 85% of the ice sheet area. Alternatively, by differencing vertical displacements between the two stations, we find a good agreement between the modeled differential SMB loading displacements and the GPS residuals after removing nonice components. Our study highlights the necessity of correcting for nonice loading contributions in the GPS measurements of crustal deformation to infer ice mass changes in Greenland at annual periods.

1387. The Gulf of Aden Intermediate Water Intrusion Regulates the Southern Red Sea Summer Phytoplankton Blooms

Authors: Denis Dreano; Dionysios E. Raitsos; John Gittings; George Krokos; Ibrahim Hoteit

Journal: PLOS ONE

Publisher: Public Library of Science (PLoS)

DOI: 10.1371/journal.pone.0168440

URL: <https://doi.org/10.1371/journal.pone.0168440>

Publication Date: 2017-01-09

Volume/Issue: Volume 11, Issue 12

References: 51

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1388. Regime shift of the dominant factor for halocline depth in the Canada Basin during 1990-2008

Authors: Longjiang Mu; Jinping Zhao; Wenli Zhong

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-016-0883-0

URL: <https://doi.org/10.1007/s13131-016-0883-0>

Publication Date: 2017-01-09

Volume/Issue: Volume 36, Issue 1

References: 37

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1389. Overlooked possibility of a collapsed Atlantic Meridional Overturning Circulation in warming climate

Authors: Wei Liu (Scripps Institution of Oceanography / University of California San Diego); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego); Zhengyu Liu (University of Wisconsin-Madison); Jiang Zhu (University of Wisconsin-Madison)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.1601666

URL: <https://doi.org/10.1126/sciadv.1601666>

Publication Date: 2017-01-05

Volume/Issue: Volume 3, Issue 1

References: 44

Cited by: 222 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, U.S. Department of Energy, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Climate models underestimate the possibility of a collapsed Atlantic Meridional Overturning Circulation under climate warming.

1390. Climate-change driven range shifts of anchovy biomass projected by bio-physical coupling individual based model in the marginal seas of East Asia

Authors: Sukgeun Jung; Ig-Chan Pang; Joon-ho Lee; Kyunghwan Lee

Journal: Ocean Science Journal

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12601-016-0055-3

URL: <https://doi.org/10.1007/s12601-016-0055-3>

Publication Date: 2016-12-30

Volume/Issue: Volume 51, Issue 4

References: 91

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1391. Decadal changes in salinity in the oceanic subtropical gyres

Authors: Bryce A. Melzer (School of the Earth, Ocean and Environment University of South Carolina Columbia South Carolina USA); Bulusu Subrahmanyam (School of the Earth, Ocean and Environment University of South Carolina Columbia South Carolina USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012243

URL: <https://doi.org/10.1002/2016jc012243>

Publication Date: 2016-12-27

Volume/Issue: Volume 122, Issue 1

References: 51

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

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License: N/A

Open Access: N/A

Abstract: We analyzed spatial and temporal salinity trends in five subtropical gyre regions over the past six decades using Simple Ocean Data Assimilation (SODA) reanalysis with a focus on the subsurface salinity of the upper 1000 m of the ocean. Our results indicate an overall salinity increase within the mixed layer, and a salinity decrease at depths greater than 200 m in the global subtropical gyres over 61 years, of which each individual gyre was analyzed in further detail. We determine that freshwater fluxes at the air-sea interface are the primary drivers of the sea surface salinity (SSS) signature over these open ocean regions by quantifying the advective contribution within the surface layer. This was demonstrated through a mixed layer salinity budget in each subtropical gyre based on the vertically integrated advection and entrainment of salt. Our analysis of decadal variability of fluxes into and out of the gyres reveals little change in the strength of the mean currents through this region despite an increase in the annual export of salt in all subtropical gyres, with the meridional component dominating the zonal. This study reveals that the salt content of E-P maximum waters advected into the subtropical gyres is increasing over time. A combination of increasing direct evaporation over the regions with increasing remote evaporation over nearby E-P maxima is believed to be the main driver in increasing salinity of the subtropical oceans, suggesting an intensification of the global water cycle over decadal time scales.

1392. Surface flux and ocean heat transport convergence contributions to seasonal and interannual variations of ocean heat content

Authors: C. D. Roberts (Met Office Hadley Centre, Met Office, European Centre for Medium-Range Weather Forecasts (ECMWF)); M. D. Palmer (Met Office Hadley Centre, Met Office); R. P. Allan (University of Reading); D.G. Desbruyeres (National Oceanography Centre); P. Hyder (Met Office Hadley Centre, Met Office); C. Liu (University of Reading); D. Smith (Met Office Hadley Centre, Met Office)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012278

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Publication Date: 2016-12-27

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References: 71

Cited by: 53 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the Joint DBEIS/Defra Met Office Hadley Centre Climate Programme, Natural Environment Research Council DEEP-C, SMURPHS

License: CC BY 4.0

Open Access: N/A

Abstract: We present an observation-based heat budget analysis for seasonal and interannual variations of ocean heat content (H) in the mixed layer (H_{mld}) and full-depth ocean (H_{tot}). Surface heat flux and ocean heat content estimates are combined using a novel Kalman smoother-based method. Regional contributions from ocean heat transport convergences are inferred as a residual and the dominant drivers of H_{mld} and H_{tot} are quantified for seasonal and interannual time scales. We find that non-Ekman ocean heat transport processes dominate H_{mld} variations in the equatorial oceans and regions of strong ocean currents and substantial eddy activity. In these locations, surface temperature anomalies generated by ocean dynamics result in turbulent flux anomalies that drive the overlying atmosphere. In addition, we find large regions of the Atlantic and Pacific oceans where heat transports combine with local air-sea fluxes to generate mixed layer temperature anomalies. In all locations, except regions of deep convection and water mass transformation, interannual variations in H_{tot} are dominated by the internal rearrangement of heat by ocean dynamics rather than the loss or addition of heat at the surface. Our analysis suggests that, even in extratropical latitudes, initialization of ocean dynamical processes could be an important source of skill for interannual predictability of H_{mld} and H_{tot}. Furthermore, we expect variations in H_{tot} (and thus thermosteric sea level) to be more predictable than near surface temperature anomalies due to the increased importance of ocean heat transport processes for full-depth heat budgets.

1393. Ice Sheet Model Intercomparison Project (ISMIP6) contribution to CMIP6

Authors: Sophie M. J. Nowicki; Anthony Payne; Eric Larour; Helene Seroussi; Heiko Goelzer; William Lipscomb; Jonathan Gregory; Ayako Abe-Ouchi; Andrew Shepherd

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-4521-2016

URL: <https://doi.org/10.5194/gmd-9-4521-2016>

Publication Date: 2016-12-21

Volume/Issue: Volume 9, Issue 12

References: 133

Cited by: 157 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Reducing the uncertainty in the past, present, and future contribution of ice sheets to sea-level change requires a coordinated effort between the climate and glaciology communities. The Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6) is the primary activity within the Coupled Model Intercomparison Project - phase 6 (CMIP6) focusing on the Greenland and Antarctic ice sheets. In this paper, we describe the framework for ISMIP6 and its relationship with other activities within CMIP6. The ISMIP6 experimental design relies on CMIP6 climate models and includes, for the first time within CMIP, coupled ice-sheet-climate models as well as standalone ice-sheet models. To facilitate analysis of the multi-model ensemble and to generate a set of standard climate inputs for standalone ice-sheet models, ISMIP6 defines a protocol for all variables related to ice sheets. ISMIP6 will provide a basis for investigating the feedbacks, impacts, and sea-level changes associated with dynamic ice sheets and for quantifying the uncertainty in ice-sheet-sourced global sea-level change.

1394. Global evaluation of new GRACE mascon products for hydrologic applications

Authors: Bridget R. Scanlon (University of Texas at Austin); Zizhan Zhang (University of Texas at Austin); Himanshu Save (University of Texas at Austin); David N. Wiese (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Felix W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Di Long (Tsinghua University); Laurent Longuevergne (Géosciences Rennes Université de Rennes Campus de Beaulieu Rennes Cedex France); Jianli Chen (University of Texas at Austin)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016wr019494

URL: <https://doi.org/10.1002/2016wr019494>

Publication Date: 2016-12-17

Volume/Issue: Volume 52, Issue 12

References: 49

Cited by: 410 publications

Language: en

Subjects: N/A

Keywords: terrestrial water storage, trends in water resources, GRACE satellites, remote sensing, global river basins

Funding: Jackson School of Geosciences

License: N/A

Open Access: N/A

Abstract: Recent developments in mascon (mass concentration) solutions for GRACE (Gravity Recovery and Climate Experiment) satellite data have significantly increased the spatial localization and amplitude of recovered terrestrial Total Water Storage anomalies (TWSA); however, land hydrology applications have been limited. Here we compare TWSA from April 2002 through March 2015 from (1) newly released GRACE mascons from the Center for Space Research (CSR-M) with (2) NASA JPL mascons (JPL-M), and with (3) CSR Tellus gridded spherical harmonics rescaled (sf) (CSRT-GSH.sf) in 176 river basins, ~60% of the global land area. Time series in TWSA mascons (CSR-M and JPL-M) and spherical harmonics are highly correlated (rank correlation coefficients mostly >0.9). The signal from long-term trends (up to +/-20 mm/yr) is much less than that from seasonal amplitudes (up to 250 mm). Net long-term trends, summed over all 176 basins, are similar for CSR and JPL mascons (66-69 km³/yr) but are lower for spherical harmonics (~14 km³/yr). Long-term TWSA declines are found mostly in irrigated basins (-41 to -69 km³/yr). Seasonal amplitudes agree among GRACE solutions, increasing confidence in GRACE-based seasonal fluctuations. Rescaling spherical harmonics significantly increases agreement with mascons for seasonal fluctuations, but less for long-term trends. Mascons provide advantages relative to spherical harmonics, including (1) reduced leakage from land to ocean increasing signal amplitude, and (2) application of geophysical data constraints during processing with little empirical postprocessing requirements, making it easier for nongeodetic users. Results of this product intercomparison should allow hydrologists to better select suitable GRACE solutions for hydrologic applications.

1395. Recent Advances in Japanese Fisheries Science in the Kuroshio-Oyashio Region through Development of the FRA-ROMS Ocean Forecast System: Overview of the Reproducibility of Reanalysis Products

Authors: Hiroshi Kuroda; Takashi Setou; Shigeho Kakehi; Shin-ichi Ito; Takeshi Taneda; Tomonori Azumaya; Denzo Inagake; Yutaka Hiroe; Kenji Morinaga; Makoto Okazaki; Takashi Yokota; Takeshi Okunishi; Kazuhiro Aoki; Yugo Shimizu; Daisuke Hasegawa; Tomowo Watanabe

Journal: Open Journal of Marine Science

Publisher: Scientific Research Publishing, Inc.

DOI: 10.4236/ojms.2017.71006

URL: <https://doi.org/10.4236/ojms.2017.71006>

Publication Date: 2016-12-13

Volume/Issue: Volume 07, Issue 01

References: 99

Cited by: 62 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1396. Oceans Melting Greenland: Early Results from NASA's Ocean-Ice Mission in Greenland

Authors: Unknown Author; Ian Fenty; Josh Willis; Ala Khazendar; Steven Dinardo; René Forsberg; Ichiro Fukumori; David Holland; Martin Jakobsson; Delwyn Moller; James Morison; Andreas Münchow; Eric Rignot; Michael Schodlok; Andrew Thompson; Kirsteen Tinto; Matthew Rutherford; Nicole Trenholm

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2016.100

URL: <https://doi.org/10.5670/oceanog.2016.100>

Publication Date: 2016-12-12

Volume/Issue: Volume 29, Issue 4

References: 47

Cited by: 81 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1397. Decade-long deep-ocean warming detected in the subtropical South Pacific

Authors: Denis L. Volkov (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami Florida USA, University of Miami); Sang-Ki Lee (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami Florida USA); Felix W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rick Lumpkin (Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami Florida USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl071661

URL: <https://doi.org/10.1002/2016gl071661>

Publication Date: 2016-12-09

Volume/Issue: Volume 44, Issue 2

References: 46

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NOAA-AOML, NASA Physical Oceanography

License: N/A

Open Access: N/A

Abstract: The persistent energy imbalance at the top of the atmosphere, inferred from satellite measurements, indicates that the Earth's climate system continues to accumulate excess heat. As only sparse and irregular measurements of ocean heat below 2000 m depth exist, one of the most challenging questions in global climate change studies is whether the excess heat has already penetrated into the deep ocean. Here we perform a comprehensive analysis of satellite and in situ measurements to report that a significant deep-ocean warming occurred in the subtropical South Pacific Ocean over the past decade (2005-2014). The local accumulation of heat accounted for up to a quarter of the global ocean heat increase, with directly and indirectly inferred deep ocean (below 2000 m) contribution of 2.4 ± 1.4 and $6.1\text{-}10.1 \pm 4.4\%$, respectively. We further demonstrate that this heat accumulation is consistent with a decade-long intensification of the subtropical convergence, possibly linked to the persistent La Niña-like state.

1398. Testing the Quality of Sea-Level Data Using the GECCO Adjoint Assimilation Approach

Authors: Martin G. Scharffenberg; Armin Köhl; Detlef Stammer

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-016-9401-3

URL: <https://doi.org/10.1007/s10712-016-9401-3>

Publication Date: 2016-11-26

Volume/Issue: Volume 38, Issue 1

References: 13

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1399. Upper oceanic response to tropical cyclone Phailin in the Bay of Bengal using a coupled atmosphere-ocean model

Authors: Kumar Ravi Prakash; Vimlesh Pant

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-016-1020-5

URL: <https://doi.org/10.1007/s10236-016-1020-5>

Publication Date: 2016-11-23

Volume/Issue: Volume 67, Issue 1

References: 87

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1400. Oceanic eddy detection and lifetime forecast using machine learning methods

Authors: Mohammad D. Ashkezari (Massachusetts Institute of Technology); Christopher N. Hill (Massachusetts Institute of Technology); Christopher N. Follett (Massachusetts Institute of Technology); Gaël Forget (Massachusetts Institute of Technology); Michael J. Follows (Massachusetts Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl071269

URL: <https://doi.org/10.1002/2016gl071269>

Publication Date: 2016-11-21

Volume/Issue: Volume 43, Issue 23

References: 24

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: eddy lifetime, remote sensing, ocean, machine learning, eddy

Funding: Simons Foundation

License: N/A

Open Access: N/A

Abstract: We report a novel altimetry-based machine learning approach for eddy identification and characterization. The machine learning models use daily maps of geostrophic velocity anomalies and are trained according to the phase angle between the zonal and meridional components at each grid point. The trained models are then used to identify the corresponding eddy phase patterns and to predict the lifetime of a detected eddy structure. The performance of the proposed method is examined at two dynamically different regions to demonstrate its robust behavior and region independency.

1401. South Atlantic interbasin exchanges of mass, heat, salt and anthropogenic carbon

Authors: G.R. Evans; E.L. McDonagh; B.A. King; H.L. Bryden; D.C.E. Bakker; P.J. Brown; U. Schuster; K.G. Speer; S.M.A.C. van Heuven

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2016.11.005

URL: <https://doi.org/10.1016/j.pocean.2016.11.005>

Publication Date: 2016-11-19

Volume/Issue: Volume 151

References: 182

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration, Natural Environment Research Council, European Commission (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1402. Optimizing estimates of annual variations and trends in geocenter motion and J2 from a combination of GRACE data and geophysical models

Authors: Yu Sun (Delft University of Technology); Riccardo Riva (Delft University of Technology); Pavel Ditmar (Delft University of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jb013073

URL: <https://doi.org/10.1002/2016jb013073>

Publication Date: 2016-11-18

Volume/Issue: Volume 121, Issue 11

References: 56

Cited by: 232 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: 973 program, Netherlands Organization for Scientific Research

License: N/A

Open Access: N/A

Abstract: The focus of the study is optimizing the technique for estimating geocenter motion and variations in J2 by combining data from the Gravity Recovery and Climate Experiment (GRACE) satellite mission with output from an Ocean Bottom Pressure model and a Glacial Isostatic Adjustment (GIA) model. First, we conduct an end-to-end numerical simulation study. We generate input time-variable gravity field observations by perturbing a synthetic Earth model with realistically simulated errors. We show that it is important to avoid large errors at short wavelengths and signal leakage from land to ocean, as well as to account for self-attraction and loading effects. Second, the optimal implementation strategy is applied to real GRACE data. We show that the estimates of annual amplitude in geocenter motion are in line with estimates from other techniques, such as satellite laser ranging (SLR) and global GPS inversion. At the same time, annual amplitudes of C10 and C11 are increased by about 50% and 20%, respectively, compared to estimates based on Swenson et al. (2008). Estimates of J2 variations are by about 15% larger than SLR results in terms of annual amplitude. Linear trend estimates are dependent on the adopted GIA model but still comparable to some SLR results.

1403. Seasonal and interannual variability of the Arctic sea ice: A comparison between AO-FVCOM and observations

Authors: Yu Zhang (Department of Marine Science and Technology, College of Marine Sciences, Shanghai Ocean University; Shanghai China, Department of Fisheries Oceanography, School for Marine Science and Technology, University of Massachusetts-Dartmouth; New Bedford Massachusetts USA); Changsheng Chen (Department of Marine Science and Technology, College of Marine Sciences, Shanghai Ocean University; Shanghai China, Department of Fisheries Oceanography, School for Marine Science and Technology, University of Massachusetts-Dartmouth; New Bedford Massachusetts USA); Robert C. Beardsley (Woods Hole Oceanographic Institution); Guoping Gao (Department of Marine Science and Technology, College of Marine Sciences, Shanghai Ocean University; Shanghai China); Jianhua Qi (Department of Fisheries Oceanography, School for Marine Science and Technology, University of Massachusetts-Dartmouth; New Bedford Massachusetts USA); Huichan Lin (Department of Fisheries Oceanography, School for Marine Science and Technology, University of Massachusetts-Dartmouth; New Bedford Massachusetts USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc011841

URL: <https://doi.org/10.1002/2016jc011841>

Publication Date: 2016-11-16

Volume/Issue: Volume 121, Issue 11

References: 61

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: sea ice extent, sea ice velocity, sea ice thickness, sea ice concentration, AO-FVCOM

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1404. Subduction of North Pacific Tropical Water and its equatorward pathways as shown by a simulated passive tracer

Authors: Xunwei Nie (Chinese Academy of Sciences); Shan Gao (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Fan Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Tangdong Qu (Joint Institute for Regional Earth System Science and Engineering University of California Los Angeles California USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012305

URL: <https://doi.org/10.1002/2016jc012305>

Publication Date: 2016-11-16

Volume/Issue: Volume 121, Issue 12

References: 64

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: mixing, North Pacific Tropical Water, resurface, pathway, subduction

Funding: Chinese Academy of Sciences, NSF

License: N/A

Open Access: N/A

Abstract: N/A

1405. Sub-basin-scale sea level budgets from satellite altimetry, Argo floats and satellite gravimetry: a case study in the North Atlantic Ocean

Authors: Marcel Kleinherenbrink; Riccardo Riva; Yu Sun

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-12-1179-2016

URL: <https://doi.org/10.5194/os-12-1179-2016>

Publication Date: 2016-11-15

Volume/Issue: Volume 12, Issue 6

References: 64

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Stichting voor de Technische Wetenschappen

License: Creative Commons

Open Access: N/A

Abstract: . In this study, for the first time, an attempt is made to close the sea level budget on a sub-basin scale in terms of trend and amplitude of the annual cycle. We also compare the residual time series after removing the trend, the semiannual and the annual signals. To obtain errors for altimetry and Argo, full variance-covariance matrices are computed using correlation functions and their errors are fully propagated. For altimetry, we apply a geographically dependent intermission bias [Ablain et al.(2015)], which leads to differences in trends up to 0.8 mm yr⁻¹. Since Argo float measurements are non-homogeneously spaced, steric sea levels are first objectively interpolated onto a grid before averaging. For the Gravity Recovery And Climate Experiment (GRACE), gravity fields full variance-covariance matrices are used to propagate errors and statistically filter the gravity fields. We use four different filtered gravity field solutions and determine which post-processing strategy is best for budget closure. As a reference, the standard 96 degree Dense Decorrelation Kernel-5 (DDK5)-filtered Center for Space Research (CSR) solution is used to compute the mass component (MC). A comparison is made with two anisotropic Wiener-filtered CSR solutions up to degree and order 60 and 96 and a Wiener-filtered 90 degree ITSG solution. Budgets are computed for 10 polygons in the North Atlantic Ocean, defined in a way that the error on the trend of the MC plus steric sea level remains within 1 mm yr⁻¹. Using the anisotropic Wiener filter on CSR gravity fields expanded up to spherical harmonic degree 96, it is possible to close the sea level budget in 9 of 10 sub-basins in terms of trend. Wiener-filtered Institute of Theoretical geodesy and Satellite Geodesy (ITSG) and the standard DDK5-filtered CSR solutions also close the trend budget if a glacial isostatic adjustment (GIA) correction error of 10-20 % is applied; however, the performance of the DDK5-filtered solution strongly depends on the orientation of the polygon due to residual striping. In 7 of 10 sub-basins, the budget of the annual cycle is closed, using the DDK5-filtered CSR or the Wiener-filtered ITSG solutions. The Wiener-filtered 60 and 96 degree CSR solutions, in combination with Argo, lack amplitude and suffer from what appears to be hydrological leakage in the Amazon and Sahel regions. After reducing the trend, the semiannual and the annual signals, 24-53 % of the residual variance in altimetry-derived sea level time series is explained by the combination of Argo steric sea levels and the Wiener-filtered ITSG

MC. Based on this, we believe that the best overall solution for the MC of the sub-basin-scale budgets is the Wiener-filtered ITSG gravity fields. The interannual variability is primarily a steric signal in the North Atlantic Ocean, so for this the choice of filter and gravity field solution is not really significant.

1406. El Niño, La Niña, and the global sea level budget

Authors: Christopher G. Piecuch; Katherine J. Quinn

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-12-1165-2016

URL: <https://doi.org/10.5194/os-12-1165-2016>

Publication Date: 2016-11-11

Volume/Issue: Volume 12, Issue 6

References: 68

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: . Previous studies show that nonseasonal variations in global-mean sea level (GMSL) are significantly correlated with El Niño-Southern Oscillation (ENSO). However, it has remained unclear to what extent these ENSO-related GMSL fluctuations correspond to steric (i.e., density) or barystatic (mass) effects. Here we diagnose the GMSL budget for ENSO events observationally using data from profiling floats, satellite gravimetry, and radar altimetry during 2005-2015. Steric and barystatic effects make comparable contributions to the GMSL budget during ENSO, in contrast to previous interpretations based largely on hydrological models, which emphasize the barystatic component. The steric contributions reflect changes in global ocean heat content, centered on the Pacific. Distributions of ocean heat storage in the Pacific arise from a mix of diabatic and adiabatic effects. Results have implications for understanding the surface warming slowdown and demonstrate the usefulness of the Global Ocean Observing System for constraining Earth's hydrological cycle and radiation imbalance.

1407. Transit Time Distribution based on the ECCO-JPL Ocean Data Assimilation

Authors: Takamitsu Ito; Ou Wang

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2016.10.015

URL: <https://doi.org/10.1016/j.jmarsys.2016.10.015>

Publication Date: 2016-11-05

Volume/Issue: Volume 167

References: 48

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. National Science Foundation, National Aeronautics and Space Administration (NASA)

License: N/A

Open Access: N/A

Abstract: N/A

1408. Four-dimensional variational ocean reanalysis: a 30-year high-resolution dataset in the western North Pacific (FORA-WNP30)

Authors: Norihisa Usui; Tsuyoshi Wakamatsu; Yusuke Tanaka; Nariaki Hirose; Takahiro Toyoda; Shiro Nishikawa; Yosuke Fujii; Yasushi Takatsuki; Hiromichi Igarashi; Haruka Nishikawa; Yoichi Ishikawa; Tsurane Kuragano; Masafumi Kamachi

Journal: Journal of Oceanography

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10872-016-0398-5

URL: <https://doi.org/10.1007/s10872-016-0398-5>

Publication Date: 2016-11-05

Volume/Issue: Volume 73, Issue 2

References: 115

Cited by: 106 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science, Ministry of Education, Culture, Sports, Science, and Technology

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1409. Brief communication: Impacts of a developing polynya off Commonwealth Bay, East Antarctica, triggered by grounding of iceberg B09B

Authors: Christopher J. Fogwill; Erik van Sebille; Eva A. Cougnon; Chris S. M. Turney; Steve R. Rintoul; Benjamin K. Galton-Fenzi; Graeme F. Clark; E. M. Marzinelli; Eleanor B. Rainsley; Lionel Carter

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-2603-2016

URL: <https://doi.org/10.5194/tc-10-2603-2016>

Publication Date: 2016-11-04

Volume/Issue: Volume 10, Issue 6

References: 19

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council

License: Creative Commons

Open Access: N/A

Abstract: . The dramatic calving of the Mertz Glacier tongue in 2010, precipitated by the movement of iceberg B09B, reshaped the oceanographic regime across the Mertz Polynya and Commonwealth Bay, regions where high-salinity shelf water (HSSW) - the precursor to Antarctic bottom water (AABW) - is formed. Here we present post-calving observations that suggest that this reconfiguration and subsequent grounding of B09B have driven the development of a new polynya and associated HSSW production off Commonwealth Bay. Supported by satellite observations and modelling, our findings demonstrate how local icescape changes may impact the formation of HSSW, with potential implications for large-scale ocean circulation.

1410. Evaluation of GRACE Mascon Gravity Solution in Relation to Interannual Oceanic Water Mass Variations

Authors: Bryce A. Melzer; Bulusu Subrahmanyam

Journal: IEEE Transactions on Geoscience and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tgrs.2016.2616760

URL: <https://doi.org/10.1109/tgrs.2016.2616760>

Publication Date: 2016-11-01

Volume/Issue: Volume 55, Issue 2

References: 38

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: Gravity Recovery and Climate Experiment (GRACE), ocean mass, GECCO2, Altimetry, water cycle

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1411. An approach to computing discrete adjoints for MPI-parallelized models applied to Ice Sheet System Model 4.11

Authors: Eric Larour; Jean Utke; Anton Bovin; Mathieu Morlighem; Gilberto Perez

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-3907-2016

URL: <https://doi.org/10.5194/gmd-9-3907-2016>

Publication Date: 2016-11-01

Volume/Issue: Volume 9, Issue 11

References: 43

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

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Open Access: N/A

Abstract: . Within the framework of sea-level rise projections, there is a strong need for hindcast validation of the evolution of polar ice sheets in a way that tightly matches observational records (from radar, gravity, and altimetry observations mainly). However, the computational requirements for making hindcast reconstructions possible are severe and rely mainly on the evaluation of the adjoint state of transient ice-flow models. Here, we look at the computation of adjoints in the context of the NASA/JPL/UCI Ice Sheet System Model (ISSM), written in C++ and designed for parallel execution with MPI. We present the adaptations required in the way the software is designed and written, but also generic adaptations in the tools facilitating the adjoint computations. We concentrate on the use of operator overloading coupled with the AdjoinableMPI library to achieve the adjoint computation of the ISSM. We present a comprehensive approach to (1) carry out type changing through the ISSM, hence facilitating operator overloading, (2) bind to external solvers such as MUMPS and GSL-LU, and (3) handle MPI-based parallelism to scale the capability. We demonstrate the success of the approach by computing sensitivities of hindcast metrics such as the misfit to observed records of surface altimetry on the northeastern Greenland Ice Stream, or the misfit to observed records of surface velocities on Upernavik Glacier, central West Greenland. We also provide metrics for the scalability of the approach, and the expected performance. This approach has the potential to enable a new generation of hindcast-validated projections that make full use of the wealth of datasets currently being collected, or already collected, in Greenland and Antarctica.

1412. Seasonality of submesoscale dynamics in the Kuroshio Extension

Authors: Cesar B. Rocha (Scripps Institution of Oceanography / University of California San Diego); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Teresa K. Chereskin (Scripps Institution of Oceanography / University of California San Diego); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl071349

URL: <https://doi.org/10.1002/2016gl071349>

Publication Date: 2016-10-25

Volume/Issue: Volume 43, Issue 21

References: 26

Cited by: 135 publications

Language: en

Subjects: N/A

Keywords: COMPIRA, inertia-gravity waves, submesoscale turbulence, Kuroshio Extension, submesoscale dynamics (and 1 more)

Funding: National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Recent studies show that the vigorous seasonal cycle of the mixed layer modulates upper ocean submesoscale turbulence. Here we provide model-based evidence that the seasonally changing upper ocean stratification in the Kuroshio Extension also modulates submesoscale (here 10-100 km) inertia-gravity waves. Summertime restratification weakens submesoscale turbulence but enhances inertia-gravity waves near the surface. Thus, submesoscale turbulence and inertia-gravity waves undergo vigorous out-of-phase seasonal cycles. These results imply a strong seasonal modulation of the accuracy of geostrophic velocity diagnosed from submesoscale sea surface height delivered by the Surface Water and Ocean Topography satellite mission.

1413. Oxygen cycling in the northern Benguela Upwelling System: Modelling oxygen sources and sinks

Authors: Martin Schmidt; Anja Eggert

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2016.09.004

URL: <https://doi.org/10.1016/j.pocean.2016.09.004>

Publication Date: 2016-10-17

Volume/Issue: Volume 149

References: 104

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: German Federal Ministry of Education and Research, NASA Ocean Vector Winds Science Team, National Oceanographic Partnership Program

License: N/A

Open Access: N/A

Abstract: N/A

1414. Comparison between the Local Ensemble Transform Kalman Filter (LETKF) and 4D-Var in atmospheric CO₂ flux inversion with the Goddard Earth Observing System-Chem model and the observation impact diagnostics from the LETKF

Authors: Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jd025100

URL: <https://doi.org/10.1002/2016jd025100>

Publication Date: 2016-10-15

Volume/Issue: Volume 121, Issue 21

References: 56

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA OCO-2 science team grant, NASA Carbon Cycle Science program, NASA Carbon Monitoring System program

License: N/A

Open Access: N/A

Abstract: Ensemble Kalman filter (EnKF) and 4D-Variational (4D-Var) are two advanced data assimilation methods that are the basis of numerical weather prediction and have been extensively used in trace gas assimilation and inverse modeling. In this study, we compare 4D-Var and the Local Ensemble Transform Kalman Filter (LETKF), one type of EnKF, in estimating CO₂ fluxes with both simulated and real satellite data from Greenhouse gases Observing Satellite (GOSAT) and propose a method to calculate flux changes and flux error reductions from assimilating each observation within the LETKF. The results show that the mean posterior flux accuracy across 11 land regions defined by the Atmospheric Tracer Transport Model Intercomparison Project is comparable between 4D-Var and the LETKF, as shown in the Observing System Simulation Experiment, but the differences between the LETKF and 4D-Var are relatively larger over data sparse regions. We show that this is most likely due to the fact that the observations from a much broader region have impact on flux estimation in 4D-Var than in the LETKF. As a result, the posterior fluxes from 4D-Var are more consistent with the atmospheric CO₂ growth rate. We find that the inversion results are less dependent on inversion methods with the increase of observations. With real GOSAT observations, we show that the posterior flux changes in 2011 relative to 2010 are more consistent between these two methods than the absolute estimates.

1415. Decomposition of thermal and dynamic changes in the South China Sea induced by boundary forcing and surface fluxes during 1970-2000

Authors: Jun Wei (Chinese Academy of Meteorological Sciences, Peking University); Paola Malanotte-Rizzoli (Massachusetts Institute of Technology); Ming-Ting Li (Peking University); Hao Wang (University of Leicester)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012078

URL: <https://doi.org/10.1002/2016jc012078>

Publication Date: 2016-10-13

Volume/Issue: Volume 121, Issue 11

References: 72

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the National Natural Science Foundation of China, the Major State Basic Research Development Program, the Strategic Priority Research Program of the Chinese Academy of Sciences

License: N/A

Open Access: N/A

Abstract: N/A

1416. Noise analysis for environmental loading effect on GPS time series

Authors: Xiaoxing He

Journal: Acta Geodynamica et Geomaterialia

Publisher: Institute of Rock Structure and Mechanics, AS CR

DOI: 10.13168/agg.2016.0034

URL: <https://doi.org/10.13168/agg.2016.0034>

Publication Date: 2016-10-13

Volume/Issue: N/A

References: 49

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1417. Evaluating mass loading products by comparison to GPS array daily solutions

Authors: Chang Xu

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggw385

URL: <https://doi.org/10.1093/gji/ggw385>

Publication Date: 2016-10-08

Volume/Issue: Volume 208, Issue 1

References: 46

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1418. Ocean forecasting for the German Bight: from regional to coastal scales

Authors: Emil V. Stanev; Johannes Schulz-Stellenfleth; Joanna Staneva; Sebastian Grayek; Sebastian Grashorn; Arno Behrens; Wolfgang Koch; Johannes Pein

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-12-1105-2016

URL: <https://doi.org/10.5194/os-12-1105-2016>

Publication Date: 2016-10-07

Volume/Issue: Volume 12, Issue 5

References: 106

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . This paper describes recent developments based on advances in coastal ocean forecasting in the fields of numerical modeling, data assimilation, and observational array design, exemplified by the Coastal Observing System for the North and Arctic Seas (COSYNA). The region of interest is the North and Baltic seas, and most of the coastal examples are for the German Bight. Several pre-operational applications are presented to demonstrate the outcome of using the best available science in coastal ocean predictions. The applications address the nonlinear behavior of the coastal ocean, which for the studied region is manifested by the tidal distortion and generation of shallow-water tides. Led by the motivation to maximize the benefits of the observations, this study focuses on the integration of observations and modeling using advanced statistical methods. Coastal and regional ocean forecasting systems do not operate in isolation but are linked, either weakly by using forcing data or interactively using two-way nesting or unstructured-grid models. Therefore, the problems of downscaling and upscaling are addressed, along with a discussion of the potential influence of the information from coastal observatories or coastal forecasting systems on the regional models. One example of coupling coarse-resolution regional models with a fine-resolution model interface in the area of straits connecting the North and Baltic seas using a two-way nesting method is presented. Illustrations from the assimilation of remote sensing, in situ and high-frequency (HF) radar data, the prediction of wind waves and storm surges, and possible applications to search and rescue operations are also presented. Concepts for seamless approaches to link coastal and regional forecasting systems are exemplified by the application of an unstructured-grid model for the Ems Estuary.

1419. Can We Infer Diapycnal Mixing Rates from the World Ocean Temperature-Salinity Distribution?

Authors: Olivier Arzel (Laboratoire d'Océanographie Physique et Spatiale, Université de Bretagne Occidentale, Brest, France); Alain Colin de Verdière (Laboratoire d'Océanographie Physique et Spatiale, Université de Bretagne Occidentale, Brest, France)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0152.1

URL: <https://doi.org/10.1175/jpo-d-16-0152.1>

Publication Date: 2016-10-05

Volume/Issue: Volume 46, Issue 12

References: 91

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The turbulent diapycnal mixing in the ocean is currently obtained from microstructure and finestructure measurements, dye experiments, and inverse models. This study presents a new method that infers the diapycnal mixing from low-resolution numerical calculations of the World Ocean whose temperatures and salinities are restored to the climatology. At the difference of robust general circulation ocean models, diapycnal diffusion is not prescribed but inferred. At steady state the buoyancy equation shows an equilibrium between the large-scale diapycnal advection and the restoring terms that take the place of the divergence of eddy buoyancy fluxes. The geography of the diapycnal flow reveals a strong regional variability of water mass transformations. Positive values of the diapycnal flow indicate an erosion of a deep-water mass and negative values indicate a creation. When the diapycnal flow is upward, a diffusion law can be fitted in the vertical and the diapycnal eddy diffusivity is obtained throughout the water column. The basin averages of diapycnal diffusivities are small in the first 1500 m [$O(10^{-5}) \text{ m}^2 \text{ s}^{-1}$] and increase downward with bottom values of about $2.5 \times 10^{-4} \text{ m}^2 \text{ s}^{-1}$ in all ocean basins, with the exception of the Southern Ocean (50 degrees-30 degreesS), where they reach $12 \times 10^{-4} \text{ m}^2 \text{ s}^{-1}$. This study confirms the small diffusivity in the thermocline and the robustness of the higher canonical Munk's value in the abyssal ocean. It indicates that the upward diapycnal transport in the Atlantic mostly takes place in the abyss and the upper ocean, supporting the quasi-adiabatic character of the middepth overturning.

1420. Spatial Patterns of Sea Level Variability Associated with Natural Internal Climate Modes

Authors: Weiqing Han; Gerald A. Meehl; Detlef Stammer; Aixue Hu; Benjamin Hamlington; Jessica Kenigson; Hindumathi Palanisamy; Philip Thompson

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-016-9386-y

URL: <https://doi.org/10.1007/s10712-016-9386-y>

Publication Date: 2016-10-04

Volume/Issue: Volume 38, Issue 1

References: 194

Cited by: 72 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, U.S. Department of Energy, U.S. Department of Energy (and 3 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1421. Inferring ocean circulation during the Last Glacial Maximum and last deglaciation using data and models

Authors: Daniel E. Amrhein

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/8428

URL: <https://doi.org/10.1575/1912/8428>

Publication Date: 2016-10-04

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: Last Glacial Maximum|Deglaciation|paleoceanography

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1422. High-resolution CSR GRACE RL05 mascons

Authors: Himanshu Save (University of Texas at Austin); Srinivas Bettadpur (University of Texas at Austin); Byron D. Tapley (University of Texas at Austin)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jb013007

URL: <https://doi.org/10.1002/2016jb013007>

Publication Date: 2016-09-23

Volume/Issue: Volume 121, Issue 10

References: 55

Cited by: 902 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA's GRACE, NASA MEaSUREs Program

License: N/A

Open Access: N/A

Abstract: The determination of the gravity model for the Gravity Recovery and Climate Experiment (GRACE) is susceptible to modeling errors, measurement noise, and observability issues. The ill-posed GRACE estimation problem causes the unconstrained GRACE RL05 solutions to have north-south stripes. We discuss the development of global equal area mascon solutions to improve the GRACE gravity information for the study of Earth surface processes. These regularized mascon solutions are developed with a 1 degrees resolution using Tikhonov regularization in a geodesic grid domain. These solutions are derived from GRACE information only, and no external model or data is used to inform the constraints. The regularization matrix is time variable and will not bias or attenuate future regional signals to some past statistics from GRACE or other models. The resulting Center for Space Research (CSR) mascon solutions have no stripe errors and capture all the signals observed by GRACE within the measurement noise level. The solutions are not tailored for specific applications and are global in nature. This study discusses the solution approach and compares the resulting solutions with postprocessed results from the RL05 spherical harmonic solutions and other global mascon solutions for studies of Arctic ice sheet processes, ocean bottom pressure variation, and land surface total water storage change. This suite of comparisons leads to the conclusion that the mascon solutions presented here are an enhanced representation of the RL05 GRACE solutions and provide accurate surface-based gridded information that can be used without further processing.

1423. Potential Seasonal Terrestrial Water Storage Monitoring from GPS Vertical Displacements: A Case Study in the Lower Three-Rivers Headwater Region, China

Authors: Bao Zhang (Wuhan University); Yibin Yao (Wuhan University); Hok Fok (Wuhan University); Yufeng Hu (Wuhan University); Qiang Chen (University of Luxembourg)

Journal: Sensors

Publisher: MDPI AG

DOI: 10.3390/s16091526

URL: <https://doi.org/10.3390/s16091526>

Publication Date: 2016-09-19

Volume/Issue: Volume 16, Issue 9

References: 60

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: This study uses the observed vertical displacements of Global Positioning System (GPS) time series obtained from the Crustal Movement Observation Network of China (CMONOC) with careful pre- and post-processing to estimate the seasonal crustal deformation in response to the hydrological loading in lower three-rivers headwater region of southwest China, followed by inferring the annual EWH changes through geodetic inversion methods. The Helmert Variance Component Estimation (HVCE) and the Minimum Mean Square Error (MMSE) criterion were successfully employed. The GPS inferred EWH changes agree well qualitatively with the Gravity Recovery and Climate Experiment (GRACE)-inferred and the Global Land Data Assimilation System (GLDAS)-inferred EWH changes, with a discrepancy of 3.2-3.9 cm and 4.8-5.2 cm, respectively. In the research areas, the EWH changes in the Lancang basin is larger than in the other regions, with a maximum of 21.8-24.7 cm and a minimum of 3.1-6.9 cm.

1424. OMIP contribution to CMIP6: experimental and diagnostic protocol for the physical component of the Ocean Model Intercomparison Project

Authors: Stephen M. Griffies; Gokhan Danabasoglu; Paul J. Durack; Alistair J. Adcroft; V. Balaji; Claus W. Böning; Eric P. Chassignet; Enrique Curchitser; Julie Deshayes; Helge Drange; Baylor Fox-Kemper; Peter J. Gleckler; Jonathan M. Gregory; Helmuth Haak; Robert W. Hallberg; Patrick Heimbach; Helene T. Hewitt; David M. Holland; Tatiana Ilyina; Johann H. Jungclaus; Yoshiki Komuro; John P. Krasting; William G. Large; Simon J. Marsland; Simona Masina; Trevor J. McDougall; A. J. George Nurser; James C. Orr; Anna Pirani; Fangli Qiao; Ronald J. Stouffer; Karl E. Taylor; Anne Marie Treguier; Hiroyuki Tsujino; Petteri Uotila; Maria Valdivieso; Qiang Wang; Michael Winton; Stephen G. Yeager

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-3231-2016

URL: <https://doi.org/10.5194/gmd-9-3231-2016>

Publication Date: 2016-09-19

Volume/Issue: Volume 9, Issue 9

References: 237

Cited by: 228 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Ocean Model Intercomparison Project (OMIP) is an endorsed project in the Coupled Model Intercomparison Project Phase 6 (CMIP6). OMIP addresses CMIP6 science questions, investigating the origins and consequences of systematic model biases. It does so by providing a framework for evaluating (including assessment of systematic biases), understanding, and improving ocean, sea-ice, tracer, and biogeochemical components of climate and earth system models contributing to CMIP6. Among the WCRP Grand Challenges in climate science (GCs), OMIP primarily contributes to the regional sea level change and near-term (climate/decadal) prediction GCs. OMIP provides (a) an experimental protocol for global ocean/sea-ice models run with a prescribed atmospheric forcing; and (b) a protocol for ocean diagnostics to be saved as part of CMIP6. We focus here on the physical component of OMIP, with a companion paper (Orr et al., 2016) detailing methods for the inert chemistry and interactive biogeochemistry. The physical portion of the OMIP experimental protocol follows the interannual Coordinated Ocean-ice Reference Experiments (CORE-II). Since 2009, CORE-I (Normal Year Forcing) and CORE-II (Interannual Forcing) have become the standard methods to evaluate global ocean/sea-ice simulations and to examine mechanisms for forced ocean climate variability. The OMIP diagnostic protocol is relevant for any ocean model component of CMIP6, including the DECK (Diagnostic, Evaluation and Characterization of Klima experiments), historical simulations, FAFMIP (Flux Anomaly Forced MIP), C4MIP (Coupled Carbon Cycle Climate MIP), DAMIP (Detection and Attribution MIP), DCPP (Decadal Climate Prediction Project), ScenarioMIP, HighResMIP (High Resolution MIP), as well as the ocean/sea-ice OMIP simulations.

1425. Data assimilation in a coupled physical-biogeochemical model of the California Current System using an incremental lognormal 4-dimensional variational approach: Part 2--Joint physical and biological data assimilation twin experiments

Authors: Hajoon Song; Christopher A. Edwards; Andrew M. Moore; Jerome Fiechter

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2016.09.003

URL: <https://doi.org/10.1016/j.ocemod.2016.09.003>

Publication Date: 2016-09-14

Volume/Issue: Volume 106

References: 79

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Gordon and Betty Moore Foundation, National Oceanic and Atmospheric Administration

License: N/A

Open Access: N/A

Abstract: N/A

1426. Interhemispheric SST Gradient Trends in the Indian Ocean prior to and during the Recent Global Warming Hiatusa

Authors: Lu Dong (NOAA/Pacific Marine Environmental Laboratory, Seattle, Washington); Michael J. McPhaden (NOAA/Pacific Marine Environmental Laboratory, Seattle, Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0130.1

URL: <https://doi.org/10.1175/jcli-d-16-0130.1>

Publication Date: 2016-09-14

Volume/Issue: Volume 29, Issue 24

References: 72

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sea surface temperatures (SSTs) have been rising for decades in the Indian Ocean in response to greenhouse gas forcing. However, this study shows that during the recent hiatus in global warming, a striking interhemispheric gradient in Indian Ocean SST trends developed around 2000, with relatively weak or little warming to the north of 10 degreesS and accelerated warming to the south of 10 degreesS. Evidence is presented from a wide variety of data sources showing that this interhemispheric gradient in SST trends is forced primarily by an increase of Indonesian Throughflow (ITF) transport from the Pacific into the Indian Ocean induced by stronger Pacific trade winds. This increased transport led to a depression of the thermocline that facilitated SST warming, presumably through a reduction in the vertical turbulent transport of heat in the southern Indian Ocean. Surface wind changes in the Indian Ocean linked to the enhanced Walker circulation also may have contributed to thermocline depth variations and associated SST changes, with downwelling-favorable wind stress curls between 10 degrees and 20 degreesS and upwelling-favorable wind stress curls between the equator and 10 degreesS. In addition, the anomalous southwesterly wind stresses off the coast of Somalia favored intensified coastal upwelling and offshore advection of upwelled water, which would have led to reduced warming of the northern Indian Ocean. Although highly uncertain, lateral heat advection associated with the ITF and surface heat fluxes may also have played a role in forming the interhemispheric SST gradient change.

1427. Quantifying and reducing leakage errors in the JPL RL05M GRACE mascon solution

Authors: David N. Wiese (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Felix W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael M. Watkins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016wr019344

URL: <https://doi.org/10.1002/2016wr019344>

Publication Date: 2016-09-12

Volume/Issue: Volume 52, Issue 9

References: 31

Cited by: 472 publications

Language: en

Subjects: N/A

Keywords: postprocessing, gain factors, leakage errors, mascon, GRACE

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Recent advances in processing data from the Gravity Recovery and Climate Experiment (GRACE) have led to a new generation of gravity solutions constrained within a Bayesian framework to remove correlated errors rather than relying on empirical filters. The JPL RL05M mascon solution is one such solution, solving for mass variations using spherical cap mass concentration elements (mascons), while relying on external information provided by near-global geophysical models to constrain the solution. This new gravity solution is fundamentally different than the traditional spherical harmonic gravity solution, and as such, requires different care when postprocessing. Here we discuss two classes of postprocessing considerations for the JPL RL05M GRACE mascon solution: (1) reducing leakage errors across land/ocean boundaries, and (2) scaling the solutions to account for leakage errors introduced through parameterizing the gravity solution in terms of mascons. A Coastline Resolution Improvement (CRI) filter is developed to reduce leakage errors across coastlines. Synthetic simulations reveal a reduction in leakage errors of ~50%, such that residual leakage errors are ~1 cm equivalent water height (EWH) averaged globally. A set of gain factors is derived to reduce leakage errors for continental hydrology applications. The combined effect of the CRI filter coupled with application of the gain factors, is shown to reduce leakage errors when determining the mass balance of large (>160,000 km²) hydrological basins from 11% to 30% (0.6-1.5 mm EWH) averaged globally, with local improvements up to 38%-81% (9-19 mm EWH).

1428. Causes of the Regional Variability in Observed Sea Level, Sea Surface Temperature and Ocean Colour Over the Period 1993-2011

Authors: B. Meyssignac; C. G. Piecuch; C. J. Merchant; M.-F. Racault; H. Palanisamy; C. MacIntosh; S. Sathyendranath; R. Brewin

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-016-9383-1

URL: <https://doi.org/10.1007/s10712-016-9383-1>

Publication Date: 2016-09-09

Volume/Issue: Volume 38, Issue 1

References: 88

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: N/A

1429. Polynomial Chaos-Based Bayesian Inference of K-Profile Parameterization in a General Circulation Model of the Tropical Pacific

Authors: Ihab Sraj (Duke University); Sarah E. Zedler (University of Texas at Austin); Omar M. Knio (Duke University); Charles S. Jackson (University of Texas at Austin); Ibrahim Hoteit (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia)

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/mwr-d-15-0394.1

URL: <https://doi.org/10.1175/mwr-d-15-0394.1>

Publication Date: 2016-08-26

Volume/Issue: Volume 144, Issue 12

References: 63

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: Ocean circulation, Spectral analysis/models/distribution, Bayesian methods

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The authors present a polynomial chaos (PC)-based Bayesian inference method for quantifying the uncertainties of the K-profile parameterization (KPP) within the MIT general circulation model (MITgcm) of the tropical Pacific. The inference of the uncertain parameters is based on a Markov chain Monte Carlo (MCMC) scheme that utilizes a newly formulated test statistic taking into account the different components representing the structures of turbulent mixing on both daily and seasonal time scales in addition to the data quality, and filters for the effects of parameter perturbations over those as a result of changes in the wind. To avoid the prohibitive computational cost of integrating the MITgcm model at each MCMC iteration, a surrogate model for the test statistic using the PC method is built. Because of the noise in the model predictions, a basis-pursuit-denoising (BPDN) compressed sensing approach is employed to determine the PC coefficients of a representative surrogate model. The PC surrogate is then used to evaluate the test statistic in the MCMC step for sampling the posterior of the uncertain parameters. Results of the posteriors indicate good agreement with the default values for two parameters of the KPP model, namely the critical bulk and gradient Richardson numbers; while the posteriors of the remaining parameters were barely informative.

1430. Forcing of recent decadal variability in the Equatorial and North Indian Ocean

Authors: P. R. Thompson (Department of Oceanography University of Hawaii at Mānoa Honolulu Hawaii USA); C. G. Piecuch (Atmospheric and Environmental Research Inc.); M. A. Merrifield (Department of Oceanography University of Hawaii at Mānoa Honolulu Hawaii USA); J. P. McCreary (International Pacific Research Center University of Hawaii at Mānoa Honolulu Hawaii USA); E. Firing (Department of Oceanography University of Hawaii at Mānoa Honolulu Hawaii USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc012132

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Publication Date: 2016-08-24

Volume/Issue: Volume 121, Issue 9

References: 34

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: sea surface height, decadal variability, satellite altimetry, ECCOV4, Indian Ocean

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License: N/A

Open Access: N/A

Abstract: Recent decadal sea surface height (SSH) variability across the Equatorial and North Indian Ocean (ENIO, north of 5 degreesS) is spatially coherent and related to a reversal in basin-scale, upper-ocean-temperature trends. Analysis of ocean and forcing fields from a data-assimilating ocean synthesis (ECCOV4) suggests that two equally important mechanisms of wind-driven heat redistribution within the Indian Ocean account for a majority of the decadal variability. The first is the Cross-Equatorial Cell (CEC) forced by zonal wind stress curl at the equator. The wind stress curl variability relates to the strength and position of the Mascarene High, which is influenced by the phase of the Indian Ocean Subtropical Dipole. The second mechanism is deep (700 m) upwelling related to zonal wind stress at the equator that causes deep, cross-equatorial overturning due to the unique geometry of the basin. The CEC acts to cool the upper ocean throughout most of the first decade of satellite altimetry, while the deep upwelling delays and then amplifies the effect of the CEC on SSH. During the subsequent decade, reversals in the forcing anomalies drive warming of the upper ocean and increasing SSH, with the effect of the deep upwelling leading the CEC.

1431. Internal tide oceanic tomography

Authors: Zhongxiang Zhao (University of Washington)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl070567

URL: <https://doi.org/10.1002/2016gl070567>

Publication Date: 2016-08-22

Volume/Issue: Volume 43, Issue 17

References: 28

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: CNES, NSF, NASA

License: N/A

Open Access: N/A

Abstract: A concept of internal tide oceanic tomography (ITOT) is proposed to monitor ocean warming on a global scale. ITOT is similar to acoustic tomography, but that work waves are internal tides. ITOT detects ocean temperature changes by precisely measuring travel time changes of long-range propagating internal tides. The underlying principle is that upper ocean warming strengthens ocean stratification and thus increases the propagation speed of internal tides. This concept is inspired by recent advances in observing internal tides by satellite altimetry. In particular, a plane wave fit method can separately resolve multiple internal tidal waves and thus accurately determines the phase of each wave. Two examples are presented to demonstrate the feasibility and usefulness of ITOT. In the eastern tropical Pacific, the yearly time series of travel time changes of the M2 internal tide is closely correlated with the El Niño-Southern Oscillation index. In the North Atlantic, significant interannual variations and bidecadal trends are observed and consistent with the changes in ocean heat content measured by Argo floats. ITOT offers a long-term, cost-effective, environmentally friendly technique for monitoring global ocean warming. Future work is needed to quantify the accuracy of this technique.

1432. Effects of surface current-wind interaction in an eddy-rich general ocean circulation simulation of the Baltic Sea

Authors: Heiner Dietze; Ulrike Löptien

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-12-977-2016

URL: <https://doi.org/10.5194/os-12-977-2016>

Publication Date: 2016-08-18

Volume/Issue: Volume 12, Issue 4

References: 50

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Deoxygenation in the Baltic Sea endangers fish yields and favours noxious algal blooms. Yet, vertical transport processes ventilating the oxygen-deprived waters at depth and replenishing nutrient-deprived surface waters (thereby fuelling export of organic matter to depth) are not comprehensively understood. Here, we investigate the effects of the interaction between surface currents and winds on upwelling in an eddy-rich general ocean circulation model of the Baltic Sea. Contrary to expectations we find that accounting for current-wind effects inhibits the overall vertical exchange between oxygenated surface waters and oxygen-deprived water at depth. At major upwelling sites, however (e.g. off the southern coast of Sweden and Finland) the reverse holds: the interaction between topographically steered surface currents with winds blowing over the sea results in a climatological sea surface temperature cooling of 0.5 K. This implies that current-wind effects drive substantial local upwelling of cold and nutrient-replete waters.

1433. Evaluation of model simulated and MODIS-Aqua retrieved sea surface chlorophyll in the eastern Arabian Sea

Authors: Kunal Chakraborty; Anubhav Gupta; Aneesh A. Lotliker; Gavin Tilstone

Journal: Estuarine, Coastal and Shelf Science

Publisher: Elsevier BV

DOI: 10.1016/j.ecss.2016.08.002

URL: <https://doi.org/10.1016/j.ecss.2016.08.002>

Publication Date: 2016-08-15

Volume/Issue: Volume 181

References: 75

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Satellite Coastal and Oceanographic Research (SATCORE), Ministry of Earth Sciences (MoES), National Network for Mathematical and Computational Biology (NNMCB)

License: N/A

Open Access: N/A

Abstract: N/A

1434. Observed and simulated variability of the Atlantic Meridional Overturning Circulation at 41 degreesN

Authors: Vladimir N. Stepanov; Doroteaciro Iovino; Simona Masina; Andrea Storto; Andrea Cipollone

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2016.08.004

URL: <https://doi.org/10.1016/j.jmarsys.2016.08.004>

Publication Date: 2016-08-11

Volume/Issue: Volume 164

References: 51

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Italian Ministry of Education, University and Research, Ministry for Environment, Land and Sea

License: N/A

Open Access: N/A

Abstract: N/A

1435. From Topographic Internal Gravity Waves to Turbulence

Authors: S. Sarkar (University of California San Diego); A. Scotti (Department of Marine Sciences, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina 27599)

Journal: Annual Review of Fluid Mechanics

Publisher: Annual Reviews

DOI: 10.1146/annurev-fluid-010816-060013

URL: <https://doi.org/10.1146/annurev-fluid-010816-060013>

Publication Date: 2016-08-02

Volume/Issue: Volume 49, Issue 1

References: 126

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Internal gravity waves are a key process linking the large-scale mechanical forcing of the oceans to small-scale turbulence and mixing. In this review, we focus on internal waves generated by barotropic tidal flow over topography. We review progress made in the past decade toward understanding the different processes that can lead to turbulence during the generation, propagation, and reflection of internal waves and how these processes affect mixing. We consider different modeling strategies and new tools that have been developed. Simulation results, the wealth of observational material collected during large-scale experiments, and new laboratory data reveal how the cascade of energy from tidal flow to turbulence occurs through a host of nonlinear processes, including intensified boundary flows, wave breaking, wave-wave interactions, and the instability of high-mode internal wave beams. The roles of various nondimensional parameters involving the ocean state, roughness geometry, and tidal forcing are described.

1436. The Curious Nature of the Hemispheric Symmetry of the Earth's Water and Energy Balances

Authors: Graeme L. Stephens; Maria Z. Hakuba; Matt Hawcroft; Jim M. Haywood; Ali Behrangi; Jennifer E. Kay; Peter J Webster

Journal: Current Climate Change Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s40641-016-0043-9

URL: <https://doi.org/10.1007/s40641-016-0043-9>

Publication Date: 2016-07-26

Volume/Issue: Volume 2, Issue 4

References: 79

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1437. Time-Dependent Eddy-Mean Energy Diagrams and Their Application to the Ocean

Authors: Ru Chen (Scripps Institution of Oceanography / University of California San Diego); Andrew F. Thompson (California Institute of Technology); Glenn R. Flierl (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0012.1

URL: <https://doi.org/10.1175/jpo-d-16-0012.1>

Publication Date: 2016-07-20

Volume/Issue: Volume 46, Issue 9

References: 70

Cited by: 23 publications

Language: N/A

Subjects: N/A

Keywords: Ocean models, Atm/Ocean Structure/ Phenomena, Southern Ocean, Circulation/ Dynamics, Models and modeling (and 6 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Insight into the global ocean energy cycle and its relationship to climate variability can be gained by examining the temporal variability of eddy-mean flow interactions. A time-dependent version of the Lorenz energy diagram is formulated and applied to energetic ocean regions from a global, eddying state estimate. The total energy in each snapshot is partitioned into three components: energy in the mean flow, energy in eddies, and energy temporal anomaly residual, whose time mean is zero. These three terms represent, respectively, correlations between mean quantities, correlations between eddy quantities, and eddy-mean correlations. Eddy-mean flow interactions involve energy exchange among these three components. The temporal coherence about energy exchange during eddy-mean flow interactions is assessed. In the Kuroshio and Gulf Stream Extension regions, a suppression relation is manifested by a reduction in the baroclinic energy pathway to the eddy kinetic energy (EKE) reservoir following a strengthening of the barotropic energy pathway to EKE; the baroclinic pathway strengthens when the barotropic pathway weakens. In the subtropical gyre and Southern Ocean, a delay in energy transfer between different reservoirs occurs during baroclinic instability. The delay mechanism is identified using a quasigeostrophic, two-layer model; part of the potential energy in large-scale eddies, gained from the mean flow, cascades to smaller scales through eddy stirring before converting to EKE. The delay time is related to this forward cascade and scales linearly with the eddy turnover time. The relation between temporal variations in wind power input and eddy-mean flow interactions is also assessed.

1438. Asymmetry of upper ocean salinity response to the Indian Ocean dipole events as seen from ECCO simulation

Authors: Ying Zhang; Yan Du; Yuhong Zhang; Shan Gao

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-016-0904-z

URL: <https://doi.org/10.1007/s13131-016-0904-z>

Publication Date: 2016-07-18

Volume/Issue: Volume 35, Issue 7

References: 53

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1439. The Mean Upper-Layer Flow in the Central Gulf of Mexico by a New Method

Authors: Wilton Sturges (Florida State University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0062.1

URL: <https://doi.org/10.1175/jpo-d-16-0062.1>

Publication Date: 2016-07-18

Volume/Issue: Volume 46, Issue 10

References: 34

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Mean flow

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Previous studies have found a puzzling disagreement between two large datasets and the results of numerical models in the central Gulf of Mexico. The observations suggest an upper-layer mean flow to the west of order 10 cm s⁻¹, while the numerical models find no such mean flow. A new technique is used here, using 23 yr of satellite-derived sea surface height data, to estimate the mean flow. This third, independent set of data yields the same westward flow found in previous studies. These findings require that there be sinking in the western Gulf. The details of the return flow remain an intriguing problem.

1440. Representation of Bay of Bengal Upper-Ocean Salinity in General Circulation Models

Authors: Jasti Chowdary; G. Srinivas; T.S. Fousiya; Anant Parekh; C. Gnanaseelan; Hyodae Seo; Jennifer MacKinnon

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2016.37

URL: <https://doi.org/10.5670/oceanog.2016.37>

Publication Date: 2016-07-14

Volume/Issue: Volume 29, Issue 2

References: 72

Cited by: 28 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1441. Barotropic kinetic energy and enstrophy transfers in the atmosphere

Authors: R. Chemke (Weizmann Institute of Science); T. Dror (Weizmann Institute of Science); Y. Kaspi (Weizmann Institute of Science)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl070350

URL: <https://doi.org/10.1002/2016gl070350>

Publication Date: 2016-07-13

Volume/Issue: Volume 43, Issue 14

References: 57

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Israeli Science Foundation, Federal German Ministry of Education and Research

License: N/A

Open Access: N/A

Abstract: The midlatitude atmosphere is characterized by turbulent eddies that act to produce a depth-independent (barotropic) mean flow. Using the NCEP (National Centers for Environmental Prediction) Reanalysis 2 data, the latitudinal dependence of barotropic kinetic energy and enstrophy are investigated. Most of the barotropization takes place in the extratropics with a maximum value at midlatitudes, due to the latitudinal variations of the static stability, tropopause height, and sphericity of the planet. Barotropic advection transfers the eddy kinetic energy to the zonal mean flow and thus maintains the barotropic component of the eddy-driven jet. The classic description of geostrophic turbulence exists only at high latitudes, where the quasi-geostrophic flow is supercritical to baroclinic instability; the eddy-eddy interactions carry both the barotropization of eddy kinetic energy upscale to the Rhines scale and the barotropization of eddy potential enstrophy downscale.

1442. The marine hydrological cycle: The ocean's floods and droughts

Authors: Arnold L. Gordon (Columbia University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl070279

URL: <https://doi.org/10.1002/2016gl070279>

Publication Date: 2016-07-12

Volume/Issue: Volume 43, Issue 14

References: 20

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: The sea surface salinity (SSS) displays fluctuations that are not solely in response to local air-sea flux of freshwater but also reflect ocean circulation and mixing processes. Ponte and Vinogradova (2016), using Estimating the Circulation and Climate of the Ocean output, estimate the relative roles of these forces for the global ocean. They find that the governing forces vary greatly across ocean regimes. Their research identifies features that will be addressed with enhanced SSS observations from orbiting satellites and in situ global arrays, which promise new insight into the marine water cycle and its place in the global hydrological system.

1443. Landfast ice thickness in the Canadian Arctic Archipelago from observations and models

Authors: Stephen E. L. Howell; Frédéric Laliberté; Ron Kwok; Chris Derksen; Joshua King

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-1463-2016

URL: <https://doi.org/10.5194/tc-10-1463-2016>

Publication Date: 2016-07-12

Volume/Issue: Volume 10, Issue 4

References: 45

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Observed and modelled landfast ice thickness variability and trends spanning more than 5 decades within the Canadian Arctic Archipelago (CAA) are summarized. The observed sites (Cambridge Bay, Resolute, Eureka and Alert) represent some of the Arctic's longest records of landfast ice thickness. Observed end-of-winter (maximum) trends of landfast ice thickness (1957-2014) were statistically significant at Cambridge Bay (-4.31 ± 1.4 cm decade⁻¹), Eureka (-4.65 ± 1.7 cm decade⁻¹) and Alert (-4.44 ± 1.6 cm decade⁻¹) but not at Resolute. Over the 50+-year record, the ice thinned by ~ 0.24 - 0.26 m at Cambridge Bay, Eureka and Alert with essentially negligible change at Resolute. Although statistically significant warming in spring and fall was present at all sites, only low correlations between temperature and maximum ice thickness were present; snow depth was found to be more strongly associated with the negative ice thickness trends. Comparison with multi-model simulations from Coupled Model Intercomparison project phase 5 (CMIP5), Ocean Reanalysis Intercomparison (ORA-IP) and Pan-Arctic Ice-Ocean Modeling and Assimilation System (PIOMAS) show that although a subset of current generation models have a "reasonable" climatological representation of landfast ice thickness and distribution within the CAA, trends are unrealistic and far exceed observations by up to 2 orders of magnitude. ORA-IP models were found to have positive correlations between temperature and ice thickness over the CAA, a feature that is inconsistent with both observations and coupled models from CMIP5.

1444. Ocean Processes Affecting the Twenty-First-Century Shift in ENSO SST Variability

Authors: Cong Guan (Chinese Academy of Sciences); Michael J. McPhaden (NOAA/Pacific Marine Environmental Laboratory, Seattle, Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0870.1

URL: <https://doi.org/10.1175/jcli-d-15-0870.1>

Publication Date: 2016-07-06

Volume/Issue: Volume 29, Issue 19

References: 52

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sea surface temperature (SST) variability associated with El Niño-Southern Oscillation (ENSO) slightly increased in the central Pacific Ocean but weakened significantly in the eastern Pacific at the beginning of twenty-first century relative to 1980-99. This decadal shift led to the greater prominence central Pacific (CP) El Niño events during the 2000s relative to the previous two decades, which were dominated by eastern Pacific (EP) events. To expand upon previous studies that have examined this shift in ENSO variability, temperature and temperature variance budgets are examined in the mixed layer of the Niño-3 (5 degreesS-5 degreesN, 150 degrees-90 degreesW) and Niño-4 (5 degreesS-5 degreesN, 160 degreesE-150 degreesW) regions from seven ocean model products spanning the period 1980-2010. This multimodel-product-based approach provides a robust assessment of dominant mechanisms that account for decadal changes in two key index regions. A temperature variance budget perspective on the role of thermocline feedbacks in the ENSO cycle based on recharge oscillator theory is also presented. As found in previous studies, thermocline and zonal advective feedbacks are the most important positive feedbacks for generating ENSO SST variance, and thermodynamic damping is the largest negative feedback for damping ENSO variance. Consistent with the shift toward more CP El Niños after 2000, thermocline feedbacks experienced a substantial reduction from 1980 to 1999 and into the 2000s, while zonal advective feedbacks were less affected. Negative feedbacks likewise weakened after 2000, particularly thermal damping in the Niño-3 region and the nonlinear sink of variance in both regions.

1445. Sub-basin scale sea level budgets from satellite altimetry, Argo floats and satellite gravimetry in the North Atlantic

Authors: Marcel Kleinherenbrink; Riccardo Riva; Yu Sun

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/os-2016-50

URL: <https://doi.org/10.5194/os-2016-50>

Publication Date: 2016-07-06

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Argo|Satellite Altimetry|Gravimetry

Funding: Stichting voor de Technische Wetenschappen

License: Creative Commons

Open Access: N/A

Abstract: . In this study for the first time an attempt is made to close the sea level budget on a sub-basin scale in terms of trend, annual amplitude and residual time series, after removing the trend, the semi-annual and annual signals. To obtain errors for altimetry and Argo full variance-covariance matrices are computed using correlation functions and their errors are fully propagated. For altimetry we apply a geographically dependent intermission bias (Ablain et al., 2015), which leads to differences in trends up to 0.8 mm yr⁻¹. Since Argo float measurements are non-homogeneously spaced, steric sea levels are first objectively interpolated onto a grid before averaging. For the Gravity Recovery And Climate Experiment (GRACE) gravity fields full variance-covariance matrices are used to propagate errors and statistically filter the gravity fields. We use four different filtered gravity field solutions and determine which post-processing strategy is best for budget closure. As a reference the standard 96-degree DDK5-filtered CSR solution is used to compute OBP. A comparison is made with two anisotropic Wiener-filtered CSR solutions up to d/o 60 and 96 and a Wiener-filtered 90-degree ITSG solution. Budgets are computed for ten polygons in the North Atlantic, defined in a way that the error on the trend of Ocean Bottom Pressure (OBP) + steric sea level remains within 1 mm yr⁻¹. Using the anisotropic Wiener filter on CSR gravity fields expanded up to spherical harmonic degree 96, it is possible to close the sea level budget in nine-out-of-ten sub-basins in terms of trend. Wiener-filtered ITSG and the standard DDK5-filtered CSR solutions also close the trend budget, if a Glacial Isostatic Adjustment (GIA) correction error of 10-20 % is applied, however the performance of the DDK5-filtered solution strongly depends on the orientation of the polygon due to residual striping. In seven-out-of-ten sub-basins the budget of the annual cycle is closed, using the DDK5-filtered CSR or the Wiener-filtered ITSG solutions. The Wiener-filtered 60- and 96-degree CSR solution in combination with Argo lack amplitude and suffer from what appears to be hydrological leakage in the Amazon and Sahel regions. After reducing the trend, semi-annual and annual signals, 24-53 % of the residual variance in altimetry-derived sea level time series is explained by the combination of Argo steric sea levels and Wiener-filtered ITSG OBP. Based on this, we believe that the best overall solution for the OBP component of the sub-basin scale budgets is the Wiener-filtered ITSG gravity fields. The interannual

variability is primarily a steric signal in the North Atlantic, so for this the choice of filter and gravity field solution is not really significant.

1446. Impacts of open-ocean deep convection in the Weddell Sea on coastal and bottom water temperature

Authors: Zhaomin Wang; Yang Wu; Xia Lin; Chengyan Liu; Zelin Xie

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-016-3244-y

URL: <https://doi.org/10.1007/s00382-016-3244-y>

Publication Date: 2016-07-04

Volume/Issue: Volume 48, Issue 9-10

References: 65

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Global Change Research Program of China, the Special Program for China Meteorology Trade (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

1447. Trends of the Pathways and Intensities of Surface Equatorial Current System in the North Pacific Ocean

Authors: Yi-Chia Hsin (Academia Sinica)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0850.1

URL: <https://doi.org/10.1175/jcli-d-15-0850.1>

Publication Date: 2016-06-30

Volume/Issue: Volume 29, Issue 18

References: 46

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An ensemble of ocean reanalysis products is utilized to quantify the long-term tendencies of pathways and along-pathway transports of the three surface equatorial currents (North Equatorial Current, North Equatorial Countercurrent, and northern branch of the South Equatorial Current) in the North Pacific Ocean during the period of the 1900s-2000s. This study uses 12 ocean reanalysis products in the ensemble for the period after the 1960s, while only 2 Simple Ocean Data Assimilation (SODA) products are taken into consideration for the period prior to 1960s. The analyses indicate that the three currents in the western (eastern) Pacific Ocean have more southern (northern) mean central positions and tend to move southward (northward) over the past 100 years. All three currents have weakening tendencies, with the exception of the North Equatorial Current having intensified in the western Pacific Ocean. The Sverdrup dynamics, which directly relates the wind-driven circulation in the interior ocean to wind stress curl and Earth rotation, can be applied to simply address the long-term changes of intensities and pathways of the three surface currents in the tropical North Pacific Ocean.

1448. Intensification and poleward shift of subtropical western boundary currents in a warming climate

Authors: Hu Yang (Alfred Wegener Institute); Gerrit Lohmann (University of Bremen, Alfred Wegener Institute); Wei Wei (Alfred Wegener Institute); Mihai Dima (Alfred Wegener Institute, University of Bucharest); Monica Ionita (Alfred Wegener Institute); Jiping Liu (Department of Atmospheric and Environmental Sciences University at Albany, State University of New York Albany, New York USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011513

URL: <https://doi.org/10.1002/2015jc011513>

Publication Date: 2016-06-27

Volume/Issue: Volume 121, Issue 7

References: 97

Cited by: 198 publications

Language: en

Subjects: N/A

Keywords: Boundary current|Forcing (mathematics)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A significant increase in sea surface temperature (SST) is observed over the midlatitude western boundary currents (WBCs) during the past century. However, the mechanism for this phenomenon remains poorly understood due to limited observations. In the present paper, several coupled parameters (i.e., sea surface temperature (SST), ocean surface heat fluxes, ocean water velocity, ocean surface winds and sea level pressure (SLP)) are analyzed to identify the dynamic changes of the WBCs. Three types of independent data sets are used, including reanalysis products, satellite-blended observations, and climate model outputs from the fifth phase of the Climate Model Intercomparison Project (CMIP5). Based on these broad ranges of data, we find that the WBCs (except the Gulf Stream) are intensifying and shifting toward the poles as long-term effects of global warming. An intensification and poleward shift of near-surface ocean winds, attributed to positive annular mode-like trends, are proposed to be the forcing of such dynamic changes. In contrast to the other WBCs, the Gulf Stream is expected to be weaker under global warming, which is most likely related to a weakening of the Atlantic Meridional Overturning Circulation (AMOC). However, we also notice that the natural variations of WBCs might conceal the long-term effect of global warming in the available observational data sets, especially over the Northern Hemisphere. Therefore, long-term observations or proxy data are necessary to further evaluate the dynamics of the WBCs.

1449. Water-mass transformation by sea ice in the upper branch of the Southern Ocean overturning

Authors: Ryan P. Abernathey; Ivana Cerovecki; Paul R. Holland; Emily Newsom; Matt Mazloff; Lynne D. Talley

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/ngeo2749

URL: <https://doi.org/10.1038/ngeo2749>

Publication Date: 2016-06-27

Volume/Issue: Volume 9, Issue 8

References: 56

Cited by: 200 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1450. Zonal Variations in the Southern Ocean Heat Budget

Authors: Veronica Tamsitt (Scripps Institution of Oceanography); Lynne D. Talley (Scripps Institution of Oceanography); Matthew R. Mazloff (Scripps Institution of Oceanography); Ivana Cerovečki (Scripps Institution of Oceanography)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0630.1

URL: <https://doi.org/10.1175/jcli-d-15-0630.1>

Publication Date: 2016-06-23

Volume/Issue: Volume 29, Issue 18

References: 50

Cited by: 47 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The spatial structure of the upper ocean heat budget in the Antarctic Circumpolar Current (ACC) is investigated using the 1/6 degrees, data-assimilating Southern Ocean State Estimate (SOSE) for 2005-10. The ACC circumpolar integrated budget shows that 0.27 PW of ocean heat gain from the atmosphere and 0.38 PW heat gain from divergence of geostrophic heat transport are balanced by -0.58 PW cooling by divergence of Ekman heat transport and -0.09 PW divergence of vertical heat transport. However, this circumpolar integrated balance obscures important zonal variations in the heat budget. The air-sea heat flux shows a zonally asymmetric pattern of ocean heat gain in the Indian and Atlantic sectors and ocean heat loss in the Pacific sector of the ACC. In the Atlantic and Indian sectors of the ACC, the surface ocean heat gain is primarily balanced by divergence of equatorward Ekman heat transport that cools the upper ocean. In the Pacific sector, surface ocean heat loss and cooling due to divergence of Ekman heat transport are balanced by warming due to divergence of geostrophic heat advection, which is similar to the dominant heat balance in the subtropical Agulhas Return Current. The divergence of horizontal and vertical eddy advection of heat is important for warming the upper ocean close to major topographic features, while the divergence of mean vertical heat advection is a weak cooling term. The results herein show that topographic steering and zonal asymmetry in air-sea exchange lead to substantial zonal asymmetries in the heat budget, which is important for understanding the upper cell of the overturning circulation.

1451. An assessment of basic processes controlling mean surface salinity over the global ocean

Authors: R. M. Ponte (Atmospheric and Environmental Research Inc.); N. T. Vinogradova (Atmospheric and Environmental Research Inc.)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl069857

URL: <https://doi.org/10.1002/2016gl069857>

Publication Date: 2016-06-22

Volume/Issue: Volume 43, Issue 13

References: 26

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: freshwater flux, surface salinity

Funding: NASA Ocean Surface Salinity

License: N/A

Open Access: N/A

Abstract: A data-constrained ocean state estimate that permits closed property budget diagnostics is used to examine the balance between surface forcing ($\bar{Q}$), advective ($\bar{Q}_a$), and diffusive ($\bar{Q}_d$) fluxes in maintaining the large-scale time-mean surface salinity. Time-mean budgets (1993-2010) are considered for the 10 m thick top layer. In general, $\bar{Q}$ tends to counteract $\bar{Q}_a$, but is important almost everywhere, and some regions show a main balance between $\bar{Q}$ and $\bar{Q}_d$ (Bay of Bengal, Arctic) or $\bar{Q}_a$ and $\bar{Q}_d$ (tropical Atlantic and Pacific). Advection tends to freshen the surface in the tropics and high latitudes, with opposite tendencies in midlatitudes. For various tropical extrema, $\bar{Q}_a$ adds to the tendencies in precipitation regions and opposes in evaporation regions. Long-term conditions thus reflect more than a simple diffusive adjustment to $\bar{Q}$, likely involving close interaction between wind- and buoyancy-driven circulation and mixing processes.

1452. A Rossby whistle: A resonant basin mode observed in the Caribbean Sea

Authors: Chris W. Hughes (University of Liverpool, National Oceanography Centre); Joanne Williams (National Oceanography Centre); Angela Hibbert (National Oceanography Centre); Carmen Boening (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); James Oram (University of Liverpool)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl069573

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Publication Date: 2016-06-20

Volume/Issue: Volume 43, Issue 13

References: 25

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

License: Creative Commons

Open Access: N/A

Abstract: We show that an important source of coastal sea level variability around the Caribbean Sea is a resonant basin mode. The mode consists of a baroclinic Rossby wave which propagates westward across the basin and is rapidly returned to the east along the southern boundary as coastal shelf waves. Almost two wavelengths of the Rossby wave fit across the basin, and it has a period of 120 days. The porous boundary of the Caribbean Sea results in this mode exciting a mass exchange with the wider ocean, leading to a dominant mode of bottom pressure variability which is almost uniform over the Grenada, Venezuela, and Colombia basins and has a sharp spectral peak at 120 day period. As the Rossby waves have been shown to be excited by instability of the Caribbean Current, this resonant mode is dynamically equivalent to the operation of a whistle.

1453. MiKlip: A National Research Project on Decadal Climate Prediction

Authors: Jochem Marotzke (Max Planck Institute for Meteorology); Wolfgang A. Müller (Max Planck Institute for Meteorology); Freja S. E. Vamborg (Max Planck Institute for Meteorology); Paul Becker (Deutscher Wetterdienst, Offenbach, Germany); Ulrich Cubasch (Institute of Meteorology, Freie Universität Berlin, Berlin, Germany); Hendrik Feldmann (Karlsruhe Institute of Technology); Frank Kaspar (Deutscher Wetterdienst, Offenbach, Germany); Christoph Kottmeier (Karlsruhe Institute of Technology); Camille Marini (University of Hamburg); Iuliia Polkova (University of Hamburg); Kerstin Prömmel (Institute of Meteorology, Freie Universität Berlin, Berlin, Germany); Henning W. Rust (Institute of Meteorology, Freie Universität Berlin, Berlin, Germany); Detlef Stammer (University of Hamburg); Uwe Ulbrich (Institute of Meteorology, Freie Universität Berlin, Berlin, Germany); Christopher Kadow (Institute of Meteorology, Freie Universität Berlin, Berlin, Germany); Armin Köhl (University of Hamburg); Jürgen Kröger (Max Planck Institute for Meteorology); Tim Kruschke (GEOMAR Helmholtz Centre for Ocean Research); Joaquim G. Pinto (University of Cologne, University of Reading); Holger Pohlmann (Max Planck Institute for Meteorology); Mark Reyers (University of Cologne); Marc Schröder (Deutscher Wetterdienst, Offenbach, Germany); Frank Sienz (Max Planck Institute for Meteorology); Claudia Timmreck (Max Planck Institute for Meteorology); Markus Ziese (Deutscher Wetterdienst, Offenbach, Germany)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/bams-d-15-00184.1

URL: <https://doi.org/10.1175/bams-d-15-00184.1>

Publication Date: 2016-06-17

Volume/Issue: Volume 97, Issue 12

References: 83

Cited by: 70 publications

Language: en

Subjects: N/A

Keywords: Hindcast|Initialization|Extratropical cyclone|Earth system science

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Mittelfristige Klimaprognose (MiKlip), an 8-yr German national research project on decadal climate prediction, is organized around a global prediction system comprising the Max Planck Institute Earth System Model (MPI-ESM) together with an initialization procedure and a model evaluation system. This paper summarizes the lessons learned from MiKlip so far; some are purely scientific, others concern strategies and structures of research that target future operational use. Three prediction system generations have been constructed, characterized by alternative initialization strategies; the later generations show a marked improvement in hindcast skill for surface temperature. Hindcast skill is also identified for multiyear-mean European summer surface temperatures, extratropical cyclone tracks, the quasi-biennial oscillation, and ocean carbon uptake, among others. Regionalization maintains or slightly enhances the skill in European surface temperature inherited from the global model and also displays hindcast skill for wind energy output. A new volcano code package permits rapid modification of the predictions in response to a future eruption. MiKlip has demonstrated the

efficacy of subjecting a single global prediction system to a major research effort. The benefits of this strategy include the rapid cycling through the prediction system generations, the development of a sophisticated evaluation package usable by all MiKlip researchers, and regional applications of the global predictions. Open research questions include the optimal balance between model resolution and ensemble size, the appropriate method for constructing a prediction ensemble, and the decision between full-field and anomaly initialization. Operational use of the MiKlip system is targeted for the end of the current decade, with a recommended generational cycle of 2-3 years.

1454. On the Minimum Potential Energy State and the Eddy Size-Constrained APE Density

Authors: Zhan Su (California Institute of Technology); Andrew P. Ingersoll (California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0074.1

URL: <https://doi.org/10.1175/jpo-d-16-0074.1>

Publication Date: 2016-06-13

Volume/Issue: Volume 46, Issue 9

References: 49

Cited by: 27 publications

Language: N/A

Subjects: N/A

Keywords: Kinetic energy, Baroclinic flows, Circulation/ Dynamics, Models and modeling, Physical Meteorology and Climatology (and 3 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Exactly solving the absolute minimum potential energy state (Lorenz reference state) is a difficult problem because of the nonlinear nature of the equation of state of seawater. This problem has been solved recently but the algorithm comes at a high computational cost. As the first part of this study, the authors develop an algorithm that is ~103-105 times faster, making it useful for energy diagnosis in ocean models. The second part of this study shows that the global patterns of Lorenz available potential energy (APE) density are distinct from those of eddy kinetic energy (EKE). This is because the Lorenz APE density is based on the entire domainwide parcel rearrangement, while mesoscale eddies, if related to baroclinic instability, are typically generated through local parcel rearrangement approximately around the eddy size. Inspired by this contrast, this study develops a locally defined APE framework: the eddy size-constrained APE density based on the strong constraint that the parcel rearrangement/displacement to achieve the minimum potential energy state should not exceed the local eddy size horizontally. This concept typically identifies baroclinically unstable regions. It is shown to be helpful to detect individual eddies/vortices and local EKE patterns, for example, around the Southern Ocean fronts and subtropical western boundary currents. This is consistent with the physical picture that mesoscale eddies are associated with a strong signature in both the velocity field (i.e., EKE) and the stratification (i.e., local APE). The new APE concept may be useful in parameterizing mesoscale eddies in ocean models.

1455. Quantifying the drivers of ocean-atmosphere CO₂ fluxes

Authors: Jonathan M. Lauderdale (Massachusetts Institute of Technology); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Richard G. Williams (University of Liverpool); Michael J. Follows (Massachusetts Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gb005400

URL: <https://doi.org/10.1002/2016gb005400>

Publication Date: 2016-06-11

Volume/Issue: Volume 30, Issue 7

References: 45

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: carbon dioxide, regional drivers, air-sea flux

Funding: J.M.L., S.D., and MJF, U.S. NSF, UK NERC

License: CC BY 4.0

Open Access: N/A

Abstract: A mechanistic framework for quantitatively mapping the regional drivers of air-sea CO₂ fluxes at a global scale is developed. The framework evaluates the interplay between (1) surface heat and freshwater fluxes that influence the potential saturated carbon concentration, which depends on changes in sea surface temperature, salinity and alkalinity, (2) a residual, disequilibrium flux influenced by upwelling and entrainment of remineralized carbon- and nutrient-rich waters from the ocean interior, as well as rapid subduction of surface waters, (3) carbon uptake and export by biological activity as both soft tissue and carbonate, and (4) the effect on surface carbon concentrations due to freshwater precipitation or evaporation. In a steady state simulation of a coarse-resolution ocean circulation and biogeochemistry model, the sum of the individually determined components is close to the known total flux of the simulation. The leading order balance, identified in different dynamical regimes, is between the CO₂ fluxes driven by surface heat fluxes and a combination of biologically driven carbon uptake and disequilibrium-driven carbon outgassing. The framework is still able to reconstruct simulated fluxes when evaluated using monthly averaged data and takes a form that can be applied consistently in models of different complexity and observations of the ocean. In this way, the framework may reveal differences in the balance of drivers acting across an ensemble of climate model simulations or be applied to an analysis and interpretation of the observed, real-world air-sea flux of CO₂.

1456. Southern Ocean biological impacts on global ocean oxygen

Authors: David P. Keller (GEOMAR Helmholtz Centre for Ocean Research); Iris Kriest (GEOMAR Helmholtz Centre for Ocean Research); Wolfgang Koeve (GEOMAR Helmholtz Centre for Ocean Research); Andreas Oschlies (GEOMAR Helmholtz Centre for Ocean Research)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl069630

URL: <https://doi.org/10.1002/2016gl069630>

Publication Date: 2016-06-10

Volume/Issue: Volume 43, Issue 12

References: 55

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: DFG, North-German Supercomputing Alliance (HLRN), BMBF BIOACID Program

License: N/A

Open Access: N/A

Abstract: Southern Ocean (SO) physical and biological processes are known to have a large impact on global biogeochemistry. However, the role that SO biology plays in determining ocean oxygen concentrations is not completely understood. These dynamics are investigated here by shutting off SO biology in two marine biogeochemical models. The results suggest that SO biological processes reduce the ocean's oxygen content, mainly in the deep ocean, by 14 to 19%. However, since these processes also trap nutrients that would otherwise be transported northward to fuel productivity and subsequent organic matter export, consumption, and the accompanying oxygen consumption in midlatitude to low-latitude waters, SO biology helps to maintain higher oxygen concentrations in these subsurface waters. Thereby, SO biology can influence the size of the tropical oxygen minimum zones. As a result of ocean circulation the link between SO biological processes and remote oxygen changes operates on decadal to centennial time scales.

1457. The eddy kinetic energy budget in the Red Sea

Authors: Peng Zhan (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology Thuwal Saudi Arabia); Aneesh C. Subramanian (University of Oxford); Fengchao Yao (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology Thuwal Saudi Arabia); Aditya R. Kartadikaria (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology Thuwal Saudi Arabia, Study Program of Oceanography Bandung Institute of Technology Bandung Indonesia); Daquan Guo (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology Thuwal Saudi Arabia); Ibrahim Hoteit (Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology Thuwal Saudi Arabia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011589

URL: <https://doi.org/10.1002/2015jc011589>

Publication Date: 2016-06-09

Volume/Issue: Volume 121, Issue 7

References: 50

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: King Abdullah University of Science and Technology (KAUST), Saudi Arabia and the Saudi ARAMCO-KAUST Marine Environmental Research Center

License: N/A

Open Access: N/A

Abstract: The budget of eddy kinetic energy (EKE) in the Red Sea, including the sources, redistributions, and sink, is examined using a high-resolution eddy-resolving ocean circulation model. A pronounced seasonally varying EKE is identified, with its maximum intensity occurring in winter, and the strongest EKE is captured mainly in the central and northern basins within the upper 200 m. Eddies acquire kinetic energy from conversion of eddy available potential energy (EPE), from transfer of mean kinetic energy (MKE), and from direct generation due to time-varying (turbulent) wind stress, the first of which contributes predominantly to the majority of the EKE. The EPE-to-EKE conversion occurs almost in the entire basin, while the MKE-to-EKE transfer appears mainly along the shelf boundary of the basin (200 m isobath) where high horizontal shear interacts with topography. The EKE generated by the turbulent wind stress is relatively small and limited to the southern basin. All these processes are intensified during winter, when the rate of energy conversion is about 4-5 times larger than that in summer. The EKE is redistributed by the vertical and horizontal divergence of energy flux and the advection of the mean flow. As a main sink of EKE, dissipation processes is ubiquitously found in the basin. The seasonal variability of these energy conversion terms can explain the significant seasonality of eddy activities in the Red Sea.

1458. Ice flow dynamics and mass loss of Totten Glacier, East Antarctica, from 1989 to 2015

Authors: Xin Li (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jeremie Mouginot (University of California Irvine); Bernd Scheuchl (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl069173

URL: <https://doi.org/10.1002/2016gl069173>

Publication Date: 2016-05-30

Volume/Issue: Volume 43, Issue 12

References: 48

Cited by: 70 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency

License: N/A

Open Access: N/A

Abstract: Totten Glacier has the largest ice discharge in East Antarctica and a basin grounded mostly below sea level. Satellite altimetry data have revealed ice thinning in areas of fast flow. Here we present a time series of ice velocity measurements spanning from 1989 to 2015 using Landsat and interferometric synthetic-aperture radar data, combined with ice thickness from Operation IceBridge, and surface mass balance from Regional Atmospheric Climate Model. We find that the glacier speed exceeded its balance speed in 1989-1996, slowed down by 11 +/- 12% in 2000 to bring its ice flux in balance with accumulation (65 +/- 4 Gt/yr), then accelerated by 18 +/- 3% until 2007, and remained constant thereafter. The average ice mass loss (7 +/- 2 Gt/yr) is dominated by ice dynamics (73%). Its acceleration (0.6 +/- 0.3 Gt/yr²) is dominated by surface mass balance (80%). Ice velocity apparently increased when ocean temperature was warmer, which suggests a linkage between ice dynamics and ocean temperature.

1459. Internal Variability Versus Anthropogenic Forcing on Sea Level and Its Components

Authors: Marta Marcos; Ben Marzeion; Sönke Dangendorf; Aimée B. A. Slangen; Hindumathi Palanisamy; Luciana Fenoglio-Marc

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-016-9373-3

URL: <https://doi.org/10.1007/s10712-016-9373-3>

Publication Date: 2016-05-30

Volume/Issue: Volume 38, Issue 1

References: 95

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1460. Modeling of ocean-induced ice melt rates of five west Greenland glaciers over the past two decades

Authors: E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Xu (University of California Irvine); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Mouginot (University of California Irvine); B. Scheuchl (University of California Irvine); X. Li (University of California Irvine); M. Morlighem (University of California Irvine); H. Seroussi (University of California Irvine); M. van den Broeke (Utrecht University); I. Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. Cai (University of California Irvine); L. An (University of California Irvine); B. de Fleurian (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl068784

URL: <https://doi.org/10.1002/2016gl068784>

Publication Date: 2016-05-30

Volume/Issue: Volume 43, Issue 12

References: 43

Cited by: 93 publications

Language: en

Subjects: N/A

Keywords: Greenland ice sheet|Future sea level|Glacier mass balance|Glacier morphology

Funding: Gordon and Betty Moore Foundation

License: N/A

Open Access: N/A

Abstract: High-resolution, three-dimensional simulations from the Massachusetts Institute of Technology general circulation model ocean model are used to calculate the subaqueous melt rate of the calving faces of Umiamak, Rinks, Kangerdlugssup, Store, and Kangilerngata glaciers, west Greenland, from 1992 to 2015. Model forcing is from monthly reconstructions of ocean state and ice sheet runoff. Results are analyzed in combination with observations of bathymetry, bed elevation, ice front retreat, and glacier speed. We calculate that subaqueous melt rates are 2-3 times larger in summer compared to winter and doubled in magnitude since the 1990s due to enhanced subglacial runoff and 1.6 +/- 0.3 degreesC warmer ocean temperature. Umiamak and Kangilerngata retreated rapidly in the 2000s when subaqueous melt rates exceeded the calving rates and ice front retreated to deeper bed elevation. In contrast, Store, Kangerdlugssup, and Rinks have remained stable because their subaqueous melt rates are 3-4 times lower than their calving rates, i.e., the glaciers are dominated by calving processes.

1461. Impact of in-consistency between the climate model and its initial conditions on climate prediction

Authors: Xueyuan Liu; Armin Köhl; Detlef Stammer; Shuhei Masuda; Yoichi Ishikawa; Takashi Mochizuki

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-016-3194-4

URL: <https://doi.org/10.1007/s00382-016-3194-4>

Publication Date: 2016-05-28

Volume/Issue: Volume 49, Issue 3

References: 65

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Union 7th Framework Programme

License: N/A

Open Access: N/A

Abstract: N/A

1462. A conventional value for the geoid reference potential W_0

Authors: L. Sánchez; R. ■underlík; N. Dayoub; K. Mikula; Z. Minarechová; Z. Šíma; V. Vatrť; M. Vojtíšková

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-016-0913-x

URL: <https://doi.org/10.1007/s00190-016-0913-x>

Publication Date: 2016-05-24

Volume/Issue: Volume 90, Issue 9

References: 101

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsches Geodätisches Forschungsinstitut, Technische Universität München, München, Germany, Department of Mathematics and Descriptive Geometry, Faculty of Civil Engineering, Slovak University of Technology in Bratislava, Slovakia, Department of Mathematics and Descriptive Geometry, Faculty of Civil Engineering, Slovak University of Technology in Bratislava, Slovakia (and 5 more)

License: N/A

Open Access: N/A

Abstract: N/A

1463. Noble gas tracers of ventilation during deep-water formation in the Weddell Sea

Authors: D P Nicholson; S Khatiwala; P Heimbach

Journal: IOP Conference Series: Earth and Environmental Science

Publisher: IOP Publishing

DOI: 10.1088/1755-1315/35/1/012019

URL: <https://doi.org/10.1088/1755-1315/35/1/012019>

Publication Date: 2016-05-21

Volume/Issue: Volume 35

References: 36

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1464. An optimized treatment for algorithmic differentiation of an important glaciological fixed-point problem

Authors: Daniel N. Goldberg; Sri Hari Krishna Narayanan; Laurent Hascoet; Jean Utke

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-1891-2016

URL: <https://doi.org/10.5194/gmd-9-1891-2016>

Publication Date: 2016-05-21

Volume/Issue: Volume 9, Issue 5

References: 54

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

License: Creative Commons

Open Access: N/A

Abstract: . We apply an optimized method to the adjoint generation of a time-evolving land ice model through algorithmic differentiation (AD). The optimization involves a special treatment of the fixed-point iteration required to solve the nonlinear stress balance, which differs from a straightforward application of AD software, and leads to smaller memory requirements and in some cases shorter computation times of the adjoint. The optimization is done via implementation of the algorithm of Christianson (1994) for reverse accumulation of fixed-point problems, with the AD tool OpenAD. For test problems, the optimized adjoint is shown to have far lower memory requirements, potentially enabling larger problem sizes on memory-limited machines. In the case of the land ice model, implementation of the algorithm allows further optimization by having the adjoint model solve a sequence of linear systems with identical (as opposed to varying) matrices, greatly improving performance. The methods introduced here will be of value to other efforts applying AD tools to ice models, particularly ones which solve a hybrid shallow ice/shallow shelf approximation to the Stokes equations.

1465. neXtSIM: a new Lagrangian sea ice model

Authors: Pierre Rampal; Sylvain Bouillon; Einar Ólason; Mathieu Morlighem

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-1055-2016

URL: <https://doi.org/10.5194/tc-10-1055-2016>

Publication Date: 2016-05-20

Volume/Issue: Volume 10, Issue 3

References: 84

Cited by: 94 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Arctic sea ice cover has changed drastically over the last decades. Associated with these changes is a shift in dynamical regime seen by an increase of extreme fracturing events and an acceleration of sea ice drift. The highly non-linear dynamical response of sea ice to external forcing makes modelling these changes and the future evolution of Arctic sea ice a challenge for current models. It is, however, increasingly important that this challenge be better met, both because of the important role of sea ice in the climate system and because of the steady increase of industrial operations in the Arctic. In this paper we present a new dynamical/thermodynamical sea ice model called neXtSIM that is designed to address this challenge. neXtSIM is a continuous and fully Lagrangian model, whose momentum equation is discretised with the finite-element method. In this model, sea ice physics are driven by the combination of two core components: a model for sea ice dynamics built on a mechanical framework using an elasto-brittle rheology, and a model for sea ice thermodynamics providing damage healing for the mechanical framework. The evaluation of the model performance for the Arctic is presented for the period September 2007 to October 2008 and shows that observed multi-scale statistical properties of sea ice drift and deformation are well captured as well as the seasonal cycles of ice volume, area, and extent. These results show that neXtSIM is an appropriate tool for simulating sea ice over a wide range of spatial and temporal scales.

1466. Impact of Synoptic Atmospheric Forcing on the Mean Ocean Circulation

Authors: Yang Wu (Nanjing University); Xiaoming Zhai (University of East Anglia); Zhaomin Wang (Nanjing University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0819.1

URL: <https://doi.org/10.1175/jcli-d-15-0819.1>

Publication Date: 2016-05-18

Volume/Issue: Volume 29, Issue 16

References: 72

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The impact of synoptic atmospheric forcing on the mean ocean circulation is investigated by comparing simulations of a global eddy-permitting ocean-sea ice model forced with and without synoptic atmospheric phenomena. Consistent with previous studies, transient atmospheric motions such as weather systems are found to contribute significantly to the time-mean wind stress and surface heat loss at mid- and high latitudes owing to the nonlinear nature of air-sea turbulent fluxes. Including synoptic atmospheric forcing in the model has led to a number of significant changes. For example, wind power input to the ocean increases by about 50%, which subsequently leads to a similar percentage increase in global eddy kinetic energy. The wind-driven subtropical gyre circulations are strengthened by about 10%-15%, whereas even greater increases in gyre strength are found in the subpolar oceans. Deep convection in the northern North Atlantic becomes significantly more vigorous, which in turn leads to an increase in the Atlantic meridional overturning circulation (AMOC) by as much as 55%. As a result of the strengthened horizontal gyre circulations and the AMOC, the maximum global northward heat transport increases by almost 50%. Results from this study show that synoptic atmospheric phenomena such as weather systems play a vital role in driving the global ocean circulation and heat transport, and therefore should be properly accounted for in paleo- and future climate studies.

1467. Estimating planktonic diversity through spatial dominance patterns in a model ocean

Authors: Alice Soccodato; Francesco d'Ovidio; Marina Lévy; Oliver Jahn; Michael J. Follows; Silvia De Monte

Journal: Marine Genomics

Publisher: Elsevier BV

DOI: 10.1016/j.margen.2016.04.015

URL: <https://doi.org/10.1016/j.margen.2016.04.015>

Publication Date: 2016-05-17

Volume/Issue: Volume 29

References: 50

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: EMERGENCE EME1111 UPMC

License: N/A

Open Access: N/A

Abstract: N/A

1468. Multidecadal freshening and lightening in the deep waters of the Bransfield Strait, Antarctica

Authors: Tiago S. Dotto (University of Southampton, Laboratório de Estudos dos Oceanos e Clima, Instituto de Oceanografia Universidade Federal do Rio Grande Rio Grande Brazil, National Oceanography Centre); Rodrigo Kerr (Laboratório de Estudos dos Oceanos e Clima, Instituto de Oceanografia Universidade Federal do Rio Grande Rio Grande Brazil); Mauricio M. Mata (Laboratório de Estudos dos Oceanos e Clima, Instituto de Oceanografia Universidade Federal do Rio Grande Rio Grande Brazil); Carlos A. E. Garcia (Laboratório de Estudos dos Oceanos e Clima, Instituto de Oceanografia Universidade Federal do Rio Grande Rio Grande Brazil)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011228

URL: <https://doi.org/10.1002/2015jc011228>

Publication Date: 2016-05-09

Volume/Issue: Volume 121, Issue 6

References: 59

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the International Polar Year SOS-CLIMATE project, the Brazilian National Institute of Science and Technology of Cryosphere, POLARCANION, NAUTILUS, and CAPES/CMAR2 projects

License: N/A

Open Access: N/A

Abstract: The deep waters of the Bransfield Strait receive considerable amounts of water from the Weddell Sea continental shelf. The restricted connections to the surrounding ocean and relatively easier access makes the Bransfield Strait an important proxy region for monitoring changes in the dense Weddell Sea shelf water masses, which are an important precursor of Antarctic Bottom Water (AABW). Long-term hydrographic data from the period 1960s-2010s showed freshening and lightening of the deep water masses of the Bransfield Strait, which was likely caused by large freshwater inputs originating from the western shelf of the Weddell Sea. The rates of freshening and lightening were $-0.0010 \pm 0.0005 \text{ yr}^{-1}$ and $-0.0016 \pm 0.0014 \text{ kg m}^{-3} \text{ yr}^{-1}$ for the central basin, respectively, and $-0.0010 \pm 0.0006 \text{ yr}^{-1}$ and $-0.0029 \pm 0.0013 \text{ kg m}^{-3} \text{ yr}^{-1}$ for the eastern basin, respectively. The deep waters showed a high degree of interannual thermohaline variability, which appeared to be caused by changes in the proportions of source water mass mixing between the years. Statistically significant negative correlations between salinity/neutral density fields and the Southern Annular Mode (SAM) were observed (-0.56 and -0.62 for the central basin, respectively, and -0.58 and -0.68 for the eastern basin, respectively) between 1980 and 2014. During SAM positive phases, communication between the Weddell Sea and the Bransfield Strait is reduced, which leads to less saline and lighter water masses in the Bransfield Strait; however, the opposite trends are observed during SAM negative phases.

1469. An Approach to Computing Discrete Adjoint for MPI-Parallelized Models Applied to the Ice Sheet System Model

Authors: Eric Larour; Jean Utke; Anton Bovin; Mathieu Morlighem; Gilberto Perez

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

Subjects: N/A

Keywords: Hindcast|Greenland ice sheet|Operator (biology)|Ice-sheet model

Funding: Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: . Within the framework of sea-level rise projections, there is a strong need for hindcast validation of the evolution of polar ice sheets in a way that tightly matches observational records (from radar and altimetry observations mainly). However, the computational requirements for making hindcast reconstructions possible are severe and rely mainly on the evaluation of the adjoint state of transient ice-flow models. Here, we look at the computation of adjoints in the context of the NASA/JPL/UCI Ice Sheet System Model, written in C++ and designed for parallel execution with MPI. We present the adaptations required in the way the software is designed and written but also generic adaptations in the tools facilitating the adjoint computations. We concentrate on the use of operator overloading coupled with the AdjoinableMPI library to achieve the adjoint computation of ISSM. We present a comprehensive approach to 1) carry out type changing through ISSM, hence facilitating operator overloading, 2) bind to external solvers such as MUMPS and GSL-LU and 3) handle MPI-based parallelism to scale the capability. We demonstrate the success of the approach by computing sensitivities of hindcast metrics such as the misfit to observed records of surface altimetry on the North-East Greenland Ice Stream, or the misfit to observed records of surface velocities on Upernavik Glacier, Central West Greenland. We also provide metrics for the scalability of the approach, and the expected performance. This approach has the potential of enabling a new generation of hindcast-validated projections that make full use of the wealth of datasets currently being collected, or already collected in Greenland and Antarctica, such as surface altimetry, surface velocities, and/or gravity measurements.

1470. Ocean Forecasting: From Regional to Coastal Scales

Authors: Emil V. Stanev; Johannes Schulz-Stellenfleth; Joanna Staneva; Sebastian Grayek; Sebastian Grashorn; Arno Behrens; Wolfgang Koch; Johannes Pein

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/os-2016-25

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Ocean observations|Forcing (mathematics)

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . In the past years, the Helmholtz Zentrum Geesthacht put in place the Coastal Observing System for the North and Arctic Seas (COSYNA) in the frame of which different aspects of forecasting the marine environment have been developed. This paper describes these developments, which are based on recent advances in coastal ocean forecasting in the field of numerical modelling, data assimilation and observational array design. The region of interest is the North and Baltic Sea; most of the coastal examples discussed in the paper are for the German Bight. Several pre-operational applications are presented exemplifying the outcome of using the best available science in coastal ocean predictions. They help to identify new challenges; most of them are associated with resolving the non-linear behavior of coastal ocean, which for the studied region, is manifested by the tidal distortion and generation of shallow-water tides. Led by the motivation to maximize the benefit from observations, the authors focus on the integration of observations and modelling by using advanced statistical methods. The coastal and regional ocean forecasting systems do not run in isolation, but are linked, either weakly by just using forcing data, or interactively by using two-way nesting or unstructured-grid models. Therefore the problem of downscaling and upscaling, which currently attracts much attention, is also addressed. One example shown is the coupling of the coarse-resolution regional models by using a two-way nesting method with fine resolution in the area of connecting straits. The major part of the paper presents illustrations from assimilation of remote sensing, in situ and HF radar data, prediction of wind waves and storm surges, as well as possible applications to search and rescue operations, and modelling support for assessing the environmental impact of wind parks. Concepts for seamless approaches to link coastal and regional forecasting systems are also presented and the two examples given illustrate (1) an application of unstructured-grid model for the Ems Estuary, and (2) the potential influence of the information from coastal observatories or coastal forecasting systems on the regional models.

1471. Use of ocean model forecasts during acoustic experiments and naval missions

Authors: Kevin D. Heaney (OASIS Inc)

Journal: Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

DOI: 10.1121/1.4949973

URL: <https://doi.org/10.1121/1.4949973>

Publication Date: 2016-05-06

Volume/Issue: Volume 139, Issue 4_Supplement

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean acoustic propagation is critically dependent upon the spatial (and to a lesser extent the temporal) scales of the ocean. With the advent of high-fidelity data-assimilative ocean models, forecasts are available that can provide an improved level of confidence for acoustic propagation modeling in support of sea-tests and naval exercises. In this paper, a set of experiments will be presented where ocean model forecasts were used within an acoustic model to help inform the experiment planning. Tests include shallow water tests (Key West), continental shelf-break environments (Shallow Water 2006, Quantifying and Predicting Uncertainty) and deep water environments (Philippine sea 2009/2010). Models used include the Navy Coastal Ocean Model (NCOM), the MSEAS model (MIT, Pierre Lermusiaux), the ECCO2 model state estimation and the Scripps Institution of Oceanography version of the ROMS (Bruce Cornuelle) model run. In-situ assimilation of acoustic signals was performed for geo-acoustic information (in shallow water) and not for ocean model forecast updates. Our conclusion for over 10 years of use of models in real-time experiments is that ocean model forecasting does provide useful information for mission planning and experiment design, particularly when combined with on-site measurements.

1472. Modelling of global mass effects in hydrology, atmosphere and oceans on surface gravity

Authors: M. Mikolaj; B. Meurers; A. Güntner

Journal: Computers & Geosciences

Publisher: Elsevier BV

DOI: 10.1016/j.cageo.2016.04.014

URL: <https://doi.org/10.1016/j.cageo.2016.04.014>

Publication Date: 2016-05-03

Volume/Issue: Volume 93

References: 51

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1473. High-resolution modeling of coastal freshwater discharge and glacier mass balance in the Gulf of Alaska watershed

Authors: J. P. Beamer (Oregon State University); D. F. Hill (Oregon State University); A. Arendt (University of Washington); G. E. Liston (Colorado State University)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015wr018457

URL: <https://doi.org/10.1002/2015wr018457>

Publication Date: 2016-04-28

Volume/Issue: Volume 52, Issue 5

References: 79

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: computational hydrology, streamflow, snow and ice, energy balance

Funding: the Oil Spill Recovery Institute, NASA Cryospheric Sciences

License: N/A

Open Access: N/A

Abstract: A comprehensive study of the Gulf of Alaska (GOA) drainage basin was carried out to improve understanding of the coastal freshwater discharge (FWD) and glacier volume loss (GVL). Hydrologic processes during the period 1980-2014 were modeled using a suite of physically based, spatially distributed weather, energy-balance snow/ice melt, soil water balance, and runoff routing models at a high-resolution (1 km horizontal grid; daily time step). Meteorological forcing was provided by the North American Regional Reanalysis (NARR), Modern Era Retrospective Analysis for Research and Applications (MERRA), and Climate Forecast System Reanalysis (CFSR) data sets. Streamflow and glacier mass balance modeled using MERRA and CFSR compared well with observations in four watersheds used for calibration in the study domain. However, only CFSR produced regional seasonal and long-term trends in water balance that compared favorably with independent Gravity Recovery and Climate Experiment (GRACE) and airborne altimetry data. Mean annual runoff using CFSR was 760 km³ yr⁻¹, 8% of which was derived from the long-term removal of stored water from glaciers (glacier volume loss). The annual runoff from CFSR was partitioned into 63% snowmelt, 17% glacier ice melt, and 20% rainfall. Glacier runoff, taken as the sum of rainfall, snow, and ice melt occurring each season on glacier surfaces, was 38% of the total seasonal runoff, with the remaining runoff sourced from nonglacier surfaces. Our simulations suggests that existing GRACE solutions, previously reported to represent glacier mass balance alone, are actually measuring the full water budget of land and ice surfaces.

1474. Forum for Arctic Modeling and Observational Synthesis (FAMOS): Past, current, and future activities

Authors: A. Proshutinsky (Woods Hole Oceanographic Institution); M. Steele (University of Washington); M.-L. Timmermans (Yale University Department of Geology & Geophysics, New Haven Connecticut USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc011898

URL: <https://doi.org/10.1002/2016jc011898>

Publication Date: 2016-04-27

Volume/Issue: Volume 121, Issue 6

References: 156

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: introduction, modeling, Arctic

Funding: the National Science Foundation Office of Polar Programs

License: Creative Commons

Open Access: N/A

Abstract: The overall goal of the Forum for Arctic Modeling and Observational Synthesis (FAMOS) community activities reported in this special issue is to enhance understanding of processes and mechanisms driving Arctic Ocean marine and sea ice changes, and the consequences of those changes especially in biogeochemical and ecosystem studies. Major 2013-2015 FAMOS accomplishments to date are: identification of consistent errors across Arctic regional models; approaches to reduce these errors, and recommendations for the most effective coupled sea ice-ocean models for use in fully coupled regional and global climate models. 2013-2015 FAMOS coordinated analyses include many process studies, using models together with observations to investigate: dynamics and mechanisms responsible for drift, deformation and thermodynamics of sea ice; pathways and mechanisms driving variability of the Atlantic, Pacific and river waters in the Arctic Ocean; processes of freshwater accumulation and release in the Beaufort Gyre; the fate of melt water from Greenland; characteristics of ocean eddies; biogeochemistry and ecosystem processes and change, climate variability, and predictability. Future FAMOS collaborations will focus on employing models and conducting observations at high and very high spatial and temporal resolution to investigate the role of subgrid-scale processes in regional Arctic Ocean and coupled ice-ocean and atmosphere-ice-ocean models.

1475. Estimates of Eddy Heat Flux Crossing the Antarctic Circumpolar Current from Observations in Drake Passage

Authors: D. Randolph Watts (University of Rhode Island); Karen L. Tracey (University of Rhode Island); Kathleen A. Donohue (University of Rhode Island); Teresa K. Chereskin (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-16-0029.1

URL: <https://doi.org/10.1175/jpo-d-16-0029.1>

Publication Date: 2016-04-25

Volume/Issue: Volume 46, Issue 7

References: 49

Cited by: 16 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The 4-yr measurements by current- and pressure-recording inverted echo sounders in Drake Passage produced statistically stable eddy heat flux estimates. Horizontal currents in the Antarctic Circumpolar Current (ACC) turn with depth when a depth-independent geostrophic current crosses the upper baroclinic zone. The dynamically important divergent component of eddy heat flux is calculated. Whereas full eddy heat fluxes differ greatly in magnitude and direction at neighboring locations within the local dynamics array (LDA), the divergent eddy heat fluxes are poleward almost everywhere. Case studies illustrate baroclinic instability events that cause meanders to grow rapidly. In the southern passage, where eddy variability is weak, heat fluxes are weak and not statistically significant. Vertical profiles of heat flux are surface intensified with ~50% above 1000 m and uniformly distributed with depth below. Summing poleward transient eddy heat transport across the LDA of -0.010 ± 0.005 PW with the stationary meander contribution of -0.004 ± 0.001 PW yields -0.013 ± 0.005 PW. A comparison metric, -0.4 PW, represents the total oceanic heat loss to the atmosphere south of 60 degreesS. Summed along the circumpolar ACC path, if the LDA heat flux occurred at six "hot spots" spanning similar or longer path segments, this could account for 20%-70% of the metric, that is, up to -0.28 PW. The balance of ocean poleward heat transport along the remaining ACC path should come from weak eddy heat fluxes plus mean cross-front temperature transports. Alternatively, the metric -0.4 PW, having large uncertainty, may be high.

1476. Annual variation detected by GPS, GRACE and loading models

Authors: Weiwei Li; Tonie van Dam; Zhao Li; Yunzhong Shen

Journal: Studia Geophysica et Geodaetica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11200-016-0205-1

URL: <https://doi.org/10.1007/s11200-016-0205-1>

Publication Date: 2016-04-22

Volume/Issue: Volume 60, Issue 4

References: 30

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1477. Vertical heat flux in the ocean: Estimates from observations and from a coupled general circulation model

Authors: Patrick F. Cummins (Institute of Ocean Sciences, Fisheries and Oceans Canada Sidney British Columbia Canada); Diane Masson (Institute of Ocean Sciences, Fisheries and Oceans Canada Sidney British Columbia Canada); Oleg A. Saenko (Canadian Centre for Climate Modelling and Analysis, Environment and Climate Change Canada)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jc011647

URL: <https://doi.org/10.1002/2016jc011647>

Publication Date: 2016-04-22

Volume/Issue: Volume 121, Issue 6

References: 33

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The net heat uptake by the ocean in a changing climate involves small imbalances between the advective and diffusive processes that transport heat vertically. Generally, it is necessary to rely on global climate models to study these processes in detail. In the present study, it is shown that a key component of the vertical heat flux, namely that associated with the large-scale mean vertical circulation, can be diagnosed over extra-tropical regions from global observational data sets. This component is estimated based on the vertical velocity obtained from the geostrophic vorticity balance, combined with estimates of absolute geostrophic flow. Results are compared with the output of a non-eddy resolving, coupled atmosphere-ocean general circulation model. Reasonable agreement is found in the latitudinal distribution of the vertical heat flux, as well as in the area-integrated flux below about 250 m depth. The correspondence with the coupled model deteriorates sharply at depths shallower than 250 m due to the omission of equatorial regions from the calculation. The vertical heat flux due to the mean circulation is found to be dominated globally by the downward contribution from the Southern Hemisphere, in particular the Southern Ocean. This is driven by the Ekman vertical velocity which induces an upward transport of seawater that is cold relative to the horizontal average at a given depth. The results indicate that the dominant characteristics of the vertical transport of heat due to the mean circulation can be inferred from simple linear vorticity dynamics over much of the ocean.

1478. The timescales of global surface-ocean connectivity

Authors: Bror F. Jönsson; James R. Watson

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/ncomms11239

URL: <https://doi.org/10.1038/ncomms11239>

Publication Date: 2016-04-19

Volume/Issue: Volume 7, Issue 1

References: 40

Cited by: 96 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Planktonic communities are shaped through a balance of local evolutionary adaptation and ecological succession driven in large part by migration. The timescales over which these processes operate are still largely unresolved. Here we use Lagrangian particle tracking and network theory to quantify the timescale over which surface currents connect different regions of the global ocean. We find that the fastest path between two patches--each randomly located anywhere in the surface ocean--is, on average, less than a decade. These results suggest that marine planktonic communities may keep pace with climate change--increasing temperatures, ocean acidification and changes in stratification over decadal timescales--through the advection of resilient types.

1479. Circulation and Stirring in the Southeast Pacific Ocean and the Scotia Sea Sectors of the Antarctic Circumpolar Current

Authors: Dhruv Balwada (Florida State University); Kevin G. Speer (Florida State University); Joseph H. LaCasce (University of Oslo); W. Brechner Owens (Woods Hole Oceanographic Institution); John Marshall (Massachusetts Institute of Technology); Raffaele Ferrari (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-15-0207.1

URL: <https://doi.org/10.1175/jpo-d-15-0207.1>

Publication Date: 2016-04-19

Volume/Issue: Volume 46, Issue 7

References: 58

Cited by: 21 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The large-scale middepth circulation and eddy diffusivities in the southeast Pacific Ocean and Scotia Sea sectors between 110 degrees and 45 degreesW of the Antarctic Circumpolar Current (ACC) are described based on a subsurface quasi-isobaric RAFOS-float-based Lagrangian dataset. These RAFOS float data were collected during the Diapycnal and Isopycnal Mixing Experiment in the Southern Ocean (DIMES). The mean flow, adjusted to a common 1400-m depth, shows the presence of jets in the time-averaged sense with speeds of 6 cm s⁻¹ in the southeast Pacific Ocean and upward of 13 cm s⁻¹ in the Scotia Sea. These jets appear to be locked to topography in the Scotia Sea but, aside from negotiating a seamount chain, are mostly free of local topographic constraints in the southeast Pacific Ocean. The eddy kinetic energy (EKE) is higher than the mean kinetic energy everywhere in the sampled domain by about 50%. The magnitude of the EKE increases drastically (by a factor of 2 or more) as the current crosses over the Hero and Shackleton fracture zones into the Scotia Sea. The meridional isopycnal stirring shows lateral and vertical variations with local eddy diffusivities as high as 2800 +/- 600 m² s⁻¹ at 700 m decreasing to 990 +/- 200 m² s⁻¹ at 1800 m in the southeast Pacific Ocean. However, the cross-ACC diffusivity in the southeast Pacific Ocean is significantly lower, with values of 690 +/- 150 and 1000 +/- 200 m² s⁻¹ at shallow and deep levels, respectively, due to the action of jets. The cross-ACC diffusivity in the Scotia Sea is about 1200 +/- 500 m² s⁻¹.

1480. Dynamic Mapping of Along-Track Ocean Altimetry: Method and Performance from Observing System Simulation Experiments

Authors: Clement Ubelmann; Bruce Cornuelle; Lee-Lueng Fu

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-15-0163.1

URL: <https://doi.org/10.1175/jtech-d-15-0163.1>

Publication Date: 2016-04-13

Volume/Issue: Volume 33, Issue 8

References: 12

Cited by: 30 publications

Language: N/A

Subjects: N/A

Keywords: Eddies, Interpolation schemes, Mathematical and statistical techniques, Inverse methods, Nonlinear dynamics (and 2 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Simulated along-track ocean altimetry data were used to implement the use of a nonlinear dynamic propagator to perform three-dimensional (time and 2D space) interpolation of mesoscale sea surface height (SSH). The method is an inverse approach to processing altimetry data unevenly sampled in time and space into high-level gridded altimetry maps. The inverse approach, similar to the standard objective mapping, contains some correction terms to the innovation vectors to account for nonlinear dynamics. Another key improvement is to solve for the covariance functions through a Green's function approach. From the Observing System Simulation Experiments carried out to simulate a three-satellite constellation over the Gulf Stream region, the new method can significantly reduce mapping errors and improve the resolving capabilities compared to the standard linear objective analysis such as that used by the AVISO gridding.

1481. Experimental and diagnostic protocol for the physical component of the CMIP6 Ocean Model Intercomparison Project (OMIP)

Authors: Stephen M. Griffies; Gokhan Danabasoglu; Paul J. Durack; Alistair J. Adcroft; V. Balaji; Claus W. Böning; Eric P. Chassignet; Enrique Curchitser; Julie Deshayes; Helge Drange; Baylor Fox-Kemper; Peter J. Gleckler; Jonathan M. Gregory; Helmuth Haak; Robert W. Hallberg; Helene T. Hewitt; David M. Holland; Tatiana Ilyina; Johann H. Jungclaus; Yoshiki Komuro; John P. Krasting; William G. Large; Simon J. Marsland; Simona Masina; Trevor J. McDougall; A. J. George Nurser; James C. Orr; Anna Pirani; Fangli Qiao; Ronald J. Stouffer; Karl E. Taylor; Anne Marie Treguier; Hiroyuki Tsujino; Petteri Uotila; Maria Valdivieso; Michael Winton; Stephen G. Yeager

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2016-77

URL: <https://doi.org/10.5194/gmd-2016-77>

Publication Date: 2016-04-12

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References: N/A

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Ocean Model Intercomparison Project (OMIP) aims to provide a framework for evaluating, understanding, and improving the ocean and sea-ice components of global climate and earth system models contributing to the Coupled Model Intercomparison Project Phase 6 (CMIP6). OMIP addresses these aims in two complementary manners: (A) by providing an experimental protocol for global ocean/sea-ice models run with a prescribed atmospheric forcing, (B) by providing a protocol for ocean diagnostics to be saved as part of CMIP6. We focus here on the physical component of OMIP, with a companion paper (Orr et al., 2016) offering details for the inert chemistry and interactive biogeochemistry. The physical portion of the OMIP experimental protocol follows that of the interannual Coordinated Ocean-ice Reference Experiments (CORE-II). Since 2009, CORE-I (Normal Year Forcing) and CORE-II have become the standard method to evaluate global ocean/sea-ice simulations and to examine mechanisms for forced ocean climate variability. The OMIP diagnostic protocol is relevant for any ocean model component of CMIP6, including the DECK (Diagnostic, Evaluation and Characterization of Klima experiments), historical simulations, FAFMIP (Flux Anomaly Forced MIP), C4MIP (Coupled Carbon Cycle Climate MIP), DAMIP (Detection and Attribution MIP), DCP (Decadal Climate Prediction Project), ScenarioMIP (Scenario MIP), as well as the ocean-sea ice OMIP simulations. The bulk of this paper offers scientific rationale for saving these diagnostics.

1482. Rapid variability of Antarctic Bottom Water transport into the Pacific Ocean inferred from GRACE

Authors: Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Carmen Boening (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl068474

URL: <https://doi.org/10.1002/2016gl068474>

Publication Date: 2016-04-10

Volume/Issue: Volume 43, Issue 8

References: 28

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Air-ice-ocean interactions in the Antarctic lead to formation of the densest waters on Earth. These waters convect and spread to fill the global abyssal oceans. The heat and carbon storage capacity of these water masses, combined with their abyssal residence times that often exceed centuries, makes this circulation pathway the most efficient sequestering mechanism on Earth. Yet monitoring this pathway has proven challenging due to the nature of the formation processes and the depth of the circulation. The Gravity Recovery and Climate Experiment (GRACE) gravity mission is providing a time series of ocean mass redistribution and offers a transformative view of the abyssal circulation. Here we use the GRACE measurements to infer, for the first time, a 2003-2014 time series of Antarctic Bottom Water export into the South Pacific. We find this export highly variable, with a standard deviation of 1.87 sverdrup (Sv) and a decorrelation timescale of less than 1 month. A significant trend is undetectable.

1483. A new dipole index of the salinity anomalies of the tropical Indian Ocean

Authors: Junde Li; Chujin Liang; Youmin Tang; Changming Dong; Dake Chen; Xiaohui Liu; Weifang Jin

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/srep24260

URL: <https://doi.org/10.1038/srep24260>

Publication Date: 2016-04-07

Volume/Issue: Volume 6, Issue 1

References: 32

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: With the increased interest in studying the sea surface salinity anomaly (SSSA) of the tropical Indian Ocean during the Indian Ocean Dipole (IOD), an index describing the dipole variability of the SSSA has been pursued recently. In this study, we first use a regional ocean model with a high spatial resolution to produce a high-quality salinity simulation during the period from 1982 to 2014, from which the SSSA dipole structure is identified for boreal autumn. On this basis, by further analysing the observed data, we define a dipole index of the SSSA between the central equatorial Indian Ocean (CEIO: 70 degreesE-90 degreesE, 5 degreesS-5 degreesN) and the region off the Sumatra-Java coast (SJC: 100 degreesE-110 degreesE, 13 degreesS-3 degreesS). Compared with previous SSSA dipole indices, this index has advantages in detecting the dipole signals and in characterizing their relationship to the sea surface temperature anomaly (SSTA) dipole variability. Finally, the mechanism of the SSSA dipole is investigated by dynamical diagnosis. It is found that anomalous zonal advection dominates the SSSA in the CEIO region, whereas the SSSA in the SJC region are mainly influenced by the anomalous surface freshwater flux. This SSSA dipole provides a positive feedback to the formation of the IOD events.

1484. Annual Sea Level Changes on the North American Northeast Coast: Influence of Local Winds and Barotropic Motions

Authors: Christopher G. Piecuch (Atmospheric and Environmental Research Inc.); Sönke Dangendorf (Research Institute for Water and Environment, University of Siegen, Siegen, Germany); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Marta Marcos (Mediterranean Institute for Advanced Studies, UIB-CSIC, Esporles, Spain)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-16-0048.1

URL: <https://doi.org/10.1175/jcli-d-16-0048.1>

Publication Date: 2016-04-06

Volume/Issue: Volume 29, Issue 13

References: 80

Cited by: 62 publications

Language: en

Subjects: N/A

Keywords: Atm/ocean structure/ phenomena, Interannual variability, Sea level, Circulation/ dynamics, North Atlantic Ocean (and 5 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Understanding the relationship between coastal sea level and the variable ocean circulation is crucial for interpreting tide gauge records and projecting sea level rise. In this study, annual sea level records (adjusted for the inverted barometer effect) from tide gauges along the North American northeast coast over 1980-2010 are compared to a set of data-assimilating ocean reanalysis products as well as a global barotropic model solution forced with wind stress and barometric pressure. Correspondence between models and data depends strongly on model and location. At sites north of Cape Hatteras, the barotropic model shows as much (if not more) skill than ocean reanalyses, explaining about 50% of the variance in the adjusted annual tide gauge sea level records. Additional numerical experiments show that annual sea level changes along this coast from the barotropic model are driven by local wind stress over the continental shelf and slope. This result is interpreted in the light of a simple dynamic framework, wherein bottom friction balances surface wind stress in the alongshore direction and geostrophy holds in the across-shore direction. Results highlight the importance of barotropic dynamics on coastal sea level changes on interannual and decadal time scales; they also have implications for diagnosing the uncertainties in current ocean reanalyses, using tide gauge records to infer past changes in ocean circulation, and identifying the physical mechanisms responsible for projected future regional sea level rise.

1485. A comparison of interannual hydrological polar motion excitation from GRACE and geodetic observations

Authors: T. Meyrath; T. van Dam

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2016.03.011

URL: <https://doi.org/10.1016/j.jog.2016.03.011>

Publication Date: 2016-03-31

Volume/Issue: Volume 99

References: 73

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1486. An Equatorial-Extratropical Dipole Structure of the Atlantic Niño

Authors: Hyacinth C. Nnamchi (Department of Geography, University of Nigeria, Nsukka, Nigeria); Jianping Li (Beijing Normal University); Fred Kucharski (Earth System Physics Section, Abdus Salam International Centre for Theoretical Physics, Trieste, Italy); In-Sik Kang (Center of Excellence for Climate Change Research, King Abdulaziz University, Jeddah, Saudi Arabia, Seoul National University); Noel S. Keenlyside (University of Bergen, Bjerknes Centre for Climate Research); Ping Chang (Ocean University of China, Department of Oceanography, Texas A&M University, College Station, Texas); Riccardo Farneti (Earth System Physics Section, Abdus Salam International Centre for Theoretical Physics, Trieste, Italy)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0894.1

URL: <https://doi.org/10.1175/jcli-d-15-0894.1>

Publication Date: 2016-03-29

Volume/Issue: Volume 29, Issue 20

References: 56

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

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License: N/A

Open Access: N/A

Abstract: Equatorial Atlantic variability is dominated by the Atlantic Niño peaking during the boreal summer. Studies have shown robust links of the Atlantic Niño to fluctuations of the St. Helena subtropical anticyclone and Benguela Niño events. Furthermore, the occurrence of opposite sea surface temperature (SST) anomalies in the eastern equatorial and southwestern extratropical South Atlantic Ocean (SAO), also peaking in boreal summer, has recently been identified and termed the SAO dipole (SAOD). However, the extent to which and how the Atlantic Niño and SAOD are related remain unclear. Here, an analysis of historical observations reveals the Atlantic Niño as a possible intrinsic equatorial arm of the SAOD. Specifically, the observed sporadic equatorial warming characteristic of the Atlantic Niño (~ 0.4 K) is consistently linked to southwestern cooling (~ -0.4 K) of the Atlantic Ocean during the boreal summer. Heat budget calculations show that the SAOD is largely driven by the surface net heat flux anomalies while ocean dynamics may be of secondary importance. Perturbations of the St. Helena anticyclone appear to be the dominant mechanism triggering the surface heat flux anomalies. A weakening of the anticyclone will tend to weaken the prevailing northeasterlies and enhance evaporative cooling over the southwestern Atlantic Ocean. In the equatorial region, the southeast trade winds weaken, thereby suppressing evaporation and leading to net surface warming. Thus, it is hypothesized that the wind-evaporation-SST feedback may be responsible for the growth of the SAOD events linking southern extratropics and equatorial Atlantic variability via surface net heat flux anomalies.

1487. Eddy Phase Speeds in a Two-Layer Model of Quasigeostrophic Baroclinic Turbulence with Applications to Ocean Observations

Authors: Lei Wang (University of Chicago); Malte Jansen (University of Chicago); Ryan Abernathey (Columbia University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-15-0192.1

URL: <https://doi.org/10.1175/jpo-d-15-0192.1>

Publication Date: 2016-03-29

Volume/Issue: Volume 46, Issue 6

References: 62

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The phase speed spectrum of ocean mesoscale eddies is fundamental to understanding turbulent baroclinic flows. Since eddy phase propagation has been shown to modulate eddy fluxes, an understanding of eddy phase speeds is also of practical importance for the development of improved eddy parameterizations for coarse resolution ocean models. However, it is not totally clear whether and how linear Rossby wave theory can be used to explain the phase speed spectra in various weakly turbulent flow regimes. Using linear analysis, theoretical constraints are identified that control the eddy phase speed in a two-layer quasigeostrophic (QG) model. These constraints are then verified in a series of nonlinear two-layer QG simulations, spanning a range of parameters with potential relevance to the ocean. In the two-layer QG model, the strength of the inverse cascade exerts an important control on the eddy phase speed. If the inverse cascade is weak, the phase speed spectrum is reasonably well approximated by the phase speed of the linearly most unstable mode. A significant inverse cascade instead leads to barotropization, which in turn leads to mean phase speeds closer to those of barotropic-mode Rossby waves. The two-layer QG results are qualitatively consistent with the observed eddy phase speed spectra in the Antarctic Circumpolar Current and may also shed light on the interpretation of phase speed spectra observed in other regions.

1488. Decomposition of Random Errors Inherent to HOAPS-3.2 Near-Surface Humidity Estimates Using Multiple Triple Collocation Analysis

Authors: Julian Kinzel (Satellite-based Climate Monitoring, Deutscher Wetterdienst, Offenbach Germany); Karsten Fennig (Satellite-based Climate Monitoring, Deutscher Wetterdienst, Offenbach Germany); Marc Schröder (Satellite-based Climate Monitoring, Deutscher Wetterdienst, Offenbach Germany); Axel Andersson (Satellite-based Climate Monitoring, Deutscher Wetterdienst, Offenbach Germany); Karl Bumke (GEOMAR Helmholtz Centre for Ocean Research); Rainer Hollmann (Satellite-based Climate Monitoring, Deutscher Wetterdienst, Offenbach, Germany)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-15-0122.1

URL: <https://doi.org/10.1175/jtech-d-15-0122.1>

Publication Date: 2016-03-29

Volume/Issue: Volume 33, Issue 7

References: 61

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Latent heat fluxes (LHF) play an essential role in the global energy budget and are thus important for understanding the climate system. Satellite-based remote sensing permits a large-scale determination of LHF, which, among others, are based on near-surface specific humidity. However, the random retrieval error (ϵ) remains unknown. Here, a novel approach is presented to quantify the error contributions to pixel-level of the Hamburg Ocean Atmosphere Parameters and Fluxes from Satellite Data, version 3.2 (HOAPS, version 3.2), dataset. The methodology makes use of multiple triple collocation (MTC) analysis between 1995 and 2008 over the global ice-free oceans. Apart from satellite records, these datasets include selected ship records extracted from the Seewetteramt Hamburg (SWA) archive and the International Comprehensive Ocean-Atmosphere Data Set (ICOADS), serving as the in situ ground reference. The MTC approach permits the derivation of ϵ as the sum of model uncertainty and sensor noise, while random uncertainties due to in situ measurement errors ($\epsilon_{\text{in situ}}$) and collocation ($\epsilon_{\text{collocation}}$) are isolated concurrently. Results show an average of $1.1 \pm 0.3 \text{ g kg}^{-1}$, whereas the mean ($\bar{\epsilon}$) is in the order of $0.5 \pm 0.1 \text{ g kg}^{-1}$ ($0.5 \pm 0.3 \text{ g kg}^{-1}$). Regional analyses indicate a maximum of exceeding 1.5 g kg^{-1} within humidity regimes of $12\text{--}17 \text{ g kg}^{-1}$, associated with the single-parameter, multilinear retrieval applied in HOAPS. Multidimensional bias analysis reveals that global maxima are located off the Arabian Peninsula.

1489. Interannual variability of the Indonesian Throughflow: The salinity effect

Authors: Shijian Hu (Qingdao National Laboratory for Marine Science and Technology, Scripps Institution of Oceanography / University of California San Diego, Chinese Academy of Sciences); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011495

URL: <https://doi.org/10.1002/2015jc011495>

Publication Date: 2016-03-19

Volume/Issue: Volume 121, Issue 4

References: 59

Cited by: 64 publications

Language: en

Subjects: N/A

Keywords: interannual variability, ENSO, Indonesian Throughflow, salinity effect, IOD

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: The Indonesian Throughflow (ITF) region possesses strong mixing and experiences significant freshwater input, but the role of salinity variability in the Indonesian Seas remains unclear. The goal of this study is to understand how salinity variability influences the ITF transport on interannual time scales. The ITF transport is calculated using observations and assimilation data sets and verified using direct ITF transport estimates. We find that the halosteric component of the ITF transport contributes (36 $\pm$ 7)% of the total ITF variability, in contrast to (63 $\pm$ 6)% by the thermosteric component. Thus, while not dominant, this result nonetheless implies that the salinity variability in the Indonesian Seas is of remarkable importance in determining the interannual variability of ITF transport. Correlation analysis indicates that the interannual variability of the total ITF transport is mainly influenced by the El Niño-Southern Oscillation (ENSO) rather than the Indian Ocean Dipole. Under the ENSO cycle, the Walker Circulation shifts longitudinally resulting in fluctuations in precipitation over the Indonesian Seas that modulates salinity and subsequently influences the interannual variability of ITF transport. This result signals the importance of precipitation and the subsequent salinity effect in determining the interannual variability of the ITF transport. The role of wind forcing and oceanic planetary waves is also revisited using this newly calculated ITF transport series. ENSO-related wind forcing is found to modulate the ITF transport via Rossby waves through the wave guide in the Indonesian Seas, which is in agreement with previous studies.

1490. ISSM-SESAW v1.0: mesh-based computation of gravitationally consistent sea-level and geodetic signatures caused by cryosphere and climate driven mass change

Authors: Surendra Adhikari; Erik R. Ivins; Eric Larour

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-1087-2016

URL: <https://doi.org/10.5194/gmd-9-1087-2016>

Publication Date: 2016-03-18

Volume/Issue: Volume 9, Issue 3

References: 92

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: . A classical Green's function approach for computing gravitationally consistent sea-level variations associated with mass redistribution on the earth's surface employed in contemporary sea-level models naturally suits the spectral methods for numerical evaluation. The capability of these methods to resolve high wave number features such as small glaciers is limited by the need for large numbers of pixels and high-degree (associated Legendre) series truncation. Incorporating a spectral model into (components of) earth system models that generally operate on a mesh system also requires repetitive forward and inverse transforms. In order to overcome these limitations, we present a method that functions efficiently on an unstructured mesh, thus capturing the physics operating at kilometer scale yet capable of simulating geophysical observables that are inherently of global scale with minimal computational cost. The goal of the current version of this model is to provide high-resolution solid-earth, gravitational, sea-level and rotational responses for earth system models operating in the domain of the earth's outer fluid envelope on timescales less than about 1 century when viscous effects can largely be ignored over most of the globe. The model has numerous important geophysical applications. For example, we compute time-varying computations of global geodetic and sea-level signatures associated with recent ice-sheet changes that are derived from space gravimetry observations. We also demonstrate the capability of our model to simultaneously resolve kilometer-scale sources of the earth's time-varying surface mass transport, derived from high-resolution modeling of polar ice sheets, and predict the corresponding local and global geodetic signatures.

1491. Assessment of intraseasonal variabilities in China Ocean Reanalysis (CORA)

Authors: Min Zhang; Lei Zhou; Hongli Fu; Lianghong Jiang; Xiangming Zhang

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-016-0820-2

URL: <https://doi.org/10.1007/s13131-016-0820-2>

Publication Date: 2016-03-14

Volume/Issue: Volume 35, Issue 3

References: 48

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1492. Stochastic filtering for determining gravity variations for decade-long time series of GRACE gravity

Authors: Lei Wang (Massachusetts Institute of Technology, Columbia University); James L. Davis (Columbia University); Emma M. Hill (Nanyang Technological University, Earth Observatory of Singapore); Mark E. Tamisiea (National Oceanography Centre)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jb012650

URL: <https://doi.org/10.1002/2015jb012650>

Publication Date: 2016-03-14

Volume/Issue: Volume 121, Issue 4

References: 48

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: We present a new stochastic filter technique for statistically rigorous separation of gravity signals and correlated "stripe" noises in a series of monthly gravitational spherical harmonic coefficients (SHCs) produced by the Gravity Recovery and Climate Experiment (GRACE) satellite mission. Unlike the standard destriping process that removes the stripe contamination empirically, the stochastic approach simultaneously estimates gravity signals and correlated noises relying on covariance information that reflects both the spatial spectral features and temporal correlations among them. A major benefit of the technique is that by estimating the stripe noise in a Bayesian framework, we are able to propagate statistically rigorous covariances for the destriped GRACE SHCs, i.e. incorporating the impact of the destriping on the SHC uncertainties. The Bayesian approach yields a natural resolution for the gravity signal that reflects the correlated stripe noise, and thus achieve a kind of spatial smoothing in and of itself. No spatial Gaussian smoothing is formally required although it might be useful for some circumstances. Using the stochastic filter, we process a decade-length series of GRACE monthly gravity solutions, and compare the results with GRACE Tellus data products that are processed using the "standard" destriping procedure. The results show that the stochastic filter is able to remove the correlated stripe noise to a remarkable degree even without an explicit smoothing step. The estimates from the stochastic filter for each destriped GRACE field are suitable for Bayesian integration of GRACE with other geodetic measurements and models, and the statistically rigorous estimation of the time-varying rates and seasonal cycles in GRACE time series.

1493. Contrasting Effects of Historical Sea Level Rise and Contemporary Ocean Currents on Regional Gene Flow of *Rhizophora racemosa* in Eastern Atlantic Mangroves

Authors: Magdalene N. Ngeve; Tom Van der Stocken; Dimitris Menemenlis; Nico Koedam; Ludwig Triest

Journal: PLOS ONE

Publisher: Public Library of Science (PLoS)

DOI: 10.1371/journal.pone.0150950

URL: <https://doi.org/10.1371/journal.pone.0150950>

Publication Date: 2016-03-10

Volume/Issue: Volume 11, Issue 3

References: 116

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1494. Geophysical fluids from different data sources, geomagnetic jerks, and their impact on Earths orientation

Authors: Jan Vondrak

Journal: Acta Geodynamica et Geomaterialia

Publisher: Institute of Rock Structure and Mechanics, AS CR

DOI: 10.13168/agg.2016.0005

URL: <https://doi.org/10.13168/agg.2016.0005>

Publication Date: 2016-03-09

Volume/Issue: N/A

References: 26

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1495. Stochastic secular trends in sea level rise

Authors: Victor Ocaña (Massachusetts Institute of Technology); Eduardo Zorita (Institut für Küstenforschung, Helmholtz-Zentrum Geesthacht Geesthacht Germany); Patrick Heimbach (University of Texas at Austin)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011301

URL: <https://doi.org/10.1002/2015jc011301>

Publication Date: 2016-03-07

Volume/Issue: Volume 121, Issue 4

References: 101

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF

License: N/A

Open Access: N/A

Abstract: Global mean sea level (GMSL) has been rising since (at least) the nineteenth century and the rate of rise may be increasing. Several studies that attempt to explain the long-term trend of GMSL during the instrumental record share the common assumption that this trend is deterministic in nature and different from natural variations. Here we show that the trend can alternatively be explained, at least in part, as being caused by random variations within the coupled ocean-atmosphere-cryosphere system, and hence not having a deterministic origin. These random trends, which add to externally forced changes (e.g., through anthropogenic climate change), are a consequence of the integrated character of GMSL, which is the cumulative addition of temporal contributions that exhibit random character, and whose integration results in GMSL variations with persistence on decadal-centennial time scales. The generation of trends by integration of random stationary noise (i.e., even in a constant climate) is a robust and fundamental feature of stochastically forced systems with memory. The integrated character of GMSL results in an intrinsic difficulty in distinguishing internal from externally forced trends.

1496. The latitudinal dependence of the oceanic barotropic eddy kinetic energy and macroturbulence energy transport

Authors: R. Chemke (Weizmann Institute of Science); Y. Kaspi (Weizmann Institute of Science)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl067847

URL: <https://doi.org/10.1002/2016gl067847>

Publication Date: 2016-03-06

Volume/Issue: Volume 43, Issue 6

References: 69

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: oceanic barotropization, barotropic inverse energy cascade, eddy-eddy interactions

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Geostrophic turbulence theory predicted already a few decades ago an inverse energy cascade in the barotropic mode, yet there has been limited evidence for it in the ocean. In this study, the latitudinal behavior of the oceanic barotropic energy balance and macroturbulent scales is studied using the ECCO2 (Estimating the Circulation and Climate of the Ocean) state estimate, which synthesizes satellite data and in situ measurements with a high-resolution general circulation model containing realistic bathymetry and wind forcing. It is found that inverse energy cascade occurs at high latitudes, as eddy-eddy interactions spread the conversion of eddy kinetic energy from the baroclinic to the barotropic mode, both upscale and downscale. At these latitudes, the conversion scale of baroclinic eddy kinetic energy and the energy-containing scale follow the most unstable and Rhines scales, respectively. Even though an inverse energy cascade occurs at high latitudes, the energy spectrum follows a steeper slope than the $-5/3$ slope. Different than classic arguments, the Rossby deformation radius does not follow the baroclinic conversion and most unstable scales.

1497. Sampling uncertainty in gridded sea surface temperature products and Advanced Very High Resolution Radiometer (AVHRR) Global Area Coverage (GAC) data

Authors: C.E. Bulgin; O. Embury; C.J. Merchant

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2016.02.021

URL: <https://doi.org/10.1016/j.rse.2016.02.021>

Publication Date: 2016-03-05

Volume/Issue: Volume 177

References: 20

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency Sea Surface Temperature Climate Change Initiative

License: N/A

Open Access: N/A

Abstract: N/A

1498. Variations of the Global Net Air-Sea Heat Flux during the "Hiatus" Period (2001-10)

Authors: Xinfeng Liang (University of South Florida); Lisan Yu (Woods Hole Oceanographic Institution)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0626.1

URL: <https://doi.org/10.1175/jcli-d-15-0626.1>

Publication Date: 2016-03-03

Volume/Issue: Volume 29, Issue 10

References: 32

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: Seasonal variability, Surface fluxes, Interannual variability, Models and modeling, Physical Meteorology and Climatology (and 4 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An assessment is made of the mean and variability of the net air-sea heat flux, Q_{net} , from four products (ECCO, OAFlux-CERES, ERA-Interim, and NCEP1) over the global ice-free ocean from January 2001 to December 2010. For the 10-yr "hiatus" period, all products agree on an overall net heat gain over the global ice-free ocean, but the magnitude varies from 1.7 to 9.5 W m⁻². The differences among products are particularly large in the Southern Ocean, where they cannot even agree on whether the region gains or loses heat on the annual mean basis. Decadal trends of Q_{net} differ significantly between products. ECCO and OAFlux-CERES show almost no trend, whereas ERA-Interim suggests a downward trend and NCEP1 shows an upward trend. Therefore, numerical simulations utilizing different surface flux forcing products will likely produce diverged trends of the ocean heat content during this period. The downward trend in ERA-Interim started from 2006, driven by a peculiar pattern change in the tropical regions. ECCO, which used ERA-Interim as initial surface forcings and is constrained by ocean dynamics and ocean observations, corrected the pattern. Among the four products, ECCO and OAFlux-CERES show great similarities in the examined spatial and temporal patterns. Given that the two estimates were obtained using different approaches and based on largely independent observations, these similarities are encouraging and instructive. It is more likely that the global net air-sea heat flux does not change much during the so-called hiatus period.

1499. Source waters for the highly productive Patagonian shelf in the southwestern Atlantic

Authors: Hajoon Song; John Marshall; Michael J. Follows; Stephanie Dutkiewicz; Gaël Forget

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2016.02.009

URL: <https://doi.org/10.1016/j.jmarsys.2016.02.009>

Publication Date: 2016-03-03

Volume/Issue: Volume 158

References: 59

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1500. Broadband assessment of degree-2 gravitational changes from GRACE and other estimates, 2002-2015

Authors: J. L. Chen (University of Texas at Austin); C. R. Wilson (University of Texas at Austin); J. C. Ries (University of Texas at Austin)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jb012708

URL: <https://doi.org/10.1002/2015jb012708>

Publication Date: 2016-03-02

Volume/Issue: Volume 121, Issue 3

References: 61

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA GRACE Science Program, NASA ESI Program, NSF OPP Program

License: N/A

Open Access: N/A

Abstract: Space geodetic measurements, including the Gravity Recovery and Climate Experiment (GRACE), satellite laser ranging (SLR), and Earth rotation provide independent and increasingly accurate estimates of variations in Earth's gravity field Stokes coefficients ΔC_{21} , ΔS_{21} , and ΔC_{20} . Mass redistribution predicted by climate models provides another independent estimate of air and water contributions to these degree-2 changes. SLR has been a successful technique in measuring these low-degree gravitational changes. Broadband comparisons of independent estimates of ΔC_{21} , ΔS_{21} , and ΔC_{20} from GRACE, SLR, Earth rotation, and climate models during the GRACE era from April 2002 to April 2015 show that the current GRACE release 5 solutions of ΔC_{21} and ΔS_{21} provided by the Center for Space Research (CSR) are greatly improved over earlier solutions and agree remarkably well with other estimates, especially on ΔS_{21} estimates. GRACE and Earth rotation ΔS_{21} agreement is exceptionally good across a very broad frequency band from intraseasonal, seasonal, to interannual and decadal periods. SLR ΔC_{20} estimates remain superior to GRACE and Earth rotation estimates, due to the large uncertainty in GRACE ΔC_{20} solutions and particularly high sensitivity of Earth rotation ΔC_{20} estimates to errors in the wind fields. With several estimates of ΔC_{21} , ΔS_{21} , and ΔC_{20} variations, it is possible to estimate broadband noise variance and noise power spectra in each, given reasonable assumptions about noise independence. The GRACE CSR release 5 solutions clearly outperform other estimates of ΔC_{21} and ΔS_{21} variations with the lowest noise levels over a broad band of frequencies.

1501. Constraints on oceanic meridional heat transport from combined measurements of oxygen and carbon

Authors: L. Resplandy; R. F. Keeling; B. B. Stephens; J. D. Bent; A. Jacobson; C. Rödenbeck; S. Khatiwala

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-016-3029-3

URL: <https://doi.org/10.1007/s00382-016-3029-3>

Publication Date: 2016-02-27

Volume/Issue: Volume 47, Issue 9-10

References: 81

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Climate Program Office, National Science Foundation, National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1502. Sea Ice Deformation in a Coupled Ocean-Sea Ice Model and in Satellite Remote Sensing Data

Authors: Gunnar Spreen; Ron Kwok; Dimitris Menemenlis; An T. Nguyen

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2016-13

URL: <https://doi.org/10.5194/tc-2016-13>

Publication Date: 2016-02-22

Volume/Issue: N/A

References: N/A

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Fast ice|Sea ice concentration

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A realistic representation of sea ice deformation in models is important for accurate simulation of the sea ice mass balance. In this study, model ice strength sensitivity experiments show an increase in Arctic Basin sea ice volume of 7 % and 35 % for a decrease in ice strength of, respectively, 30 % and 70 %, after 8 years of model integration. This volume increase is caused by a combination of dynamic and thermodynamic processes. On the one hand, a weaker ice cover initially produces more ice due to increased deformation and new ice growth. The thickening of the ice, on the other hand, increases the ice strength and decreases the sea ice volume export out of the Arctic Basin. The balance of these processes leads to a new equilibrium Arctic Basin ice volume. Simulated sea ice deformation strain rates from model simulations with 4.5, 9, and 18-km horizontal grid spacing are compared with synthetic aperture radar satellite observations (RGPS). All three model simulations can reproduce the large-scale ice deformation patterns but they do not reproduce all aspects of the observed deformation rates. The overall sea ice deformation rate is about 50 % lower in all model solutions than in the satellite observations, especially in the seasonal sea ice zone. Small scale sea ice deformation and linear kinematic features are not adequately reproduced. A decrease in model grid spacing, however, produces a higher density and more localized ice deformation features. Overall, the 4.5-km simulation produces the lowest misfits in divergence, vorticity, and shear when compared with RGPS data. Not addressed in this study is whether the differences between simulated and observed deformation rates are an intrinsic limitation of the viscous-plastic sea ice rheology that was used in the sensitivity experiments, or whether it indicates a lack of adjustment of existing model parameters to better represent these processes. Either way, this study provides new quantitative metrics for existing and new sea ice rheologies to strive for.

1503. Suitability of satellite sea surface salinity data for use in assessing and correcting ocean forecasts

Authors: Matthew J. Martin

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2016.02.004

URL: <https://doi.org/10.1016/j.rse.2016.02.004>

Publication Date: 2016-02-15

Volume/Issue: Volume 180

References: 31

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1504. Dynamical Attribution of Recent Variability in Atlantic Overturning

Authors: Helen R. Pillar (University of Oxford); Patrick Heimbach (Massachusetts Institute of Technology); Helen L. Johnson (University of Oxford); David P. Marshall (University of Oxford)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0727.1

URL: <https://doi.org/10.1175/jcli-d-15-0727.1>

Publication Date: 2016-02-15

Volume/Issue: Volume 29, Issue 9

References: 63

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: Mass fluxes/transport, Dynamics, Ocean circulation, Meridional overturning circulation, Circulation/dynamics (and 2 more)

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Attributing observed variability of the Atlantic meridional overturning circulation (AMOC) to past changes in surface forcing is challenging but essential for detecting any influence of anthropogenic forcing and reducing uncertainty in future climate predictions. Here, quantitative estimates of separate contributions from wind and buoyancy forcing to AMOC variations at 25 degreesN are obtained. These estimates are achieved by projecting observed atmospheric anomalies onto model-based dynamical patterns of AMOC sensitivity to surface wind, thermal, and freshwater forcing over the preceding 15 years. Local wind forcing is shown to dominate AMOC variability on short time scales, whereas subpolar heat fluxes dominate on decadal time scales. The reconstructed transport time series successfully reproduces most of the interannual variability observed by RAPID-MOCHA. However, the apparent decadal trend in the RAPID-MOCHA time series is not captured, requiring improved model representation of ocean adjustment to subpolar heat fluxes over at least the past two decades and highlighting the importance of sustained monitoring of the high-latitude North Atlantic.

1505. Impact of uncertainties in atmospheric boundary conditions on ocean model solutions

Authors: Ayan H. Chaudhuri; Rui M. Ponte; Gael Forget

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2016.02.003

URL: <https://doi.org/10.1016/j.ocemod.2016.02.003>

Publication Date: 2016-02-14

Volume/Issue: Volume 100

References: 61

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF, NASA

License: N/A

Open Access: N/A

Abstract: N/A

1506. Sea Surface Height Variability in Drake Passage

Authors: Kathleen A. Donohue (University of Rhode Island); Maureen A. Kennelly (University of Rhode Island); Amy Cutting (Naval Undersea Warfare Center, Newport, Rhode Island)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-15-0249.1

URL: <https://doi.org/10.1175/jtech-d-15-0249.1>

Publication Date: 2016-02-11

Volume/Issue: Volume 33, Issue 4

References: 33

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: Observational techniques and algorithms, Anomalies, In situ oceanic observations, Satellite observations, Variability (and 2 more)

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License: N/A

Open Access: N/A

Abstract: Intercomparisons between altimeter sea surface height (SSH) and open-ocean in situ observations have been limited owing to sparse available datasets. Here, SSH anomaly (SSHA) determined from current and pressure recording inverted echo sounders (CPIES) from the cDrake experiment were compared with an up-to-date AVISO-mapped product. Meandering Antarctic Circumpolar Current (ACC) fronts in the passage interior elevated SSHA variance; south of the Shackleton Fracture Zone and along the northern continental slope, the variance decreased by factors between 6 and 10. In situ analysis focused on the two constituents of SSHA, SSHA_{ref} determined from bottom pressure and SSHA_{bc} calculated from geopotential height referenced to the bottom. The peak variance of both SSHA_{bc} and SSHA_{ref} occurred in the energetic region between the Subantarctic Front and the Polar Front. The contribution of SSHA_{bc} to total SSHA variance was greater than 40% at all sites and averaged over all sites it was 73%. For most sites, high-frequency (>1/20 cpd) SSHA_{bc} signals dominated total high-frequency variance. Aliasing of high-frequency signals resulting from 10-day altimeter sampling was assessed. The fraction of aliased energy at frequencies longer than 1/50 cpd for sites at and north of the Shackleton Fracture Zone approached 0.25 and approached 0.50 for southern sites. CPIES and mapped altimeter SSHA agreed well. The mean correlation coefficient was 0.82 and the mean RMS difference was 0.075 m. Correlations between CPIES and AVISO were notably poorer at the northern and southern boundaries. RMS differences increased as a function of CPIES high-frequency SSHA variance because the mapped altimetry product does not resolve these frequencies.

1507. Ocean calibration approach for data from the GRACE Follow-On mission

Authors: Peter L. Bender (University of Colorado Boulder); Casey R. Betts (University of Colorado Boulder)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jb012433

URL: <https://doi.org/10.1002/2015jb012433>

Publication Date: 2016-02-07

Volume/Issue: Volume 121, Issue 2

References: 21

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: mass distribution variations, GRACE Follow-On mission, geopotential variations at satellite altitude, ECCO-JPL ocean model

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Gravity Recovery and Climate Experiment (GRACE) mission has been providing valuable new information on time variations in the Earth's gravity field since 2002. In addition, the GRACE Follow-On mission is scheduled to be flown soon after the end of life of the GRACE mission in order to minimize the loss of valuable data on the Earth's gravity field changes. In view of the major benefits to hydrology and oceanography, as well as to other fields, it is desirable to investigate the fundamental limits to monitoring the time variations in the Earth's gravity field during GRACE-type missions. A simplified model is presented in this paper for making estimates of the effect of differential spurious accelerations of the satellites during times when four successive revolutions cross the Pacific Ocean. The analysis approach discussed is to make use of changes in the satellite separation observed during passages across low-latitude regions of the Pacific and of other oceans to correct for spurious accelerations of the satellites. The low-latitude regions of the Pacific and of other oceans are the extended regions where the a priori uncertainties in the time variations of the geopotential heights due to mass distribution changes are known best. In addition, advantage can be taken of the repeated crossings of the South Pole and the North Pole, since the uncertainties in changes in the geopotential heights at the poles during the time required for four orbit revolutions are likely to be small.

1508. The role of local atmospheric forcing on the modulation of the ocean mixed layer depth in reanalyses and a coupled single column ocean model

Authors: Byju Pookkandy; Dietmar Dommenges; Nicholas Klingaman; Scott Wales; Christine Chung; Claudia Frauen; Holger Wolff

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-016-3009-7

URL: <https://doi.org/10.1007/s00382-016-3009-7>

Publication Date: 2016-02-03

Volume/Issue: Volume 47, Issue 9-10

References: 44

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1509. Loading-Induced Deformation Due to Atmosphere, Ocean and Hydrology: Model Comparisons and the Impact on Global SLR, VLBI and GNSS Solutions

Authors: O. Roggenbuck; D. Thaller; G. Engelhardt; S. Franke; R. Dach; P. Steigenberger

Journal: International Association of Geodesy Symposia

Publisher: Springer International Publishing

DOI: 10.1007/1345_2015_214

URL: https://doi.org/10.1007/1345_2015_214

Publication Date: 2016-02-03

Volume/Issue: N/A

References: 13

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Very-long-baseline interferometry|Satellite laser ranging

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1510. Marine mixotrophy increases trophic transfer efficiency, mean organism size, and vertical carbon flux

Authors: Ben A. Ward (Laboratoire des Sciences de l'Environnement Marin, Institut Universitaire Européen de la Mer, Technopole Brest Iroise, 29280 Plouzané, France; University of Bristol); Michael J. Follows (Massachusetts Institute of Technology)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.1517118113

URL: <https://doi.org/10.1073/pnas.1517118113>

Publication Date: 2016-02-02

Volume/Issue: Volume 113, Issue 11

References: 51

Cited by: 269 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: SignificanceMarine plankton commonly combine the autotrophic use of light and inorganic resources with the heterotrophic ingestion of prey. These mixotrophs blur the strict boundary between producers and consumers and allow energy and biomass to enter the food web across multiple trophic levels. Incorporating this flexibility into a global simulation of the surface ocean food web reveals that mixotrophy enhances the transfer of biomass to larger organisms at higher trophic levels, which in turn increases the efficiency of oceanic carbon storage through the production of larger, faster-sinking, and carbon-enriched organic detritus.

1511. Assessment of the Global and Regional Land Hydrosphere and Its Impact on the Balance of the Geophysical Excitation Function of Polar Motion

Authors: Małgorzata Wińska; Jolanta Nastula; Barbara Kończak

Journal: Acta Geophysica

Publisher: Springer Science and Business Media LLC

DOI: 10.1515/acgeo-2015-0041

URL: <https://doi.org/10.1515/acgeo-2015-0041>

Publication Date: 2016-02-02

Volume/Issue: Volume 64, Issue 1

References: 41

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1512. Intercomparison of the Arctic sea ice cover in global ocean-sea ice reanalyses from the ORA-IP project

Authors: Matthieu Chevallier; Gregory C. Smith; Frédéric Dupont; Jean-François Lemieux; Gael Forget; Yosuke Fujii; Fabrice Hernandez; Rym Msadek; K. Andrew Peterson; Andrea Storto; Takahiro Toyoda; Maria Valdivieso; Guillaume Vernieres; Hao Zuo; Magdalena Balmaseda; You-Soon Chang; Nicolas Ferry; Gilles Garric; Keith Haines; Sarah Keeley; Robin M. Kovach; Tsurane Kuragano; Simona Masina; Yongming Tang; Hiroyuki Tsujino; Xiaochun Wang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-016-2985-y

URL: <https://doi.org/10.1007/s00382-016-2985-y>

Publication Date: 2016-01-20

Volume/Issue: Volume 49, Issue 3

References: 102

Cited by: 92 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1513. Taking into Account Atmospheric Uncertainty Improves Sequential Assimilation of SMOS Sea Ice Thickness Data in an Ice-Ocean Model

Authors: Qinghua Yang (Alfred Wegener Institute, National Marine Environmental Forecasting Center); Martin Losch (Alfred Wegener Institute); Svetlana N. Losa (Alfred Wegener Institute); Thomas Jung (University of Bremen, Alfred Wegener Institute); Lars Nerger (Alfred Wegener Institute)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-15-0176.1

URL: <https://doi.org/10.1175/jtech-d-15-0176.1>

Publication Date: 2016-01-15

Volume/Issue: Volume 33, Issue 3

References: 39

Cited by: 16 publications

Language: N/A

Subjects: N/A

Keywords: Forecasting, Numerical weather prediction/forecasting, Models and modeling, Data assimilation, Probability forecasts/models/distribution

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The sensitivity of assimilating sea ice thickness data to uncertainty in atmospheric forcing fields is examined using ensemble-based data assimilation experiments with the Massachusetts Institute of Technology General Circulation Model (MITgcm) in the Arctic Ocean during November 2011-January 2012 and the Met Office (UKMO) ensemble atmospheric forecasts. The assimilation system is based on a local singular evolutive interpolated Kalman (LSEIK) filter. It combines sea ice thickness data derived from the European Space Agency's (ESA) Soil Moisture Ocean Salinity (SMOS) satellite and Special Sensor Microwave Imager/Sounder (SSMIS) sea ice concentration data with the numerical model. The effect of representing atmospheric uncertainty implicit in the ensemble forcing is assessed by three different assimilation experiments. The first two experiments use a single deterministic forcing dataset and a different forgetting factor to inflate the ensemble spread. The third experiment uses 23 members of the UKMO atmospheric ensemble prediction system. It avoids additional ensemble inflation and is hence easier to implement. As expected, the model-data misfits are substantially reduced in all three experiments, but with the ensemble forcing the errors in the forecasts of sea ice concentration and thickness are smaller compared to the experiments with deterministic forcing. This is most likely because the ensemble forcing results in a more plausible spread of the model state ensemble, which represents model uncertainty and produces a better forecast.

1514. Modeling of present-day atmosphere and ocean non-tidal de-aliasing errors for future gravity mission simulations

Authors: Henryk Dobslaw; Inga Bergmann-Wolf; Ehsan Forootan; Christoph Dahle; Torsten Mayer-Gürr; Jürgen Kusche; Frank Flechtner

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-015-0884-3

URL: <https://doi.org/10.1007/s00190-015-0884-3>

Publication Date: 2016-01-08

Volume/Issue: Volume 90, Issue 5

References: 46

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency

License: N/A

Open Access: N/A

Abstract: N/A

1515. Earth orientation and its excitations by atmosphere, oceans, and geomagnetic jerks

Authors: J. Vondrák (Czech Academy of Sciences); C. Ron (Czech Academy of Sciences)

Journal: Serbian Astronomical Journal

Publisher: National Library of Serbia

DOI: 10.2298/saj1591059v

URL: <https://doi.org/10.2298/saj1591059v>

Publication Date: 2016-01-04

Volume/Issue: Issue 191

References: N/A

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: Nutation|Precession

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: In addition to torques exerted by the Moon, Sun, and planets, changes of the Earth orientation parameters (EOP) are known to be caused also by excitations by the atmosphere and oceans. Recently appeared studies, hinting that geomagnetic jerks (GMJ, rapid changes of geomagnetic field) might be associated with sudden changes of phase and amplitude of EOP (Holme and de Viron 2005, 2013, Gibert and Le Mouél 2008, Malkin 2013). We (Ron et al. 2015) used additional excitations applied at the epochs of GMJ to derive its influence on motion of the spin axis of the Earth in space (precession-nutation). We demonstrated that this effect, if combined with the influence of the atmosphere and oceans, improves substantially the agreement with celestial pole offsets observed by Very Long-Baseline Interferometry. Here we concentrate our efforts to study possible influence of GMJ on temporal changes of all five Earth orientation parameters defining the complete Earth orientation in space. Numerical integration of Brzeziński's broad-band Liouville equations (Brzeziński 1994) with atmospheric and oceanic excitations, combined with expected GMJ effects, is used to derive EOP and compare them with their observed values. We demonstrate that the agreement between all five Earth orientation parameters integrated by this method and those observed by space geodesy is improved substantially if the influence of additional excitations at GMJ epochs is added to excitations by the atmosphere and oceans.

1516. Three-Dimensional Distribution of Turbulent Mixing in the South China Sea

Authors: Qingxuan Yang (Ocean University of China); Wei Zhao (Ocean University of China); Xinfeng Liang (Massachusetts Institute of Technology); Jiwei Tian (Ocean University of China)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0220.1

URL: <https://doi.org/10.1175/jpo-d-14-0220.1>

Publication Date: 2015-12-29

Volume/Issue: Volume 46, Issue 3

References: 55

Cited by: 88 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A three-dimensional distribution of turbulent mixing in the South China Sea (SCS) is obtained for the first time, using the Gregg-Henyey-Polzin parameterization and hydrographic observations from 2005 to 2012. Results indicate that turbulent mixing generally increases with depth in the SCS, reaching the order of $10^{-2} \text{ m}^2 \text{ s}^{-1}$ at depth. In the horizontal direction, turbulence is more active in the northern SCS than in the south and is more active in the east than the west. Two mixing "hotspots" are identified in the bottom water of the Luzon Strait and Zhongsha Island Chain area, where diapycnal diffusivity values are around $3 \times 10^{-2} \text{ m}^2 \text{ s}^{-1}$. Potential mechanisms responsible for these spatial patterns are discussed, which include internal tide, bottom bathymetry, and near-inertial energy.

1517. Ice shelf basal melt rates around Antarctica from simulations and observations

Authors: M. P. Schodlok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. J. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011117

URL: <https://doi.org/10.1002/2015jc011117>

Publication Date: 2015-12-26

Volume/Issue: Volume 121, Issue 2

References: 75

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: ice shelf basal melt, Southern Ocean, Antarctica, numerical modeling, ice shelf ocean interaction

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We introduce an explicit representation of Antarctic ice shelf cavities in the Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2) ocean retrospective analysis; and compare resulting basal melt rates and patterns to independent estimates from satellite observations. Two simulations are carried out: the first is based on the original ECCO2 vertical discretization; the second has higher vertical resolution particularly at the depth range of ice shelf cavities. The original ECCO2 vertical discretization produces higher than observed melt rates and leads to a misrepresentation of Southern Ocean water mass properties and transports. In general, thicker levels at the base of the ice shelves lead to increased melting because of their larger heat capacity. This strengthens horizontal gradients and circulation within and outside the cavities and, in turn, warm water transports from the shelf break to the ice shelves. The simulation with more vertical levels produces basal melt rates (1735 +/- 164 Gt/a) and patterns that are in better agreement with observations. Thinner levels in the sub-ice-shelf cavities improve the representation of a fresh/cold layer at the ice shelf base and of warm/salty water near the bottom, leading to a sharper pycnocline and reduced vertical mixing underneath the ice shelf. Improved water column properties lead to more accurate melt rates and patterns, especially for melt/freeze patterns under large cold-water ice shelves. At the 18 km grid spacing of the ECCO2 model configuration, the smaller, warm-water ice shelves cannot be properly represented, with higher than observed melt rates in both simulations.

1518. Observations, inferences, and mechanisms of the Atlantic Meridional Overturning Circulation: A review

Authors: Martha W. Buckley (Oceanography Research Group Atmospheric and Environmental Research Lexington Massachusetts USA, George Mason University); John Marshall (Massachusetts Institute of Technology)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015rg000493

URL: <https://doi.org/10.1002/2015rg000493>

Publication Date: 2015-12-23

Volume/Issue: Volume 54, Issue 1

References: 558

Cited by: 503 publications

Language: en

Subjects: N/A

Keywords: Atlantic Meridional Overturning Circulation, climate variability

Funding: National Oceanic and Atmospheric Administration, National Science Foundation, National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: This is a review about the Atlantic Meridional Overturning Circulation (AMOC), its mean structure, temporal variability, controlling mechanisms, and role in the coupled climate system. The AMOC plays a central role in climate through its heat and freshwater transports. Northward ocean heat transport achieved by the AMOC is responsible for the relative warmth of the Northern Hemisphere compared to the Southern Hemisphere and is thought to play a role in setting the mean position of the Intertropical Convergence Zone north of the equator. The AMOC is a key means by which heat anomalies are sequestered into the ocean's interior and thus modulates the trajectory of climate change. Fluctuations in the AMOC have been linked to low-frequency variability of Atlantic sea surface temperatures with a host of implications for climate variability over surrounding landmasses. On intra-annual timescales, variability in AMOC is large and primarily reflects the response to local wind forcing; meridional coherence of anomalies is limited to that of the wind field. On interannual to decadal timescales, AMOC changes are primarily geostrophic and related to buoyancy anomalies on the western boundary. A pacemaker region for decadal AMOC changes is located in a western "transition zone" along the boundary between the subtropical and subpolar gyres. Decadal AMOC anomalies are communicated meridionally from this region. AMOC observations, as well as the expanded ocean observational network provided by the Argo array and satellite altimetry, are inspiring efforts to develop decadal predictability systems using coupled atmosphere-ocean models initialized by ocean data.

1519. The Pacific North Equatorial Current: New Insights from the Origins of the Kuroshio and Mindanao Currents (OKMC) Project

Authors: Bo Qiu; Daniel Rudnick; Ivana Cerovecki; Bruce Cornuelle; Shuiming Chen; Martha Schönau; Julie McClean; Ganesh Gopalakrishnan

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2015.78

URL: <https://doi.org/10.5670/oceanog.2015.78>

Publication Date: 2015-12-22

Volume/Issue: Volume 28, Issue 4

References: N/A

Cited by: 56 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1520. The Mindanao Current: Mean Structure and Connectivity

Authors: Martha Schönau; Daniel Rudnick; Ivana Cerovecki; Ganesh Gopalakrishnan; Bruce Cornuelle; Julie McClean; Bo Qiu

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2015.79

URL: <https://doi.org/10.5670/oceanog.2015.79>

Publication Date: 2015-12-22

Volume/Issue: Volume 28, Issue 4

References: N/A

Cited by: 31 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1521. Mesoscale to Submesoscale Wavenumber Spectra in Drake Passage

Authors: Cesar B. Rocha (Scripps Institution of Oceanography / University of California San Diego); Teresa K. Chereskin (Scripps Institution of Oceanography / University of California San Diego); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-15-0087.1

URL: <https://doi.org/10.1175/jpo-d-15-0087.1>

Publication Date: 2015-12-22

Volume/Issue: Volume 46, Issue 2

References: 52

Cited by: 190 publications

Language: N/A

Subjects: N/A

Keywords: Observational techniques and algorithms, Atm/Ocean Structure/Phenomena, Ship observations, Eddies, Tides (and 2 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study discusses the upper-ocean (0-200 m) horizontal wavenumber spectra in the Drake Passage from 13 yr of shipboard ADCP measurements, altimeter data, and a high-resolution numerical simulation. At scales between 10 and 200 km, the ADCP kinetic energy spectra approximately follow k^{-3} power law. The observed flows are more energetic at the surface, but the shape of the kinetic energy spectra is independent of depth. These characteristics resemble predictions of isotropic interior quasigeostrophic turbulence. The ratio of across-track to along-track kinetic energy spectra, however, significantly departs from the expectation of isotropic interior quasigeostrophic turbulence. The inconsistency is dramatic at scales smaller than 40 km. A Helmholtz decomposition of the ADCP spectra and analyses of synthetic and numerical model data show that horizontally divergent, ageostrophic flows account for the discrepancy between the observed spectra and predictions of isotropic interior quasigeostrophic turbulence. In Drake Passage, ageostrophic motions appear to be dominated by inertia-gravity waves and account for about half of the near-surface kinetic energy at scales between 10 and 40 km. Model results indicate that ageostrophic flows imprint on the sea surface, accounting for about half of the sea surface height variance between 10 and 40 km.

1522. Low-frequency variability of terrestrial water budget in China using GRACE satellite measurements from 2003 to 2010

Authors: Kaixuan Kang; Hui Li; Peng Peng; Zhengbo Zou

Journal: Geodesy and Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.geog.2015.12.001

URL: <https://doi.org/10.1016/j.geog.2015.12.001>

Publication Date: 2015-12-18

Volume/Issue: Volume 6, Issue 6

References: 30

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: China National Science Funds

License: N/A

Open Access: N/A

Abstract: N/A

1523. Interannual Variability of the Mindanao Current/Undercurrent in Direct Observations and Numerical Simulations

Authors: Shijian Hu (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Dunxin Hu (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Cong Guan (Chinese Academy of Sciences); Fan Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Linlin Zhang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Fujun Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences); Qingye Wang (Qingdao National Laboratory for Marine Science and Technology, Chinese Academy of Sciences)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-15-0092.1

URL: <https://doi.org/10.1175/jpo-d-15-0092.1>

Publication Date: 2015-12-17

Volume/Issue: Volume 46, Issue 2

References: 45

Cited by: 29 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The interannual variability of the boundary currents east of the Mindanao Island, including the Mindanao Current/Undercurrent (MC/MUC), is investigated using moored acoustic Doppler current profiler (ADCP) measurements combined with a series of numerical experiments. The ADCP mooring system was deployed east of the Mindanao Island at 7 degrees59'N, 127 degrees3'E during December 2010-August 2014. Depth-dependent interannual variability is detected in the two western boundary currents: strong and lower-frequency variability dominates the upper-layer MC, while weaker and higher-frequency fluctuation controls the subsurface MUC. Throughout the duration of mooring measurements, the weakest MC was observed in June 2012, in contrast to the maximum peaks in December 2010 and June 2014, while in the deeper layer the MUC shows speed peaks circa December 2010, January 2011, April 2013, and July 2014 and valleys circa June 2011, August 2012, and November 2013. Diagnostic analysis and numerical sensitivity experiments using a 2.5-layer reduced-gravity model indicate that wind forcing in the western Pacific Ocean is a driving agent in conditioning the interannual variability of MC and MUC. Results suggest that westward-propagating Rossby waves that generate in the western Pacific Ocean (roughly 150 degrees-180 degreesE) are of much significance in the interannual variability of the two boundary currents. Fluctuation of Ekman pumping due to local wind stress curl anomaly in the far western Pacific Ocean (roughly 120 degrees-150 degreesE) also plays a role in the interannual variability of the MC. The relationship between the MC/MUC and El Niño is discussed.

1524. Monitoring Atlantic overturning circulation and transport variability with GRACE-type ocean bottom pressure observations - a sensitivity study

Authors: K. Bentel; F. W. Landerer; C. Boening

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-11-953-2015

URL: <https://doi.org/10.5194/os-11-953-2015>

Publication Date: 2015-12-13

Volume/Issue: Volume 11, Issue 6

References: 32

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Atlantic Meridional Overturning Circulation (AMOC) is a key mechanism for large-scale northward heat transport and thus plays an important role for global climate. Relatively warm water is transported northward in the upper layers of the North Atlantic Ocean and, after cooling at subpolar latitudes, sinks down and is transported back south in the deeper limb of the AMOC. The utility of in situ ocean bottom pressure (OBP) observations to infer AMOC changes at single latitudes has been characterized in the recent literature using output from ocean models. We extend the analysis and examine the utility of space-based observations of time-variable gravity and the inversion for ocean bottom pressure to monitor AMOC changes and variability between 20 and 60 degrees N. Consistent with previous results, we find a strong correlation between the AMOC signal and OBP variations, mainly along the western slope of the Atlantic Basin. We then use synthetic OBP data - smoothed and filtered to resemble the resolution of the GRACE (Gravity Recovery and Climate Experiment) gravity mission, but without errors - and reconstruct geostrophic AMOC transport. Due to the coarse resolution of GRACE-like OBP fields, we find that leakage of signal across the step slopes of the ocean basin is a significant challenge at certain latitudes. Transport signal rms is of a similar order of magnitude as error rms for the reconstructed time series. However, the interannual AMOC anomaly time series can be recovered from 20 years of monthly GRACE-like OBP fields with errors less than 1 sverdrup in many locations.

1525. Earth rotation and geodynamics

Authors: Janusz Bogusz; Aleksander Brzezinski; Wieslaw Kosek; Jolanta Nastula

Journal: Geodesy and Cartography

Publisher: Walter de Gruyter GmbH

DOI: 10.1515/geocart-2015-0013

URL: <https://doi.org/10.1515/geocart-2015-0013>

Publication Date: 2015-12-11

Volume/Issue: Volume 64, Issue 2

References: 44

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: This paper presents the summary of research activities carried out in Poland in 2011-2014 in the field of Earth rotation and geodynamics by several Polish research institutions. It contains a summary of works on Earth rotation, including evaluation and prediction of its parameters and analysis of the related excitation data as well as research on associated geodynamic phenomena such as geocentre motion, global sea level change and hydrological processes. The second part of the paper deals with monitoring of geodynamic phenomena. It contains analysis of geodynamic networks of local, and regional scale using space (GNSS and SLR) techniques, Earth tides monitoring with gravimeters and water-tube hydrostatic clinometer, and the determination of secular variation of the Earth' magnetic field.

1526. Data-based estimates of the ocean carbon sink variability - first results of the Surface Ocean pCO₂ Mapping intercomparison (SOCOM)

Authors: C. Rödenbeck; D. C. E. Bakker; N. Gruber; Y. Iida; A. R. Jacobson; S. Jones; P. Landschützer; N. Metz; S. Nakaoka; A. Olsen; G.-H. Park; P. Peylin; K. B. Rodgers; T. P. Sasse; U. Schuster; J. D. Shutler; V. Valsala; R. Wanninkhof; J. Zeng

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-12-7251-2015

URL: <https://doi.org/10.5194/bg-12-7251-2015>

Publication Date: 2015-12-11

Volume/Issue: Volume 12, Issue 23

References: 68

Cited by: 174 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Using measurements of the surface-ocean CO₂ partial pressure (pCO₂) and 14 different pCO₂ mapping methods recently collated by the Surface Ocean pCO₂ Mapping intercomparison (SOCOM) initiative, variations in regional and global sea-air CO₂ fluxes are investigated. Though the available mapping methods use widely different approaches, we find relatively consistent estimates of regional pCO₂ seasonality, in line with previous estimates. In terms of interannual variability (IAV), all mapping methods estimate the largest variations to occur in the eastern equatorial Pacific. Despite considerable spread in the detailed variations, mapping methods that fit the data more closely also tend to agree more closely with each other in regional averages. Encouragingly, this includes mapping methods belonging to complementary types - taking variability either directly from the pCO₂ data or indirectly from driver data via regression. From a weighted ensemble average, we find an IAV amplitude of the global sea-air CO₂ flux of 0.31 PgC yr⁻¹ (standard deviation over 1992-2009), which is larger than simulated by biogeochemical process models. From a decadal perspective, the global ocean CO₂ uptake is estimated to have gradually increased since about 2000, with little decadal change prior to that. The weighted mean net global ocean CO₂ sink estimated by the SOCOM ensemble is -1.75 PgC yr⁻¹ (1992-2009), consistent within uncertainties with estimates from ocean-interior carbon data or atmospheric oxygen trends.

1527. North Pacific Subtropical Mode Water Volume Decrease in 2006-09 Estimated from Argo Observations: Influence of Surface Formation and Basin-Scale Oceanic Variability

Authors: Ivana Cerovečki (Scripps Institution of Oceanography / University of California San Diego);
Donata Giglio (University of Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0179.1

URL: <https://doi.org/10.1175/jcli-d-15-0179.1>

Publication Date: 2015-12-10

Volume/Issue: Volume 29, Issue 6

References: 39

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: Variability, Atm/ocean structure/ phenomena, Diabatic heating, Water masses, Interannual variability (and 3 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Analysis of Argo temperature and salinity profiles (gridded at 0.5 degrees x 0.5 degrees resolution for 2005-12) shows a strong North Pacific Subtropical Mode Water (NPSTMW) volume and density decrease during 2006-09. In this time period, upper-ocean temperature, stratification, and potential vorticity (PV) all increased within the region in and around the NPSTMW low-PV pool, contributing to the NPSTMW volume decrease in two ways: (i) the volume of water satisfying the low-PV constraint that is part of the "mode water" definition decreased, and (ii) some water that was initially in the NPSTMW density range $\sigma_\theta = 25.0\text{--}25.5 \text{ kg m}^{-3}$ was transformed into lighter water. Both changes in density and in PV in the NPSTMW region were a manifestation of basinwide changes. A positive PV anomaly started to propagate westward from the central Pacific in 2005, followed by a negative density anomaly in 2007, which caused a dramatic NPSTMW volume and density decrease. A Walin estimate of surface formation in the NPSTMW density range accounted better (although not entirely) for the interannual variability of the volume of water in the NPSTMW density range without imposing the $PV < 2 \times 10^{-10} \text{ m}^{-1} \text{ s}^{-1}$ constraint than did the same estimate with the PV constraint imposed. This underlines the importance of the PV constraint in identifying the mode water. The mode water evolution cannot be fully described from a density budget alone; rather, the PV budget must be considered simultaneously.

1528. Assessment of conservative force models from GRACE accelerometers and precise orbit determination

Authors: Andrés Calabia; Shuanggen Jin

Journal: Aerospace Science and Technology

Publisher: Elsevier BV

DOI: 10.1016/j.ast.2015.11.034

URL: <https://doi.org/10.1016/j.ast.2015.11.034>

Publication Date: 2015-12-02

Volume/Issue: Volume 49

References: 47

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: MOST, National Natural Science Foundation of China, Shanghai Science and Technology Commission Project

License: N/A

Open Access: N/A

Abstract: N/A

1529. GODAE inter-comparisons in the Tasman and Coral Seas

Authors: P R Oke (Commonwealth Scientific and Industrial Research Organisation (CSIRO)); G B Brassington (CAWCR Bureau of Meteorology (BoM), Melbourne Australia); J Cummings (Naval Research Laboratory (NRL), Monterey USA); M Martin (Met Office); F Hernandez (Mercator Ocean, Ramonville France)

Journal: Journal of Operational Oceanography

Publisher: Informa UK Limited

DOI: 10.1080/1755876x.2012.11020135

URL: <https://doi.org/10.1080/1755876x.2012.11020135>

Publication Date: 2015-11-29

Volume/Issue: Volume 5, Issue 2

References: 40

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1530. Crustal vertical deformation response to different spatial scales of GRACE and GCMs surface loading

Authors: Haoming Yan; Wu Chen; Linguo Yuan

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggv385

URL: <https://doi.org/10.1093/gji/ggv385>

Publication Date: 2015-11-26

Volume/Issue: Volume 204, Issue 1

References: 49

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1531. Decadal trends of ocean and land carbon fluxes from a regional joint ocean-atmosphere inversion

Authors: K. Steinkamp (Now at National Institute of Water and Atmospheric Research Wellington New Zealand, ETH Zurich); N. Gruber (ETH Zurich)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gb004907

URL: <https://doi.org/10.1002/2014gb004907>

Publication Date: 2015-11-25

Volume/Issue: Volume 29, Issue 12

References: 78

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Institute of Water and Atmospheric Research

License: N/A

Open Access: N/A

Abstract: From 1980 until 2010, the combined CO₂ sink strengths of ocean and land increased by nearly 50% (-0.55 Pg C yr⁻¹ decade⁻¹), but the spatial distribution of this trend is not well known. We address this by performing a joint cyclostationary ocean-atmosphere inversion for the three decades 1980-1989, 1990-1999, and 2000-2008, using only carbon data from the ocean and atmosphere as constraints, i.e., without applying any prior information about the land fluxes. We find that in the inversion, most of the 30 year sink trend stems from the ocean (-0.44 Pg C yr⁻¹ decade⁻¹). The contribution of the terrestrial biosphere is commensurably smaller but has more decadal variability. First, the land sink strength intensified in the 1990s by 0.4 (+/-0.3) Pg C yr⁻¹ compared to the 1980s but then weakened slightly by 0.2 (+/-0.4) Pg C yr⁻¹ in the 2000s. The different land regions contributed very variedly to these global trends. While the northern extratropical land acted as an increasing carbon sink throughout the examined period primarily driven by boreal regions, the tropical land is estimated to have acted as an increasing source of CO₂, with source magnitude and trend dominated by enhanced release in tropical America during the Amazon mean wet season. This pattern is largely unchanged if the oceanic inversion constraint, which is based on a stationary ocean circulation, is replaced by an estimate based on simulation results from an ocean biogeochemical general circulation model that includes year-to-year variability in the air-sea CO₂ fluxes and also has a trend (-0.07 Pg C yr⁻¹ decade⁻¹) that is at the very low end of current estimates. However, the land/ocean partitioning of the trend contribution is adjusted accordingly. Oceanic carbon data has a major impact on carbon exchange for all tropical regions and southern Africa but also for observationally better constrained regions in North America and temperate Asia. The European trend exhibits a strong sensitivity to the choice of the atmospheric CO₂ network.

1532. Short-term sea ice forecasting: An assessment of ice concentration and ice drift forecasts using the U.S. Navy's Arctic Cap Nowcast/Forecast System

Authors: David A. Hebert (Naval Research Laboratory Stennis Space Center Mississippi USA); Richard A. Allard (Naval Research Laboratory Stennis Space Center Mississippi USA); E. Joseph Metzger (Naval Research Laboratory Stennis Space Center Mississippi USA); Pamela G. Posey (Naval Research Laboratory Stennis Space Center Mississippi USA); Ruth H. Preller (Naval Research Laboratory Stennis Space Center Mississippi USA); Alan J. Wallcraft (Naval Research Laboratory Stennis Space Center Mississippi USA); Michael W. Phelps (Jacobs Technology Inc., Stennis Space Center Mississippi USA); Ole Martin Smedstad (Vencore Services and Solutions, Inc. Stennis Space Center Mississippi USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011283

URL: <https://doi.org/10.1002/2015jc011283>

Publication Date: 2015-11-23

Volume/Issue: Volume 120, Issue 12

References: 75

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In this study the forecast skill of the U.S. Navy operational Arctic sea ice forecast system, the Arctic Cap Nowcast/Forecast System (ACNFS), is presented for the period February 2014 to June 2015. ACNFS is designed to provide short term, 1-7 day forecasts of Arctic sea ice and ocean conditions. Many quantities are forecast by ACNFS; the most commonly used include ice concentration, ice thickness, ice velocity, sea surface temperature, sea surface salinity, and sea surface velocities. Ice concentration forecast skill is compared to a persistent ice state and historical sea ice climatology. Skill scores are focused on areas where ice concentration changes by +/-5% or more, and are therefore limited to primarily the marginal ice zone. We demonstrate that ACNFS forecasts are skilful compared to assuming a persistent ice state, especially beyond 24 h. ACNFS is also shown to be particularly skilful compared to a climatologic state for forecasts up to 102 h. Modeled ice drift velocity is compared to observed buoy data from the International Arctic Buoy Programme. A seasonal bias is shown where ACNFS is slower than IABP velocity in the summer months and faster in the winter months. In February 2015, ACNFS began to assimilate a blended ice concentration derived from Advanced Microwave Scanning Radiometer 2 (AMSR2) and the Interactive Multisensor Snow and Ice Mapping System (IMS). Preliminary results show that assimilating AMSR2 blended with IMS improves the short-term forecast skill and ice edge location compared to the independently derived National Ice Center Ice Edge product.

1533. A Direct Determination of the World Ocean Barotropic Circulation

Authors: Alain Colin de Verdière (Laboratoire de Physique des Océans, Université de Bretagne Occidentale, Brest, France); Michel Ollitrault (Laboratoire de Physique des Océans, IFREMER, Centre de Brest, Plouzané, France)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-15-0046.1

URL: <https://doi.org/10.1175/jpo-d-15-0046.1>

Publication Date: 2015-11-13

Volume/Issue: Volume 46, Issue 1

References: 110

Cited by: 40 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The time-mean Argo float displacements and the World Ocean Atlas 2009 temperature-salinity climatology are used to obtain the total, top to bottom, mass transports. Outside of an equatorial band, the total transports are the sum of the vertical integrals of geostrophic- and wind-driven Ekman currents. However, these transports are generally divergent, and to obtain a mass conserving circulation, a Poisson equation is solved for the streamfunction with Dirichlet boundary conditions at solid boundaries. The value of the streamfunction on islands is also part of the unknowns. This study presents and discusses an energetic circulation in three basins: the North Atlantic, the North Pacific, and the Southern Ocean. This global method leads to new estimations of the time-mean western Eulerian boundary current transports maxima of 97 Sverdrups (Sv; $1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) at 60 degreesW for the Gulf Stream, 84 Sv at 157 degreesE for the Kuroshio, 80 Sv for the Agulhas Current between 32 degrees and 36 degreesS, and finally 175 Sv for the Antarctic Circumpolar Current at Drake Passage. Although the large-scale structure and boundary of the interior gyres is well predicted by the Sverdrup relation, the transports derived from the wind stress curl are lower than the observed transports in the interior by roughly a factor of 2, suggesting an important contribution of the bottom torques. With additional Argo displacement data, the errors caused by the presence of remaining transient terms at the 1000-db reference level will continue to decrease, allowing this method to produce increasingly accurate results in the future.

1534. Elliptic polarisation of the polar motion excitation

Authors: Christian Bizouard

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-015-0864-7

URL: <https://doi.org/10.1007/s00190-015-0864-7>

Publication Date: 2015-11-12

Volume/Issue: Volume 90, Issue 2

References: 17

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: Polar motion

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1535. Sensitivity of contemporary sea level trends in a global ocean state estimate to effects of geothermal fluxes

Authors: Christopher G. Piecuch; Patrick Heimbach; Rui M. Ponte; Gaël Forget

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2015.10.008

URL: <https://doi.org/10.1016/j.ocemod.2015.10.008>

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Volume/Issue: Volume 96

References: 49

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1536. Characterizing seawater oxygen isotopic variability in a regional ocean modeling framework: Implications for coral proxy records

Authors: S. Stevenson (Department of Oceanography University of Hawaii Honolulu Hawaii USA, National Center for Atmospheric Research); B. S. Powell (Department of Oceanography University of Hawaii Honolulu Hawaii USA); M. A. Merrifield (Department of Oceanography University of Hawaii Honolulu Hawaii USA); K. M. Cobb (Georgia Institute of Technology); J. Nusbaumer (University of Colorado Boulder); D. Noone (Oregon State University)

Journal: Paleoceanography

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015pa002824

URL: <https://doi.org/10.1002/2015pa002824>

Publication Date: 2015-11-06

Volume/Issue: Volume 30, Issue 11

References: 78

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF Ocean Sciences, NSF Paleoclimate, Climate and Large-scale Dynamics

License: N/A

Open Access: N/A

Abstract: Reconstructions of the El Niño-Southern Oscillation (ENSO) are often created using the oxygen isotopic ratio in tropical coral skeletons ($\delta^{18}\text{O}$). However, coral $\delta^{18}\text{O}$ can be difficult to interpret quantitatively, as it reflects changes in both temperature and the $\delta^{18}\text{O}$ value of seawater. Small-scale (10-100 km) processes affecting local temperature and seawater $\delta^{18}\text{O}$ are also poorly quantified and contribute an unknown amount to intercoral $\delta^{18}\text{O}$ offsets. A new version of the Regional Ocean Modeling System capable of directly simulating seawater $\delta^{18}\text{O}$ (isoROMS) is therefore presented to address these issues. The model is used to simulate $\delta^{18}\text{O}$ variations over the 1979-2009 period throughout the Pacific at coarse ($O(50\text{ km})$) resolution, in addition to 10 km downscaling experiments covering the central equatorial Pacific Line Islands, a preferred site for paleo-ENSO reconstruction from corals. A major impact of downscaling at the Line Islands is the ability to resolve fronts associated with tropical instability waves (TIWs), which generate large excursions in both temperature and seawater $\delta^{18}\text{O}$ at Palmyra Atoll (5.9 degreesN, 162.1 degreesW). TIW-related sea surface temperature gradients are smaller at neighboring Christmas Island (1.9 degreesN, 157.5 degreesW), but the interaction of mesoscale features with the steep island topography nonetheless generates cross-island temperature differences of up to 1 degreesC. These nonlinear processes alter the slope of the salinity:seawater $\delta^{18}\text{O}$ relationship at Palmyra and Christmas, as well as affect the relation between coral $\delta^{18}\text{O}$ and indices of ENSO variability. Consideration of the full physical oceanographic context of reef environments is therefore crucial for improving $\delta^{18}\text{O}$ -based ENSO reconstructions.

1537. The Spatiotemporal Structure of Diabatic Processes Governing the Evolution of Subantarctic Mode Water in the Southern Ocean

Authors: Ivana Cerovečki (Scripps Institution of Oceanography); Matthew R. Mazloff (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0243.1

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References: 57

Cited by: 28 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A coupled ice-ocean eddy-permitting Southern Ocean State Estimate (SOSE) for 2008-10 is used to describe and quantify the processes forming and destroying water in the Subantarctic Mode Water (SAMW) density range ($\sigma_\theta = 26.7\text{--}27.2 \text{ kg m}^{-3}$). All the terms in the temperature and salinity equations have been diagnosed to obtain a three-dimensional and time-varying volume budget for individual isopycnal layers. This study finds that air-sea buoyancy fluxes, diapycnal mixing, advection, and storage are all important to the SAMW volume budget. The formation and destruction of water in the SAMW density range occurs over a large latitude range because of the seasonal migration of the outcrop window. The strongest formation is by wintertime surface ocean heat loss occurring equatorward of the Subantarctic Front. Spring and summertime formation occur in the polar gyres through the freshening of water with $\sigma_\theta > 27.2 \text{ kg m}^{-3}$, with an important contribution from sea ice melt. Further buoyancy gain by heating is accomplished only after these waters have already been transformed into the SAMW density range. The spatially integrated and time-averaged SAMW formation rate in the ocean surface layer is 7.9 Sverdrups (Sv; $1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) by air-sea buoyancy fluxes and 8.8 Sv by diapycnal mixing, and it is balanced by advective export into the interior ocean. Maps show that these average rates are the result of highly variable processes with strong cancellation in both space and time, revealing the complexity of water mass transformation in the three-dimensional Southern Ocean overturning circulation.

1538. Weakening of subduction in the Subtropical Mode Water formation region observed during 2003-2013

Authors: Ran Wang (Chinese Academy of Sciences); Fei Yu (Chinese Academy of Sciences); Feng Nan (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc010967

URL: <https://doi.org/10.1002/2015jc010967>

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Volume/Issue: Volume 120, Issue 11

References: 36

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Subduction plays an important role in the formation of the Subtropical Mode Water (STMW). Variation of subduction in the western North Pacific was studied using gridded monthly data from Argo float profiles. The results revealed that there exists a weakening trend of subduction in the STMW formation region (28 degreesN-35 degreesN, 142 degreesE-175 degreesW) due to decreasing winter mixed layer depth (MLD) during 2003-2013. In the STMW formation region, the mean subduction rate was about 64 m yr⁻¹ and showed a decreasing trend at -3.44 +/- 2.47 m yr⁻² during 2003-2013. Meanwhile, the late winter (March) MLD showed a decreasing trend at -4.02 +/- 2.41 m yr⁻¹. Associated with the weakening subduction, the STMW volume had a similar decreasing trend in late winter. Diagnostic calculation indicated that change of the mixed layer temperature (MLT) is the key factor in determining the MLD variations in the STMW formation region. It is demonstrated that the increasing MLT tends to decrease oceanic density and stabilize the upper ocean. This oceanic processes act to weaken the vertical mixing and decrease the MLD, resulting in the weakening of subduction.

1539. Decadal Patterns of Westerly Winds, Temperatures, Ocean Gyre Circulations and Fish Abundance: A Review

Authors: Candace Oviatt (University of Rhode Island); Leslie Smith (Your Ocean Consulting, LLC, 9924 Rainbow Dr., Knoxville, TN 37921, USA); M. McManus (RPS ASA, 55 Village Square Drive, South Kingston, RI 02879, USA, University of Rhode Island); Kimberly Hyde (National Oceanic and Atmospheric Administration, National Marine Fisheries Service, Northeast Fisheries Science Center, Narragansett Laboratory, 28 Tarzwell Drive, Narragansett, RI 02882, USA)

Journal: Climate

Publisher: MDPI AG

DOI: 10.3390/cli3040833

URL: <https://doi.org/10.3390/cli3040833>

Publication Date: 2015-10-20

Volume/Issue: Volume 3, Issue 4

References: 107

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The purpose of this review is to describe the global scope of the multidecadal climate oscillations that go back at least, through several hundred years. Literature, historic data, satellite data and global circulation model output have been used to provide evidence for the zonal and meridional jet stream patterns. These patterns were predominantly zonal from the 1970s to 1990s and switched since the 1990s to a meridional wind phase, with weakening jet streams forming Rossby waves in the northern and southern hemispheres. A weakened northern jet stream has allowed northerly winds to flow down over the continents in the northern hemisphere during the winter period, causing some harsh winters and slowing anthropogenic climate warming regionally. Wind oscillations impact ocean gyre circulation affecting upwelling strength and pelagic fish abundance with synchronous behavior in sub Arctic gyres during phases of the oscillation and asynchronous behavior in subtropical gyres between the Atlantic and Pacific oceans.

1540. On the observability of turbulent transport rates by Argo: supporting evidence from an inversion experiment

Authors: G. Forget; D. Ferreira; X. Liang

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-11-839-2015

URL: <https://doi.org/10.5194/os-11-839-2015>

Publication Date: 2015-10-19

Volume/Issue: Volume 11, Issue 5

References: 74

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Oceanic and Atmospheric Administration

License: Creative Commons

Open Access: N/A

Abstract: . Although estimation of turbulent transport parameters using inverse methods is not new, there is little evaluation of the method in the literature. Here, it is shown that extended observation of the broad-scale hydrography by Argo provides a path to improved estimates of regional turbulent transport rates. Results from a 20-year ocean state estimate produced with the ECCO v4 (Estimating the Circulation and Climate of the Ocean, version 4) non-linear inverse modeling framework provide supporting evidence. Turbulent transport parameter maps are estimated under the constraints of fitting the extensive collection of Argo profiles collected through 2011. The adjusted parameters dramatically reduce misfits to in situ profiles as compared with earlier ECCO solutions. They also yield a clear reduction in the model drift away from observations over multi-century-long simulations, both for assimilated variables (temperature and salinity) and independent variables (biogeochemical tracers). Despite the minimal constraints imposed specifically on the estimated parameters, their geography is physically plausible and exhibits close connections with the upper-ocean stratification as observed by Argo. The estimated parameter adjustments furthermore have first-order impacts on upper-ocean stratification and mixed layer depths over 20 years. These results identify the constraint of fitting Argo profiles as an effective observational basis for regional turbulent transport rate inversions. Uncertainties and further improvements of the method are discussed.

1541. Ocean Data Assimilation in Support of Climate Applications: Status and Perspectives

Authors: D. Stammer (University of Hamburg); M. Balmaseda (European Centre for Medium-Range Weather Forecasts (ECMWF)); P. Heimbach (Massachusetts Institute of Technology, University of Texas at Austin); A. Köhl (University of Hamburg); A. Weaver (Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS))

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

DOI: 10.1146/annurev-marine-122414-034113

URL: <https://doi.org/10.1146/annurev-marine-122414-034113>

Publication Date: 2015-10-16

Volume/Issue: Volume 8, Issue 1

References: 144

Cited by: 76 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean data assimilation brings together observations with known dynamics encapsulated in a circulation model to describe the time-varying ocean circulation. Its applications are manifold, ranging from marine and ecosystem forecasting to climate prediction and studies of the carbon cycle. Here, we address only climate applications, which range from improving our understanding of ocean circulation to estimating initial or boundary conditions and model parameters for ocean and climate forecasts. Because of differences in underlying methodologies, data assimilation products must be used judiciously and selected according to the specific purpose, as not all related inferences would be equally reliable. Further advances are expected from improved models and methods for estimating and representing error information in data assimilation systems. Ultimately, data assimilation into coupled climate system components is needed to support ocean and climate services. However, maintaining the infrastructure and expertise for sustained data assimilation remains challenging.

1542. Arctic pathways of Pacific Water: Arctic Ocean Model Intercomparison experiments

Authors: Yevgeny Aksenov (National Oceanography Centre); Michael Karcher (Alfred Wegener Institute); Andrey Proshutinsky (Woods Hole Oceanographic Institution); Rüdiger Gerdes (Alfred Wegener Institute); Beverly de Cuevas (National Oceanography Centre); Elena Golubeva (Novosibirsk State University, Russian Academy of Sciences); Frank Kauker (Alfred Wegener Institute); An T. Nguyen (Massachusetts Institute of Technology); Gennady A. Platov (Novosibirsk State University, Russian Academy of Sciences); Martin Wadley (University of East Anglia); Eiji Watanabe (Japan Agency for Marine-Earth Science and Technology Kanagawa Japan); Andrew C. Coward (National Oceanography Centre); A. J. George Nurser (National Oceanography Centre)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011299

URL: <https://doi.org/10.1002/2015jc011299>

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Cited by: 81 publications

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Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, NASA Advanced Supercomputing (NAS) Division, JPL Supercomputing and Visualization Facility (and 3 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Pacific Water (PW) enters the Arctic Ocean through Bering Strait and brings in heat, fresh water, and nutrients from the northern Bering Sea. The circulation of PW in the central Arctic Ocean is only partially understood due to the lack of observations. In this paper, pathways of PW are investigated using simulations with six state-of-the art regional and global Ocean General Circulation Models (OGCMs). In the simulations, PW is tracked by a passive tracer, released in Bering Strait. Simulated PW spreads from the Bering Strait region in three major branches. One of them starts in the Barrow Canyon, bringing PW along the continental slope of Alaska into the Canadian Straits and then into Baffin Bay. The second begins in the vicinity of the Herald Canyon and transports PW along the continental slope of the East Siberian Sea into the Transpolar Drift, and then through Fram Strait and the Greenland Sea. The third branch begins near the Herald Shoal and the central Chukchi shelf and brings PW into the Beaufort Gyre. In the models, the wind, acting via Ekman pumping, drives the seasonal and interannual variability of PW in the Canadian Basin of the Arctic Ocean. The wind affects the simulated PW pathways by changing the vertical shear of the relative vorticity of the ocean flow in the Canada Basin.

1543. Sea-surface temperature and salinity product comparison against external in situ data in the Arctic Ocean

Authors: J. N. Stroh (University of Alaska Fairbanks); Gleb Panteleev (University of Alaska Fairbanks, Institute of Natural Resources, National Research Tomsk, Polytechnic University Tomsk Russia); Sergey Kirillov (University of Manitoba); Mikhail Makhotin (Arctic and Antarctic Research Institute Saint Petersburg Russia); Natalia Shakhova (University of Alaska Fairbanks, Institute of Natural Resources, National Research Tomsk, Polytechnic University Tomsk Russia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc011005

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References: 33

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Language: en

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License: N/A

Open Access: N/A

Abstract: Sea-surface temperature and salinity (SST/S) in the Arctic Ocean (AO) are largely governed by sea-ice and continental runoff rather than evaporation and precipitation as in lower latitude oceans, and global satellite analyses and models which incorporate remotely observed SST/S may be inaccurate in the AO due to lack of direct measurements for calibrating satellite data. For this reason, we are motivated to validate several satellite sea-surface temperature (SST) data products and SST/S models by comparing gridded data in the AO with oceanographic records from 2006 to 2013. Statistical analysis of product-minus-observation differences reveals that the satellite SST products considered have a temperature bias magnitude of less than 0.5C compared to ship-based CTD measurements, and most of these biases are negative in sign. SST/S models also show an overall negative temperature bias, but no common sign or magnitude of salinity bias against CTD data. Ice tethered profiler (ITP) near-surface data span the seasons of several years, and these measurements reflect a sea-ice dominated region where the ocean surface cannot be remotely observed. Against this data, many of the considered models and products show large errors with detectable seasonal differences in SST bias. Possible sources of these errors are discussed, and two adjustments of product SST on the basis of sea-ice concentration are suggested for reducing bias to within less than 0.01C of ITP near-surface temperatures.

1544. Status and future of global and regional ocean prediction systems

Authors: Marina Tonani (Istituto Nazionale di Geofisica e Vulcanologia, Bologna, Italy); Magdalena Balmaseda (European Centre for Medium Range Weather Forecasts, Shinfield Park, Reading, UK); Laurent Bertino (Nansen Environmental and Remote Sensing Center); Ed Blockley (Met Office); Gary Brassington (Bureau of Meteorology, Melbourne, Victoria, Australia); Fraser Davidson (Northwest Atlantic Fisheries Centre, Fisheries and Oceans Canada, Canada); Yann Drillet (Mercator-Ocean, Toulouse, France); Pat Hogan (NRL, Stennis Space Center, USA); Tsurane Kuragano (Meteorological Research Institute); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Avichal Mehra (NOAA/National Centers for Environmental Prediction, College Park, Maryland, USA); Francis Paranathara (Indian National Centre for Ocean Information Services, India); Clemente A.S. Tanajura (Physics Institute, Federal University of Bahia (UFBA), Salvador, Brazil); Hui Wang (National Marine Environmental Forecasting Center)

Journal: Journal of Operational Oceanography

Publisher: Informa UK Limited

DOI: 10.1080/1755876x.2015.1049892

URL: <https://doi.org/10.1080/1755876x.2015.1049892>

Publication Date: 2015-10-14

Volume/Issue: Volume 8, Issue sup2

References: 47

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1545. Precession-driven changes in Iceland-Scotland Overflow Water penetration and bottom water circulation on Gardar Drift since ~ 200 ka

Authors: A.C. Elmore; J.D. Wright; T.B. Chalk

Journal: Palaeogeography, Palaeoclimatology, Palaeoecology

Publisher: Elsevier BV

DOI: 10.1016/j.palaeo.2015.09.042

URL: <https://doi.org/10.1016/j.palaeo.2015.09.042>

Publication Date: 2015-10-09

Volume/Issue: Volume 440

References: 107

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1546. ECCO version 4: an integrated framework for non-linear inverse modeling and global ocean state estimation

Authors: G. Forget; J.-M. Campin; P. Heimbach; C. N. Hill; R. M. Ponte; C. Wunsch

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-8-3071-2015

URL: <https://doi.org/10.5194/gmd-8-3071-2015>

Publication Date: 2015-10-06

Volume/Issue: Volume 8, Issue 10

References: 144

Cited by: 452 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Oceanic and Atmospheric Administration

License: Creative Commons

Open Access: N/A

Abstract: . This paper presents the ECCO v4 non-linear inverse modeling framework and its baseline solution for the evolving ocean state over the period 1992-2011. Both components are publicly available and subjected to regular, automated regression tests. The modeling framework includes sets of global conformal grids, a global model setup, implementations of data constraints and control parameters, an interface to algorithmic differentiation, as well as a grid-independent, fully capable Matlab toolbox. The baseline ECCO v4 solution is a dynamically consistent ocean state estimate without unidentified sources of heat and buoyancy, which any interested user will be able to reproduce accurately. The solution is an acceptable fit to most data and has been found to be physically plausible in many respects, as documented here and in related publications. Users are being provided with capabilities to assess model-data misfits for themselves. The synergy between modeling and data synthesis is asserted through the joint presentation of the modeling framework and the state estimate. In particular, the inverse estimate of parameterized physics was instrumental in improving the fit to the observed hydrography, and becomes an integral part of the ocean model setup available for general use. More generally, a first assessment of the relative importance of external, parametric and structural model errors is presented. Parametric and external model uncertainties appear to be of comparable importance and dominate over structural model uncertainty. The results generally underline the importance of including turbulent transport parameters in the inverse problem.

1547. Atmosphere-ocean coupled processes in the Madden-Julian oscillation

Authors: Charlotte A. DeMott (Colorado State University); Nicholas P. Klingaman (University of Reading); Steven J. Woolnough (University of Reading)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014rg000478

URL: <https://doi.org/10.1002/2014rg000478>

Publication Date: 2015-10-06

Volume/Issue: Volume 53, Issue 4

References: 390

Cited by: 195 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Colorado State University, Natural Environment Research Council

License: Creative Commons

Open Access: N/A

Abstract: The Madden-Julian oscillation (MJO) is a convectively coupled 30-70 day (intraseasonal) tropical atmospheric mode that drives variations in global weather but which is poorly simulated in most atmospheric general circulation models. Over the past two decades, field campaigns and modeling experiments have suggested that tropical atmosphere-ocean interactions may sustain or amplify the pattern of enhanced and suppressed atmospheric convection that defines the MJO and encourage its eastward propagation through the Indian and Pacific Oceans. New observations collected during the past decade have advanced our understanding of the ocean response to atmospheric MJO forcing and the resulting intraseasonal sea surface temperature fluctuations. Numerous modeling studies have revealed a considerable impact of the mean state on MJO ocean-atmosphere coupled processes, as well as the importance of resolving the diurnal cycle of atmosphere-upper ocean interactions. New diagnostic methods provide insight to atmospheric variability and physical processes associated with the MJO but offer limited insight on the role of ocean feedbacks. Consequently, uncertainty remains concerning the role of the ocean in MJO theory. Our understanding of how atmosphere-ocean coupled processes affect the MJO can be improved by collecting observations in poorly sampled regions of MJO activity, assessing oceanic and atmospheric drivers of surface fluxes, improving the representation of upper ocean mixing in coupled model simulations, designing model experiments that minimize mean state differences, and developing diagnostic tools to evaluate the nature and role of coupled ocean-atmosphere processes over the MJO cycle.

1548. Influence of assimilating transports and in situ data from the Rapid-MOCHA array into the GECCO2 ocean synthesis

Authors: Armin Köhl

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2015.09.001

URL: <https://doi.org/10.1016/j.dynatmoce.2015.09.001>

Publication Date: 2015-10-01

Volume/Issue: Volume 72

References: 34

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission, Natural Environment Research Council

License: N/A

Open Access: N/A

Abstract: N/A

1549. Seasonality of tropical Pacific decadal trends associated with the 21st century global warming hiatus

Authors: Dillon J. Amaya (Scripps Institution of Oceanography / University of California San Diego); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego); Arthur J. Miller (Scripps Institution of Oceanography / University of California San Diego); Michael J. McPhaden (NOAA/Pacific Marine Environmental Laboratory Seattle Washington USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc010906

URL: <https://doi.org/10.1002/2015jc010906>

Publication Date: 2015-10-01

Volume/Issue: Volume 120, Issue 10

References: 47

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NOAA's National Climate Data Center, the NSF

License: N/A

Open Access: N/A

Abstract: Equatorial Pacific changes during the transition from a nonhiatus period (pre-1999) to the present global warming hiatus period (post-1999) are identified using a combination of reanalysis and observed data sets. Results show increased surface wind forcing has excited significant changes in wind-driven circulation. Over the last two decades, the core of the Equatorial Undercurrent intensified at a rate of 6.9 cm s⁻¹ decade⁻¹. Similarly, equatorial upwelling associated with the shallow meridional overturning circulation increased at a rate of 2.0 × 10⁻⁴ cm s⁻¹ decade⁻¹ in the central Pacific. Further, a seasonal dependence is identified in the sea surface temperature trends and in subsurface dynamics. Seasonal variations are evident in reversals of equatorial surface flow trends, changes in subsurface circulation, and seasonal deepening/shoaling of the thermocline. Anomalous westward surface flow drives cold-water zonal advection from November to February, leading to surface cooling from December through May. Conversely, eastward surface current anomalies in June-July drive warm-water zonal advection producing surface warming from July to November. An improved dynamical understanding of how the tropical Pacific Ocean responds during transitions into hiatus events, including its seasonal structure, may help to improve future predictability of decadal climate variations.

1550. Can organic matter flux profiles be diagnosed using remineralisation rates derived from observed tracers and modelled ocean transport rates?

Authors: J. D. Wilson; A. Ridgwell; S. Barker

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-12-5547-2015

URL: <https://doi.org/10.5194/bg-12-5547-2015>

Publication Date: 2015-09-30

Volume/Issue: Volume 12, Issue 18

References: 53

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, European Research Council

License: Creative Commons

Open Access: N/A

Abstract: . The average depth in the ocean at which the majority of sinking organic matter particles remineralise is a fundamental parameter in the ocean's role in regulating atmospheric CO₂. Observed spatial patterns in sinking fluxes and relationships between the fluxes of different particles in the modern ocean have widely been used to invoke controlling mechanisms with important implications for CO₂ regulation. However, such analyses are limited by the sparse spatial sampling of the available sediment trap data. Here we explore whether model ocean circulation rates, in the form of a transport matrix, can be used to derive remineralisation rates and infer sinking particle flux curves from the much more highly resolved observations of dissolved nutrient concentrations. Initially we show an example of the method using a transport matrix from the MITgcm model and demonstrate that there are a number of potential uncertainties associated with the method. We then use the Earth system model GENIE to generate a synthetic tracer data set to explore the method and its sensitivity to key sources of uncertainty arising from errors in the tracer observations and in the model circulation. We use a 54-member ensemble of different, but plausible, estimates of the modern circulation to explore errors associated with model transport rates. We find that reconstructed re-mineralisation rates are very sensitive to both errors in observations and model circulation rates, such that a simple inversion cannot provide a robust estimate of particulate flux profiles. Estimated remineralisation rates are particularly sensitive to differences between the "observed" and modelled circulation because remineralisation rates are 3-4 magnitudes smaller than transport rates. We highlight a potential method of constraining the uncertainty associated with using modelled circulation rates, although its success is limited by the observations currently available. Finally, we show that there are additional uncertainties when inferring particle flux curves from reliable estimates of remineralisation rates due to processes that are not restricted to the vertical water column transport, such as the cycling of dissolved organic matter.

1551. MOPS-1.0: towards a model for the regulation of the global oceanic nitrogen budget by marine biogeochemical processes

Authors: I. Kriest; A. Oschlies

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-8-2929-2015

URL: <https://doi.org/10.5194/gmd-8-2929-2015>

Publication Date: 2015-09-23

Volume/Issue: Volume 8, Issue 9

References: 82

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft

License: Creative Commons

Open Access: N/A

Abstract: . Global models of the oceanic nitrogen cycle are subject to many uncertainties regarding the representation of the relevant biogeochemical processes and of the feedbacks between nitrogen sources and sinks that determine space- and timescales on which the global nitrogen budget is regulated. We investigate these aspects using a global model of ocean biogeochemistry that explicitly considers phosphorus and nitrogen, including pelagic denitrification and nitrogen fixation as sink and source terms of fixed nitrogen, respectively. The model explores different parameterizations of organic matter sinking speed, oxidant affinity of oxic and suboxic remineralization, and regulation of nitrogen fixation by temperature and different stoichiometric ratios. Examination of the initial transient behavior of different model setups initialized from observed biogeochemical tracer distributions reveal changes in simulated nitrogen inventories and fluxes particularly during the first centuries. Millennial timescales have to be resolved in order to bring all biogeochemical and physical processes into a dynamically consistent steady state. Analysis of global properties suggests that not only particularly particle sinking speed but also the parameterization of denitrification determine the extent of oxygen minimum zones, global nitrogen fluxes, and hence the oceanic nitrogen inventory. However, the ways and directions in which different parameterizations of particle sinking, nitrogen fixation, and denitrification affect the global diagnostics are different suggesting that these may, in principle, be constrained independently from each other. Analysis of the model misfit with respect to observed biogeochemical tracer distributions and fluxes suggests a particle flux profile close to the one suggested by Martin et al. (1987). Simulated pelagic denitrification best agrees with the lower values between 59 and 84 Tg N yr⁻¹ recently estimated by other authors.

1552. The Benguela Upwelling System: Quantifying the Sensitivity to Resolution and Coastal Wind Representation in a Global Climate Model*

Authors: R. Justin Small (National Center for Atmospheric Research); Enrique Curchitser (Department of Environmental Sciences, Rutgers, The State University of New Jersey, New Brunswick, New Jersey); Katherine Hedstrom (University of Alaska Fairbanks); Brian Kauffman (National Center for Atmospheric Research); William G. Large (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-15-0192.1

URL: <https://doi.org/10.1175/jcli-d-15-0192.1>

Publication Date: 2015-09-21

Volume/Issue: Volume 28, Issue 23

References: 77

Cited by: 111 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Of all the major coastal upwelling systems in the world's oceans, the Benguela, located off southwest Africa, is the one that climate models find hardest to simulate well. This paper investigates the sensitivity of upwelling processes, and of sea surface temperature (SST), in this region to resolution of the climate model and to the offshore wind structure. The Community Climate System Model (version 4) is used here, together with the Regional Ocean Modeling System. The main result is that a realistic wind stress curl at the eastern boundary, and a high-resolution ocean model, are required to well simulate the Benguela upwelling system. When the wind stress curl is too broad (as with a 1 degrees atmosphere model or coarser), a Sverdrup balance prevails at the eastern boundary, implying southward ocean transport extending as far as 30 degreesS and warm advection. Higher atmosphere resolution, up to 0.5 degrees, does bring the atmospheric jet closer to the coast, but there can be too strong a wind stress curl. The most realistic representation of the upwelling system is found by adjusting the 0.5 degrees atmosphere model wind structure near the coast toward observations, while using an eddy-resolving ocean model. A similar adjustment applied to a 1 degrees ocean model did not show such improvement. Finally, the remote equatorial Atlantic response to restoring SST in a broad region offshore of Benguela is substantial; however, there is not a large response to correcting SST in the narrow coastal upwelling zone alone.

1553. Comparison of the Atlantic meridional overturning circulation between 1960 and 2007 in six ocean reanalysis products

Authors: A. R. Karspeck; D. Stammer; A. Köhl; G. Danabasoglu; M. Balmaseda; D. M. Smith; Y. Fujii; S. Zhang; B. Giese; H. Tsujino; A. Rosati

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-015-2787-7

URL: <https://doi.org/10.1007/s00382-015-2787-7>

Publication Date: 2015-09-19

Volume/Issue: Volume 49, Issue 3

References: 107

Cited by: 81 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1554. Formation and variability of the Lofoten basin vortex in a high-resolution ocean model

Authors: Denis L. Volkov; Arseny A. Kubryakov; Rick Lumpkin

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2015.09.001

URL: <https://doi.org/10.1016/j.dsr.2015.09.001>

Publication Date: 2015-09-14

Volume/Issue: Volume 105

References: 51

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Physical Oceanography program, NOAA Atlantic Oceanographic and Meteorological Laboratory (USA), Saint Petersburg State University (Russia)

License: N/A

Open Access: N/A

Abstract: N/A

1555. An ocean data assimilation system in the Indian Ocean and west Pacific Ocean

Authors: Changxiang Yan; Jiang Zhu; Jiping Xie

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-015-4121-z

URL: <https://doi.org/10.1007/s00376-015-4121-z>

Publication Date: 2015-09-08

Volume/Issue: Volume 32, Issue 11

References: 55

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1556. Multiple Equilibria as a Possible Mechanism for Decadal Variability in the North Atlantic Ocean

Authors: Andreas Born (University of Bern); Juliette Mignot (University of Bern, Institut de Recherche pour le Développement (IRD)); Thomas F. Stocker (University of Bern)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-14-00813.1

URL: <https://doi.org/10.1175/jcli-d-14-00813.1>

Publication Date: 2015-09-08

Volume/Issue: Volume 28, Issue 22

References: 77

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Decadal climate variability in the North Atlantic has received increased attention in recent years, because modeling results suggest predictability of heat content and circulation indices several years ahead. However, determining the applicability of these results in the real world is challenging because of an incomplete understanding of the underlying mechanisms. Here, the authors show that recent attempts to reconstruct the decadal variations in one of the dominant circulation systems of the region, the subpolar gyre (SPG), are not always consistent. A coherent picture is partly recovered by a simple conceptual model solely forced by reanalyzed surface air temperatures. This confirms that surface heat flux indeed plays a leading role for this type of variability, as has been suggested in previous studies. The results further suggest that large variations in the SPG correspond to the crossing of a bifurcation point that is predicted from idealized experiments and an analytical solution of the model used herein. Performance of this conceptual model is tested against a statistical stochastic model. Hysteresis and the existence of two stable modes of the SPG circulation shape its response to forcing by atmospheric temperatures. The identification of the essential dynamics and the reduction to a minimal model of SPG variability provide a quantifiable basis and a framework for future studies on decadal climate variability and predictability.

1557. Global coupled sea ice-ocean state estimation

Authors: Ian Fenty; Dimitris Menemenlis; Hong Zhang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-015-2796-6

URL: <https://doi.org/10.1007/s00382-015-2796-6>

Publication Date: 2015-09-05

Volume/Issue: Volume 49, Issue 3

References: 93

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: Adjoint, Sea ice concentration data, State estimation, 4DVAR, ECCO2 (and 3 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1558. Analysis of systematic differences from GPS-measured and GRACE-modeled deformation in Central Valley, California

Authors: Weijie Tan; Danan Dong; Junping Chen; Bin Wu

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2015.08.034

URL: <https://doi.org/10.1016/j.asr.2015.08.034>

Publication Date: 2015-09-04

Volume/Issue: Volume 57, Issue 1

References: 50

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National High Technology Research and Development Program of China, Shanghai Committee of Science and Technology

License: N/A

Open Access: N/A

Abstract: N/A

1559. Quantifying the resolution level where the GRACE satellites can separate Greenland's glacial mass balance from surface mass balance

Authors: J. A. Bonin; D. P. Chambers

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-9-1761-2015

URL: <https://doi.org/10.5194/tc-9-1761-2015>

Publication Date: 2015-09-04

Volume/Issue: Volume 9, Issue 5

References: 15

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: . Mass change over Greenland can be caused by either changes in the glacial dynamic mass balance (DMB) or the surface mass balance (SMB). The GRACE satellite gravity mission cannot directly separate the two physical causes because it measures the sum of the entire mass column with limited spatial resolution. We demonstrate one theoretical way to indirectly separate cumulative SMB from DMB with GRACE, using a least squares inversion technique with knowledge of the location of the glaciers. However, we find that the limited 60 x 60 spherical harmonic representation of current GRACE data does not provide sufficient resolution to adequately accomplish the task. We determine that at a maximum degree/order of 90 x 90 or above, a noise-free gravity measurement could theoretically separate the SMB from DMB signals. However, current GRACE satellite errors are too large at present to separate the signals. A noise reduction of a factor of 10 at a resolution of 90 x 90 would provide the accuracy needed for the interannual cumulative SMB and DMB to be accurately separated.

1560. Global Ocean Integrals and Means, with Trend Implications

Authors: Carl Wunsch (Harvard University)

Journal: Annual Review of Marine Science

Publisher: Annual Reviews

DOI: 10.1146/annurev-marine-122414-034040

URL: <https://doi.org/10.1146/annurev-marine-122414-034040>

Publication Date: 2015-09-02

Volume/Issue: Volume 8, Issue 1

References: 125

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Understanding the ocean requires determining and explaining global integrals and equivalent average values of temperature (heat), salinity (freshwater and salt content), sea level, energy, and other properties. Attempts to determine means, integrals, and climatologies have been hindered by thinly and poorly distributed historical observations in a system in which both signals and background noise are spatially very inhomogeneous, leading to potentially large temporal bias errors that must be corrected at the 1% level or better. With the exception of the upper ocean in the current altimetric-Argo era, no clear documentation exists on the best methods for estimating means and their changes for quantities such as heat and freshwater at the levels required for anthropogenic signals. Underestimates of trends are as likely as overestimates; for example, recent inferences that multidecadal oceanic heat uptake has been greatly underestimated are plausible. For new or augmented observing systems, calculating the accuracies and precisions of global, multidecadal sampling densities for the full water column is necessary to avoid the irrecoverable loss of scientifically essential information.

1561. An estimate of tidal and non-tidal modulations of plate subduction speed in the transition zone in the Tokai district

Authors: Yoshiyuki Tanaka; Suguru Yabe; Satoshi Ide

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/s40623-015-0311-2

URL: <https://doi.org/10.1186/s40623-015-0311-2>

Publication Date: 2015-09-01

Volume/Issue: Volume 67, Issue 1

References: 61

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1562. Decadal and quadratic variations of Earth's oblateness and polar ice mass balance from 1979 to 2010

Authors: Ki-Weon Seo; Jianli Chen; Clark R. Wilson; Choon-Ki Lee

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggv312

URL: <https://doi.org/10.1093/gji/ggv312>

Publication Date: 2015-08-28

Volume/Issue: Volume 203, Issue 1

References: 32

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1563. A study of reduced-order 4DVAR with a finite element shallow water model

Authors: M. U. Altaf (Division of Environmental Sciences King Abdullah University of Science and Technology Saudi Arabia, Division of Earth Sciences King Abdullah University of Science and Technology Saudi Arabia); M. Ambrozic (Delft University of Technology); M. F. McCabe (Division of Environmental Sciences King Abdullah University of Science and Technology Saudi Arabia); I. Hoteit (Division of Earth Sciences King Abdullah University of Science and Technology Saudi Arabia)

Journal: International Journal for Numerical Methods in Fluids

Publisher: Wiley

DOI: 10.1002/fld.4167

URL: <https://doi.org/10.1002/fld.4167>

Publication Date: 2015-08-25

Volume/Issue: Volume 80, Issue 11

References: 33

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: King Abdullah University of Science and Technology

License: N/A

Open Access: N/A

Abstract: SummaryFour-dimensional variational data assimilation (4DVAR) is frequently used to improve model forecasting skills. This method improves a model consistency with available data by minimizing a cost function measuring the model-data misfit with respect to some model inputs and parameters. Associated with this type of method, however, are difficulties related to the coding of the adjoint model, which is needed to compute the gradient of the 4DVAR cost function. Proper orthogonal decomposition (POD) is a model reduction method that can be used to approximate the gradient calculation in 4DVAR. In this work, two ways of using POD in 4DVAR are presented, namely model-reduced 4DVAR and reduced adjoint 4DVAR (RA-4DVAR). Both techniques employ POD to obtain a reduced-order approximation of the forward linear tangent operator. The difference between the two methods lies in the treatment of the forward model. Model-reduced 4DVAR performs minimization entirely in the POD-reduced space, thereby achieving very low computational costs, but sacrificing accuracy of the end result. On the other hand, the RA-4DVAR uses POD to approximate only the adjoint model. The main contribution of this study is a comparative performance analysis of these 4DVAR methodologies on a nonlinear finite element shallow water model. The sensitivity of the methods to perturbations in observations and the number of observation points is examined. The results from twin experiments suggest that the RA-4DVAR method is easy to implement and computationally efficient and provides a robust approach for achieving reasonable results in the context of variational data assimilation. Copyright © 2015 John Wiley & Sons, Ltd.

1564. The DTU13 MSS (Mean Sea Surface) and MDT (Mean Dynamic Topography) from 20 Years of Satellite Altimetry

Authors: Ole Andersen; Per Knudsen; Lars Stenseng

Journal: International Association of Geodesy Symposia

Publisher: Springer International Publishing

DOI: 10.1007/1345_2015_182

URL: https://doi.org/10.1007/1345_2015_182

Publication Date: 2015-08-12

Volume/Issue: N/A

References: 23

Cited by: 68 publications

Language: en

Subjects: N/A

Keywords: Satellite Altimetry|Sea-surface height|Ocean surface topography

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1565. Enhancement of the southward return flow of the Atlantic Meridional Overturning Circulation by data assimilation and its influence in an assimilative ocean simulation forced by CORE-II atmospheric forcing

Authors: Yosuke Fujii; Hiroyuki Tsujino; Takahiro Toyoda; Hideyuki Nakano

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-015-2780-1

URL: <https://doi.org/10.1007/s00382-015-2780-1>

Publication Date: 2015-08-11

Volume/Issue: Volume 49, Issue 3

References: 66

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Meteorological Agency/ Meteorological Research Institute

License: N/A

Open Access: N/A

Abstract: N/A

1566. A global assessment of accelerations in surface mass transport

Authors: Xiaoping Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael B. Heflin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015gl064941

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References: 19

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration (NASA), NASA's GRACE Science Team and Interdisciplinary Research in Earth Science programs

License: N/A

Open Access: N/A

Abstract: Water mass transport in the Earth's dynamic surface layer of atmosphere, cryosphere, and hydrosphere driven by various global change processes has complex spatiotemporal patterns. Here we determine global patterns and regional mean values of accelerations in surface mass variations during the Gravity Recovery and Climate Experiment (GRACE) mission's data span from 2002.2 to 2015.0. GRACE gravity data are supplemented by surface deformation from 607 Global Navigation Satellite System stations, an ocean bottom pressure model, satellite laser ranging, and loose a priori knowledge on mass variation regimes incorporating high-resolution geographic boundaries. While Greenland and West Antarctica have strong negative accelerations, Alaska and the Arctic Ocean show significant positive accelerations. In addition, the accelerations are not constant in time with some regions showing considerable variability due to irregular interannual changes. No evidence of significant nonsteric mean sea level acceleration has been found, but the uncertainty is quite large.

1567. Predictive Skill for Regional Interannual Steric Sea Level and Mechanisms for Predictability*

Authors: Iuliia Polkova (University of Hamburg); Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-14-00811.1

URL: <https://doi.org/10.1175/jcli-d-14-00811.1>

Publication Date: 2015-07-30

Volume/Issue: Volume 28, Issue 18

References: 60

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Based on decadal hindcasts initialized every five years over the period 1960-2000, the predictive skill of annual-mean regional steric sea level and associated mechanisms are investigated. Predictive skill for steric sea level is found over large areas of the World Ocean, notably over the subtropical Atlantic and Pacific Oceans, along the path of the North Atlantic Current, and over the Indian and Southern Oceans. Mechanisms for the predictability of the thermosteric and halosteric contributions to the steric signal are studied by separating these components into signals originating from processes within and beneath the mixed layer. Contributions originating from below the mixed layer are further decomposed into density-related (isopycnal motion term) and density-compensated (spice term) changes. In regions of the subtropical Pacific and Atlantic Oceans, predictive skill results from the interannual variability associated with the contribution from isopycnal motion to thermosteric sea level. Skill related to thermosteric mixed layer processes is found to be important in the subtropical Atlantic, while the spice contribution shows skill over the subpolar North Atlantic. In the subtropics, the high predictive skill can be rationalized in terms of westward-propagating baroclinic Rossby waves for a lead time of 2-5 yr, as demonstrated using an initialized Rossby wave model. Because of the low Rossby wave speed in high latitudes, this process is not separable from the persistence there.

1568. Capturing optically important constituents and properties in a marine biogeochemical and ecosystem model

Authors: S. Dutkiewicz; A. E. Hickman; O. Jahn; W. W. Gregg; C. B. Mouw; M. J. Follows

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-12-4447-2015

URL: <https://doi.org/10.5194/bg-12-4447-2015>

Publication Date: 2015-07-29

Volume/Issue: Volume 12, Issue 14

References: 130

Cited by: 134 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Natural Environment Research Council

License: Creative Commons

Open Access: N/A

Abstract: . We present a numerical model of the ocean that couples a three-stream radiative transfer component with a marine biogeochemical-ecosystem component in a dynamic three-dimensional physical framework. The radiative transfer component resolves the penetration of spectral irradiance as it is absorbed and scattered within the water column. We explicitly include the effect of several optically important water constituents (different phytoplankton functional types; detrital particles; and coloured dissolved organic matter, CDOM). The model is evaluated against in situ-observed and satellite-derived products. In particular we compare to concurrently measured biogeochemical, ecosystem, and optical data along a meridional transect of the Atlantic Ocean. The simulation captures the patterns and magnitudes of these data, and estimates surface upwelling irradiance analogous to that observed by ocean colour satellite instruments. We find that incorporating the different optically important constituents explicitly and including spectral irradiance was crucial to capture the variability in the depth of the subsurface chlorophyll a (Chl a) maximum. We conduct a series of sensitivity experiments to demonstrate, globally, the relative importance of each of the water constituents, as well as the crucial feedbacks between the light field, the relative fitness of phytoplankton types, and the biogeochemistry of the ocean. CDOM has proportionally more importance at attenuating light at short wavelengths and in more productive waters, phytoplankton absorption is relatively more important at the subsurface Chl a maximum, and water molecules have the greatest contribution when concentrations of other constituents are low, such as in the oligotrophic gyres. Scattering had less effect on attenuation, but since it is important for the amount and type of upwelling irradiance, it is crucial for setting sea surface reflectance. Strikingly, sensitivity experiments in which absorption by any of the optical constituents was increased led to a decrease in the size of the oligotrophic regions of the subtropical gyres: lateral nutrient supplies were enhanced as a result of decreasing high-latitude productivity. This new model that captures bio-optical feedbacks will be important for improving our understanding of the role of light and optical constituents on ocean biogeochemistry, especially in a changing environment. Further, resolving surface upwelling irradiance will make it easier to connect to satellite-derived products in the future.

1569. Quality Evaluation of the Weekly Vertical Loading Effects Induced from Continental Water Storage Models

Authors: Z. Li; T. van Dam; X. Collilieux; Z. Altamimi; P. Rebischung; S. Nahmani

Journal: International Association of Geodesy Symposia

Publisher: Springer International Publishing

DOI: 10.1007/1345_2015_174

URL: https://doi.org/10.1007/1345_2015_174

Publication Date: 2015-07-29

Volume/Issue: N/A

References: 19

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1570. Time-Space SST Variability in the Atlantic during 2013: Seasonal Cycle

Authors: Liyan Liu (I. M. Systems Group, and Environmental Modeling Center, NOAA/NWS/NCEP, College Park, Maryland); Carlos Lozano (Environmental Modeling Center, NOAA/NWS/NCEP, College Park, Maryland); Dan Iredell (I. M. Systems Group, and Environmental Modeling Center, NOAA/NWS/NCEP, College Park, Maryland)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-15-0028.1

URL: <https://doi.org/10.1175/jtech-d-15-0028.1>

Publication Date: 2015-07-28

Volume/Issue: Volume 32, Issue 9

References: 31

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Seasonal cycle, Seasonal variability, Anomalies, Annual variations, Regression analysis (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A 2-yr-long daily gridded field of sea surface temperature (SST) in the Atlantic centered for the year 2013 is projected onto orthogonal components: its mean, six harmonics of the year cycle, the slow-varying contribution, and the fast-varying contribution. The periodic function defined by the year harmonics, referred to here as the seasonal harmonic, contains most of the year variability in 2013. The seasonal harmonic is examined in its spatial and temporal distribution by describing the amplitude and phase of its maxima and minima, and other associated parameters. In the seasonal harmonic, the ratio of the duration of warming period to cooling period ranges from 0.2 to 2.0. There are also differences in the spatial patterns and dominance of the year harmonics--in general associated with regions with different insolation, oceanic, and atmospheric regimes. Empirical orthogonal functions (EOFs) of the seasonal harmonic allow for a succinct description of the seasonal evolution for the Atlantic and its subdomains. The decomposition can be applied to model output, allowing for a more incisive model validation and data assimilation. The decorrelation time scale of the rapidly varying signal is found to be nearly independent of the time of the year once four or more harmonics are used. The decomposition algorithm, here implemented for a single year cycle, can be applied to obtain a multiyear average of the seasonal harmonic.

1571. Using Green's Functions to initialize and adjust a global, eddying ocean biogeochemistry general circulation model

Authors: H. Brix; D. Menemenlis; C. Hill; S. Dutkiewicz; O. Jahn; D. Wang; K. Bowman; H. Zhang

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2015.07.008

URL: <https://doi.org/10.1016/j.ocemod.2015.07.008>

Publication Date: 2015-07-26

Volume/Issue: Volume 95

References: 86

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1572. An Ocean Reanalysis System for the Joining Area of Asia and Indian-Pacific Ocean

Authors: Yan Chang-Xiang; Zhu Jiang; Xie Ji-Ping

Journal: Atmospheric and Oceanic Science Letters

Publisher: Elsevier BV

DOI: 10.1080/16742834.2010.11446848

URL: <https://doi.org/10.1080/16742834.2010.11446848>

Publication Date: 2015-07-23

Volume/Issue: Volume 3, Issue 2

References: 3

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1573. 14C-age tracers in global ocean circulation models

Authors: W. Koeve; H. Wagner; P. Kähler; A. Oschlies

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-8-2079-2015

URL: <https://doi.org/10.5194/gmd-8-2079-2015>

Publication Date: 2015-07-16

Volume/Issue: Volume 8, Issue 7

References: 74

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung, Seventh Framework Programme, Deutsche Forschungsgemeinschaft

License: Creative Commons

Open Access: N/A

Abstract: . The natural abundance of ^{14}C in total CO_2 dissolved in seawater (DIC) is a property applied to evaluate the water age structure and circulation in the ocean and in ocean models. In this study we use three different representations of the global ocean circulation augmented with a suite of idealised tracers to study the potential and limitations of using natural ^{14}C to determine water age, which is the time elapsed since a body of water has been in contact with the atmosphere. We find that, globally, bulk ^{14}C -age is dominated by two equally important components, one associated with ageing, i.e. the time component of circulation, and one associated with a "preformed ^{14}C -age". The latter quantity exists because of the slow and incomplete atmosphere-ocean equilibration of ^{14}C particularly in high latitudes where many water masses form. In the ocean's interior, preformed ^{14}C -age behaves like a passive tracer. The relative contribution of the preformed component to bulk ^{14}C -age varies regionally within a given model, but also between models. Regional variability in the Atlantic Ocean is associated with the mixing of waters with very different end members of preformed ^{14}C -age. Here, variations in the preformed component over space and time mask the circulation component to an extent that its patterns are not detectable from bulk ^{14}C -age. Between models, the variability of preformed ^{14}C -age can also be considerable (factor of 2), related to the combination of physical model parameters, which influence circulation dynamics or gas exchange. The preformed component was found to be very sensitive to gas exchange and moderately sensitive to ice cover. In our model evaluation, the choice of the gas-exchange constant from within the currently accepted range of uncertainty had such a strong influence on preformed and bulk ^{14}C -age that if model evaluation would be based on bulk ^{14}C -age, it could easily impair the evaluation and tuning of a model's circulation on global and regional scales. Based on the results of this study, we propose that considering preformed ^{14}C -age is critical for a correct assessment of circulation in ocean models.

1574. Monthly maps of optimally interpolated in situ hydrography in the North Sea from 1948 to 2013

Authors: Ismael Núñez-Riboni; Anna Akimova

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2015.06.003

URL: <https://doi.org/10.1016/j.jmarsys.2015.06.003>

Publication Date: 2015-07-04

Volume/Issue: Volume 151

References: 65

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1575. Impact of assimilating temperature and salinity measurements by animal-borne sensors on FOAM ocean model fields

Authors: Fiona Carse (Met Office); Matthew J. Martin (Met Office); Alistair Sellar (Met Office); Edward W. Blockley (Met Office)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2613

URL: <https://doi.org/10.1002/qj.2613>

Publication Date: 2015-06-27

Volume/Issue: Volume 141, Issue 693

References: 26

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Temperature and salinity profiles are being obtained from instrumented marine mammals in near real-time. The mammals, mostly elephant seals, sample to depths of up to 2000 m in high-latitude regions where there are very few other in situ observations. This study investigates the impact of mammal temperature and salinity observations on the UK Met Office's global ocean forecasting model, focussing on impacts in the Southern Ocean region. Three reanalyses were performed, each assimilating a varying amount of seal data whilst holding all other conditions constant. Assimilation of both temperature and salinity profiles from mammals has a detrimental effect on the model's temperature and salinity fields due to a high bias in the salinity values. Restricting assimilation to temperature profiles only has benefits for both temperature and salinity fields in regions of the ocean where seals are sampling, improving root-mean-square error statistics by between 1 and 6% compared to the case where no mammal data were assimilated. In addition, the assimilation of seal temperature profiles alters the mixed-layer depth and repositions and increases gradients of some fronts compared to the no-seals case.

1576. Geographical distribution and anisotropy of the inverse kinetic energy cascade, and its role in the eddy equilibrium processes

Authors: Shihong Wang (Chinese Academy of Sciences); Zhiliang Liu (Chinese Academy of Sciences); Chongguang Pang (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010476

URL: <https://doi.org/10.1002/2014jc010476>

Publication Date: 2015-06-23

Volume/Issue: Volume 120, Issue 7

References: 54

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: anisotropy, satellite altimeters, multiple zonal jets, inverse kinetic energy cascades

Funding: National Natural Science Foundation of China, Special Fund for strategic pilot technology Chinese Academy of Sciences

License: N/A

Open Access: N/A

Abstract: N/A

1577. Mesoscale and intraseasonal air-sea CO₂ exchanges in the western Arabian Sea during boreal summer

Authors: Vinu Valsala; Raghu Murtugudde

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2015.06.001

URL: <https://doi.org/10.1016/j.dsr.2015.06.001>

Publication Date: 2015-06-13

Volume/Issue: Volume 103

References: 56

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Earth Sciences (MoES)

License: N/A

Open Access: N/A

Abstract: N/A

1578. The Ocean Reanalyses Intercomparison Project (ORA-IP)

Authors: M.A. Balmaseda (European Centre for Medium-Range Weather Forecasts (ECMWF)); F. Hernandez (Mercator Océan, Ramonville Saint-Agne, France, Institut de Recherche pour le Développement (IRD)); A. Storto (Centro Euro mediterraneo sui Cambiamenti Climatici (CMCC), Bologna, Italy); M.D. Palmer (Met Office); O. Alves (Met Office); L. Shi (Centre for Australian Weather and Climate Research, Bureau of Meteorology (BOM), Melbourne, Australia); G.C. Smith (Environment Canada, Québec, Canada); T. Toyoda (Meteorological Research Institute); M. Valdivieso (University of Reading); B. Barnier (Centre National de Recherche Scientifique (CNRS), Laboratoire de Glaciologie et Géophysique de l'Environnement (LGGE), Grenoble, France); D. Behringer (Climate Prediction Center, NOAA/NWS/NCEP, Camp Springs, Maryland, USA); T. Boyer (NOAA/NODC, College Park, Maryland); Y-S. Chang (Kongju National University, Geophysical Fluid Dynamics Laboratory, National Oceanic and Atmospheric Administration (GFDL/NOAA), Princeton, New Jersey); G.A. Chepurin (University of Maryland); N. Ferry (Mercator Océan, Ramonville Saint-Agne, France); G. Forget (Massachusetts Institute of Technology); Y. Fujii (Meteorological Research Institute); S. Good (Met Office); S. Guinehut (Collecte Localisation Satellites (CLS), Ramonville Saint-Agne, France); K. Haines (University of Reading); Y. Ishikawa (Center for Earth Information Science and Technology, Japan Agency of Marine-Earth Science and Technology (CEIST/JAMSTEC), Yokohama, Japan); S. Keeley (European Centre for Medium-Range Weather Forecasts (ECMWF)); A. Köhl (University of Hamburg); T. Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M.J. Martin (Met Office); S. Masina (Istituto Nazionale di Geofisica e Vulcanologia, Bologna, Italy, Centro Euro mediterraneo sui Cambiamenti Climatici (CMCC), Bologna, Italy); S. Masuda (Research and Development Center for Global Change (RCGC), JAMSTEC, Yokosuka, Japan); B. Meyssignac (Centre National d'Études Spatiales (CNES)); K. Mogensen (European Centre for Medium-Range Weather Forecasts (ECMWF)); L. Parent (Mercator Océan, Ramonville Saint-Agne, France); K.A. Peterson (Met Office); Y.M. Tang (European Centre for Medium-Range Weather Forecasts (ECMWF)); Y. Yin (Centre for Australian Weather and Climate Research, Bureau of Meteorology (BOM), Melbourne, Australia); G. Vernieres (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); X. Wang (University of California Los Angeles); J. Waters (Met Office); R. Wedd (Centre for Australian Weather and Climate Research, Bureau of Meteorology (BOM), Melbourne, Australia); O. Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Xue (Climate Prediction Center, NOAA/NWS/NCEP, Camp Springs, Maryland, USA); M. Chevallier (CNRM-GAME, Météo-France, CNRS UMR3589, Toulouse, France); J-F. Lemieux (Environment Canada, Québec, Canada); F. Dupont (Environment Canada, Québec, Canada); T. Kuragano (Meteorological Research Institute); M. Kamachi (Meteorological Research Institute); T. Awaji (Center for Earth Information Science and Technology, Japan Agency of Marine-Earth Science and Technology (CEIST/JAMSTEC), Yokohama, Japan); A. Caltabiano (International CLIVAR Global Project Office, First Institute of Oceanography, State Oceanic Administration, China.); K. Wilmer-Becker (Met Office); F. Gaillard (Laboratoire de Physique des Océans (LPO/IFREMER), France)

Journal: Journal of Operational Oceanography

Publisher: Informa UK Limited

DOI: 10.1080/1755876x.2015.1022329

URL: <https://doi.org/10.1080/1755876x.2015.1022329>

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References: 87

Cited by: 170 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1579. The partition of regional sea level variability

Authors: Gaël Forget; Rui M. Ponte

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2015.06.002

URL: <https://doi.org/10.1016/j.pocean.2015.06.002>

Publication Date: 2015-06-06

Volume/Issue: Volume 137

References: 89

Cited by: 90 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Oceanic and Atmospheric Administration

License: N/A

Open Access: N/A

Abstract: N/A

1580. Biogeochemical drivers of the fate of riverine mercury discharged to the global and Arctic oceans

Authors: Yanxu Zhang (Harvard John A. Paulson School of Engineering and Applied Science Cambridge Massachusetts USA); Daniel J. Jacob (Harvard John A. Paulson School of Engineering and Applied Science Cambridge Massachusetts USA); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Helen M. Amos (Department of Environmental Health Harvard T. H. Chan School of Public Health Boston Massachusetts USA); Michael S. Long (Harvard John A. Paulson School of Engineering and Applied Science Cambridge Massachusetts USA); Elsie M. Sunderland (Department of Environmental Health Harvard T. H. Chan School of Public Health Boston Massachusetts USA, Harvard John A. Paulson School of Engineering and Applied Science Cambridge Massachusetts USA)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015gb005124

URL: <https://doi.org/10.1002/2015gb005124>

Publication Date: 2015-06-03

Volume/Issue: Volume 29, Issue 6

References: 83

Cited by: 108 publications

Language: en

Subjects: N/A

Keywords: DARWIN project, Arctic, ECCO-GODAE, ocean, Mercury (and 1 more)

Funding: U.S. National Science Foundation, Electric Power Research Institute (EPRI)

License: N/A

Open Access: N/A

Abstract: Rivers discharge 28 ± 13 Mmol yr⁻¹ of mercury (Hg) to ocean margins, an amount comparable to atmospheric deposition to the global oceans. Most of the Hg discharged by rivers is sequestered by burial of benthic sediment in estuaries or the coastal zone, but some is evaded to the atmosphere and some is exported to the open ocean. We investigate the fate of riverine Hg by developing a new global 3-D simulation for Hg in the Massachusetts Institute of Technology ocean general circulation model. The model includes plankton dynamics and carbon respiration (DARWIN project model) coupled to inorganic Hg chemistry. Results are consistent with observed spatial patterns and magnitudes of surface ocean Hg concentrations. We use observational constraints on seawater Hg concentrations and evasion to infer that most Hg from rivers is sorbed to refractory organic carbon and preferentially buried. Only 6% of Hg discharged by rivers (1.8 Mmol yr⁻¹) is transported to the open ocean on a global basis. This fraction varies from a low of 2.6% in East Asia due to the barrier imposed by the Korean Peninsula and Japanese archipelago, up to 25% in eastern North America facilitated by the Gulf Stream. In the Arctic Ocean, low tributary particle loads and efficient degradation of particulate organic carbon by deltaic microbial communities favor a more labile riverine Hg pool. Evasion of Hg to the Arctic atmosphere is indirectly enhanced by heat transport during spring freshet that accelerates sea ice melt and ice rafting. Discharges of 0.23 Mmol Hg yr⁻¹ from Arctic rivers can explain the observed summer maximum in the Arctic atmosphere, and this magnitude of releases is consistent with recent observations. Our work indicates that rivers are major contributors to Hg loads in the Arctic Ocean.

1581. Geodetic secular velocity errors due to interannual surface loading deformation

Authors: Alvaro Santamaría-Gómez (LIENSs, Université de La Rochelle/CNRS, La Rochelle, France. E-mail: alvaro.santamaria@univ-lr.fr, University of Tasmania); Anthony Mémin (University of Tasmania, Institut de Recherche pour le Développement (IRD))

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggv190

URL: <https://doi.org/10.1093/gji/ggv190>

Publication Date: 2015-05-30

Volume/Issue: Volume 202, Issue 2

References: 34

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Geodetic vertical velocities derived from data as short as 3 yr are often assumed to be representative of linear deformation over past decades to millennia. We use two decades of surface loading deformation predictions due to variations of atmospheric, oceanic and continental water mass to assess the effect on secular velocities estimated from short time-series. The interannual deformation is time-correlated at most locations over the globe, with the level of correlation depending mostly on the chosen continental water model. Using the most conservative loading model and 5-yr-long time-series, we found median vertical velocity errors of 0.5 mm yr⁻¹ over the continents (0.3 mm yr⁻¹ globally), exceeding 1 mm yr⁻¹ in regions around the southern Tropic. Horizontal velocity errors were seven times smaller. Unless an accurate loading model is available, a decade of continuous data is required in these regions to mitigate the impact of the interannual loading deformation on secular velocities.

1582. The Contribution of Striations to the Eddy Energy Budget and Mixing: Diagnostic Frameworks and Results in a Quasigeostrophic Barotropic System with Mean Flow

Authors: Ru Chen (Scripps Institution of Oceanography / University of California San Diego); Glenn R. Flierl (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0199.1

URL: <https://doi.org/10.1175/jpo-d-14-0199.1>

Publication Date: 2015-05-29

Volume/Issue: Volume 45, Issue 8

References: 60

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: Quasigeostrophic models, Mixing, Eddies, Jets, Models and modeling (and 4 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Low-frequency oceanic motions have banded structures termed "striations." Since these striations embedded in large-scale gyre flows can have large amplitudes, the authors investigated the effect of mean flow on their directions as well as their contribution to energetics and mixing using a beta-plane, barotropic, quasigeostrophic ocean model. In spite of the model simplicity, striations are always found to exist regardless of the imposed barotropic mean flow. However, their properties are sensitive to the mean flow. Rhines jets move with the mean flow and are not necessarily striations. If the meridional component of the mean flow is large, Rhines jets become high-frequency motions; low-frequency striations still exist, but they are nonzonal, have small magnitudes, and contribute little to energetics and mixing. Otherwise, striations are zonal, dominated by Rhines jets, and contribute significantly to energetics and mixing. This study extends the theory of beta-plane, barotropic turbulence, driven by white noise forcing at small scales, to include the effect of a constant mean flow. Theories developed in this study, based upon the Galilean invariance property, illustrate that the barotropic mean flow has no effect on total mixing rates, but does affect the energy cascades in the frequency domain. Diagnostic frameworks developed here can be useful to quantify the striations' contribution to energetics and mixing in the ocean and more realistic models. A novel diagnostic formula is applied to estimating eddy diffusivities.

1583. Intercomparison and validation of the mixed layer depth fields of global ocean syntheses

Authors: Takahiro Toyoda; Yosuke Fujii; Tsurane Kuragano; Masafumi Kamachi; Yoichi Ishikawa; Shuhei Masuda; Kanako Sato; Toshiyuki Awaji; Fabrice Hernandez; Nicolas Ferry; Stéphanie Guinehut; Matthew J. Martin; K. Andrew Peterson; Simon A. Good; Maria Valdivieso; Keith Haines; Andrea Storto; Simona Masina; Armin Köhl; Hao Zuo; Magdalena Balmaseda; Yonghong Yin; Li Shi; Oscar Alves; Gregory Smith; You-Soon Chang; Guillaume Vernieres; Xiaochun Wang; Gael Forget; Patrick Heimbach; Ou Wang; Ichiro Fukumori; Tong Lee

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-015-2637-7

URL: <https://doi.org/10.1007/s00382-015-2637-7>

Publication Date: 2015-05-27

Volume/Issue: Volume 49, Issue 3

References: 59

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1584. Environmental characteristics of Agulhas rings affect interocean plankton transport

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Abstract: Agulhas rings provide the principal route for ocean waters to circulate from the Indo-Pacific to the Atlantic basin. Their influence on global ocean circulation is well known, but their role in plankton transport is largely unexplored. We show that, although the coarse taxonomic structure of plankton communities is continuous across the Agulhas choke point, South Atlantic plankton diversity is altered compared with Indian Ocean source populations. Modeling and in situ sampling of a young Agulhas ring indicate that strong vertical mixing drives complex nitrogen cycling, shaping community metabolism and biogeochemical signatures as the ring and associated plankton transit westward. The peculiar local environment inside Agulhas rings may provide a selective mechanism contributing to the limited dispersal of Indian Ocean plankton populations into the Atlantic.

1585. Impact of the North Atlantic Oscillation on Southern Europe Water Distribution: Insights from Geodetic Data

Authors: Pierre Valty (Institut National de l'Information Géographique et Forestière/Laboratoire de Recherche en Géodésie, Université Paris-Diderot, Paris, France); Olivier de Viron (Laboratoire Littoral Environnement et Sociétés, Université de la Rochelle, La Rochelle, France); Isabelle Panet (Institut National de l'Information Géographique et Forestière/Laboratoire de Recherche en Géodésie, Université Paris-Diderot, Paris, France); Xavier Collilieux (Institut National de l'Information Géographique et Forestière/Laboratoire de Recherche en Géodésie, Université Paris-Diderot, Paris, France)

Journal: Earth Interactions

Publisher: American Meteorological Society

DOI: 10.1175/ei-d-14-0028.1

URL: <https://doi.org/10.1175/ei-d-14-0028.1>

Publication Date: 2015-05-20

Volume/Issue: Volume 19, Issue 10

References: 73

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: From space gravity and station position data over southern Europe from 2002 to 2010, this study investigates the interannual mass redistributions using principal component analysis. The dominant mode, which appears both in gravity and positioning, results from the North Atlantic Oscillation (NAO). This analysis allows us to isolate and characterize the NAO impact on the mass distribution, which appears centered over the Black Sea and its two main catchment basins, the Danube and Dnieper.

1586. Estimates of Ocean Macroturbulence: Structure Function and Spectral Slope from Argo Profiling Floats

Authors: Katherine McCaffrey (University of Colorado Boulder); Baylor Fox-Kemper (Brown University); Gael Forget (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0023.1

URL: <https://doi.org/10.1175/jpo-d-14-0023.1>

Publication Date: 2015-05-19

Volume/Issue: Volume 45, Issue 7

References: 48

Cited by: 19 publications

Language: N/A

Subjects: N/A

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Funding: N/A

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Open Access: N/A

Abstract: The Argo profiling float network has repeatedly sampled much of the World Ocean. This study uses Argo temperature and salinity data to form the tracer structure function of ocean variability at the macroscale (10-1000 km, mesoscale and above). Here, second-order temperature and salinity structure functions over horizontal separations are calculated along either pressure or potential density surfaces, which allows analysis of both active and passive tracer structure functions. Using Argo data, a map of global variance is created from the climatological average and each datum. When turbulence is homogeneous, the structure function slope from Argo can be related to the wavenumber spectrum slope in ocean temperature or salinity variability. This first application of structure function techniques to Argo data gives physically meaningful results based on bootstrapped confidence intervals, showing geographical dependence of the structure functions with slopes near $2/3$ on average, independent of depth.

1587. On the seasonal variations of salinity of the tropical Atlantic mixed layer

Authors: I. Camara (LPAO-SF, ESP, UCAD Dakar Senegal, Institut de Recherche pour le Développement (IRD)); Nicolas Kolodziejczyk (Institut de Recherche pour le Développement (IRD)); Juliette Mignot (University of Bern, LPAO-SF, ESP, UCAD Dakar Senegal, Institut de Recherche pour le Développement (IRD)); Alban Lazar (LPAO-SF, ESP, UCAD Dakar Senegal, Institut de Recherche pour le Développement (IRD)); Amadou T. Gaye (LPAO-SF, ESP, UCAD Dakar Senegal)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc010865

URL: <https://doi.org/10.1002/2015jc010865>

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Volume/Issue: Volume 120, Issue 6

References: 92

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Union Seventh Framework Programme

License: N/A

Open Access: N/A

Abstract: The physical processes controlling the mixed layer salinity (MLS) seasonal budget in the tropical Atlantic Ocean are investigated using a regional configuration of an ocean general circulation model. The analysis reveals that the MLS cycle is generally weak in comparison of individual physical processes entering in the budget because of strong compensation. In evaporative regions, around the surface salinity maxima, the ocean acts to freshen the mixed layer against the action of evaporation. Poleward of the southern SSS maxima, the freshening is ensured by geostrophic advection, the vertical salinity diffusion and, during winter, a dominant contribution of the convective entrainment. On the equatorward flanks of the SSS maxima, Ekman transport mainly contributes to supply freshwater from ITCZ regions while vertical salinity diffusion adds on the effect of evaporation. All these terms are phase locked through the effect of the wind. Under the seasonal march of the ITCZ and in coastal areas affected by river (7 degreesS:15 degreesN), the upper ocean freshening by precipitations and/or runoff is attenuated by vertical salinity diffusion. In the eastern equatorial regions, seasonal cycle of wind forced surface currents advect freshwaters, which are mixed with subsurface saline water because of the strong vertical turbulent diffusion. In all these regions, the vertical diffusion presents an important contribution to the MLS budget by providing, in general, an upwelling flux of salinity. It is generally due to vertical salinity gradient and mixing due to winds. Furthermore, in the equator where the vertical shear, associated to surface horizontal currents, is developed, the diffusion depends also on the sheared flow stability.

1588. Freshwater transport in the coupled ocean-atmosphere system: a passive ocean

Authors: David Ferreira; John Marshall

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-015-0846-6

URL: <https://doi.org/10.1007/s10236-015-0846-6>

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References: 39

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1589. Seasonal Variability of Salinity and Salt Transport in the Northern Indian Ocean

Authors: Joseph M. D'Addezio (Marine Science Program, University of South Carolina, Columbia, South Carolina); Bulusu Subrahmanyam (Marine Science Program, and Department of Earth and Ocean Sciences, University of South Carolina, Columbia, South Carolina); Ebenezer S. Nyadjro (Department of Physics, University of New Orleans, New Orleans, Louisiana); V. S. N. Murty (Council of Scientific and Industrial Research, National Institute of Oceanography Regional Centre, Visakhapatnam, India)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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Publication Date: 2015-05-14

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References: 49

Cited by: 29 publications

Language: N/A

Subjects: N/A

Keywords: Observational techniques and algorithms, Circulation/ Dynamics, Models and modeling, Transport, Satellite observations (and 5 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Analyses using a suite of observational datasets (Aquarius and Argo) and model simulations are carried out to examine the seasonal variability of salinity in the northern Indian Ocean (NIO). The model simulations include Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2), the European Centre for Medium-Range Weather Forecasts-Ocean Reanalysis System 4 (ECMWF-ORAS4), Simple Ocean Data Assimilation (SODA) reanalysis, and the Hybrid Coordinate Ocean Model (HYCOM). The analyses of salinity at the surface and at depths up to 200 m, surface salt transport in the top 5-m layer, and depth-integrated salt transports revealed different salinity processes in the NIO that are dominantly related to the semiannual monsoons. Aquarius proves a useful tool for observing this dynamic region and reveals some aspects of sea surface salinity (SSS) variability that Argo cannot resolve. The study revealed large disagreement between surface salt transports derived from observed- and analysis-derived salinity fields. Although differences in SSS between the observations and the model solutions are small, model simulations provide much greater spatial variability of surface salt transports due to finer detailed current structure. Meridional depth-integrated salt transports along 6 degreesN revealed dominant advective processes from the surface toward near-bottom depths. In the Arabian Sea (Bay of Bengal), the net monthly mean maximum northward (southward) salt transport of $\sim 50 \times 10^6 \text{ kg s}^{-1}$ occurs in July, and annual-mean salt transports across this section are about $-2.5 \times 10^6 \text{ kg s}^{-1}$ ($3 \times 10^6 \text{ kg s}^{-1}$).

1590. Long-Period Tides in an Atmospherically Driven, Stratified Ocean

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.); Ayan H. Chaudhuri (Atmospheric and Environmental Research Inc.); Sergey V. Vinogradov (Earth Resources Technology, Inc./National Oceanic and Atmospheric Administration, Silver Spring, Maryland)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

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References: 30

Cited by: 11 publications

Language: N/A

Subjects: N/A

Keywords: Atm/Ocean Structure/ Phenomena, Tides, Circulation/ Dynamics, Models and modeling, Sea level (and 2 more)

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License: N/A

Open Access: N/A

Abstract: Long-period tides (LPT) are studied using a stratified, primitive equation model on a global domain and in the presence of a fully developed, atmospherically forced ocean general circulation. The major LPT constituents, from termensual to nodal (18.6 yr) periods, are examined. Ocean circulation variability can overwhelm the longest tide signals and make inferring LPT from data difficult, but model results suggest that bottom pressure offers cleaner signal-to-noise ratios than sea level, particularly at low latitudes where atmospherically driven variability is substantially stronger at the surface than at the bottom. Most tides exhibit a significant large-scale dynamic response, with the tendency for weaker nonequilibrium signals in the Atlantic compared to the Pacific as seen in previous studies. However, across most tidal lines, the largest dynamic signals tend to occur in the Arctic and Nordic Seas and also in Hudson Bay. Bathymetry and coastal geometry contribute to the modeled nonequilibrium behavior. Baroclinic effects tend to increase with the tidal period. Apart from short spatial-scale modulations associated with topographic interactions, the excitation of various propagating baroclinic wave modes is clearly part of the modeled LPT, particularly at tropical latitudes, for fortnightly and longer-period tides.

1591. Hydrological Excitations of Polar Motion from GRACE Gravity Field Solutions

Authors: J. Nastula; D. A. Salstein; W. Popiński

Journal: International Association of Geodesy Symposia

Publisher: Springer International Publishing

DOI: 10.1007/1345_2015_85

URL: https://doi.org/10.1007/1345_2015_85

Publication Date: 2015-05-14

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References: 25

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: Polar motion

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1592. Evaluation of the Tropical Pacific Observing System from the ocean data assimilation perspective

Authors: Yosuke Fujii (Meteorological Research Institute); James Cummings (Naval Research Laboratory Monterey CA USA); Yan Xue (Climate Prediction Center NOAA/NWS/NCEP College Park MD USA); Andreas Schiller (CSIRO Oceans and Atmospheric Flagship Hobart Tasmania Australia); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Magdalena Alonso Balmaseda (European Centre for Medium-Range Weather Forecasts (ECMWF)); Elisabeth Rémy (Mercator Océan Toulouse France); Shuhei Masuda (Research and Development Center for Global Change Japan Agency for Marine-Earth Science and Technology Yokosuka Japan); Gary Brassington (Centre for Australian Weather and Climate Research Bureau of Meteorology Melbourne Victoria Australia); Oscar Alves (Centre for Australian Weather and Climate Research Bureau of Meteorology Melbourne Victoria Australia); Bruce Cornuelle (Scripps Institution of Oceanography); Matthew Martin (Met Office); Peter Oke (CSIRO Oceans and Atmospheric Flagship Hobart Tasmania Australia); Gregory Smith (Environment Canada Dorval Québec Canada); Xiaosong Yang (Geophysical Fluid Dynamics Laboratory NOAA Princeton NJ USA)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

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Publication Date: 2015-05-11

Volume/Issue: Volume 141, Issue 692

References: 86

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Meteorological Agency/ Meteorological Research Institute

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Abstract: The drastic reduction in the number of observation data from the Tropical Atmospheric Ocean (TAO)/Triangle Trans-Ocean Buoy Network (TRITON) array since 2012 has given rise to a need to assess the impact of those data in ocean data assimilation (DA) systems. This article provides a review of existing studies evaluating the impacts of data from the TAO/TRITON array and other components of the Tropical Pacific Observing System (TPOS) on current ocean DA systems used for a variety of operational and research applications. It can be considered as background information that can guide the evaluation exercise of TPOS. Temperature data from TAO/TRITON array are assimilated in most ocean DA systems which cover the tropical Pacific in order to constrain the ocean heat content, stratification, and circulation. It is shown that the impacts of observation data depend considerably on the system and application. The presence of model error often makes the results difficult to interpret. Nevertheless there is consensus that the data from TAO/TRITON generally have positive impacts complementary to Argo floats. In the equatorial Pacific, the impacts are generally around the same level or larger than those of Argo. We therefore conclude that, with the current configuration of TPOS, the

loss of the TAO/TRITON data is having a significant detrimental impact on many applications based on ocean DA systems. This conclusion needs to be kept under review because the equatorial coverage by Argo is expected to improve in the future.

1593. Using ocean bottom pressure from the gravity recovery and climate experiment (GRACE) to estimate transport variability in the southern Indian Ocean

Authors: Jessica K. Makowski (University of South Florida); Don P. Chambers (University of South Florida); Jennifer A. Bonin (University of South Florida)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010575

URL: <https://doi.org/10.1002/2014jc010575>

Publication Date: 2015-05-05

Volume/Issue: Volume 120, Issue 6

References: 63

Cited by: 18 publications

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Keywords: Southern Ocean, GRACE, Antarctic circumpolar current, ocean bottom pressure

Funding: NASA

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Open Access: N/A

Abstract: Previous studies have suggested that ocean bottom pressure (OBP) from the Gravity Recovery and Climate Experiment (GRACE) can be used to measure the depth-averaged, or barotropic, transport variability of the Antarctic Circumpolar Current (ACC). Here, we use GRACE OBP observations to calculate transport variability in a region of the southern Indian Ocean encompassing the major fronts of the ACC. We use a statistical analysis of a simulated GRACE-like data set to determine the uncertainty of the estimated transport for the 2003.0-2013.0 time period. We find that when the transport is averaged over 60 degrees of longitude, the uncertainty (one standard error) is close to 1 Sv (1 Sv = 10^6 m³ s⁻¹) for low-pass filtered transport, which is significantly smaller than the signal and lower than previous studies have found. The interannual variability is correlated with the Southern Annual mode (SAM) (0.61), but more highly correlated with circumpolar zonally averaged winds between 45 degreesS and 65 degreesS (0.88). GRACE transport reflects significant changes in transport between 2007 and 2009 that is observed in the zonal wind variations but not in the SAM index. We also find a statistically significant trend in transport (-1.0 ± 0.4 Sv yr⁻¹, 90% confidence) that is correlated with a local deceleration in zonal winds related to an asymmetry in the SAM on multidecadal periods.

1594. Second-degree Stokes coefficients from multi-satellite SLR

Authors: Mathis Bloßfeld; Horst Müller; Michael Gerstl; Vojtěch Štefka; Johannes Bouman; Franziska Göttl; Martin Horwath

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

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URL: <https://doi.org/10.1007/s00190-015-0819-z>

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References: 56

Cited by: 33 publications

Language: en

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1595. Estimating uncertainties on a Gulf Stream mixed-layer heat budget from stochastic modeling

Authors: Nadia K. Ayoub; Marc Lucas; Pierre De Mey

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2015.04.008

URL: <https://doi.org/10.1016/j.jmarsys.2015.04.008>

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Volume/Issue: Volume 150

References: 41

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Service Hydrographique de la Marine, Centre National des Etudes Spatiales (Centre National des Etudes Spatiales, CNES), Groupe Mission Mercator Coriolis (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

1596. The Effect of Atmospheric Forcing Resolution on Delivery of Ocean Heat to the Antarctic Floating Ice Shelves*,+

Authors: Michael S. Dinniman (Old Dominion University); John M. Klinck (Old Dominion University); Le-Sheng Bai (Ohio State University); David H. Bromwich (Ohio State University); Keith M. Hines (Ohio State University); David M. Holland (Courant Institute of Mathematical Sciences, New York University, New York, New York)

Journal: Journal of Climate

Publisher: American Meteorological Society

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References: 116

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Ice shelves, Southern Ocean, Antarctica, Continental shelf/slope (and 1 more)

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License: N/A

Open Access: N/A

Abstract: Oceanic melting at the base of the floating Antarctic ice shelves is now thought to be a more significant cause of mass loss for the Antarctic ice sheet than iceberg calving. In this study, a 10-km horizontal-resolution circum-Antarctic ocean-sea ice-ice shelf model [based on the Regional Ocean Modeling System (ROMS)] is used to study the delivery of ocean heat to the base of the ice shelves. The atmospheric forcing comes from the ERA-Interim reanalysis (~80-km resolution) and from simulations using the polar-optimized Weather Research and Forecasting Model (30-km resolution), where the upper atmosphere was relaxed to the ERA-Interim reanalysis. The modeled total basal ice shelf melt is low compared to observational estimates but increases by 14% with the higher-resolution winds and just 3% with both the higher-resolution winds and atmospheric surface temperatures. The higher-resolution winds lead to more heat being delivered to the ice shelf cavities from the adjacent ocean and an increase in the efficiency of heat transfer between the water and the ice. The higher-resolution winds also lead to changes in the heat delivered from the open ocean to the continental shelves as well as changes in the heat lost to the atmosphere over the shelves, and the sign of these changes varies regionally. Addition of the higher-resolution temperatures to the winds results in lowering, primarily during summer, the wind-driven increase in heat advected into the ice shelf cavities due to colder summer air temperatures near the coast.

1597. Eddy stirring and horizontal diffusivity from Argo float observations: Geographic and depth variability

Authors: Sylvia T. Cole (Woods Hole Oceanographic Institution); Cimarron Wortham (University of Washington); Eric Kunze (Northwest Research Associates Redmond Washington USA); W. Brechner Owens (Woods Hole Oceanographic Institution)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015gl063827

URL: <https://doi.org/10.1002/2015gl063827>

Publication Date: 2015-04-29

Volume/Issue: Volume 42, Issue 10

References: 55

Cited by: 82 publications

Language: en

Subjects: N/A

Keywords: mixing length, diffusivity, Argo, eddy stirring, spice

Funding: National Science Foundation, Office of Naval Research

License: N/A

Open Access: N/A

Abstract: Stirring along isopycnals is a significant factor in determining the distribution of tracers within the ocean. Salinity anomalies on density surfaces from Argo float profiles are used to investigate horizontal stirring and estimate eddy mixing lengths. Eddy mixing length and velocity fluctuations from the ECCO2 global state estimate are used to estimate horizontal diffusivity at a 300 km scale in the upper 2000 m with near-global coverage. Diffusivity varies by over two orders of magnitude with latitude, longitude, and depth. In all basins, diffusivity is elevated in zonal bands corresponding to strong current regions, including western boundary current extension regions, the Antarctic Circumpolar Current, and equatorial current systems. The estimated mixing lengths and diffusivities provide an observationally based data set that can be used to test and constrain predictions and parameterizations of eddy stirring.

1598. Source-receptor relationships of column-average CO₂ and implications for the impact of observations on flux inversions

Authors: Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Daven K. Henze (University of Colorado Boulder)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jd022914

URL: <https://doi.org/10.1002/2014jd022914>

Publication Date: 2015-04-28

Volume/Issue: Volume 120, Issue 10

References: 47

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Source-receptor relationships are the fundamental quantities used in atmosphere CO₂ flux inversions. In this study, we systematically investigate the global source-receptor relationships of column CO₂ (XCO₂) in 12 continental-scale receptors in terms of transport and local versus nonlocal flux contributions using the GEOS-Chem adjoint model. Using simulated Greenhouse gases Observing Satellite (GOSAT) XCO₂, we quantify the impact of inclusion (add-on) or exclusion of observations (data-denial) within a receptor region on flux inversion. We discuss the connections between the observation impact and the underlying source-receptor relationships. The strong sensitivity of XCO₂ to nonlocal fluxes makes the XCO₂ observations have strong impact on nonlocal flux estimation. On an annual mean, the impact of GOSAT XCO₂ over Europe on North America flux estimation is 10%-39% of the full observation impact. Because the Southern Hemisphere midlatitude XCO₂ are most sensitive to local and tropical fluxes, the mean impact of GOSAT XCO₂ over Southern South America on Northern South America (N-S-America) flux estimation is 30%-59% of the full observation impact. Because of the strong sensitivity of the XCO₂ over N-S-America to Central Africa (C-Africa) fluxes, the mean impact on C-Africa flux estimation is between 14% and 31% of the full observation impact in spite of the sparse observation coverage. The results also show that XCO₂ have similar sensitivity to local and nonlocal fluxes occurring 3 months before, which indicates that XCO₂ observations cannot differentiate any fluxes occurring beyond 3 months.

1599. A Multiwavenumber Theory for Eddy Diffusivities and Its Application to the Southeast Pacific (DIMES) Region

Authors: Ru Chen (Scripps Institution of Oceanography / University of California San Diego); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Julie L. McClean (Scripps Institution of Oceanography / University of California San Diego); Glenn R. Flierl (Massachusetts Institute of Technology); Alexa Griesel (University of Hamburg)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0229.1

URL: <https://doi.org/10.1175/jpo-d-14-0229.1>

Publication Date: 2015-04-28

Volume/Issue: Volume 45, Issue 7

References: 59

Cited by: 21 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A multiwavenumber theory is formulated to represent eddy diffusivities. It expands on earlier single-wavenumber theories and includes the wide range of wavenumbers encompassed in eddy motions. In the limiting case in which ocean eddies are only composed of a single wavenumber, the multiwavenumber theory is equivalent to the single-wavenumber theory and both show mixing suppression by the eddy propagation relative to the mean flow. The multiwavenumber theory was tested in a region of the Southern Ocean (70 degrees-45 degreesS, 110 degrees-20 degreesW) that covers the Drake Passage and includes the tracer/float release locations during the Diapycnal and Isopycnal Mixing Experiment in the Southern Ocean (DIMES). Cross-stream eddy diffusivities and mixing lengths were estimated in this region from the single-wavenumber theory, from the multiwavenumber theory, and from floats deployed in a global degrees Parallel Ocean Program (POP) simulation. Compared to the single-wavenumber theory, the horizontal structures of cross-stream mixing lengths from the multiwavenumber theory agree better with the simulated float-based estimates at almost all depth levels. The multiwavenumber theory better represents the vertical structure of cross-stream mixing lengths both inside and outside the Antarctica Circumpolar Current (ACC). Both the single-wavenumber and multiwavenumber theories represent the horizontal structures of cross-stream diffusivities, which resemble the eddy kinetic energy patterns.

1600. Role of tides on the formation of the Antarctic Slope Front at the Weddell-Scotia Confluence

Authors: M. M. Flexas (Previously at IMEDEA, UIB-CSIC Spain, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. P. Schodlok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); L. Padman (Earth & Space Research Corvallis Oregon USA); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. H. Orsi (Texas A&M; University Department of Oceanography College Station Texas USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010372

URL: <https://doi.org/10.1002/2014jc010372>

Publication Date: 2015-04-24

Volume/Issue: Volume 120, Issue 5

References: 72

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: numerical model, Antarctic Slope Front, Antarctic Slope Current, SASSI, Scotia sea (and 1 more)

Funding: Spanish Research and Innovation (I+D+i) National Plan, National Science Foundation, National Aeronautics and Space Administration (and 1 more)

License: N/A

Open Access: N/A

Abstract: The structure of the Antarctic Slope Front (ASF) and the associated Antarctic Slope Current (ASC) on the Scotia Sea side of the Weddell-Scotia Confluence (WSC) is described using data from a hydrographic survey and three 1 year long moorings across the continental slope. The ASC in this region flows westward along isobaths with an annual mean speed of ~ 0.2 m s⁻¹, with time variability dominated by the K1 and O1 tidal diurnal constituents, a narrowband oscillation with ~ 2 -week period attributable to the spring/neap tidal cycle, and seasonal variability. Realistic and idealized high-resolution numerical simulations are used to determine the contribution of tides to the structure of the ASF and the speed of the ASC. Two simulations forced by realistic atmospheric forcing and boundary conditions integrated with and without tidal forcing show that tidal forcing is essential to reproduce the measured ASF/ASC cross-slope structure, the time variability at our moorings, and the reduced stratification within the WSC. Two idealized simulations run with tide-only forcing, one with a homogeneous ocean and the other with initial vertical stratification that is laterally homogeneous, show that tides can generate the ASC and ASF through volume flux convergence along the slope initiated by effects including the Lagrangian component of tidal rectification and mixing at the seabed and in the stratified ocean interior. Climate models that exclude the effects of tides will not correctly represent the ASF and ASC or their influence on the injection of intermediate and dense waters from the WSC to the deep ocean.

1601. Tropical Cyclone-Induced Ocean Response: A Comparative Study of the South China Sea and Tropical Northwest Pacific*,+

Authors: Wei Mei (Scripps Institution of Oceanography / University of California San Diego); Chun-Chi Lien (National Taiwan University); I.-I. Lin (National Taiwan University); Shang-Ping Xie (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-14-00651.1

URL: <https://doi.org/10.1175/jcli-d-14-00651.1>

Publication Date: 2015-04-24

Volume/Issue: Volume 28, Issue 15

References: 62

Cited by: 78 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The thermocline shoals in the South China Sea (SCS) relative to the tropical northwest Pacific Ocean (NWP), as required by geostrophic balance with the Kuroshio. The present study examines the effect of this difference in ocean state on the response of sea surface temperature (SST) and chlorophyll concentration to tropical cyclones (TCs), using both satellite-derived measurements and three-dimensional numerical simulations. In both regions, TC-produced SST cooling strongly depends on TC characteristics (including intensity as measured by the maximum surface wind speed, translation speed, and size). When subject to identical TC forcing, the SST cooling in the SCS is more than 1.5 times that in the NWP, which may partially explain weaker TC intensity on average observed in the SCS. Both a shallower mixed layer and stronger subsurface thermal stratification in the SCS contribute to this regional difference in SST cooling. The mixed layer effect dominates when TCs are weak, fast-moving, and/or small; and for strong and slow-moving TCs or strong and large TCs, both factors are equally important. In both regions, TCs tend to elevate surface chlorophyll concentration. For identical TC forcing, the surface chlorophyll increase in the SCS is around 10 times that in the NWP, a difference much stronger than that in SST cooling. This large regional difference in the surface chlorophyll response is at least partially due to a shallower nutricline and stronger vertical nutrient gradient in the SCS. The effect of regional difference in upper-ocean density stratification on the surface nutrient response is negligible. The total annual primary production increase associated with the TC passage estimated using the vertically generalized production model in the SCS is nearly 3 times that in the NWP (i.e., $6.4 \pm 0.4 \times 10^{12}$ versus $2.2 \pm 0.2 \times 10^{12}$ g C), despite the weaker TC activity in the SCS.

1602. Alternative validation method of satellite gradiometric data by integral transform of satellite altimetry data

Authors: Michal Šprlák; Eliška Hamáková; Pavel Novák

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-015-0813-5

URL: <https://doi.org/10.1007/s00190-015-0813-5>

Publication Date: 2015-04-23

Volume/Issue: Volume 89, Issue 8

References: 63

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1603. Seasonal Variation of the Pacific South Equatorial Current Bifurcation

Authors: Zhaohui Chen (Ocean University of China); Lixin Wu (Ocean University of China)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0085.1

URL: <https://doi.org/10.1175/jpo-d-14-0085.1>

Publication Date: 2015-04-22

Volume/Issue: Volume 45, Issue 6

References: 35

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: Seasonal variability, Circulation/ Dynamics, Kelvin waves, Ocean circulation, Variability (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The seasonal variation of the South Equatorial Current (SEC) bifurcation off the Australian coast in the South Pacific (SP) is investigated with observations and a nonlinear, reduced-gravity, primitive equation model of the upper ocean. The mean SEC bifurcation latitude (SBL) integrated over the upper thermocline is around 17.5 degreesS, almost 2 degrees south of the position predicted by Sverdrup theory. For its seasonal variation, the SBL reaches its southernmost position in June/July and its northernmost position in November/December. The south-north migration of 2.7 degrees is twice as large as its counterpart in the North Pacific. It is found that the large seasonal amplitude of the SBL results from the combined effect of Low-Lat-SP and Non-Low-Lat-SP processes. The Low-Lat-SP process (referred to as the Rossby wave dynamics forced by the wind stress curl over the low-latitude SP) accounts for almost 2/3 of the SBL seasonal variability, and the Non-Low-Lat-SP processes account for 1/3. Both of these processes are responsible for its south-north migration but in different ways. The Low-Lat-SP wind forcing determines the offshore upper-layer thickness (ULT) via Rossby wave propagation, while the Non-Low-Lat-SP wind forcing determines the alongshore ULT via coastal Kelvin wave propagation. A simple bifurcation model is proposed under the framework of linear Rossby wave dynamics. It is found that the seasonal bifurcation latitude is predominantly determined by the spatial pattern of the wind and baroclinic Rossby wave propagation. This model explains the roles of local/remote wind forcing and baroclinic adjustment in the south-north migration and peak seasons of the bifurcation latitude.

1604. User awareness concerning feedback data and input observations used in reanalysis systems

Authors: H. Gregow; P. Poli; H. M. Mäkelä; K. Jylhä; A. K. Kaiser-Weiss; A. Obregon; D. G. H. Tan; S. Kekki; F. Kaspar

Journal: Advances in Science and Research

Publisher: Copernicus GmbH

DOI: 10.5194/asr-12-63-2015

URL: <https://doi.org/10.5194/asr-12-63-2015>

Publication Date: 2015-04-22

Volume/Issue: Volume 12, Issue 1

References: 6

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A web-based survey to assess the benefits and gaps in reanalyses as part of growing climate services was carried out in 2013-2014. The survey elicited responses from about 2500 users of climate information. One of the eleven survey points specifically addressed the observations used in reanalysis, with a multiple-choice question "Have you used reanalysis input observations and feedback data?". Almost half of the respondents admitted to not knowing what such data were about. Among the others, specific queries asked for these observations to be made available more openly. This paper summarizes the main findings in regard to use of existing reanalyses as well as user awareness and needs in regard to reanalysis feedback data and input observations. In the future, the information obtained via the survey makes it possible to perform various statistically robust analyses addressing different aspects of the use of reanalysis data.

1605. Estimation of the time series of the meridional heat transport across 15 degreesN in the Pacific Ocean from Argo and satellite data

Authors: Tingting Yang (Chinese Academy of Sciences); Yongsheng Xu (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jc010752

URL: <https://doi.org/10.1002/2015jc010752>

Publication Date: 2015-04-10

Volume/Issue: Volume 120, Issue 4

References: 42

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: meridional heat transport, altimeter, Argo, North Pacific

Funding: CAS (Chinese Academy of Sciences) "100 Talent" Program, China 973 Project, National Natural Science Foundation of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: The time series of the net meridional heat transport (MHT) at 15 degreesN in the Pacific Ocean from 2003 to 2012 is estimated by combining the Argo profiles with the satellite altimeter and scatterometer data. The estimate is validated against the climatological ocean data and the ECCO2 products, and is demonstrated to be reasonable. The MHT has a high degree of variability with a temporal mean of 0.70 +/- 0.31 PW, which is concentrated in the upper 800 dbar. The time series of the MHT and Ekman temperature transport have a significant annual cycle which peaks near April and December, whereas the time series of the geostrophic temperature transport have a subannual cycle. The results are consistent with previous estimates and lower than the ECCO2 estimate, which may mainly be caused by the different data sources and processings. The correlation between the air-sea flux and MHT is 0.50 with a 3 month delay. This report describes the first such attempt at a continuous transport of heat at 15 degreesN in the Pacific Ocean from in situ observations.

1606. On the origin of the seasonal and interannual T-S variability of the inflow through the Strait of Gibraltar

Authors: Jesús García-Lafuente; Cristina Naranjo; Ricardo Sánchez-Leal; Simone Sammartino; Maria J. Bellanco; Jose C. Sánchez-Garrido; Javier Soto-Navarro

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2015.03.005

URL: <https://doi.org/10.1016/j.dsr.2015.03.005>

Publication Date: 2015-04-10

Volume/Issue: Volume 101

References: 49

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Oceanographic Partnership Program for making ECCO, Spanish Ministry of Economy Competitiveness under Research Project INGRES-3

License: N/A

Open Access: N/A

Abstract: N/A

1607. KALREF--A Kalman filter and time series approach to the International Terrestrial Reference Frame realization

Authors: Xiaoping Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Claudio Abbondanza (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Zuheir Altamimi (Institut National de l'Information Géographique et Forestière Paris France); T. Mike Chin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Xavier Collilieux (Institut National de l'Information Géographique et Forestière Paris France); Richard S. Gross (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael B. Heflin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yan Jiang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Now at Geological Survey of Canada Sidney British Columbia Canada); Jay W. Parker (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jb011622

URL: <https://doi.org/10.1002/2014jb011622>

Publication Date: 2015-04-07

Volume/Issue: Volume 120, Issue 5

References: 53

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration (NASA)

License: N/A

Open Access: N/A

Abstract: The current International Terrestrial Reference Frame is based on a piecewise linear site motion model and realized by reference epoch coordinates and velocities for a global set of stations. Although linear motions due to tectonic plates and glacial isostatic adjustment dominate geodetic signals, at today's millimeter precisions, nonlinear motions due to earthquakes, volcanic activities, ice mass losses, sea level rise, hydrological changes, and other processes become significant. Monitoring these (sometimes rapid) changes desires consistent and precise realization of the terrestrial reference frame (TRF) quasi-instantaneously. Here, we use a Kalman filter and smoother approach to combine time series from four space geodetic techniques to realize an experimental TRF through weekly time series of geocentric coordinates. In addition to secular, periodic, and stochastic components for station coordinates, the Kalman filter state variables also include daily Earth orientation parameters and transformation parameters from input data frames to the combined TRF. Local tie measurements among colocated stations are used at their known or nominal epochs of observation, with comotion constraints applied to almost all colocated stations. The filter/smoother approach unifies different geodetic time series in a single geocentric frame. Fragmented and multitechnique tracking records at

colocation sites are bridged together to form longer and coherent motion time series. While the time series approach to TRF reflects the reality of a changing Earth more closely than the linear approximation model, the filter/smoothing is computationally powerful and flexible to facilitate incorporation of other data types and more advanced characterization of stochastic behavior of geodetic time series.

1608. Development of an Ocean Current Forecast System for the South China Sea

Authors: YonggangWang; Zexun Wei; Zhan Lian; Yongzeng Yang

Journal: Aquatic Procedia

Publisher: Elsevier BV

DOI: 10.1016/j.aqpro.2015.02.206

URL: <https://doi.org/10.1016/j.aqpro.2015.02.206>

Publication Date: 2015-04-06

Volume/Issue: Volume 3

References: 19

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1609. Water mass pathways to the North Atlantic oxygen minimum zone

Authors: Jesús Peña-Izquierdo (Departament d'Oceanografia Física i Tecnològica, Institut de Ciències del Mar, ICM-CSIC Barcelona Spain); Erik van Sebille (Imperial College London, University of New South Wales); Josep L. Pelegrí (Departament d'Oceanografia Física i Tecnològica, Institut de Ciències del Mar, ICM-CSIC Barcelona Spain); Janet Sprintall (Scripps Institution of Oceanography / University of California San Diego); Evan Mason (Instituto Mediterráneo de Estudios Avanzados, IMEDEA-CSIC Esporles Spain); Pedro J. Llanillo (Universidad de Santiago de Chile, Departament d'Oceanografia Física i Tecnològica, Institut de Ciències del Mar, ICM-CSIC Barcelona Spain); Francisco Machín (InvestigAdHoc S.A. Telde Spain)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010557

URL: <https://doi.org/10.1002/2014jc010557>

Publication Date: 2015-04-03

Volume/Issue: Volume 120, Issue 5

References: 63

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Spanish Ministerio de Economía y Competitividad through projects MOC2, TIC-MOC

License: N/A

Open Access: N/A

Abstract: The water mass pathways to the North Atlantic Oxygen Minimum Zone (naOMZ) are traditionally sketched within the cyclonic tropical circulation via the poleward branching from the eastward flowing jets that lie south of 10 degreesN. However, our water mass analysis of historic hydrographic observations together with numerical Lagrangian experiments consistently reveal that the potential density level of $\sigma_\theta = 26.8 \text{ kg m}^{-3}$ ($\sigma_{26.8}$, approximately 300 m depth) separates two distinct regimes of circulation within the Central Water (CW) stratum of the naOMZ. In the upper CW (above $\sigma_{26.8}$), and in agreement with previous studies, the supply of water mainly comes from the south with a predominant contribution of South Atlantic CW. In the lower CW (below $\sigma_{26.8}$), where minimal oxygen content is found, the tropical pathway is instead drastically weakened in favor of a subtropical pathway. More than two thirds of the total water supply to this lower layer takes place north of 10 degreesN, mainly via an eastward flow at 14 degreesN and northern recirculations from the northern subtropical gyre. The existence of these northern jets explains the greater contribution of North Atlantic CW observed in the lower CW, making up to 50% of the water mass at the naOMZ core. The equatorward transfer of mass from the well-ventilated northern subtropical gyre emerges as an essential part of the ventilation of the naOMZ.

1610. Evaluation of a regional ocean reanalysis system for the East Asian Marginal Seas based on the ensemble Kalman filter

Authors: Gwang-Ho Seo; Byoung-Ju Choi; Yang-Ki Cho; Young Ho Kim; Sangil Kim

Journal: Ocean Science Journal

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12601-015-0003-7

URL: <https://doi.org/10.1007/s12601-015-0003-7>

Publication Date: 2015-04-02

Volume/Issue: Volume 50, Issue 1

References: 53

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1611. Steric sea level variability (1993-2010) in an ensemble of ocean reanalyses and objective analyses

Authors: Andrea Storto; Simona Masina; Magdalena Balmaseda; Stéphanie Guinehut; Yan Xue; Tanguy Szekely; Ichiro Fukumori; Gael Forget; You-Soon Chang; Simon A. Good; Armin Köhl; Guillaume Vernieres; Nicolas Ferry; K. Andrew Peterson; David Behringer; Masayoshi Ishii; Shuhei Masuda; Yosuke Fujii; Takahiro Toyoda; Yonghong Yin; Maria Valdivieso; Bernard Barnier; Tim Boyer; Tony Lee; Jérôme Gourrion; Ou Wang; Patrick Heimback; Anthony Rosati; Robin Kovach; Fabrice Hernandez; Matthew J. Martin; Masafumi Kamachi; Tsurane Kuragano; Kristian Mogensen; Oscar Alves; Keith Haines; Xiaochun Wang

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-015-2554-9

URL: <https://doi.org/10.1007/s00382-015-2554-9>

Publication Date: 2015-03-29

Volume/Issue: Volume 49, Issue 3

References: 73

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1612. Impacts of regional mixing on the temperature structure of the equatorial Pacific Ocean. Part 2: Depth-dependent vertical diffusion

Authors: Yanli Jia; Ryo Furue; Julian P. McCreary

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2015.02.007

URL: <https://doi.org/10.1016/j.ocemod.2015.02.007>

Publication Date: 2015-03-26

Volume/Issue: Volume 91

References: 29

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA, Japan Agency for Marine-Earth Science and Technology, NOAA

License: N/A

Open Access: N/A

Abstract: N/A

1613. The reduction of hydrology-induced gravity variations at sites with insufficient hydrological instrumentation

Authors: Michal Mikolaj; Bruno Meurers; Marcel Mojzeš

Journal: Studia Geophysica et Geodaetica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11200-014-0232-8

URL: <https://doi.org/10.1007/s11200-014-0232-8>

Publication Date: 2015-03-24

Volume/Issue: Volume 59, Issue 3

References: 35

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1614. Global climate simulations at 3000-year intervals for the last 21 000 years with the GENMOM coupled atmosphere-ocean model

Authors: J. R. Alder; S. W. Hostetler

Journal: Climate of the Past

Publisher: Copernicus GmbH

DOI: 10.5194/cp-11-449-2015

URL: <https://doi.org/10.5194/cp-11-449-2015>

Publication Date: 2015-03-17

Volume/Issue: Volume 11, Issue 3

References: 117

Cited by: 41 publications

Language: en

Subjects: N/A

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Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We apply GENMOM, a coupled atmosphere-ocean climate model, to simulate eight equilibrium time slices at 3000-year intervals for the past 21 000 years forced by changes in Earth-Sun geometry, atmospheric greenhouse gases (GHGs), continental ice sheets, and sea level. Simulated global cooling during the Last Glacial Maximum (LGM) is 3.8 degreesC and the rate of post-glacial warming is in overall agreement with recently published temperature reconstructions. The greatest rate of warming occurs between 15 and 12 ka (2.4 degreesC over land, 0.7 degreesC over oceans, and 1.4 degreesC globally) in response to changes in radiative forcing from the diminished extent of the Northern Hemisphere (NH) ice sheets and increases in GHGs and NH summer insolation. The modeled LGM and 6 ka temperature and precipitation climatologies are generally consistent with proxy reconstructions, the PMIP2 and PMIP3 simulations, and other paleoclimate data-model analyses. The model does not capture the mid-Holocene "thermal maximum" and gradual cooling to preindustrial (PI) global temperature found in the data. Simulated monsoonal precipitation in North Africa peaks between 12 and 9 ka at values ~ 50% greater than those of the PI, and Indian monsoonal precipitation peaks at 12 and 9 ka at values ~ 45% greater than the PI. GENMOM captures the reconstructed LGM extent of NH and Southern Hemisphere (SH) sea ice. The simulated present-day Antarctica Circumpolar Current (ACC) is ~ 48% weaker than the observed (62 versus 119 Sv). The simulated present-day Atlantic Meridional Overturning Circulation (AMOC) of 19.3 +/- 1.4 Sv on the Bermuda Rise (33 degrees N) is comparable with observed value of 18.7 +/- 4.8 Sv. AMOC at 33 degrees N is reduced by ~ 15% during the LGM, and the largest post-glacial increase (~ 11%) occurs during the 15 ka time slice.

1615. Strengthening Kuroshio observed at its origin during November 2010 to October 2012

Authors: Zhaohui Chen (Ocean University of China); Lixin Wu (Ocean University of China); Bo Qiu (University of Hawaii at Manoa); Lei Li (Ocean University of China); Dunxin Hu (Chinese Academy of Sciences); Chengyan Liu (Nanjing University); Fan Jia (Chinese Academy of Sciences); Xi Liang (National Marine Environmental Forecasting Center)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010590

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Publication Date: 2015-03-13

Volume/Issue: Volume 120, Issue 4

References: 26

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: origin, increase, Kuroshio

Funding: National Science Foundation of China, National Basic Research Program of China, Global change project

License: N/A

Open Access: N/A

Abstract: Direct measurements of Kuroshio at its origin (18 degreesN, east of the Luzon Island) are conducted from November 2010 to October 2012. It is found that the depth-averaged Kuroshio between 200 and 700 m has increased over 15 cm s⁻¹ during the 2 year observational period and it is accompanied by the pronounced southward shift of the North Equatorial Current (NEC) bifurcation. Further analysis indicates that the Kuroshio's strengthening is confined to the upstream segment east of the Luzon Island while the Kuroshio decreased as it passed the Luzon Strait due to a dipole-like sea surface height (SSH) trend between 15 degreesN and 23 degreesN. It is demonstrated that the 2 year strengthening of the Kuroshio, as well as the dipole-like SSH trend can be adequately reproduced by a 1.5 layer nonlinear reduced gravity model, suggesting an important role of upper ocean response to low-frequency wind forcing in the western Pacific. Salinity at 500 m depth is also found to increase during the concurrent 2 years. This subthermocline salinity increase is a combined outcome of vertical (basin-scale isopycnal surface movement) and horizontal advections (i.e., strengthened Kuroshio) due to changes in the large-scale wind-driven ocean circulation.

1616. Vertical Redistribution of Oceanic Heat Content

Authors: Xinfeng Liang (Massachusetts Institute of Technology); Carl Wunsch (Harvard University); Patrick Heimbach (Massachusetts Institute of Technology); Gael Forget (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-14-00550.1

URL: <https://doi.org/10.1175/jcli-d-14-00550.1>

Publication Date: 2015-03-10

Volume/Issue: Volume 28, Issue 9

References: 27

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Estimated values of recent oceanic heat uptake are on the order of a few tenths of a W m^{-2} , and are a very small residual of air-sea exchanges, with annual average regional magnitudes of hundreds of W m^{-2} . Using a dynamically consistent state estimate, the redistribution of heat within the ocean is calculated over a 20-yr period. The 20-yr mean vertical heat flux shows strong variations in both the lateral and vertical directions, consistent with the ocean being a dynamically active and spatially complex heat exchanger. Between mixing and advection, the two processes determining the vertical heat transport in the deep ocean, advection plays a more important role in setting the spatial patterns of vertical heat exchange and its temporal variations. The global integral of vertical heat flux shows an upward heat transport in the deep ocean, suggesting a cooling trend in the deep ocean. These results support an inference that the near-surface thermal properties of the ocean are a consequence, at least in part, of internal redistributions of heat, some of which must reflect water that has undergone long trajectories since last exposure to the atmosphere. The small residual heat exchange with the atmosphere today is unlikely to represent the interaction with an ocean that was in thermal equilibrium at the start of global warming. An analogy is drawn with carbon-14 "reservoir ages," which range from over hundreds to a thousand years.

1617. Available Potential Energy and the General Circulation: Partitioning Wind, Buoyancy Forcing, and Diapycnal Mixing

Authors: Varvara E. Zemskova (Marine Sciences Department, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina); Brian L. White (Marine Sciences Department, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina); Alberto Scotti (Marine Sciences Department, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0043.1

URL: <https://doi.org/10.1175/jpo-d-14-0043.1>

Publication Date: 2015-03-10

Volume/Issue: Volume 45, Issue 6

References: 45

Cited by: 24 publications

Language: N/A

Subjects: N/A

Keywords: Buoyancy, Circulation/ Dynamics, Models and modeling, Diapycnal mixing, Ocean circulation (and 3 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The ocean energy cycle is calculated using a new available potential energy (APE) decomposition, which partitions adiabatic buoyancy fluxes from diapycnal mixing, applied to results from the Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2), eddy-permitting ocean state estimate and observed surface buoyancy fluxes from the WHOI OAFlux project. Compared with the traditional Lorenz energy cycle, this framework provides a more accurate estimate of the background potential energy (PE) of the global oceans and the surface generation and interior fluxes of APE. Calculations of the global energy budget using 16 yr of ECCO2 output suggest that the adiabatic portion of the general circulation is maintained by a balance between the mean wind-driven upwelling that increases APE (+0.27 TW) and time-fluctuating processes, including mesoscale eddies, which release APE (-0.27 TW). The APE generated by surface buoyancy fluxes (0.46 TW) is comparable to the generation by the mean winds. The global rate of irreversible mixing (0.46 TW), which balances surface APE generation, is consistent with previous estimates of the diapycnal fluxes associated with maintaining deep stratification (see Munk and Wunsch) and a global diapycnal diffusivity of $O(1 \times 10^{-4})$ m² s⁻¹. However, the net contribution of diapycnal mixing to the total potential energy is negligible, which suggests that mixing, contrary to one current paradigm, does not place a global demand on kinetic energy dissipation. However, there are regions where mixing is significant, for example, between 3000 and 5000 m (in ECCO2), where mixing increases PE by 0.1 TW. The work provides a new framework for separating adiabatic-diabatic fluxes and for monitoring the global rate of diapycnal mixing rate using measurable surface properties such as SST and heat flux.

1618. Optimizing velocities and transports for complex coastal regions and archipelagos

Authors: Patrick J. Haley; Arpit Agarwal; Pierre F.J. Lermusiaux

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2015.02.005

URL: <https://doi.org/10.1016/j.ocemod.2015.02.005>

Publication Date: 2015-03-06

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References: 95

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Naval Research, Naval Research Laboratory

License: N/A

Open Access: N/A

Abstract: N/A

1619. Bottom-pressure signature of annual baroclinic Rossby waves in the northeast tropical Pacific Ocean

Authors: Christopher G. Piecuch (Atmospheric and Environmental Research Inc.)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010667

URL: <https://doi.org/10.1002/2014jc010667>

Publication Date: 2015-03-02

Volume/Issue: Volume 120, Issue 4

References: 36

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: annual cycle, GRACE, tropical Pacific, ocean mass

Funding: NASA

License: N/A

Open Access: N/A

Abstract: The annual cycle in bottom pressure (pb) in the northeast tropical Pacific Ocean (NTPO) is studied. Focus is on a zonal section along 12 degreesN between 105 degreesW and 145 degreesW that is characterized by a strong annual cycle in sea level from satellite altimetry. Estimates of pb from the Gravity Recovery and Climate Experiment (GRACE), a state estimate produced by the Estimating the Circulation and Climate of the Ocean (ECCO) consortium, and a linear Rossby wave model (LRWM) are used. The GRACE NTPO pb annual cycle shows amplitudes as large as 1 cm water equivalent. The GRACE data also evidence westward propagation that is consistent with the behavior of long mode-1 Rossby waves at this latitude, with phase increasing from east to west at a rate of ~ 0.34 m s⁻¹. The ECCO and LRWM pb estimates corroborate the notion that GRACE reveals the pb signature of annual Rossby waves driven by interior wind stress curl and possibly damped by frictional processes. Results have implications for attempts to constrain global ocean mass using a single point mooring.

1620. Applications and Current Challenges of Supercomputing across Multiple Domains of Computational Sciences

Authors: Neha Gupta (Northeastern University, USA & Osmania University, India)

Journal: Advances in Systems Analysis, Software Engineering, and High Performance Computing

Publisher: IGI Global

DOI: 10.4018/978-1-4666-7461-5.ch003

URL: <https://doi.org/10.4018/978-1-4666-7461-5.ch003>

Publication Date: 2015-02-27

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References: 29

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Supercomputing is a contemporary solution to enhance the speed of calculations in nanoseconds. Presently, there are different aspects of supercomputing like Cloud Computing, High Performance Computing, Grid Computing, etc. provided by companies like Amazon (Amazon Web Services), Windows (Azure), Google (Google Cloud Platform). Supercomputers play an important role in the field of Computer Science and are used for a wide range of computationally intensive tasks across domains like Bioinformatics, Computational Earth and Atmospheric Sciences, Computational Materials Sciences and Engineering, Computational Chemistry, Computational Fluid Dynamics, Computational Physics, Computational and Data Enabled Social Sciences, Aerospace, Manufacturing, Industrial Applications, Computational Medicine, and Biomedical Engineering. However, there are a lot of issues that need to be solved to develop next generation supercomputers. In this chapter, the potential applications and current challenges of supercomputing across these domains are explained in detail. The current status of supercomputing and limitations are discussed, which forms the basis for future work in these areas. The future ideas that can be applied efficiently with the availability of good computing resources are explained coherently in this chapter.

1621. Improved methods for observing Earth's time variable mass distribution with GRACE using spherical cap mascons

Authors: Michael M. Watkins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David N. Wiese (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dah-Ning Yuan (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Carmen Boening (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Felix W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jb011547

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Publication Date: 2015-02-26

Volume/Issue: Volume 120, Issue 4

References: 75

Cited by: 877 publications

Language: en

Subjects: N/A

Keywords: mass concentrations, GRACE, time variable gravity, mascons

Funding: Jet Propulsion Laboratory, California Institute of Technology, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: We discuss several classes of improvements to gravity solutions from the Gravity Recovery and Climate Experiment (GRACE) mission. These include both improvements in background geophysical models and orbital parameterization leading to the unconstrained spherical harmonic solution JPL RL05, and an alternate JPL RL05M mass concentration (mascon) solution benefitting from those same improvements but derived in surface spherical cap mascons. The mascon basis functions allow for convenient application of a priori information derived from near-global geophysical models to prevent striping in the solutions. The resulting mass flux solutions are shown to suffer less from leakage errors than harmonic solutions, and do not necessitate empirical filters to remove north-south stripes, lowering the dependence on using scale factors (the global mean scale factor decreases by 0.17) to gain accurate mass estimates. Ocean bottom pressure (OBP) time series derived from the mascon solutions are shown to have greater correlation with in situ data than do spherical harmonic solutions (increase in correlation coefficient of 0.08 globally), particularly in low-latitude regions with small signal power (increase in correlation coefficient of 0.35 regionally), in addition to reducing the error RMS with respect to the in situ data (reduction of 0.37 cm globally, and as much as 1 cm regionally). Greenland and Antarctica mass balance estimates derived from the mascon solutions agree within formal uncertainties with previously published results. Computing basin averages for hydrology applications shows general agreement between harmonic and mascon solutions for large basins; however, mascon solutions typically have greater resolution for smaller spatial regions, in particular when studying secular signals.

1622. Electromagnetic Coupling of Ocean Flow with the Earth System

Authors: Robert Tyler

Journal: Terrestrial, Atmospheric and Oceanic Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.3319/tao.2014.08.19.04(grt)

URL: [https://doi.org/10.3319/tao.2014.08.19.04\(grt\)](https://doi.org/10.3319/tao.2014.08.19.04(grt))

Publication Date: 2015-02-26

Volume/Issue: Volume 26, Issue 1

References: N/A

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: Earth system science|Momentum (technical analysis)|Momentum transfer

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1623. New Magnetic Anomaly Map of the East Asia with Some Preliminary Tectonic Interpretations

Authors: Wen-Bin Doo; Shu-Kun Hsu; Leo Armada

Journal: Terrestrial, Atmospheric and Oceanic Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.3319/tao.2014.08.19.07(grt)

URL: [https://doi.org/10.3319/tao.2014.08.19.07\(grt\)](https://doi.org/10.3319/tao.2014.08.19.07(grt))

Publication Date: 2015-02-26

Volume/Issue: Volume 26, Issue 1

References: N/A

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: Anomaly (physics)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1624. Carbon dynamics of the Weddell Gyre, Southern Ocean

Authors: Peter J. Brown (University of East Anglia, National Oceanography Centre, British Antarctic Survey Cambridge UK); Loïc Jullion (University of Southampton, Institut de Recherche pour le Développement (IRD), National Oceanography Centre); Peter Landschützer (University of East Anglia, ETH Zurich); Dorothee C. E. Bakker (University of East Anglia); Alberto C. Naveira Garabato (University of Southampton, National Oceanography Centre); Michael P. Meredith (Scottish Association for Marine Science Oban UK, British Antarctic Survey Cambridge UK); Sinhue Torres-Valdés (National Oceanography Centre); Andrew J. Watson (University of Exeter); Mario Hoppema (Alfred Wegener Institute); Brice Loose (University of Rhode Island); Elizabeth M. Jones (University of Groningen, Royal Netherlands Institute for Sea Research, Royal Netherlands Institute for Sea Research (NIOZ)); Maciej Telszewski (International Ocean Carbon Coordination Project Institute of Oceanology of Polish Academy of Sciences Sopot Poland); Steve D. Jones (University of East Anglia); Rik Wanninkhof (Ocean Chemistry Division, Atlantic Oceanographic and Meteorological Laboratory National Oceanic and Atmospheric Administration Miami Florida USA)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gb005006

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Publication Date: 2015-02-19

Volume/Issue: Volume 29, Issue 3

References: 92

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: air-sea exchange of CO₂, observations, Southern Ocean, Weddell Gyre, neural network (and 1 more)

Funding: Natural Environment Research Council, Sixth Framework Programme, Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: The accumulation of carbon within the Weddell Gyre and its exchanges across the gyre boundaries are investigated with three recent full-depth oceanographic sections enclosing this climatically important region. The combination of carbon measurements with ocean circulation transport estimates from a box inverse analysis reveals that deepwater transports associated with Warm Deep Water (WDW) and Weddell Sea Deep Water dominate the gyre's carbon budget, while a dual-cell vertical overturning circulation leads to both upwelling and the delivery of large quantities of carbon to the deep ocean. Historical sea surface pCO₂ observations, interpolated using a neural network technique, confirm the net summertime sink of 0.044 to 0.058 +/- 0.010 Pg C yr⁻¹ derived from the inversion. However, a wintertime outgassing signal similar in size results in a statistically insignificant annual air-to-sea CO₂ flux of 0.002 +/- 0.007 Pg C yr⁻¹ (mean 1998-2011) to 0.012 +/- 0.024 Pg C yr⁻¹ (mean 2008-2010) to be diagnosed for the Weddell Gyre. A surface layer carbon balance, independently derived from in situ biogeochemical measurements, reveals that freshwater inputs and biological drawdown decrease surface ocean inorganic carbon levels more than they are increased by

WDW entrainment, resulting in an estimated annual carbon sink of $0.033 \pm 0.021 \text{ Pg C yr}^{-1}$. Although relatively less efficient for carbon uptake than the global oceans, the summertime Weddell Gyre suppresses the winter outgassing signal, while its biological pump and deepwater formation act as key conduits for transporting natural and anthropogenic carbon to the deep ocean where they can reside for long time scales.

1625. Primary discussion of a carbon sink in the oceans

Authors: Caihua Ma; Kui You; Dechun Ji; Weiwei Ma; Fengqi Li

Journal: Journal of Ocean University of China

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11802-015-2548-6

URL: <https://doi.org/10.1007/s11802-015-2548-6>

Publication Date: 2015-02-19

Volume/Issue: Volume 14, Issue 2

References: 40

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1626. A near-uniform fluctuation of ocean bottom pressure and sea level across the deep ocean basins of the Arctic Ocean and the Nordic Seas

Authors: Ichiro Fukumori; Ou Wang; William Llovel; Ian Fenty; Gael Forget

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2015.01.013

URL: <https://doi.org/10.1016/j.pocean.2015.01.013>

Publication Date: 2015-02-18

Volume/Issue: Volume 134

References: 49

Cited by: 52 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1627. Eddy-driven sediment transport in the Argentine Basin: Is the height of the Zapiola Rise hydrodynamically controlled?

Authors: Wilbert Weijer (Los Alamos National Laboratory); Mathew E. Maltrud (Los Alamos National Laboratory); William B. Homoky (University of Oxford); Kurt L. Polzin (Woods Hole Oceanographic Institution); Leo R. M. Maas (Royal Netherlands Institute for Sea Research)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010573

URL: <https://doi.org/10.1002/2014jc010573>

Publication Date: 2015-02-18

Volume/Issue: Volume 120, Issue 3

References: 82

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Nuclear Security Administration, U.S. Department of Energy

License: Creative Commons

Open Access: N/A

Abstract: In this study, we address the question whether eddy-driven transports in the Argentine Basin can be held responsible for enhanced sediment accumulation over the Zapiola Rise, hence accounting for the existence and growth of this sediment drift. To address this question, we perform a 6 year simulation with a strongly eddying ocean model. We release two passive tracers, with settling velocities that are consistent with silt and clay size particles. Our experiments show contrasting behavior between the silt fraction and the lighter clay. Due to its larger settling velocity, the silt fraction reaches a quasisteady state within a few years, with abyssal sedimentation rates that match net input. In contrast, clay settles only slowly, and its distribution is heavily stratified, being transported mainly along isopycnals. Yet, both size classes display a significant and persistent concentration minimum over the Zapiola Rise. We show that the Zapiola Anticyclone, a strong eddy-driven vortex that circulates around the Zapiola Rise, is a barrier to sediment transport, and hence prevents significant accumulation of sediments on the Rise. We conclude that sediment transport by the turbulent circulation in the Argentine Basin alone cannot account for the preferred sediment accumulation over the Rise. We speculate that resuspension is a critical process in the formation and maintenance of the Zapiola Rise.

1628. Regionally coupled atmosphere-ocean-sea ice-marine biogeochemistry model ROM: 1. Description and validation

Authors: Dmitry V. Sein (Alfred Wegener Institute); Uwe Mikolajewicz (Max Planck Institute for Meteorology); Matthias Gröger (Swedish Meteorological and Hydrological Institute Norrköping Sweden); Irina Fast (German Climate Computing Center Hamburg Germany); William Cabos (Department of Physics University of Alcala Madrid Spain); Joaquim G. Pinto (University of Cologne, University of Reading); Stefan Hagemann (Max Planck Institute for Meteorology); Tido Semmler (Alfred Wegener Institute); Alfredo Izquierdo (University of Cadiz, Marine Science and Technological Center of Andalusia Cadiz Spain); Daniela Jacob (Climate Service Center 2.0 Hamburg Germany)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014ms000357

URL: <https://doi.org/10.1002/2014ms000357>

Publication Date: 2015-02-12

Volume/Issue: Volume 7, Issue 1

References: 117

Cited by: 124 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: German Federal Ministry of Education and Research, NORDATLANTIK, MiKlip-DEPARTURE (and 2 more)

License: Creative Commons

Open Access: N/A

Abstract: The general circulation models used to simulate global climate typically feature resolution too coarse to reproduce many smaller-scale processes, which are crucial to determining the regional responses to climate change. A novel approach to downscale climate change scenarios is presented which includes the interactions between the North Atlantic Ocean and the European shelves as well as their impact on the North Atlantic and European climate. The goal of this paper is to introduce the global ocean-regional atmosphere coupling concept and to show the potential benefits of this model system to simulate present-day climate. A global ocean-sea ice-marine biogeochemistry model (MPIOM/HAMOCC) with regionally high horizontal resolution is coupled to an atmospheric regional model (REMO) and global terrestrial hydrology model (HD) via the OASIS coupler. Moreover, results obtained with ROM using NCEP/NCAR reanalysis and ECHAM5/MPIOM CMIP3 historical simulations as boundary conditions are presented and discussed for the North Atlantic and North European region. The validation of all the model components, i.e., ocean, atmosphere, terrestrial hydrology, and ocean biogeochemistry is performed and discussed. The careful and detailed validation of ROM provides evidence that the proposed model system improves the simulation of many aspects of the regional climate, remarkably the ocean, even though some biases persist in other model components, thus leaving potential for future improvement. We conclude that ROM is a powerful tool to estimate possible impacts of climate change on the regional scale.

1629. A new river discharge and river temperature climatology data set for the pan-Arctic region

Authors: Jonathan Whitefield; Peter Winsor; James McClelland; Dimitris Menemenlis

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2014.12.012

URL: <https://doi.org/10.1016/j.ocemod.2014.12.012>

Publication Date: 2015-02-07

Volume/Issue: Volume 88

References: 50

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NPRB Graduate Student Research Award, UAF Center for Global Change Student Research Grant, HPC

License: N/A

Open Access: N/A

Abstract: N/A

1630. Potential impact of climate change on the Intra-Americas Sea: Part 2. Implications for Atlantic bluefin tuna and skipjack tuna adult and larval habitats

Authors: Barbara A. Muhling; Yanyun Liu; Sang-Ki Lee; John T. Lamkin; Mitchell A. Roffer; Frank Muller-Karger; John F. Walter

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2015.01.010

URL: <https://doi.org/10.1016/j.jmarsys.2015.01.010>

Publication Date: 2015-02-07

Volume/Issue: Volume 148

References: 67

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: the National Aeronautics and Space Administration (NASA), NOAA Atlantic Oceanographic and Meteorological Laboratory (AOML)

License: N/A

Open Access: N/A

Abstract: N/A

1631. Determining the Origins of Advective Heat Transport Convergence Variability in the North Atlantic

Authors: Martha W. Buckley (Atmospheric and Environmental Research, Lexington, Massachusetts); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts); Gaël Forget (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-14-00579.1

URL: <https://doi.org/10.1175/jcli-d-14-00579.1>

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Volume/Issue: Volume 28, Issue 10

References: 32

Cited by: 44 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A recent state estimate covering the period 1992-2010 from the Estimating the Circulation and Climate of the Ocean (ECCO) project is utilized to quantify the roles of air-sea heat fluxes and advective heat transport convergences in setting upper-ocean heat content anomalies H in the North Atlantic Ocean on monthly to interannual time scales. Anomalies in (linear) advective heat transport convergences, as well as Ekman and geostrophic contributions, are decomposed into parts that are due to velocity variability, temperature variability, and their covariability. Ekman convergences are generally dominated by variability in Ekman mass transports, which reflect the instantaneous response to local wind forcing, except in the tropics, where variability in the temperature field plays a significant role. In contrast, both budget analyses and simple dynamical arguments demonstrate that geostrophic heat transport convergences that are due to temperature and velocity variability are anticorrelated, and thus their separate treatment is not insightful. In the interior of the subtropical gyre, the sum of air-sea heat fluxes and Ekman heat transport convergences is a reasonable measure of local atmospheric forcing, and such forcing explains the majority of H variability on all time scales resolved by ECCO. In contrast, in the Gulf Stream region and subpolar gyre, ocean dynamics are found to be important in setting H on interannual time scales. Air-sea heat fluxes damp anomalies created by the ocean and thus are not set by local atmospheric variability.

1632. The tilt of mean sea level along the east coast of North America

Authors: S. Higginson (Bedford Institute of Oceanography); K. R. Thompson (Dalhousie University); P. L. Woodworth (National Oceanography Centre); C. W. Hughes (University of Liverpool, National Oceanography Centre)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015gl063186

URL: <https://doi.org/10.1002/2015gl063186>

Publication Date: 2015-02-06

Volume/Issue: Volume 42, Issue 5

References: 46

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: sea level tilt, tide gauges, ocean model, geoid

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Open Access: N/A

Abstract: The tilt of mean sea level along the North American east coast has been a subject of debate for many decades. Improvements in geoid and ocean circulation models, and GPS positioning of tide gauge benchmarks, provide an opportunity to produce new tilt estimates. Tilts estimated using tide gauge measurements referenced to high-resolution geoid models (the geodetic approach) and ocean circulation models (the ocean approach) are compared. The geodetic estimates are broadly similar, with tilts downward to the north through the Florida Straits and at Cape Hatteras. Estimates from the ocean approach show good agreement with the geodetic estimates, indicating a convergence of the two approaches and resolving the long standing debate as to the sign of the tilt. These tilts differ from those used by Yin and Goddard (2013) to support a link between changing ocean circulation and coastal sea level rise.

1633. A wind-driven nonseasonal barotropic fluctuation of the Canadian inland seas

Authors: C. G. Piecuch; R. M. Ponte

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-11-175-2015

URL: <https://doi.org/10.5194/os-11-175-2015>

Publication Date: 2015-02-06

Volume/Issue: Volume 11, Issue 1

References: 58

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Division of Ocean Sciences, National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: . A wind-driven, spatially coherent mode of nonseasonal, depth-independent variability in the Canadian inland seas (i.e., the collective of Hudson Bay, James Bay, and Foxe Basin) is identified based on Gravity Recovery and Climate Experiment (GRACE) retrievals, a tide-gauge record, and a barotropic model over 2003-2013. This dominant mode of nonseasonal variability is correlated with the North Atlantic Oscillation and is associated with net flows into and out of the Canadian inland seas; the anomalous inflows and outflows, which are reflected in mean sea level and bottom pressure changes, are driven by wind stress anomalies over Hudson Strait, probably related to wind setup, as well as over the northern North Atlantic Ocean, possibly mediated by various wave mechanisms. The mode is also associated with mass redistribution within the Canadian inland seas, reflecting linear response to local wind stress variations under the combined influences of rotation, gravity, and variable bottom topography. Results exemplify the usefulness of GRACE for studying regional ocean circulation and climate.

1634. Roles of initial ocean surface and subsurface states on successfully predicting 2006-2007 El Niño with an intermediate coupled model

Authors: F. Zheng; J. Zhu

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-11-187-2015

URL: <https://doi.org/10.5194/os-11-187-2015>

Publication Date: 2015-02-06

Volume/Issue: Volume 11, Issue 1

References: 44

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The 2006-2007 El Niño event, an unusually weak event, was predicted by most models only after the warming in the eastern Pacific had commenced. In this study, on the basis of an El Niño prediction system, roles of the initial ocean surface and subsurface states on predicting the 2006-2007 El Niño event are investigated to determine conditions favorable for predicting El Niño growth and are isolated in three sets of hindcast experiments. The hindcast is initialized through assimilation of only the sea surface temperature (SST) observations to optimize the initial surface condition, only the sea level (SL) data to update the initial subsurface state, or both the SST and SL data. Results highlight that the hindcasts with three different initial states can all successfully predict the 2006-2007 El Niño event 1 year in advance and that the hindcast initialized by both the SST and SL data performs best. A comparison between the various sets of hindcast results further demonstrates that successful prediction is more significantly affected by the initial subsurface state than by the initial surface condition. The accurate initial surface state can trigger the easier prediction of the 2006-2007 El Niño, whereas a more reasonable initial subsurface state can contribute to improving the prediction in the growth of the warm event.

1635. Impact of assimilating physical oceanographic data on modeled ecosystem dynamics in the California Current System

Authors: Kaustubha Raghukumar; Christopher A. Edwards; Nicole L. Goebel; Gregoire Broquet; Milena Veneziani; Andrew M. Moore; Jon P. Zehr

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2015.01.004

URL: <https://doi.org/10.1016/j.pocean.2015.01.004>

Publication Date: 2015-02-04

Volume/Issue: Volume 138

References: 53

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Gordon and Betty Moore Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1636. Internal and external forcing of sea level variability in the Black Sea

Authors: Denis L. Volkov; Felix W. Landerer

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-015-2498-0

URL: <https://doi.org/10.1007/s00382-015-2498-0>

Publication Date: 2015-02-02

Volume/Issue: Volume 45, Issue 9-10

References: 34

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1637. The role of atmospheric uncertainty in Arctic summer sea ice data assimilation and prediction

Authors: Qinghua Yang (National Marine Environmental Forecasting Center, Alfred Wegener Institute); Svetlana N. Losa (Alfred Wegener Institute); Martin Losch (Alfred Wegener Institute); Thomas Jung (University of Bremen, Alfred Wegener Institute); Lars Nerger (Alfred Wegener Institute)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2523

URL: <https://doi.org/10.1002/qj.2523>

Publication Date: 2015-01-28

Volume/Issue: Volume 141, Issue 691

References: 42

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: German Federal Ministry of Education and Research, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: The role of atmospheric uncertainty for the assimilation and prediction of Arctic sea ice is explored by running the Massachusetts Institute of Technology general circulation model (MITgcm) in data assimilation (DA) and prediction mode for summer 2010. The atmospheric ensemble forcing is taken from the UK Met Office (UKMO) system available through the TIGGE (THORPEX Interactive Grand Global Ensemble) database. The DA system is based on a Local Singular Evolutive Interpolated Kalman (LSEIK) filter, and Special Sensor Microwave Imager/Sounder (SSMIS) sea ice concentration operational products from the National Snow and Ice Data Center (NSIDC) are assimilated. Two kinds of experiments are carried out differing in the LSEIK configuration and forcing used: the first one uses a single deterministic control forcing and a forgetting factor necessary to inflate the ensemble spread in the DA phase; the second one uses 23 members from the UKMO atmospheric ensemble prediction system, thereby avoiding any additional ensemble inflation and making further tuning unnecessary. With both systems the model data misfit improves as expected, but the ensemble approach outperforms the deterministic filter. The ice concentration of 24 h forecasts is consistently closer to observations with the ensemble approach, because a larger and more realistic ensemble spread, representing model uncertainty, leads to a better adjustment. Fifteen-day forecasts are also better with ensemble forcing than with deterministic forcing, because of both the larger spread and the better initial state in the ensemble forced system. The ensemble forcing can also improve poor initial states obtained with the deterministic control forcing, because the ensemble forcing introduces a larger spread that spans a larger range of model simulations. Ice thickness forecasts cannot be significantly improved with the ensemble forcing.

1638. Investigating Solution Convergence in a Global Ocean Model Using a 2048-Processor Cluster of Distributed Shared Memory Machines

Authors: Chris Hill; Dimitris Menemenlis; Bob Ciotti; Chris Henze

Journal: Scientific Programming

Publisher: Wiley

DOI: 10.1155/2007/458463

URL: <https://doi.org/10.1155/2007/458463>

Publication Date: 2015-01-12

Volume/Issue: Volume 15, Issue 2

References: N/A

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: Up to 1920 processors of a cluster of distributed shared memory machines at the NASA Ames Research Center are being used to simulate ocean circulation globally at horizontal resolutions of 1/4, 1/8, and 1/16-degree with the Massachusetts Institute of Technology General Circulation Model, a finite volume code that can scale to large numbers of processors. The study aims to understand physical processes responsible for skill improvements as resolution is increased and to gain insight into what resolution is sufficient for particular purposes. This paper focuses on the computational aspects of reaching the technical objective of efficiently performing these global eddy-resolving ocean simulations. At 1/16-degree resolution the model grid contains 1.2 billion cells. At this resolution it is possible to simulate approximately one month of ocean dynamics in about 17 hours of wallclock time with a model timestep of two minutes on a cluster of four 512-way NUMA Altix systems. The Altix systems' large main memory and I/O subsystems allow computation and disk storage of rich sets of diagnostics during each integration, supporting the scientific objective to develop a better understanding of global ocean circulation model solution convergence as model resolution is increased.

1639. Impact of Anomalous Northward Oceanic Heat Transport on Global Climate in a Slab Ocean Setting

Authors: Blandine L'Hévéder (Laboratoire de Météorologie Dynamique, L'Institut Pierre-Simon Laplace, Centre National de la Recherche Scientifiques, Ecole Normale Supérieure, and Université Pierre et Marie Curie, Paris, France); Francis Codron (Laboratoire de Météorologie Dynamique, L'Institut Pierre-Simon Laplace, Centre National de la Recherche Scientifiques, Ecole Normale Supérieure, and Université Pierre et Marie Curie, Paris, France); Michael Ghil (University of California Los Angeles)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-14-00377.1

URL: <https://doi.org/10.1175/jcli-d-14-00377.1>

Publication Date: 2015-01-12

Volume/Issue: Volume 28, Issue 7

References: 41

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper explores the impact of anomalous northward oceanic heat transport on global climate in a slab ocean setting. To that end, the GCM LMDZ5A of the Laboratoire de Météorologie Dynamique is coupled to a slab ocean, with realistic zonal asymmetries and seasonal cycle. Two simulations with different anomalous surface heating are imposed: 1) uniform heating over the North Atlantic basin and 2) concentrated heating in the Gulf Stream region, with a compensating uniform cooling in the Southern Ocean in both cases. The magnitudes of the heating and of the implied northward interhemispheric heat transport are within the range of current natural variability. Both simulations show global effects that are particularly strong in the tropics, with a northward shift of the intertropical convergence zone (ITCZ) toward the heating anomalies. This shift is accompanied by a northward shift of the storm tracks in both hemispheres. From the comparison between the two simulations with different anomalous surface heating in the North Atlantic, it emerges that the global climate response is nearly insensitive to the spatial distribution of the heating. The cloud response acts as a large positive feedback on the oceanic forcing, mainly because of the low-cloud-induced shortwave anomalies in the extratropics. While previous literature has speculated that the extratropical Q flux may impact the tropics by the way of the transient eddy fluxes, it is explicitly demonstrated here. In the midlatitudes, the authors find a systematic northward shift of the jets, as well as of the associated Ferrel cells, storm tracks, and precipitation bands.

1640. The effect of Mediterranean exchange flow on European time mean sea level

Authors: Chris W. Hughes (University of Liverpool, National Oceanography Centre); Rory J. Bingham (University of Bristol); Vassil Roussenov (University of Liverpool); Joanne Williams (National Oceanography Centre); Philip L. Woodworth (National Oceanography Centre)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gl062654

URL: <https://doi.org/10.1002/2014gl062654>

Publication Date: 2015-01-11

Volume/Issue: Volume 42, Issue 2

References: 37

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NERC, European Space Agency, National Oceanographic Partnership Program (NOPP)

License: CC BY 4.0

Open Access: N/A

Abstract: Using a suite of ocean model simulations and a set of dedicated twin experiments, we show that the exchange flow between the Mediterranean and the North Atlantic leads to a drop in time mean European coastal sea level along the Atlantic coast north of Gibraltar. The drop is about 7 cm along the Portuguese coast and remains apparent (though reduced) as far north as the Norwegian coast. We also show that Mediterranean time and spatial mean sea level is about 9 cm lower than it would be without the exchange flow (but assuming a small supply from the Atlantic to balance evaporation). Each of these relationships makes possible an estimate of the magnitude of the exchange flow based on sea level measurements, and estimates of 0.8 and 0.91 sverdrups are made consistent with previous determinations based mainly on current measurements in the Strait of Gibraltar.

1641. Accuracy enhancement of GPS time series using principal component analysis and block spatial filtering

Authors: Xiaoxing He; Xianghong Hua; Kegen Yu; Wei Xuan; Tieding Lu; W. Zhang; X. Chen

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2014.12.016

URL: <https://doi.org/10.1016/j.asr.2014.12.016>

Publication Date: 2015-01-08

Volume/Issue: Volume 55, Issue 5

References: 48

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Key Laboratory of Precise Engineering and Industry Surveying of National Administration of Surveying Mapping and Geoinformation

License: N/A

Open Access: N/A

Abstract: N/A

1642. Spatial trend patterns in the Pacific Ocean sea level during the altimetry era: the contribution of thermocline depth change and internal climate variability

Authors: H. Palanisamy; A. Cazenave; T. Delcroix; B. Meyssignac

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-014-0805-7

URL: <https://doi.org/10.1007/s10236-014-0805-7>

Publication Date: 2015-01-07

Volume/Issue: Volume 65, Issue 3

References: 64

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1643. Operational protocol for the sighting and tracking of Portuguese man-of-war in the southeastern Bay of Biscay: Observations and modeling

Authors: L. Ferrer; N. Zaldúa-Mendizabal; A. Del Campo; J. Franco; J. Mader; U. Cotano; I. Fraile; A. Rubio; Ad. Uriarte; A. Caballero

Journal: Continental Shelf Research

Publisher: Elsevier BV

DOI: 10.1016/j.csr.2014.12.011

URL: <https://doi.org/10.1016/j.csr.2014.12.011>

Publication Date: 2014-12-31

Volume/Issue: Volume 95

References: 54

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Department of Education, Linguistic Policy and Culture of the Basque Government, Department of Industry, Innovation, Commerce and Tourism of the Basque Government

License: N/A

Open Access: N/A

Abstract: N/A

1644. Natural variability of surface oceanographic conditions in the offshore Gulf of Mexico

Authors: Frank E. Muller-Karger; Joseph P. Smith; Sandra Werner; Robert Chen; Mitchell Roffer; Yanyun Liu; Barbara Muhling; David Lindo-Atichati; John Lamkin; Sergio Cerdeira-Estrada; David B. Enfield

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2014.12.007

URL: <https://doi.org/10.1016/j.pocean.2014.12.007>

Publication Date: 2014-12-29

Volume/Issue: Volume 134

References: 144

Cited by: 130 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Ocean Partnership Program

License: N/A

Open Access: N/A

Abstract: N/A

1645. Representations of the Nordic Seas overflows and their large scale climate impact in coupled models

Authors: He Wang; Sonya A. Legg; Robert W. Hallberg

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2014.12.005

URL: <https://doi.org/10.1016/j.ocemod.2014.12.005>

Publication Date: 2014-12-29

Volume/Issue: Volume 86

References: 46

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1646. A reanalysis dataset of the South China Sea

Authors: Xuezhi Zeng; Shiqiu Peng; Zhijin Li; Yiquan Qi; Rongyu Chen

Journal: Scientific Data

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/sdata.2014.52

URL: <https://doi.org/10.1038/sdata.2014.52>

Publication Date: 2014-12-22

Volume/Issue: Volume 1, Issue 1

References: 34

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Ocean reanalysis provides a temporally continuous and spatially gridded four-dimensional estimate of the ocean state for a better understanding of the ocean dynamics and its spatial/temporal variability. Here we present a 19-year (1992-2010) high-resolution ocean reanalysis dataset of the upper ocean in the South China Sea (SCS) produced from an ocean data assimilation system. A wide variety of observations, including in-situ temperature/salinity profiles, ship-measured and satellite-derived sea surface temperatures, and sea surface height anomalies from satellite altimetry, are assimilated into the outputs of an ocean general circulation model using a multi-scale incremental three-dimensional variational data assimilation scheme, yielding a daily high-resolution reanalysis dataset of the SCS. Comparisons between the reanalysis and independent observations support the reliability of the dataset. The presented dataset provides the research community of the SCS an important data source for studying the thermodynamic processes of the ocean circulation and meso-scale features in the SCS, including their spatial and temporal variability.

1647. Separation of atmospheric, oceanic and hydrological polar motion excitation mechanisms based on a combination of geometric and gravimetric space observations

Authors: F. Göttl; M. Schmidt; F. Seitz; M. Bloßfeld

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-014-0782-0

URL: <https://doi.org/10.1007/s00190-014-0782-0>

Publication Date: 2014-12-17

Volume/Issue: Volume 89, Issue 4

References: 58

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1648. Vertical Structure of Ocean Pressure Variations with Application to Satellite-Gravimetric Observations

Authors: Christopher G. Piecuch (Atmospheric and Environmental Research Inc.); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-14-00156.1

URL: <https://doi.org/10.1175/jtech-d-14-00156.1>

Publication Date: 2014-12-16

Volume/Issue: Volume 32, Issue 3

References: 59

Cited by: 16 publications

Language: N/A

Subjects: N/A

Keywords: Abyssal circulation, Planetary waves, Satellite observations, Topographic effects, General circulation models (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The nature of ocean bottom pressure () variability is considered on large spatial scales and long temporal scales. Monthly gridded estimates from the Gravity Recovery and Climate Experiment (GRACE) Release-05 and the new version 4 bi-decadal ocean state estimate of the Consortium for Estimating the Circulation and Climate of the Ocean (ECCO) are used. Estimates of from GRACE and ECCO are generally in good agreement, providing an independent measure of the quality of both products. Diagnostic fields from the state estimate are used to compute barotropic (depth independent) and baroclinic (depth dependent) components. The relative roles of baroclinic and barotropic processes are found to vary with latitude and time scale: variations in at higher latitudes and shorter periods are affected by barotropic processes, whereas fluctuations at lower latitudes and longer periods can be influenced by baroclinic effects, broadly consistent with theoretical scaling arguments. Wind-driven Rossby waves and coupling of baroclinic and barotropic modes due to flow-topography interactions appear to be important influences on the baroclinic variability. Decadal simulations of monthly variability based on purely barotropic frameworks are expected to be in error by about 30% on average (in the tropical ocean and at higher latitudes). Results have implications for applying GRACE observations to problems such as estimating transports of the Antarctic Circumpolar Current.

1649. Dynamic Adjustment of the Ocean Circulation to Self-Attraction and Loading Effects

Authors: Nadya T. Vinogradova (Atmospheric and Environmental Research, Lexington, Massachusetts); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts); Katherine J. Quinn (Atmospheric and Environmental Research, Lexington, Massachusetts); Mark E. Tamisiea (National Oceanography Centre); Jean-Michel Campin (Massachusetts Institute of Technology); James L. Davis (Columbia University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0150.1

URL: <https://doi.org/10.1175/jpo-d-14-0150.1>

Publication Date: 2014-12-12

Volume/Issue: Volume 45, Issue 3

References: 40

Cited by: 20 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The oceanic response to surface loading, such as that related to atmospheric pressure, freshwater exchange, and changes in the gravity field, is essential to our understanding of sea level variability. In particular, so-called self-attraction and loading (SAL) effects caused by the redistribution of mass within the land-atmosphere-ocean system can have a measurable impact on sea level. In this study, the nature of SAL-induced variability in sea level is examined in terms of its equilibrium (static) and nonequilibrium (dynamic) components, using a general circulation model that implicitly includes the physics of SAL. The additional SAL forcing is derived by decomposing ocean mass anomalies into spherical harmonics and then applying Love numbers to infer associated crustal displacements and gravitational shifts. This implementation of SAL physics incurs only a relatively small computational cost. Effects of SAL on sea level amount to about 10% of the applied surface loading on average but depend strongly on location. The dynamic component exhibits large-scale basinwide patterns, with considerable contributions from subweekly time scales. Departures from equilibrium decrease toward longer time scales but are not totally negligible in many places. Ocean modeling studies should benefit from using a dynamical implementation of SAL as used here.

1650. Sensitivity of the Arctic sea ice concentration forecasts to different atmospheric forcing: a case study

Authors: Qinghua Yang; Jiping Liu; Zhanhai Zhang; Cuijuan Sui; Jianyong Xing; Ming Li; Chunhua Li; Jiechen Zhao; Lin Zhang

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-014-0566-7

URL: <https://doi.org/10.1007/s13131-014-0566-7>

Publication Date: 2014-12-10

Volume/Issue: Volume 33, Issue 12

References: 30

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1651. Variability in subtropical-tropical cells drives oxygen levels in the tropical Pacific Ocean

Authors: Olaf Duteil (GEOMAR Helmholtz Centre for Ocean Research); Claus W. Böning (GEOMAR Helmholtz Centre for Ocean Research); Andreas Oschlies (GEOMAR Helmholtz Centre for Ocean Research)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gl061774

URL: <https://doi.org/10.1002/2014gl061774>

Publication Date: 2014-12-08

Volume/Issue: Volume 41, Issue 24

References: 33

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: oxygen minimum zones, Pacific Ocean, tropical variability, subtropical-tropical cells, biogeochemical modeling

Funding: SFB754 supported by the Deutsche Forschungsgemeinschaft

License: N/A

Open Access: N/A

Abstract: Previous studies found a negative trend in oxygen concentrations in tropical regions during the last decades. Employing a biogeochemical circulation model, we highlight the importance of wind-driven ocean transport associated with the Subtropical-Tropical Cells (STCs) in setting the oxygen levels in the tropical ocean. The observed and simulated slowdown of the STCs by 30% from the 1960s to the 1990s caused a decrease in oxygen transport to the tropics. Transport of phosphate was similarly reduced, decreasing export production and respiration. The effects of physical transport and biological consumption partly compensate, damping oxygen interannual and decadal variability. Our results suggest that the observed residual oxygen trend in the tropical Pacific is mainly driven by changes in oxygen transport. Accordingly, the observed recent strengthening of the STCs leads us to expect a pause in the oxygen decrease or even an increase of tropical Pacific oxygen values in the near future.

1652. Assessing the magnitude of CO₂ flux uncertainty in atmospheric CO₂ records using products from NASA's Carbon Monitoring Flux Pilot Project

Authors: Lesley E. Ott (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Steven Pawson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); George J. Collatz (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Watson W. Gregg (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Holger Brix (Now at Helmholtz-Zentrum Geesthacht Institute of Coastal Research Geesthacht Germany, University of California Los Angeles); Cecile S. Rousseaux (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Annmarie Eldering (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael R. Gunson (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Stephan R. Kawa (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jd022411

URL: <https://doi.org/10.1002/2014jd022411>

Publication Date: 2014-12-03

Volume/Issue: Volume 120, Issue 2

References: 110

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA's CMS FPP and Phase One activities

License: N/A

Open Access: N/A

Abstract: NASA's Carbon Monitoring System Flux Pilot Project (FPP) was designed to better understand contemporary carbon fluxes by bringing together state-of-the art models with remote sensing data sets. Here we report on simulations using NASA's Goddard Earth Observing System Model, version 5 (GEOS-5) which was used to evaluate the consistency of two different sets of observationally informed land and ocean fluxes with atmospheric CO₂ records. Despite the observation inputs, the average difference in annual terrestrial biosphere flux between the two land (NASA Ames Carnegie-Ames-Stanford-Approach (CASA) and CASA-Global Fire Emissions Database version 3 (GFED)) models is 1.7 Pg C for 2009-2010. Ocean models (NASA's Ocean Biogeochemical Model (NOBM) and Estimating the Circulation and Climate of the Ocean Phase II (ECCO2)-Darwin) differ by 35% in their global estimates of carbon flux with particularly strong disagreement in high latitudes. Based upon combinations of terrestrial and ocean fluxes, GEOS-5 reasonably simulated the seasonal

cycle observed at Northern Hemisphere surface sites and by the Greenhouse gases Observing SATellite (GOSAT) while the model struggled to simulate the seasonal cycle at Southern Hemisphere surface locations. Though GEOS-5 was able to reasonably reproduce the patterns of XCO₂ observed by GOSAT, it struggled to reproduce these aspects of Atmospheric Infrared Sounder observations. Despite large differences between land and ocean flux estimates, resulting differences in atmospheric mixing ratio were small, typically less than 5 ppm at the surface and 3 ppm in the XCO₂ column. A statistical analysis based on the variability of observations shows that flux differences of these magnitudes are difficult to distinguish from inherent measurement variability, regardless of the measurement platform.

1653. Hydrodynamic processes and their impacts on the mud deposit in the Southern Yellow Sea

Authors: Chunyan Zhou; Ping Dong; Guangxue Li

Journal: Marine Geology

Publisher: Elsevier BV

DOI: 10.1016/j.margeo.2014.11.012

URL: <https://doi.org/10.1016/j.margeo.2014.11.012>

Publication Date: 2014-11-30

Volume/Issue: Volume 360

References: 95

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Science Foundation of China, NSFC, Chinese Scholarship Council (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1654. Quantifying the processes controlling intraseasonal mixed-layer temperature variability in the tropical Indian Ocean

Authors: D. J. Halkides (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California at Los Angeles-Joint Institute For Regional Earth System Science and Engineering (JIFRESSE) Los Angeles California USA); Duane E. Waliser (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California at Los Angeles-Joint Institute For Regional Earth System Science and Engineering (JIFRESSE) Los Angeles California USA); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bin Guan (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California at Los Angeles-Joint Institute For Regional Earth System Science and Engineering (JIFRESSE) Los Angeles California USA)

Journal: Journal of Geophysical Research: Oceans

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References: 73

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: MJO, ECCO, mixed-layer, Indian Ocean, intraseasonal (and 1 more)

Funding: National Aeronautics and Space Administration (NASA), Estimating the Circulation and Climate of the Ocean (ECCO), National Oceanographic Partnership Program (NOPP)

License: N/A

Open Access: N/A

Abstract: Spatial and temporal variation of processes that determine ocean mixed-layer (ML) temperature (MLT) variability on the timescale of the Madden-Julian Oscillation (MJO) in the Tropical Indian Ocean (TIO) are examined in a heat-conserving ocean state estimate for years 1993-2011. We introduce a new metric for representing spatial variability of the relative importance of processes. In general, horizontal advection is most important at the Equator. Subsurface processes and surface heat flux are more important away from the Equator, with surface heat flux being the more dominant factor. Analyses at key sites are discussed in the context of local dynamics and literature. At 0 degrees, 80.5 degreesE, for MLT events > 2 standard deviations, ocean dynamics account for more than two thirds of the net tendency during cooling and warming phases. Zonal advection alone accounts for ~40% of the net tendency. Moderate events (1-2 standard deviations) show more differences between events, and some are dominated by surface heat flux. At 8 degreesS, 67 degreesE in the Seychelles-Chagos Thermocline Ridge (SCTR) area, surface heat flux accounts for ~70% of the tendency during strong cooling and warming phases; subsurface processes linked to ML depth (MLD) deepening (shoaling) during cooling (warming) account for ~30%. MLT is more sensitive to subsurface processes in the

SCTR, due to the thin MLD, thin barrier layer and raised thermocline. Results for 8 degreesS, 67 degreesE support assertions by Vialard et al. (2008) not previously confirmed due to measurement error that prevented budget closure and the small number of events studied. The roles of MLD, barrier layer thickness, and thermocline depth on different timescales are examined.

1655. Improving the geoid: Combining altimetry and mean dynamic topography in the California coastal ocean

Authors: Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Bruce Cornuelle (Scripps Institution of Oceanography / University of California San Diego)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gl062402

URL: <https://doi.org/10.1002/2014gl062402>

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References: 28

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Geospatial-Intelligence Agency (NGA), NGA University Research Initiative (NURI), Climate Observations and Monitoring Program, National Oceanic and Atmospheric Administration, U.S. Department of Commerce

License: N/A

Open Access: N/A

Abstract: Satellite gravity mapping missions, altimeters, and other platforms have allowed the Earth's geoid to be mapped over the ocean to a horizontal resolution of approximately 100 km with an uncertainty of less than 10 cm. At finer resolution this uncertainty increases to greater than 10 cm. Achieving greater accuracy requires accurate estimates of the dynamic ocean topography (DOT). In this study two DOT estimates for the California Current System with uncertainties less than 10 cm are used to solve for a geoid correction field. The derived field increases the consistency between the DOTs and along-track altimetric observations, suggesting it is a useful correction to the gravitational field. The correction is large compared to the dynamic ocean topography, with a magnitude of 15 cm and significant structure, especially near the coast. The results are evidence that modern high-resolution dynamic ocean topography products can be used to improve estimates of the geoid.

1656. Mean sea-level variability along the northeast American Atlantic coast and the roles of the wind and the overturning circulation

Authors: Philip L. Woodworth (National Oceanography Centre); Miguel Á. Morales Maqueda (National Oceanography Centre); Vassil M. Roussenov (University of Liverpool); Richard G. Williams (University of Liverpool); Chris W. Hughes (University of Liverpool, National Oceanography Centre)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010520

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Publication Date: 2014-11-24

Volume/Issue: Volume 119, Issue 12

References: 61

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: Tide gauge|Forcing (mathematics)|Circulation (fluid dynamics)|Cape|Mode (computer interface)

Funding: Durham and York Universities in the UK Natural Environment Research Council (NERC) project "North Atlantic sea-level change and climate in the last 500 years", The NERC project "Climate variability in the North Atlantic Ocean"

License: N/A

Open Access: N/A

Abstract: The variability in mean sea level (MSL) during 1950-2009 along the northeast American Atlantic coast north of Cape Hatteras has been studied, using data from tide gauges and satellite altimetry and information from the Liverpool/Hadley Centre (LHC) ocean model, thereby providing new insights into the spatial and temporal scales of the variability. Although a relationship between sea level and the overturning circulation can be identified (an increase of approximately 1.5 cm in MSL for a decrease of 1 Sv in overturning transport), it is the effect of the nearshore wind forcing on the shelf that is found to dominate the interannual sea-level variability. In particular, winds are found to be capable of producing low-frequency changes in MSL ("accelerations") in a narrow coastal band, comparable to those observed by the tide gauges. Evidence is presented supporting the idea of a "common mode" of spatially coherent low-frequency MSL variability, both to the north and south of Cape Hatteras and throughout the northwest Atlantic, which is associated with large spatial-scale density changes from year to year.

1657. Assessment of the representation of Antarctic Bottom Water properties in the ECCO2 reanalysis

Authors: M. Azaneu; R. Kerr; M. M. Mata

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-10-923-2014

URL: <https://doi.org/10.5194/os-10-923-2014>

Publication Date: 2014-11-21

Volume/Issue: Volume 10, Issue 6

References: 83

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We analyzed the ability of the Estimating the Circulation and Climate of the Ocean - Phase II (ECCO2) reanalysis to represent the hydrographic properties and variability of Antarctic Bottom Water (AABW) in the Southern Ocean. We used a 20-year (1992-2011) observational database to perform comparisons of hydrographic properties and reanalysis output for the same time period. Four case studies based on current meter data and the AABW volume transport estimates previously reported in the literature were also evaluated. The opening and maintenance of an oceanic polynya in the Weddell Sea sector is observed after 2004 in the reanalysis product. Moreover, intense deep water production due to deep convection occurs, which leads to a scenario in which the Weddell Sea is flooded with AABW. For this reason, our analyses focused on the period that was identified as more reliable (1992-2004). The main Southern Ocean oceanographic features, as well as the characteristic shape of the regional potential temperature-salinity (θ -S) diagrams, are coincident with observations. However, the reanalysis output produces surface waters that are generally denser than observations due to the reproduction of waters that are generally saltier than expected, which probably resulted from the strong seasonality of sea ice concentrations. Bottom waters are warmer and less dense, while intermediate waters are statistically closest to the observations. The differences in bottom water properties are partially due to the inability of the reanalysis to properly reproduce the formation and export of dense waters from the shelf and the consequent absence of the densest AABW variety for most of the analyzed period. Despite differences in the absolute values, the upper AABW limit ($\gamma_{\sigma} \geq 28.27 \text{ kg m}^{-3}$) and AABW occupied area estimates are coincident with the observations in the World Ocean Circulation Experiment (WOCE) repeat sections SR2 and SR4. Moreover, the AABW volume export and current velocity variability are correlated with the observed time series in the most important region of dense water export (i.e., the Weddell Sea). Despite the consistency in terms of variability, the absolute volume transport and velocity estimates are underrepresented in all cases.

1658. Estimating mean dynamic topography in boundary currents and the use of Argo trajectories

Authors: Melissa Bowen (School of Environment, University of Auckland Auckland New Zealand); Philip Sutton (National Institute of Water and Atmospheric Research Wellington New Zealand); Dean Roemmich (Scripps Institution of Oceanography)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010281

URL: <https://doi.org/10.1002/2014jc010281>

Publication Date: 2014-11-21

Volume/Issue: Volume 119, Issue 12

References: 33

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A Mean Dynamic Topography (MDT) is required to estimate mean transport in the ocean, to combine with altimetry to derive instantaneous geostrophic surface velocities, and to estimate transport from shipboard hydrography. A number of MDTs are now available globally but differ most markedly in boundary currents and the Antarctic Circumpolar Current. We evaluate several MDTs in two boundary currents off New Zealand (in the subtropical western boundary current system east of the country and in the Subantarctic Front to the south) using satellite, hydrographic, and Argo observations near two altimeter tracks. Argo float trajectories are combined with estimates of shear to produce new MDTs along both altimeter tracks: sufficiently high numbers of Argo floats travel in both boundary currents to allow a useful estimate of the mean flow at 1000 m depth and conservation of potential vorticity is used to account for more realistic flow paths. In finding a MDT, we show the uncertainties in the estimates of velocity differences between 1000 m and the surface from density climatologies, while often not estimated, need to be considered. The MDT computed from the Argo trajectories is generally consistent with the CLS09 MDT in both boundary currents and, in some locations, distinctly different from the MDT using a "level of no motion" assumption. The comparison suggests velocities from Argo trajectories can be usefully combined with other observations to improve estimates of flows and MDT in boundary currents.

1659. Spectral-finite element approach to post-seismic relaxation in a spherical compressible Earth: application to gravity changes due to the 2004 Sumatra-Andaman earthquake

Authors: Y. Tanaka; T. Hasegawa; H. Tsuruoka; V. Klemann; Z. Martinec

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggu391

URL: <https://doi.org/10.1093/gji/ggu391>

Publication Date: 2014-11-18

Volume/Issue: Volume 200, Issue 1

References: 74

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1660. A method for reconstructing global ocean-induced surface displacements from land-based in-situ stations

Authors: Julian Kuhlmann; Irina Rogozhina; Robert Dill; Inga Bergmann-Wolf; Maik Thomas

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2014.10.003

URL: <https://doi.org/10.1016/j.jog.2014.10.003>

Publication Date: 2014-11-17

Volume/Issue: Volume 83

References: 31

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1661. Impacts of regional mixing on the temperature structure of the equatorial Pacific Ocean. Part 1: Vertically uniform vertical diffusion

Authors: Ryo Furue; Yanli Jia; Julian P. McCreary; Niklas Schneider; Kelvin J. Richards; Peter Müller; Bruce D. Cornuelle; Nidia Martínez Avellaneda; Detlef Stammer; Chuanyu Liu; Armin Köhl

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2014.10.002

URL: <https://doi.org/10.1016/j.ocemod.2014.10.002>

Publication Date: 2014-11-16

Volume/Issue: Volume 91

References: 57

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1662. Mixed layer salinity budget in the tropical Pacific Ocean estimated by a global GCM

Authors: Shan Gao (Chinese Academy of Sciences); Tangdong Qu (University of Hawaii at Manoa); Xunwei Nie (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010336

URL: <https://doi.org/10.1002/2014jc010336>

Publication Date: 2014-11-08

Volume/Issue: Volume 119, Issue 12

References: 56

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: ECCO, tropical Pacific, warm pool, salinity budget, ENSO (and 1 more)

Funding: Chinese Academy of Sciences, National Natural Science Foundation of China, National Basic Research Program of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: The mixed layer salinity (MLS) budget of the tropical Pacific is investigated using results from a model of the Consortium for Estimating the Circulation and Climate of the Ocean (ECCO). The results focusing on the western Pacific freshwater pool indicate that the long-term averaged surface freshwater flux is well balanced by ocean dynamics, in which the subsurface processes account for the major part. The MLS budget shows significant seasonal and interannual variability, as a consequence of interplay among surface freshwater flux, advection, mixing, and vertical entrainment. On seasonal time scale, both the MLS and mixed layer depth are largely controlled by surface freshwater flux. The opposite phase between the subsurface processes and the barrier layer thickness confirms the important influence of the barrier layer on vertical mixing and entrainment from below. On interannual time scale, all the MLS budget terms show significant ENSO signal, which in turn is highly correlated with the salinity front and barrier layer thickness in the equatorial Pacific.

1663. On the decorrelation filtering of RL05 GRACE data for global applications

Authors: Santiago Belda; David García-García; José M. Ferrándiz

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggu386

URL: <https://doi.org/10.1093/gji/ggu386>

Publication Date: 2014-11-08

Volume/Issue: Volume 200, Issue 1

References: 21

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1664. Quantifying and Interpreting Striations in a Subtropical Gyre: A Spectral Perspective

Authors: Ru Chen (Massachusetts Institute of Technology, Woods Hole Oceanographic Institution); Glenn R. Flierl (Massachusetts Institute of Technology); Carl Wunsch (Harvard University, Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0038.1

URL: <https://doi.org/10.1175/jpo-d-14-0038.1>

Publication Date: 2014-11-07

Volume/Issue: Volume 45, Issue 2

References: 76

Cited by: 11 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The amplitude, origin, and direction of striations in the subtropical gyre are investigated using simulated and analytical multidimensional spectra. Striations, defined as banded structures in the low-frequency motions, account for a noticeable percentage of zonal velocity variability in the east North Pacific (ENP: 25 degrees-42 degreesN, 150 degrees-130 degreesW) and central North Pacific (CNP: 10 degrees-22 degreesN, 132 degreesE-162 degreesW) regions in an eddying global ocean model. Thus, they likely are nonnegligible in mixing and transport processes. Striations in the ENP region are nonzonal and are embedded in the nonzonal gyre flow, whereas striations in the CNP region are more zonal, as are the mean gyre flows. An idealized 1.5-layer model shows the gyre flow partially determines their directions, which qualitatively resemble those in the global eddying model. In the linear limit, structures are quasi-stationary (frequency $\omega \rightarrow 0$) linear Rossby waves and the gyre flow influences the direction by influencing the nature of the zero Rossby wave frequency curve. In the nonlinear regime, striations are consistent with the nondispersively propagating eddies, whose low-frequency component has banded structures. The gyre flow influences the striation direction by changing the eddy propagation direction. Their origin in the nonlinear regime is consistent with the existence of a nondispersive line in the frequency-wavenumber spectra. This study does not exclude other striation mechanisms from literature, considering that the interpretations here are based on an idealized model and only from a spectral perspective.

1665. The imprint of Southern Ocean overturning on seasonal water mass variability in Drake Passage

Authors: Dafydd Gwyn Evans (University of Southampton, National Oceanography Centre); Jan D. Zika (University of Southampton, National Oceanography Centre); Alberto C. Naveira Garabato (University of Southampton, National Oceanography Centre); A. J. George Nurser (University of Southampton, National Oceanography Centre)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc010097

URL: <https://doi.org/10.1002/2014jc010097>

Publication Date: 2014-10-31

Volume/Issue: Volume 119, Issue 11

References: 42

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council studentship award at the University of Southampton, Natural Environment Research Council Postdoctoral Fellowship Award

License: N/A

Open Access: N/A

Abstract: N/A

1666. A Hessian-Based Method for Uncertainty Quantification in Global Ocean State Estimation

Authors: Alexander G. Kalmikov; Patrick Heimbach

Journal: SIAM Journal on Scientific Computing

Publisher: Society for Industrial & Applied Mathematics (SIAM)

DOI: 10.1137/130925311

URL: <https://doi.org/10.1137/130925311>

Publication Date: 2014-10-30

Volume/Issue: Volume 36, Issue 5

References: 34

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1667. Cyclonic Eddies in the Gulf of Mexico: Observations by Underwater Gliders and Simulations by Numerical Model

Authors: Daniel L. Rudnick (Scripps Institution of Oceanography); Ganesh Gopalakrishnan (Scripps Institution of Oceanography); Bruce D. Cornuelle (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0138.1

URL: <https://doi.org/10.1175/jpo-d-14-0138.1>

Publication Date: 2014-10-29

Volume/Issue: Volume 45, Issue 1

References: 47

Cited by: 47 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Circulation in the Gulf of Mexico (GoM) is dominated by the Loop Current (LC) and by Loop Current eddies (LCEs) that form at irregular multimonth intervals by separation from the LC. Comparatively small cyclonic eddies (CEs) are thought to have a controlling influence on the LCE, including its separation from the LC. Because the CEs are so dynamic and short-lived, lasting only a few weeks, they have proved a challenge to observe. This study addresses that challenge using underwater gliders. These gliders' data and satellite sea surface height (SSH) are used in a four-dimensional variational (4DVAR) assimilation in the Massachusetts Institute of Technology (MIT) general circulation model (MITgcm). The model serves two purposes: first, the model's estimate of ocean state allows the analysis of four-dimensional fields, and second, the model forecasts are examined to determine the value of glider data. CEs have a Rossby number of about 0.2, implying that the effects of flow curvature, cyclostrophy, to modify the geostrophic momentum balance are slight. The velocity field in CEs is nearly depth independent, while LCEs are more baroclinic, consistent with the CEs origin on the less stratified, dense side of the LCE. CEs are formed from water in the GoM, rather than the Atlantic water that distinguishes the LCE. Model forecasts are improved by glider data, using a quality metric based on satellite SSH, with the best 2-month GoM forecast rivaling the accuracy of a global hindcast.

1668. Observed interannual variability of zonal currents in the equatorial Indian Ocean thermocline and their relation to Indian Ocean Dipole

Authors: Dongxiao Zhang (University of Washington); Michael J. McPhaden (Pacific Marine Environmental Laboratory NOAA Seattle Washington USA); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gl061449

URL: <https://doi.org/10.1002/2014gl061449>

Publication Date: 2014-10-27

Volume/Issue: Volume 41, Issue 22

References: 46

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Joint Institute for the Study of the Atmosphere and Ocean

License: N/A

Open Access: N/A

Abstract: Interannual variability of zonal currents in the eastern equatorial Indian Ocean thermocline is significantly correlated with sea surface temperature (SST) in the eastern pole of the Indian Ocean Dipole (IOD). This relationship is more significant than for zonal currents and the Dipole Mode Index, the latter of which measures zonal SST anomaly gradient. The variability of zonal currents in the thermocline is consistent with the variations of eastward pressure gradient force in the thermocline and the equatorward geostrophic thermocline flows associated with the shallow meridional overturning circulations in the eastern tropical Indian Ocean. Our analysis suggests a positive feedback between zonal currents in the thermocline and SST in the eastern pole of IOD. The combination of equatorial wind and off-equatorial wind stress curls associated with SST anomalies in the eastern pole of the IOD drives interannual variability in equatorial zonal thermocline flows and its covariability with the shallow meridional overturning cells.

1669. The relation of wind stress curl and meridional transport in the Benguela upwelling system

Authors: Tim Junker; Martin Schmidt; Volker Mohrholz

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2014.10.006

URL: <https://doi.org/10.1016/j.jmarsys.2014.10.006>

Publication Date: 2014-10-22

Volume/Issue: Volume 143

References: 25

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1670. The Persistence of Winter Sea Surface Temperature in the North Atlantic

Authors: No authors listed

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/1520-0442(2003)16<1364:tpowss>2.0.co;2

URL: [https://doi.org/10.1175/1520-0442\(2003\)16<1364:tpowss>2.0.co;2](https://doi.org/10.1175/1520-0442(2003)16<1364:tpowss>2.0.co;2)

Publication Date: 2014-10-14

Volume/Issue: Volume 16, Issue 9

References: N/A

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Anomaly (physics)|Mixed layer|Forcing (mathematics)|Persistence (discontinuity)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1671. Impact of Data Assimilation on ECCO2 Equatorial Undercurrent and North Equatorial Countercurrent in the Pacific Ocean

Authors: David Halpern (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Xiaochun Wang (University of California Los Angeles)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-14-00025.1

URL: <https://doi.org/10.1175/jtech-d-14-00025.1>

Publication Date: 2014-10-13

Volume/Issue: Volume 32, Issue 1

References: 38

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The impact of data assimilation on the transports of eastward-flowing Equatorial Undercurrent (EUC) and North Equatorial Countercurrent (NECC) in the Pacific Ocean from 145 degreesE to 95 degreesW during 2004-05 and 2009-11 was assessed. Two Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2), solutions were analyzed: one with data assimilation and one without. Assimilated data included satellite observations of sea surface temperature and ocean surface topography, in which the sampling patterns were approximately uniform over the 5 years, and in situ measurements of subsurface salinity and temperature profiles, in which the sampling patterns varied considerably in space and time throughout the 5 years. Velocity measurements were not assimilated. The impact of data assimilation was considered significant when the difference between the transports computed with and without data assimilation was greater than $5.5 \times 10^6 \text{ m}^3 \text{ s}^{-1}$ (or 5.5 Sv; $1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) for the EUC and greater than 5.0 Sv for the NECC. In addition, the difference of annual-mean transports computed from 3-day-averaged data was statistically significant at the 95% level. The impact of data assimilation ranged from no impact to very substantial impact when data assimilation increased the EUC transport and decreased the NECC transport. The study's EUC results had some correspondence with other studies and no simple agreement or disagreement pattern emerged among all studies of the impact of data assimilation. No comparable study of the impact of data assimilation on the NECC has been made.

1672. Spatial and seasonal variability of the air-sea equilibration timescale of carbon dioxide

Authors: Daniel C. Jones (Georgia Institute of Technology, British Antarctic Survey NERC Cambridge UK); Takamitsu Ito (Georgia Institute of Technology); Yohei Takano (Georgia Institute of Technology); Wei-Ching Hsu (Georgia Institute of Technology, Department of Oceanography Texas A&M; University College Station Texas USA)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gb004813

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Publication Date: 2014-10-08

Volume/Issue: Volume 28, Issue 11

References: 50

Cited by: 96 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF, NOAA USA, NERC UK

License: N/A

Open Access: N/A

Abstract: The exchange of carbon dioxide between the ocean and the atmosphere tends to bring waters within the mixed layer toward equilibrium by reducing the partial pressure gradient across the air-water interface. However, the equilibration process is not instantaneous; in general, there is a lag between forcing and response. The timescale of air-sea equilibration depends on several factors involving the depth of the mixed layer, wind speed, and carbonate chemistry. We use a suite of observational data sets to generate climatological and seasonal composite maps of the air-sea equilibration timescale. The relaxation timescale exhibits considerable spatial and seasonal variations that are largely set by changes in mixed layer depth and wind speed. The net effect is dominated by the mixed layer depth; the gas exchange velocity and carbonate chemistry parameters only provide partial compensation. Broadly speaking, the adjustment timescale tends to increase with latitude. We compare the observationally derived air-sea gas exchange timescale with a model-derived surface residence time and a data-derived horizontal transport timescale, which allows us to define two nondimensional metrics of equilibration efficiency. These parameters highlight the tropics, subtropics, and northern North Atlantic as regions of inefficient air-sea equilibration where carbon anomalies are relatively likely to persist. The efficiency parameters presented here can serve as simple tools for understanding the large-scale persistence of air-sea disequilibrium of CO₂ in both observations and models.

1673. Basal mass budget of Ross and Filchner-Ronne ice shelves, Antarctica, derived from Lagrangian analysis of ICESat altimetry

Authors: Geir Moholdt (Norwegian Polar Institute, Scripps Institution of Oceanography); Laurie Padman (Earth & Space Research Corvallis Oregon USA); Helen Amanda Fricker (Scripps Institution of Oceanography)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jf003171

URL: <https://doi.org/10.1002/2014jf003171>

Publication Date: 2014-10-01

Volume/Issue: Volume 119, Issue 11

References: 90

Cited by: 88 publications

Language: en

Subjects: N/A

Keywords: Mass balance, Ice shelf, Basal melting, Antarctica, Icesat (and 1 more)

Funding: NASA

License: N/A

Open Access: N/A

Abstract: Traditional methods of deriving temporal variability of Antarctic ice-shelf elevation from satellite altimetry use a fixed ("Eulerian") reference frame, where the measured changes include advection of ice thickness gradients between measurement epochs. We present a new method which removes advection effects by using an independent velocity field to compare elevations in a moving ("Lagrangian") reference frame. Applying the technique to ICESat laser altimetry for the period 2003-2009 over the two largest Antarctic ice shelves, Ross and Filchner-Ronne, we show that the Lagrangian approach reduces the variability of derived elevation changes by about 50% compared to the Eulerian approach and reveals clearer spatial patterns of elevation change. The method simplifies the process of estimating basal mass budget from the residual of all other processes that contribute to ice-shelf elevation changes. We use field data and ICESat measurements over ice rises and the grounded ice sheet to account for surface accumulation and changes in firn air content, and remove the effect of ice-flow divergence using surface velocity and ice thickness data. The results show highest basal melt rates ($>5 \text{ m a}^{-1}$) near the deep grounding lines of major ice streams, but smaller melt rates ($<5 \text{ m a}^{-1}$) near the ice-shelf fronts are equally important to total meltwater production since they occur over larger areas. Integrating over the entire ice-shelf areas, we obtain basal mass budgets of $-50 \pm 64 \text{ Gt a}^{-1}$ for Ross and $-124 \pm 66 \text{ Gt a}^{-1}$ for Filchner-Ronne, with changes in firn air content as the largest error source.

1674. Representation errors and retrievals in linear and nonlinear data assimilation

Authors: Peter Jan van Leeuwen (University of Reading)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2464

URL: <https://doi.org/10.1002/qj.2464>

Publication Date: 2014-09-30

Volume/Issue: Volume 141, Issue 690

References: 28

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NERC

License: CC BY 4.0

Open Access: N/A

Abstract: This article shows how one can formulate the representation problem starting from Bayes' theorem. The purpose of this article is to raise awareness of the formal solutions, so that approximations can be placed in a proper context. The representation errors appear in the likelihood, and the different possibilities for the representation of reality in model and observations are discussed, including nonlinear representation probability density functions. Specifically, the assumptions needed in the usual procedure to add a representation error covariance to the error covariance of the observations are discussed, and it is shown that, when several sub-grid observations are present, their mean still has a representation error; so-called 'superobbing' does not resolve the issue. Connection is made to the off-line or on-line retrieval problem, providing a new simple proof of the equivalence of assimilating linear retrievals and original observations. Furthermore, it is shown how nonlinear retrievals can be assimilated without loss of information. Finally we discuss how errors in the observation operator model can be treated consistently in the Bayesian framework, connecting to previous work in this area.

1675. Ocean dynamics and tropical Pacific climate change in ocean reanalyses and coupled climate models

Authors: Chunxue Yang (Centro Euro-Mediterraneo per i Cambiamenti Climatici Bologna Italy, Department of Oceanography Texas A&M; University College Station Texas USA); Benjamin S. Giese (Department of Oceanography Texas A&M; University College Station Texas USA); Lixin Wu (Ocean University of China)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc009979

URL: <https://doi.org/10.1002/2014jc009979>

Publication Date: 2014-09-25

Volume/Issue: Volume 119, Issue 10

References: 57

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NOAA, NSF

License: N/A

Open Access: N/A

Abstract: The role of ocean dynamics in tropical Pacific climate change is studied using an ensemble run of Simple Ocean Data Assimilation--sparse input version 1 (SODAsi.1) and the Coupled Model Intercomparison Project Phase 5 (CMIP5) historical runs. An eight-member ensemble of ocean reanalyses (SODAsi.1) from 1871 to 2008 is produced by using forcing from eight ensemble members of an atmospheric reanalysis. The long-term trends of tropical Pacific surface temperature, wind stress, subsurface temperature, and strength of the subtropical cells (STCs) are analyzed. The ensemble reanalysis shows that there is a slight cooling trend of surface and subsurface temperature in the central tropical Pacific due to enhanced tropical Pacific circulation. The STCs, which consist of equatorial upwelling, Ekman transport, extratropical subduction, and pycnocline transport from the subtropical to the tropical region, strengthen from 1900 to 2008. When the STCs are accelerated, equatorial upwelling increases bringing cold water from the subsurface that cools the surface. An increasing trend in convergence transport is mainly from the Southern Hemisphere. In contrast with the reanalysis most of the CMIP5 models have warming trends at the surface and the transport of the STCs has a decreasing trend. The CMIP5 models also underestimate tropical Pacific Ocean circulation relative to the reanalysis that is mostly due to differences in wind forcing.

1676. Assimilating SMOS sea ice thickness into a coupled ice-ocean model using a local SEIK filter

Authors: Qinghua Yang (National Marine Environmental Forecasting Center, Alfred Wegener Institute); Svetlana N. Losa (Alfred Wegener Institute); Martin Losch (Alfred Wegener Institute); Xiangshan Tian-Kunze (University of Hamburg); Lars Nerger (Alfred Wegener Institute); Jiping Liu (Department of Atmospheric and Environmental Sciences University at Albany, State University of New York Albany New York USA); Lars Kaleschke (University of Hamburg); Zhanhai Zhang (Polar Research Institute of China)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc009963

URL: <https://doi.org/10.1002/2014jc009963>

Publication Date: 2014-09-19

Volume/Issue: Volume 119, Issue 10

References: 39

Cited by: 70 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: BMBF (Federal Ministry of Education and Research, Germany), SOA (State Oceanic Administration, China), Ocean Public Welfare Project of China (and 2 more)

License: N/A

Open Access: N/A

Abstract: The impact of assimilating sea ice thickness data derived from ESA's Soil Moisture and Ocean Salinity (SMOS) satellite together with Special Sensor Microwave Imager/Sounder (SSMIS) sea ice concentration data of the National Snow and Ice Data Center (NSIDC) in a coupled sea ice-ocean model is examined. A period of 3 months from 1 November 2011 to 31 January 2012 is selected to assess the forecast skill of the assimilation system. The 24 h forecasts and longer forecasts are based on the Massachusetts Institute of Technology general circulation model (MITgcm), and the assimilation is performed by a localized Singular Evolutive Interpolated Kalman (LSEIK) filter. For comparison, the assimilation is repeated only with the SSMIS sea ice concentrations. By running two different assimilation experiments, and comparing with the unassimilated model, independent satellite-derived data, and in situ observation, it is shown that the SMOS ice thickness assimilation leads to improved thickness forecasts. With SMOS thickness data, the sea ice concentration forecasts also agree better with observations, although this improvement is smaller.

1677. Interpretation of coastal wind transfer functions with momentum balances derived from idealized numerical model simulations

Authors: Sung Yong Kim; Ganesh Gopalakrishnan; Aurelien Ponte

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-014-0766-x

URL: <https://doi.org/10.1007/s10236-014-0766-x>

Publication Date: 2014-09-19

Volume/Issue: Volume 65, Issue 1

References: 87

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1678. Sensitivity of the dynamics of Pine Island Glacier, West Antarctica, to climate forcing for the next 50 years

Authors: H. Seroussi; M. Morlighem; E. Rignot; J. Mouginot; E. Larour; M. Schodlok; A. Khazendar

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-8-1699-2014

URL: <https://doi.org/10.5194/tc-8-1699-2014>

Publication Date: 2014-09-17

Volume/Issue: Volume 8, Issue 5

References: 45

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Pine Island Glacier, a major contributor to sea level rise in West Antarctica, has been undergoing significant changes over the last few decades. Here, we employ a three-dimensional, higher-order model to simulate its evolution over the next 50 yr in response to changes in its surface mass balance, the position of its calving front and ocean-induced ice shelf melting. Simulations show that the largest climatic impact on ice dynamics is the rate of ice shelf melting, which rapidly affects the glacier speed over several hundreds of kilometers upstream of the grounding line. Our simulations show that the speedup observed in the 1990s and 2000s is consistent with an increase in sub-ice-shelf melting. According to our modeling results, even if the grounding line stabilizes for a few decades, we find that the glacier reaction can continue for several decades longer. Furthermore, Pine Island Glacier will continue to change rapidly over the coming decades and remain a major contributor to sea level rise, even if ocean-induced melting is reduced.

1679. Requirements Analysis for Future Satellite Gravity Mission Improved-GRACE

Authors: Wei Zheng; Houtse Hsu; Min Zhong; Meijuan Yun

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-014-9306-y

URL: <https://doi.org/10.1007/s10712-014-9306-y>

Publication Date: 2014-09-09

Volume/Issue: Volume 36, Issue 1

References: 49

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1680. The ocean's role in the transient response of climate to abrupt greenhouse gas forcing

Authors: John Marshall; Jeffery R. Scott; Kyle C. Armour; J.-M. Campin; Maxwell Kelley; Anastasia Romanou

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-014-2308-0

URL: <https://doi.org/10.1007/s00382-014-2308-0>

Publication Date: 2014-09-03

Volume/Issue: Volume 44, Issue 7-8

References: 42

Cited by: 158 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1681. A Southern Indian Ocean database of hydrographic profiles obtained with instrumented elephant seals

Authors: Fabien Roquet; Guy Williams; Mark A. Hindell; Rob Harcourt; Clive McMahon; Christophe Guinet; Jean-Benoit Charrassin; Gilles Reverdin; Lars Boehme; Phil Lovell; Mike Fedak

Journal: Scientific Data

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/sdata.2014.28

URL: <https://doi.org/10.1038/sdata.2014.28>

Publication Date: 2014-09-01

Volume/Issue: Volume 1, Issue 1

References: 47

Cited by: 113 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The instrumentation of southern elephant seals with satellite-linked CTD tags has offered unique temporal and spatial coverage of the Southern Indian Ocean since 2004. This includes extensive data from the Antarctic continental slope and shelf regions during the winter months, which is outside the conventional areas of Argo autonomous floats and ship-based studies. This landmark dataset of around 75,000 temperature and salinity profiles from 20-140 degreesE, concentrated on the sector between the Kerguelen Islands and Prydz Bay, continues to grow through the coordinated efforts of French and Australian marine research teams. The seal data are quality controlled and calibrated using delayed-mode techniques involving comparisons with other existing profiles as well as cross-comparisons similar to established protocols within the Argo community, with a resulting accuracy of ± 0.03 degreesC in temperature and ± 0.05 in salinity or better. The data offer invaluable new insights into the water masses, oceanographic processes and provides a vital tool for oceanographers seeking to advance our understanding of this key component of the global ocean climate.

1682. Elemental Mercury Concentrations and Fluxes in the Tropical Atmosphere and Ocean

Authors: Anne L. Soerensen (Department of Environmental Health, Harvard School of Public Health, Boston, Massachusetts 02215, United States, Harvard University); Robert P. Mason (Department of Marine Sciences, University of Connecticut, 1080 Sennecossett Road, Groton, Connecticut 06340, United States); Prentiss H. Balcom (Department of Marine Sciences, University of Connecticut, 1080 Sennecossett Road, Groton, Connecticut 06340, United States); Daniel J. Jacob (Harvard University); Yanxu Zhang (Harvard University); Joachim Kuss (Leibniz Institute for Baltic Sea Research); Elsie M. Sunderland (Department of Environmental Health, Harvard School of Public Health, Boston, Massachusetts 02215, United States, Harvard University)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

DOI: 10.1021/es503109p

URL: <https://doi.org/10.1021/es503109p>

Publication Date: 2014-08-29

Volume/Issue: Volume 48, Issue 19

References: 59

Cited by: 78 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Division of Ocean Sciences, Carlsbergfondet

License: N/A

Open Access: N/A

Abstract: N/A

1683. Observed seasonal and intraseasonal variability of the East India Coastal Current on the continental slope

Authors: A Mukherjee; D Shankar; V Fernando; P Amol; S G Aparna; R Fernandes; G S Michael; S T Khalap; N P Satelkar; Y Agarvadekar; M G Gaonkar; A P Tari; A Kankonkar; S Vernekar

Journal: Journal of Earth System Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12040-014-0471-7

URL: <https://doi.org/10.1007/s12040-014-0471-7>

Publication Date: 2014-08-28

Volume/Issue: Volume 123, Issue 6

References: 135

Cited by: 66 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1684. Sustaining observations of the unsteady ocean circulation

Authors: E. Frajka-Williams (University of Southampton, National Oceanography Centre)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rsta.2013.0335

URL: <https://doi.org/10.1098/rsta.2013.0335>

Publication Date: 2014-08-26

Volume/Issue: Volume 372, Issue 2025

References: 69

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sustained observations of ocean properties reveal a global warming trend and rising sea levels. These changes have been documented by traditional ship-based measurements of ocean properties, whereas more recent Argo profiling floats and satellite records permit estimates of ocean changes on a near real-time basis. Through these and newer methods of observing the oceans, scientists are moving from quantifying the 'state of the ocean' to monitoring its variability, and distinguishing the physical processes bringing signals of change. In this paper, I give a brief overview of the UK contributions to the physical oceanographic observations, and the role they have played in the wider global observing systems. While temperature and salinity are the primary measurements of physical oceanography, new transbasin mooring arrays also resolve changes in ocean circulation on daily timescales. Emerging technologies permit routine observations at higher-than-ever spatial resolutions. Following this, I then give a personal perspective on the future of sustained observations. New measurement techniques promise exciting discoveries concerning the role of smaller scales and boundary processes in setting the large-scale ocean circulation and the ocean's role in climate. The challenges now facing the scientific community include sustaining critical observations in the case of funding system changes or shifts in government priorities. These long records will enable a determination of the role and response of the ocean to climate change.

1685. Southern Ocean mass variation studies using GRACE and satellite altimetry

Authors: Chung-Yen Kuo; C. K. Shum; Jun-yi Guo; Yuchan Yi; Alexander Braun; Ichiro Fukumori; Koji Matsumoto; Tadahiro Sato; Kazuo Shibuya

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/bf03352814

URL: <https://doi.org/10.1186/bf03352814>

Publication Date: 2014-08-15

Volume/Issue: Volume 60, Issue 5

References: 38

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: Altimetry, GRACE, Southern Ocean

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The Southern Ocean is a major link between the world oceans via complicated processes associated with the melting and accumulation of the vast Antarctic ice sheets and the surrounding sea ice. The Southern Ocean sea level is poorly observed except from recent near-polar orbiting space geodetic satellites. In this study, the Southern Ocean mass variations at the seasonal scale are compared using three independent data sets: (1) the Gravity Recovery And Climate Recovery Experiment (GRACE) observed ocean bottom pressure (OBP), (2) steric-corrected satellite altimetry (ENVISAT) and, (3) the Estimating the Circulation and Climate of the Ocean (ECCO) model OBP data. The height difference between sea level derived from altimetry and steric sea level contains the vertical displacement of the Earth surface due to elastic loading. Here we provide a formulation of this loading term which has not been considered previously in other studies and demonstrate that it is not negligible, especially for regional studies. In this study, we first conduct a global comparison using steric-corrected JASON-1 altimetry with GRACE to validate our technique and to compare with recent studies. The global ocean mass variation comparison shows excellent agreement with high correlation (~ 0.81) and with discrepancies at 3-5 mm RMS. However, the discrepancies in the Southern Ocean are much larger at 12-17 mm RMS. The mis-modeling of geocenter variations and the second degree zonal harmonic (J 2) degrade the accuracy of GRACE-derived mass variations, and the choice of ocean temperature data sets and neglecting the loading correction on altimetry affect the OBP comparisons between GRACE and altimetry. This study indicates that the satellite observations (GRACE and ENVISAT) are capable of providing an improved constraint of oceanic mass variations in the Southern Ocean.

1686. Mass-redistribution-induced crustal deformation of global satellite laser ranging stations due to non-tidal ocean and land water circulation

Authors: Hiroshi Takiguchi; Toshimichi Otsubo; Yoichi Fukuda

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/bf03353365

URL: <https://doi.org/10.1186/bf03353365>

Publication Date: 2014-08-15

Volume/Issue: Volume 58, Issue 2

References: 19

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: Topex/Poseidon, crustal deformation, Satellite laser ranging, ECCO, continental water load (and 1 more)

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The effect of the non-tidal ocean load (NTOL) and the continental water load (CWL) on crustal deformation are calculated for global satellite laser ranging (SLR) stations and on 4 degreesx4 degrees grids (only over the land). For the regions most severely affected, the peak-to-peak displacements due to the NTOL are found to be as large as 3 mm for the horizontal components and 10 mm for the vertical component. The peak-to-peak displacements due to the CWL reach 3 mm for the horizontal components and 15 mm for the vertical component. We apply the time series of NTOL and CWL to precise SLR analysis. The LAGEOS orbit analysis reveals that the Estimating the Circulation and Climate of the Ocean (ECCO) model makes the root mean square (RMS) of the range residual 0.2% smaller, and that the CWL makes it 0.8% smaller, compared with the case where loading displacement is neglected. On the other hand, with the NTOL derived from Topex/Poseidon altimetry data, the SLR orbit fit is not improved.

1687. Ocean circulation generated magnetic signals

Authors: C. Manoj; A. Kuvshinov; S. Maus; H. Lühr

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.1186/bf03351939

URL: <https://doi.org/10.1186/bf03351939>

Publication Date: 2014-08-15

Volume/Issue: Volume 58, Issue 4

References: 33

Cited by: 83 publications

Language: en

Subjects: N/A

Keywords: Ocean flow, Geomagnetic field, Satellite measurements

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Conducting ocean water, as it flows through the Earth's magnetic field, generates secondary electric and magnetic fields. An assessment of the ocean-generated magnetic fields and their detectability may be of importance for geomagnetism and oceanography. Motivated by the clear identification of ocean tidal signatures in the CHAMP magnetic field data we estimate the ocean magnetic signals of steady flow using a global 3-D EM numerical solution. The required velocity data are from the ECCO ocean circulation experiment and alternatively from the OCCAM model for higher resolution. We assume an Earth's conductivity model with a surface thin shell of variable conductance with a realistic 1D mantle underneath. Simulations using both models predict an amplitude range of ± 2 nT at Swarm altitude (430 km). However at sea level, the higher resolution simulation predicts a higher strength of the magnetic field, as compared to the ECCO simulation. Besides the expected signatures of the global circulation patterns, we find significant seasonal variability of ocean magnetic signals in the Indian and Western Pacific Oceans. Compared to seasonal variation, interannual variations produce weaker signals.

1688. On the Early Response of the Climate System to a Meltwater Input from Greenland

Authors: Neeraj Agarwal (Max Planck Institute for Meteorology); Armin Köhl (University of Hamburg); Carlos Roberto Mechoso (University of California Los Angeles); Detlaf Stammer (University of Hamburg)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00762.1

URL: <https://doi.org/10.1175/jcli-d-13-00762.1>

Publication Date: 2014-08-13

Volume/Issue: Volume 27, Issue 21

References: 70

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: Atlantic Ocean, Arctic Oscillation, Freshwater, Teleconnections, Ice sheets (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The early response of the atmosphere-ocean system to meltwater runoff originating from the Greenland ice sheet is studied using a coupled atmosphere-ocean general circulation model (AOGCM). For this purpose, AOGCM ensemble simulations without and with associated ocean freshening around Greenland are compared. For freshwater perturbations initiated in northern winter, the mean response for the first three months shows the emergence of negative sea surface temperature (SST) anomalies in the Denmark Strait, in association with enhanced oceanic advection by the East Greenland Current. The response also shows negative SST anomalies in the North Atlantic associated with enhanced westerlies at the ocean surface. Additionally, the baroclinic atmospheric cyclonic circulation east of Greenland intensifies, and anticyclonic circulations with equivalent barotropic structures develop over western Europe and the North Pacific Ocean. Simulations by the atmospheric component of the AOGCM indicate that atmosphere-ocean interactions contribute significantly to enhance the response. The sensitivity of the coupled system response to the timing of freshwater perturbation is also studied. For freshwater perturbations initialized in northern summer, the response during the following winter is similar, but stronger in magnitude. In the Northern Hemisphere, the atmospheric response resembles the Arctic Oscillation (AO) mode of variability. The association between anomalies in the Denmark Strait SSTs and in the atmosphere east of Greenland is consistent with that observed during previous great salinity anomaly (GSA) events. The results obtained highlight the importance of atmosphere-ocean interaction in the early climate response to Greenland melting, the teleconnections with the North Pacific and the contribution of GSA events to North Atlantic Oscillation (NAO) variability.

1689. Recent variability of the global ocean carbon sink

Authors: P. Landschützer (University of East Anglia, ETH Zurich); N. Gruber (ETH Zurich); D. C. E. Bakker (University of East Anglia); U. Schuster (University of Exeter)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gb004853

URL: <https://doi.org/10.1002/2014gb004853>

Publication Date: 2014-08-13

Volume/Issue: Volume 28, Issue 9

References: 94

Cited by: 349 publications

Language: en

Subjects: N/A

Keywords: Sink (geography)|Carbon sink

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We present a new observation-based estimate of the global oceanic carbon dioxide (CO₂) sink and its temporal variation on a monthly basis from 1998 through 2011 and at a spatial resolution of 1 degreesx1 degrees. This sink estimate rests upon a neural network-based mapping of global surface ocean observations of the partial pressure of CO₂ (pCO₂) from the Surface Ocean CO₂ Atlas database. The resulting pCO₂ has small biases when evaluated against independent observations in the different ocean basins, but larger randomly distributed differences exist particularly in high latitudes. The seasonal climatology of our neural network-based product agrees overall well with the Takahashi et al. (2009) climatology, although our product produces a stronger seasonal cycle at high latitudes. From our global pCO₂ product, we compute a mean net global ocean (excluding the Arctic Ocean and coastal regions) CO₂ uptake flux of -1.42 +/- 0.53 Pg C yr⁻¹, which is in good agreement with ocean inversion-based estimates. Our data indicate a moderate level of interannual variability in the ocean carbon sink (+/-0.12 Pg C yr⁻¹, 1sigma) from 1998 through 2011, mostly originating from the equatorial Pacific Ocean, and associated with the El Niño-Southern Oscillation. Accounting for steady state riverine and Arctic Ocean carbon fluxes our estimate further implies a mean anthropogenic CO₂ uptake of -1.99 +/- 0.59 Pg C yr⁻¹ over the analysis period. From this estimate plus the most recent estimates for fossil fuel emissions and atmospheric CO₂ accumulation, we infer a mean global land sink of -2.82 +/- 0.85 Pg C yr⁻¹ over the 1998 through 2011 period with strong interannual variation.

1690. Weighing the ocean with bottom-pressure sensors: robustness of the ocean mass annual cycle estimate

Authors: Joanne Williams; C. W. Hughes; M. E. Tamisiea; S. D. P. Williams

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-10-701-2014

URL: <https://doi.org/10.5194/os-10-701-2014>

Publication Date: 2014-08-11

Volume/Issue: Volume 10, Issue 4

References: 30

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We use ocean bottom-pressure measurements from 17 tropical sites to determine the annual cycle of ocean mass. We show that such a calculation is robust, and use three methods to estimate errors in the mass determination. Our final best estimate, using data from the best sites and two ocean models, is that the annual cycle has an amplitude of 0.85 mbar (equivalent to 8.4 mm of sea level, or 3100 Gt of water), with a 95% chance of lying within the range 0.61-1.17 mbar. The time of the peak in ocean mass is 10 October, with 95% chance of occurring between 21 September and 25 October. The simultaneous fitting of annual ocean mass also improves the fitting of bottom-pressure instrument drift.

1691. Spatial and Temporal Scales of Sverdrup Balance*

Authors: Matthew D. Thomas (University of East Anglia); Agatha M. De Boer (Stockholm University); Helen L. Johnson (University of Oxford); David P. Stevens (University of East Anglia)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-0192.1

URL: <https://doi.org/10.1175/jpo-d-13-0192.1>

Publication Date: 2014-08-07

Volume/Issue: Volume 44, Issue 10

References: 45

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sverdrup balance underlies much of the theory of ocean circulation and provides a potential tool for describing the interior ocean transport from only the wind stress. Using both a model state estimate and an eddy-permitting coupled climate model, this study assesses to what extent and over what spatial and temporal scales Sverdrup balance describes the meridional transport. The authors find that Sverdrup balance holds to first order in the interior subtropical ocean when considered at spatial scales greater than approximately 5 degrees. Outside the subtropics, in western boundary currents and at short spatial scales, significant departures occur due to failures in both the assumptions that there is a level of no motion at some depth and that the vorticity equation is linear. Despite the ocean transport adjustment occurring on time scales consistent with the basin-crossing times for Rossby waves, as predicted by theory, Sverdrup balance gives a useful measure of the subtropical circulation after only a few years. This is because the interannual transport variability is small compared to the mean transports. The vorticity input to the deep ocean by the interaction between deep currents and topography is found to be very large in both models. These deep transports, however, are separated from upper-layer transports that are in Sverdrup balance when considered over large scales.

1692. Interannual variability of the Arabian Sea Warm Pool: observations and governing mechanisms

Authors: R. R. Rao; V. Jitendra; M. S. GirishKumar; M. Ravichandran; S. S. V. S. Ramakrishna

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-014-2243-0

URL: <https://doi.org/10.1007/s00382-014-2243-0>

Publication Date: 2014-08-04

Volume/Issue: Volume 44, Issue 7-8

References: 134

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1693. On Pacific Subtropical Cell Variability over the Second Half of the Twentieth Century

Authors: Riccardo Farneti (Earth System Physics Section, Abdus Salam International Centre for Theoretical Physics, Trieste, Italy); Suneet Dwivedi (University of Allahabad); Fred Kucharski (Earth System Physics Section, Abdus Salam International Centre for Theoretical Physics, Trieste, Italy, and Center of Excellence for Climate Change Research/Department of Meteorology, King Abdulaziz University, Jeddah, Saudi Arabia); Franco Molteni (European Centre for Medium-Range Weather Forecasts (ECMWF)); Stephen M. Griffies (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00707.1

URL: <https://doi.org/10.1175/jcli-d-13-00707.1>

Publication Date: 2014-07-25

Volume/Issue: Volume 27, Issue 18

References: 41

Cited by: 24 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The evolution of the Pacific subtropical cells (STC) is presented for the period 1948-2007. Using ocean models of different resolutions forced with interannually varying atmospheric forcing datasets, the mechanisms responsible for the observed STC weakening and late recovery during the period of study are analyzed. As a result of the STC weakening (strengthening), warming (cooling) trends are found in the equatorial Pacific sea surface temperatures (SSTs). Model results agree well with observed estimates of STC transport, STC convergence, and equatorial SST anomalies. It is shown that subtropical atmospheric variability is the primary driver of the STC and equatorial SST low-frequency evolution and is responsible for both the slowdown during the second half of the twentieth century and the rebound at the end of the century. Subtropically forced STC variability is identified as a major player in the generation of equatorial Pacific decadal SST anomalies, pacing tropical Pacific natural climate variability on interdecadal time scales, as observed in historical records. The natural mode of variability has implications for the evolution of equatorial SST in the coming decades under the concomitant effects of climate change.

1694. Coherent sea level variability on the North Atlantic western boundary

Authors: P. R. Thompson (University of South Florida, Department of Oceanography University of Hawai'i at Mānoa Honolulu Hawaii USA); G. T. Mitchum (University of South Florida)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc009999

URL: <https://doi.org/10.1002/2014jc009999>

Publication Date: 2014-07-25

Volume/Issue: Volume 119, Issue 9

References: 32

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: decadal, GECCO, North Atlantic, sea level

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Interannual to decadal sea level variability on the North Atlantic western boundary is surprisingly coherent over substantial distances stretching from the Caribbean to Nova Scotia. The physical mechanisms responsible for this basin-scale, low-frequency coherence are explored in a diagnosis of simulated ocean fields from GECCO, which reproduces the observations to good approximation. Coastal sea level variability on the western boundary is known to be influenced by meridional divergence in the boundary current resulting in a geostrophic tilting of the sea surface. This mechanism is found to be of leading order along some stretches of the boundary, but it does not account for the coherence spanning the western North Atlantic. Instead, the coherence along the entire boundary is accounted for by vertical divergence resulting in the uniform rise and fall of the sea surface west of the 295 degreesE meridian. The vertical divergence is found to be due to net vertically integrated zonal transport across this meridian resulting from meridional variation in the Sverdrup transport over the basin interior.

1695. Investigation of the causes of historical changes in the subsurface salinity minimum of the South Atlantic

Authors: Marlos Goes (University of Miami); Ilana Wainer (University of São Paulo); Natalia Signorelli (University of São Paulo)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc009812

URL: <https://doi.org/10.1002/2014jc009812>

Publication Date: 2014-07-24

Volume/Issue: Volume 119, Issue 9

References: 78

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Antarctic Intermediate Water, Southern Annular Mode, South Atlantic, Agulhas leakage

Funding: NOAA/AOML and NOAA's Climate Program Office, São Paulo Research Foundation (FAPESP)

License: N/A

Open Access: N/A

Abstract: In this study, we investigate the subsurface salinity changes on decadal timescales across the subtropical South Atlantic Ocean using two ocean reanalysis products, the latest version of the Simple Ocean Data Assimilation and the Estimating the Circulation and Climate of the Ocean, Phase II, as well as using additional climate model experiments. Results show that there is a recent significant salinity increase at the core of the salinity minimum at intermediate levels. The main underlying mechanism for this subsurface salinity increase is the lateral advective (gyre) changes due to the Southern Annular Mode variability, which conditions an increased contribution from the Indian Ocean high salinity waters into the Atlantic. The global warming signal has a secondary but complementary contribution. Latitudinal differences at intermediate depth in response to large-scale forcing are in part caused by local variation of westward propagation features, and by compensating contributions of salinity and temperature to density changes.

1696. Dispersion and fate of 90Sr in the Northwestern Pacific and adjacent seas: Global fallout and the Fukushima Dai-ichi accident

Authors: V. Maderich; K.T. Jung; R. Bezhenar; G. de With; F. Qiao; N. Casacuberta; P. Masque; Y.H. Kim

Journal: Science of The Total Environment

Publisher: Elsevier BV

DOI: 10.1016/j.scitotenv.2014.06.136

URL: <https://doi.org/10.1016/j.scitotenv.2014.06.136>

Publication Date: 2014-07-21

Volume/Issue: Volume 494-495

References: 55

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministerio de Ciencia e Innovación of Spain, Generalitat de Catalunya

License: N/A

Open Access: N/A

Abstract: N/A

1697. Assimilating summer sea-ice concentration into a coupled ice-ocean model using a LSEIK filter

Authors: Qinghua Yang; Svetlana N. Losa; Martin Losch; Jiping Liu; Zhanhai Zhang; Lars Nerger; Hu Yang

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2015aog69a740

URL: <https://doi.org/10.3189/2015aog69a740>

Publication Date: 2014-07-16

Volume/Issue: Volume 56, Issue 69

References: 41

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The decrease in summer sea-ice extent in the Arctic Ocean opens shipping routes and creates potential for many marine operations. For these activities accurate predictions of sea-ice conditions are required to maintain marine safety. In an attempt at Arctic sea-ice prediction, the summer of 2010 is selected to implement an Arctic sea-ice data assimilation (DA) study. The DA system is based on a regional Arctic configuration of the Massachusetts Institute of Technology general circulation model (MITgcm) and a local singular evolutive interpolated Kalman (LSEIK) filter to assimilate Special Sensor Microwave Imager/Sounder (SSMIS) sea-ice concentration operational products from the US National Snow and Ice Data Center (NSIDC). Based on comparisons with both the assimilated NSIDC SSMIS concentration and concentration data from the Ocean and Sea Ice Satellite Application Facility, the forecasted sea-ice edge and concentration improve upon simulations without data assimilation. By the nature of the assimilation algorithm with multivariate covariance between ice concentration and thickness, sea-ice thickness fields are also updated, and the evaluation with in situ observation shows some improvement compared to the forecast without data assimilation.

1698. Estimating satellite salinity errors for assimilation of Aquarius and SMOS data into climate models

Authors: Nadya T. Vinogradova (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jc009906

URL: <https://doi.org/10.1002/2014jc009906>

Publication Date: 2014-07-15

Volume/Issue: Volume 119, Issue 8

References: 28

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: Middle latitudes

Funding: NASA's Physical Oceanography Program and the Ocean Surface Salinity Project through contract, NASA's Physical Oceanography Distributed Active Archive, Barcelona Expert (and 4 more)

License: N/A

Open Access: N/A

Abstract: Constraining dynamical systems with new information from ocean measurements, including observations of sea surface salinity (SSS) from Aquarius and SMOS, requires careful consideration of data errors that are used to determine the importance of constraints in the optimization. Here such errors are derived by comparing satellite SSS observations from Aquarius and SMOS with ocean model output and in situ data. The associated data error variance maps have a complex spatial pattern, ranging from less than 0.05 in the open ocean to 1-2 (units of salinity variance) along the coasts and high latitude regions. Comparing the data-model misfits to the data errors indicates that the Aquarius and SMOS constraints could potentially affect estimated SSS values in several ocean regions, including most tropical latitudes. In reference to the Aquarius error budget, derived errors are less than the total allocation errors for the Aquarius mission accuracy requirements in low and midlatitudes, but exceed allocation errors in high latitudes.

1699. Observed intraseasonal and seasonal variability of the West India Coastal Current on the continental slope

Authors: P Amol; D Shankar; V Fernando; A Mukherjee; S G Aparna; R Fernandes; G S Michael; S T Khalap; N P Satelkar; Y Agarvadekar; M G Gaonkar; A P Tari; A Kankonkar; S P Vernekar

Journal: Journal of Earth System Science

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12040-014-0449-5

URL: <https://doi.org/10.1007/s12040-014-0449-5>

Publication Date: 2014-07-12

Volume/Issue: Volume 123, Issue 5

References: 131

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1700. Global Biogeochemical Implications of Mercury Discharges from Rivers and Sediment Burial

Authors: Helen M. Amos (Department of Environmental Health, Harvard School of Public Health, Boston, Massachusetts 02115, United States); Daniel J. Jacob; David Kocman (Department of Environmental Sciences, Jožef Stefan Institute, 1000 Ljubljana, Slovenia); Hannah M. Horowitz; Yanxu Zhang; Stephanie Dutkiewicz (Massachusetts Institute of Technology); Milena Horvat (Department of Environmental Sciences, Jožef Stefan Institute, 1000 Ljubljana, Slovenia); Elizabeth S. Corbitt; David P. Krabbenhoft (U.S. Geological Survey, Middleton, Wisconsin 53562, United States); Elsie M. Sunderland (Department of Environmental Health, Harvard School of Public Health, Boston, Massachusetts 02115, United States)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

DOI: 10.1021/es502134t

URL: <https://doi.org/10.1021/es502134t>

Publication Date: 2014-07-09

Volume/Issue: Volume 48, Issue 16

References: 80

Cited by: 255 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission, National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

1701. Assessing the horizontal refraction of ocean acoustic tomography signals using high-resolution ocean state estimates

Authors: Brian D. Dushaw (University of Washington)

Journal: The Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

DOI: 10.1121/1.4881928

URL: <https://doi.org/10.1121/1.4881928>

Publication Date: 2014-07-03

Volume/Issue: Volume 136, Issue 1

References: 30

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: ONR

License: N/A

Open Access: N/A

Abstract: The analysis of signals for acoustic tomography sent between a source and a receiver most often uses the unrefracted geodesic path, an approximation that is justified from theoretical considerations, relying on estimates of horizontal gradients of sound speed, or on simple theoretical models. To quantify the effects of horizontal refraction caused by a realistic ocean environment, horizontal refractions of long-range signals were computed using global ocean state estimates for 2004 from the Estimating the Circulation and Climate of the Ocean (ECCO2) project. Basin-scale paths in the eastern North Pacific Ocean and regional-scale paths in the Philippine Sea were used as examples. At O(5 Mm) basin scales, refracted geodesic and geodesic paths differed by only about 5 km. Gyre-scale features had the greatest refractive influence, but the precise refractive effects depended on the path geometry with respect to oceanographic features. Refraction decreased travel times by 5-10 ms and changed azimuthal angles by about 0.2 degrees. At O(500 km) regional scales, paths deviated from the geodesic by only 250 m, and travel times deviated by less than 0.5 ms. Such effects are of little consequence in the analysis of tomographic data. Refraction details depend only slightly on mode number and frequency.

1702. Non-linearity of geocentre motion and its impact on the origin of the terrestrial reference frame

Authors: Danan Dong; Weijing Qu; Peng Fang; Dongju Peng

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggu187

URL: <https://doi.org/10.1093/gji/ggu187>

Publication Date: 2014-06-27

Volume/Issue: Volume 198, Issue 2

References: 64

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1703. How important is diversity for capturing environmental-change responses in ecosystem models?

Authors: A. E. F. Prowe; M. Pahlow; S. Dutkiewicz; A. Oschlies

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-11-3397-2014

URL: <https://doi.org/10.5194/bg-11-3397-2014>

Publication Date: 2014-06-27

Volume/Issue: Volume 11, Issue 12

References: 36

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Marine ecosystem models used to investigate how global change affects ocean ecosystems and their functioning typically omit pelagic plankton diversity. Diversity, however, may affect functions such as primary production and their sensitivity to environmental changes. Here we use a global ocean ecosystem model that explicitly resolves phytoplankton diversity by defining subtypes within four phytoplankton functional types (PFTs). We investigate the model's ability to capture diversity effects on primary production under environmental change. An idealized scenario with a sudden reduction in vertical mixing causes diversity and primary-production changes that turn out to be largely independent of the number of coexisting phytoplankton subtypes. The way diversity is represented in the model provides a small number of niches with respect to nutrient use in accordance with the PFTs defined in the model. Increasing the number of phytoplankton subtypes increases the resolution within the niches. Diversity effects such as niche complementarity operate between, but not within PFTs, and are constrained by the variety of traits and trade-offs resolved in the model. The number and nature of the niches formulated in the model, for example via trade-offs or different PFTs, thus determines the diversity effects on ecosystem functioning captured in ocean ecosystem models.

1704. A numerical study on seasonal variations of the thermocline in the South China Sea based on the ROMS

Authors: Wei Fan; Jinbao Song; Shuang Li

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-014-0504-8

URL: <https://doi.org/10.1007/s13131-014-0504-8>

Publication Date: 2014-06-26

Volume/Issue: Volume 33, Issue 7

References: 22

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1705. Can tidal perturbations associated with sea level variations in the western Pacific Ocean be used to understand future effects of tidal evolution?

Authors: Adam T. Devlin; David A. Jay; Stefan A. Talke; Edward Zaron

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-014-0741-6

URL: <https://doi.org/10.1007/s10236-014-0741-6>

Publication Date: 2014-06-24

Volume/Issue: Volume 64, Issue 8

References: 79

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1706. Assessment of the structure and variability of Weddell Sea water masses in distinct ocean reanalysis products

Authors: T. S. Dotto; R. Kerr; M. M. Mata; M. Azaneu; I. Wainer; E. Fahrbach; G. Rohardt

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-10-523-2014

URL: <https://doi.org/10.5194/os-10-523-2014>

Publication Date: 2014-06-23

Volume/Issue: Volume 10, Issue 3

References: 82

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We assessed and evaluated the performance of five ocean reanalysis products in reproducing essential hydrographic properties and their associated temporal variability for the Weddell Sea, Antarctica. The products used in this assessment were ECMWF ORAS4 (European Centre for Medium-Range Weather Forecasts Ocean Reanalysis System 4), CFSR (Climate Forecast System Reanalysis), MyOcean UR025.4 (University of Reading), ECCO2 (Estimating the Circulation and Climate of the Ocean, Phase II) and SODA (Simple Ocean Data Assimilation). The present study focuses on the Weddell Sea deep layer, which is composed of the following three main water masses: Warm Deep Water (WDW), Weddell Sea Deep Water (WSDW) and Weddell Sea Bottom Water (WSBW). The MyOcean UR025.4 product provided the most accurate representation of the structure and thermohaline properties of the Weddell Sea water masses when compared with observations. All the ocean reanalysis products analyzed exhibited limited capabilities in representing the surface water masses in the Weddell Sea. The CFSR and ECCO2 products were not able to represent deep water masses with a neutral density $\geq 28.40 \text{ kg m}^{-3}$, which was considered the WSBW's upper limit throughout the simulation period. The expected WDW warming was only reproduced by the SODA product, whereas the ECCO2 product was able to represent the trends in the WSDW's hydrographic properties. All the assessed ocean reanalyses were able to represent the decrease in the WSBW's density, except the SODA product in the inner Weddell Sea. Improvements in parameterization may have as much impact on the reanalyses assessed as improvements in horizontal resolution primarily because the Southern Ocean lacks in situ data, and the data that are currently available are summer-biased. The choice of the reanalysis product should be made carefully, taking into account the performance, the parameters of interest, and the type of physical processes to be evaluated.

1707. Horizontal eddy energy flux in the world oceans diagnosed from altimetry data

Authors: Chi Xu; Xiao-Dong Shang; Rui Xin Huang

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/srep05316

URL: <https://doi.org/10.1038/srep05316>

Publication Date: 2014-06-17

Volume/Issue: Volume 4, Issue 1

References: 35

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: N/A

1708. Bathymetric control of tidewater glacier mass loss in northwest Greenland

Authors: David F. Porter; Kirsty J. Tinto; Alexandra Boghosian; James R. Cochran; Robin E. Bell; Serdar S. Manizade; John G. Sonntag

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2014.05.058

URL: <https://doi.org/10.1016/j.epsl.2014.05.058>

Publication Date: 2014-06-17

Volume/Issue: Volume 401

References: 46

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA

License: N/A

Open Access: N/A

Abstract: N/A

1709. The oceanic anthropogenic CO₂ sink: Storage, air-sea fluxes, and transports over the industrial era

Authors: Tim DeVries (University of California Los Angeles)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013gb004739

URL: <https://doi.org/10.1002/2013gb004739>

Publication Date: 2014-06-16

Volume/Issue: Volume 28, Issue 7

References: 63

Cited by: 230 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study presents a new estimate of the oceanic anthropogenic CO₂(Cant) sink over the industrial era (1780 to present), from assimilation of potential temperature, salinity, radiocarbon, and CFC-11 observations in a global steady state ocean circulation inverse model (OCIM). This study differs from previous data-based estimates of the oceanic Cant sink in that dynamical constraints on ocean circulation are accounted for, and the ocean circulation is explicitly modeled, allowing the calculation of oceanic Cant storage, air-sea fluxes, and transports in a consistent manner. The resulting uncertainty of the OCIM-estimated Cant uptake, transport, and storage is significantly smaller than the comparable uncertainty from purely data-based or model-based estimates. The OCIM-estimated oceanic Cant storage is 160-166 PgC in 2012, and the oceanic Cant uptake rate averaged over the period 2000-2010 is 2.6 PgC yr⁻¹ or about 30% of current anthropogenic CO₂ emissions. This result implies a residual (primarily terrestrial) Cant sink of about 1.6 PgC yr⁻¹ for the same period. The Southern Ocean is the primary conduit for Cant entering the ocean, taking up about 1.1 PgC yr⁻¹ in 2012, which represents about 40% of the contemporary oceanic Cant uptake. It is suggested that the most significant source of remaining uncertainty in the oceanic Cant sink is due to potential variability in the ocean circulation over the industrial era.

1710. A Description of Local and Nonlocal Eddy-Mean Flow Interaction in a Global Eddy-Permitting State Estimate

Authors: Ru Chen (Scripps Institution of Oceanography / University of California San Diego); Glenn R. Flierl (Massachusetts Institute of Technology); Carl Wunsch (Harvard University, Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-14-0009.1

URL: <https://doi.org/10.1175/jpo-d-14-0009.1>

Publication Date: 2014-06-13

Volume/Issue: Volume 44, Issue 9

References: 78

Cited by: 93 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The assumption that local baroclinic instability dominates eddy-mean flow interactions is tested on a global scale using a dynamically consistent eddy-permitting state estimate. Interactions are divided into local and nonlocal. If all the energy released from the mean flow through eddy-mean flow interaction is used to support eddy growth in the same region, or if all the energy released from eddies through eddy-mean flow interaction is used to feed back to the mean flow in the same region, eddy-mean flow interaction is local; otherwise, it is nonlocal. Different regions have different characters: in the subtropical region studied in detail, interactions are dominantly local. In the Southern Ocean and Kuroshio and Gulf Stream Extension regions, they are mainly nonlocal. Geographical variability of dominant eddy-eddy and eddy-mean flow processes is a dominant factor in understanding ocean energetics.

1711. Non-linear motions of Australian geodetic stations induced by non-tidal ocean loading and the passage of tropical cyclones

Authors: A. Mémin; C. Watson; I. D. Haigh; L. MacPherson; P. Tregoning

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-014-0734-8

URL: <https://doi.org/10.1007/s00190-014-0734-8>

Publication Date: 2014-06-11

Volume/Issue: Volume 88, Issue 10

References: 37

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1712. On the Flow Through Bering Strait: A Synthesis of Model Results and Observations

Authors: Jaclyn Clement Kinney; Wieslaw Maslowski; Yevgeny Aksenov; Beverly de Cuevas; Jaromir Jakacki; An Nguyen; Robert Osinski; Michael Steele; Rebecca A. Woodgate; Jinlun Zhang

Journal: The Pacific Arctic Region

Publisher: Springer Netherlands

DOI: 10.1007/978-94-017-8863-2_7

URL: https://doi.org/10.1007/978-94-017-8863-2_7

Publication Date: 2014-06-10

Volume/Issue: N/A

References: 79

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: Pacific water, Bering strait, Ocean modeling, Numerical modeling

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1713. Eddies in the Red Sea: A statistical and dynamical study

Authors: Peng Zhan (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia); Aneesh C. Subramanian (Scripps Institution of Oceanography / University of California San Diego); Fengchao Yao (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia); Ibrahim Hoteit (Division of Physical Sciences and Engineering King Abdullah University of Science and Technology Thuwal Saudi Arabia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009563

URL: <https://doi.org/10.1002/2013jc009563>

Publication Date: 2014-06-06

Volume/Issue: Volume 119, Issue 6

References: 59

Cited by: 110 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: King Abdullah University of Science and Technology (KAUST), AVISO

License: N/A

Open Access: N/A

Abstract: Sea level anomaly (SLA) data spanning 1992-2012 were analyzed to study the statistical properties of eddies in the Red Sea. An algorithm that identifies winding angles was employed to detect 4998 eddies propagating along 938 unique eddy tracks. Statistics suggest that eddies are generated across the entire Red Sea but that they are prevalent in certain regions. A high number of eddies is found in the central basin between 18 degreesN and 24 degreesN. More than 87% of the detected eddies have a radius ranging from 50 to 135 km. Both the intensity and relative vorticity scale of these eddies decrease as the eddy radii increase. The averaged eddy lifespan is approximately 6 weeks. AEs and cyclonic eddies (CEs) have different deformation features, and those with stronger intensities are less deformed and more circular. Analysis of long-lived eddies suggests that they are likely to appear in the central basin with AEs tending to move northward. In addition, their eddy kinetic energy (EKE) increases gradually throughout their lifespans. The annual cycles of CE and AE differ, although both exhibit significant seasonal cycles of intensity with the winter and summer peaks appearing in February and August, respectively. The seasonal cycle of EKE is negatively correlated with stratification but positively correlated with vertical shear of horizontal velocity and eddy growth rate, suggesting that the generation of baroclinic instability is responsible for the activities of eddies in the Red Sea.

1714. The ocean's role in polar climate change: asymmetric Arctic and Antarctic responses to greenhouse gas and ozone forcing

Authors: John Marshall (Massachusetts Institute of Technology); Kyle C. Armour (Massachusetts Institute of Technology); Jeffery R. Scott (Massachusetts Institute of Technology); Yavor Kostov (Massachusetts Institute of Technology); Ute Hausmann (Massachusetts Institute of Technology); David Ferreira (University of Reading); Theodore G. Shepherd (University of Reading); Cecilia M. Bitz (University of Washington)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rsta.2013.0040

URL: <https://doi.org/10.1098/rsta.2013.0040>

Publication Date: 2014-06-03

Volume/Issue: Volume 372, Issue 2019

References: 72

Cited by: 111 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In recent decades, the Arctic has been warming and sea ice disappearing. By contrast, the Southern Ocean around Antarctica has been (mainly) cooling and sea-ice extent growing. We argue here that interhemispheric asymmetries in the mean ocean circulation, with sinking in the northern North Atlantic and upwelling around Antarctica, strongly influence the sea-surface temperature (SST) response to anthropogenic greenhouse gas (GHG) forcing, accelerating warming in the Arctic while delaying it in the Antarctic. Furthermore, while the amplitude of GHG forcing has been similar at the poles, significant ozone depletion only occurs over Antarctica. We suggest that the initial response of SST around Antarctica to ozone depletion is one of cooling and only later adds to the GHG-induced warming trend as upwelling of sub-surface warm water associated with stronger surface westerlies impacts surface properties. We organize our discussion around 'climate response functions' (CRFs), i.e. the response of the climate to 'step' changes in anthropogenic forcing in which GHG and/or ozone-hole forcing is abruptly turned on and the transient response of the climate revealed and studied. Convolutions of known or postulated GHG and ozone-hole forcing functions with their respective CRFs then yield the transient forced SST response (implied by linear response theory), providing a context for discussion of the differing warming/cooling trends in the Arctic and Antarctic. We speculate that the period through which we are now passing may be one in which the delayed warming of SST associated with GHG forcing around Antarctica is largely cancelled by the cooling effects associated with the ozone hole. By mid-century, however, ozone-hole effects may instead be adding to GHG warming around Antarctica but with diminished amplitude as the ozone hole heals. The Arctic, meanwhile, responding to GHG forcing but in a manner amplified by ocean heat transport, may continue to warm at an accelerating rate.

1715. Limitations of the potential predictability of meridional mass and heat transports in the North Atlantic

Authors: Bente Tiedje (University of Hamburg); Johanna Baehr (University of Hamburg)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gl060174

URL: <https://doi.org/10.1002/2014gl060174>

Publication Date: 2014-06-01

Volume/Issue: Volume 41, Issue 12

References: 34

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The climatological relevance of the Atlantic meridional overturning circulation (AMOC) stems from its capability to transport heat meridionally. The variability of AMOC and the meridional heat transport (MHT) are intimately linked, but not interchangeable. Here we ask to what extent AMOC's predictability may be used as a proxy for MHT's predictability. We find that the geostrophic parts of MHT and AMOC show similar potential predictability (0°-45°N), which originates from the potential predictability of the density field at the western boundary (above 1700 m). Maximum timescales of potential predictability are found for both MHT's and AMOC's geostrophic parts between the subpolar and the subtropical gyre, while meridionally propagating density anomalies at the western boundary limit the potential predictability north and south of the gyre boundary. The inference of MHT predictability from AMOC predictability is thus limited to the geostrophic parts of MHT and AMOC, and particularly adequate at the gyre boundary.

1716. How Much Has the North Atlantic Ocean Overturning Circulation Changed in the Last 50 Years?

Authors: Simon F. B. Tett (University of Edinburgh); Toby J. Sherwin (Scottish Association for Marine Science, Scottish Marine Institute, Oban, United Kingdom); Amrita Shrivastava (University of Edinburgh); Oliver Browne (University of Edinburgh)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00095.1

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Publication Date: 2014-05-29

Volume/Issue: Volume 27, Issue 16

References: 47

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Volume transports from six ocean reanalyses are compared with four sets of in situ observations: across the Greenland-Scotland ridge (GSR), in the Labrador Sea boundary current, in the deep western boundary current at 43 degreesN, and in the Atlantic meridional overturning circulation (AMOC) at 26 degreesN in the North Atlantic. The higher-resolution reanalyses (on the order of 1/4 degrees x 1/4 degrees) are better at reproducing the circulation pattern in the subpolar gyre than those with lower resolution (on the order of 1 degrees). Simple Ocean Data Assimilation (SODA) and Estimating the Circulation and Climate of the Ocean (ECCO)-Jet Propulsion Laboratory (JPL) produce transports at 26 degreesN that are close to those observed [17 Sv (1 Sv = 106 m³ s⁻¹)]. ECCO, version 2, and SODA produce northward transports across the GSR (observed transport of 8.2 Sv) that are 22% and 29% too big, respectively. By contrast, the low-resolution reanalyses have transports that are either too small [by 31% for ECCO-JPL and 49% for Ocean Reanalysis, system 3 (ORA-S3)] or much too large [Decadal Prediction System (DePreSys)]. SODA had the best simulations of mixed layer depth and with two coarse grid long-term reanalyses (DePreSys and ORA-S3) is used to examine changes in North Atlantic circulation from 1960 to 2008. Its results suggest that the AMOC increased by about 20% at 26 degreesN while transport across the GSR hardly altered. The other (less reliable) long-term reanalyses also had small changes across the GSR but changes of +10% and -20%, respectively, at 26 degreesN. Thus, it appears that changes in the overturning circulation at 26 degreesN are decoupled from the flow across the GSR. It is recommended that transport observations should not be assimilated in ocean reanalyses but used for validation instead.

1717. Data assimilation: making sense of Earth Observation

Authors: William A. Lahoz; Philipp Schneider

Journal: Frontiers in Environmental Science

Publisher: Frontiers Media SA

DOI: 10.3389/fenvs.2014.00016

URL: <https://doi.org/10.3389/fenvs.2014.00016>

Publication Date: 2014-05-28

Volume/Issue: Volume 2

References: 11

Cited by: 129 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1718. Seaglider observations of equatorial Indian Ocean Rossby waves associated with the Madden-Julian Oscillation

Authors: Benjamin G. M. Webber (University of East Anglia); Adrian J. Matthews (University of East Anglia); Karen J. Heywood (University of East Anglia); Jan Kaiser (University of East Anglia); Sunke Schmidtke (GEOMAR Helmholtz Centre for Ocean Research, University of East Anglia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009657

URL: <https://doi.org/10.1002/2013jc009657>

Publication Date: 2014-05-26

Volume/Issue: Volume 119, Issue 6

References: 44

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: MJO, ocean dynamics, equatorial ocean Rossby waves, DYNAMO/CINDY/AMIE/LASP, AUV

Funding: European Union 7th Framework Programme

License: N/A

Open Access: N/A

Abstract: During the CINDY-DYNAMO field campaign of September 2011-January 2012, a Seaglider was deployed at 80 degreesE and completed 10 north-south sections between 3 and 4 degreesS, measuring temperature, salinity, dissolved oxygen concentration, and chlorophyll fluorescence. These high-resolution subsurface observations provide insight into equatorial ocean Rossby wave activity forced by three Madden-Julian Oscillation (MJO) events during this time period. These Rossby waves generate variability in temperature $O(1 \text{ degreesC})$, salinity $O(0.2 \text{ g kg}^{-1})$, density $O(0.2 \text{ kg m}^{-3})$, and oxygen concentration $O(10 \text{ mumol kg}^{-1})$, associated with 10 m vertical displacements of the thermocline. The variability extends down to 1000 m, the greatest depth of the Seaglider observations, highlighting the importance of surface forcing for the deep equatorial ocean. The temperature variability observed by the Seaglider is greater than that simulated in the ECCO-JPL reanalysis, especially at depth. There is also marked variability in chlorophyll fluorescence at the surface and at the depth of the chlorophyll maximum. Upwelling from Rossby waves and local wind stress curl leads to an enhanced shoaling of the chlorophyll maximum by 10-25 m in response to the increased availability of nutrients and light. This influence of the MJO on primary production via equatorial ocean Rossby waves has not previously been recognized.

1719. Process-based analysis of climate model ENSO simulations: Intermodel consistency and compensating errors

Authors: Marianna Linz (Massachusetts Institute of Technology); Eli Tziperman (Harvard University); Douglas G. MacMartin (California Institute of Technology)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jd021415

URL: <https://doi.org/10.1002/2013jd021415>

Publication Date: 2014-05-26

Volume/Issue: Volume 119, Issue 12

References: 32

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1720. Linkage between the Pacific Decadal Oscillation and the low frequency variability of the Pacific Subtropical Cell

Authors: Lingya Hong (Ocean University of China); Liping Zhang (Princeton University, NOAA/Geophysical Fluid Dynamics Laboratory Princeton New Jersey USA); Zhaohui Chen (Ocean University of China); Lixin Wu (Ocean University of China)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009650

URL: <https://doi.org/10.1002/2013jc009650>

Publication Date: 2014-05-24

Volume/Issue: Volume 119, Issue 6

References: 37

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: decadal variability, STC, PDO, SODA, buoyancy forced adjustment (and 1 more)

Funding: National Natural Science Foundation of China (NSFC), China National Global Change Major Research Project, NSFC Innovation Project (and 1 more)

License: N/A

Open Access: N/A

Abstract: The decadal variability of Pacific Subtropical Cell (STC) and its linkages with the Pacific Decadal Oscillation (PDO) are investigated in the present study based on a Simple Ocean Data Assimilation (SODA 2.2.4). It is found that, on decadal time scales, the western boundary and interior pycnocline transports are anticorrelated and the variation of the interior component is more significant, which is consistent with previous studies. The decadal variability of STC in the Northern Hemisphere is found to be strongly associated with PDO. Associated with a positive (negative) phase of PDO, the relaxation (acceleration) of the northeast trades slows down (spins up) the STC within a few years through baroclinic adjustment in conjunction with the subduction of the cold (warm) mixed-layer anomalies in the extratropics. The cold (warm) water is then injected into the thermocline and advected further southwestward to the tropics along the isopycnal surfaces, leading to the slowdown (spin-up) of STC due to zonal pressure gradient change at low latitude. Along with the STC weakening (strengthening), a significant warming (cold) anomaly appears in the tropics and it is advected to the midlatitude by the Kuroshio and North Pacific currents, thus feeding back to the atmosphere over the North Pacific. In contrast to the Northern Hemisphere, it is found the STC in the south only passively responds to the PDO. The mechanism found here highlights the role of the STC advection of extratropical anomalies to the tropics and horizontal gyre advection of the tropical anomalies to the extratropics in decadal variability of the STC and PDO.

1721. Isopycnal Eddy Diffusivities and Critical Layers in the Kuroshio Extension from an Eddying Ocean Model

Authors: Ru Chen (Scripps Institution of Oceanography / University of California San Diego); Julie L. McClean (Scripps Institution of Oceanography / University of California San Diego); Sarah T. Gille (Scripps Institution of Oceanography / University of California San Diego); Alexa Griesel (University of Hamburg)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-0258.1

URL: <https://doi.org/10.1175/jpo-d-13-0258.1>

Publication Date: 2014-05-23

Volume/Issue: Volume 44, Issue 8

References: 84

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: Mixing, Atm/Ocean Structure/ Phenomena, Eddies, Circulation/ Dynamics, Mesoscale processes

Funding: N/A

License: N/A

Open Access: N/A

Abstract: High spatial resolution isopycnal diffusivities are estimated in the Kuroshio Extension (KE) region (28 degrees-40 degreesN, 120 degrees-190 degreesE) from a global degrees Parallel Ocean Program (POP) simulation. The numerical float tracks are binned using a clustering approach. The number of tracks in each bin is thus roughly the same leading to diffusivity estimates that converge better than those in bins defined by a regular geographic grid. Cross-stream diffusivities are elevated in the southern recirculation gyre region, near topographic obstacles and downstream in the KE jet, where the flow has weakened. Along-stream diffusivities, which are much larger than cross-stream diffusivities, correlate well with the magnitudes of eddy velocity. The KE jet suppresses cross-stream mixing only in some longitude ranges. This study estimates the critical layer depth both from linear local baroclinic instability analysis and from eddy phase speeds in the POP model using the Radon transform. The latter is a better predictor of large mixing length in the cross-stream direction. Critical layer theory is most applicable in the intense jet regions away from topography.

1722. Interannual variation of the southern limit in the Yellow Sea Bottom Cold Water and its causes

Authors: Hee-Won Yang; Yang-Ki Cho; Gwang-Ho Seo; Sung Hyup You; Jang-Won Seo

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2014.05.007

URL: <https://doi.org/10.1016/j.jmarsys.2014.05.007>

Publication Date: 2014-05-20

Volume/Issue: Volume 139

References: 27

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Korea Meteorological Administration Research and Development Program

License: N/A

Open Access: N/A

Abstract: N/A

1723. Antarctic ice sheet fertilises the Southern Ocean

Authors: R. Death; J. L. Wadham; F. Monteiro; A. M. Le Brocq; M. Tranter; A. Ridgwell; S. Dutkiewicz; R. Raiswell

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-11-2635-2014

URL: <https://doi.org/10.5194/bg-11-2635-2014>

Publication Date: 2014-05-19

Volume/Issue: Volume 11, Issue 10

References: 60

Cited by: 87 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Southern Ocean (SO) marine primary productivity (PP) is strongly influenced by the availability of iron in surface waters, which is thought to exert a significant control upon atmospheric CO₂ concentrations on glacial/interglacial timescales. The zone bordering the Antarctic Ice Sheet exhibits high PP and seasonal plankton blooms in response to light and variations in iron availability. The sources of iron stimulating elevated SO PP are in debate. Established contributors include dust, coastal sediments/upwelling, icebergs and sea ice. Subglacial meltwater exported at the ice margin is a more recent suggestion, arising from intense iron cycling beneath the ice sheet. Icebergs and subglacial meltwater may supply a large amount of bioavailable iron to the SO, estimated in this study at 0.07-0.2 Tg yr⁻¹. Here we apply the MIT global ocean model (Follows et al., 2007) to determine the potential impact of this level of iron export from the ice sheet upon SO PP. The export of iron from the ice sheet raises modelled SO PP by up to 40%, and provides one plausible explanation for seasonally very high in situ measurements of PP in the near-coastal zone. The impact on SO PP is greatest in coastal regions, which are also areas of high measured marine PP. These results suggest that the export of Antarctic runoff and icebergs may have an important impact on SO PP and should be included in future biogeochemical modelling.

1724. Carbon monitoring system flux estimation and attribution: impact of ACOS-GOSAT XCO₂ sampling on the inference of terrestrial biospheric sources and sinks

Authors: Junjie Liu; Kevin W. Bowman; Meemong Lee; Daven K. Henze; Nicolas Bousserez; Holger Brix; G. James Collatz; Dimitris Menemenlis; Lesley Ott; Steven Pawson; Dylan Jones; Ray Nassar

Journal: Tellus B: Chemical and Physical Meteorology

Publisher: Stockholm University Press

DOI: 10.3402/tellusb.v66.22486

URL: <https://doi.org/10.3402/tellusb.v66.22486>

Publication Date: 2014-05-09

Volume/Issue: Volume 66, Issue 1

References: 66

Cited by: 94 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1725. Simulated melt rates for the Totten and Dalton ice shelves

Authors: D. E. Gwyther; B. K. Galton-Fenzi; J. R. Hunter; J. L. Roberts

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-10-267-2014

URL: <https://doi.org/10.5194/os-10-267-2014>

Publication Date: 2014-05-06

Volume/Issue: Volume 10, Issue 3

References: 63

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Totten Glacier is rapidly losing mass. It has been suggested that this mass loss is driven by changes in oceanic forcing; however, the details of the ice-ocean interaction are unknown. Here we present results from an ice shelf-ocean model of the region that includes the Totten, Dalton and Moscow University ice shelves, based on the Regional Oceanic Modeling System for the period 1992-2007. Simulated area-averaged basal melt rates (net basal mass loss) for the Totten and Dalton ice shelves are 9.1 m ice yr⁻¹ (44.5 Gt ice yr⁻¹) and 10.1 m ice yr⁻¹ (46.6 Gt ice yr⁻¹), respectively. The melting of the ice shelves varies strongly on seasonal and interannual timescales. Basal melting (mass loss) from the Totten ice shelf spans a range of 5.7 m ice yr⁻¹ (28 Gt ice yr⁻¹) on interannual timescales and 3.4 m ice yr⁻¹ (17 Gt ice yr⁻¹) on seasonal timescales. This study links basal melt of the Totten and Dalton ice shelves to warm water intrusions across the continental shelf break and atmosphere-ocean heat exchange. Totten ice shelf melting is high when the nearby Dalton polynya interannual strength is below average, and vice versa. Melting of the Dalton ice shelf is primarily controlled by the strength of warm water intrusions across the Dalton rise and into the ice shelf cavity. During periods of strong westward coastal current flow, Dalton melt water flows directly under the Totten ice shelf further reducing melting. This is the first such modelling study of this region to provide a valuable framework for directing future observational and modelling efforts.

1726. Metric of the 2-6day sea-surface temperature response to wind stress in the Tropical Pacific and its sensitivity to the K-Profile Parameterization of vertical mixing

Authors: Benjamin M. Wagman; Charles S. Jackson; Fengchao Yao; Sarah E. Zedler; Ibrahim Hoteit

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2014.04.003

URL: <https://doi.org/10.1016/j.ocemod.2014.04.003>

Publication Date: 2014-05-05

Volume/Issue: Volume 79

References: 30

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: The King Abdullah University of Science and Technology

License: N/A

Open Access: N/A

Abstract: N/A

1727. Quasi-biennial oscillation and solar cycle influences on winter Arctic total ozone

Authors: King-Fai Li (University of Washington, University Corporation for Atmospheric Research Boulder Colorado USA); Ka-Kit Tung (University of Washington)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jd021065

URL: <https://doi.org/10.1002/2013jd021065>

Publication Date: 2014-04-30

Volume/Issue: Volume 119, Issue 10

References: 61

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The total column ozone (TCO) observed from satellites and assimilated in the European Centre for Medium-Range Weather Forecasts since 1979 is used as an atmospheric tracer to study the modulations of the winter Arctic stratosphere by the quasi-biennial oscillation (QBO) and the solar cycle. It is found that both the QBO and solar forcings in low latitudes can perturb the late winter polar vortex, likely via planetary wave divergence, causing an early breakdown of the vortex in the form of sudden stratospheric warming. As a result, TCO within the vortex in late winter can increase by ~60 Dobson unit during either a solar maximum or an easterly phase of the QBO, or both, relative to the least perturbed state when the solar cycle is minimum and the QBO is in the westerly phase. In addition, from the solar maximum to the solar minimum during the QBO easterly phase, the change in TCO is found to be statistically insignificant. Therefore, the "reversal" of the Holton-Tan effect, reported in some previous studies using lower stratospheric temperature, is not evident in the TCO behavior of both observation and assimilation.

1728. Assessment of the capabilities of the temporal and spatiotemporal ICA method for geophysical signal separation in GRACE data

Authors: Eva Boergens (University of Bonn); Elena Rangelova (University of Calgary); Michael G. Sideris (University of Calgary); Juergen Kusche (University of Bonn)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jb010452

URL: <https://doi.org/10.1002/2013jb010452>

Publication Date: 2014-04-22

Volume/Issue: Volume 119, Issue 5

References: 63

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We investigate the potential of two independent component analysis (ICA) methods, i.e., the temporal and spatiotemporal ICA, for separating geophysical signals in Gravity Recovery and Climate Experiment (GRACE) data. These methods are based on the assumption of the statistical independence of the signals and thus separate the GRACE-observed mass changes into maximal independent signals. These two ICA methods are compared to the conventional principal component analysis (PCA) method. We test the three methods with respect to their ability to separate a periodic hydrological signal from a trend signal originating in the solid Earth or the cryosphere with simulated and Center for Space Research GRACE mass changes for the time period of January 2003 to December 2010. In addition, we investigate whether the methods are capable of separating hydrological annual and semiannual mass variations. It is shown that both ICA methods are superior to PCA when non-Gaussian mass variations are analyzed. Furthermore, the spatiotemporal ICA resolves successfully the lack of full temporal and spatial independence of the geophysical signals observed by GRACE both in global and regional simulation scenarios. Although the temporal and spatiotemporal ICA are nearly equivalent, both superior to PCA in the global GRACE analysis, the spatiotemporal ICA proves to be more efficient in regional applications by recover more reliably the postglacial rebound trend in North America and the bimodal total water storage variability in Africa.

1729. Evaluation of altimetry-derived surface current products using Lagrangian drifter trajectories in the eastern Gulf of Mexico

Authors: Yonggang Liu (University of South Florida); Robert H. Weisberg (University of South Florida); Stefano Vignudelli (Consiglio Nazionale delle Ricerche; Area Ricerca CNR; Pisa Italy); Gary T. Mitchum (University of South Florida)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009710

URL: <https://doi.org/10.1002/2013jc009710>

Publication Date: 2014-04-21

Volume/Issue: Volume 119, Issue 5

References: 106

Cited by: 81 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1730. Deep Meridional Overturning Circulation in the Indian Ocean and Its Relation to Indian Ocean Dipole

Authors: Weiqiang Wang (Chinese Academy of Sciences); Xiuhua Zhu (University of Hamburg); Chunzai Wang (NOAA/Atlantic Oceanographic and Meteorological Laboratory, Miami, Florida); Armin Köhl (University of Hamburg)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00472.1

URL: <https://doi.org/10.1175/jcli-d-13-00472.1>

Publication Date: 2014-04-04

Volume/Issue: Volume 27, Issue 12

References: 55

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: Atmosphere-ocean interaction, Streamfunction, Climate variability, Interannual variability, Oceanic variability, Seasonal variability

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper uses the 42-yr German Estimating the Circulation and Climate of the Ocean (GECCO) synthesis data to analyze and examine the relationship of the Indian Ocean deep meridional overturning circulation (DMOC) with the Indian Ocean dipole mode (IOD). Contributions of various dynamical processes are assessed by decomposing the DMOC into the Ekman and geostrophic transport, the external mode, and a residual term. The first three terms successfully describe the DMOC with a marginal residual term. The following conclusions are obtained. First, the seasonal cycle of the DMOC is mainly determined by the Ekman component. The exception is during the transitional seasons (March-April and September-October) in the northern Indian Ocean Basin, where the geostrophic component dominates. Second, at the beginning phase of the IOD (May-June), the Ekman component dominates the DMOC structure; at and after the peak phase of the IOD (September-December), the DMOC structure is primarily determined by the geostrophic component in correspondence with the well-developed sea surface temperature anomalies, while the wind (and thus the Ekman component) plays a secondary role south of 10 degreesS and contributes negatively within the zonal band of 10 degrees on both sides of the equator. Therefore, there exists a surface to deep-ocean connection through which IOD-related surface wind and ocean temperature anomalies are transferred down to the deep ocean. Westward-propagating signals are observed even in the deep ocean, suggesting possible roles of Rossby waves in transferring the surface signal to the deep ocean.

1731. Wind-Driven Coastal Sea Level Variability in the Northeast Pacific

Authors: Philip R. Thompson (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Mark A. Merrifield (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Judith R. Wells (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii); Chantel M. Chang (Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, Hawaii)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00225.1

URL: <https://doi.org/10.1175/jcli-d-13-00225.1>

Publication Date: 2014-04-03

Volume/Issue: Volume 27, Issue 12

References: 69

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: Southern Oscillation, Sea level

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The rate of coastal sea level change in the northeast Pacific (NEP) has decreased in recent decades. The relative contributions to the decreased rate from remote equatorial wind stress, local longshore wind stress, and local windstress curl are examined. Regressions of sea level onto wind stress time series and comparisons between NEP and Fremantle sea levels suggest that the decreased rate in the NEP is primarily due to oceanic adjustment to strengthened trade winds along the equatorial and coastal waveguides. When taking care to account for correlations between the various wind stress time series, the roles of longshore wind stress and local windstress curl are found to be of minor importance in comparison to equatorial forcing. The predictability of decadal sea level change rates along the NEP coastline is therefore largely determined by tropical variability. In addition, the importance of accounting for regional, wind-driven sea level variations when attempting to calculate accelerations in the long-term rate of sea level rise is demonstrated.

1732. Low-Frequency SST and Upper-Ocean Heat Content Variability in the North Atlantic

Authors: Martha W. Buckley (Atmospheric and Environmental Research, Lexington, Massachusetts); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts); Gaël Forget (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00316.1

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Publication Date: 2014-04-01

Volume/Issue: Volume 27, Issue 13

References: 68

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: Sea surface temperature, Interannual variability, North Atlantic Ocean

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A recent state estimate covering the period 1992-2010 from the Estimating the Circulation and Climate of the Ocean (ECCO) project is utilized to quantify the upper-ocean heat budget in the North Atlantic on monthly to interannual time scales (seasonal cycle removed). Three novel techniques are introduced: 1) the heat budget is integrated over the maximum climatological mixed layer depth (integral denoted as H), which gives results that are relevant for explaining SST while avoiding strong contributions from vertical diffusion and entrainment; 2) advective convergences are separated into Ekman and geostrophic parts, a technique that is successful away from ocean boundaries; and 3) air-sea heat fluxes and Ekman advection are combined into one local forcing term. The central results of our analysis are as follows: 1) In the interior of subtropical gyre, local forcing explains the majority of H variance on all time scales resolved by the ECCO estimate. 2) In the Gulf Stream region, low-frequency H anomalies are forced by geostrophic convergences and damped by air-sea heat fluxes. 3) In the interior of the subpolar gyre, diffusion and bolus transports play a leading order role in H variability, and these transports are correlated with low-frequency variability in wintertime mixed layer depths.

1733. Rationalizing the Spatial Distribution of Mesoscale Eddy Diffusivity in Terms of Mixing Length Theory

Authors: Michael Bates (Massachusetts Institute of Technology); Ross Tulloch (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology); Raffaele Ferrari (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-0130.1

URL: <https://doi.org/10.1175/jpo-d-13-0130.1>

Publication Date: 2014-03-27

Volume/Issue: Volume 44, Issue 6

References: 74

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: Circulation/Dynamics, Mixing, Eddies, Subgrid-scale processes, Models and modeling (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Observations and theory suggest that lateral mixing by mesoscale ocean eddies only reaches its maximum potential at steering levels, surfaces at which the propagation speed of eddies approaches that of the mean flow. Away from steering levels, mixing is strongly suppressed because the mixing length is smaller than the eddy scale, thus reducing the mixing rates. The suppression is particularly pronounced in strong currents where mesoscale eddies are most energetic. Here, a framework for parameterizing eddy mixing is explored that attempts to capture this suppression. An expression of the surface eddy diffusivity proposed by Ferrari and Nikurashin is evaluated using observations of eddy kinetic energy, eddy scale, and eddy propagation speed. The resulting global maps of eddy diffusivity have a broad correspondence with recent estimates of diffusivity based on the rate at which tracer contours are stretched by altimetric-derived surface currents. Finally, the expression for the eddy diffusivity is extrapolated in the vertical to infer the eddy-induced meridional heat transport and the overturning streamfunction.

1734. Computing transport budgets along the shelf and across the shelf edge in the northern Benguela during summer (DJF) and winter (JJA)

Authors: Muller A.A.; C.J.C. Reason; M. Schmidt; V. Mohrholz; A. Eggert

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2014.02.007

URL: <https://doi.org/10.1016/j.jmarsys.2014.02.007>

Publication Date: 2014-03-27

Volume/Issue: Volume 140

References: 40

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung

License: N/A

Open Access: N/A

Abstract: N/A

1735. Bidecadal Thermal Changes in the Abyssal Ocean

Authors: Carl Wunsch (Harvard University); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-096.1

URL: <https://doi.org/10.1175/jpo-d-13-096.1>

Publication Date: 2014-03-20

Volume/Issue: Volume 44, Issue 8

References: 42

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A dynamically consistent state estimate is used for the period 1992-2011 to describe the changes in oceanic temperatures and heat content, with an emphasis on determining the noise background in the abyssal (below 2000 m) depths. Interpretation requires close attention to the long memory of the deep ocean, implying that meteorological forcing of decades to thousands of years ago should still be producing trendlike changes in abyssal heat content. Much of the deep-ocean volume remained unobserved. At the present time, warming is seen in the deep western Atlantic and Southern Oceans, roughly consistent with those regions of the ocean expected to display the earliest responses to surface disturbances. Parts of the deeper ocean, below 3600 m, show cooling. Most of the variation in the abyssal Pacific Ocean is comparatively featureless, consistent with the slow, diffusive approach to a steady state expected there. In the global average, changes in heat content below 2000 m are roughly 10% of those inferred for the upper ocean over the 20-yr period. A useful global observing strategy for detecting future change has to be designed to account for the different time and spatial scales manifested in the observed changes. If the precision estimates of heat content change are independent of systematic errors, determining oceanic heat uptake values equivalent to 0.1 W m^{-2} is possibly attainable over future bidecadal periods.

1736. Seasonal overturning circulation in the Red Sea: 1. Model validation and summer circulation

Authors: Fengchao Yao (Division of Physical Sciences and Engineering; King Abdullah University of Science and Technology; Thuwal Saudi Arabia); Ibrahim Hoteit (Division of Physical Sciences and Engineering; King Abdullah University of Science and Technology; Thuwal Saudi Arabia); Larry J. Pratt (Woods Hole Oceanographic Institution); Amy S. Bower (Woods Hole Oceanographic Institution); Ping Zhai (Woods Hole Oceanographic Institution); Armin Köhl (University of Hamburg); Ganesh Gopalakrishnan (Scripps Institution of Oceanography)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009004

URL: <https://doi.org/10.1002/2013jc009004>

Publication Date: 2014-03-17

Volume/Issue: Volume 119, Issue 4

References: 45

Cited by: 120 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1737. Interannual Bottom Pressure Signals in the Australian-Antarctic and Bellingshausen Basins

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.); Christopher G. Piecuch (Atmospheric and Environmental Research Inc.)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-0223.1

URL: <https://doi.org/10.1175/jpo-d-13-0223.1>

Publication Date: 2014-03-12

Volume/Issue: Volume 44, Issue 5

References: 40

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Analyses of large-scale (>750 km) ocean bottom pressure pb fields, derived from the Gravity Recovery and Climate Experiment (GRACE) and from an Estimating the Circulation & Climate of the Ocean (ECCO) state estimate, reveal enhanced interannual variability, partially connected to the Antarctic Oscillation, in regions of the Australian-Antarctic Basin and the Bellingshausen Basin, with pb magnitudes comparable to those of sea level and good correlation between the GRACE and ECCO pb series. Consistent with the theory of Gill and Niiler, the patterns of stronger pb variability are partly related to enhanced local wind curl forcing and weakened gradients in H/f , where H is ocean depth and f is the Coriolis parameter. Despite weaker H/f gradients, motions against them are sufficiently strong to play a role in balancing the local wind input. Topographic effects are as or more important than changes in f . Additionally, and contrary to the dominance of barotropic processes at subannual time scales, baroclinic effects are not negligible when balancing wind input at periods of a few years. Results highlight the emerging capability to accurately observe and estimate interannual changes in large-scale pb over the Southern Ocean, with implications for the interpretation of low-frequency variability in sea level in terms of steric height and heat content.

1738. Energetics of lateral eddy diffusion/advection: Part II. Numerical diffusion/diffusivity and gravitational potential energy change due to isopycnal diffusion

Authors: Rui Xin Huang

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-014-0410-0

URL: <https://doi.org/10.1007/s13131-014-0410-0>

Publication Date: 2014-03-10

Volume/Issue: Volume 33, Issue 3

References: 24

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1739. Exploring the Red Sea seasonal ecosystem functioning using a three-dimensional biophysical model

Authors: G. Triantafyllou (Hellenic Centre for Marine Research); F. Yao (Earth Sciences and Engineering, King Abdullah University of Science and Technology; Thuwal Saudi Arabia); G. Petihakis (Hellenic Centre for Marine Research); K. P. Tsiaras (Hellenic Centre for Marine Research); D. E. Raitsos (Plymouth Marine Laboratory); I. Hoteit (Earth Sciences and Engineering, King Abdullah University of Science and Technology; Thuwal Saudi Arabia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009641

URL: <https://doi.org/10.1002/2013jc009641>

Publication Date: 2014-02-19

Volume/Issue: Volume 119, Issue 3

References: 62

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1740. Evaluation of the GECCO2 ocean synthesis: transports of volume, heat and freshwater in the Atlantic

Authors: Armin Köhl (University of Hamburg)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2347

URL: <https://doi.org/10.1002/qj.2347>

Publication Date: 2014-02-12

Volume/Issue: Volume 141, Issue 686

References: 86

Cited by: 127 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Forschung und Wissenschaft

License: N/A

Open Access: N/A

Abstract: We present results from a new synthesis (GECCO2) which covers the years 1948 to 2011 employing a similar configuration of the Massachusetts Institute of Technology general circulation model as the previous 50-year (1952 to 2001) GECCO synthesis. In GECCO2, the resolution was increased; it now includes the Arctic Ocean and a dynamic/thermodynamic sea ice model. The synthesis uses the adjoint method to bring the model into consistency with available hydrographic and satellite data as well as prior estimates of surface fluxes. In comparison to GECCO, GECCO2 provides a better agreement with the assimilated data, however the estimated flux adjustments remain similar to GECCO. Global heat content changes are in agreement with recent observational estimates and the estimate of the global heat flux is close to a radiative forcing estimate. Both show a clear effect of the radiative forcing from volcanic eruptions and a weak relation to ENSO events. In contrast to GECCO, the importance of the Denmark Strait overflow for the variability of the Atlantic Meridional Overturning Circulation (AMOC) is replaced in GECCO2 by water mass transformation in the subpolar gyre, which is shown to be part of the thermohaline circulation if the overturning is defined as a function of density. Heat and freshwater transport estimates in the Atlantic are more consistent with previous estimates than the unconstrained run. Decomposing heat and freshwater transports into overturning and gyre components by averaging on density coordinates demonstrates that in these coordinates the contribution from the gyre circulation largely disappears for heat transport and is reduced for the freshwater transport.

1741. Modeled Trends in Antarctic Sea Ice Thickness

Authors: Paul R. Holland (British Antarctic Survey, Cambridge, United Kingdom); Nicolas Bruneau (British Antarctic Survey, Cambridge, United Kingdom); Clare Enright (University of East Anglia); Martin Losch (Alfred Wegener Institute); Nathan T. Kurtz (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Ron Kwok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00301.1

URL: <https://doi.org/10.1175/jcli-d-13-00301.1>

Publication Date: 2014-02-10

Volume/Issue: Volume 27, Issue 10

References: 78

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Unlike the rapid sea ice losses reported in the Arctic, satellite observations show an overall increase in Antarctic sea ice concentration over recent decades. However, observations of decadal trends in Antarctic ice thickness, and hence ice volume, do not currently exist. In this study a model of the Southern Ocean and its sea ice, forced by atmospheric reanalyses, is used to assess 1992-2010 trends in ice thickness and volume. The model successfully reproduces observations of mean ice concentration, thickness, and drift, and decadal trends in ice concentration and drift, imparting some confidence in the hindcasted trends in ice thickness. The model suggests that overall Antarctic sea ice volume has increased by approximately 30 km³ yr⁻¹ (0.4% yr⁻¹) as an equal result of areal expansion (20 × 10³ km² yr⁻¹ or 0.2% yr⁻¹) and thickening (1.5 mm yr⁻¹ or 0.2% yr⁻¹). This ice volume increase is an order of magnitude smaller than the Arctic decrease, and about half the size of the increased freshwater supply from the Antarctic Ice Sheet. Similarly to the observed ice concentration trends, the small overall increase in modeled ice volume is actually the residual of much larger opposing regional trends. Thickness changes near the ice edge follow observed concentration changes, with increasing concentration corresponding to increased thickness. Ice thickness increases are also found in the inner pack in the Amundsen and Weddell Seas, where the model suggests that observed ice-drift trends directed toward the coast have caused dynamical thickening in autumn and winter. Modeled changes are predominantly dynamic in origin in the Pacific sector and thermodynamic elsewhere.

1742. Decadal variations of Pacific North Equatorial Current bifurcation from multiple ocean products

Authors: Fangguo Zhai (Chinese Academy of Sciences, Third Institute of Oceanography State Oceanic Administration Xiamen China); Qingye Wang (Chinese Academy of Sciences); Fujun Wang (Chinese Academy of Sciences); Dunxin Hu (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009692

URL: <https://doi.org/10.1002/2013jc009692>

Publication Date: 2014-02-08

Volume/Issue: Volume 119, Issue 2

References: 60

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Project of State Strategic Program of Global Change

License: N/A

Open Access: N/A

Abstract: In this study, we examine the decadal variations of the Pacific North Equatorial Current (NEC) bifurcation latitude (NBL) averaged over upper 100 m and underlying dynamics over the past six decades using 11 ocean products, including seven kinds of ocean reanalyzes based on ocean data assimilation systems, two kinds of numerical simulations without assimilating observations and two kinds of objective analyzes based on in situ observations only. During the period of 1954-2007, the multiproduct mean of decadal NBL anomalies shows maxima around 1965/1966, 1980/1981, 1995/1996, and 2003/2004, and minima around 1958, 1971/1972, 1986/1987, and 2000/2001, respectively. The NBL decadal variations are related to the first Empirical Orthogonal Function mode of decadal anomalies of sea surface height (SSH) in the northwestern tropical Pacific Ocean, which shows spatially coherent variation over the whole region and explains most of the total variance. Further regression and composite analyzes indicate that northerly/southerly NBL corresponds to negative/positive SSH anomalies and cyclonic/anticyclonic gyre anomalies in the northwestern tropical Pacific Ocean. These decadal circulation variations and thus the decadal NBL variations are governed mostly by the first two vertical modes and attribute the most to the first baroclinic mode. The NBL decadal variation is highly positively correlated with the tropical Pacific decadal variability (TPDV) around the zero time lag. With a lead of about half the decadal cycle the NBL displays closer but negative relationship to TPDV in four ocean products, possibly manifesting the dynamical role of the circulation in the northwestern tropical Pacific in the phase-shifting of TPDV.

1743. Weighing the ocean with bottom-pressure sensors: robustness of the ocean mass annual cycle estimate

Authors: Joanne Williams; C. W. Hughes; M. E. Tamisiea; S. D. P. Williams

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/osd-11-453-2014

URL: <https://doi.org/10.5194/osd-11-453-2014>

Publication Date: 2014-02-05

Volume/Issue: N/A

References: 30

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Annual cycle|Robustness|Ocean tide|Ocean bottom

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We use ocean bottom pressure measurements from 17 tropical sites to determine the annual cycle of ocean mass. We show that such a calculation is robust, and use three methods to estimate errors in the mass determination. Our final best estimate, using data from the best sites and two ocean models, is that the annual cycle has an amplitude of 0.85 mbar (equivalent to 8.4 mm of sea level, or 3100 Gt of water), with a 95% chance of lying within the range 0.61-1.17 mbar. The time of the peak in ocean mass is 10 October, with 95% chance of occurring between 21 September and 25 October. The simultaneous fitting of annual ocean mass also improves the fitting of bottom pressure instrument drift.

1744. Large-scale impact of Saharan dust on the North Atlantic Ocean circulation

Authors: N. Serra (University of Hamburg); N. Martínez Avellaneda (Scripps Institution of Oceanography / University of California San Diego); D. Stammer (University of Hamburg)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009274

URL: <https://doi.org/10.1002/2013jc009274>

Publication Date: 2014-02-04

Volume/Issue: Volume 119, Issue 2

References: 40

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The potential for a dynamical impact of Saharan mineral dust on the North Atlantic Ocean large-scale circulation is investigated. To this end, an ocean general circulation model forced by atmospheric fluxes is perturbed by an idealized, seasonally varying, net shortwave flux anomaly, as it results from remote sensing observations of aerosol optical thickness representing Saharan dust load in the atmosphere. The dust dynamical impact on the circulation is assessed through a comparison between perturbed and an unperturbed run. Results suggest that, following the dust-induced shortwave irradiance anomaly, a buoyancy anomaly is created in the Atlantic offshore the African coast, which over the course of the time propagates westward into the interior Atlantic while progressively subducting. Changes in the large-scale barotropic and overturning circulations are significant after 3 years, which coincides with the elapsed time required by the bulk of the buoyancy perturbation to reach the western boundary of the North Atlantic. Although small in amplitude, the changes in the meridional overturning are of the same order as interannual-to-decadal variability. Variations in the amplitude of the forcing lead to changes in the amplitude of the response, which is almost linear during the first 3 years. In addition, a fast, but dynamically insignificant, response can be observed to propagate poleward along the eastern boundary of the North Atlantic, which contributes to a nonlinear response in the subpolar region north of 40 degreesN.

1745. A Global Analysis of Sverdrup Balance Using Absolute Geostrophic Velocities from Argo

Authors: Alison R. Gray (University of Washington); Stephen C. Riser (University of Washington)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-0206.1

URL: <https://doi.org/10.1175/jpo-d-12-0206.1>

Publication Date: 2014-01-28

Volume/Issue: Volume 44, Issue 4

References: 38

Cited by: 78 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Using observations from the Argo array of profiling floats, the large-scale circulation of the upper 2000 decibars (db) of the global ocean is computed for the period from December 2004 to November 2010. The geostrophic velocity relative to a reference level of 900 db is estimated from temperature and salinity profiles, and the absolute geostrophic velocity at the reference level is estimated from the trajectory data provided by the floats. Combining the two gives the absolute geostrophic velocity on 29 pressure surfaces spanning the upper 2000 db of the global ocean. These velocities, together with satellite observations of wind stress, are then used to evaluate Sverdrup balance, the simple canonical theory relating meridional geostrophic transport to wind forcing. Observed transports agree well with predictions based on the wind field over large areas, primarily in the tropics and subtropics. Elsewhere, especially at higher latitudes and in boundary regions, Sverdrup balance does not accurately describe meridional geostrophic transports, possibly due to the increased importance of the barotropic flow, nonlinear dynamics, and topographic influence. Thus, while it provides an effective framework for understanding the zero-order wind-driven circulation in much of the global ocean, Sverdrup balance should not be regarded as axiomatic.

1746. Evaluating Greenland glacial isostatic adjustment corrections using GRACE, altimetry and surface mass balance data

Authors: Tyler C Sutterley; Isabella Velicogna; Beata Csatho; Michiel van den Broeke; Soroush Rezvan-Behbahani; Greg Babonis

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/9/1/014004

URL: <https://doi.org/10.1088/1748-9326/9/1/014004>

Publication Date: 2014-01-17

Volume/Issue: Volume 9, Issue 1

References: 63

Cited by: 19 publications

Language: N/A

Subjects: N/A

Keywords: Post-glacial rebound|Greenland ice sheet|Groenlandia|Glacier mass balance

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1747. A 4DVAR System for the Navy Coastal Ocean Model. Part I: System Description and Assimilation of Synthetic Observations in Monterey Bay*

Authors: Hans Ngodock (Naval Research Laboratory, Stennis Space Center, Mississippi); Matthew Carrier (Naval Research Laboratory, Stennis Space Center, Mississippi)

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/mwr-d-13-00221.1

URL: <https://doi.org/10.1175/mwr-d-13-00221.1>

Publication Date: 2014-01-16

Volume/Issue: Volume 142, Issue 6

References: 51

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A 4D variational data assimilation system was developed for assimilating ocean observations with the Navy Coastal Ocean Model. It is described in this paper, along with initial assimilation experiments in Monterey Bay using synthetic observations. The assimilation system is tested in a series of twin data experiments to assess its ability to fit assimilated and independent observations by controlling the initial conditions and/or the external forcing while assimilating surface and/or subsurface observations. In all strong and weak constraint experiments, the minimization of the cost function is done with both the gradient descent method (in the control space) and the representer method (observation space). The accuracy of the forecasts following the analysis and the relevance of the retrieved forcing correction in the case of weak constraints are evaluated. It is shown that the assimilation system generally fits the assimilated and nonassimilated observations well in all experiments, yielding lower forecast errors.

1748. Variations of the North Pacific Subtropical Mode Water from Direct Observations

Authors: Luc Rainville (University of Washington); Steven R. Jayne (Woods Hole Oceanographic Institution); Meghan F. Cronin (NOAA/Pacific Marine Environmental Laboratory, Seattle, Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00227.1

URL: <https://doi.org/10.1175/jcli-d-13-00227.1>

Publication Date: 2014-01-16

Volume/Issue: Volume 27, Issue 8

References: 46

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Mooring measurements from the Kuroshio Extension System Study (June 2004-June 2006) and from the ongoing Kuroshio Extension Observatory (June 2004-present) are combined with float measurements of the Argo network to study the variability of the North Pacific Subtropical Mode Water (STMW) across the entire gyre, on time scales from days, to seasons, to a decade. The top of the STMW follows a seasonal cycle, although observations reveal that it primarily varies in discrete steps associated with episodic wind events. The variations of the STMW bottom depth are tightly related to the sea surface height (SSH), reflecting mesoscale eddies and large-scale variations of the Kuroshio Extension and recirculation gyre systems. Using the observed relationship between SSH and STMW, gridded SSH products and in situ estimates from floats are used to construct weekly maps of STMW thickness, providing nonbiased estimates of STMW total volume, annual formation and erosion volumes, and seasonal and interannual variability for the past decade. Year-to-year variations are detected, particularly a significant decrease of STMW volume in 2007-10 primarily attributable to a smaller volume formed. Variability of the heat content in the mode water region is dominated by the seasonal cycle and mesoscale eddies; there is only a weak link to STMW on interannual time scales, and no long-term trends in heat content and STMW thickness between 2002 and 2011 are detected. Weak lagged correlations among air-sea fluxes, oceanic heat content, and STMW thickness are found when averaged over the northwestern Pacific recirculation gyre region.

1749. Antipodal acoustic thermometry: 1960, 2004

Authors: Brian D. Dushaw; Dimitris Menemenlis

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2013.12.008

URL: <https://doi.org/10.1016/j.dsr.2013.12.008>

Publication Date: 2014-01-09

Volume/Issue: Volume 86

References: 64

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Office of Naval Research

License: N/A

Open Access: N/A

Abstract: N/A

1750. Observations on the flow structures and transport in a simulated warm-core ring in the Gulf of Mexico

Authors: Doug Lipinski; Kamran Mohseni

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-013-0674-5

URL: <https://doi.org/10.1007/s10236-013-0674-5>

Publication Date: 2013-12-26

Volume/Issue: Volume 64, Issue 1

References: 46

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1751. Seasonal barotropic sea surface height fluctuation in relation to regional ocean mass variation

Authors: Tsurane Kuragano; Yosuke Fujii; Takahiro Toyoda; Norihisa Usui; Koji Ogawa; Masafumi Kamachi

Journal: Journal of Oceanography

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10872-013-0211-7

URL: <https://doi.org/10.1007/s10872-013-0211-7>

Publication Date: 2013-12-26

Volume/Issue: Volume 70, Issue 1

References: 30

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1752. Effect of river runoff on sea level from in-situ measurements and numerical models in the Bay of Biscay

Authors: Irene Laiz; Luis Ferrer; Theocharis A. Plomaritis; Guillaume Charria

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2013.12.013

URL: <https://doi.org/10.1016/j.dsr2.2013.12.013>

Publication Date: 2013-12-24

Volume/Issue: Volume 106

References: 81

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministerio de Ciencia e Innovación (Spain), EU FP7 Environment Research Programme through Project "MICORE"

License: N/A

Open Access: N/A

Abstract: N/A

1753. Swept under the carpet: organic matter burial decreases global ocean biogeochemical model sensitivity to remineralization length scale

Authors: I. Kriest; A. Oschlies

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-10-8401-2013

URL: <https://doi.org/10.5194/bg-10-8401-2013>

Publication Date: 2013-12-20

Volume/Issue: Volume 10, Issue 12

References: 64

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission

License: Creative Commons

Open Access: N/A

Abstract: . Although of substantial importance for marine tracer distributions and eventually global carbon, oxygen, and nitrogen fluxes, the interaction between sinking and remineralization of organic matter, benthic fluxes and burial is not always represented consistently in global biogeochemical models. We here aim to investigate the relationships between these processes with a suite of global biogeochemical models, each simulated over millennia, and compared against observed distributions of pelagic tracers and benthic and pelagic fluxes. We concentrate on the representation of sediment-water interactions in common numerical models, and investigate their potential impact on simulated global sediment-water fluxes and nutrient and oxygen distributions. We find that model configurations with benthic burial simulate global oxygen well over a wide range of possible sinking flux parameterizations, making the model more robust with regard to uncertainties about the remineralization length scale. On a global scale, burial mostly affects oxygen in the meso- to bathypelagic zone. While all model types show an almost identical fit to observed pelagic particle flux, and the same sensitivity to particle sinking speed, comparison to observational estimates of benthic fluxes reveals a more complex pattern, but definite interpretation is not straightforward because of heterogeneous data distribution and methodology. Still, evaluating model results against observed pelagic and benthic fluxes of organic matter can complement model assessments based on more traditional tracers such as nutrients or oxygen. Based on a combined metric of dissolved tracers and biogeochemical fluxes, we here identify two model descriptions of burial as suitable candidates for further experiments and eventual model refinements.

1754. Assessing the Twenty-First-Century Shift in ENSO Variability in Terms of the Bjerknes Stability Index*

Authors: Joke F. Lübbecke (NOAA/Pacific Marine Environmental Laboratory, Seattle, Washington); Michael J. McPhaden (NOAA/Pacific Marine Environmental Laboratory, Seattle, Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00438.1

URL: <https://doi.org/10.1175/jcli-d-13-00438.1>

Publication Date: 2013-12-19

Volume/Issue: Volume 27, Issue 7

References: 51

Cited by: 102 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A decadal change in the character of ENSO was observed around year 2000 toward weaker-amplitude, higher-frequency events with an increased occurrence of central Pacific El Niños. Here these changes are assessed in terms of the Bjerknes stability index (BJ index), which is a measure of the growth rate of ENSO-related SST anomalies. The individual terms of the index are calculated from ocean reanalysis products separately for the time periods 1980-99 and 2000-10. The spread between the products is large, but they show a robust weakening of the thermocline feedback due to a reduced thermocline slope response to anomalous zonal wind stress as well as a weakened wind stress response to eastern equatorial Pacific SST anomalies. These changes are consistent with changes in the background state of the tropical Pacific: cooler mean SST in the eastern and central equatorial Pacific results in reduced convection there together with a westward shift in the ascending branch of the Walker circulation. This shift leads to a weakening in the relationship between eastern Pacific SST and longitudinally averaged equatorial zonal wind stress. Also, despite a steeper mean thermocline slope in the more recent period, the thermocline slope response to wind stress anomalies weakened due to a smaller zonal wind fetch that results from ENSO-related wind anomalies being more confined to the western basin. As a result, the total BJ index is more negative, corresponding to a more strongly damped system in the past decade compared to the 1980s and 1990s.

1755. A novel transport assimilation method for the Atlantic meridional overturning circulation at 26 degreesN

Authors: Leon Hermanson (Met Office); Nick Dunstone (Met Office); Keith Haines (University of Reading); Jon Robson (University of Reading); Doug Smith (Met Office); Rowan Sutton (University of Reading)

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2321

URL: <https://doi.org/10.1002/qj.2321>

Publication Date: 2013-12-19

Volume/Issue: Volume 140, Issue 685

References: 30

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

License: N/A

Open Access: N/A

Abstract: One of the prerequisites for achieving skill in decadal climate prediction is to initialize and predict the circulation in the Atlantic Ocean successfully. The RAPID array measures the Atlantic Meridional Overturning Circulation (MOC) at 26 degreesN. Here we develop a method to include these observations in the Met Office Decadal Prediction System (DePreSys). The proposed method uses covariances of overturning transport anomalies at 26 degreesN with ocean temperature and salinity anomalies throughout the ocean to create the density structure necessary to reproduce the observed transport anomaly. Assimilating transport alone in this way effectively reproduces the observed transport anomalies at 26 degreesN and is better than using basin-wide temperature and salinity observations alone. However, when the transport observations are combined with in situ temperature and salinity observations in the analysis, the transport is not currently reproduced so well. The reasons for this are investigated using pseudo-observations in a twin experiment framework. Sensitivity experiments show that the MOC on monthly time-scales, at least in the HadCM3 model, is modulated by a mechanism where non-local density anomalies appear to be more important for transport variability at 26 degreesN than local density gradients.

1756. Sea level in ocean reanalyses and tide gauges

Authors: Gennady A. Chepurin (University of Maryland); James A. Carton (University of Maryland); Eric Leuliette (NOAA Laboratory for Satellite Altimetry College Park Maryland USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009365

URL: <https://doi.org/10.1002/2013jc009365>

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References: 43

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Previous studies have noted the presence of interannual to multidecadal variability in tide gauge sea level records which is correlated with meteorological variability and which can overwhelm the signal associated with global sea level rise. This study examines the usefulness of using a set of seven ocean reanalysis and synthesis products in studies of sea level variability by comparing the tide gauges and reanalysis products at a representative set of 87 tide gauge station locations. The comparison is carried out for both a half-century base period and a century long-extended period. Treating the set of products as an ensemble of realizations obtained using different techniques, the results show generally good agreement for the half-century period with ensemble average correlations of 0.57 and RMS differences of 2.2 cm, reducing to a correlation of 0.5 for the extended period. A significant fraction of the difference between tide gauge sea level and product sea level is associated with meteorological forcing. These results support the conclusion that much of the interannual to multidecadal variability that appears in the tide gauge records is meteorologically driven. This suggests that ocean products have potential to be used to isolate this variability from the signal associated with the underlying global sea level rise.

1757. Creep deformation and buttressing capacity of damaged ice shelves: theory and application to Larsen C ice shelf

Authors: C. P. Borstad; E. Rignot; J. Mouginot; M. P. Schodlok

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-1931-2013

URL: <https://doi.org/10.5194/tc-7-1931-2013>

Publication Date: 2013-12-18

Volume/Issue: Volume 7, Issue 6

References: 82

Cited by: 84 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Around the perimeter of Antarctica, much of the ice sheet discharges to the ocean through floating ice shelves. The buttressing provided by ice shelves is critical for modulating the flux of ice into the ocean, and the presently observed thinning of ice shelves is believed to be reducing their buttressing capacity and contributing to the acceleration and thinning of the grounded ice sheet. However, relatively little attention has been paid to the role that fractures play in the ability of ice shelves to sustain and transmit buttressing stresses. Here, we present a new framework for quantifying the role that fractures play in the creep deformation and buttressing capacity of ice shelves. We apply principles of continuum damage mechanics to derive a new analytical relation for the creep of an ice shelf that accounts for the softening influence of fractures on longitudinal deformation using a state damage variable. We use this new analytical relation, combined with a temperature calculation for the ice, to partition an inverse method solution for ice shelf rigidity into independent solutions for softening damage and stabilizing backstress. Using this new approach, field and remote sensing data can be utilized to monitor the structural integrity of ice shelves, their ability to buttress the flow of ice at the grounding line, and thus their indirect contribution to ice sheet mass balance and global sea level. We apply this technique to the Larsen C ice shelf using remote sensing and Operation IceBridge data, finding damage in areas with known crevasses and rifts. Backstress is highest near the grounding line and upstream of ice rises, in agreement with patterns observed on other ice shelves. The ice in contact with the Bawden ice rise is weakened by fractures, and additional damage or thinning in this area could diminish the backstress transmitted upstream. We model the consequences for the ice shelf if it loses contact with this small ice rise, finding that flow speeds would increase by 25% or more over an area the size of the former Larsen B ice shelf. Such a perturbation could potentially destabilize the northern part of Larsen C along pre-existing lines of weakness, highlighting the importance of the feedback between buttressing and fracturing in an ice shelf.

1758. Evaluating seasonal loading models and their impact on global and regional reference frame alignment

Authors: Rong Zou (University of Alaska Fairbanks, Institute of Geophysics and Geomatics; China University of Geosciences; Wuhan Hubei China); Jeffrey T. Freymueller (University of Alaska Fairbanks); Kaihua Ding (Institute of Geophysics and Geomatics; China University of Geosciences; Wuhan Hubei China); Shaomin Yang (China Earthquake Administration); Qi Wang (Institute of Geophysics and Geomatics; China University of Geosciences; Wuhan Hubei China)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jb010186

URL: <https://doi.org/10.1002/2013jb010186>

Publication Date: 2013-12-18

Volume/Issue: Volume 119, Issue 2

References: 46

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

1759. Subsurface and deeper ocean remote sensing from satellites: An overview and new results

Authors: Victor Klemas; Xiao-Hai Yan

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2013.11.010

URL: <https://doi.org/10.1016/j.pocean.2013.11.010>

Publication Date: 2013-12-16

Volume/Issue: Volume 122

References: 93

Cited by: 90 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1760. Detecting Processes Contributing to Interannual Halosteric and Thermosteric Sea Level Variability

Authors: Armin Köhl (University of Hamburg)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00412.1

URL: <https://doi.org/10.1175/jcli-d-13-00412.1>

Publication Date: 2013-12-06

Volume/Issue: Volume 27, Issue 6

References: 34

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: On interannual time scales, regional sea level variability is largely determined by changes in the steric component. The relation between the thermosteric and halosteric components is studied by separating the components into contributions from the mixed layer and, below the mixed layer, into the part that is related to isopycnal motion and that contributes to the steric sea level and the inactive part related to changes of spiciness. The decomposition provides a simple diagnostic to detect and understand physical mechanisms leading to regional sea level changes. In most areas of the world's oceans, steric sea level variability is dominated by the contribution from isopycnal motion to the thermosteric sea level while halosteric variability relates more to spiciness. Because of the salinity minimum at middepth, different spatial salinity gradients above and below the minimum lead to opposing contributions and thus to a small contribution from isopycnal motion to the halosteric sea level. In nonpolar regions, both active components oppose each other, rendering the thermosteric variability larger than the steric variability. In the Arctic, the variability of both components is governed by spiciness in the Eurasian Basin and isopycnal motion in the Amerasian Basin.

1761. Observed thinning of Totten Glacier is linked to coastal polynya variability

Authors: A. Khazendar; M.P. Schodlok; I. Fenty; S.R.M. Ligtenberg; E. Rignot; M.R. van den Broeke

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/ncomms3857

URL: <https://doi.org/10.1038/ncomms3857>

Publication Date: 2013-12-05

Volume/Issue: Volume 4, Issue 1

References: 58

Cited by: 81 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1762. North Atlantic warming and the retreat of Greenland's outlet glaciers

Authors: Fiammetta Straneo; Patrick Heimbach

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/nature12854

URL: <https://doi.org/10.1038/nature12854>

Publication Date: 2013-12-03

Volume/Issue: Volume 504, Issue 7478

References: 100

Cited by: 323 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1763. A neural network-based estimate of the seasonal to inter-annual variability of the Atlantic Ocean carbon sink

Authors: P. Landschützer; N. Gruber; D. C. E. Bakker; U. Schuster; S. Nakaoka; M. R. Payne; T. P. Sasse; J. Zeng

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-10-7793-2013

URL: <https://doi.org/10.5194/bg-10-7793-2013>

Publication Date: 2013-11-29

Volume/Issue: Volume 10, Issue 11

References: 71

Cited by: 180 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission

License: Creative Commons

Open Access: N/A

Abstract: . The Atlantic Ocean is one of the most important sinks for atmospheric carbon dioxide (CO₂), but this sink has been shown to vary substantially in time. Here we use surface ocean CO₂ observations to estimate this sink and the temporal variability from 1998 through 2007 in the Atlantic Ocean. We benefit from (i) a continuous improvement of the observations, i.e. the Surface Ocean CO₂ Atlas (SOCAT) v1.5 database and (ii) a newly developed technique to interpolate the observations in space and time. In particular, we use a two-step neural network approach to reconstruct basin-wide monthly maps of the sea surface partial pressure of CO₂ (pCO₂) at a resolution of 1 degrees x 1 degrees. From those, we compute the air-sea CO₂ flux maps using a standard gas exchange parameterization and high-resolution wind speeds. The neural networks fit the observed pCO₂ data with a root mean square error (RMSE) of about 10 μatm and with almost no bias. A check against independent time-series data and new data from SOCAT v2 reveals a larger RMSE of 22.8 μatm for the entire Atlantic Ocean, which decreases to 16.3 μatm for data south of 40 degrees N. We estimate a decadal mean uptake flux of $-0.45 \pm 0.15 \text{ Pg C yr}^{-1}$ for the Atlantic between 44 degrees S and 79 degrees N, representing the sum of a strong uptake north of 18 degrees N ($-0.39 \pm 0.10 \text{ Pg C yr}^{-1}$), outgassing in the tropics (18 degrees S-18 degrees N, $0.11 \pm 0.07 \text{ Pg C yr}^{-1}$), and uptake in the subtropical/temperate South Atlantic south of 18 degrees S ($-0.16 \pm 0.06 \text{ Pg C yr}^{-1}$), consistent with recent studies. The strongest seasonal variability of the CO₂ flux occurs in the temperature-driven subtropical North Atlantic, with uptake in winter and outgassing in summer. The seasonal cycle is antiphased in the subpolar latitudes relative to the subtropics largely as a result of the biologically driven winter-to-summer drawdown of CO₂. Over the 10 yr analysis period (1998 through 2007), sea surface pCO₂ increased faster than that of the atmosphere in large areas poleward of 40 degrees N, while in other regions of the North Atlantic the sea surface pCO₂ increased at a slower rate, resulting in a barely changing Atlantic carbon sink north of the Equator ($-0.01 \pm 0.02 \text{ Pg C yr}^{-1} \text{ decade}^{-1}$). Surface ocean pCO₂ increased at a slower rate relative to atmospheric CO₂ over most of the Atlantic south of the Equator, leading to a substantial trend toward a stronger CO₂ sink for the entire South Atlantic

(-0.14 ± 0.02 Pg C yr⁻¹ decade⁻¹). In contrast to the 10 yr trends, the Atlantic Ocean carbon sink varies relatively little on inter-annual timescales (± 0.04 Pg C yr⁻¹; 1 sigma).

1764. A novel estimate of ocean oxygen utilisation points to a reduced rate of respiration in the ocean interior

Authors: O. Duteil; W. Koeve; A. Oschlies; D. Bianchi; E. Galbraith; I. Kriest; R. Matear

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-10-7723-2013

URL: <https://doi.org/10.5194/bg-10-7723-2013>

Publication Date: 2013-11-28

Volume/Issue: Volume 10, Issue 11

References: 46

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Apparent Oxygen Utilisation (AOU) is a classical measure of the amount of oxygen respired in the ocean's interior. We show that AOU systematically overestimates True Oxygen Utilisation (TOU) in 6 coupled circulation-biogeochemical ocean models. This is due to atmosphere-ocean oxygen disequilibria in the subduction regions, consistent with previous work. We develop a simple, new, observationally-based approach which we call Evaluated Oxygen Utilisation (EOU). In this approach, we take into account the impact of the upper ocean oxygen disequilibria into the interior, considering that transport takes place predominantly along isopycnal surfaces. The EOU approximates the TOU with less than half of the bias of AOU in all 6 models despite large differences in the physical and biological components of the models. Applying the EOU approach to a global observational dataset yields an oxygen consumption rate 25% lower than that derived from AOU-based estimates, for a given ventilation rate.

1765. A Multidimensional Spectral Description of Ocean Variability

Authors: Cimarron Wortham (University of Washington); Carl Wunsch (Harvard University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-0113.1

URL: <https://doi.org/10.1175/jpo-d-13-0113.1>

Publication Date: 2013-11-27

Volume/Issue: Volume 44, Issue 3

References: 69

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An updated empirical, analytical model for the frequency and wavenumber distribution of balanced motion in the ocean is presented. The spectrum model spans periods longer than the inertial but shorter than a decade and wavelengths between 100 and 10 000 km. Assuming geostrophic dynamics, a spectrum model for the streamfunction is constructed to be consistent with a range of observations, including sea surface height from satellite altimetry, velocity from moored and shipboard instruments, and temperature from moorings. First-order characteristics of the observed spectra, including amplitude and spectral moments, vary slowly geographically. The spectrum model is horizontally anisotropic, accommodating observations that zonal wavenumber-frequency spectra are dominated by a "nondispersive line." Qualitative and quantitative agreement is found with one-dimensional frequency and wavenumber spectra and observed vertical profiles of variance. Illustrative application is made of the model spectrum to observing-system design, data mapping, and uncertainty estimation for trends.

1766. Near cessation of Eighteen Degree Water renewal in the western North Atlantic in the warm winter of 2011-2012

Authors: Sam Billheimer (Scripps Institution of Oceanography / University of California San Diego); Lynne D. Talley (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009024

URL: <https://doi.org/10.1002/2013jc009024>

Publication Date: 2013-11-26

Volume/Issue: Volume 118, Issue 12

References: 43

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF Ocean Sciences

License: N/A

Open Access: N/A

Abstract: The winter of 2011-2012 was a particularly weak season for the renewal of "Eighteen Degree Water" (EDW), the Subtropical Mode Water of the western North Atlantic, as demonstrated by Argo and repeat hydrography. Weak, late winter buoyancy forcing produced shallower than usual winter mixed layers throughout the subtropical gyre, failing to thoroughly ventilate the underlying mode water, and can likely be attributed to the coinciding high, positive phase of the North Atlantic Oscillation (NAO). The only region where EDW was renewed was in the far northeastern Sargasso Sea where it is understood that the Gulf Stream plays a central role in formation; no EDW formed over the large regions of the gyre where deep winter mixed layers driven by surface buoyancy loss normally create EDW. The present investigation evaluates 2011-2012 winter buoyancy content anomalies, surface buoyancy fluxes, and advection of buoyancy via the Gulf Stream and compares them with the previous seven winters that exhibited more vigorous EDW formation. The weak 2011-2012 formation did not result from increased Gulf Stream heat advection, and was also not driven by preconditioning as the buoyancy content of the region prior to the onset of winter forcing was not unusually high. Rather, the weak formation resulted from climatologically weak surface cooling late in winter. The winter of 2007-2008 also experienced particularly weak EDW formation under similar conditions, including a high NAO and weak late winter surface cooling.

1767. Estimates of the Southern Ocean general circulation improved by animal-borne instruments

Authors: Fabien Roquet (Stockholm University); Carl Wunsch (Harvard University); Gael Forget (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology); Christophe Guinet (Centre d'Etudes Biologiques de Chizé, Centre National de la Recherche Scientifique Villiers en Bois France); Gilles Reverdin (Laboratoire d'Océanographie et du Climat: Expérimentation et Approches Numériques Paris France); Jean-Benoit Charrassin (Laboratoire d'Océanographie et du Climat: Expérimentation et Approches Numériques Paris France); Frederic Bailleul (Centre d'Etudes Biologiques de Chizé, Centre National de la Recherche Scientifique Villiers en Bois France); Daniel P. Costa (University of California Santa Cruz); Luis A. Huckstadt (University of California Santa Cruz); Kimberly T. Goetz (University of California Santa Cruz); Kit M. Kovacs (Norwegian Polar Institute); Christian Lydersen (Norwegian Polar Institute); Martin Biuw (Norwegian Polar Institute); Ole A. Nøst (Norwegian Polar Institute); Horst Bornemann (Alfred Wegener Institute); Joachim Ploetz (Alfred Wegener Institute); Marthan N. Bester (Mammal Research Institute, Department of Zoology and Entomology University of Pretoria Pretoria South Africa); Trevor McIntyre (Mammal Research Institute, Department of Zoology and Entomology University of Pretoria Pretoria South Africa); Monica C. Muelbert (Instituto de Oceanografia Universidade Federal do Rio Grande Porto Alegre Brazil); Mark A. Hindell (University of Tasmania); Clive R. McMahon (University of Tasmania); Guy Williams (University of Tasmania); Robert Harcourt (Macquarie University); Iain C. Field (Macquarie University); Leon Chafik (Stockholm University); Keith W. Nicholls (British Antarctic Survey Natural Environment Research Council Cambridge UK); Lars Boehme (University of St Andrews); Mike A. Fedak (University of St Andrews)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013gl058304

URL: <https://doi.org/10.1002/2013gl058304>

Publication Date: 2013-11-19

Volume/Issue: Volume 40, Issue 23

References: 19

Cited by: 109 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Over the last decade, several hundred seals have been equipped with conductivity-temperature-depth sensors in the Southern Ocean for both biological and physical oceanographic studies. A calibrated collection of seal-derived hydrographic data is now available, consisting of more than 165,000 profiles. The value of these hydrographic data within the existing Southern Ocean observing system is demonstrated herein by conducting two state estimation experiments, differing only in the use or not of seal data to constrain the system. Including seal-derived data substantially modifies the estimated surface mixed-layer properties and circulation patterns within

and south of the Antarctic Circumpolar Current. Agreement with independent satellite observations of sea ice concentration is improved, especially along the East Antarctic shelf. Instrumented animals efficiently reduce a critical observational gap, and their contribution to monitoring polar climate variability will continue to grow as data accuracy and spatial coverage increase.

1768. Dispersal, eddies, and the diversity of marine phytoplankton

Authors: Sophie Clayton; Stephanie Dutkiewicz; Oliver Jahn; Michael J. Follows

Journal: Limnology and Oceanography: Fluids and Environments

Publisher: Wiley

DOI: 10.1215/21573689-2373515

URL: <https://doi.org/10.1215/21573689-2373515>

Publication Date: 2013-11-09

Volume/Issue: Volume 3, Issue 1

References: 57

Cited by: 65 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Lay AbstractPhytoplankton are the "plants" of the ocean; they fix carbon dioxide from the atmosphere and make up the base of the oceanic food chain. Phytoplankton are incredibly diverse, but the reason for their high biodiversity is not well understood. Generally speaking, phytoplankton might be present in a given location either because they are "happy" and able to grow there, or if they are unable to grow there, because they have been transported in from elsewhere (i.e., via dispersal). Previous studies have suggested that "hotspots" of high phytoplankton diversity found in the most dynamic oceanic regions can be explained by dispersal. We explored this question by using a set of computer simulations of phytoplankton ecology coupled to a realistic ocean circulation. We show that dispersal can only partly explain diversity hotspots. The mixing of populations by ocean currents and eddies is important, but a plentiful supply of nutrients and variable environmental conditions also enhance phytoplankton diversity. This work has generated hypotheses that will stimulate new projects to test them in the field.

1769. Nonseasonal fluctuations of the Arctic Ocean mass observed by the GRACE satellites

Authors: Denis L. Volkov (University of Miami); Felix W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009341

URL: <https://doi.org/10.1002/2013jc009341>

Publication Date: 2013-11-09

Volume/Issue: Volume 118, Issue 12

References: 34

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: sea level, Arctic Ocean, ECCO2, ocean mass, GRACE (and 1 more)

Funding: The ECCO2 model runs have been carried out at Jet Propulsion Laboratory, California Institute of Technology, NASA MEASURES Program, NASA Physical Oceanography program grant NNX11AE27G

License: N/A

Open Access: N/A

Abstract: Time variable gravity observations from the GRACE satellites reveal strong nonseasonal fluctuations of bottom pressure in the Arctic Ocean on the time scales from 2 to 6 months and a record-high bottom pressure anomaly in February of 2011. Here, we examine the nature and driving forces behind those fluctuations. We find that the nonseasonal variability of the Arctic Ocean mass is strongly coupled to wind forcing. The zonal wind pattern is correlated with a dipole pattern of Arctic Ocean mass changes. Westerly wind intensification over the North Atlantic at about 60 degreesN as well as over the Russian Arctic continental shelf break cause the ocean mass to decrease in the Nordic seas and in the central Arctic, and to increase over the Russian Arctic shelf. Basin-wide Arctic Ocean mass fluctuations are correlated with northward wind anomalies over the northeastern North Atlantic and Nordic seas, and over the Bering Sea. We show that positive (negative) Arctic Ocean mass anomalies are associated with anticyclonic (cyclonic) anomalies of the large-scale ocean circulation pattern. Based on ocean model simulations, we conclude that the observed nonseasonal Arctic Ocean mass variability is mostly explained by the net horizontal wind-driven transports, and the contribution of fresh water fluxes is negligible. We demonstrate that transport anomalies across both the Atlantic and Pacific gateways were equally important for generating large Arctic Ocean mass anomalies in 2011.

1770. North Atlantic simulations in Coordinated Ocean-ice Reference Experiments phase II (CORE-II). Part I: Mean states

Authors: Gokhan Danabasoglu; Steve G. Yeager; David Bailey; Erik Behrens; Mats Bentsen; Daohua Bi; Arne Biastoch; Claus Böning; Alexandra Bozec; Vittorio M. Canuto; Christophe Cassou; Eric Chassignet; Andrew C. Coward; Sergey Danilov; Nikolay Diansky; Helge Drange; Riccardo Farneti; Elodie Fernandez; Pier Giuseppe Fogli; Gael Forget; Yosuke Fujii; Stephen M. Griffies; Anatoly Gusev; Patrick Heimbach; Armando Howard; Thomas Jung; Maxwell Kelley; William G. Large; Anthony Leboissetier; Jianhua Lu; Gurvan Madec; Simon J. Marsland; Simona Masina; Antonio Navarra; A.J. George Nurser; Anna Pirani; David Salas y Mélia; Bonita L. Samuels; Markus Scheinert; Dmitry Sidorenko; Anne-Marie Treguier; Hiroyuki Tsujino; Petteri Uotila; Sophie Valcke; Aurore Voldoire; Qiang Wang

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2013.10.005

URL: <https://doi.org/10.1016/j.ocemod.2013.10.005>

Publication Date: 2013-11-08

Volume/Issue: Volume 73

References: 193

Cited by: 323 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U. S. National Science Foundation (NSF), NSF, U. S. Department of Energy (and 8 more)

License: N/A

Open Access: N/A

Abstract: N/A

1771. The Coherence and Impact of Meridional Heat Transport Anomalies in the Atlantic Ocean Inferred from Observations*

Authors: Kathryn A. Kelly (University of Washington); LuAnne Thompson (University of Washington); John Lyman (NOAA/Pacific Marine Environmental Laboratory, Seattle, Washington, and Joint Institute for Marine and Atmospheric Research, University of Hawai'i at Mānoa, Honolulu, Hawaii)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00131.1

URL: <https://doi.org/10.1175/jcli-d-12-00131.1>

Publication Date: 2013-11-07

Volume/Issue: Volume 27, Issue 4

References: 47

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Observations of thermosteric sea level (TSL) from hydrographic data, equivalent water thickness (EWT) from the Gravity Recovery and Climate Experiment (GRACE), and altimetric sea surface height (SSH) are used to infer meridional heat transport (MHT) anomalies for the Atlantic Ocean. An "unknown control" version of a Kalman filter in each of eight regions extracts smooth estimates of heat transport convergence (HTC) from discrepancies between the response to monthly surface heat and freshwater fluxes and observed mass and heat content. Two models are used: model A using only the heat budget for 1993-2010 and model B using both heat and mass budgets for 2003-10. Based on the small contributions of mass to SSH, model A is rerun using SSH in place of TSL to improve temporal resolution and data consistency. Estimates of MHT are derived by summing the HTC from north to south assuming either negligible anomalies at 67 degreesN or setting MHT to observed values near 40 degreesN. Both methods show that MHT is highly coherent between 35 degreesS and 40 degreesN. The former method gives a large drop in coherence north of 40 degreesN while the latter method gives a less dramatic drop. Estimated anomalies in MHT comparable to or larger than that recently observed at the Rapid Climate Change and Meridional Overturning Circulation and Heatflux Array (RAPID/MOCHA) line at 26.5 degreesN have occurred multiple times in this 18-yr period. Positive anomalies in coherent MHT correspond to increased heat loss in the North Atlantic subtropical gyre demonstrating the feedback of oceanic heat transport anomalies on air-sea fluxes. A correlation of MHT with the Antarctic Oscillation suggests a southern source for the coherent MHT anomalies.

1772. Parameter and state estimation with a time-dependent adjoint marine ice sheet model

Authors: D. N. Goldberg; P. Heimbach

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-1659-2013

URL: <https://doi.org/10.5194/tc-7-1659-2013>

Publication Date: 2013-11-06

Volume/Issue: Volume 7, Issue 6

References: 82

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . To date, assimilation of observations into large-scale ice models has consisted predominantly of time-independent inversions of surface velocities for basal traction, bed elevation, or ice stiffness, and has relied primarily on analytically derived adjoints of glaciological stress balance models. To overcome limitations of such "snapshot" inversions - i.e., their inability to assimilate time-dependent data for the purpose of constraining transient flow states, or to produce initial states with minimum artificial drift and suitable for time-dependent simulations - we have developed an adjoint of a time-dependent parallel glaciological flow model. The model implements a hybrid shallow shelf-shallow ice stress balance, solves the continuity equation for ice thickness evolution, and can represent the floating, fast-sliding, and frozen bed regimes of a marine ice sheet. The adjoint is generated by a combination of analytic methods and the use of algorithmic differentiation (AD) software. Several experiments are carried out with idealized geometries and synthetic observations, including inversion of time-dependent surface elevations for past thicknesses, and simultaneous retrieval of basal traction and topography from surface data. Flexible generation of the adjoint for a range of independent uncertain variables is exemplified through sensitivity calculations of grounded ice volume to changes in basal melting of floating and basal sliding of grounded ice. The results are encouraging and suggest the feasibility, using real observations, of improved ice sheet state estimation and comprehensive transient sensitivity assessments.

1773. Arctic Ocean Circulation Patterns Revealed by GRACE

Authors: Cecilia Peralta-Ferriz (University of Washington); James H. Morison (University of Washington); John M. Wallace (University of Washington); Jennifer A. Bonin (University of South Florida); Jinlun Zhang (University of Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00013.1

URL: <https://doi.org/10.1175/jcli-d-13-00013.1>

Publication Date: 2013-11-05

Volume/Issue: Volume 27, Issue 4

References: 57

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Measurements of ocean bottom pressure (OBP) anomalies from the satellite mission Gravity Recovery and Climate Experiment (GRACE), complemented by information from two ocean models, are used to investigate the variations and distribution of the Arctic Ocean mass from 2002 through 2011. The forcing and dynamics associated with the observed OBP changes are explored. Major findings are the identification of three primary temporal-spatial modes of OBP variability at monthly-to-interannual time scales with the following characteristics. Mode 1 (50% of the variance) is a wintertime basin-coherent Arctic mass change forced by southerly winds through Fram Strait, and to a lesser extent through Bering Strait. These winds generate northward geostrophic current anomalies that increase the mass in the Arctic Ocean. Mode 2 (20%) reveals a mass change along the Siberian shelves, driven by surface Ekman transport and associated with the Arctic Oscillation. Mode 3 (10%) reveals a mass dipole, with mass decreasing in the Chukchi, East Siberian, and Laptev Seas, and mass increasing in the Barents and Kara Seas. During the summer, the mass decrease on the East Siberian shelves is due to the basin-scale anticyclonic atmospheric circulation that removes mass from the shelves via Ekman transport. During the winter, the forcing mechanisms include a large-scale cyclonic atmospheric circulation in the eastern-central Arctic that produces mass divergence into the Canada Basin and the Barents Sea. In addition, strengthening of the Beaufort high tends to remove mass from the East Siberian and Chukchi Seas. Supporting previous modeling results, the month-to-month variability in OBP associated with each mode is predominantly of barotropic character.

1774. Modelling spatial and temporal patterns in size-structured marine plankton communities: top-down and bottom-up controls

Authors: Ben A. Ward; Stephanie Dutkiewicz; Michael J. Follows

Journal: Journal of Plankton Research

Publisher: Oxford University Press (OUP)

DOI: 10.1093/plankt/fbt097

URL: <https://doi.org/10.1093/plankt/fbt097>

Publication Date: 2013-11-01

Volume/Issue: Volume 36, Issue 1

References: 57

Cited by: 137 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1775. Ocean acoustic thermometry and the seasonal cycle of temperature in the North Pacific Ocean

Authors: Brian D. Dushaw (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105-6698); R. Andrew (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105-6698); B. Howe (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105-6698); J. Mercer (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105-6698); B. Cornuelle (Scripps Institution of Oceanography); M. Dzieciuch (Scripps Institution of Oceanography); W. Munk (Scripps Institution of Oceanography); P. Worcester (Scripps Institution of Oceanography); T. Birdsall (Univ. of Michigan, Ann Arbor, MI); K. Metzger (Univ. of Michigan, Ann Arbor, MI); D. Menemenlis (California Institute of Technol., Pasadena, CA); R. Spindel (California Institute of Technol., Pasadena, CA)

Journal: The Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

DOI: 10.1121/1.4787091

URL: <https://doi.org/10.1121/1.4787091>

Publication Date: 2013-10-31

Volume/Issue: Volume 120, Issue 5_Supplement

References: N/A

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: With several years of long-range (several Mm) acoustic propagation data obtained during the Acoustic Thermometry of Ocean Climate (ATOC) and North Pacific Acoustic Laboratory (NPAL) projects, the seasonal cycle of ocean temperature in the North Pacific can be examined. Acoustic transmissions have been made from a source located off the northern Californian coast and from a source located north of Kauai, HI to several receivers of opportunity located in the North Pacific Basin. The acoustic data are a high signal-to-noise measure of large-scale temperature with excellent temporal resolution. Although only a few realizations of the seasonal cycle are available, it is clear that inter- and intraannual variabilities are large contributions to the time series, with signal amplitudes comparable to the seasonal cycle. Such variabilities are likely advective in origin. The time scales for some of the changes in temperature are short, sometimes of order weeks. Not all available acoustic paths are suitable for assessing the seasonal cycle, however. Near Hawaii, the acoustic sampling does not extend to the near-surface waters, so seasonal variations there are not measured. The acoustic results will be compared to measures of the seasonal cycle by satellite altimetry, profiling floats, and the ECCO numerical ocean model.

1776. Monitoring of a quasi-stationary eddy in the Bay of Biscay by means of satellite, in situ and model results

Authors: Ainhoa Caballero; Luis Ferrer; Anna Rubio; Guillaume Charria; Benjamin H. Taylor; Nicolas Grima

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2013.09.029

URL: <https://doi.org/10.1016/j.dsr2.2013.09.029>

Publication Date: 2013-10-26

Volume/Issue: Volume 106

References: 59

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Spanish Ministry of Science and Innovation (National R&D;&I; Plan, ESTIBB CTM 2009-12339 project), Department of Environment, Regional Planning, Agriculture and Fisheries of the Basque Government (Marco Program), Torres Quevedo Grant (Spanish Ministry of Science and Innovation (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

1777. A decade of acoustic thermometry in the North Pacific Ocean: Using long-range acoustic travel times to test gyre-scale temperature variability derived from other observations and ocean models

Authors: Brian Dushaw (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105); R. Andrew (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105); B. Howe (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105); J. Mercer (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105); R. Spindel (Appl. Phys. Lab., Univ. of Washington, 1013 N.E. 40th St., Seattle, WA 98105); P. Worcester (Scripps Institution of Oceanography); B. Cornuelle (Scripps Institution of Oceanography); M. Dzieciuch (Scripps Institution of Oceanography); T. Birdsall (Univ. of Michigan, Ann Arbor, MI); K. Metzger (Univ. of Michigan, Ann Arbor, MI); D. Menemenlis (J.P.L., Calif. Inst. Tech., Pasadena, CA)

Journal: The Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

DOI: 10.1121/1.4781783

URL: <https://doi.org/10.1121/1.4781783>

Publication Date: 2013-10-25

Volume/Issue: Volume 121, Issue 5_Supplement

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Large-scale temperatures in the North Pacific were measured by long-range acoustic transmissions from 1996-2006. Acoustic sources off California and Kauai transmitted to receivers distributed throughout the North Pacific from 1996-1999. Kauai transmissions continued from 2002-2006. Acoustic travel-time data are inherently integrating. This averaging suppresses mesoscale variability and provides an accurate measure of large-scale temperature, subject to the limitations of the ray path sampling. At basin scales, the ocean is highly variable, with significant changes occurring at time scales from weeks to years. The interannual variability is large compared to trends in the data. Willis, et al. used objective mapping techniques applied to satellite altimetry and hydrography to derive 0-750 m temperature fields for the global ocean. Travel times equivalent to the measured travel times can be calculated using these fields. The measured and calculated travel times are similar, but also show significant differences. Similar comparisons using travel times derived from the "Estimating the Circulation and Climate of the Ocean" (ECCO) model and a high-resolution Parallel Ocean Program (POP) model also show similarities and differences. The ECCO model was constrained by altimetric and profile data by data assimilation, suggesting that the acoustic travel times provide meaningful additional constraints on model behavior.

1778. Variability of the oxygen minimum zone in the tropical North Pacific during the late twentieth century

Authors: Takamitsu Ito (Georgia Institute of Technology); Curtis Deutsch (University of Washington)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013gb004567

URL: <https://doi.org/10.1002/2013gb004567>

Publication Date: 2013-10-25

Volume/Issue: Volume 27, Issue 4

References: 41

Cited by: 56 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF

License: N/A

Open Access: N/A

Abstract: The oxygen minimum zones (OMZ) are especially sensitive to ocean deoxygenation due to the nonlinear dependence of their size on the oxygen inventory. Significant decadal variability has been observed in the extent and intensity of the tropical Pacific OMZ. Here we diagnose the physical and biogeochemical mechanisms behind the interannual and decadal variability of oxygen in the tropical Pacific thermocline using a three-dimensional ocean biogeochemistry model that reproduces the expansion of the tropical Pacific OMZ since the 1980s. On interannual time scales, heat content and respiration rates are strongly influenced by El Niño-Southern Oscillation cycles and the associated changes in upwelling. The resulting changes in solubility and apparent oxygen utilization tend to compensate one another, thus damping the magnitude of oxygen variability. Regional oxygen budget reveals the subtle balance between the lateral and vertical ocean circulation in controlling the physical oxygen supply to the eastern tropical Pacific. Spectral analysis shows that the tropical Pacific oxygen has a stronger variance in decadal time scales than its physical and biological drivers. Our results suggest that the physical oxygen supply and biological oxygen loss are integrated through the finite memory of thermocline waters to produce the multidecadal variability of OMZ.

1779. Dynamically and Kinematically Consistent Global Ocean Circulation and Ice State Estimates

Authors: Carl Wunsch; Patrick Heimbach

Journal: International Geophysics

Publisher: Elsevier

DOI: 10.1016/b978-0-12-391851-2.00021-0

URL: <https://doi.org/10.1016/b978-0-12-391851-2.00021-0>

Publication Date: 2013-10-23

Volume/Issue: N/A

References: 153

Cited by: 93 publications

Language: N/A

Subjects: N/A

Keywords: State estimation, Circulation (fluid dynamics), Ice-ocean interactions, Data assimilation, General circulation (and 4 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1780. Global Distribution and Formation of Mode Waters

Authors: Kevin Speer; Gael Forget

Journal: International Geophysics

Publisher: Elsevier

DOI: 10.1016/b978-0-12-391851-2.00009-x

URL: <https://doi.org/10.1016/b978-0-12-391851-2.00009-x>

Publication Date: 2013-10-23

Volume/Issue: N/A

References: 81

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: Argo|Mode water|Mode (computer interface)|Stratification (seeds)|Water cycle

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1781. Variations of heat transport in the northwestern Pacific marginal seas inferred from high-resolution reanalysis

Authors: Gwang-Ho Seo; Yang-Ki Cho; Byoung-Ju Choi

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2013.10.005

URL: <https://doi.org/10.1016/j.pocean.2013.10.005>

Publication Date: 2013-10-21

Volume/Issue: Volume 121

References: 54

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Research Foundation of Korea, Ministry of Land, Transport and Maritime Affairs, Ministry of Education

License: N/A

Open Access: N/A

Abstract: N/A

1782. On the wind mechanical forcing of the ocean general circulation

Authors: Xiaoming Zhai (University of East Anglia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc009086

URL: <https://doi.org/10.1002/2013jc009086>

Publication Date: 2013-10-21

Volume/Issue: Volume 118, Issue 12

References: 43

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The wind mechanical forcing of the ocean general circulation and its seasonal variations are examined using available observational products, focusing on the role of the mean and fluctuating winds. It is found that including wind fluctuations in the stress calculation produces a qualitative change in the estimates of the mean and seasonal wind stress, particularly at mid and high latitudes where the synoptic wind variability is large. This effect of wind fluctuations on air-sea momentum exchange has immediate dynamical consequences for the large-scale ocean circulation. For example, power input to the ocean general circulation and subtropical gyre transport can be underestimated by more than 50% if the fluctuating winds are not taken into account. However, the impact of including wind fluctuations depends strongly on the presence of the mean winds. If the mean winds are ignored in the stress calculation, the net effect of the fluctuating winds is to take energy out of the ocean owing to the skewness of the near-surface wind field. Furthermore, covariances of wind fluctuations are found to explain most of the effect of the fluctuating winds, while the variable drag coefficient makes a non-negligible contribution in the Southern Ocean. These results imply that paleo and future climate studies need to take into account the changes of the large-scale low-frequency wind field as well as the synoptic weather systems.

1783. A global ocean reanalysis product in the China Ocean Reanalysis (CORA) project

Authors: Guijun Han; Hongli Fu; Xuefeng Zhang; Wei Li; Xinrong Wu; Xidong Wang; Lianxin Zhang

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-013-2198-9

URL: <https://doi.org/10.1007/s00376-013-2198-9>

Publication Date: 2013-10-19

Volume/Issue: Volume 30, Issue 6

References: 28

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1784. Modeling interannual dense shelf water export in the region of the Mertz Glacier Tongue (1992-2007)

Authors: E. A. Cougnon (Commonwealth Scientific and Industrial Research Organisation; Marine and Atmospheric Research; Hobart Australia, University of Tasmania); B. K. Galton-Fenzi (University of Tasmania, Australian Antarctic Division; Kingston Tasmania Australia); A. J. S. Meijers (British Antarctic Survey; Cambridge UK); B. Legrésy (University of Tasmania, Institut de Recherche pour le Développement (IRD))

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jc008790

URL: <https://doi.org/10.1002/2013jc008790>

Publication Date: 2013-10-19

Volume/Issue: Volume 118, Issue 10

References: 54

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: dense shelf water, numerical modeling, Mertz glacier tongue, Antarctic bottom water

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1785. Impact of initialization procedures on the predictive skill of a coupled ocean-atmosphere model

Authors: Iuliia Polkova; Armin Köhl; Detlef Stammer

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-013-1969-4

URL: <https://doi.org/10.1007/s00382-013-1969-4>

Publication Date: 2013-10-17

Volume/Issue: Volume 42, Issue 11-12

References: 46

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1786. Changes in the Arctic Ocean CO₂ sink (1996-2007): A regional model analysis

Authors: M. Manizza (Scripps Institution of Oceanography / University of California San Diego, Massachusetts Institute of Technology); M. J. Follows (Massachusetts Institute of Technology); S. Dutkiewicz (Massachusetts Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. N. Hill (Massachusetts Institute of Technology); R. M. Key (Princeton University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2012gb004491

URL: <https://doi.org/10.1002/2012gb004491>

Publication Date: 2013-10-16

Volume/Issue: Volume 27, Issue 4

References: 52

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The rapid recent decline of Arctic Ocean sea ice area increases the flux of solar radiation available for primary production and the area of open water for air-sea gas exchange. We use a regional physical-biogeochemical model of the Arctic Ocean, forced by the National Centers for Environmental Prediction/National Center for Atmospheric Research atmospheric reanalysis, to evaluate the mean present-day CO₂ sink and its temporal evolution. During the 1996-2007 period, the model suggests that the Arctic average sea surface temperature warmed by 0.04 degreesC a⁻¹, that sea ice area decreased by ~0.1 x 10⁶ km² a⁻¹, and that the biological drawdown of dissolved inorganic carbon increased. The simulated 1996-2007 time-mean Arctic Ocean CO₂ sink is 58 +/- 6 Tg C a⁻¹. The increase in ice-free ocean area and consequent carbon drawdown during this period enhances the CO₂ sink by ~1.4 Tg C a⁻¹, consistent with estimates based on extrapolations of sparse data. A regional analysis suggests that during the 1996-2007 period, the shelf regions of the Laptev, East Siberian, Chukchi, and Beaufort Seas experienced an increase in the efficiency of their biological pump due to decreased sea ice area, especially during the 2004-2007 period, consistent with independently published estimates of primary production. In contrast, the CO₂ sink in the Barents Sea is reduced during the 2004-2007 period due to a dominant control by warming and decreasing solubility. Thus, the effect of decreasing sea ice area and increasing sea surface temperature partially cancel, though the former is dominant.

1787. Iron, phosphorus, and nitrogen supply ratios define the biogeography of nitrogen fixation

Authors: Ben A. Ward; Stephanie Dutkiewicz; C. Mark Moore; Michael J. Follows

Journal: Limnology and Oceanography

Publisher: Wiley

DOI: 10.4319/lo.2013.58.6.2059

URL: <https://doi.org/10.4319/lo.2013.58.6.2059>

Publication Date: 2013-10-16

Volume/Issue: Volume 58, Issue 6

References: N/A

Cited by: 156 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We present a unified conceptual framework describing the competition between diazotrophs and non-nitrogen-fixing marine plankton and their interaction with three essential nutrient elements: nitrogen (N), phosphorus (P), and iron (Fe). The theory explains the global biogeography of diazotrophs and the observed large-scale variations in surface ocean nutrient concentrations. The ratios in which N, P, and Fe are delivered to the surface ocean, relative to the demands of the phytoplankton community, define several biogeochemical provinces in terms of the limiting nutrients and the presence or absence of diazotrophs. Nutrient supply ratios provided by a global ecosystem model support the theoretical view that diazotroph biogeography is dominated by the Fe : N supply ratio, with the P : N supply ratio taking an important secondary role. The theory yields robust predictions for which strong empirical support is found in global observations of surface nutrient concentrations and diazotroph abundance.

1788. Aquarius and SMOS detect effects of an extreme Mississippi River flooding event in the Gulf of Mexico

Authors: Michelle M. Gierach (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jorge Vazquez-Cuervo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Vardis M. Tsonos (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/grl.50995

URL: <https://doi.org/10.1002/grl.50995>

Publication Date: 2013-10-12

Volume/Issue: Volume 40, Issue 19

References: 19

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sea surface salinity (SSS) measurements from the Aquarius/Satélite de Aplicaciones Científicas (SAC)-D satellite and Soil Moisture and Ocean Salinity (SMOS) mission were used to document the freshening associated with the record 2011 Mississippi River flooding event in the Gulf of Mexico (GoM). Assessment of the salinity response was aided by additional satellite observations, including chlorophyll-a (chl-a) and ocean surface currents, and a passive tracer simulation. Low SSS values associated with the spreading of the river plume were observed 1-3 months after peak river discharge which then receded and became unidentifiable from satellite observations 5 months after maximum discharge. The seasonal wind pattern and general circulation of the GoM dramatically impacted the observed salinity response, transporting freshwater eastward along the Gulf coast and entraining low salinity waters into the open GoM. The observed salinity response from Aquarius was consistent with SMOS SSS, chl-a concentrations, and the passive tracer simulation in terms of the pathway and transit time of the river plume spreading. This study is the first successful application of satellite SSS to study salinity variation in marginal seas.

1789. Wind-Driven Sea Level Variability on the California Coast: An Adjoint Sensitivity Analysis

Authors: Ariane Verdy (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego); Bruce D. Cornuelle (Scripps Institution of Oceanography / University of California San Diego); Sung Yong Kim (Division of Ocean Systems Engineering, School of Mechanical, Aerospace and Systems Engineering, Korea Advanced Institute of Science and Technology, Daejeon, South Korea)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-018.1

URL: <https://doi.org/10.1175/jpo-d-13-018.1>

Publication Date: 2013-10-04

Volume/Issue: Volume 44, Issue 1

References: 43

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Effects of atmospheric forcing on coastal sea surface height near Port San Luis, central California, are investigated using a regional state estimate and its adjoint. The physical pathways for the propagation of nonlocal [O(100 km)] wind stress effects are identified through adjoint sensitivity analyses, with a cost function that is localized in space so that the adjoint shows details of the propagation of sensitivities. Transfer functions between wind stress and SSH response are calculated and compared to previous work. It is found that (i) the response to local alongshore wind stress dominates on short time scales of O(1 day); (ii) the effect of nonlocal winds dominates on longer time scales and is carried by coastally trapped waves, as well as inertia-gravity waves for offshore wind stress; and (iii) there are significant seasonal variations in the sensitivity of SSH to wind stress due to changes in stratification. In a more stratified ocean, the damping of sensitivities to local and offshore winds is reduced, allowing for a larger and longer-lasting SSH response to wind stress.

1790. Intensification of decadal and multi-decadal sea level variability in the western tropical Pacific during recent decades

Authors: Weiqing Han; Gerald A. Meehl; Aixue Hu; Michael A. Alexander; Toshio Yamagata; Dongliang Yuan; Masayoshi Ishii; Philip Pegion; Jian Zheng; Benjamin D. Hamlington; Xiao-Wei Quan; Robert R. Leben

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-013-1951-1

URL: <https://doi.org/10.1007/s00382-013-1951-1>

Publication Date: 2013-10-01

Volume/Issue: Volume 43, Issue 5-6

References: 96

Cited by: 163 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1791. Ensemble-based global ocean data assimilation

Authors: Balasubramanya T. Nadiga; W. Riley Casper; Philip W. Jones

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2013.09.002

URL: <https://doi.org/10.1016/j.ocemod.2013.09.002>

Publication Date: 2013-09-27

Volume/Issue: Volume 72

References: 54

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1792. Understanding the Hydrosphere Using Satellite Observations

Authors: Felix W. Landerer; Carmen Boening; Michael Watkins

Journal: Eos, Transactions American Geophysical Union

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013eo380004

URL: <https://doi.org/10.1002/2013eo380004>

Publication Date: 2013-09-24

Volume/Issue: Volume 94, Issue 38

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Using satellite observations from the joint NASA/DLR Gravity Recovery And Climate Experiment (GRACE) mission to monitor, simulate, and understand ongoing changes in the Earth's hydrosphere and water cycle was the topic of a recent workshop sponsored by the NASA Jet Propulsion Laboratory's Center for Climate Sciences and GRACE team, with support from the Global Energy and Water Exchanges Project (GEWEX). The meeting brought together more than 60 international scientists with backgrounds in hydrology, oceanography, geodesy, and climate modeling to discuss the best uses of GRACE data in these fields. Since 2002, the GRACE mission has provided global observations of variations in the Earth's gravity field in unprecedented detail, revealing key insights into global water storage and transport.

1793. Down-scaled regional ocean modeling system (ROMS) for high-resolution coastal hydrodynamics in Korea

Authors: Hak-Soo Lim; Chang S. Kim; Kwang-Soon Park; Jae Seol Shim; Insik Chun

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-013-0352-y

URL: <https://doi.org/10.1007/s13131-013-0352-y>

Publication Date: 2013-09-20

Volume/Issue: Volume 32, Issue 9

References: 22

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1794. Mechanisms of Global-Mean Steric Sea Level Change

Authors: Christopher G. Piecuch (Atmospheric and Environmental Research, Lexington, Massachusetts); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00373.1

URL: <https://doi.org/10.1175/jcli-d-13-00373.1>

Publication Date: 2013-09-18

Volume/Issue: Volume 27, Issue 2

References: 41

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Global-mean sea level change partly reflects volumetric expansion of the oceans because of density change, otherwise known as global-mean steric sea level change. Owing to nonlinearities in the equation of state of seawater, the nature of processes contributing to recent observed global-mean steric sea level changes has not been well understood. Using a data-constrained ocean state estimate, global-mean steric sea level change over 1993-2003 is revisited, and contributions from ocean transports and surface exchanges are quantified using closed potential temperature and salinity budgets. Analyses demonstrate that estimated decadal global-mean steric sea level change results mainly from a slight, time-mean imbalance between atmospheric forcing and ocean transports over the integration period: surface heat and freshwater exchanges produce a trend in global-mean steric sea level that is mainly offset by the redistribution of potential temperature and salinity through small-scale diffusion and large-scale advection. A set of numerical experiments demonstrates that global-mean steric sea level changes simulated by ocean general circulation models are sensitive to the regional distribution of ocean heat and freshwater content changes.

1795. Dynamical Reconstruction of Upper-Ocean Conditions in the Last Glacial Maximum Atlantic

Authors: Holly Dail (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-13-00211.1

URL: <https://doi.org/10.1175/jcli-d-13-00211.1>

Publication Date: 2013-09-17

Volume/Issue: Volume 27, Issue 2

References: 65

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Proxies indicate that the Last Glacial Maximum (LGM) Atlantic Ocean was marked by increased meridional and zonal near sea surface temperature gradients relative to today. Using a least squares fit of a full general circulation and sea ice model to upper-ocean proxy data with specified error estimates, a seasonally varying reconstruction is sought of the Atlantic Ocean state that is consistent with both the known dynamics and the data. With reasonable uncertainty assumptions for the observations and the adjustable (control) variables, a consistent LGM ocean state is found, one not radically different from the modern one. Inferred changes include a strengthening of the easterly and westerly winds, leading to strengthened subtropical and subpolar gyres, and increased upwelling favorable winds off the coast of Africa, leading to particularly cold SSTs in those regions.

1796. A Puzzling Disagreement between Observations and Numerical Models in the Central Gulf of Mexico

Authors: Wilton Sturges (Florida State University); Alexandra Bozec (Florida State University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-081.1

URL: <https://doi.org/10.1175/jpo-d-13-081.1>

Publication Date: 2013-09-13

Volume/Issue: Volume 43, Issue 12

References: 29

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Two large, independent sets of direct observations in the central Gulf of Mexico show a mean near-surface flow of ~ 10 cm s⁻¹ to the west, concentrated in the northern and southern Gulf. Numerical models that the authors have examined do not produce this mean westward flow. The observed speeds appear to be almost an order of magnitude larger than the estimated errors; this paper studies the observations to estimate carefully the possible errors involved and compares the observations with model results. The flow to the west in the southern Gulf is presumably wind driven on the shallow parts of the shelf, and, in slightly deeper water at the outer edges of the shelf, is possibly the result of southward Sverdrup interior flow driven by the negative curl of the wind stress. In another possibly related issue, long-term deep current-meter observations in the northern Gulf at ~ 1000 m and below find flow to the west, whereas some models find flow to the east. The flow proposed here assumes a mean flow to the west above roughly 300 m, with a required return flow in deep water. The difference between the deep observations and the models will produce a slope of pressure surfaces of the opposite sign below 1000 m, reversing the direction of upper-layer geostrophic flow in the models.

1797. Decadal Evolution of Ocean Thermal Anomalies in the North Atlantic: The Effects of Ekman, Overturning, and Horizontal Transport

Authors: Richard G. Williams (University of Liverpool); Vassil Roussenov (University of Liverpool); Doug Smith (Hadley Centre, Exeter, United Kingdom); M. Susan Lozier (Duke University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00234.1

URL: <https://doi.org/10.1175/jcli-d-12-00234.1>

Publication Date: 2013-09-13

Volume/Issue: Volume 27, Issue 2

References: 37

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Basin-scale thermal anomalies in the North Atlantic, extending to depths of 1-2 km, are more pronounced than the background warming over the last 60 years. A dynamical analysis based on reanalyses of historical data from 1965 to 2000 suggests that these thermal anomalies are formed by ocean heat convergences, augmented by the poorly known air-sea fluxes. The heat convergence is separated into contributions from the horizontal circulation and the meridional overturning circulation (MOC), the latter further separated into Ekman and MOC transport minus Ekman transport (MOC-Ekman) cells. The subtropical thermal anomalies are mainly controlled by wind-induced changes in the Ekman heat convergence, while the subpolar thermal anomalies are controlled by the MOC-Ekman heat convergence; the horizontal heat convergence is generally weaker, only becoming significant within the subpolar gyre. These thermal anomalies often have an opposing sign between the subtropical and subpolar gyres, associated with opposing changes in the meridional volume transport driving the Ekman and MOC-Ekman heat convergences. These changes in gyre-scale convergences in heat transport are probably induced by the winds, as they correlate with the zonal wind stress at gyre boundaries.

1798. Ocean state estimation from hydrography and velocity observations during EIFEX with a regional biogeochemical ocean circulation model

Authors: Martin Losch; Volker Strass; Boris Cisewski; Christine Klaas; Richard G.J. Bellerby

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2013.09.003

URL: <https://doi.org/10.1016/j.jmarsys.2013.09.003>

Publication Date: 2013-09-13

Volume/Issue: Volume 129

References: 56

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1799. Observational evidence supports the role of tropical cyclones in regulating climate

Authors: Ryan L. Sriver (University of Illinois)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.1314721110

URL: <https://doi.org/10.1073/pnas.1314721110>

Publication Date: 2013-09-07

Volume/Issue: Volume 110, Issue 38

References: 20

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1800. Impact of the Indonesian Throughflow on Agulhas leakage

Authors: D. Le Bars; H. A. Dijkstra; W. P. M. De Ruijter

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-9-773-2013

URL: <https://doi.org/10.5194/os-9-773-2013>

Publication Date: 2013-09-05

Volume/Issue: Volume 9, Issue 5

References: 43

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Using ocean models of different complexity we show that opening the Indonesian Passage between the Pacific and the Indian oceans increases the input of Indian Ocean water into the South Atlantic via the Agulhas leakage. In a strongly eddying global ocean model this response results from an increased Agulhas Current transport and a constant proportion of Agulhas retroflexion south of Africa. The leakage increases through an increased frequency of ring shedding events. In an idealized two-layer and flat-bottom eddy resolving model, the proportion of the Agulhas Current transport that retroflects is (for a wide range of wind stress forcing) not affected by an opening of the Indonesian Passage. Using a comparison with a linear model and previous work on the retroflexion problem, the result is explained as a balance between two mechanisms: decrease retroflexion due to large-scale momentum balance and increase due to local barotropic/baroclinic instabilities.

1801. Reanalyzing temperature and salinity on decadal time scales using the ensemble optimal interpolation data assimilation method and a 3D ocean circulation model of the Baltic Sea

Authors: Ye Liu (Swedish Meteorological and Hydrological Institute; Folkborgsvägen 17 Norrköping Sweden); H. E. Markus Meier (Stockholm University, Swedish Meteorological and Hydrological Institute; Folkborgsvägen 17 Norrköping Sweden); Lars Axell (Swedish Meteorological and Hydrological Institute; Folkborgsvägen 17 Norrköping Sweden)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20384

URL: <https://doi.org/10.1002/jgrc.20384>

Publication Date: 2013-09-04

Volume/Issue: Volume 118, Issue 10

References: 81

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Community's Seventh Framework Programme, joint Baltic Sea research and development programme, Swedish Environmental Protection Agency (and 5 more)

License: N/A

Open Access: N/A

Abstract: N/A

1802. Impact of ocean warm layer thickness on the intensity of hurricane Katrina in a regional coupled model

Authors: Hyodae Seo; Shang-Ping Xie

Journal: Meteorology and Atmospheric Physics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00703-013-0275-3

URL: <https://doi.org/10.1007/s00703-013-0275-3>

Publication Date: 2013-09-02

Volume/Issue: Volume 122, Issue 1-2

References: 76

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1803. Numerical simulations of global-scale high-resolution hydrological crustal deformations

Authors: R. Dill (Deutsches GeoForschungsZentrum Section 1.3: Earth System Modelling Potsdam Germany); H. Dobslaw (Deutsches GeoForschungsZentrum Section 1.3: Earth System Modelling Potsdam Germany)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrb.50353

URL: <https://doi.org/10.1002/jgrb.50353>

Publication Date: 2013-08-28

Volume/Issue: Volume 118, Issue 9

References: 36

Cited by: 204 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: High-resolution load-induced crustal deformations calculated from numerical models are tested for their ability to predict hydrologically-induced station height variability, as they are known to be large enough to affect epoch-wise parameters obtained from the analysis of global geodetic networks. Loading contributions due to terrestrial water storage as given by global hydrological models are calculated on a 0.5 degrees global regular grid with daily temporal resolution. Apart from the dominant seasonal variations, the hydrological loading signal discloses also rapid changes exceeding 1 mm in several regions that can be associated with major precipitation events and river floods. Locally strong loading signals with exceptionally high amplitudes, in many cases even with nonseasonal nature, occur along the major river channels. Only high-resolution loading calculations considering also the water mass anomalies stored in the model river flow can resolve the correct amplitudes in the surrounded area up to 100 km distance. The comparison of the modeled hydrological surface deformation with GPS station time series shows that high-resolution hydrological loading estimates based on global-scale models are able to explain a considerable fraction (up to 54%) of the observed vertical station movements caused by continental water storage variations.

1804. Observing and understanding the Earth system variations from space geodesy

Authors: Shuanggen Jin; Tonie van Dam; Shimon Wdowinski

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2013.08.001

URL: <https://doi.org/10.1016/j.jog.2013.08.001>

Publication Date: 2013-08-16

Volume/Issue: Volume 72

References: 65

Cited by: 86 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1805. An assessment of degree-2 Stokes coefficients from Earth rotation data

Authors: T. Meyrath; T. van Dam; M. Weigelt; M. Cheng

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggt263

URL: <https://doi.org/10.1093/gji/ggt263>

Publication Date: 2013-08-03

Volume/Issue: Volume 195, Issue 1

References: 60

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1806. Polar motion excitations for an Earth model with frequency-dependent responses: 2. Numerical tests of the meteorological excitations

Authors: Wei Chen (Wuhan University, State Key Laboratory of Geodesy and Earth's Dynamics Institute of Geodesy and Geophysics, Chinese Academy of Science Wuhan China); Jim Ray (National Oceanic and Atmospheric Administration Silver Spring Maryland USA); WenBin Shen (Wuhan University); ChengLi Huang (Shanghai Astronomical Observatory, Chinese Academy of Science Shanghai China)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrb.50313

URL: <https://doi.org/10.1002/jgrb.50313>

Publication Date: 2013-08-03

Volume/Issue: Volume 118, Issue 9

References: 40

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: Polar motion

Funding: the National 973 Project, China, the National Natural Science Foundation, China

License: N/A

Open Access: N/A

Abstract: Polar motion excitation involves mass redistributions and motions of the Earth system relative to the mantle, as well as the frequency-dependent rheology of the Earth, where the latter has recently been modeled in the form of frequency-dependent Love numbers and polar motion transfer functions. At seasonal and intraseasonal time scales, polar motions are dominated by angular momentum fluctuations due to mass redistributions and relative motions in the atmosphere, oceans, and continental water, snow, and ice. In this study, we compare the geophysical excitations derived from various global atmospheric, oceanic, and hydrological models (NCEP, ECCO, ERA40, ERAinterim, and ECMWF operational products), and construct two model sets LDC1 and LDC2 by combining the above models with a least difference method. Comparisons between the geodetic excitation (derived from the polar motion series IERS EOP 08 C04) and the geophysical excitations (based on those meteorological models) imply that the atmospheric models are the most reliable while the hydrological ones are the most inaccurate; that the ERAinterim is, in general, the best model set among the original ones, but the combined models LDC1 and LDC2 are much better than ERAinterim; and that applying the frequency-dependent transfer functions to LDC1 and LDC2 improves their agreements with the geodetic excitation. Thus, we conclude that the combined models LDC1 and LDC2 are reliable, and the frequency-dependent Love numbers and polar motion transfer functions are well modeled.

1807. Towards worldwide height system unification using ocean information

Authors: P.L. Woodworth (National Oceanography Centre); C.W. Hughes (National Oceanography Centre); R.J. Bingham (Newcastle University); T. Gruber (3Institut für Astronomische und Physikalische Geodäsie, Technische Universität München, Arcisstrasse 21, 80333 München, Germany)

Journal: Journal of Geodetic Science

Publisher: Walter de Gruyter GmbH

DOI: 10.2478/v10156-012-0004-8

URL: <https://doi.org/10.2478/v10156-012-0004-8>

Publication Date: 2013-08-02

Volume/Issue: Volume 2, Issue 4

References: N/A

Cited by: 67 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We describe the application of ocean levelling to worldwide height system unification. The study involves a comparison of 'geodetic' and 'ocean' approaches to determination of the mean dynamic topography (MDT) at the coast, from which confidence in the accuracy of state-of-the-art ocean and geoid models can be obtained. We conclude that models are consistent at the sub-decimetre level for the regions that we have studied (North Atlantic coastlines and islands, North American Pacific coast and Mediterranean). That level of consistency provides an estimate of the accuracy of using the ocean models to provide an MDT correction to the national datums of countries with coastlines, and thereby of achieving unification. It also provides a validation of geoid model accuracy for application to height system unification in general. We show how our methods can be applied worldwide, as long as the necessary data sets are available, and explain why such an extension of the present study is necessary if worldwide height system unification is to be realised.

1808. Referencing regional geoid-based vertical datums to national tide gauge networks

Authors: D. Bolkas (University of Texas at Dallas); G. Fotopoulos (University of Texas at Dallas); M. G. Sideris (University of Calgary)

Journal: Journal of Geodetic Science

Publisher: Walter de Gruyter GmbH

DOI: 10.2478/v10156-011-0050-7

URL: <https://doi.org/10.2478/v10156-011-0050-7>

Publication Date: 2013-08-02

Volume/Issue: Volume 2, Issue 4

References: N/A

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The objective of this study is to investigate the best means for referencing a regional geoid-based vertical datum to a network of tide gauges. In this study, a network of 27 tide gauge stations scattered along the coasts of Canada are used in order to assess the replacement of the conventionally derived Canadian Geodetic Vertical Datum of 1928 with a geoid-based datum. This is in-line with the future implementation plan of Canada's geoid-based vertical height system. A mixed least-squares adjustment was performed for various scenarios, including satellite-only global geoid models, combined global geoid models and regional geoid models. In addition, various sea surface topography and vertical ground motion models were tested for estimating orthometric heights. The resulting approximation of a local equipotential surface is compared to previously published values and considerations for referencing a geoid-based vertical datum to tide gauge networks are emphasized.

1809. Towards a vertical datum standardisation under the umbrella of Global Geodetic Observing System

Authors: L. Sánchez (1Deutsches Geodätisches Forschungsinstitut (DGFI), Munich, Germany)

Journal: Journal of Geodetic Science

Publisher: Walter de Gruyter GmbH

DOI: 10.2478/v10156-012-0002-x

URL: <https://doi.org/10.2478/v10156-012-0002-x>

Publication Date: 2013-08-02

Volume/Issue: Volume 2, Issue 4

References: N/A

Cited by: 34 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Most of the existing height systems refer to local sea surface levels, are stationary (do not consider variations in time), and realise different physical height types (orthometric, normal, normal-orthometric, etc.). In general, their accuracy is about two orders of magnitude less than that of the realisation of geometric reference systems (sub millimetre level). The Global Geodetic Observing System (GGOS) of the International Association of Geodesy (IAG), taking care of providing a precise geodetic infrastructure for monitoring the system Earth, promotes the standardisation of height systems worldwide. The main objectives are: (1) to provide a reliable frame for consistent analysis and modelling of global phenomena and processes affecting the Earth's gravity field and the Earth's surface geometry; and (2) to support the precise combination of physical and geometric heights in order to exploit at a maximum the advantages of satellite geodesy (e.g. combination of satellite positioning and gravity field models for worldwide unified precise height determination). According to this, the GGOS Theme 1 "Unified Height System" was established in February 2010 with the purpose to bring together existing initiatives and to address the activities to be faced. Starting point are the results delivered by the IAG Inter-Commission Project 1.2 "Vertical Reference Frames" during the period 2003-2011. The present actions related to the vertical datum homogenisation are being coordinated by the working group "Vertical Datum Standardisation", which directly depends on the GGOS Theme 1 and is supported by the IAG Commissions 1 (Reference Frames) and 2 (Gravity Field), as well as by the International Gravity Field Service (IGFS). This paper discusses some aspects to take into consideration for the realisation of a standardised globally unified vertical reference system.

1810. An estimate of the influence of loading effects on tectonic velocities in the Pyrenees

Authors: Marcell Ferenc; Joelle Nicolas; Tonie van Dam; Laurent Polidori; Alexis Rigo; Philippe Vernant

Journal: Studia Geophysica et Geodaetica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11200-012-0458-2

URL: <https://doi.org/10.1007/s11200-012-0458-2>

Publication Date: 2013-08-01

Volume/Issue: Volume 58, Issue 1

References: 37

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1811. Freshening of Subsurface Waters in the Northwest Pacific Subtropical Gyre: Observations and Dynamics

Authors: Youfang Yan (Chinese Academy of Sciences); Eric P. Chassignet (Florida State University); Yiquan Qi (Chinese Academy of Sciences); William K. Dewar (Florida State University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-13-03.1

URL: <https://doi.org/10.1175/jpo-d-13-03.1>

Publication Date: 2013-07-31

Volume/Issue: Volume 43, Issue 12

References: 60

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Subsurface salinity anomalies propagating between mid- and low latitudes along isopycnal surfaces have been shown to play an important role in modulating ocean and climate variability. In this study, a sustained freshening and southwestward propagation of subsurface salinity anomalies in the northwest Pacific subtropical gyre and its dynamical mechanism are investigated using observations, numerical outputs, and a predictive model. Analyses of the observations show a pronounced subsurface freshening with salinity decreasing about 0.25 PSU near the 24.5-sigma θ surface in the northwest Pacific subtropical gyre during 2003-11. This freshening is found to be related to the surface forcing of salinity anomalies in the outcrop zone (25 degrees-35 degreesN, 130 degrees-160 degreesE). A predictive model based on the assumption of salinity conservation along the outcrop isopycnals is derived and used to examine this surface-forcing mechanism. The resemblance between the spatial structures of the subsurface salinity derived from the predictive model and from observations and numerical outputs suggests that subsurface salinity anomalies are ventilated over the outcrop zone. A salinity anomaly with an amplitude of about 0.25 PSU generated by the surface forcing is subducted in the outcrop zone and then propagates southwestward, accompanied by potential vorticity anomalies, to the east of Luzon Strait (~15 degreesN) in roughly one year. When the anomalies reach 15 degreesN, they turn and move gradually eastward toward the central Pacific, associated with an eastward countercurrent on the southern subtropical gyre.

1812. On the time-mean state of ocean models and the properties of long range acoustic propagation

Authors: B. D. Dushaw (University of Washington); P. F. Worcester (Scripps Institution of Oceanography / University of California San Diego); M. A. Dzieciuch (Scripps Institution of Oceanography / University of California San Diego); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20325

URL: <https://doi.org/10.1002/jgrc.20325>

Publication Date: 2013-07-29

Volume/Issue: Volume 118, Issue 9

References: 58

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Receptions on three vertical hydrophone arrays from basin-scale acoustic transmissions in the North Pacific during 1996 and 1998 are used to test the time-mean sound-speed properties of the World Ocean Atlas 2005 (WOA05), of an eddying unconstrained simulation of the Parallel Ocean Program (POP), and of three data-constrained solutions provided by the estimating the circulation and climate of the ocean (ECCO) project: a solution based on an approximate Kalman filter from the Jet Propulsion Laboratory (ECCO-JPL), a solution based on the adjoint method from the Massachusetts Institute of Technology (ECCO-MIT), and an eddying solution based on a Green's function approach from ECCO, Phase II (ECCO2). Predictions for arrival patterns using annual average WOA05 fields match observations to within small travel time offsets (0.3-1.0 s). Predictions for arrival patterns from the models differ substantially from the measured arrival patterns, from the WOA05 climatology, and from each other, both in terms of travel time and in the structure of the arrival patterns. The acoustic arrival patterns are sensitive to the vertical gradients of sound speed that govern acoustic propagation. Basin-scale acoustic transmissions, therefore, provide stringent tests of the vertical temperature structure of ocean state estimates. This structure ultimately influences the mixing between the surface waters and the ocean interior. The relatively good agreement of the acoustic data with the more recent ECCO solutions indicates that numerical ocean models have reached a level of accuracy where the acoustic data can provide useful additional constraints for ocean state estimation.

1813. The genesis of sea level variability in the Barents Sea

Authors: Denis L. Volkov; Felix W. Landerer; Sergey A. Kirillov

Journal: Continental Shelf Research

Publisher: Elsevier BV

DOI: 10.1016/j.csr.2013.07.007

URL: <https://doi.org/10.1016/j.csr.2013.07.007>

Publication Date: 2013-07-27

Volume/Issue: Volume 66

References: 33

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1814. Data assimilation and prognostic whole ice sheet modelling with the variationally derived, higher order, open source, and fully parallel ice sheet model VarGlaS

Authors: D. J. Brinkerhoff; J. V. Johnson

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-1161-2013

URL: <https://doi.org/10.5194/tc-7-1161-2013>

Publication Date: 2013-07-25

Volume/Issue: Volume 7, Issue 4

References: 62

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We introduce a novel, higher order, finite element ice sheet model called VarGlaS (Variational Glacier Simulator), which is built on the finite element framework FEniCS. Contrary to standard procedure in ice sheet modelling, VarGlaS formulates ice sheet motion as the minimization of an energy functional, conferring advantages such as a consistent platform for making numerical approximations, a coherent relationship between motion and heat generation, and implicit boundary treatment. VarGlaS also solves the equations of enthalpy rather than temperature, avoiding the solution of a contact problem. Rather than include a lengthy model spin-up procedure, VarGlaS possesses an automated framework for model inversion. These capabilities are brought to bear on several benchmark problems in ice sheet modelling, as well as a 500 yr simulation of the Greenland ice sheet at high resolution. VarGlaS performs well in benchmarking experiments and, given a constant climate and a 100 yr relaxation period, predicts a mass evolution of the Greenland ice sheet that matches present-day observations of mass loss. VarGlaS predicts a thinning in the interior and thickening of the margins of the ice sheet.

1815. Ocean bottom pressure seasonal cycles and decadal trends from GRACE Release-05: Ocean circulation implications

Authors: Gregory C. Johnson (Pacific Marine Environmental Laboratory NOAA Seattle Washington USA); Don P. Chambers (University of South Florida)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20307

URL: <https://doi.org/10.1002/jgrc.20307>

Publication Date: 2013-07-17

Volume/Issue: Volume 118, Issue 9

References: 52

Cited by: 92 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean mass variations are important for diagnosing sea level budgets, the hydrological cycle, the global energy budget, and ocean circulation variability. Here seasonal cycles and decadal trends of ocean mass from January 2003 to December 2012, both global and regional, are analyzed using GRACE Release-05 data. The trend of global flux of mass into the ocean approaches 2 cm decade⁻¹ in equivalent sea level rise. Regional trends are of similar magnitude, with the North Pacific, South Atlantic, and South Indian oceans generally gaining mass and other regions losing mass. These trends suggest a spin-down of the North Pacific western boundary current extension and the Antarctic Circumpolar Current in the South Atlantic and South Indian oceans. The global average seasonal cycle of ocean mass is about 1 cm in amplitude, with a maximum in early October and volume fluxes in and out of the ocean reaching 0.5 Sv (1 Sv = 1 × 10⁶ m³ s⁻¹) when integrated over the area analyzed here. Regional patterns of seasonal ocean mass change have typical amplitudes of 1-4 cm, and include maxima in the subtropics and minima in the subpolar regions in hemispheric winters. The subtropical mass gains and subpolar mass losses in the winter spin-up both subtropical and subpolar gyres, hence the western boundary current extensions. Seasonal variations in these currents are order 10 Sv, but since the associated depth-averaged current variations are only order 0.1 cm s⁻¹, they would be difficult to detect using in situ oceanographic instruments.

1816. Challenges to Understanding the Dynamic Response of Greenland's Marine Terminating Glaciers to Oceanic and Atmospheric Forcing

Authors: Fiammetta Straneo (Woods Hole Oceanographic Institution); Patrick Heimbach (Massachusetts Institute of Technology); Olga Sergienko (Princeton University); Gordon Hamilton (University of Maine); Ginny Catania (University of Texas at Austin); Stephen Griffies (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); Robert Hallberg (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); Adrian Jenkins (British Antarctic Survey, Cambridge, United Kingdom); Ian Joughin (University of Washington); Roman Motyka (University of Alaska Fairbanks); W. Tad Pfeffer (University of Colorado Boulder); Stephen F. Price (Los Alamos National Laboratory); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Ted Scambos (University of Colorado Boulder); Martin Truffer (University of Alaska Fairbanks); Andreas Vieli (Durham University, Durham, United Kingdom)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/bams-d-12-00100.1

URL: <https://doi.org/10.1175/bams-d-12-00100.1>

Publication Date: 2013-07-15

Volume/Issue: Volume 94, Issue 8

References: 104

Cited by: 122 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The recent retreat and speedup of outlet glaciers, as well as enhanced surface melting around the ice sheet margin, have increased Greenland's contribution to sea level rise to 0.6 ± 0.1 mm yr⁻¹ and its discharge of freshwater into the North Atlantic. The widespread, near-synchronous glacier retreat, and its coincidence with a period of oceanic and atmospheric warming, suggests a common climate driver. Evidence points to the marine margins of these glaciers as the region from which changes propagated inland. Yet, the forcings and mechanisms behind these dynamic responses are poorly understood and are either missing or crudely parameterized in climate and ice sheet models. Resulting projected sea level rise contributions from Greenland by 2100 remain highly uncertain. This paper summarizes the current state of knowledge and highlights key physical aspects of Greenland's coupled ice sheet-ocean-atmosphere system. Three research thrusts are identified to yield fundamental insights into ice sheet, ocean, sea ice, and atmosphere interactions, their role in Earth's climate system, and probable trajectories of future changes: 1) focused process studies addressing critical glacier, ocean, atmosphere, and coupled dynamics; 2) sustained observations at key sites; and 3) inclusion of relevant dynamics in Earth system models. Understanding the dynamic response of Greenland's glaciers to climate forcing constitutes both a scientific and technological frontier, given the challenges of obtaining the appropriate measurements from the glaciers' marine termini and the complexity of the

dynamics involved, including the coupling of the ocean, atmosphere, glacier, and sea ice systems. Interdisciplinary and international cooperation are crucial to making progress on this novel and complex problem.

1817. Modeled diversity effects on microbial ecosystem functions of primary production, nutrient uptake, and remineralization

Authors: Nicole L. Goebel; Christopher A. Edwards; Michael J. Follows; Jonathan P. Zehr

Journal: Ecology

Publisher: Wiley

DOI: 10.1890/13-0421.1

URL: <https://doi.org/10.1890/13-0421.1>

Publication Date: 2013-07-09

Volume/Issue: Volume 95, Issue 1

References: 51

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ecosystem-wide primary productivity generally increases with primary producer diversity, emphasizing the importance of diversity for ecosystem function. However, most studies that demonstrate this positive relationship have focused on terrestrial and aquatic benthic systems, with little attention to the diverse marine pelagic primary producers that play an important role in regulating global climate. Here we show how phytoplankton biodiversity enhances overall marine ecosystem primary productivity and other ecosystem functions using a self-organizing ecosystem model. Diversity manipulation numerical experiments reveal positive, asymptotically saturating relationships between ecosystem-wide phytoplankton diversity and functions of productivity, nutrient uptake, remineralization, and diversity metrics used to identify mechanisms shaping these relationships. Increase in productivity with increasing diversity improves modeled ecosystem stability and model robustness and leads to productivity rates that exceed expected yields primarily through niche complementarity and facilitative interactions between coexisting phytoplankton types; the composition of traits in assemblages determines the magnitude of complementarity and selection effects. While findings based on these aggregate measures of diversity effects parallel those from the majority of experimental outcomes of terrestrial and benthic biodiversity-ecosystem function studies, we combine analyses of community diversity effects and investigations of the underlying interactions among phytoplankton types to demonstrate how an increase in recycled production of non-diatoms through an increase in new production of diatoms drives this diversity-ecosystem function response. We demonstrate the important role that facilitation plays in the modeled marine plankton and how this facilitative interaction could amplify future climate-driven changes in ocean ecosystem productivity.

1818. Variability in Southern Hemisphere Ocean Circulation from the 1980s to the 2000s

Authors: K. Katsumata (RIGC JAMSTEC, Yokosuka, Japan); S. Masuda (RIGC JAMSTEC, Yokosuka, Japan)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-0209.1

URL: <https://doi.org/10.1175/jpo-d-12-0209.1>

Publication Date: 2013-07-08

Volume/Issue: Volume 43, Issue 9

References: 84

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Interannual-to-decadal variability of ocean circulation in the Southern Hemisphere was examined using data from the 1980s to the 2000s in a box inverse model to estimate transport across hydrographic sections and three ocean general circulation models (OGCMs). The westerly wind stress over the OGCM Southern Ocean showed a steady increase of 5%-8% decade⁻¹. The meridional overturning circulation was quantified by the transport across 30 degreesS. The OGCMs suggested a slight strengthening [from 0.2 +/- 1.0 to 0.8 +/- 1.3 Sv decade⁻¹ (1 Sv ≡ 106 m³ s⁻¹)] of the upper meridional cell (Deacon cell) and two OGCMs showed a weakening (-0.8 +/- 0.6 and -1.0 +/- 0.3 Sv decade⁻¹) of the lower meridional [Antarctic Bottom Water (AABW)] cell, partly explained by contraction of the AABW volume. The box inverse estimates did not contradict these two findings. For Antarctic Circumpolar Current transport, quantified by zonal transport across four key sections, the box inverse model estimated a decrease of 5-21 Sv. Decomposition of the decrease into baroclinic transport by the Subantarctic and Polar Fronts, barotropic transport, and others shows that the decrease is mostly due to barotropic transport and transport carried by the flow north of the Subantarctic Front and south of the Polar Front. In the OGCMs, the variability of transport across key sections is often correlated with transport carried by a flow south of the Polar Front and with the southern annular mode index. In all models, then, the transport of the Antarctic Circumpolar Current, defined as the transport carried by the fronts, has not decreased significantly over the study period.

1819. On the Connection between Continental-Scale Land Surface Processes and the Tropical Climate in a Coupled Ocean-Atmosphere-Land System

Authors: Hsi-Yen Ma (Lawrence Livermore National Laboratory); C. Roberto Mechoso (University of California Los Angeles); Yongkang Xue (University of California Los Angeles); Heng Xiao (@Atmospheric Sciences and Global Change Division, Pacific North National Laboratory, Richland, Washington); J. David Neelin (University of California Los Angeles); Xuan Ji (University of California Los Angeles)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00819.1

URL: <https://doi.org/10.1175/jcli-d-12-00819.1>

Publication Date: 2013-07-05

Volume/Issue: Volume 26, Issue 22

References: 52

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: Atmospheric circulation, Atmosphere-ocean interaction, Convection, Teleconnections, Climate models (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An evaluation is presented of the impact on tropical climate of continental-scale perturbations given by different representations of land surface processes (LSPs) in a general circulation model that includes atmosphere-ocean interactions. One representation is a simple land scheme, which specifies climatological albedos and soil moisture availability. The other representation is the more comprehensive Simplified Simple Biosphere Model, which allows for interactive soil moisture and vegetation biophysical processes. The results demonstrate that such perturbations have strong impacts on the seasonal mean states and seasonal cycles of global precipitation, clouds, and surface air temperature. The impact is especially significant over the tropical Pacific Ocean. To explore the mechanisms for such impact, model experiments are performed with different LSP representations confined to selected continental-scale regions where strong interactions of climate-vegetation biophysical processes are present. The largest impact found over the tropical Pacific is mainly from perturbations in the tropical African continent where convective heating anomalies associated with perturbed surface heat fluxes trigger global teleconnections through equatorial wave dynamics. In the equatorial Pacific, the remote impacts of the convection anomalies are further enhanced by strong air-sea coupling between surface wind stress and upwelling, as well as by the effects of ocean memory. LSP perturbations over South America and Asia-Australia have much weaker global impacts. The results further suggest that correct representations of LSP, land use change, and associated changes in the deep convection over tropical Africa are crucial to reducing the uncertainty of future climate projections with global climate models under various climate change scenarios.

1820. The ability of the adjoint technique to recover decadal variability of the North Atlantic circulation

Authors: Michael Brüdgam; Carsten Eden; Lars Czeschel; Johanna Baehr

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2013.06.006

URL: <https://doi.org/10.1016/j.ocemod.2013.06.006>

Publication Date: 2013-06-28

Volume/Issue: Volume 69

References: 34

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1821. Analysis of deep-layer and bottom circulations in the South China Sea based on eight quasi-global ocean model outputs

Authors: Qiang Xie; JinGen Xiao; DongXiao Wang; YongQiang Yu

Journal: Chinese Science Bulletin

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11434-013-5791-5

URL: <https://doi.org/10.1007/s11434-013-5791-5>

Publication Date: 2013-06-26

Volume/Issue: Volume 58, Issue 32

References: 49

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1822. Subduction of South Pacific Tropical Water and Its Equatorward Pathways as Shown by a Simulated Passive Tracer

Authors: Tangdong Qu; Shan Gao; Rana A. Fine

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-0180.1

URL: <https://doi.org/10.1175/jpo-d-12-0180.1>

Publication Date: 2013-06-12

Volume/Issue: Volume 43, Issue 8

References: 44

Cited by: 54 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study investigates the subduction of South Pacific Tropical Water (SPTW) and its equatorward pathways using a simulated passive tracer of the consortium Estimating the Circulation & Climate of the Ocean (ECCO). The results show that approximately 5.8 Sv ($1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) of the SPTW is formed in the subtropical South Pacific Ocean within the density range between 24.0 and 25.0 kg m^{-3} , of which about 87% is due to vertical pumping and 13% is due to lateral induction, comparing reasonably well with estimates from climatological data. Once subducted, most SPTW spreads in the subtropical South Pacific. Because of the presence of mixing, some portion of the water is transformed, and its tracer-weighted density steadily increases from an initial value of 24.4 to nearly 25.0 kg m^{-3} after 13 years of integration. Approximately 42% of the water makes its way into the equatorial Pacific, either through the western boundary or interior pathway. The two equatorward pathways are essentially of equal importance. A large (~70%) portion of the SPTW entering the equatorial region resurfaces in the central equatorial Pacific. The potential impacts of the resurfacing SPTW on the equatorial thermocline and surface stratification are discussed.

1823. Sea-state-dependent wind work on the oceanic general circulation

Authors: Guoqiang Liu (Bedford Institute of Oceanography); William Perrie (Bedford Institute of Oceanography)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/grl.50624

URL: <https://doi.org/10.1002/grl.50624>

Publication Date: 2013-06-05

Volume/Issue: Volume 40, Issue 12

References: 33

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We examine sea-state-dependent wind work on the oceanic general circulation, using a wave hindcast dataset, QuikSCAT winds, and geostrophic and total ocean surface currents from (1) AVISO and (2) ECCO2 model products. For wind work on surface geostrophic currents estimated from AVISO or ECCO2, sea-state-dependent wind stress increases an average of 24% (0.17TW) or 23% (0.15TW), compared with estimates that exclude sea state effects. For wind work on the total surface currents, the sea-state-dependent wind stress increases the wind work by about 24% (0.4TW). In terms of spatial distribution, the increase in wind power input occurs mainly in high-wind tropical regions and the mid-latitude storm track regions, like the Antarctic Circumpolar Current region, where the relatively rough ocean surface is characterized by young waves and high winds. By comparison, in some regions with relatively low winds and mature ocean waves, there is a slight reduction in estimated wind power input.

1824. Satellite Gravimetry: Mass Transport and Redistribution in the Earth System

Authors: Shuanggen Jin

Journal: Geodetic Sciences - Observations, Modeling and Applications

Publisher: InTech

DOI: 10.5772/51698

URL: <https://doi.org/10.5772/51698>

Publication Date: 2013-06-03

Volume/Issue: N/A

References: N/A

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: Gravimetry|Redistribution

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1825. Assessing the precision in loading estimates by geodetic techniques in Southern Europe

Authors: Pierre Valtý; Olivier de Viron; Isabelle Panet; Michel Van Camp; Juliette Legrand

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggt173

URL: <https://doi.org/10.1093/gji/ggt173>

Publication Date: 2013-06-01

Volume/Issue: Volume 194, Issue 3

References: 59

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1826. Non-Tidal Oceanic Contribution to the Variation of the Earth's Oblateness

Authors: No authors listed

Journal: Chinese Journal of Geophysics

Publisher: Wiley

DOI: 10.1002/cjg2.511

URL: <https://doi.org/10.1002/cjg2.511>

Publication Date: 2013-05-31

Volume/Issue: Volume 47, Issue 3

References: 9

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Cox and Chao obtained time series of the Earth's dynamic oblateness J_2 by analyzing Satellite Laser Ranging data for 25 years and found that since 1997/1998, the J_2 variation shows dramatical abnormal changes after removing seasonal signals. The long-term change of J_2 before 1996 can be well explained by the theory of mantle rebound since the last ice age but the change since 1997/1998 is inconsistent with the theory. It was suspected that the abnormal change was the result of geophysical fluid motions and the atmosphere has been excluded as a possible source. In this study, we investigate two ocean general circulation models to conclude that the mass transfer within the oceans, specifically within the Pacific ocean, contributes to at least part of the observed variation since 1997/1998. The identification of geophysical source helps verify the SLR results of Cox and Chao. Different from previous work on the interpretation of J_2 variation, our study uses two ocean models and the altimetry data from TOPEX/Poseidon for model validation. Our conclusion is that part of the observed abnormal J_2 variation since 1997/1998 may be from variations of steric effects, which are the result of thermal and saline variations, and might not be easily discerned from the sea surface height variations observed with the T/P altimetry.

1827. An Adjoint-Based Adaptive Ensemble Kalman Filter

Authors: Hajoon Song (University of California Santa Cruz); Ibrahim Hoteit (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia); Bruce D. Cornuelle (Scripps Institution of Oceanography / University of California San Diego); Xiaodong Luo (King Abdullah University of Science and Technology, Thuwal, Saudi Arabia, and International Research Institute of Stavanger, Bergen, Norway); Aneesh C. Subramanian (Scripps Institution of Oceanography / University of California San Diego)

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/mwr-d-12-00244.1

URL: <https://doi.org/10.1175/mwr-d-12-00244.1>

Publication Date: 2013-05-30

Volume/Issue: Volume 141, Issue 10

References: 46

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A new hybrid ensemble Kalman filter/four-dimensional variational data assimilation (EnKF/4D-VAR) approach is introduced to mitigate background covariance limitations in the EnKF. The work is based on the adaptive EnKF (AEnKF) method, which bears a strong resemblance to the hybrid EnKF/three-dimensional variational data assimilation (3D-VAR) method. In the AEnKF, the representativeness of the EnKF ensemble is regularly enhanced with new members generated after back projection of the EnKF analysis residuals to state space using a 3D-VAR [or optimal interpolation (OI)] scheme with a preselected background covariance matrix. The idea here is to reformulate the transformation of the residuals as a 4D-VAR problem, constraining the new member with model dynamics and the previous observations. This should provide more information for the estimation of the new member and reduce dependence of the AEnKF on the assumed stationary background covariance matrix. This is done by integrating the analysis residuals backward in time with the adjoint model. Numerical experiments are performed with the Lorenz-96 model under different scenarios to test the new approach and to evaluate its performance with respect to the EnKF and the hybrid EnKF/3D-VAR. The new method leads to the least root-mean-square estimation errors as long as the linear assumption guaranteeing the stability of the adjoint model holds. It is also found to be less sensitive to choices of the assimilation system inputs and parameters.

1828. Comparative analysis of different environmental loading methods and their impacts on the GPS height time series

Authors: Weiping Jiang; Zhao Li; Tonie van Dam; Wenwu Ding

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-013-0642-3

URL: <https://doi.org/10.1007/s00190-013-0642-3>

Publication Date: 2013-05-25

Volume/Issue: Volume 87, Issue 7

References: 47

Cited by: 91 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1829. Coupled climate impacts of the Drake Passage and the Panama Seaway

Authors: Simon Yang; Eric Galbraith; Jaime Palter

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-013-1809-6

URL: <https://doi.org/10.1007/s00382-013-1809-6>

Publication Date: 2013-05-24

Volume/Issue: Volume 43, Issue 1-2

References: 72

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1830. State estimates and forecasts of the loop current in the Gulf of Mexico using the MITgcm and its adjoint

Authors: Ganesh Gopalakrishnan (Scripps Institution of Oceanography); Bruce D. Cornuelle (Scripps Institution of Oceanography); Ibrahim Hoteit (Department of Earth Sciences and Engineering, King Abdullah University of Science and Technology; Jeddah Kingdom of Saudi Arabia); Daniel L. Rudnick (Scripps Institution of Oceanography); W. Brechner Owens (Woods Hole Oceanographic Institution)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20239

URL: <https://doi.org/10.1002/jgrc.20239>

Publication Date: 2013-05-23

Volume/Issue: Volume 118, Issue 7

References: 58

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1831. Marine Phytoplankton Temperature versus Growth Responses from Polar to Tropical Waters - Outcome of a Scientific Community-Wide Study

Authors: Philip W. Boyd; Tatiana A. Rynearson; Evelyn A. Armstrong; Feixue Fu; Kendra Hayashi; Zhangxi Hu; David A. Hutchins; Raphael M. Kudela; Elena Litchman; Margaret R. Mulholland; Uta Passow; Robert F. Strzepek; Kerry A. Whittaker; Elizabeth Yu; Mridul K. Thomas

Journal: PLoS ONE

Publisher: Public Library of Science (PLoS)

DOI: 10.1371/journal.pone.0063091

URL: <https://doi.org/10.1371/journal.pone.0063091>

Publication Date: 2013-05-21

Volume/Issue: Volume 8, Issue 5

References: 106

Cited by: 271 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

1832. Adjoint sensitivity studies of loop current and eddy shedding in the Gulf of Mexico

Authors: Ganesh Gopalakrishnan (Scripps Institution of Oceanography); Bruce D. Cornuelle (Scripps Institution of Oceanography); Ibrahim Hoteit (Department of Earth Sciences and Engineering, King Abdullah University of Science and Technology; Kingdom of Saudi Arabia Thuwal Saudi Arabia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20240

URL: <https://doi.org/10.1002/jgrc.20240>

Publication Date: 2013-05-21

Volume/Issue: Volume 118, Issue 7

References: 68

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1833. Evaluation of the Global Ocean Data Assimilation System at INCOIS: The Tropical Indian Ocean

Authors: M. Ravichandran; D. Behringer; S. Sivareddy; M.S. Girishkumar; Neethu Chacko; R. Harikumar

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2013.05.003

URL: <https://doi.org/10.1016/j.ocemod.2013.05.003>

Publication Date: 2013-05-16

Volume/Issue: Volume 69

References: 85

Cited by: 76 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1834. GLACIAL CLIMATES | Thermohaline Circulation

Authors: S. Rahmstorf

Journal: Encyclopedia of Quaternary Science

Publisher: Elsevier

DOI: 10.1016/b978-0-444-53643-3.00020-0

URL: <https://doi.org/10.1016/b978-0-444-53643-3.00020-0>

Publication Date: 2013-05-14

Volume/Issue: N/A

References: 29

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Deglaciation, Antarctic Bottom Water, North Atlantic Deep Water, Upwelling, Dansgaard-Oeschger event (and 6 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1835. Rapid ice melting drives Earth's pole to the east

Authors: J. L. Chen (University of Texas at Austin); C. R. Wilson (University of Texas at Austin); J. C. Ries (University of Texas at Austin); B. D. Tapley (University of Texas at Austin)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/grl.50552

URL: <https://doi.org/10.1002/grl.50552>

Publication Date: 2013-05-13

Volume/Issue: Volume 40, Issue 11

References: 25

Cited by: 71 publications

Language: en

Subjects: N/A

Keywords: Polar motion|Ice caps

Funding: NASA GRACE Science Program, NASA ESI Program, NSF OPP Program

License: N/A

Open Access: N/A

Abstract: Space geodetic observations of polar motion show that around 2005, the average annual pole position began drifting toward the east, an abrupt departure from the drift direction seen over the past century. Satellite gravity measurements from the Gravity Recovery and Climate Experiment (GRACE) show that about 90% of this change is due to accelerated melting of polar ice sheets and mountain glaciers and related sea level rise. The close relationship between long-term polar motion and climate-related mass redistribution established using GRACE data indicates that accurately measured polar motion data offer an additional tool for monitoring global-scale ice melting and sea level rise and should be useful in bridging the anticipated gap between GRACE and follow-on satellite gravity missions.

1836. Antarctic contribution to sea level rise observed by GRACE with improved GIA correction

Authors: Erik R. Ivins (California Institute of Technology); Thomas S. James (School of Earth and Ocean Sciences University of Victoria Victoria British Columbia Canada, Geological Survey of Canada Sidney British Columbia Canada); John Wahr (University of Colorado Boulder); Ernst J. O. Schrama (Delft University of Technology); Felix W. Landerer (California Institute of Technology); Karen M. Simon (School of Earth and Ocean Sciences University of Victoria Victoria British Columbia Canada, Geological Survey of Canada Sidney British Columbia Canada)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrb.50208

URL: <https://doi.org/10.1002/jgrb.50208>

Publication Date: 2013-05-09

Volume/Issue: Volume 118, Issue 6

References: 95

Cited by: 202 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA

License: N/A

Open Access: N/A

Abstract: Antarctic volume changes during the past 21 thousand years are smaller than previously thought, and here we construct an ice sheet history that drives a forward model prediction of the glacial isostatic adjustment (GIA) gravity signal. The new model, in turn, should give predictions that are constrained with recent uplift data. The impact of the GIA signal on a Gravity Recovery and Climate Experiment (GRACE) Antarctic mass balance estimate depends on the specific GRACE analysis method used. For the method described in this paper, the GIA contribution to the apparent surface mass change is re-evaluated to be $+55 \pm 13$ Gt/yr by considering a revised ice history model and a parameter search for vertical motion predictions that best fit the GPS observations at 18 high-quality stations. Although the GIA model spans a range of possible Earth rheological structure values, the data are not yet sufficient for solving for a preferred value of upper and lower mantle viscosity nor for a preferred lithospheric thickness. GRACE monthly solutions from the Center for Space Research Release 04 (CSR-RL04) release time series from January 2003 to the beginning of January 2012, uncorrected for GIA, yield an ice mass rate of $+2.9 \pm 29$ Gt/yr. The new GIA correction increases the solved-for ice mass imbalance of Antarctica to -57 ± 34 Gt/yr. The revised GIA correction is smaller than past GRACE estimates by about 50 to 90 Gt/yr. The new upper bound to the sea level rise from the Antarctic ice sheet, averaged over the time span 2003.0-2012.0, is about 0.16 ± 0.09 mm/yr.

1837. An Ensemble Adjustment Kalman Filter for the CCSM4 Ocean Component

Authors: Alicia R. Karspeck (National Center for Atmospheric Research); Steve Yeager (National Center for Atmospheric Research); Gokhan Danabasoglu (National Center for Atmospheric Research); Tim Hoar (National Center for Atmospheric Research); Nancy Collins (National Center for Atmospheric Research); Kevin Raeder (National Center for Atmospheric Research); Jeffrey Anderson (National Center for Atmospheric Research); Joseph Tribbia (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00402.1

URL: <https://doi.org/10.1175/jcli-d-12-00402.1>

Publication Date: 2013-05-08

Volume/Issue: Volume 26, Issue 19

References: 66

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Forecasting, Kalman filters, Data assimilation, Ensembles (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The authors report on the implementation and evaluation of a 48-member ensemble adjustment Kalman filter (EAKF) for the ocean component of the Community Climate System Model, version 4 (CCSM4). The ocean assimilation system described was developed to support the eventual generation of historical ocean-state estimates and ocean-initialized climate predictions with the CCSM4 and its next generation, the Community Earth System Model (CESM). In this initial configuration of the system, daily subsurface temperature and salinity data from the 2009 World Ocean Database are assimilated into the ocean model from 1 January 1998 to 31 December 2005. Each ensemble member of the ocean is forced by a member of an independently generated CCSM4 atmospheric EAKF analysis, making this a loosely coupled framework. Over most of the globe, the time-mean temperature and salinity fields are improved relative to an identically forced ocean model simulation without assimilation. This improvement is especially notable in strong frontal regions such as the western and eastern boundary currents. The assimilation system is most effective in the upper 1000 m of the ocean, where the vast majority of in situ observations are located. Because of the shortness of this experiment, ocean variability is not discussed. Challenges that arise from using an ocean model with strong regional biases, coarse resolution, and low internal variability to assimilate real observations are discussed, and areas of ongoing improvement for the assimilation system are outlined.

1838. Estimation of the drag coefficient from the upper ocean response to a hurricane: A variational data assimilation approach

Authors: Sarah E. Zedler; Guido Kanschä; Ibrahim Hoteit; Robert Korty

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2013.04.004

URL: <https://doi.org/10.1016/j.ocemod.2013.04.004>

Publication Date: 2013-05-04

Volume/Issue: Volume 68

References: 62

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1839. Evaluation of a near-global eddy-resolving ocean model

Authors: P. R. Oke; D. A. Griffin; A. Schiller; R. J. Matear; R. Fiedler; J. Mansbridge; A. Lenton; M. Cahill; M. A. Chamberlain; K. Ridgway

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-6-591-2013

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References: 90

Cited by: 125 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Analysis of the variability of the last 18 yr (1993-2012) of a 32 yr run of a new near-global, eddy-resolving ocean general circulation model coupled with biogeochemistry is presented. Comparisons between modelled and observed mean sea level (MSL), mixed layer depth (MLD), sea level anomaly (SLA), sea surface temperature (SST), and $\{chla\}$ indicate that the model variability is realistic. We find some systematic errors in the modelled MLD, with the model generally deeper than observations, which results in errors in the $\{chla\}$, owing to the strong biophysical coupling. We evaluate several other metrics in the model, including the zonally averaged seasonal cycle of SST, meridional overturning, volume transports through key straits and passages, zonally averaged temperature and salinity, and El Niño-related SST indices. We find that the modelled seasonal cycle in SST is 0.5-1.5 degreesC weaker than observed; volume transports of the Antarctic Circumpolar Current, the East Australian Current, and Indonesian Throughflow are in good agreement with observational estimates; and the correlation between the modelled and observed NINO SST indices exceeds 0.91. Most aspects of the model circulation are realistic. We conclude that the model output is suitable for broader analysis to better understand upper ocean dynamics and ocean variability at mid- and low latitudes. The new model is intended to underpin a future version of Australia's operational short-range ocean forecasting system.

1840. Temporal trends in biodiversity of the middle-slope assemblages in Sardinian seas (Central-Western Mediterranean)

Authors: Andrea Sabatini; Ivan Locci; Anna M. Deiana; Maria C. Follesa; Alice Gastoni; Antonio A. Pendugiu; Paola Pesci; Angelo Cau

Journal: Journal of the Marine Biological Association of the United Kingdom

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/s0025315413000258

URL: <https://doi.org/10.1017/s0025315413000258>

Publication Date: 2013-05-02

Volume/Issue: Volume 93, Issue 7

References: 49

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: It has been amply demonstrated that trawl fishing affects overall biomass, size and species structure of demersal fish communities, and an increasing number of studies are proving that this could have even greater unexpected effects on biological diversity. The aims of this study are: (1) to examine the temporal trends of several ecological diversity indices for middle-slope communities in specific locations within the Sardinian seas, which have been recently subject to increasing fishing capacity; and (2) to simulate the Shannon's entropy (H') temporal trend, using a multiple linear regression, in order to investigate about the relationships between fishing effort and species diversity. The data come from 11 MEDITS trawl surveys (1994 to 2004). Temporal trends in fishing effort were analysed and ecological diversity indices were measured. Analysis was conducted for four areas (NW, NE, SE, SSW). Significant variations over time in both fishing pressure and ecological diversity indices were found for the southern zones. More precisely, the SE zone showed an increase in fishing effort (+120% in 2004), and a decrease of Shannon-Weiner's diversity. The SSW zone exhibited a 22.6% increase of fishing effort and a decrease of species richness. The best model of H' incorporates species richness and fishing effort expressed as number of boats per trawling area. Our results seem to indicate that fishing pressure affected the species richness and the abundance of middle-slope species and that variation in ecological diversity indices differed with different levels of fishing effort.

1841. Clarifying the link between surface salinity and freshwater fluxes on monthly to interannual time scales

Authors: Nadya T. Vinogradova (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20200

URL: <https://doi.org/10.1002/jgrc.20200>

Publication Date: 2013-04-30

Volume/Issue: Volume 118, Issue 6

References: 68

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA's Physical Oceanography Program, Ocean Surface Salinity Project

License: N/A

Open Access: N/A

Abstract: Freshwater fluxes (F) between the ocean and the atmosphere and land, comprised of evaporation, precipitation and terrestrial runoff, are an essential component of the Earth's climate system. However, direct observations of F and its components are sparse and available estimates have substantial uncertainties. In this study we investigate if measurements of sea-surface salinity (S) can provide an alternative indirect method for estimating F . We examine the relationship between F and oceanic fluxes from surface advection and mixing processes, on time scales from months to years, using a consistent estimate of the ocean/atmosphere state obtained from model/data synthesis produced by the ECCO (Estimating Circulation and Climate of the Ocean) consortium. ECCO salinity averaged over the mixed layer is used as an estimate of S . Budget analysis shows that variability in tendencies can be attributed to both F and oceanic fluxes, demonstrating the importance of the ocean's role in evolution of F , for both local and global mean fields. Regression analysis of the 13 year long ECCO fields shows that there are only a few regions (e.g., subtropical gyres) where S can be used as a proxy for using linear models, and only at monthly to annual time scales. Results are similar over a range of spatial scales from ~ 100 to 2000 km. Findings are discussed in the context of the general sensitivities of F to atmospheric and oceanic processes and the potential of satellite salinity measurements to constrain estimates of F .

1842. Development of a variational data assimilation system for the diurnal cycle of sea surface temperature

Authors: J. While (Met Office); M. Martin (Met Office)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20215

URL: <https://doi.org/10.1002/jgrc.20215>

Publication Date: 2013-04-30

Volume/Issue: Volume 118, Issue 6

References: 41

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: UK Ministry of Defense science program

License: N/A

Open Access: N/A

Abstract: A variational data assimilation system based on an incremental 4D-Var approach is proposed for use with a zero-dimensional model of the diurnal cycle of sea surface temperature (SST). Traditional 4D-Var, which seeks to find the initial state of a system, is not appropriate for diurnal SST which is a wind and heat flux driven system that has only a limited memory of its prior state. Instead the proposed assimilation system corrects both the initial SST and the heat and wind fluxes applied throughout the day. The assimilation system is tested using ensembles in a set of idealized twin experiments. In these tests controlling parameters are varied around reasonable "default" values with the quality of the analyses assessed against a known "truth". Within our tests data assimilation is shown to improve diurnal SST under most circumstances. Analyzed heat fluxes are also sometimes improved, although the improvement is much less than that observed for diurnal SST. The system was not found to improve the wind stress. The only circumstances where diurnal SST was not found to be improved by the assimilation were where either observational errors were large (greater than 0.5 K in our tests), or biases in the observations were too big (less than -0.3 K or greater than 0.2 K). The non-Gaussian behavior of the wind stress was found to have an impact on the assimilation in low-wind conditions and under these conditions the best analyses were obtained by artificially inflating the observation error.

1843. Interannual sea level variability in the tropical Pacific Ocean from 1993 to 2006

Authors: Qing Lu; Juncheng Zuo; Yanfang Li; Meixiang Chen

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2013.04.008

URL: <https://doi.org/10.1016/j.gloplacha.2013.04.008>

Publication Date: 2013-04-26

Volume/Issue: Volume 107

References: 49

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1844. Formation of salinity maximum water and its contribution to the overturning circulation in the North Atlantic as revealed by a global general circulation model

Authors: Tangdong Qu (International Pacific Research Center, SOEST University of Hawai'i at Mānoa Honolulu Hawaii USA); Shan Gao (International Pacific Research Center, SOEST University of Hawai'i at Mānoa Honolulu Hawaii USA, Chinese Academy of Sciences); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20152

URL: <https://doi.org/10.1002/jgrc.20152>

Publication Date: 2013-04-23

Volume/Issue: Volume 118, Issue 4

References: 47

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The formation of salinity maximum water in the North Atlantic is investigated using a simulated passive tracer and its adjoint. The results reveal that most salinity maximum water in the North Atlantic comes from the northwestern part of the subtropical gyre, and direct contribution from the evaporation-precipitation maximum region via the surface Ekman current is minor. Water originating from the evaporation-precipitation maximum region has to recirculate in the subtropical gyre before entering the sea surface salinity maximum region from the northwest. Once subducted, some portion (~10%) of the salinity maximum water enters the equatorial region in the shallow subtropical cell, but most (~70%) of it appears to turn northward to join the North Atlantic Deep Water. The latter pathway involves a three-dimensional circulation. When the warm, fresh surface water flows northward along the western boundary, it turns eastward in the northern subtropical gyre. As a result of the large excess of evaporation over precipitation, this water gradually gains its salinity on the route, until it reaches the sea surface salinity maximum region in the central subtropical gyre. From there, the salinity maximum water is subducted and flows back to the western boundary in the depth range of the thermocline. With its high-salinity nature, a major portion of this water penetrates into the subpolar region and directly contributes to the deep thermohaline circulation.

1845. Seasonal variability of submarine melt rate and circulation in an East Greenland fjord

Authors: R. Sciascia (Massachusetts Institute of Technology, Institute of Atmospheric Sciences and Climate CNR Torino Italy, Doctorate program in Fluid Dynamics Politecnico di Torino Italy); F. Straneo (Woods Hole Oceanographic Institution); C. Cenedese (Woods Hole Oceanographic Institution); P. Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20142

URL: <https://doi.org/10.1002/jgrc.20142>

Publication Date: 2013-04-23

Volume/Issue: Volume 118, Issue 5

References: 68

Cited by: 148 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The circulation in a glacial fjord driven by a large tidewater glacier is investigated using a nonhydrostatic ocean general circulation model with a melt rate parameterization at the vertical glacier front. The model configuration and water properties are based on data collected in Sermilik Fjord near Helheim Glacier, a major Greenland outlet glacier. The approximately two-layer stratification of the fjord's ambient waters causes the meltwater plume at the glacier front to drive a "double cell" circulation with two distinct outflows, one at the free surface and one at the layers' interface. In summer, the discharge of surface runoff at the base of the glacier (subglacial discharge) causes the circulation to be much more vigorous and associated with a larger melt rate than in winter. The simulated "double cell" circulation is consistent, in both seasons, with observations from Sermilik Fjord. Seasonal differences are also present in the vertical structure of the melt rate, which is maximum at the base of the glacier in summer and at the layers' interface in winter. Simulated submarine melt rates are strongly sensitive to the amount of subglacial discharge, to changes in water temperature, and to the height of the layers. They are also consistent with those inferred from simplified one-dimensional models based on the theory of buoyant plumes. Our results also indicate that to correctly represent the dynamics of the meltwater plume, care must be taken in the choice of viscosity and diffusivity values in the model.

1846. Global surface eddy diffusivities derived from satellite altimetry

Authors: R. P. Abernathey (Scripps Institution of Oceanography, Massachusetts Institute of Technology); J. Marshall (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrc.20066

URL: <https://doi.org/10.1002/jgrc.20066>

Publication Date: 2013-04-22

Volume/Issue: Volume 118, Issue 2

References: 74

Cited by: 129 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Velocities derived from AVISO sea-surface height observations, adjusted to be nondivergent, are used to simulate the evolution of passive tracers at the ocean surface. Eddy mixing rates are derived from the tracer fields in two ways. First, the method of Nakamura is applied to a sector in the East Pacific. Second, the Osborn-Cox diffusivity is calculated globally to yield estimates of diffusivity in two dimensions. The results from the East Pacific show weak meridional mixing at the surface in the Southern Ocean ($<1000 \text{ m}^2 \text{ s}^{-1}$, consistent with previous results) but higher mixing rates ($\sim 3000\text{--}5000 \text{ m}^2 \text{ s}^{-1}$) in the tropical ocean. The Osborn-Cox diagnostic provides a global picture of mixing rates and agrees reasonably well with the results from the East Pacific. It also shows extremely high mixing rates ($\sim 10^4 \text{ m}^2 \text{ s}^{-1}$) in western boundary current regions. The Osborn-Cox diffusivity is sensitive to the tracer initialization, which we attribute to the presence of anisotropic mixing processes. The mixing rates are strongly influenced by the presence of a mean flow nearly everywhere, as shown by comparison with an eddy-only calculation, with the mean flow absent. Finally, results are compared with other recent estimates of mixing rates using Lagrangian and inverse methods.

1847. Abyssal connections of Antarctic Bottom Water in a Southern Ocean State Estimate

Authors: Erik van Sebille (University of New South Wales); Paul Spence (University of New South Wales); Matthew R. Mazloff (Scripps Institution of Oceanography); Matthew H. England (University of New South Wales); Stephen R. Rintoul (Wealth From Oceans National Research Flagship, CSIRO Hobart Tas Australia, Antarctic Climate and Ecosystems Cooperative Research Centre, Centre for Australian Weather and Climate Research Hobart Tas Australia, University of Tasmania); Oleg A. Saenko (Canadian Centre for Climate Modelling and Analysis, University of New South Wales)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/grl.50483

URL: <https://doi.org/10.1002/grl.50483>

Publication Date: 2013-04-19

Volume/Issue: Volume 40, Issue 10

References: 23

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: Antarctic Bottom Water|Abyssal zone|Circumpolar deep water|Circumpolar star|Antarctic Intermediate Water|Abyssal plain|Oceanic basin|Temperature salinity diagrams

Funding: Australian Research Council

License: N/A

Open Access: N/A

Abstract: Antarctic Bottom Water (AABW) is formed in a few locations around the Antarctic continent, each source with distinct temperature and salinity. After formation, the different AABW varieties cross the Southern Ocean and flow into the subtropical abyssal basins. It is shown here, using the analysis of Lagrangian trajectories within the Southern Ocean State Estimate (SOSE) model, that the pathways of the different sources of AABW have to a large extent amalgamated into one pathway by the time it reaches 31 degreesS in the deep subtropical basins. The Antarctic Circumpolar Current appears to play an important role in the amalgamation, as 70% of the AABW completes at least one circumpolar loop before reaching the subtropical basins. This amalgamation of AABW pathways suggests that on decadal to centennial time scales, changes to properties and formation rates in any of the AABW source regions will be conveyed to all three subtropical abyssal basins.

1848. Uncertainty estimates of a GRACE inversion modelling technique over Greenland using a simulation

Authors: Jennifer Bonin; Don Chambers

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggt091

URL: <https://doi.org/10.1093/gji/ggt091>

Publication Date: 2013-04-17

Volume/Issue: Volume 194, Issue 1

References: 28

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1849. Towards a dynamically balanced eddy-resolving ocean reanalysis: BRAN3

Authors: Peter R. Oke; Pavel Sakov; Madeleine L. Cahill; Jeff R. Dunn; Russell Fiedler; David A. Griffin; Jim V. Mansbridge; Ken R. Ridgway; Andreas Schiller

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2013.03.008

URL: <https://doi.org/10.1016/j.ocemod.2013.03.008>

Publication Date: 2013-04-16

Volume/Issue: Volume 67

References: 78

Cited by: 120 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1850. A MITgcm/DART ensemble analysis and prediction system with application to the Gulf of Mexico

Authors: Ibrahim Hoteit; Tim Hoar; Ganesh Gopalakrishnan; Nancy Collins; Jeffrey Anderson; Bruce Cornuelle; Armin Köhl; Patrick Heimbach

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2013.03.002

URL: <https://doi.org/10.1016/j.dynatmoce.2013.03.002>

Publication Date: 2013-04-06

Volume/Issue: Volume 63

References: 55

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1851. Latitudinal shifts in the distribution of exploited fishes in Korean waters during the last 30 years: a consequence of climate change

Authors: Sukgeun Jung; Ig-Chan Pang; Joon-ho Lee; Iisu Choi; Hyung Kee Cha

Journal: Reviews in Fish Biology and Fisheries

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11160-013-9310-1

URL: <https://doi.org/10.1007/s11160-013-9310-1>

Publication Date: 2013-04-04

Volume/Issue: Volume 24, Issue 2

References: 94

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1852. Global ocean storage of anthropogenic carbon

Authors: S. Khatiwala; T. Tanhua; S. Mikaloff Fletcher; M. Gerber; S. C. Doney; H. D. Graven; N. Gruber; G. A. McKinley; A. Murata; A. F. Ríos; C. L. Sabine

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-10-2169-2013

URL: <https://doi.org/10.5194/bg-10-2169-2013>

Publication Date: 2013-04-02

Volume/Issue: Volume 10, Issue 4

References: N/A

Cited by: 374 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission

License: Creative Commons

Open Access: N/A

Abstract: . The global ocean is a significant sink for anthropogenic carbon (Cant), absorbing roughly a third of human CO₂ emitted over the industrial period. Robust estimates of the magnitude and variability of the storage and distribution of Cant in the ocean are therefore important for understanding the human impact on climate. In this synthesis we review observational and model-based estimates of the storage and transport of Cant in the ocean. We pay particular attention to the uncertainties and potential biases inherent in different inference schemes. On a global scale, three data-based estimates of the distribution and inventory of Cant are now available. While the inventories are found to agree within their uncertainty, there are considerable differences in the spatial distribution. We also present a review of the progress made in the application of inverse and data assimilation techniques which combine ocean interior estimates of Cant with numerical ocean circulation models. Such methods are especially useful for estimating the air-sea flux and interior transport of Cant, quantities that are otherwise difficult to observe directly. However, the results are found to be highly dependent on modeled circulation, with the spread due to different ocean models at least as large as that from the different observational methods used to estimate Cant. Our review also highlights the importance of repeat measurements of hydrographic and biogeochemical parameters to estimate the storage of Cant on decadal timescales in the presence of the variability in circulation that is neglected by other approaches. Data-based Cant estimates provide important constraints on forward ocean models, which exhibit both broad similarities and regional errors relative to the observational fields. A compilation of inventories of Cant gives us a "best" estimate of the global ocean inventory of anthropogenic carbon in 2010 of 155 ± 31 PgC (±20% uncertainty). This estimate includes a broad range of values, suggesting that a combination of approaches is necessary in order to achieve a robust quantification of the ocean sink of anthropogenic CO₂.

1853. Uncertainty quantification in ocean state estimation

Authors: Alexander G. Kalmikov

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/5818

URL: <https://doi.org/10.1575/1912/5818>

Publication Date: 2013-03-21

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1854. Two Decades of the Atlantic Meridional Overturning Circulation: Anatomy, Variations, Extremes, Prediction, and Overcoming Its Limitations

Authors: Carl Wunsch (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00478.1

URL: <https://doi.org/10.1175/jcli-d-12-00478.1>

Publication Date: 2013-03-20

Volume/Issue: Volume 26, Issue 18

References: 56

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The zonally integrated meridional volume transport in the North Atlantic [Atlantic meridional overturning circulation (AMOC)] is described in a 19-yr-long ocean-state estimate, one consistent with a diverse global dataset. Apart from a weak increasing trend at high northern latitudes, the AMOC appears statistically stable over the last 19 yr with fluctuations indistinguishable from those of a stationary Gaussian stochastic process. This characterization makes it possible to study (using highly developed tools) extreme values, predictability, and the statistical significance of apparent trends. Gaussian behavior is consistent with the central limit theorem for a process arising from numerous independent disturbances. In this case, generators include internal instabilities, changes in wind and buoyancy forcing fields, boundary waves, the Gulf Stream and deep western boundary current transports, the interior fraction in Sverdrup balance, and all similar phenomena arriving as summation effects from long distances and times. As a zonal integral through the sum of the large variety of physical processes in the three-dimensional ocean circulation, understanding of the AMOC, if it is of central climate importance, requires breaking it down into its unintegrated components over the entire basin.

1855. Structure and Evolution of the Cold Dome off Northeastern Taiwan: A Numerical Study

Authors: Ganesh Gopalakrishnan; Bruce Cornuelle; Glen Gawarkiewicz; Julie McClean

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2013.06

URL: <https://doi.org/10.5670/oceanog.2013.06>

Publication Date: 2013-03-13

Volume/Issue: Volume 26, Issue 1

References: N/A

Cited by: 17 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1856. Energy pathways and structures of oceanic eddies from the ECCO2 state estimate and simplified models

Authors: Ru Chen

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/5793

URL: <https://doi.org/10.1575/1912/5793>

Publication Date: 2013-03-01

Volume/Issue: N/A

References: N/A

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Eddy|Mean flow|Energy Flow|Energy exchange

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1857. Multidecadal Ocean Temperature and Salinity Variability in the Tropical North Atlantic: Linking with the AMO, AMOC, and Subtropical Cell

Authors: Chunzai Wang (NOAA/Atlantic Oceanographic and Meteorological Laboratory, Miami, Florida); Liping Zhang (University of Miami)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00721.1

URL: <https://doi.org/10.1175/jcli-d-12-00721.1>

Publication Date: 2013-02-22

Volume/Issue: Volume 26, Issue 16

References: 49

Cited by: 58 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Atlantic multidecadal oscillation (AMO) is characterized by the sea surface warming (cooling) of the entire North Atlantic during its warm (cold) phase. Both observations and most of the phase 5 of the Coupled Model Intercomparison Project (CMIP5) models also show that the warm (cold) phase of the AMO is associated with a surface warming (cooling) and a subsurface cooling (warming) in the tropical North Atlantic (TNA). It is further shown that the warm phase of the AMO corresponds to a strengthening of the Atlantic meridional overturning circulation (AMOC) and a weakening of the Atlantic subtropical cell (STC), which both induce an anomalous northward current in the TNA subsurface ocean. Because the mean meridional temperature gradient of the subsurface ocean is positive because of the temperature dome around 9 degreesN, the advection by the anomalous northward current cools the TNA subsurface ocean during the warm phase of the AMO. The opposite is true during the cold phase of the AMO. It is concluded that the anticorrelated ocean temperature variation in the TNA associated with the AMO is caused by the meridional current variation induced by variability of the AMOC and STC, but the AMOC plays a more important role than the STC. Observations do not seem to show an obvious anticorrelated salinity relation between the TNA surface and subsurface oceans, but most of CMIP5 models simulate an out-of-phase salinity variation. Similar to the temperature variation, the mechanism is the salinity advection by the meridional current variation induced by the AMOC and STC associated with the AMO.

1858. Role of wind stress in causing maximum transport through the Korea Strait in autumn

Authors: Yang-Ki Cho; Gwang-Ho Seo; Chang-Sin Kim; Byoung-Ju Choi; Dinesh Chandra Shaha

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2013.02.002

URL: <https://doi.org/10.1016/j.jmarsys.2013.02.002>

Publication Date: 2013-02-15

Volume/Issue: Volume 115-116

References: 30

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1859. Objective Determination of Feature Resolution in Two Sea Surface Temperature Analyses

Authors: Richard W. Reynolds (North Carolina State University); Dudley B. Chelton (Oregon State University); Jonah Roberts-Jones (Met Office); Matthew J. Martin (Met Office); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christopher John Merchant (University of Edinburgh)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00787.1

URL: <https://doi.org/10.1175/jcli-d-12-00787.1>

Publication Date: 2013-02-13

Volume/Issue: Volume 26, Issue 8

References: 19

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Considerable effort is presently being devoted to producing high-resolution sea surface temperature (SST) analyses with a goal of spatial grid resolutions as low as 1 km. Because grid resolution is not the same as feature resolution, a method is needed to objectively determine the resolution capability and accuracy of SST analysis products. Ocean model SST fields are used in this study as simulated "true" SST data and subsampled based on actual infrared and microwave satellite data coverage. The subsampled data are used to simulate sampling errors due to missing data. Two different SST analyses are considered and run using both the full and the subsampled model SST fields, with and without additional noise. The results are compared as a function of spatial scales of variability using wavenumber auto- and cross-spectral analysis. The spectral variance at high wavenumbers (smallest wavelengths) is shown to be attenuated relative to the true SST because of smoothing that is inherent to both analysis procedures. Comparisons of the two analyses (both having grid sizes of roughly) show important differences. One analysis tends to reproduce small-scale features more accurately when the high-resolution data coverage is good but produces more spurious small-scale noise when the high-resolution data coverage is poor. Analysis procedures can thus generate small-scale features with and without data, but the small-scale features in an SST analysis may be just noise when high-resolution data are sparse. Users must therefore be skeptical of high-resolution SST products, especially in regions where high-resolution (~5 km) infrared satellite data are limited because of cloud cover.

1860. Origins of heat and freshwater anomalies underlying regional decadal sea level trends

Authors: Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/grl.50164

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Publication Date: 2013-02-13

Volume/Issue: Volume 40, Issue 3

References: 24

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Regional sea level changes often differ from global mean changes due to geographic variations in surface fluxes and to changes in ocean circulation. Here we study such regional sea level trends from 1993 to 2004 using a synthesis of observations and an ocean general circulation model. Unlike the global mean, steric changes dominate regional trends with negligible contributions from column-integrated mass variations. Regional heat and freshwater anomalies underlying steric changes are in turn distinguished between redistribution of pre-existing anomalies within the ocean and contributions from additional surface fluxes external to the ocean. Internal redistribution accounts for most regional trends but exceptions are found, most notably in the western tropical Pacific Ocean where a warming of external origin dominates the trend. On average, external thermosteric sea level trends are found to be positive in temperate regions while negative at higher latitudes with opposite trends found in halosteric anomalies of external origin.

1861. The anatomy of recent large sea level fluctuations in the Mediterranean Sea

Authors: Felix W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Denis L. Volkov (UCLA/Joint Institute for Regional Earth System Science and Engineering California USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/grl.50140

URL: <https://doi.org/10.1002/grl.50140>

Publication Date: 2013-02-13

Volume/Issue: Volume 40, Issue 3

References: 35

Cited by: 65 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: During the boreal winter months of 2009/2010 and 2010/2011, Mediterranean mean sea level rose 10 cm above the average monthly climatological values. The non-seasonal anomalies were observed in sea surface height (from altimetry), as well as ocean mass (from gravimetry), indicating they were mostly of barotropic nature. These relatively rapid basin-wide fluctuations occurred over time scales of 1-5 months. Here we use observations and re-analysis data to attribute the non-seasonal sea level and ocean mass fluctuations in the Mediterranean Sea to concurrent wind stress anomalies over the adjacent subtropical Northeast Atlantic Ocean, just west of the Strait of Gibraltar, and extending into the strait itself. The observed Mediterranean sea level fluctuations are strongly anti-correlated with the monthly North-Atlantic-Oscillation (NAO) index.

1862. Effect of Langmuir circulation on upper ocean mixing in the South China Sea

Authors: Shuang Li; Jinbao Song; Wei Fan

Journal: Acta Oceanologica Sinica

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s13131-013-0285-5

URL: <https://doi.org/10.1007/s13131-013-0285-5>

Publication Date: 2013-02-08

Volume/Issue: Volume 32, Issue 3

References: 25

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1863. Calibration procedures and first dataset of Southern Ocean chlorophyll a profiles collected by elephant seals equipped with a newly developed CTD-fluorescence tags

Authors: C. Guinet; X. Xing; E. Walker; P. Monestiez; S. Marchand; B. Picard; T. Jaud; M. Authier; C. Cotté; A. C. Dragon; E. Diamond; D. Antoine; P. Lovell; S. Blain; F. D'Ortenzio; H. Claustre

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-5-15-2013

URL: <https://doi.org/10.5194/essd-5-15-2013>

Publication Date: 2013-02-04

Volume/Issue: Volume 5, Issue 1

References: 59

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . In situ observation of the marine environment has traditionally relied on ship-based platforms. The obvious consequence is that physical and biogeochemical properties have been dramatically undersampled, especially in the remote Southern Ocean (SO). The difficulty in obtaining in situ data represents the major limitations to our understanding, and interpretation of the coupling between physical forcing and the biogeochemical response. Southern elephant seals (*Mirounga leonina*) equipped with a new generation of oceanographic sensors can measure ocean structure in regions and seasons rarely observed with traditional oceanographic platforms. Over the last few years, seals have allowed for a considerable increase in temperature and salinity profiles from the SO, but we were still lacking information on the spatiotemporal variation of phytoplankton concentration. This information is critical to assess how the biological productivity of the SO, with direct consequences on the amount of CO₂ "fixed" by the biological pump, will respond to global warming. In this research programme, we use an innovative sampling fluorescence approach to quantify phytoplankton concentration at sea. For the first time, a low energy consumption fluorometer was added to Argos CTD-SRDL tags, and these novel instruments were deployed on 27 southern elephant seals between 25 December 2007 and the 4 February 2011. As many as 3388 fluorescence profiles associated with temperature and salinity measurements were thereby collected from a vast sector of the Southern Indian Ocean. This paper addresses the calibration issue of the fluorometer before being deployed on elephant seals and presents the first results obtained for the Indian sector of the Southern Ocean. This in situ system is implemented in synergy with satellite ocean colour radiometry. Satellite-derived data is limited to the surface layer and is restricted over the SO by extensive cloud cover. However, with the addition of these new tags, we are able to assess the 3-dimension distribution of phytoplankton concentration by foraging southern elephant seals. This approach reveals that for the Indian sector of the SO, the surface chlorophyll a (chl a) concentrations provided by MODIS were underestimated by a

factor 2 compared to chl a concentrations estimated from HPLC corrected in situ fluorescence measurements. The scientific outcomes of this programme include an improved understanding of both the present state and variability in ocean biology, and the accompanying biogeochemistry, as well as the delivery of real-time and open-access data to scientists (doi:10.7491/MEMO.1).

1864. Role of the Seasonal Cycle in the Subduction Rates of Upper-Southern Ocean Waters

Authors: Eun Young Kwon (University of California Los Angeles); Stephanie M. Downes (Australian National University); Jorge L. Sarmiento (Princeton University); Riccardo Farneti (Earth System Physics Section, ICTP, Trieste, Italy); Curtis Deutsch (University of California Los Angeles)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-060.1

URL: <https://doi.org/10.1175/jpo-d-12-060.1>

Publication Date: 2013-02-01

Volume/Issue: Volume 43, Issue 6

References: 41

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A kinematic approach is used to diagnose the subduction rates of upper-Southern Ocean waters across seasonally migrating density outcrops at the base of the mixed layer. From an Eulerian viewpoint, the term representing the temporal change in the mixed layer depth (which is labeled as the temporal induction in this study; i.e., $\text{Stemp} = \partial h / \partial t$ where h is the mixed layer thickness, and t is time) vanishes over several annual cycles. Following seasonally migrating density outcrops, however, the temporal induction is attributed partly to the temporal change in the mixed layer thickness averaged over a density outcrop following its seasonally varying position and partly to the lateral movement of the outcrop position intersecting the sloping mixed layer base. Neither the temporal induction following an outcrop nor its integral over the outcrop area vanishes over several annual cycles. Instead, the seasonal eddy subduction, which arises primarily because of the subannual correlations between the seasonal cycles of the mixed layer depth and the outcrop area, explains the key mechanism by which mode waters are transferred from the mixed layer to the underlying pycnocline. The time-mean exchange rate of waters across the base of the mixed layer is substantially different from the exchange rate of waters across the fixed winter mixed layer base in mode water density classes. Nearly 40% of the newly formed Southern Ocean mode waters appear to be diapycnally transformed within the seasonal pycnocline before either being subducted into the main pycnocline or entrained back to the mixed layer through lighter density classes.

1865. Predictability of the mid-latitude Atlantic meridional overturning circulation in a multi-model system

Authors: Holger Pohlmann; Doug M. Smith; Magdalena A. Balmaseda; Noel S. Keenlyside; Simona Masina; Daniela Matei; Wolfgang A. Müller; Philippe Rogel

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-013-1663-6

URL: <https://doi.org/10.1007/s00382-013-1663-6>

Publication Date: 2013-01-31

Volume/Issue: Volume 41, Issue 3-4

References: 71

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1866. Seasonal probability forecasts of december-january-february precipitation in northern Uruguay and Rio Grande do Sul obtained with the coupled forecast system v2 of NOAA and statistical downscaling

Authors: Gabriel Cazes Boezio (Universidad de la República, Uruguay); Stefanie Talento (Universidad de la República, Uruguay); Gabriel Jorge Pisciotano Jalabert (Universidad de la República, Uruguay)

Journal: Revista Brasileira de Meteorologia

Publisher: FapUNIFESP (SciELO)

DOI: 10.1590/s0102-77862012000400001

URL: <https://doi.org/10.1590/s0102-77862012000400001>

Publication Date: 2013-01-30

Volume/Issue: Volume 27, Issue 4

References: 34

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The present work analyses the inter-seasonal predictability of precipitation during the austral summer in a subregion of Southeastern South America that includes Rio Grande do Sul and Northern Uruguay (RGS-NU), and proposes a methodology to produce probabilistic precipitation forecasts for this region, based on the use of NOAA CFS v2. It is found that the correlation between ENSO and the precipitation over RGS-NU during December-January-February is statistically significant after the late 70's, but not before. Considering that this relationship changes in different multidecadal periods, it is useful to explore a forecast system based on numerical models. We studied the hindcasts from NOAA CFS v2 initialized during October of the years 1983 to 2009, and found that the hindcasts of meridional wind at 850 hPa averaged over certain region of South America have statistically significant skill, in terms of correlation, to predict the observed precipitation over RGS-NU. The proposed forecasts are based on this relationship. The CFS v2 hindcasts also show realistic anomalous circulation patterns associated with the anomalous precipitation in the region and the season considered. This suggests that the forecasts may be further improved by using regional models combined with the CFS v2 outputs.

1867. The production of temperature and salinity variance and covariance : implications for mixing

Authors: Julian J. Schanze

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/5741

URL: <https://doi.org/10.1575/1912/5741>

Publication Date: 2013-01-25

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1868. Jet Jumping: Low-Frequency Variability in the Southern Ocean

Authors: Christopher C. Chapman (Australian National University); Andrew McC. Hogg (Australian National University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-0123.1

URL: <https://doi.org/10.1175/jpo-d-12-0123.1>

Publication Date: 2013-01-24

Volume/Issue: Volume 43, Issue 5

References: 22

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The authors study intrinsic variability in the position of jets in a beta-plane channel ocean with simple topography using a quasigeostrophic numerical model. This study links the variability in jet position with abyssal anticyclones that form as a result of interaction of mesoscale eddies and subsurface topography, reminiscent of such flows as the Zapiola anticyclone. A simple dynamical framework explaining this behavior is developed. In this framework, this study shows that the topographic anticyclones form closed regions of homogenized yet time-varying potential vorticity. Neighboring topographic anticyclones are coupled by eddy fluxes. Interaction of a baroclinic jet with these two (or more) anticyclones can drive variability in local jet strength. Predictions of the dynamical framework are then compared with the results of the numerical model, and it is demonstrated that this model has merit in explaining the observed model variability. This study argues that this simple mode of variability has relevance for the ocean.

1869. The Atlantic Meridional Heat Transport at 26.5 degreesN and Its Relationship with the MOC in the RAPID Array and the GFDL and NCAR Coupled Models

Authors: Rym Msadek (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); William E. Johns (Rosenstiel School of Marine and Atmospheric Science, Miami, Florida); Stephen G. Yeager (National Center for Atmospheric Research); Gokhan Danabasoglu (National Center for Atmospheric Research); Thomas L. Delworth (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); Anthony Rosati (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00081.1

URL: <https://doi.org/10.1175/jcli-d-12-00081.1>

Publication Date: 2013-01-22

Volume/Issue: Volume 26, Issue 12

References: 39

Cited by: 58 publications

Language: en

Subjects: N/A

Keywords: Mass fluxes/transport, North Atlantic Ocean, Meridional overturning circulation, Climate models, Ocean dynamics

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The link at 26.5 degreesN between the Atlantic meridional heat transport (MHT) and the Atlantic meridional overturning circulation (MOC) is investigated in two climate models, the GFDL Climate Model version 2.1 (CM2.1) and the NCAR Community Climate System Model version 4 (CCSM4), and compared with the recent observational estimates from the Rapid Climate Change-Meridional Overturning Circulation and Heatflux Array (RAPID-MOCHA) array. Despite a stronger-than-observed MOC magnitude, both models underestimate the mean MHT at 26.5 degreesN because of an overly diffuse thermocline. Biases result from errors in both overturning and gyre components of the MHT. The observed linear relationship between MHT and MOC at 26.5 degreesN is realistically simulated by the two models and is mainly due to the overturning component of the MHT. Fluctuations in overturning MHT are dominated by Ekman transport variability in CM2.1 and CCSM4, whereas baroclinic geostrophic transport variability plays a larger role in RAPID. CCSM4, which has a parameterization of Nordic Sea overflows and thus a more realistic North Atlantic Deep Water (NADW) penetration, shows smaller biases in the overturning heat transport than CM2.1 owing to deeper NADW at colder temperatures. The horizontal gyre heat transport and its sensitivity to the MOC are poorly represented in both models. The wind-driven gyre heat transport is northward in observations at 26.5 degreesN, whereas it is weakly southward in both models, reducing the total MHT. This study emphasizes model biases that are responsible for the too-weak MHT, particularly at the western boundary. The use of direct MHT observations through RAPID allows for identification of the source of

the too-weak MHT in the two models, a bias shared by a number of Coupled Model Intercomparison Project phase 5 (CMIP5) coupled models.

1870. Profiling tropospheric CO₂ using Aura TES and TCCON instruments

Authors: L. Kuai; J. Worden; S. Kulawik; K. Bowman; M. Lee; S. C. Biraud; J. B. Abshire; S. C. Wofsy; V. Natraj; C. Frankenberg; D. Wunch; B. Connor; C. Miller; C. Roehl; R.-L. Shia; Y. Yung

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

DOI: 10.5194/amt-6-63-2013

URL: <https://doi.org/10.5194/amt-6-63-2013>

Publication Date: 2013-01-10

Volume/Issue: Volume 6, Issue 1

References: 63

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Monitoring the global distribution and long-term variations of CO₂ sources and sinks is required for characterizing the global carbon budget. Total column measurements are useful for estimating regional-scale fluxes; however, model transport remains a significant error source, particularly for quantifying local sources and sinks. To improve the capability of estimating regional fluxes, we estimate lower tropospheric CO₂ concentrations from ground-based near-infrared (NIR) measurements with space-based thermal infrared (TIR) measurements. The NIR measurements are obtained from the Total Carbon Column Observing Network (TCCON) of solar measurements, which provide an estimate of the total CO₂ column amount. Estimates of tropospheric CO₂ that are co-located with TCCON are obtained by assimilating Tropospheric Emission Spectrometer (TES) free tropospheric CO₂ estimates into the GEOS-Chem model. We find that quantifying lower tropospheric CO₂ by subtracting free tropospheric CO₂ estimates from total column estimates is a linear problem, because the calculated random uncertainties in total column and lower tropospheric estimates are consistent with actual uncertainties as compared to aircraft data. For the total column estimates, the random uncertainty is about 0.55 ppm with a bias of -5.66 ppm, consistent with previously published results. After accounting for the total column bias, the bias in the lower tropospheric CO₂ estimates is 0.26 ppm with a precision (one standard deviation) of 1.02 ppm. This precision is sufficient for capturing the winter to summer variability of approximately 12 ppm in the lower troposphere; double the variability of the total column. This work shows that a combination of NIR and TIR measurements can profile CO₂ with the precision and accuracy needed to quantify lower tropospheric CO₂ variability.

1871. Evaluation of the ECMWF ocean reanalysis system ORAS4

Authors: Magdalena Alonso Balmaseda; Kristian Mogensen; Anthony T. Weaver

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2063

URL: <https://doi.org/10.1002/qj.2063>

Publication Date: 2013-01-01

Volume/Issue: Volume 139, Issue 674

References: 67

Cited by: 844 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A new operational ocean reanalysis system (ORAS4) has been implemented at ECMWF. It spans the period 1958 to the present. This article describes its main components and evaluates its quality. The adequacy of ORAS4 for the initialization of seasonal forecasts is discussed, along with the robustness of some prominent climate signals. ORAS4 has been evaluated using different metrics, including comparison with observed ocean currents, RAPID-derived transports, sea-level gauges, and GRACE-derived bottom pressure. Compared to a control ocean model simulation, ORAS4 improves the fit to observations, the interannual variability, and seasonal forecast skill. Some problems have been identified, such as the underestimation of meridional overturning at 26 degreesN, the magnitude of which is shown to be sensitive to the treatment of the coastal observations. ORAS4 shows a clear and robust shallowing trend of the Pacific Equatorial thermocline. It also shows a clear and robust nonlinear trend in the 0-700 m ocean heat content, consistent with other observational estimates. Some aspects of these climate signals are sensitive to the choice of sea-surface temperature product and the specification of the observation-error variances. The global sea-level trend is consistent with the altimeter estimate, but the partition into volume and mass variations is more debatable, as inferred by discrepancies in the trend between ORAS4- and GRACE-derived bottom pressure.

1872. Ensemble spread-based assessment of observation impact: application to a global ocean analysis system

Authors: Andrea Storto; Simona Masina; Srdjan Dobricic

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.2071

URL: <https://doi.org/10.1002/qj.2071>

Publication Date: 2012-12-27

Volume/Issue: Volume 139, Issue 676

References: 68

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This article explores an ensemble strategy for evaluating the impact of different observing networks. The impact is represented by the relative ensemble spread increase, in model space, of data-denial ensemble simulations with respect to an 'all-observation' ensemble experiment, evaluated independently for each observing network. The forecast-error covariance intercomparison reduces to the ensemble spread intercomparison; thus, the method can be applied to any assimilation system and requires only the proper construction of an ensemble system, although the impact assessment results depend on the specific configuration of the investigated analysis system. Our approach allows us to determine the impact of the observing networks in model space (unlike Observing System Experiments) and for different forecast ranges of the ocean general circulation model. No tangent-linear and adjoint coding of the ocean model are required. The method is applied for demonstration to a large-scale global ocean variational analysis system. The ensemble members are generated by (i) perturbing the observations within the 3D-Var assimilation scheme, (ii) perturbing the surface forcing, and (iii) stochastically perturbing the ocean model parametrisation tendencies. The impact is calculated for CTDs, XBTs, moorings, Argo, sea-level anomaly observations and sea-surface temperature measurements from space-borne microwave instruments within the three-year period from January 2003 to December 2005. It turns out, on the global scale, that altimetry exhibits the largest impact on near-surface temperature and sea-surface height. In contrast, deep-ocean impacts are led by the Argo float network. As expected, space-borne observations (sea-level anomaly and sea-surface temperature observations) increase their impact in the Southern Ocean, due to the lack of a robust network of in situ observations. The results of the impact on the salinity indicate the great importance of Argo floats, especially in the northern Extratropics.

1873. A multi-dimensional spectral description of ocean variability with applications

Authors: Cimarron J. L. Wortham

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/5662

URL: <https://doi.org/10.1575/1912/5662>

Publication Date: 2012-12-21

Volume/Issue: N/A

References: N/A

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Wavenumber|Empirical orthogonal functions|Stochastic modelling|Electromagnetic spectrum|Sea-surface height

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1874. Global Surface Geostrophic Ocean Currents Derived from Satellite Altimetry and GOCE Geoid

Authors: J. M. Sánchez-Reales (University of Alicante); M. I. Vigo (University of Alicante); S. Jin (b Chinese Academy of Science , Shanghai , China); B. F. Chao (c Academic Sinica , Taipei , Taiwan)

Journal: Marine Geodesy

Publisher: Informa UK Limited

DOI: 10.1080/01490419.2012.718696

URL: <https://doi.org/10.1080/01490419.2012.718696>

Publication Date: 2012-12-18

Volume/Issue: Volume 35, Issue sup1

References: 35

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1875. On the Variability of Wind Power Input to the Oceans with a Focus on the Subpolar North Atlantic

Authors: Xiaoming Zhai (University of East Anglia); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00472.1

URL: <https://doi.org/10.1175/jcli-d-12-00472.1>

Publication Date: 2012-12-17

Volume/Issue: Volume 26, Issue 11

References: 41

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Variations in power input to the ocean using a recent global "reanalysis" extending back to 1871 show a strong trend in the net power input since then, a trend dominated by the Southern Ocean region. This trend is interpreted as a spurious result of the changing observational system. Focusing therefore on the North Atlantic Ocean, where the database is somewhat more secure, it is found that the input power in the subpolar North Atlantic varies significantly in time, showing a strong relationship to the North Atlantic Oscillation (NAO). During positive NAO index years, power input is greater owing to enhanced synoptic activity. Furthermore, cumulative power input to the subpolar North Atlantic is found to correlate significantly with both the eddy kinetic energy there and the Atlantic multidecadal oscillation (AMO), although the physical mechanism at work remains unclear. The assumption that the changing ocean can be neglected relative to the changing atmosphere in calculating the power input is found to be a usefully accurate approximation over the two decades for which changing ocean state estimates are available. Strong dependence on synoptic weather systems of monthly-mean stress distributions implies that past and future climate simulations must account properly for changes in weather systems, not just the large-scale variations.

1876. Atlantic meridional heat transports in two ocean reanalyses evaluated against the RAPID array

Authors: Keith Haines (University of Reading); Vladimir N. Stepanov (University of Reading); Maria Valdivieso (University of Reading); Hao Zuo (European Centre for Medium-Range Weather Forecasts (ECMWF))

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl054581

URL: <https://doi.org/10.1029/2012gl054581>

Publication Date: 2012-12-14

Volume/Issue: Volume 40, Issue 2

References: 34

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Atlantic meridional overturning circulation in two versions of the NEMO degrees global ocean model has been compared with the RAPID transport array at 26 degreesN. Both model versions reproduce the mean MOC strength well although the Florida Straits flows differ because the pathway of the Gulf Stream is not strongly constrained at this resolution. Both models however have a mean meridional heat transport of 1.07PW, much lower than the 1.35PW from RAPID observations in April 2004-October 2007. Much of the heat transport discrepancy is due to lower transports in summer across the MidOcean (Bahamas-Africa) section, due to stronger southward geostrophic flows in the top 100 m where the water is warmest. Seasonal thermocline changes increase temperature differences across the basin driving stronger geostrophic shear, but this effect is much weaker in the top 100 m of the RAPID velocity data. The effect accounts for a reduction of 1.1Sv in MOC and 0.1PW in heat transports. The rest of the discrepancy comes from lower Ekman transports from using ERAInterim winds instead of QuikSCAT, a smaller zonally varying "Eddy" heat transport component, estimated from repeat XBT sections in the observations, and the southward throughflow in the model. Other differences in depth structure of the model MOC and RAPID observations are described but have much less impact on heat transports.

1877. Modeled phytoplankton diversity and productivity in the California Current System

Authors: N.L. Goebel; C.A. Edwards; J.P. Zehr; M.J. Follows; S.G. Morgan

Journal: Ecological Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ecolmodel.2012.11.008

URL: <https://doi.org/10.1016/j.ecolmodel.2012.11.008>

Publication Date: 2012-12-11

Volume/Issue: Volume 264

References: 60

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1878. A Review of the Quality Control of Global Ocean Temperature and Salinity Data

Authors: You-Soon Chang

Journal: Journal of the Korean earth science society

Publisher: The Korean Earth Science Society

DOI: 10.5467/jkess.2012.33.6.554

URL: <https://doi.org/10.5467/jkess.2012.33.6.554>

Publication Date: 2012-12-06

Volume/Issue: Volume 33, Issue 6

References: 60

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1879. Coupled Sea Ice-Ocean-State Estimation in the Labrador Sea and Baffin Bay

Authors: Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-065.1

URL: <https://doi.org/10.1175/jpo-d-12-065.1>

Publication Date: 2012-11-26

Volume/Issue: Volume 43, Issue 5

References: 90

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: Coupled models, Sea ice, Interannual variability, Data assimilation, North Atlantic Ocean (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sea ice variability in the Labrador Sea is of climatic interest because of its relationship to deep convection, mode-water formation, and the North Atlantic atmospheric circulation. Historically, quantifying the relationship between sea ice and ocean variability has been difficult because of in situ observation paucity and technical challenges associated with synthesizing observations with numerical models. Here the relationship between ice and ocean variability is explored by analyzing new estimates of the ocean-ice state in the northwest North Atlantic. The estimates are syntheses of in situ and satellite hydrographic and ice data with a regional 1/3 degrees coupled ocean-sea ice model. The synthesis of sea ice data is achieved with an improved adjoint of a thermodynamic ice model. Model and data are made consistent, in a least squares sense, by iteratively adjusting control variables, including ocean initial and lateral boundary conditions and the atmospheric state, to minimize an uncertainty-weighted model-data misfit cost function. The utility of the state estimate is demonstrated in an analysis of energy and buoyancy budgets in the marginal ice zone (MIZ). In mid-March the system achieves a state of quasi-equilibrium during which net ice growth and melt approaches zero; newly formed ice diverges from coastal areas and converges via wind and ocean forcing in the MIZ. The convergence of ice mass in the MIZ is ablated primarily by turbulent ocean-ice enthalpy fluxes. The primary source of the enthalpy required for sustained MIZ ice ablation is the sensible heat reservoir of the subtropical-origin subsurface waters.

1880. Hydrographic Preconditioning for Seasonal Sea Ice Anomalies in the Labrador Sea

Authors: Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-064.1

URL: <https://doi.org/10.1175/jpo-d-12-064.1>

Publication Date: 2012-11-26

Volume/Issue: Volume 43, Issue 5

References: 54

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study investigates the hydrographic processes involved in setting the maximum wintertime sea ice (SI) extent in the Labrador Sea and Baffin Bay. The analysis is based on an ocean and sea ice state estimate covering the summer-to-summer 1996/97 annual cycle. The estimate is a synthesis of in situ and satellite hydrographic and ice data with a regional coupled 1/3 degrees ocean-sea ice model. SI advective processes are first demonstrated to be required to reproduce the observed ice extent. With advection, the marginal ice zone (MIZ) location stabilizes where ice melt balances ice mass convergence, a quasi-equilibrium condition achieved via the convergence of warm subtropical-origin subsurface waters into the mixed layer seaward of the MIZ. An analysis of ocean surface buoyancy fluxes reveals a critical role of low-salinity upper ocean (100 m) anomalies for the advancement of SI seaward of the Arctic Water-Irminger Water Thermohaline Front. Anomalous low-salinity waters slow the rate of buoyancy loss-driven mixed layer deepening, shielding an advancing SI pack from the warm subsurface waters, and are conducive to a positive surface meltwater stabilization enhancement (MESEM) feedback driven by SI meltwater release. The low-salinity upper-ocean hydrographic conditions in which the MESEM efficiently operates are termed sea ice-preconditioned waters (SIPW). The SI extent seaward of the Thermohaline Front is shown to closely correspond to the distribution of SIPW. The analysis of two additional state estimates (1992/93, 2003/04) suggests that interannual hydrographic variability provides a first-order explanation for SI maximum extent anomalies in the region.

1881. On the processes controlling the seasonal cycles of the air-sea fluxes of O₂ and N₂O: A modelling study

Authors: Manfredi Manizza; Ralph F. Keeling; Cynthia D. Nevison

Journal: Tellus B: Chemical and Physical Meteorology

Publisher: Stockholm University Press

DOI: 10.3402/tellusb.v64i0.18429

URL: <https://doi.org/10.3402/tellusb.v64i0.18429>

Publication Date: 2012-11-20

Volume/Issue: Volume 64, Issue 1

References: 53

Cited by: 31 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1882. On the drivers of phytoplankton blooms in the Antarctic marginal ice zone: A modeling approach

Authors: Marc H. Taylor (Alfred Wegener Institute); Martin Losch (Alfred Wegener Institute); Astrid Bracher (University of Bremen, Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc008418

URL: <https://doi.org/10.1029/2012jc008418>

Publication Date: 2012-11-15

Volume/Issue: Volume 118, Issue 1

References: 85

Cited by: 70 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The pelagic province of the Southern Ocean generally has low levels of primary production attributable to a short growing season in the higher latitudes, a deep mixed layer, and iron limitation. Exceptions include phytoplankton blooms in the marginal ice zone (MIZ) during spring and summer sea-ice retreat. The prevailing hypothesis as to the drivers of the blooms is that sea-ice retreat increases the vertical stability of the water column through the production of melt water and provides shelter from wind mixing in areas of partial sea-ice coverage. These conditions are favorable to phytoplankton growth by allowing them to maintain their position in the upper reaches of the water column. This work investigates the drivers of MIZ blooms using a biochemically coupled global circulation model. Results support the hypothesis in that physical conditions related to a shallow, vertically stable water column (e.g., mixed layer depth and available light) were the most significant predictors of bloom dynamics, while nutrient limitation was of lesser importance. We estimate that MIZ blooms account for 15% of yearly net primary production in the Southern Ocean and that the earlier phases of the MIZ bloom, occurring under partial ice coverage and invisible to remote sensing, account for about two thirds of this production. MIZ blooms were not found to enhance depth-integrated net primary production when compared to similar ecological provinces outside of the MIZ, although the elevated phytoplankton concentrations in surface waters are hypothesized to provide important feeding habitats for grazing organisms such as krill.

1883. Atlantic Ocean circulation at the Last Glacial Maximum : inferences from data and models

Authors: Holly J. Dail

Journal: Unknown Journal

Publisher: Massachusetts Institute of Technology and Woods Hole Oceanographic Institution

DOI: 10.1575/1912/5542

URL: <https://doi.org/10.1575/1912/5542>

Publication Date: 2012-11-13

Volume/Issue: N/A

References: N/A

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: Circulation (fluid dynamics)|Last Glacial Maximum

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1884. Dynamical Potential Energy: A New Approach to Ocean Energetics

Authors: Fabien Roquet (Stockholm University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-098.1

URL: <https://doi.org/10.1175/jpo-d-12-098.1>

Publication Date: 2012-11-06

Volume/Issue: Volume 43, Issue 2

References: 36

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The concept of available potential energy is supposed to indicate which part of the potential energy is available to transform into kinetic energy. Yet it is impossible to obtain a unique definition of available potential energy for the real ocean because of nonlinearities of the equation of state, rendering its usefulness largely hypothetical. In this paper, the conservation of energy is first reformulated in terms of horizontal anomalies of density and pressure for a simplified ocean model using the Boussinesq and hydrostatic approximations. This framework introduces the concept of "dynamical potential energy," defined as the horizontal anomaly of potential energy, to replace available potential energy. Modified conservation equations are derived that make it much simpler to identify oceanic power input by buoyancy and mechanical forces. Closed budgets of energy are presented for idealized circulations obtained with a general circulation model, comparing spatial patterns of power inputs generated by wind and thermal forcings. Finally, a generalization of the framework to compressible fluids is presented, opening the way to applications in atmosphere energetics.

1885. A size-structured food-web model for the global ocean

Authors: B. A. Ward; S. Dutkiewicz; O. Jahn; M. J. Follows

Journal: Limnology and Oceanography

Publisher: Wiley

DOI: 10.4319/lo.2012.57.6.1877

URL: <https://doi.org/10.4319/lo.2012.57.6.1877>

Publication Date: 2012-11-06

Volume/Issue: Volume 57, Issue 6

References: N/A

Cited by: 271 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We present a model of diverse phytoplankton and zooplankton populations embedded in a global ocean circulation model. Physiological and ecological traits of the organisms are constrained by relationships with cell size. The model qualitatively reproduces global distributions of nutrients, biomass, and primary productivity, and captures the power-law relationship between cell size and numerical density, which has realistic slopes of between -1.3 and -0.8. We use the model to explore the global structure of marine ecosystems, highlighting the importance of both nutrient and grazer controls. The model suggests that zooplankton : phytoplankton (Z : P) biomass ratios may vary from an order of 0.1 in the oligotrophic gyres to an order of 10 in upwelling and high-latitude regions. Global estimates of the strength of bottom-up and top-down controls within plankton size classes suggest that these large-scale gradients in Z : P ratios are driven by a shift from strong bottom-up, nutrient limitation in the oligotrophic gyres to the dominance of top-down, grazing controls in more productive regions.

1886. Fundamentals of climate change science

Authors: I. Colin Prentice; Peter G. Baines; Marko Scholze; Martin J. Wooster

Journal: Understanding the Earth System

Publisher: Cambridge University Press

DOI: 10.1017/cbo9780511921155.005

URL: <https://doi.org/10.1017/cbo9780511921155.005>

Publication Date: 2012-11-05

Volume/Issue: N/A

References: 169

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1887. Near 13 day barotropic ocean response to the atmospheric forcing in the North Pacific

Authors: Hanna Na; Jae-Hun Park; D. Randolph Watts; Kathleen A. Donohue; Ho Jin Lee

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc008211

URL: <https://doi.org/10.1029/2012jc008211>

Publication Date: 2012-10-31

Volume/Issue: Volume 117, Issue C12

References: 32

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In the Kuroshio Extension System Study (KESS) east of Japan, bottom pressure observations over the 2 year study period exhibit strong high-frequency variability near 13 days. The first cyclostationary empirical orthogonal function mode for the band-pass-filtered KESS bottom pressure explains about 57% of the near 13 day variance and exhibits almost in-phase variability in space with a hint of westward propagation. The 13 day variability is strong during the winter and is driven by the large-scale wind stress curl over a broad region of the North Pacific. Modeling results over the North Pacific closely follow the observations and indicate that topography confines the barotropic response to the west of Emperor Seamount Chain and slows the westward propagation of the near 13 day bottom pressure variability.

1888. Buoyancy-Driven Interannual Sea Level Changes in the Tropical South Atlantic

Authors: Christopher G. Piecuch (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-12-093.1

URL: <https://doi.org/10.1175/jpo-d-12-093.1>

Publication Date: 2012-10-30

Volume/Issue: Volume 43, Issue 3

References: 29

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Linear models of dynamical ocean adjustment to wind field changes, local atmospheric driving, and eastern boundary forcing are often invoked to explain observed patterns of interannual regional sea level variability. While skillful in some regions, these processes alone cannot explain low levels of interannual sea level variability observed in the tropical Atlantic. In this study, through a set of modeling approaches, interannual sea level changes in the tropical South Atlantic are attributed and the dynamical influence of buoyancy forcing is elucidated. Similar to recent findings in the southeast tropical Pacific, sea level patterns in the tropical South Atlantic (as estimated from a data-constrained ocean general circulation model) are found to result from the action of both surface wind and buoyancy forcing; in addition to static local effects, the buoyancy-driven changes comprise important nonlocal ocean dynamical processes. It is shown that the buoyancy-driven sea level changes can be understood within the framework of a linear first baroclinic mode Rossby wave model forced by atmospheric fields and variability along the eastern boundary. To lowest order, the linear model framework also reproduces qualitative patterns of basinwide compensation between wind- and buoyancy-driven sea level changes, which are mostly tied to the anticorrelation of both surface and boundary forcing. Results suggest that the ocean's dynamical adjustment to buoyancy forcing exerts an important influence on interannual sea level changes across all tropical oceans.

1889. Impact of tropical cyclones on the intensity and phase propagation of fall Wyrтки jets

Authors: P. Sreenivas; J. S. Chowdary; C. Gnanaseelan

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl053974

URL: <https://doi.org/10.1029/2012gl053974>

Publication Date: 2012-10-26

Volume/Issue: Volume 39, Issue 22

References: 26

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Observations and model simulations are used to study the impact of tropical cyclones (TC) on the fall Wyrтки jets (WJ). These strong narrow equatorial currents peak during November and play a vital role in the energy and mass transport in the tropical Indian Ocean (TIO). Maximum number of TCs is observed over TIO during November with longer than normal life span (8-15 days). These TCs enhance equatorial westerly winds (surface) and amplify monthly mean WJs (both at surface and subsurface) by 0.4 ms^{-1} (anomalies exceed 0.7 ms^{-1} during TC), which is about half of the climatological amplitude. Intensified WJs increase the heat content of eastern TIO and modulate air-sea interaction. It is also shown that movement of TCs is mainly responsible for the westward phase propagation of WJs, a previously unexplored mechanism. These features are evident in ECCO2 simulations as well.

1890. A reduced adjoint approach to variational data assimilation

Authors: M.U. Altaf; M. El Gharamti; A.W. Heemink; I. Hoteit

Journal: Computer Methods in Applied Mechanics and Engineering

Publisher: Elsevier BV

DOI: 10.1016/j.cma.2012.10.003

URL: <https://doi.org/10.1016/j.cma.2012.10.003>

Publication Date: 2012-10-26

Volume/Issue: Volume 254

References: 33

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1891. Evaluation of Release-05 GRACE time-variable gravity coefficients over the ocean

Authors: D. P. Chambers; J. A. Bonin

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-8-859-2012

URL: <https://doi.org/10.5194/os-8-859-2012>

Publication Date: 2012-10-12

Volume/Issue: Volume 8, Issue 5

References: 35

Cited by: 135 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The latest release of GRACE (Gravity Recovery and Climate Experiment) gravity field coefficients (Release-05, or RL05) are evaluated for ocean applications. Data have been processed using the current methodology for Release-04 (RL04) coefficients, and have been compared to output from two different ocean models. Results indicate that RL05 data from the three Science Data Centers - the Center for Space Research (CSR), GeoForschungsZentrum (GFZ), and Jet Propulsion Laboratory (JPL) - are more consistent among themselves than the previous RL04 data. Moreover, the variance of residuals with the output of an ocean model is 50-60% lower for RL05 data than for RL04 data. A more optimized destriping algorithm is also tested, which improves the results slightly. By comparing the GRACE maps with two different ocean models, we can better estimate the uncertainty in the RL05 maps. We find the standard error to be about 1 cm (equivalent water thickness) in the low- and mid-latitudes, and between 1.5 and 2 cm in the polar and subpolar oceans, which is comparable to estimated uncertainty for the output from the ocean models.

1892. Sensitivity of Global Tropical Climate to Land Surface Processes: Mean State and Interannual Variability

Authors: Hsi-Yen Ma (Lawrence Livermore National Laboratory); Heng Xiao (Atmospheric Sciences and Global Change Division, Pacific North National Laboratory, Richland, Washington); C. Roberto Mechoso (University of California Los Angeles); Yongkang Xue (University of California Los Angeles)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00142.1

URL: <https://doi.org/10.1175/jcli-d-12-00142.1>

Publication Date: 2012-10-08

Volume/Issue: Volume 26, Issue 5

References: 70

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: Land surface model, Seasonal cycle, ENSO, Convection, Atmosphere-ocean interaction (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study examines the sensitivity of the global climate to land surface processes (LSP) using an atmospheric general circulation model both uncoupled (with prescribed SSTs) and coupled to an oceanic general circulation model. The emphasis is on the interactive soil moisture and vegetation biophysical processes, which have first-order influence on the surface energy and water budgets. The sensitivity to those processes is represented by the differences between model simulations, in which two land surface schemes are considered: 1) a simple land scheme that specifies surface albedo and soil moisture availability and 2) the Simplified Simple Biosphere Model (SSiB), which allows for consideration of interactive soil moisture and vegetation biophysical process. Observational datasets are also employed to assess the extent to which results are realistic. The mean state sensitivity to different LSP is stronger in the coupled mode, especially in the tropical Pacific. Furthermore, the seasonal cycle of SSTs in the equatorial Pacific, as well as the ENSO frequency, amplitude, and locking to the seasonal cycle of SSTs, is significantly modified and more realistic with SSiB. This outstanding sensitivity of the atmosphere-ocean system develops through changes in the intensity of equatorial Pacific trades modified by convection over land. The results further demonstrate that the direct impact of land-atmosphere interactions on the tropical climate is modified by feedbacks associated with perturbed oceanic conditions ("indirect effect" of LSP). The magnitude of such an indirect effect is strong enough to suggest that comprehensive studies on the importance of LSP on the global climate have to be made in a system that allows for atmosphere-ocean interactions.

1893. The Effects of Thermohaline Circulation on Wind-Driven Circulation in the South China Sea

Authors: Guihua Wang (Second Institute of Oceanography); Rui Xin Huang (Woods Hole Oceanographic Institution); Jilan Su (Second Institute of Oceanography); Dake Chen (Second Institute of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-11-0227.1

URL: <https://doi.org/10.1175/jpo-d-11-0227.1>

Publication Date: 2012-09-24

Volume/Issue: Volume 42, Issue 12

References: 32

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The dynamic influence of thermohaline circulation on wind-driven circulation in the South China Sea (SCS) is studied using a simple reduced gravity model, in which the upwelling driven by mixing in the abyssal ocean is treated in terms of an upward pumping distributed at the base of the upper layer. Because of the strong upwelling of deep water, the cyclonic gyre in the northern SCS is weakened, but the anticyclonic gyre in the southern SCS is intensified in summer, while cyclonic gyres in both the southern and northern SCS are weakened in winter. For all seasons, the dynamic influence of thermohaline circulation on wind-driven circulation is larger in the northern SCS than in the southern SCS. Analysis suggests that the upwelling associated with the thermohaline circulation in the deep ocean plays a crucial role in regulating the wind-driven circulation in the upper ocean.

1894. The interannual variability of potential temperature in the central Labrador Sea

Authors: Hao Luo; Annalisa Bracco; Igor Yashayaev; Emanuele Di Lorenzo

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc007988

URL: <https://doi.org/10.1029/2012jc007988>

Publication Date: 2012-09-20

Volume/Issue: Volume 117, Issue C10

References: 80

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The interannual variability of potential temperature in the central Labrador Sea is studied with a suite of numerical simulations with an eddy-resolving regional ocean model and compared with available observations. The model successfully reproduces the observed variations in potential temperature at depths comprised between 150 and 2000 m over the period 1980-2009, capturing also the warming trend of the last decade and the deep water formation event in 2008. The suite of experiments allows for quantifying the contribution from the physical forcings responsible for the interannual variability of potential temperature in the region. The local atmospheric forcing drives the interannual signal by driving convection, while the incoming current system along the east coast of Greenland is responsible for about half of the warming trend (~ 0.3 - 0.4 degreesC) during the last decade through restratification process. The lateral transport of Irminger water in the convective region into the central Labrador Sea is further analyzed integrating a passive tracer. It is found that the overall amount of Irminger water transported in the convective region of the Labrador Sea is directly correlated with the amount of vertical convective mixing. In the last decade, following the decrease in convective activity, the model reveals a substantial decrease in concentration of Irminger Current water below 500 m in the Labrador Sea interior: by 2010 the overall amount is less than half than in the previous 20 years.

1895. Combination of gravimetric and altimetric space observations for estimating oceanic polar motion excitations

Authors: F. Göttl; M. Schmidt; R. Heinkelmann; R. Savcenko; J. Bouman

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc007915

URL: <https://doi.org/10.1029/2012jc007915>

Publication Date: 2012-09-20

Volume/Issue: Volume 117, Issue C10

References: 48

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Global dynamic processes cause variations in the Earth's rotation, which are monitored by various geometric observation techniques such as Global Navigation Satellite Systems (GNSS), Satellite Laser Ranging (SLR), and Very Long Baseline Interferometry (VLBI) with millimeter accuracy. The integral effect on Earth rotation of mass displacements and motion is therefore precisely known, but the separation of contributions from particular geodynamic processes remains a challenge. Here we show that the oceanic mass effect on Earth rotation can be derived from both time variable gravity field solutions from the Gravity Recovery And Climate Experiment (GRACE) and sea level anomalies (SLA) observed from satellite altimeter missions. The GRACE solutions require filtering and the application of an ocean mask, whereas the SLA need to be corrected for the steric effect as polar motion is only affected by mass redistributions. We assess the accuracy of our oceanic polar motion excitations by using GRACE and SLA solutions from different processing centers. In addition, we compare polar motion excitations from GRACE, satellite altimeter data and their combinations with excitations estimated from ocean models. We show that the combination of gravimetric and altimetric solutions reduces systematic errors of the individual solutions. The combined solutions are about 2 times more accurate than ocean model results and about 3 times more accurate than the so-called reduced geodetic excitation functions. We anticipate our analysis to be valuable input for improved modeling of oceanic mass redistributions.

1896. Detecting storm surge loading deformations around the southern North Sea using subdaily GPS

Authors: Jianghui Geng; Simon D.P. Williams; Felix N. Teferle; Alan H. Dodson

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2012.05656.x

URL: <https://doi.org/10.1111/j.1365-246x.2012.05656.x>

Publication Date: 2012-09-15

Volume/Issue: Volume 191, Issue 2

References: 40

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: Storm Surge

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1897. Changing controls on oceanic radiocarbon: New insights on shallow-to-deep ocean exchange and anthropogenic CO₂ uptake

Authors: H. D. Graven; N. Gruber; R. Key; S. Khatiwala; X. Giraud

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc008074

URL: <https://doi.org/10.1029/2012jc008074>

Publication Date: 2012-09-10

Volume/Issue: Volume 117, Issue C10

References: 116

Cited by: 95 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The injection of radiocarbon (¹⁴C) into the atmosphere by nuclear weapons testing in the 1950s and 1960s has provided a powerful tracer to investigate ocean physical and chemical processes. While the oceanic uptake of bomb-derived ¹⁴C was primarily controlled by air-sea exchange in the early decades after the bomb spike, we demonstrate that changes in oceanic ¹⁴C are now primarily controlled by shallow-to-deep ocean exchange, i.e., the same mechanism that governs anthropogenic CO₂ uptake. This is a result of accumulated bomb ¹⁴C uptake that has rapidly decreased the air-sea gradient of ¹⁴C/C ($\Delta^{14}\text{C}$) and shifted the main reservoir of bomb ¹⁴C from the atmosphere to the upper ocean. The air-sea $\Delta^{14}\text{C}$ gradient, reduced further by fossil fuel dilution, is now weaker than before weapons testing in most regions. Oceanic ¹⁴C, and particularly its temporal change, can now be used to study the oceanic uptake of anthropogenic CO₂. We examine observed changes in oceanic $\Delta^{14}\text{C}$ between the WOCE/SAVE (1988-1995) and the CLIVAR (2001-2007) eras and simulations with two ocean general circulation models, the Community Climate System Model (CCSM) and the Estimating the Circulation and Climate of the Ocean Model (ECCO). Observed oceanic $\Delta^{14}\text{C}$ and its changes between the 1980s-90s and 2000s indicate that shallow-to-deep exchange is too efficient in ECCO and too sluggish in CCSM. These findings suggest that mean global oceanic uptake of anthropogenic CO₂ between 1990 and 2007 is bounded by the ECCO-based estimate of 2.3 Pg C yr⁻¹ and the CCSM-based estimate of 1.7 Pg C yr⁻¹.

1898. Consistent near-surface ocean warming since 1900 in two largely independent observing networks

Authors: Viktor Gouretski; John Kennedy; Tim Boyer; Armin Köhl

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl052975

URL: <https://doi.org/10.1029/2012gl052975>

Publication Date: 2012-09-06

Volume/Issue: Volume 39, Issue 19

References: 31

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We compare historical global temperature time series, based on bias-adjusted sea-surface temperatures with independent temperature time series, for the upper 20 meter layer of the ocean based on the latest update of an historical hydrographic profile data set. Despite the two underlying data sets being different in number of data points, instrumentation and applied adjustments, both of the time series are consistent in showing an overall warming since 1900. We also extend records of temperature change in the upper 400 m back to 1900. Noting that the geographic coverage is limited prior to 1950, the temperature change in the 0-400 m layer is characterized by two periods of temperature increase between 1900 and 1940-45 and between 1970 and 2003, separated by a period of little change.

1899. A Fully Conserved Minimal Adjustment Scheme with (T, S) Coherency for Stabilization of Hydrographic Profiles

Authors: Xidong Wang (Key Laboratory of Marine Environmental Information Technology, State Oceanic Administration, National Marine Data and Information Service, Tianjin, China); Peter C. Chu (Naval Ocean Analysis and Prediction Laboratory, Department of Oceanography, Naval Postgraduate School, Monterey, California); Guijun Han (Key Laboratory of Marine Environmental Information Technology, State Oceanic Administration, National Marine Data and Information Service, Tianjin, China); Wei Li (Key Laboratory of Marine Environmental Information Technology, State Oceanic Administration, National Marine Data and Information Service, Tianjin, China); Xuefeng Zhang (Key Laboratory of Marine Environmental Information Technology, State Oceanic Administration, National Marine Data and Information Service, Tianjin, China); Dong Li (Key Laboratory of Marine Environmental Information Technology, State Oceanic Administration, National Marine Data and Information Service, Tianjin, China)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-12-00025.1

URL: <https://doi.org/10.1175/jtech-d-12-00025.1>

Publication Date: 2012-09-05

Volume/Issue: Volume 29, Issue 12

References: 20

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A new, fully conserved minimal adjustment scheme with temperature and salinity (T, S) coherency is presented for eliminating false static instability generated from analyzing and assimilating stable ocean (T, S) profiles data, that is, from generalized averaging over purely observed data (data analysis) or over modeled/observed data (data assimilation). This approach consists of a variational method with (a) fully (heat, salt, and potential energy) conserved conditions, (b) minimal adjustment, and (c) (T, S) coherency. Comparison with three existing schemes (minimal adjustment, conserved minimal adjustment, and convective adjustment) using observational profiles and a simple one-dimensional ocean mixed layer model shows the superiority of this new scheme.

1900. Response of Freshwater Flux and Sea Surface Salinity to Variability of the Atlantic Warm Pool

Authors: Chunzai Wang (NOAA/Atlantic Oceanographic and Meteorological Laboratory, Miami, Florida); Liping Zhang (Ocean University of China, University of Miami); Sang-Ki Lee (University of Miami)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00284.1

URL: <https://doi.org/10.1175/jcli-d-12-00284.1>

Publication Date: 2012-09-01

Volume/Issue: Volume 26, Issue 4

References: 34

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The response of freshwater flux and sea surface salinity (SSS) to the Atlantic warm pool (AWP) variations from seasonal to multidecadal time scales is investigated by using various reanalysis products and observations. All of the datasets show a consistent response for all time scales: A large (small) AWP is associated with a local freshwater gain (loss) to the ocean, less (more) moisture transport across Central America, and a local low (high) SSS. The moisture budget analysis demonstrates that the freshwater change is dominated by the atmospheric mean circulation dynamics, while the effect of thermodynamics is of secondary importance. Further decomposition points out that the contribution of the mean circulation dynamics primarily arises from its divergent part, which mainly reflects the wind divergent change in the low level as a result of SST change. In association with a large (small) AWP, warmer (colder) than normal SST over the tropical North Atlantic can induce anomalous low-level convergence (divergence), which favors anomalous ascent (descent) and thus generates more (less) precipitation. On the other hand, a large (small) AWP weakens (strengthens) the trade wind and its associated westward moisture transport to the eastern North Pacific across Central America, which also favors more (less) moisture residing in the Atlantic and hence more (less) precipitation. The results imply that variability of freshwater flux and ocean salinity in the North Atlantic associated with the AWP may have the potential to affect the Atlantic meridional overturning circulation.

1901. Sensitivity of Western Boundary Transport at the Mean North Equatorial Current Bifurcation Latitude to Wind Forcing

Authors: Xuebin Zhang (Centre for Australian Weather and Climate Research, CSIRO Wealth from Oceans National Research Flagship, CSIRO Marine and Atmospheric Research, Hobart, Tasmania, Australia); Bruce Cornuelle (Scripps Institution of Oceanography / University of California San Diego); Dean Roemmich (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-11-0229.1

URL: <https://doi.org/10.1175/jpo-d-11-0229.1>

Publication Date: 2012-08-27

Volume/Issue: Volume 42, Issue 11

References: 45

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: Pacific Ocean, Numerical analysis/modeling, Boundary currents, General circulation models, Rossby waves

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The bifurcation of the North Equatorial Current (NEC) plays an important role in the heat and water mass exchanges between the tropical and subtropical gyres in the Pacific Ocean. The variability of western boundary transport (WBT) east of the Philippine coast at the mean NEC bifurcation latitude (12 degreesN) is examined here. A tropical Pacific regional model is set up based on the Massachusetts Institute of Technology general circulation model and its adjoint, which calculates the sensitivities of a defined meridional transport to atmospheric forcing fields and ocean state going backward in time. The adjoint-derived sensitivity of the WBT at the mean NEC bifurcation latitude to surface wind stress is dominated by curl-like patterns that are located farther eastward and southward with increasing time lag. The temporal evolution of the adjoint sensitivity of the WBT to wind stress resembles wind-forced Rossby wave dynamics but propagating with speeds determined by the background stratification and current, suggesting that wind-forced Rossby waves are the underlying mechanism. Interannual-to-decadal variations of the WBT can be hindcast well by multiplying the adjoint sensitivity and the time-lagged wind stress over the whole model domain and summing over time lags. The analysis agrees with previous findings that surface wind stress (especially zonal wind stress in the western subtropical Pacific) largely determines the WBT east of the Philippines, and with a time lag based on Rossby wave propagation. This adjoint sensitivity study quantifies the contribution of wind stress at all latitudes and longitudes and provides a novel perspective to understand the relationship between the WBT and wind forcing over the Pacific Ocean.

1902. Modeling the basal melting and marine ice accretion of the Amery Ice Shelf

Authors: B. K. Galton-Fenzi; J. R. Hunter; R. Coleman; S. J. Marsland; R. C. Warner

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc008214

URL: <https://doi.org/10.1029/2012jc008214>

Publication Date: 2012-08-23

Volume/Issue: Volume 117, Issue C9

References: 85

Cited by: 120 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The basal mass balance of the Amery Ice Shelf (AIS) in East Antarctica is investigated using a numerical ocean model. The main improvements of this model over previous studies are the inclusion of frazil formation and dynamics, tides and the use of the latest estimate of the sub-ice-shelf cavity geometry. The model produces a net basal melt rate of 45.6 Gt year⁻¹ (0.74 m ice year⁻¹) which is in good agreement with reviewed observations. The melting at the base of the ice shelf is primarily due to interaction with High Salinity Shelf Water created from the surface sea-ice formation in winter. The temperature difference between the coldest waters created in the open ocean and their in situ freezing point of ocean water in contact with the deepest part of the AIS drives a melt rate that can exceed 30 m of ice year⁻¹. The inclusion of frazil dynamics is shown to be important for both melting and marine ice accretion (refreezing). Frazil initially forms in the supercooled water layer adjacent to the base of the ice shelf. The net accretion of marine ice is 5.3 Gt year⁻¹, comprised of 3.7 Gt year⁻¹ of frazil accretion and 1.6 Gt year⁻¹ of direct basal refreezing.

1903. Argon supersaturation indicates low decadal-scale vertical mixing in the ocean thermocline

Authors: Steven Emerson; Taka Ito; Roberta C. Hamme

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl053054

URL: <https://doi.org/10.1029/2012gl053054>

Publication Date: 2012-08-23

Volume/Issue: Volume 39, Issue 18

References: 24

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The rate of vertical mixing in the ocean's stratified waters limits the uptake of anthropogenic CO₂, influences the strength of the overturning circulation, and regulates the transport of nutrients to the lighted surface waters, controlling global biological production. Despite this fundamental importance, there is a long-standing conundrum in oceanography that experimentally-measured rates of turbulent mixing across density surfaces (diapycnal mixing) in the main thermocline cannot support sufficient nutrient fluxes from below to explain rates of biological production measured in the subtropical euphotic zone. Possible solutions to this problem are transport mechanisms that occur intermittently on short time and space scales that would be difficult to observe in tracer- release experiments and are not resolved in large-scale ocean models. We tested this hypothesis by measuring highly-accurate argon profiles from the subtropical thermocline in the North Pacific Ocean. It has been shown theoretically that the change in argon supersaturation along density surfaces is a measure of diapycnal mixing averaged over the decadal time-scale of thermocline ventilation. Two different model interpretations of our data indicate that the mean rate of diapycnal mixing on density surfaces between $\sigma_\theta = 26.4 - 26.7$ (depths 150-600 m) is no more than $0.2 \times 10^{-4} \text{ m}^2 \text{ s}^{-1}$. This supports low diapycnal mixing rates even on decadal time-scales and rules out enhancement of diapycnal mixing on this density interval by intermittent mixing or mixing at boundaries that propagates into the ocean interior.

1904. The deep meridional overturning circulation in the Indian Ocean inferred from the GECCO synthesis

Authors: Weiqiang Wang; Armin Köhl; Detlef Stammer

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2012.08.001

URL: <https://doi.org/10.1016/j.dynatmoce.2012.08.001>

Publication Date: 2012-08-22

Volume/Issue: Volume 58

References: 52

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: BMBF, Deutsche Forschungsgemeinschaft

License: N/A

Open Access: N/A

Abstract: N/A

1905. Observations of Freshening in the Northwest Pacific Subtropical Gyre near Luzon Strait

Authors: Youfang Yan (Chinese Academy of Sciences); Dazhi Xu (Chinese Academy of Sciences); Yiquan Qi (Chinese Academy of Sciences); Zijun Gan (Chinese Academy of Sciences)

Journal: Atmosphere-Ocean

Publisher: Informa UK Limited

DOI: 10.1080/07055900.2012.715078

URL: <https://doi.org/10.1080/07055900.2012.715078>

Publication Date: 2012-08-20

Volume/Issue: Volume 50, Issue sup1

References: 48

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1906. Mechanisms of decadal sea level variability in the eastern North Atlantic and the Mediterranean Sea

Authors: F. M. Calafat; D. P. Chambers; M. N. Tsimplis

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc008285

URL: <https://doi.org/10.1029/2012jc008285>

Publication Date: 2012-08-13

Volume/Issue: Volume 117, Issue C9

References: 54

Cited by: 101 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Decadal sea level variations from tide gauge records along the western European coast and in the Mediterranean Sea commencing in the late 19th and early 20th centuries are examined relative to large-scale atmospheric forcing. Recent studies have provided evidence for a link between sea level in the eastern North Atlantic and atmospheric forcing, however the nature of this relationship is still unclear. Here the outputs of a regional barotropic model and a nearly global baroclinic model are used in conjunction with wind stress and heat flux data to explore the physical mechanisms responsible for the observed sea level variability. All tide gauge records show significant decadal variability (up to 15 cm) and are highly correlated with the NAO and among themselves at decadal periods. There is a coherent sea level signal that affects the eastern boundary of the North Atlantic northward of 25 degreesN and is limited to a narrow band of the order of a few hundred kilometers along the coast. This band tends to become narrower towards higher latitudes. We find that longshore wind and wave propagation along the boundary are the major contributors to coastal sea level variability but no significant contribution from mass redistribution linked to changes in the strength of the subtropical gyre is observed. The mass component dominates sea level in the Mediterranean and is mainly driven by mass exchanges with the Atlantic, which explains the correlation between both regions. Southward of 25 degreesN, sea level changes are mainly driven by heat advection through Ekman fluxes.

1907. Tracer transport timescales and the observed Atlantic-Pacific lag in the timing of the Last Termination

Authors: Geoffrey Gebbie

Journal: Paleoceanography

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011pa002273

URL: <https://doi.org/10.1029/2011pa002273>

Publication Date: 2012-08-02

Volume/Issue: Volume 27, Issue 3

References: 62

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The midpoint of the Last Termination occurred 4,000 years earlier in the deep Atlantic than the deep Pacific according to a pair of benthic foraminiferal $\delta^{18}\text{O}$ records, seemingly implying an internal circulation shift because the lag is much longer than the deep radiocarbon age. Here a scenario where the lag is instead caused by regional surface boundary condition changes, delays due to oceanic transit timescales, and the interplay between temperature and seawater $\delta^{18}\text{O}$ ($\delta^{18}\text{O}_w$) is quantified with a tracer transport model of the modern-day ocean circulation. Using an inverse method with individual Green functions for 2,806 surface sources, a time history of surface temperature and $\delta^{18}\text{O}_w$ is reconstructed for the last 30,000 years that is consistent with the foraminiferal oxygen-isotope data, Mg/Ca-derived deep temperature, and glacial pore water records. Thus, in the case that the ocean circulation was relatively unchanged between glacial and modern times, the interbasin lag could be explained by the relatively late local glacial maximum around Antarctica where surface $\delta^{18}\text{O}_w$ continues to rise even after the North Atlantic $\delta^{18}\text{O}_w$ falls. The arrival of the signal of the Termination is delayed at the Pacific core site due to the destructive interference of the still-rising Antarctic signal and the falling North Atlantic signal. This scenario is only possible because the ocean is not a single conveyor belt where all waters at the Pacific core site previously passed the Atlantic core site, but instead the Pacific core site is bathed more prominently by waters with a direct Antarctic source.

1908. Determination of the Earth's pole tide Love number k_2 from observations of polar motion using an adaptive Kalman filter approach

Authors: F. Seitz; S. Kirschner; D. Neubersch

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jb009296

URL: <https://doi.org/10.1029/2012jb009296>

Publication Date: 2012-08-01

Volume/Issue: Volume 117, Issue B9

References: 44

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The geophysical interpretation of observed time series of Earth rotation parameters (ERP) is commonly based on numerical models that describe and balance variations of angular momentum in various subsystems of the Earth. Naturally, models are dependent on geometrical, rheological and physical parameters. Many of these are weakly determined from other models or observations. In our study we present an adaptive Kalman filter approach for the improvement of parameters of the dynamic Earth system model DyMEG which acts as a simulator of ERP. In particular we focus on the improvement of the pole tide Love number k_2 . In the frame of a sensitivity analysis k_2 has been identified as one of the most crucial parameters of DyMEG since it directly influences the modeled Chandler oscillation. At the same time k_2 is one of the most uncertain parameters in the model. Our simulations with DyMEG cover a period of 60 years after which a steady state of k_2 is reached. The estimate for k_2 , accounting for the anelastic response of the Earth's mantle and the ocean, is $0.3531 + 0.0030i$. We demonstrate that the application of the improved parameter k_2 in DyMEG leads to significantly better results for polar motion than the original value taken from the Conventions of the International Earth Rotation and Reference Systems Service (IERS).

1909. Weighing the ocean: Using a single mooring to measure changes in the mass of the ocean

Authors: Chris W. Hughes; Mark E. Tamisiea; Rory J. Bingham; Joanne Williams

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl052935

URL: <https://doi.org/10.1029/2012gl052935>

Publication Date: 2012-07-30

Volume/Issue: Volume 39, Issue 17

References: 34

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Combining ocean and earth models, we show that there is a region in the central Pacific ocean where ocean bottom pressure is a direct measure of interannual changes in ocean mass, with a noise level for annual means below 3 mm water equivalent, and a trend error below 1 mm/yr. We demonstrate this concept using existing ocean bottom pressure measurements from the region, from which we extract the annual cycle of ocean mass (amplitude 8.5 mm, peaking in late September), which is in agreement with previous determinations based on complex combinations of global data sets. This method sidesteps a number of limitations in satellite gravity-based calculations, but its direct implementation is currently limited by the precision of pressure sensors, which suffer from significant drift. Development of a low-drift method to measure ocean bottom pressure at a few sites could provide an important geodetic constraint on the earth system.

1910. The impact of the Indonesian Throughflow and tidal mixing on the summertime sea surface temperature in the western Indonesian Seas

Authors: Shinichiro Kida; Susan Wijffels

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc008162

URL: <https://doi.org/10.1029/2012jc008162>

Publication Date: 2012-07-27

Volume/Issue: Volume 117, Issue C9

References: 44

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: KAKENHI

License: N/A

Open Access: N/A

Abstract: A numerical model is used to investigate how the Indonesian Throughflow and tidal mixing are affecting the seasonal cycle of the sea surface temperature (SST) in the Indonesian Seas. The SST in these seas is considered to play a major role on the development of the Australian Summer Monsoon. Based on a quantitative assessment of the heat budget, the Indonesian Throughflow is found to affect the SST in the western Indonesian Seas primarily during Austral summer. The Throughflow advects the warm water from the Pacific and maintains the warm SST when the Northwestern Monsoonal wind induces coastal upwelling along the northern side of the Nusa Tenggara and cools the SST. Such balance is supported by observations. The hydrographic sections show the isotherms tilting upward toward the northern coast of the Nusa Tenggara when satellite observations show slight decrease of the SST in the region. Tidal mixing is found to cool the SST during summer the most. This is because the Northwest Monsoonal wind induces coastal upwelling near where strong tidal mixing above seamount occurs and brings the tidally well-mixed upper thermocline water to the surface. The surface Ekman flow also spreads this cool water around the Banda Sea where tidal mixing does not occur. The impact of tidal mixing on the SST is also found to come largely from that occurring above seamounts. The impact of tidal mixing on the continental shelves is limited to shelf-breaks because cold subsurface water is necessary for enhanced vertical mixing to cool the SST.

1911. Covariances and linear predictability of the Atlantic Ocean

Authors: Carl Wunsch

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2012.07.015

URL: <https://doi.org/10.1016/j.dsr2.2012.07.015>

Publication Date: 2012-07-25

Volume/Issue: Volume 85

References: 52

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1912. Seasonal thermal structure and exchange in Baker Channel, Chile

Authors: Christopher M. Aiken

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2012.07.001

URL: <https://doi.org/10.1016/j.dynatmoce.2012.07.001>

Publication Date: 2012-07-21

Volume/Issue: Volume 58

References: 36

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1913. The impact of animal platforms on polar ocean observation

Authors: M.A. Fedak

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2012.07.007

URL: <https://doi.org/10.1016/j.dsr2.2012.07.007>

Publication Date: 2012-07-16

Volume/Issue: Volume 88-89

References: 39

Cited by: 58 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1914. Potential Predictability of the North Atlantic Heat Transport Based on an Oceanic State Estimate

Authors: Bente Tiedje (University of Hamburg); Armin Köhl (University of Hamburg); Johanna Baehr (University of Hamburg)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-11-00606.1

URL: <https://doi.org/10.1175/jcli-d-11-00606.1>

Publication Date: 2012-07-09

Volume/Issue: Volume 25, Issue 24

References: 38

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Data assimilation, Ocean circulation, Meridional overturning circulation, Ensembles, Hindcasts (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper investigates the potential predictability of the meridional heat transport (MHT) in the North Atlantic on interannual time scales using hindcast ensembles based on an oceanic data assimilation product. The work analyzes the prognostic potential predictability (PPP), using the ocean synthesis of the German partner of the consortium for Estimating the Circulation and Climate of the Ocean (GECCO) as initial conditions and as boundary conditions. The PPP of the MHT varies with latitude: local maxima are apparent within the subpolar and the subtropical gyres, and a minimum is apparent at the boundary between the gyres. This PPP minimum can also be seen in the PPP structure of the Atlantic meridional overturning circulation (AMOC), although it is considerably less pronounced. The decomposition of the MHT shows that within the subpolar gyre, the gyre component of the MHT influences the PPP structure of the MHT. Within the subtropical gyre, the overturning component of the MHT characterizes the PPP structure of the MHT. At the boundary between the subpolar and the subtropical gyres, the dynamics of the Ekman heat transport limit the predictable lead times of the MHT. At most latitudes, variations in the velocity field control the PPP structure of the MHT. The PPP structure of the AMOC can also be classified into gyre and gyre-boundary regimes, but the predictable lead times within the gyres are only similar to those of the overturning component of the MHT. Overall, the analysis provides a reference point for the latitude dependence of the MHT's PPP structure and relates it to the latitude dependence of the AMOC's PPP structure.

1915. Incremental four-dimensional variational data assimilation of positive-definite oceanic variables using a logarithm transformation

Authors: Hajoon Song; Christopher A. Edwards; Andrew M. Moore; Jerome Fiechter

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2012.06.001

URL: <https://doi.org/10.1016/j.ocemod.2012.06.001>

Publication Date: 2012-07-07

Volume/Issue: Volume 54-55

References: 60

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1916. A Comparison of Atmospheric Reanalysis Surface Products over the Ocean and Implications for Uncertainties in Air-Sea Boundary Forcing

Authors: Ayan H. Chaudhuri (Atmospheric and Environmental Research, Lexington, Massachusetts); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts); Gael Forget (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00090.1

URL: <https://doi.org/10.1175/jcli-d-12-00090.1>

Publication Date: 2012-06-30

Volume/Issue: Volume 26, Issue 1

References: 71

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper investigates the uncertainties related to atmospheric fields from reanalysis products used in forcing ocean models. Four reanalysis products, namely from 1) the interim ECMWF Re-Analysis (ERA-Interim), 2) version 2 of the Common Reference Ocean-Ice Experiments (CORE2), 3) the 25-Year Japanese Reanalysis Project (JRA-25), and 4) NCEP-NCAR, are evaluated against satellite-derived observations for eight different fields (zonal and meridional winds, precipitation, specific humidity, continental discharge, surface air temperature, and downwelling longwave and shortwave radiation fluxes). No single product is found to agree better in all fields with satellite-derived observations. Reanalysis products are mostly comparable to each other because of their similar physical assumptions and assimilation of common observations. Adjusted atmospheric fields from the Estimating the Circulation and Climate of the Ocean (ECCO) optimizations are also in agreement with other reanalysis products. Time-mean and time-variable errors are estimated separately and mapped globally in space, based on 14-day average fields to focus on monthly to interannual periods. Time-variable errors are larger in comparison to the signal than time-mean errors for most fields, thus justifying the need to separate them for studying uncertainties as well as formulating optimization procedures. Precipitation and wind stress fields show significant time-mean and time-variable errors whereas downwelling radiation, air temperature, and humidity fields show small time-mean errors but large time-variable errors, particularly in the tropics. Uncertainties based on evaluating multiple products presented here are considerably larger than uncertainties based on single product pairs.

1917. Baroclinic Motions and Energetics as Measured by Altimeters

Authors: Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech-d-12-00035.1

URL: <https://doi.org/10.1175/jtech-d-12-00035.1>

Publication Date: 2012-06-30

Volume/Issue: Volume 30, Issue 1

References: 32

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Eddies, Baroclinic models, Tides, Satellite observations (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Small surface displacements appearing in tide gauge and altimetric records are used to detect hydrostatic baroclinic modes in the ocean. Those deflections are a small fraction of the interior isopycnal vertical displacements and are dependent directly upon the in situ stratification. Conversion of surface height to interior amplitudes and energies encounters significant spatial and seasonal shifts that need to be accounted for in quantitative use. This technical article analyzes the global-scale spatial variations in the relationship between surface deflections and interior motions. Similar considerations make it possible to use altimetric data to estimate the deep interior temperature variability as a function of position, calculations having a strong influence on abyssal trend determination in the presence of eddies.

1918. Two Tales of Initializing Decadal Climate Prediction Experiments with the ECHAM5/MPI-OM Model

Authors: Daniela Matei (Max Planck Institute for Meteorology); Holger Pohlmann (Max Planck Institute for Meteorology); Johann Jungclaus (Max Planck Institute for Meteorology); Wolfgang Müller (Max Planck Institute for Meteorology); Helmuth Haak (Max Planck Institute for Meteorology); Jochem Marotzke (Max Planck Institute for Meteorology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-11-00633.1

URL: <https://doi.org/10.1175/jcli-d-11-00633.1>

Publication Date: 2012-06-30

Volume/Issue: Volume 25, Issue 24

References: 51

Cited by: 125 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper investigates the impact of different ocean initialization strategies on the forecast skill of decadal prediction experiments performed with the ECHAM5/Max Planck Institute Ocean Model (MPI-OM) coupled model. The ocean initializations assimilate three-dimensional temperature and salinity anomalies from two different ocean state estimates, the ocean reanalysis of the German contribution to Estimating the Circulation and Climate of the Ocean (GECCO) and an ensemble of MPI-OM ocean experiments forced with the NCEP-NCAR atmospheric reanalysis. The results show that North Atlantic and Mediterranean sea surface temperature (SST) variations can be skillfully predicted up to a decade ahead and with greater skill than by both uninitialized simulations and persistence forecasts. The regional distribution of SST predictive skill is similar in both initialization approaches; however, higher skill is found for the NCEP hindcasts than for the GECCO hindcasts when a combination of predictive skill measures is used. Skillful predictions of surface air temperature are obtained over northwestern Europe, northern Africa, and central-eastern Asia. The North Atlantic subpolar gyre region stands out as the region with the highest predictive skill beyond the warming trend, in both SST and upper-ocean heat-content predictions. Here the NCEP hindcasts deliver the best results due to a more accurate initialization of the observed variability. The dominant mechanism for North Atlantic climate predictability is of dynamical origin and can be attributed to the initialization of the Atlantic meridional overturning circulation, thus explaining the reoccurrence of high predictive skill within the second pentad of the hindcasts experiments. The results herein demonstrate that ocean experiments forced with the observed history of the atmospheric state constitute a simple but successful alternative strategy for the initialization of skillful climate predictions over the next decade.

1919. The role of ocean cooling in setting glacial southern source bottom water salinity

Authors: M. D. Miller; J. F. Adkins; D. Menemenlis; M. P. Schodlok

Journal: Paleoceanography

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012pa002297

URL: <https://doi.org/10.1029/2012pa002297>

Publication Date: 2012-06-27

Volume/Issue: Volume 27, Issue 3

References: 64

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

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License: N/A

Open Access: N/A

Abstract: At the Last Glacial Maximum (LGM), the salinity contrast between northern source deep water and southern source bottom water was reversed with respect to the contrast today. Additionally, Glacial Southern Source Bottom Water (GSSBW) was saltier than Antarctic Bottom Water (AABW), over and above the difference implied by the mean sea level change. This study examines to what extent cold temperatures, through their effect on ice formation and melting, could have caused these differences. Computational sensitivity experiments using a coupled ice shelf cavity-sea ice-ocean model are performed in a Weddell Sea domain, as a representative case study for bottom water formation originating from Antarctic continental shelves. Ocean temperatures at the domain open boundaries are systematically lowered to determine the sensitivity of Weddell Sea water mass properties to a range of cool ocean temperatures. The steady state salinities differ between experiments due to temperature-induced responses of ice shelf and sea ice melting and freezing, evaporation and open boundary fluxes. The results of the experiments indicate that reduced ocean temperature can explain up to 30% of the salinity difference between GSSBW and AABW, primarily due to decreased ice shelf melting. The smallest and most exposed ice shelves, which abut narrow continental shelves, have the greatest sensitivity to the ocean temperature changes, suggesting that at the LGM there could have been a shift in geographical site dominance in bottom water formation. More sea ice is formed and exported in the cold ocean experiments, but the effect of this on salinity is negated by an equal magnitude reduction in evaporation.

1920. Arctic Ocean freshwater: How robust are model simulations?

Authors: A. Jahn; Y. Aksenov; B. A. de Cuevas; L. de Steur; S. Häkkinen; E. Hansen; C. Herbaut; M.-N. Houssais; M. Karcher; F. Kauker; C. Lique; A. Nguyen; P. Pemberton; D. Worthen; J. Zhang

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc007907

URL: <https://doi.org/10.1029/2012jc007907>

Publication Date: 2012-06-27

Volume/Issue: Volume 117, Issue C8

References: 110

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Arctic freshwater (FW) has been the focus of many modeling studies, due to the potential impact of Arctic FW on the deep water formation in the North Atlantic. A comparison of the hindcasts from ten ocean-sea ice models shows that the simulation of the Arctic FW budget is quite different in the investigated models. While they agree on the general sink and source terms of the Arctic FW budget, the long-term means as well as the variability of the FW export vary among models. The best model-to-model agreement is found for the interannual and seasonal variability of the solid FW export and the solid FW storage, which also agree well with observations. For the interannual and seasonal variability of the liquid FW export, the agreement among models is better for the Canadian Arctic Archipelago (CAA) than for Fram Strait. The reason for this is that models are more consistent in simulating volume flux anomalies than salinity anomalies and volume-flux anomalies dominate the liquid FW export variability in the CAA but not in Fram Strait. The seasonal cycle of the liquid FW export generally shows a better agreement among models than the interannual variability, and compared to observations the models capture the seasonality of the liquid FW export rather well. In order to improve future simulations of the Arctic FW budget, the simulation of the salinity field needs to be improved, so that model results on the variability of the liquid FW export and storage become more robust.

1921. Spreading of warm ocean waters around Greenland as a possible cause for glacier acceleration

Authors: E. Rignot; I. Fenty; D. Menemenlis; Y. Xu

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2012aog60a136

URL: <https://doi.org/10.3189/2012aog60a136>

Publication Date: 2012-06-27

Volume/Issue: Volume 53, Issue 60

References: 50

Cited by: 88 publications

Language: en

Subjects: N/A

Keywords: Fjord|Sill|Greenland ice sheet

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We examine the pattern of spreading of warm subtropical-origin waters around Greenland for the years 1992-2009 using a high-resolution (4km horizontal grid) coupled ocean and sea-ice simulation. The simulation, provided by the Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2) project, qualitatively reproduces the observed warming of subsurface waters in the subpolar gyre associated with changes of the North Atlantic atmospheric state that occurred in the mid-1990s. The modeled subsurface ocean temperature warmed by 1.5 °C in southeast and southwest Greenland during 1994-2005 and subsequently cooled by 0.5 °C; modeled subsurface ocean temperature increased by 2-2.5 °C in central and then northwest Greenland during 1997-2005 and stabilized thereafter, while it increased after 2005 by <0.5 °C in north Greenland. Comparisons with in situ measurements off the continental shelf in the Labrador and Irminger Seas indicate that the model initial conditions were 0.4 °C too warm in the south but the simulated warming is correctly reproduced; while measurements from eastern Baffin Bay reveal that the model initial conditions were 1.0 °C too cold in the northwest but the simulated ocean warming brought modeled temperature closer to observations, i.e. the simulated warming is 1.0 °C too large. At several key locations, the modeled oceanic changes off the shelf and below the seasonal mixed layer were rapidly transmitted to the shelf within troughs towards (model-unresolved) fjords. Unless blocked in the fjords by shallow sills, these warm subsurface waters had potential to propagate down the fjords and melt the glacier fronts. Based on model sensitivity simulations from an independent study (Xu and others, 2012), we show that the oceanic changes have very likely increased the subaqueous melt rates of the glacier fronts, and in turn impacted the rates of glacier flow.

1922. Response of the Benguela upwelling systems to spatial variations in the wind stress

Authors: Wolfgang Fennel; Tim Junker; Martin Schmidt; Volker Mohrholz

Journal: Continental Shelf Research

Publisher: Elsevier BV

DOI: 10.1016/j.csr.2012.06.004

URL: <https://doi.org/10.1016/j.csr.2012.06.004>

Publication Date: 2012-06-19

Volume/Issue: Volume 45

References: 17

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1923. Inverse methods and assimilation techniques

Authors: No authors listed

Journal: Modeling Methods for Marine Science

Publisher: Cambridge University Press

DOI: 10.1017/cbo9780511975721.019

URL: <https://doi.org/10.1017/cbo9780511975721.019>

Publication Date: 2012-06-18

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Inverse method

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1924. Modeling Methods for Marine Science

Authors: David M. Glover; William J. Jenkins; Scott C. Doney

Journal: Unknown Journal

Publisher: Cambridge University Press

DOI: 10.1017/cbo9780511975721

URL: <https://doi.org/10.1017/cbo9780511975721>

Publication Date: 2012-06-18

Volume/Issue: N/A

References: N/A

Cited by: 116 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This advanced textbook on modeling, data analysis and numerical techniques for marine science has been developed from a course taught by the authors for many years at the Woods Hole Oceanographic Institute. The first part covers statistics: singular value decomposition, error propagation, least squares regression, principal component analysis, time series analysis and objective interpolation. The second part deals with modeling techniques: finite differences, stability analysis and optimization. The third part describes case studies of actual ocean models of ever increasing dimensionality and complexity, starting with zero-dimensional models and finishing with three-dimensional general circulation models. Throughout the book hands-on computational examples are introduced using the MATLAB programming language and the principles of scientific visualization are emphasised. Ideal as a textbook for advanced students of oceanography on courses in data analysis and numerical modeling, the book is also an invaluable resource for a broad range of scientists undertaking modeling in chemical, biological, geological and physical oceanography.

1925. Ocean Heat Transport and Its Projected Change in CanESM2

Authors: Duo Yang (Canadian Centre for Climate Modelling and Analysis); Oleg A. Saenko (Canadian Centre for Climate Modelling and Analysis)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-11-00715.1

URL: <https://doi.org/10.1175/jcli-d-11-00715.1>

Publication Date: 2012-06-14

Volume/Issue: Volume 25, Issue 23

References: 64

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: Ocean models, Coupled models, Seasonal cycle, Model evaluation/performance, Ocean circulation (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The meridional ocean heat transport (MOHT), its seasonal variability, and projected changes simulated by the second generation Canadian Earth System Model (CanESM2) are presented. The global mean MOHT is within the uncertainty of the observational estimates. However, a correct simulation of the MOHT for individual ocean basins is more challenging, and the Atlantic MOHT south of 30 degreesN is underestimated. The partitioning of the MOHT into the overturning and gyre components is generally consistent with such partitioning in an observationally optimized ocean model. At low latitudes, the time-mean MOHT is dominated by its overturning component, whereas in the Southern Ocean and, especially, in the subpolar North Atlantic, it is the gyre component that plays a more important role. In the projected warmer climates, CanESM2 simulates a weakening of the poleward MOHT essentially in both hemispheres. The projected MOHT changes are largely determined by the overturning component, except in the subpolar Atlantic where it is dominated by the gyre component. Consistent with (the limited number of) previous studies, the seasonal variability of the MOHT is large and is mostly driven by the seasonal variability of the meridional Ekman transport. In the simulated warmer climates, the seasonal cycle of the MOHT is projected to change, mostly in the tropics and also in the Southern Hemisphere midlatitudes. The eddy contribution to the MOHT is broadly consistent with that in the observationally optimized eddy-permitting model. However, in the tropics a significant fraction of the eddy energy is converted back to the mean circulation, and the heat transports due to the parameterized and permitted eddies differ.

1926. Multidecadal sea level anomalies and trends in the western tropical Pacific

Authors: Mark A. Merrifield; Philip R. Thompson; Mark Lander

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl052032

URL: <https://doi.org/10.1029/2012gl052032>

Publication Date: 2012-06-12

Volume/Issue: Volume 39, Issue 13

References: 26

Cited by: 144 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NOAA, Office of Climate Observations

License: N/A

Open Access: N/A

Abstract: Tide gauge data are used to relate low frequency sea level changes over the past 60 years in the western tropical Pacific, including a significant positive trend over the past two decades, to Pacific climate indices. Five-year averages of western tropical Pacific sea level are well explained by global sea-level rise and a combination of slowly varying trade wind fluctuations captured by dominant climate indices for the tropical Pacific. Implications for multidecadal sea level variations in the region are considered using reconstructions based on the climate indices.

1927. A western boundary upwelling system response to recent climate variation (1960-2006)

Authors: Jian Su; Mingquan Xu; Thomas Pohlmann; Dongfeng Xu; Daoru Wang

Journal: Continental Shelf Research

Publisher: Elsevier BV

DOI: 10.1016/j.csr.2012.05.010

URL: <https://doi.org/10.1016/j.csr.2012.05.010>

Publication Date: 2012-05-26

Volume/Issue: Volume 57

References: 37

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1928. Assimilation of RAPID array observations into an ocean model

Authors: Vladimir N. Stepanov; Keith Haines; Gregory C. Smith

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.1945

URL: <https://doi.org/10.1002/qj.1945>

Publication Date: 2012-05-11

Volume/Issue: Volume 138, Issue 669

References: 51

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Experiments assimilating the RAPID dataset of deep temperature and salinity profiles at 26.5 degreesN on the western and eastern Atlantic boundaries into a 1 degrees global NEMO ocean model have been performed. The meridional overturning circulation (MOC) is then assessed against the transports calculated directly from observations. The best initialization found for this short period was obtained by assimilating the EN3 upper-ocean hydrography database prior to 2004, after which different methods of assimilating 5-day average RAPID profiles at the western boundary were tested. The model MOC is strengthened by ~ 2 Sv giving closer agreement with the RAPID array transports, when the western boundary profiles are assimilated only below 900 m (the approximate depth of the Florida Straits, which are not well resolved) and when the T,S observations are spread meridionally from 10 to 35 degreesN along the deep western boundary. The use of boundary-focused covariances has the largest impact on the assimilation results, otherwise using more conventional Gaussian covariances has a very local impact on the MOC at 26 degreesN with strong adverse impacts on the MOC stream function at higher and lower latitudes. Even using boundary-focused covariances only enables the MOC to be strengthened for ~ 2 years, after which the increased transport of warm waters leads to a negative feedback on water formation in the subpolar gyre which then reduces the MOC. This negative feedback can be mitigated if EN3 hydrography data continue to be assimilated along with the RAPID array boundary data. Copyright © 2012 Royal Meteorological Society and Crown in the right of Canada.

1929. Nontidal ocean loading: amplitudes and potential effects in GPS height time series

Authors: T. van Dam; X. Collilieux; J. Wuite; Z. Altamimi; J. Ray

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-012-0564-5

URL: <https://doi.org/10.1007/s00190-012-0564-5>

Publication Date: 2012-05-05

Volume/Issue: Volume 86, Issue 11

References: 64

Cited by: 98 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1930. Long-term change of the Pacific North Equatorial Current bifurcation in SODA

Authors: Zhaohui Chen; Lixin Wu

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007814

URL: <https://doi.org/10.1029/2011jc007814>

Publication Date: 2012-05-03

Volume/Issue: Volume 117, Issue C6

References: 38

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The long-term change of the North Equatorial Current (NEC) bifurcation in the Pacific Ocean is assessed based on the recently developed Simple Ocean Data Assimilation (SODA, version 2.2.4). It is found that the NEC bifurcation latitude (NBL) has shifted southward over the past 60 years, although it displayed a slight northward migration from 1970 to 1992. This southward shift of the bifurcation latitude is associated with changes in the wind stress curl over the tropical Pacific Ocean between 10 degreesN and 20 degreesN, leading to the strengthening of the Kuroshio at its origin. The conclusion is further supported by simulations of Intergovernmental Panel on Climate Change models. It is demonstrated that the long-term change of the seasonal south-north migration of the bifurcation is modulated by the southward shift of the mean position. Over the past 6 decades, the phase speed of first-mode baroclinic Rossby waves (CR) at the latitude of the bifurcation increases from 13 cm s⁻¹ in 1950 to 18 cm s⁻¹ in 2005, and the corresponding seasonal amplitude increases (decreases) before (after) the mid-1980s. Using a linear vorticity model, it is found that the long-term modulation of the NBL seasonal migration amplitude is associated with the increase of CR in responses to the southward shift of the mean NBL. It is expected that the seasonal amplitude will decrease moderately in the following decades if the ocean continues warming.

1931. Model Thermohaline Trends in the Mediterranean Sea during the Last Years: A Change with Respect to the Last Decades?

Authors: F. Javier Soto-Navarro (Physical Oceanography Group, Department of Applied Physics II, University of Málaga, 29071 Málaga, Spain); Francisco Criado-Aldeanueva (Physical Oceanography Group, Department of Applied Physics II, University of Málaga, 29071 Málaga, Spain)

Journal: The Scientific World Journal

Publisher: Hindawi Limited

DOI: 10.1100/2012/365698

URL: <https://doi.org/10.1100/2012/365698>

Publication Date: 2012-04-30

Volume/Issue: Volume 2012

References: 27

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Conserjería de Innovación Ciencia y Empresa, Junta de Andalucía, Spain

License: Creative Commons

Open Access: N/A

Abstract: Temperature and salinity outputs from ECCO (years 93-09) and GLORYS (years 03-09) models have been used to compute the thermohaline and steric sea level trends in the surface (0-150 m), intermediate (150 m-600 m), and deep (600 m-bottom) layers of the Mediterranean Sea. Some changes with respect to the second half of the 20th century have been observed: the cooling of the upper waters of the entire eastern basin since 1950 seems to have vanished; the warming of WMDW historically reported for the second half of the last century could have reversed, although there is no agreement between both models at this point (trends of different sign are predicted); the salinification of WMDW reported for the previous decades is not observed in the south-westernmost area in the period 93-09, and a clear change from positive to negative in the steric sea level trend with respect to the period 93-05 is detected due to the sharp decreasing steric sea level of years 02-06.

1932. Performance of GOCE and GRACE-derived mean dynamic topographies in resolving Antarctic Circumpolar Current fronts

Authors: Denis L. Volkov; Victor Zlotnicki

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-012-0541-9

URL: <https://doi.org/10.1007/s10236-012-0541-9>

Publication Date: 2012-04-28

Volume/Issue: Volume 62, Issue 6

References: 20

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1933. Water mass variation in the Mediterranean and Black Seas

Authors: L. Fenoglio-Marc; R. Rietbroek; S. Grayek; M. Becker; J. Kusche; E. Stanev

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2012.04.001

URL: <https://doi.org/10.1016/j.jog.2012.04.001>

Publication Date: 2012-04-24

Volume/Issue: Volume 59-60

References: 52

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1934. Sensitivity of the ice-shelf/ocean system to the sub-ice-shelf cavity shape measured by NASA IceBridge in Pine Island Glacier, West Antarctica

Authors: Michael P. Schodlok; Dimitris Menemenlis; Eric Rignot; Michael Studinger

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2012aog60a073

URL: <https://doi.org/10.3189/2012aog60a073>

Publication Date: 2012-04-24

Volume/Issue: Volume 53, Issue 60

References: 34

Cited by: 104 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Two high-resolution (1 km grid) numerical model simulations of the Amundsen Sea, West Antarctica, are used to study the role of the ocean in the mass loss and grounding line retreat of Pine Island Glacier. The first simulation uses BEDMAP bathymetry under the Pine Island ice shelf, and the second simulation uses NASA IceBridge-derived bathymetry. The IceBridge data reveal the existence of a trough from the ice-shelf edge to the grounding line, enabling warm Circumpolar Deep Water to penetrate to the grounding line, leading to higher melt rates than previously estimated. The mean melt rate for the simulation with NASA IceBridge data is 28 ma⁻¹, much higher than previous model estimates but closer to estimates from remote sensing. Although the mean melt rate is 25% higher than in the simulation with BEDMAP bathymetry, the temporal evolution remains unchanged between the two simulations. This indicates that temporal variability of melting is mostly driven by processes outside the cavity. Spatial melt rate patterns of BEDMAP and IceBridge simulations differ significantly, with the latter in closer agreement with satellite-derived melt rate estimates of ~50ma⁻¹ near the grounding line. Our simulations confirm that knowledge of the cavity shape and its time evolution are essential to accurately capture basal mass loss of Antarctic ice shelves.

1935. Estimates of Analysis and Forecast Error Variances Derived from the Adjoint of 4D-Var

Authors: Andrew M. Moore (University of California Santa Cruz); Hernan G. Arango (Rutgers University); Gregoire Broquet (Laboratoire des Sciences du Climat et de l'Environnement, CEA, Gif-Sur-Yvette, France)

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/mwr-d-11-00141.1

URL: <https://doi.org/10.1175/mwr-d-11-00141.1>

Publication Date: 2012-04-23

Volume/Issue: Volume 140, Issue 10

References: 48

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A method is presented in which the adjoint of a four-dimensional variational data assimilation system (4D-Var) was used to compute the expected analysis and forecast error variances of linear functions of the ocean state vector. The power and utility of the approach are demonstrated using the Regional Ocean Modeling System configured for the California Current system. Linear functions of the ocean state vector were considered in the form of indices that characterize various aspects of the coastal upwelling circulation. It was found that for configurations of 4D-Var typically used in ocean models, reliable estimates of the expected analysis error variances can be obtained both for variables that are observed and unobserved. In addition, the contribution of uncertainties in the model control variables to the forecast error variance was also quantified. One particularly powerful and illuminating aspect of the adjoint 4D-Var approach to the forecast problem is that the contribution of individual observations to the predictability of the circulation can be readily computed. An important finding of the work presented here is that despite the plethora of available satellite observations, the relatively modest fraction of in situ subsurface observations sometimes exerts a significant influence on the predictability of the coastal ocean. Independent checks of the analysis and forecast error variances are also presented, which provide a direct test of the hypotheses that underpin the prior error and observation error estimates used during 4D-Var.

1936. Ocean State Estimation and Prediction in Support of Oceanographic Research

Authors: Detlef Stammer; Eric Chassignet

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2000.34

URL: <https://doi.org/10.5670/oceanog.2000.34>

Publication Date: 2012-04-21

Volume/Issue: Volume 13, Issue 2

References: N/A

Cited by: 16 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1937. Eddy-Resolving Model Estimate of the Cabbeling Effect on the Water Mass Transformation in the Southern Ocean

Authors: L. Shogo Urakawa (University of Tokyo); Hiroyasu Hasumi (University of Tokyo)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-11-0173.1

URL: <https://doi.org/10.1175/jpo-d-11-0173.1>

Publication Date: 2012-04-20

Volume/Issue: Volume 42, Issue 8

References: 49

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Cabbeling effect on the water mass transformation in the Southern Ocean is investigated with the use of an eddy-resolving Southern Ocean model. A significant amount of water is densified by cabbeling: water mass transformation rates are about 4 Sv ($1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) for transformation from surface/thermocline water to Subantarctic Mode Water (SAMW), about 7 Sv for transformation from SAMW to Antarctic Intermediate Water (AAIW), and about 5 Sv for transformation from AAIW to Upper Circumpolar Deep Water. These diapycnal volume transports occur around the Antarctic Circumpolar Current (ACC), where mesoscale eddies are active. The water mass transformation by cabbeling in this study is also characterized by a large amount of densification of Lower Circumpolar Deep Water (LCDW) into Antarctic Bottom Water (AABW) (about 9 Sv). Large diapycnal velocity is found not only along the ACC but also along the coast of Antarctica at the boundary between LCDW and AABW. It is found that about 3 Sv of LCDW is densified into AABW by cabbeling on the continental slopes of Antarctica in this study. This densification is not small compared with observational and numerical estimates on the AABW formation rate, which ranges from 10 to 20 Sv.

1938. Biological response to the 1997-98 and 2009-10 El Niño events in the equatorial Pacific Ocean

Authors: Michelle M. Gierach; Tong Lee; Daniela Turk; Michael J. McPhaden

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl051103

URL: <https://doi.org/10.1029/2012gl051103>

Publication Date: 2012-04-20

Volume/Issue: Volume 39, Issue 10

References: 30

Cited by: 66 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Changes in the physical environment associated with eastern Pacific (EP)-El Niño and central Pacific (CP)-El Niño events affect the biological response in the equatorial Pacific Ocean differently. However, such responses have not been adequately investigated, especially in terms of the relevant physical processes. This paper addresses the mechanistic differences in the biological response of the equatorial Pacific Ocean during the strongest CP- and EP-El Niño to date (i.e., 1997-98 EP-El Niño and 2009-10 CP-El Niño) using satellite data and water mass pathway analysis based on an ocean reanalysis product. The 1997-98 EP-El Niño was associated with a larger reduction of chlorophyll-a (chl-a) in the eastern equatorial Pacific (EEP) and the 2009-10 CP-El Niño was associated with a larger reduction of chl-a in the central equatorial Pacific (CEP). These biological responses were dependent on the strength and extent of westerly wind anomalies and their impact on horizontal and vertical processes. Horizontal advection was the primary contributor to differences in chl-a between the two El Niño events in the CEP, whereas vertical advection and mixing were the dominant processes in the EEP.

1939. Short-term transport variability of the Antarctic Circumpolar Current from satellite gravity observations

Authors: I. Bergmann; H. Dobslaw

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jc007872

URL: <https://doi.org/10.1029/2012jc007872>

Publication Date: 2012-04-13

Volume/Issue: Volume 117, Issue C5

References: 46

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean bottom pressure gradients deduced from the satellite gravity mission Gravity Recovery and Climate Experiment (GRACE) were previously shown to provide barotropic transport variations of the Antarctic Circumpolar Current (ACC) with up to monthly resolution. Here, bottom pressure distributions from GRACE with monthly (GFZ RL04) and higher temporal resolution (CNES/GRGS with 10 days, ITG-GRACE2010 with daily resolution) are evaluated over the ACC area. Even on time scales shorter than 10 days, correlations with in situ bottom pressure records frequently exceed 0.6 with positive explained variances, giving evidence that high-frequency nontidal ocean mass variability is captured by the daily ITG-GRACE2010 solutions not already included in the applied background models. Bottom pressure is subsequently taken to calculate the barotropic component of the ACC transport variability across Drake Passage. For periods longer than 30 days, transport shows high correlations between 0.4 and 0.5 with several tide gauge records along the coast of Antarctica. Still significant correlations around 0.25 are obtained even for variability with periods shorter than 10 days. Since transport variations are predominantly affected by time-variable surface winds, GRACE-based transports are contrasted against an atmospheric index of the Southern Annular Mode (SAM), which represents the Southern Hemispheric wind variability. Correlations between the SAM and GRACE-based transports are consistently higher than correlations between any of the available sea level records in all frequency bands considered, indicating that GRACE is indeed able to accurately observe a hemispherically consistent pattern of bottom pressure (and hence ACC transport) variability that is otherwise at least partially masked in tide gauge records due to local weather effects, sea ice presence and steric signals.

1940. Sea level variations in the South China Sea inferred from satellite gravity, altimetry, and oceanographic data

Authors: Wei Feng; Min Zhong; HouZe Xu

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-012-4394-3

URL: <https://doi.org/10.1007/s11430-012-4394-3>

Publication Date: 2012-04-10

Volume/Issue: Volume 55, Issue 10

References: 37

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1941. Climate Variability of the Northern Argentinean Shelf Circulation: Impact on *Engraulis Anchoita*

Authors: Guillermo Auad (Scripps Institution of Oceanography); Patricia Martos (Instituto Nacional de Investigación y Desarrollo Pesquero, Mar del Plata, Argentina (*)Now at Bureau of Ocean Energy Management, Herndon, Virginia, Universidad Nacional de Mar del Plata, Argentina)

Journal: The International Journal of Ocean and Climate Systems

Publisher: SAGE Publications

DOI: 10.1260/1759-3131.3.1.17

URL: <https://doi.org/10.1260/1759-3131.3.1.17>

Publication Date: 2012-04-04

Volume/Issue: Volume 3, Issue 1

References: 98

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A high-resolution ocean model and hydrographic observations are used to characterize the shelf circulation of the northern Argentinean shelf during the study period (1993-2008) and ultimately to explore possible linkages among atmospheric, oceanic, and biological climatic variability. Abundance of larvae and eggs of the local anchovy species, *Engraulis anchoita*, exhibit a spatial and temporal variability similar to those stocks found in other parts of the world and that we interpret in the context of the particularities of the local circulation and hydrography. Two (statistically) coupled modes of wind stress-surface velocity are described and interpreted in terms of historical and new information. A complex picture emerges in which the intensity of both a thermal shelf front, the alongshore flow, and larvae abundance would be connected and forced by local wind stresses. For all areas examined on the shelf, the larvae/egg abundance would not be very sensitive to short-lived climatic fluctuations (e.g., year-to-year) but they would be indeed to regime shifts. The shallow shelf area bounded by the 39 degreesS and 41 degreesS parallels would expose a clearer linkage between physical and biological variables than that north of 39 degreesS. We attribute this fact to the particular physical conditions found in the southernmost area, which would favor an increased habitat quality for *Engraulis anchoita*.

1942. Adjoint-Based Estimation of Eddy-Induced Tracer Mixing Parameters in the Global Ocean

Authors: Chuanyu Liu (University of Hamburg); Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-11-0162.1

URL: <https://doi.org/10.1175/jpo-d-11-0162.1>

Publication Date: 2012-04-02

Volume/Issue: Volume 42, Issue 7

References: 75

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: Data assimilation, Mixing

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Using the German Estimating the Circulation and Climate of the Ocean (GECCO) synthesis framework, four separate eddy tracer mixing coefficients are adjusted jointly with external forcing fields, such as to reduce a global misfit between the model simulations and ocean observations over a single 10-yr period and weighted by uncertainties. The suite of the adjusted eddy tracer mixing coefficients includes the vertical diffusivity k_z , the along-isopycnal surface diffusivity k_{redi} , the isopycnal layer thickness diffusivity k_{gm} , and the along-iso-thickness advection coefficient k_{gmskew} . Large and geographically varying adjustments are found in all four parameters, which all together lead to an additional 10% reduction of the total cost function, as compared to using only surface flux parameters. However, their relative contribution to the cost reduction varies from 1% to 50% among the four coefficients, with the adjusted k_{gm} contributing most. Regionally, the estimated k_{gm} ranges from less than -800 to about 2500 $\text{m}^2 \text{ s}^{-1}$. Largest adjustments in k_{gm} reside in the vicinity of large isopycnal slopes and support a mixing length hypothesis; they also likewise support the hypothesis of a critical layer enhancement and high potential density gradient suppression. In a few occasions, resulting negative net k_{gm} values can be found in the core of main currents, suggesting the potential for an inverse energy cascade transfer there. Large adjustments of k_{redi} and k_{gmskew} are found in the vicinity of isopycnal slopes. The adjustments of k_z in the tropical thermoclines suggest deficiencies of the mixed layer parameterization.

1943. Origin and pathway of the Luzon Undercurrent identified by a simulated adjoint tracer

Authors: Shan Gao; Tangdong Qu; Dunxin Hu

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007748

URL: <https://doi.org/10.1029/2011jc007748>

Publication Date: 2012-04-02

Volume/Issue: Volume 117, Issue C5

References: 31

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The origin and pathway of Luzon Undercurrent (LUC) water are investigated using a simulated adjoint tracer, based on circulation estimates of a global general circulation model. The results demonstrate that most of the LUC water comes from the subtropical North Pacific, confirming the earlier hypothesis on its relation to the northward shift of the North Equatorial Current (NEC) bifurcation at subsurface. Of the total volume of initially tracer-tagged LUC water, approximately 41% is traced back to the winter mixed layer in the Kuroshio extension after 50 years of integration, coinciding with the formation of subtropical mode waters, while the rest is trapped in the northern subtropical gyre with its density $>26.8 \text{ kg m}^{-3}$, indicative of a significant South Pacific origin. As these waters move toward the western boundary, they get mixed and finally reach the density range of the LUC water. This result provides quantitative evidence for the dramatic impact of mixing on the route of subtropical water to becoming the LUC water.

1944. Impact of assimilating bottom pressure anomalies from GRACE on ocean circulation estimates

Authors: Armin Köhl; Frank Siegismund; Detlef Stammer

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007623

URL: <https://doi.org/10.1029/2011jc007623>

Publication Date: 2012-03-27

Volume/Issue: Volume 117, Issue C4

References: 58

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The impact of assimilating global ocean bottom pressure (OBP) information from the Gravity Recovery and Climate Experiment (GRACE) gravity anomalies on the circulation estimate was investigated. For this an estimate of the ocean circulation is being inferred by extending the 50-year-long German part of the Estimating the Circulation and Climate of the Ocean (GECCO) ocean synthesis into recent years. The assimilation system is an improved version of the previous GECCO optimization, which now includes a sea ice model, has enhanced resolution on a truly global domain including the Arctic Ocean. By analyzing differences to a synthesis that additionally assimilated OBP, the GRACE data was found to provide complementary information to standard ocean data sets including satellite altimetry when assimilated. Although in principle standard ocean data sets include the OBP information, the reason why this cannot be extracted is the much larger prior errors for hydrographic and altimeter data in comparison to OBP data owing to the fact that only the former two need to include the unresolved eddy signal. The largest impact of gravity data is found to be on the barotropic circulation, particularly in the subtropical gyres and the polar latitudes. Remaining differences between the simulated and observed OBP information are associated with meridional stripes in the GRACE gravity maps and with the leakage of terrestrial hydrological information into the ocean. Additional differences close to the continental boundaries are related to the self-attraction and loading, processes that are not included in the models.

1945. Agreement between Earth's rotation and mass displacement as detected by GRACE

Authors: L. Seoane; R. Biancale; D. Gambis

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2012.02.008

URL: <https://doi.org/10.1016/j.jog.2012.02.008>

Publication Date: 2012-03-23

Volume/Issue: Volume 62

References: 28

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1946. Adjoint sensitivities of sub-ice-shelf melt rates to ocean circulation under the Pine Island Ice Shelf, West Antarctica

Authors: Patrick Heimbach; Martin Losch

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2012/aog60a025

URL: <https://doi.org/10.3189/2012/aog60a025>

Publication Date: 2012-03-13

Volume/Issue: Volume 53, Issue 60

References: 64

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We investigate the sensitivity of sub-ice-shelf melt rates under Pine Island Ice Shelf, West Antarctica, to changes in the oceanic state using an adjoint ocean model that is capable of representing the flow in sub-ice-shelf cavities. The adjoint code is based on algorithmic differentiation (AD) of the Massachusetts Institute of Technology's ocean general circulation model (MITgcm). The adjoint model was extended by adding into the AD process the corresponding sub-ice-shelf cavity code, which implements a three-equation thermodynamic melt-rate parameterization to infer heat and freshwater fluxes at the ice-shelf/ocean boundary. The inferred sensitivities reveal dominant timescales of 30-60 days over which the shelf exit is connected to the deep interior via advective processes. They exhibit rich three-dimensional time-evolving patterns that can be understood in terms of a combination of the buoyancy forcing by inflowing water masses, the cavity geometry and the effect of rotation and topography in steering the flow in the presence of prominent features in the bedrock bathymetry. Dominant sensitivity pathways are found over a sill, as well as 'shadow regions' of very low sensitivities. To the extent that these transient patterns are robust they carry important information for decision-making in observation deployment and monitoring.

1947. The Gauss-Listing geopotential value W_0 and its rate from altimetric mean sea level and GRACE

Authors: N. Dayoub; S. J. Edwards; P. Moore

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-012-0547-6

URL: <https://doi.org/10.1007/s00190-012-0547-6>

Publication Date: 2012-03-09

Volume/Issue: Volume 86, Issue 9

References: 51

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: Geopotential

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1948. The development and evaluation of the Earth Gravitational Model 2008 (EGM2008)

Authors: Nikolaos K. Pavlis; Simon A. Holmes; Steve C. Kenyon; John K. Factor

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jb008916

URL: <https://doi.org/10.1029/2011jb008916>

Publication Date: 2012-03-04

Volume/Issue: Volume 117, Issue B4

References: 77

Cited by: 1677 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: EGM2008 is a spherical harmonic model of the Earth's gravitational potential, developed by a least squares combination of the ITG-GRACE03S gravitational model and its associated error covariance matrix, with the gravitational information obtained from a global set of area-mean free-air gravity anomalies defined on a 5 arc-minute equiangular grid. This grid was formed by merging terrestrial, altimetry-derived, and airborne gravity data. Over areas where only lower resolution gravity data were available, their spectral content was supplemented with gravitational information implied by the topography. EGM2008 is complete to degree and order 2159, and contains additional coefficients up to degree 2190 and order 2159. Over areas covered with high quality gravity data, the discrepancies between EGM2008 geoid undulations and independent GPS/Leveling values are on the order of +/-5 to +/-10 cm. EGM2008 vertical deflections over USA and Australia are within +/-1.1 to +/-1.3 arc-seconds of independent astrogeodetic values. These results indicate that EGM2008 performs comparably with contemporary detailed regional geoid models. EGM2008 performs equally well with other GRACE-based gravitational models in orbit computations. Over EGM96, EGM2008 represents improvement by a factor of six in resolution, and by factors of three to six in accuracy, depending on gravitational quantity and geographic area. EGM2008 represents a milestone and a new paradigm in global gravity field modeling, by demonstrating for the first time ever, that given accurate and detailed gravimetric data, a single global model may satisfy the requirements of a very wide range of applications.

1949. Impact of different ocean reanalyses on decadal climate prediction

Authors: Jürgen Kröger; Wolfgang A. Müller; Jin-Song von Storch

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-012-1310-7

URL: <https://doi.org/10.1007/s00382-012-1310-7>

Publication Date: 2012-03-01

Volume/Issue: Volume 39, Issue 3-4

References: 34

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1950. The effect of Coriolis-Stokes forcing on upper ocean circulation in a two-way coupled wave-current model

Authors: Zeng'an Deng; Li'an Xie; Guijun Han; Xuefeng Zhang; Kejian Wu

Journal: Chinese Journal of Oceanology and Limnology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00343-012-1069-z

URL: <https://doi.org/10.1007/s00343-012-1069-z>

Publication Date: 2012-02-29

Volume/Issue: Volume 30, Issue 2

References: 38

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1951. Buoyancy-driven interannual sea level changes in the southeast tropical Pacific

Authors: Christopher G. Piecuch; Rui M. Ponte

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl051130

URL: <https://doi.org/10.1029/2012gl051130>

Publication Date: 2012-02-21

Volume/Issue: Volume 39, Issue 5

References: 22

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: It is commonly held that interannual-to-decadal sea level variability patterns mainly represent the ocean's response to wind forcing. This view is based in part on modeling studies of wind-driven sea level changes along the tropical Pacific. However, because buoyancy forcing (and other generating mechanisms) are usually ignored, this paradigm may overemphasize the role of winds. Focusing on the southeast tropical Pacific, we use a data-constrained ocean state estimate to demonstrate that distinct mechanisms--including the ocean's response to buoyancy forcing as well as nonlinear processes--can also contribute to interannual sea level variability. Contrary to the notion that buoyancy-driven sea level changes are dynamically passive, such changes exhibit a strongly nonlocal, dynamically active character, made manifest in westward propagating waves. As similar findings apply elsewhere, accurate modeling of interannual-to-decadal regional sea level changes requires consideration of a variety of forcing mechanisms, including, but not limited to, the winds.

1952. Temperature correction and usefulness of ocean bottom pressure data from cabled seafloor observatories around Japan for analyses of tsunamis, ocean tides, and low-frequency geophysical phenomena

Authors: Daisuke Inazu; Ryota Hino

Journal: Earth, Planets and Space

Publisher: Springer Science and Business Media LLC

DOI: 10.5047/eps.2011.07.014

URL: <https://doi.org/10.5047/eps.2011.07.014>

Publication Date: 2012-02-14

Volume/Issue: Volume 63, Issue 11

References: 74

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1953. Fortnightly Earth rotation, ocean tides and mantle anelasticity

Authors: Richard D. Ray; Gary D. Egbert

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2012.05351.x

URL: <https://doi.org/10.1111/j.1365-246x.2012.05351.x>

Publication Date: 2012-02-13

Volume/Issue: Volume 189, Issue 1

References: 68

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1954. Predictability of Mesoscale Variability in the East Australian Current Given Strong-Constraint Data Assimilation

Authors: Javier Zavala-Garay (Institute of Marine and Coastal Sciences, Rutgers, The State University of New Jersey, New Brunswick, New Jersey); J. L. Wilkin (Institute of Marine and Coastal Sciences, Rutgers, The State University of New Jersey, New Brunswick, New Jersey); H. G. Arango (Institute of Marine and Coastal Sciences, Rutgers, The State University of New Jersey, New Brunswick, New Jersey)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-11-0168.1

URL: <https://doi.org/10.1175/jpo-d-11-0168.1>

Publication Date: 2012-02-10

Volume/Issue: Volume 42, Issue 9

References: 58

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: One of the many applications of data assimilation is the estimation of adequate initial conditions for model forecasts. In this work, the authors evaluate to what extent the incremental, strong-constraint, four-dimensional variational data assimilation (IS4DVAR) can improve prediction of mesoscale variability in the East Australian Current (EAC) using the Regional Ocean Modeling System (ROMS). The observations considered in the assimilation experiments are daily composites of satellite sea surface temperature (SST), 7-day average reanalysis of satellite altimeter sea level anomalies, and subsurface temperature profiles from high-resolution expendable bathythermograph (XBT). Considering all available observations for years 2001 and 2002, ROMS forecast initial conditions are generated every week by assimilating the available observations from the 7 days prior to the forecast initial time. It is shown that assimilation of surface information only [SST and sea surface height (SSH)] results in poor estimates of the true subsurface ocean state (as depicted by the XBTs) and therefore poor forecast skill of subsurface conditions. Including the XBTs in the assimilation experiments improves the ocean state estimation in the vicinity of the XBT transects. By introducing subsurface pseudo-observations (which are called synthetic CTD) based on an empirical relationship between satellite surface observations and subsurface variability, the authors find a significant improvement in ocean state estimates that leads to skillful forecasts for up to 2 weeks in the domain considered.

1955. Impact of combining GRACE and GOCE gravity data on ocean circulation estimates

Authors: T. Janjić; J. Schröter; R. Savcenko; W. Bosch; A. Albertella; R. Rummel; O. Klatt

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-8-65-2012

URL: <https://doi.org/10.5194/os-8-65-2012>

Publication Date: 2012-02-08

Volume/Issue: Volume 8, Issue 1

References: 58

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . With the focus on the Southern Ocean circulation, results of assimilation of multi-mission-altimeter data and the GRACE/GOCE gravity data into the finite element ocean model (FEOM) are investigated. We use the geodetic method to obtain the dynamical ocean topography (DOT). This method combines the multi-mission-altimeter sea surface height and the GRACE/GOCE gravity field. Using the profile approach, the spectral consistency of both fields is achieved by filtering the sea surface height and the geoid. By combining the GRACE and GOCE data, a considerably shorter filter length can be used, which results in more DOT details. We show that this increase in resolution of measured DOT carries onto the results of data assimilation for the surface data. By assimilating only absolute dynamical topography data using the ensemble Kalman filter, we were able to improve modeled fields. Results are closer to observations which were not used for assimilation and lie outside the area covered by altimetry in the Southern Ocean (e.g. temperature of surface drifters or deep temperatures in the Weddell Sea area at 800 m depth derived from Argo composite.)

1956. Detection of the Earth rotation response to a rapid fluctuation of Southern Ocean circulation in November 2009

Authors: S. L. Marcus; J. O. Dickey; I. Fukumori; O. de Viron

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl050671

URL: <https://doi.org/10.1029/2011gl050671>

Publication Date: 2012-02-01

Volume/Issue: Volume 39, Issue 4

References: 23

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: At seasonal and shorter periods the solid Earth and its overlying geophysical fluids form a closed dynamical system, which (except for tidal forcing) conserves its total angular momentum. While atmospheric effects dominate changes in the Earth's rate of rotation and hence length-of-day (LOD) on these time scales, the addition of oceanic angular momentum (OAM) estimates has been shown to improve closure of the LOD budget in a statistical sense. Here we demonstrate, for the first time, the signature of a specific, sub-monthly ocean current fluctuation on the Earth's rotation rate, coinciding with recently-reported anomalies which developed in southeast Pacific surface temperature and bottom pressure fields during late 2009. Our results show that concurrent variations in the Antarctic Circumpolar Current (ACC), which saw a sharp drop and recovery in zonal transport during a two-week period in November, were strong enough to cause a detectable change in LOD following the removal of atmospheric angular momentum (AAM) computed from the Modern Era Retrospective Analysis for Research and Applications (MERRA) database. The strong OAM variations driving the LOD-AAM changes were diagnosed from ocean state estimates of the Consortium for Estimating the Circulation and Climate of the Ocean (ECCO) and involved roughly equal contributions from the current and pressure terms, with in situ confirmation for the latter provided by tide-corrected bottom pressure recorder data from the South Drake Passage site of the Antarctic Circumpolar Current Levels by Altimetry and Island Measurements (ACCLAIM) network.

1957. Quality assessment of a satellite altimetry data product in the Nordic, Barents, and Kara seas

Authors: Denis L. Volkov; M.-Isabelle Pujol

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007557

URL: <https://doi.org/10.1029/2011jc007557>

Publication Date: 2012-01-31

Volume/Issue: Volume 117, Issue C3

References: 45

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA

License: N/A

Open Access: N/A

Abstract: Satellite altimetry provides high-quality sea surface height data that have been successfully used to study the variability of sea level and surface geostrophic circulation at different spatial and temporal scales. However, the high-latitude regions have traditionally been avoided due to the persistent sea ice cover. Most of the validation studies have focused on the areas below the polar circles. In this paper we examine the quality and performance of a gridded satellite altimetry product in the Nordic, Barents, and Kara seas. The altimetric sea level in coastal areas is validated using available tide gauge records. We show that at most locations in the Nordic seas the altimetry and tide gauge measurements are in a good agreement in terms of the root-mean square differences and the amplitudes and phases of the seasonal cycle. The agreement deteriorates in the shallow areas of the Barents and Kara seas subject to the seasonal presence of sea ice, and where the altimetry data are contaminated by the residual aliasing of unresolved high-frequency signals. The comparison of linear trends at the locations of tide gauges reveals discrepancies that need to be taken into account when interpreting long-term changes of sea level in the region. Away from the coast the altimetry data are compared to drifter trajectories, corrected for Ekman currents. The drifter trajectories are found consistent with the mesoscale variability of the altimetric sea level. This study provides the first comprehensive validation of a gridded satellite altimetry data product in the high-latitude seas.

1958. Assessment of terrestrial water contributions to polar motion from GRACE and hydrological models

Authors: S.G. Jin; A.A. Hassan; G.P. Feng

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2012.01.009

URL: <https://doi.org/10.1016/j.jog.2012.01.009>

Publication Date: 2012-01-30

Volume/Issue: Volume 62

References: 29

Cited by: 58 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1959. Geocenter motion and its geodetic and geophysical implications

Authors: Xiaoping Wu; Jim Ray; Tonie van Dam

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2012.01.007

URL: <https://doi.org/10.1016/j.jog.2012.01.007>

Publication Date: 2012-01-30

Volume/Issue: Volume 58

References: 104

Cited by: 120 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1960. Source and Pathway of the Western Arctic Upper Halocline in a Data-Constrained Coupled Ocean and Sea Ice Model

Authors: An T. Nguyen (Massachusetts Institute of Technology); Ronald Kwok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-11-040.1

URL: <https://doi.org/10.1175/jpo-d-11-040.1>

Publication Date: 2012-01-30

Volume/Issue: Volume 42, Issue 5

References: 45

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A coupled ocean and sea ice model is used to investigate dense water (DW) formation in the Chukchi and Bering shelves and the pathways by which this water feeds the upper halocline. Two 1992-2008 data-constrained solutions at 9- and 4-km horizontal grid spacing show that 1) winter sea ice growth results in brine rejection and DW formation; 2) the DW flows primarily down Barrow and Central-Herald Canyons in the form of bottom-trapped, intermittent currents to depths of 50-150 m from the late winter to late summer seasons; and 3) eddies with diameters ~ 30 km carry the cold DW from the shelf break into the Canada Basin interior at depths of 50-150 m. The 4-km data-constrained solution does not show eddy transport across the Chukchi Shelf at shallow depths; instead, advection of DW downstream of polynya regions is driven by a strong (~ 0.1 m s⁻¹) mean current on the Chukchi Shelf. Upper halocline water (UHW) formation rate was obtained from two methods: one is based on satellite data and on a simple parameterized approach, and the other is computed from the authors' model solution. The two methods yield 5740 ± 1420 km³ yr⁻¹ and 4190 - 4860 ± 1440 km³ yr⁻¹, respectively. These rates imply a halocline replenishment period of 10-21 yr. Passive tracers also show that water with highest density forms in the Gulf of Anadyr and along the eastern Siberian coast immediately north of the Bering Strait. These results provide a coherent picture of the seasonal development of UHW at high spatial and temporal resolutions and serve as a guide for improving understanding of water-mass formation in the western Arctic Ocean.

1961. Geophysical Excitation of the Chandler Wobble Revisited

Authors: Aleksander Brzeziński; Henryk Dobslaw; Robert Dill; Maik Thomas

Journal: International Association of Geodesy Symposia

Publisher: Springer Berlin Heidelberg

DOI: 10.1007/978-3-642-20338-1_60

URL: https://doi.org/10.1007/978-3-642-20338-1_60

Publication Date: 2012-01-24

Volume/Issue: N/A

References: 16

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: Polar motion|Speed wobble

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1962. Evaluation of EGM2008 Within Geopotential Space from GPS, Tide Gauges and Altimetry

Authors: N. Dayoub; P. Moore; N. T. Penna; S. J. Edwards

Journal: International Association of Geodesy Symposia

Publisher: Springer Berlin Heidelberg

DOI: 10.1007/978-3-642-20338-1_39

URL: https://doi.org/10.1007/978-3-642-20338-1_39

Publication Date: 2012-01-24

Volume/Issue: N/A

References: 20

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Geopotential|Tide gauge|Levelling|Geopotential height|Post-glacial rebound

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1963. Evaluation of Arctic sea ice thickness simulated by Arctic Ocean Model Intercomparison Project models

Authors: Mark Johnson; Andrey Proshutinsky; Yevgeny Aksenov; An T. Nguyen; Ron Lindsay; Christian Haas; Jinlun Zhang; Nikolay Diansky; Ron Kwok; Wieslaw Maslowski; Sirpa Häkkinen; Igor Ashik; Beverly de Cuevas

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007257

URL: <https://doi.org/10.1029/2011jc007257>

Publication Date: 2012-01-23

Volume/Issue: Volume 117, Issue C8

References: 128

Cited by: 65 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Six Arctic Ocean Model Intercomparison Project model simulations are compared with estimates of sea ice thickness derived from pan-Arctic satellite freeboard measurements (2004-2008); airborne electromagnetic measurements (2001-2009); ice draft data from moored instruments in Fram Strait, the Greenland Sea, and the Beaufort Sea (1992-2008) and from submarines (1975-2000); and drill hole data from the Arctic basin, Laptev, and East Siberian marginal seas (1982-1986) and coastal stations (1998-2009). Despite an assessment of six models that differ in numerical methods, resolution, domain, forcing, and boundary conditions, the models generally overestimate the thickness of measured ice thinner than ~2 m and underestimate the thickness of ice measured thicker than about ~2 m. In the regions of flat immobile landfast ice (shallow Siberian Seas with depths less than 25-30 m), the models generally overestimate both the total observed sea ice thickness and rates of September and October ice growth from observations by more than 4 times and more than one standard deviation, respectively. The models do not reproduce conditions of fast ice formation and growth. Instead, the modeled fast ice is replaced with pack ice which drifts, generating ridges of increasing ice thickness, in addition to thermodynamic ice growth. Considering all observational data sets, the better correlations and smaller differences from observations are from the Estimating the Circulation and Climate of the Ocean, Phase II and Pan-Arctic Ice Ocean Modeling and Assimilation System models.

1964. An assessment of deep steric height variability over the global ocean

Authors: Rui M. Ponte

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl050681

URL: <https://doi.org/10.1029/2011gl050681>

Publication Date: 2012-01-23

Volume/Issue: Volume 39, Issue 4

References: 16

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An ocean state estimate constrained by most available data is explored to assess characteristics of variability in deep steric height--a mostly unobserved quantity, yet important for understanding the relation between sea level, heat content and other ocean climate parameters. Results are based on monthly-averaged steric height anomalies, vertically integrated over the "unobserved" deep ocean (below ~1700 m). Excluding linear trends, variability in deep steric height is typically 10-20% of that in the upper ocean, with larger values seen in extensive regions. Enhanced deep variability, at monthly to interannual time scales, occurs in areas of strong eddy energy. Deep signals are mostly thermosteric in nature, with halosteric contributions tightly correlated and generally compensating in the Atlantic and Indian oceans and adding in the Pacific. Potential inference of deep signals from knowledge of the upper ocean is hampered by poor correlations, and regressions based on upper ocean steric height fail to represent the estimated deep variability. Monthly sampling at ~2 degrees scales would allow for best determination of deep variability and long term trends.

1965. Polar motion excitation - A broad-band perspective

Authors: Clark R. Wilson; Jianli Chen

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2012.01.001

URL: <https://doi.org/10.1016/j.jog.2012.01.001>

Publication Date: 2012-01-21

Volume/Issue: Volume 62

References: 13

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Polar motion

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1966. Oceanic excitation of polar motion: Identification of specific oceanic areas important for polar motion excitation

Authors: Jolanta Nastula; Richard Gross; David A. Salstein

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2012.01.002

URL: <https://doi.org/10.1016/j.jog.2012.01.002>

Publication Date: 2012-01-11

Volume/Issue: Volume 62

References: 31

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1967. The influences of ENSO on tropical cyclone activity in the Bay of Bengal during October-December

Authors: M. S. Girishkumar; M. Ravichandran

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007417

URL: <https://doi.org/10.1029/2011jc007417>

Publication Date: 2012-01-04

Volume/Issue: Volume 117, Issue C2

References: 72

Cited by: 144 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The El Niño-Southern Oscillation (ENSO) influence on tropical cyclone (TC) activity (frequency, genesis location, and intensity) in the Bay of Bengal (BoB) during the primary TC peak season (October-December) are studied for the period of 1993-2010. The study shows that during primary TC peak season, accumulated cyclone energy in the BoB is negatively correlated with Niño3.4 sea surface temperature anomaly. Under La Niña regime number of extreme TC cases (wind speed >64 kt) increases significantly in the BoB during the primary TC peak season. The analysis further shows that negative Indian Ocean dipole year is also favorable for extreme TC activity in the BoB during the primary TC peak season. The existence of low-level cyclonic (anticyclonic) vorticity, enhanced (suppressed) convection, and high (low) tropical cyclonic heat potential (TCHP) in the BoB provides favorable (unfavorable) conditions for the TC activity under La Niña (El Niño) regimes together with weak vertical wind shear and high sea surface temperature (SST). The genesis location of TC shifts to the east (west) of 87 degreesE in the BoB during La Niña (El Niño) regime due to the variability in convective activity. The probable reason for the intense TC during a La Niña regime is likely explained in terms of longer track for TCs over warm SST and high TCHP due to eastward shifting of genesis location together with other favorable conditions. The variability of Madden-Julian Oscillation and its influence on TC activity in the BoB during La Niña and El Niño regime are also examined.

1968. The Impact of High-End Computing on NASA Missions

Authors: Rupak Biswas; Jill Dunbar; John Hardman; F. Ron Bailey; Lorien Wheeler; Stuart Rogers

Journal: IT Professional

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/mitp.2011.110

URL: <https://doi.org/10.1109/mitp.2011.110>

Publication Date: 2011-12-20

Volume/Issue: Volume 14, Issue 2

References: 12

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1969. Seasonal excitation of polar motion

Authors: J.L. Chen; C.R. Wilson; Y.H. Zhou

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2011.12.002

URL: <https://doi.org/10.1016/j.jog.2011.12.002>

Publication Date: 2011-12-14

Volume/Issue: Volume 62

References: 35

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1970. Top-down control of marine phytoplankton diversity in a global ecosystem model

Authors: A.E. Friederike Prowe; Markus Pahlow; Stephanie Dutkiewicz; Michael Follows; Andreas Oschlies

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2011.11.016

URL: <https://doi.org/10.1016/j.pocean.2011.11.016>

Publication Date: 2011-12-14

Volume/Issue: Volume 101, Issue 1

References: 61

Cited by: 116 publications

Language: en

Subjects: N/A

Keywords: Marine ecosystem

Funding: Kiel Cluster of Excellence "The Future Ocean", Gordon and Betty Moore Foundation, National Oceanic and Atmospheric Administration (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

1971. Effect of global mass conservation among geophysical fluids on the seasonal length of day variation

Authors: Haoming Yan; Benjamin F. Chao

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jb008788

URL: <https://doi.org/10.1029/2011jb008788>

Publication Date: 2011-12-12

Volume/Issue: Volume 117, Issue B2

References: 29

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Knowledge Innovation Program of the Chinese Academy of Sciences, National Natural Science Foundation of China, National Science Council of Taiwan

License: N/A

Open Access: N/A

Abstract: Geophysical fluids models for the atmosphere, ocean, and land hydrology usually do not properly enforce the conservation of mass either individually or collectively. Called the global mass balance (GMB) effect, these errors, if not corrected, will significantly affect the determination of the effect of geophysical fluids on the variation in the Earth's rotation rate or length of day (LOD). Here we analyze the seasonal budget of the excitation sources for LOD variation in comparison with corresponding observations. We find that (1) the combined mass-induced excitations of LOD variation by geophysical fluids are brought to much better agreement with the observed upon accounting for the GMB effect; (2) the above can be further improved to almost perfect closure if the motion term of the atmospheric angular momentum (to be removed from the observed LOD variation in obtaining the mass-induced LOD variation) is magnified by 7%, corroborating the finding of Chao and Yan (2010); and (3) the above quantities all agree remarkably well with the equivalent LOD variation determined via J2 variation from satellite laser ranging observations, indicating the usefulness of the latter as an independent measurement for mass transports in the geophysical fluids.

1972. Possible Spreadings of Buoyant Plumes and Local Coastline Sensitivities Using Flow Syntheses From 1992 to 2007

Authors: Ross Tulloch; Chris Hill; Oliver Jahn

Journal: Geophysical Monograph Series

Publisher: American Geophysical Union

DOI: 10.1029/2011gm001125

URL: <https://doi.org/10.1029/2011gm001125>

Publication Date: 2011-12-05

Volume/Issue: N/A

References: 93

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Spreading, Adjoint, Oil spill, Tracer, Loop Current (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1973. Interconnection of nitrogen fixers and iron in the Pacific Ocean: Theory and numerical simulations

Authors: S. Dutkiewicz; B. A. Ward; F. Monteiro; M. J. Follows

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gb004039

URL: <https://doi.org/10.1029/2011gb004039>

Publication Date: 2011-11-28

Volume/Issue: Volume 26, Issue 1

References: 51

Cited by: 99 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We examine the interplay between iron supply, iron concentrations and phytoplankton communities in the Pacific Ocean. We present a theoretical framework which considers the competition for iron and nitrogen resources between phytoplankton to explain where nitrogen fixing autotrophs (diazotrophs, which require higher iron quotas, and have slower maximum growth) can co-exist with other phytoplankton. The framework also indicates that iron and fixed nitrogen concentrations can be strongly controlled by the local phytoplankton community. Together with results from a three-dimensional numerical model, we characterize three distinct biogeochemical provinces: 1) where iron supply is very low diazotrophs are excluded, and iron-limited nondiazotrophic phytoplankton control the iron concentrations; 2) a transition region where nondiazotrophic phytoplankton are nitrogen limited and control the nitrogen concentrations, but the iron supply is still too low relative to nitrate to support diazotrophy; 3) where iron supplies increase further relative to the nitrogen source, diazotrophs and other phytoplankton coexist; nitrogen concentrations are controlled by nondiazotrophs and iron concentrations are controlled by diazotrophs. The boundaries of these three provinces are defined by the rate of supply of iron relative to the supply of fixed nitrogen. The numerical model and theory provide a useful tool to understand the state of, links between, and response to changes in iron supply and phytoplankton community structure that have been suggested by observations.

1974. Midlatitude North Atlantic heat transport: A time series based on satellite and drifter data

Authors: Will R. Hobbs; Joshua K. Willis

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007039

URL: <https://doi.org/10.1029/2011jc007039>

Publication Date: 2011-11-16

Volume/Issue: Volume 117, Issue C1

References: 48

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Using temperature, salinity, and displacement data from Argo floats combined with satellite sea surface height, a time series of the Atlantic meridional heat transport from January 2002 to August 2010 has been estimated for 41 degreesN. The calculation method is validated against hydrographic climatologies and output from the ECCO2 ocean data assimilation model, and the assumptions are shown to be reasonable; the greatest source of error is from the sparse distribution of Argo floats. The mean heat transport is 0.50 ± 0.1 PW, which is consistent with previous estimates made using surface flux data but is low compared estimates from hydrographic cruise data. Consistent with results from the RAPID array, the heat transport has a significant annual cycle and high degree of subannual variability, indicating that statistical uncertainty in previous calculations may have been underestimated. There is little evidence of a trend over the short period of available data. Correlations with sea surface temperature suggest clear physical relationships between heat transport and SST, even on the short time scales of available data.

1975. Observation of decadal change in the Atlantic meridional overturning circulation using 10 years of continuous transport data

Authors: Uwe Send; Matthias Lankhorst; Torsten Kanzow

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl049801

URL: <https://doi.org/10.1029/2011gl049801>

Publication Date: 2011-11-14

Volume/Issue: Volume 38, Issue 24

References: 23

Cited by: 76 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1976. Comparison of Earth rotation excitation in data-constrained and unconstrained atmosphere models

Authors: Lisa J. Neef; Katja Matthes

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jd016555

URL: <https://doi.org/10.1029/2011jd016555>

Publication Date: 2011-11-14

Volume/Issue: Volume 117, Issue D2

References: 48

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Helmholtz Association

License: N/A

Open Access: N/A

Abstract: Changes in Earth rotation are strongly related to fluctuations in the angular momentum of the atmosphere, and therefore contain integral information about the atmospheric state. Here we investigate the extent to which observed Earth rotation parameters can be used to evaluate and potentially constrain atmospheric models. This is done by comparing the atmospheric excitation function, computed geophysically from reanalysis data and climate model simulations constrained only by boundary forcings, to the excitation functions inferred from geodetic monitoring data. Model differences are assessed for subseasonal variations, the annual and semiannual cycles, interannual variations, and decadal-scale variations. Observed length-of-day anomalies on the subseasonal timescale are simulated well by the simulations that are constrained by meteorological data only, whereas the annual cycle in length-of-day is simulated well by all models. Interannual length-of-day variations are captured fairly well as long as a model has realistic, time-varying SST boundary conditions and QBO forcing. Observations of polar motion are most clearly relatable to atmospheric dynamics on subseasonal to annual timescales, though angular momentum budget closure is difficult to achieve even for data-constrained atmospheric simulations. Closure of the angular momentum budget on decadal timescales is difficult and strongly dependent on estimates of angular momentum fluctuations due to core-mantle interactions in the solid Earth.

1977. Dynamical Ocean Forcing of the Madden-Julian Oscillation at Lead Times of up to Five Months

Authors: Benjamin G. M. Webber (University of East Anglia); David P. Stevens (University of East Anglia); Adrian J. Matthews (University of East Anglia); Karen J. Heywood (University of East Anglia)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-11-00268.1

URL: <https://doi.org/10.1175/jcli-d-11-00268.1>

Publication Date: 2011-11-11

Volume/Issue: Volume 25, Issue 8

References: 58

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The authors show that a simple three-dimensional ocean model linearized about a resting basic state can accurately simulate the dynamical ocean response to wind forcing by the Madden-Julian oscillation (MJO). This includes the propagation of equatorial waves in the Indian Ocean, from the generation of oceanic equatorial Kelvin waves to the arrival of downwelling oceanic equatorial Rossby waves in the western Indian Ocean, where they have been shown to trigger MJO convective activity. Simulations with idealized wind forcing suggest that the latitudinal width of this forcing plays a crucial role in determining the potential for such feedbacks. Forcing the model with composite MJO winds accurately captures the global ocean response, demonstrating that the observed ocean dynamical response to the MJO can be interpreted as a linear response to surface wind forcing. The model is then applied to study "primary" Madden-Julian events, which are not immediately preceded by any MJO activity or by any apparent atmospheric triggers, but have been shown to coincide with the arrival of downwelling oceanic equatorial Rossby waves. Case study simulations show how this oceanic equatorial Rossby wave activity is partly forced by reflection of an oceanic equatorial Kelvin wave triggered by a westerly wind burst 140 days previously, and partly directly forced by easterly wind stress anomalies around 40 days prior to the event. This suggests predictability for primary Madden-Julian events on times scales of up to five months, following the reemergence of oceanic anomalies forced by winds almost half a year earlier.

1978. Numerical modeling of the effects of current and sea level variation on wave height and propagation

Authors: Sung Hyup You

Journal: KSCE Journal of Civil Engineering

Publisher: Elsevier BV

DOI: 10.1007/s12205-011-1105-0

URL: <https://doi.org/10.1007/s12205-011-1105-0>

Publication Date: 2011-11-04

Volume/Issue: Volume 15, Issue 8

References: 22

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1979. A model of the Arctic Ocean carbon cycle

Authors: M. Manizza; M. J. Follows; S. Dutkiewicz; D. Menemenlis; J. W. McClelland; C. N. Hill; B. J. Peterson; R. M. Key

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc006998

URL: <https://doi.org/10.1029/2011jc006998>

Publication Date: 2011-10-20

Volume/Issue: Volume 116, Issue C12

References: 266

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1980. Propagation pathways of classical Labrador Sea water from its source region to 26 degreesN

Authors: Erik van Sebille; Molly O. Baringer; William E. Johns; Christopher S. Meinen; Lisa M. Beal; M. Femke de Jong; Hendrik M. van Aken

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007171

URL: <https://doi.org/10.1029/2011jc007171>

Publication Date: 2011-10-17

Volume/Issue: Volume 116, Issue C12

References: 216

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1981. Regional sea level trends due to a Pacific trade wind intensification

Authors: Mark A. Merrifield (Department of Oceanography; University of Hawaii'i at Mānoa; Honolulu; Hawaii; USA); Mathew E. Maltrud (Los Alamos National Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl049576

URL: <https://doi.org/10.1029/2011gl049576>

Publication Date: 2011-10-13

Volume/Issue: Volume 38, Issue 21

References: 33

Cited by: 119 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1982. Progress and Prospects of U.S. Data Assimilation in Ocean Research

Authors: Pierre Lermusiaux; Paola Malanotte-Rizzoli; Detlef Stammer; James Carton; James Cummings; Andrew Moore

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2006.102

URL: <https://doi.org/10.5670/oceanog.2006.102>

Publication Date: 2011-10-11

Volume/Issue: Volume 19, Issue 1

References: N/A

Cited by: 27 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1983. Ocean Rossby waves as a triggering mechanism for primary Madden-Julian events

Authors: Benjamin G. M. Webber; Adrian J. Matthews; Karen J. Heywood; David P. Stevens

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.936

URL: <https://doi.org/10.1002/qj.936>

Publication Date: 2011-10-04

Volume/Issue: Volume 138, Issue 663

References: 49

Cited by: 62 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Madden-Julian Oscillation (MJO) is sporadic, with episodes of cyclical activity interspersed with inactive periods. However, it remains unclear what may trigger a Madden-Julian (MJ) event which is not immediately preceded by any MJO activity: a 'primary' MJ event. A combination of case-studies and composite analysis is used to examine the extent to which the triggering of primary MJ events might occur in response to ocean dynamics. The case-studies show that such events can be triggered by the arrival of a downwelling oceanic equatorial Rossby wave, which is shown to be associated with a deepening of the mixed layer and positive sea-surface temperature (SST) anomalies of the order of 0.5-1 degreesC. These SST anomalies are not attributable to forcing by surface fluxes which are weak for the case-studies analysed. Furthermore, composite analysis suggests that such forcing is consistently important for triggering primary events. The relationship is much weaker for successive events, due to the many other triggering mechanisms which operate during periods of cyclical MJO activity. This oceanic feedback mechanism is a viable explanation for the sporadic and broadband nature of the MJO. Additionally, it provides hope for forecasting MJ events during periods of inactivity, when MJO forecasts generally exhibit low skill. Copyright © 2011 Royal Meteorological Society

1984. Upper-ocean singular vectors of the North Atlantic climate with implications for linear predictability and variability

Authors: L. Zanna; P. Heimbach; A. M. Moore; E. Tziperman

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.937

URL: <https://doi.org/10.1002/qj.937>

Publication Date: 2011-10-03

Volume/Issue: Volume 138, Issue 663

References: 52

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The limits of predictability of the meridional overturning circulation (MOC) and upper-ocean temperatures due to errors in ocean initial conditions and model parametrizations are investigated in an idealized configuration of an ocean general circulation model (GCM). Singular vectors (optimal perturbations) are calculated using the GCM, its tangent linear and adjoint models to determine an upper bound on the predictability of North Atlantic climate. The maximum growth time-scales of MOC and upper-ocean temperature anomalies, excited by the singular vectors, are 18.5 and 13 years respectively and in part explained by the westward propagation of upper-ocean anomalies against the mean flow. As a result of the linear interference of non-orthogonal eigenmodes of the non-normal dynamics, the ocean dynamics are found to actively participate in the significant growth of the anomalies. An initial density perturbation of merely 0.02 kg m^{-3} is found to lead to a 1.7 Sv MOC anomaly after 18.5 years. In addition, Northern Hemisphere upper-ocean temperature perturbations can be amplified by a factor of 2 after 13 years. The growth of upper-ocean temperature and MOC anomalies is slower and weaker when excited by the upper-ocean singular vectors than when the deep ocean is perturbed. This leads to the conclusion that predictability experiments perturbing only the atmospheric initial state may overestimate the predictability time. Interestingly, optimal MOC and upper-ocean temperature excitations are only weakly correlated, thus limiting the utility of SST observations to infer MOC variability. The excitation of anomalies in this model might have a crucial impact on the variability and predictability of Atlantic climate. The limit of predictability of the MOC is found to be different from that of the upper-ocean heat content, emphasizing that errors in ocean initial conditions will affect various measures differently and such uncertainties should be carefully considered in decadal prediction experiments. Copyright © 2011 Royal Meteorological Society

1985. Ocean bottom pressure variability in the Antarctic Divergence Zone off Lützow-Holm Bay, East Antarctica

Authors: Hideaki Hayakawa; Kazuo Shibuya; Yuichi Aoyama; Yoshifumi Nogi; Koichiro Doi

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2011.09.005

URL: <https://doi.org/10.1016/j.dsr.2011.09.005>

Publication Date: 2011-09-29

Volume/Issue: Volume 60

References: 28

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1986. Interannual to Interdecadal Salinity Variations Observed Near Hawaii: Local and Remote Forcing by Surface Freshwater Fluxes

Authors: Roger Lukas; Fernando Santiago-Mandujano

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2008.66

URL: <https://doi.org/10.5670/oceanog.2008.66>

Publication Date: 2011-09-21

Volume/Issue: Volume 21, Issue 1

References: N/A

Cited by: 23 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1987. Circulation and deep water export of the subpolar North Atlantic during the 1990's

Authors: Friedrich A. Schott; Peter Brandt

Journal: Geophysical Monograph Series

Publisher: American Geophysical Union

DOI: 10.1029/173gm08

URL: <https://doi.org/10.1029/173gm08>

Publication Date: 2011-09-09

Volume/Issue: N/A

References: 99

Cited by: 14 publications

Language: N/A

Subjects: N/A

Keywords: Meridional, Circulation, Overturning

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1988. Isotropic Image Analysis for Improving CBR Forecasting

Authors: Aitor Mata; M. Dolores Muñoz; Emilio Corchado; Juan M. Corchado

Journal: Journal of Mathematical Imaging and Vision

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10851-011-0315-x

URL: <https://doi.org/10.1007/s10851-011-0315-x>

Publication Date: 2011-08-31

Volume/Issue: Volume 42, Issue 2-3

References: 54

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1989. Recent Advances in Arctic Ocean Studies Employing Models from the Arctic Ocean Model Intercomparison Project

Authors: Andrey Proshutinsky; Yevgeny Aksenov; Jaclyn Clement Kinney; Rüdiger Gerdes; Elena Golubeva; David Holland; Greg Holloway; Alexandra Jahn; Mark Johnson; Ekaterina Popova; Mike Steele; Eiji Watanabe

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2011.61

URL: <https://doi.org/10.5670/oceanog.2011.61>

Publication Date: 2011-08-26

Volume/Issue: Volume 24, Issue 3

References: N/A

Cited by: 46 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1990. Oceanic origin of a recent La Niña-like trend in the tropical Pacific

Authors: Liping Zhang; Lixin Wu; Lisan Yu

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-010-0129-6

URL: <https://doi.org/10.1007/s00376-010-0129-6>

Publication Date: 2011-08-18

Volume/Issue: Volume 28, Issue 5

References: 40

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1991. Seasonal and Interannual Variations of the SST above the Seychelles Dome

Authors: Takaaki Yokoi (University of Tokyo); Tomoki Tozuka (University of Tokyo); Toshio Yamagata (University of Tokyo)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-10-05001.1

URL: <https://doi.org/10.1175/jcli-d-10-05001.1>

Publication Date: 2011-08-12

Volume/Issue: Volume 25, Issue 2

References: 49

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The seasonal and interannual variations of the sea surface temperature (SST) above the Seychelles Dome (SD) are investigated using outputs from an OGCM. The SST warms from August to April and cools from May to July. The surface heat flux plays the most important role in the seasonal variation, and it is mostly due to shortwave radiation. The horizontal advection tends to warm the SST in austral winter owing to the southward Ekman heat transport associated with the Indian summer monsoon. The cooling by the vertical turbulent diffusion becomes most effective in austral summer owing to the thin mixed layer during that time. On the interannual time scale, the SST becomes anomalously warm (cool) when the SD is weak (strong). In contrast to the seasonal variation, the vertical diffusion plays the most important role and causes anomalous warming (cooling). This warming (cooling) is due to the anomalously warm (cold) water below the mixed layer as a result of the deeper (shallower) thermocline in response to ocean dynamics. Also, the cooling by the vertical diffusion becomes less (more) efficient, because the mixed layer is anomalously thick (thin). The horizontal advection contributes to the anomalous warming (cooling) due to the anomalous southward (northward) Ekman heat transport. On the other hand, the anomalous surface heat flux tends to cool (warm) the mixed layer, because the warming of the mixed layer by the shortwave radiation becomes less (more) efficient due to the anomalously thick (thin) mixed layer.

1992. A Shift in Western Tropical Pacific Sea Level Trends during the 1990s

Authors: Mark A. Merrifield (University of Hawaii at Manoa)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2011jcli3932.1

URL: <https://doi.org/10.1175/2011jcli3932.1>

Publication Date: 2011-08-11

Volume/Issue: Volume 24, Issue 15

References: 45

Cited by: 184 publications

Language: en

Subjects: N/A

Keywords: Pacific Ocean, Tides, ENSO, Sea level, Satellite observations

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Pacific Ocean sea surface height trends from satellite altimeter observations for 1993-2009 are examined in the context of longer tide gauge records and wind stress patterns. The dominant regional trends are high rates in the western tropical Pacific and minimal to negative rates in the eastern Pacific, particularly off North America. Interannual sea level variations associated with El Niño-Southern Oscillation events do not account for these trends. In the western tropical Pacific, tide gauge records indicate that the recent high rates represent a significant trend increase in the early 1990s relative to the preceding 40 years. This sea level trend shift in the western Pacific corresponds to an intensification of the easterly trade winds across the tropical Pacific. The wind change appears to be distinct from climate variations centered in the North Pacific, such as the Pacific decadal oscillation. In the eastern Pacific, tide gauge records exhibit higher-amplitude decadal fluctuations than in the western tropical Pacific, and the recent negative sea level trends are indistinguishable from these fluctuations. The shifts in trade wind strength and western Pacific sea level rate resemble changes in dominant global modes of outgoing longwave radiation and sea surface temperature. It is speculated that the western Pacific sea level response indicates a general strengthening of the atmospheric circulation over the tropical Pacific since the early 1990s that has developed in concert with recent warming trends.

1993. The Present and Future System for Measuring the Atlantic Meridional Overturning Circulation and Heat Transport

Authors: Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart
Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart
Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart
Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart Cunningham; Stuart
Cunningham; Stuart Cunningham; Stuart Cunningham

Journal: Proceedings of OceanObs'09: Sustained Ocean Observations and Information for Society

Publisher: European Space Agency

DOI: 10.5270/oceanobs09.cwp.21

URL: <https://doi.org/10.5270/oceanobs09.cwp.21>

Publication Date: 2011-08-10

Volume/Issue: N/A

References: N/A

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1994. Predicting habitat preferences for *Anthometrina adriani* (Echinodermata) on the East Antarctic continental shelf

Authors: LG Hemery; B Galton-Fenzi; N Améziane; MJ Riddle; SR Rintoul; RJ Beaman; AL Post; M Eléaume

Journal: Marine Ecology Progress Series

Publisher: Inter-Research Science Center

DOI: 10.3354/meps09330

URL: <https://doi.org/10.3354/meps09330>

Publication Date: 2011-08-10

Volume/Issue: Volume 441

References: 62

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Crinoid|Marine ecosystem

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1995. Effects of parameterized eddy stress on volume, heat, and freshwater transports through Fram Strait

Authors: Zeliang Wang; Greg Holloway; Charles Hannah

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006871

URL: <https://doi.org/10.1029/2010jc006871>

Publication Date: 2011-08-09

Volume/Issue: Volume 116

References: 41

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1996. Mechanisms controlling mixed-layer temperature variability in the eastern tropical Pacific on the intraseasonal timescale

Authors: D. J. Halkides; Lisanne E. Lucas (University Corporation for Atmospheric Research, Climate Program Office; NOAA; Silver Spring; Maryland; USA); Duane E. Waliser (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Raghu Murtugudde (University of Maryland)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl048545

URL: <https://doi.org/10.1029/2011gl048545>

Publication Date: 2011-08-08

Volume/Issue: Volume 38, Issue 17

References: 26

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1997. Oceans and ocean models as seen by current meters

Authors: Greg Holloway; An Nguyen; Zeliang Wang

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007044

URL: <https://doi.org/10.1029/2011jc007044>

Publication Date: 2011-08-05

Volume/Issue: Volume 116

References: 30

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1998. Self-attraction and loading effects on ocean mass redistribution at monthly and longer time scales

Authors: N. T. Vinogradova; R. M. Ponte; M. E. Tamisiea; K. J. Quinn; E. M. Hill; J. L. Davis

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007037

URL: <https://doi.org/10.1029/2011jc007037>

Publication Date: 2011-08-05

Volume/Issue: Volume 116, Issue C8

References: 36

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

1999. Temporal and spatial variability of noble gas tracers in the North Pacific

Authors: Taka Ito; Roberta C. Hamme; Steve Emerson

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006828

URL: <https://doi.org/10.1029/2010jc006828>

Publication Date: 2011-08-05

Volume/Issue: Volume 116, Issue C8

References: 38

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2000. Improving mass redistribution estimates by modeling ocean bottom pressure uncertainties

Authors: S.-E. Brunnabend; R. Rietbroek; R. Timmermann; J. Schröter; J. Kusche

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006617

URL: <https://doi.org/10.1029/2010jc006617>

Publication Date: 2011-08-05

Volume/Issue: Volume 116, Issue C8

References: 46

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2001. IPCC climate models do not capture Arctic sea ice drift acceleration: Consequences in terms of projected sea ice thinning and decline

Authors: P. Rampal; J. Weiss; C. Dubois; J.-M. Campin

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011jc007110

URL: <https://doi.org/10.1029/2011jc007110>

Publication Date: 2011-08-05

Volume/Issue: Volume 116

References: 42

Cited by: 112 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2002. A global relationship between the ocean water cycle and near-surface salinity

Authors: Lisan Yu

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006937

URL: <https://doi.org/10.1029/2010jc006937>

Publication Date: 2011-08-05

Volume/Issue: Volume 116, Issue C10

References: 59

Cited by: 189 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2003. Mechanisms of interannual steric sea level variability

Authors: C. G. Piecuch (Atmospheric and Environmental Research Inc.); R. M. Ponte (Atmospheric and Environmental Research Inc.)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl048440

URL: <https://doi.org/10.1029/2011gl048440>

Publication Date: 2011-08-04

Volume/Issue: Volume 38, Issue 15

References: 20

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Processes contributing to interannual steric sea level variability are studied over the period 1993-2004 using an observationally-constrained ocean state estimate produced by the ECCO ("Estimating the Circulation and Climate of the Ocean") consortium. The estimate's dynamical consistency allows for the comprehensive attribution of steric changes in terms of advection, diffusion, and surface buoyancy exchange processes. Steric variations are found to be owing more to oceanic transports than to local surface buoyancy fluxes. Advection is responsible for steric variability in the tropical Indian and Pacific oceans. At extratropical latitudes, advection and diffusion appear to be equally important. Local surface buoyancy fluxes can contribute in some regions (e.g., the tropical Atlantic). Analysis of the anomalous wind stress curl shows that extra-equatorial vertical advection is driven primarily by Ekman pumping. The complexity of the interannual steric budget suggests that anomalous sea level is probably not predictable on the basis of ocean memory alone. Furthermore, proper parameterizations of mixing processes and good estimates of wind-driven transports both appear to be very important to reliable projections of interannual sea level.

2004. Dynamically and Observationally Constrained Estimates of Water-Mass Distributions and Ages in the Global Ocean

Authors: Tim DeVries (University of California Los Angeles); François Primeau (University of California Irvine)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-10-05011.1

URL: <https://doi.org/10.1175/jpo-d-10-05011.1>

Publication Date: 2011-08-04

Volume/Issue: Volume 41, Issue 12

References: 58

Cited by: 157 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A data-constrained ocean circulation model is used to characterize the distribution of water masses and their ages in the global ocean. The model is constrained by the time-averaged temperature, salinity, and radiocarbon distributions in the ocean, as well as independent estimates of the mean sea surface height and sea surface heat and freshwater fluxes. The data-constrained model suggests that the interior ocean is ventilated primarily by water masses forming in the Southern Ocean. Southern Ocean waters, including those waters forming in the Antarctic and subantarctic regions, make up about 55% of the interior ocean volume and an even larger percentage of the deep-ocean volume. In the deep North Pacific, the ratio of Southern Ocean to North Atlantic waters is almost 3:1. Approximately 65% of interior ocean waters make first contact with the atmosphere in the Southern Ocean, further emphasizing the central role played by the Southern Ocean in the regulation of the earth's climate. Results of the age analysis suggest that the mean ventilation age of deep waters is greater than 1000 yr throughout most of the Indian and Pacific Oceans, reaching a maximum of about 1400-1500 yr in the middepth North Pacific. The mean time for deep waters to be reexposed at the surface also reaches a maximum of about 1400-1500 yr in the deep North Pacific. Together these findings suggest that the deep North Pacific can be characterized as a "holding pen" of stagnant and recirculating waters.

2005. The transient response of the Southern Ocean pycnocline to changing atmospheric winds

Authors: Daniel C. Jones (Colorado State University); Takamitsu Ito (Colorado State University); Nicole S. Lovenduski (Department of Atmospheric and Oceanic Sciences and Institute of Arctic and Alpine Research University of Colorado at Boulder Boulder Colorado USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl048145

URL: <https://doi.org/10.1029/2011gl048145>

Publication Date: 2011-08-04

Volume/Issue: Volume 38, Issue 15

References: 25

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The vertical density structure of the Southern Ocean is dynamically linked to wind stress at the surface, but the nature of this coupling is not fully understood. Observations from the last several decades show a significant increase in the strength of westerly winds over the Southern Ocean, but an appreciable change in the tilt of constant density surfaces (isopycnals) has not yet been detected there. Using a combination of theory and idealized numerical simulations, we show that the response of the density structure occurs on centennial timescales, making it difficult to detect significant changes with a few decades of hydrographic observations. Dynamic coupling between the circumpolar current and northern basins regulates the slow adjustment of the density structure. Our results provide a new interpretation for recent observations and highlight the importance of the interaction between regional Southern Ocean dynamics and global ocean circulation.

2006. Response of a Coupled Ocean-Atmosphere Model to Greenland Ice Melting

Authors: D. Stammer; N. Agarwal; P. Herrmann; A. Köhl; C. R. Mechoso

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-011-9142-2

URL: <https://doi.org/10.1007/s10712-011-9142-2>

Publication Date: 2011-08-02

Volume/Issue: Volume 32, Issue 4-5

References: 32

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2007. Resolving sea level contributions by identifying fingerprints in time-variable gravity and altimetry

Authors: R. Rietbroek; S.-E. Brunnabend; J. Kusche; J. Schröter

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2011.06.007

URL: <https://doi.org/10.1016/j.jog.2011.06.007>

Publication Date: 2011-07-29

Volume/Issue: Volume 59-60

References: 43

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2008. Ocean bottom pressure variations estimated from gravity, nonsteric sea surface height and hydrodynamic model simulations

Authors: F. Siegismund (University of Hamburg); V. Romanova (Potsdam Institut für Klimafolgenforschung Potsdam Germany, University of Hamburg); A. Köhl (University of Hamburg); D. Stammer (University of Hamburg)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006727

URL: <https://doi.org/10.1029/2010jc006727>

Publication Date: 2011-07-26

Volume/Issue: Volume 116, Issue C7

References: 51

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean bottom pressure variability is analyzed from three monthly products available from (1) the Gravity Recovery and Climate Experiment (GRACE), (2) sterically corrected altimetry, and (3) from a forward run of the German part of the Estimating the Circulation and Climate of the Ocean (GECCO-2) model. Results lead to an approximate error estimate for each of the ocean bottom pressure (OBP) maps under the assumption of noncorrelated errors among the three products. The estimated error maps are consistent with the misfits of individual fields against OBP sensor data, with the caveat that a general underestimation of the signal strength, as a common, correlated error in all products, cannot be recovered by the method. The signal-to-noise ratio (SNR) increases in all products, when a 3 month running mean filter is applied. Using this filter, we estimate globally averaged errors of 8.6, 11.1, and 5.7 mm of equivalent water height for GRACE, nonsteric altimetry, and GECCO2, respectively. Based on resulting uncertainties, a new OBP product is being produced by merging all three data sets. When validated with bottom pressure observations this new OBP product has a 20% increased SNR compared to the best individual product (GECCO2-ref). Estimated total ocean mass variations explain a considerable part of OBP variability with a SNR above 1 in most of the ocean. In some regions the nonuniform part is weaker than the estimated error. However, most dynamic ocean models are designed to reproduce only the nonuniform, dynamic, OBP variability, but do not accurately describe total mass variability.

2009. Numerical modeling with application to tracking marine debris

Authors: James T. Potemra

Journal: Marine Pollution Bulletin

Publisher: Elsevier BV

DOI: 10.1016/j.marpolbul.2011.06.026

URL: <https://doi.org/10.1016/j.marpolbul.2011.06.026>

Publication Date: 2011-07-25

Volume/Issue: Volume 65, Issue 1-3

References: 11

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2010. The Global General Circulation of the Ocean Estimated by the ECCO-Consortium

Authors: Carl Wunsch; Patrick Heimbach; Rui Ponte; Ichiro Fukumori

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.41

URL: <https://doi.org/10.5670/oceanog.2009.41>

Publication Date: 2011-07-21

Volume/Issue: Volume 22, Issue 2

References: N/A

Cited by: 160 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2011. Assimilation of geodetic dynamic ocean topography using ensemble based Kalman filter

Authors: Tijana Janjić; Jens Schröter; Alberta Albertella; Wolfgang Bosch; Reiner Rummel; Roman Savcenko; Joachim Schwabe; Mirko Scheinert

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2011.07.001

URL: <https://doi.org/10.1016/j.jog.2011.07.001>

Publication Date: 2011-07-21

Volume/Issue: Volume 59-60

References: 44

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2012. The National Oceanographic Partnership Program: A Decade of Impacts on Oceanography

Authors: Cynthia Decker; Colin Reed

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.50

URL: <https://doi.org/10.5670/oceanog.2009.50>

Publication Date: 2011-07-21

Volume/Issue: Volume 22, Issue 2

References: N/A

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2013. GODAE Systems in Operation

Authors: Eric Dombrowsky; Laurent Bertino; Gary Brassington; Eric Chassignet; Fraser Davidson; Harley Hurlburt; Masafumi Kamachi; Tong Lee; Matthew Martin; Shan Mei; Marina Tonani

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.68

URL: <https://doi.org/10.5670/oceanog.2009.68>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 79 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2014. APPLICATIONS IN COASTAL MODELING AND FORECASTING

Authors: Pierre De Mey; Peter Craig; Fraser Davidson; Christopher Edwards; Yoichi Ishikawa; John Kindle; Roger Proctor; Keith Thompson; Jiang Zhu

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.79

URL: <https://doi.org/10.5670/oceanog.2009.79>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 12 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2015. Ocean Data Assimilation Systems for GODAE

Authors: James Cummings; Laurent Bertino; Pierre Brasseur; Ichiro Fukumori; Masafumi Kamachi; Matthew Martin; Kristian Mogensen; Peter Oke; Charles Testut; Jacques Verron; Anthony Weaver

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.69

URL: <https://doi.org/10.5670/oceanog.2009.69>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 70 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2016. Ocean State Estimation for Climate Research

Authors: Tong Lee; Toshiyuki Awaji; Magdalena Balmaseda; Eric Greiner; Detlef Stammer

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.74

URL: <https://doi.org/10.5670/oceanog.2009.74>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 39 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2017. Ocean Initialization for Seasonal Forecasts

Authors: Magdalena Balmaseda; Oscar Alves; Alberto Arribas; Toshiyuki Awaji; David Behringer; Nicolas Ferry; Yosuke Fujii; Tong Lee; Michele Rienecker; Tony Rosati; Detlef Stammer

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.73

URL: <https://doi.org/10.5670/oceanog.2009.73>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 58 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2018. Integrating Biogeochemistry and Ecology Into Ocean Data Assimilation Systems

Authors: Pierre Brasseur; Nicolas Gruber; Rosa Barciela; Keith Brander; Maeva Doron; Abdelali El Moussaoui; Alistair Hobday; Martin Huret; Anee-Sophie Kremer; Patrick Lehodey; Richard Matear; Cyril Moulin; Ragu Murtugudde; Inna Senina; Einar Svendsen

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.80

URL: <https://doi.org/10.5670/oceanog.2009.80>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 61 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2019. GODAE: The Global Ocean Data Assimilation Experiment

Authors: Michael Bell; Michel Lefèvre; Pierre-Yves Le Traon; Neville Smith; Kirsten Wilmer-Becker

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.62

URL: <https://doi.org/10.5670/oceanog.2009.62>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 72 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2020. Serving GODAE Data and Products to the Ocean Community

Authors: Jon Blower; Frederique Blanc; Mike Clancy; Peter Cornillon; Craig Donlon; Peter Hacker; Keith Haines; Steve Hankin; Thomas Loubrieu; Sylvie Pouliquen; Martin Price; Tim Pugh; Ashwanth Srinivasan

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2009.67

URL: <https://doi.org/10.5670/oceanog.2009.67>

Publication Date: 2011-07-20

Volume/Issue: Volume 22, Issue 3

References: N/A

Cited by: 11 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2021. On the Patterns of Wind-Power Input to the Ocean Circulation

Authors: Fabien Roquet (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology); Gurvan Madec (Laboratoire d'Océanographie Dynamique et de Climatologie, Paris, France)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo-d-11-024.1

URL: <https://doi.org/10.1175/jpo-d-11-024.1>

Publication Date: 2011-07-19

Volume/Issue: Volume 41, Issue 12

References: 22

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Pathways of wind-power input into the ocean general circulation are analyzed using Ekman theory. Direct rates of wind work can be calculated through the wind stress acting on the surface geostrophic flow. However, because that energy is transported laterally in the Ekman layer, the injection into the geostrophic interior is actually controlled by Ekman pumping, with a pattern determined by the wind curl rather than the wind itself. Regions of power injection into the geostrophic interior are thus generally shifted poleward compared to regions of direct wind-power input, most notably in the Southern Ocean, where on average energy enters the interior 10 degrees south of the Antarctic Circumpolar Current core. An interpretation of the wind-power input to the interior is proposed, expressed as a downward flux of pressure work. This energy flux is a measure of the work done by the Ekman pumping against the surface elevation pressure, helping to maintain the observed anomaly of sea surface height relative to the global-mean sea level.

2022. High frequency variability of the Atlantic meridional overturning circulation

Authors: B. Balan Sarojini; J. M. Gregory; R. Tailleux; G. R. Bigg; A. T. Blaker; D. R. Cameron; N. R. Edwards; A. P. Megann; L. C. Shaffrey; B. Sinha

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-7-471-2011

URL: <https://doi.org/10.5194/os-7-471-2011>

Publication Date: 2011-07-18

Volume/Issue: Volume 7, Issue 4

References: 54

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We compare the variability of the Atlantic meridional overturning circulation (AMOC) as simulated by the coupled climate models of the RAPID project, which cover a wide range of resolution and complexity, and observed by the RAPID/MOCHA array at about 26 degrees N. We analyse variability on a range of timescales, from five-daily to interannual. In models of all resolutions there is substantial variability on timescales of a few days; in most AOGCMs the amplitude of the variability is of somewhat larger magnitude than that observed by the RAPID array, while the time-mean is within about 10 % of the observational estimate. The amplitude of the simulated annual cycle is similar to observations, but the shape of the annual cycle shows a spread among the models. A dynamical decomposition shows that in the models, as in observations, the AMOC is predominantly geostrophic (driven by pressure and sea-level gradients), with both geostrophic and Ekman contributions to variability, the latter being exaggerated and the former underrepresented in models. Other ageostrophic terms, neglected in the observational estimate, are small but not negligible. The time-mean of the western boundary current near the latitude of the RAPID/MOCHA array has a much wider model spread than the AMOC does, indicating large differences among models in the simulation of the wind-driven gyre circulation, and its variability is unrealistically small in the models. In many RAPID models and in models of the Coupled Model Intercomparison Project Phase 3 (CMIP3), interannual variability of the maximum of the AMOC wherever it lies, which is a commonly used model index, is similar to interannual variability in the AMOC at 26 degrees N. Annual volume and heat transport timeseries at the same latitude are well-correlated within 15-45 degrees N, indicating the climatic importance of the AMOC. In the RAPID and CMIP3 models, we show that the AMOC is correlated over considerable distances in latitude, but not the whole extent of the North Atlantic; consequently interannual variability of the AMOC at 50 degrees N, where it is particularly relevant to European climate, is not well-correlated with that of the AMOC at 26 degrees N, where it is monitored by the RAPID/MOCHA array.

2023. Seasonal cycle of near-surface freshwater budget in the western tropical Atlantic

Authors: V. Romanova (University of Hamburg, Potsdam Institute for Climate Impact Research Potsdam Germany); A. Köhl (University of Hamburg); D. Stammer (University of Hamburg)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006650

URL: <https://doi.org/10.1029/2010jc006650>

Publication Date: 2011-07-12

Volume/Issue: Volume 116, Issue C7

References: 28

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We investigate differences of the ocean response in the Amazon domain to the seasonal variability of the river discharge that are either introduced via assimilating climatological temperature and salinity or by specifying seasonally varying river runoff. The role of the seasonal cycle of the Amazon freshwater discharge for the evolution of the barrier layer (BL) in the western tropical Atlantic and on the freshwater budget is estimated. During the experiments, three different runoff fields are being applied, including a time-mean runoff, a seasonally varying runoff, and one that results from the GECCO assimilation approach. The simulation forced with a seasonal Amazon discharge appears to be closer to the constrained solution and moves away from the run with a constant runoff, demonstrating that the seasonal variability of the Amazon is an essential contributor in the freshwater forcing of the western tropical Atlantic. The modeled time-mean BL thickness seems to be overestimated by the model relative to the data. On the seasonal timescale, the simulated spatial mean BL is found to vary between 13 and 30 m, with a maximum occurring in July, following the Amazon high discharge period in May. Analyzing the freshwater content balance, we find integrated near-surface freshwater import from the western tropical Atlantic interior of around 0.20 Sv in October-November at 38 degreesW and cumulative freshwater export out of the domain with a maximum of around 0.4 Sv in June as an effect of the Amazon flood in May.

2024. Importance of Circulation Changes to Atlantic Heat Storage Rates on Seasonal and Interannual Time Scales

Authors: Christopher G. Piecuch (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-11-00123.1

URL: <https://doi.org/10.1175/jcli-d-11-00123.1>

Publication Date: 2011-07-11

Volume/Issue: Volume 25, Issue 1

References: 56

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: Optimization, Atlantic Ocean, Climate variability, Meridional overturning circulation, Energy transport

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean heat budgets and transports are diagnosed to elucidate the importance of general circulation changes to Atlantic Ocean heat storage rates. The focus is on low- and midlatitude regions and on seasonal and interannual time scales. An estimate of the ocean state over 1993-2004, produced by a coarse-resolution general circulation model fit to observations via the method of Lagrange multipliers, is used. Meridional heat transports are first decomposed into contributions from time-mean and time-variable velocity and temperature and second from zonally symmetric baroclinic (overturning, including Ekman) and zonally asymmetric (gyre and other spatially correlated) circulations. Heat storage rates are then ascribed to ocean-atmosphere heat exchanges, diffusive mixing, and advective processes related to the various components of the meridional heat transport. Results show that seasonal heat storage changes generally represent a local response to surface heat inputs, but seasonal advective changes are also important near the equator. Interannual heat storage rate anomalies are mostly due to advection in tropical regions, whereas both surface heat fluxes and advection contribute at higher latitudes. Low-latitude advection can be primarily attributed to zonally symmetric baroclinic circulations, but temperature variations and zonally asymmetric flows can contribute elsewhere. A relationship between interannual heat storage rates in the equatorial Atlantic's top 100 m and meridional heat transport associated with the zonally symmetric baroclinic flow is observed; however, due in part to the role of shallow advective processes at these latitudes, any direct relationship between sea surface temperature variability and heat transport changes associated with intermediate or deep meridional overturning circulations is not clear.

2025. Deciphering the ocean bottom pressure variation in the Logatchev hydrothermal field at the eastern flank of the Mid-Atlantic Ridge

Authors: Hans-Hermann Gennerich (University of Bremen); Heinrich Villinger (University of Bremen)

Journal: Geochemistry, Geophysics, Geosystems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gc003441

URL: <https://doi.org/10.1029/2010gc003441>

Publication Date: 2011-07-08

Volume/Issue: Volume 12, Issue 7

References: 64

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: seafloor deformation, gravity waves, ocean bottom pressure, Logatchev hydrothermal field, surface waves (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2026. An Investigation of the Initial Development of the Double-ITCZ Warm SST Biases in the CCSM

Authors: Hailong Liu (Chinese Academy of Sciences); Minghua Zhang (Institute for Terrestrial and Planetary Atmospheres, Stony Brook University, Stony Brook, New York); Wuyin Lin (Institute for Terrestrial and Planetary Atmospheres, Stony Brook University, Stony Brook, New York)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2011jcli4001.1

URL: <https://doi.org/10.1175/2011jcli4001.1>

Publication Date: 2011-07-08

Volume/Issue: Volume 25, Issue 1

References: 34

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper investigates the initial development of the double ITCZ in the Community Climate System Model version 3 (CCSM3) in the central Pacific. Starting from a resting initial condition of the ocean in January, the model developed a warm bias of sea surface temperature (SST) in the central Pacific from 5 degreesS to 10 degreesS in the first three months. This initial bias is caused by excessive surface shortwave radiation that is also present in the stand-alone atmospheric model. The initial bias is further amplified by biases in both surface latent heat flux and horizontal heat transport in the upper ocean. These biases are caused by the responses of surface winds to SST bias and the thermocline structure to surface wind curls. This study also showed that the warming biases in surface solar radiation and latent heat fluxes are seasonally offset by cooling biases from reduced solar radiation after the austral summer due to cloud responses and in the austral fall due to enhanced evaporation when the maximum SST is closest to the equator. The warming biases from the dynamic heat transport by ocean currents however stay throughout all seasons once they are developed, which are eventually balanced by enhanced energy exchange and penetration of solar radiation below the mixed layer. It was also shown that the equatorial cold tongue develops after the warm biases in the south-central Pacific, and the overestimation of surface shortwave radiation recurs in the austral summer in each year. The results provide a case study on the physical processes leading to the development of the double ITCZ. Applicability of the results in other models is discussed.

2027. Mixed layer modeling in the East Pacific warm pool during 2002

Authors: Luke P. Van Roekel; Eric D. Maloney

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-011-1130-1

URL: <https://doi.org/10.1007/s00382-011-1130-1>

Publication Date: 2011-07-08

Volume/Issue: Volume 38, Issue 11-12

References: 32

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2028. Accuracy of the International Terrestrial Reference Frame origin and Earth expansion

Authors: X. Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); X. Collilieux (Institut Géographique National; Champs-sur-Marne; France); Z. Altamimi (Institut Géographique National; Champs-sur-Marne; France); B. L. A. Vermeersen (Delft University of Technology); R. S. Gross (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); I. Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl047450

URL: <https://doi.org/10.1029/2011gl047450>

Publication Date: 2011-07-07

Volume/Issue: Volume 38, Issue 13

References: 22

Cited by: 53 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2029. Source water of two-pronged northward flow in the southern Taiwan Strait in summer

Authors: Huasheng Hong; Chen-Tung Arthur Chen; Yuwu Jiang; Jiann-Yuh Lou; Zhaozhang Chen; Jia Zhu

Journal: Journal of Oceanography

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10872-011-0036-1

URL: <https://doi.org/10.1007/s10872-011-0036-1>

Publication Date: 2011-07-04

Volume/Issue: Volume 67, Issue 4

References: 20

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2030. Ocean Temperature and Salinity Contributions to Global and Regional Sea-Level Change

Authors: John A. Church; Dean Roemmich; Catia M. Domingues; Josh K. Willis; Neil J. White; John E. Gilson; Detlef Stammer; Armin Köhl; Don P. Chambers; Felix W. Landerer; Jochem Marotzke; Jonathan M. Gregory; Tatsuo Suzuki; Anny Cazenave; Pierre-Yves Le Traon

Journal: Understanding Sea-Level Rise and Variability

Publisher: Wiley

DOI: 10.1002/9781444323276.ch6

URL: <https://doi.org/10.1002/9781444323276.ch6>

Publication Date: 2011-06-23

Volume/Issue: N/A

References: 118

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: Oceans, central component of climate systems - storing, transporting quantities of heat, Gravity Recovery and Climate Experiment (GRACE) satellite mission, 20th-century climate and project 21st-century climate simulations - using coupled AOGCMs, Expendable bathythermograph (XBT) - and upper-ocean temperature profiles, Estimating steric sea-level change - using ocean syntheses (and 5 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2031. A Comparison of Southern Ocean Air-Sea Buoyancy Flux from an Ocean State Estimate with Five Other Products

Authors: Ivana Cerovečki (Scripps Institution of Oceanography / University of California San Diego); Lynne D. Talley (Scripps Institution of Oceanography / University of California San Diego); Matthew R. Mazloff (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2011jccli3858.1

URL: <https://doi.org/10.1175/2011jccli3858.1>

Publication Date: 2011-06-22

Volume/Issue: Volume 24, Issue 24

References: 75

Cited by: 64 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The authors have intercompared the following six surface buoyancy flux estimates, averaged over the years 2005-07: two reanalyses [the recent ECMWF reanalysis (ERA-Interim; hereafter ERA), and the National Centers for Environmental Prediction (NCEP)-NCAR reanalysis 1 (hereafter NCEP1)], two recent flux products developed as an improvement of NCEP1 [the flux product by Large and Yeager and the Southern Ocean State Estimate (SOSE)], and two ad hoc air-sea flux estimates that are obtained by combining the NCEP1 or ERA net radiative fluxes with turbulent flux estimates using the Coupled Ocean-Atmosphere Response Experiment (COARE) 3.0 bulk formulas with NCEP1 or ERA input variables. The accuracy of SOSE adjustments of NCEP1 atmospheric fields (which SOSE uses as an initial guess and a constraint) was assessed by verification that SOSE reduces the biases in the NCEP1 fluxes as diagnosed by the Working Group on Air-Sea Fluxes (Taylor), suggesting that oceanic observations may be a valuable constraint to improve atmospheric variables. Compared with NCEP1, both SOSE and Large and Yeager increase the net ocean heat loss in high latitudes, decrease ocean heat loss in the subtropical Indian Ocean, decrease net evaporation in the subtropics, and decrease net precipitation in polar latitudes. The large-scale pattern of SOSE and Large and Yeager turbulent heat flux adjustment is similar, but the magnitude of SOSE adjustments is significantly larger. Their radiative heat flux adjustments patterns differ. Turbulent heat fluxes determined by combining COARE bulk formulas with NCEP1 or ERA should not be combined with unmodified NCEP1 or ERA radiative fluxes as the net ocean heat gain poleward of 25 degreesS becomes unrealistically large. The other surface flux products (i.e., NCEP1, ERA, Large and Yeager, and SOSE) balance more closely. Overall, the statistical estimates of the differences between the various air-sea heat flux products tend to be largest in regions with strong ocean mesoscale activity such as the Antarctic Circumpolar Current and the western boundary currents.

2032. Diagnosing the Observed Seasonal Cycle of Atlantic Subtropical Mode Water Using Potential Vorticity and Its Attendant Theorems

Authors: Guillaume Maze (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2011jpo4576.1

URL: <https://doi.org/10.1175/2011jpo4576.1>

Publication Date: 2011-06-22

Volume/Issue: Volume 41, Issue 10

References: 29

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Analyzed fields of ocean circulation and the flux form of the potential vorticity equation are used to map the creation and subsequent circulation of low potential vorticity waters known as subtropical mode water (STMW) in the North Atlantic. Novel mapping techniques are applied to (i) render the seasonal cycle and annual-mean mixed layer vertical flux of potential vorticity (PV) through outcrops and (ii) visualize the extraction of PV from the mode water layer in winter, over and to the south of the Gulf Stream. Both buoyancy loss and wind forcing contribute to the extraction of PV, but the authors find that the former greatly exceeds the latter. The subsequent path of STMW is also mapped using Bernoulli contours on isopycnal surfaces.

2033. The Regional Ocean Modeling System (ROMS) 4-dimensional variational data assimilation systems

Authors: Andrew M. Moore; Hernan G. Arango; Gregoire Broquet; Chris Edwards; Milena Veneziani; Brian Powell; Dave Foley; James D. Doyle; Dan Costa; Patrick Robinson

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2011.05.005

URL: <https://doi.org/10.1016/j.pocean.2011.05.005>

Publication Date: 2011-06-19

Volume/Issue: Volume 91, Issue 1

References: 33

Cited by: 79 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2034. Hydrological Excitation of Polar Motion Derived from GRACE Gravity Field Solutions

Authors: L. Seoane (CNRS/UMR8630, Observatoire de Paris, 61 avenue de l'Observatoire, 75014 Paris, France); J. Nastula (Space Research Center, Polish Academy of Sciences, Bartycka 18 A, 00-716 Warsaw, Poland); C. Bizouard (CNRS/UMR8630, Observatoire de Paris, 61 avenue de l'Observatoire, 75014 Paris, France); D. Gambis (CNRS/UMR8630, Observatoire de Paris, 61 avenue de l'Observatoire, 75014 Paris, France)

Journal: International Journal of Geophysics

Publisher: Wiley

DOI: 10.1155/2011/174396

URL: <https://doi.org/10.1155/2011/174396>

Publication Date: 2011-06-19

Volume/Issue: Volume 2011

References: 27

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Observatoire de Paris

License: Creative Commons

Open Access: N/A

Abstract: The influence of the continental water storage on the polar motion is not well known. Different models have been developed to evaluate these effects and compared to geodetic observations. However, previous studies have shown large discrepancies mainly attributed to the lack of global measurements of related hydrological parameters. Now, from the observations of the GRACE mission, we can estimate the polar motion excitation due to the global hydrology. Data processing of GRACE data is carried out by several centers of analysis, we focus on the new solution computed by the Groupe de Recherche de Géodésie Spatiale. At annual scales, excitations derived from GRACE data are in better agreement with geodetic observations than models estimates. The main contribution to the hydrological excitation comes from the monsoon climates regions where GRACE and models estimates are in a very good agreement. Still, the effect of the north high latitudes regions, where the principal areas of snow cover are found, cannot be neglected. At these regions, GRACE and models estimated contributions to polar motion excitations show significant discrepancies. Finally, GRACE-based excitations reveal the possible influence of water storage variations in exciting polar motion around the frequency of 3 cycles per year.

2035. The Regional Ocean Modeling System (ROMS) 4-dimensional variational data assimilation systems

Authors: Andrew M. Moore; Hernan G. Arango; Gregoire Broquet; Chris Edwards; Milena Veneziani; Brian Powell; Dave Foley; James D. Doyle; Dan Costa; Patrick Robinson

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2011.05.003

URL: <https://doi.org/10.1016/j.pocean.2011.05.003>

Publication Date: 2011-06-18

Volume/Issue: Volume 91, Issue 1

References: 47

Cited by: 126 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2036. The Regional Ocean Modeling System (ROMS) 4-dimensional variational data assimilation systems

Authors: Andrew M. Moore; Hernan G. Arango; Gregoire Broquet; Brian S. Powell; Anthony T. Weaver; Javier Zavala-Garay

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2011.05.004

URL: <https://doi.org/10.1016/j.pocean.2011.05.004>

Publication Date: 2011-06-18

Volume/Issue: Volume 91, Issue 1

References: 71

Cited by: 218 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2037. Why is Indian Ocean warming consistently?

Authors: Suryachandra A. Rao; Ashish R. Dhakate; Subodh K. Saha; Somnath Mahapatra; Hemantkumar S. Chaudhari; Samir Pokhrel; Sobhan K. Sahu

Journal: Climatic Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10584-011-0121-x

URL: <https://doi.org/10.1007/s10584-011-0121-x>

Publication Date: 2011-06-09

Volume/Issue: Volume 110, Issue 3-4

References: 33

Cited by: 129 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2038. Strategies to mitigate aliasing of loading signals while estimating GPS frame parameters

Authors: Xavier Collilieux; Tonie van Dam; Jim Ray; David Coulot; Laurent Métivier; Zuheir Altamimi

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-011-0487-6

URL: <https://doi.org/10.1007/s00190-011-0487-6>

Publication Date: 2011-06-07

Volume/Issue: Volume 86, Issue 1

References: 27

Cited by: 87 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2039. Can sea surface height be used to estimate oceanic transport variability?

Authors: V. O. Ivchenko; D. Sidorenko (Alfred Wegener Institute); S. Danilov (Alfred Wegener Institute); M. Losch (Alfred Wegener Institute); J. Schröter (Alfred Wegener Institute)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl047387

URL: <https://doi.org/10.1029/2011gl047387>

Publication Date: 2011-06-01

Volume/Issue: Volume 38, Issue 11

References: 16

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2040. Australian water mass variations from GRACE data linked to Indo-Pacific climate variability

Authors: David García-García; Caroline C. Ummenhofer; Victor Zlotnicki

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2011.04.007

URL: <https://doi.org/10.1016/j.rse.2011.04.007>

Publication Date: 2011-05-23

Volume/Issue: Volume 115, Issue 9

References: 45

Cited by: 51 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2041. Numerical effects on organic-matter sedimentation and remineralization in biogeochemical ocean models

Authors: I. Kriest; A. Oschlies

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2011.05.001

URL: <https://doi.org/10.1016/j.ocemod.2011.05.001>

Publication Date: 2011-05-21

Volume/Issue: Volume 39, Issue 3-4

References: 34

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2042. Altimetry Combined with Hydrography for Ocean Transport Estimation

Authors: Claire Gourcuff (Institut de Recherche pour le Développement (IRD)); Pascale Lherminier (Institut de Recherche pour le Développement (IRD)); Herlé Mercier (Institut de Recherche pour le Développement (IRD)); Pierre Yves Le Traon (Laboratoire d'Océanographie Spatiale, IFREMER, Brest, France)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2011jtecho818.1

URL: <https://doi.org/10.1175/2011jtecho818.1>

Publication Date: 2011-05-20

Volume/Issue: Volume 28, Issue 10

References: 44

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: Mass fluxes/transport, Sea level, Satellite observations

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A method to estimate mass and heat transports across hydrographic sections using hydrography together with altimetry data in a geostrophic inverse box model is presented. Absolute surface velocities computed from Archiving, Validation, and Interpretation of Satellite Oceanographic data (AVISO) altimetry products made up of a combination of sea surface height measurements and geoid estimate are first compared to ship acoustic Doppler current profiler (S-ADCP) measurements of the Observatoire de la Variabilité Interannuelle et Décennale (OVIDE) project along hydrographic sections repeated every 2 yr in summer from Portugal to Greenland. The RMS difference between S-ADCP and altimetry velocities averaged on distances of about 100 km accounts for 3.3 cm s⁻¹. Considering that the uncertainty of S-ADCP velocities is found at 1.5 cm s⁻¹, altimetry errors are estimated at 3 cm s⁻¹. Transports across OVIDE sections previously obtained using S-ADCP data to constrain the geostrophic inverse box model are used as a reference. The new method is found useful to estimate absolute transports across the sections, as well as part of their variability. Despite associated uncertainties that are about 50% larger than when S-ADCP is used, the results for the North Atlantic Current and heat transports, with uncertainties of 10%-15%, reproduce the already observed variability. The largest uncertainties are found in the estimates of the East Greenland Irminger Current (EGIC) transport (30%), induced by larger uncertainties associated with altimetry data at the western boundary.

2043. Ocean variations associated with fishing conditions for yellowfin tuna (*Thunnus albacares*) in the equatorial Atlantic Ocean

Authors: Kuo-Wei Lan (National Taiwan Ocean University); Ming-An Lee (National Taiwan Ocean University); Hsueh-Jung Lu (National Taiwan Ocean University); Wei-Juan Shieh (National Taiwan Ocean University, Taiwan Ocean Research Institute, National Applied Research Laboratories, 3F, 106, Ho-Ping E. Road, Section 2, Taipei 106, Taiwan, ROC); Wei-Kuan Lin (National Taiwan Ocean University); Szu-Chia Kao (National Taiwan Ocean University)

Journal: ICES Journal of Marine Science

Publisher: Oxford University Press (OUP)

DOI: 10.1093/icesjms/fsr045

URL: <https://doi.org/10.1093/icesjms/fsr045>

Publication Date: 2011-05-18

Volume/Issue: Volume 68, Issue 6

References: 52

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Lan, K-W., Lee, M-A., Lu, H-J., Shieh, W-J., Lin, W-K., and Kao, S-C. 2011. Ocean variations associated with fishing conditions for yellowfin tuna (*Thunnus albacares*) in the equatorial Atlantic Ocean. - ICES Journal of Marine Science, 68: 1063-1071. In this study, the Taiwanese longline (LL) fishery data were divided into two types: regular LL and deep LL. Furthermore, we collected environmental variables, such as sea surface temperature (SST), subsurface temperature, chlorophyll a concentration, net primary productivity, windspeed, and the north tropical Atlantic SST index (NTA) during the period 1998-2007 to investigate the relationship between LL catch data and oceanic environmental factors using principal component analysis (PCA). After the daily LL was separated into two types of LL, the results indicated that the deep LL was the major fishery catching yellowfin tuna (YFT) in the equatorial Atlantic Ocean. In 2003-2005, especially in 2005, the monthly catch by deep LL was double those of other years. The spatial distribution of the nominal catch per unit effort (cpue) by deep LL showed the maximum aggregation of YFT in waters with temperature above 24-25 degreesC. The YFT mainly aggregated in the equatorial Atlantic, extending east in the first and second quarters of the year. In the third quarter of the year, the SST decreased off West Africa and the YFT migrated westwards to 15 degreesW. Results of PCA indicated that higher subsurface water temperatures resulted in a deeper thermocline and caused a higher cpue of YFT, but the influence of NTA on the cpue of YFT seemed to be insignificant.

2044. Impact of sea surface salinity assimilation on coupled forecasts in the tropical Pacific

Authors: Eric Hackert; Joaquim Ballabrera-Poy; Antonio J. Busalacchi; Rong-Hua Zhang; Raghu Murtugudde

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006708

URL: <https://doi.org/10.1029/2010jc006708>

Publication Date: 2011-05-13

Volume/Issue: Volume 116, Issue C5

References: 95

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2045. Non-tidal ocean loading effects on geodetic GPS heights

Authors: S. D. P. Williams (National Oceanography Centre); N. T. Penna (Newcastle University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl046940

URL: <https://doi.org/10.1029/2011gl046940>

Publication Date: 2011-05-10

Volume/Issue: Volume 38, Issue 9

References: 28

Cited by: 90 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: GPS observations used in geophysical studies are not usually corrected for non-tidal ocean loading (NTOL) displacement. Here we investigate NTOL effects on 3-4 year GPS height time series from 17 sites around the southern North Sea, and compute the NTOL displacement according to two ocean models; the global ECCO model and a high resolution regional model, POLSSM, which covers the northwest European continental shelf. To assess the susceptibility of GPS height estimates to NTOL, reprocessed GIPSY PPP daily solutions are corrected for the resulting displacement, together with atmospheric pressure loading (ATML). We find that the displacements due to NTOL are comparable in size to ATML and the combined correction reduces the variance by 12-22 mm² (20-30% reduction in RMS). We find that POLSSM outperforms ECCO, with around an 11% greater variance reduction, and hence high resolution models are recommended to correct GPS heights for NTOL.

2046. Models: Tools for Synthesis in International Oceanographic Research Programs

Authors: Dennis McGillicuddy; Brad de Young; Scott Doney; Patricia Glibert; Detlef Stammer; Francisco Werner

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2010.28

URL: <https://doi.org/10.5670/oceanog.2010.28>

Publication Date: 2011-05-05

Volume/Issue: Volume 23, Issue 3

References: N/A

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2047. A TOGA Retrospective

Authors: Michael McPhaden; Antonio Busalacchi; David Anderson

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2010.26

URL: <https://doi.org/10.5670/oceanog.2010.26>

Publication Date: 2011-05-05

Volume/Issue: Volume 23, Issue 3

References: N/A

Cited by: 74 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2048. A regional ocean reanalysis system for coastal waters of China and adjacent seas

Authors: Guijun Han; Wei Li; Xuefeng Zhang; Dong Li; Zhongjie He; Xidong Wang; Xinrong Wu; Ting Yu; Jirui Ma

Journal: Advances in Atmospheric Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00376-010-9184-2

URL: <https://doi.org/10.1007/s00376-010-9184-2>

Publication Date: 2011-05-04

Volume/Issue: Volume 28, Issue 3

References: 25

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2049. Water Mass Exchange in the Southern Ocean in Coupled Climate Models

Authors: Stephanie M. Downes (Princeton University); Anand Gnanadesikan (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); Stephen M. Griffies (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); Jorge L. Sarmiento (Princeton University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2011jpo4586.1

URL: <https://doi.org/10.1175/2011jpo4586.1>

Publication Date: 2011-04-29

Volume/Issue: Volume 41, Issue 9

References: 63

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: Southern Ocean, Coupled models, Climate models

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The authors estimate water mass transformation rates resulting from surface buoyancy fluxes and interior diapycnal fluxes in the region south of 30 degreesS in the Estimating the Circulation and Climate of the Ocean (ECCO) model-based state estimation and three free-running coupled climate models. The meridional transport of deep and intermediate waters across 30 degreesS agrees well between models and observationally based estimates in the Atlantic Ocean but not in the Indian and Pacific, where the model-based estimates are much smaller. Associated with this, in the models about half the southward-flowing deep water is converted into lighter waters and half is converted to denser bottom waters, whereas the observationally based estimates convert most of the inflowing deep water to bottom waters. In the models, both Antarctic Intermediate Water (AAIW) and Antarctic Bottom Water (AABW) are formed primarily via an interior diapycnal transformation rather than being transformed at the surface via heat or freshwater fluxes. Given the small vertical diffusivity specified in the models in this region, the authors conclude that other processes such as cabbeling and thermobaricity must be playing an important role in water mass transformation. Finally, in the models, the largest contribution of the surface buoyancy fluxes in the Southern Ocean is to convert Upper Circumpolar Deep Water (UCDW) and AAIW into lighter Subantarctic Mode Water (SAMW).

2050. Estimating high frequency ocean bottom pressure variability

Authors: Katherine J. Quinn (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gl046537

URL: <https://doi.org/10.1029/2010gl046537>

Publication Date: 2011-04-27

Volume/Issue: Volume 38, Issue 8

References: 17

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2051. Arctic ice-ocean simulation with optimized model parameters: Approach and assessment

Authors: An T. Nguyen; Dimitris Menemenlis; Ronald Kwok

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006573

URL: <https://doi.org/10.1029/2010jc006573>

Publication Date: 2011-04-26

Volume/Issue: Volume 116, Issue C4

References: 83

Cited by: 126 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2052. Ocean Measurements from Space in 2025

Authors: Anthony Freeman; Victor Zlotnicki; Tim Liu; Ben Holt; Ron Kwok; Simon Yueh; Jorge Vazquez; Dave Siegel; Gary Lagerloef

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2010.12

URL: <https://doi.org/10.5670/oceanog.2010.12>

Publication Date: 2011-04-26

Volume/Issue: Volume 23, Issue 4

References: N/A

Cited by: 10 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2053. Sensitivity of superconducting gravimeters in central Europe on variations in regional river and drainage basins

Authors: C. Kroner; A. Weise

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-011-0471-1

URL: <https://doi.org/10.1007/s00190-011-0471-1>

Publication Date: 2011-04-25

Volume/Issue: Volume 85, Issue 10

References: 41

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: Gravimeter

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2054. Changes in upwelling and its water sources in the California Current System driven by different wind forcing

Authors: Hajoon Song; Arthur J. Miller; Bruce D. Cornuelle; Emanuele Di Lorenzo

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2011.03.001

URL: <https://doi.org/10.1016/j.dynatmoce.2011.03.001>

Publication Date: 2011-04-15

Volume/Issue: Volume 52, Issue 1-2

References: 41

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2055. Mechanisms controlling seasonal mixed layer temperature and salinity in the Southwestern Tropical Indian Ocean

Authors: Daria Halkides; Tong Lee

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2011.03.002

URL: <https://doi.org/10.1016/j.dynatmoce.2011.03.002>

Publication Date: 2011-04-14

Volume/Issue: Volume 51, Issue 3

References: 64

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2056. Multidecadal variability of the North Brazil Current and its connection to the Atlantic meridional overturning circulation

Authors: Dongxiao Zhang; Rym Msadek; Michael J. McPhaden; Tom Delworth

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006812

URL: <https://doi.org/10.1029/2010jc006812>

Publication Date: 2011-04-14

Volume/Issue: Volume 116, Issue C4

References: 52

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2057. What governs the North Atlantic salinity maximum in a global GCM?

Authors: Tangdong Qu (University of Hawaii at Manoa); Shan Gao (Chinese Academy of Sciences); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2011gl046757

URL: <https://doi.org/10.1029/2011gl046757>

Publication Date: 2011-04-07

Volume/Issue: Volume 38, Issue 7

References: 28

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2058. Will climate change mathematics (?)

Authors: C. K. R. T. Jones

Journal: IMA Journal of Applied Mathematics

Publisher: Oxford University Press (OUP)

DOI: 10.1093/imamat/hxr018

URL: <https://doi.org/10.1093/imamat/hxr018>

Publication Date: 2011-04-02

Volume/Issue: Volume 76, Issue 3

References: N/A

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2059. Comparison of the geophysical excitations of polar motion from the period: 1980.0-2009.0

Authors: Jolanta Nastula; Małgorzata Pańnicka; Barbara Kołaczek

Journal: Acta Geophysica

Publisher: Springer Science and Business Media LLC

DOI: 10.2478/s11600-011-0008-2

URL: <https://doi.org/10.2478/s11600-011-0008-2>

Publication Date: 2011-03-31

Volume/Issue: Volume 59, Issue 3

References: 25

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2060. Vorticity Structures in the Tropical Pacific from a Numerical Simulation

Authors: Clement Ubelmann (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2011jpo4507.1

URL: <https://doi.org/10.1175/2011jpo4507.1>

Publication Date: 2011-03-31

Volume/Issue: Volume 41, Issue 8

References: 25

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: Pacific Ocean, Vorticity, Numerical analysis/modeling, Tropics, Ocean modeling

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The small-scale variability of the tropical Pacific is studied with the simulations from a numerical model in terms of vorticity structures. A Lagrangian method based on the Okubo-Weiss parameter is used to identify the structures and track their main characteristics. Between 8 degreesS and 8 degreesN, the structure characteristics are spatially inhomogeneous compared to higher latitudes. They can be grouped into three categories: anticyclonic and cyclonic structures off the equator and the equatorial structures between 2 degreesS and 2 degreesN. They all have a strong annual cycle with maximum presence from September to March, except during strong El Niño years, when the number of structures becomes very low. Off the equator from 2 degrees to 8 degrees, the anticyclonic structures dominate, but with drastically different characteristics north and south of the equator. In the north, large nonlinear vortices develop (known as the tropical instability vortices) in phase with the 30-35-day oscillation related to an unstable first-meridional-mode Rossby waves. In the south, mostly fragmentary linear structures are present, with lower propagation speeds. The equatorial structures are mostly counterclockwise. The larger ones tend to be linear and are clearly associated with Yanai waves. The large majority of the cyclonic structures off the equator are also quite linear and smaller and less numerous than the anticyclonic structures. However, some of them are nonlinear with vorticity values higher than 2 times the Coriolis parameter.

2061. Adjoint Sensitivity of the Niño-3 Surface Temperature to Wind Forcing

Authors: Xuebin Zhang (Scripps Institution of Oceanography); Bruce Cornuelle (Scripps Institution of Oceanography); Dean Roemmich (Scripps Institution of Oceanography)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2011jcli3917.1

URL: <https://doi.org/10.1175/2011jcli3917.1>

Publication Date: 2011-03-23

Volume/Issue: Volume 24, Issue 16

References: 54

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Wind, ENSO, Regression analysis, Surface temperature, Forcing

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The evolution of sea surface temperature (SST) over the eastern equatorial Pacific plays a significant role in the intense tropical air-sea interaction there and is of central importance to the El Niño-Southern Oscillation (ENSO) phenomenon. Effects of atmospheric fields (especially wind stress) and ocean state on the eastern equatorial Pacific SST variations are investigated using the Massachusetts Institute of Technology general circulation model (MITgcm) and its adjoint model, which can calculate the sensitivities of a cost function (in this case the averaged 0-30-m temperature in the Niño-3 region during an ENSO event peak) to previous atmospheric forcing fields and ocean state going backward in time. The sensitivity of the Niño-3 surface temperature to monthly zonal wind stress in preceding months can be understood by invoking mixed layer heat balance, ocean dynamics, and especially linear equatorial wave dynamics. The maximum positive sensitivity of the Niño-3 surface temperature to local wind forcing usually happens ~1-2 months before the peak of the ENSO event and is hypothesized to be associated with the Ekman pumping mechanism. In model experiments, its magnitude is closely related to the subsurface vertical temperature gradient, exhibiting strong event-to-event differences with strong (weak) positive sensitivity during La Niña (strong El Niño) events. The adjoint sensitivity to remote wind forcing in the central and western equatorial Pacific is consistent with the standard hypothesis that the remote wind forcing affects the Niño-3 surface temperature indirectly by exciting equatorial Kelvin and Rossby waves and modulating thermocline depth in the Niño-3 region. The current adjoint sensitivity study is consistent with a previous regression-based sensitivity study derived from perturbation experiments. Finally, implication for ENSO monitoring and prediction is also discussed.

2062. Upper ocean state estimation in the Southern Ocean Gas Exchange Experiment region using the four-dimensional variational technique

Authors: S. Dwivedi; T. W. N. Haine; C. E. Del Castillo

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005615

URL: <https://doi.org/10.1029/2009jc005615>

Publication Date: 2011-03-05

Volume/Issue: Volume 116

References: 66

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2063. AMOC variations in 1979-2008 simulated by NCEP operational ocean data assimilation system

Authors: Boyin Huang; Yan Xue; Arun Kumar; David W. Behringer

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-011-1035-z

URL: <https://doi.org/10.1007/s00382-011-1035-z>

Publication Date: 2011-03-01

Volume/Issue: Volume 38, Issue 3-4

References: 43

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2064. Global surface mass from a new combination of GRACE, modelled OBP and reprocessed GPS data

Authors: R. Rietbroek; M. Fritsche; S.-E. Brunnabend; I. Daras; J. Kusche; J. Schröter; F. Flechtner; R. Dietrich

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2011.02.003

URL: <https://doi.org/10.1016/j.jog.2011.02.003>

Publication Date: 2011-02-24

Volume/Issue: Volume 59-60

References: 35

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft

License: N/A

Open Access: N/A

Abstract: N/A

2065. An ocean modelling and assimilation guide to using GOCE geoid products

Authors: K. Haines; J. A. Johannessen; P. Knudsen; D. Lea; M.-H. Rio; L. Bertino; F. Davidson; F. Hernandez

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-7-151-2011

URL: <https://doi.org/10.5194/os-7-151-2011>

Publication Date: 2011-02-23

Volume/Issue: Volume 7, Issue 1

References: 51

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We review the procedures and challenges that must be considered when using geoid data derived from the Gravity and steady-state Ocean Circulation Explorer (GOCE) mission in order to constrain the circulation and water mass representation in an ocean general circulation model. It covers the combination of the geoid information with time-mean sea level information derived from satellite altimeter data, to construct a mean dynamic topography (MDT), and considers how this complements the time-varying sea level anomaly, also available from the satellite altimeter. We particularly consider the compatibility of these different fields in their spatial scale content, their temporal representation, and in their error covariances. These considerations are very important when the resulting data are to be used to estimate ocean circulation and its corresponding errors. We describe the further steps needed for assimilating the resulting dynamic topography information into an ocean circulation model using three different operational forecasting and data assimilation systems. We look at methods used for assimilating altimeter anomaly data in the absence of a suitable geoid, and then discuss different approaches which have been tried for assimilating the additional geoid information. We review the problems that have been encountered and the lessons learned in order to help future users. Finally we present some results from the use of GRACE geoid information in the operational oceanography community and discuss the future potential gains that may be obtained from a new GOCE geoid.

2066. Mechanisms controlling the seasonal mixed-layer temperature and salinity of the Indonesian seas

Authors: Daria Halkides; Tong Lee; Shinichiro Kida

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-010-0374-3

URL: <https://doi.org/10.1007/s10236-010-0374-3>

Publication Date: 2011-02-17

Volume/Issue: Volume 61, Issue 4

References: 40

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2067. Advection and starvation cause krill (*Euphausia pacifica*) decreases in 2005 Northern California coastal populations: Implications from a model study

Authors: Jeffrey G. Dorman (University of California Berkeley); Thomas M. Powell (University of California Berkeley); William J. Sydeman (Farallon Institute for Advanced Ecosystem Research; Petaluma; California; USA); Steven J. Bograd (Southwest Fisheries Science Center, Environmental Research Division; NOAA; Pacific Grove; California; USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gl046245

URL: <https://doi.org/10.1029/2010gl046245>

Publication Date: 2011-02-16

Volume/Issue: Volume 38, Issue 4

References: 37

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2068. Poleward flows in the southern California Current System: Glider observations and numerical simulation

Authors: Robert E. Todd; Daniel L. Rudnick; Matthew R. Mazloff; Russ E. Davis; Bruce D. Cornuelle

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006536

URL: <https://doi.org/10.1029/2010jc006536>

Publication Date: 2011-02-16

Volume/Issue: Volume 116, Issue C2

References: 47

Cited by: 91 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2069. A subtropical fate awaited freshwater discharged from glacial Lake Agassiz

Authors: Alan Condron (University of Massachusetts Amherst); Peter Winsor (University of Alaska Fairbanks)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gl046011

URL: <https://doi.org/10.1029/2010gl046011>

Publication Date: 2011-02-09

Volume/Issue: Volume 38, Issue 3

References: 24

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2070. Mediterranean Sea level variations: Analysis of the satellite altimetric data, 1992-2008

Authors: M.I. Vigo; J.M. Sánchez-Reales; M. Trottini; B.F. Chao

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2011.02.002

URL: <https://doi.org/10.1016/j.jog.2011.02.002>

Publication Date: 2011-02-09

Volume/Issue: Volume 52, Issue 3-4

References: 30

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: Eddy

Funding: Ministerio de Educación, Cultura y Deporte, National Science Council

License: N/A

Open Access: N/A

Abstract: N/A

2071. Timescales and regions of the sensitivity of Atlantic meridional volume and heat transport: Toward observing system design

Authors: Patrick Heimbach; Carl Wunsch; Rui M. Ponte; Gael Forget; Chris Hill; Jean Utke

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2010.10.065

URL: <https://doi.org/10.1016/j.dsr2.2010.10.065>

Publication Date: 2011-02-07

Volume/Issue: Volume 58, Issue 17-18

References: 69

Cited by: 56 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2072. Seasonal variations in sea level induced by continental water mass: First results from GRACE

Authors: B. Wouters (Netherlands Royal Meteorological Institute; De Bilt Netherlands); R. E. M. Riva (Delft University of Technology); D. A. Lavallée (Delft University of Technology); J. L. Bamber (Bristol Glaciology Centre, School of Geographical Science; Bristol UK)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gl046128

URL: <https://doi.org/10.1029/2010gl046128>

Publication Date: 2011-02-03

Volume/Issue: Volume 38, Issue 3

References: 16

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2073. Evaluation of a present-day climate simulation with a new coupled atmosphere-ocean model GENMOM

Authors: J. R. Alder; S. W. Hostetler; D. Pollard; A. Schmittner

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-4-69-2011

URL: <https://doi.org/10.5194/gmd-4-69-2011>

Publication Date: 2011-02-02

Volume/Issue: Volume 4, Issue 1

References: 65

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We present a new, non-flux corrected AOGCM, GENMOM, that combines the GENESIS version 3 atmospheric GCM (Global Environmental and Ecological Simulation of Interactive Systems) and MOM2 (Modular Ocean Model version 2) nominally at T31 resolution. We evaluate GENMOM by comparison with reanalysis products (e.g., NCEP2) and three models used in the IPCC AR4 assessment. GENMOM produces a global temperature bias of 0.6 degreesC. Atmospheric features such as the jet stream structure and major semi-permanent sea level pressure centers are well simulated as is the mean planetary-scale wind structure that is needed to produce the correct position of stormtracks. Most ocean surface currents are reproduced except where they are not resolvable at T31 resolution. Overall, GENMOM captures reasonably well the observed gradients and spatial distributions of annual surface temperature and precipitation and the simulations are on par with other AOGCMs. Deficiencies in the GENMOM simulations include a warm bias in the surface temperature over the southern oceans, a split in the ITCZ and weaker-than-observed overturning circulation.

2074. Varied representation of the Atlantic Meridional Overturning across multidecadal ocean reanalyses

Authors: Ernesto Munoz; Ben Kirtman; Wilbert Weijer

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2010.10.064

URL: <https://doi.org/10.1016/j.dsr2.2010.10.064>

Publication Date: 2011-02-01

Volume/Issue: Volume 58, Issue 17-18

References: 38

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2075. Monitoring the Atlantic meridional overturning circulation

Authors: Darren Rayner; Joël J.-M. Hirschi; Torsten Kanzow; William E. Johns; Paul G. Wright; Eleanor Frajka-Williams; Harry L. Bryden; Christopher S. Meinen; Molly O. Baringer; Jochem Marotzke; Lisa M. Beal; Stuart A. Cunningham

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2010.10.056

URL: <https://doi.org/10.1016/j.dsr2.2010.10.056>

Publication Date: 2011-01-28

Volume/Issue: Volume 58, Issue 17-18

References: 64

Cited by: 128 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2076. Variations of the Earth's figure axis from satellite laser ranging and GRACE

Authors: Minkang Cheng; John C. Ries; Byron D. Tapley

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jb000850

URL: <https://doi.org/10.1029/2010jb000850>

Publication Date: 2011-01-26

Volume/Issue: Volume 116, Issue B1

References: 46

Cited by: 184 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2077. The 4-D structure of upwelling and Pearl River plume in the northern South China Sea during summer 2008 revealed by a data assimilation model

Authors: Yeqiang Shu; Dongxiao Wang; Jiang Zhu; Shiqiu Peng

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2011.01.002

URL: <https://doi.org/10.1016/j.ocemod.2011.01.002>

Publication Date: 2011-01-24

Volume/Issue: Volume 36, Issue 3-4

References: 39

Cited by: 70 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2078. Realistic test cases for limited area ocean modelling

Authors: M. Herzfeld; M. Schmidt; S.M. Griffies; Z. Liang

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2010.12.008

URL: <https://doi.org/10.1016/j.ocemod.2010.12.008>

Publication Date: 2011-01-19

Volume/Issue: Volume 37, Issue 1-2

References: 92

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2079. Decadal changes in the South Pacific western boundary current system revealed in observations and ocean state estimates

Authors: K. L. Hill; S. R. Rintoul; K. R. Ridgway; P. R. Oke

Journal: Journal of Geophysical Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005926

URL: <https://doi.org/10.1029/2009jc005926>

Publication Date: 2011-01-19

Volume/Issue: Volume 116, Issue C1

References: 30

Cited by: 68 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2080. Oceanic tracer and proxy time scales revisited

Authors: C. Siberlin; C. Wunsch

Journal: Climate of the Past

Publisher: Copernicus GmbH

DOI: 10.5194/cp-7-27-2011

URL: <https://doi.org/10.5194/cp-7-27-2011>

Publication Date: 2011-01-11

Volume/Issue: Volume 7, Issue 1

References: 29

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Quantifying time-responses of the ocean to tracer input is important to the interpretation of paleodata from sediment cores - because surface-injected tracers do not instantaneously spread throughout the ocean. To obtain insights into the time response, a computationally efficient state-transition matrix method is demonstrated and used to compute successive states of passive tracer concentrations in the global ocean. Times to equilibrium exceed a thousand years for regions of the global ocean outside of the injection and convective areas and concentration gradients give time-lags from hundreds to thousands of years between the Atlantic and Pacific abyss, depending on the injection region and the nature of the boundary conditions employed. Equilibrium times can be much longer than radiocarbon ages - both because the latter are strongly biased towards the youngest fraction of fluid captured in a sample, and because they represent distinct physical properties. Use of different boundary conditions - concentration, or flux - produces varying response times, with the latter depending directly upon pulse duration. With pulses, the sometimes very different transient approach to equilibrium in various parts of the ocean generates event identification problems.

2081. The understanding of length-of-day variations from satellite gravity and laser ranging measurements

Authors: Shuanggen Jin; L. J. Zhang; B. D. Tapley

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2010.04869.x

URL: <https://doi.org/10.1111/j.1365-246x.2010.04869.x>

Publication Date: 2011-01-10

Volume/Issue: Volume 184, Issue 2

References: 39

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: Earth rotation variations, Satellite geodesy, Time series analysis, Time variable gravity

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2082. Sea surface salinity variability in the tropical Indian Ocean

Authors: Bulusu Subrahmanyam; V.S.N. Murty; David M. Heffner

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2010.12.004

URL: <https://doi.org/10.1016/j.rse.2010.12.004>

Publication Date: 2011-01-08

Volume/Issue: Volume 115, Issue 3

References: 41

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2083. Assimilation of sea surface temperature in the Northwest Pacific Ocean and its marginal seas using the ensemble Kalman filter

Authors: Gwang-Ho Seo; Byoung-Ju Choi; Yang-Ki Cho; Young Ho Kim; Sangil Kim

Journal: Ocean Science Journal

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12601-010-0021-4

URL: <https://doi.org/10.1007/s12601-010-0021-4>

Publication Date: 2011-01-04

Volume/Issue: Volume 45, Issue 4

References: 45

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2084. CROS: A Contingency Response multi-agent system for Oil Spills situations

Authors: Aitor Mata; Juan M. Corchado; Dante I. Tapia

Journal: Applied Soft Computing

Publisher: Elsevier BV

DOI: 10.1016/j.asoc.2010.12.017

URL: <https://doi.org/10.1016/j.asoc.2010.12.017>

Publication Date: 2010-12-31

Volume/Issue: Volume 11, Issue 3

References: 48

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2085. Observed circulation in the Solomon Sea from SADCP data

Authors: Sophie Cravatte; Alexandre Ganachaud; Quoc-Phi Duong; William S. Kessler; Gérard Eldin; Pierre Dutrieux

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2010.12.015

URL: <https://doi.org/10.1016/j.pocean.2010.12.015>

Publication Date: 2010-12-29

Volume/Issue: Volume 88, Issue 1-4

References: 67

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2086. Testing an eddy-permitting model of the Southern Ocean carbon cycle against observations

Authors: M. Woloszyn; M. Mazloff; T. Ito

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2010.12.004

URL: <https://doi.org/10.1016/j.ocemod.2010.12.004>

Publication Date: 2010-12-24

Volume/Issue: Volume 39, Issue 1-2

References: 42

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2087. New estimates of the inertia tensor and rotation of the triaxial nonrigid Earth

Authors: Wei Chen (Wuhan University, State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing Wuhan China); WenBin Shen (Wuhan University, Key Laboratory of Geospace Environment and Geodesy Ministry of Education Wuhan China, State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing Wuhan China)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jb007094

URL: <https://doi.org/10.1029/2009jb007094>

Publication Date: 2010-12-22

Volume/Issue: Volume 115, Issue B12

References: 82

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Earth's rotation is perturbed by mass redistributions and relative motions within the Earth system, as well as by the torques from both the internal Earth and celestial bodies. The present study aims to establish a theory to incorporate all these factors perturbing the rotation state of the triaxial Earth, just like the traditional rotation theory of the axial-symmetric Earth. First of all, we reestimate the Earth's inertia tensor on the basis of two new gravity models, EIGEN-GL05C and EGM2008. Then we formulate the dynamic equations and obtain their normal modes for an Earth model with a triaxial anelastic mantle, a triaxial fluid core, and dissipative oceans. The periods of the Chandler wobble and the free core nutation are successfully recovered, being ~ 433 and ~ 430 mean solar days, respectively. Further, the Liouville equations and their general solutions for that triaxial nonrigid Earth are deduced. The Liouville equations are characterized by the complex frequency-dependent transfer functions, which incorporate the effects of triaxialities and deformations of both the mantle and the core, as well as the effects of the mantle anelasticity, the equilibrium, and dissipative ocean tides. Complex transfer functions just reflect the fact that decays and phase lags exist in the Earth's response to the periodic forcing. Our theory reduces to the traditional rotation theory of the axial-symmetric Earth when assuming rotational symmetry of the inertia tensor. Finally, the present theory is applied to the case of atmospheric-oceanic excitation. The effective atmospheric-oceanic angular momentum function $(AMF)\chi_{eff} = \chi_{eff1} + i\chi_{eff2}$ for the present theory is compared with the $AMF\chi_{effsym} = \chi_{eff1sym} + i\chi_{eff2sym}$ for the traditional theory and the observed $AMF\chi_{obs} = \chi_{1obs} + i\chi_{2obs}$; we find that the difference between χ_{eff} and χ_{effsym} is of a few milliseconds of arc (mas) and can sometimes exceed 10 mas. In addition, spectrum analyses indicate that χ_{eff} is in good agreement with χ_{effsym} and, further, show an increase of coherency with χ_{obs} especially in the low-frequency band. The obvious advantage of χ_{eff} in the low-frequency band with respect to χ_{effsym} is the critical support of the present theory. However, still better performance of our theory can be expected if the models of the

mantle anelasticity and oceanic dynamics were improved. Thus we conclude that the traditional Earth rotation theory should be revised and upgraded to include the effects of the Earth's triaxiality, the mantle anelasticity, and oceanic dynamics. The theory presented in this study might be more appropriate to describe the rotation of the triaxial Earth (or other triaxial celestial bodies such as Mars), though further studies are needed to incorporate the effects of the solid inner core and other possible influences.

2088. Quality Assessment of a 1985-2007 Mediterranean Sea Reanalysis

Authors: Mario Adani (Gruppo Nazionale di Oceanografia Operativa, INGV, Sezione di Bologna, Bologna, Italy); Srdjan Dobricic (Centro EuroMediterraneo per i Cambiamenti Climatici, Bologna, Italy); Nadia Pinardi (Department of Environmental Sciences, University of Bologna, Ravenna, Italy)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2010jtecho798.1

URL: <https://doi.org/10.1175/2010jtecho798.1>

Publication Date: 2010-12-15

Volume/Issue: Volume 28, Issue 4

References: 46

Cited by: 53 publications

Language: en

Subjects: N/A

Keywords: Variational analysis, Salinity, Temperature, Data assimilation, Mediterranean Sea

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A simulation and two reanalyses from 1985 to 2007 have been produced for the Mediterranean Sea using different assimilation schemes: a reduced-order optimal interpolation [System for Ocean Forecast and Analysis (SOFA)] and a three-dimensional variational scheme (OceanVar). The observational dataset consists of vertical temperature and salinity in situ profiles and along-track satellite sea level anomalies; daily mean fields of satellite sea surface temperature are used for correcting the air-sea fluxes. This paper assesses the quality of the reanalyses with respect to observations and the simulation. Both the SOFA and OceanVar schemes give very similar root-mean-square errors and biases for temperature and salinity fields compared with the assimilated observations. The largest errors are at the thermocline level and in regions of large eddy field variability. However, OceanVar gives 20% better results for sea level anomaly root-mean-square error.

2089. Steric and mass-induced sea level variations in the Mediterranean Sea revisited

Authors: D. García-García (University of Alicante); B. F. Chao (Academia Sinica); J.-P. Boy (University of Strasbourg, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005928

URL: <https://doi.org/10.1029/2009jc005928>

Publication Date: 2010-12-12

Volume/Issue: Volume 115, Issue C12

References: 52

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The total sea level variation (SLV) is the combination of steric and mass-induced SLV, whose exact shares are key to understanding the oceanic response to climate system changes. Total SLV can be observed by radar altimetry satellites such as TOPEX/POSEIDON and Jason 1/2. The steric SLV can be computed through temperature and salinity profiles from in situ measurements or from ocean general circulation models (OGCM), which can assimilate the said observations. The mass-induced SLV can be estimated from its time-variable gravity (TVG) signals. We revisit this problem in the Mediterranean Sea estimating the observed, steric, and mass-induced SLV, for the latter we analyze the latest TVG data set from the GRACE (Gravity Recovery and Climate Experiment) satellite mission launched in 2002, which is 3.5 times longer than in previous studies, with the application of a two-stage anisotropic filter to reduce the noise in high-degree and -order spherical harmonic coefficients. We confirm that the intra-annual total SLV are only produced by water mass changes, a fact explained in the literature as a result of the wind field around the Gibraltar Strait. The steric SLV estimated from the residual of "altimetry minus GRACE" agrees in phase with that estimated from OGCMs and in situ measurements, although showing a higher amplitude. The net water fluxes through both the straits of Gibraltar and Sicily have also been estimated accordingly.

2090. The Earth's variable Chandler wobble

Authors: C. Bizouard; F. Remus; S. B. Lambert; L. Seoane; D. Gambis

Journal: Astronomy & Astrophysics

Publisher: EDP Sciences

DOI: 10.1051/0004-6361/201015894

URL: <https://doi.org/10.1051/0004-6361/201015894>

Publication Date: 2010-12-09

Volume/Issue: Volume 526

References: 25

Cited by: 22 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2091. Effects of Antenna Cross-Polarization Coupling on the Brightness Temperature Retrieval at L-Band

Authors: Seung-Bum Kim; Frank J. Wentz; Gary S. E. Lagerloef

Journal: IEEE Transactions on Geoscience and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tgrs.2010.2087028

URL: <https://doi.org/10.1109/tgrs.2010.2087028>

Publication Date: 2010-11-30

Volume/Issue: Volume 49, Issue 5

References: 28

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2092. J2: An evaluation of new estimates from GPS, GRACE, and load models compared to SLR

Authors: D. A. Lavallée (Newcastle University, Delft University of Technology); P. Moore (Newcastle University); P. J. Clarke (Newcastle University); E. J. Petrie (Newcastle University); T. van Dam (University of Luxembourg); M. A. King (Newcastle University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gl045229

URL: <https://doi.org/10.1029/2010gl045229>

Publication Date: 2010-11-18

Volume/Issue: Volume 37, Issue 22

References: 24

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Changes in J2, resulting from past and present changes in Earth's climate, are traditionally observed by Satellite Laser ranging (SLR). Assuming an elastic Earth, it is possible to infer changes in J2 from changes in Earth's shape observed by GPS. We compare estimates of non-secular J2 changes from GPS, SLR, GRACE, and a load model. The GPS and SLR annual signals agree but are different (16%) to the load model. Subtraction of the load model removes the annual variation from GPS, SLR, and GRACE, and the semi-annual variation in GPS. The GPS and SLR long-term signals are highly correlated, but GPS is better correlated with the loading model. Subtraction of the load model removes the 1998 anomaly from the GPS J2 series but not completely from the SLR J2 series, suggesting that the SLR anomaly may not be entirely due to mass re-distribution as has been presumed.

2093. Towards Prediction of Decadal Climate Variability and Change

Authors: J. Murphy; V. Kattsov; N. Keenlyside; M. Kimoto; G. Meehl; V. Mehta; H. Pohlmann; A. Scaife; D. Smith

Journal: Procedia Environmental Sciences

Publisher: Elsevier BV

DOI: 10.1016/j.proenv.2010.09.018

URL: <https://doi.org/10.1016/j.proenv.2010.09.018>

Publication Date: 2010-11-18

Volume/Issue: Volume 1

References: 142

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2094. Understanding and Predicting Seasonal-to-Interannual Climate Variability - The Producer Perspective

Authors: T.N. Stockdale; O. Alves; G. Boer; M. Deque; Y. Ding; A. Kumar; K. Kumar; W. Landman; S. Mason; P. Nobre; A. Scaife; O. Tomoaki; W.T. Yun

Journal: Procedia Environmental Sciences

Publisher: Elsevier BV

DOI: 10.1016/j.proenv.2010.09.006

URL: <https://doi.org/10.1016/j.proenv.2010.09.006>

Publication Date: 2010-11-18

Volume/Issue: Volume 1

References: 107

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2095. Polar motion excited by atmosphere and ocean in multi-frequency bands

Authors: Wei Chen; Wenbin Shen

Journal: Geo-spatial Information Science

Publisher: Informa UK Limited

DOI: 10.1007/s11806-010-0405-8

URL: <https://doi.org/10.1007/s11806-010-0405-8>

Publication Date: 2010-11-11

Volume/Issue: Volume 13, Issue 4

References: 15

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Polar motion|Radio spectrum

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2096. Fueling export production: nutrient return pathways from the deep ocean and their dependence on the Meridional Overturning Circulation

Authors: J. B. Palter; J. L. Sarmiento; A. Gnanadesikan; J. Simeon; R. D. Slater

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-7-3549-2010

URL: <https://doi.org/10.5194/bg-7-3549-2010>

Publication Date: 2010-11-10

Volume/Issue: Volume 7, Issue 11

References: 66

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . In the Southern Ocean, mixing and upwelling in the presence of heat and freshwater surface fluxes transform subpycnocline water to lighter densities as part of the upward branch of the Meridional Overturning Circulation (MOC). One hypothesized impact of this transformation is the restoration of nutrients to the global pycnocline, without which biological productivity at low latitudes would be significantly reduced. Here we use a novel set of modeling experiments to explore the causes and consequences of the Southern Ocean nutrient return pathway. Specifically, we quantify the contribution to global productivity of nutrients that rise from the ocean interior in the Southern Ocean, the northern high latitudes, and by mixing across the low latitude pycnocline. In addition, we evaluate how the strength of the Southern Ocean winds and the parameterizations of subgridscale processes change the dominant nutrient return pathways in the ocean. Our results suggest that nutrients upwelled from the deep ocean in the Antarctic Circumpolar Current and subducted in Subantarctic Mode Water support between 33 and 75% of global export production between 30 degrees S and 30 degrees N. The high end of this range results from an ocean model in which the MOC is driven primarily by wind-induced Southern Ocean upwelling, a configuration favored due to its fidelity to tracer data, while the low end results from an MOC driven by high diapycnal diffusivity in the pycnocline. In all models, nutrients exported in the SAMW layer are utilized and converted rapidly (in less than 40 years) to remineralized nutrients, explaining previous modeling results that showed little influence of the drawdown of SAMW surface nutrients on atmospheric carbon concentrations.

2097. Effects of mixing on the subduction of South Pacific waters identified by a simulated passive tracer and its adjoint

Authors: Shan Gao; Tangdong Qu; Ichiro Fukumori

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2010.10.002

URL: <https://doi.org/10.1016/j.dynatmoce.2010.10.002>

Publication Date: 2010-11-05

Volume/Issue: Volume 51, Issue 1-2

References: 29

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2098. How well do coupled models replicate ocean energetics relevant to ENSO?

Authors: Jaclyn N. Brown; Alexey V. Fedorov; Eric Guilyardi

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-010-0926-8

URL: <https://doi.org/10.1007/s00382-010-0926-8>

Publication Date: 2010-11-04

Volume/Issue: Volume 36, Issue 11-12

References: 56

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2099. ESTIMATION OF THE EQUATORIAL PACIFIC SALINITY FIELD USING OCEAN DATA ASSIMILATION SYSTEMS

Authors: Y. FUJII; S. MATSUMOTO; M. KAMACHI; S. ISHIZAKI

Journal: Advances in Geosciences - A 6-Volume Set

Publisher: World Scientific Publishing Co. Pte. Ltd.

DOI: 10.1142/9789812838148_0012

URL: https://doi.org/10.1142/9789812838148_0012

Publication Date: 2010-11-04

Volume/Issue: N/A

References: N/A

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2100. Effects of Hydrographic Variability on the Spatial, Seasonal and Diel Diving Patterns of Southern Elephant Seals in the Eastern Weddell Sea

Authors: Martin Biuw; Ole Anders Nøst; Audun Stien; Qin Zhou; Christian Lydersen; Kit M. Kovacs

Journal: PLoS ONE

Publisher: Public Library of Science (PLoS)

DOI: 10.1371/journal.pone.0013816

URL: <https://doi.org/10.1371/journal.pone.0013816>

Publication Date: 2010-11-03

Volume/Issue: Volume 5, Issue 11

References: 101

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

2101. Corrections to ocean surface forcing in the California Current System using 4D variational data assimilation

Authors: G. Broquet; A.M. Moore; H.G. Arango; C.A. Edwards

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2010.10.005

URL: <https://doi.org/10.1016/j.ocemod.2010.10.005>

Publication Date: 2010-11-01

Volume/Issue: Volume 36, Issue 1-2

References: 51

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2102. Super-parameterization in ocean modeling: Application to deep convection

Authors: Jean-Michel Campin; Chris Hill; Helen Jones; John Marshall

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2010.10.003

URL: <https://doi.org/10.1016/j.ocemod.2010.10.003>

Publication Date: 2010-10-26

Volume/Issue: Volume 36, Issue 1-2

References: 40

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2103. Relation between length-of-day variation and angular momentum of geophysical fluids

Authors: Benjamin F. Chao (National Central University); Haoming Yan (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jb007024

URL: <https://doi.org/10.1029/2009jb007024>

Publication Date: 2010-10-20

Volume/Issue: Volume 115, Issue B10

References: 30

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The remarkable relation well observed between variation in the length of day (ΔLOD) and variation in the axial atmospheric angular momentum (ΔAAM) (plus the oceanic counterpart, to a much lesser extent) is a consequence of the conservation of angular momentum on planet Earth. Quantification of the exact ΔLOD - ΔAAM relation, which we seek in the present study, depends significantly on the extent to which the core participates, or is dynamically coupled with the mantle, in transmission of the axial ΔAAM from the mantle to the core. If, after conversion to the equivalent ΔLOD assuming core-mantle decoupling (as in current standard practice), the calculated axial ΔAAM (according to atmospheric general circulation models [GCMs]) is systematically greater than the observed ΔLOD , then we can conclude the presence, and, furthermore, estimate the strength, of the said dynamic core-mantle coupling. However, in this study we find the opposite instead, that the calculated ΔAAM (plus the small oceanic counterpart) is smaller than the observed ΔLOD by 10%-20% consistently across the intraseasonal and seasonal time scales. Our main logical conclusion is that the atmospheric GCMs in general underpredict the ΔAAM , by at least 10%-20% at the mentioned time scales, a fact of importance with respect to the assessment of GCMs. Therefore, the systematic discrepancy found between the ΔAAM -predicted and the observed ΔLOD masks the relevant information on core-mantle coupling that we seek.

2104. Estimation of the Atlantic inflow through the Strait of Gibraltar from climatological and in situ data

Authors: Javier Soto-Navarro (Physical Oceanography Group, Department of Applied Physics II University of Málaga Málaga Spain); Francisco Criado-Aldeanueva (Physical Oceanography Group, Department of Applied Physics II University of Málaga Málaga Spain); Jesús García-Lafuente (Physical Oceanography Group, Department of Applied Physics II University of Málaga Málaga Spain); Antonio Sánchez-Román (Physical Oceanography Group, Department of Applied Physics II University of Málaga Málaga Spain)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010jc006302

URL: <https://doi.org/10.1029/2010jc006302>

Publication Date: 2010-10-11

Volume/Issue: Volume 115, Issue C10

References: 33

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Atmospheric data from reanalysis, satellite, and experimental observations have been combined to calculate a four-year time series of the Atlantic inflow through the Strait of Gibraltar. The net flow through the strait, estimated from the Mediterranean water budget, and the Mediterranean outflow, estimated from currentmeter observations in Espartel sill (western Strait of Gibraltar) from October 2004 to January 2009, made it possible to estimate the Atlantic inflow as the sum of both of them. The obtained mean net flow is 0.038 ± 0.007 Sv, with a seasonal cycle of 0.042 ± 0.018 Sv annual amplitude and maximum in September. The Mediterranean outflow shows a seasonal signal with annual amplitude of 0.027 ± 0.015 Sv peaking in April (in absolute value), and a mean value of -0.78 ± 0.05 Sv. The resulting Atlantic inflow has a mean value of 0.81 ± 0.06 Sv and a seasonal cycle with annual amplitude of 0.034 ± 0.011 Sv, peaking in September, and high interannual variability. The inflow seasonal cycle is the result of a barotropic forcing associated with the cycle of the net flow, driven by the evaporative cycle, and a baroclinic forcing linked to the seasonal cycle of the reduced gravity that drives the exchange.

2105. Seasonal polar motion excitation from numerical models of atmosphere, ocean, and continental hydrosphere

Authors: H. Dobslaw (Section 1.3: Earth System Modeling Deutsches GeoForschungsZentrum Potsdam Germany); R. Dill (Section 1.3: Earth System Modeling Deutsches GeoForschungsZentrum Potsdam Germany); A. Grötzsch (Section 1.3: Earth System Modeling Deutsches GeoForschungsZentrum Potsdam Germany); A. Brzeziński (Warsaw University of Technology, Space Research Centre Polish Academy of Sciences Warsaw Poland); M. Thomas (Section 1.3: Earth System Modeling Deutsches GeoForschungsZentrum Potsdam Germany)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jb007127

URL: <https://doi.org/10.1029/2009jb007127>

Publication Date: 2010-10-07

Volume/Issue: Volume 115, Issue B10

References: 31

Cited by: 91 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Effective angular momentum functions from atmosphere, oceans, and terrestrial water storage are obtained from European Centre for Medium-Range Weather Forecasts atmospheric data and corresponding simulations with the Ocean Model for Circulation and Tides and the Land Surface and Discharge Model (LSDM). Mass exchanges among the subsystems are realized by means of freshwater fluxes, causing the total ocean mass to vary predominantly annually. Variations in total ocean mass affect the oceanic excitations of the annual wobble by almost 1 milliarc second (mas) for both prograde and retrograde components, whereas the motion term contributions of terrestrial water flow derived from LSDM are found to be 3 orders of magnitude smaller. Since differences to geodetic excitations are not substantially reduced and regional decompositions demonstrate the large spatial variability of contributions to seasonal polar motion excitation that compensate each other when integrated globally, it is concluded that the closure of the seasonal excitation budget is still inhibited by remaining model errors in all subsystems.

2106. Sensitivities of an adjoint, unstructured mesh, tidal model on the European Continental Shelf

Authors: Silvia Maßmann

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-010-0347-6

URL: <https://doi.org/10.1007/s10236-010-0347-6>

Publication Date: 2010-10-06

Volume/Issue: Volume 60, Issue 6

References: 53

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2107. Atmospheric angular momentum variations of Earth, Mars and Venus at seasonal time scales

Authors: Ö. Karatekin; O. de Viron; S. Lambert; V. Dehant; P. Rosenblatt; T. Van Hoolst; S. Le Maistre

Journal: Planetary and Space Science

Publisher: Elsevier BV

DOI: 10.1016/j.pss.2010.09.010

URL: <https://doi.org/10.1016/j.pss.2010.09.010>

Publication Date: 2010-09-30

Volume/Issue: Volume 59, Issue 10

References: 76

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency, Belgian Federal Science Policy Office

License: N/A

Open Access: N/A

Abstract: N/A

2108. Reduction of the usable wind-work on the general circulation by forced symmetric instability

Authors: L. N. Thomas (Department of Environmental Earth System Science Stanford University Stanford California USA); J. R. Taylor (Massachusetts Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gl044680

URL: <https://doi.org/10.1029/2010gl044680>

Publication Date: 2010-09-29

Volume/Issue: Volume 37, Issue 18

References: 19

Cited by: 82 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Winds aligned with geostrophic currents input energy into the ocean circulation. For baroclinic, surface-intensified currents, winds of this orientation (i.e., down-front winds) reduce the potential vorticity in the mixed layer, making the flow susceptible to symmetric instability (SI). SI is a submesoscale shear instability that draws its energy from baroclinic, geostrophic currents. High-resolution numerical simulations are used to demonstrate how forced SI driven by down-front winds extracts energy from the geostrophic flow at a rate proportional to the Ekman buoyancy flux, the dot product of the Ekman transport and the surface buoyancy gradient. The kinetic energy (KE) going into SI is converted to 3D turbulence through a secondary shear instability and ultimately dissipated at small scales. Thus under the conditions where winds input KE to the circulation, some of the KE of baroclinic currents is dissipated by this mechanism. The net result of this submesoscale sink of KE is to reduce the usable wind-work, i.e., the wind-induced rate of KE increase, by an amount that depends on the wind-stress and the change in geostrophic velocity across the mixed layer.

2109. Mixed Layer Lateral Eddy Fluxes Mediated by Air-Sea Interaction

Authors: Emily Shuckburgh (British Antarctic Survey, Cambridge, United Kingdom); Guillaume Maze (Laboratoire de Physique des Océans, IFREMER, Plouzané, France); David Ferreira (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology); Helen Jones (Massachusetts Institute of Technology); Chris Hill (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2010jpo4429.1

URL: <https://doi.org/10.1175/2010jpo4429.1>

Publication Date: 2010-09-22

Volume/Issue: Volume 41, Issue 1

References: 42

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The modulation of air-sea heat fluxes by geostrophic eddies due to the stirring of temperature at the sea surface is discussed and quantified. It is argued that the damping of eddy temperature variance by such air-sea fluxes enhances the dissipation of surface temperature fields. Depending on the time scale of damping relative to that of the eddying motions, surface eddy diffusivities can be significantly enhanced over interior values. The issues are explored and quantified in a controlled setting by driving a tracer field, a proxy for sea surface temperature, with surface altimetric observations in the Antarctic Circumpolar Current (ACC) of the Southern Ocean. A new, tracer-based diagnostic of eddy diffusivity is introduced, which is related to the Nakamura effective diffusivity. Using this, the mixed layer lateral eddy diffusivities associated with (i) eddy stirring and small-scale mixing and (ii) surface damping by air-sea interaction is quantified. In the ACC, a diffusivity associated with surface damping of a comparable magnitude to that associated with eddy stirring ($\sim 500 \text{ m}^2 \text{ s}^{-1}$) is found. In frontal regions prevalent in the ACC, an augmentation of surface lateral eddy diffusivities of this magnitude is equivalent to an air-sea flux of 100 W m^{-2} acting over a mixed layer depth of 100 m, a very significant effect. Finally, the implications for other tracer fields such as salinity, dissolved gases, and chlorophyll are discussed. Different tracers are found to have surface eddy diffusivities that differ significantly in magnitude.

2110. Optimal Excitation of Interannual Atlantic Meridional Overturning Circulation Variability

Authors: Laure Zanna (Harvard University); Patrick Heimbach (Massachusetts Institute of Technology); Andrew M. Moore (University of California Santa Cruz); Eli Tziperman (Harvard University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2010jcli3610.1

URL: <https://doi.org/10.1175/2010jcli3610.1>

Publication Date: 2010-09-22

Volume/Issue: Volume 24, Issue 2

References: 74

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The optimal excitation of Atlantic meridional overturning circulation (MOC) anomalies is investigated in an ocean general circulation model with an idealized configuration. The optimal three-dimensional spatial structure of temperature and salinity perturbations, defined as the leading singular vector and generating the maximum amplification of MOC anomalies, is evaluated by solving a generalized eigenvalue problem using tangent linear and adjoint models. Despite the stable linearized dynamics, a large amplification of MOC anomalies, mostly due to the interference of nonnormal modes, is initiated by the optimal perturbations. The largest amplification of MOC anomalies, found to be excited by high-latitude deep density perturbations in the northern part of the basin, is achieved after about 7.5 years. The anomalies grow as a result of a conversion of mean available potential energy into potential and kinetic energy of the perturbations, reminiscent of baroclinic instability. The time scale of growth of MOC anomalies can be understood by examining the time evolution of deep zonal density gradients, which are related to the MOC via the thermal wind relation. The velocity of propagation of the density anomalies, found to depend on the horizontal component of the mean flow velocity and the mean density gradient, determines the growth time scale of the MOC anomalies and therefore provides an upper bound on the MOC predictability time. The results suggest that the nonnormal linearized ocean dynamics can give rise to enhanced MOC variability if, for instance, overflows, eddies, and/or deep convection can excite high-latitude density anomalies in the ocean interior with a structure resembling that of the optimal perturbations found in this study. The findings also indicate that errors in ocean initial conditions or in model parameterizations or processes, particularly at depth, may significantly reduce the Atlantic MOC predictability time to less than a decade.

2111. Simulation of Mixed Layer Depth in the Northeast Pacific Utilizing Spectral Nudging

Authors: Shawn M. Donohue (Department of Physics, Royal Military College of Canada, Kingston, Ontario, Canada); Michael W. Stacey (Department of Physics, Royal Military College of Canada, Kingston, Ontario, Canada)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2010jpo4499.1

URL: <https://doi.org/10.1175/2010jpo4499.1>

Publication Date: 2010-09-22

Volume/Issue: Volume 41, Issue 3

References: 22

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A numerical model, the Parallel Ocean Program (POP), is used to run a 46-yr simulation of the North Pacific Ocean beginning in January 1960. The model has 0.25 degrees horizontal resolution and 28 vertical levels, and it employs spectral nudging, which, unlike standard nudging, nudges only specific frequency and wavenumber bands. This simulation is nudged to the mean and annual Levitus climatological potential temperature and salinity. The model was forced with National Centers for Environmental Prediction (NCEP) mean monthly winds, sea level pressure, net heat flux, and rain rate. The simulated mixed layer depths (MLD) suggest significant shoaling of the MLD between 1970 and 2006, with faster rates in the northern Gulf of Alaska and slower rates to the west and south of Line Papa. The rates are of similar magnitude to those found in past studies and are consistent with the observed freshening and warming of the upper waters in the Gulf of Alaska. The rates are not spatially uniform, and the simulated MLD in the northeast Pacific actually deepens with time at some locations. These regions of increase form zonal bands in the simulation. The simulated MLD at Ocean Weather Station Papa (OWSP) shoals on average, but it is located close to one of these deepening bands. On average, the simulated, low-frequency MLD at OWSP gives a good indication of the MLD along Line Papa and in the Gulf of Alaska near Line Papa's latitude. The correlation coefficient between the MLD at OWSP and the latitudinal average of the MLD within the greater Gulf of Alaska (with OWSP removed) is 0.7 at zero lag. The correlation coefficient between the MLD at OWSP and the latitudinal average along Line Papa alone (with OWSP removed) is 0.6 at zero lag. Observed variability of the MLD along Line Papa and at OWSP is reproduced by the model. However, there is considerable spatial variability in the simulated MLD in the Gulf of Alaska as a whole, so MLD variability at OWSP is not necessarily a good indicator of the MLD variability throughout the region. Over the span of the simulation, the low-frequency MLD variability in the Gulf of Alaska is better correlated to the North Pacific Gyre Oscillation (NPGO) than to either the Pacific decadal oscillation (PDO) or Southern Oscillation index (SOI).

2112. Quality assessment of GPS reprocessed terrestrial reference frame

Authors: Xavier Collilieux; Laurent Métivier; Zuheir Altamimi; Tonie van Dam; Jim Ray

Journal: GPS Solutions

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10291-010-0184-6

URL: <https://doi.org/10.1007/s10291-010-0184-6>

Publication Date: 2010-09-15

Volume/Issue: Volume 15, Issue 3

References: 42

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2113. Numerical Investigations of Seasonal and Interannual Variability of North Pacific Subtropical Mode Water and Its Implications for Pacific Climate Variability

Authors: Xujing Jia Davis (Woods Hole Oceanographic Institution); Lewis M. Rothstein (University of Rhode Island); William K. Dewar (Florida State University); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2010jcli3435.1

URL: <https://doi.org/10.1175/2010jcli3435.1>

Publication Date: 2010-09-13

Volume/Issue: Volume 24, Issue 11

References: 58

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: Seasonal variability, Interannual variability, North Pacific Ocean, Climate variability, Pacific decadal oscillation (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: North Pacific Subtropical Mode Water (NPSTMW) is an essential feature of the North Pacific subtropical gyre imparting significant influence on regional SST evolution on seasonal and longer time scales and, as such, is an important component of basin-scale North Pacific climate variability. This study examines the seasonal-to-interannual variability of NPSTMW, the physical processes responsible for this variability, and the connections between NPSTMW and basin-scale climate signals using an eddy-permitting 1979-2006 ocean simulation made available by the Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2). The monthly mean seasonal cycle of NPSTMW in the simulation exhibits three distinct phases: (i) formation during November-March, (ii) isolation during March-June, and (iii) dissipation during June-November--each corresponding to significant changes in upper-ocean structure. An interannual signal is also evident in NPSTMW volume and other characteristic properties with volume minima occurring in 1979, 1988, and 1999. This volume variability is correlated with the Pacific decadal oscillation (PDO) with zero time lag. Further analyses demonstrate the connection of NPSTMW to the basin-scale ocean circulation. With this, modulations of upper-ocean structure driven by the varying strength and position of the westerlies as well as the regional air-sea heat flux pattern are seen to contribute to the variability of NPSTMW volume on interannual time scales.

2114. Opposing decadal changes for the North Atlantic meridional overturning circulation

Authors: M. Susan Lozier; Vassil Roussenov; Mark S. C. Reed; Richard G. Williams

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/ngeo947

URL: <https://doi.org/10.1038/ngeo947>

Publication Date: 2010-09-13

Volume/Issue: Volume 3, Issue 10

References: 33

Cited by: 104 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2115. Distribution of diverse nitrogen fixers in the global ocean

Authors: F. M. Monteiro (University of Bristol, Massachusetts Institute of Technology); M. J. Follows (Massachusetts Institute of Technology); S. Dutkiewicz (Massachusetts Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009gb003731

URL: <https://doi.org/10.1029/2009gb003731>

Publication Date: 2010-09-09

Volume/Issue: Volume 24, Issue 3

References: 99

Cited by: 102 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We employ a global three-dimensional model to simulate diverse phytoplanktonic diazotrophs (nitrogen fixers) in the oceans. In the model, the structure of the marine phytoplankton community self-assembles from a large number of potentially viable physiologies. Amongst them, analogs of *Trichodesmium*, unicellular diazotrophs and diatom-diazotroph associations (DDA) are successful and abundant. The simulated biogeography and nitrogen fixation rates of the modeled diazotrophs compare favorably with a compilation of published observations, which includes both traditional and molecular measurements of abundance and activity of marine diazotrophs. In the model, the diazotroph analogs occupy warm subtropical and tropical waters, with higher concentrations and nitrogen fixation rates in the tropical Atlantic Ocean and the Arabian Sea/Northern Indian Ocean, and lower values in the tropical and subtropical South Pacific Ocean. The three main diazotroph types typically co-exist in the model, although *Trichodesmium* analogs dominate the diazotroph population in much of the North and tropical Atlantic Ocean and the Arabian Sea, while unicellular-diazotroph analogs dominate in the South Atlantic, Pacific and Indian oceans. This pattern reflects the relative degree of nutrient limitation by iron or phosphorus. The model suggests in addition that unicellular diazotrophs could add as much new nitrogen to the global ocean as *Trichodesmium*.

2116. Estimated Seasonal Cycle of North Atlantic Eighteen Degree Water Volume

Authors: Gaël Forget (Massachusetts Institute of Technology); Guillaume Maze (Massachusetts Institute of Technology); Martha Buckley (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2010jpo4257.1

URL: <https://doi.org/10.1175/2010jpo4257.1>

Publication Date: 2010-08-24

Volume/Issue: Volume 41, Issue 2

References: 21

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The seasonal cycle in the volume and formation rate of Eighteen Degree Water (EDW) in the North Atlantic is quantified over the 3-yr period from 2004 to 2006. The EDW layer is defined as all waters that have a temperature between 17 degrees and 19 degreesC. The study is facilitated by a synthesis of various observations--principally Argo profiles of temperature and salinity, sea surface temperature, and altimetry--using a general circulation model as an interpolation tool. The winter increase in EDW volume is most pronounced in February, peaking at about 8.6 Svy, where 1 Svy $\approx 3.15 \times 10^{13}$ m³ corresponding to a 1 Sv ($\text{Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) flow sustained for one year. This largely reflects winter EDW formation due to air-sea heat fluxes. Over the remainder of the year, newly created EDW is consumed by air-sea heat fluxes and ocean mixing, which roughly contribute 2/3 and 1/3, respectively. The authors estimate a net annual volume increase of 1.4 Svy, averaged over the 3-yr period. It is small compared to the amplitude of the seasonal cycle (8.6 Svy) and annual formation due to air-sea fluxes (4.6 Svy). The overall EDW layer volume thus appears to fluctuate around a stable point during the study period. An estimate of the full EDW volume budget is provided along with an uncertainty estimate of 1.8 Svy, and largely resolves apparent conflicts between previous estimates.

2117. Sea State Bias in Radar Altimetry Revisited

Authors: Jessica Hausman; Victor Zlotnicki

Journal: Marine Geodesy

Publisher: Informa UK Limited

DOI: 10.1080/01490419.2010.487804

URL: <https://doi.org/10.1080/01490419.2010.487804>

Publication Date: 2010-08-10

Volume/Issue: Volume 33, Issue sup1

References: 16

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2118. Impact of hydrographic data assimilation on the modelled Atlantic meridional overturning circulation

Authors: G. C. Smith; K. Haines; T. Kanzow; S. Cunningham

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-6-761-2010

URL: <https://doi.org/10.5194/os-6-761-2010>

Publication Date: 2010-08-05

Volume/Issue: Volume 6, Issue 3

References: 45

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Here we make an initial step toward the development of an ocean assimilation system that can constrain the modelled Atlantic Meridional Overturning Circulation (AMOC) to support climate predictions. A detailed comparison is presented of 1 degrees and 1/4 degrees resolution global model simulations with and without sequential data assimilation, to the observations and transport estimates from the RAPID mooring array across 26.5 degrees N in the Atlantic. Comparisons of modelled water properties with the observations from the merged RAPID boundary arrays demonstrate the ability of in situ data assimilation to accurately constrain the east-west density gradient between these mooring arrays. However, the presence of an unconstrained "western boundary wedge" between Abaco Island and the RAPID mooring site WB2 (16 km offshore) leads to the intensification of an erroneous southwards flow in this region when in situ data are assimilated. The result is an overly intense southward upper mid-ocean transport (0-1100 m) as compared to the estimates derived from the RAPID array. Correction of upper layer zonal density gradients is found to compensate mostly for a weak subtropical gyre circulation in the free model run (i.e. with no assimilation). Despite the important changes to the density structure and transports in the upper layer imposed by the assimilation, very little change is found in the amplitude and sub-seasonal variability of the AMOC. This shows that assimilation of upper layer density information projects mainly on the gyre circulation with little effect on the AMOC at 26 degrees N due to the absence of corrections to density gradients below 2000 m (the maximum depth of Argo). The sensitivity to initial conditions was explored through two additional experiments using a climatological initial condition. These experiments showed that the weak bias in gyre intensity in the control simulation (without data assimilation) develops over a period of about 6 months, but does so independently from the overturning, with no change to the AMOC. However, differences in the properties and volume transport of North Atlantic Deep Water (NADW) persisted throughout the 3 year simulations resulting in a difference of 3 Sv in AMOC intensity. The persistence of these dense water anomalies and their influence on the AMOC is promising for the development of decadal forecasting capabilities. The results suggest that the deeper waters must be accurately reproduced in order to constrain the AMOC.

2119. Estimates of global ocean volume transports during 1960 through 2001

Authors: Weiqiang Wang (University of Hamburg); Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2010gl043949

URL: <https://doi.org/10.1029/2010gl043949>

Publication Date: 2010-08-02

Volume/Issue: Volume 37, Issue 15

References: 22

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An estimate of global time-varying ocean volume transports is provided for the period 1960-2001 as it results from the German ECCO (GECCO) synthesis. Results confirm previously discussed mean-state ocean circulation systems, encompassing an upper meridional cell in the Atlantic with sinking in the North Atlantic connected to a lower meridional cell with sinking around Antarctica in the Pacific and Atlantic. Decadal to interdecadal variability as well as long-term trends of global ocean transport, analyzed over the course of the 42 years, appear as a reorganization of time-mean circulation structures, including the layer transports in the Antarctic Circumpolar Current (ACC) and the circulation involving the Indo-Pacific exchange.

2120. Effects of self-attraction and loading on annual variations of ocean bottom pressure

Authors: Nadya T. Vinogradova (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Mark E. Tamisiea (National Oceanography Centre); James L. Davis (Harvard-Smithsonian Center for Astrophysics Cambridge Massachusetts USA); Emma M. Hill (Harvard-Smithsonian Center for Astrophysics Cambridge Massachusetts USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005783

URL: <https://doi.org/10.1029/2009jc005783>

Publication Date: 2010-06-28

Volume/Issue: Volume 115, Issue C6

References: 23

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The impact of self-attraction and loading (SAL) on ocean bottom pressure ξ , an effect not previously considered, is analyzed in terms of the mean annual cycle based on decade-long estimates of changes in land hydrology, atmospheric pressure, and oceanic circulation. The SAL-related changes in ξ occur as a result of deformation of the crust due to loading and self-gravitation of the variable fluid loads. In the absence of SAL, net freshwater input and changes in mean atmospheric pressure over the ocean give rise to a spatially constant ξ annual cycle with an amplitude ~ 1 -2 cm in equivalent water thickness. Consideration of SAL physics introduces spatial variations that can be significant, particularly around continental boundaries, where the amplitude of deviations can exceed 1 cm. For the spatial variability induced by SAL effects, changes in both land hydrology and atmospheric pressure are important. Effects related to the changing ocean circulation are relatively weaker, apart from a few shallow coastal regions. Comparisons with a few in situ, deep ocean observations indicate that for the most accurate ξ estimates, one needs to consider spatially varying SAL-related signals, along with the effects of mean atmospheric pressure and net freshwater input into the oceans. Nevertheless, the most complete estimates, including also effects of ocean circulation, are able to account for only $\sim 1/3$ of the observed annual variances. Sources of the remaining contribution remain unclear.

2121. On the Reasons for the Formation and Variability of the Azores Current

Authors: Denis L. Volkov (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2010jpo4326.1

URL: <https://doi.org/10.1175/2010jpo4326.1>

Publication Date: 2010-06-22

Volume/Issue: Volume 40, Issue 10

References: 61

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Recent studies have shown that the formation of the well-defined, zonally oriented Azores Current may be the result of water mass transformation associated with the Mediterranean outflow in the Gulf of Cadiz. As the denser Mediterranean water descends down the continental slope, it entrains overlying North Atlantic Central Water. It is believed that the Azores Current then forms as part of the horizontal recirculating gyre generated through the beta-plume mechanism. In this study, the authors further explore this hypothesis by performing a series of numerical experiments. These experiments are based on a high-resolution general circulation model that includes the Mediterranean Sea and that realistically simulates the water mass exchange through the Strait of Gibraltar and the transport and variability of the Azores Current. The authors show that the divergence of the relative vorticity flux and the planetary vorticity flux, associated with planetary waves, are the main factors determining the variability of the Azores Current. It is shown experimentally that the closure of the Strait of Gibraltar leads to a complete disappearance of the Azores Current. On the other hand, with the open Strait of Gibraltar, the Azores Current persists even when the wind forcing over the region is turned off. The atmospheric forcing is thus not responsible for the formation of the Azores Current, but it affects the variability of the current with a minor effect on its magnitude. Numerical experiments suggest that the strength and the variability of the Azores Current depend on the magnitude of the water mass exchange through the Strait of Gibraltar but not on its seasonal variability.

2122. Mass contribution to Mediterranean Sea level variability for the period 1948-2000

Authors: F.M. Calafat; M. Marcos; D. Gomis

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2010.06.002

URL: <https://doi.org/10.1016/j.gloplacha.2010.06.002>

Publication Date: 2010-06-11

Volume/Issue: Volume 73, Issue 3-4

References: 30

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: Tide gauge|Mediterranean Basin

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2123. Thickness sensitivities in the CICE sea ice model

Authors: Elizabeth C. Hunke

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2010.05.004

URL: <https://doi.org/10.1016/j.ocemod.2010.05.004>

Publication Date: 2010-06-10

Volume/Issue: Volume 34, Issue 3-4

References: 39

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2124. The Obduction of Equatorial 13 degreesC Water in the Pacific Identified by a Simulated Passive Tracer*

Authors: Tangdong Qu (University of Hawaii at Manoa); Shan Gao (Chinese Academy of Sciences); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rana A. Fine (University of Miami); Eric J. Lindstrom (Science Mission Directorate, National Aeronautics and Space Administration, Washington, D.C)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2010jpo4358.1

URL: <https://doi.org/10.1175/2010jpo4358.1>

Publication Date: 2010-06-07

Volume/Issue: Volume 40, Issue 10

References: 42

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: Pacific ocean, Sea surface temperature, Ekman pumping/transport, Tracers, Mixed layer (and 2 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The obduction of equatorial 13 degreesC Water in the Pacific is investigated using a simulated passive tracer of the Consortium for Estimating the Circulation and Climate of the Ocean (ECCO). The result shows that the 13 degreesC Water initialized in the region 8 degreesN-8 degreesS, 130 degrees-90 degreesW enters the surface mixed layer in the eastern tropical Pacific, mainly through upwelling near the equator, in the Costa Rica Dome, and along the coast of Peru. Approximately two-thirds of this obduction occurs within 10 years after the 13 degreesC Water being initialized, with the upper portion of the water mass reaching the surface mixed layer in only about a month. The obduction of the 13 degreesC Water helps to maintain a cool sea surface temperature year-round, equivalent to a surface heat flux of about -6.0 W m⁻² averaged over the eastern tropical Pacific (15 degreesS-15 degreesN, 130 degreesW-eastern boundary) for the period of integration (1993-2006). During El Niño years, when the thermocline deepens as a consequence of the easterly wind weakening, the obduction of the 13 degreesC Water is suppressed, and the reduced vertical entrainment generates a warming anomaly of up to 10 W m⁻² in the eastern tropical Pacific and in particular along the coast of Peru, providing explanations for the warming of sea surface temperature that cannot be accounted for by local winds alone. The situation is reversed during La Niña years.

2125. Coupled assimilation for an intermediated coupled ENSO prediction model

Authors: Fei Zheng; Jiang Zhu

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-010-0307-1

URL: <https://doi.org/10.1007/s10236-010-0307-1>

Publication Date: 2010-06-07

Volume/Issue: Volume 60, Issue 5

References: 51

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2126. Toward a Midlatitude Ocean Frequency-Wavenumber Spectral Density and Trend Determination

Authors: Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2010jpo4376.1

URL: <https://doi.org/10.1175/2010jpo4376.1>

Publication Date: 2010-06-07

Volume/Issue: Volume 40, Issue 10

References: 70

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The time- and space-scale descriptive power of two-dimensional Fourier analysis is exploited to reanalyze the behavior of midlatitude variability as seen in altimetric data. These data are used to construct a purely empirical and analytical frequency-zonal wavenumber spectrum of ocean variability for periods between about 20 days and 15 yr and on spatial scales of about 200-10 000 km. The spectrum is dominated by motions along a "nondispersive" line, which is a robust feature of the data but for whose prominence a complete theoretical explanation is not available. The estimated spectrum also contains significant energy at all frequencies and wavenumbers in this range, including eastward-propagating motions, which are likely some combination of nonlinear spectral cascades, wave propagation, and wind-forced motions. The spectrum can be used to calculate statistical expectations of spatial average sea level and transport variations. However, because the statistics of trend determination in quantities such as sea level and volume transports depend directly upon the spectral limit of the frequency approaching zero, the appropriate significance calculations remain beyond reach, because low-frequency variability is indistinguishable from trends already present in the data.

2127. On the use of discrete adjoints in goal error estimation for shallow water equations

Authors: Florian Rauser; Jan Riehme; Klaus Leppkes; Peter Korn; Uwe Naumann

Journal: Procedia Computer Science

Publisher: Elsevier BV

DOI: 10.1016/j.procs.2010.04.013

URL: <https://doi.org/10.1016/j.procs.2010.04.013>

Publication Date: 2010-06-07

Volume/Issue: Volume 1, Issue 1

References: 17

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: International Max Planck Research School for Advanced Methods in Process and Systems Engineering, Engineering and Physical Sciences Research Council

License: N/A

Open Access: N/A

Abstract: N/A

2128. The Surface Water and Ocean Topography Mission: Observing Terrestrial Surface Water and Oceanic Submesoscale Eddies

Authors: Michael Durand; Lee-Lueng Fu; Dennis P. Lettenmaier; Douglas E. Alsdorf; Ernesto Rodriguez; Daniel Esteban-Fernandez

Journal: Proceedings of the IEEE

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/jproc.2010.2043031

URL: <https://doi.org/10.1109/jproc.2010.2043031>

Publication Date: 2010-05-12

Volume/Issue: Volume 98, Issue 5

References: 57

Cited by: 294 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2129. An emergent community ecosystem model applied to the California Current System

Authors: Nicole L. Goebel; Christopher A. Edwards; Jonathan P. Zehr; Michael J. Follows

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2010.05.002

URL: <https://doi.org/10.1016/j.jmarsys.2010.05.002>

Publication Date: 2010-05-12

Volume/Issue: Volume 83, Issue 3-4

References: 82

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2130. Towards an assessment of simple global marine biogeochemical models of different complexity

Authors: I. Kriest; S. Khatiwala; A. Oschlies

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2010.05.002

URL: <https://doi.org/10.1016/j.pocean.2010.05.002>

Publication Date: 2010-05-12

Volume/Issue: Volume 86, Issue 3-4

References: 53

Cited by: 97 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2131. Observed and simulated estimates of the meridional overturning circulation at 26.5 degrees N in the Atlantic

Authors: J. Baehr; S. Cunningham; H. Haak; P. Heimbach; T. Kanzow; J. Marotzke

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-5-575-2009

URL: <https://doi.org/10.5194/os-5-575-2009>

Publication Date: 2010-04-29

Volume/Issue: Volume 5, Issue 4

References: 37

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Daily timeseries of the meridional overturning circulation (MOC) estimated from the UK/US RAPID/MOCHA array at 26.5 degrees N in the Atlantic are used to evaluate the MOC as simulated in two global circulation models: (I) an 8-member ensemble of the coupled climate model ECHAM5/MPI-OM, and (II) the ECCO-GODAE state estimate. In ECHAM5/MPI-OM, we find that the observed and simulated MOC have a similar variability and time-mean within the 99% confidence interval. In ECCO-GODAE, we find that the observed and simulated MOC show a significant correlation within the 99% confidence interval. To investigate the contribution of the different transport components, the MOC is decomposed into Florida Current, Ekman and mid-ocean transports. In both models, the mid-ocean transport is closely approximated by the residual of the MOC minus Florida Current and Ekman transports. As the models conserve volume by definition, future comparisons of the RAPID/MOCHA mid-ocean transport should be done against the residual transport in the models. The similarity in the variance and the correlation between the RAPID/MOCHA, and respectively ECHAM5/MPI-OM and ECCO-GODAE MOC estimates at 26.5 degrees N is encouraging in the context of estimating (natural) variability in climate simulations and its use in climate change signal-to-noise detection analyses. Enhanced confidence in simulated hydrographic and transport variability will require longer observational time series.

2132. Detecting unstable structures and controlling error growth by assimilation of standard and adaptive observations in a primitive equation ocean model

Authors: F. Uboldi; A. Trevisan

Journal: Nonlinear Processes in Geophysics

Publisher: Copernicus GmbH

DOI: 10.5194/npg-13-67-2006

URL: <https://doi.org/10.5194/npg-13-67-2006>

Publication Date: 2010-04-29

Volume/Issue: Volume 13, Issue 1

References: 41

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Oceanic and atmospheric prediction is based on cyclic analysis-forecast systems that assimilate new observations as they become available. In such observationally forced systems, errors amplify depending on their components along the unstable directions; these can be estimated by Breeding on the Data Assimilation System (BDAS). Assimilation in the Unstable Subspace (AUS) uses the available observations to estimate the amplitude of the unstable structures (computed by BDAS), present in the forecast error field, in order to eliminate them and to control the error growth. For this purpose, it is crucial that the observational network can detect the unstable structures that are active in the system. These concepts are demonstrated here by twin experiments with a large state dimension, primitive equation ocean model and an observational network having a fixed and an adaptive component. The latter consists of observations taken each time at different locations, chosen to target the estimated instabilities, whose positions and features depend on the dynamical characteristics of the flow. The adaptive placement and the dynamically consistent assimilation of observations (both relying upon the estimate of the unstable directions of the data-forced system), allow to obtain a remarkable reduction of errors with respect to a non-adaptive setting. The space distribution of the positions chosen for the observations allows to characterize the evolution of instabilities, from deep layers in western boundary current regions, to near-surface layers in the eastward jet area.

2133. A comparison of the performance of 4D-Var in an explicit and implicit version of a nonlinear barotropic ocean model

Authors: A. D. Terwisscha van Scheltinga; H. A. Dijkstra

Journal: Nonlinear Processes in Geophysics

Publisher: Copernicus GmbH

DOI: 10.5194/npg-14-763-2007

URL: <https://doi.org/10.5194/npg-14-763-2007>

Publication Date: 2010-04-29

Volume/Issue: Volume 14, Issue 6

References: 29

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A comparison is made of the performance of the four-dimensional variational data assimilation (4D-Var) method in an explicit and implicit version of a barotropic quasi-geostrophic model of the wind-driven double-gyre ocean circulation. As is well known, implicit methods have the advantage that relatively large time steps can be taken with respect to explicit methods, but the computational costs of each time step is larger. We focus here on two issues: (i) the computational efficiency in the range of time steps where the chosen explicit method is still numerically stable and (ii) the performance of 4D-Var in the implicit model for time steps out of reach for the explicit model. For the same time step Δt and the same number of points n per assimilation interval, the analyses in the implicit model is always more accurate than that in the explicit model. Due to this property the use of 4D-Var combined with the implicit model can be computationally more efficient than its use in the explicit model.

2134. Variability in the subtropical-tropical cells and its effect on near-surface temperature of the equatorial Pacific: a model study

Authors: J. F. Lübbecke; C. W. Böning; A. Biastoch

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-4-73-2008

URL: <https://doi.org/10.5194/os-4-73-2008>

Publication Date: 2010-04-29

Volume/Issue: Volume 4, Issue 1

References: 47

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A set of experiments utilizing different implementations of the global ORCA-LIM model with horizontal resolutions of 2 degrees, 0.5 degrees and 0.25 degrees is used to investigate tropical and extra-tropical influences on equatorial Pacific SST variability at interannual to decadal time scales. The model experiments use a bulk forcing methodology building on the global forcing data set for 1958 to 2000 developed by Large and Yeager (2004) that is based on a blend of atmospheric reanalysis data and satellite products. Whereas representation of the mean structure and transports of the (sub-) tropical Pacific current fields is much improved with the enhanced horizontal resolution, there is only little difference in the simulation of the interannual variability in the equatorial regime between the 0.5 degrees and 0.25 degrees model versions, with both solutions capturing the observed SST variability in the Niño3-region. The question of remotely forced oceanic contributions to the equatorial variability, in particular, the role of low-frequency changes in the transports of the Subtropical Cells (STCs), is addressed by a sequence of perturbation experiments using different combinations of fluxes. The solutions show the near-surface temperature variability to be governed by wind-driven changes in the Equatorial Undercurrent. The relative contributions of equatorial and off-equatorial atmospheric forcing differ between interannual and longer, (multi-) decadal timescales: for the latter there is a significant impact of changes in the equatorward transport of subtropical thermocline water associated with the lower branches of the STCs, related to variations in the off-equatorial trade winds. A conspicuous feature of the STC variability is that the equatorward transports in the interior and along the western boundary partially compensate each other at both decadal and interannual time scales, with the strongest transport extrema occurring during El Niño episodes. The behaviour is rationalized in terms of a wobbling in the poleward extents of the tropical gyres, which is manifested also in a meridional shifting of the bifurcation latitudes of the North and South Equatorial Current systems.

2135. Sediment $^{231}\text{Pa}/^{230}\text{Th}$ as a recorder of the rate of the Atlantic meridional overturning circulation: insights from a 2-D model

Authors: Y. Luo; R. Francois; S. E. Allen

Journal: Ocean Science

Publisher: Copernicus GmbH

DOI: 10.5194/os-6-381-2010

URL: <https://doi.org/10.5194/os-6-381-2010>

Publication Date: 2010-04-29

Volume/Issue: Volume 6, Issue 1

References: 54

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A two dimensional scavenging model is used to investigate the patterns of sediment $^{231}\text{Pa}/^{230}\text{Th}$ generated by the Atlantic Meridional Overturning Circulation (AMOC) and further advance the application of this proxy for ocean paleocirculation studies. The scavenging parameters and the geometry of the overturning circulation cell have been chosen so that the model generates meridional sections of dissolved ^{230}Th and ^{231}Pa consistent with published water column profiles and an additional 12 previously unpublished profiles measured in the North and Equatorial Atlantic. The processes that generate the meridional sections of dissolved and particulate ^{230}Th , dissolved and particulate ^{231}Pa , dissolved and particulate $^{231}\text{Pa}/^{230}\text{Th}$, and sediment $^{231}\text{Pa}/^{230}\text{Th}$ are discussed in detail. The results indicate that the relationship between sediment $^{231}\text{Pa}/^{230}\text{Th}$ at any given site and the overturning circulation is very complex. They clearly show that constraining past changes in the strength and geometry of the AMOC requires an extensive data set and they suggest strategies to maximize information from a limited number of samples.

2136. Detection of anthropogenic climate change in satellite records of ocean chlorophyll and productivity

Authors: S. A. Henson; J. L. Sarmiento; J. P. Dunne; L. Bopp; I. Lima; S. C. Doney; J. John; C. Beaulieu

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-7-621-2010

URL: <https://doi.org/10.5194/bg-7-621-2010>

Publication Date: 2010-04-29

Volume/Issue: Volume 7, Issue 2

References: 94

Cited by: 356 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Global climate change is predicted to alter the ocean's biological productivity. But how will we recognise the impacts of climate change on ocean productivity? The most comprehensive information available on its global distribution comes from satellite ocean colour data. Now that over ten years of satellite-derived chlorophyll and productivity data have accumulated, can we begin to detect and attribute climate change-driven trends in productivity? Here we compare recent trends in satellite ocean colour data to longer-term time series from three biogeochemical models (GFDL, IPSL and NCAR). We find that detection of climate change-driven trends in the satellite data is confounded by the relatively short time series and large interannual and decadal variability in productivity. Thus, recent observed changes in chlorophyll, primary production and the size of the oligotrophic gyres cannot be unequivocally attributed to the impact of global climate change. Instead, our analyses suggest that a time series of ~40 years length is needed to distinguish a global warming trend from natural variability. In some regions, notably equatorial regions, detection times are predicted to be shorter (~20-30 years). Analysis of modelled chlorophyll and primary production from 2001-2100 suggests that, on average, the climate change-driven trend will not be unambiguously separable from decadal variability until ~2055. Because the magnitude of natural variability in chlorophyll and primary production is larger than, or similar to, the global warming trend, a consistent, decades-long data record must be established if the impact of climate change on ocean productivity is to be definitively detected.

2137. Phytoplankton blooms induced/sustained by cyclonic eddies during the Indian Ocean Dipole event of 1997 along the southern coasts of Java and Sumatra

Authors: P. Rahul Chand Reddy; P. S. Salvekar

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/bgd-5-3905-2008

URL: <https://doi.org/10.5194/bgd-5-3905-2008>

Publication Date: 2010-04-29

Volume/Issue: N/A

References: 26

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Archipelago|SeaWiFS|Eddy

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Indonesian archipelago is the gateway in the tropics connecting two oceans (Pacific and the Indian Ocean) and two continents (Asia and Australia). During the Indian Ocean Dipole 1997, record anomalous and unanticipated upwelling had occurred along the southern coasts of Java and Sumatra causing massive phytoplankton blooms. But the method/mode/process for such anomalous upwelling was not known. Using monthly SeaWifs chlorophyll-a anomalies, TOPEX Sea Surface Height (SSH) anomalies, Sea Surface Temperatures (SST) and currents from a state-of-the-art OGCM, we report the presence of a series of cyclonic eddies along southern coasts of Sumatra and Java during November, December 1997 and January 1998. Upwelling caused by these cyclonic eddies, as also supported by the SSH and SST anomalies, has been responsible for the phytoplankton blooms to persist and dissipate during the 3 months (November, December 1997 and January 1998).

2138. Mechanisms of the meridional heat transport in the Southern Ocean

Authors: Denis L. Volkov; Lee-Lueng Fu; Tong Lee

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-010-0288-0

URL: <https://doi.org/10.1007/s10236-010-0288-0>

Publication Date: 2010-04-28

Volume/Issue: Volume 60, Issue 4

References: 44

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2139. Determining Ocean Circulation and Sea Level from Satellite Altimetry: Progress and Challenges

Authors: Lee-Lueng Fu

Journal: Oceanography from Space

Publisher: Springer Netherlands

DOI: 10.1007/978-90-481-8681-5_9

URL: https://doi.org/10.1007/978-90-481-8681-5_9

Publication Date: 2010-04-23

Volume/Issue: N/A

References: 63

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2140. Observing and modeling changes in the Atlantic MOC

Authors: Stuart A. Cunningham; Robert Marsh

Journal: WIREs Climate Change

Publisher: Wiley

DOI: 10.1002/wcc.22

URL: <https://doi.org/10.1002/wcc.22>

Publication Date: 2010-03-30

Volume/Issue: Volume 1, Issue 2

References: 79

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Atlantic Meridional Overturning Circulation (MOC) is defined, and our present understanding of MOC driving mechanisms is summarized. Evidence for the changing MOC is reviewed, covering recent developments in observing and modeling the MOC, and the climatic consequences of MOC variability. On a timescale of the next 5-10 years, further developments in MOC monitoring, modeling and prediction are both anticipated and recommended. In the context of what is presently known about the MOC, the evidence for a recent slowing trend is considered. Copyright © 2010 John Wiley & Sons, Inc. This article is categorized under: Paleoclimates and Current Trends > Paleoclimate

2141. The NCEP GODAS Ocean Analysis of the Tropical Pacific Mixed Layer Heat Budget on Seasonal to Interannual Time Scales

Authors: Boyin Huang (NOAA/Climate Prediction Center, Camp Springs, Maryland); Yan Xue (NOAA/Climate Prediction Center, Camp Springs, Maryland); Dongxiao Zhang (NOAA/Pacific Marine and Environmental Laboratory, Seattle, Washington); Arun Kumar (NOAA/Climate Prediction Center, Camp Springs, Maryland); Michael J. McPhaden (NOAA/Pacific Marine and Environmental Laboratory, Seattle, Washington)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2010jcli3373.1

URL: <https://doi.org/10.1175/2010jcli3373.1>

Publication Date: 2010-03-29

Volume/Issue: Volume 23, Issue 18

References: 63

Cited by: 98 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The mixed layer heat budget in the tropical Pacific is diagnosed using pentad (5 day) averaged outputs from the Global Ocean Data Assimilation System (GODAS), which is operational at the National Centers for Environmental Prediction (NCEP). The GODAS is currently used by the NCEP Climate Prediction Center (CPC) to monitor and to understand El Niño and La Niña in near real time. The purpose of this study is to assess the feasibility of using an operational ocean data assimilation system to understand SST variability. The climatological mean and seasonal cycle of mixed layer heat budgets derived from GODAS agree reasonably well with previous observational and model-based estimates. However, significant differences and biases were noticed. Large biases were found in GODAS zonal and meridional currents, which contributed to biases in the annual cycle of zonal and meridional advective heat fluxes. The warming due to tropical instability waves in boreal fall is severely underestimated owing to use of a 4-week data assimilation window. On interannual time scales, the GODAS heat budget closure is good for weak-to-moderate El Niños. A composite for weak-to-moderate El Niños suggests that zonal and meridional temperature advection and vertical entrainment/diffusion all contributed to the onset of the event and that zonal advection played the dominant role during decay of the event and the transition to La Niña. The net surface heat flux acts as a damping during the development stage, but plays a critical role in the decay of El Niño and the transition to the following La Niña. The GODAS heat budget closure is generally poor for strong La Niñas. Despite the biases, the GODAS heat budget analysis tool is useful in monitoring and understanding the physical processes controlling SST variability associated with ENSO. Therefore, it has been implemented operationally at CPC in support of NOAA's ENSO forecasting.

2142. Automatic Generation of Efficient Adjoint Code for a Parallel Navier-Stokes Solver

Authors: Patrick Heimbach; Chris Hill; Ralf Giering

Journal: Lecture Notes in Computer Science

Publisher: Springer Berlin Heidelberg

DOI: 10.1007/3-540-46080-2_107

URL: https://doi.org/10.1007/3-540-46080-2_107

Publication Date: 2010-03-29

Volume/Issue: N/A

References: 17

Cited by: 19 publications

Language: N/A

Subjects: N/A

Keywords: Solver|Automatic differentiation|Code (set theory)|Control flow|Complement

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2143. Sea surface freshwater flux estimates from GECCO, HOAPS and NCEP

Authors: V. ROMANOVA; A. KÖHL; D. STAMMER; C. KLEPP; A. ANDERSSON; S. BAKAN

Journal: Tellus A

Publisher: Stockholm University Press

DOI: 10.1111/j.1600-0870.2010.00447.x

URL: <https://doi.org/10.1111/j.1600-0870.2010.00447.x>

Publication Date: 2010-03-25

Volume/Issue: Volume 62, Issue 4

References: 61

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2144. Quasi-Biennial Modulation of the Southern Ocean Coherent Mode

Authors: Angela Hibbert; Harry Leach; Philip Woodworth; Chris W Hughes; Vassil Roussenov

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1002/qj.581

URL: <https://doi.org/10.1002/qj.581>

Publication Date: 2010-03-10

Volume/Issue: Volume 136, Issue 648

References: 53

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Empirical Orthogonal Function (EOF) analysis of two independent Southern Ocean sea-level datasets for the period 1992-2002 shows that the coherent oceanic mode-a zonally symmetric counterpart to the atmospheric Southern Annular Mode (SAM)-consistently emerges as the predominant mode of variability and is revealed to be a reliable proxy for circumpolar transport upon subannual to interannual time-scales. Cross-validation of the coherent mode from each dataset and that of a global barotropic model demonstrates a high degree of correspondence between sea-level observations and model estimations and also indicates that the coherent oceanic mode is largely barotropically driven upon subannual time-scales, whilst baroclinic oceanic effects predominate at longer periods. Both the coherent mode and transport are shown to be modified by the stratospheric Quasi-Biennial Oscillation (QBO) and a mechanism is described by which QBO-dependent planetary wave activity may be guided from the stratosphere towards the surface by high-latitude easterly and westerly jets, inducing wind anomalies that force sea-level variability through equatorward Ekman transport. A longer (18-year) tide-gauge record from the Vernadsky/Faraday station and a 22-year time series of sea-level estimations from an ocean general circulation model are used to show that QBO-induced sea-level and transport variability may be modulated upon longer time-scales by a quasi-decadal cycle of solar activity. Copyright © 2010 Royal Meteorological Society

2145. An eddy-permitting, dynamically consistent adjoint-based assimilation system for the tropical Pacific: Hindcast experiments in 2000

Authors: I. Hoteit (King Abdullah University of Science and Technology Thuwal Saudi Arabia); B. Cornuelle (Scripps Institution of Oceanography / University of California San Diego); P. Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005437

URL: <https://doi.org/10.1029/2009jc005437>

Publication Date: 2010-03-01

Volume/Issue: Volume 115, Issue C3

References: 69

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An eddy-permitting adjoint-based assimilation system has been implemented to estimate the state of the tropical Pacific Ocean. The system uses the Massachusetts Institute of Technology's general circulation model and its adjoint. The adjoint method is used to adjust the model to observations by controlling the initial temperature and salinity; temperature, salinity, and horizontal velocities at the open boundaries; and surface fluxes of momentum, heat, and freshwater. The model is constrained with most of the available data sets in the tropical Pacific, including Tropical Atmosphere and Ocean, ARGO, expendable bathythermograph, and satellite SST and sea surface height data, and climatologies. Results of hindcast experiments in 2000 suggest that the iterated adjoint-based descent is able to significantly improve the model consistency with the multivariate data sets, providing a dynamically consistent realization of the tropical Pacific circulation that generally matches the observations to within specified errors. The estimated model state is evaluated both by comparisons with observations and by checking the controls, the momentum balances, and the representation of small-scale features that were not well sampled by the observations used in the assimilation. As part of these checks, the estimated controls are smoothed and applied in independent model runs to check that small changes in the controls do not greatly change the model hindcast. This is a simple ensemble-based uncertainty analysis. In addition, the original and smoothed controls are applied to a version of the model with doubled horizontal resolution resulting in a broadly similar "downscaled" hindcast, showing that the adjustments are not tuned to a single configuration (meaning resolution, topography, and parameter settings). The time-evolving model state and the adjusted controls should be useful for analysis or to supply the forcing, initial, and boundary conditions for runs of other models.

2146. A Conserved Minimal Adjustment Scheme for Stabilization of Hydrographic Profiles

Authors: Peter C. Chu (Department of Oceanography, Naval Postgraduate School, Monterey, California); Chenwu Fan (Department of Oceanography, Naval Postgraduate School, Monterey, California)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2010jtecho742.1

URL: <https://doi.org/10.1175/2010jtecho742.1>

Publication Date: 2010-02-18

Volume/Issue: Volume 27, Issue 6

References: 17

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean (T, S) data analysis/assimilation, conducted in the three-dimensional physical space, is a generalized average of purely observed data (data analysis) or of modeled/observed data (data assimilation). Because of the high nonlinearity of the equation of the state of the seawater and nonuniform vertical distribution of the observational profile data, false static instability may be generated. A new analytical conserved adjustment scheme has been developed on the base of conservation of heat, salt, and static stability for the whole water column with predetermined (T, S) adjustment ratios. A set of well-posed combined linear and nonlinear algebraic equations has been established and is solved using Newton's method. This new scheme can be used for ocean hydrographic data analysis and data assimilation.

2147. Denoising Satellite Gravity Signals by Independent Component Analysis

Authors: F. Frappart; G. Ramillien; P. Maisongrande; M.-P. Bonnet

Journal: IEEE Geoscience and Remote Sensing Letters

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/lgrs.2009.2037837

URL: <https://doi.org/10.1109/lgrs.2009.2037837>

Publication Date: 2010-02-12

Volume/Issue: Volume 7, Issue 3

References: 20

Cited by: 41 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2148. On the formulation of sea-ice models. Part 2: Lessons from multi-year adjoint sea-ice export sensitivities through the Canadian Arctic Archipelago

Authors: Patick Heimbach; Dimitris Menemenlis; Martin Losch; Jean-Michel Campin; Chris Hill

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2010.02.002

URL: <https://doi.org/10.1016/j.ocemod.2010.02.002>

Publication Date: 2010-02-09

Volume/Issue: Volume 33, Issue 1-2

References: 49

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2149. Closure in the Earth's angular momentum budget observed from subseasonal periods down to four days: No core effects needed

Authors: Jean O. Dickey (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Steven L. Marcus (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Olivier de Viron (Géophysique Spatiale et Planétaire Institute de Physique du Globe de Paris Paris France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009gl041118

URL: <https://doi.org/10.1029/2009gl041118>

Publication Date: 2010-02-09

Volume/Issue: Volume 37, Issue 3

References: 16

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Short period variations in the Earth's rotation rate, length-of-day (LOD), are driven mainly by the atmosphere with smaller contributions by the oceans. Previous studies have noted a lag of atmospheric angular momentum (AAM) with LOD that would imply another source. We examine AAM from the European Centre for Medium-Range Weather Forecasts (ECMWF) and the National Centers for Environmental Prediction (NCEP) reanalysis series, along with oceanic angular momentum (OAM) from the ECCO consortium; land hydrological effects made no discernible impact. The NCEP reanalysis together with OAM produces a significant lag with LOD, while the ECMWF reanalysis AAM with OAM shows no phase lag. We find significant coherence with LOD variations down to periods of 4 days; coherence losses at shorter periods likely arise from the inverted barometer assumption and unmodeled dynamical processes. Thus the inclusion of core effects is not needed to balance the axial angular momentum budget on sub-seasonal time scales.

2150. Seasonal versus permanent thermocline warming by tropical cyclones

Authors: Malte F. Jansen (Massachusetts Institute of Technology); Raffaele Ferrari (Massachusetts Institute of Technology); Todd A. Mooring (Massachusetts Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009gl041808

URL: <https://doi.org/10.1029/2009gl041808>

Publication Date: 2010-02-09

Volume/Issue: Volume 37, Issue 3

References: 18

Cited by: 72 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Recent studies suggest that the enhanced upper ocean mixing caused by tropical cyclones significantly contributes to the ocean heat transport. However, existing studies that try to quantify this contribution make the assumption that all heat pumped below the mixed layer by tropical cyclones is finally released in higher latitudes. Tropical cyclones occur primarily during summer and early fall, when the ocean mixed layer is generally shallow. As the mixed layer deepens in the following winter, any warm anomaly deposited within the seasonal thermocline will be reabsorbed by the mixed layer and lost to the atmosphere. Analysis of satellite sea surface temperature and sea surface height data, together with climatological subsurface ocean data, suggests that only about one quarter of the heat that is mixed downward by tropical cyclones eventually makes it into the permanent thermocline.

2151. Hydrological and oceanic effects on polar motion from GRACE and models

Authors: Shuanggen Jin (Chinese Academy of Sciences, University of Texas at Austin); Don P. Chambers (University of South Florida); Byron D. Tapley (University of Texas at Austin)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jb006635

URL: <https://doi.org/10.1029/2009jb006635>

Publication Date: 2010-02-05

Volume/Issue: Volume 115, Issue B2

References: 49

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Terrestrial water storage (TWS) and ocean bottom pressure (OBP) are major contributors to the observed polar motion excitations, second only to atmospheric mass movement. However, quantitative assessment of the hydrological and oceanic effects on polar motion remains unclear because of the lack of global observations. In this paper, hydrological and oceanic mass excitations to polar motion are investigated using monthly TWS and OBP derived from the Gravity Recovery and Climate Experiment (GRACE) for January 2003 until December 2008. The results from this analysis are compared with hydrological model excitations from the European Center for Medium-Range Weather Forecasts (ECMWF) and oceanic model excitations obtained from the Jet Propulsion Laboratory (JPL) using Estimating the Circulation and Climate of the Ocean (ECCO). Results show that the GRACE-derived OBP and TWS better explain the geodetic residual polar motion excitations for the Px component at the annual period, while the GRACE OBP and ECMWF hydrological angular momentum agree better with the geodetic residuals for the annual Py excitation. GRACE ocean and hydrology excitations better explain the geodetic residuals for the semiannual Py excitation. However, the JPL ECCO and ECMWF models better explain the intraseasonal geodetic residual of polar motion excitation in the Px and Py components. The GRACE data demonstrate much higher intraseasonal variability than either the models or the geodetic observations.

2152. Comparison between observations and models of the Mozambique Channel transport: Seasonal cycle and eddy frequencies

Authors: P. M. van der Werf (Utrecht University); P. J. van Leeuwen (Utrecht University, University of Reading); H. Ridderinkhof (Royal Netherlands Institute for Sea Research); W. P. M. de Ruijter (Utrecht University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005633

URL: <https://doi.org/10.1029/2009jc005633>

Publication Date: 2010-02-02

Volume/Issue: Volume 115, Issue C2

References: 42

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A time series of the observed transport through an array of moorings across the Mozambique Channel is compared with that of six model runs with ocean general circulation models. In the observations, the seasonal cycle cannot be distinguished from red noise, while this cycle is dominant in the transport of the numerical models. It is found, however, that the seasonal cycles of the observations and numerical models are similar in strength and phase. These cycles have an amplitude of 5 Sv and a maximum in September, and can be explained by the yearly variation of the wind forcing. The seasonal cycle in the models is dominant because the spectral density at other frequencies is underrepresented. Main deviations from the observations are found at depths shallower than 1500 m and in the 5/y-6/y frequency range. Nevertheless, the structure of eddies in the models is close to the observed eddy structure. The discrepancy is found to be related to the formation mechanism and the formation position of the eddies. In the observations, eddies are frequently formed from an overshooting current near the mooring section, as proposed by Ridderinkhof and de Ruijter (2003) and Harlander et al. (2009). This causes an alternation of events at the mooring section, varying between a strong southward current, and the formation and passing of an eddy. This results in a large variation of transport in the frequency range of 5/y-6/y. In the models, the eddies are formed further north and propagate through the section. No alternation similar to the observations is observed, resulting in a more constant transport.

2153. Subtropics-Related Interannual Sea Surface Temperature Variability in the Central Equatorial Pacific

Authors: Jin-Yi Yu (University of California Irvine); Hsun-Ying Kao (University of California Irvine); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2010jcli3171.1

URL: <https://doi.org/10.1175/2010jcli3171.1>

Publication Date: 2010-02-02

Volume/Issue: Volume 23, Issue 11

References: 36

Cited by: 253 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Interannual sea surface temperature (SST) variability in the central equatorial Pacific consists of a component related to eastern Pacific SST variations (called Type-1 SST variability) and a component not related to them (called Type-2 SST variability). Lead-lagged regression and ocean surface-layer temperature balance analyses were performed to contrast their control mechanisms. Type-1 variability is part of the canonical, which is characterized by SST anomalies extending from the South American coast to the central Pacific, is coupled with the Southern Oscillation, and is associated with basinwide subsurface ocean variations. This type of variability is dominated by a major 4-5-yr periodicity and a minor biennial (2-2.5 yr) periodicity. In contrast, Type-2 variability is dominated by a biennial periodicity, is associated with local air-sea interactions, and lacks a basinwide anomaly structure. In addition, Type-2 SST variability exhibits a strong connection to the subtropics of both hemispheres, particularly the Northern Hemisphere. Type-2 SST anomalies appear first in the northeastern subtropical Pacific and later spread toward the central equatorial Pacific, being generated in both regions by anomalous surface heat flux forcing associated with wind anomalies. The SST anomalies undergo rapid intensification in the central equatorial Pacific through ocean advection processes, and eventually decay as a result of surface heat flux damping and zonal advection. The southward spreading of trade wind anomalies within the northeastern subtropics-to-central tropics pathway of Type-2 variability is associated with intensity variations of the subtropical high. Type-2 variability is found to become stronger after 1990, associated with a concurrent increase in the subtropical variability. It is concluded that Type-2 interannual variability represents a subtropical-excited phenomenon that is different from the conventional ENSO Type-1 variability.

2154. Transient upwelling hot spots in the oligotrophic North Pacific

Authors: P. H. R. Calil (University of Hawaii at Manoa); K. J. Richards (University of Hawaii at Manoa)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005360

URL: <https://doi.org/10.1029/2009jc005360>

Publication Date: 2010-02-02

Volume/Issue: Volume 115, Issue C2

References: 55

Cited by: 88 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Vertical exchange between deep, nutrient-rich and nutrient-depleted surface waters controls the physical supply of nutrients into the euphotic zone in the oligotrophic waters of the subtropical North Pacific Ocean. Sea surface temperature and ocean color satellite images are used in this study to show that submesoscale chlorophyll patches are locally intensified. These patches are associated with surface temperature fronts and regions of strong stretching of the eddying flow. The latter are identified by calculating the finite-size Lyapunov exponents of the geostrophic flow derived from the sea surface height field from altimetry. Surface frontogenesis and nonlinear Ekman pumping (NLEP) arise as natural candidates for the sporadic generation of large vertical velocities that could favor the formation of these features. A high-resolution regional numerical model simulation confirms that large vertical velocities are indeed associated with strong density fronts and gradients of relative vorticity. Using the Q vector divergence as a proxy for frontogenetically generated vertical velocities yields good agreement with the model vertical velocities, particularly in regions where they are large. The pattern and magnitude of NLEP suggests that it is also contributing to the strong vertical motions, although NLEP and model vertical velocities are not necessarily in phase. Surface frontogenesis and NLEP provide a possible explanation for the episodic injection of nutrients at the submesoscale, which are then horizontally stirred and modulated by the unstable manifolds of the flow field. Given that these surface-intensified processes would be biogeochemically more effective when the pycnocline is shallow, they could also serve as one possible explanation for the late summer chlorophyll blooms that are known to occur in the region.

2155. On the formulation of sea-ice models. Part 1: Effects of different solver implementations and parameterizations

Authors: Martin Losch; Dimitris Menemenlis; Jean-Michel Campin; Patick Heimbach; Chris Hill

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2009.12.008

URL: <https://doi.org/10.1016/j.ocemod.2009.12.008>

Publication Date: 2010-01-19

Volume/Issue: Volume 33, Issue 1-2

References: 75

Cited by: 310 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2156. Influence of the 26 degreesN RAPID-MOCHA Array and Florida Current Cable Observations on the ECCO-GODAE State Estimate

Authors: Johanna Baehr (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2009jpo4118.1

URL: <https://doi.org/10.1175/2009jpo4118.1>

Publication Date: 2010-01-19

Volume/Issue: Volume 40, Issue 5

References: 31

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: In situ observations, Ocean models, Ocean circulation, North Atlantic Ocean, Meridional overturning circulation (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The incorporation of local temperature and salinity observations from the Rapid Climate Change-Meridional Overturning Circulation and Heatflux Array (RAPID-MOCHA), as well as the cable estimates of volume transport in the Florida Current (FC), is tested in the Estimating the Circulation and Climate of the Ocean-Global Ocean Data Assimilation Experiment (ECCO-GODAE) estimation system for their impact on the estimate of the meridional overturning circulation (MOC) and the meridional heat transport in the Atlantic. An experimental setup covering the first deployment period of RAPID-MOCHA from March 2004 to March 2005 is used to test different strategies for incorporating these datasets. Incorporating both monthly means of the FC data and monthly means of the RAPID-MOCHA temperature and salinity measurements at the eastern and western boundaries of the basin as an observational constraint in a 1-yr experiment results in an adjustment to the reference estimate, which does not include these datasets, of approximately 1 Sv ($1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) in the MOC at 26 degreesN and the adjacent latitudes (approximately ± 15 degrees), with a larger northward branch of the MOC above 1000 m, compensated by a larger flow in the southward branch of the MOC between approximately 2000 and 3000 m. The meridional heat transport from 26 degreesN to near 40 degreesN is approximately 0.05 PW larger than in the reference experiment.

2157. A forecasting solution to the oil spill problem based on a hybrid intelligent system

Authors: Bruno Baruque; Emilio Corchado; Aitor Mata; Juan M. Corchado

Journal: Information Sciences

Publisher: Elsevier BV

DOI: 10.1016/j.ins.2009.12.032

URL: <https://doi.org/10.1016/j.ins.2009.12.032>

Publication Date: 2010-01-15

Volume/Issue: Volume 180, Issue 10

References: 61

Cited by: 76 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministerio de Ciencia e Innovación, Ministerio de Ciencia y Tecnología

License: N/A

Open Access: N/A

Abstract: N/A

2158. An Eddy-Permitting Southern Ocean State Estimate

Authors: Matthew R. Mazloff (Scripps Institution of Oceanography); Patrick Heimbach (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2009jpo4236.1

URL: <https://doi.org/10.1175/2009jpo4236.1>

Publication Date: 2010-01-12

Volume/Issue: Volume 40, Issue 5

References: 72

Cited by: 334 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An eddy-permitting general circulation model of the Southern Ocean is fit by constrained least squares to a large observational dataset during 2005-06. Data used include Argo float profiles, CTD synoptic sections, Southern Elephant Seals as Oceanographic Samplers (SEaOS) instrument-mounted seal profiles, XBTs, altimetric observations [Envisat, Geosat, Jason-1, and Ocean Topography Experiment (TOPEX)/Poseidon], and infrared and microwave radiometer observed sea surface temperature. An adjoint model is used to determine descent directions in minimizing a misfit function, each of whose elements has been weighted by an estimate of the observational plus model error. The model is brought into near agreement with the data by adjusting its control vector, here consisting of initial and meteorological boundary conditions. Although total consistency has not yet been achieved, the existing solution is in good agreement with the great majority of the 2005 and 2006 Southern Ocean observations and better represents these data than does the World Ocean Atlas 2001 (WOA01) climatological product. The estimate captures the oceanic temporal variability and in this respect represents a major improvement upon earlier static inverse estimates. During the estimation period, the Drake Passage volume transport is 153 ± 5 Sv ($1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$). The Ross and Weddell polar gyre transports are 20 ± 5 Sv and 40 ± 8 Sv, respectively. Across 32 degreesS there is a surface meridional overturning cell of 12 ± 12 Sv, an intermediate cell of 17 ± 12 Sv, and an abyssal cell of 13 ± 6 Sv. The northward heat and freshwater anomaly transports across 30 degreesS are -0.3 PW and 0.7 Sv, with estimated uncertainties of 0.5 PW and 0.2 Sv. The net rate of wind work is 2.1 ± 1.1 TW. Southern Ocean theories involving short temporal- and spatial-scale dynamics may now be tested with a dynamically and thermodynamically realistic general circulation model solution that is known to be compatible with the modern observational datasets.

2159. Interannual variability in North Pacific heat and freshwater budgets

Authors: Elizabeth Douglass; Dean Roemmich; Detlef Stammer

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2010.01.001

URL: <https://doi.org/10.1016/j.dsr2.2010.01.001>

Publication Date: 2010-01-12

Volume/Issue: Volume 57, Issue 13-14

References: 39

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2160. Heat Content Change in the Pacific Ocean between the 1990s and 2000s

Authors: Takeshi Kawano; Toshimasa Doi; Hiroshi Uchida; Shinya Kouketsu; Masao Fukasawa; Yoshimi Kawai; Katsuro Katsumata

Journal: Deep Sea Research Part II: Topical Studies in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.dsr2.2009.12.003

URL: <https://doi.org/10.1016/j.dsr2.2009.12.003>

Publication Date: 2010-01-12

Volume/Issue: Volume 57, Issue 13-14

References: 38

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2161. Consistency and fidelity of Indonesian-throughflow total volume transport estimated by 14 ocean data assimilation products

Authors: Tong Lee; Toshiyuki Awaji; Magdalena Balmaseda; Nicolas Ferry; Yosuke Fujii; Ichiro Fukumori; Benjamin Giese; Patrick Heimbach; Armin Köhl; Simona Masina; Elisabeth Remy; Anthony Rosati; Michael Schodlok; Detlef Stammer; Anthony Weaver

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2009.12.004

URL: <https://doi.org/10.1016/j.dynatmoce.2009.12.004>

Publication Date: 2010-01-07

Volume/Issue: Volume 50, Issue 2

References: 55

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2162. Variability of the Indo-Pacific Ocean exchanges

Authors: Carl Wunsch

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2009.12.001

URL: <https://doi.org/10.1016/j.dynatmoce.2009.12.001>

Publication Date: 2009-12-18

Volume/Issue: Volume 50, Issue 2

References: 26

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2163. A multi-subwaveform parametric retracker of the radar satellite altimetric waveform and recovery of gravity anomalies over coastal oceans

Authors: JinYun Guo; YongGang Gao; CheinWay Hwang; JiaLong Sun

Journal: Science China Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-009-0171-3

URL: <https://doi.org/10.1007/s11430-009-0171-3>

Publication Date: 2009-12-18

Volume/Issue: Volume 53, Issue 4

References: 26

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2164. Earth oblateness changes reveal land ice contribution to interannual sea level variability

Authors: Steven L. Marcus (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jean O. Dickey (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Florian Seitz (Earth Oriented Space Science and Technology Technische Universität München Munich Germany)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009gl041130

URL: <https://doi.org/10.1029/2009gl041130>

Publication Date: 2009-12-08

Volume/Issue: Volume 36, Issue 23

References: 28

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A broad consensus has emerged regarding the importance of both ocean mass and heat content changes for global sea level (GSL) rise, but their respective contributions to interannual GSL variability are less certain. Here, we use changes in the Earth's dynamic oblateness (J_2) to infer land ice contributions to GSL variability during a 10.5-year period encompassing the intense 1997-98 ENSO event. By accounting for heat content and water mass distribution changes in the oceans, atmosphere and land hydrology using observational and model results, we isolate unexplained residuals in J_2 and GSL that are well-correlated with each other and with melt-season temperature and mass balance anomalies in Alaska, a particularly active and well-studied glaciated region. The close agreement between residual GSL and oblateness variations found in our results indicates that the latter can provide a useful proxy for changes in high-latitude land ice when corrections for other sources are applied.

2165. Role of a Cyclonic Eddy in the 7000-Year-Old Mentawai Coral Reef Death During the 1997 Indian Ocean Dipole Event

Authors: P R C Rahul; P S Salvekar; B K Sahu; Shailesh Nayak; T Srinivas Kumar

Journal: IEEE Geoscience and Remote Sensing Letters

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/lgrs.2009.2033950

URL: <https://doi.org/10.1109/lgrs.2009.2033950>

Publication Date: 2009-11-23

Volume/Issue: Volume 7, Issue 2

References: 49

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Mesoscale circulation, Satellite data, Coral reefs

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2166. Heat Budget of the Upper Ocean in the South-Central Equatorial Pacific

Authors: Hailong Liu (Chinese Academy of Sciences); Wuyin Lin (Institute for Terrestrial and Planetary Atmospheres, Stony Brook University, Stony Brook, New York); Minghua Zhang (Institute for Terrestrial and Planetary Atmospheres, Stony Brook University, Stony Brook, New York)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2009jcli3135.1

URL: <https://doi.org/10.1175/2009jcli3135.1>

Publication Date: 2009-11-19

Volume/Issue: Volume 23, Issue 7

References: 55

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The double intertropical convergence zone (ITCZ) over the tropical Pacific, with a spurious band of maximum annual sea surface temperature (SST) south of the equator between 5 degreesS and 10 degreesS, is a chronic bias in coupled ocean-atmosphere models. This study focuses on a region of the double ITCZ in the central Pacific from 5 degreesS to 10 degreesS and 170 degreesE to 150 degreesW, where coupled models display the largest biases in precipitation, by deriving a best estimate of the mixed layer heat budget for the region. Seven global datasets of objectively analyzed surface energy fluxes and four ocean assimilation products are first compared and then evaluated against field measurements in adjacent regions. It was shown that the global datasets differ greatly in their net downward surface energy flux in this region, but they fall broadly into two categories: one with net downward heat flux of about 30 W m⁻² and the other around 10 W m⁻². Measurements from the adjacent Manus and Nauru sites of the Atmospheric Radiation Measurement Program (ARM), the Tropical Atmosphere Ocean (TAO) buoys, and the Tropical Ocean and Global Atmosphere Coupled Ocean-Atmosphere Response Experiment (TOGA COARE) are then used to show that the smaller value is more realistic. An energy balance of the mixed layer is finally presented for the region as primarily between warming from surface heat flux of 7 W m⁻² and horizontal advective cooling in the zonal direction of about 5 W m⁻², with secondary contributions from meridional and vertical advections, heat storage, and subgrid-scale mixing. The 7 W m⁻² net surface heat flux consists of warming of 210 W m⁻² from solar radiation and cooling of 53, 141, and 8 W m⁻², respectively, from longwave radiation, latent heat flux, and sensible heat flux. These values provide an observational basis to further study the initial development of excessive precipitation in coupled climate models in the central Pacific.

2167. A Vector Geometry-Based Eddy Detection Algorithm and Its Application to a High-Resolution Numerical Model Product and High-Frequency Radar Surface Velocities in the Southern California Bight

Authors: Francesco Nencioli (University of California Santa Barbara); Changming Dong (University of California Los Angeles); Tommy Dickey (University of California Santa Barbara); Libe Washburn (University of California Santa Barbara); James C. McWilliams (University of California Los Angeles)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2009jtecho725.1

URL: <https://doi.org/10.1175/2009jtecho725.1>

Publication Date: 2009-11-19

Volume/Issue: Volume 27, Issue 3

References: 27

Cited by: 396 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Automated eddy detection methods are fundamental tools to analyze eddy activity from the large datasets derived from satellite measurements and numerical model simulations. Existing methods are either based on the distribution of physical parameters usually computed from velocity derivatives or on the geometry of velocity streamlines around minima or maxima of sea level anomaly. A new algorithm was developed based exclusively on the geometry of the velocity vectors. Four constraints characterizing the spatial distribution of the velocity vectors around eddy centers were derived from the general features associated with velocity fields in the presence of eddies. The grid points in the domain for which these four constraints are satisfied are detected as eddy centers. Eddy sizes are computed from closed contours of the streamfunction field, and eddy tracks are retrieved by comparing the distribution of eddy centers at successive time steps. The results were validated against manually derived eddy fields. Two parameters in the algorithm can be modified by the users to optimize its performance. The algorithm is applied to both a high-resolution model product and high-frequency radar surface velocity fields in the Southern California Bight.

2168. Pattern and velocity of propagation of the global ocean eddy variability

Authors: Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005349

URL: <https://doi.org/10.1029/2009jc005349>

Publication Date: 2009-11-13

Volume/Issue: Volume 114, Issue C11

References: 33

Cited by: 94 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Satellite altimeter data are used to study the characteristics of the horizontal propagation of eddy variability of the global oceans. Decade-long time series of sea surface height is analyzed for finding the maximum cross correlation with neighboring time series within a window of space and time lags. The space and time lags corresponding to the maximum correlation allow an estimate of the propagation velocity of the eddy variability that dominates the variance of sea surface height anomalies. The method cannot distinguish the various forms of eddy variability: isolated eddies and fronts, the meandering of ocean currents, or planetary waves. However, the results provide, at a given location of the global oceans, a uniquely determined propagation velocity that represents a time-averaged description of the tendency of the movement of the local eddy variability. The propagation velocity is highly inhomogeneous in space. Outside the equatorial zone, the zonal propagation is intrinsically westward, modified by ocean currents, which could reverse the zonal propagation to eastward in regions like the Gulf Stream and the Antarctic Circumpolar Current. At midlatitudes and high latitudes, the propagation pattern is highly affected by the path of ocean currents and the shape of bottom topography. At tropical latitudes, the meridional propagation is convergent toward the equator in the western basins and divergent away from the equator in the eastern basins. Comparison with the simulations of an eddy-permitting ocean general circulation model shows overall agreement, especially in the latitudinal variation of the zonal propagation velocity. The result suggests that the model has captured the essence of the dynamics governing the propagation of ocean eddy variability.

2169. Improved modeling of the Arctic halocline with a subgrid-scale brine rejection parameterization

Authors: A. T. Nguyen (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); R. Kwok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc005121

URL: <https://doi.org/10.1029/2008jc005121>

Publication Date: 2009-11-10

Volume/Issue: Volume 114, Issue C11

References: 38

Cited by: 81 publications

Language: en

Subjects: N/A

Keywords: Halocline|Mixed layer|Brine

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The halocline in the Arctic Ocean plays an important role in regulating heat exchange at the bottom of the mixed layer and it has a direct effect on the ocean sea ice energy balance and sea ice mass balance. Modeling the halocline, however, remains a challenge in current state-of-the-art coupled ocean sea ice models including those that participated in the Arctic Ocean Model Intercomparison Project. In this study, we successfully reproduce a cold halocline in the Canada Basin by implementing a subgrid-scale brine rejection parameterization in an ocean general circulation model. The brine rejection scheme improves the solution by redistributing surface salts rejected during sea ice formation to their neutral buoyancy depths. The depths are based on salt plume physics and published laboratory and numerical experiments. Compared with hydrographic data from 1993 to 2004, distribution of most of the rejected salt to the bottom of the mixed layer seems to yield the lowest model-data misfits. We also show that the model's mixed layer depth is sensitive to the background diffusivity ν used in the k-profile parameterization vertical mixing scheme. A background diffusivity of 10^{-6} m²/s in combination with brine rejection scheme described herein yield the best simulation of the Arctic halocline.

2170. Changes in total ocean mass derived from GRACE, GPS, and ocean modeling with weekly resolution

Authors: R. Rietbroek (Helmholtz Centre Potsdam, Institute of Geodesy and Geoinformation Bonn Germany); S.-E. Brunnabend (Alfred Wegener Institute); C. Dahle (Helmholtz Centre Potsdam); J. Kusche (Helmholtz Centre Potsdam, Institute of Geodesy and Geoinformation Bonn Germany); F. Flechtner (Helmholtz Centre Potsdam); J. Schröter (Alfred Wegener Institute); R. Timmermann (Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jc005449

URL: <https://doi.org/10.1029/2009jc005449>

Publication Date: 2009-11-02

Volume/Issue: Volume 114, Issue C11

References: 42

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We derive changes in ocean bottom pressure (OBP) and ocean mass by combining modeled ocean bottom pressure, weekly GRACE-derived models of gravity change, and large-scale deformation patterns sensed by a global network of GPS stations in a joint least squares inversion. The weekly combination allows a consistent estimation of geocenter motion, loading mass harmonics up to degree 30, and a spatially uniform mass correction term, which serves as a correction for forcing of the ocean model. We provide maps and time series of ocean mass and bottom pressure variations. Furthermore, we discuss the estimated geocenter motion and the estimated model correction. Our results indicate that the total ocean mass change is predominantly annual, with a maximum amplitude corresponding to 7.4 mm in October, which is in line with earlier work. The mean ocean bottom pressure (i.e., ocean plus atmospheric mass) shows an annual amplitude of 8.7 mm and is shifted forward by about 1.5 months. In addition, the solution exhibits typical autocorrelation times of about 2 weeks. A comparison with in situ bottom pressure time series in the southern Indian Ocean shows a good agreement, with correlations of 0.7-0.8. Based on these comparisons, we see that our results monitor realistic submonthly variations, which are strongest at high latitudes. The addition of GRACE data in the inversion is found to improve these high-latitude variations and enables better separability of the geocenter motion from other unknowns. Increasing the OBP model error from 3 cm to 4.8 cm affects mainly the higher-degree coefficients.

2171. Modeling the coupling of ocean ecology and biogeochemistry

Authors: S. Dutkiewicz (Massachusetts Institute of Technology); M. J. Follows (Massachusetts Institute of Technology); J. G. Bragg (Massachusetts Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008gb003405

URL: <https://doi.org/10.1029/2008gb003405>

Publication Date: 2009-11-02

Volume/Issue: Volume 23, Issue 4

References: 31

Cited by: 188 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We examine the interplay between ecology and biogeochemical cycles in the context of a global three-dimensional ocean model where self-assembling phytoplankton communities emerge from a wide set of potentially viable cell types. We consider the complex model solutions in the light of resource competition theory. The emergent community structures and ecological regimes vary across different physical environments in the model ocean: Strongly seasonal, high-nutrient regions are dominated by fast growing bloom specialists, while stable, low-seasonality regions are dominated by organisms that can grow at low nutrient concentrations and are suited to oligotrophic conditions. In the latter regions, the framework of resource competition theory provides a useful qualitative and quantitative diagnostic tool with which to interpret the outcome of competition between model organisms, their regulation of the resource environment, and the sensitivity of the system to changes in key physiological characteristics of the cells.

2172. Excitation of Chandler Wobble by Pacific, Indian and Atlantic Oceans from 1980 to 2005

Authors: Jin Ma; Yong-Hong Zhou; De-Chun Liao; Jian-Li Chen

Journal: Chinese Astronomy and Astrophysics

Publisher: Elsevier BV

DOI: 10.1016/j.chinastron.2009.09.007

URL: <https://doi.org/10.1016/j.chinastron.2009.09.007>

Publication Date: 2009-10-31

Volume/Issue: Volume 33, Issue 4

References: 21

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2173. Greenland ice-sheet volume sensitivity to basal, surface and initial conditions derived from an adjoint model

Authors: Patrick Heimbach; Véronique Bugnion

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/172756409789624256

URL: <https://doi.org/10.3189/172756409789624256>

Publication Date: 2009-10-15

Volume/Issue: Volume 50, Issue 52

References: 51

Cited by: 52 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We extend the application of control methods to a comprehensive three-dimensional thermomechanical ice-sheet model, SICOPOLIS (Simulation COde for POLythermal Ice Sheets). Lagrange multipliers, i.e. sensitivities, are computed with an exact, efficient adjoint model that has been generated from SICOPOLIS by rigorous application of automatic differentiation. The case study uses the adjoint model to determine the sensitivity of the total Greenland ice volume to various control variables over a 100 year period. The control space has of the order 1.2×10^6 elements, consisting of spatial fields of basal flow parameters, surface and basal forcings and initial conditions. Reliability of the adjoint model was tested through finite-difference perturbation calculations for various control variables and perturbation regions, ascertaining quantitative inferences of the adjoint model. As well as confirming qualitative aspects of ice-sheet sensitivities (e.g. expected regional variations), we detect regions where model sensitivities are seemingly unexpected or counter-intuitive, albeit 'real' in the sense of actual model behavior. An example is inferred regions where sensitivities of ice-sheet volume to basal sliding coefficient are positive, i.e. where a local increase in basal sliding parameter increases the ice-sheet volume. Similarly, positive (generally negative) ice temperature sensitivities in certain parts of the ice sheet are found, the detection of which seems highly unlikely if only conventional perturbation experiments had been used. The object of this paper is largely a proof of concept. Available adjoint-code generation tools now open up a variety of novel model applications, notably with regard to sensitivity and uncertainty analyses and ice-sheet state estimation or data assimilation.

2174. Increasing rates of ice mass loss from the Greenland and Antarctic ice sheets revealed by GRACE

Authors: I. Velicogna (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009gl040222

URL: <https://doi.org/10.1029/2009gl040222>

Publication Date: 2009-10-12

Volume/Issue: Volume 36, Issue 19

References: 23

Cited by: 527 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We use monthly measurements of time-variable gravity from the GRACE (Gravity Recovery and Climate Experiment) satellite gravity mission to determine the ice mass-loss for the Greenland and Antarctic Ice Sheets during the period between April 2002 and February 2009. We find that during this time period the mass loss of the ice sheets is not a constant, but accelerating with time, i.e., that the GRACE observations are better represented by a quadratic trend than by a linear one, implying that the ice sheets contribution to sea level becomes larger with time. In Greenland, the mass loss increased from 137 Gt/yr in 2002-2003 to 286 Gt/yr in 2007-2009, i.e., an acceleration of -30 ± 11 Gt/yr² in 2002-2009. In Antarctica the mass loss increased from 104 Gt/yr in 2002-2006 to 246 Gt/yr in 2006-2009, i.e., an acceleration of -26 ± 14 Gt/yr² in 2002-2009. The observed acceleration in ice sheet mass loss helps reconcile GRACE ice mass estimates obtained for different time periods.

2175. Modeling transport and fate of riverine dissolved organic carbon in the Arctic Ocean

Authors: M. Manizza (Massachusetts Institute of Technology); M. J. Follows (Massachusetts Institute of Technology); S. Dutkiewicz (Massachusetts Institute of Technology); J. W. McClelland (University of Texas at Austin); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. N. Hill (Massachusetts Institute of Technology); A. Townsend-Small (University of California Irvine); B. J. Peterson (Marine Biological Laboratory Woods Hole Massachusetts USA)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008gb003396

URL: <https://doi.org/10.1029/2008gb003396>

Publication Date: 2009-10-06

Volume/Issue: Volume 23, Issue 4

References: 50

Cited by: 62 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The spatial distribution and fate of riverine dissolved organic carbon (DOC) in the Arctic may be significant for the regional carbon cycle but are difficult to fully characterize using the sparse observations alone. Numerical models of the circulation and biogeochemical cycles of the region can help to interpret and extrapolate the data and may ultimately be applied in global change sensitivity studies. Here we develop and explore a regional, three-dimensional model of the Arctic Ocean in which, for the first time, we explicitly represent the sources of riverine DOC with seasonal discharge based on climatological field estimates. Through a suite of numerical experiments, we explore the distribution of DOC-like tracers with realistic riverine sources and a simple linear decay to represent remineralization through microbial degradation. The model reproduces the slope of the DOC-salinity relationship observed in the eastern and western Arctic basins when the DOC tracer lifetime is about 10 years, consistent with published inferences from field data. The new empirical parameterization of riverine DOC and the regional circulation and biogeochemical model provide new tools for application in both regional and global change studies.

2176. Local and global hydrological contributions to gravity variations observed in Strasbourg

Authors: Laurent Longuevergne; J.P. Boy; Nicolas Florsch; Daniel Viville; Gilbert Ferhat; Patrice Ulrich; Bernard Luck; Jacques Hinderer

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2009.09.008

URL: <https://doi.org/10.1016/j.jog.2009.09.008>

Publication Date: 2009-10-05

Volume/Issue: Volume 48, Issue 3-5

References: 37

Cited by: 55 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2177. Seasonal effects of non-tidal oceanic mass shifts in observations with superconducting gravimeters

Authors: C. Kroner; M. Thomas; H. Dobslaw; M. Abe; A. Weise

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2009.09.009

URL: <https://doi.org/10.1016/j.jog.2009.09.009>

Publication Date: 2009-10-01

Volume/Issue: Volume 48, Issue 3-5

References: 31

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2178. Ocean tidal effects on Earth rotation

Authors: Richard S. Gross

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2009.09.016

URL: <https://doi.org/10.1016/j.jog.2009.09.016>

Publication Date: 2009-10-01

Volume/Issue: Volume 48, Issue 3-5

References: 26

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2179. Seasonal excitation of polar motion estimated from recent geophysical models and observations

Authors: Aleksander Brzeziński; Jolanta Nastula; Barbara Kołaczek

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2009.09.021

URL: <https://doi.org/10.1016/j.jog.2009.09.021>

Publication Date: 2009-09-29

Volume/Issue: Volume 48, Issue 3-5

References: 19

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2180. How Much Energy Is Transferred from the Winds to the Thermocline on ENSO Time Scales?

Authors: Jaclyn N. Brown (Department of Geology and Geophysics, Yale University, New Haven, Connecticut); Alexey V. Fedorov (Department of Geology and Geophysics, Yale University, New Haven, Connecticut)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2009jcli2914.1

URL: <https://doi.org/10.1175/2009jcli2914.1>

Publication Date: 2009-09-21

Volume/Issue: Volume 23, Issue 6

References: 45

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The dynamics of El Niño-Southern Oscillation (ENSO) are studied in terms of the balance between energy input from the winds (via wind power) and changes in the storage of available potential energy in the tropical ocean. Presently, there are broad differences in the way global general circulation models simulate the dynamics, magnitude, and phase of ENSO events; hence, there is a need for simple, physically based metrics to allow for model evaluation. This energy description is a basinwide, integral, quantitative approach, ideal for intermodel comparison, that assesses model behavior in the subsurface ocean. Here it is applied to a range of ocean models and data assimilations within ENSO spatial and temporal scales. The onset of an El Niño is characterized by a decrease in wind power that leads to a decrease in available potential energy, and hence a flatter thermocline. In contrast, La Niña events are preceded by an increase in wind power that leads to an increase in the available potential energy and a steeper thermocline. The wind power alters the available potential energy via buoyancy power, associated with vertical mass fluxes that modify the slope of the isopycnals. Only a fraction of wind power is converted to buoyancy power. The efficiency of this conversion γ is estimated in this study at 50%-60%. Once the energy is delivered to the thermocline it is subject to small, but important, diffusive dissipation. It is estimated that this dissipation sets the e-folding damping rate α for the available potential energy on the order of 1 yr⁻¹. The authors propose to use the efficiency γ and the damping rate α as two energy-based metrics for evaluating dissipative properties of the ocean component of general circulation models, providing a simple method for understanding subsurface ENSO dynamics and a diagnostic tool for exploring differences between the models.

2181. The WOCE-era 3-D Pacific Ocean circulation and heat budget

Authors: A.M. Macdonald; S. Mecking; P.E. Robbins; J.M. Toole; G.C. Johnson; L. Talley; M. Cook; S.E. Wijffels

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2009.08.002

URL: <https://doi.org/10.1016/j.pocean.2009.08.002>

Publication Date: 2009-09-21

Volume/Issue: Volume 82, Issue 4

References: 134

Cited by: 55 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2182. An observationally based energy balance for the Earth since 1950

Authors: D. M. Murphy (Chemical Sciences Division, Earth System Research Laboratory NOAA Boulder Colorado USA); S. Solomon (Chemical Sciences Division, Earth System Research Laboratory NOAA Boulder Colorado USA); R. W. Portmann (Chemical Sciences Division, Earth System Research Laboratory NOAA Boulder Colorado USA); K. H. Rosenlof (Chemical Sciences Division, Earth System Research Laboratory NOAA Boulder Colorado USA); P. M. Forster (University of Leeds); T. Wong (National Aeronautics and Space Administration (NASA) Langley Research Center)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009jd012105

URL: <https://doi.org/10.1029/2009jd012105>

Publication Date: 2009-09-08

Volume/Issue: Volume 114, Issue D17

References: 64

Cited by: 166 publications

Language: en

Subjects: N/A

Keywords: Forcing (mathematics)|Cloud forcing|Earth's energy budget|Greenhouse effect

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We examine the Earth's energy balance since 1950, identifying results that can be obtained without using global climate models. Important terms that can be constrained using only measurements and radiative transfer models are ocean heat content, radiative forcing by long-lived trace gases, and radiative forcing from volcanic eruptions. We explicitly consider the emission of energy by a warming Earth by using correlations between surface temperature and satellite radiant flux data and show that this term is already quite significant. About 20% of the integrated positive forcing by greenhouse gases and solar radiation since 1950 has been radiated to space. Only about 10% of the positive forcing (about 1/3 of the net forcing) has gone into heating the Earth, almost all into the oceans. About 20% of the positive forcing has been balanced by volcanic aerosols, and the remaining 50% is mainly attributable to tropospheric aerosols. After accounting for the measured terms, the residual forcing between 1970 and 2000 due to direct and indirect forcing by aerosols as well as semidirect forcing from greenhouse gases and any unknown mechanism can be estimated as $-1.1 \pm 0.4 \text{ W m}^{-2}$ (1sigma). This is consistent with the Intergovernmental Panel on Climate Change's best estimates but rules out very large negative forcings from aerosol indirect effects. Further, the data imply an increase from the 1950s to the 1980s followed by constant or slightly declining aerosol forcing into the 1990s, consistent with estimates of trends in global sulfate emissions. An apparent increase in residual forcing in the late 1990s is discussed.

2183. Impact of loading effects on determination of the International Terrestrial Reference Frame

Authors: Xavier Collilieux; Zuheir Altamimi; David Coulot; Tonie van Dam; Jim Ray

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2009.08.024

URL: <https://doi.org/10.1016/j.asr.2009.08.024>

Publication Date: 2009-09-02

Volume/Issue: Volume 45, Issue 1

References: 40

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2184. Atmospheric and oceanic forcing of the rapid polar motion

Authors: Christian Bizouard; L. Seoane

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-009-0341-2

URL: <https://doi.org/10.1007/s00190-009-0341-2>

Publication Date: 2009-09-02

Volume/Issue: Volume 84, Issue 1

References: 33

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2185. Changes in the flow of energy through the Earth's climate system

Authors: Kevin E. Trenberth; John T. Fasullo

Journal: Meteorologische Zeitschrift

Publisher: Schweizerbart

DOI: 10.1127/0941-2948/2009/0388

URL: <https://doi.org/10.1127/0941-2948/2009/0388>

Publication Date: 2009-09-01

Volume/Issue: Volume 18, Issue 4

References: N/A

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Energy Flow|Earth system science|Energy system|Transient climate simulation|Climate system

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2186. High-order regridding-remapping schemes for continuous isopycnal and generalized coordinates in ocean models

Authors: Laurent White; Alistair Adcroft; Robert Hallberg

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2009.08.016

URL: <https://doi.org/10.1016/j.jcp.2009.08.016>

Publication Date: 2009-08-29

Volume/Issue: Volume 228, Issue 23

References: 27

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2187. A spectral barotropic model of the wind-driven world ocean

Authors: Thomas Frisius; Klaus Fraedrich; Xiuhua Zhu; Weiqiang Wang

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2009.07.008

URL: <https://doi.org/10.1016/j.ocemod.2009.07.008>

Publication Date: 2009-08-13

Volume/Issue: Volume 30, Issue 4

References: 39

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2188. A decade of acoustic thermometry in the North Pacific Ocean

Authors: B. D. Dushaw (University of Washington); P. F. Worcester (Scripps Institution of Oceanography / University of California San Diego); W. H. Munk (Scripps Institution of Oceanography / University of California San Diego); R. C. Spindel (University of Washington); J. A. Mercer (University of Washington); B. M. Howe (University of Washington); K. Metzger (University of Michigan); T. G. Birdsall (University of Michigan); R. K. Andrew (University of Washington); M. A. Dzieciuch (Scripps Institution of Oceanography / University of California San Diego); B. D. Cornuelle (Scripps Institution of Oceanography / University of California San Diego); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc005124

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Publication Date: 2009-07-18

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References: 76

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Over the decade 1996-2006, acoustic sources located off central California (1996-1999) and north of Kauai (1997-1999, 2002-2006) transmitted to receivers distributed throughout the northeast and north central Pacific. The acoustic travel times are inherently spatially integrating, which suppresses mesoscale variability and provides a precise measure of ray-averaged temperature. Daily average travel times at 4-day intervals provide excellent temporal resolution of the large-scale thermal field. The interannual, seasonal, and shorter-period variability is large, with substantial changes sometimes occurring in only a few weeks. Linear trends estimated over the decade are small compared to the interannual variability and inconsistent from path to path, with some acoustic paths warming slightly and others cooling slightly. The measured travel times are compared with travel times derived from four independent estimates of the North Pacific: (1) climatology, as represented by the World Ocean Atlas 2005 (WOA05); (2) objective analysis of the upper-ocean temperature field derived from satellite altimetry and in situ profiles; (3) an analysis provided by the Estimating the Circulation and Climate of the Ocean project, as implemented at the Jet Propulsion Laboratory (JPL-ECCO); and (4) simulation results from a high-resolution configuration of the Parallel Ocean Program (POP) model. The acoustic data show that WOA05 is a better estimate of the time mean hydrography than either the JPL-ECCO or the POP estimates, both of which proved incapable of reproducing the observed acoustic arrival patterns. The comparisons of time series provide a stringent test of the large-scale temperature variability in the models. The differences are sometimes substantial, indicating that acoustic thermometry data can provide significant additional constraints for numerical ocean models.

2189. High-resolution regional ocean dynamics simulation in the southwestern tropical Atlantic

Authors: Marcus Silva; Moacyr Araujo; Jacques Servain; Pierrick Penven; Carlos A.D. Lentini

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2009.07.002

URL: <https://doi.org/10.1016/j.ocemod.2009.07.002>

Publication Date: 2009-07-18

Volume/Issue: Volume 30, Issue 4

References: 74

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2190. Enhancement of Mesoscale Eddy Stirring at Steering Levels in the Southern Ocean

Authors: Ryan Abernathey (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology); Matt Mazloff (Massachusetts Institute of Technology); Emily Shuckburgh (British Antarctic Survey, Cambridge, United Kingdom)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2009jpo4201.1

URL: <https://doi.org/10.1175/2009jpo4201.1>

Publication Date: 2009-07-13

Volume/Issue: Volume 40, Issue 1

References: 55

Cited by: 104 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Meridional cross sections of effective diffusivity in the Southern Ocean are presented and discussed. The effective diffusivity, K_{eff} , characterizes the rate at which mesoscale eddies stir properties on interior isopycnal surfaces and laterally at the sea surface. The distributions are obtained by monitoring the rate at which eddies stir an idealized tracer whose initial distribution varies monotonically across the Antarctic Circumpolar Current (ACC). In the absence of observed maps of eddy currents in the interior ocean, the advecting velocity field is taken from an eddy-permitting state estimate of the Southern Ocean (SOSE). A three-dimensional advection-diffusion equation is solved and the diffusivity diagnosed by applying the Nakamura technique on both horizontal and isopycnal surfaces. The resulting meridional sections of K_{eff} reveal intensified isopycnal eddy stirring (reaching values of $\sim 2000 \text{ m}^2 \text{ s}^{-1}$) in a layer deep beneath the ACC but rising toward the surface on the equatorward flank. Lower effective diffusivity values ($\sim 500 \text{ m}^2 \text{ s}^{-1}$) are found near the surface where the mean flow of the ACC is strongest. It is argued that K_{eff} is enhanced in the vicinity of the steering level of baroclinic waves, which is deep along the axis of the ACC but shallows on the equatorial flank. Values of K_{eff} are also found to be spatially correlated with gradients of potential vorticity on isopycnal surfaces and are large where those gradients are weak and vice versa, as expected from simple dynamical arguments. Finally, implications of the spatial distributions of K_{eff} for the dynamics of the ACC and its overturning circulation are discussed.

2191. Bottom pressure changes around Antarctica and wind-driven meridional flows

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.); Katherine J. Quinn (Atmospheric and Environmental Research Inc.)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2009gl039060

URL: <https://doi.org/10.1029/2009gl039060>

Publication Date: 2009-07-02

Volume/Issue: Volume 36, Issue 13

References: 14

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Spatially-averaged bottom pressure anomalies near Antarctica (south of 60 degreesS) calculated from GRACE data are well correlated with those produced by the ECCO project using least-squares optimization methods to fit an ocean model to most available data. Both GRACE and ECCO results indicate mass exchange primarily between the Southern Ocean and the Pacific and the importance of zonal wind stress to this exchange. The ECCO flow fields show that the near-surface meridional Ekman transport, directly driven by the zonal wind stress, is nearly balanced by return flows below the shallowest topography at 60 degreesS (~1300 m at this Drake Passage latitude), with the return transport being slightly lagged in time relative to the Ekman transport. Such time lags, which may result from geostrophic adjustment at depth, cause the small associated net transport across 60 degreesS to be ~90 degrees out of phase with the wind. This in turn can explain why zonal wind stress and bottom pressure anomalies around Antarctica tend to be anticorrelated in both GRACE and ECCO results.

2192. Toward a global ocean data assimilation system based on ensemble optimum interpolation: altimetry data assimilation experiment

Authors: Weiwei Fu; Jiang Zhu; Changxiang Yan; Hailong Liu

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-009-0206-5

URL: <https://doi.org/10.1007/s10236-009-0206-5>

Publication Date: 2009-07-01

Volume/Issue: Volume 59, Issue 4

References: 45

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2193. Data Sensitivity of the ECCO State Estimate in a Regional Setting

Authors: Elizabeth Douglass (Scripps Institution of Oceanography); Dean Roemmich (Scripps Institution of Oceanography); Detlef Stammer (University of Hamburg)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2009jtecho641.1

URL: <https://doi.org/10.1175/2009jtecho641.1>

Publication Date: 2009-06-29

Volume/Issue: Volume 26, Issue 11

References: 27

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Estimating the Circulation and Climate of the Ocean (ECCO) consortium provides a framework in which the adjoint method of data assimilation is applied to a general circulation model to provide a dynamically self-consistent estimate of the time-varying ocean state, which is constrained by observations. In this study, the sensitivity of the solution to the constraints provided by various datasets is investigated in a regional setting in the North Pacific. Four assimilation experiments are performed, which vary by the data used as constraints and the relative weights associated with these data. The resulting estimates are compared to two of the assimilated datasets as well as to data from two time series stations not used as constraints. These comparisons demonstrate that increasing the weights of the subsurface data provides overall improvement in the model-data consistency of the estimate of the state of the North Pacific Ocean. However, some elements of the solution are degraded. This could result from incompatibility between datasets, possibly because of hidden biases, or from errors in the model physics made more evident by the increased weight on subsurface data. The adjustments to the control parameters of surface forcing and initial conditions necessary to obtain the more accurate fit to the data are found to be within prior error bars.

2194. Tilt and strain deformation induced by hydrologically active natural fractures: application to the tiltmeters installed in Sainte-Croix-aux-Mines observatory (France)

Authors: Laurent Longuevergne; Nicolas Florsch; Frédéric Boudin; Ludovic Oudin; Christian Camerlynck

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2009.04197.x

URL: <https://doi.org/10.1111/j.1365-246x.2009.04197.x>

Publication Date: 2009-06-22

Volume/Issue: Volume 178, Issue 2

References: 51

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2195. Connectivity among straits of the northwest Pacific marginal seas

Authors: Yang-Ki Cho (Chonnam National University); Gwang-Ho Seo (Chonnam National University); Byoung-Ju Choi (Department of Oceanography Kunsan National University Gunsan South Korea); Sangil Kim (Oregon State University); Young-Gyu Kim (Agency for Defense Development); Yong-Hoon Youn (Korea Meteorological Agency Seoul South Korea); Edward P. Dever (Oregon State University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc005218

URL: <https://doi.org/10.1029/2008jc005218>

Publication Date: 2009-06-18

Volume/Issue: Volume 114, Issue C6

References: 29

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The connectivity among straits of the northwest Pacific marginal seas is investigated with a primitive-equation ocean circulation model simulated for 10 years from 1994 to 2003. Over the simulation interval the temporal and spatial means and variations of the model sea surface temperature are comparable to those of the satellite sea surface temperature. The model transport through the straits shows good agreement with the available observations and a high seasonality in the Taiwan Strait, the Korea Strait, and the Soya Strait but relatively low seasonality in the Tsugaru Strait. The Kuroshio and Taiwan Warm Current (TWC) are two sources of water flowing through the Korea Strait. The volume transport in the Korea Strait is dominated by the Kuroshio in winter (83%) and by the TWC in summer (66%). Relative to the transport through the Korea Strait, the transport percentages of the Tsugaru Strait connecting to the northwest Pacific Ocean are 79% in winter and 65% in summer. The seasonality of the Korea Strait transport is positively correlated with the cross-strait wind stress. The drifter experiments show that it takes about 4 months for most of the drifters deployed in the Taiwan Strait to enter the Korea Strait and more than 2 months to travel from the Korea Strait to the Tsugaru and Soya straits.

2196. Secular variation of the Earth's magnetic field: induced by the ocean flow?

Authors: Gregory Ryskin

Journal: New Journal of Physics

Publisher: IOP Publishing

DOI: 10.1088/1367-2630/11/6/063015

URL: <https://doi.org/10.1088/1367-2630/11/6/063015>

Publication Date: 2009-06-13

Volume/Issue: Volume 11, Issue 6

References: 33

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2197. Effects of biologically induced differential heating in an eddy-permitting coupled ocean-ecosystem model

Authors: Ulrike Löptien (Ozeanzirkulation und Klimadynamik Leibniz-Institut für Meereswissenschaften Kiel Germany); Carsten Eden (Ozeanzirkulation und Klimadynamik Leibniz-Institut für Meereswissenschaften Kiel Germany); Axel Timmermann (International Pacific Research Center, School of Ocean and Earth Science and Technology University of Hawai'i at Manoa Honolulu Hawaii USA); Heiner Dietze (Marine Biogeochemie Leibniz-Institut für Meereswissenschaften Kiel Germany)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004936

URL: <https://doi.org/10.1029/2008jc004936>

Publication Date: 2009-06-10

Volume/Issue: Volume 114, Issue C6

References: 52

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: On the basis of integrations of an eddy-permitting coupled physical-biological model of the tropical Pacific we explore changes in the simulated mean circulation as well as its intraseasonal to interannual variability driven by the biologically modulated vertical absorption profiles of solar radiation. Three sensitivity ocean hind-cast experiments, covering the period from 1948 to 2003, are performed. In the first one, simulated chlorophyll affects the attenuation of light in the water column, while in the second experiment, the chlorophyll concentration is kept constant in time by prescribing an empirically derived spatial pattern. The third experiment uses a spatially and temporally constant value for the attenuation depth. The biotically induced differential heating is generated by increased absorption of light in the surface layers, leading to a surface warming and subsurface cooling. The effect is largest in the eastern equatorial Pacific. However, the initial vertical redistribution of heat leads to considerable changes of the near-surface ocean circulation subsequently influencing the near-surface temperature structure. In general, including biophysical coupling improves the model performance in terms of temperature and ocean circulation patterns. In particular, the upwelling in the eastern equatorial Pacific is enhanced, the mixed layer becomes shallower, the warm bias in the eastern Pacific is reduced, and the zonal temperature gradient increases. This leads to stronger La Niña events and an associated increase in the variability of the Niño3 SSTA time series. Furthermore, the eddy kinetic energy (EKE) associated with mesoscale eddies in the eastern equatorial Pacific increases by almost 100% because of enhanced EKE production due to enhanced horizontal and vertical shear of the mean currents.

2198. Mapping Ocean Observations in a Dynamical Framework: A 2004-06 Ocean Atlas

Authors: Gaël Forget (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2009jpo4043.1

URL: <https://doi.org/10.1175/2009jpo4043.1>

Publication Date: 2009-05-15

Volume/Issue: Volume 40, Issue 6

References: 43

Cited by: 96 publications

Language: en

Subjects: N/A

Keywords: Sea surface temperature, Databases, Salinity

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper exploits a new observational atlas for the near-global ocean for the best-observed 3-yr period from December 2003 through November 2006. The atlas consists of mapped observations and derived quantities. Together they form a full representation of the ocean state and its seasonal cycle. The mapped observations are primarily altimeter data, satellite SST, and Argo profiles. GCM interpolation is used to synthesize these datasets, and the resulting atlas is a fairly close fit to each one of them. For observed quantities especially, the atlas is a practical means to evaluate free-running GCM simulations and to put field experiments into a broader context. The atlas-derived quantities include the middepth dynamic topography, as well as ocean fluxes of heat and salt-freshwater. The atlas is publicly available online (www.ecco-group.org). This paper provides insight into two oceanographic problems that are the subject of vigorous ongoing research. First, regarding ocean circulation estimates, it can be inferred that the RMS uncertainty in modern surface dynamic topography (SDT) estimates is only on the order of 3.5 cm at scales beyond 300 km. In that context, it is found that assumptions of "reference-level" dynamic topography may yield significant errors (of order 2.2 cm or more) in SDT estimates using in situ data. Second, in the perspective of mode water investigations, it is estimated that ocean fluxes (advection plus mixing) largely contribute to the seasonal fluctuation in heat content and freshwater/salt content. Hence, representing the seasonal cycle as a simple interplay of air-sea flux and ocean storage would not yield a meaningful approximation. For the salt-freshwater seasonal cycle especially, contributions from ocean fluxes usually exceed direct air-sea flux contributions.

2199. Central South Pacific thermocline water circulation from a high-resolution ocean model validated against satellite data: Seasonal variability and El Niño 1997-1998 influence

Authors: Elodie Martinez (Laboratoire Terre-Océan Université de la Polynésie Française Faaa French Polynesia); Alexandre Ganachaud (Institut de Recherche pour le Développement (IRD)); Jerome Lefevre (Institut de Recherche pour le Développement (IRD)); Keitapu Maamaatuaiahutapu (Laboratoire Terre-Océan Université de la Polynésie Française Faaa French Polynesia)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004824

URL: <https://doi.org/10.1029/2008jc004824>

Publication Date: 2009-05-12

Volume/Issue: Volume 114, Issue C5

References: 54

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The oceanic circulation in French Polynesia, central South Pacific, is investigated based on a high-resolution model validated with satellite data. We focus on the upper 500 m, where ocean current variability has been barely documented due to a lack of in situ measurements. Seasonal and El Niño-Southern Oscillation 1997-1999 variability of surface and subsurface currents are characterized. In the north, the South Equatorial Current (SEC) follows a seasonal cycle with winter enhancement corresponding to trade wind strengthening. During summer, the SEC slows and the eastward South Equatorial Countercurrent (SECC) appears across the domain north of an upward Ekman pumping region related to the South Pacific Convergence Zone development in the area. Around the Marquesas Islands, a wind curl dipole is suspected to create westward and eastward (Marquesas Countercurrent, MCC) jets. The eastward Subtropical Countercurrent (STCC) is found south of 20 degreesS. It is disturbed by eddies and broken into westward and eastward branches. During El Niño 1997-1998, due to the reversed winds in the equatorial band, the SEC weakened, while eastward countercurrents appeared. The SECC reinforced and moved northeastward in summer 1998. During La Niña 1998-1999, following the trade wind enhancement, the SEC strengthened. The countercurrents weakened or disappeared and the SECC moved southwestward in summer 1999. During both El Niño and La Niña events, the STCC eddy activity strengthened, while the MCC and associated westward jet were absent. These variations and their dynamical origins are discussed, and a coherent picture of the French Polynesian current system and its variations is proposed.

2200. Oceanic time variability near a large scale topographic circulation

Authors: Sebastien Bigorre; William K. Dewar

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2009.04.004

URL: <https://doi.org/10.1016/j.ocemod.2009.04.004>

Publication Date: 2009-05-05

Volume/Issue: Volume 29, Issue 3

References: 30

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2201. The use of gravimetric data from GRACE mission in the understanding of polar motion variations

Authors: L. Seoane; J. Nastula; C. Bizouard; D. Gambis

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2009.04181.x

URL: <https://doi.org/10.1111/j.1365-246x.2009.04181.x>

Publication Date: 2009-04-30

Volume/Issue: Volume 178, Issue 2

References: 23

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2202. A central California coastal ocean modeling study: 2. Adjoint sensitivities to local and remote forcing mechanisms

Authors: M. Veneziani (University of California Santa Cruz); C. A. Edwards (University of California Santa Cruz); A. M. Moore (University of California Santa Cruz)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004775

URL: <https://doi.org/10.1029/2008jc004775>

Publication Date: 2009-04-28

Volume/Issue: Volume 114, Issue C4

References: 43

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The sensitivity of particular features within the California Current System to remote and local forcing is numerically investigated using the adjoint model version of the Regional Ocean Modeling System. We define three integral measures of nearshore surface temperature, cross-shore transport, and sea surface height as metrics that characterize important aspects of the coastal circulation. We determine the metrics' relative sensitivity to different components of the surface external forcing and model initial condition fields. Calculations are performed for 14-d intervals over the 5-year period from 2000 to 2004. Sensitivities are ranked in terms of their relative importance in the annual average, and their seasonal cycle is examined. Contributions from different portions of the domain are identified to determine the importance of local versus remote forcing. Local forcing, particularly wind stress and, at times, heat flux, are the dominant forcing contributors to temperature and transport sensitivities, although model initial conditions can contribute quantitatively on the same order as surface fluxes. The sea surface height metric reveals important nonlocal effects, particularly along the open boundary of this regional simulation.

2203. On the Relationship between the North Atlantic Meridional Overturning Circulation and the Surface-Forced Overturning Streamfunction

Authors: Jeremy P. Grist (National Oceanography Centre); Robert Marsh (University of Southampton, National Oceanography Centre); Simon A. Josey (National Oceanography Centre)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2009jcli2574.1

URL: <https://doi.org/10.1175/2009jcli2574.1>

Publication Date: 2009-04-27

Volume/Issue: Volume 22, Issue 19

References: 37

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The influence of surface thermohaline forcing on the variability of the Atlantic meridional overturning circulation (MOC) at mid-high latitudes is investigated using output from three Intergovernmental Panel on Climate Change (IPCC) coupled climate models. The method employed is an extension of the surface-forced streamfunction approach, based on water mass transformation theory, used in an earlier study by Marsh (2000). The maximum value of the MOC at 48 degreesN is found to have a significant lagged relationship with the maximum surface-forced streamfunction in the region north of 48 degreesN with a surface density greater than $\sigma_0 = 27.5 \text{ kg m}^{-3}$. This correlation peaks when the index of the surface-forced streamfunction leads the MOC by 2-4 yr, depending on the coupled model considered. A method for estimating the MOC variability solely from the surface forcing fields is developed and found to be in good agreement with the actual model MOC variability in all three of the models considered when a past averaging window of 10 yr is employed. This method is then applied with NCEP-NCAR reanalysis surface flux fields for the period 1949-2007 to reconstruct MOC strength over 1958-2007. The reconstructed MOC shows considerable multidecadal variability but no discernible trend over the modern observational era.

2204. A central California coastal ocean modeling study: 1. Forward model and the influence of realistic versus climatological forcing

Authors: M. Veneziani (University of California Santa Cruz); C. A. Edwards (University of California Santa Cruz); J. D. Doyle (Naval Research Laboratory Monterey California USA); D. Foley (Environmental Research Division NOAA Southwest Fisheries Science Center Pacific Grove California USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004774

URL: <https://doi.org/10.1029/2008jc004774>

Publication Date: 2009-04-24

Volume/Issue: Volume 114, Issue C4

References: 72

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We report on a numerical simulation of the California Current circulation using the Regional Ocean Modeling System model, focusing on the region of northern and central California during the 5-year period from 2000 to 2004. Unlike previous model studies of the California Current System, the present configuration is characterized by both realistic external forcing and a spatial domain covering most of the North American West Coast. Specifically, this configuration is driven at the surface by high-resolution meteorological fields from the Coupled Ocean-Atmosphere Mesoscale Prediction System and at the lateral open boundaries by output from the project Estimating the Circulation and Climate of the Ocean supported by the Global Ocean Data Assimilation Experiment. The simulation is evaluated favorably through quantitative comparisons with the California Cooperative Fisheries Investigations data set, satellite-derived sea surface temperature, and surface drifters-derived eddy kinetic energy. The impact of adopting realistic versus climatological surface forcing is demonstrated by comparing mean and mesoscale circulation characteristics. Realistic surface forcing qualitatively alters the seasonal cycle of the mean alongshore jet and better reproduces the summer spatial structure and intensity of the eddy kinetic energy field along the central California coast.

2205. Ocean's response to Hurricane Frances and its implications for drag coefficient parameterization at high wind speeds

Authors: S. E. Zedler (Scripps Institution of Oceanography); P. P. Niiler (Scripps Institution of Oceanography); D. Stammer (Institut fuer Meereskunde, Zentrum fuer Meeres- und Klimaforschung Universitaet Hamburg Hamburg Germany); E. Terrill (Scripps Institution of Oceanography); J. Morzel (Rosetta Consulting Boulder Colorado USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc005205

URL: <https://doi.org/10.1029/2008jc005205>

Publication Date: 2009-04-24

Volume/Issue: Volume 114, Issue C4

References: 40

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The drag coefficient parameterization of wind stress is investigated for tropical storm conditions using model sensitivity studies. The Massachusetts Institute of Technology (MIT) Ocean General Circulation Model was run in a regional setting with realistic stratification and forcing fields representing Hurricane Frances, which in early September 2004 passed east of the Caribbean Leeward Island chain. The model was forced with a NOAA-HWIND wind speed product after converting it to wind stress using four different drag coefficient parameterizations. Respective model results were tested against in situ measurements of temperature profiles and velocity, available from an array of 22 surface drifters and 12 subsurface floats. Changing the drag coefficient parameterization from one that saturated at a value of 2.3×10^{-3} to a constant drag coefficient of 1.2×10^{-3} reduced the standard deviation difference between the simulated minus the measured sea surface temperature change from 0.8 degreesC to 0.3 degreesC. Additionally, the standard deviation in the difference between simulated minus measured high pass filtered 15-m current speed reduced from 15 cm/s to 5 cm/s. The maximum difference in sea surface temperature response when two different turbulent mixing parameterizations were implemented was 0.3 degreesC, i.e., only 11% of the maximum change of sea surface temperature caused by the storm.

2206. The sensitivity of the Seychelles-Chagos thermocline ridge to large-scale wind anomalies

Authors: Juliet C. Hermes (South African Environmental Observation Network, Private Bag X2, Roggebaai 8012, South Africa); Chris J. C. Reason (University of Cape Town)

Journal: ICES Journal of Marine Science

Publisher: Oxford University Press (OUP)

DOI: 10.1093/icesjms/fsp074

URL: <https://doi.org/10.1093/icesjms/fsp074>

Publication Date: 2009-04-09

Volume/Issue: Volume 66, Issue 7

References: 41

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Hermes, J. C., and Reason, C. J. C. 2009. The sensitivity of the Seychelles-Chagos thermocline ridge to large-scale wind anomalies. - ICES Journal of Marine Science, 66: 1455-1466. The Seychelles-Chagos thermocline ridge (SCTR) in the southwest tropical Indian Ocean is important for regional climate, the Madden-Julian Oscillation, as well as upper-ocean nutrients and related phytoplankton and zooplankton densities. Subsurface variability in this region has been proved to influence the overlying sea surface temperatures, which in turn can influence eastern African rainfall. There is evidence that austral summers with a deeper (shallower) SCTR tend to have more (less) tropical cyclone (TC) days in the Southwest Indian Ocean. The importance of this relationship was underlined during the 2006/2007 austral summer, when areas of Madagascar and central Mozambique experienced devastating floods, because of ten named tropical storms, including several intense TCs, effecting on these areas. At the same time, the SCTR during this season was anomalously deep, partly because of a downwelling Rossby wave that propagated across the South Indian Ocean during the previous austral winter/spring. In this paper, a regional ocean model is used to investigate the effect of remote forcing on this region and to study the sensitivity of the SCTR to changes in the large-scale winds over the South Indian Ocean, with a particular focus on the events of the 2006/2007 austral summer.

2207. Seawater density variations in the North Atlantic and the Atlantic meridional overturning circulation

Authors: Chunzai Wang; Shenfu Dong; Ernesto Munoz

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-009-0560-5

URL: <https://doi.org/10.1007/s00382-009-0560-5>

Publication Date: 2009-04-07

Volume/Issue: Volume 34, Issue 7-8

References: 46

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2208. Application of 4D-Variational data assimilation to the California Current System

Authors: G. Broquet; C.A. Edwards; A.M. Moore; B.S. Powell; M. Veneziani; J.D. Doyle

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2009.03.001

URL: <https://doi.org/10.1016/j.dynatmoce.2009.03.001>

Publication Date: 2009-04-06

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References: 48

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2209. Effect of the satellite laser ranging network distribution on geocenter motion estimation

Authors: X. Collilieux (Laboratoire de Recherche en Géodésie Institut Géographique National Marne-La-Vallée France); Z. Altamimi (Laboratoire de Recherche en Géodésie Institut Géographique National Marne-La-Vallée France); J. Ray (NOAA National Geodetic Survey Silver Spring Maryland USA); T. van Dam (University of Luxembourg); X. Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jb005727

URL: <https://doi.org/10.1029/2008jb005727>

Publication Date: 2009-04-06

Volume/Issue: Volume 114, Issue B4

References: 48

Cited by: 65 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: SLR network translations estimated between a quasi-instantaneous station position set, theoretically expressed with respect to the center of mass of the Earth (CM), and a secular reference frame are the signature of the motion of the CM with respect to the Earth crust. Geocenter motion is defined here to be the motion of the CM with respect to the geometric center of the solid Earth surface (CF). SLR translational variations cannot be rigorously interpreted as identical to geocenter motion due to the sparse and nonuniform distribution of the SLR network. Their difference is called the network effect, which should be dominated at subdecadal timescales by loading signals. We have computed translation time series of the SLR network using two independent geophysically based loading models. One is a displacement model estimated from surface fluid data (Green's function approach), called forward model, and the other is a displacement model estimated from GPS and ocean bottom pressure (OBP) data, called inverse model. The translation models have been subtracted from their respective geocenter motion models computed from degree-1 mass load coefficients in order to evaluate their network effect biases. Scatter due to the SLR network effect is at the level of 1.5 mm RMS. It could slightly shift the phase of the annual SLR geocenter motion estimate by less than 1 month and could affect X and Z annual geocenter motion amplitudes at the 1-mm level, which is about one third of the expected signal. Two distinct methods are suggested to account for network effect when comparing SLR translations to geocenter motion models. The first is to add the network effect term predicted by a displacement model to the geocenter motion loading model. The second relies on an adequate combination of SLR and GPS products to estimate SLR translation that could be better compared with geocenter motion.

2210. Observation and origin of an interannual salinity anomaly in the Mozambique Channel

Authors: P. M. van der Werf (Utrecht University); M. W. Schouten (Royal Netherlands Institute for Sea Research); P. J. van Leeuwen (Utrecht University); H. Ridderinkhof (Royal Netherlands Institute for Sea Research); W. P. M. de Ruijter (Utrecht University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004911

URL: <https://doi.org/10.1029/2008jc004911>

Publication Date: 2009-03-23

Volume/Issue: Volume 114, Issue C3

References: 37

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A positive salinity anomaly of 0.2 PSU was observed between 50 and 200 m over the years 2000-2001 across the Mozambique Channel at a section at 17 degreesS which was repeated in 2003, 2005, 2006, and 2008. Meanwhile, a moored array is continued from 2003 to 2008. This anomaly was most distinct showing an interannual but nonseasonal variation. The possible origin of the anomaly is investigated using output from three ocean general circulation models (Estimating the Circulation and Climate of the Ocean, Ocean Circulation and Climate Advanced Modeling, and Parallel Ocean Program). The most probable mechanism for the salinity anomaly is the anomalous inflow of subtropical waters caused by a weakening of the northern part of the South Equatorial Current by weaker trade winds. This mechanism was found in all three numerical models. In addition, the numerical models indicate a possible salinization of one of the source water masses to the Mozambique Channel as an additional cause of the anomaly. The anomaly propagated southward into the Agulhas Current and northward along the African coast.

2211. Improved inverse and probabilistic methods for geophysical applications of GRACE gravity data

Authors: Xiaoping Wu; Ronald G. Blom; Erik R. Ivins; Fabiano A. Oyafuso; Min Zhong

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2009.04141.x

URL: <https://doi.org/10.1111/j.1365-246x.2009.04141.x>

Publication Date: 2009-03-19

Volume/Issue: Volume 177, Issue 3

References: 33

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2212. Rate of Work Done by Atmospheric Pressure on the Ocean General Circulation and Tides

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2008jpo4034.1

URL: <https://doi.org/10.1175/2008jpo4034.1>

Publication Date: 2009-03-06

Volume/Issue: Volume 39, Issue 2

References: 17

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Quantitative analysis of the energetics of the ocean is crucial for understanding its circulation and mixing. The power input by fluctuations in atmospheric pressure p_a resulting from the S1 and S2 air tides and the stochastic continuum is analyzed here, with a focus on globally integrated, time-mean values. Results are based on available 1 degrees x 1 degrees near-global p_a and sea level fields and are intended as mainly order-of-magnitude estimates. The rate of work done on the radiational and gravitational components of the S2 ocean tide is estimated at 14 and -60 GW, respectively, mostly occurring at low latitudes. The net extraction of energy at a rate of -46 GW is about 10% of available estimates of the work rates by gravity on the S2 tide. For the mainly radiational S1 tide, the power input by p_a is much weaker (0.25 GW). Based on daily mean quantities, the stochastic p_a continuum contributes ~3 GW to the nontidal circulation, with substantial power input being associated with the p_a -driven dynamic response in the Southern Ocean at submonthly time scales. Missing contributions from nontidal variability at the shortest periods (≤ 2 days) may be substantial, but the rate of work done by p_a on the general circulation is likely to remain $< 1\%$ of the available wind input estimates. The importance of p_a effects when considering local, time-variable energetics remains a possibility, however.

2213. Validating global hydrological models by ground and space gravimetry

Authors: JiangCun Zhou; HePing Sun; JianQiao Xu

Journal: Science Bulletin

Publisher: Elsevier BV

DOI: 10.1007/s11434-009-0006-9

URL: <https://doi.org/10.1007/s11434-009-0006-9>

Publication Date: 2009-03-04

Volume/Issue: Volume 54, Issue 9

References: 28

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2214. Absolute flow field estimation for the Nordic seas from combined gravimetric, altimetric, and in situ data

Authors: A. Hunegnaw (University of Edinburgh); F. Siegismund (Nansen Environmental and Remote Sensing Center); R. Hipkin (University of Edinburgh); K. A. Mork (Bjerknes Centre for Climate Research)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004797

URL: <https://doi.org/10.1029/2008jc004797>

Publication Date: 2009-02-26

Volume/Issue: Volume 114, Issue C2

References: 49

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In the Nordic seas, we combine a computation of absolute surface current flow derived from geodetic data with in situ historical hydrographic data to estimate the absolute volume, heat, and salt transports as a function of depth. Our mean dynamic topography (MDT) is calculated from marine, airborne and satellite gravimetry, combined with satellite altimetry, using a new algorithm called the iterative combination method (ICM). Residual noise in the gravimetric geoid is the limit on MDT resolution and is suppressed using a Gaussian filter with a width at half-peak amplitude of 59 km. Detailed and coherent flow paths for surface geostrophic currents are clearly identified. ICM MDT was used as fixed boundary condition to transform historical hydrography into absolute estimates of volume, heat, and salt transport, replacing the assumption of an isobaric surface at a predetermined depth. For the inflow of Atlantic Water (potential temperature $\Theta > 6$ degreesC) through the Faroe-Shetland Channel into the Nordic seas, we obtain time-averaged fluxes between 1993 and 1996 of 3.5 Sv (volume), 121 TW (heat), and 124×10^6 kg s⁻¹ (salt), very close to reported observations from acoustic Doppler current profiler moorings and conductivity-temperature-depth data. For the Svinøy section, we obtain a northward transport of Atlantic Water ($S > 35.0$, $T > 5.0$ degreesC) of 3.9 Sv in the eastern branch of the Norwegian Atlantic Current comparable with reported measurements of 4.2 Sv. Similarly good agreement is found for the Hornbanki and Iceland-Faroe Ridge sections and for monitoring Atlantic Water outflow across the Barents Sea Opening to the Arctic shelf.

2215. Initializing Decadal Climate Predictions with the GECCO Oceanic Synthesis: Effects on the North Atlantic

Authors: Holger Pohlmann (Max-Planck-Institut für Meteorologie, Hamburg, Germany); Johann H. Jungclaus (Max-Planck-Institut für Meteorologie, Hamburg, Germany); Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg); Jochem Marotzke (Max-Planck-Institut für Meteorologie, Hamburg, Germany)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2009jcli2535.1

URL: <https://doi.org/10.1175/2009jcli2535.1>

Publication Date: 2009-02-23

Volume/Issue: Volume 22, Issue 14

References: 49

Cited by: 232 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This study aims at improving the forecast skill of climate predictions through the use of ocean synthesis data for initial conditions of a coupled climate model. For this purpose, the coupled model of the Max Planck Institute (MPI) for Meteorology, which consists of the atmosphere model ECHAM5 and the MPI Ocean Model (MPI-OM), is initialized with oceanic synthesis fields available from the German contribution to Estimating the Circulation and Climate of the Ocean (GECCO) project. The use of an anomaly coupling scheme during the initialization avoids the main problems with drift in the climate predictions. Thus, the coupled model is continuously forced to follow the density anomalies of the GECCO synthesis over the period 1952-2001. Hindcast experiments are initialized from this experiment at constant intervals. The results show predictive skill through the initialization up to the decadal time scale, particularly over the North Atlantic. Viewed over the time scales analyzed here (annual, 5-yr, and 10-yr mean), greater skill for the North Atlantic sea surface temperature (SST) is obtained in the hindcast experiments than in either a damped persistence or trend forecast. The Atlantic meridional overturning circulation hindcast closely follows that of the GECCO oceanic synthesis. Hindcasts of global-mean temperature do not obtain greater skill than either damped persistence or a trend forecast, owing to the SST errors in the GECCO synthesis, outside the North Atlantic. An ensemble of forecast experiments is subsequently performed over the period 2002-11. North Atlantic SST from the forecast experiment agrees well with observations until the year 2007, and it is higher than if simulated without the oceanic initialization (averaged over the forecast period). The results confirm that both the initial and the boundary conditions must be accounted for in decadal climate predictions.

2216. Oceanic sources, sinks, and transport of atmospheric CO₂

Authors: Nicolas Gruber (ETH Zurich); Manuel Gloor (University of Leeds); Sara E. Mikaloff Fletcher (Princeton University); Scott C. Doney (Woods Hole Oceanographic Institution); Stephanie Dutkiewicz (Massachusetts Institute of Technology); Michael J. Follows (Massachusetts Institute of Technology); Markus Gerber (University of Bern); Andrew R. Jacobson (NOAA Earth System Research Laboratory Global Monitoring Division Boulder Colorado USA); Fortunat Joos (University of Bern); Keith Lindsay (National Center for Atmospheric Research); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anne Mouchet (University of Liège); Simon A. Müller (University of Bern); Jorge L. Sarmiento (Princeton University); Taro Takahashi (Columbia University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008gb003349

URL: <https://doi.org/10.1029/2008gb003349>

Publication Date: 2009-02-18

Volume/Issue: Volume 23, Issue 1

References: 111

Cited by: 465 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We synthesize estimates of the contemporary net air-sea CO₂ flux on the basis of an inversion of interior ocean carbon observations using a suite of 10 ocean general circulation models (Mikaloff Fletcher et al., 2006, 2007) and compare them to estimates based on a new climatology of the air-sea difference of the partial pressure of CO₂ (pCO₂) (Takahashi et al., 2008). These two independent flux estimates reveal a consistent description of the regional distribution of annual mean sources and sinks of atmospheric CO₂ for the decade of the 1990s and the early 2000s with differences at the regional level of generally less than 0.1 Pg C a⁻¹. This distribution is characterized by outgassing in the tropics, uptake in midlatitudes, and comparatively small fluxes in the high latitudes. Both estimates point toward a small (~ -0.3 Pg C a⁻¹) contemporary CO₂ sink in the Southern Ocean (south of 44 degreesS), a result of the near cancellation between a substantial outgassing of natural CO₂ and a strong uptake of anthropogenic CO₂. A notable exception in the generally good agreement between the two estimates exists within the Southern Ocean: the ocean inversion suggests a relatively uniform uptake, while the pCO₂-based estimate suggests strong uptake in the region between 58 degreesS and 44 degreesS, and a source in the region south of 58 degreesS. Globally and for a nominal period between 1995 and 2000, the contemporary net air-sea flux of CO₂ is estimated to be -1.7 +/- 0.4 Pg C a⁻¹ (inversion) and -1.4 +/- 0.7 Pg C a⁻¹ (pCO₂-climatology), respectively, consisting of an outgassing flux of river-derived carbon of ~+0.5 Pg C a⁻¹, and an uptake flux of anthropogenic carbon of -2.2 +/- 0.3 Pg C a⁻¹ (inversion) and -1.9 +/- 0.7 Pg C a⁻¹ (pCO₂-climatology). The two flux estimates also imply a consistent description of the contemporary meridional transport of carbon with

southward ocean transport throughout most of the Atlantic basin, and strong equatorward convergence in the Indo-Pacific basins. Both transport estimates suggest a small hemispheric asymmetry with a southward transport of between -0.2 and -0.3 Pg C a^{-1} across the equator. While the convergence of these two independent estimates is encouraging and suggests that it is now possible to provide relatively tight constraints for the net air-sea CO_2 fluxes at the regional basis, both studies are limited by their lack of consideration of long-term changes in the ocean carbon cycle, such as the recent possible stalling in the expected growth of the Southern Ocean carbon sink.

2217. Mechanisms controlling seasonal-to-interannual mixed layer temperature variability in the southeastern tropical Indian Ocean

Authors: D. J. Halkides (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004949

URL: <https://doi.org/10.1029/2008jc004949>

Publication Date: 2009-02-16

Volume/Issue: Volume 114, Issue C2

References: 72

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We use an Estimating the Circulation and Climate of the Ocean assimilation product to investigate seasonal-interannual mixed layer temperature (MLT) budgets in the southeastern tropical Indian Ocean (SETIO) during 1993-2006. We examine spatial inhomogeneity of the SETIO MLT budget, contrasting three subregions with different forcing/circulation characteristics to better understand the area mean budget over the full SETIO. The subregions are the equatorial zone (box 1), the Sumatra-Java upwelling zone (box 2), and east of the thermocline ridge (box 3). Seasonally, surface heat flux dominates MLT in all regions; advection and subsurface processes generally play secondary roles. On interannual scales, surface heat flux makes major contributions in all three boxes to termination of SETIO cooling associated with the Indian Ocean Zonal/Dipole Mode. Ocean dynamics show vital differences between regions: Subsurface processes cool box 1 and 2 but warm box 3. Horizontal advection warms box 1 but cools box 2 and 3. Averaging the MLT budget over the SETIO obscures regional physics. We explain spatial variations of the SETIO MLT budget in terms of differences in forcing, circulation, MLT distribution, and mixed layer and barrier layer thicknesses. We also examine SETIO MLT budget differences during 1994, 1997, and 2006, years exhibiting notable SETIO cooling events. In box 1, horizontal advection dominates warming after the 1994 and 2006 coolings, while in 1997, surface heat flux dominates warming. In box 2, cooling peaks earlier in 1994 than in 1997 and 2006 because of subsurface processes. Last, we show that the MLT budget is very different from heat budgets for fixed depth layers (e.g., the top 50-60 m).

2218. The impact of GRACE, GPS and OBP data on estimates of global mass redistribution

Authors: M. J. F. Jansen; B. C. Gunter; J. Kusche

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2008.04031.x

URL: <https://doi.org/10.1111/j.1365-246x.2008.04031.x>

Publication Date: 2009-02-16

Volume/Issue: Volume 177, Issue 1

References: 23

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2219. Using Transformation and Formation Maps to Study the Role of Air-Sea Heat Fluxes in North Atlantic Eighteen Degree Water Formation

Authors: Guillaume Maze (Massachusetts Institute of Technology); Gael Forget (Massachusetts Institute of Technology); Martha Buckley (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology); Ivana Cerovecki (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2009jpo3985.1

URL: <https://doi.org/10.1175/2009jpo3985.1>

Publication Date: 2009-02-13

Volume/Issue: Volume 39, Issue 8

References: 25

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Walin water mass framework quantifies the rate at which water is transformed from one temperature class to another by air-sea heat fluxes (transformation). The divergence of the transformation rate yields the rate at which a given temperature range is created or destroyed by air-sea heat fluxes (formation). Walin's framework provides a precise integral statement at the expense of losing spatial information. In this study the integrand of Walin's expression to yield transformation and formation maps is plotted and used to study the role of air-sea heat fluxes in the cycle of formation-destruction of the 18 degrees +/- 1 degreesC layer in the North Atlantic. Using remotely sensed sea surface temperatures and air-sea heat flux estimates based on both analyzed meteorological fields and ocean data-model syntheses for the 3-yr period from 2004 to 2006, the authors find that Eighteen Degree Water (EDW) is formed by air-sea heat fluxes in the western part of the subtropical gyre, just south of the Gulf Stream. The formation rate peaks in February when the EDW layer is thickened by convection owing to buoyancy loss. EDW is destroyed by air-sea heat fluxes from spring to summer over the entire subtropical gyre. In the annual mean there is net EDW formation in the west to the south of the Gulf Stream, and net destruction over the eastern part of the gyre. Results suggest that annual mean formation rates of EDW associated with air-sea fluxes are in the range from 3 to 5 Sv ($\text{Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$). Finally, error estimates are computed from sea surface temperature and heat flux data using an ensemble perturbation method. The transformation/formation patterns are found to be robust and errors mostly affect integral quantities.

2220. Origin and Pathway of Equatorial 13 degreesC Water in the Pacific Identified by a Simulated Passive Tracer and Its Adjoint*

Authors: Tangdong Qu (University of Hawaii at Manoa); Shan Gao (University of Hawaii at Manoa, Chinese Academy of Sciences); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Rana A. Fine (University of Miami); Eric J. Lindstrom (Science Mission Directorate, National Aeronautics and Space Administration, Washington, D.C)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2009jpo4045.1

URL: <https://doi.org/10.1175/2009jpo4045.1>

Publication Date: 2009-02-13

Volume/Issue: Volume 39, Issue 8

References: 56

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The origin and pathway of the thermostad water in the eastern equatorial Pacific Ocean, often referred to as the equatorial 13 degreesC Water, are investigated using a simulated passive tracer and its adjoint, based on circulation estimates of a global general circulation model. Results demonstrate that the source region of the 13 degreesC Water lies well outside the tropics. In the South Pacific, some 13 degreesC Water is formed northeast of New Zealand, confirming an earlier hypothesis on the water's origin. The South Pacific origin of the 13 degreesC Water is also related to the formation of the Eastern Subtropical Mode Water (ESTMW) and the Sub-Antarctic Mode Water (SAMW). The portion of the ESTMW and SAMW that eventually enters the density range of the 13 degreesC Water ($25.8 < \sigma_{\theta} < 26.6 \text{ kg m}^{-3}$) does so largely by mixing. Water formed in the subtropics enters the equatorial region predominantly through the western boundary, while its interior transport is relatively small. The fresher North Pacific ESTMW and Central Mode Water (CMW) are also important sources of the 13 degreesC Water. The ratio of the southern versus the northern origins of the water mass is about 2 to 1 and tends to increase with time elapsed from its origin. Of the total volume of initially tracer-tagged water in the eastern equatorial Pacific, approximately 47.5% originates from depths above $\sigma_{\theta} = 25.8 \text{ kg m}^{-3}$ and 34.6% from depths below $\sigma_{\theta} = 26.6 \text{ kg m}^{-3}$, indicative of a dramatic impact of mixing on the route of subtropical water to becoming the 13 degreesC Water. Still only a small portion of the water formed in the subtropics reaches the equatorial region, because most of the water is trapped and recirculates in the subtropical gyre.

2221. Labrador Sea convection and subpolar North Atlantic Deep Water export in the SODA assimilation model

Authors: Friedrich A. Schott; Lothar Stramma; Benjamin S. Giese; Rainer Zantopp

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2009.01.001

URL: <https://doi.org/10.1016/j.dsr.2009.01.001>

Publication Date: 2009-01-27

Volume/Issue: Volume 56, Issue 6

References: 38

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2222. Separation of coseismic and postseismic gravity changes for the 2004 Sumatra-Andaman earthquake from 4.6 yr of GRACE observations and modelling of the coseismic change by normal-modes summation

Authors: Caroline de Linage; Luis Rivera; Jacques Hinderer; Jean-Paul Boy; Yves Rogister; Sophie Lambotte; Richard Biancale

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2008.04025.x

URL: <https://doi.org/10.1111/j.1365-246x.2008.04025.x>

Publication Date: 2009-01-20

Volume/Issue: Volume 176, Issue 3

References: 45

Cited by: 89 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2223. The role of salinity in the decadal variability of the North Atlantic meridional overturning circulation

Authors: Claude Frankignoul; Julie Deshayes; Ruth Curry

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-008-0523-2

URL: <https://doi.org/10.1007/s00382-008-0523-2>

Publication Date: 2009-01-09

Volume/Issue: Volume 33, Issue 6

References: 62

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2224. Assessing 4D-VAR for dynamical mapping of coastal high-frequency radar in San Diego

Authors: I. Hoteit; B. Cornuelle; S.Y. Kim; G. Forget; A. Köhl; E. Terrill

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2008.11.005

URL: <https://doi.org/10.1016/j.dynatmoce.2008.11.005>

Publication Date: 2008-12-28

Volume/Issue: Volume 48, Issue 1-3

References: 44

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2225. On the Influence of Random Wind Stress Errors on the Four-Dimensional, Midlatitude Ocean Inverse Problem

Authors: Tsuyoshi Wakamatsu (Fisheries and Oceans Canada, Institute of Ocean Sciences, Sidney, British Columbia, Canada); Michael G. G. Foreman (Fisheries and Oceans Canada, Institute of Ocean Sciences, Sidney, British Columbia, Canada); Patrick F. Cummins (Fisheries and Oceans Canada, Institute of Ocean Sciences, Sidney, British Columbia, Canada); Josef Y. Cherniawsky (Fisheries and Oceans Canada, Institute of Ocean Sciences, Sidney, British Columbia, Canada)

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/2008mwr2621.1

URL: <https://doi.org/10.1175/2008mwr2621.1>

Publication Date: 2008-12-15

Volume/Issue: Volume 137, Issue 6

References: 33

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The effects of the parameterized wind stress error covariance function on the a priori error covariance of an ocean general circulation model (OGCM) are examined. These effects are diagnosed by computing the projection of the a priori model state error covariance matrix to sea surface height (SSH). The sensitivities of the a priori error covariance to the wind stress curl error are inferred from the a priori SSH error covariance and are shown to differ between the subpolar and subtropical gyres because of different contributions from barotropic and baroclinic ocean dynamics. The spatial structure of the SSH error covariance due to the wind stress error indicates that the a priori model state error is determined indirectly by the wind stress curl error. The impact of this sensitivity on the solution of a four-dimensional inverse problem is inferred.

2226. Estimating weights for the use of time-dependent gravity recovery and climate experiment data in constraining ocean models

Authors: Katherine J. Quinn (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004903

URL: <https://doi.org/10.1029/2008jc004903>

Publication Date: 2008-12-15

Volume/Issue: Volume 113, Issue C12

References: 34

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Using Gravity Recovery And Climate Experiment (GRACE) data to constrain ocean general circulation models requires quantitative knowledge of the errors in GRACE-derived estimates of ocean bottom pressure (pb) change, which for our purposes include not only instrument noise but also variability not represented in the models (e.g., post-glacial rebound and self-gravitation effects). We attempt a spatial mapping of these errors by comparing several GRACE data products to pb simulations from an ocean model. Uncertainties in the global ocean mean, partly related to the net freshwater flux into the ocean, and in the regional pb anomalies about that mean are considered separately. The resultant regional error estimates (~1-3 cm), when zonally averaged, are comparable to the calibrated errors provided by the GRACE processing centers, except for enhanced errors near some continental regions with high seasonal hydrology signals or large mass trends. Errors in the GRACE-derived values estimated from model-data differences (~0.2 cm) are also comparable with those from the calibrated errors. For both pb and estimates, accounting for the effects of geocenter noise is important. Replacing the C20 harmonic term in the GRACE data with estimates derived from satellite laser ranging results in significantly lower errors in the Southern Ocean. We also find lower errors at high latitudes when the variability of the atmospheric pressure over the land is removed from the data. Given the estimated errors and model-data comparisons, GRACE data should be useful for constraining estimates of, particularly at interannual periods, but less so when considering regional pb variability.

2227. Low-frequency variations of the large-scale ocean circulation and heat transport in the North Atlantic from 1955-1998 in situ temperature and salinity data

Authors: T. Huck; A. Colin de Verdière; P. Estrade; R. Schopp

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008gl035635

URL: <https://doi.org/10.1029/2008gl035635>

Publication Date: 2008-12-12

Volume/Issue: Volume 35, Issue 23

References: 32

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Low-frequency variations of the large-scale ocean circulation in the Atlantic are reconstructed from NODC pentadal anomalies of temperature and salinity from 1955 to 1998 based on hydrographic data, in addition to atmospheric reanalysis surface forcing. Diagnostic ocean circulations are estimated from simple methods using dynamical model integrations: namely diagnostic, robust diagnostic, and short prognostic. Mean transports of heat and mass are sensitive to the method and model configuration, but their decadal variability is much more coherent and does not depend explicitly on the variations of the surface forcing, its influence being imprinted in the thermohaline structure. Multidecadal variations are of the order of 20%, with large transports in the subpolar gyre in the early 1960's and mid 1990's, and low values in the mid 1970's. By reducing the influence of subgrid-scale parameterizations and surface forcings, these methods offer alternatives to exhaustive GCM simulations.

2228. Regional patterns of observed sea level change: insights from a 1/4 degrees global ocean/sea-ice hindcast

Authors: Alix Lombard; Gilles Garric; Thierry Penduff

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-008-0161-6

URL: <https://doi.org/10.1007/s10236-008-0161-6>

Publication Date: 2008-12-03

Volume/Issue: Volume 59, Issue 3

References: 50

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2229. Calculating trends from GRACE in the presence of large changes in continental ice storage and ocean mass

Authors: Don P. Chambers

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2008.04012.x

URL: <https://doi.org/10.1111/j.1365-246x.2008.04012.x>

Publication Date: 2008-11-28

Volume/Issue: Volume 176, Issue 2

References: 27

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2230. Estimating the rates of mass change, ice volume change and snow volume change in Greenland from ICESat and GRACE data

Authors: D. C. Slobbe; P. Ditmar; R. C. Lindenbergh

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2008.03978.x

URL: <https://doi.org/10.1111/j.1365-246x.2008.03978.x>

Publication Date: 2008-11-25

Volume/Issue: Volume 176, Issue 1

References: 31

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2231. The circulation of the ocean and its variability

Authors: Carl Wunsch (Massachusetts Institute of Technology)

Journal: Progress in Physical Geography: Earth and Environment

Publisher: SAGE Publications

DOI: 10.1177/0309133308096753

URL: <https://doi.org/10.1177/0309133308096753>

Publication Date: 2008-11-18

Volume/Issue: Volume 32, Issue 4

References: 47

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2232. The role of vorticity fluxes in the dynamics of the Zapiola Anticyclone

Authors: Denis L. Volkov (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004841

URL: <https://doi.org/10.1029/2008jc004841>

Publication Date: 2008-11-17

Volume/Issue: Volume 113, Issue C11

References: 24

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Argentine Basin in the South Atlantic Ocean is one of the most energetic regions in the ocean with complicated dynamics, which plays an important role in the global climate. A number of observations have discovered an intense anticyclonic gyre of barotropic circulation around the Zapiola Rise in the center of the basin. Theoretical studies have shown that the Zapiola Anticyclone represents an eddy-driven flow controlled by bottom friction. Recent advances in high-resolution global-ocean data syntheses, performed using NASA supercomputing facilities, provide realistic simulations of the circulation and the variability in the Argentine Basin. Using these simulations and satellite altimeter observations, we analyzed the vorticity balance of the Zapiola Anticyclone. Our results suggest the dominance of vorticity fluxes and the advection of the potential vorticity over a nonuniform bottom topography in determining the variability of the gyre, while the impact of the local wind stress is small. The divergence of the relative vorticity anomaly advection by eddies is found to be the most important contributor to the relative vorticity flux divergence influencing the variability of the Zapiola Anticyclone. Our results demonstrate that the relative vorticity influencing the variability of the anticyclone is mainly advected from the south where the northern branch of the Antarctic Circumpolar Current at the Subpolar Front is located.

2233. Variability of sea ice simulations assessed with RGPS kinematics

Authors: R. Kwok (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. C. Hunke (Los Alamos National Laboratory); W. Maslowski (Department of Oceanography Graduate School of Engineering and Applied Sciences, Naval Postgraduate School Monterey California USA); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Zhang (University of Washington)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004783

URL: <https://doi.org/10.1029/2008jc004783>

Publication Date: 2008-11-11

Volume/Issue: Volume 113, Issue C11

References: 62

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sea ice drift and deformation from coupled ice-ocean models are compared with high-resolution ice motion from the RADARSAT Geophysical Processor System (RGPS). In contrast to buoy drift, the density and extent of the RGPS coverage allows a more extensive assessment and understanding of model simulations at spatial scales from ~10 km to near basin scales and from days to seasonal timescales. This work illustrates the strengths of the RGPS data set as a basis for examining model ice drift and its gradients. As it is not our intent to assess relative performance, we have selected four models with a range of attributes and grid resolution. Model fields are examined in terms of ice drift, export, deformation, deformation-related ice production, and spatial deformation patterns. Even though the models are capable of reproducing large-scale drift patterns, variability among model behavior is high. When compared to the RGPS kinematics, the characteristics shared by the models are (1) ice drift along coastal Alaska and Siberia is slower, (2) the skill in explaining the time series of regional divergence of the ice cover is poor, and (3) the deformation-related volume production is consistently lower. Attribution of some of these features to specific causes is beyond our current scope because of the complex interplay between model processes, parameters, and forcing. The present work suggests that high-resolution ice drift observations, like those from the RGPS, would be essential for future model developments, improvements, intercomparisons, and especially for evaluation of the small-scale behavior of models with finer grid spacing.

2234. Incorporating a semi-stochastic model of ocean-modulated westerly wind bursts into an ENSO prediction model

Authors: Geoffrey Gebbie; Eli Tziperman

Journal: Theoretical and Applied Climatology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00704-008-0069-6

URL: <https://doi.org/10.1007/s00704-008-0069-6>

Publication Date: 2008-11-06

Volume/Issue: Volume 97, Issue 1-2

References: 27

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2235. Super Cyclone Induces a Mesoscale Phytoplankton Bloom in the Bay of Bengal

Authors: P. Rahul Chand Reddy; P. S. Salvekar; Shailesh Nayak

Journal: IEEE Geoscience and Remote Sensing Letters

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/lgrs.2008.2000650

URL: <https://doi.org/10.1109/lgrs.2008.2000650>

Publication Date: 2008-11-06

Volume/Issue: Volume 5, Issue 4

References: 34

Cited by: 16 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2236. Analysis of large-scale ocean bottom pressure variability in the North Pacific

Authors: Don P. Chambers (University of Texas at Austin); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004930

URL: <https://doi.org/10.1029/2008jc004930>

Publication Date: 2008-11-03

Volume/Issue: Volume 113, Issue C11

References: 39

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We use the leading empirical orthogonal functions (EOFs) of ocean bottom pressure (OBP) derived from an ocean model and the technique of EOF reconstruction to reduce noise in the large-scale OBP variations derived from the Gravity Recovery and Climate Experiment (GRACE). The subsequent OBP variations from the model and GRACE are then examined in the North Pacific between January 2003 and May 2007. Although annual and semiannual variations are similar, GRACE observes large interannual fluctuations poleward of 30 degrees, where OBP increases from a low of nearly 3 cm below normal in early 2003 to normal throughout 2004 and 2005, then an increase of nearly the same magnitude in 2006. These fluctuations have also been observed in OBP inferred from satellite altimetry corrected for steric variations computed from Argo float data. Since GRACE and steric-corrected altimetry are completely independent observations of OBP, we conclude that the model has errors or deficiencies in predicting the interannual OBP fluctuations in the North Pacific.

2237. An improved empirical model for the effect of long-period ocean tides on polar motion

Authors: Richard S. Gross

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-008-0277-y

URL: <https://doi.org/10.1007/s00190-008-0277-y>

Publication Date: 2008-10-27

Volume/Issue: Volume 83, Issue 7

References: 25

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2238. Simulations of Hydrographic Properties in the Northwestern North Atlantic Ocean in Coupled Climate Models

Authors: M. F. de Jong (Royal Netherlands Institute for Sea Research); S. S. Drijfhout (Royal Netherlands Meteorological Institute); W. Hazeleger (Royal Netherlands Meteorological Institute); H. M. van Aken (Royal Netherlands Institute for Sea Research); C. A. Severijns (Royal Netherlands Meteorological Institute)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2008jcli2448.1

URL: <https://doi.org/10.1175/2008jcli2448.1>

Publication Date: 2008-10-22

Volume/Issue: Volume 22, Issue 7

References: 46

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The performance of coupled climate models (CCMs) in simulating the hydrographic structure and variability of the northwestern North Atlantic Ocean, in particular the Labrador and Irminger Seas, has been assessed. This area plays an important role in the meridional overturning circulation. Hydrographic properties of the preindustrial run of eight CCMs used in the Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report (AR4) are compared with observations from the World Ocean Circulation Experiment Repeat section 7 (WOCE AR7). The mean and standard deviation of 20 yr of simulated data are compared in three layers, representing the surface waters, intermediate waters, and deep waters. Two models simulate an extremely cold, fresh surface layer with model biases down to -1.7 psu and -4.0 degreesC, much larger than the observed ranges of variability. The intermediate and deep layers are generally too warm and saline, with biases up to 0.7 psu and 2.8 degreesC. An analysis of the maximum mixed layer depth shows that the low surface salinity is related to a convective regime restricted to the upper 500 dbar. Thus, intermediate water formed by convection is partly replaced by warmer water from the south. Model biases seem to be caused by the coupling to the atmospheric component of the CCM. Model drift during long spinup periods allows the initially small biases in water mass characteristics to become significant. Biases that develop in the control run are carried over to the twentieth-century runs, which are initialized from the control runs.

2239. Forecasting the probability of finding oil slicks using a CBR system

Authors: Aitor Mata; Juan Manuel Corchado

Journal: Expert Systems with Applications

Publisher: Elsevier BV

DOI: 10.1016/j.eswa.2008.10.003

URL: <https://doi.org/10.1016/j.eswa.2008.10.003>

Publication Date: 2008-10-22

Volume/Issue: Volume 36, Issue 4

References: 25

Cited by: 53 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2240. Development of a four-dimensional variational coupled data assimilation system for enhanced analysis and prediction of seasonal to interannual climate variations

Authors: Nozomi Sugiura (Frontier Research Center for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Toshiyuki Awaji (Kyoto University, Frontier Research Center for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Shuhei Masuda (Frontier Research Center for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Takashi Mochizuki (Frontier Research Center for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Takahiro Toyoda (Frontier Research Center for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Toru Miyama (Frontier Research Center for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Hiromichi Igarashi (Marine-Earth Data and Information Department Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Yoichi Ishikawa (Kyoto University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008jc004741

URL: <https://doi.org/10.1029/2008jc004741>

Publication Date: 2008-10-21

Volume/Issue: Volume 113, Issue C10

References: 88

Cited by: 91 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A four-dimensional variational (4D-VAR) data assimilation system using a coupled ocean-atmosphere global model has been successfully developed with the aim of better defining the dynamical states of the global climate on seasonal to interannual scales. The application of this system to state estimations of climate processes during the 1996-1998 period shows, in particular, that the representations of structures associated with several key events in the tropical Pacific and Indian Ocean sector (such as the El Niño, the Indian Ocean dipole, and the Asian summer monsoon) are significantly improved. This fact suggests that our 4D-VAR coupled data assimilation (CDA) approach has the potential to correct the initial location of the model climate attractor on the basis of observational data. In addition, the coupling parameters that control the air-sea exchange fluxes of mass, momentum, and heat become well adjusted. Such an initialization using the 4D-VAR CDA approach allows us to make a roughly 1.5-year lead time prediction of the 1997-1998 El Niño event. These results demonstrate that our 4D-VAR CDA system has the ability to enhance forecast potential for seasonal to interannual phenomena.

2241. Eddy-induced meridional heat transport in the ocean

Authors: Denis L. Volkov (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008gl035490

URL: <https://doi.org/10.1029/2008gl035490>

Publication Date: 2008-10-15

Volume/Issue: Volume 35, Issue 20

References: 13

Cited by: 114 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A global ocean data synthesis product at eddy-permitting resolution from Estimating the Circulation and Climate of the Ocean, Phase II (ECCO2) project are used to estimate the oceanic eddy heat transport. We show that in a number of locations the time-mean eddy heat transport constitutes a considerable portion of the total time-mean heat transport, in particular, in the tropics, in the Southern Ocean and in the Kuroshio Current. This research demonstrates that the variability of the eddy heat transport is a significant contributor to the variability of the total heat transport and globally it explains about 1/3 of its variance. Eddies are also found to explain a significant portion of the seasonal-interannual heat transport variance.

2242. 3-D Global Induction in the Oceans and Solid Earth: Recent Progress in Modeling Magnetic and Electric Fields from Sources of Magnetospheric, Ionospheric and Oceanic Origin

Authors: A. V. Kuvshinov

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-008-9045-z

URL: <https://doi.org/10.1007/s10712-008-9045-z>

Publication Date: 2008-10-09

Volume/Issue: Volume 29, Issue 2

References: 87

Cited by: 120 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2243. The design of an optimal filter for monthly GRACE gravity models

Authors: R. Klees; E. A. Revtova; B. C. Gunter; P. Ditmar; E. Oudman; H. C. Winsemius; H. H. G. Savenije

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2008.03922.x

URL: <https://doi.org/10.1111/j.1365-246x.2008.03922.x>

Publication Date: 2008-10-09

Volume/Issue: Volume 175, Issue 2

References: 25

Cited by: 129 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2244. Ocean cooling: Constraints from changes in Earth's dynamic oblateness (J2) and altimetry

Authors: J. O. Dickey (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); S. L. Marcus (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008gl035115

URL: <https://doi.org/10.1029/2008gl035115>

Publication Date: 2008-09-23

Volume/Issue: Volume 35, Issue 18

References: 14

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Dynamic oblateness (J2) observations reflect changes in the latitudinal distribution of mass within the Earth system, and hence can give insights into processes involving water transport and sea level rise. We seek to use records of J2 variations deduced from satellite laser ranging, together with altimetry results, to provide an independent geodetic constraint on changes in ocean heat content (OHC). We show that the size and signature of the J2 signal presented here do not support a recently published report of rapid oceanic heat loss beginning in 2003, which was subsequently attributed to Argo and other instrumental bias effects. Our results are consistent with recent findings of a flattening of the OHC trend at this time inferred from the bias-corrected Argo data, and demonstrate that J2 observations, in combination with other observational and model data types, can provide useful constraints for monitoring and validating ongoing changes in the Earth system.

2245. The Global Zonally Integrated Ocean Circulation, 1992-2006: Seasonal and Decadal Variability

Authors: Carl Wunsch (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2008jpo4012.1

URL: <https://doi.org/10.1175/2008jpo4012.1>

Publication Date: 2008-09-12

Volume/Issue: Volume 39, Issue 2

References: 42

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The zonally integrated meridional and vertical velocities as well as the enthalpy transports and fluxes in a least squares adjusted general circulation model are used to estimate the top-to-bottom oceanic meridional overturning circulation (MOC) and its variability from 1992 to 2006. A variety of simple theories all produce time scales suggesting that the mid- and high-latitude oceans should respond to atmospheric driving only over several decades. In practice, little change is seen in the MOC and associated heat transport except very close to the sea surface, at depth near the equator, and in parts of the Southern Ocean. Variability in meridional transports in both volume and enthalpy is dominated by the annual cycle and secondarily by the semiannual cycle, particularly in the Southern Ocean. On time scales longer than a year, the solution exhibits small trends with complicated global spatial patterns. Apart from a net uptake of heat from the atmosphere (forced by the NCEP-NCAR reanalysis, which produces net ocean heating), the origins of the meridional transport trends are not distinguishable and are likely a combination of model disequilibrium, shifts in the observing system, other trends (real or artificial) in the meteorological fields, and/or true oceanic secularities. None of the results, however, supports an inference of oceanic circulation shifts taking the system out of the range in which changes are more than small perturbations. That the oceanic observations do not conflict with an apparent excess heat uptake from the atmosphere implies a continued undersampling of the global ocean, even in the upper layers.

2246. Understanding El Niño in Ocean-Atmosphere General Circulation Models: Progress and Challenges

Authors: Eric Guilyardi; Andrew Wittenberg; Alexey Fedorov; Mat Collins; Chunzai Wang; Antonietta Capotondi; Geert Jan van Oldenborgh; Tim Stockdale

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/2008bams2387.1

URL: <https://doi.org/10.1175/2008bams2387.1>

Publication Date: 2008-08-27

Volume/Issue: Volume 90, Issue 3

References: N/A

Cited by: 445 publications

Language: en

Subjects: N/A

Keywords: Teleconnection|Centennial

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2247. Estimating geocenter variations from a combination of GRACE and ocean model output

Authors: Sean Swenson (National Center for Atmospheric Research); Don Chambers (Center for Space Research University of Texas Austin Texas USA); John Wahr (University of Colorado Boulder)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007jb005338

URL: <https://doi.org/10.1029/2007jb005338>

Publication Date: 2008-08-18

Volume/Issue: Volume 113, Issue B8

References: 36

Cited by: 653 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In this study, we estimate a time series of geocenter anomalies from a combination of data from the Gravity Recovery and Climate Experiment (GRACE) satellite mission and the output from ocean models. A matrix equation is derived relating total geocenter variations to the GRACE coefficients of degrees two and higher and to the oceanic component of the degree one coefficients. We estimate the oceanic component from two state-of-the-art ocean models. Results are compared to independent estimates of geocenter derived from other satellite data, such as satellite laser ranging and GPS. Finally, we compute degree one coefficients that are consistent with the processing applied to the GRACE Level-2 gravity field coefficients. The estimated degree one coefficients can be used to improve estimates of mass variability from GRACE, which alone cannot provide them directly.

2248. 4DVAR data assimilation in the Intra-Americas Sea with the Regional Ocean Modeling System (ROMS)

Authors: B.S. Powell; H.G. Arango; A.M. Moore; E. Di Lorenzo; R.F. Milliff; D. Foley

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2008.08.002

URL: <https://doi.org/10.1016/j.ocemod.2008.08.002>

Publication Date: 2008-08-16

Volume/Issue: Volume 25, Issue 3-4

References: 92

Cited by: 52 publications

Language: en

Subjects: N/A

Keywords: Ocean observations|Sea-surface height

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2249. On the variability of the deep meridional transports in the tropical North Atlantic

Authors: T. Kanzow; U. Send; M. McCartney

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2008.07.011

URL: <https://doi.org/10.1016/j.dsr.2008.07.011>

Publication Date: 2008-08-13

Volume/Issue: Volume 55, Issue 12

References: 44

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2250. Ocean Water Clarity and the Ocean General Circulation in a Coupled Climate Model

Authors: Anand Gnanadesikan (Princeton University); Whit G. Anderson (Princeton University)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2008jpo3935.1

URL: <https://doi.org/10.1175/2008jpo3935.1>

Publication Date: 2008-08-12

Volume/Issue: Volume 39, Issue 2

References: 47

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean water clarity affects the distribution of shortwave heating in the water column. In a one-dimensional time-mean sense, increased clarity would be expected to cool the surface and heat subsurface depths as shortwave radiation penetrates deeper into the water column. However, wind-driven upwelling, boundary currents, and the seasonal cycle of mixing can bring water heated at depth back to the surface. This warms the equator and cools the subtropics throughout the year while reducing the amplitude of the seasonal cycle of temperature in polar regions. This paper examines how these changes propagate through the climate system in a coupled model with an isopycnal ocean component focusing on the different impacts associated with removing shading from different regions. Increasing shortwave penetration along the equator causes warming to the south of the equator. Increasing it in the relatively clear gyres off the equator causes the Hadley cells to strengthen and the subtropical gyres to shift equatorward. Increasing shortwave penetration in the less clear regions overlying the oxygen minimum zones causes the cold tongue to warm and the Walker circulation to weaken. Increasing shortwave penetration in the high-latitude Southern Ocean causes an increase in the formation of mode water from subtropical water. The results suggest that more attention be paid to the processes distributing heat below the mixed layer.

2251. A method of error adjustment for marine gravity with application to Mean Dynamic Topography in the northern North Atlantic

Authors: A. Hunegnaw; R. G. Hipkin; J. Edwards

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-008-0249-2

URL: <https://doi.org/10.1007/s00190-008-0249-2>

Publication Date: 2008-07-30

Volume/Issue: Volume 83, Issue 2

References: 43

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2252. Mixed-layer heat budget associated with ENSO, IOD, and PDO inferred from satellite ocean data assimilation

Authors: T. Lee

Journal: Oceans 2003. Celebrating the Past ... Teaming Toward the Future (IEEE Cat. No.03CH37492)

Publisher: IEEE

DOI: 10.1109/oceans.2003.178434

URL: <https://doi.org/10.1109/oceans.2003.178434>

Publication Date: 2008-07-18

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Mixed layer|Ocean observations

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2253. Contrasting Eastern-Pacific and Central-Pacific Types of ENSO

Authors: Hsun-Ying Kao (University of California Irvine); Jin-Yi Yu (University of California Irvine)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2008jcli2309.1

URL: <https://doi.org/10.1175/2008jcli2309.1>

Publication Date: 2008-07-11

Volume/Issue: Volume 22, Issue 3

References: 43

Cited by: 1327 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Surface observations and subsurface ocean assimilation datasets are examined to contrast two distinct types of El Niño-Southern Oscillation (ENSO) in the tropical Pacific: an eastern-Pacific (EP) type and a central-Pacific (CP) type. An analysis method combining empirical orthogonal function (EOF) analysis and linear regression is used to separate these two types. Correlation and composite analyses based on the principal components of the EOF were performed to examine the structure, evolution, and teleconnection of these two ENSO types. The EP type of ENSO is found to have its SST anomaly center located in the eastern equatorial Pacific attached to the coast of South America. This type of ENSO is associated with basinwide thermocline and surface wind variations and shows a strong teleconnection with the tropical Indian Ocean. In contrast, the CP type of ENSO has most of its surface wind, SST, and subsurface anomalies confined in the central Pacific and tends to onset, develop, and decay in situ. This type of ENSO appears less related to the thermocline variations and may be influenced more by atmospheric forcing. It has a stronger teleconnection with the southern Indian Ocean. Phase-reversal signatures can be identified in the anomaly evolutions of the EP-ENSO but not for the CP-ENSO. This implies that the CP-ENSO may occur more as events or epochs than as a cycle. The EP-ENSO has experienced a stronger interdecadal change with the dominant period of its SST anomalies shifted from 2 to 4 yr near 1976/77, while the dominant period for the CP-ENSO stayed near the 2-yr band. The different onset times of these two types of ENSO imply that the difference between the EP and CP types of ENSO could be caused by the timing of the mechanisms that trigger the ENSO events.

2254. Delayed baroclinic response of the Antarctic circumpolar current to surface wind stress

Authors: XiaoYi Yang; RuiXin Huang; Jia Wang; DongXiao Wang

Journal: Science in China Series D: Earth Sciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11430-008-0074-8

URL: <https://doi.org/10.1007/s11430-008-0074-8>

Publication Date: 2008-07-01

Volume/Issue: Volume 51, Issue 7

References: 28

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2255. Pacific Sea Surface Temperatures in the Twentieth Century: An Evolution-Centric Analysis of Variability and Trend

Authors: Bin Guan (University of Maryland); Sumant Nigam (University of Maryland)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2007jcli2076.1

URL: <https://doi.org/10.1175/2007jcli2076.1>

Publication Date: 2008-07-01

Volume/Issue: Volume 21, Issue 12

References: 53

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A consistent analysis of natural variability and secular trend in Pacific SSTs in the twentieth century is presented. By focusing on spatial and temporal recurrence, but without imposition of periodicity constraints, this single analysis discriminates between biennial, ENSO, and decadal variabilities, leading to refined evolutionary descriptions, and between these natural variability modes and secular trend, all without advance filtering (and potential aliasing) of the SST record. SST anomalies of all four seasons are analyzed together using the extended-EOF technique. Canonical ENSO variability is encapsulated in two modes that depict the growth (east-to-west along the equator) and decay (near-simultaneous amplitude loss across the basin) phases. Another interannual mode, energetic in recent decades, is shown linked to the west-to-east SST development seen in post-climate shift ENSOs: the noncanonical ENSO mode. The mode is closely related to Chiang and Vimont's meridional mode, and leads to some reduction in canonical ENSO's oscillatory tendency. Pacific decadal variability is characterized by two modes: the Pan-Pacific mode has a horseshoe structure with the closed end skirting the North American coast, and a quiescent eastern equatorial Pacific. The mode exhibits surprising connections to the tropical/subtropical Atlantic, with correlations there resembling the Atlantic multidecadal oscillation. The second decadal mode--the North Pacific mode--captures the 1976/77 climate shift and is closer to Mantua's Pacific decadal oscillation. This analysis shows, perhaps for the first time, the striking links of the North Pacific mode to the western tropical Pacific and Indian Ocean SSTs. The physicality of both modes is assessed from correlations with the Pacific biological time series. Finally, the secular trend is characterized: implicit accommodation of natural variability leads to a nonstationary SST trend, including midcentury cooling. The SST trend is remarkably similar to the global surface air temperature trend. Geographically, a sliver of cooling is found in the central equatorial Pacific in the midst of widespread but nonuniform warming in all basins. An extensive suite of sensitivity tests, including counts of the number of observational analogs of the modes in test analyses, supports the robustness of this analysis.

2256. A near uniform basin-wide sea level fluctuation over the Japan/East Sea: A semienclosed sea with multiple straits

Authors: Seung-Bum Kim (Remote Sensing Systems Santa Rosa California USA); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007jc004409

URL: <https://doi.org/10.1029/2007jc004409>

Publication Date: 2008-06-27

Volume/Issue: Volume 113, Issue C6

References: 24

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Sea level of the Japan/East Sea observed by the TOPEX/Poseidon (T/P) satellite altimeter is analyzed using a 1/4 degrees-resolution ocean general circulation model. A significant fraction of the Japan/East Sea sea level variability is found to be spatially uniform with periods ranging from 20 d to a year. The model simulation is consistent with T/P records in terms of the basin-wide sea level fluctuation's spectral energy and coherence. The simulation indicates that the changes are barotropic in nature and controlled, notably at high frequencies, by the net mass transport through the straits of the Japan/East Sea driven by winds in the vicinity of the Korea/Tsushima and Soya Straits. A series of barotropic simulations suggest that the sea level fluctuations are the result of a dynamic balance at the straits among near-strait winds, friction, and geostrophic control. The basin-wide sea level response is a linear superposition of changes due to winds near the individual straits. In particular, a basin-wide sea level response can be established by winds near either one of the straits alone. For the specific geometry and winds, winds near the Soya Strait have a larger impact on the Japan/East Sea mean sea level than those near the Korea/Tsushima Strait.

2257. Response of the global ocean to Greenland and Antarctic ice melting

Authors: D. Stammer (University of Hamburg)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006jc004079

URL: <https://doi.org/10.1029/2006jc004079>

Publication Date: 2008-06-23

Volume/Issue: Volume 113, Issue C6

References: 44

Cited by: 94 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We investigate the transient response of the global ocean circulation to enhanced freshwater forcing associated with melting of the Greenland and Antarctic ice sheets. Increased freshwater runoff from Greenland results in a basin-wide response of the North Atlantic on timescales of a few years, communicated via boundary waves, equatorial Kelvin waves, and westward propagating Rossby waves. In addition, modified air-sea interaction plays a fundamental role in setting up the basin-scale response of the Atlantic circulation in its subpolar and subtropical gyres. In particular, the modified ocean dynamics and thermodynamics lead to a depression in the central North and South Atlantic that would not be expected from linear wave dynamics. Moreover, the heat content increases on basin and global scales in response to anomalous freshwater forcing from Greenland, suggesting that the ocean's response to enhanced freshwater forcing would be a coupled problem. Other parts of the world ocean experience a much slower adjustment in response to Greenland freshwater forcing, communicated via planetary waves, but also involving advective/diffusive processes, especially in the Southern Ocean. Over the 50 years considered here, most of the sea level increase associated with freshwater input from Greenland remains in the Atlantic Ocean. In contrast, ice melting around Antarctica has a much reduced effect on the global ocean. In both cases, none of the basins came to a stationary state during the 50-year experiment.

2258. Trends of sea level variations in the Indo-Pacific warm pool

Authors: Xuhua Cheng; Yiquan Qi; Wen Zhou

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2008.06.001

URL: <https://doi.org/10.1016/j.gloplacha.2008.06.001>

Publication Date: 2008-06-16

Volume/Issue: Volume 63, Issue 1

References: 53

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2259. Low degree gravity changes from GRACE, Earth rotation, geophysical models, and satellite laser ranging

Authors: J. L. Chen (Center for Space Research University of Texas Austin Texas USA); C. R. Wilson (Center for Space Research University of Texas Austin Texas USA, Department of Geological Sciences University of Texas Austin Texas USA, University of Texas at Austin)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007jb005397

URL: <https://doi.org/10.1029/2007jb005397>

Publication Date: 2008-06-10

Volume/Issue: Volume 113, Issue B6

References: 41

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Several independent time series of variations ΔC_{21} , ΔS_{21} , and ΔC_{20} in Earth's gravity field are compared for the period April 2002 to May 2007. We examine estimates from the Gravity Recovery and Climate Experiment (GRACE), Earth rotation variations, climate models, and satellite laser ranging (SLR). Recently released GRACE solutions show significant improvement relative to earlier results, especially for ΔC_{21} and ΔS_{21} . At the annual period, all estimates agree remarkably well, and good correlation is found among time series at intraseasonal periods. In general, Earth rotation values for ΔC_{21} and ΔS_{21} , and SLR values for ΔC_{20} agree best with GRACE estimates. GRACE ΔC_{20} time series are contaminated by aliased ocean tide model errors. SLR ΔC_{21} and ΔS_{21} time series have been reported without an ocean pole tide (OPT) correction and with an older Solid Earth Pole Tide (SEPT) model. After correcting for OPT and SEPT deficiencies, SLR ΔC_{21} and ΔS_{21} time series show improved agreement with other estimates.

2260. Modeling the distribution of Nd isotopes in the oceans using an ocean general circulation model

Authors: Kevin M. Jones; Samar P. Khatiwala; Steven L. Goldstein; Sidney R. Hemming; Tina van de Flierdt

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2008.05.027

URL: <https://doi.org/10.1016/j.epsl.2008.05.027>

Publication Date: 2008-06-06

Volume/Issue: Volume 272, Issue 3-4

References: 53

Cited by: 70 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2261. Representation Error of Oceanic Observations for Data Assimilation

Authors: Peter R. Oke (CSIRO Marine and Atmospheric Research, and Wealth from Oceans Flagship Program, Hobart, Tasmania, Australia); Pavel Sakov (CSIRO Marine and Atmospheric Research, and Wealth from Oceans Flagship Program, Hobart, Tasmania, Australia)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2007jtecho558.1

URL: <https://doi.org/10.1175/2007jtecho558.1>

Publication Date: 2008-06-05

Volume/Issue: Volume 25, Issue 6

References: 39

Cited by: 105 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A simple approach to the estimation of representation error (RE) of sea level (η), temperature (T), and salinity (S) observations for ocean data assimilation is described. It is assumed that the main source of RE is due to unresolved processes and scales in the model. Therefore, RE is calculated as a function of model resolution. The methods described here exploit the availability of mapped sea level anomalies (mSLAs) and along-track sea level anomalies (atSLAs). The mSLA fields or atSLA observations are regarded as the true ocean state. Here, they are averaged according to the resolution of the model grid, and the averaged field is taken as a representation of the true state on the given grid. The difference between the original data and the averaged field is then regarded as the RE for η . Subsequently, the RE is projected for η over depth using a standard technique, giving an estimate of the RE for T and S. Examples of RE estimates for an intermediate- and high-resolution global grid are presented. It is found that there is significant spatial variability in the RE for η , T, and S, with values that are typically greater than or comparable to measurement error, particularly in regions of strong mesoscale variability.

2262. A 170-year Sr/Ca and Ba/Ca coral record from the western Pacific warm pool: 2. A window into variability of the New Ireland Coastal Undercurrent

Authors: Chantal Alibert (Australian National University); Leslie Kinsley (Australian National University)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007jc004263

URL: <https://doi.org/10.1029/2007jc004263>

Publication Date: 2008-06-05

Volume/Issue: Volume 113, Issue C6

References: 69

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A Porites coral from New Ireland (2.5 degreesS, 150.5 degreesE), at the heart of the Pacific warm pool, records variations of Ba/Ca back to the early 1820s. The New Ireland Coastal Undercurrent, which flows along the north coast and transports high-nutrient thermocline waters, is thought to be the main source of the Ba enrichment observed in the coral during El Niño. Between the 1850s and the 1960s, frequent large Ba peaks indicate that nutrients were available during both phases of El Niño-Southern Oscillation (ENSO). During La Niña, Ba could be advected along the South Equatorial Current. Also during this period, clusters of Ba/Ca peaks at nearly decadal timescale generally coincide with the return time of a strong El Niño, in accord with the high decadal variance of NINO3 sea surface temperatures. Ba enrichment in this coral primarily reflects the stratification in the thermocline that controls vertical mixing. The coral records long-term changes in those properties affecting nutrients in surface waters, with reduced Ba after the 1960s, and even less between 1823 and 1850. These reductions are tentatively attributed to the high rate of warming at the end of the Dalton Minimum and since the mid-20th century. An ensuing weakening of the trade winds may have produced a more stratified equatorial thermocline, hindering the transport of Ba and nutrients from the undercurrent to the shallow coastal waters north of New Ireland.

2263. Pacific Subtropical Cell variability in the SODA 2.0.2/3 assimilation

Authors: Friedrich A. Schott (GEOMAR Helmholtz Centre for Ocean Research); Lothar Stramma (GEOMAR Helmholtz Centre for Ocean Research); Weiqiang Wang (GEOMAR Helmholtz Centre for Ocean Research); Benjamin S. Giese (Department of Oceanography Texas A&M; University College Station Texas USA); Rainer Zantopp (GEOMAR Helmholtz Centre for Ocean Research)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2008gl033757

URL: <https://doi.org/10.1029/2008gl033757>

Publication Date: 2008-05-28

Volume/Issue: Volume 35, Issue 10

References: 10

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A new version of SODA, which covers the time period 1958-2005, is used to analyze decadal variability of the Pacific Subtropical Cell (STC) circulation. The analysis is based on transport time series across 9 degreesS and 9 degreesN. At the interannual time scale, STC convergence anomalies decrease during El Niños and increase during La Niñas through Sverdrup transport convergence changes. At decadal time scales, the assimilation shows a reduction of interior STC convergence of about 8 Sv from the 1960s to the 1990s and a subsequent rebound into the early 2000s by a similar amount, in agreement with the STC tendencies reported earlier from geostrophic section analysis, and associated with the occurrence and intensity of ENSO events among the decades analyzed. The results are compared with, and differ significantly from, those obtained by the German ECCO (GECCO) assimilation.

2264. Sea ice-ocean coupling using a rescaled vertical coordinate z^*

Authors: Jean-Michel Campin; John Marshall; David Ferreira

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2008.05.005

URL: <https://doi.org/10.1016/j.ocemod.2008.05.005>

Publication Date: 2008-05-22

Volume/Issue: Volume 24, Issue 1-2

References: 33

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2265. 4DVAR data assimilation in the Intra-Americas Sea with the Regional Ocean Modeling System (ROMS)

Authors: B.S. Powell; H.G. Arango; A.M. Moore; E. Di Lorenzo; R.F. Milliff; D. Foley

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2008.04.008

URL: <https://doi.org/10.1016/j.ocemod.2008.04.008>

Publication Date: 2008-05-18

Volume/Issue: Volume 23, Issue 3-4

References: 92

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2266. Fast spin up of Ocean biogeochemical models using matrix-free Newton-Krylov

Authors: Samar Khatiwala

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2008.05.002

URL: <https://doi.org/10.1016/j.ocemod.2008.05.002>

Publication Date: 2008-05-16

Volume/Issue: Volume 23, Issue 3-4

References: 43

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2267. A high-order finite volume remapping scheme for nonuniform grids: The piecewise quartic method (PQM)

Authors: Laurent White; Alistair Adcroft

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2008.04.026

URL: <https://doi.org/10.1016/j.jcp.2008.04.026>

Publication Date: 2008-05-08

Volume/Issue: Volume 227, Issue 15

References: 21

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2268. Impact of ECCO Ocean-State Estimates on the Initialization of Seasonal Climate Forecasts

Authors: Gabriel Cazes-Boezio (University of California Los Angeles); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Carlos R. Mechoso (University of California Los Angeles)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2007jcli1574.1

URL: <https://doi.org/10.1175/2007jcli1574.1>

Publication Date: 2008-04-29

Volume/Issue: Volume 21, Issue 9

References: 56

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The impact of ocean-state estimates generated by the consortium for Estimating the Circulation and Climate of the Ocean (ECCO) on the initialization of a coupled general circulation model (CGCM) for seasonal climate forecasts is examined. The CGCM consists of the University of California, Los Angeles, Atmospheric GCM (UCLA AGCM) and an ECCO ocean configuration of the Massachusetts Institute of Technology GCM (MITgcm). The forecasts correspond to ensemble seasonal hindcasts for the period 1993-2001. For the forecasts, the ocean component of the CGCM is initialized in either early March or in early June using ocean states provided either by an unconstrained forward ocean integration of the MITgcm (the "baseline" hindcasts) or by data-constrained ECCO results (the "ECCO" hindcasts). Forecast skill for both the baseline and the ECCO hindcasts is significantly higher than persistence and compares well with the skill of other state-of-the art CGCM forecast systems. For March initial conditions, the standard errors of sea surface temperature (SST) anomalies in ECCO hindcasts (relative to observed anomalies) are up to 1 degreesC smaller than in the baseline hindcasts over the central and eastern equatorial Pacific (150 degrees-120 degreesW). For June initial conditions, the errors of ECCO hindcasts are up to 0.5 degreesC smaller than in the baseline hindcasts. The smaller standard error of the ECCO hindcasts is, in part, due to a more realistic equatorial thermocline structure of the ECCO initial conditions. This study confirms the value of physically consistent ocean-state estimation for the initialization of seasonal climate forecasts.

2269. Decadal Sea Level Changes in the 50-Year GECCO Ocean Synthesis

Authors: Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2007jcli2081.1

URL: <https://doi.org/10.1175/2007jcli2081.1>

Publication Date: 2008-04-29

Volume/Issue: Volume 21, Issue 9

References: 51

Cited by: 95 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An estimate of the time-varying ocean circulation, obtained over the period 1952-2001, is analyzed here with respect to its decadal and longer-term changes in sea level. The estimate results from a synthesis of most of the ocean datasets available during this 50-yr period with the Estimating the Circulation and Climate of the Ocean/Massachusetts Institute of Technology (ECCO/MIT) ocean circulation model. Over the period 1992 through 2001, the increase in thermosteric sea level rise on average amounts to 1.2 mm yr⁻¹ over the top 750 m and 1.8 mm yr⁻¹ over the total water column. This corresponds to an increase in upper-ocean heat content of 1.5×10^{22} J yr⁻¹ and is in agreement with the estimates of Willis et al. However, over the period 1962 through 2001 the global net thermosteric sea level rise is estimated as 0.66 mm yr⁻¹ over the top 750 m, which is twice the recent estimate from Antonov et al. (0.33 mm yr⁻¹). The corresponding trend over the total water column of 0.92 mm yr⁻¹ is also about twice their value for the layer of 0-3000 m (0.40 mm yr⁻¹). For the last decade, the global heat flux into the ocean of 1.5 W m⁻² is twice as large as the recent estimate by Willis et al. due to the heat content change in deeper layers. Regional changes in sea level are predominantly associated with an intensification of the subtropical gyre circulation and a corresponding redistribution of heat. The horizontal advection of heat due to an increase in wind stress curl is found to explain a major fraction of the estimated regional sea level trends over the last 40 years. However, the mechanisms appear different during the last decade when in some regions changes in surface heat flux may explain as much as 50% of the sea level changes.

2270. North Pacific Gyre Oscillation links ocean climate and ecosystem change

Authors: E. Di Lorenzo (Georgia Institute of Technology); N. Schneider (University of Hawaii at Manoa); K. M. Cobb (Georgia Institute of Technology); P. J. S. Franks (Scripps Institution of Oceanography / University of California San Diego); K. Chhak (Georgia Institute of Technology); A. J. Miller (Scripps Institution of Oceanography / University of California San Diego); J. C. McWilliams (University of California Los Angeles); S. J. Bograd (Southwest Fisheries Science Center, NMFS, NOAA Pacific Grove California USA); H. Arango (Rutgers University); E. Curchitser (Rutgers University); T. M. Powell (University of California Berkeley); P. Rivière (LEMAR, IUEM Technopole Brest-Iroise Plouzane France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007gl032838

URL: <https://doi.org/10.1029/2007gl032838>

Publication Date: 2008-04-29

Volume/Issue: Volume 35, Issue 8

References: 28

Cited by: 885 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Decadal fluctuations in salinity, nutrients, chlorophyll, a variety of zooplankton taxa, and fish stocks in the Northeast Pacific are often poorly correlated with the most widely-used index of large-scale climate variability in the region - the Pacific Decadal Oscillation (PDO). We define a new pattern of climate change, the North Pacific Gyre Oscillation (NPGO) and show that its variability is significantly correlated with previously unexplained fluctuations of salinity, nutrients and chlorophyll. Fluctuations in the NPGO are driven by regional and basin-scale variations in wind-driven upwelling and horizontal advection - the fundamental processes controlling salinity and nutrient concentrations. Nutrient fluctuations drive concomitant changes in phytoplankton concentrations, and may force similar variability in higher trophic levels. The NPGO thus provides a strong indicator of fluctuations in the mechanisms driving planktonic ecosystem dynamics. The NPGO pattern extends beyond the North Pacific and is part of a global-scale mode of climate variability that is evident in global sea level trends and sea surface temperature. Therefore the amplification of the NPGO variance found in observations and in global warming simulations implies that the NPGO may play an increasingly important role in forcing global-scale decadal changes in marine ecosystems.

2271. How long to oceanic tracer and proxy equilibrium?

Authors: Carl Wunsch; Patrick Heimbach

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2008.01.006

URL: <https://doi.org/10.1016/j.quascirev.2008.01.006>

Publication Date: 2008-04-21

Volume/Issue: Volume 27, Issue 7-8

References: 27

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2272. The mean seasonal cycle in sea level estimated from a data-constrained general circulation model

Authors: Sergey V. Vinogradov (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Patrick Heimbach (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007jc004496

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Publication Date: 2008-03-29

Volume/Issue: Volume 113, Issue C3

References: 43

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A near-global ocean state estimate over the period 1992-2004 is used to study the mean seasonal cycle in sea level ζ . The state estimate combines most available observations, including all the altimetric missions, with a general circulation model in an optimization procedure. The annual cycle tends to be larger than the semi-annual one, except in tropical regions. For global mean ζ , annual thermosteric and freshwater terms are nearly out-of-phase and lead to an annual cycle of only a few mm in amplitude. Regionally, surface wind stress and heat flux are the primary drivers for seasonal ζ variations in the tropics and midlatitudes, respectively, with both mechanisms playing a role at high latitudes. A substantial part of the annual ζ variability can be assigned to changes in thermosteric height in the upper 100 m in midlatitudes and 200 m in the tropics. Bottom pressure variability is larger at high latitudes, and also in some regions in the Southern Ocean and North Pacific. Apparent nonlinear rectification processes lead to a noticeable impact of submonthly forcing on the annual cycle in the western North Atlantic and North Pacific. Other features include the substantial ζ gradients associated with strong spatial variability in seasonal surface heat flux in some western boundary regions, the damping effects of surface heat flux on the seasonal cycle in the tropics, and the importance of wind driving and bottom pressure in shallow regions, which can cause differences in the seasonal cycle in some coastal and contiguous deep-ocean regions.

2273. A reduced-order simulated annealing approach for four-dimensional variational data assimilation in meteorology and oceanography

Authors: I. Hoteit

Journal: International Journal for Numerical Methods in Fluids

Publisher: Wiley

DOI: 10.1002/fld.1794

URL: <https://doi.org/10.1002/fld.1794>

Publication Date: 2008-03-27

Volume/Issue: Volume 58, Issue 11

References: 49

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Four-dimensional variational data assimilation in meteorology and oceanography suffers from the presence of local minima in the cost function. These local minima arise when the system under study is strongly nonlinear. The number of local minima further dramatically increases with the length of the assimilation period and often renders the solution to the problem intractable. Global optimization methods are therefore needed to resolve this problem. However, the huge computational burden makes the application of these sophisticated techniques unfeasible for large variational data assimilation systems. In this study, a Simulated Annealing (SA) algorithm, complemented with an order-reduction of the control vector, is used to tackle this problem. SA is a very powerful tool of combinatorial minimization in the presence of several local minima at the cost of increasing the execution time. Order-reduction is then used to reduce the dimension of the search space in order to speed up the convergence rate of the SA algorithm. This is achieved through a proper orthogonal decomposition. The new approach was implemented with a realistic eddy-permitting configuration of the Massachusetts Institute of Technology general circulation model (MITgcm) of the tropical Pacific Ocean. Numerical results indicate that the reduced-order SA approach was able to efficiently reduce the cost function with a reasonable number of function evaluations. Copyright © 2008 John Wiley & Sons, Ltd.

2274. Evaluation of a 3dVAR system for the South China Sea

Authors: Xianjun Xiao; Dongxiao Wang; Changxiang Yan; Jiang Zhu

Journal: Progress in Natural Science

Publisher: Elsevier BV

DOI: 10.1016/j.pnsc.2007.12.007

URL: <https://doi.org/10.1016/j.pnsc.2007.12.007>

Publication Date: 2008-03-19

Volume/Issue: Volume 18, Issue 5

References: 22

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2275. Mechanisms of Interannual Variations of the Meridional Overturning Circulation of the North Atlantic Ocean

Authors: Cécile Cabanes (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2007jpo3726.1

URL: <https://doi.org/10.1175/2007jpo3726.1>

Publication Date: 2008-02-27

Volume/Issue: Volume 38, Issue 2

References: 27

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The authors investigate the nature of the interannual variability of the meridional overturning circulation (MOC) of the North Atlantic Ocean using an Estimating the Circulation and Climate of the Ocean (ECCO) assimilation product for the period of 1993-2003. The time series of the first empirical orthogonal function of the MOC is found to be correlated with the North Atlantic Oscillation (NAO) index, while the associated circulation anomalies correspond to cells extending over the full ocean depth. Model sensitivity experiments suggest that the wind is responsible for most of this interannual variability, at least south of 40 degreesN. A dynamical decomposition of the meridional streamfunction allows a further look into the mechanisms. In particular, the contributions associated with 1) the Ekman flow and its depth-independent compensation, 2) the vertical shear flow, and 3) the barotropic gyre flowing over zonally varying topography are examined. Ekman processes are found to dominate the shorter time scales (1.5-3 yr), while for longer time scales (3-10 yr) the MOC variations associated with vertical shear flow are of greater importance. The latter is primarily caused by heaving of the pycnocline in the western subtropics associated with the stronger wind forcing. Finally, how these changes in the MOC affect the meridional heat transport (MHT) is examined. It is found that overall, Ekman processes explain a larger part of interannual variability (3-10 yr) for MHT (57%) than for the MOC (33%).

2276. Variability of the Meridional Overturning in the North Atlantic from the 50-Year GECCO State Estimation

Authors: Armin Köhl (University of Hamburg); Detlef Stammer (University of Hamburg)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2008jpo3775.1

URL: <https://doi.org/10.1175/2008jpo3775.1>

Publication Date: 2008-02-19

Volume/Issue: Volume 38, Issue 9

References: 59

Cited by: 126 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The German partner of the consortium for Estimating the Circulation and Climate of the Ocean (GECCO) provided a dynamically consistent estimate of the time-varying ocean circulation over the 50-yr period 1952-2001. The GECCO synthesis combines most of the data available during the entire estimation period with the ECCO-Massachusetts Institute of Technology (MIT) ocean circulation model using its adjoint. This GECCO estimate is analyzed here for the period 1962-2001 with respect to decadal and longer-term changes of the meridional overturning circulation (MOC) of the North Atlantic. A special focus is on the maximum MOC values at 25 degreesN. Over this period, the dynamically self-consistent synthesis stays within the error bars of H. L. Bryden et al., but reveals a general increase of the MOC strength. The variability on decadal and longer time scales is decomposed into contributions from different processes. Changes in the model's MOC strength are strongly influenced by the southward communication of density anomalies along the western boundary originating from the subpolar North Atlantic, which are related to changes in the Denmark Strait overflow but are only marginally influenced by water mass formation in the Labrador Sea. The influence of density anomalies propagating along the southern edge of the subtropical gyre associated with baroclinically unstable Rossby waves is found to be equally important. Wind-driven processes such as local Ekman transport explain a smaller fraction of the variability on those long time scales.

2277. Causes for Large-Scale Hydrographic Changes at the Hawaii Ocean Time Series Station

Authors: D. Stammer (University of Hamburg); S. Park (Korean Meteorological Administration, Seoul, Korea); A. Köhl (University of Hamburg); R. Lukas (University of Hawaii at Manoa); F. Santiago-Mandujano (University of Hawaii at Manoa)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2008jpo3751.1

URL: <https://doi.org/10.1175/2008jpo3751.1>

Publication Date: 2008-02-11

Volume/Issue: Volume 38, Issue 9

References: 45

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Results from Estimating the Circulation and Climate of the Ocean (ECCO)-Scripps Institution of Oceanography (SIO) global ocean state estimate, available over the 11-yr period 1992 through 2002, are compared with independent observations available at the Hawaii Ocean time series station ALOHA. The comparison shows that at this position, the estimated temporal variability has some skill in simulating observed ocean variability and that the quality of future syntheses could benefit from additional information available from the Argo network and from the time series observations themselves. On a decadal time scale, the influence radius of the station ALOHA T-S time series covers large parts of the tropical and subtropical Pacific Ocean and reaches even into the Indian Ocean through the Indonesian Throughflow. Estimated changes in sea surface height (SSH) result largely from thermosteric changes; however, nonsteric (barotropic) variations on the order of 1-2 cm also contribute to SSH changes at station ALOHA. Moreover, changes of similar magnitude can be caused by changes in the salinity field because of a quasi-biennial oscillation in the horizontal flow structure and heaving of the mean salinity structure on seasonal and interannual time scales. The adjoint modeling framework confirms westward-propagating Rossby waves (due to wind forcing) and subduction of water-mass anomalies (due to surface buoyancy forcing) as the primary mechanisms leading to observed changes of T-S structures at station ALOHA. Specifically, the analysis identifies surface freshwater fluxes along the wintertime outcrop of intermediate waters as a primary cause for salinity changes at station ALOHA and wind stress forcing east of the station position as another forcing mechanism of salinity variations around the Hawaiian Archipelago.

2278. The Inverse Ocean Modeling System. Part I: Implementation

Authors: A. F. Bennett (Oregon State University); B. S. Chua (Oregon State University); B. L. Pflaum (Oregon State University); M. Erwig (Oregon State University); Z. Fu (Microsoft Corporation, Redmond, Washington); R. D. Loft (National Center for Atmospheric Research); J. C. Muccino (Arizona State University)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2008jtecho519.1

URL: <https://doi.org/10.1175/2008jtecho519.1>

Publication Date: 2008-02-06

Volume/Issue: Volume 25, Issue 9

References: 53

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Inverse Ocean Modeling (IOM) system constructs and runs weak-constraint, four-dimensional variational data assimilation (W4DVAR) for any dynamical model and any observing array. The dynamics and the observing algorithms may be nonlinear but must be functionally smooth. The user need only provide the model and the observing algorithms, together with an interpolation scheme that relates the model numerics to the observer's coordinates. All other model-dependent elements of the Inverse Ocean Modeling assimilation algorithm (see both Chua and Bennett), including adjoint generators and Monte Carlo estimates of posteriors, have been derived and coded as templates in Parametric FORTRAN (Erwig et al.). This language has been developed for the IOM but has wider application in scientific programming. Guided by the Parametric FORTRAN templates, and by model information entered via a graphical user interface (GUI), the IOM generates conventional FORTRAN code for each of the many algorithm elements, customized to the user's model. The IOM also runs the various W4DVAR assimilations, which are monitored by the GUI. The system is supported by a Web site that includes interactive tutorials for the assimilation algorithm.

2279. Excitation of Annual Polar Wobble by Global Oceans

Authors: Min Zhong; Hao-ming Yan

Journal: Chinese Astronomy and Astrophysics

Publisher: Elsevier BV

DOI: 10.1016/j.chinastron.2008.01.009

URL: <https://doi.org/10.1016/j.chinastron.2008.01.009>

Publication Date: 2008-02-06

Volume/Issue: Volume 32, Issue 1

References: 8

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2280. Mean Circulation and Variability of the Tropical Atlantic during 1952-2001 in the GECCO Assimilation Fields

Authors: Benjamin Rabe (GEOMAR Helmholtz Centre for Ocean Research); Friedrich A. Schott (GEOMAR Helmholtz Centre for Ocean Research); Armin Köhl (University of Hamburg)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2007jpo3541.1

URL: <https://doi.org/10.1175/2007jpo3541.1>

Publication Date: 2008-02-01

Volume/Issue: Volume 38, Issue 1

References: 45

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The shallow subtropical-tropical cells (STC) of the Atlantic Ocean have been studied from the output fields of a 50-yr run of the German partner of the Estimating the Circulation and Climate of the Ocean (GECCO) consortium assimilation model. Comparison of GECCO with time-mean observational estimates of density and meridional currents at 10 degreesS and 10 degreesN, which represent the boundaries between the tropics and subtropics in GECCO, shows good agreement in transports of major currents. The variability of the GECCO wind stress in the interior at 10 degreesS and 10 degreesN remains consistent with the NCEP forcing, although temporary changes can be large. On pentadal and longer time scales, an STC loop response is found between the poleward Ekman divergence and STC-layer convergence at 10 degreesS and 10 degreesN via the Equatorial Undercurrent (EUC) at 23 degreesW, where the divergence leads the EUC and the convergence, suggesting a "pulling" mechanism via equatorial upwelling. The divergence is also associated with changes in the eastern equatorial upper-ocean heat content. Within the STC layer, partial compensation of the western boundary current (WBC) and the interior occurs at 10 degreesS and 10 degreesN. For the meridional overturning circulation (MOC) at 10 degreesS it is found that more than one-half of the variability in the upper limb can be explained by the WBC. The explained MOC variance can be increased to 85% by including the geostrophic (Sverdrup) part of the wind-driven transports.

2281. The relationship between sea-level and bottom pressure variability in an eddy permitting ocean model

Authors: Rory J. Bingham (Proudman Oceanographic Laboratory Liverpool UK); Chris W. Hughes (Proudman Oceanographic Laboratory Liverpool UK)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007gl032662

URL: <https://doi.org/10.1029/2007gl032662>

Publication Date: 2008-02-01

Volume/Issue: Volume 35, Issue 3

References: 10

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We investigate the relationship between sea-level (after application of an inverse-barometer correction) and ocean bottom pressure, in an eddy-permitting ocean model. We find the presence of eddies can disrupt this relationship even on timescales as short as 10-20 days, but only in the regions of most energetic eddy variability. Away from eddies, the relationship is similar to that seen in a coarser-resolution model, with a tight relationship between sea-level and bottom pressure at high frequencies, but with significant correlations between sea-level and bottom pressure at interannual timescales seen only in shelf sea regions. In the deep ocean, regions where sea-level and bottom pressure remain related out to the longest timescales are in the Arctic Ocean and regions of the Southern Ocean, where particularly large amplitude barotropic fluctuations are found but where the mesoscale signal is weak.

2282. The ECPC Coupled Prediction Model

Authors: E. Yulaeva (Scripps Institution of Oceanography / University of California San Diego); M. Kanamitsu (Scripps Institution of Oceanography / University of California San Diego); J. Roads (Scripps Institution of Oceanography / University of California San Diego)

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/2007mwr1929.1

URL: <https://doi.org/10.1175/2007mwr1929.1>

Publication Date: 2008-01-31

Volume/Issue: Volume 136, Issue 1

References: 64

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper presents a new Experimental Climate Prediction Center (ECPC) Coupled Prediction Model (ECPM). The ECPM includes the Jet Propulsion Laboratory (JPL) version of the Massachusetts Institute of Technology (MIT) ocean model coupled to the ECPC version of the National Centers for Environmental Prediction (NCEP) Atmospheric Global Spectral Model (GSM). The adjoint and forward versions of the MIT model forced with the NCEP atmospheric analyses are routinely used at JPL for ocean state assimilation. An earlier version of the GSM was used for the NCEP-Department of Energy reanalysis-2 project and for operational seasonal forecasts at NCEP. The ECPM climatology and internal variability derived from a 56-yr-long coupled integration are compared with the observations and reanalysis data. Though the ECPM exhibits climatological biases, these biases are relatively small and comparable to the systematic errors produced by other well-known coupled models, including the recent NCEP Climate Forecast System. The internal variability of the model resembles the observations. ECPM simulates both seasonal and interannual variability in the tropical Pacific reasonably well. The model has good skill in reproducing the mechanism of ENSO evolution as well as ENSO teleconnection patterns (including the Indian monsoon-ENSO relationship). The skill of the ECPM in predicting 1994-2006 SST anomalies over the Niño-3.4 region is shown to be comparable to other coupled models. These retrospective forecasts were used for deriving a model climatology for real-time seasonal forecasts that are currently produced and displayed at ECPC.

2283. Impact of Resolution and Optimized ECCO Forcing on Simulations of the Tropical Pacific

Authors: I. Hoteit (Scripps Institution of Oceanography / University of California San Diego); B. Cornuelle (Scripps Institution of Oceanography / University of California San Diego); V. Thierry (Laboratoire de Physique des Océans, IFREMER, Brest, France); D. Stammer (Institut für Meereskunde, Hamburg, Germany)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/2007jtecho528.1

URL: <https://doi.org/10.1175/2007jtecho528.1>

Publication Date: 2008-01-24

Volume/Issue: Volume 25, Issue 1

References: 48

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The sensitivity of the dynamics of a tropical Pacific Massachusetts Institute of Technology (MIT) general circulation model (MITgcm) to the surface forcing fields and to the horizontal resolution is analyzed. During runs covering the period 1992-2002, two different sets of surface forcing boundary conditions are used, obtained 1) from the National Centers for Environmental Prediction/National Center for Atmospheric Research (NCEP/NCAR) reanalysis project and 2) from the Estimating the Circulation and Climate of the Ocean (ECCO) assimilation consortium. The "ECCO forcing" is the "NCEP forcing" adjusted by a state estimation procedure using the MITgcm with a 1 degrees x 1 degrees global grid and the adjoint method assimilating a multivariate global ocean dataset. The skill of the model is evaluated against ocean observations available in situ and from satellites. The model domain is limited to the tropical Pacific, with open boundaries located along 26 degreesS, 26 degreesN, and in the Indonesian throughflow. To account for large-scale changes of the ocean circulation, the model is nested in the global time-varying ocean state provided by the ECCO consortium on a 1 degrees grid. Increasing the spatial resolution to 1/3 degrees and using the ECCO forcing fields significantly improves many aspects of the circulation but produces overly strong currents in the western model domain. Increasing the resolution to 1/6 degrees does not yield further improvements of model results. Using the ECCO heat and freshwater fluxes in place of NCEP products leads to improved time-mean model skill (i.e., reduced biases) over most of the model domain, underlining the important role of adjusted heat and freshwater fluxes for improving model representations of the tropical Pacific. Combinations of ECCO and NCEP wind forcing fields can improve certain aspects of the model solutions, but neither ECCO nor NCEP winds show clear overall superiority.

2284. Excitation of Annual Polar Motion by the Pacific, Atlantic and Indian Oceans

Authors: Jin Ma; Yong-Hong Zhou; De-Chun Liao; Jian-Li Chen

Journal: Chinese Journal of Astronomy and Astrophysics

Publisher: IOP Publishing

DOI: 10.1088/1009-9271/7/6/11

URL: <https://doi.org/10.1088/1009-9271/7/6/11>

Publication Date: 2008-01-08

Volume/Issue: Volume 7, Issue 6

References: 22

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: Polar motion|Polar night|Physical oceanography|Gulf Stream

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2285. The Bluelink ocean data assimilation system (BODAS)

Authors: Peter R. Oke; Gary B. Brassington; David A. Griffin; Andreas Schiller

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2007.11.002

URL: <https://doi.org/10.1016/j.ocemod.2007.11.002>

Publication Date: 2008-01-04

Volume/Issue: Volume 21, Issue 1-2

References: 59

Cited by: 324 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2286. Decadal Trends in Sea Level Patterns: 1993-2004

Authors: Carl Wunsch (Massachusetts Institute of Technology); Rui M. Ponte (Atmospheric and Environmental Research, Lexington, Massachusetts); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/2007jcli1840.1

URL: <https://doi.org/10.1175/2007jcli1840.1>

Publication Date: 2008-01-03

Volume/Issue: Volume 20, Issue 24

References: 70

Cited by: 151 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Estimates of regional patterns of global sea level change are obtained from a 1 degrees horizontal resolution general circulation model constrained by least squares to about 100 million ocean observations and many more meteorological estimates during the period 1993-2004. The data include not only altimetric variability, but most of the modern hydrography, Argo float profiles, sea surface temperature, and other observations. Spatial-mean trends in altimetric data are explicitly suppressed to isolate global average long-term changes required by the in situ data alone. On large scales, some regions display strong signals although few individual points have statistically significant trends. In the regional patterns, thermal, salinity, and mass redistribution contributions are all important, showing that regional sea level change is tied directly to the general circulation. Contributions below about 900 m are significant, but not dominant, and are expected to grow with time as the abyssal ocean shifts. Estimates made here produce a global mean of about 1.6 mm yr⁻¹, or about 60% of the pure altimetric estimate, of which about 70% is from the addition of freshwater. Interannual global variations may be dominated by the freshwater changes rather than by heating changes. The widely quoted altimetric global average values may well be correct, but the accuracies being inferred in the literature are not testable by existing in situ observations. Useful estimation of the global averages is extremely difficult given the realities of space-time sampling and model approximations. Systematic errors are likely to dominate most estimates of global average change: published values and error bars should be used very cautiously.

2287. Reply to comment by L. Fenoglio-Marc et al. on "On the steric and mass-induced contributions to the annual sea level variations in the Mediterranean Sea"

Authors: David García (University of Alicante); Benjamin F. Chao (National Central University); Jorge Del Río (Department of Applied Physics II University of Malaga Malaga Spain); Isabel Vigo (University of Alicante); Jesús García-Lafuente (Department of Applied Physics II University of Malaga Malaga Spain)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007jc004465

URL: <https://doi.org/10.1029/2007jc004465>

Publication Date: 2007-12-26

Volume/Issue: Volume 112, Issue C12

References: 8

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2288. Historical reconstruction of the Atlantic Meridional Overturning Circulation from the ECMWF operational ocean reanalysis

Authors: Magdalena A. Balmaseda (European Centre for Medium Range Weather Forecasts Reading UK); Gregory C. Smith (Environmental Systems Science Centre Reading University Reading UK); Keith Haines (Environmental Systems Science Centre Reading University Reading UK); David Anderson (European Centre for Medium Range Weather Forecasts Reading UK); T. N. Palmer (European Centre for Medium Range Weather Forecasts Reading UK); Arthur Vidard (L'Institut National de Recherche en Informatique et en Automatique Grenoble France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007gl031645

URL: <https://doi.org/10.1029/2007gl031645>

Publication Date: 2007-12-13

Volume/Issue: Volume 34, Issue 23

References: 21

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A reconstruction of the Atlantic Meridional Overturning Circulation (MOC) for the period 1959-2006 has been derived from the ECMWF operational ocean reanalysis. The reconstruction shows a wide range of time-variability, including a downward trend. At 26N, both the MOC intensity and changes in its vertical structure are in good agreement with previous estimates based on trans-Atlantic surveys. At 50N, the MOC and strength of the subpolar gyre are correlated at interannual time scales, but show opposite secular trends. Heat transport variability is highly correlated with the MOC but shows a smaller trend due to the warming of the upper ocean, which partially compensates for the weakening of the circulation. Results from sensitivity experiments show that although the time-varying upper boundary forcing provides useful MOC information, the sequential assimilation of ocean data further improves the MOC estimation by increasing both the mean and the time variability.

2289. Generation and Stability of a Quasi-Permanent Vortex in the Lofoten Basin

Authors: Armin Köhl (University of Hamburg)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2007jpo3694.1

URL: <https://doi.org/10.1175/2007jpo3694.1>

Publication Date: 2007-12-13

Volume/Issue: Volume 37, Issue 11

References: 34

Cited by: 72 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In the Nordic seas the Lofoten Basin is a region of high mesoscale activity. The generation mechanism and the conditions for the stability of a quasi-permanent vortex in the center of the Lofoten Basin are studied with a high-resolution ocean circulation model and altimeter data. The vortex and its generation mechanism manifest themselves by a pronounced sea surface height (SSH) signature and variability, which are found to be in agreement with altimeter data. The vortex results primarily from anticyclonic eddies shed from the eastern branch of the Norwegian Atlantic Current, which propagate southwestward. The large-scale bottom depression of the Lofoten Basin plays a crucial role for attracting anticyclones into the trough and for enabling the dynamical stability of the vortex. The water mass characteristics of the anticyclone lead to enhanced atmospheric interaction (heat loss) during wintertime. The cold water trapped in the upper part of the vortex preconditions convection in the following winter. This positive feedback mechanism tends to deepen convection progressively within the upper part of the vortex.

2290. Testing the use of adjoints for parameter estimation in a simple GCM on climate time-scales

Authors: Thomas Kaminski; Simon Blessing; Ralf Giering; Marko Scholze; Michael Vossbeck

Journal: Meteorologische Zeitschrift

Publisher: Schweizerbart

DOI: 10.1127/0941-2948/2007/0259

URL: <https://doi.org/10.1127/0941-2948/2007/0259>

Publication Date: 2007-12-13

Volume/Issue: Volume 16, Issue 6

References: N/A

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2291. The North Indian Ocean circulation and its variability as seen in a numerical hindcast of the years 1993-2004

Authors: Lakshmi Kantha; Thaned Rojsiraphisal; Joseph Lopez

Journal: Progress in Oceanography

Publisher: Elsevier BV

DOI: 10.1016/j.pocean.2007.05.006

URL: <https://doi.org/10.1016/j.pocean.2007.05.006>

Publication Date: 2007-12-04

Volume/Issue: Volume 76, Issue 1

References: 86

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2292. GRACE's spatial aliasing error

Authors: Ki-Weon Seo; Clark R. Wilson; Jianli Chen; Duane E. Waliser

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2007.03611.x

URL: <https://doi.org/10.1111/j.1365-246x.2007.03611.x>

Publication Date: 2007-11-27

Volume/Issue: Volume 172, Issue 1

References: 22

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2293. Emergent Biogeography of Microbial Communities in a Model Ocean

Authors: Michael J. Follows (Department of Oceanography, University of Hawaii, 1000 Pope Road, Honolulu, HI 96822, USA., Massachusetts Institute of Technology); Stephanie Dutkiewicz (Department of Oceanography, University of Hawaii, 1000 Pope Road, Honolulu, HI 96822, USA., Massachusetts Institute of Technology); Scott Grant (Department of Oceanography, University of Hawaii, 1000 Pope Road, Honolulu, HI 96822, USA., Massachusetts Institute of Technology); Sallie W. Chisholm (Department of Oceanography, University of Hawaii, 1000 Pope Road, Honolulu, HI 96822, USA., Massachusetts Institute of Technology)

Journal: Science

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/science.1138544

URL: <https://doi.org/10.1126/science.1138544>

Publication Date: 2007-11-12

Volume/Issue: Volume 315, Issue 5820

References: 28

Cited by: 660 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A marine ecosystem model seeded with many phytoplankton types, whose physiological traits were randomly assigned from ranges defined by field and laboratory data, generated an emergent community structure and biogeography consistent with observed global phytoplankton distributions. The modeled organisms included types analogous to the marine cyanobacterium *Prochlorococcus*. Their emergent global distributions and physiological properties simultaneously correspond to observations. This flexible representation of community structure can be used to explore relations between ecosystems, biogeochemical cycles, and climate change.

2294. Estimation of the surface and mid-depth currents from Argo floats in the Pacific and error analysis

Authors: Jiping Xie; Jiang Zhu

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2007.09.001

URL: <https://doi.org/10.1016/j.jmarsys.2007.09.001>

Publication Date: 2007-11-09

Volume/Issue: Volume 73, Issue 1-2

References: 44

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2295. The Discovery of Western Oceania: A New Perspective

Authors: Chris Avis; Álvaro Montenegro; Andrew Weaver

Journal: The Journal of Island and Coastal Archaeology

Publisher: Informa UK Limited

DOI: 10.1080/15564890701518557

URL: <https://doi.org/10.1080/15564890701518557>

Publication Date: 2007-10-30

Volume/Issue: Volume 2, Issue 2

References: 23

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2296. The GeoForschungsZentrum Potsdam/Groupe de Recherche de Gèodésie Spatiale satellite-only and combined gravity field models: EIGEN-GL04S1 and EIGEN-GL04C

Authors: Christoph Förste; Roland Schmidt; Richard Stubenvoll; Frank Flechtner; Ulrich Meyer; Rolf König; Hans Neumayer; Richard Biancale; Jean-Michel Lemoine; Sean Bruinsma; Sylvain Loyer; Franz Barthelmes; Saskia Esselborn

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-007-0183-8

URL: <https://doi.org/10.1007/s00190-007-0183-8>

Publication Date: 2007-10-22

Volume/Issue: Volume 82, Issue 6

References: 43

Cited by: 178 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2297. Basis functions for the consistent and accurate representation of surface mass loading

Authors: Peter J. Clarke; David A. Lavallée; Geoff Blewitt; Tonie van Dam

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2007.03493.x

URL: <https://doi.org/10.1111/j.1365-246x.2007.03493.x>

Publication Date: 2007-10-10

Volume/Issue: Volume 171, Issue 1

References: 21

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2298. The assimilation experiment in the southwestern South China Sea in summer 2000

Authors: Xianjun Xiao; Dongxiao Wang; Jianjun Xu

Journal: Chinese Science Bulletin

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11434-006-9031-4

URL: <https://doi.org/10.1007/s11434-006-9031-4>

Publication Date: 2007-09-22

Volume/Issue: Volume 51, Issue S2

References: 15

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2299. Software tools for pre- and post-processing of oceanic regional simulations

Authors: Pierrick Penven; Patrick Marchesiello; Laurent Debreu; Jérôme Lefèvre

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2007.07.004

URL: <https://doi.org/10.1016/j.envsoft.2007.07.004>

Publication Date: 2007-09-17

Volume/Issue: Volume 23, Issue 5

References: 5

Cited by: 163 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2300. Estimated Global Hydrographic Variability

Authors: Gaël Forget (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3072.1

URL: <https://doi.org/10.1175/jpo3072.1>

Publication Date: 2007-09-13

Volume/Issue: Volume 37, Issue 8

References: 28

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An estimate is made of the three-dimensional global oceanic temperature and salinity variability, omitting the seasonal cycle, both as a major descriptive element of the ocean circulation and for use in the error estimates of state estimation. Historical hydrography, recent data from the World Ocean Circulation Experiment, and Argo profile data are all used. Root-mean-square vertical displacements in the upper 300 m of the ocean are generally smaller than 50 m, except in energetic boundary currents and in the North Atlantic subpolar gyre. Variability in temperature and salinity is strongly correlated below the top 100 m. Salinity contributions to sea surface height variability appear more significant at low latitudes than expected, possibly resulting from advective and diffusive processes. Results are generally consistent with altimetric variability under two simple kinematic hypotheses, and much of the observed structure coincides with known dynamical features. A large fraction of the sea surface height variability is consistent with the hypothesis of dominance of the first baroclinic mode.

2301. Impact of Accurate Geoid Fields on Estimates of the Ocean Circulation

Authors: Detlef Stammer (University of Hamburg); Armin Köhl (University of Hamburg); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech2044.1

URL: <https://doi.org/10.1175/jtech2044.1>

Publication Date: 2007-08-28

Volume/Issue: Volume 24, Issue 8

References: 29

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The impact of new geoid height models on estimates of the ocean circulation, now available from the Gravity Recovery and Climate Experiment (GRACE) spacecraft, is assessed, and the implications of far more accurate geoids, anticipated from the European Space Agency's (ESA) Gravity and Ocean Circulation Explorer (GOCE) mission, are explored. The study is based on several circulation estimates obtained over the period 1992-2002 by combining most of the available ocean datasets with a global general circulation model on a 1 degrees horizontal grid and by exchanging only the EGM96 geoid model with two different geoid models available from GRACE. As compared to the EGM96-based solution, the GRACE geoid leads to an estimate of the ocean circulation that is more consistent with the Levitus temperature and salinity climatology. While not a formal proof, this finding supports the inference of a substantially improved GRACE geoid skill. However, oceanographic implications of the GRACE model are only modest compared to what can be obtained from ocean observations alone. To understand the extent to which this is merely a consequence of a not-optimally converged solution or if a much more accurate geoid field could in principle play a profound role in the ocean estimation procedure, an additional experiment was performed in which the geoid error was artificially reduced relative to all other datasets. Adjustments occur then in all elements of the ocean circulation, including 10% changes in the meridional overturning circulation and the corresponding meridional heat transport in the Atlantic. For an optimal use of new geoid fields, improved error information is required. The error budget of existing time-mean dynamic topography estimates may now be dominated by residual errors in time-mean altimetric corrections. Both these and the model errors need to be better understood before improved geoid estimates can be fully exploited. As is commonly found, the Southern Ocean is of particular concern.

2302. Mechanisms Controlling the Interannual Variation of Mixed Layer Temperature Averaged over the Niño-3 Region

Authors: Seung-Bum Kim (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli4206.1

URL: <https://doi.org/10.1175/jcli4206.1>

Publication Date: 2007-08-16

Volume/Issue: Volume 20, Issue 15

References: 39

Cited by: 73 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Processes controlling the interannual variation of mixed layer temperature (MLT) averaged over the Niño-3 domain (5 degreesN-5 degreesS, 150 degrees-90 degreesW) are studied using an ocean data assimilation product that covers the period of 1993-2003. The overall balance is such that surface heat flux opposes the MLT change but horizontal advection and subsurface processes assist the change. Advective tendencies are estimated here as the temperature fluxes through the domain's boundaries, with the boundary temperature referenced to the domain-averaged temperature to remove the dependence on temperature scale. This allows the authors to characterize external advective processes that warm or cool the water within the domain as a whole. The zonal advective tendency is caused primarily by large-scale advection of warm-pool water through the western boundary of the domain. The meridional advective tendency is contributed to mostly by Ekman current advecting large-scale temperature anomalies through the southern boundary of the domain. Unlike many previous studies, the subsurface processes that consist of vertical mixing and entrainment are explicitly evaluated. In particular, a rigorous method to estimate entrainment allows an exact budget closure. The vertical mixing across the mixed layer (ML) base has a contribution in phase with the MLT change. The entrainment tendency due to the temporal change in ML depth is negligible compared to other subsurface processes. The entrainment tendency by vertical advection across the ML base is dominated by large-scale changes in upwelling and the temperature of upwelling water. Tropical instability waves (TIWs) result in smaller-scale vertical advection that warms the domain during La Niña cooling events. However, such a warming tendency is overwhelmed by the cooling tendency associated with the large-scale upwelling by a factor of 2. In summary, all the balance terms are important in the MLT budget except the entrainment due to lateral induction and temporal variation in ML depth. All three advective tendencies are primarily caused by large-scale and low-frequency

processes, and they assist the Niño-3 MLT change. When the advective tendencies are evaluated by spatially averaging the conventional local advection of temperature, the apparent effects of currents with spatial scales smaller than the domain (such as TIWs) become very important as they redistribute heat within the Niño-3 domain. As a result, for example, the averaged zonal advective tendency counteracts rather than assists the Niño-3 MLT change. However, such internal redistribution of heat does not represent external processes that control the domain-averaged MLT.

2303. Steric and mass-induced Mediterranean sea level trends from 14 years of altimetry data

Authors: Francisco Criado-Aldeanueva; Jorge Del Río Vera; Jesús García-Lafuente

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2007.07.003

URL: <https://doi.org/10.1016/j.gloplacha.2007.07.003>

Publication Date: 2007-08-15

Volume/Issue: Volume 60, Issue 3-4

References: 46

Cited by: 90 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2304. A computational framework for simulation of biogeochemical tracers in the ocean

Authors: Samar Khatiwala (Columbia University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007gb002923

URL: <https://doi.org/10.1029/2007gb002923>

Publication Date: 2007-07-09

Volume/Issue: Volume 21, Issue 3

References: 52

Cited by: 114 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A novel computational framework is introduced for the efficient simulation of chemical and biological tracers in ocean models. The framework is based on the "transport matrix" formulation, a scheme for capturing the complex three-dimensional transport of tracers in a general circulation model (GCM) as a sparse matrix, thus reducing the task of simulating tracers to a sequence of simple matrix-vector products. The principal advantages of this formulation are efficiency and convenience. It is many orders of magnitude more efficient than GCMs, allowing us to address problems that are currently either difficult or unaffordable with GCMs. The scheme also allows us to quickly "prototype" new biogeochemical parameterizations or "plug in" existing ones. This paper describes the key features and advantages of the transport matrix method, and illustrates its application to a series of realistic problems in chemical and biological oceanography. The examples range from simulation of a transient tracer (SF₆) to adjoint sensitivity of a complex coupled biogeochemical model. Finally, the paper describes an efficient, portable, and freely available implementation of this computational scheme that provides the necessary framework for simulating any biogeochemical tracer.

2305. Ocean mass variations from GRACE and tsunami gauges

Authors: H. Munekane (Geographical Survey Institute Tsukuba, Ibaraki Japan)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006jb004618

URL: <https://doi.org/10.1029/2006jb004618>

Publication Date: 2007-07-06

Volume/Issue: Volume 112, Issue B7

References: 30

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We compared the mass variations from Gravity Recovery and Climate Experiment (GRACE) with those measured by bottom pressure recorders (BPRs) deployed as tsunami gauges. Prior to the comparison, we estimated the degree-1 Stokes coefficients by the degree-1 loading method using both the GPS and GRACE data. We successfully recovered the degree-1 Stokes coefficients that are consistent with those from the satellite laser ranging analysis and from the loading model. We found that the use of the GRACE data in the degree-1 loading method significantly mitigates the aliasing effect from the unmodeled higher-degree loading deformations. Then, we compared the mass variations from GRACE with those from the five BPRs that have been deployed as part of the Deep-ocean Assessment and Reporting of Tsunamis (DART) system. We observed a fairly good (over 0.4) correlation except for one station that is close to the U.S. mainland. However, we found that the GRACE-derived mass variations underestimate those from BPRs by nearly a factor of 2. Through numerical simulation, we found that the artificial leakage from the landmass is responsible for the low correlation observed at the station, and the spatial averaging in the GRACE processing may be responsible for the underestimation of the mass variations from BPRs by GRACE. These results illustrate the importance of performing a simulation for these effects with numerical models before comparing the GRACE-derived mass variations with those from pointwise measurements.

2306. Oceanic changes in the Sargasso Sea and declines in recruitment of the European eel

Authors: Kevin D. Friedland (National Marine Fisheries Service, 28 Tarzwell Drive, Narragansett, RI 02882, USA); Michael J. Miller (University of Tokyo); Brian Knights (University of Westminster, 115 New Cavendish St., London W1W 6UW, UK)

Journal: ICES Journal of Marine Science

Publisher: Oxford University Press (OUP)

DOI: 10.1093/icesjms/fsm022

URL: <https://doi.org/10.1093/icesjms/fsm022>

Publication Date: 2007-07-06

Volume/Issue: Volume 64, Issue 3

References: 79

Cited by: 174 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Friedland, K. D., Miller, M. J., and Knights, B. 2007. Oceanic changes in the Sargasso Sea and declines in recruitment of the European eel. - ICES Journal of Marine Science, 64: 519-530. Anguillid eel recruitment in the North Atlantic has declined in recent decades, raising concerns that climatic changes in the Sargasso Sea may be influencing oceanic reproduction and larval survival. There is a significant negative correlation between the North Atlantic Oscillation and long-term variations in catches of glass eel stages of the European eel *Anguilla anguilla* recorded by the fishery independent Den Oever recruitment index (DOI) in the Netherlands, lagged by one year. Ocean-atmospheric changes in the Sargasso Sea may affect the location of spawning areas by silver eels and the survival of leptocephali during the key period when they are transported towards the Gulf Stream. A northward shift in a key isotherm (22.5 degreesC), defining the northern boundary of the spawning area, a declining trend in winds and transport conditions in larval transport areas, and a shallowing of the mixed layer depth could affect primary productivity in areas where leptocephali feed. The relationships between these ocean parameters and the DOI suggest that these changing ocean conditions could be contributing to declining recruitment of the European eel and probably also of the American eel (*A. rostrata*), but anthropogenic factors during their continental life stage must also be considered.

2307. Combining Argo profiles with a general circulation model in the North Atlantic. Part 2: Realistic transports and improved hydrography, between spring 2002 and spring 2003

Authors: Gaël Forget; Herlé Mercier; Bruno Ferron

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2007.06.002

URL: <https://doi.org/10.1016/j.ocemod.2007.06.002>

Publication Date: 2007-06-29

Volume/Issue: Volume 20, Issue 1

References: 26

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2308. Causes of Changes in the Denmark Strait Overflow

Authors: Armin Köhl (University of Hamburg); Rolf H. Käse (University of Hamburg); Detlef Stammer (University of Hamburg); Nuno Serra (Instituto de Oceanografia, Faculdade de Ciências da Universidade de Lisboa, Lisbon, Portugal)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3080.1

URL: <https://doi.org/10.1175/jpo3080.1>

Publication Date: 2007-06-26

Volume/Issue: Volume 37, Issue 6

References: 48

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The warming Nordic seas potentially tend to decrease the overflow across the Greenland-Iceland-Scotland Ridge (GISR) system. Recent observations by Macranders et al. document a significant drop in the intensity of outflowing Denmark Strait Overflow Water of more than 20% over 3 yr and a simultaneous increase in the temperature of the bottom layers of more than 0.4 degreesC. A simulation of the exchange across the GISR with a regional ocean circulation model is used here to identify possible mechanisms that control changes in the Denmark Strait overflow and its relations to changed forcing condition. On seasonal and longer time scales, the authors establish links of the overflow anomalies to a decreasing capacity of the dense water reservoir caused by a change of circulation pattern north of the sill. On annual and shorter time scales, the wind stress curl around Iceland determines the barotropic circulation around the island and thus the barotropic flow through Denmark Strait. For the overlapping time scales, the barotropic and overflow component interactively determine transport variations. Last, a relation between sea surface height and reservoir height changes upstream of the sill is used to predict the overflow variability from altimeter data. Estimated changes are in agreement with other recent transport estimates based on current-meter arrays.

2309. Yearly impact of submonthly rain fluctuations on the Indian Ocean salinity

Authors: Serena Illig (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Claire Perigaud (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007gl029655

URL: <https://doi.org/10.1029/2007gl029655>

Publication Date: 2007-06-26

Volume/Issue: Volume 34, Issue 12

References: 10

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: Forcing (mathematics)|BENGAL

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We investigate the impact of submonthly rain fluctuations on the daily-to-interannual variations of the salinity in the tropical Indian Ocean. A three-year record of daily observed precipitation and wind are used to force an Indian Ocean model in two experiments that differ only by their rain forcing, which is either daily- (Exp_day) or monthly- (Exp_month) averaged. Results show that submonthly precipitations significantly impact the Surface Layer Salinity (SLS) in the northern and eastern Bay of Bengal during the wet season from May to November. In September when the Bay reaches its SLS minimum, the Andaman Sea is saltier by as much as 1.3 psu for Exp_day. The frequency shift between the submonthly rain forcing and the yearly salinity response is due to accumulation in time of nonlinear mechanisms: sudden rain deficits increase the SLS by entraining salty waters from below, whereas rain excesses are inefficient to decrease the salinity of the thick surface layer.

2310. Effects of sub-seasonal variability on seasonal-to-interannual Indian Ocean meridional heat transport

Authors: D. J. Halkides (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Weiqing Han (University of Colorado Boulder); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yukio Masumoto (Institute of Observational Research for Global Change Japan Agency for Marine-Earth Science and Technology Yokosuka Japan, University of Tokyo)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007gl030150

URL: <https://doi.org/10.1029/2007gl030150>

Publication Date: 2007-06-21

Volume/Issue: Volume 34, Issue 12

References: 31

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Numerical experiments using the Hybrid Coordinate Ocean Model are performed to examine atmospheric intraseasonal oscillation (ISO) effects on seasonal-to-interannual meridional heat transport (Q_v) in the tropical Indian Ocean. Analysis focuses on the equator and 14.5 degreesS, latitudes associated with the cross-equatorial and southern subtropical meridional cells. ISOs alter seasonal Q_v anomaly (Q_v^*) at these latitudes by a-few-hundredths PW up to 2-3 tenths PW. On interannual timescales, heat transport anomaly (Q'_v) induced by ISOs tends to reinforce (correlates positively with) the total Q'_v signal. ISO winds can induce interannual Q'_v up to 0.15 PW, comparable to the maximum amplitude of the Q'_v signal. Net effects of atmospheric ISOs can account for ~30% of Q'_v in some years; in others, ISO contributions are insignificant. Wind stress (via momentum flux) and wind speed (via turbulent heat fluxes and vertical mixing) play notable roles in ISO effects on Q_v^* and in some years on Q'_v .

2311. Spatial Mapping of Time-Variable Errors in Jason-1 and TOPEX/Poseidon Sea Surface Height Measurements

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.); Carl Wunsch (Massachusetts Institute of Technology); Detlef Stammer (University of Hamburg)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech2029.1

URL: <https://doi.org/10.1175/jtech2029.1>

Publication Date: 2007-06-14

Volume/Issue: Volume 24, Issue 6

References: 22

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Fitting ocean models to altimeter sea surface height (SSH) measurements requires knowledge of instrument noise (radar noise, sea state bias, path delay corrections, and orbit errors) and "representation" errors related to SSH signals (e.g., tidal or pressure driven) not computed in the models. Comparisons between the independent Ocean Topography Experiment (TOPEX)/Poseidon and Jason-1 altimetric missions when they were in identical orbits show that point by point the data are consistent within the mission specifications of about 3-cm rms, but large-scale dependences exist in the data differences, and these are both poorly known and capable of introducing major errors into oceanic state estimates. Here the authors focus on the time-variable component of the spatially dependent errors. The analysis reveals errors ranging from 2 cm in the Tropics to 4 cm at mid- and high latitudes and roughly consistent with a dependence of instrument noise on significant wave height. Analysis of the representation errors suggests that, over the deep ocean, uncertainties associated with the simplifying assumption of an inverted barometer response to pressure loading are larger than the remaining errors in modeling the large-scale tides. Over extensive regions, however, errors associated with eddy signals missing in coarse resolution models dominate. Obtaining a more quantitative estimate of the latter errors remains a challenge.

2312. Does eddy subduction matter in the northeast Atlantic Ocean?

Authors: Geoffrey Gebbie (Massachusetts Institute of Technology, Woods Hole Oceanographic Institution)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006jc003568

URL: <https://doi.org/10.1029/2006jc003568>

Publication Date: 2007-06-11

Volume/Issue: Volume 112, Issue C6

References: 43

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Mesoscale eddies are an important contributor to subduction in the Gulf Stream region and the Antarctic Circumpolar Current, but is eddy subduction also important in the relatively quiescent interior of the world's subtropical gyres? Observations from the Subduction Experiment of the northeast Atlantic do not have the spatial resolution necessary to calculate eddy subduction and answer this question. Regional numerical models can diagnose subduction, but their representativeness is unknown. Furthermore, water mass budgets in an open-ocean domain show that the simulated properties of subducted water directly depend upon uncertain open-boundary conditions and surface fluxes. To remedy these problems, a state estimate of the ocean circulation is formed by constraining an eddy-permitting general circulation model to observations by adjusting the model parameters within their uncertainty. The resulting estimate is self-consistent with the equations of motion and has the necessary resolution for diagnosing subduction. In the northeast Atlantic during 1991-1993, the time-variable circulation contributes less than 1 Sv of net subduction, while the total subduction is 4 Sv. Eddy volume fluxes of 40 m³/yr in the North Equatorial Current and the Azores Current, however, are significant and rival the subduction by Ekman pumping locally. Furthermore, a state estimate at 1/6 degrees resolution has 2-3 Sv more subduction in the density bands centered around $\sigma = 24.0$ kg/m³ and $\sigma = 26.0$ kg/m³ than a 2 degrees state estimate. This result implies that the inability to accurately simulate mesoscale phenomena and surface fluxes in climate models would lead to an accumulation of errors in water mass properties over 10-20 years, even in the interior of the subtropical gyre.

2313. Carbon dioxide and oxygen fluxes in the Southern Ocean: Mechanisms of interannual variability

Authors: A. Verdy (Massachusetts Institute of Technology); S. Dutkiewicz (Massachusetts Institute of Technology); M. J. Follows (Massachusetts Institute of Technology); J. Marshall (Massachusetts Institute of Technology); A. Czaja (Imperial College London)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gb002916

URL: <https://doi.org/10.1029/2006gb002916>

Publication Date: 2007-05-30

Volume/Issue: Volume 21, Issue 2

References: 26

Cited by: 51 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We analyze the variability of air-sea fluxes of carbon dioxide and oxygen in the Southern Ocean during the period 1993-2003 in a biogeochemical and physical simulation of the global ocean. Our results suggest that the nonseasonal variability is primarily driven by changes in entrainment of carbon-rich, oxygen-poor waters into the mixed layer during winter convection episodes. The Southern Annular Mode (SAM), known to impact the variability of air-sea fluxes of carbon dioxide, is also found to affect oxygen fluxes. We find that El Niño-Southern Oscillation (ENSO) also plays an important role in generating interannual variability in air-sea fluxes of carbon and oxygen. Anomalies driven by SAM and ENSO constitute a significant fraction of the simulated variability; the two climate indices are associated with surface heat fluxes, which control the modeled mixed layer depth variability. We adopt a Lagrangian view of tracers advected along the Antarctic Circumpolar Current (ACC) to highlight the importance of convective mixing in inducing anomalous air-sea fluxes of carbon dioxide and oxygen. The idealized Lagrangian model captures the principal features of the variability simulated by the more complex model, suggesting that knowledge of entrainment, temperature, and mean geostrophic flow in the mixed layer is sufficient to obtain a first-order description of the large-scale variability in air-sea transfer of soluble gases. Distinct spatial and temporal patterns arise from the different equilibration timescales of the two gases.

2314. Modeling the prehistoric arrival of the sweet potato in Polynesia

Authors: Álvaro Montenegro; Chris Avis; Andrew Weaver

Journal: Journal of Archaeological Science

Publisher: Elsevier BV

DOI: 10.1016/j.jas.2007.04.004

URL: <https://doi.org/10.1016/j.jas.2007.04.004>

Publication Date: 2007-05-25

Volume/Issue: Volume 35, Issue 2

References: 45

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2315. A comparison of model and GRACE estimates of the large-scale seasonal cycle in ocean bottom pressure

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.); Katherine J. Quinn (Atmospheric and Environmental Research Inc.); Carl Wunsch (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2007gl029599

URL: <https://doi.org/10.1029/2007gl029599>

Publication Date: 2007-05-10

Volume/Issue: Volume 34, Issue 9

References: 21

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Seasonal variability in ocean bottom pressure p_b is analyzed using GRACE (Gravity Recovery and Climate Experiment) data products and an optimized model solution obtained by fitting most available ocean data in a least-squares sense. The annual cycle in the spatial mean is a substantial part of the observed seasonal p_b variability; net freshwater input and atmospheric pressure effects are both important. For the residual spatially-varying patterns, GRACE and model results agree well over the Southern Ocean where strongest variability at both annual and semiannual periods is present. Phase patterns tend to match well, although model amplitudes are generally weaker. Considerable uncertainty remains in both GRACE and model p_b fields, judging from the spread among available estimates. Improving the p_b estimates requires removal of data noise from aliasing and leakage of land hydrology signals, and further optimization of the ocean model, including possible use of GRACE data to constrain the solution.

2316. Uncertainties in the mean ocean dynamic topography before the launch of the Gravity Field and Steady-State Ocean Circulation Explorer (GOCE)

Authors: Femke C. Vossepoel (Institute of Marine and Atmospheric Research Utrecht (IMAU) and SRON Netherlands Institute for Space Research Utrecht Netherlands)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006jc003891

URL: <https://doi.org/10.1029/2006jc003891>

Publication Date: 2007-05-03

Volume/Issue: Volume 112, Issue C5

References: 109

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In anticipation of the future observations of the gravity mission Gravity Field and Steady-State Ocean Circulation Explorer (GOCE), the present-day accuracy of mean dynamic topography (MDT) is estimated from both observations and models. A comparison of five observational estimates illustrates that RMS differences in MDT vary from 4.2 to 10.5 cm after low-pass filtering the fields with a Hamming window with wavenumber $N=120$ (corresponding to an effective resolution of 167 km). RMS differences in observational MDT reduce to 2.4-8.3 cm for $N=15$ (1334 km). Differences in data sources (geoid model, in situ data) are mostly visible in the small-scale oceanic features, while differences in processing (filtering, inverse modeling techniques) are reflected at larger scales. A comparison of seven different numerical ocean models demonstrates that model estimates differ mostly in western boundary currents and in the Antarctic Circumpolar Current. RMS differences between modeled and observed MDT are at best 8.8 cm for $N=120$, and reduce to 6.4 cm for $N=15$. For models with data assimilation, a minimal RMS difference of 6.6 cm ($N=120$) to 3.4 cm ($N=15$) is obtained with respect to the observational MDTs. The reduction of differences between MDTs with increasing filtering scales is smaller than expected. While it is expected that GOCE will improve MDT estimates at small spatial scales, improvement of mean sea surface estimates from satellite altimetry may be needed to improve MDT estimates at larger scales.

2317. Effects of vertical variations of thickness diffusivity in an ocean general circulation model

Authors: Gokhan Danabasoglu; John Marshall

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2007.03.006

URL: <https://doi.org/10.1016/j.ocemod.2007.03.006>

Publication Date: 2007-05-02

Volume/Issue: Volume 18, Issue 2

References: 56

Cited by: 112 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2318. Dynamic and regression modeling of ocean variability in the tide-gauge record at seasonal and longer periods

Authors: Emma M. Hill (Harvard-Smithsonian Center for Astrophysics Cambridge Massachusetts USA); Rui M. Ponte (Atmospheric and Environmental Research Inc.); James L. Davis (Harvard-Smithsonian Center for Astrophysics Cambridge Massachusetts USA)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006jc003745

URL: <https://doi.org/10.1029/2006jc003745>

Publication Date: 2007-04-30

Volume/Issue: Volume 112, Issue C5

References: 36

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Comparison of monthly mean tide-gauge time series to corresponding model time series based on a static inverted barometer (IB) for pressure-driven fluctuations and a ocean general circulation model (OM) reveals that the combined model successfully reproduces seasonal and interannual changes in relative sea level at many stations. Removal of the OM and IB from the tide-gauge record produces residual time series with a mean global variance reduction of 53%. The OM is mis-scaled for certain regions, and 68% of the residual time series contain a significant seasonal variability after removal of the OM and IB from the tide-gauge data. Including OM admittance parameters and seasonal coefficients in a regression model for each station, with IB also removed, produces residual time series with mean global variance reduction of 71%. Examination of the regional improvement in variance caused by scaling the OM, including seasonal terms, or both, indicates weakness in the model at predicting sea-level variation for constricted ocean regions. The model is particularly effective at reproducing sea-level variation for stations in North America, Europe, and Japan. The RMS residual for many stations in these areas is 25-35 mm. The production of "cleaner" tide-gauge time series, with oceanographic variability removed, is important for future analysis of nonsecular and regionally differing sea-level variations. Understanding the ocean model's strengths and weaknesses will allow for future improvements of the model.

2319. Validation and Error Analysis of OSCAR Sea Surface Currents

Authors: Eric S. Johnson (Earth and Space Research, Seattle, Washington); Fabrice Bonjean (Earth and Space Research, Seattle, Washington); Gary S. E. Lagerloef (Earth and Space Research, Seattle, Washington); John T. Gunn (Earth and Space Research, Seattle, Washington); Gary T. Mitchum (University of South Florida)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech1971.1

URL: <https://doi.org/10.1175/jtech1971.1>

Publication Date: 2007-04-16

Volume/Issue: Volume 24, Issue 4

References: 19

Cited by: 155 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Comparisons of OSCAR satellite-derived sea surface currents with in situ data from moored current meters, drifters, and shipboard current profilers indicate that OSCAR presently provides accurate time means of zonal and meridional currents, and in the near-equatorial region reasonably accurate time variability (correlation = 0.5-0.8) of zonal currents at periods as short as 40 days and meridional wavelengths as short as 8 degrees. At latitudes higher than 10 degrees the zonal current correlation remains respectable, but OSCAR amplitudes diminish unrealistically. Variability of meridional currents is poorly reproduced, with severely diminished amplitudes and reduced correlations relative to those for zonal velocity on the equator. OSCAR's RMS differences from drifter velocities are very similar to those experienced by the ECCO (Estimating the Circulation and Climate of the Ocean) data-assimilating models, but OSCAR generally provides a larger ocean-correlated signal, which enhances its ratio of estimated signal over noise. Several opportunities exist for modest improvements in OSCAR fidelity even with presently available datasets.

2320. A Seasonal Heat Budget across the Extent of the California Current

Authors: Kathleen A. Edwards (University of Washington); Kathryn A. Kelly (University of Washington)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2990.1

URL: <https://doi.org/10.1175/jpo2990.1>

Publication Date: 2007-04-13

Volume/Issue: Volume 37, Issue 3

References: 39

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A seasonal heat budget is based on observations that span the broad California Current (CC) region. Budget terms are estimated from satellite data (oceanic heat advection), repeat ship transects (heat storage rate), and the Comprehensive Ocean-Atmosphere Data Set (COADS) (surface heat flux). The balance between terms differs with distance from shore. Offshore, a local balance between the heat storage rate and net heat flux (Q_0) holds; the latter is dominated by its shortwave component QSW. Shoreward of ~500 km, oceanic heat advection shifts the phase of the heat storage rate to earlier in the year and partially offsets an increase in Q_0 due to cloud clearing. During the summer maximum of Q_0 , the ~500-km-wide CC region loses heat to alongshore geostrophic transport, offshore Ekman transport, and, to a lesser degree, cross-shore geostrophic transport and eddy transport. The advective heat loss is neither uniform in space nor temporal phase; instead, the region of geostrophic and eddy heat loss expands cross shore with the annual widening of the California Current to ~500 km. This expansion begins in spring with the onset of equatorward winds. A region of relatively positive wind stress curl widens at the same gradual rate as the CC, suggesting a coupling mechanism between the two.

2321. Effects of inter-annual water storage variations on polar motion

Authors: L. I. Fernández; H. Schuh; M. Schmidt; F. Seitz

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2006.03304.x

URL: <https://doi.org/10.1111/j.1365-246x.2006.03304.x>

Publication Date: 2007-03-26

Volume/Issue: Volume 169, Issue 1

References: 28

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2322. A comparison of annual vertical crustal displacements from GPS and Gravity Recovery and Climate Experiment (GRACE) over Europe

Authors: T. van Dam (University of Luxembourg); J. Wahr (University of Colorado Boulder); David Lavallée (University of Newcastle upon Tyne)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006jb004335

URL: <https://doi.org/10.1029/2006jb004335>

Publication Date: 2007-03-14

Volume/Issue: Volume 112, Issue B3

References: 36

Cited by: 180 publications

Language: en

Subjects: N/A

Keywords: Spurious relationship|Barotropic fluid|SIGNAL (programming language)|Post-glacial rebound

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We compare approximately 3 years of GPS height residuals (with respect to the International Terrestrial Reference Frame) with predictions of vertical surface displacements derived from the Gravity Recovery and Climate Experiment (GRACE) gravity fields for stations in Europe. An annual signal fit to the residual monthly heights, corrected for atmospheric pressure and barotropic ocean loading effects, should primarily represent surface displacements due to long-wavelength variations in water storage. A comparison of the annual height signal from GPS and GRACE over Europe indicates that at most sites, the annual signals do not agree in amplitude or phase. We find that unlike the annual signal predicted from GRACE, the annual signal in the GPS heights is not coherent over the region, displaying significant variability from site to site. Confidence in the GRACE data and the unlikely possibility of large-amplitude small-scale features in the load field not captured by the GRACE data leads us to conclude that some of the discrepancy between the GPS and GRACE observations is due to technique errors in the GPS data processing. This is evidenced by the fact that the disagreement between GPS and GRACE is largest at coastal sites, where mismodeling of the semidiurnal ocean tidal loading signal can result in spurious annual signals.

2323. GLACIAL CLIMATES | Thermohaline Circulation

Authors: S. Rahmstorf

Journal: Encyclopedia of Quaternary Science

Publisher: Elsevier

DOI: 10.1016/b0-44-452747-8/00014-4

URL: <https://doi.org/10.1016/b0-44-452747-8/00014-4>

Publication Date: 2007-03-09

Volume/Issue: N/A

References: 29

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: Younger dryas, Deglaciation, Antarctic bottom water, Upwelling, Dansgaard oeschger event (and 6 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2324. Variability of Pacific subtropical cells in the 50-year ECCO assimilation

Authors: Friedrich A. Schott (GEOMAR Helmholtz Centre for Ocean Research); Weiqiang Wang (GEOMAR Helmholtz Centre for Ocean Research); Detlef Stammer (University of Hamburg)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gl028478

URL: <https://doi.org/10.1029/2006gl028478>

Publication Date: 2007-03-03

Volume/Issue: Volume 34, Issue 5

References: 14

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Pacific Subtropical Cell (STC) circulation is being analyzed from transport time series across 9 degreesS and 9 degreesN, obtained from the German ECCO (GECCO) assimilation results for the period 1952-2002. In this estimate, the interior Pacific STC convergence shows significantly less decadal slowdown from the 1960's to the 1990's (~5Sv), than in previous estimates based on hydrographic sections. In the GECCO results, about half of this STC convergence decrease is compensated by an increase in the equatorward transport of the western boundary currents. Overall, the STC varies primarily on interannual time scale, with relatively short time lags between STC convergence and transport variations of the Equatorial Undercurrent at 140 degreesW.

2325. Interannual to Decadal Changes in the ECCO Global Synthesis

Authors: A. Köhl (University of Hamburg); D. Stammer (University of Hamburg); B. Cornuelle (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3014.1

URL: <https://doi.org/10.1175/jpo3014.1>

Publication Date: 2007-02-28

Volume/Issue: Volume 37, Issue 2

References: 56

Cited by: 113 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An estimate of the time-varying global ocean circulation for the period 1992-2002 was obtained by combining most of the World Ocean Circulation Experiment (WOCE) ocean datasets with a general circulation model on a 1 degrees horizontal grid. The estimate exactly satisfies the model equations without artificial sources or sinks of momentum, heat, and freshwater. To bring the model into agreement with observations, its initial temperature and salinity conditions were permitted to change, as were the time-dependent surface fluxes of momentum, heat, and freshwater. The estimation of these "control variables" is largely consistent with accepted uncertainties in the hydrographic climatology and meteorological analyses. The estimated time-mean horizontal transports of volume, heat, and freshwater, which were largely underestimated in the previous 2 degrees optimization performed by Stammer et al., have converged with time-independent estimates from box inversions over most parts of the World Ocean. Trends in the model's heat content are 7% larger than those reported by Levitus and correspond to a global net heat uptake of about 1.1 W m⁻² over the model domain. The associated model trend in sea surface height over the estimation period resembles the observations from Ocean Topography Experiment (TOPEX)/Poseidon over most of the global ocean. Sea surface height changes in the model are primarily steric but show contributions from mass redistributions from the subpolar North Atlantic Ocean and the Southern Ocean to the subtropical Pacific Ocean gyres. Steric contributions are primarily temperature based but are partly compensated by salt variation. However, the North Atlantic and the Southern Ocean reveal a clear contribution of salt to large-scale sea level variations.

2326. Antarctic Circumpolar Current Transport Variability during 2003-05 from GRACE

Authors: Victor Zlotnicki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John Wahr (University of Colorado Boulder); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yuhe T. Song (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3009.1

URL: <https://doi.org/10.1175/jpo3009.1>

Publication Date: 2007-02-28

Volume/Issue: Volume 37, Issue 2

References: 59

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Gravity Recovery and Climate Experiment (GRACE) gravity data spanning January 2003–November 2005 are used as proxies for ocean bottom pressure (BP) averaged over 1 month, spherical Gaussian caps 500 km in radius, and along paths bracketing the Antarctic Circumpolar Current's various fronts. The GRACE BP signals are compared with those derived from the Estimating the Circulation and Climate of the Ocean (ECCO) ocean modeling-assimilation system, and to a non-Boussinesq version of the Regional Ocean Model System (ROMS). The discrepancy found between GRACE and the models is 1.7 cmH₂O (1 cmH₂O ~ 1 hPa), slightly lower than the 1.9 cmH₂O estimated by the authors independently from propagation of GRACE errors. The northern signals are weak and uncorrelated among basins. The southern signals are strong, with a common seasonality. The seasonal cycle GRACE data observed in the Pacific and Indian Ocean sectors of the ACC are consistent, with annual and semiannual amplitudes of 3.6 and 0.6 cmH₂O (1.1 and 0.6 cmH₂O with ECCO), the average over the full southern path peaks (stronger ACC) in the southern winter, on days of year 197 and 97 for the annual and semiannual components, respectively; the Atlantic Ocean annual peak is 20 days earlier. An approximate conversion factor of 3.1 Sv (Sv ≡ 10⁶ m³ s⁻¹) of barotropic transport variability per cmH₂O of BP change is estimated. Wind stress data time series from the Quick Scatterometer (QuikSCAT), averaged monthly, zonally, and over the latitude band 40 degrees–65 degreesS, are also constructed and subsampled at the same months as with the GRACE data. The annual and semiannual harmonics of the wind stress peak on days 198 and 82, respectively. A decreasing trend over the 3 yr is observed in the three data types.

2327. Adjoint Sensitivity of an Ocean General Circulation Model to Bottom Topography

Authors: Martin Losch (Alfred Wegener Institute); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3017.1

URL: <https://doi.org/10.1175/jpo3017.1>

Publication Date: 2007-02-28

Volume/Issue: Volume 37, Issue 2

References: 78

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Bottom topography, or more generally the geometry of the ocean basins, is an important ingredient in numerical ocean modeling. With the help of an adjoint model, it is shown that scalar diagnostics or objective functions in a coarse-resolution model, such as the transport through Drake Passage, the strength of the Atlantic Ocean meridional overturning circulation, the Deacon cell, and the meridional heat transport across 32 degreesS, are sensitive to bottom topography as much as they are to surface boundary conditions. For example, adjoint topography sensitivities of the transport through Drake Passage are large in choke-point areas such as the Crozet-Kerguelen Plateau and south of New Zealand; the Atlantic meridional overturning circulation is sensitive to topography in the western boundary region of the North Atlantic Ocean and along the Scotland-Iceland Ridge. Many sensitivities are connected to steep topography and can be interpreted in terms of bottom form stress, that is, the product of bottom pressure and topography gradient. The adjoint sensitivities are found to agree with direct perturbation methods with deviations smaller than 30% for significant perturbations on time scales of 100 yr, so that the assumption of quasi linearity that is implicit in the adjoint method holds. The horizontal resolution of the numerical model affects the sensitivities to bottom topography, but large-scale patterns and the overall impact of changes in topography appear to be robust. The relative impact of changes in topography and surface boundary conditions on the model circulation is estimated by multiplying the adjoint sensitivities with assumed uncertainties. If the uncertainties are correlated in space, changing the surface boundary conditions has a larger impact on the scalar diagnostics than topography does, but the effects can locally be on the same order of magnitude if uncorrelated uncertainties are assumed. In either case, bottom topography variations within their prior uncertainties affect the solution of an ocean circulation model. To this extent, including topography in the control vector can be expected to compensate for identifiable model errors and, thus, to improve the solutions of estimation problems.

2328. Effects of Stratification on the Large-Scale Ocean Response to Barometric Pressure

Authors: Rui M. Ponte (Atmospheric and Environmental Research Inc.); Sergey V. Vinogradov (Atmospheric and Environmental Research Inc.)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3010.1

URL: <https://doi.org/10.1175/jpo3010.1>

Publication Date: 2007-02-28

Volume/Issue: Volume 37, Issue 2

References: 35

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Single-layer (barotropic) models have been commonly used in studies of the inverted barometer effect and the oceanic response to atmospheric pressure loading. The potential effects of stratification on this response are explored here using a general circulation model in a near-global domain with realistic coasts and bathymetry. Periodic forcing by the diurnal and semidiurnal atmospheric tides and 6-hourly stochastic forcing from weather center analyses are both examined. A global dynamic response (i.e., departures from inverted barometer behavior) is clear in the response to atmospheric tides; for stochastic forcing, the largest dynamic signals occur in shallow and semienclosed regions and at mid- and high latitudes. The influence of stratification in the dynamics is assessed by comparing surface and bottom pressure signals. Baroclinic effects are generally weak, particularly in the response to the large-scale atmospheric tides. Under stochastic forcing, largest differences between surface and bottom pressure signals reach 10%-20% of the surface signals and tend to occur in regions of enhanced topographic gradients. Bottom-intensified, localized interactions with topography seem to be involved. Enhanced baroclinicity is also seen at low latitudes, where stratification effects are also felt in the upper ocean. General implications for modeling the ocean response to high-frequency atmospheric and tidal forcing are discussed.

2329. A Near-Uniform Basin-Wide Sea Level Fluctuation of the Mediterranean Sea

Authors: Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3016.1

URL: <https://doi.org/10.1175/jpo3016.1>

Publication Date: 2007-02-28

Volume/Issue: Volume 37, Issue 2

References: 54

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A new basin-wide oscillation of the Mediterranean Sea is identified and analyzed using sea level observations from the Ocean Topography Experiment (TOPEX)/Poseidon satellite altimeter and a numerical ocean circulation model. More than 50% of the large-scale, nontidal, and non-pressure-driven variance of sea level can be attributed to this oscillation, which is nearly uniform in phase and amplitude across the entire basin. The oscillation has periods ranging from 10 days to several years and has a magnitude as large as 10 cm. The model suggests that the fluctuations are driven by winds at the Strait of Gibraltar and its neighboring region, including the Alboran Sea and a part of the Atlantic Ocean immediately to the west of the strait. Winds in this region force a net mass flux through the Strait of Gibraltar to which the Mediterranean Sea adjusts almost uniformly across its entire basin with depth-independent pressure perturbations. The wind-driven response can be explained in part by wind setup; a near-stationary balance is established between the along-strait wind in this forcing region and the sea level difference between the Mediterranean Sea and the Atlantic Ocean. The amplitude of this basin-wide wind-driven sea level fluctuation is inversely proportional to the setup region's depth but is insensitive to its width including that of Gibraltar Strait. The wind-driven fluctuation is coherent with atmospheric pressure over the basin and contributes to the apparent deviation of the Mediterranean Sea from an inverse barometer response.

2330. Atlantic to Mediterranean Sea Level Difference Driven by Winds near Gibraltar Strait

Authors: Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3015.1

URL: <https://doi.org/10.1175/jpo3015.1>

Publication Date: 2007-02-28

Volume/Issue: Volume 37, Issue 2

References: 40

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Observations and numerical simulations show that winds near Gibraltar Strait cause an Atlantic Ocean to Mediterranean Sea sea level difference of 20 cm peak to peak with a 3-cm standard deviation for periods of days to years. Theoretical arguments and numerical experiments establish that this wind-driven sea level difference is caused in part by storm surges due to alongshore winds near the North African coastline on the Atlantic side of Gibraltar. The fraction of the Moroccan coastal current offshore of the 284-m isobath is deflected across Gibraltar Strait, west of Camarinal Sill, resulting in a geostrophic surface pressure gradient that contributes to a sea level difference at the stationary limit. The sea level difference is also caused in part by the along-strait wind setup, with a contribution proportional to the along-strait wind stress and to the length of Gibraltar Strait and adjoining regions and inversely proportional to its depth. In the 20-360-day band, average transfer coefficients between the Atlantic-Alboran sea level difference and surface wind stress at 36 degreesN, 6.5 degreesW, estimated from barometrically corrected Ocean Topography Experiment (TOPEX)/Poseidon data and NCEP-NCAR reanalysis data, are 0.10 +/- 0.04 m Pa⁻¹ with 1 +/- 5-day lag and 0.19 +/- 0.08 m Pa⁻¹ with 5 +/- 4-day lag for the zonal and meridional wind stresses, respectively. This transfer function is consistent with equivalent estimates derived from a 1992-2003 high-resolution barotropic simulation forced by the NCEP-NCAR wind stress. The barotropic simulation explains 29% of the observed Atlantic-Alboran sea level difference in the 20-360-day band. In turn, the Alboran and Mediterranean mean sea level time series are highly correlated, $\rho = 0.7$ in the observations and $\rho = 0.8$ in the barotropic simulation, hence providing a pathway for winds near Gibraltar Strait to affect the mean sea level of the entire Mediterranean.

2331. Intraseasonal Variability of the Equatorial Indian Ocean Observed from Sea Surface Height, Wind, and Temperature Data

Authors: Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo3006.1

URL: <https://doi.org/10.1175/jpo3006.1>

Publication Date: 2007-02-28

Volume/Issue: Volume 37, Issue 2

References: 32

Cited by: 49 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The forcing of the equatorial Indian Ocean by the highly periodic monsoon wind cycle creates many interesting intraseasonal variabilities. The frequency spectrum of the wind stress observations from the European Remote Sensing Satellite scatterometers reveals peaks at the seasonal cycle and its higher harmonics at 180, 120, 90, and 75 days. The observations of sea surface height (SSH) from the Jason and Ocean Topography Experiment (TOPEX)/Poseidon radar altimeters are analyzed to study the ocean's response. The focus of the study is on the intraseasonal periods shorter than the annual period. The semiannual SSH variability is characterized by a basin mode involving Rossby waves and Kelvin waves traveling back and forth in the equatorial Indian Ocean between 10 degreesS and 10 degreesN. However, the interference of these waves with each other masks the appearance of individual Kelvin and Rossby waves, leading to a nodal point (amphidrome) of phase propagation on the equator at the center of the basin. The characteristics of the mode correspond to a resonance of the basin according to theoretical models. For the semiannual period and the size of the basin, the resonance involves the second baroclinic vertical mode of the ocean. The theory also calls for similar modes at 90 and 60 days. These modes are found only in the eastern part of the basin, where the wind forcing at these periods is primarily located. The western parts of the theoretical modal patterns are not observed, probably because of the lack of wind forcing. There is also similar SSH variability at 120 and 75 days. The 120-day variability, with spatial patterns resembling the semiannual mode, is close to a resonance involving the first baroclinic vertical mode. The 75-day variability, although not a resonant basin mode in theory, exhibits properties similar to the 60- and 90-day variabilities with energy confined to the eastern basin, where the SSH variability seems in resonance with the local wind forcing. The time it takes an oceanic signal to travel eastward as Kelvin waves from the forcing location along the equator and back as Rossby waves off the equator roughly corresponds to the period of the wind forcing. The SSH variability at 60-90 days is coherent with sea surface temperature (SST) with a near-zero phase difference, showing the effects of the time-varying thermocline depth on SST, which may affect the wind in an ocean-atmosphere coupled process

governing the intraseasonal variability.

2332. On the usefulness of atmospheric and oceanic angular momentum in recovering polar motion and gravity field variations in a unified process

Authors: C.M. Hancock; P. Moore

Journal: Advances in Space Research

Publisher: Elsevier BV

DOI: 10.1016/j.asr.2007.02.051

URL: <https://doi.org/10.1016/j.asr.2007.02.051>

Publication Date: 2007-02-25

Volume/Issue: Volume 39, Issue 10

References: 33

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2333. Fast ensemble smoothing

Authors: Sai Ravela; Dennis McLaughlin

Journal: Ocean Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10236-006-0098-6

URL: <https://doi.org/10.1007/s10236-006-0098-6>

Publication Date: 2007-02-15

Volume/Issue: Volume 57, Issue 2

References: 21

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2334. The Scripps Coupled Ocean-Atmosphere Regional (SCOAR) Model, with Applications in the Eastern Pacific Sector

Authors: Hyodae Seo (Scripps Institution of Oceanography); Arthur J. Miller (Scripps Institution of Oceanography); John O. Roads (Scripps Institution of Oceanography)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli4016.1

URL: <https://doi.org/10.1175/jcli4016.1>

Publication Date: 2007-02-09

Volume/Issue: Volume 20, Issue 3

References: 114

Cited by: 98 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A regional coupled ocean-atmosphere model is introduced. It is designed to admit the air-sea feedbacks arising in the presence of an oceanic mesoscale eddy field. It consists of the Regional Ocean Modeling System (ROMS) and the Regional Spectral Model (RSM). Large-scale forcing is provided by NCEP/DOE reanalysis fields, which have physics consistent with the RSM. Coupling allows the sea surface temperature (SST) to influence the stability of the atmospheric boundary layer and, hence, the surface wind stress and heat flux fields. The system is denominated the Scripps Coupled Ocean-Atmosphere Regional (SCOAR) Model. The model is tested in three scenarios in the eastern Pacific Ocean sector: tropical instability waves of the eastern tropical Pacific, mesoscale eddies and fronts of the California Current System, and gap winds of the Central American coast. Recent observational evidence suggests air-sea interactions involving the oceanic mesoscale in these three regions. Evolving SST fronts are shown to drive an unambiguous response of the atmospheric boundary layer in the coupled model. This results in significant model anomalies of wind stress curl, wind stress divergence, surface heat flux, and precipitation that resemble the observations and substantiate the importance of ocean-atmosphere feedbacks involving the oceanic mesoscale.

2335. Inverse estimates of the oceanic sources and sinks of natural CO₂ and the implied oceanic carbon transport

Authors: S. E. Mikaloff Fletcher (University of California Los Angeles); N. Gruber (University of California Los Angeles); A. R. Jacobson (Princeton University); M. Gloor (Princeton University); S. C. Doney (Woods Hole Oceanographic Institution); S. Dutkiewicz (Massachusetts Institute of Technology); M. Gerber (University of Bern); M. Follows (Massachusetts Institute of Technology); F. Joos (University of Bern); K. Lindsay (National Center for Atmospheric Research); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); A. Mouchet (University of Liège); S. A. Müller (University of Bern); J. L. Sarmiento (Princeton University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gb002751

URL: <https://doi.org/10.1029/2006gb002751>

Publication Date: 2007-02-09

Volume/Issue: Volume 21, Issue 1

References: 76

Cited by: 158 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We use an inverse method to estimate the global-scale pattern of the air-sea flux of natural CO₂, i.e., the component of the CO₂ flux due to the natural carbon cycle that already existed in preindustrial times, on the basis of ocean interior observations of dissolved inorganic carbon (DIC) and other tracers, from which we estimate ΔC_{gasex} , i.e., the component of the observed DIC that is due to the gas exchange of natural CO₂. We employ a suite of 10 different Ocean General Circulation Models (OGCMs) to quantify the error arising from uncertainties in the modeled transport required to link the interior ocean observations to the surface fluxes. The results from the contributing OGCMs are weighted using a model skill score based on a comparison of each model's simulated natural radiocarbon with observations. We find a pattern of air-sea flux of natural CO₂ characterized by outgassing in the Southern Ocean between 44 degreesS and 59 degreesS, vigorous uptake at midlatitudes of both hemispheres, and strong outgassing in the tropics. In the Northern Hemisphere and the tropics, the inverse estimates generally agree closely with the natural CO₂ flux results from forward simulations of coupled OGCM-biogeochemistry models undertaken as part of the second phase of the Ocean Carbon Model Intercomparison Project (OCMIP-2). The OCMIP-2 simulations find far less air-sea exchange than the inversion south of 20 degreesS, but more recent forward OGCM studies are in better agreement with the inverse estimates in the Southern Hemisphere. The strong source and sink pattern south of 20 degreesS was not apparent in an earlier inversion study, because the choice of region boundaries led to a partial cancellation of the sources and sinks. We show that the inversely estimated flux pattern is clearly traceable to gradients in the observed ΔC_{gasex} , and that it is relatively insensitive to the choice of OGCM or potential biases in ΔC_{gasex} . Our inverse estimates

imply a southward interhemispheric transport of $0.31 \pm 0.02 \text{ Pg C yr}^{-1}$, most of which occurs in the Atlantic. This is considerably smaller than the 1 Pg C yr^{-1} of Northern Hemisphere uptake that has been inferred from atmospheric CO_2 observations during the 1980s and 1990s, which supports the hypothesis of a Northern Hemisphere terrestrial sink.

2336. Relation between sea level and bottom pressure and the vertical dependence of oceanic variability

Authors: Nadya T. Vinogradova (Atmospheric and Environmental Research Inc.); Rui M. Ponte (Atmospheric and Environmental Research Inc.); Detlef Stammer (University of Hamburg)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gl028588

URL: <https://doi.org/10.1029/2006gl028588>

Publication Date: 2007-02-07

Volume/Issue: Volume 34, Issue 3

References: 19

Cited by: 81 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The relation between large-scale sea level and bottom pressure variability is studied using long (50-yr) simulations of a general circulation model under realistic forcing. Admittance and coherence analyses are used to characterize the bottom pressure and sea level relationship as a function of period, horizontal spatial scale, and location. At the model grid scale (1 degrees), bottom pressure is found to be essentially equivalent to sea level at periods <30 days, except in the tropics. This equivalence still holds for longer periods (~100 days), but only at high latitudes (>60 degrees) and in shallow depths (<200 m). Elsewhere, bottom pressure and sea level fields can differ significantly. Results indicate an increase of the importance of baroclinic signals with decreasing latitude and spatial scale, with significant baroclinic signals at intra-seasonal and longer periods present in many subtropical and mid-latitude regions. Variability is clearly baroclinic at inter-annual periods, regardless of location and spatial scale. Results have broad implications for the interpretation and processing of both satellite altimetry and gravity data and for their assimilation into numerical models.

2337. Trends of sea level variations in the South China Sea from merged altimetry data

Authors: Xuhua Cheng; Yiquan Qi

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2007.01.005

URL: <https://doi.org/10.1016/j.gloplacha.2007.01.005>

Publication Date: 2007-02-05

Volume/Issue: Volume 57, Issue 3-4

References: 43

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2338. Variability in the Indian Ocean circulation and salinity and its impact on SST anomalies during dipole events

Authors: Bijoy Thompson; C. Gnanaseelan; P. S. Salvekar

Journal: Journal of Marine Research

Publisher: Journal of Marine Research/Yale

DOI: 10.1357/002224006779698350

URL: <https://doi.org/10.1357/002224006779698350>

Publication Date: 2007-01-28

Volume/Issue: Volume 64, Issue 6

References: N/A

Cited by: 115 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2339. Acoustic remote sensing of large-scale temperature variability in the North Pacific Ocean

Authors: Worcester; Cornuelle; Dushaw; Dzieciuch; Howe; Menemenlis; Mercer; Munk; Spindel; Stammer

Journal: Oceans 2003. Celebrating the Past ... Teaming Toward the Future (IEEE Cat. No.03CH37492)

Publisher: IEEE

DOI: 10.1109/oceans.2003.178563

URL: <https://doi.org/10.1109/oceans.2003.178563>

Publication Date: 2007-01-19

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Argo

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2340. Estimated Decadal Changes in the North Atlantic Meridional Overturning Circulation and Heat Flux 1993-2004

Authors: Carl Wunsch (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2957.1

URL: <https://doi.org/10.1175/jpo2957.1>

Publication Date: 2006-12-06

Volume/Issue: Volume 36, Issue 11

References: 30

Cited by: 98 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Results from a global 1 degrees model constrained by least squares to a multiplicity of datasets over the interval 1992-2004 are used to describe apparent changes in the North Atlantic Ocean meridional overturning circulation and associated heat fluxes at 26 degreesN. The least squares fit is both stable and adequately close to the data to make the analysis worthwhile. Changes over the 12 yr are spatially and temporally complex. A weak statistically significant trend is found in net North Atlantic volume flux above about 1200 m, which drops slightly (-0.19 ± 0.05 Sv yr⁻¹; 1 Sv $\equiv 10^6$ m³ s⁻¹) but with a corresponding strengthening of the outflow of North Atlantic Deep Water and inflow of abyssal waters. The slight associated trend in meridional heat flux is very small and not statistically significant. The month-to-month variability implies that single-section determinations of heat and volume flux are subject to serious aliasing errors.

2341. The Mercator global ocean operational analysis system: Assessment and validation of an 11-year reanalysis

Authors: Nicolas Ferry; Elisabeth Rémy; Pierre Brasseur; Christophe Maes

Journal: Journal of Marine Systems

Publisher: Elsevier BV

DOI: 10.1016/j.jmarsys.2005.08.004

URL: <https://doi.org/10.1016/j.jmarsys.2005.08.004>

Publication Date: 2006-11-29

Volume/Issue: Volume 65, Issue 1-4

References: 52

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2342. Concurrent Visualization in a Production Supercomputing Environment

Authors: D. Ellsworth; B. Green; C. Henze; P. Moran; T. Sandstrom

Journal: IEEE Transactions on Visualization and Computer Graphics

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tvcg.2006.128

URL: <https://doi.org/10.1109/tvcg.2006.128>

Publication Date: 2006-11-22

Volume/Issue: Volume 12, Issue 5

References: 22

Cited by: 29 publications

Language: N/A

Subjects: N/A

Keywords: supercomputing, high temporal resolution visualization, GEOS4 global climate model, interactive visual computing, time-varying data (and 4 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2343. Seasonal effect on vertical positioning by Satellite Laser Ranging and Global Positioning System and on absolute gravity at the OCA geodetic station, Grasse, France

Authors: J. Nicolas; J.-M. Nocquet; M. Van Camp; T. van Dam; J.-P. Boy; J. Hinderer; P. Gegout; E. Calais; M. Amalvict

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2006.03205.x

URL: <https://doi.org/10.1111/j.1365-246x.2006.03205.x>

Publication Date: 2006-11-17

Volume/Issue: Volume 167, Issue 3

References: 60

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2344. NASA supercomputer improves prospects for ocean climate research

Authors: D. Menemenlis; C. Hill; A. Adcroft; J.-M. Campin; B. Cheng; B. Ciotti; I. Fukumori; P. Heimbach; C. Henze; A. Köhl; T. Lee; D. Stammer; J. Taft; J. Zhang

Journal: Eos, Transactions American Geophysical Union

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005eo090002

URL: <https://doi.org/10.1029/2005eo090002>

Publication Date: 2006-10-19

Volume/Issue: Volume 86, Issue 9

References: 16

Cited by: 108 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Estimates of ocean circulation constrained by in situ and remotely sensed observations have become routinely available during the past five years, and they are being applied to myriad scientific and operational problems [Stammer et al., 2002]. Under the Global Ocean Data Assimilation Experiment (GODAE), several regional and global estimates have evolved for applications in climate research, seasonal forecasting, naval operations, marine safety, fisheries, the offshore oil industry coastal management, and other areas. This article reports on recent progress by one effort, the consortium for Estimating the Circulation and Climate of the Ocean (ECCO), toward a next-generation synthesis of ocean and sea-ice data that is global, that covers the full ocean depth, and that permits eddies.

2345. State estimation improves prospects for ocean research

Authors: Detlef Stammer; C. Wunsch; I. Fukumori; J. Marshall

Journal: Eos, Transactions American Geophysical Union

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2002eo000207

URL: <https://doi.org/10.1029/2002eo000207>

Publication Date: 2006-10-19

Volume/Issue: Volume 83, Issue 27

References: 13

Cited by: 72 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Rigorous global ocean state estimation methods can now be used to produce dynamically consistent time-varying model/data syntheses, the results of which are being used to study a variety of important scientific problems. Figure 1 shows a schematic of a complete ocean observing and synthesis system that includes global observations and state-of-the-art ocean general circulation models (OGCM) run on modern computer platforms. A global observing system is described in detail in Smith and Koblinsky [2001], and the present status of ocean modeling and anticipated improvements are addressed by Griffies et al. [2001]. Here, the focus is on the third component of state estimation: the synthesis of the observations and a model into a unified, dynamically consistent estimate.

2346. Strategies for nested and eddy-permitting state estimation

Authors: Geoffrey Gebbie (Massachusetts Institute of Technology, Woods Hole Oceanographic Institution); Patrick Heimbach (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jc003094

URL: <https://doi.org/10.1029/2005jc003094>

Publication Date: 2006-10-11

Volume/Issue: Volume 111, Issue C10

References: 49

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Both ocean process and prediction studies increasingly rely on state estimation (i.e., data assimilation) to provide the most complete representation of how the ocean circulates. This study applies the formalism and methodology of state estimation, recently developed for the global, coarse-resolution problem, to eddy-permitting state estimation in regional domains. Two major challenges exist for a state estimate that is nested inside a global state estimate: (1) estimation of open-boundary conditions consistent with information interior and exterior to the regional domain and (2) estimation with the higher-resolution models of regional studies. Here a least squares cost function defines the problem of minimizing the misfit between a North Atlantic regional general circulation model and actual observations, including those of the Subduction Experiment. A first experiment, using a novel "multiscale" method to constrain the large-scale regional circulation, shows that the use of the adjoint of both an eddy-permitting model and its coarse-resolution twin leads to a solution of the least squares problem in a computationally practical way. Therefore no fundamental obstacle exists to constraining the large-scale regional circulation nested within a global circulation. A second experiment in the North Atlantic shows that the model circulation can be constrained to the full observational signal, including eddy variability, as observed at selected point locations. Both experiments in this study produce eddy-permitting state estimates which are exactly self-consistent with the equations of motion as embodied by a general circulation model. Therefore dynamical balances can be diagnosed and easily interpreted; in particular, a companion paper uses the state estimates to determine eddy subduction rates in the North Atlantic.

2347. Mass variation in the Mediterranean Sea from GRACE and its validation by altimetry, steric and hydrologic fields

Authors: L. Fenoglio-Marc (Institut für Physikalische Geodäsie Technische Universität Darmstadt Darmstadt Germany); J. Kusche (Delft University of Technology); M. Becker (Institut für Physikalische Geodäsie Technische Universität Darmstadt Darmstadt Germany)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gl026851

URL: <https://doi.org/10.1029/2006gl026851>

Publication Date: 2006-10-05

Volume/Issue: Volume 33, Issue 19

References: 22

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: Seasonality

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The seasonal seawater mass variation in the Mediterranean Sea is estimated between April 2002 and July 2004 from GRACE and altimetry data and from hydrologic and oceanographic models. A smoothed spatial averaging kernel is applied to each field, in order to obtain comparable basin averages. The GRACE seawater mass corrected for the leakage of continental hydrology and the filtered steric-corrected altimeter sea level have similar annual amplitude and phase. To restore the magnitude of the GRACE-derived water mass signal we apply a scaling factor to the smoothed annual amplitude. The estimated scaled mass signal has an annual amplitude of 52 +/- 17 mm peaking in November. We combine the seawater mass variation with the Mediterranean freshwater deficit and obtain a net flow at the Strait of Gibraltar with annual amplitude of 60 +/- 25 mm/month (0.06 Sv) and maximum in September.

2348. On the steric and mass-induced contributions to the annual sea level variations in the Mediterranean Sea

Authors: David García (University of Alicante); Benjamin F. Chao (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Jorge Del Río (Department of Applied Physics II University of Malaga Malaga Spain); Isabel Vigo (University of Alicante); Jesús García-Lafuente (Department of Applied Physics II University of Malaga Malaga Spain)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jc002956

URL: <https://doi.org/10.1029/2005jc002956>

Publication Date: 2006-09-25

Volume/Issue: Volume 111, Issue C9

References: 29

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The sea level variation (SLV_{total}) is the sum of two major contributions: steric and mass-induced. The steric SLV_{steric} is that resulting from the thermal and salinity changes in a given water column. It only involves volume change, hence has no gravitational effect. The mass-induced SLV_{mass}, on the other hand, arises from adding or subtracting water mass to or from the water column and has direct gravitational signature. We examine the closure of the seasonal SLV budget and estimate the relative importance of the two contributions in the Mediterranean Sea as a function of time. We use ocean altimetry data (from TOPEX/Poseidon, Jason 1, ERS, and ENVISAT missions) to estimate SLV_{total}, temperature, and salinity data (from the Estimating the Circulation and Climate of the Ocean ocean model) to estimate SLV_{steric}, and time variable gravity data (from Gravity Recovery and Climate Experiment (GRACE) Project, April 2002 to July 2004) to estimate SLV_{mass}. We find that the annual cycle of SLV_{total} in the Mediterranean is mainly driven by SLV_{steric} but moderately offset by SLV_{mass}. The agreement between the seasonal SLV_{mass} estimations from SLV_{total} - SLV_{steric} and from GRACE is quite remarkable; the annual cycle reaches the maximum value in mid-February, almost half a cycle later than SLV_{total} or SLV_{steric}, which peak by mid-October and mid-September, respectively. Thus, when sea level is rising (falling), the Mediterranean Sea is actually losing (gaining) mass. Furthermore, as SLV_{mass} is balanced by vertical (precipitation minus evaporation, P-E) and horizontal (exchange of water with the Atlantic, Black Sea, and river runoff) mass fluxes, we compared it with the P-E determined from meteorological data to estimate the annual cycle of the horizontal flux.

2349. Assimilating long-term hydrographic information into an eddy-permitting model of the North Atlantic

Authors: D. G. Wright (Bedford Institute of Oceanography); K. R. Thompson (Dalhousie University); Y. Lu (Bedford Institute of Oceanography)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jc003200

URL: <https://doi.org/10.1029/2005jc003200>

Publication Date: 2006-09-14

Volume/Issue: Volume 111, Issue C9

References: 48

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The utility of a new technique for assimilating long-term hydrographic information into eddy-permitting ocean models is demonstrated using the Parallel Ocean Program (POP) applied to the North Atlantic. Robust diagnostic and standard prognostic simulations of the North Atlantic yield results similar to those obtained in earlier studies and the differences between them emphasize the need for data assimilation. The basic idea of the new technique is to add correction terms to the model equations that directly influence the model solution only in prescribed frequency and wave number bands, leaving the variations outside of these bands free to evolve prognostically. For this reason the technique is referred to as spectral nudging. We consider two approaches for constraining eddy-permitting models based on observed long-term hydrographic conditions. In the first approach, the model's temperature and salinity climatologies are spectrally nudged toward observed values using restoring terms in the tracer equations; in the second, correction terms are added to the momentum equations. Both approaches result in significant improvements in the model's climatology, as expected. Both approaches also result in more realistic meso-scale eddy fields. However, for the simulations considered here, spectral nudging in the tracer equations generally provides better results than nudging in the momentum equations. An examination of the relationship between the two approaches reveals that this result might be expected to also occur in other model simulations but this is achieved at the cost of increased constraints placed on model dynamics within the nudged frequency and wave number bands.

2350. Seasonal and interannual global surface mass variations from multisatellite geodetic data

Authors: Xiaoping Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael B. Heflin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Erik R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jb004100

URL: <https://doi.org/10.1029/2005jb004100>

Publication Date: 2006-09-05

Volume/Issue: Volume 111, Issue B9

References: 51

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Monthly global surface mass distribution changes are estimated in the spherical harmonic domain with a complete spectrum up to degree and order 50. The estimates are derived by inverting GPS displacement series measured at roughly 450 continuously tracking sites and ocean bottom pressure (OBP) estimates of a data assimilated ocean circulation model from 1993 to 2004. The inversion uses a hybrid estimator after singular value decomposition of the normalized measurement equations with reduced reliance on a priori spectral information. The results are then compared and combined with Gravity Recovery and Climate Experiment (GRACE) gravity data to provide enhanced spectral and geographic coverage. High fidelity degree-1 surface mass variation coefficients are recovered, corresponding to equivalent geocenter motion with better than 0.5 mm annual amplitude precisions in all three components. A clear annual surface mass cycle occurs in both GPS/OBP- and GRACE-derived results. There is very good agreement among the low-degree spherical harmonic coefficients and in global geographic pattern but with significant regional differences in amplitudes. Annual variations of average mass over the global oceans, Antarctica, and Greenland are then derived. Total surface mass over both Antarctica and Greenland peak in their respective summers due to increased atmospheric mass. Large interannual variations have also been found involving several continents in the 11-year GPS/OBP solution. The patterns of variation suggest that the recent satellite laser ranging (SLR) observation of a reversal in trend of the Earth's oblateness may be largely an interannual surface mass cycle with considerable contribution from the northern continents.

2351. Geocentre motion measured with DORIS and SLR, and predicted by geophysical models

Authors: M. Feissel-Vernier; K. Le Bail; P. Berio; D. Coulot; G. Ramillien; J. -J. Valette

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-006-0079-z

URL: <https://doi.org/10.1007/s00190-006-0079-z>

Publication Date: 2006-08-31

Volume/Issue: Volume 80, Issue 8-11

References: 36

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2352. Accuracy assessment of the monthly GRACE geoids based upon a simulation

Authors: E. J. O. Schrama; P. N. A. M. Visser

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-006-0085-1

URL: <https://doi.org/10.1007/s00190-006-0085-1>

Publication Date: 2006-08-28

Volume/Issue: Volume 81, Issue 1

References: 37

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2353. Efficiency of a POD-based reduced second-order adjoint model in 4D-Var data assimilation

Authors: D. N. Daescu; I. M. Navon

Journal: International Journal for Numerical Methods in Fluids

Publisher: Wiley

DOI: 10.1002/fld.1316

URL: <https://doi.org/10.1002/fld.1316>

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References: 44

Cited by: 71 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Order reduction strategies aim to alleviate the computational burden of the four-dimensional variational data assimilation by performing the optimization in a low-order control space. The proper orthogonal decomposition (POD) approach to model reduction is used to identify a reduced-order control space for a two-dimensional global shallow water model. A reduced second-order adjoint (SOA) model is developed and used to facilitate the implementation of a Hessian-free truncated-Newton (HFTN) minimization algorithm in the POD-based space. The efficiency of the SOA/HFTN implementation is analysed by comparison with the quasi-Newton BFGS and a nonlinear conjugate gradient algorithm. Several data assimilation experiments that differ only in the optimization algorithm employed are performed in the reduced control space. Numerical results indicate that first-order derivative methods are effective during the initial stages of the assimilation; in the later stages, the use of second-order derivative information is of benefit and HFTN provided significant CPU time savings when compared to the BFGS and CG algorithms. A comparison with data assimilation experiments in the full model space shows that with an appropriate selection of the basis functions the optimization in the POD space is able to provide accurate results at a reduced computational cost. The HFTN algorithm benefited most from the order reduction since computational savings were achieved both in the outer and inner iterations of the method. Further experiments are required to validate the approach for comprehensive global circulation models. Copyright © 2006 John Wiley & Sons, Ltd.

2354. Treating strong adjoint sensitivities in tropical eddy-permitting variational data assimilation

Authors: I. Hoteit; B. Cornuelle; A. Köhl; D. Stammer

Journal: Quarterly Journal of the Royal Meteorological Society

Publisher: Wiley

DOI: 10.1256/qj.05.97

URL: <https://doi.org/10.1256/qj.05.97>

Publication Date: 2006-08-17

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References: 53

Cited by: 52 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A variational data assimilation system has been implemented for the tropical Pacific Ocean using an eddy-permitting regional implementation of the MITgcm. The adjoint assimilation system was developed by the Estimation of the Circulation and the Climate of the Ocean consortium, and has been extended to deal with open boundaries. This system is used to adjust the model to match observations in the tropical Pacific region using control parameters which include initial conditions, open boundaries and time-dependent surface fluxes. This paper focuses on problems related to strong adjoint sensitivities that may impede the model fit to the observations. A decomposition of the velocities at the open boundaries into barotropic and baroclinic modes is introduced to deal with very strong sensitivities of the model sea surface height to the barotropic component of the inflow. Increased viscosity and diffusivity terms are used in the adjoint model to reduce exponentially growing sensitivities in the backward run associated with nonlinearity of the forward model. Simplified experiments in which the model was constrained with Levitus temperature and salinity data, Reynolds sea surface temperature data and TOPEX/POSEIDON altimeter data were performed to demonstrate the controllability of this assimilation system and to study its sensitivity to the starting guesses for forcing and initial conditions. Copyright © 2005 Royal Meteorological Society

2355. Ocean bottom pressure observation off Sanriku and comparison with ocean tide models, altimetry, and barotropic signals from ocean models

Authors: K. Matsumoto (Division of Radio Astronomy National Astronomical Observatory of Japan Oshu Japan); T. Sato (Division of Radio Astronomy National Astronomical Observatory of Japan Oshu Japan); H. Fujimoto (Tohoku University); Y. Tamura (Division of Radio Astronomy National Astronomical Observatory of Japan Oshu Japan); M. Nishino (Tohoku University); R. Hino (Tohoku University); T. Higashi (Kyoto University); T. Kanazawa (University of Tokyo)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gl026706

URL: <https://doi.org/10.1029/2006gl026706>

Publication Date: 2006-08-15

Volume/Issue: Volume 33, Issue 16

References: 16

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Barotropic fluid|Ocean tide|Internal tide|Tidal Model

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We discuss tidal and non-tidal signals which are obtained from ocean bottom pressure (OBP) measurements at two sites off Sanriku, Japan. Comparison between the observed tidal component and five ocean tide models shows that recent models have accuracy better than 1.3 cm in terms of root sum square of vector differences for eight principal constituents. The empirical estimates of non-tidal signal from TOPEX/POSEIDON (T/P) and JASON-1 deviate from barotropic signals observed by OBP recorders, which can be attributed to baroclinic component in the altimeter data. We also compared the non-tidal barotropic signal derived from the observation with those from ECCO model (wind-driven) and Kyushu University model (pressure-driven). The comparison indicates that the pressure-driven model increases explained variance and it is suggested to combine the wind-driven model and the pressure-driven model in order to describe non-tidal ocean mass variability in particular at period shorter than 30 days.

2356. The rotational and gravitational signature of the December 26, 2004 Sumatran earthquake

Authors: Richard S. Gross; Benjamin F. Chao

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-006-9008-1

URL: <https://doi.org/10.1007/s10712-006-9008-1>

Publication Date: 2006-08-11

Volume/Issue: Volume 27, Issue 6

References: 43

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: Polar motion|Longitude

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2357. Satellite altimetry and ocean dynamics

Authors: Lee Lueng Fu; Pierre-Yves Le Traon

Journal: Comptes Rendus. Géoscience

Publisher: Cellule MathDoc/Centre Mersenne

DOI: 10.1016/j.crte.2006.05.015

URL: <https://doi.org/10.1016/j.crte.2006.05.015>

Publication Date: 2006-08-02

Volume/Issue: Volume 338, Issue 14-15

References: 85

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2358. Weak influences of the Earth's magnetic field on ocean circulation

Authors: R. H. Tyler (University of Washington)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gl026372

URL: <https://doi.org/10.1029/2006gl026372>

Publication Date: 2006-07-27

Volume/Issue: Volume 33, Issue 14

References: 10

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: In this work, models for the Earth's magnetic field and global ocean circulation are combined to produce the first quantitative estimates of the electromagnetic Lorentz forces which couple abyssal ocean layers with the energetic surface layers. Under present conditions these coupling forces are very weak: typical time scales for the transfer of momentum to the lower layers are about three thousand years. But they are worth describing quantitatively because of the following: 1) The coupling rapidly increases in proportion with the square of the Earth's magnetic field strength; 2) This process is intrinsically efficient at accelerating slow or stagnant abyssal ocean layers; 3) These time scales, while long, are similar to those of rapid climate variability which are expected to be associated with past variability in ocean circulation, overturning and heat transport.

2359. Efficiency of reduced-order, time-dependent adjoint data assimilation approaches

Authors: Ibrahim Hoteit; Armin Köhl

Journal: Journal of Oceanography

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10872-006-0074-2

URL: <https://doi.org/10.1007/s10872-006-0074-2>

Publication Date: 2006-07-15

Volume/Issue: Volume 62, Issue 4

References: 31

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2360. Ocean mixed layer depth: A subsurface proxy of ocean-atmosphere variability

Authors: K. Lorbacher (Scripps Institution of Oceanography); D. Dommenges (Scripps Institution of Oceanography); P. P. Niiler (Scripps Institution of Oceanography); A. Köhl (Scripps Institution of Oceanography)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003jc002157

URL: <https://doi.org/10.1029/2003jc002157>

Publication Date: 2006-07-11

Volume/Issue: Volume 111, Issue C7

References: 29

Cited by: 167 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A new criterion, based on the shallowest extreme curvature of near surface layer density or temperature profiles, is established for demarking the mixed layer depth, h_{mix} . Using historical global hydrographic profile data, including conductivity-temperature-depth and expendable bathythermograph data obtained during World Ocean Circulation Experiment, its seasonal variability and monthly to interannual anomalies are computed. Unlike the more commonly used Δ criterion, the new criterion is able to deal with both different vertical resolutions of the data set and a large variety of observed stratification profiles. For about two thirds of the profiles our algorithm produces an h_{mix}/c that is more reliable than the one of the Δ criterion. The uncertainty for h_{mix}/c is ± 5 m for high- (< 5 m) and ± 8 m for low- (< 20 m) resolution profiles. A quality index, Q_{lmix} , which compares the variance of a profile above h_{mix} to the variance to a depth of $1.5 \times h_{mix}$, shows that for the 70% of the profile data for which a clearly recognizable well-mixed zone exists near the surface, our criterion identifies the depth of the well-mixed zone in all cases. The standard deviation of anomalous monthly h_{mix}/c is typically 20-70% of the long-term mean h_{mix}/c . In the tropical Pacific the monthly mean anomalies of h_{mix}/c are not well correlated with anomalies of sea surface temperature, which indicate that a variety of turbulent processes, other than surface heat fluxes, are important in the upper ocean there. Comparisons between observed h_{mix}/c and Massachusetts Institute of Technology/ocean general circulation model/Estimating the Circulation and Climate of the Ocean model simulated mixed layer depth indicate that the KPP algorithm captures in general a 30% smaller mixed layer depth than observed.

2361. What Is Data Assimilation Really Solving, and How Is the Calculation Actually Done?

Authors: Ichiro Fukumori

Journal: Ocean Weather Forecasting

Publisher: Springer-Verlag

DOI: 10.1007/1-4020-4028-8_11

URL: https://doi.org/10.1007/1-4020-4028-8_11

Publication Date: 2006-07-07

Volume/Issue: N/A

References: 19

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Data assimilation, smoother, ECCO, consistency, Kalman filter

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2362. The Closure of the Ocean Mixed Layer Temperature Budget Using Level-Coordinate Model Fields

Authors: Seung-Bum Kim (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/jtech1883.1

URL: <https://doi.org/10.1175/jtech1883.1>

Publication Date: 2006-07-06

Volume/Issue: Volume 23, Issue 6

References: 29

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Entrainment is an important element of the mixed layer mass, heat, and temperature budgets. Conventional procedures to estimate entrainment heat advection often do not permit the closure of heat and temperature budgets because of inaccuracies in its formulation. In this study a rigorous approach to evaluate the effect of entrainment using the output of a general circulation model (GCM) that does not have an explicit prognostic mixed layer model is described. The integral elements of the evaluation are 1) the rigorous estimates of the temperature difference between mixed layer water and entrained water at each horizontal grid point, 2) the formulation of the temperature difference such that the budget closes over a volume greater than one horizontal grid point, and 3) the apparent warming of the mixed layer during the mixed layer shoaling to account for the weak vertical temperature gradient within the mixed layer. This evaluation of entrainment heat advection is compared with the estimates by other commonly used ad hoc formulations by applying them in three regions: the north-central Pacific, the Kuroshio Extension, and the Niño-3 areas in the tropical Pacific. In all three areas the imbalance in the mixed layer temperature budget by the ad hoc estimates is significant, reaching a maximum of about 4 K yr⁻¹.

2363. Rapid variations in polar motion during the 2005-2006 winter season

Authors: S. B. Lambert (Royal Observatory of Belgium); C. Bizouard (Earth Orientation Center International Earth Rotation and Reference Systems Service, Paris Observatory Paris France); V. Dehant (Royal Observatory of Belgium)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2006gl026422

URL: <https://doi.org/10.1029/2006gl026422>

Publication Date: 2006-07-02

Volume/Issue: Volume 33, Issue 13

References: 8

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The terrestrial motion of the Earth's rotation pole shows mainly two large oscillations: the Chandler and the annual wobbles driven by mass redistributions in surface geophysical fluids. These contributions interfere destructively every 6.4 years, slowing down the pole at that period of time. This is the case for the recent period from November 2005 till February 2006. Due to the high precision of Earth orientation data of these days and for the first time in the history of polar motion observation, very small structures of the motion are observed. We compared the observed polar motion with the contribution of atmosphere and oceans predicted from global atmospheric and oceanic analyses and models. We clearly see that centimeter level polar motion displacements during the 2005-2006 winter season are almost fully explained by major pressure events on the continents and on the ocean, especially a depression over Northern Europe in-phase with similar events over North America.

2364. Optimized smoothing of Gravity Recovery and Climate Experiment (GRACE) time-variable gravity observations

Authors: J. L. Chen (University of Texas at Austin); C. R. Wilson (University of Texas at Austin); K.-W. Seo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jb004064

URL: <https://doi.org/10.1029/2005jb004064>

Publication Date: 2006-06-26

Volume/Issue: Volume 111, Issue B6

References: 19

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: High-degree and high-order spherical harmonics of time-variable gravity fields observed by the Gravity Recovery and Climate Experiment (GRACE) gravity mission are dominated by noise. We develop two smoothing methods that suppress these high-degree and high-order errors with results superior to more commonly used Gaussian smoothing. These optimized smoothing methods considerably improve signal-to-noise levels of GRACE terrestrial water storage estimates relative to residual signal and noise over the oceans and show significantly better spatial resolution and lower leakage error. On the basis of analysis using an advanced land surface model, the equivalent spatial resolution from these optimized smoothing estimates is about 500 km, compared to the roughly 800-1000 km Gaussian smoothing that is required to suppress high-degree noise in the GRACE fields.

2365. The variations of earth rotation excited by the atmosphere -- A further analysis of the method and data

Authors: Yan Hao-ming; Zhong Min; Zhu Yao-zhong

Journal: Chinese Astronomy and Astrophysics

Publisher: Elsevier BV

DOI: 10.1016/j.chinastron.2006.01.002

URL: <https://doi.org/10.1016/j.chinastron.2006.01.002>

Publication Date: 2006-06-12

Volume/Issue: Volume 30, Issue 1

References: 18

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2366. Terrestrial water mass load changes from Gravity Recovery and Climate Experiment (GRACE)

Authors: K.-W. Seo (University of Texas at Austin); C. R. Wilson (University of Texas at Austin); J. S. Famiglietti (University of California Irvine); J. L. Chen (University of Texas at Austin); M. Rodell (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005wr004255

URL: <https://doi.org/10.1029/2005wr004255>

Publication Date: 2006-05-15

Volume/Issue: Volume 42, Issue 5

References: 17

Cited by: 82 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Recent studies show that data from the Gravity Recovery and Climate Experiment (GRACE) is promising for basin- to global-scale water cycle research. This study provides varied assessments of errors associated with GRACE water storage estimates. Thirteen monthly GRACE gravity solutions from August 2002 to December 2004 are examined, along with synthesized GRACE gravity fields for the same period that incorporate simulated errors. The synthetic GRACE fields are calculated using numerical climate models and GRACE internal error estimates. We consider the influence of measurement noise, spatial leakage error, and atmospheric and ocean dealiasing (AOD) model error as the major contributors to the error budget. Leakage error arises from the limited range of GRACE spherical harmonics not corrupted by noise. AOD model error is due to imperfect correction for atmosphere and ocean mass redistribution applied during GRACE processing. Four methods of forming water storage estimates from GRACE spherical harmonics (four different basin filters) are applied to both GRACE and synthetic data. Two basin filters use Gaussian smoothing, and the other two are dynamic basin filters which use knowledge of geographical locations where water storage variations are expected. Global maps of measurement noise, leakage error, and AOD model errors are estimated for each basin filter. Dynamic basin filters yield the smallest errors and highest signal-to-noise ratio. Within 12 selected basins, GRACE and synthetic data show similar amplitudes of water storage change. Using 53 river basins, covering most of Earth's land surface excluding Antarctica and Greenland, we document how error changes with basin size, latitude, and shape. Leakage error is most affected by basin size and latitude, and AOD model error is most dependent on basin latitude.

2367. Geocenter motions from GPS: A unified observation model

Authors: David A. Lavallée (University of Newcastle upon Tyne); Tonie van Dam (European Center for Geodynamics and Seismology); Geoffrey Blewitt (University of Nevada); Peter J. Clarke (University of Newcastle upon Tyne)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jb003784

URL: <https://doi.org/10.1029/2005jb003784>

Publication Date: 2006-05-06

Volume/Issue: Volume 111, Issue B5

References: 51

Cited by: 74 publications

Language: en

Subjects: N/A

Keywords: Satellite laser ranging|Realization (probability)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We test a unified observation model for estimating surface-loading-induced geocenter motion using GPS. In principle, this model is more complete than current methods, since both the translation and deformation of the network are modeled in a frame at the center of mass of the entire Earth system. Real and synthetic data for six different GPS analyses over the period 1997.25-2004.25 are used to (1) build a comprehensive appraisal of the errors and (2) compare this unified approach with the alternatives. The network shift approach is found to perform particularly poorly with GPS. Furthermore, erroneously estimating additional scale changes with this approach can suggest an apparently significant seasonal variation which is due to real loading. An alternative to the network shift approach involves modeling degree-1 and possibly higher-degree deformations of the solid Earth in a realization of the center of figure frame. This approach is shown to be more robust for unevenly distributed networks. We find that a unified approach gives the lowest formal error of geocenter motion, smaller differences from the true value when using synthetic data, the best agreement between five different GPS analyses, and the closest (submillimeter) agreement with the geocenter motion predicted from loading models and estimated using satellite laser ranging. For five different GPS analyses, best estimates of annual geocenter motion have a weighted root-mean-square agreement of 0.6, 0.6, and 0.8 mm in amplitude and 21 degrees, 22 degrees, and 22 degrees in phase for x, y, and z, respectively.

2368. Estimating Large-Scale Precipitation Minus Evapotranspiration from GRACE Satellite Gravity Measurements

Authors: Sean Swenson (University of Colorado Boulder); John Wahr (University of Colorado Boulder)

Journal: Journal of Hydrometeorology

Publisher: American Meteorological Society

DOI: 10.1175/jhm478.1

URL: <https://doi.org/10.1175/jhm478.1>

Publication Date: 2006-04-27

Volume/Issue: Volume 7, Issue 2

References: 40

Cited by: 101 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Currently, observations of key components of the earth's large-scale water and energy budgets are sparse or even nonexistent. One key component, precipitation minus evapotranspiration (P - ET), remains largely unmeasured due to the absence of observations of ET. Precipitation minus evapotranspiration describes the flux of water between the atmosphere and the earth's surface, and therefore provides important information regarding the interaction of the atmosphere with the land surface. In this paper, large-scale changes in continental water storage derived from satellite gravity data from the Gravity Recovery and Climate Experiment (GRACE) project are combined with river discharge data to obtain estimates of areally averaged P - ET. After constructing an equation describing the large-scale terrestrial water balance reflecting the temporal sampling of GRACE water storage estimates, GRACE-derived P - ET estimates are compared to those obtained from a reanalysis dataset [NCEP/Department of Energy (DOE) reanalysis-2] and a land surface model driven with observation-based forcing [Global Land Data Assimilation System (GLDAS)/Noah] for two large U.S. river basins. GRACE-derived P - ET compares quite favorably with the reanalysis-2 output, while P - ET from the Noah model shows significant differences. Because the uncertainties in the GRACE results can be computed rigorously, this comparison may be considered as a validation of the models. In addition to showing how GRACE P - ET estimates may be used to validate model output, the accuracy of GRACE estimates of both the seasonal cycle and the monthly averaged rate of P - ET is examined. Finally, the potential for estimating seasonal evapotranspiration is demonstrated by combining GRACE seasonal P - ET estimates with independent estimates of the seasonal cycle of precipitation.

2369. Post-processing removal of correlated errors in GRACE data

Authors: Sean Swenson (University of Colorado Boulder); John Wahr (University of Colorado Boulder)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005gl025285

URL: <https://doi.org/10.1029/2005gl025285>

Publication Date: 2006-04-25

Volume/Issue: Volume 33, Issue 8

References: 17

Cited by: 1178 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Gravity fields produced by the Gravity Recovery and Climate Experiment (GRACE) satellite mission require smoothing to reduce the effects of errors present in short wavelength components. As the smoothing radius decreases, these errors manifest themselves in maps of surface mass variability as long, linear features generally oriented north to south (i.e., stripes). The presence of stripes implies correlations in the gravity field coefficients. Here we examine the spectral signature of these correlated errors, and present a method to remove them. Finally, we apply the filter to a model of surface-mass variability to show that the filter has relatively little degradation of the underlying geophysical signals we seek to recover.

2370. Temporal Changes in Ocean Eddy Transports

Authors: Detlef Stammer (University of Hamburg); Carl Wunsch (Massachusetts Institute of Technology); Kyozi Ueyoshi (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2858.1

URL: <https://doi.org/10.1175/jpo2858.1>

Publication Date: 2006-04-24

Volume/Issue: Volume 36, Issue 3

References: 25

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: New estimates from 11 yr of altimetric data are made of the global time-average variability kinetic energy and its decadal-scale variability. Making the approximation that the variability reflects primarily eddy motions, a time-mean, but spatially varying, eddy mixing coefficient is then estimated along with its changes over the last decade. With a record length more than 2 times that previously available, the time-mean variability kinetic energy KE is statistically more reliable and smoother in its spatial pattern. Minimum values of KE are present in the subpolar North Pacific Ocean and in the eastern South Pacific (both less than 100 cm² s⁻²). In contrast to the North Pacific, the subpolar North Atlantic Ocean shows relatively enhanced KE. Eddy kinetic energy and eddy mixing appear to have declined during the last decade over large parts of the western Pacific Ocean, in some regions by as much as 50% of the time-mean value. Increased eddy variability can be found in the Kuroshio and Gulf Stream regions, as well as in the Agulhas region, east of Australia, and at several locations along the Antarctic Circumpolar Current. Somewhat enhanced eddy variability and eddy mixing are also apparent in the eastern tropical Pacific. A numerical simulation of the ocean circulation at 1 degrees spatial resolution over a 10-yr period suggests that variations in eddy mixing of this order of magnitude measurably affect the deep temperature field in the vicinity of permanent frontal structures on a time scale of less than 4 yr. The meridional overturning circulation also reacts on these time scales. If persistent over longer periods in the ocean, these effects would be important for climate simulations.

2371. Observing seasonal bottom pressure variability in the North Pacific with GRACE

Authors: R. J. Bingham (Proudman Oceanographic Laboratory Liverpool UK); C. W. Hughes (Proudman Oceanographic Laboratory Liverpool UK)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005gl025489

URL: <https://doi.org/10.1029/2005gl025489>

Publication Date: 2006-04-22

Volume/Issue: Volume 33, Issue 8

References: 14

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Using monthly gravity field solutions from GRACE together with output from an ocean model we assess the ability of GRACE to observe seasonal ocean bottom pressure variability in the North Pacific. First, the spatial structure and temporal evolution of the long-term mean seasonal bottom pressure cycle in the North Pacific is characterised in the context of the ocean model. It is then shown that a mean seasonal cycle based on a limited subset of the model output corresponding to the intervals over which the GRACE monthly solutions were formed is representative of this long-term seasonal cycle. It is therefore concluded that the significant features of the seasonal cycle should be present in the GRACE data. Agreement is found between the ocean model and GRACE estimates of the bottom pressure seasonal cycle, thus demonstrating the worth of GRACE as a tool for studying ocean dynamics.

2372. Interannual variability in northeast Pacific circulation

Authors: Elizabeth Douglass (Scripps Institution of Oceanography); Dean Roemmich (Scripps Institution of Oceanography); Detlef Stammer (Institut für Meereskunde, Zentrum fuer Meeres-und Klimaforschung Universitaet Hamburg Hamburg Germany)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jc003015

URL: <https://doi.org/10.1029/2005jc003015>

Publication Date: 2006-04-04

Volume/Issue: Volume 111, Issue C4

References: 20

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Interannual variability of the circulation in the northeast Pacific Ocean is explored through a joint analysis of expendable bathythermograph (XBT) and expendable conductivity-temperature-depth (XCTD) data, satellite altimetry, and output from a model that was constrained by ocean data. XBT temperature profiles with high spatial resolution are available in the eastern North Pacific along two repeated transects. These ship tracks, along with the coast of North America, define a closed "box" which is used to study the time-mean circulation and its variability on interannual timescales. Geostrophic velocities from XBT data are compared with geostrophic velocities from model output as well as the full model velocity fields. Correlations in variability on interannual timescales between transport in the subpolar gyre and in the subtropical gyre are present in both model output and data. The nature of the variability, and its relation to the changes of the strength of the North Pacific Current (NPC), which supplies the water for both gyres, are explored. Interannual variability in gyre transport is found to be related to both the bifurcation of the NPC, resulting in an anticorrelation in transport between the two gyres, and to variations in NPC strength, resulting in simultaneous changes in the two gyres. The dominant signal is found to be a long-term increase in the NPC, which results in a strengthening of the subtropical gyre. Possible connections with local-scale wind stress changes and with the El Niño/Southern Oscillation phenomenon are also explored.

2373. Inverse estimates of anthropogenic CO₂ uptake, transport, and storage by the ocean

Authors: S. E. Mikaloff Fletcher (University of California Los Angeles); N. Gruber (University of California Los Angeles); A. R. Jacobson (Princeton University); S. C. Doney (Woods Hole Oceanographic Institution); S. Dutkiewicz (Massachusetts Institute of Technology); M. Gerber (University of Bern); M. Follows (Massachusetts Institute of Technology); F. Joos (University of Bern); K. Lindsay (National Center for Atmospheric Research); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); A. Mouchet (University of Liège); S. A. Müller (University of Bern); J. L. Sarmiento (Princeton University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005gb002530

URL: <https://doi.org/10.1029/2005gb002530>

Publication Date: 2006-04-04

Volume/Issue: Volume 20, Issue 2

References: 54

Cited by: 330 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Regional air-sea fluxes of anthropogenic CO₂ are estimated using a Green's function inversion method that combines data-based estimates of anthropogenic CO₂ in the ocean with information about ocean transport and mixing from a suite of Ocean General Circulation Models (OGCMs). In order to quantify the uncertainty associated with the estimated fluxes owing to modeled transport and errors in the data, we employ 10 OGCMs and three scenarios representing biases in the data-based anthropogenic CO₂ estimates. On the basis of the prescribed anthropogenic CO₂ storage, we find a global uptake of 2.2 ± 0.25 Pg C yr⁻¹, scaled to 1995. This error estimate represents the standard deviation of the models weighted by a CFC-based model skill score, which reduces the error range and emphasizes those models that have been shown to reproduce observed tracer concentrations most accurately. The greatest anthropogenic CO₂ uptake occurs in the Southern Ocean and in the tropics. The flux estimates imply vigorous northward transport in the Southern Hemisphere, northward cross-equatorial transport, and equatorward transport at high northern latitudes. Compared with forward simulations, we find substantially more uptake in the Southern Ocean, less uptake in the Pacific Ocean, and less global uptake. The large-scale spatial pattern of the estimated flux is generally insensitive to possible biases in the data and the models employed. However, the global uptake scales approximately linearly with changes in the global anthropogenic CO₂ inventory. Considerable uncertainties remain in some regions, particularly the Southern Ocean.

2374. Mean dynamic topography: intercomparisons and errors

Authors: Rory J Bingham (University of Reading); Keith Haines (University of Reading)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rsta.2006.1745

URL: <https://doi.org/10.1098/rsta.2006.1745>

Publication Date: 2006-03-16

Volume/Issue: Volume 364, Issue 1841

References: 18

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Knowledge of the ocean dynamic topography, defined as the height of the sea surface above its rest-state (the geoid), would allow oceanographers to study the absolute circulation of the ocean and determine the associated geostrophic surface currents that help to regulate the Earth's climate. Here a novel approach to computing a mean dynamic topography (MDT), together with an error field, is presented for the northern North Atlantic. The method uses an ensemble of MDTs, each of which has been produced by the assimilation of hydrographic data into a numerical ocean model, to form a composite MDT, and uses the spread within the ensemble as a measure of the error on this MDT. The r.m.s. error for the composite MDT is 3.2 cm, and for the associated geostrophic currents the r.m.s. error is 2.5 cm s⁻¹. Taylor diagrams are used to compare the composite MDT with several MDTs produced by a variety of alternative methods. Of these, the composite MDT is found to agree remarkably well with an MDT based on the GRACE geoid GGM01C. It is shown how the composite MDT and its error field are useful validation products against which other MDTs and their error fields can be compared.

2375. Constraints on mantle anelasticity from geodetic observations, and implications for the J2 anomaly

Authors: David Benjamin; John Wahr; Richard D. Ray; Gary D. Egbert; Shailen D. Desai

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2006.02915.x

URL: <https://doi.org/10.1111/j.1365-246x.2006.02915.x>

Publication Date: 2006-03-09

Volume/Issue: Volume 165, Issue 1

References: 38

Cited by: 70 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2376. Monitoring the integrated deep meridional flow in the tropical North Atlantic: Long-term performance of a geostrophic array

Authors: Torsten Kanzow; Uwe Send; Walter Zenk; Alan D. Chave; Monika Rhein

Journal: Deep Sea Research Part I: Oceanographic Research Papers

Publisher: Elsevier BV

DOI: 10.1016/j.dsr.2005.12.007

URL: <https://doi.org/10.1016/j.dsr.2005.12.007>

Publication Date: 2006-03-08

Volume/Issue: Volume 53, Issue 3

References: 37

Cited by: 68 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2377. The relation of meridional pressure gradients to North Atlantic deep water volume transport in an ocean general circulation model

Authors: Alexa Griesel; Miguel Angel Morales Maqueda

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-006-0122-z

URL: <https://doi.org/10.1007/s00382-006-0122-z>

Publication Date: 2006-03-06

Volume/Issue: Volume 26, Issue 7-8

References: 45

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2378. Circulation of the North Atlantic Ocean from altimetry and the Gravity Recovery and Climate Experiment geoid

Authors: Steven R. Jayne (Woods Hole Oceanographic Institution)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jc003128

URL: <https://doi.org/10.1029/2005jc003128>

Publication Date: 2006-03-03

Volume/Issue: Volume 111, Issue C3

References: 43

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We discuss the ocean circulation derived from the temporally averaged sea surface height, which is referenced to the recently released geoid from the Gravity Recovery and Climate Experiment (GRACE) mission (GRACE Gravity Model 02 (GGM02)). The creation of a precise, independent geoid allows for the calculation of the reference gravitational potential undulation surface, which is associated with the resting ocean surface height. This reference height is then removed from the temporally averaged sea surface height, leaving the dynamic ocean topography. At its most basic level the dynamic ocean topography can be related to the ocean's surface circulation through geostrophy. This has previously been impracticable because of large uncertainties in previous estimates of the Earth's geoid. Prior geoids included the temporally averaged sea surface from altimeters as a proxy for the geoid and therefore were unsuitable for calculations of the ocean's circulation. Geoid undulations are calculated from the GRACE geoid and compared to those from the NASA Goddard Space Flight Center and National Imagery and Mapping Agency Joint Earth Geopotential Model (EGM96) geoid. Error estimates are made to assess the accuracy of the new geoid. The deep ocean pressure field is also estimated by combining the calculated dynamic ocean topography with hydrography. Finally, the derived circulation is compared to independent observations of the circulation from sea surface drifters and subsurface floats. It is shown that the GGM02 geoid is significantly more accurate for use in estimating the ocean's circulation.

2379. GRACE observations of changes in continental water storage

Authors: R. Schmidt; P. Schwintzer; F. Flechtner; Ch. Reigber; A. Güntner; P. Döll; G. Ramillien; A. Cazenave; S. Petrovic; H. Jochmann; J. Wunsch

Journal: Global and Planetary Change

Publisher: Elsevier BV

DOI: 10.1016/j.gloplacha.2004.11.018

URL: <https://doi.org/10.1016/j.gloplacha.2004.11.018>

Publication Date: 2006-03-01

Volume/Issue: Volume 50, Issue 1-2

References: 44

Cited by: 194 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Helmholtz-Zentrum Potsdam - Deutsches GeoForschungsZentrum GFZ, Bundesministerium für Bildung und Forschung

License: N/A

Open Access: N/A

Abstract: N/A

2380. Nontidal oceanic contribution to length-of-day changes estimated from two ocean models during 1992-2001

Authors: Haoming Yan (Chinese Academy of Sciences); Min Zhong (Chinese Academy of Sciences); Yaozhong Zhu (Chinese Academy of Sciences); Lintao Liu (Chinese Academy of Sciences); Xianhe Cao (University of Maryland)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jb003538

URL: <https://doi.org/10.1029/2004jb003538>

Publication Date: 2006-02-22

Volume/Issue: Volume 111, Issue B2

References: 31

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Contributions of ocean bottom pressure and currents to changes in the length of day (LOD) during 1992-2001 are evaluated at seasonal timescales by using two different ocean general circulation model products developed at the Scripps Institution of Oceanography and the University of Maryland. The annual amplitudes for the two bottom pressure terms of oceanic angular momentum (OAM) differ by approximately a factor of 3, while their annual phases are similar. Comparing the gravity-derived nonatmosphere LOD changes with the two OAM bottom pressure terms at annual frequency, we find obvious discrepancies in phase. This result indicates that excitation of LOD changes by other sources, such as hydrology, is also important. The annual amplitudes and phases of the two OAM current terms compare well with those of the observed LOD minus gravity-derived LOD changes and axial atmosphere angular momentum wind term but only if wind effects between 10 and 0.3 hPa levels are not included in the latter. The two OAM series have similar seasonal amplitudes and phases, but they differ substantially when examined in different latitudinal bands. The annual amplitudes of two OAM-induced LOD changes are comparable with that of the nonatmosphere LOD changes, but the semiannual amplitudes are far smaller. Our study indicates that the oceanic effects on LOD should continue to be examined. In addition, the effects of zonal winds (10-0.3 hPa) on LOD are more significant than that of atmospheric surface pressures at semiannual frequency, while they play almost the same roles for annual term.

2381. Monthly spherical harmonic gravity field solutions determined from GRACE inter-satellite range-rate data alone

Authors: S. B. Luthcke (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); D. D. Rowlands (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); F. G. Lemoine (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); S. M. Klosko (Science Division, SGT Inc. Greenbelt Maryland USA); D. Chinn (Science Division, SGT Inc. Greenbelt Maryland USA); J. J. McCarthy (Science Division, SGT Inc. Greenbelt Maryland USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005gl024846

URL: <https://doi.org/10.1029/2005gl024846>

Publication Date: 2006-01-25

Volume/Issue: Volume 33, Issue 2

References: 13

Cited by: 93 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: New monthly estimates of the Earth's gravity field determined solely from GRACE inter-satellite range-rate measurements using an improved method of accelerometer calibration and the use of a baseline state parameterization are presented in this paper. Our methodology exploits the inherent power of the inter-satellite range-rate data at the expense of the GPS data, which are used solely for establishing an accurate orbital reference and for calibrating accelerometers. Resulting gravity solutions show significantly less error than previously published GRACE solutions, especially for spherical harmonic terms of degree 2 and terms of order 15,16.

2382. Modelling pre-historic transoceanic crossings into the Americas

Authors: Álvaro Montenegro; Renéé Hetherington; Michael Eby; Andrew J. Weaver

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2005.11.008

URL: <https://doi.org/10.1016/j.quascirev.2005.11.008>

Publication Date: 2006-01-19

Volume/Issue: Volume 25, Issue 11-12

References: N/A

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2383. Space-borne gravimetry: determination of the time variable gravity field

Authors: P.N.A.M. Visser; E.J.O. Schrama

Journal: International Association of Geodesy Symposia

Publisher: Springer-Verlag

DOI: 10.1007/3-540-26932-0_2

URL: https://doi.org/10.1007/3-540-26932-0_2

Publication Date: 2005-12-23

Volume/Issue: N/A

References: 15

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: Atmosphere, Ocean models, Low-low satellite-to-satellite tracking, Tides, Temporal gravity (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2384. Study of the seasonal gravity signal in superconducting gravimeter data

Authors: Jean-Paul Boy; Jacques Hinderer

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2005.08.035

URL: <https://doi.org/10.1016/j.jog.2005.08.035>

Publication Date: 2005-12-16

Volume/Issue: Volume 41, Issue 1-3

References: 22

Cited by: 89 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2385. Variability of the ocean-induced magnetic field predicted at sea surface and at satellite altitudes

Authors: Roman E. Glazman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Yury N. Golubev (Raytheon ITSS)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jc002926

URL: <https://doi.org/10.1029/2005jc002926>

Publication Date: 2005-12-12

Volume/Issue: Volume 110, Issue C12

References: 22

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Spatial and temporal variability of the magnetic field component induced by ocean circulation is investigated on the basis of a standard thin-shell approximation of electro- and magneto-static equations. Well-known difficulties of numerical solution of the governing equations are resolved by reducing the problem to an equation for the electric field potential, Φ , as opposed to a more conventional approach focused on the vertical jump, ψ , of the magnetic field potential across a combined ocean/marine-sediment-layer spherical shell. The present formulation permits using more realistic input data on ocean currents and ultimately yields much greater (by at least an order of magnitude) values of the magnetic field at sea surface than predicted in earlier studies. Such large values are comparable to, and in some cases exceed, magnetic field variations caused by lithospheric and ionospheric sources on monthly to interannual timescales. At the 400-km altitude (of CHAMP satellite), the field attains 6 nT. The model predictions show favorable comparisons with some in situ measurements as well as with Challenging Minisatellite Payload (CHAMP) satellite magnetometer data.

2386. Non-tidal oceanic contribution to polar wobble estimated from two oceanic assimilation data sets

Authors: Min Zhong; Haoming Yan; Xiaoping Wu; Jianbin Duan; Yaozhong Zhu

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2005.10.011

URL: <https://doi.org/10.1016/j.jog.2005.10.011>

Publication Date: 2005-12-03

Volume/Issue: Volume 41, Issue 1-3

References: 20

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2387. The Total Meridional Heat Flux and Its Oceanic and Atmospheric Partition

Authors: Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli3539.1

URL: <https://doi.org/10.1175/jcli3539.1>

Publication Date: 2005-11-28

Volume/Issue: Volume 18, Issue 21

References: 23

Cited by: 115 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Atmospheric meridional heat transport is inferred as a residual from the Earth Radiation Budget Experiment (ERBE) data and in situ oceanic estimates. Reversing the conventional approach of computing the ocean as an atmospheric model residual is done to permit calculation of a preliminary uncertainty estimate for the atmospheric flux. The structure of the ERBE errors is itself an important uncertainty. Total energy transport is almost indistinguishable from a hemispherically antisymmetric analytic function, despite the great asymmetry of the oceanic heat fluxes. ERBE data appear sufficiently noisy so that a considerable range of atmospheric transports remains possible: the maximum atmospheric value lies between 3 and 5 PW in the Northern Hemisphere, at one standard deviation, although the values are sensitive to the noise assumptions made here. The Northern Hemisphere ocean and atmosphere carry comparable poleward heat fluxes to about 28 degreesN where the oceanic flux drops rapidly, but does not actually vanish until the oceanic surface area goes to zero. Within the estimated error bars, there is a remarkable antisymmetry about the equator of the combined ocean and atmospheric transports, despite the marked oceanic transport asymmetry.

2388. Atmospheric and oceanic contributions to Chandler wobble excitation determined by wavelet filtering

Authors: F. Seitz (Deutsches Geodätisches Forschungsinstitut Munich Germany); M. Schmidt (Deutsches Geodätisches Forschungsinstitut Munich Germany)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005jb003826

URL: <https://doi.org/10.1029/2005jb003826>

Publication Date: 2005-11-17

Volume/Issue: Volume 110, Issue B11

References: 37

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Fluctuations of Earth rotation are associated with the redistribution and motion of mass elements in the Earth system. On seasonal to interannual timescales, the largest effects are due to mass redistributions within atmosphere and oceans. In order to study the Earth's reaction on geophysical excitations, the dynamic Earth system model DyMEG has been developed. It is based on the balance of angular momentum in the Earth system. The model is forced by consistent atmospheric and oceanic angular momenta from reanalysis data and a global ocean circulation model. As rotational deformations of the Earth are regarded, forced variations of Earth rotation due to atmospheric and oceanic excitations influence the free polar motion (Chandler wobble) of DyMEG. The comparison of the numerical model results with geodetic observations reveals a good agreement in both the annual and the Chandler frequency band. In order to support the presumption that the excitation energy of atmospheric and oceanic angular momenta within a spectral band around the Chandler frequency is sufficient to reproduce the observed Chandler wobble, the excitation series are band-pass filtered. For consistent filtering and signal analysis, wavelet techniques based on the Morlet function are applied. When DyMEG is forced with the filtered excitations, the resulting polar motion resembles the actually observed Chandler oscillation, which is determined from geodetic observations applying the same filtering method. Between 1980 and 2002 the correlation coefficient between the unconstrained model result and the observed Chandler wobble amounts to 0.99 and the root-mean-square difference is 16 milliarc seconds.

2389. Water Mass Transformation and Subduction in the South Atlantic

Authors: J. Donners (Royal Netherlands Meteorological Institute); S. S. Drijfhout (Royal Netherlands Meteorological Institute); W. Hazeleger (Royal Netherlands Meteorological Institute)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2782.1

URL: <https://doi.org/10.1175/jpo2782.1>

Publication Date: 2005-11-17

Volume/Issue: Volume 35, Issue 10

References: 46

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The transformation of water masses induced by air-sea fluxes in the South Atlantic Ocean is calculated with a global ocean model, Ocean Circulation and Climate Advanced Modeling (OCCAM), and has been compared with several observational datasets. Air-sea interaction supplies buoyancy to the ocean at almost all density levels. The uncertainty of the estimates of water mass transformations is at least 10 Sv ($\text{Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$), largely caused by the uncertainties in heat fluxes. Further analysis of the buoyancy budget of the mixed layer in the OCCAM model shows that diffusion extracts buoyancy from the water column at all densities. In agreement with observations, water mass formation of surface water by air-sea interaction is completely balanced by consumption from diffusion. There is a large interocean exchange with the Indian and Pacific Oceans. Intermediate water is imported from the Pacific, and light surface water is imported from the Indian Ocean. South Atlantic Central Water and denser water masses are exported to the Indian Ocean. The air-sea formation rate is only a qualitative estimate of the sum of subduction and interocean exchange. Subduction generates teleconnections between the South Atlantic and remote areas where these water masses reemerge in the mixed layer. Therefore, the subduction is analyzed with a Lagrangian trajectory analysis. Surface water obducts in the South Atlantic, while all other water masses experience net subduction. The subducted Antarctic Intermediate Water and Subantarctic Mode Water reemerge mainly in the Antarctic Circumpolar Current farther downstream. Lighter waters reemerge in the eastern tropical Atlantic. As a result, the extratropical South Atlantic has a strong link with the tropical Atlantic basin and only a weak direct link with the extratropical North Atlantic. The impact of the South Atlantic on the upper branch of the thermohaline circulation is indirect: water is significantly transformed by air-sea fluxes and mixing in the South Atlantic, but most of it reemerges and subducts again farther downstream.

2390. Estimating Eddy Stresses by Fitting Dynamics to Observations Using a Residual-Mean Ocean Circulation Model and Its Adjoint

Authors: David Ferreira (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology); Patrick Heimbach (Massachusetts Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2785.1

URL: <https://doi.org/10.1175/jpo2785.1>

Publication Date: 2005-11-17

Volume/Issue: Volume 35, Issue 10

References: 48

Cited by: 147 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A global ocean circulation model is formulated in terms of the "residual mean" and used to study eddy-mean flow interaction. Adjoint techniques are used to compute the three-dimensional eddy stress field that minimizes the departure of the coarse-resolution model from climatological observations of temperature. The resulting 3D maps of eddy stress and residual-mean circulation yield a wealth of information about the role of eddies in large-scale ocean circulation. In eddy-rich regions such as the Southern Ocean, the Kuroshio, and the Gulf Stream, eddy stresses have an amplitude comparable to the wind stress, of order 0.2 N m^{-2} , and carry momentum from the surface down to the bottom, where they are balanced by mountain form drag. From the optimized eddy stress, 3D maps of horizontal eddy diffusivity κ are inferred. The diffusivities have a well-defined large-scale structure whose prominent features are 1) large values of κ (up to $4000 \text{ m}^2 \text{ s}^{-1}$) in the western boundary currents and on the equatorial flank of the Antarctic Circumpolar Current and 2) a surface intensification of κ , suggestive of a dependence on the stratification N^2 . It is shown that implementation of an eddy parameterization scheme in which the eddy diffusivity has an N^2 dependence significantly improves the climatology of the ocean model state relative to that obtained using a spatially uniform diffusivity.

2391. The gravity field and GGOS

Authors: R. Forsberg; M.G. Sideris; C.K. Shum

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2005.06.014

URL: <https://doi.org/10.1016/j.jog.2005.06.014>

Publication Date: 2005-10-28

Volume/Issue: Volume 40, Issue 4-5

References: 7

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2392. Constraining hydrological and cryospheric mass flux in southeastern Alaska using space-based gravity measurements

Authors: M. E. Tamisiea (Harvard-Smithsonian Center for Astrophysics Cambridge Massachusetts USA); E. W. Leuliette (University of Colorado Boulder); J. L. Davis (Harvard-Smithsonian Center for Astrophysics Cambridge Massachusetts USA); J. X. Mitrovica (University of Toronto)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005gl023961

URL: <https://doi.org/10.1029/2005gl023961>

Publication Date: 2005-10-20

Volume/Issue: Volume 32, Issue 20

References: 16

Cited by: 86 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Watersheds draining into the Gulf of Alaska (GoA) experience large seasonal and inter-annual variations of water in the form of rain, snow, and ice, but accurate constraints on these variations have been difficult to obtain. Over larger geographic regions, water variations can be inferred directly from the Gravity Recovery and Climate Experiment (GRACE) data. However, because GoA variations occur over such a small region, the inferred average value of water flux increases as the applied smoothing of the GRACE data decreases. We use this observed scaling together with scaling results obtained from forward models to infer a seasonal amplitude of 115 ± 20 km³ of water and an average contribution to sea level rise over the two years of data of 0.31 ± 0.09 mm/yr. These results suggest that accelerated melting that began in the late 1990s, as inferred from altimetry, continues unabated.

2393. Greenland mass balance from GRACE

Authors: Isabella Velicogna (University of Colorado Boulder); John Wahr (University of Colorado Boulder)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005gl023955

URL: <https://doi.org/10.1029/2005gl023955>

Publication Date: 2005-09-29

Volume/Issue: Volume 32, Issue 18

References: 23

Cited by: 109 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We use 22 monthly GRACE (Gravity Recovery and Climate Experiment) gravity fields to estimate the linear trend in Greenland ice mass during 2002-2004. We recover a decrease in total ice mass of 82 ± 28 km³ of ice per year, consistent with estimates from other techniques. Our uncertainty estimate is dominated by the effects of GRACE measurement errors and errors in our post glacial rebound (PG) correction. The main advantages of GRACE are that it is sensitive to the entire ice sheet, and that it provides mass estimates with only minimal use of supporting physical assumptions or ancillary data.

2394. Surface mass redistribution inversion from global GPS deformation and Gravity Recovery and Climate Experiment (GRACE) gravity data

Authors: J. Kusche (Delft University of Technology); E. J. O. Schrama (Delft University of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jb003556

URL: <https://doi.org/10.1029/2004jb003556>

Publication Date: 2005-09-19

Volume/Issue: Volume 110, Issue B9

References: 22

Cited by: 129 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Monitoring hydrological redistributions through their integrated gravitational effect is the primary aim of the Gravity Recovery and Climate Experiment (GRACE) mission. Time-variable gravity data from GRACE can be uniquely inverted to hydrology, since mass transfers located at or near the Earth's surface are much larger on shorter timescales than those taking place within the deeper Earth and because one can remove the contribution of atmospheric masses from air pressure data. Yet it has been proposed that at larger scales this may be achieved independently by measuring and inverting the elastic loading associated with redistributing masses, e.g., with the global network of the International GPS Service (IGS). This is particularly interesting as long as GRACE monthly gravity solutions do not (yet) match the targeted baseline accuracies at the lower spherical harmonic degrees. In this contribution (1) we describe and investigate an inversion technique which can deal jointly with GPS data and monthly GRACE solutions. (2) Previous studies with GPS data have used least squares estimators and impose solution constraints through low-degree spherical harmonic series truncation. Here we introduce a physically motivated regularization method that guarantees a stable inversion up to higher degrees, while seeking to avoid spatial aliasing. (3) We apply this technique to GPS data provided by the IGS service covering recent years. We can show that after removing the contribution ascribed to atmospheric pressure loading, estimated annual variations of continental-scale mass redistribution exhibit pattern similar to those observed with GRACE and predicted by a global hydrology model, although systematic differences appear to be present. (4) We compute what the relative contribution of GRACE and GPS would be in a joint inversion: Using current error estimates, GPS could contribute with up to 60% to degree 2 till 4 spherical harmonic coefficients and up to 30% for higher-degree coefficients.

2395. Anomalies of Meridional Overturning: Mechanisms in the North Atlantic

Authors: Armin Köhl (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2767.1

URL: <https://doi.org/10.1175/jpo2767.1>

Publication Date: 2005-09-16

Volume/Issue: Volume 35, Issue 8

References: 50

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Optimal observations are used to investigate the overturning streamfunction in the North Atlantic at 30 degreesN and 900-m depth. Those observations are designed to impact the meridional overturning circulation (MOC) in numerical models maximally when assimilated and therefore establish the most efficient observation network for studying changes in the MOC. They are also ideally suited for studying the related physical mechanisms in a general circulation model. Optimal observations are evaluated here in the framework of a global 1 degrees model over a 10-yr period. Hydrographic observations useful to monitor the MOC are primarily located along the western boundary north of 30 degreesN and along the eastern boundary south of 30 degreesN. Additional locations are in the Labrador, Irminger, and Iberian Seas. On time scales of less than a year, variations in MOC are mainly wind driven and are made up through changes in Ekman transport and coastal up- and downwelling. Only a small fraction is buoyancy driven and constitutes a slow response, acting on time scales of a few years, to primarily wintertime anomalies in the Labrador and Irminger Seas. Those anomalies are communicated southward along the west coast by internal Kelvin waves at the depth level of Labrador Sea Water. They primarily set the conditions at the northern edge of the MOC anomaly. The southern edge is mainly altered through Rossby waves of the advective type, which originate from temperature and salinity anomalies in the Canary Basin. Those anomalies are amplified on their way westward in the baroclinic unstable region of the subtropical gyre. The exact meridional location of the maximum MOC response is therefore set by the ratio of the strength of these two signals.

2396. Non-isotropic filtering of GRACE temporal gravity for geophysical signal enhancement

Authors: Shin-Chan Han; C. K. Shum; Christopher Jekeli; Chung-Yen Kuo; Clark Wilson; Ki-Weon Seo

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2005.02756.x

URL: <https://doi.org/10.1111/j.1365-246x.2005.02756.x>

Publication Date: 2005-09-02

Volume/Issue: Volume 163, Issue 1

References: 12

Cited by: 119 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2397. Atmospheric and oceanic excitation of decadal-scale Earth orientation variations

Authors: Richard S. Gross (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jb003565

URL: <https://doi.org/10.1029/2004jb003565>

Publication Date: 2005-09-02

Volume/Issue: Volume 110, Issue B9

References: 57

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The contribution of atmospheric wind and surface pressure and oceanic current and bottom pressure variations during 1949-2002 to exciting changes in the Earth's orientation on decadal timescales is investigated using an atmospheric angular momentum series computed from the National Centers for Environmental Prediction/National Center for Atmospheric Research (NCEP/NCAR) reanalysis project and an oceanic angular momentum series computed from a near-global ocean model that was forced by surface fluxes from the NCEP/NCAR reanalysis project. Not surprisingly, since decadal-scale variations in the length of day are caused mainly by interactions between the mantle and core, the effect of the atmosphere and oceans is found to be only about 14% of that observed. More surprisingly, it is found that the effect of atmospheric and oceanic processes on decadal-scale changes in polar motion is also only about 20% (xcomponent) and 38% (ycomponent) of that observed. Therefore redistribution of mass within the atmosphere and oceans does not appear to be the main cause of the Markowitz wobble. It is also found that on timescales between 10 days and 4 years the atmospheric and oceanic angular momentum series used here have very little skill in explaining Earth orientation variations before the mid to late 1970s. This is attributed to errors in both the Earth orientation observations prior to 1976 when measurements from the accurate space-geodetic techniques became available and to errors in the modeled atmospheric fields prior to 1979 when the satellite era of global weather observing systems began.

2398. Seasonal variation of ocean bottom pressure derived from Gravity Recovery and Climate Experiment (GRACE): Local validation and global patterns

Authors: Torsten Kanzow (GEOMAR Helmholtz Centre for Ocean Research); Frank Flechtner (Department 1 (Geodesy and Remote Sensing) GeoForschungsZentrum Potsdam (GFZ) Potsdam Germany); Alan Chave (Woods Hole Oceanographic Institution); Roland Schmidt (Department 1 (Geodesy and Remote Sensing) GeoForschungsZentrum Potsdam (GFZ) Potsdam Germany); Peter Schwintzer (Department 1 (Geodesy and Remote Sensing) GeoForschungsZentrum Potsdam (GFZ) Potsdam Germany); Uwe Send (GEOMAR Helmholtz Centre for Ocean Research)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jc002772

URL: <https://doi.org/10.1029/2004jc002772>

Publication Date: 2005-09-01

Volume/Issue: Volume 110, Issue C9

References: 32

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Gravity Recovery and Climate Experiment (GRACE) processing centers at the GeoForschungsZentrum Potsdam (GFZ) and the University of Texas Center for Space Research (UTCSR) provide time series of monthly gravity field solutions covering the period since mission launch in March 2002. Although the achieved accuracy still remains an order of magnitude below the mission's baseline goal, these time series have successfully been used to study terrestrial phenomena such as water storage variations. Over the oceans, the monthly gravity field solutions can be converted into estimates of the fluctuating ocean bottom pressure (OBP), which is the sum of atmospheric and oceanic mass variations. The GRACE products may be validated against in situ OBP observations which are available from a ground truth site in the tropical northwest Atlantic Ocean. Large differences are observed between the in situ and GRACE-derived OBP which are investigated by comparing the tidal and nontidal ocean models used at GFZ and UTCSR for dealiasing short-term (<2 months) mass variations from satellite measurements. Results show that the barotropic nontidal and tide models need improvement at periods shorter than 1 day and longer than 2 weeks. On a global scale the monthly OBP fields from GRACE generally overestimate the variability compared to ocean general circulation models, especially in tropical regions. This may be attributed to continuing deficiencies in GRACE data processing. Nevertheless, there is some initial evidence that GRACE possesses the potential to observe large-scale averages of bottom pressure fluctuations.

2399. Spatial sensitivity of the Gravity Recovery and Climate Experiment (GRACE) time-variable gravity observations

Authors: J. L. Chen (University of Texas at Austin); C. R. Wilson (University of Texas at Austin); J. S. Famiglietti (University of California Irvine); M. Rodell (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jb003536

URL: <https://doi.org/10.1029/2004jb003536>

Publication Date: 2005-08-15

Volume/Issue: Volume 110, Issue B8

References: 16

Cited by: 76 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We analyze the spatial sensitivities of terrestrial water storage and geoid height changes determined from the time-variable gravity observed by the Gravity Recovery and Climate Experiment (GRACE) twin satellite mission. On the basis of 15 GRACE monthly gravity solutions, covering the period April 2002 to December 2003, we examine the effects of spatial smoothing at radii varying from 400 to 2000 km and conclude that a 800 km Gaussian smoothing radius is effective for GRACE-derived terrestrial water storage and produces the minimum RMS residuals over the land of the differences between GRACE results and estimated water storage change from a global land data assimilation system. For GRACE estimated geoid height change, the effective smoothing radius can go down to 600 km. When the annual (e.g., the sine and cosine) components are the primary concern, the effective spatial resolution can reach 600 and 400 km for GRACE estimated terrestrial water storage or geoid height change, respectively.

2400. Global mass balance and the length-of-day variation

Authors: Jianli Chen (University of Texas at Austin)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jb003474

URL: <https://doi.org/10.1029/2004jb003474>

Publication Date: 2005-08-05

Volume/Issue: Volume 110, Issue B8

References: 33

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Variations of atmospheric angular momentum are dominated driving forces behind the length-of-day change, while mass redistribution in the oceans and hydrosphere is expected to cause the length-of-day to change as well. We estimate atmospheric, oceanic, and hydrological contributions to the length-of-day variations during 1993-2004 using the National Centers for Environmental Prediction reanalysis atmospheric model, the Estimating the Circulation and Climate of the Ocean consortium's data assimilating ocean general circulation model, and the U.S. Climate Prediction Center's land data assimilation system model. A coherent mass balance among the atmosphere, ocean, and continental water is implemented. At seasonal timescales the mass balance among the atmosphere, land, and oceans is very important and nearly cancels all the contributions to the length-of-day variation predicted by the oceanic and hydrological models, although at intraseasonal timescales, good correlations exist between observed length-of-day variations unaccounted for by the atmosphere and contributions from the ocean and continental water. This study indicates that in a fully mass balanced Earth system the combined seasonal oceanic and hydrological contributions to the length-of-day variation are too small to explain the residual length-of-day variations unaccounted for by the atmosphere. The discrepancies between observed length-of-day variations and atmospheric contributions appear more likely caused by the errors of the atmospheric models, in particular of the wind fields.

2401. Bubble finite elements for the primitive equations of the ocean

Authors: F Guillén-González; D Rodríguez-Gómez

Journal: Numerische Mathematik

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00211-005-0626-1

URL: <https://doi.org/10.1007/s00211-005-0626-1>

Publication Date: 2005-08-01

Volume/Issue: Volume 101, Issue 4

References: 38

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2402. Estimation of boundary values in a North Atlantic circulation model using an adjoint method

Authors: Nadia Ayoub

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2005.06.003

URL: <https://doi.org/10.1016/j.ocemod.2005.06.003>

Publication Date: 2005-07-28

Volume/Issue: Volume 12, Issue 3-4

References: 38

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2403. Oceanic excitations on polar motion: a cross comparison among models

Authors: Y. H. Zhou; J. L. Chen; X. H. Liao; C. R. Wilson

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2005.02694.x

URL: <https://doi.org/10.1111/j.1365-246x.2005.02694.x>

Publication Date: 2005-07-20

Volume/Issue: Volume 162, Issue 2

References: 35

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: Excitation, Polar motion, Ocean general circulation model (OGCM), Oceanic angular momentum (OAM)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2404. Adjusting Internal Model Errors through Ocean State Estimation

Authors: Detlef Stammer (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2733.1

URL: <https://doi.org/10.1175/jpo2733.1>

Publication Date: 2005-07-12

Volume/Issue: Volume 35, Issue 6

References: 19

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Oceanic state estimation is a powerful tool to estimate internal model parameters simultaneously with the model's initial conditions and surface forcing field that jointly would bring a model into consistency with time-varying large-scale ocean observations. Here an attempt to estimate geographically varying fields of horizontal and vertical viscosity and diffusivity within a 9-yr-long estimation procedure is presented. The estimated coefficients are highly efficient in preserving watermass characteristics and frontal structures by reducing the model temperature and salinity drift, especially around the Southern Ocean. The estimated mean circulation results in stronger transports of western boundary currents and of the Antarctic Circumpolar Current. Moreover, an increase of about 10% in the strength of the meridional overturning circulation and in the poleward heat transport can be found. Estimated changes in the horizontal mixing coefficients seem to agree with the notion that diapycnal mixing is superficially high with Laplacian mixing formulations, especially close to frontal structures in the ocean. In comparison with adjustments in tracer diffusivities (vertically and horizontally), adjustments of viscosity coefficients are fairly minor outside lateral boundary regions, suggesting that state estimation attempts might be most successful in providing enhanced insight into tracer mixing.

2405. Intermediate-Depth Circulation of the Indian and South Pacific Oceans Measured by Autonomous Floats

Authors: Russ E. Davis (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/jpo2702.1

URL: <https://doi.org/10.1175/jpo2702.1>

Publication Date: 2005-06-20

Volume/Issue: Volume 35, Issue 5

References: 31

Cited by: 117 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: As part of the World Ocean Circulation Experiment, 306 autonomous floats were deployed in the tropical and South Pacific Ocean and 228 were deployed in the Indian Ocean to observe the basinwide circulation near 900-m depth. Mean velocities, seasonal variability, and lateral eddy diffusivity from the resultant 2583 float-years of data are presented. Area averages, local function fits, and a novel application of objective mapping are used to estimate the mean circulation. Patterns of mean circulation resemble those at the surface in both basins. Well-developed subtropical gyres, twice as strong in the Indian Ocean as in the Pacific, feed western boundary currents. Tropical gyres are separated by eastward flow along the equator in both hemispheres of both basins, although the Indian subcontinent splits the north Indian tropical gyre. The Antarctic Circumpolar Current (ACC) and west wind drifts are prominent in both basins, generally tending slightly southward but deviating to the north behind the Del Cano, Kerguelen, and Campbell Plateaus and, of course, South America. Remarkably, the eastern boundaries of the southern subtropical gyres in all three basins apparently occur in the ocean interior, away from land. The Indian Ocean's subtropical gyre, and perhaps part of the South Atlantic's, reaches east to a retroflexion just upstream of the Campbell Plateau south of New Zealand. Seasonal variability at 900 m is focused around the equator with weaker variability found near certain bathymetric features. There is a remarkable agreement between the observed seasonable variability and that predicted by the Jet Propulsion Laboratory (JPL)-Estimating the Circulation and Climate of the Ocean (ECCO) data-assimilating numerical model. Aside from seasonal effects, eddy variability is greatest along the equator, in tropical and subtropical western basins, and along the ACC. Integrals of velocity across regional passages (Tasman Sea, Mozambique Channel) provide useful reference for hydrographic analyses of transport. Across whole ocean basins, however, the uncertainty associated with the appropriate continuity relation for horizontal flow (e.g., geostrophy vs nondivergence) is comparable to the mean flow.

2406. Using Green's Functions to Calibrate an Ocean General Circulation Model

Authors: Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/mwr2912.1

URL: <https://doi.org/10.1175/mwr2912.1>

Publication Date: 2005-06-06

Volume/Issue: Volume 133, Issue 5

References: 33

Cited by: 174 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Green's functions provide a simple yet effective method to test and to calibrate general circulation model (GCM) parameterizations, to study and to quantify model and data errors, to correct model biases and trends, and to blend estimates from different solutions and data products. The method is applied to an ocean GCM, resulting in substantial improvements of the solution relative to observations when compared to prior estimates: overall model bias and drift are reduced and there is a 10%-30% increase in explained variance. Within the context of this optimization, the following new estimates for commonly used ocean GCM parameters are obtained. Background vertical diffusivity is $(15.1 \pm 0.1) \times 10^{-6} \text{ m}^2 \text{ s}^{-2}$. Background vertical viscosity is $(18 \pm 3) \times 10^{-6} \text{ m}^2 \text{ s}^{-2}$. The critical bulk Richardson number, which sets boundary layer depth, is $Ric = 0.354 \pm 0.004$. The threshold gradient Richardson number for shear instability vertical mixing is $Ri_0 = 0.699 \pm 0.008$. The estimated isopycnal diffusivity coefficient ranges from 550 to 1350 $\text{m}^2 \text{ s}^{-2}$, with the largest values occurring at depth in regions of increased mesoscale eddy activity. Surprisingly, the estimated isopycnal diffusivity exhibits a 5%-35% decrease near the surface. Improved estimates of initial and boundary conditions are also obtained. The above estimates are the backbone of a quasi-operational, global-ocean circulation analysis system.

2407. Time variations of land water storage from an inversion of 2 years of GRACE geoids

Authors: G RAMILLIEN; F FRAPPART; A CAZENAVE; A GUNTNER

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2005.04.005

URL: <https://doi.org/10.1016/j.epsl.2005.04.005>

Publication Date: 2005-06-02

Volume/Issue: Volume 235, Issue 1-2

References: N/A

Cited by: 168 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2408. Effect of gravitational consistency and mass conservation on seasonal surface mass loading models

Authors: P. J. Clarke (University of Newcastle upon Tyne); D. A. Lavallée (University of Newcastle upon Tyne); G. Blewitt (University of Nevada); T. M. van Dam (European Center for Geodynamics and Seismology); J. M. Wahr (University of Colorado Boulder)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2005gl022441

URL: <https://doi.org/10.1029/2005gl022441>

Publication Date: 2005-04-18

Volume/Issue: Volume 32, Issue 8

References: 17

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Increasingly, models of surface mass loads are used either to correct geodetic time coordinates by removing seasonal and other "noise", or for comparison with other geodetic parameters. However, models of surface loading obtained by simply combining the mass redistribution due to individual phenomena will not in general be self-consistent, in that (i) the implied global water budget will not be mass-conserving, and (ii) the modelled sea level will not be an equipotential surface of Earth's total gravity field. We force closure of the global water budget by allowing the "passive" ocean to change in mass. This medium-term passive ocean response will not be a uniform change in non-steric ocean surface height, but must necessarily be spatially variable to keep the "passive" ocean surface on an equipotential. Using existing load models, we demonstrate the effects of our consistency theory. Geocenter motion is amplified significantly, by up to 43%.

2409. Multi-reference evaluation of uncertainty in earth orientation parameter measurements

Authors: T. M. Chin; R. S. Gross; J. O. Dickey

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-005-0439-0

URL: <https://doi.org/10.1007/s00190-005-0439-0>

Publication Date: 2005-04-16

Volume/Issue: Volume 79, Issue 1-3

References: N/A

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2410. Interannual variability of low-degree gravitational change, 1980?2002

Authors: J.L. Chen; C.R. Wilson; B.D. Tapley

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-004-0417-y

URL: <https://doi.org/10.1007/s00190-004-0417-y>

Publication Date: 2005-03-21

Volume/Issue: Volume 78, Issue 9

References: N/A

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2411. Acoustic remote sensing of large-scale temperature variability in the North Pacific Ocean

Authors: B.M. Howe; B.D. Cornuelle; B.D. Dushaw; M.A. Dzieciuch; D. Menemenlis; J.A. Mercer; W.H. Munk; R.C. Spindel; D. Stammer; P.F. Worcester; M.R. Zarnetske

Journal: Oceans '04 MTS/IEEE Techno-Ocean '04 (IEEE Cat. No.04CH37600)

Publisher: IEEE

DOI: 10.1109/oceans.2004.1406343

URL: <https://doi.org/10.1109/oceans.2004.1406343>

Publication Date: 2005-03-21

Volume/Issue: Volume 3

References: 6

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Argo

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2412. Mode coherence at megameter ranges in the North Pacific Ocean

Authors: Kathleen E. Wage (George Mason University); Matthew A. Dzieciuch (Scripps Institution of Oceanography); Peter F. Worcester (Scripps Institution of Oceanography); Bruce M. Howe (University of Washington); James A. Mercer (University of Washington)

Journal: The Journal of the Acoustical Society of America

Publisher: Acoustical Society of America (ASA)

DOI: 10.1121/1.1854551

URL: <https://doi.org/10.1121/1.1854551>

Publication Date: 2005-03-12

Volume/Issue: Volume 117, Issue 3

References: 44

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This article analyzes the coherence of low-mode signals at ranges of 3515 and 5171 km using data from the Acoustic Thermometry of Ocean Climate (ATOC) and Alternate Source Test (AST) experiments. Vertical line arrays at Hawaii and Kiritimati received M-sequences transmitted from two sources: the 75-Hz bottom-mounted ATOC source on Pioneer Seamount and the near-axial dual-frequency (28/84 Hz) AST source deployed nearby. This study demonstrates that the characteristics of the mode signals at 5171-km range are quite similar to those at 3515-km range. At 75 Hz the mode time spreads are on the order of 1.5 s, implying a coherence bandwidth of 0.67 Hz. The time spread of the 28-Hz signals is somewhat lower, but these signals show significantly less frequency-selective fading than the 75-Hz signals, suggesting that at the lower frequency the multipaths are temporally resolvable. Coherence times for mode 1 at 75 Hz are on the order of 8 min for the 3515-km range and 6 min for 5171-km range. At 28 Hz mode 1 is much more stable, with a magnitude-squared coherence of greater than 0.6 for the 20-min transmission period.

2413. Hydrological excitations of polar motion, 1993-2002

Authors: J. L. Chen; C. R. Wilson

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2005.02522.x

URL: <https://doi.org/10.1111/j.1365-246x.2005.02522.x>

Publication Date: 2005-03-11

Volume/Issue: Volume 160, Issue 3

References: 24

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2414. Simulated estimation of hydrological loads from GRACE

Authors: K.-W. Seo; C. R. Wilson

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-004-0410-5

URL: <https://doi.org/10.1007/s00190-004-0410-5>

Publication Date: 2005-03-05

Volume/Issue: Volume 78, Issue 7-8

References: N/A

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2415. Short term mass variability in Greenland, from GRACE

Authors: Isabella Velicogna (University of Colorado Boulder); John Wahr (University of Colorado Boulder); Edward Hanna (University of Sheffield); Philippe Huybrechts (University of Brussels, Alfred Wegener Institute)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl021948

URL: <https://doi.org/10.1029/2004gl021948>

Publication Date: 2005-02-28

Volume/Issue: Volume 32, Issue 5

References: 13

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: Groenlandia|Proxy (statistics)|Glacier mass balance|Greenland ice sheet

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We use twenty-two monthly GRACE (Gravity Recovery and Climate Experiment) gravity fields to recover nonsecular mass change in Greenland. The results show large seasonal variability. We compare with modeled precipitation, evaporation, and runoff derived from ERA40 (the 40-year ECMWF Re-Analysis of the global atmosphere). The model's seasonal amplitude is controlled by runoff and agrees reasonably well with GRACE. Both GRACE and the model show an April/May maximum. But the GRACE results show a delayed minimum relative to the model. This difference is probably associated with omissions in the runoff model, ice discharge, subglacial hydrology, mass loss by blowing-snow, and hydrology in ice-free regions. The discrepancy is smaller, but still significant, for south Greenland alone. When we include a proxy for ice discharge the agreement is improved.

2416. ITG-CHAMP01: a CHAMP gravity field model from short kinematic arcs over a one-year observation period

Authors: T. Mayer-Gürr; K.H. Ilk; A. Eicker; M. Feuchtinger

Journal: Journal of Geodesy

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00190-004-0413-2

URL: <https://doi.org/10.1007/s00190-004-0413-2>

Publication Date: 2005-02-22

Volume/Issue: Volume 78, Issue 7-8

References: N/A

Cited by: 57 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2417. Geomagnetic jerks and a high-resolution length-of-day profile for core studies

Authors: R. Holme; O. de Viron

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2004.02510.x

URL: <https://doi.org/10.1111/j.1365-246x.2004.02510.x>

Publication Date: 2005-02-17

Volume/Issue: Volume 160, Issue 2

References: 33

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: Day length|Momentum (technical analysis)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2418. Velocity Statistics Inferred from the TOPEX/Poseidon-Jason-1 Tandem Mission Data

Authors: D. STAMMER; J. THEISS

Journal: Marine Geodesy

Publisher: Informa UK Limited

DOI: 10.1080/01490410490902052

URL: <https://doi.org/10.1080/01490410490902052>

Publication Date: 2005-02-09

Volume/Issue: Volume 27, Issue 3-4

References: 23

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2419. Meridional component of oceanic Rossby wave propagation

Authors: Roman E. Glazman; Peter B. Weichman

Journal: Dynamics of Atmospheres and Oceans

Publisher: Elsevier BV

DOI: 10.1016/j.dynatmoce.2004.11.002

URL: <https://doi.org/10.1016/j.dynatmoce.2004.11.002>

Publication Date: 2005-02-08

Volume/Issue: Volume 38, Issue 3-4

References: 31

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2420. Improved near-term Earth rotation predictions using atmospheric angular momentum analysis and forecasts

Authors: Thomas J. Johnson; Brian J. Luzum; Jim R. Ray

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2004.10.004

URL: <https://doi.org/10.1016/j.jog.2004.10.004>

Publication Date: 2005-01-20

Volume/Issue: Volume 39, Issue 3

References: 17

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2421. An efficient exact adjoint of the parallel MIT General Circulation Model, generated via automatic differentiation

Authors: Patrick Heimbach; Chris Hill; Ralf Giering

Journal: Future Generation Computer Systems

Publisher: Elsevier BV

DOI: 10.1016/j.future.2004.11.010

URL: <https://doi.org/10.1016/j.future.2004.11.010>

Publication Date: 2004-12-29

Volume/Issue: Volume 21, Issue 8

References: 31

Cited by: 96 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2422. Generating efficient derivative code with TAF

Authors: R. Giering; T. Kaminski; T. Slawig

Journal: Future Generation Computer Systems

Publisher: Elsevier BV

DOI: 10.1016/j.future.2004.11.003

URL: <https://doi.org/10.1016/j.future.2004.11.003>

Publication Date: 2004-12-24

Volume/Issue: Volume 21, Issue 8

References: 58

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2423. Atmosphere-Ocean Modeling Exploiting Fluid Isomorphisms

Authors: John Marshall; Alistair Adcroft; Jean-Michel Campin; Chris Hill; Andy White

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/mwr2835.1

URL: <https://doi.org/10.1175/mwr2835.1>

Publication Date: 2004-12-17

Volume/Issue: Volume 132, Issue 12

References: 18

Cited by: 53 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2424. Observing and assessing nontidal ocean loading using ocean, continuous GPS and gravity data in the Adriatic area

Authors: S. Zerbini (Dipartimento di Fisica University of Bologna Bologna Italy); F. Matonti (Dipartimento di Fisica University of Bologna Bologna Italy); F. Raicich (Sezione di Oceanografia Chimica e Fisica Istituto di Scienze Marine, Consiglio Nazionale delle Ricerche Trieste Italy); B. Richter (Bundesamt fuer Kartographie und Geodaesie Frankfurt am Main Germany); T. van Dam (European Center for Geodynamics and Seismology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl021185

URL: <https://doi.org/10.1029/2004gl021185>

Publication Date: 2004-12-09

Volume/Issue: Volume 31, Issue 23

References: 17

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: 1208 Geodesy and Gravity: Crustal movements - Interplate (8110), 1243 Geodesy and Gravity: Space geodetic surveys, 1223 Geodesy and Gravity: Ocean/Earth/atmosphere interactions (3339), 1829 Hydrology: Groundwater hydrology, 1299 Geodesy and Gravity: General or miscellaneous

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The effect of nontidal ocean loading (NTOL) is observed in the height series of four permanent GPS stations in the northern Adriatic. A validation of the ECCO model is performed by comparing model estimates of sea-level anomalies from tide-gauges with TOPEX/POSEIDON data, and ECCO model estimates of bottom pressure with those derived from temperature and salinity observations. The amplitudes of the ECCO sea-level anomaly are found to be 1.4 times smaller than observations; bottom pressure is 2 times smaller. Using a Green's functions approach to determine elastic deformations, the ECCO ocean bottom pressure is used to estimate surface displacements at the GPS sites. Model results were compared with the height series and with the observed NTOL effect. The height series and the predicted NTOL are highly correlated at all four stations. The analysis performed on superconducting gravimeter data at the Medicina station also shows high correlation.

2425. The interannual variability of the North Atlantic Ocean revealed by combined data from TOPEX/Poseidon and Jason altimetric measurements

Authors: Lee-Lueng Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl021200

URL: <https://doi.org/10.1029/2004gl021200>

Publication Date: 2004-12-07

Volume/Issue: Volume 31, Issue 23

References: 16

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A decade-long record of sea surface height from combined altimeter data taken by the TOPEX/Poseidon and Jason satellites was analyzed for studying the interannual variability of the North Atlantic Ocean. On time scales of 5-6 years, variations of sea surface height have maximum amplitudes in three areas: the subpolar gyre, the Gulf Stream gyre (the Gulf Stream and its recirculation), and the subtropical region. The variation of the subpolar gyre is 180 degrees out of phase with that of the Gulf Stream gyre at its eastern end. The variation of the Gulf Stream gyre at its western end is connected to that of the subtropical region, exhibiting phase propagation from the subtropics all the way to the eastern end of the Gulf Stream gyre. The patterns of phase change suggest possible roles of Rossby waves in the dynamics of the basin-wide variability.

2426. Eastward propagating surface anomalies at ocean gyre boundaries

Authors: Bablu Sinha (University of Southampton); Brenda Topliss (Bedford Institute of Oceanography); James Harle (University of Southampton)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jc002393

URL: <https://doi.org/10.1029/2004jc002393>

Publication Date: 2004-12-03

Volume/Issue: Volume 109, Issue C12

References: 39

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Climate model, Sea surface temperature, Ocean-atmosphere coupling, Air sea interaction, Ocean dynamics

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Inspection of animations and time-distance (Hovmoller) plots of unfiltered data from a state-of-the-art coupled climate model (HadCM3) reveals eastward propagating surface temperature-salinity anomalies at subpolar/subtropical gyre boundaries throughout the world ocean. Maximum anomalies are ~4 degreesC and ~0.3 practical salinity units. Horizontal length scales are ~1000 km. The anomalies travel at a speed consistent with advection by mean near-surface currents. Generation regions are located at western boundaries and possess an order of magnitude higher variability than all other regions. In some cases, surface heat fluxes are shown to cause anomaly generation; in other cases, surface current anomalies are likely to be responsible. Eastward anomaly propagation also occurs in the Estimating the Circulation and Climate of the Ocean model. The coarser resolution Fast Ocean Rapid Troposphere Experiment (FORTE) coupled model shows no propagating anomalies. Sea surface anomalies in the instrumental record are investigated, using ship-based sea surface temperature (SST) and heat fluxes, and satellite SST and altimetry. Unlike the models, Hovmoller plots and animations of the unfiltered observational data showed no evidence of eastward propagation. By applying a decadal-timescale band-pass filter to the SST time series, we identify underlying propagation of basin-scale anomalies in both HadCM3 and observations. Maximum anomalies are ~1 degreesC and length scales are ~2500 km. These anomalies travel at speeds significantly slower than mean near-surface currents. Ten anomalies are identified during the period 1870-2002, mostly in the eastern Atlantic basin (east of 45 degreesW). HadCM3 displayed only one propagating anomaly of this type in a typical 100 years of data. We discuss possible reasons for the range of results in models and observations.

2427. Assessing ENSO Simulations and Predictions Using Adjoint Ocean State Estimation

Authors: Dietmar Dommenget (Scripps Institution of Oceanography / University of California San Diego); Detlef Stammer (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/3211.1

URL: <https://doi.org/10.1175/3211.1>

Publication Date: 2004-11-29

Volume/Issue: Volume 17, Issue 22

References: 21

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Simulations and seasonal forecasts of tropical Pacific SST and subsurface fields that are based on the global Consortium for Estimating the Circulation and Climate of the Ocean (ECCO) ocean-state estimation procedure are investigated. As compared to similar results from a traditional ENSO simulation and forecast procedure, the hindcast of the constrained ocean state is significantly closer to observed surface and subsurface conditions. The skill of the 12-month lead SST forecast in the equatorial Pacific is comparable in both approaches. The optimization appears to have better skill in the SST anomaly correlations, suggesting that the initial ocean conditions and forcing corrections calculated by the ocean-state estimation do have a positive impact on the predictive skill. However, the optimized forecast skill is currently limited by the low quality of the statistical atmosphere. Progress is expected from optimizing a coupled model over a longer time interval with the coupling statistics being part of the control vector.

2428. Low degree gravitational changes from GRACE: Validation and interpretation

Authors: J. L. Chen (Center for Space Research University of Texas Austin Texas USA); C. R. Wilson (Center for Space Research University of Texas Austin Texas USA); B. D. Tapley (Center for Space Research University of Texas Austin Texas USA); J. C. Ries (Center for Space Research University of Texas Austin Texas USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl021670

URL: <https://doi.org/10.1029/2004gl021670>

Publication Date: 2004-11-24

Volume/Issue: Volume 31, Issue 22

References: 13

Cited by: 73 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We examine low degree gravitational variations ΔC_{21} , ΔS_{21} , and ΔC_{20} observed by the Gravity Recovery and Climate Experiment (GRACE) satellites during the first 2 years of this gravity mission. The GRACE observations are compared with independent estimates from accurately measured Earth rotational changes and predictions from atmospheric, oceanic, and hydrological models. The 18 GRACE monthly gravity solutions, covering the period April 2002 to March 2004, show strong seasonal variability in the ΔC_{21} , ΔS_{21} , and ΔC_{20} time series, and generally agree with Earth rotation-derived changes and geophysical model estimates, in particular for ΔS_{21} and ΔC_{20} . The reason for the poorer agreement between the GRACE results and the Earth rotation-derived estimates for ΔC_{21} is unclear. We demonstrate that the omission of the ocean pole tide in the GRACE data processing does have significant effects on the estimated ΔC_{21} and ΔS_{21} .

2429. Isolation of atmospheric effects on rapid polar motion through Wiener filtering

Authors: Susan M. di Leonardo; S. R. Dickman

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2004.02445.x

URL: <https://doi.org/10.1111/j.1365-246x.2004.02445.x>

Publication Date: 2004-11-19

Volume/Issue: Volume 159, Issue 3

References: 32

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2430. The 1997-1999 abrupt change of the upper ocean temperature in the north central Pacific

Authors: Seung-Bum Kim (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl021142

URL: <https://doi.org/10.1029/2004gl021142>

Publication Date: 2004-11-18

Volume/Issue: Volume 31, Issue 22

References: 27

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The abrupt warming of the north central Pacific Ocean from 1997 to 1999 is studied using an ocean data assimilation product. During this period, the average mixed-layer temperature in the region of 170-210 degreesE, 25-40 degreesN rises by 1.8 K. The major contributors to the warming are surface heat flux (1.3 K), geostrophic advection (0.7 K), and entrainment (0.7 K). For the geostrophic advection, the contributions by the zonal, meridional, and vertical components are 0.4, -0.1 and 0.3 K, respectively. Mixing and meridional Ekman advection have cooling effect. The significance of the geostrophic advection indicates the importance of ocean dynamics in controlling the abrupt warming tendency during the 1997-99 period and the inadequacy of a slab mixed-layer model in simulating such warming tendency.

2431. Shallow overturning circulation of the Western Indian Ocean

Authors: Friedrich A. Schott (GEOMAR Helmholtz Centre for Ocean Research)

Journal: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences

Publisher: The Royal Society

DOI: 10.1098/rsta.2004.1483

URL: <https://doi.org/10.1098/rsta.2004.1483>

Publication Date: 2004-11-03

Volume/Issue: Volume 363, Issue 1826

References: 17

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Indian Ocean differs from the other two oceans in not possessing an eastern equatorial upwelling regime. Instead, the upwelling occurs dominantly in the northwestern Arabian Sea and, to a lesser degree, around the Indian subcontinent. Subduction, on the other hand, occurs dominantly in the Southern Hemisphere. The result is a shallow Cross-Equatorial Cell connecting both regimes. The northward flow at thermocline levels occurs as part of the Somali Current and the southward upper-layer return flow is carried by the Ekman transports that are directed southward in both hemispheres. The main forcing is by the Southwest Monsoon that overwhelms the effects of the Northeast Monsoon and is the cause for the annual mean Northern Hemisphere upwelling and southward Ekman transports. In the Southern Hemisphere, the annual mean upwelling at the northern rim of the Southeast Trades causes a zonally extended open-ocean upwelling regime that is apparent in isopycnal doming in the 3-12° S band; it drives a shallow Subtropical Cell.

2432. Long-term crustal deformation monitored by gravity and space techniques at Medicina, Italy and Wettzell, Germany

Authors: B. Richter; S. Zerbini; F. Matonti; D. Simon

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2004.07.013

URL: <https://doi.org/10.1016/j.jog.2004.07.013>

Publication Date: 2004-11-03

Volume/Issue: Volume 38, Issue 3-5

References: 12

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2433. Seasonal and long-term dynamics of the upper ocean carbon cycle at Station ALOHA near Hawaii

Authors: Charles D. Keeling (Scripps Institution of Oceanography / University of California San Diego); Holger Brix (University of California Los Angeles); Nicolas Gruber (University of California Los Angeles)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gb002227

URL: <https://doi.org/10.1029/2004gb002227>

Publication Date: 2004-10-14

Volume/Issue: Volume 18, Issue 4

References: 94

Cited by: 156 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Long-term trends and average seasonal variability in the upper ocean carbon cycle are investigated at Station ALOHA, the site of the U.S. JGOFS Hawaii Ocean Time series program (HOT), on the basis of a 14-year time series (1988-2002) of dissolved inorganic carbon (DIC), alkalinity, and $^{13}\text{C}/^{12}\text{C}$ ratio of DIC data. Salinity-normalized DIC (sDIC) and computed oceanic pCO_2 show distinct upward trends of $1.2 \pm 0.1 \text{ } \mu\text{mol kg}^{-1} \text{ yr}^{-1}$ and $2.5 \pm 0.1 \text{ } \mu\text{atm yr}^{-1}$, respectively, while the $^{13}\text{C}/^{12}\text{C}$ isotopic ratio of DIC (expressed as $\delta^{13}\text{C}_{\text{DIC}}$) decreases at a mean rate of $-0.027 \pm 0.001\text{‰ yr}^{-1}$. More than half of the rates of change in sDIC and oceanic pCO_2 , and most of the change in $^{13}\text{C}/^{12}\text{C}$, are attributed to the uptake of isotopically light anthropogenic CO_2 from the atmosphere. The residual trends appear to be caused mainly by a regional change in the net freshwater budget, perhaps associated with a regime change of the North Pacific climate system near 1997. Computed oceanic pCO_2 is below atmospheric pCO_2 for nearly the entire year, leading to an annual mean surface ocean pCO_2 undersaturation of about 18 μatm , and to an annual uptake of CO_2 from the atmosphere, which we compute to be $1.0 \pm 0.1 \text{ mol m}^{-2} \text{ yr}^{-1}$. We estimate that about 30% of this flux relates to the uptake of anthropogenic CO_2 , and the remainder to biologically mediated export of organic carbon. Using a modified version of the diagnostic model of Gruber et al. [1998], constrained by $\delta^{13}\text{C}_{\text{DIC}}$, we infer net community production of organic carbon (NCP) to be the dominant process generating the observed seasonal variability in sDIC. The annual integral of NCP, $2.3 \pm 0.8 \text{ mol m}^{-2} \text{ yr}^{-1}$, is comparable to previous estimates of biological production in the subtropical North Pacific. Annually integrated fluxes of air-sea gas exchange and NCP at Station ALOHA are each about two thirds of those computed for the upper ocean near Bermuda using similar methods of estimation [Gruber et al., 1998, 2002]. However, the seasonal amplitudes of sDIC and $\delta^{13}\text{C}_{\text{DIC}}$ near Hawaii are only half as large as near Bermuda, because air-sea gas exchange and NCP tend to oppose each other near Hawaii, but reinforce each other near Bermuda.

2434. Tropical Cyclone Eline and Its Unusual Penetration and Impacts over the Southern African Mainland

Authors: C. J. C. Reason; A. Keibel

Journal: Weather and Forecasting

Publisher: American Meteorological Society

DOI: 10.1175/1520-0434(2004)019<0789:tceaiu>2.0.co;2

URL: [https://doi.org/10.1175/1520-0434\(2004\)019<0789:tceaiu>2.0.co;2](https://doi.org/10.1175/1520-0434(2004)019<0789:tceaiu>2.0.co;2)

Publication Date: 2004-10-13

Volume/Issue: Volume 19, Issue 5

References: 15

Cited by: 128 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2435. An Earth gravity field model complete to degree and order 150 from GRACE: EIGEN-GRACE02S

Authors: Christoph Reigber; Roland Schmidt; Frank Flechtner; Rolf König; Ulrich Meyer; Karl-Hans Neumayer; Peter Schwintzer; Sheng Yuan Zhu

Journal: Journal of Geodynamics

Publisher: Elsevier BV

DOI: 10.1016/j.jog.2004.07.001

URL: <https://doi.org/10.1016/j.jog.2004.07.001>

Publication Date: 2004-10-12

Volume/Issue: Volume 39, Issue 1

References: 21

Cited by: 256 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung

License: N/A

Open Access: N/A

Abstract: N/A

2436. Variations in the Earth's oblateness during the past 28 years

Authors: Minkang Cheng (University of Texas at Austin); Byron D. Tapley (University of Texas at Austin)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004jb003028

URL: <https://doi.org/10.1029/2004jb003028>

Publication Date: 2004-09-15

Volume/Issue: Volume 109, Issue B9

References: 29

Cited by: 370 publications

Language: en

Subjects: N/A

Keywords: Secular Variation|Variation (astronomy)|Annual cycle|Satellite laser ranging

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Analysis of satellite laser ranging (SLR) data indicates that the Earth's dynamic oblateness (J_2) has undergone significant variations during the past 28 years. The dominant signatures in the observed variations in J_2 are (1) a secular decrease with a rate of approximately $-2.75 \times 10^{-11} \text{ yr}^{-1}$, (2) seasonal annual variations with a mean amplitude of 2.9×10^{-10} , (3) significant interannual variations with timescales of 4-6 years, and (4) a variation with period of ~ 21 years and an amplitude of $\sim 1.4 \times 10^{-10}$ with minimum in December 1988. Two large interannual variations are related to the strong El Niño-Southern Oscillation events during the periods of 1986-1991 and 1996-2002, and it appears that another interannual cycle may have started in late 2002. The superposition of the decadal variation on the interannual signal makes the J_2 fluctuation appear to be anomalously large during the 1996-2002 period. Contemporary models of the mass redistributions in the atmosphere, ocean, and surface water can explain a major part of the 4- to 6-year fluctuations. However, the cause of the decadal variation remains unknown.

2437. Three-dimensional modelling of ocean electrodynamics using gauged potentials

Authors: Robert H. Tyler; Frédéric Vivier; Shenghui Li

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2004.02318.x

URL: <https://doi.org/10.1111/j.1365-246x.2004.02318.x>

Publication Date: 2004-08-20

Volume/Issue: Volume 158, Issue 3

References: 26

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2438. On the cause of eastern equatorial Pacific Ocean T-S variations associated with El Niño

Authors: Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Benny Cheng (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl020188

URL: <https://doi.org/10.1029/2004gl020188>

Publication Date: 2004-08-13

Volume/Issue: Volume 31, Issue 15

References: 17

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The nature of observed variations in temperature-salinity (T-S) relationship between El Niño and non-El Niño years in the pycnocline of the eastern equatorial Pacific Ocean (NINO3 region, 5 degreesS-5 degreesN, 150 degreesW-90 degreesW) is investigated using an ocean general circulation model. The origin of the subject water mass is identified using the adjoint of a simulated passive tracer. The higher salinity during El Niño is attributed to larger convergence of saltier water from the Southern Hemisphere and smaller convergence of fresher water from the Northern Hemisphere.

2439. Temperature Advection: Internal versus External Processes

Authors: Tong Lee; Ichiro Fukumori; Benyang Tang

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/1520-0485(2004)034<1936:taivep>2.0.co;2

URL: [https://doi.org/10.1175/1520-0485\(2004\)034<1936:taivep>2.0.co;2](https://doi.org/10.1175/1520-0485(2004)034<1936:taivep>2.0.co;2)

Publication Date: 2004-08-06

Volume/Issue: Volume 34, Issue 8

References: 12

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2440. Time-variable gravity from GRACE: First results

Authors: John Wahr (University of Colorado Boulder); Sean Swenson (University of Colorado Boulder); Victor Zlotnicki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Isabella Velicogna (University of Colorado Boulder)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl019779

URL: <https://doi.org/10.1029/2004gl019779>

Publication Date: 2004-06-08

Volume/Issue: Volume 31, Issue 11

References: 10

Cited by: 644 publications

Language: en

Subjects: N/A

Keywords: Water storage|Smoothing

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Eleven monthly GRACE gravity field solutions are now available for analyses. We show those fields can be used to recover monthly changes in water storage, both on land and in the ocean, to accuracies of 1.5 cm of water thickness when smoothed over 1000 km. The amplitude of the annually varying signal can be determined to 1.0 cm. Results are 30% better for a 1500 km smoothing radius, and 40% worse for a 750 km radius. We estimate the annually varying component of water storage for three large drainage basins (the Mississippi, the Amazon, and a region draining into the Bay of Bengal), to accuracies of 1.0-1.5 cm.

2441. Ocean dynamics associated with rapid J2 fluctuations: Importance of circumpolar modes and identification of a coherent Arctic mode

Authors: C. W. Hughes (Proudman Oceanographic Laboratory Liverpool UK); V. N. Stepanov (Proudman Oceanographic Laboratory Liverpool UK)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003jc002176

URL: <https://doi.org/10.1029/2003jc002176>

Publication Date: 2004-06-01

Volume/Issue: Volume 109, Issue C6

References: 25

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Fluctuations of the Earth's dynamic oblateness J2 are a measure of global redistributions of mass between equatorial and polar regions, sensitive to the component which is symmetric about the equator. Cox and Chao recently published a time series of monthly values of J2 based on satellite orbit determination, which showed rapid fluctuations from month to month, as well as a large change in J2 in 1997/1998. We here investigate the ocean dynamics associated with these rapid fluctuations using a barotropic ocean model forced by atmospheric wind stress and pressure. We find that the model has significant skill in reproducing the observations, though not the change in 1997/1998. Much of the model variability is related to coherent circumpolar fluctuations in Arctic and Antarctic bottom pressure which, despite their apparent symmetry, are forced by quite different mechanisms. The Antarctic mode is forced by circumpolar winds, whereas the Arctic mode is forced by northward winds over the entrances to the polar sea. Comparison with in situ observations shows that the model has skill in reproducing the polar modes. Arctic tide gauge records confirm the presence of a coherent mode at short periods, and at long periods over at least the Eurasian Arctic shelf. If barotropic, and representative of the whole Arctic, this would be large enough to make a major contribution to measured J2 fluctuations, but does not account for the 1997/1998 event.

2442. Mechanisms of air-sea CO₂ flux variability in the equatorial Pacific and the North Atlantic

Authors: Galen A. McKinley (Massachusetts Institute of Technology); Michael J. Follows (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003gb002179

URL: <https://doi.org/10.1029/2003gb002179>

Publication Date: 2004-05-27

Volume/Issue: Volume 18, Issue 2

References: 59

Cited by: 135 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A global ocean general circulation model is used to estimate the magnitude of interannual variability in air-sea fluxes of CO₂ and O₂ from 1980-1998 and to examine the controlling mechanisms. The global variability in the air-sea flux of carbon ($\pm 0.5 \times 10^{15}$ grams Carbon yr⁻¹ (PgC yr⁻¹)) is forced by changes of $\Delta p\text{CO}_2$ and wind speeds related to the El Niño/Southern Oscillation (ENSO) cycle in the equatorial Pacific. In contrast the air-sea O₂ flux is controlled by two regions: the equatorial Pacific and North Atlantic. The model captures much of the interannual variability of the CO₂ flux observed at Bermuda, with some correlation with the North Atlantic Oscillation (NAO) index. However, basin-scale air-sea CO₂ flux anomalies are not correlated with the NAO due to a rapid neutralization of entrained DIC anomalies by biological uptake and export production in the subpolar gyre. CO₂ flux variability estimates from our ocean model and the mean atmospheric inversion results of Bousquet et al. [2000] are in broad agreement in the equatorial Pacific, but not in the North Atlantic. This model suggests that the projection of air-sea flux anomalies onto the large-scale, mean air-sea flux pattern in atmospheric inversions may lead to an overestimate of the flux variability in the extra-tropics where the patterns of variability do not correspond to those of the mean flux.

2443. Simulations of magnetic fields generated by the Antarctic Circumpolar Current at satellite altitude: Can geomagnetic measurements be used to monitor the flow?

Authors: Frédéric Vivier (Laboratoire d'Océanographie Dynamique et de Climatologie, CNRS UMR 7617 Université Pierre et Marie Curie Paris France); Ernst Maier-Reimer (Max Planck Institute for Meteorology); Robert H. Tyler (University of Washington)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl019804

URL: <https://doi.org/10.1029/2004gl019804>

Publication Date: 2004-05-26

Volume/Issue: Volume 31, Issue 10

References: 22

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: With a volume transport of $\sim 134 \times 10^6 \text{ m}^3/\text{s}$ at the Drake Passage, the Antarctic Circumpolar Current (ACC) is the strongest ocean current. In the interest of estimating the secondary magnetic fields generated by the magnetohydrodynamic interaction of this flow with Earth's main field, we compare numerical results for the magnetic fields obtained using flow from three different ocean general circulation models. These simulations all expect detectable ocean signals in the magnetic records at ground and satellite altitude (400 km). The variability of this contribution is highly correlated with the ACC transport, a very important variable for climate studies. Observed magnetic fields could then be used, in principle, to derive an index of variability of the ACC. However given its small amplitude compared with other magnetic contributions, extracting the ocean's signal from observations remains a challenge at this time.

2444. Estimating air-sea fluxes of heat, freshwater, and momentum through global ocean data assimilation

Authors: D. Stammer (Scripps Institution of Oceanography); K. Ueyoshi (Scripps Institution of Oceanography); A. Köhl (Scripps Institution of Oceanography); W. G. Large (National Center for Atmospheric Research); S. A. Josey (James Rennell Division Southampton Oceanography Centre Southampton UK); C. Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003jc002082

URL: <https://doi.org/10.1029/2003jc002082>

Publication Date: 2004-05-20

Volume/Issue: Volume 109, Issue C5

References: 50

Cited by: 146 publications

Language: en

Subjects: N/A

Keywords: Data assimilation, Air-sea fluxes, Ocean synthesis

Funding: N/A

License: N/A

Open Access: N/A

Abstract: ECCO state estimation results from 10 years during the World Ocean Circulation Experiment are used to assess the quality of surface flux adjustments made to the initial NCEP re-analysis-1 products. During the state estimation procedure, surface fluxes are adjusted together with initial temperature and salinity conditions so that the model simulation becomes consistent with ocean observations. Independent estimates of the adjustments from bulk formula and regional field observations are also employed to evaluate the results. Buoyancy flux adjustments are found to be within the crude prior error bars on these fields. Outside the boundary current regions, they are consistent with known large-scale deficiencies in the NCEP products. Wind stress adjustments are also everywhere within the prior error bars, but exhibit regional small-scale features that reflect ocean model failures to resolve intense boundary currents. On large scales, the inferred adjustments to NCEP wind stress fields are consistent with inferences made from satellite wind stress measurements. Further improvements in the surface flux estimates obtained through state estimation procedures are anticipated as the estimation procedure becomes more complete by including the use of improved prior error covariance information, and as the ocean model becomes more skillful, for example, in simulating boundary currents by increasing its resolution.

2445. Acoustic thermometry time series in the North Pacific

Authors: B.M. Howe; B.D. Dushaw; J.A. Mercer; P.F. Worcester; J.A. Colosi; B.C. Cornuelle; M.A. Dzieciuch; R.C. Spindel

Journal: 2003 International Conference Physics and Control. Proceedings (Cat. No.03EX708)

Publisher: IEEE

DOI: 10.1109/ssc.2003.1224123

URL: <https://doi.org/10.1109/ssc.2003.1224123>

Publication Date: 2004-04-23

Volume/Issue: N/A

References: 18

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Argo|Float (project management)|Mode (computer interface)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2446. Circulation and Deep-Water Export at the Western Exit of the Subpolar North Atlantic

Authors: Friedrich A. Schott; Rainer Zantopp; Lothar Stramma; Marcus Dengler; Jürgen Fischer; Mathieu Wibaux

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/1520-0485(2004)034<0817:cadeat>2.0.co;2

URL: [https://doi.org/10.1175/1520-0485\(2004\)034<0817:cadeat>2.0.co;2](https://doi.org/10.1175/1520-0485(2004)034<0817:cadeat>2.0.co;2)

Publication Date: 2004-04-21

Volume/Issue: Volume 34, Issue 4

References: 27

Cited by: 120 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2447. Consistent atmospheric and oceanic excitation of the Earth's free polar motion

Authors: Florian Seitz; Jochen Stuck; Maik Thomas

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1111/j.1365-246x.2004.02208.x

URL: <https://doi.org/10.1111/j.1365-246x.2004.02208.x>

Publication Date: 2004-04-02

Volume/Issue: Volume 157, Issue 1

References: 45

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2448. Degree-2 harmonics of the Earth's mass load estimated from GPS and Earth rotation data

Authors: Richard S. Gross (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Geoffrey Blewitt (School of Civil Engineering and Geosciences University of Newcastle Newcastle-upon-Tyne UK, University of Nevada); Peter J. Clarke (School of Civil Engineering and Geosciences University of Newcastle Newcastle-upon-Tyne UK); David Lavallée (University of Nevada)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2004gl019589

URL: <https://doi.org/10.1029/2004gl019589>

Publication Date: 2004-03-31

Volume/Issue: Volume 31, Issue 7

References: 16

Cited by: 31 publications

Language: en

Subjects: N/A

Keywords: Mass distribution

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A fluid, mobile atmosphere and oceans surrounds the solid Earth and upon its land surface lays a continually changing distribution of ice, snow, and ground water. The changing distribution of mass associated with the motion of these surficial fluids changes the Earth's rotation by changing its inertia tensor and changes the Earth's shape by changing the load on the solid Earth. It has recently been demonstrated that large-scale changes of the Earth's shape, and hence of the mass load causing the Earth's shape to change, can be measured using the global network of GPS receivers. Here, the degree-2 mass load coefficients determined from GPS data are compared with those obtained from Earth orientation observations from which the effects of tides, winds, and currents have been removed. Good agreement is found between these two estimates of the degree-2 mass load, particularly at seasonal frequencies.

2449. Vorticity Balance in Coarse-Resolution Global Ocean Simulations

Authors: Youyu Lu (Scripps Institution of Oceanography / University of California San Diego); Detlef Stammer (Scripps Institution of Oceanography / University of California San Diego)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2504.1

URL: <https://doi.org/10.1175/2504.1>

Publication Date: 2004-03-17

Volume/Issue: Volume 34, Issue 3

References: 29

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The vorticity budget of the vertically integrated circulation from two global ocean simulations is analyzed using a horizontal spacing of 2 degrees x 2 degrees in longitude/latitude. The two simulations differ in their initial hydrographic conditions and surface wind and buoyancy forcing. The constrained simulation obtains optimal initial condition and surface forcing through assimilating observational data using the model's adjoint, whereas the unconstrained simulation uses Levitus climatological conditions for initialization and is driven by NCEP-NCAR reanalysis forcing, plus restoring to the monthly surface temperature and salinity climatological conditions. The goal is to examine the dynamics that sets the time-mean circulation and to understand the differences between the constrained and unconstrained simulations. It is found that, similar to eddy-permitting simulations, the bottom pressure torque (BPT) in coarse-resolution models plays an important role in the western boundary currents and in the Southern Ocean, and largely balances the difference between wind stress curl and βV for the depth-integrated flow. BPT is a controlling factor of the interior abyssal flow. The geostrophic vorticity relation holds in the interior basins in intermediate and deep layers and breaks down in the upper ocean toward the surface. In the upper layer of the interior basins, the model simulations show statistically significant deviation from the Sverdrup balance. In the subtropical gyre regions, the deviation from Sverdrup balance is confined to zonal bands that are balanced by the curls of lateral friction and nonlinear advection. The differences between the constrained and unconstrained simulations are significant in vorticity terms. The adjustment to Levitus hydrographic climatological data as the model's initial condition causes the most significant changes in BPT, which is the main reason for changes in abyssal flow. The analysis also points to needs for further improvement of models and controlling the influence of data errors in ocean state estimation.

2450. Optimal Observations for Variational Data Assimilation

Authors: Armin Köhl (Scripps Institution of Oceanography); Detlef Stammer (Scripps Institution of Oceanography)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2513.1

URL: <https://doi.org/10.1175/2513.1>

Publication Date: 2004-03-17

Volume/Issue: Volume 34, Issue 3

References: 39

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An important part of ocean state estimation is the design of an observing system that allows for the efficient study of climate related questions in the ocean. A solution to the design problem is presented here in terms of optimal observations that emerge as singular vectors of the modified data resolution matrix. The actual computation is feasible only for scalar quantities and in the limit of large observational errors. Identical twin experiments performed in the framework of a 1 degrees North Atlantic primitive equation model demonstrate that such optimal observations, when applied to determining the heat transport across the Greenland-Scotland ridge, perform significantly better than traditional section data. On seasonal to interannual time scales, optimal observations are located primarily along the continental shelf and information about heat transport, wind stress, and stratification is being communicated through boundary waves and advective processes. On time scales of about 1 month, sea surface height observations appear to be more efficient in reconstructing the cross-ridge heat transport than hydrographic observations. Optimal observations also provide a tool for understanding changes of ocean state associated with anomalies of integral quantities such as meridional heat transport.

2451. The Origin, Pathway, and Destination of Niño-3 Water Estimated by a Simulated Passive Tracer and Its Adjoint

Authors: Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Benny Cheng (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/2515.1

URL: <https://doi.org/10.1175/2515.1>

Publication Date: 2004-03-17

Volume/Issue: Volume 34, Issue 3

References: 47

Cited by: 111 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The nature of subtropical-tropical water mass exchange in the Pacific Ocean is investigated, focusing on the origin, pathway, and destination of water occupying the surface layer of the eastern equatorial Pacific Ocean (Niño-3 region; 5 degreesS-5 degreesN, 150 degrees-90 degreesW). Simulated passive tracers and their adjoint are employed to explicitly follow the circulation of specific water masses accounting for advective and diffusive effects and their time variabilities. The evolution of the forward passive tracer and adjoint passive tracer can be identified as describing where the tracer-tagged water mass goes and from where it comes, respectively. Over 10 years on average, water mass of the Niño-3 region can be traced back to eastern subtropical thermocline waters of the Northern (27%) and Southern Hemispheres (39%). The Niño-3 water subsequently returns to these subtropical latitudes in the upper ocean. In contrast to the hypothesized "subtropical cell," however, this circulation is an open circuit with water returning to the western regions of the two hemispheres (subtropical gyres) and to the Indian Ocean, instead of returning to its origins. The representative transit time scale from the subtropics to the Tropics is 10-15 yr. Temporal variability causes the tropical circulation inferred from a time-mean state to differ significantly from the average circulation. In particular, stirring due to nonseasonal, intra-annual variability significantly enhances the transport magnitude of the so-called interior pathways relative to that of the circuitous low-latitude western boundary pathways. Such short-circuit in the subtropical-tropical exchange may help better to explain tracer distributions, such as the observed midbasin tritium maximum in the equatorial Pacific Ocean. Significant differences in circulation pathways are also identified that are associated with El Niño and La Niña events. The strength of the subtropical-tropical water mass exchange is estimated to have weakened during the 1990s.

2452. Eastern equatorial Pacific Ocean T-S variations with El Niño

Authors: Ou Wang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Tong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Gregory C. Johnson (NOAA/Pacific Marine Environmental Laboratory Seattle Washington USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003gl019087

URL: <https://doi.org/10.1029/2003gl019087>

Publication Date: 2004-02-25

Volume/Issue: Volume 31, Issue 4

References: 16

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Temperature-Salinity (T-S) relationship variability in the pycnocline of the eastern equatorial Pacific Ocean (NINO3 region, 5 degreesS-5 degreesN, 150 degreesW-90 degreesW) over the last two decades is investigated using observational data and model simulation. A numerical model simulation using the MITgcm (Massachusetts Institute of Technology General Circulation Model) suggests that, during El Niño years, the water in the eastern equatorial Pacific Ocean becomes saltier (by 0.1 to 0.2) and warmer (by 0.5 to 1 degreesC) on density surfaces within the pycnocline. This simulation is consistent with Conductivity-Temperature-Depth (CTD) data collected mostly during Tropical Atmosphere Ocean (TAO) mooring maintenance cruises.

2453. Oceanic effects on polar motion determined from an ocean model and satellite altimetry: 1993-2001

Authors: J.-L. Chen (University of Texas at Austin); C. R. Wilson (University of Texas at Austin); X.-G. Hu (Chinese Academy of Sciences); Y.-H. Zhou (Chinese Academy of Sciences); B. D. Tapley (University of Texas at Austin)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003jb002664

URL: <https://doi.org/10.1029/2003jb002664>

Publication Date: 2004-02-25

Volume/Issue: Volume 109, Issue B2

References: 30

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Mass redistribution and motion in the oceans are major driving forces of geodetic variations, including polar motion, length of day, geocenter, and gravity field changes. We examine oceanic contribution to polar motion using estimates from a data-assimilating ocean general circulation model and satellite radar altimeter observations. The data include model estimates of variations in oceanic mass (OBP) and meridional and zonal velocities. Sea level anomalies from TOPEX/Poseidon (T/P) satellite altimeter measurements and steric sea surface height changes deduced from the model are also used to estimate OBP effects. Estimated oceanic contributions from both the model and T/P show considerably better agreement with polar motion observations compared with results from previous studies. The improvement is particularly significant at intraseasonal timescales. Both OBP and ocean current variations provide important contributions to polar motion. At intraseasonal timescales the oceans appear to be a dominant contributor to residual polar motion not accounted for by the atmosphere. The oceans also play an important role in seasonal excitation. Combined OBP and ocean current contributions explain much of the residual semiannual variability.

2454. Measuring gravity field variability, the geoid, ocean bottom pressure fluctuations, and their dynamical implications

Authors: Francis Condi (Massachusetts Institute of Technology); Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2002jc001727

URL: <https://doi.org/10.1029/2002jc001727>

Publication Date: 2004-02-13

Volume/Issue: Volume 109, Issue C2

References: 54

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: To obtain insight into bottom pressure variability and requirements for assimilating gravity data into an ocean general circulation model, we use recent results from a constrained global model. Bottom pressure variability, as contrasted to surface elevation variability, is examined at the annual, semiannual, Chandler-wobble, and 4-month periods. Surface elevation and bottom pressure variability occurs on all spatial and temporal scales resolvable by our analysis, and patterns can be quite different. A barotropic index, developed to quantify variability character, suggests the importance of steric effects at longer periods. Variability decomposition into standing, eastward, and westward traveling waves indicates a strong zonal component for both surface elevation and bottom pressure, with bottom pressure showing a greater overall tendency toward eastward propagation, and surface elevation showing a greater overall tendency toward westward propagation. Bottom pressure variability signals are estimated to be recoverable from space to spherical harmonic degree 27 for the annual cycle and 14 for the semiannual and Chandler-wobble cycles. Mass exchange, arising from model approximations, is detectable to about degree 7. Polar motion excitation by ocean seafloor loading is found to be quantitatively important. Spurious oscillations encountered in using spherical harmonics to represent oceanic fields are suppressed using Lanczos factors.

2455. On the mid-depth circulation in the Labrador and Irminger Seas

Authors: R. H. Käse; A. Biastoch; D. B. Stammer

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2001gl013192

URL: <https://doi.org/10.1029/2001gl013192>

Publication Date: 2004-02-05

Volume/Issue: Volume 28, Issue 18

References: 16

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A numerical circulation model with 1/6 degrees resolution and an accurate topography formulation explains details of the observed circulation in the Irminger and Labrador Seas that were recently revealed by Lavender et al. [2000]. We show that the recirculation pattern is established through a locally wind induced flow controlled by the bottom topography and enhanced through remote baroclinic forcing by the dense plume of Denmark Strait overflow water. The basic circulation is a robust feature in a hierarchy of model setups. It exists in the purely barotropic case driven by steady winds and is even maintained when realistic daily forcing is added. The narrow recirculation zone is manifested by a sea level depression spanning from the Denmark Strait across the Irminger into the Labrador Sea.

2456. Improving ocean angular momentum estimates using a model constrained by data

Authors: Rui M. Ponte; Detlef Stammer; Carl Wunsch

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2000gl011671

URL: <https://doi.org/10.1029/2000gl011671>

Publication Date: 2004-02-05

Volume/Issue: Volume 28, Issue 9

References: 9

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ocean angular momentum (OAM) calculations using forward model runs without any data constraints have recently revealed the effects of OAM variability on the Earth's rotation. Here we use an ocean model and its adjoint to estimate OAM values by constraining the model to available oceanic data. The optimization procedure yields substantial changes in OAM, related to adjustments in both motion and mass fields, as well as in the wind stress torques acting on the ocean. Constrained and unconstrained OAM values are discussed in the context of closing the planet's angular momentum budget. The estimation procedure yields noticeable improvements in the agreement with the observed Earth rotation parameters, particularly at the seasonal timescale. The comparison with Earth rotation measurements provides an independent consistency check on the estimated ocean state and underlines the importance of ocean state estimation for quantitative studies of the variable large-scale oceanic mass and circulation fields, including studies of OAM.

2457. The ACPI Project, Element 1: Initializing a Coupled Climate Model from Observed Conditions

Authors: David W. Pierce; Tim P. Barnett; Robin Tokmakian; Albert Semtner; Mathew Maltrud; JoAnne Lysne; Anthony Craig

Journal: Climatic Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1023/b:clim.0000013676.42672.23

URL: <https://doi.org/10.1023/b:clim.0000013676.42672.23>

Publication Date: 2004-01-30

Volume/Issue: Volume 62, Issue 1-3

References: N/A

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2458. U.S. Climate Sensitivity Simulated with the NCEP Regional Spectral Model

Authors: Jongil Han; John O. Roads

Journal: Climatic Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1023/b:clim.0000013675.66917.15

URL: <https://doi.org/10.1023/b:clim.0000013675.66917.15>

Publication Date: 2004-01-30

Volume/Issue: Volume 62, Issue 1-3

References: N/A

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2459. How Sensitive Are Coarse General Circulation Models to Fundamental Approximations in the Equations of Motion?

Authors: Martin Losch; Alistair Adcroft; Jean-Michel Campin

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/1520-0485(2004)034<0306:hsacgc>2.0.co;2

URL: [https://doi.org/10.1175/1520-0485\(2004\)034<0306:hsacgc>2.0.co;2](https://doi.org/10.1175/1520-0485(2004)034<0306:hsacgc>2.0.co;2)

Publication Date: 2004-01-27

Volume/Issue: Volume 34, Issue 1

References: 23

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2460. Transient Diapycnal Mixing and the Meridional Overturning Circulation

Authors: William R. Boos; Jeffery R. Scott; Kerry A. Emanuel

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/1520-0485(2004)034<0334:tdmatm>2.0.co;2

URL: [https://doi.org/10.1175/1520-0485\(2004\)034<0334:tdmatm>2.0.co;2](https://doi.org/10.1175/1520-0485(2004)034<0334:tdmatm>2.0.co;2)

Publication Date: 2004-01-27

Volume/Issue: Volume 34, Issue 1

References: 43

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2461. Comparison of the nonlocal transport characteristics of a series of one-dimensional oceanic boundary layer models

Authors: A.James Kettle

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/j.ocemod.2004.01.002

URL: <https://doi.org/10.1016/j.ocemod.2004.01.002>

Publication Date: 2004-01-25

Volume/Issue: Volume 8, Issue 4

References: 67

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2462. Atmospheric and oceanic excitation of length-of-day variations during 1980-2000

Authors: Richard S. Gross (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Pascal Gegout (Institut de Physique du Globe)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003jb002432

URL: <https://doi.org/10.1029/2003jb002432>

Publication Date: 2004-01-14

Volume/Issue: Volume 109, Issue B1

References: 60

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Although nontidal changes in the Earth's length-of-day on timescales of a few days to a few years are primarily caused by changes in the angular momentum of the zonal winds, other processes can be expected to cause the length-of-day to change as well. Here the relative contribution of upper atmospheric winds, surface pressure, oceanic currents, and ocean-bottom pressure to changing the length-of-day during 1980-2000 is evaluated using estimates of atmospheric angular momentum from the National Centers for Environmental Prediction/National Center for Atmospheric Research reanalysis project, estimates of the angular momentum of the zonal winds in the upper atmosphere from the United Kingdom Meteorological Office, and estimates of oceanic angular momentum from the Estimating the Circulation and Climate of the Ocean consortium's simulation of the general circulation of the oceans. On intraseasonal timescales, atmospheric surface pressure, oceanic currents, and ocean-bottom pressure are found to be about equally important in causing the length-of-day to change, while upper atmospheric winds are found to be less important than these mechanisms. On seasonal timescales, the upper atmospheric winds are more important than the sum of currents and bottom pressure in causing the length-of-day to change and, except at the annual frequency, are even more important than surface pressure changes. On interannual timescales, oceanic currents and ocean-bottom pressure are found to be only marginally effective in causing the length-of-day to change.

2463. Westward propagating twin gyres in the equatorial Indian Ocean

Authors: P. Rahul Chand Reddy (Indian Institute of Tropical Meteorology); P. S. Salvekar (Indian Institute of Tropical Meteorology); A. A. Deo (Indian Institute of Tropical Meteorology); D. W. Ganer (Indian Institute of Tropical Meteorology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003gl018615

URL: <https://doi.org/10.1029/2003gl018615>

Publication Date: 2004-01-05

Volume/Issue: Volume 31, Issue 1

References: 19

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A reduced-gravity (1-layer) model forced by daily climatological winds simulates twin, anticyclonic gyres, which propagate westward on either side of the equator. The gyres form at the beginning of both the Southwest Monsoon and the Northeast monsoon in the equatorial eastern Indian Ocean, and subsequently propagate across the basin. Their existence is supported by velocity observations taken during WOCE in 1995 and by TOPEX/Poseidon sea-level observations during 1993. They are also present in the ECCO model/data product. They form at the front of a Rossby-wave packet generated by the reflection of the equatorial jet (EJ) from the eastern boundary of the basin. They are likely either Rossby solitons or result from the nonlinear interaction between the EJ and the Rossby-wave front.

2464. Low degree gravitational changes from earth rotation and geophysical models

Authors: J. L. Chen (Center for Space Research University of Texas Austin Texas USA); C. R. Wilson (Center for Space Research University of Texas Austin Texas USA, Department of Geological Sciences University of Texas Austin Texas USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003gl018688

URL: <https://doi.org/10.1029/2003gl018688>

Publication Date: 2003-12-18

Volume/Issue: Volume 30, Issue 24

References: 15

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Spherical harmonic degree 2 gravitational variations ΔC_{21} , ΔS_{21} , and ΔC_{20} are estimated from accurately measured Earth rotational changes and compared with predictions from atmospheric, oceanic, and hydrological models. Earth rotation-derived changes agree very well with model predictions over a broad frequency band, and particularly well at intraseasonal and seasonal time scales. The agreement is significantly better compared to previous studies, due mainly to improved oceanic and hydrological models. An independent determination of degree 2 changes serves as an important constraint for satellite-based estimates such as those of the Gravity Recovery and Climate Experiment (GRACE) mission.

2465. Interannual-to-Decadal Variations of Tropical-Subtropical Exchange in the Pacific Ocean: Boundary versus Interior Pycnocline Transports

Authors: Tong Lee; Ichiro Fukumori

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/1520-0442(2003)016<4022:ivotei>2.0.co;2

URL: [https://doi.org/10.1175/1520-0442\(2003\)016<4022:ivotei>2.0.co;2](https://doi.org/10.1175/1520-0442(2003)016<4022:ivotei>2.0.co;2)

Publication Date: 2003-12-05

Volume/Issue: Volume 16, Issue 24

References: 39

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2466. Time-variable gravity signal of an anomalous redistribution of water mass in the extratropic Pacific during 1998-2002

Authors: B. F. Chao (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); A. Y. Au (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Raytheon ITSS); J.-P. Boy (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); C. M. Cox (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Raytheon ITSS)

Journal: Geochemistry, Geophysics, Geosystems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003gc000589

URL: <https://doi.org/10.1029/2003gc000589>

Publication Date: 2003-11-13

Volume/Issue: Volume 4, Issue 11

References: 33

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: J_2 anomaly, Time-variable gravity, Water mass distribution, Extratropic pacific

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Cox and Chao [2002] reported the detection of a large anomaly in the time series of Earth's dynamic oblateness J_2 , the lowest-degree gravity spatial harmonic, in the form of a positive jump since 1998 overshadowing the decreasing secular trend in J_2 caused primarily by the postglacial rebound (PGR). Here we report that recent data show that J_2 has been rapidly returning toward "normal" (with PGR considered) since early 2001. In search of the geophysical and climatic causes for this "1998-2002 J_2 anomaly," we report an oceanographic event that took place in the extratropic north and south Pacific basins that was found to match remarkably well with the time evolution of the anomaly. We examine the leading (nonseasonal, extratropic Pacific) Empirical Orthogonal Function/Principal Component modes in the sea-surface height (SSH) data from TOPEX/Poseidon, sea surface temperature (SST) data from the National Center for Environmental Predictions, and output fields of the Estimating the Circulation and the Climate of the Ocean (ECCO) ocean general circulation model (OGCM), including ocean bottom pressure (OBP) and temperature and salinity profiles. The phenomenon appears to be part of the Pacific Decadal Oscillation, and temporal correlations are made. However, quantitatively, the OBP field of the ECCO model predicts a J_2 anomaly that is smaller in magnitude than the observed by a factor of about 3. We discuss various possibilities for reconciling this discrepancy in terms of inadequacies of present OGCMs and considering other geophysical contributions; a complete resolution of the J_2 enigma awaits further studies.

2467. Bottom Topography as a Control Variable in an Ocean Model

Authors: Martin Losch; Carl Wunsch

Journal: Journal of Atmospheric and Oceanic Technology

Publisher: American Meteorological Society

DOI: 10.1175/1520-0426(2003)020<1685:btaacv>2.0.co;2

URL: [https://doi.org/10.1175/1520-0426\(2003\)020<1685:btaacv>2.0.co;2](https://doi.org/10.1175/1520-0426(2003)020<1685:btaacv>2.0.co;2)

Publication Date: 2003-10-29

Volume/Issue: Volume 20, Issue 11

References: 24

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2468. The Sensitivity of the Greenland-Scotland Ridge Overflow to Forcing Changes

Authors: Arne Biastoch; Rolf H. Käse; Detlef B. Stammer

Journal: Journal of Physical Oceanography

Publisher: American Meteorological Society

DOI: 10.1175/1520-0485(2003)033<2307:tsotgr>2.0.co;2

URL: [https://doi.org/10.1175/1520-0485\(2003\)033<2307:tsotgr>2.0.co;2](https://doi.org/10.1175/1520-0485(2003)033<2307:tsotgr>2.0.co;2)

Publication Date: 2003-10-22

Volume/Issue: Volume 33, Issue 11

References: 18

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2469. Large-scale mass redistribution in the oceans, 1993-2001

Authors: J. L. Chen (Center for Space Research University of Texas Austin Texas USA); C. R. Wilson (Center for Space Research University of Texas Austin Texas USA, Department of Geological Sciences University of Texas Austin Texas USA); X. G. Hu (Chinese Academy of Sciences); B. D. Tapley (Center for Space Research University of Texas Austin Texas USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003gl018048

URL: <https://doi.org/10.1029/2003gl018048>

Publication Date: 2003-10-16

Volume/Issue: Volume 30, Issue 20

References: 16

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A recent study of satellite laser ranging measurements by Cox and Chao [2002] indicates that large-scale mass redistribution has caused a rapid change in Earth's dynamic oblateness (J_2) since 1998. Using satellite altimeter observations and two runs of an ocean general circulation models (OGCM), we examine mass variations in the oceans and their estimated effect on J_2 . Combined altimeter and OGCM results indicate variations similar to the J_2 changes in 1998, and predict considerably larger oceanic effects on the J_2 anomaly than purely OGCM estimates, suggesting that the oceans should not be ruled out by any means as a significant source of this interesting geodetic anomaly, until they have been studied further.

2470. Atmospheric and oceanic excitation of the Earth's wobbles during 1980-2000

Authors: Richard S. Gross (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2002jb002143

URL: <https://doi.org/10.1029/2002jb002143>

Publication Date: 2003-08-08

Volume/Issue: Volume 108, Issue B8

References: 72

Cited by: 146 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Because of the action of various geophysical excitation mechanisms, the Earth does not rotate about its figure axis, so it wobbles as it rotates. Here, the effectiveness of atmospheric and oceanic processes in exciting the Earth's wobbles during 1980-2000 is evaluated using estimates of atmospheric angular momentum from the National Centers for Environmental Prediction/National Center for Atmospheric Research (NCEP/NCAR) reanalysis project and estimates of oceanic angular momentum from the Estimating the Circulation and Climate of the Ocean (ECCO) consortium's simulation of the general circulation of the oceans. On intraseasonal timescales, atmospheric surface pressure changes are found to be the single most effective process exciting the Earth's wobbles, explaining about twice as much of the observed variance as do either atmospheric wind or ocean bottom pressure changes and nearly 4 times as much of the observed variance as do oceanic currents. However, on interannual timescales, ocean bottom pressure changes are found to be the single most effective process exciting the Earth's wobbles, explaining more than 5 times as much of the observed variance as do atmospheric wind and pressure changes combined, and more than twice as much of the observed variance as do oceanic currents. Within the Chandler band it is found that during 1980-2000 atmospheric and oceanic processes have enough power to excite the Chandler wobble and are significantly coherent with it. The single most important mechanism exciting the Chandler wobble is found to be ocean bottom pressure variations. Atmospheric and oceanic processes do not appear to have enough power to excite the Earth's wobbles to their observed levels on pentadal and longer timescales, although series longer than the 21-yearlong series used here need to be studied in order to obtain greater statistical significance of this result.

2471. Large-scale global surface mass variations inferred from GPS measurements of load-induced deformation

Authors: Xiaoping Wu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael B. Heflin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Erik R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Donald F. Argus (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Frank H. Webb (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2003gl017546

URL: <https://doi.org/10.1029/2003gl017546>

Publication Date: 2003-07-24

Volume/Issue: Volume 30, Issue 14

References: 12

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Detrended crustal deformation measurements from a global network of up to 200 continuous Global Positioning System (GPS) tracking sites are inverted for low degree and order ($n, m \leq 6$) spherical harmonic series of surface mass variations. An independent geophysical model of the atmosphere, oceans, and water and snow on land is also used to guide the spherical harmonic truncation and to evaluate aliasing errors in the results. Inversion uncertainties decrease significantly as spatial coverage and data quality improve with time, especially in the last few years. Consequently, GPS inverted seasonal geocenter and low degree zonal harmonics converge to those determined by measurements from satellite laser ranging (SLR). Significant $n > 1$ and non-zonal variations have also been found.

2472. Volume, heat, and freshwater transports of the global ocean circulation 1993-2000, estimated from a general circulation model constrained by World Ocean Circulation Experiment (WOCE) data

Authors: D. Stammer (Scripps Institution of Oceanography / University of California San Diego); C. Wunsch (Massachusetts Institute of Technology); R. Giering (FastOpt, GBR Hamburg Germany); C. Eckert (Massachusetts Institute of Technology); P. Heimbach (Massachusetts Institute of Technology); J. Marotzke (University of Southampton); A. Adcroft (Massachusetts Institute of Technology); C. N. Hill (Massachusetts Institute of Technology); J. Marshall (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2001jc001115

URL: <https://doi.org/10.1029/2001jc001115>

Publication Date: 2003-05-23

Volume/Issue: Volume 108, Issue C1

References: 44

Cited by: 120 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: An analysis of ocean volume, heat, and freshwater transports from a fully constrained general circulation model (GCM) is described. Output from a data synthesis, or state estimation, method is used by which the model was forced to large-scale, time-varying global ocean data sets over 1993 through 2000. Time-mean horizontal transports, estimated from this fully time-dependent circulation, have converged with independent time-independent estimates from box inversions over most parts of the world ocean but especially in the southern hemisphere. However, heat transport estimates differ substantially in the North Atlantic where our estimates result in only 1/2 previous results. The models drift over the estimation period is consistent with observations from TOPEX/Poseidon in their spatial pattern, but smaller in their amplitudes by about a factor of 2. Associated temperature and salinity changes are complex, and both point toward air-sea interaction over water mass formation regions as the primary source for changes in the deep ocean. The estimated mean circulation around Australia involves a net volume transport of 11 Sv through the Indonesian Throughflow and the Mozambique Channel. In addition, we show that this flow regime exists on all timescales above 1 month, rendering the variability in the South Pacific strongly coupled to the Indian Ocean. Moreover, the dynamically consistent variations in the model show temporal variability of oceanic heat transports, heat storage, and atmospheric exchanges that are complex and with a strong dependence upon location, depth, and timescale. Our results demonstrate the great potential of an ocean state estimation system to provide a dynamical description of the time-dependent observed heat transport and heat content changes and their relation to air-sea interactions.

2473. On the efficiency of statistical assimilation techniques in the presence of model and data error

Authors: Peter D. Killworth (Southampton Oceanography Centre Southampton UK); Jian-Guo Li (Southampton Oceanography Centre Southampton UK); David Smeed (Southampton Oceanography Centre Southampton UK)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2002jc001444

URL: <https://doi.org/10.1029/2002jc001444>

Publication Date: 2003-05-23

Volume/Issue: Volume 108, Issue C4

References: 15

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Statistically based assimilation methods make use of error statistics. When these statistics apply to data, some aspects of them are known. However, model error statistics are at best poorly known. The variability of these error fields causes the strength of the assimilated signal to vary. This (spatial) variability, plus diffusion, can cause the assimilated signal to rapidly disappear, decreasing the impact of the data. Our demonstration uses a statistical method [from Oschlies and Willebrand, 1996], which assimilates sea surface height using vertical regressions. An idealized Gaussian depression of sea surface height about 10 degrees in diameter was assimilated, using both the original statistical scheme and a scheme with a horizontally uniform regression coefficient, to test the effects of variability in model errors. Results indicated that the uniform scheme is more efficient in assimilation of sea surface height than the original scheme (in that assimilated information is lost less rapidly because of small-scale mixing) for the simplified Gaussian problem, but predictive skill was not improved when assimilating over the full North Atlantic. The influence of interpolated satellite observation errors caused by limitations of satellite track coverage was also assessed, using a chessboard sea surface depression that would be induced from the (statistically) optimal combination of model and data. Artificial eddies on the scale of the satellite track intervals are induced by including the satellite errors in the ocean model. These eddies dissipate rapidly in the presence of mixing, taking with them much of the assimilated signal. We suggest a possible solution.

2474. Interannual variability of air-sea O₂ fluxes and the determination of CO₂ sinks using atmospheric O₂/N₂

Authors: Galen A. McKinley (Massachusetts Institute of Technology); Michael J. Follows (Massachusetts Institute of Technology); John Marshall (Massachusetts Institute of Technology); Song-Miao Fan (Princeton University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2002gl016044

URL: <https://doi.org/10.1029/2002gl016044>

Publication Date: 2003-05-23

Volume/Issue: Volume 30, Issue 3

References: 25

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Motivated by the use of atmospheric O₂/N₂ to determine CO₂ sinks under the assumption of negligible interannual variability in air-sea O₂ fluxes, we examine interannual fluctuations of the global air-sea flux of O₂ during the period 1980-1998 using a global ocean circulation and biogeochemistry model along with an atmospheric transport model. It is found that both the El Niño/Southern Oscillation (ENSO) cycle and wintertime convection in the North Atlantic are primary drivers of global air-sea oxygen flux interannual variability. Model estimated extremes of O₂ flux variability are -70/+100 × 10¹² mol/yr (Tmol/yr), where positive fluxes are to the atmosphere. O₂/N₂ variability could cause an up to +/-1.0 PgC/yr error in estimates of interannual variability in land and ocean CO₂ sinks derived from atmospheric O₂/N₂ observations.

2475. VERTICAL MIXING, ENERGY, AND THE GENERAL CIRCULATION OF THE OCEANS

Authors: Carl Wunsch (Massachusetts Institute of Technology)

Journal: Annual Review of Fluid Mechanics

Publisher: Annual Reviews

DOI: 10.1146/annurev.fluid.36.050802.122121

URL: <https://doi.org/10.1146/annurev.fluid.36.050802.122121>

Publication Date: 2003-04-08

Volume/Issue: Volume 36, Issue 1

References: 122

Cited by: 1126 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: ■ Abstract The coexistence in the deep ocean of a finite, stable stratification, a strong meridional overturning circulation, and mesoscale eddies raises complex questions concerning the circulation energetics. In particular, small-scale mixing processes are necessary to resupply the potential energy removed in the interior by the overturning and eddy-generating process. A number of lines of evidence, none complete, suggest that the oceanic general circulation, far from being a heat engine, is almost wholly governed by the forcing of the wind field and secondarily by deep water tides. In detail however, the budget of mechanical energy input into the ocean is poorly constrained. The now inescapable conclusion that over most of the ocean significant "vertical" mixing is confined to topographically complex boundary areas implies a potentially radically different interior circulation than is possible with uniform mixing. Whether ocean circulation models, either simple box or full numerical ones, neither explicitly accounting for the energy input into the system nor providing for spatial variability in the mixing, have any physical relevance under changed climate conditions is at issue.

2476. Conservation of properties in a free-surface model

Authors: Jean-Michel Campin; Alistair Adcroft; Chris Hill; John Marshall

Journal: Ocean Modelling

Publisher: Elsevier BV

DOI: 10.1016/s1463-5003(03)00009-x

URL: [https://doi.org/10.1016/s1463-5003\(03\)00009-x](https://doi.org/10.1016/s1463-5003(03)00009-x)

Publication Date: 2003-03-04

Volume/Issue: Volume 6, Issue 3-4

References: 16

Cited by: 74 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

2477. Recent Earth Oblateness Variations: Unraveling Climate and Postglacial Rebound Effects

Authors: Jean O. Dickey (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Steven L. Marcus (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Olivier de Viron (Royal Observatory of Belgium); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Science

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/science.1077777

URL: <https://doi.org/10.1126/science.1077777>

Publication Date: 2002-12-05

Volume/Issue: Volume 298, Issue 5600

References: 29

Cited by: 85 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Earth's dynamic oblateness (J_2) has been decreasing due to postglacial rebound (PGR). However, J_2 began to increase in 1997, indicating a pronounced global-scale mass redistribution within Earth's system. We have determined that the observed increases in J_2 are caused primarily by a recent surge in subpolar glacial melting and by mass shifts in the Southern, Pacific, and Indian oceans. When these effects are removed, the residual trend in J_2 ($-2.9 \times 10^{-11} \text{ year}^{-1}$) becomes consistent with previous estimates of PGR from satellite and eclipse data. The climatic significance of these rapid shifts in glacial and oceanic mass, however, remains to be investigated.

2478. Global ocean circulation during 1992-1997, estimated from ocean observations and a general circulation model

Authors: D. Stammer (Scripps Institution of Oceanography); C. Wunsch (Massachusetts Institute of Technology); R. Giering (FastOpt Hamburg Germany); C. Eckert (Massachusetts Institute of Technology); P. Heimbach (Massachusetts Institute of Technology); J. Marotzke (University of Southampton); A. Adcroft (Massachusetts Institute of Technology); C. N. Hill (Massachusetts Institute of Technology); J. Marshall (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2001jc000888

URL: <https://doi.org/10.1029/2001jc000888>

Publication Date: 2002-10-27

Volume/Issue: Volume 107, Issue C9

References: 36

Cited by: 272 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: A three-dimensional oceanic state is estimated for the period 1992-1997 as it results from combining large-scale ocean data sets with a general circulation model. At the cost of increased computational load, the estimation (assimilation) method is chosen specifically so that the resulting state estimate is consistent with the model equations, having no artificial sources or sinks. To bring the model into close agreement with observations, its initial temperature and salinity conditions are permitted to change, as are the time-dependent surface fluxes of momentum, heat and freshwater. Resulting changes of these "control vectors" are largely consistent with accepted uncertainties in the hydrographic climatology and meteorological analyses. The assimilation procedure is able to correct for many of the traditional shortcomings of the flow field by changing the surface boundary conditions. Changes in the resulting flow field are predominantly on the gyre scale and affect many features that are often poorly simulated in traditional numerical simulations, such as the strengths of the Gulf Stream and its extension, the Azores Current and the anticyclonic circulation associated with the Labrador Sea. Tests of the results and their consistency with prior error assumptions show that the constrained model has moved considerably closer to the observations imposed as constraints, but has also moved closer to independent data from the World Ocean Circulation Experiment not used in the assimilation procedure. In some regions where the comparisons remain indeterminate, not enough ocean observations are available, and it is difficult to ascribe the residuals to either the model or the observations. Although problems remain, a useful first solution to the global time-dependent ocean state estimation problem has been found. The estimates will continue to improve through the evolution of numerical models, computer power increases, more data, and more efficient estimation methods.

2479. Oceanic age and transient tracers: Analytical and numerical solutions

Authors: Carl Wunsch (Massachusetts Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2001jc000797

URL: <https://doi.org/10.1029/2001jc000797>

Publication Date: 2002-10-27

Volume/Issue: Volume 107, Issue C6

References: 54

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Transient tracers and the closely related "age" tracers exhibit a rich physical and mathematical structure even for problems of one space dimension. This richness tends to make interpretation of observations, which are inevitably thin in both space and time, difficult, in contrast to the situation in modeling studies. At least six different timescales and corresponding space scales can appear in one-dimensional problems. In higher dimensions the number of scales increases. Several examples of analytical and numerical solutions are explored for the light they cast on understanding a fluid flow. Boundary Green functions emerge as the fundamental physical/mathematical link between interior tracer distributions and surface and other boundary variations. With transient tracers in inverse calculations one should normally use the underlying tracer distributions to attempt to solve for fundamental fluid properties, such as the mixing coefficients, rather than ambiguous "ventilation" times, which among other problems, may be determined only by the detection threshold and are often mainly functions of the tracer decay constant rather than of fluid properties. Tracers that are transient only through stochastic boundary conditions show that large-scale space/time patterns can emerge in the tracer field, having little or no clear connection to the underlying fluid flow.

2480. A Partitioned Kalman Filter and Smoother

Authors: Ichiro Fukumori

Journal: Monthly Weather Review

Publisher: American Meteorological Society

DOI: 10.1175/1520-0493(2002)130<1370:apkfas>2.0.co;2

URL: [https://doi.org/10.1175/1520-0493\(2002\)130<1370:apkfas>2.0.co;2](https://doi.org/10.1175/1520-0493(2002)130<1370:apkfas>2.0.co;2)

Publication Date: 2002-07-27

Volume/Issue: Volume 130, Issue 5

References: 18

Cited by: 139 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A